

Comparativa de Semánticas Formales (WHILE Enriquecido)

1. Semántica Natural (NS - Big-Step)

$$\mathcal{T}_3 = \frac{\langle y := y + 1, \sigma_2 \rangle \longrightarrow \sigma_3 [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_2 \rangle \longrightarrow \sigma_3} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_5 = \frac{\langle y := y + 1, \sigma_4 \rangle \longrightarrow \sigma_5 [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_4 \rangle \longrightarrow \sigma_5} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_7 = \frac{\langle y := y + 1, \sigma_6 \rangle \longrightarrow \sigma_7 [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_6 \rangle \longrightarrow \sigma_7} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_9 = \frac{\langle y := y + 1, \sigma_8 \rangle \longrightarrow \sigma_9 [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_8 \rangle \longrightarrow \sigma_9} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{11} = \frac{\langle y := y + 1, \sigma_{10} \rangle \longrightarrow \sigma_{11} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{10} \rangle \longrightarrow \sigma_{11}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{13} = \frac{\langle y := y + 1, \sigma_{12} \rangle \longrightarrow \sigma_{13} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{12} \rangle \longrightarrow \sigma_{13}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{15} = \frac{\langle y := y + 1, \sigma_{14} \rangle \longrightarrow \sigma_{15} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{14} \rangle \longrightarrow \sigma_{15}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{17} = \frac{\langle y := y + 1, \sigma_{16} \rangle \longrightarrow \sigma_{17} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{16} \rangle \longrightarrow \sigma_{17}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{19} = \frac{\langle y := y + 1, \sigma_{18} \rangle \longrightarrow \sigma_{19} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{18} \rangle \longrightarrow \sigma_{19}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{21} = \frac{\langle y := y + 1, \sigma_{20} \rangle \longrightarrow \sigma_{21} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{20} \rangle \longrightarrow \sigma_{21}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{23} = \frac{\langle y := y + 1, \sigma_{22} \rangle \longrightarrow \sigma_{23} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{22} \rangle \longrightarrow \sigma_{23}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{25} = \frac{\langle y := y + 1, \sigma_{24} \rangle \longrightarrow \sigma_{25} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{24} \rangle \longrightarrow \sigma_{25}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{27} = \frac{\langle y := y + 1, \sigma_{26} \rangle \longrightarrow \sigma_{27} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{26} \rangle \longrightarrow \sigma_{27}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{29} = \frac{\langle y := y + 1, \sigma_{28} \rangle \longrightarrow \sigma_{29} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{28} \rangle \longrightarrow \sigma_{29}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{31} = \frac{\langle y := y + 1, \sigma_{30} \rangle \longrightarrow \sigma_{31} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{30} \rangle \longrightarrow \sigma_{31}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{33} = \frac{\langle y := y + 1, \sigma_{32} \rangle \longrightarrow \sigma_{33} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{32} \rangle \longrightarrow \sigma_{33}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{35} = \frac{\langle y := y + 1, \sigma_{34} \rangle \longrightarrow \sigma_{35} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{34} \rangle \longrightarrow \sigma_{35}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{37} = \frac{\langle y := y + 1, \sigma_{36} \rangle \longrightarrow \sigma_{37} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{36} \rangle \longrightarrow \sigma_{37}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{39} = \frac{\langle y := y + 1, \sigma_{38} \rangle \longrightarrow \sigma_{39} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{38} \rangle \longrightarrow \sigma_{39}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{41} = \frac{\langle y := y + 1, \sigma_{40} \rangle \longrightarrow \sigma_{41} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{40} \rangle \longrightarrow \sigma_{41}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{43} = \frac{\langle y := y + 1, \sigma_{42} \rangle \longrightarrow \sigma_{43} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{42} \rangle \longrightarrow \sigma_{43}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{45} = \frac{\langle y := y + 1, \sigma_{44} \rangle \longrightarrow \sigma_{45} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{44} \rangle \longrightarrow \sigma_{45}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{47} = \frac{\langle y := y + 1, \sigma_{46} \rangle \longrightarrow \sigma_{47} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{46} \rangle \longrightarrow \sigma_{47}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{49} = \frac{\langle y := y + 1, \sigma_{48} \rangle \longrightarrow \sigma_{49} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{48} \rangle \longrightarrow \sigma_{49}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{51} = \frac{\langle y := y + 1, \sigma_{50} \rangle \longrightarrow \sigma_{51} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{50} \rangle \longrightarrow \sigma_{51}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{53} = \frac{\langle y := y + 1, \sigma_{52} \rangle \longrightarrow \sigma_{53} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{52} \rangle \longrightarrow \sigma_{53}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{55} = \frac{\langle y := y + 1, \sigma_{54} \rangle \longrightarrow \sigma_{55} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{54} \rangle \longrightarrow \sigma_{55}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{57} = \frac{\langle y := y + 1, \sigma_{56} \rangle \longrightarrow \sigma_{57} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{56} \rangle \longrightarrow \sigma_{57}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{59} = \frac{\langle y := y + 1, \sigma_{58} \rangle \longrightarrow \sigma_{59} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{58} \rangle \longrightarrow \sigma_{59}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{61} = \frac{\langle y := y + 1, \sigma_{60} \rangle \longrightarrow \sigma_{61} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{60} \rangle \longrightarrow \sigma_{61}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{63} = \frac{\langle y := y + 1, \sigma_{62} \rangle \longrightarrow \sigma_{63} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{62} \rangle \longrightarrow \sigma_{63}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{65} = \frac{\langle y := y + 1, \sigma_{64} \rangle \longrightarrow \sigma_{65} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{64} \rangle \longrightarrow \sigma_{65}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{67} = \frac{\langle y := y + 1, \sigma_{66} \rangle \longrightarrow \sigma_{67} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{66} \rangle \longrightarrow \sigma_{67}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{69} = \frac{\langle y := y + 1, \sigma_{68} \rangle \longrightarrow \sigma_{69} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{68} \rangle \longrightarrow \sigma_{69}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{71} = \frac{\langle y := y + 1, \sigma_{70} \rangle \longrightarrow \sigma_{71} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{70} \rangle \longrightarrow \sigma_{71}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{73} = \frac{\langle y := y + 1, \sigma_{72} \rangle \longrightarrow \sigma_{73} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{72} \rangle \longrightarrow \sigma_{73}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{75} = \frac{\langle y := y + 1, \sigma_{74} \rangle \longrightarrow \sigma_{75} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{74} \rangle \longrightarrow \sigma_{75}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{77} = \frac{\langle y := y + 1, \sigma_{76} \rangle \longrightarrow \sigma_{77} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{76} \rangle \longrightarrow \sigma_{77}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{79} = \frac{\langle y := y + 1, \sigma_{78} \rangle \longrightarrow \sigma_{79} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{78} \rangle \longrightarrow \sigma_{79}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{81} = \frac{\langle y := y + 1, \sigma_{80} \rangle \longrightarrow \sigma_{81} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{80} \rangle \longrightarrow \sigma_{81}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{83} = \frac{\langle y := y + 1, \sigma_{82} \rangle \longrightarrow \sigma_{83} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{82} \rangle \longrightarrow \sigma_{83}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{85} = \frac{\langle y := y + 1, \sigma_{84} \rangle \longrightarrow \sigma_{85} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{84} \rangle \longrightarrow \sigma_{85}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{87} = \frac{\langle y := y + 1, \sigma_{86} \rangle \longrightarrow \sigma_{87} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{86} \rangle \longrightarrow \sigma_{87}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{89} = \frac{\langle y := y + 1, \sigma_{88} \rangle \longrightarrow \sigma_{89} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{88} \rangle \longrightarrow \sigma_{89}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{91} = \frac{\langle y := y + 1, \sigma_{90} \rangle \longrightarrow \sigma_{91} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{90} \rangle \longrightarrow \sigma_{91}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{93} = \frac{\langle y := y + 1, \sigma_{92} \rangle \longrightarrow \sigma_{93} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{92} \rangle \longrightarrow \sigma_{93}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{95} = \frac{\langle y := y + 1, \sigma_{94} \rangle \longrightarrow \sigma_{95} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{94} \rangle \longrightarrow \sigma_{95}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{97} = \frac{\langle y := y + 1, \sigma_{96} \rangle \longrightarrow \sigma_{97} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{96} \rangle \longrightarrow \sigma_{97}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{99} = \frac{\langle y := y + 1, \sigma_{98} \rangle \longrightarrow \sigma_{99} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{98} \rangle \longrightarrow \sigma_{99}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{101} = \frac{\langle y := y + 1, \sigma_{100} \rangle \longrightarrow \sigma_{101} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{100} \rangle \longrightarrow \sigma_{101}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{103} = \frac{\langle y := y + 1, \sigma_{102} \rangle \longrightarrow \sigma_{103} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{102} \rangle \longrightarrow \sigma_{103}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{105} = \frac{\langle y := y + 1, \sigma_{104} \rangle \longrightarrow \sigma_{105} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{104} \rangle \longrightarrow \sigma_{105}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{107} = \frac{\langle y := y + 1, \sigma_{106} \rangle \longrightarrow \sigma_{107} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{106} \rangle \longrightarrow \sigma_{107}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{109} = \frac{\langle y := y + 1, \sigma_{108} \rangle \longrightarrow \sigma_{109} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{108} \rangle \longrightarrow \sigma_{109}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{111} = \frac{\langle y := y + 1, \sigma_{110} \rangle \longrightarrow \sigma_{111} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{110} \rangle \longrightarrow \sigma_{111}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{113} = \frac{\langle y := y + 1, \sigma_{112} \rangle \longrightarrow \sigma_{113} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{112} \rangle \longrightarrow \sigma_{113}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{115} = \frac{\langle y := y + 1, \sigma_{114} \rangle \longrightarrow \sigma_{115} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{114} \rangle \longrightarrow \sigma_{115}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{117} = \frac{\langle y := y + 1, \sigma_{116} \rangle \longrightarrow \sigma_{117} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{116} \rangle \longrightarrow \sigma_{117}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{119} = \frac{\langle y := y + 1, \sigma_{118} \rangle \longrightarrow \sigma_{119} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{118} \rangle \longrightarrow \sigma_{119}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{121} = \frac{\langle y := y + 1, \sigma_{120} \rangle \longrightarrow \sigma_{121} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{120} \rangle \longrightarrow \sigma_{121}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{123} = \frac{\langle y := y + 1, \sigma_{122} \rangle \longrightarrow \sigma_{123} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{122} \rangle \longrightarrow \sigma_{123}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{125} = \frac{\langle y := y + 1, \sigma_{124} \rangle \longrightarrow \sigma_{125} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{124} \rangle \longrightarrow \sigma_{125}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{127} = \frac{\langle y := y + 1, \sigma_{126} \rangle \longrightarrow \sigma_{127} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{126} \rangle \longrightarrow \sigma_{127}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{129} = \frac{\langle y := y + 1, \sigma_{128} \rangle \longrightarrow \sigma_{129} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{128} \rangle \longrightarrow \sigma_{129}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{131} = \frac{\langle y := y + 1, \sigma_{130} \rangle \longrightarrow \sigma_{131} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{130} \rangle \longrightarrow \sigma_{131}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{133} = \frac{\langle y := y + 1, \sigma_{132} \rangle \longrightarrow \sigma_{133} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{132} \rangle \longrightarrow \sigma_{133}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{135} = \frac{\langle y := y + 1, \sigma_{134} \rangle \longrightarrow \sigma_{135} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{134} \rangle \longrightarrow \sigma_{135}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{137} = \frac{\langle y := y + 1, \sigma_{136} \rangle \longrightarrow \sigma_{137} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{136} \rangle \longrightarrow \sigma_{137}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{139} = \frac{\langle y := y + 1, \sigma_{138} \rangle \longrightarrow \sigma_{139} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{138} \rangle \longrightarrow \sigma_{139}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{141} = \frac{\langle y := y + 1, \sigma_{140} \rangle \longrightarrow \sigma_{141} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{140} \rangle \longrightarrow \sigma_{141}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{143} = \frac{\langle y := y + 1, \sigma_{142} \rangle \longrightarrow \sigma_{143} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{142} \rangle \longrightarrow \sigma_{143}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{145} = \frac{\langle y := y + 1, \sigma_{144} \rangle \longrightarrow \sigma_{145} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{144} \rangle \longrightarrow \sigma_{145}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{147} = \frac{\langle y := y + 1, \sigma_{146} \rangle \longrightarrow \sigma_{147} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{146} \rangle \longrightarrow \sigma_{147}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{149} = \frac{\langle y := y + 1, \sigma_{148} \rangle \longrightarrow \sigma_{149} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{148} \rangle \longrightarrow \sigma_{149}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{151} = \frac{\langle y := y + 1, \sigma_{150} \rangle \longrightarrow \sigma_{151} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{150} \rangle \longrightarrow \sigma_{151}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{153} = \frac{\langle y := y + 1, \sigma_{152} \rangle \longrightarrow \sigma_{153} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{152} \rangle \longrightarrow \sigma_{153}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{155} = \frac{\langle y := y + 1, \sigma_{154} \rangle \longrightarrow \sigma_{155} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{154} \rangle \longrightarrow \sigma_{155}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{157} = \frac{\langle y := y + 1, \sigma_{156} \rangle \longrightarrow \sigma_{157} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{156} \rangle \longrightarrow \sigma_{157}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{159} = \frac{\langle y := y + 1, \sigma_{158} \rangle \longrightarrow \sigma_{159} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{158} \rangle \longrightarrow \sigma_{159}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{161} = \frac{\langle y := y + 1, \sigma_{160} \rangle \longrightarrow \sigma_{161} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{160} \rangle \longrightarrow \sigma_{161}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{163} = \frac{\langle y := y + 1, \sigma_{162} \rangle \longrightarrow \sigma_{163} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{162} \rangle \longrightarrow \sigma_{163}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{165} = \frac{\langle y := y + 1, \sigma_{164} \rangle \longrightarrow \sigma_{165} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{164} \rangle \longrightarrow \sigma_{165}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{167} = \frac{\langle y := y + 1, \sigma_{166} \rangle \longrightarrow \sigma_{167} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{166} \rangle \longrightarrow \sigma_{167}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{169} = \frac{\langle y := y + 1, \sigma_{168} \rangle \longrightarrow \sigma_{169} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{168} \rangle \longrightarrow \sigma_{169}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{171} = \frac{\langle y := y + 1, \sigma_{170} \rangle \longrightarrow \sigma_{171} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{170} \rangle \longrightarrow \sigma_{171}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{173} = \frac{\langle y := y + 1, \sigma_{172} \rangle \longrightarrow \sigma_{173} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{172} \rangle \longrightarrow \sigma_{173}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{175} = \frac{\langle y := y + 1, \sigma_{174} \rangle \longrightarrow \sigma_{175} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{174} \rangle \longrightarrow \sigma_{175}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{177} = \frac{\langle y := y + 1, \sigma_{176} \rangle \longrightarrow \sigma_{177} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{176} \rangle \longrightarrow \sigma_{177}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{179} = \frac{\langle y := y + 1, \sigma_{178} \rangle \longrightarrow \sigma_{179} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{178} \rangle \longrightarrow \sigma_{179}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{181} = \frac{\langle y := y + 1, \sigma_{180} \rangle \longrightarrow \sigma_{181} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{180} \rangle \longrightarrow \sigma_{181}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{183} = \frac{\langle y := y + 1, \sigma_{182} \rangle \longrightarrow \sigma_{183} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{182} \rangle \longrightarrow \sigma_{183}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{185} = \frac{\langle y := y + 1, \sigma_{184} \rangle \longrightarrow \sigma_{185} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{184} \rangle \longrightarrow \sigma_{185}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{187} = \frac{\langle y := y + 1, \sigma_{186} \rangle \longrightarrow \sigma_{187} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{186} \rangle \longrightarrow \sigma_{187}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{189} = \frac{\langle y := y + 1, \sigma_{188} \rangle \longrightarrow \sigma_{189} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{188} \rangle \longrightarrow \sigma_{189}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{191} = \frac{\langle y := y + 1, \sigma_{190} \rangle \longrightarrow \sigma_{191} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{190} \rangle \longrightarrow \sigma_{191}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{193} = \frac{\langle y := y + 1, \sigma_{192} \rangle \longrightarrow \sigma_{193} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{192} \rangle \longrightarrow \sigma_{193}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{195} = \frac{\langle y := y + 1, \sigma_{194} \rangle \longrightarrow \sigma_{195} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{194} \rangle \longrightarrow \sigma_{195}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{197} = \frac{\langle y := y + 1, \sigma_{196} \rangle \longrightarrow \sigma_{197} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{196} \rangle \longrightarrow \sigma_{197}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{199} = \frac{\langle y := y + 1, \sigma_{198} \rangle \longrightarrow \sigma_{199} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{198} \rangle \longrightarrow \sigma_{199}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{201} = \frac{\langle y := y + 1, \sigma_{200} \rangle \longrightarrow \sigma_{201} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{200} \rangle \longrightarrow \sigma_{201}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{203} = \frac{\langle y := y + 1, \sigma_{202} \rangle \longrightarrow \sigma_{203} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{202} \rangle \longrightarrow \sigma_{203}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{205} = \frac{\langle y := y + 1, \sigma_{204} \rangle \longrightarrow \sigma_{205} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{204} \rangle \longrightarrow \sigma_{205}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{207} = \frac{\langle y := y + 1, \sigma_{206} \rangle \longrightarrow \sigma_{207} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{206} \rangle \longrightarrow \sigma_{207}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{209} = \frac{\langle y := y + 1, \sigma_{208} \rangle \longrightarrow \sigma_{209} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{208} \rangle \longrightarrow \sigma_{209}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{211} = \frac{\langle y := y + 1, \sigma_{210} \rangle \longrightarrow \sigma_{211} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{210} \rangle \longrightarrow \sigma_{211}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{213} = \frac{\langle y := y + 1, \sigma_{212} \rangle \longrightarrow \sigma_{213} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{212} \rangle \longrightarrow \sigma_{213}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{215} = \frac{\langle y := y + 1, \sigma_{214} \rangle \longrightarrow \sigma_{215} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{214} \rangle \longrightarrow \sigma_{215}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{217} = \frac{\langle y := y + 1, \sigma_{216} \rangle \longrightarrow \sigma_{217} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{216} \rangle \longrightarrow \sigma_{217}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{219} = \frac{\langle y := y + 1, \sigma_{218} \rangle \longrightarrow \sigma_{219} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{218} \rangle \longrightarrow \sigma_{219}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{221} = \frac{\langle y := y + 1, \sigma_{220} \rangle \longrightarrow \sigma_{221} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{220} \rangle \longrightarrow \sigma_{221}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{223} = \frac{\langle y := y + 1, \sigma_{222} \rangle \longrightarrow \sigma_{223} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{222} \rangle \longrightarrow \sigma_{223}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{225} = \frac{\langle y := y + 1, \sigma_{224} \rangle \longrightarrow \sigma_{225} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{224} \rangle \longrightarrow \sigma_{225}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{227} = \frac{\langle y := y + 1, \sigma_{226} \rangle \longrightarrow \sigma_{227} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{226} \rangle \longrightarrow \sigma_{227}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{229} = \frac{\langle y := y + 1, \sigma_{228} \rangle \longrightarrow \sigma_{229} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{228} \rangle \longrightarrow \sigma_{229}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{231} = \frac{\langle y := y + 1, \sigma_{230} \rangle \longrightarrow \sigma_{231} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{230} \rangle \longrightarrow \sigma_{231}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{233} = \frac{\langle y := y + 1, \sigma_{232} \rangle \longrightarrow \sigma_{233} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{232} \rangle \longrightarrow \sigma_{233}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{235} = \frac{\langle y := y + 1, \sigma_{234} \rangle \longrightarrow \sigma_{235} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{234} \rangle \longrightarrow \sigma_{235}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{237} = \frac{\langle y := y + 1, \sigma_{236} \rangle \longrightarrow \sigma_{237} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{236} \rangle \longrightarrow \sigma_{237}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{239} = \frac{\langle y := y + 1, \sigma_{238} \rangle \longrightarrow \sigma_{239} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{238} \rangle \longrightarrow \sigma_{239}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{241} = \frac{\langle y := y + 1, \sigma_{240} \rangle \longrightarrow \sigma_{241} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{240} \rangle \longrightarrow \sigma_{241}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{243} = \frac{\langle y := y + 1, \sigma_{242} \rangle \longrightarrow \sigma_{243} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{242} \rangle \longrightarrow \sigma_{243}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{245} = \frac{\langle y := y + 1, \sigma_{244} \rangle \longrightarrow \sigma_{245} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{244} \rangle \longrightarrow \sigma_{245}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{247} = \frac{\langle y := y + 1, \sigma_{246} \rangle \longrightarrow \sigma_{247} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{246} \rangle \longrightarrow \sigma_{247}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{249} = \frac{\langle y := y + 1, \sigma_{248} \rangle \longrightarrow \sigma_{249} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{248} \rangle \longrightarrow \sigma_{249}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{251} = \frac{\langle y := y + 1, \sigma_{250} \rangle \longrightarrow \sigma_{251} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{250} \rangle \longrightarrow \sigma_{251}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{253} = \frac{\langle y := y + 1, \sigma_{252} \rangle \longrightarrow \sigma_{253} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{252} \rangle \longrightarrow \sigma_{253}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{255} = \frac{\langle y := y + 1, \sigma_{254} \rangle \longrightarrow \sigma_{255} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{254} \rangle \longrightarrow \sigma_{255}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{257} = \frac{\langle y := y + 1, \sigma_{256} \rangle \longrightarrow \sigma_{257} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{256} \rangle \longrightarrow \sigma_{257}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{259} = \frac{\langle y := y + 1, \sigma_{258} \rangle \longrightarrow \sigma_{259} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{258} \rangle \longrightarrow \sigma_{259}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{261} = \frac{\langle y := y + 1, \sigma_{260} \rangle \longrightarrow \sigma_{261} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{260} \rangle \longrightarrow \sigma_{261}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{263} = \frac{\langle y := y + 1, \sigma_{262} \rangle \longrightarrow \sigma_{263} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{262} \rangle \longrightarrow \sigma_{263}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{265} = \frac{\langle y := y + 1, \sigma_{264} \rangle \longrightarrow \sigma_{265} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{264} \rangle \longrightarrow \sigma_{265}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{267} = \frac{\langle y := y + 1, \sigma_{266} \rangle \longrightarrow \sigma_{267} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{266} \rangle \longrightarrow \sigma_{267}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{269} = \frac{\langle y := y + 1, \sigma_{268} \rangle \longrightarrow \sigma_{269} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{268} \rangle \longrightarrow \sigma_{269}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{271} = \frac{\langle y := y + 1, \sigma_{270} \rangle \longrightarrow \sigma_{271} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{270} \rangle \longrightarrow \sigma_{271}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{273} = \frac{\langle y := y + 1, \sigma_{272} \rangle \longrightarrow \sigma_{273} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{272} \rangle \longrightarrow \sigma_{273}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{275} = \frac{\langle y := y + 1, \sigma_{274} \rangle \longrightarrow \sigma_{275} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{274} \rangle \longrightarrow \sigma_{275}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{277} = \frac{\langle y := y + 1, \sigma_{276} \rangle \longrightarrow \sigma_{277} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{276} \rangle \longrightarrow \sigma_{277}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{279} = \frac{\langle y := y + 1, \sigma_{278} \rangle \longrightarrow \sigma_{279} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{278} \rangle \longrightarrow \sigma_{279}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{281} = \frac{\langle y := y + 1, \sigma_{280} \rangle \longrightarrow \sigma_{281} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{280} \rangle \longrightarrow \sigma_{281}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{283} = \frac{\langle y := y + 1, \sigma_{282} \rangle \longrightarrow \sigma_{283} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{282} \rangle \longrightarrow \sigma_{283}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{285} = \frac{\langle y := y + 1, \sigma_{284} \rangle \longrightarrow \sigma_{285} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{284} \rangle \longrightarrow \sigma_{285}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{287} = \frac{\langle y := y + 1, \sigma_{286} \rangle \longrightarrow \sigma_{287} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{286} \rangle \longrightarrow \sigma_{287}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{289} = \frac{\langle y := y + 1, \sigma_{288} \rangle \longrightarrow \sigma_{289} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{288} \rangle \longrightarrow \sigma_{289}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{291} = \frac{\langle y := y + 1, \sigma_{290} \rangle \longrightarrow \sigma_{291} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{290} \rangle \longrightarrow \sigma_{291}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{293} = \frac{\langle y := y + 1, \sigma_{292} \rangle \longrightarrow \sigma_{293} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{292} \rangle \longrightarrow \sigma_{293}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{295} = \frac{\langle y := y + 1, \sigma_{294} \rangle \longrightarrow \sigma_{295} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{294} \rangle \longrightarrow \sigma_{295}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{297} = \frac{\langle y := y + 1, \sigma_{296} \rangle \longrightarrow \sigma_{297} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{296} \rangle \longrightarrow \sigma_{297}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{299} = \frac{\langle y := y + 1, \sigma_{298} \rangle \longrightarrow \sigma_{299} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{298} \rangle \longrightarrow \sigma_{299}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{301} = \frac{\langle y := y + 1, \sigma_{300} \rangle \longrightarrow \sigma_{301} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{300} \rangle \longrightarrow \sigma_{301}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{303} = \frac{\langle y := y + 1, \sigma_{302} \rangle \longrightarrow \sigma_{303} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{302} \rangle \longrightarrow \sigma_{303}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{305} = \frac{\langle y := y + 1, \sigma_{304} \rangle \longrightarrow \sigma_{305} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{304} \rangle \longrightarrow \sigma_{305}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{307} = \frac{\langle y := y + 1, \sigma_{306} \rangle \longrightarrow \sigma_{307} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{306} \rangle \longrightarrow \sigma_{307}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{309} = \frac{\langle y := y + 1, \sigma_{308} \rangle \longrightarrow \sigma_{309} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{308} \rangle \longrightarrow \sigma_{309}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{311} = \frac{\langle y := y + 1, \sigma_{310} \rangle \longrightarrow \sigma_{311} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{310} \rangle \longrightarrow \sigma_{311}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{313} = \frac{\langle y := y + 1, \sigma_{312} \rangle \longrightarrow \sigma_{313} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{312} \rangle \longrightarrow \sigma_{313}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{315} = \frac{\langle y := y + 1, \sigma_{314} \rangle \longrightarrow \sigma_{315} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{314} \rangle \longrightarrow \sigma_{315}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{317} = \frac{\langle y := y + 1, \sigma_{316} \rangle \longrightarrow \sigma_{317} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{316} \rangle \longrightarrow \sigma_{317}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{319} = \frac{\langle y := y + 1, \sigma_{318} \rangle \longrightarrow \sigma_{319} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{318} \rangle \longrightarrow \sigma_{319}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{321} = \frac{\langle y := y + 1, \sigma_{320} \rangle \longrightarrow \sigma_{321} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{320} \rangle \longrightarrow \sigma_{321}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{323} = \frac{\langle y := y + 1, \sigma_{322} \rangle \longrightarrow \sigma_{323} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{322} \rangle \longrightarrow \sigma_{323}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{325} = \frac{\langle y := y + 1, \sigma_{324} \rangle \longrightarrow \sigma_{325} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{324} \rangle \longrightarrow \sigma_{325}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{327} = \frac{\langle y := y + 1, \sigma_{326} \rangle \longrightarrow \sigma_{327} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{326} \rangle \longrightarrow \sigma_{327}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{329} = \frac{\langle y := y + 1, \sigma_{328} \rangle \longrightarrow \sigma_{329} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{328} \rangle \longrightarrow \sigma_{329}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{331} = \frac{\langle y := y + 1, \sigma_{330} \rangle \longrightarrow \sigma_{331} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{330} \rangle \longrightarrow \sigma_{331}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{333} = \frac{\langle y := y + 1, \sigma_{332} \rangle \longrightarrow \sigma_{333} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{332} \rangle \longrightarrow \sigma_{333}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{335} = \frac{\langle y := y + 1, \sigma_{334} \rangle \longrightarrow \sigma_{335} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{334} \rangle \longrightarrow \sigma_{335}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{337} = \frac{\langle y := y + 1, \sigma_{336} \rangle \longrightarrow \sigma_{337} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{336} \rangle \longrightarrow \sigma_{337}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{339} = \frac{\langle y := y + 1, \sigma_{338} \rangle \longrightarrow \sigma_{339} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{338} \rangle \longrightarrow \sigma_{339}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{341} = \frac{\langle y := y + 1, \sigma_{340} \rangle \longrightarrow \sigma_{341} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{340} \rangle \longrightarrow \sigma_{341}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{343} = \frac{\langle y := y + 1, \sigma_{342} \rangle \longrightarrow \sigma_{343} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{342} \rangle \longrightarrow \sigma_{343}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{345} = \frac{\langle y := y + 1, \sigma_{344} \rangle \longrightarrow \sigma_{345} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{344} \rangle \longrightarrow \sigma_{345}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{347} = \frac{\langle y := y + 1, \sigma_{346} \rangle \longrightarrow \sigma_{347} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{346} \rangle \longrightarrow \sigma_{347}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{349} = \frac{\langle y := y + 1, \sigma_{348} \rangle \longrightarrow \sigma_{349} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{348} \rangle \longrightarrow \sigma_{349}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{351} = \frac{\langle y := y + 1, \sigma_{350} \rangle \longrightarrow \sigma_{351} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{350} \rangle \longrightarrow \sigma_{351}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{353} = \frac{\langle y := y + 1, \sigma_{352} \rangle \longrightarrow \sigma_{353} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{352} \rangle \longrightarrow \sigma_{353}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{355} = \frac{\langle y := y + 1, \sigma_{354} \rangle \longrightarrow \sigma_{355} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{354} \rangle \longrightarrow \sigma_{355}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{357} = \frac{\langle y := y + 1, \sigma_{356} \rangle \longrightarrow \sigma_{357} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{356} \rangle \longrightarrow \sigma_{357}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{359} = \frac{\langle y := y + 1, \sigma_{358} \rangle \longrightarrow \sigma_{359} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{358} \rangle \longrightarrow \sigma_{359}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{361} = \frac{\langle y := y + 1, \sigma_{360} \rangle \longrightarrow \sigma_{361} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{360} \rangle \longrightarrow \sigma_{361}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{363} = \frac{\langle y := y + 1, \sigma_{362} \rangle \longrightarrow \sigma_{363} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{362} \rangle \longrightarrow \sigma_{363}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{365} = \frac{\langle y := y + 1, \sigma_{364} \rangle \longrightarrow \sigma_{365} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{364} \rangle \longrightarrow \sigma_{365}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{367} = \frac{\langle y := y + 1, \sigma_{366} \rangle \longrightarrow \sigma_{367} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{366} \rangle \longrightarrow \sigma_{367}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{369} = \frac{\langle y := y + 1, \sigma_{368} \rangle \longrightarrow \sigma_{369} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{368} \rangle \longrightarrow \sigma_{369}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{371} = \frac{\langle y := y + 1, \sigma_{370} \rangle \longrightarrow \sigma_{371} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{370} \rangle \longrightarrow \sigma_{371}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{373} = \frac{\langle y := y + 1, \sigma_{372} \rangle \longrightarrow \sigma_{373} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{372} \rangle \longrightarrow \sigma_{373}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{375} = \frac{\langle y := y + 1, \sigma_{374} \rangle \longrightarrow \sigma_{375} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{374} \rangle \longrightarrow \sigma_{375}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{377} = \frac{\langle y := y + 1, \sigma_{376} \rangle \longrightarrow \sigma_{377} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{376} \rangle \longrightarrow \sigma_{377}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{379} = \frac{\langle y := y + 1, \sigma_{378} \rangle \longrightarrow \sigma_{379} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{378} \rangle \longrightarrow \sigma_{379}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{381} = \frac{\langle y := y + 1, \sigma_{380} \rangle \longrightarrow \sigma_{381} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{380} \rangle \longrightarrow \sigma_{381}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{383} = \frac{\langle y := y + 1, \sigma_{382} \rangle \longrightarrow \sigma_{383} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{382} \rangle \longrightarrow \sigma_{383}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{385} = \frac{\langle y := y + 1, \sigma_{384} \rangle \longrightarrow \sigma_{385} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{384} \rangle \longrightarrow \sigma_{385}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{387} = \frac{\langle y := y + 1, \sigma_{386} \rangle \longrightarrow \sigma_{387} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{386} \rangle \longrightarrow \sigma_{387}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{389} = \frac{\langle y := y + 1, \sigma_{388} \rangle \longrightarrow \sigma_{389} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{388} \rangle \longrightarrow \sigma_{389}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{391} = \frac{\langle y := y + 1, \sigma_{390} \rangle \longrightarrow \sigma_{391} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{390} \rangle \longrightarrow \sigma_{391}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{393} = \frac{\langle y := y + 1, \sigma_{392} \rangle \longrightarrow \sigma_{393} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{392} \rangle \longrightarrow \sigma_{393}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{395} = \frac{\langle y := y + 1, \sigma_{394} \rangle \longrightarrow \sigma_{395} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{394} \rangle \longrightarrow \sigma_{395}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{397} = \frac{\langle y := y + 1, \sigma_{396} \rangle \longrightarrow \sigma_{397} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{396} \rangle \longrightarrow \sigma_{397}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{399} = \frac{\langle y := y + 1, \sigma_{398} \rangle \longrightarrow \sigma_{399} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{398} \rangle \longrightarrow \sigma_{399}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{401} = \frac{\langle y := y + 1, \sigma_{400} \rangle \longrightarrow \sigma_{401} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{400} \rangle \longrightarrow \sigma_{401}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{403} = \frac{\langle y := y + 1, \sigma_{402} \rangle \longrightarrow \sigma_{403} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{402} \rangle \longrightarrow \sigma_{403}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{405} = \frac{\langle y := y + 1, \sigma_{404} \rangle \longrightarrow \sigma_{405} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{404} \rangle \longrightarrow \sigma_{405}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{407} = \frac{\langle y := y + 1, \sigma_{406} \rangle \longrightarrow \sigma_{407} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{406} \rangle \longrightarrow \sigma_{407}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{409} = \frac{\langle y := y + 1, \sigma_{408} \rangle \longrightarrow \sigma_{409} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{408} \rangle \longrightarrow \sigma_{409}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{411} = \frac{\langle y := y + 1, \sigma_{410} \rangle \longrightarrow \sigma_{411} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{410} \rangle \longrightarrow \sigma_{411}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{413} = \frac{\langle y := y + 1, \sigma_{412} \rangle \longrightarrow \sigma_{413} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{412} \rangle \longrightarrow \sigma_{413}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{415} = \frac{\langle y := y + 1, \sigma_{414} \rangle \longrightarrow \sigma_{415} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{414} \rangle \longrightarrow \sigma_{415}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{417} = \frac{\langle y := y + 1, \sigma_{416} \rangle \longrightarrow \sigma_{417} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{416} \rangle \longrightarrow \sigma_{417}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{419} = \frac{\langle y := y + 1, \sigma_{418} \rangle \longrightarrow \sigma_{419} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{418} \rangle \longrightarrow \sigma_{419}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{421} = \frac{\langle y := y + 1, \sigma_{420} \rangle \longrightarrow \sigma_{421} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{420} \rangle \longrightarrow \sigma_{421}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{423} = \frac{\langle y := y + 1, \sigma_{422} \rangle \longrightarrow \sigma_{423} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{422} \rangle \longrightarrow \sigma_{423}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{425} = \frac{\langle y := y + 1, \sigma_{424} \rangle \longrightarrow \sigma_{425} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{424} \rangle \longrightarrow \sigma_{425}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{427} = \frac{\langle y := y + 1, \sigma_{426} \rangle \longrightarrow \sigma_{427} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{426} \rangle \longrightarrow \sigma_{427}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{429} = \frac{\langle y := y + 1, \sigma_{428} \rangle \longrightarrow \sigma_{429} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{428} \rangle \longrightarrow \sigma_{429}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{431} = \frac{\langle y := y + 1, \sigma_{430} \rangle \longrightarrow \sigma_{431} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{430} \rangle \longrightarrow \sigma_{431}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{433} = \frac{\langle y := y + 1, \sigma_{432} \rangle \longrightarrow \sigma_{433} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{432} \rangle \longrightarrow \sigma_{433}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{435} = \frac{\langle y := y + 1, \sigma_{434} \rangle \longrightarrow \sigma_{435} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{434} \rangle \longrightarrow \sigma_{435}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{437} = \frac{\langle y := y + 1, \sigma_{436} \rangle \longrightarrow \sigma_{437} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{436} \rangle \longrightarrow \sigma_{437}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{439} = \frac{\langle y := y + 1, \sigma_{438} \rangle \longrightarrow \sigma_{439} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{438} \rangle \longrightarrow \sigma_{439}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{441} = \frac{\langle y := y + 1, \sigma_{440} \rangle \longrightarrow \sigma_{441} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{440} \rangle \longrightarrow \sigma_{441}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{443} = \frac{\langle y := y + 1, \sigma_{442} \rangle \longrightarrow \sigma_{443} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{442} \rangle \longrightarrow \sigma_{443}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{445} = \frac{\langle y := y + 1, \sigma_{444} \rangle \longrightarrow \sigma_{445} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{444} \rangle \longrightarrow \sigma_{445}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{447} = \frac{\langle y := y + 1, \sigma_{446} \rangle \longrightarrow \sigma_{447} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{446} \rangle \longrightarrow \sigma_{447}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{449} = \frac{\langle y := y + 1, \sigma_{448} \rangle \longrightarrow \sigma_{449} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{448} \rangle \longrightarrow \sigma_{449}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{451} = \frac{\langle y := y + 1, \sigma_{450} \rangle \longrightarrow \sigma_{451} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{450} \rangle \longrightarrow \sigma_{451}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{453} = \frac{\langle y := y + 1, \sigma_{452} \rangle \longrightarrow \sigma_{453} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{452} \rangle \longrightarrow \sigma_{453}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{455} = \frac{\langle y := y + 1, \sigma_{454} \rangle \longrightarrow \sigma_{455} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{454} \rangle \longrightarrow \sigma_{455}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{457} = \frac{\langle y := y + 1, \sigma_{456} \rangle \longrightarrow \sigma_{457} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{456} \rangle \longrightarrow \sigma_{457}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{459} = \frac{\langle y := y + 1, \sigma_{458} \rangle \longrightarrow \sigma_{459} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{458} \rangle \longrightarrow \sigma_{459}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{461} = \frac{\langle y := y + 1, \sigma_{460} \rangle \longrightarrow \sigma_{461} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{460} \rangle \longrightarrow \sigma_{461}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{463} = \frac{\langle y := y + 1, \sigma_{462} \rangle \longrightarrow \sigma_{463} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{462} \rangle \longrightarrow \sigma_{463}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{465} = \frac{\langle y := y + 1, \sigma_{464} \rangle \longrightarrow \sigma_{465} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{464} \rangle \longrightarrow \sigma_{465}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{467} = \frac{\langle y := y + 1, \sigma_{466} \rangle \longrightarrow \sigma_{467} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{466} \rangle \longrightarrow \sigma_{467}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{469} = \frac{\langle y := y + 1, \sigma_{468} \rangle \longrightarrow \sigma_{469} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{468} \rangle \longrightarrow \sigma_{469}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{471} = \frac{\langle y := y + 1, \sigma_{470} \rangle \longrightarrow \sigma_{471} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{470} \rangle \longrightarrow \sigma_{471}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{473} = \frac{\langle y := y + 1, \sigma_{472} \rangle \longrightarrow \sigma_{473} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{472} \rangle \longrightarrow \sigma_{473}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{475} = \frac{\langle y := y + 1, \sigma_{474} \rangle \longrightarrow \sigma_{475} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{474} \rangle \longrightarrow \sigma_{475}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{477} = \frac{\langle y := y + 1, \sigma_{476} \rangle \longrightarrow \sigma_{477} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{476} \rangle \longrightarrow \sigma_{477}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{479} = \frac{\langle y := y + 1, \sigma_{478} \rangle \longrightarrow \sigma_{479} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{478} \rangle \longrightarrow \sigma_{479}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{481} = \frac{\langle y := y + 1, \sigma_{480} \rangle \longrightarrow \sigma_{481} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{480} \rangle \longrightarrow \sigma_{481}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{483} = \frac{\langle y := y + 1, \sigma_{482} \rangle \longrightarrow \sigma_{483} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{482} \rangle \longrightarrow \sigma_{483}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{485} = \frac{\langle y := y + 1, \sigma_{484} \rangle \longrightarrow \sigma_{485} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{484} \rangle \longrightarrow \sigma_{485}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{487} = \frac{\langle y := y + 1, \sigma_{486} \rangle \longrightarrow \sigma_{487} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{486} \rangle \longrightarrow \sigma_{487}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{489} = \frac{\langle y := y + 1, \sigma_{488} \rangle \longrightarrow \sigma_{489} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{488} \rangle \longrightarrow \sigma_{489}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{491} = \frac{\langle y := y + 1, \sigma_{490} \rangle \longrightarrow \sigma_{491} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{490} \rangle \longrightarrow \sigma_{491}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{493} = \frac{\langle y := y + 1, \sigma_{492} \rangle \longrightarrow \sigma_{493} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{492} \rangle \longrightarrow \sigma_{493}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{495} = \frac{\langle y := y + 1, \sigma_{494} \rangle \longrightarrow \sigma_{495} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{494} \rangle \longrightarrow \sigma_{495}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{497} = \frac{\langle y := y + 1, \sigma_{496} \rangle \longrightarrow \sigma_{497} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{496} \rangle \longrightarrow \sigma_{497}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{499} = \frac{\langle y := y + 1, \sigma_{498} \rangle \longrightarrow \sigma_{499} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{498} \rangle \longrightarrow \sigma_{499}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{501} = \frac{\langle y := y + 1, \sigma_{500} \rangle \longrightarrow \sigma_{501} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{500} \rangle \longrightarrow \sigma_{501}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{503} = \frac{\langle y := y + 1, \sigma_{502} \rangle \longrightarrow \sigma_{503} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{502} \rangle \longrightarrow \sigma_{503}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{505} = \frac{\langle y := y + 1, \sigma_{504} \rangle \longrightarrow \sigma_{505} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{504} \rangle \longrightarrow \sigma_{505}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{507} = \frac{\langle y := y + 1, \sigma_{506} \rangle \longrightarrow \sigma_{507} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{506} \rangle \longrightarrow \sigma_{507}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{509} = \frac{\langle y := y + 1, \sigma_{508} \rangle \longrightarrow \sigma_{509} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{508} \rangle \longrightarrow \sigma_{509}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{511} = \frac{\langle y := y + 1, \sigma_{510} \rangle \longrightarrow \sigma_{511} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{510} \rangle \longrightarrow \sigma_{511}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{513} = \frac{\langle y := y + 1, \sigma_{512} \rangle \longrightarrow \sigma_{513} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{512} \rangle \longrightarrow \sigma_{513}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{515} = \frac{\langle y := y + 1, \sigma_{514} \rangle \longrightarrow \sigma_{515} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{514} \rangle \longrightarrow \sigma_{515}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{517} = \frac{\langle y := y + 1, \sigma_{516} \rangle \longrightarrow \sigma_{517} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{516} \rangle \longrightarrow \sigma_{517}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{519} = \frac{\langle y := y + 1, \sigma_{518} \rangle \longrightarrow \sigma_{519} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{518} \rangle \longrightarrow \sigma_{519}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{521} = \frac{\langle y := y + 1, \sigma_{520} \rangle \longrightarrow \sigma_{521} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{520} \rangle \longrightarrow \sigma_{521}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{523} = \frac{\langle y := y + 1, \sigma_{522} \rangle \longrightarrow \sigma_{523} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{522} \rangle \longrightarrow \sigma_{523}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{525} = \frac{\langle y := y + 1, \sigma_{524} \rangle \longrightarrow \sigma_{525} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{524} \rangle \longrightarrow \sigma_{525}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{527} = \frac{\langle y := y + 1, \sigma_{526} \rangle \longrightarrow \sigma_{527} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{526} \rangle \longrightarrow \sigma_{527}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{529} = \frac{\langle y := y + 1, \sigma_{528} \rangle \longrightarrow \sigma_{529} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{528} \rangle \longrightarrow \sigma_{529}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{531} = \frac{\langle y := y + 1, \sigma_{530} \rangle \longrightarrow \sigma_{531} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{530} \rangle \longrightarrow \sigma_{531}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{533} = \frac{\langle y := y + 1, \sigma_{532} \rangle \longrightarrow \sigma_{533} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{532} \rangle \longrightarrow \sigma_{533}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{535} = \frac{\langle y := y + 1, \sigma_{534} \rangle \longrightarrow \sigma_{535} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{534} \rangle \longrightarrow \sigma_{535}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{537} = \frac{\langle y := y + 1, \sigma_{536} \rangle \longrightarrow \sigma_{537} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{536} \rangle \longrightarrow \sigma_{537}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{539} = \frac{\langle y := y + 1, \sigma_{538} \rangle \longrightarrow \sigma_{539} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{538} \rangle \longrightarrow \sigma_{539}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{541} = \frac{\langle y := y + 1, \sigma_{540} \rangle \longrightarrow \sigma_{541} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{540} \rangle \longrightarrow \sigma_{541}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{543} = \frac{\langle y := y + 1, \sigma_{542} \rangle \longrightarrow \sigma_{543} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{542} \rangle \longrightarrow \sigma_{543}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{545} = \frac{\langle y := y + 1, \sigma_{544} \rangle \longrightarrow \sigma_{545} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{544} \rangle \longrightarrow \sigma_{545}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{547} = \frac{\langle y := y + 1, \sigma_{546} \rangle \longrightarrow \sigma_{547} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{546} \rangle \longrightarrow \sigma_{547}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{549} = \frac{\langle y := y + 1, \sigma_{548} \rangle \longrightarrow \sigma_{549} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{548} \rangle \longrightarrow \sigma_{549}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{551} = \frac{\langle y := y + 1, \sigma_{550} \rangle \longrightarrow \sigma_{551} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{550} \rangle \longrightarrow \sigma_{551}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{553} = \frac{\langle y := y + 1, \sigma_{552} \rangle \longrightarrow \sigma_{553} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{552} \rangle \longrightarrow \sigma_{553}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{555} = \frac{\langle y := y + 1, \sigma_{554} \rangle \longrightarrow \sigma_{555} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{554} \rangle \longrightarrow \sigma_{555}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{557} = \frac{\langle y := y + 1, \sigma_{556} \rangle \longrightarrow \sigma_{557} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{556} \rangle \longrightarrow \sigma_{557}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{559} = \frac{\langle y := y + 1, \sigma_{558} \rangle \longrightarrow \sigma_{559} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{558} \rangle \longrightarrow \sigma_{559}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{561} = \frac{\langle y := y + 1, \sigma_{560} \rangle \longrightarrow \sigma_{561} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{560} \rangle \longrightarrow \sigma_{561}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{563} = \frac{\langle y := y + 1, \sigma_{562} \rangle \longrightarrow \sigma_{563} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{562} \rangle \longrightarrow \sigma_{563}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{565} = \frac{\langle y := y + 1, \sigma_{564} \rangle \longrightarrow \sigma_{565} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{564} \rangle \longrightarrow \sigma_{565}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{567} = \frac{\langle y := y + 1, \sigma_{566} \rangle \longrightarrow \sigma_{567} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{566} \rangle \longrightarrow \sigma_{567}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{569} = \frac{\langle y := y + 1, \sigma_{568} \rangle \longrightarrow \sigma_{569} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{568} \rangle \longrightarrow \sigma_{569}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{571} = \frac{\langle y := y + 1, \sigma_{570} \rangle \longrightarrow \sigma_{571} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{570} \rangle \longrightarrow \sigma_{571}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{573} = \frac{\langle y := y + 1, \sigma_{572} \rangle \longrightarrow \sigma_{573} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{572} \rangle \longrightarrow \sigma_{573}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{575} = \frac{\langle y := y + 1, \sigma_{574} \rangle \longrightarrow \sigma_{575} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{574} \rangle \longrightarrow \sigma_{575}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{577} = \frac{\langle y := y + 1, \sigma_{576} \rangle \longrightarrow \sigma_{577} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{576} \rangle \longrightarrow \sigma_{577}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{579} = \frac{\langle y := y + 1, \sigma_{578} \rangle \longrightarrow \sigma_{579} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{578} \rangle \longrightarrow \sigma_{579}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{581} = \frac{\langle y := y + 1, \sigma_{580} \rangle \longrightarrow \sigma_{581} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{580} \rangle \longrightarrow \sigma_{581}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{583} = \frac{\langle y := y + 1, \sigma_{582} \rangle \longrightarrow \sigma_{583} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{582} \rangle \longrightarrow \sigma_{583}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{585} = \frac{\langle y := y + 1, \sigma_{584} \rangle \longrightarrow \sigma_{585} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{584} \rangle \longrightarrow \sigma_{585}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{587} = \frac{\langle y := y + 1, \sigma_{586} \rangle \longrightarrow \sigma_{587} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{586} \rangle \longrightarrow \sigma_{587}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{589} = \frac{\langle y := y + 1, \sigma_{588} \rangle \longrightarrow \sigma_{589} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{588} \rangle \longrightarrow \sigma_{589}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{591} = \frac{\langle y := y + 1, \sigma_{590} \rangle \longrightarrow \sigma_{591} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{590} \rangle \longrightarrow \sigma_{591}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{593} = \frac{\langle y := y + 1, \sigma_{592} \rangle \longrightarrow \sigma_{593} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{592} \rangle \longrightarrow \sigma_{593}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{595} = \frac{\langle y := y + 1, \sigma_{594} \rangle \longrightarrow \sigma_{595} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{594} \rangle \longrightarrow \sigma_{595}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{597} = \frac{\langle y := y + 1, \sigma_{596} \rangle \longrightarrow \sigma_{597} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{596} \rangle \longrightarrow \sigma_{597}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{599} = \frac{\langle y := y + 1, \sigma_{598} \rangle \longrightarrow \sigma_{599} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{598} \rangle \longrightarrow \sigma_{599}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{601} = \frac{\langle y := y + 1, \sigma_{600} \rangle \longrightarrow \sigma_{601} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{600} \rangle \longrightarrow \sigma_{601}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{603} = \frac{\langle y := y + 1, \sigma_{602} \rangle \longrightarrow \sigma_{603} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{602} \rangle \longrightarrow \sigma_{603}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{605} = \frac{\langle y := y + 1, \sigma_{604} \rangle \longrightarrow \sigma_{605} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{604} \rangle \longrightarrow \sigma_{605}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{607} = \frac{\langle y := y + 1, \sigma_{606} \rangle \longrightarrow \sigma_{607} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{606} \rangle \longrightarrow \sigma_{607}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{609} = \frac{\langle y := y + 1, \sigma_{608} \rangle \longrightarrow \sigma_{609} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{608} \rangle \longrightarrow \sigma_{609}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{611} = \frac{\langle y := y + 1, \sigma_{610} \rangle \longrightarrow \sigma_{611} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{610} \rangle \longrightarrow \sigma_{611}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{613} = \frac{\langle y := y + 1, \sigma_{612} \rangle \longrightarrow \sigma_{613} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{612} \rangle \longrightarrow \sigma_{613}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{615} = \frac{\langle y := y + 1, \sigma_{614} \rangle \longrightarrow \sigma_{615} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{614} \rangle \longrightarrow \sigma_{615}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{617} = \frac{\langle y := y + 1, \sigma_{616} \rangle \longrightarrow \sigma_{617} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{616} \rangle \longrightarrow \sigma_{617}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{619} = \frac{\langle y := y + 1, \sigma_{618} \rangle \longrightarrow \sigma_{619} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{618} \rangle \longrightarrow \sigma_{619}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{621} = \frac{\langle y := y + 1, \sigma_{620} \rangle \longrightarrow \sigma_{621} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{620} \rangle \longrightarrow \sigma_{621}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{623} = \frac{\langle y := y + 1, \sigma_{622} \rangle \longrightarrow \sigma_{623} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{622} \rangle \longrightarrow \sigma_{623}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{625} = \frac{\langle y := y + 1, \sigma_{624} \rangle \longrightarrow \sigma_{625} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{624} \rangle \longrightarrow \sigma_{625}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{627} = \frac{\langle y := y + 1, \sigma_{626} \rangle \longrightarrow \sigma_{627} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{626} \rangle \longrightarrow \sigma_{627}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{629} = \frac{\langle y := y + 1, \sigma_{628} \rangle \longrightarrow \sigma_{629} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{628} \rangle \longrightarrow \sigma_{629}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{631} = \frac{\langle y := y + 1, \sigma_{630} \rangle \longrightarrow \sigma_{631} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{630} \rangle \longrightarrow \sigma_{631}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{633} = \frac{\langle y := y + 1, \sigma_{632} \rangle \longrightarrow \sigma_{633} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{632} \rangle \longrightarrow \sigma_{633}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{635} = \frac{\langle y := y + 1, \sigma_{634} \rangle \longrightarrow \sigma_{635} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{634} \rangle \longrightarrow \sigma_{635}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{637} = \frac{\langle y := y + 1, \sigma_{636} \rangle \longrightarrow \sigma_{637} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{636} \rangle \longrightarrow \sigma_{637}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{639} = \frac{\langle y := y + 1, \sigma_{638} \rangle \longrightarrow \sigma_{639} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{638} \rangle \longrightarrow \sigma_{639}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{641} = \frac{\langle y := y + 1, \sigma_{640} \rangle \longrightarrow \sigma_{641} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{640} \rangle \longrightarrow \sigma_{641}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{643} = \frac{\langle y := y + 1, \sigma_{642} \rangle \longrightarrow \sigma_{643} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{642} \rangle \longrightarrow \sigma_{643}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{645} = \frac{\langle y := y + 1, \sigma_{644} \rangle \longrightarrow \sigma_{645} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{644} \rangle \longrightarrow \sigma_{645}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{647} = \frac{\langle y := y + 1, \sigma_{646} \rangle \longrightarrow \sigma_{647} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{646} \rangle \longrightarrow \sigma_{647}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{649} = \frac{\langle y := y + 1, \sigma_{648} \rangle \longrightarrow \sigma_{649} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{648} \rangle \longrightarrow \sigma_{649}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{651} = \frac{\langle y := y + 1, \sigma_{650} \rangle \longrightarrow \sigma_{651} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{650} \rangle \longrightarrow \sigma_{651}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{653} = \frac{\langle y := y + 1, \sigma_{652} \rangle \longrightarrow \sigma_{653} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{652} \rangle \longrightarrow \sigma_{653}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{655} = \frac{\langle y := y + 1, \sigma_{654} \rangle \longrightarrow \sigma_{655} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{654} \rangle \longrightarrow \sigma_{655}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{657} = \frac{\langle y := y + 1, \sigma_{656} \rangle \longrightarrow \sigma_{657} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{656} \rangle \longrightarrow \sigma_{657}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{659} = \frac{\langle y := y + 1, \sigma_{658} \rangle \longrightarrow \sigma_{659} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{658} \rangle \longrightarrow \sigma_{659}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{661} = \frac{\langle y := y + 1, \sigma_{660} \rangle \longrightarrow \sigma_{661} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{660} \rangle \longrightarrow \sigma_{661}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{663} = \frac{\langle y := y + 1, \sigma_{662} \rangle \longrightarrow \sigma_{663} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{662} \rangle \longrightarrow \sigma_{663}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{665} = \frac{\langle y := y + 1, \sigma_{664} \rangle \longrightarrow \sigma_{665} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{664} \rangle \longrightarrow \sigma_{665}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{667} = \frac{\langle y := y + 1, \sigma_{666} \rangle \longrightarrow \sigma_{667} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{666} \rangle \longrightarrow \sigma_{667}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{669} = \frac{\langle y := y + 1, \sigma_{668} \rangle \longrightarrow \sigma_{669} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{668} \rangle \longrightarrow \sigma_{669}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{671} = \frac{\langle y := y + 1, \sigma_{670} \rangle \longrightarrow \sigma_{671} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{670} \rangle \longrightarrow \sigma_{671}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{673} = \frac{\langle y := y + 1, \sigma_{672} \rangle \longrightarrow \sigma_{673} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{672} \rangle \longrightarrow \sigma_{673}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{675} = \frac{\langle y := y + 1, \sigma_{674} \rangle \longrightarrow \sigma_{675} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{674} \rangle \longrightarrow \sigma_{675}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{677} = \frac{\langle y := y + 1, \sigma_{676} \rangle \longrightarrow \sigma_{677} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{676} \rangle \longrightarrow \sigma_{677}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{679} = \frac{\langle y := y + 1, \sigma_{678} \rangle \longrightarrow \sigma_{679} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{678} \rangle \longrightarrow \sigma_{679}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{681} = \frac{\langle y := y + 1, \sigma_{680} \rangle \longrightarrow \sigma_{681} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{680} \rangle \longrightarrow \sigma_{681}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{683} = \frac{\langle y := y + 1, \sigma_{682} \rangle \longrightarrow \sigma_{683} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{682} \rangle \longrightarrow \sigma_{683}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{685} = \frac{\langle y := y + 1, \sigma_{684} \rangle \longrightarrow \sigma_{685} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{684} \rangle \longrightarrow \sigma_{685}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{687} = \frac{\langle y := y + 1, \sigma_{686} \rangle \longrightarrow \sigma_{687} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{686} \rangle \longrightarrow \sigma_{687}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{689} = \frac{\langle y := y + 1, \sigma_{688} \rangle \longrightarrow \sigma_{689} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{688} \rangle \longrightarrow \sigma_{689}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{691} = \frac{\langle y := y + 1, \sigma_{690} \rangle \longrightarrow \sigma_{691} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{690} \rangle \longrightarrow \sigma_{691}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{693} = \frac{\langle y := y + 1, \sigma_{692} \rangle \longrightarrow \sigma_{693} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{692} \rangle \longrightarrow \sigma_{693}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{695} = \frac{\langle y := y + 1, \sigma_{694} \rangle \longrightarrow \sigma_{695} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{694} \rangle \longrightarrow \sigma_{695}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{697} = \frac{\langle y := y + 1, \sigma_{696} \rangle \longrightarrow \sigma_{697} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{696} \rangle \longrightarrow \sigma_{697}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{699} = \frac{\langle y := y + 1, \sigma_{698} \rangle \longrightarrow \sigma_{699} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{698} \rangle \longrightarrow \sigma_{699}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{701} = \frac{\langle y := y + 1, \sigma_{700} \rangle \longrightarrow \sigma_{701} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{700} \rangle \longrightarrow \sigma_{701}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{703} = \frac{\langle y := y + 1, \sigma_{702} \rangle \longrightarrow \sigma_{703} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{702} \rangle \longrightarrow \sigma_{703}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{705} = \frac{\langle y := y + 1, \sigma_{704} \rangle \longrightarrow \sigma_{705} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{704} \rangle \longrightarrow \sigma_{705}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{707} = \frac{\langle y := y + 1, \sigma_{706} \rangle \longrightarrow \sigma_{707} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{706} \rangle \longrightarrow \sigma_{707}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{709} = \frac{\langle y := y + 1, \sigma_{708} \rangle \longrightarrow \sigma_{709} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{708} \rangle \longrightarrow \sigma_{709}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{711} = \frac{\langle y := y + 1, \sigma_{710} \rangle \longrightarrow \sigma_{711} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{710} \rangle \longrightarrow \sigma_{711}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{713} = \frac{\langle y := y + 1, \sigma_{712} \rangle \longrightarrow \sigma_{713} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{712} \rangle \longrightarrow \sigma_{713}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{715} = \frac{\langle y := y + 1, \sigma_{714} \rangle \longrightarrow \sigma_{715} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{714} \rangle \longrightarrow \sigma_{715}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{717} = \frac{\langle y := y + 1, \sigma_{716} \rangle \longrightarrow \sigma_{717} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{716} \rangle \longrightarrow \sigma_{717}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{719} = \frac{\langle y := y + 1, \sigma_{718} \rangle \longrightarrow \sigma_{719} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{718} \rangle \longrightarrow \sigma_{719}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{721} = \frac{\langle y := y + 1, \sigma_{720} \rangle \longrightarrow \sigma_{721} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{720} \rangle \longrightarrow \sigma_{721}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{723} = \frac{\langle y := y + 1, \sigma_{722} \rangle \longrightarrow \sigma_{723} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{722} \rangle \longrightarrow \sigma_{723}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{725} = \frac{\langle y := y + 1, \sigma_{724} \rangle \longrightarrow \sigma_{725} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{724} \rangle \longrightarrow \sigma_{725}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{727} = \frac{\langle y := y + 1, \sigma_{726} \rangle \longrightarrow \sigma_{727} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{726} \rangle \longrightarrow \sigma_{727}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{729} = \frac{\langle y := y + 1, \sigma_{728} \rangle \longrightarrow \sigma_{729} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{728} \rangle \longrightarrow \sigma_{729}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{731} = \frac{\langle y := y + 1, \sigma_{730} \rangle \longrightarrow \sigma_{731} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{730} \rangle \longrightarrow \sigma_{731}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{733} = \frac{\langle y := y + 1, \sigma_{732} \rangle \longrightarrow \sigma_{733} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{732} \rangle \longrightarrow \sigma_{733}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{735} = \frac{\langle y := y + 1, \sigma_{734} \rangle \longrightarrow \sigma_{735} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{734} \rangle \longrightarrow \sigma_{735}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{737} = \frac{\langle y := y + 1, \sigma_{736} \rangle \longrightarrow \sigma_{737} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{736} \rangle \longrightarrow \sigma_{737}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{739} = \frac{\langle y := y + 1, \sigma_{738} \rangle \longrightarrow \sigma_{739} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{738} \rangle \longrightarrow \sigma_{739}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{741} = \frac{\langle y := y + 1, \sigma_{740} \rangle \longrightarrow \sigma_{741} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{740} \rangle \longrightarrow \sigma_{741}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{743} = \frac{\langle y := y + 1, \sigma_{742} \rangle \longrightarrow \sigma_{743} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{742} \rangle \longrightarrow \sigma_{743}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{745} = \frac{\langle y := y + 1, \sigma_{744} \rangle \longrightarrow \sigma_{745} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{744} \rangle \longrightarrow \sigma_{745}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{747} = \frac{\langle y := y + 1, \sigma_{746} \rangle \longrightarrow \sigma_{747} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{746} \rangle \longrightarrow \sigma_{747}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{749} = \frac{\langle y := y + 1, \sigma_{748} \rangle \longrightarrow \sigma_{749} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{748} \rangle \longrightarrow \sigma_{749}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{751} = \frac{\langle y := y + 1, \sigma_{750} \rangle \longrightarrow \sigma_{751} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{750} \rangle \longrightarrow \sigma_{751}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{753} = \frac{\langle y := y + 1, \sigma_{752} \rangle \longrightarrow \sigma_{753} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{752} \rangle \longrightarrow \sigma_{753}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{755} = \frac{\langle y := y + 1, \sigma_{754} \rangle \longrightarrow \sigma_{755} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{754} \rangle \longrightarrow \sigma_{755}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{757} = \frac{\langle y := y + 1, \sigma_{756} \rangle \longrightarrow \sigma_{757} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{756} \rangle \longrightarrow \sigma_{757}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{759} = \frac{\langle y := y + 1, \sigma_{758} \rangle \longrightarrow \sigma_{759} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{758} \rangle \longrightarrow \sigma_{759}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{761} = \frac{\langle y := y + 1, \sigma_{760} \rangle \longrightarrow \sigma_{761} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{760} \rangle \longrightarrow \sigma_{761}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{763} = \frac{\langle y := y + 1, \sigma_{762} \rangle \longrightarrow \sigma_{763} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{762} \rangle \longrightarrow \sigma_{763}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{765} = \frac{\langle y := y + 1, \sigma_{764} \rangle \longrightarrow \sigma_{765} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{764} \rangle \longrightarrow \sigma_{765}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{767} = \frac{\langle y := y + 1, \sigma_{766} \rangle \longrightarrow \sigma_{767} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{766} \rangle \longrightarrow \sigma_{767}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{769} = \frac{\langle y := y + 1, \sigma_{768} \rangle \longrightarrow \sigma_{769} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{768} \rangle \longrightarrow \sigma_{769}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{771} = \frac{\langle y := y + 1, \sigma_{770} \rangle \longrightarrow \sigma_{771} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{770} \rangle \longrightarrow \sigma_{771}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{773} = \frac{\langle y := y + 1, \sigma_{772} \rangle \longrightarrow \sigma_{773} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{772} \rangle \longrightarrow \sigma_{773}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{775} = \frac{\langle y := y + 1, \sigma_{774} \rangle \longrightarrow \sigma_{775} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{774} \rangle \longrightarrow \sigma_{775}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{777} = \frac{\langle y := y + 1, \sigma_{776} \rangle \longrightarrow \sigma_{777} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{776} \rangle \longrightarrow \sigma_{777}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{779} = \frac{\langle y := y + 1, \sigma_{778} \rangle \longrightarrow \sigma_{779} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{778} \rangle \longrightarrow \sigma_{779}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{781} = \frac{\langle y := y + 1, \sigma_{780} \rangle \longrightarrow \sigma_{781} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{780} \rangle \longrightarrow \sigma_{781}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{783} = \frac{\langle y := y + 1, \sigma_{782} \rangle \longrightarrow \sigma_{783} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{782} \rangle \longrightarrow \sigma_{783}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{785} = \frac{\langle y := y + 1, \sigma_{784} \rangle \longrightarrow \sigma_{785} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{784} \rangle \longrightarrow \sigma_{785}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{787} = \frac{\langle y := y + 1, \sigma_{786} \rangle \longrightarrow \sigma_{787} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{786} \rangle \longrightarrow \sigma_{787}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{789} = \frac{\langle y := y + 1, \sigma_{788} \rangle \longrightarrow \sigma_{789} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{788} \rangle \longrightarrow \sigma_{789}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{791} = \frac{\langle y := y + 1, \sigma_{790} \rangle \longrightarrow \sigma_{791} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{790} \rangle \longrightarrow \sigma_{791}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{793} = \frac{\langle y := y + 1, \sigma_{792} \rangle \longrightarrow \sigma_{793} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{792} \rangle \longrightarrow \sigma_{793}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{795} = \frac{\langle y := y + 1, \sigma_{794} \rangle \longrightarrow \sigma_{795} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{794} \rangle \longrightarrow \sigma_{795}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{797} = \frac{\langle y := y + 1, \sigma_{796} \rangle \longrightarrow \sigma_{797} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{796} \rangle \longrightarrow \sigma_{797}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{799} = \frac{\langle y := y + 1, \sigma_{798} \rangle \longrightarrow \sigma_{799} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{798} \rangle \longrightarrow \sigma_{799}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{801} = \frac{\langle y := y + 1, \sigma_{800} \rangle \longrightarrow \sigma_{801} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{800} \rangle \longrightarrow \sigma_{801}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{803} = \frac{\langle y := y + 1, \sigma_{802} \rangle \longrightarrow \sigma_{803} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{802} \rangle \longrightarrow \sigma_{803}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{805} = \frac{\langle y := y + 1, \sigma_{804} \rangle \longrightarrow \sigma_{805} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{804} \rangle \longrightarrow \sigma_{805}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{807} = \frac{\langle y := y + 1, \sigma_{806} \rangle \longrightarrow \sigma_{807} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{806} \rangle \longrightarrow \sigma_{807}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{809} = \frac{\langle y := y + 1, \sigma_{808} \rangle \longrightarrow \sigma_{809} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{808} \rangle \longrightarrow \sigma_{809}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{811} = \frac{\langle y := y + 1, \sigma_{810} \rangle \longrightarrow \sigma_{811} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{810} \rangle \longrightarrow \sigma_{811}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{813} = \frac{\langle y := y + 1, \sigma_{812} \rangle \longrightarrow \sigma_{813} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{812} \rangle \longrightarrow \sigma_{813}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{815} = \frac{\langle y := y + 1, \sigma_{814} \rangle \longrightarrow \sigma_{815} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{814} \rangle \longrightarrow \sigma_{815}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{817} = \frac{\langle y := y + 1, \sigma_{816} \rangle \longrightarrow \sigma_{817} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{816} \rangle \longrightarrow \sigma_{817}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{819} = \frac{\langle y := y + 1, \sigma_{818} \rangle \longrightarrow \sigma_{819} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{818} \rangle \longrightarrow \sigma_{819}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{821} = \frac{\langle y := y + 1, \sigma_{820} \rangle \longrightarrow \sigma_{821} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{820} \rangle \longrightarrow \sigma_{821}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{823} = \frac{\langle y := y + 1, \sigma_{822} \rangle \longrightarrow \sigma_{823} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{822} \rangle \longrightarrow \sigma_{823}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{825} = \frac{\langle y := y + 1, \sigma_{824} \rangle \longrightarrow \sigma_{825} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{824} \rangle \longrightarrow \sigma_{825}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{827} = \frac{\langle y := y + 1, \sigma_{826} \rangle \longrightarrow \sigma_{827} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{826} \rangle \longrightarrow \sigma_{827}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{829} = \frac{\langle y := y + 1, \sigma_{828} \rangle \longrightarrow \sigma_{829} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{828} \rangle \longrightarrow \sigma_{829}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{831} = \frac{\langle y := y + 1, \sigma_{830} \rangle \longrightarrow \sigma_{831} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{830} \rangle \longrightarrow \sigma_{831}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{833} = \frac{\langle y := y + 1, \sigma_{832} \rangle \longrightarrow \sigma_{833} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{832} \rangle \longrightarrow \sigma_{833}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{835} = \frac{\langle y := y + 1, \sigma_{834} \rangle \longrightarrow \sigma_{835} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{834} \rangle \longrightarrow \sigma_{835}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{837} = \frac{\langle y := y + 1, \sigma_{836} \rangle \longrightarrow \sigma_{837} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{836} \rangle \longrightarrow \sigma_{837}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{839} = \frac{\langle y := y + 1, \sigma_{838} \rangle \longrightarrow \sigma_{839} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{838} \rangle \longrightarrow \sigma_{839}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{841} = \frac{\langle y := y + 1, \sigma_{840} \rangle \longrightarrow \sigma_{841} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{840} \rangle \longrightarrow \sigma_{841}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{843} = \frac{\langle y := y + 1, \sigma_{842} \rangle \longrightarrow \sigma_{843} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{842} \rangle \longrightarrow \sigma_{843}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{845} = \frac{\langle y := y + 1, \sigma_{844} \rangle \longrightarrow \sigma_{845} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{844} \rangle \longrightarrow \sigma_{845}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{847} = \frac{\langle y := y + 1, \sigma_{846} \rangle \longrightarrow \sigma_{847} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{846} \rangle \longrightarrow \sigma_{847}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{849} = \frac{\langle y := y + 1, \sigma_{848} \rangle \longrightarrow \sigma_{849} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{848} \rangle \longrightarrow \sigma_{849}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{851} = \frac{\langle y := y + 1, \sigma_{850} \rangle \longrightarrow \sigma_{851} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{850} \rangle \longrightarrow \sigma_{851}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{853} = \frac{\langle y := y + 1, \sigma_{852} \rangle \longrightarrow \sigma_{853} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{852} \rangle \longrightarrow \sigma_{853}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{855} = \frac{\langle y := y + 1, \sigma_{854} \rangle \longrightarrow \sigma_{855} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{854} \rangle \longrightarrow \sigma_{855}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{857} = \frac{\langle y := y + 1, \sigma_{856} \rangle \longrightarrow \sigma_{857} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{856} \rangle \longrightarrow \sigma_{857}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{859} = \frac{\langle y := y + 1, \sigma_{858} \rangle \longrightarrow \sigma_{859} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{858} \rangle \longrightarrow \sigma_{859}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{861} = \frac{\langle y := y + 1, \sigma_{860} \rangle \longrightarrow \sigma_{861} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{860} \rangle \longrightarrow \sigma_{861}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{863} = \frac{\langle y := y + 1, \sigma_{862} \rangle \longrightarrow \sigma_{863} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{862} \rangle \longrightarrow \sigma_{863}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{865} = \frac{\langle y := y + 1, \sigma_{864} \rangle \longrightarrow \sigma_{865} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{864} \rangle \longrightarrow \sigma_{865}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{867} = \frac{\langle y := y + 1, \sigma_{866} \rangle \longrightarrow \sigma_{867} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{866} \rangle \longrightarrow \sigma_{867}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{869} = \frac{\langle y := y + 1, \sigma_{868} \rangle \longrightarrow \sigma_{869} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{868} \rangle \longrightarrow \sigma_{869}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{871} = \frac{\langle y := y + 1, \sigma_{870} \rangle \longrightarrow \sigma_{871} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{870} \rangle \longrightarrow \sigma_{871}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{873} = \frac{\langle y := y + 1, \sigma_{872} \rangle \longrightarrow \sigma_{873} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{872} \rangle \longrightarrow \sigma_{873}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{875} = \frac{\langle y := y + 1, \sigma_{874} \rangle \longrightarrow \sigma_{875} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{874} \rangle \longrightarrow \sigma_{875}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{877} = \frac{\langle y := y + 1, \sigma_{876} \rangle \longrightarrow \sigma_{877} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{876} \rangle \longrightarrow \sigma_{877}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{879} = \frac{\langle y := y + 1, \sigma_{878} \rangle \longrightarrow \sigma_{879} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{878} \rangle \longrightarrow \sigma_{879}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{881} = \frac{\langle y := y + 1, \sigma_{880} \rangle \longrightarrow \sigma_{881} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{880} \rangle \longrightarrow \sigma_{881}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{883} = \frac{\langle y := y + 1, \sigma_{882} \rangle \longrightarrow \sigma_{883} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{882} \rangle \longrightarrow \sigma_{883}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{885} = \frac{\langle y := y + 1, \sigma_{884} \rangle \longrightarrow \sigma_{885} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{884} \rangle \longrightarrow \sigma_{885}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{887} = \frac{\langle y := y + 1, \sigma_{886} \rangle \longrightarrow \sigma_{887} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{886} \rangle \longrightarrow \sigma_{887}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{889} = \frac{\langle y := y + 1, \sigma_{888} \rangle \longrightarrow \sigma_{889} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{888} \rangle \longrightarrow \sigma_{889}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{891} = \frac{\langle y := y + 1, \sigma_{890} \rangle \longrightarrow \sigma_{891} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{890} \rangle \longrightarrow \sigma_{891}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{893} = \frac{\langle y := y + 1, \sigma_{892} \rangle \longrightarrow \sigma_{893} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{892} \rangle \longrightarrow \sigma_{893}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{895} = \frac{\langle y := y + 1, \sigma_{894} \rangle \longrightarrow \sigma_{895} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{894} \rangle \longrightarrow \sigma_{895}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{897} = \frac{\langle y := y + 1, \sigma_{896} \rangle \longrightarrow \sigma_{897} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{896} \rangle \longrightarrow \sigma_{897}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{899} = \frac{\langle y := y + 1, \sigma_{898} \rangle \longrightarrow \sigma_{899} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{898} \rangle \longrightarrow \sigma_{899}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{901} = \frac{\langle y := y + 1, \sigma_{900} \rangle \longrightarrow \sigma_{901} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{900} \rangle \longrightarrow \sigma_{901}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{903} = \frac{\langle y := y + 1, \sigma_{902} \rangle \longrightarrow \sigma_{903} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{902} \rangle \longrightarrow \sigma_{903}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{905} = \frac{\langle y := y + 1, \sigma_{904} \rangle \longrightarrow \sigma_{905} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{904} \rangle \longrightarrow \sigma_{905}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{907} = \frac{\langle y := y + 1, \sigma_{906} \rangle \longrightarrow \sigma_{907} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{906} \rangle \longrightarrow \sigma_{907}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{909} = \frac{\langle y := y + 1, \sigma_{908} \rangle \longrightarrow \sigma_{909} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{908} \rangle \longrightarrow \sigma_{909}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{911} = \frac{\langle y := y + 1, \sigma_{910} \rangle \longrightarrow \sigma_{911} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{910} \rangle \longrightarrow \sigma_{911}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{913} = \frac{\langle y := y + 1, \sigma_{912} \rangle \longrightarrow \sigma_{913} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{912} \rangle \longrightarrow \sigma_{913}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{915} = \frac{\langle y := y + 1, \sigma_{914} \rangle \longrightarrow \sigma_{915} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{914} \rangle \longrightarrow \sigma_{915}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{917} = \frac{\langle y := y + 1, \sigma_{916} \rangle \longrightarrow \sigma_{917} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{916} \rangle \longrightarrow \sigma_{917}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{919} = \frac{\langle y := y + 1, \sigma_{918} \rangle \longrightarrow \sigma_{919} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{918} \rangle \longrightarrow \sigma_{919}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{921} = \frac{\langle y := y + 1, \sigma_{920} \rangle \longrightarrow \sigma_{921} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{920} \rangle \longrightarrow \sigma_{921}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{923} = \frac{\langle y := y + 1, \sigma_{922} \rangle \longrightarrow \sigma_{923} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{922} \rangle \longrightarrow \sigma_{923}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{925} = \frac{\langle y := y + 1, \sigma_{924} \rangle \longrightarrow \sigma_{925} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{924} \rangle \longrightarrow \sigma_{925}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{927} = \frac{\langle y := y + 1, \sigma_{926} \rangle \longrightarrow \sigma_{927} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{926} \rangle \longrightarrow \sigma_{927}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{929} = \frac{\langle y := y + 1, \sigma_{928} \rangle \longrightarrow \sigma_{929} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{928} \rangle \longrightarrow \sigma_{929}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{931} = \frac{\langle y := y + 1, \sigma_{930} \rangle \longrightarrow \sigma_{931} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{930} \rangle \longrightarrow \sigma_{931}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{933} = \frac{\langle y := y + 1, \sigma_{932} \rangle \longrightarrow \sigma_{933} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{932} \rangle \longrightarrow \sigma_{933}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{935} = \frac{\langle y := y + 1, \sigma_{934} \rangle \longrightarrow \sigma_{935} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{934} \rangle \longrightarrow \sigma_{935}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{937} = \frac{\langle y := y + 1, \sigma_{936} \rangle \longrightarrow \sigma_{937} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{936} \rangle \longrightarrow \sigma_{937}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{939} = \frac{\langle y := y + 1, \sigma_{938} \rangle \longrightarrow \sigma_{939} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{938} \rangle \longrightarrow \sigma_{939}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{941} = \frac{\langle y := y + 1, \sigma_{940} \rangle \longrightarrow \sigma_{941} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{940} \rangle \longrightarrow \sigma_{941}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{943} = \frac{\langle y := y + 1, \sigma_{942} \rangle \longrightarrow \sigma_{943} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{942} \rangle \longrightarrow \sigma_{943}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{945} = \frac{\langle y := y + 1, \sigma_{944} \rangle \longrightarrow \sigma_{945} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{944} \rangle \longrightarrow \sigma_{945}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{947} = \frac{\langle y := y + 1, \sigma_{946} \rangle \longrightarrow \sigma_{947} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{946} \rangle \longrightarrow \sigma_{947}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{949} = \frac{\langle y := y + 1, \sigma_{948} \rangle \longrightarrow \sigma_{949} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{948} \rangle \longrightarrow \sigma_{949}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{951} = \frac{\langle y := y + 1, \sigma_{950} \rangle \longrightarrow \sigma_{951} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{950} \rangle \longrightarrow \sigma_{951}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{953} = \frac{\langle y := y + 1, \sigma_{952} \rangle \longrightarrow \sigma_{953} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{952} \rangle \longrightarrow \sigma_{953}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{955} = \frac{\langle y := y + 1, \sigma_{954} \rangle \longrightarrow \sigma_{955} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{954} \rangle \longrightarrow \sigma_{955}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{957} = \frac{\langle y := y + 1, \sigma_{956} \rangle \longrightarrow \sigma_{957} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{956} \rangle \longrightarrow \sigma_{957}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{959} = \frac{\langle y := y + 1, \sigma_{958} \rangle \longrightarrow \sigma_{959} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{958} \rangle \longrightarrow \sigma_{959}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{961} = \frac{\langle y := y + 1, \sigma_{960} \rangle \longrightarrow \sigma_{961} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{960} \rangle \longrightarrow \sigma_{961}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{963} = \frac{\langle y := y + 1, \sigma_{962} \rangle \longrightarrow \sigma_{963} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{962} \rangle \longrightarrow \sigma_{963}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{965} = \frac{\langle y := y + 1, \sigma_{964} \rangle \longrightarrow \sigma_{965} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{964} \rangle \longrightarrow \sigma_{965}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{967} = \frac{\langle y := y + 1, \sigma_{966} \rangle \longrightarrow \sigma_{967} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{966} \rangle \longrightarrow \sigma_{967}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{969} = \frac{\langle y := y + 1, \sigma_{968} \rangle \longrightarrow \sigma_{969} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{968} \rangle \longrightarrow \sigma_{969}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{971} = \frac{\langle y := y + 1, \sigma_{970} \rangle \longrightarrow \sigma_{971} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{970} \rangle \longrightarrow \sigma_{971}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{973} = \frac{\langle y := y + 1, \sigma_{972} \rangle \longrightarrow \sigma_{973} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{972} \rangle \longrightarrow \sigma_{973}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{975} = \frac{\langle y := y + 1, \sigma_{974} \rangle \longrightarrow \sigma_{975} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{974} \rangle \longrightarrow \sigma_{975}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{977} = \frac{\langle y := y + 1, \sigma_{976} \rangle \longrightarrow \sigma_{977} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{976} \rangle \longrightarrow \sigma_{977}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{979} = \frac{\langle y := y + 1, \sigma_{978} \rangle \longrightarrow \sigma_{979} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{978} \rangle \longrightarrow \sigma_{979}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{981} = \frac{\langle y := y + 1, \sigma_{980} \rangle \longrightarrow \sigma_{981} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{980} \rangle \longrightarrow \sigma_{981}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{983} = \frac{\langle y := y + 1, \sigma_{982} \rangle \longrightarrow \sigma_{983} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{982} \rangle \longrightarrow \sigma_{983}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{985} = \frac{\langle y := y + 1, \sigma_{984} \rangle \longrightarrow \sigma_{985} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{984} \rangle \longrightarrow \sigma_{985}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{987} = \frac{\langle y := y + 1, \sigma_{986} \rangle \longrightarrow \sigma_{987} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{986} \rangle \longrightarrow \sigma_{987}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{989} = \frac{\langle y := y + 1, \sigma_{988} \rangle \longrightarrow \sigma_{989} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{988} \rangle \longrightarrow \sigma_{989}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{991} = \frac{\langle y := y + 1, \sigma_{990} \rangle \longrightarrow \sigma_{991} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{990} \rangle \longrightarrow \sigma_{991}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{993} = \frac{\langle y := y + 1, \sigma_{992} \rangle \longrightarrow \sigma_{993} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{992} \rangle \longrightarrow \sigma_{993}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{995} = \frac{\langle y := y + 1, \sigma_{994} \rangle \longrightarrow \sigma_{995} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{994} \rangle \longrightarrow \sigma_{995}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{997} = \frac{\langle y := y + 1, \sigma_{996} \rangle \longrightarrow \sigma_{997} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{996} \rangle \longrightarrow \sigma_{997}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{999} = \frac{\langle y := y + 1, \sigma_{998} \rangle \longrightarrow \sigma_{999} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{998} \rangle \longrightarrow \sigma_{999}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1001} = \frac{\langle y := y + 1, \sigma_{1000} \rangle \longrightarrow \sigma_{1001} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1000} \rangle \longrightarrow \sigma_{1001}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1003} = \frac{\langle y := y + 1, \sigma_{1002} \rangle \longrightarrow \sigma_{1003} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1002} \rangle \longrightarrow \sigma_{1003}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1005} = \frac{\langle y := y + 1, \sigma_{1004} \rangle \longrightarrow \sigma_{1005} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1004} \rangle \longrightarrow \sigma_{1005}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1007} = \frac{\langle y := y + 1, \sigma_{1006} \rangle \longrightarrow \sigma_{1007} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1006} \rangle \longrightarrow \sigma_{1007}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1009} = \frac{\langle y := y + 1, \sigma_{1008} \rangle \longrightarrow \sigma_{1009} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1008} \rangle \longrightarrow \sigma_{1009}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1011} = \frac{\langle y := y + 1, \sigma_{1010} \rangle \longrightarrow \sigma_{1011} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1010} \rangle \longrightarrow \sigma_{1011}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1013} = \frac{\langle y := y + 1, \sigma_{1012} \rangle \longrightarrow \sigma_{1013} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1012} \rangle \longrightarrow \sigma_{1013}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1015} = \frac{\langle y := y + 1, \sigma_{1014} \rangle \longrightarrow \sigma_{1015} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1014} \rangle \longrightarrow \sigma_{1015}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1017} = \frac{\langle y := y + 1, \sigma_{1016} \rangle \longrightarrow \sigma_{1017} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1016} \rangle \longrightarrow \sigma_{1017}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1019} = \frac{\langle y := y + 1, \sigma_{1018} \rangle \longrightarrow \sigma_{1019} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1018} \rangle \longrightarrow \sigma_{1019}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1047} = \frac{\langle y := y + 1, \sigma_{1046} \rangle \longrightarrow \sigma_{1047} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1046} \rangle \longrightarrow \sigma_{1047}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1049} = \frac{\langle y := y + 1, \sigma_{1048} \rangle \longrightarrow \sigma_{1049} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1048} \rangle \longrightarrow \sigma_{1049}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1051} = \frac{\langle y := y + 1, \sigma_{1050} \rangle \longrightarrow \sigma_{1051} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1050} \rangle \longrightarrow \sigma_{1051}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1053} = \frac{\langle y := y + 1, \sigma_{1052} \rangle \longrightarrow \sigma_{1053} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1052} \rangle \longrightarrow \sigma_{1053}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1055} = \frac{\langle y := y + 1, \sigma_{1054} \rangle \longrightarrow \sigma_{1055} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1054} \rangle \longrightarrow \sigma_{1055}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1057} = \frac{\langle y := y + 1, \sigma_{1056} \rangle \longrightarrow \sigma_{1057} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1056} \rangle \longrightarrow \sigma_{1057}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1059} = \frac{\langle y := y + 1, \sigma_{1058} \rangle \longrightarrow \sigma_{1059} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1058} \rangle \longrightarrow \sigma_{1059}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1061} = \frac{\langle y := y + 1, \sigma_{1060} \rangle \longrightarrow \sigma_{1061} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1060} \rangle \longrightarrow \sigma_{1061}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1063} = \frac{\langle y := y + 1, \sigma_{1062} \rangle \longrightarrow \sigma_{1063} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1062} \rangle \longrightarrow \sigma_{1063}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1065} = \frac{\langle y := y + 1, \sigma_{1064} \rangle \longrightarrow \sigma_{1065} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1064} \rangle \longrightarrow \sigma_{1065}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1067} = \frac{\langle y := y + 1, \sigma_{1066} \rangle \longrightarrow \sigma_{1067} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1066} \rangle \longrightarrow \sigma_{1067}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1069} = \frac{\langle y := y + 1, \sigma_{1068} \rangle \longrightarrow \sigma_{1069} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1068} \rangle \longrightarrow \sigma_{1069}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1097} = \frac{\langle y := y + 1, \sigma_{1096} \rangle \longrightarrow \sigma_{1097} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1096} \rangle \longrightarrow \sigma_{1097}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1099} = \frac{\langle y := y + 1, \sigma_{1098} \rangle \longrightarrow \sigma_{1099} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1098} \rangle \longrightarrow \sigma_{1099}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1101} = \frac{\langle y := y + 1, \sigma_{1100} \rangle \longrightarrow \sigma_{1101} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1100} \rangle \longrightarrow \sigma_{1101}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1103} = \frac{\langle y := y + 1, \sigma_{1102} \rangle \longrightarrow \sigma_{1103} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1102} \rangle \longrightarrow \sigma_{1103}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1105} = \frac{\langle y := y + 1, \sigma_{1104} \rangle \longrightarrow \sigma_{1105} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1104} \rangle \longrightarrow \sigma_{1105}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1107} = \frac{\langle y := y + 1, \sigma_{1106} \rangle \longrightarrow \sigma_{1107} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1106} \rangle \longrightarrow \sigma_{1107}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1109} = \frac{\langle y := y + 1, \sigma_{1108} \rangle \longrightarrow \sigma_{1109} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1108} \rangle \longrightarrow \sigma_{1109}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1111} = \frac{\langle y := y + 1, \sigma_{1110} \rangle \longrightarrow \sigma_{1111} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1110} \rangle \longrightarrow \sigma_{1111}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1113} = \frac{\langle y := y + 1, \sigma_{1112} \rangle \longrightarrow \sigma_{1113} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1112} \rangle \longrightarrow \sigma_{1113}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1115} = \frac{\langle y := y + 1, \sigma_{1114} \rangle \longrightarrow \sigma_{1115} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1114} \rangle \longrightarrow \sigma_{1115}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1117} = \frac{\langle y := y + 1, \sigma_{1116} \rangle \longrightarrow \sigma_{1117} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1116} \rangle \longrightarrow \sigma_{1117}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1119} = \frac{\langle y := y + 1, \sigma_{1118} \rangle \longrightarrow \sigma_{1119} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1118} \rangle \longrightarrow \sigma_{1119}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1147} = \frac{\langle y := y + 1, \sigma_{1146} \rangle \longrightarrow \sigma_{1147} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1146} \rangle \longrightarrow \sigma_{1147}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1149} = \frac{\langle y := y + 1, \sigma_{1148} \rangle \longrightarrow \sigma_{1149} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1148} \rangle \longrightarrow \sigma_{1149}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1151} = \frac{\langle y := y + 1, \sigma_{1150} \rangle \longrightarrow \sigma_{1151} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1150} \rangle \longrightarrow \sigma_{1151}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1153} = \frac{\langle y := y + 1, \sigma_{1152} \rangle \longrightarrow \sigma_{1153} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1152} \rangle \longrightarrow \sigma_{1153}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1155} = \frac{\langle y := y + 1, \sigma_{1154} \rangle \longrightarrow \sigma_{1155} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1154} \rangle \longrightarrow \sigma_{1155}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1157} = \frac{\langle y := y + 1, \sigma_{1156} \rangle \longrightarrow \sigma_{1157} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1156} \rangle \longrightarrow \sigma_{1157}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1159} = \frac{\langle y := y + 1, \sigma_{1158} \rangle \longrightarrow \sigma_{1159} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1158} \rangle \longrightarrow \sigma_{1159}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1161} = \frac{\langle y := y + 1, \sigma_{1160} \rangle \longrightarrow \sigma_{1161} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1160} \rangle \longrightarrow \sigma_{1161}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1163} = \frac{\langle y := y + 1, \sigma_{1162} \rangle \longrightarrow \sigma_{1163} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1162} \rangle \longrightarrow \sigma_{1163}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1165} = \frac{\langle y := y + 1, \sigma_{1164} \rangle \longrightarrow \sigma_{1165} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1164} \rangle \longrightarrow \sigma_{1165}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1167} = \frac{\langle y := y + 1, \sigma_{1166} \rangle \longrightarrow \sigma_{1167} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1166} \rangle \longrightarrow \sigma_{1167}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1169} = \frac{\langle y := y + 1, \sigma_{1168} \rangle \longrightarrow \sigma_{1169} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1168} \rangle \longrightarrow \sigma_{1169}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1197} = \frac{\langle y := y + 1, \sigma_{1196} \rangle \longrightarrow \sigma_{1197} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1196} \rangle \longrightarrow \sigma_{1197}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1199} = \frac{\langle y := y + 1, \sigma_{1198} \rangle \longrightarrow \sigma_{1199} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1198} \rangle \longrightarrow \sigma_{1199}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1201} = \frac{\langle y := y + 1, \sigma_{1200} \rangle \longrightarrow \sigma_{1201} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1200} \rangle \longrightarrow \sigma_{1201}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1203} = \frac{\langle y := y + 1, \sigma_{1202} \rangle \longrightarrow \sigma_{1203} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1202} \rangle \longrightarrow \sigma_{1203}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1205} = \frac{\langle y := y + 1, \sigma_{1204} \rangle \longrightarrow \sigma_{1205} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1204} \rangle \longrightarrow \sigma_{1205}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1207} = \frac{\langle y := y + 1, \sigma_{1206} \rangle \longrightarrow \sigma_{1207} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1206} \rangle \longrightarrow \sigma_{1207}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1209} = \frac{\langle y := y + 1, \sigma_{1208} \rangle \longrightarrow \sigma_{1209} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1208} \rangle \longrightarrow \sigma_{1209}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1211} = \frac{\langle y := y + 1, \sigma_{1210} \rangle \longrightarrow \sigma_{1211} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1210} \rangle \longrightarrow \sigma_{1211}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1213} = \frac{\langle y := y + 1, \sigma_{1212} \rangle \longrightarrow \sigma_{1213} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1212} \rangle \longrightarrow \sigma_{1213}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1215} = \frac{\langle y := y + 1, \sigma_{1214} \rangle \longrightarrow \sigma_{1215} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1214} \rangle \longrightarrow \sigma_{1215}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1217} = \frac{\langle y := y + 1, \sigma_{1216} \rangle \longrightarrow \sigma_{1217} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1216} \rangle \longrightarrow \sigma_{1217}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1219} = \frac{\langle y := y + 1, \sigma_{1218} \rangle \longrightarrow \sigma_{1219} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1218} \rangle \longrightarrow \sigma_{1219}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1247} = \frac{\langle y := y + 1, \sigma_{1246} \rangle \longrightarrow \sigma_{1247} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1246} \rangle \longrightarrow \sigma_{1247}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1249} = \frac{\langle y := y + 1, \sigma_{1248} \rangle \longrightarrow \sigma_{1249} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1248} \rangle \longrightarrow \sigma_{1249}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1251} = \frac{\langle y := y + 1, \sigma_{1250} \rangle \longrightarrow \sigma_{1251} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1250} \rangle \longrightarrow \sigma_{1251}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1253} = \frac{\langle y := y + 1, \sigma_{1252} \rangle \longrightarrow \sigma_{1253} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1252} \rangle \longrightarrow \sigma_{1253}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1255} = \frac{\langle y := y + 1, \sigma_{1254} \rangle \longrightarrow \sigma_{1255} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1254} \rangle \longrightarrow \sigma_{1255}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1257} = \frac{\langle y := y + 1, \sigma_{1256} \rangle \longrightarrow \sigma_{1257} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1256} \rangle \longrightarrow \sigma_{1257}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1259} = \frac{\langle y := y + 1, \sigma_{1258} \rangle \longrightarrow \sigma_{1259} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1258} \rangle \longrightarrow \sigma_{1259}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1261} = \frac{\langle y := y + 1, \sigma_{1260} \rangle \longrightarrow \sigma_{1261} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1260} \rangle \longrightarrow \sigma_{1261}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1263} = \frac{\langle y := y + 1, \sigma_{1262} \rangle \longrightarrow \sigma_{1263} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1262} \rangle \longrightarrow \sigma_{1263}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1265} = \frac{\langle y := y + 1, \sigma_{1264} \rangle \longrightarrow \sigma_{1265} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1264} \rangle \longrightarrow \sigma_{1265}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1267} = \frac{\langle y := y + 1, \sigma_{1266} \rangle \longrightarrow \sigma_{1267} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1266} \rangle \longrightarrow \sigma_{1267}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1269} = \frac{\langle y := y + 1, \sigma_{1268} \rangle \longrightarrow \sigma_{1269} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1268} \rangle \longrightarrow \sigma_{1269}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1297} = \frac{\langle y := y + 1, \sigma_{1296} \rangle \longrightarrow \sigma_{1297} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1296} \rangle \longrightarrow \sigma_{1297}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1299} = \frac{\langle y := y + 1, \sigma_{1298} \rangle \longrightarrow \sigma_{1299} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1298} \rangle \longrightarrow \sigma_{1299}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1301} = \frac{\langle y := y + 1, \sigma_{1300} \rangle \longrightarrow \sigma_{1301} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1300} \rangle \longrightarrow \sigma_{1301}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1303} = \frac{\langle y := y + 1, \sigma_{1302} \rangle \longrightarrow \sigma_{1303} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1302} \rangle \longrightarrow \sigma_{1303}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1305} = \frac{\langle y := y + 1, \sigma_{1304} \rangle \longrightarrow \sigma_{1305} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1304} \rangle \longrightarrow \sigma_{1305}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1307} = \frac{\langle y := y + 1, \sigma_{1306} \rangle \longrightarrow \sigma_{1307} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1306} \rangle \longrightarrow \sigma_{1307}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1309} = \frac{\langle y := y + 1, \sigma_{1308} \rangle \longrightarrow \sigma_{1309} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1308} \rangle \longrightarrow \sigma_{1309}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1311} = \frac{\langle y := y + 1, \sigma_{1310} \rangle \longrightarrow \sigma_{1311} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1310} \rangle \longrightarrow \sigma_{1311}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1313} = \frac{\langle y := y + 1, \sigma_{1312} \rangle \longrightarrow \sigma_{1313} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1312} \rangle \longrightarrow \sigma_{1313}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1315} = \frac{\langle y := y + 1, \sigma_{1314} \rangle \longrightarrow \sigma_{1315} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1314} \rangle \longrightarrow \sigma_{1315}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1317} = \frac{\langle y := y + 1, \sigma_{1316} \rangle \longrightarrow \sigma_{1317} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1316} \rangle \longrightarrow \sigma_{1317}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1319} = \frac{\langle y := y + 1, \sigma_{1318} \rangle \longrightarrow \sigma_{1319} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1318} \rangle \longrightarrow \sigma_{1319}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1347} = \frac{\langle y := y + 1, \sigma_{1346} \rangle \longrightarrow \sigma_{1347} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1346} \rangle \longrightarrow \sigma_{1347}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1349} = \frac{\langle y := y + 1, \sigma_{1348} \rangle \longrightarrow \sigma_{1349} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1348} \rangle \longrightarrow \sigma_{1349}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1351} = \frac{\langle y := y + 1, \sigma_{1350} \rangle \longrightarrow \sigma_{1351} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1350} \rangle \longrightarrow \sigma_{1351}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1353} = \frac{\langle y := y + 1, \sigma_{1352} \rangle \longrightarrow \sigma_{1353} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1352} \rangle \longrightarrow \sigma_{1353}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1355} = \frac{\langle y := y + 1, \sigma_{1354} \rangle \longrightarrow \sigma_{1355} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1354} \rangle \longrightarrow \sigma_{1355}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1357} = \frac{\langle y := y + 1, \sigma_{1356} \rangle \longrightarrow \sigma_{1357} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1356} \rangle \longrightarrow \sigma_{1357}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1359} = \frac{\langle y := y + 1, \sigma_{1358} \rangle \longrightarrow \sigma_{1359} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1358} \rangle \longrightarrow \sigma_{1359}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1361} = \frac{\langle y := y + 1, \sigma_{1360} \rangle \longrightarrow \sigma_{1361} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1360} \rangle \longrightarrow \sigma_{1361}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1363} = \frac{\langle y := y + 1, \sigma_{1362} \rangle \longrightarrow \sigma_{1363} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1362} \rangle \longrightarrow \sigma_{1363}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1365} = \frac{\langle y := y + 1, \sigma_{1364} \rangle \longrightarrow \sigma_{1365} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1364} \rangle \longrightarrow \sigma_{1365}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1367} = \frac{\langle y := y + 1, \sigma_{1366} \rangle \longrightarrow \sigma_{1367} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1366} \rangle \longrightarrow \sigma_{1367}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1369} = \frac{\langle y := y + 1, \sigma_{1368} \rangle \longrightarrow \sigma_{1369} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1368} \rangle \longrightarrow \sigma_{1369}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1397} = \frac{\langle y := y + 1, \sigma_{1396} \rangle \longrightarrow \sigma_{1397} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1396} \rangle \longrightarrow \sigma_{1397}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1399} = \frac{\langle y := y + 1, \sigma_{1398} \rangle \longrightarrow \sigma_{1399} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1398} \rangle \longrightarrow \sigma_{1399}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1401} = \frac{\langle y := y + 1, \sigma_{1400} \rangle \longrightarrow \sigma_{1401} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1400} \rangle \longrightarrow \sigma_{1401}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1403} = \frac{\langle y := y + 1, \sigma_{1402} \rangle \longrightarrow \sigma_{1403} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1402} \rangle \longrightarrow \sigma_{1403}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1405} = \frac{\langle y := y + 1, \sigma_{1404} \rangle \longrightarrow \sigma_{1405} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1404} \rangle \longrightarrow \sigma_{1405}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1407} = \frac{\langle y := y + 1, \sigma_{1406} \rangle \longrightarrow \sigma_{1407} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1406} \rangle \longrightarrow \sigma_{1407}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1409} = \frac{\langle y := y + 1, \sigma_{1408} \rangle \longrightarrow \sigma_{1409} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1408} \rangle \longrightarrow \sigma_{1409}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1411} = \frac{\langle y := y + 1, \sigma_{1410} \rangle \longrightarrow \sigma_{1411} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1410} \rangle \longrightarrow \sigma_{1411}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1413} = \frac{\langle y := y + 1, \sigma_{1412} \rangle \longrightarrow \sigma_{1413} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1412} \rangle \longrightarrow \sigma_{1413}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1415} = \frac{\langle y := y + 1, \sigma_{1414} \rangle \longrightarrow \sigma_{1415} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1414} \rangle \longrightarrow \sigma_{1415}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1417} = \frac{\langle y := y + 1, \sigma_{1416} \rangle \longrightarrow \sigma_{1417} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1416} \rangle \longrightarrow \sigma_{1417}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1419} = \frac{\langle y := y + 1, \sigma_{1418} \rangle \longrightarrow \sigma_{1419} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1418} \rangle \longrightarrow \sigma_{1419}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1445} = \frac{\langle y := y + 1, \sigma_{1444} \rangle \longrightarrow \sigma_{1445} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1444} \rangle \longrightarrow \sigma_{1445}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1447} = \frac{\langle y := y + 1, \sigma_{1446} \rangle \longrightarrow \sigma_{1447} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1446} \rangle \longrightarrow \sigma_{1447}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1449} = \frac{\langle y := y + 1, \sigma_{1448} \rangle \longrightarrow \sigma_{1449} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1448} \rangle \longrightarrow \sigma_{1449}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1451} = \frac{\langle y := y + 1, \sigma_{1450} \rangle \longrightarrow \sigma_{1451} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1450} \rangle \longrightarrow \sigma_{1451}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1453} = \frac{\langle y := y + 1, \sigma_{1452} \rangle \longrightarrow \sigma_{1453} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1452} \rangle \longrightarrow \sigma_{1453}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1455} = \frac{\langle y := y + 1, \sigma_{1454} \rangle \longrightarrow \sigma_{1455} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1454} \rangle \longrightarrow \sigma_{1455}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1457} = \frac{\langle y := y + 1, \sigma_{1456} \rangle \longrightarrow \sigma_{1457} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1456} \rangle \longrightarrow \sigma_{1457}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1459} = \frac{\langle y := y + 1, \sigma_{1458} \rangle \longrightarrow \sigma_{1459} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1458} \rangle \longrightarrow \sigma_{1459}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1461} = \frac{\langle y := y + 1, \sigma_{1460} \rangle \longrightarrow \sigma_{1461} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1460} \rangle \longrightarrow \sigma_{1461}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1463} = \frac{\langle y := y + 1, \sigma_{1462} \rangle \longrightarrow \sigma_{1463} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1462} \rangle \longrightarrow \sigma_{1463}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1465} = \frac{\langle y := y + 1, \sigma_{1464} \rangle \longrightarrow \sigma_{1465} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1464} \rangle \longrightarrow \sigma_{1465}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1467} = \frac{\langle y := y + 1, \sigma_{1466} \rangle \longrightarrow \sigma_{1467} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1466} \rangle \longrightarrow \sigma_{1467}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1469} = \frac{\langle y := y + 1, \sigma_{1468} \rangle \longrightarrow \sigma_{1469} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1468} \rangle \longrightarrow \sigma_{1469}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1497} = \frac{\langle y := y + 1, \sigma_{1496} \rangle \longrightarrow \sigma_{1497} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1496} \rangle \longrightarrow \sigma_{1497}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1499} = \frac{\langle y := y + 1, \sigma_{1498} \rangle \longrightarrow \sigma_{1499} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1498} \rangle \longrightarrow \sigma_{1499}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1501} = \frac{\langle y := y + 1, \sigma_{1500} \rangle \longrightarrow \sigma_{1501} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1500} \rangle \longrightarrow \sigma_{1501}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1503} = \frac{\langle y := y + 1, \sigma_{1502} \rangle \longrightarrow \sigma_{1503} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1502} \rangle \longrightarrow \sigma_{1503}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1505} = \frac{\langle y := y + 1, \sigma_{1504} \rangle \longrightarrow \sigma_{1505} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1504} \rangle \longrightarrow \sigma_{1505}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1507} = \frac{\langle y := y + 1, \sigma_{1506} \rangle \longrightarrow \sigma_{1507} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1506} \rangle \longrightarrow \sigma_{1507}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1509} = \frac{\langle y := y + 1, \sigma_{1508} \rangle \longrightarrow \sigma_{1509} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1508} \rangle \longrightarrow \sigma_{1509}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1511} = \frac{\langle y := y + 1, \sigma_{1510} \rangle \longrightarrow \sigma_{1511} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1510} \rangle \longrightarrow \sigma_{1511}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1513} = \frac{\langle y := y + 1, \sigma_{1512} \rangle \longrightarrow \sigma_{1513} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1512} \rangle \longrightarrow \sigma_{1513}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1515} = \frac{\langle y := y + 1, \sigma_{1514} \rangle \longrightarrow \sigma_{1515} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1514} \rangle \longrightarrow \sigma_{1515}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1517} = \frac{\langle y := y + 1, \sigma_{1516} \rangle \longrightarrow \sigma_{1517} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1516} \rangle \longrightarrow \sigma_{1517}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1519} = \frac{\langle y := y + 1, \sigma_{1518} \rangle \longrightarrow \sigma_{1519} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1518} \rangle \longrightarrow \sigma_{1519}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1545} = \frac{\langle y := y + 1, \sigma_{1544} \rangle \longrightarrow \sigma_{1545} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1544} \rangle \longrightarrow \sigma_{1545}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1547} = \frac{\langle y := y + 1, \sigma_{1546} \rangle \longrightarrow \sigma_{1547} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1546} \rangle \longrightarrow \sigma_{1547}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1549} = \frac{\langle y := y + 1, \sigma_{1548} \rangle \longrightarrow \sigma_{1549} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1548} \rangle \longrightarrow \sigma_{1549}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1551} = \frac{\langle y := y + 1, \sigma_{1550} \rangle \longrightarrow \sigma_{1551} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1550} \rangle \longrightarrow \sigma_{1551}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1553} = \frac{\langle y := y + 1, \sigma_{1552} \rangle \longrightarrow \sigma_{1553} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1552} \rangle \longrightarrow \sigma_{1553}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1555} = \frac{\langle y := y + 1, \sigma_{1554} \rangle \longrightarrow \sigma_{1555} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1554} \rangle \longrightarrow \sigma_{1555}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1557} = \frac{\langle y := y + 1, \sigma_{1556} \rangle \longrightarrow \sigma_{1557} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1556} \rangle \longrightarrow \sigma_{1557}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1559} = \frac{\langle y := y + 1, \sigma_{1558} \rangle \longrightarrow \sigma_{1559} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1558} \rangle \longrightarrow \sigma_{1559}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1561} = \frac{\langle y := y + 1, \sigma_{1560} \rangle \longrightarrow \sigma_{1561} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1560} \rangle \longrightarrow \sigma_{1561}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1563} = \frac{\langle y := y + 1, \sigma_{1562} \rangle \longrightarrow \sigma_{1563} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1562} \rangle \longrightarrow \sigma_{1563}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1565} = \frac{\langle y := y + 1, \sigma_{1564} \rangle \longrightarrow \sigma_{1565} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1564} \rangle \longrightarrow \sigma_{1565}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1567} = \frac{\langle y := y + 1, \sigma_{1566} \rangle \longrightarrow \sigma_{1567} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1566} \rangle \longrightarrow \sigma_{1567}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1569} = \frac{\langle y := y + 1, \sigma_{1568} \rangle \longrightarrow \sigma_{1569} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1568} \rangle \longrightarrow \sigma_{1569}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1597} = \frac{\langle y := y + 1, \sigma_{1596} \rangle \longrightarrow \sigma_{1597} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1596} \rangle \longrightarrow \sigma_{1597}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1599} = \frac{\langle y := y + 1, \sigma_{1598} \rangle \longrightarrow \sigma_{1599} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1598} \rangle \longrightarrow \sigma_{1599}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1601} = \frac{\langle y := y + 1, \sigma_{1600} \rangle \longrightarrow \sigma_{1601} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1600} \rangle \longrightarrow \sigma_{1601}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1603} = \frac{\langle y := y + 1, \sigma_{1602} \rangle \longrightarrow \sigma_{1603} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1602} \rangle \longrightarrow \sigma_{1603}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1605} = \frac{\langle y := y + 1, \sigma_{1604} \rangle \longrightarrow \sigma_{1605} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1604} \rangle \longrightarrow \sigma_{1605}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1607} = \frac{\langle y := y + 1, \sigma_{1606} \rangle \longrightarrow \sigma_{1607} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1606} \rangle \longrightarrow \sigma_{1607}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1609} = \frac{\langle y := y + 1, \sigma_{1608} \rangle \longrightarrow \sigma_{1609} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1608} \rangle \longrightarrow \sigma_{1609}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1611} = \frac{\langle y := y + 1, \sigma_{1610} \rangle \longrightarrow \sigma_{1611} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1610} \rangle \longrightarrow \sigma_{1611}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1613} = \frac{\langle y := y + 1, \sigma_{1612} \rangle \longrightarrow \sigma_{1613} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1612} \rangle \longrightarrow \sigma_{1613}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1615} = \frac{\langle y := y + 1, \sigma_{1614} \rangle \longrightarrow \sigma_{1615} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1614} \rangle \longrightarrow \sigma_{1615}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1617} = \frac{\langle y := y + 1, \sigma_{1616} \rangle \longrightarrow \sigma_{1617} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1616} \rangle \longrightarrow \sigma_{1617}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1619} = \frac{\langle y := y + 1, \sigma_{1618} \rangle \longrightarrow \sigma_{1619} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1618} \rangle \longrightarrow \sigma_{1619}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1645} = \frac{\langle y := y + 1, \sigma_{1644} \rangle \longrightarrow \sigma_{1645} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1644} \rangle \longrightarrow \sigma_{1645}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1647} = \frac{\langle y := y + 1, \sigma_{1646} \rangle \longrightarrow \sigma_{1647} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1646} \rangle \longrightarrow \sigma_{1647}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1649} = \frac{\langle y := y + 1, \sigma_{1648} \rangle \longrightarrow \sigma_{1649} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1648} \rangle \longrightarrow \sigma_{1649}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1651} = \frac{\langle y := y + 1, \sigma_{1650} \rangle \longrightarrow \sigma_{1651} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1650} \rangle \longrightarrow \sigma_{1651}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1653} = \frac{\langle y := y + 1, \sigma_{1652} \rangle \longrightarrow \sigma_{1653} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1652} \rangle \longrightarrow \sigma_{1653}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1655} = \frac{\langle y := y + 1, \sigma_{1654} \rangle \longrightarrow \sigma_{1655} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1654} \rangle \longrightarrow \sigma_{1655}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1657} = \frac{\langle y := y + 1, \sigma_{1656} \rangle \longrightarrow \sigma_{1657} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1656} \rangle \longrightarrow \sigma_{1657}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1659} = \frac{\langle y := y + 1, \sigma_{1658} \rangle \longrightarrow \sigma_{1659} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1658} \rangle \longrightarrow \sigma_{1659}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1661} = \frac{\langle y := y + 1, \sigma_{1660} \rangle \longrightarrow \sigma_{1661} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1660} \rangle \longrightarrow \sigma_{1661}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1663} = \frac{\langle y := y + 1, \sigma_{1662} \rangle \longrightarrow \sigma_{1663} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1662} \rangle \longrightarrow \sigma_{1663}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1665} = \frac{\langle y := y + 1, \sigma_{1664} \rangle \longrightarrow \sigma_{1665} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1664} \rangle \longrightarrow \sigma_{1665}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1667} = \frac{\langle y := y + 1, \sigma_{1666} \rangle \longrightarrow \sigma_{1667} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1666} \rangle \longrightarrow \sigma_{1667}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1669} = \frac{\langle y := y + 1, \sigma_{1668} \rangle \longrightarrow \sigma_{1669} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1668} \rangle \longrightarrow \sigma_{1669}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1671} = \frac{\langle y := y + 1, \sigma_{1670} \rangle \longrightarrow \sigma_{1671} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1670} \rangle \longrightarrow \sigma_{1671}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1673} = \frac{\langle y := y + 1, \sigma_{1672} \rangle \longrightarrow \sigma_{1673} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1672} \rangle \longrightarrow \sigma_{1673}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1675} = \frac{\langle y := y + 1, \sigma_{1674} \rangle \longrightarrow \sigma_{1675} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1674} \rangle \longrightarrow \sigma_{1675}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1677} = \frac{\langle y := y + 1, \sigma_{1676} \rangle \longrightarrow \sigma_{1677} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1676} \rangle \longrightarrow \sigma_{1677}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1679} = \frac{\langle y := y + 1, \sigma_{1678} \rangle \longrightarrow \sigma_{1679} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1678} \rangle \longrightarrow \sigma_{1679}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1681} = \frac{\langle y := y + 1, \sigma_{1680} \rangle \longrightarrow \sigma_{1681} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1680} \rangle \longrightarrow \sigma_{1681}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1683} = \frac{\langle y := y + 1, \sigma_{1682} \rangle \longrightarrow \sigma_{1683} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1682} \rangle \longrightarrow \sigma_{1683}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1685} = \frac{\langle y := y + 1, \sigma_{1684} \rangle \longrightarrow \sigma_{1685} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1684} \rangle \longrightarrow \sigma_{1685}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1687} = \frac{\langle y := y + 1, \sigma_{1686} \rangle \longrightarrow \sigma_{1687} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1686} \rangle \longrightarrow \sigma_{1687}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1689} = \frac{\langle y := y + 1, \sigma_{1688} \rangle \longrightarrow \sigma_{1689} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1688} \rangle \longrightarrow \sigma_{1689}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1691} = \frac{\langle y := y + 1, \sigma_{1690} \rangle \longrightarrow \sigma_{1691} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1690} \rangle \longrightarrow \sigma_{1691}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1693} = \frac{\langle y := y + 1, \sigma_{1692} \rangle \longrightarrow \sigma_{1693} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1692} \rangle \longrightarrow \sigma_{1693}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1695} = \frac{\langle y := y + 1, \sigma_{1694} \rangle \longrightarrow \sigma_{1695} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1694} \rangle \longrightarrow \sigma_{1695}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1697} = \frac{\langle y := y + 1, \sigma_{1696} \rangle \longrightarrow \sigma_{1697} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1696} \rangle \longrightarrow \sigma_{1697}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1699} = \frac{\langle y := y + 1, \sigma_{1698} \rangle \longrightarrow \sigma_{1699} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1698} \rangle \longrightarrow \sigma_{1699}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1701} = \frac{\langle y := y + 1, \sigma_{1700} \rangle \longrightarrow \sigma_{1701} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1700} \rangle \longrightarrow \sigma_{1701}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1703} = \frac{\langle y := y + 1, \sigma_{1702} \rangle \longrightarrow \sigma_{1703} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1702} \rangle \longrightarrow \sigma_{1703}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1705} = \frac{\langle y := y + 1, \sigma_{1704} \rangle \longrightarrow \sigma_{1705} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1704} \rangle \longrightarrow \sigma_{1705}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1707} = \frac{\langle y := y + 1, \sigma_{1706} \rangle \longrightarrow \sigma_{1707} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1706} \rangle \longrightarrow \sigma_{1707}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1709} = \frac{\langle y := y + 1, \sigma_{1708} \rangle \longrightarrow \sigma_{1709} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1708} \rangle \longrightarrow \sigma_{1709}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1711} = \frac{\langle y := y + 1, \sigma_{1710} \rangle \longrightarrow \sigma_{1711} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1710} \rangle \longrightarrow \sigma_{1711}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1713} = \frac{\langle y := y + 1, \sigma_{1712} \rangle \longrightarrow \sigma_{1713} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1712} \rangle \longrightarrow \sigma_{1713}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1715} = \frac{\langle y := y + 1, \sigma_{1714} \rangle \longrightarrow \sigma_{1715} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1714} \rangle \longrightarrow \sigma_{1715}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1717} = \frac{\langle y := y + 1, \sigma_{1716} \rangle \longrightarrow \sigma_{1717} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1716} \rangle \longrightarrow \sigma_{1717}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1719} = \frac{\langle y := y + 1, \sigma_{1718} \rangle \longrightarrow \sigma_{1719} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1718} \rangle \longrightarrow \sigma_{1719}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1747} = \frac{\langle y := y + 1, \sigma_{1746} \rangle \longrightarrow \sigma_{1747} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1746} \rangle \longrightarrow \sigma_{1747}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1749} = \frac{\langle y := y + 1, \sigma_{1748} \rangle \longrightarrow \sigma_{1749} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1748} \rangle \longrightarrow \sigma_{1749}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1751} = \frac{\langle y := y + 1, \sigma_{1750} \rangle \longrightarrow \sigma_{1751} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1750} \rangle \longrightarrow \sigma_{1751}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1753} = \frac{\langle y := y + 1, \sigma_{1752} \rangle \longrightarrow \sigma_{1753} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1752} \rangle \longrightarrow \sigma_{1753}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1755} = \frac{\langle y := y + 1, \sigma_{1754} \rangle \longrightarrow \sigma_{1755} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1754} \rangle \longrightarrow \sigma_{1755}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1757} = \frac{\langle y := y + 1, \sigma_{1756} \rangle \longrightarrow \sigma_{1757} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1756} \rangle \longrightarrow \sigma_{1757}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1759} = \frac{\langle y := y + 1, \sigma_{1758} \rangle \longrightarrow \sigma_{1759} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1758} \rangle \longrightarrow \sigma_{1759}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1761} = \frac{\langle y := y + 1, \sigma_{1760} \rangle \longrightarrow \sigma_{1761} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1760} \rangle \longrightarrow \sigma_{1761}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1763} = \frac{\langle y := y + 1, \sigma_{1762} \rangle \longrightarrow \sigma_{1763} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1762} \rangle \longrightarrow \sigma_{1763}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1765} = \frac{\langle y := y + 1, \sigma_{1764} \rangle \longrightarrow \sigma_{1765} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1764} \rangle \longrightarrow \sigma_{1765}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1767} = \frac{\langle y := y + 1, \sigma_{1766} \rangle \longrightarrow \sigma_{1767} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1766} \rangle \longrightarrow \sigma_{1767}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1769} = \frac{\langle y := y + 1, \sigma_{1768} \rangle \longrightarrow \sigma_{1769} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1768} \rangle \longrightarrow \sigma_{1769}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1797} = \frac{\langle y := y + 1, \sigma_{1796} \rangle \longrightarrow \sigma_{1797} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1796} \rangle \longrightarrow \sigma_{1797}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1799} = \frac{\langle y := y + 1, \sigma_{1798} \rangle \longrightarrow \sigma_{1799} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1798} \rangle \longrightarrow \sigma_{1799}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1801} = \frac{\langle y := y + 1, \sigma_{1800} \rangle \longrightarrow \sigma_{1801} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1800} \rangle \longrightarrow \sigma_{1801}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1803} = \frac{\langle y := y + 1, \sigma_{1802} \rangle \longrightarrow \sigma_{1803} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1802} \rangle \longrightarrow \sigma_{1803}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1805} = \frac{\langle y := y + 1, \sigma_{1804} \rangle \longrightarrow \sigma_{1805} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1804} \rangle \longrightarrow \sigma_{1805}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1807} = \frac{\langle y := y + 1, \sigma_{1806} \rangle \longrightarrow \sigma_{1807} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1806} \rangle \longrightarrow \sigma_{1807}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1809} = \frac{\langle y := y + 1, \sigma_{1808} \rangle \longrightarrow \sigma_{1809} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1808} \rangle \longrightarrow \sigma_{1809}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1811} = \frac{\langle y := y + 1, \sigma_{1810} \rangle \longrightarrow \sigma_{1811} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1810} \rangle \longrightarrow \sigma_{1811}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1813} = \frac{\langle y := y + 1, \sigma_{1812} \rangle \longrightarrow \sigma_{1813} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1812} \rangle \longrightarrow \sigma_{1813}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1815} = \frac{\langle y := y + 1, \sigma_{1814} \rangle \longrightarrow \sigma_{1815} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1814} \rangle \longrightarrow \sigma_{1815}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1817} = \frac{\langle y := y + 1, \sigma_{1816} \rangle \longrightarrow \sigma_{1817} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1816} \rangle \longrightarrow \sigma_{1817}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1819} = \frac{\langle y := y + 1, \sigma_{1818} \rangle \longrightarrow \sigma_{1819} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1818} \rangle \longrightarrow \sigma_{1819}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1845} = \frac{\langle y := y + 1, \sigma_{1844} \rangle \longrightarrow \sigma_{1845} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1844} \rangle \longrightarrow \sigma_{1845}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1847} = \frac{\langle y := y + 1, \sigma_{1846} \rangle \longrightarrow \sigma_{1847} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1846} \rangle \longrightarrow \sigma_{1847}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1849} = \frac{\langle y := y + 1, \sigma_{1848} \rangle \longrightarrow \sigma_{1849} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1848} \rangle \longrightarrow \sigma_{1849}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1851} = \frac{\langle y := y + 1, \sigma_{1850} \rangle \longrightarrow \sigma_{1851} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1850} \rangle \longrightarrow \sigma_{1851}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1853} = \frac{\langle y := y + 1, \sigma_{1852} \rangle \longrightarrow \sigma_{1853} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1852} \rangle \longrightarrow \sigma_{1853}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1855} = \frac{\langle y := y + 1, \sigma_{1854} \rangle \longrightarrow \sigma_{1855} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1854} \rangle \longrightarrow \sigma_{1855}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1857} = \frac{\langle y := y + 1, \sigma_{1856} \rangle \longrightarrow \sigma_{1857} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1856} \rangle \longrightarrow \sigma_{1857}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1859} = \frac{\langle y := y + 1, \sigma_{1858} \rangle \longrightarrow \sigma_{1859} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1858} \rangle \longrightarrow \sigma_{1859}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1861} = \frac{\langle y := y + 1, \sigma_{1860} \rangle \longrightarrow \sigma_{1861} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1860} \rangle \longrightarrow \sigma_{1861}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1863} = \frac{\langle y := y + 1, \sigma_{1862} \rangle \longrightarrow \sigma_{1863} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1862} \rangle \longrightarrow \sigma_{1863}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1865} = \frac{\langle y := y + 1, \sigma_{1864} \rangle \longrightarrow \sigma_{1865} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1864} \rangle \longrightarrow \sigma_{1865}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1867} = \frac{\langle y := y + 1, \sigma_{1866} \rangle \longrightarrow \sigma_{1867} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1866} \rangle \longrightarrow \sigma_{1867}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1869} = \frac{\langle y := y + 1, \sigma_{1868} \rangle \longrightarrow \sigma_{1869} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1868} \rangle \longrightarrow \sigma_{1869}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1897} = \frac{\langle y := y + 1, \sigma_{1896} \rangle \longrightarrow \sigma_{1897} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1896} \rangle \longrightarrow \sigma_{1897}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1899} = \frac{\langle y := y + 1, \sigma_{1898} \rangle \longrightarrow \sigma_{1899} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1898} \rangle \longrightarrow \sigma_{1899}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1901} = \frac{\langle y := y + 1, \sigma_{1900} \rangle \longrightarrow \sigma_{1901} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1900} \rangle \longrightarrow \sigma_{1901}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1903} = \frac{\langle y := y + 1, \sigma_{1902} \rangle \longrightarrow \sigma_{1903} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1902} \rangle \longrightarrow \sigma_{1903}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1905} = \frac{\langle y := y + 1, \sigma_{1904} \rangle \longrightarrow \sigma_{1905} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1904} \rangle \longrightarrow \sigma_{1905}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1907} = \frac{\langle y := y + 1, \sigma_{1906} \rangle \longrightarrow \sigma_{1907} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1906} \rangle \longrightarrow \sigma_{1907}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1909} = \frac{\langle y := y + 1, \sigma_{1908} \rangle \longrightarrow \sigma_{1909} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1908} \rangle \longrightarrow \sigma_{1909}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1911} = \frac{\langle y := y + 1, \sigma_{1910} \rangle \longrightarrow \sigma_{1911} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1910} \rangle \longrightarrow \sigma_{1911}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1913} = \frac{\langle y := y + 1, \sigma_{1912} \rangle \longrightarrow \sigma_{1913} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1912} \rangle \longrightarrow \sigma_{1913}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1915} = \frac{\langle y := y + 1, \sigma_{1914} \rangle \longrightarrow \sigma_{1915} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1914} \rangle \longrightarrow \sigma_{1915}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1917} = \frac{\langle y := y + 1, \sigma_{1916} \rangle \longrightarrow \sigma_{1917} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1916} \rangle \longrightarrow \sigma_{1917}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1919} = \frac{\langle y := y + 1, \sigma_{1918} \rangle \longrightarrow \sigma_{1919} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1918} \rangle \longrightarrow \sigma_{1919}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1945} = \frac{\langle y := y + 1, \sigma_{1944} \rangle \longrightarrow \sigma_{1945} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1944} \rangle \longrightarrow \sigma_{1945}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1947} = \frac{\langle y := y + 1, \sigma_{1946} \rangle \longrightarrow \sigma_{1947} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1946} \rangle \longrightarrow \sigma_{1947}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1949} = \frac{\langle y := y + 1, \sigma_{1948} \rangle \longrightarrow \sigma_{1949} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1948} \rangle \longrightarrow \sigma_{1949}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1951} = \frac{\langle y := y + 1, \sigma_{1950} \rangle \longrightarrow \sigma_{1951} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1950} \rangle \longrightarrow \sigma_{1951}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1953} = \frac{\langle y := y + 1, \sigma_{1952} \rangle \longrightarrow \sigma_{1953} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1952} \rangle \longrightarrow \sigma_{1953}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1955} = \frac{\langle y := y + 1, \sigma_{1954} \rangle \longrightarrow \sigma_{1955} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1954} \rangle \longrightarrow \sigma_{1955}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1957} = \frac{\langle y := y + 1, \sigma_{1956} \rangle \longrightarrow \sigma_{1957} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1956} \rangle \longrightarrow \sigma_{1957}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1959} = \frac{\langle y := y + 1, \sigma_{1958} \rangle \longrightarrow \sigma_{1959} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1958} \rangle \longrightarrow \sigma_{1959}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1961} = \frac{\langle y := y + 1, \sigma_{1960} \rangle \longrightarrow \sigma_{1961} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1960} \rangle \longrightarrow \sigma_{1961}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1963} = \frac{\langle y := y + 1, \sigma_{1962} \rangle \longrightarrow \sigma_{1963} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1962} \rangle \longrightarrow \sigma_{1963}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1965} = \frac{\langle y := y + 1, \sigma_{1964} \rangle \longrightarrow \sigma_{1965} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1964} \rangle \longrightarrow \sigma_{1965}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1967} = \frac{\langle y := y + 1, \sigma_{1966} \rangle \longrightarrow \sigma_{1967} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1966} \rangle \longrightarrow \sigma_{1967}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1969} = \frac{\langle y := y + 1, \sigma_{1968} \rangle \longrightarrow \sigma_{1969} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1968} \rangle \longrightarrow \sigma_{1969}} [\text{assert} - \text{tt}_{\text{ns}}]$$

$$\begin{aligned}
\mathcal{T}_{1997} &= \frac{\langle y := y + 1, \sigma_{1996} \rangle \longrightarrow \sigma_{1997} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1996} \rangle \longrightarrow \sigma_{1997}} [\text{assert} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1999} &= \frac{\langle y := y + 1, \sigma_{1998} \rangle \longrightarrow \sigma_{1999} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{1998} \rangle \longrightarrow \sigma_{1999}} [\text{assert} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{2001} &= \frac{\langle y := y + 1, \sigma_{2000} \rangle \longrightarrow \sigma_{2001} [\text{ass}_{\text{ns}}]}{\langle S_1, \sigma_{2000} \rangle \longrightarrow \sigma_{2001}} [\text{assert} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{2002} &= \frac{\langle x := x + 1, \sigma_{2001} \rangle \longrightarrow \sigma_{2002} [\text{ass}_{\text{ns}}] \quad \langle S_1, \sigma_{2002} \rangle \longrightarrow \perp [\text{assert} - \text{ff}_{\text{ns}}]}{\langle S_2, \sigma_{2001} \rangle \longrightarrow \perp} [\text{for} - \text{abort}_{\text{ns}}] \\
\mathcal{T}_{2000} &= \frac{\langle x := x + 1, \sigma_{1999} \rangle \longrightarrow \sigma_{2000} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{2001} \quad \mathcal{T}_{2002}}{\langle S_2, \sigma_{1999} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1998} &= \frac{\langle x := x + 1, \sigma_{1997} \rangle \longrightarrow \sigma_{1998} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1999} \quad \mathcal{T}_{2000}}{\langle S_2, \sigma_{1997} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1996} &= \frac{\langle x := x + 1, \sigma_{1995} \rangle \longrightarrow \sigma_{1996} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1997} \quad \mathcal{T}_{1998}}{\langle S_2, \sigma_{1995} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1994} &= \frac{\langle x := x + 1, \sigma_{1993} \rangle \longrightarrow \sigma_{1994} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1995} \quad \mathcal{T}_{1996}}{\langle S_2, \sigma_{1993} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1992} &= \frac{\langle x := x + 1, \sigma_{1991} \rangle \longrightarrow \sigma_{1992} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1993} \quad \mathcal{T}_{1994}}{\langle S_2, \sigma_{1991} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1990} &= \frac{\langle x := x + 1, \sigma_{1989} \rangle \longrightarrow \sigma_{1990} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1991} \quad \mathcal{T}_{1992}}{\langle S_2, \sigma_{1989} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1988} &= \frac{\langle x := x + 1, \sigma_{1987} \rangle \longrightarrow \sigma_{1988} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1989} \quad \mathcal{T}_{1990}}{\langle S_2, \sigma_{1987} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1986} &= \frac{\langle x := x + 1, \sigma_{1985} \rangle \longrightarrow \sigma_{1986} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1987} \quad \mathcal{T}_{1988}}{\langle S_2, \sigma_{1985} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]
\end{aligned}$$

$$\mathcal{T}_{1984} = \frac{\langle x := x + 1, \sigma_{1983} \rangle \longrightarrow \sigma_{1984} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1985} \quad \mathcal{T}_{1986}}{\langle S_2, \sigma_{1983} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1982} = \frac{\langle x := x + 1, \sigma_{1981} \rangle \longrightarrow \sigma_{1982} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1983} \quad \mathcal{T}_{1984}}{\langle S_2, \sigma_{1981} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1980} = \frac{\langle x := x + 1, \sigma_{1979} \rangle \longrightarrow \sigma_{1980} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1981} \quad \mathcal{T}_{1982}}{\langle S_2, \sigma_{1979} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1978} = \frac{\langle x := x + 1, \sigma_{1977} \rangle \longrightarrow \sigma_{1978} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1979} \quad \mathcal{T}_{1980}}{\langle S_2, \sigma_{1977} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1976} = \frac{\langle x := x + 1, \sigma_{1975} \rangle \longrightarrow \sigma_{1976} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1977} \quad \mathcal{T}_{1978}}{\langle S_2, \sigma_{1975} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1974} = \frac{\langle x := x + 1, \sigma_{1973} \rangle \longrightarrow \sigma_{1974} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1975} \quad \mathcal{T}_{1976}}{\langle S_2, \sigma_{1973} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1972} = \frac{\langle x := x + 1, \sigma_{1971} \rangle \longrightarrow \sigma_{1972} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1973} \quad \mathcal{T}_{1974}}{\langle S_2, \sigma_{1971} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1970} = \frac{\langle x := x + 1, \sigma_{1969} \rangle \longrightarrow \sigma_{1970} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1971} \quad \mathcal{T}_{1972}}{\langle S_2, \sigma_{1969} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1968} = \frac{\langle x := x + 1, \sigma_{1967} \rangle \longrightarrow \sigma_{1968} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1969} \quad \mathcal{T}_{1970}}{\langle S_2, \sigma_{1967} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1966} = \frac{\langle x := x + 1, \sigma_{1965} \rangle \longrightarrow \sigma_{1966} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1967} \quad \mathcal{T}_{1968}}{\langle S_2, \sigma_{1965} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1964} = \frac{\langle x := x + 1, \sigma_{1963} \rangle \longrightarrow \sigma_{1964} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1965} \quad \mathcal{T}_{1966}}{\langle S_2, \sigma_{1963} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1962} = \frac{\langle x := x + 1, \sigma_{1961} \rangle \longrightarrow \sigma_{1962} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1963} \quad \mathcal{T}_{1964}}{\langle S_2, \sigma_{1961} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1960} = \frac{\langle x := x + 1, \sigma_{1959} \rangle \longrightarrow \sigma_{1960} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1961} \quad \mathcal{T}_{1962}}{\langle S_2, \sigma_{1959} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1958} = \frac{\langle x := x + 1, \sigma_{1957} \rangle \longrightarrow \sigma_{1958} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1959} \quad \mathcal{T}_{1960}}{\langle S_2, \sigma_{1957} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1956} = \frac{\langle x := x + 1, \sigma_{1955} \rangle \longrightarrow \sigma_{1956} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1957} \quad \mathcal{T}_{1958}}{\langle S_2, \sigma_{1955} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1954} = \frac{\langle x := x + 1, \sigma_{1953} \rangle \longrightarrow \sigma_{1954} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1955} \quad \mathcal{T}_{1956}}{\langle S_2, \sigma_{1953} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1952} = \frac{\langle x := x + 1, \sigma_{1951} \rangle \longrightarrow \sigma_{1952} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1953} \quad \mathcal{T}_{1954}}{\langle S_2, \sigma_{1951} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1950} = \frac{\langle x := x + 1, \sigma_{1949} \rangle \longrightarrow \sigma_{1950} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1951} \quad \mathcal{T}_{1952}}{\langle S_2, \sigma_{1949} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1948} = \frac{\langle x := x + 1, \sigma_{1947} \rangle \longrightarrow \sigma_{1948} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1949} \quad \mathcal{T}_{1950}}{\langle S_2, \sigma_{1947} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1946} = \frac{\langle x := x + 1, \sigma_{1945} \rangle \longrightarrow \sigma_{1946} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1947} \quad \mathcal{T}_{1948}}{\langle S_2, \sigma_{1945} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1944} = \frac{\langle x := x + 1, \sigma_{1943} \rangle \longrightarrow \sigma_{1944} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1945} \quad \mathcal{T}_{1946}}{\langle S_2, \sigma_{1943} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1942} = \frac{\langle x := x + 1, \sigma_{1941} \rangle \longrightarrow \sigma_{1942} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1943} \quad \mathcal{T}_{1944}}{\langle S_2, \sigma_{1941} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1940} = \frac{\langle x := x + 1, \sigma_{1939} \rangle \longrightarrow \sigma_{1940} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1941} \quad \mathcal{T}_{1942}}{\langle S_2, \sigma_{1939} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1938} = \frac{\langle x := x + 1, \sigma_{1937} \rangle \longrightarrow \sigma_{1938} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1939} \quad \mathcal{T}_{1940}}{\langle S_2, \sigma_{1937} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1936} = \frac{\langle x := x + 1, \sigma_{1935} \rangle \longrightarrow \sigma_{1936} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1937} \quad \mathcal{T}_{1938}}{\langle S_2, \sigma_{1935} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\begin{aligned}
\mathcal{T}_{1934} &= \frac{\langle x := x + 1, \sigma_{1933} \rangle \longrightarrow \sigma_{1934} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1935} \quad \mathcal{T}_{1936}}{\langle S_2, \sigma_{1933} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1932} &= \frac{\langle x := x + 1, \sigma_{1931} \rangle \longrightarrow \sigma_{1932} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1933} \quad \mathcal{T}_{1934}}{\langle S_2, \sigma_{1931} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1930} &= \frac{\langle x := x + 1, \sigma_{1929} \rangle \longrightarrow \sigma_{1930} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1931} \quad \mathcal{T}_{1932}}{\langle S_2, \sigma_{1929} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1928} &= \frac{\langle x := x + 1, \sigma_{1927} \rangle \longrightarrow \sigma_{1928} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1929} \quad \mathcal{T}_{1930}}{\langle S_2, \sigma_{1927} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1926} &= \frac{\langle x := x + 1, \sigma_{1925} \rangle \longrightarrow \sigma_{1926} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1927} \quad \mathcal{T}_{1928}}{\langle S_2, \sigma_{1925} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1924} &= \frac{\langle x := x + 1, \sigma_{1923} \rangle \longrightarrow \sigma_{1924} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1925} \quad \mathcal{T}_{1926}}{\langle S_2, \sigma_{1923} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1922} &= \frac{\langle x := x + 1, \sigma_{1921} \rangle \longrightarrow \sigma_{1922} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1923} \quad \mathcal{T}_{1924}}{\langle S_2, \sigma_{1921} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1920} &= \frac{\langle x := x + 1, \sigma_{1919} \rangle \longrightarrow \sigma_{1920} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1921} \quad \mathcal{T}_{1922}}{\langle S_2, \sigma_{1919} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1918} &= \frac{\langle x := x + 1, \sigma_{1917} \rangle \longrightarrow \sigma_{1918} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1919} \quad \mathcal{T}_{1920}}{\langle S_2, \sigma_{1917} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1916} &= \frac{\langle x := x + 1, \sigma_{1915} \rangle \longrightarrow \sigma_{1916} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1917} \quad \mathcal{T}_{1918}}{\langle S_2, \sigma_{1915} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1914} &= \frac{\langle x := x + 1, \sigma_{1913} \rangle \longrightarrow \sigma_{1914} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1915} \quad \mathcal{T}_{1916}}{\langle S_2, \sigma_{1913} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1912} &= \frac{\langle x := x + 1, \sigma_{1911} \rangle \longrightarrow \sigma_{1912} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1913} \quad \mathcal{T}_{1914}}{\langle S_2, \sigma_{1911} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_{1910} &= \frac{\langle x := x + 1, \sigma_{1909} \rangle \longrightarrow \sigma_{1910} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1911} \quad \mathcal{T}_{1912}}{\langle S_2, \sigma_{1909} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]
\end{aligned}$$

$$\mathcal{T}_{1908} = \frac{\langle x := x + 1, \sigma_{1907} \rangle \longrightarrow \sigma_{1908} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1909} \quad \mathcal{T}_{1910}}{\langle S_2, \sigma_{1907} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1906} = \frac{\langle x := x + 1, \sigma_{1905} \rangle \longrightarrow \sigma_{1906} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1907} \quad \mathcal{T}_{1908}}{\langle S_2, \sigma_{1905} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1904} = \frac{\langle x := x + 1, \sigma_{1903} \rangle \longrightarrow \sigma_{1904} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1905} \quad \mathcal{T}_{1906}}{\langle S_2, \sigma_{1903} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1902} = \frac{\langle x := x + 1, \sigma_{1901} \rangle \longrightarrow \sigma_{1902} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1903} \quad \mathcal{T}_{1904}}{\langle S_2, \sigma_{1901} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1900} = \frac{\langle x := x + 1, \sigma_{1899} \rangle \longrightarrow \sigma_{1900} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1901} \quad \mathcal{T}_{1902}}{\langle S_2, \sigma_{1899} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1898} = \frac{\langle x := x + 1, \sigma_{1897} \rangle \longrightarrow \sigma_{1898} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1899} \quad \mathcal{T}_{1900}}{\langle S_2, \sigma_{1897} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1896} = \frac{\langle x := x + 1, \sigma_{1895} \rangle \longrightarrow \sigma_{1896} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1897} \quad \mathcal{T}_{1898}}{\langle S_2, \sigma_{1895} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1894} = \frac{\langle x := x + 1, \sigma_{1893} \rangle \longrightarrow \sigma_{1894} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1895} \quad \mathcal{T}_{1896}}{\langle S_2, \sigma_{1893} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1892} = \frac{\langle x := x + 1, \sigma_{1891} \rangle \longrightarrow \sigma_{1892} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1893} \quad \mathcal{T}_{1894}}{\langle S_2, \sigma_{1891} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1890} = \frac{\langle x := x + 1, \sigma_{1889} \rangle \longrightarrow \sigma_{1890} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1891} \quad \mathcal{T}_{1892}}{\langle S_2, \sigma_{1889} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1888} = \frac{\langle x := x + 1, \sigma_{1887} \rangle \longrightarrow \sigma_{1888} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1889} \quad \mathcal{T}_{1890}}{\langle S_2, \sigma_{1887} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1886} = \frac{\langle x := x + 1, \sigma_{1885} \rangle \longrightarrow \sigma_{1886} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1887} \quad \mathcal{T}_{1888}}{\langle S_2, \sigma_{1885} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1884} = \frac{\langle x := x + 1, \sigma_{1883} \rangle \longrightarrow \sigma_{1884} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1885} \quad \mathcal{T}_{1886}}{\langle S_2, \sigma_{1883} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1882} = \frac{\langle x := x + 1, \sigma_{1881} \rangle \longrightarrow \sigma_{1882} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1883} \quad \mathcal{T}_{1884}}{\langle S_2, \sigma_{1881} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1880} = \frac{\langle x := x + 1, \sigma_{1879} \rangle \longrightarrow \sigma_{1880} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1881} \quad \mathcal{T}_{1882}}{\langle S_2, \sigma_{1879} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1878} = \frac{\langle x := x + 1, \sigma_{1877} \rangle \longrightarrow \sigma_{1878} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1879} \quad \mathcal{T}_{1880}}{\langle S_2, \sigma_{1877} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1876} = \frac{\langle x := x + 1, \sigma_{1875} \rangle \longrightarrow \sigma_{1876} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1877} \quad \mathcal{T}_{1878}}{\langle S_2, \sigma_{1875} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1874} = \frac{\langle x := x + 1, \sigma_{1873} \rangle \longrightarrow \sigma_{1874} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1875} \quad \mathcal{T}_{1876}}{\langle S_2, \sigma_{1873} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1872} = \frac{\langle x := x + 1, \sigma_{1871} \rangle \longrightarrow \sigma_{1872} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1873} \quad \mathcal{T}_{1874}}{\langle S_2, \sigma_{1871} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1870} = \frac{\langle x := x + 1, \sigma_{1869} \rangle \longrightarrow \sigma_{1870} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1871} \quad \mathcal{T}_{1872}}{\langle S_2, \sigma_{1869} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1868} = \frac{\langle x := x + 1, \sigma_{1867} \rangle \longrightarrow \sigma_{1868} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1869} \quad \mathcal{T}_{1870}}{\langle S_2, \sigma_{1867} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1866} = \frac{\langle x := x + 1, \sigma_{1865} \rangle \longrightarrow \sigma_{1866} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1867} \quad \mathcal{T}_{1868}}{\langle S_2, \sigma_{1865} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1864} = \frac{\langle x := x + 1, \sigma_{1863} \rangle \longrightarrow \sigma_{1864} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1865} \quad \mathcal{T}_{1866}}{\langle S_2, \sigma_{1863} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1862} = \frac{\langle x := x + 1, \sigma_{1861} \rangle \longrightarrow \sigma_{1862} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1863} \quad \mathcal{T}_{1864}}{\langle S_2, \sigma_{1861} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1860} = \frac{\langle x := x + 1, \sigma_{1859} \rangle \longrightarrow \sigma_{1860} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1861} \quad \mathcal{T}_{1862}}{\langle S_2, \sigma_{1859} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1858} = \frac{\langle x := x + 1, \sigma_{1857} \rangle \longrightarrow \sigma_{1858} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1859} \quad \mathcal{T}_{1860}}{\langle S_2, \sigma_{1857} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1856} = \frac{\langle x := x + 1, \sigma_{1855} \rangle \longrightarrow \sigma_{1856} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1857} \quad \mathcal{T}_{1858}}{\langle S_2, \sigma_{1855} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1854} = \frac{\langle x := x + 1, \sigma_{1853} \rangle \longrightarrow \sigma_{1854} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1855} \quad \mathcal{T}_{1856}}{\langle S_2, \sigma_{1853} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1852} = \frac{\langle x := x + 1, \sigma_{1851} \rangle \longrightarrow \sigma_{1852} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1853} \quad \mathcal{T}_{1854}}{\langle S_2, \sigma_{1851} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1850} = \frac{\langle x := x + 1, \sigma_{1849} \rangle \longrightarrow \sigma_{1850} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1851} \quad \mathcal{T}_{1852}}{\langle S_2, \sigma_{1849} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1848} = \frac{\langle x := x + 1, \sigma_{1847} \rangle \longrightarrow \sigma_{1848} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1849} \quad \mathcal{T}_{1850}}{\langle S_2, \sigma_{1847} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1846} = \frac{\langle x := x + 1, \sigma_{1845} \rangle \longrightarrow \sigma_{1846} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1847} \quad \mathcal{T}_{1848}}{\langle S_2, \sigma_{1845} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1844} = \frac{\langle x := x + 1, \sigma_{1843} \rangle \longrightarrow \sigma_{1844} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1845} \quad \mathcal{T}_{1846}}{\langle S_2, \sigma_{1843} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1842} = \frac{\langle x := x + 1, \sigma_{1841} \rangle \longrightarrow \sigma_{1842} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1843} \quad \mathcal{T}_{1844}}{\langle S_2, \sigma_{1841} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1840} = \frac{\langle x := x + 1, \sigma_{1839} \rangle \longrightarrow \sigma_{1840} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1841} \quad \mathcal{T}_{1842}}{\langle S_2, \sigma_{1839} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1838} = \frac{\langle x := x + 1, \sigma_{1837} \rangle \longrightarrow \sigma_{1838} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1839} \quad \mathcal{T}_{1840}}{\langle S_2, \sigma_{1837} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1836} = \frac{\langle x := x + 1, \sigma_{1835} \rangle \longrightarrow \sigma_{1836} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1837} \quad \mathcal{T}_{1838}}{\langle S_2, \sigma_{1835} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1834} = \frac{\langle x := x + 1, \sigma_{1833} \rangle \longrightarrow \sigma_{1834} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1835} \quad \mathcal{T}_{1836}}{\langle S_2, \sigma_{1833} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1832} = \frac{\langle x := x + 1, \sigma_{1831} \rangle \longrightarrow \sigma_{1832} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1833} \quad \mathcal{T}_{1834}}{\langle S_2, \sigma_{1831} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1830} = \frac{\langle x := x + 1, \sigma_{1829} \rangle \longrightarrow \sigma_{1830} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1831} \quad \mathcal{T}_{1832}}{\langle S_2, \sigma_{1829} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1828} = \frac{\langle x := x + 1, \sigma_{1827} \rangle \longrightarrow \sigma_{1828} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1829} \quad \mathcal{T}_{1830}}{\langle S_2, \sigma_{1827} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1826} = \frac{\langle x := x + 1, \sigma_{1825} \rangle \longrightarrow \sigma_{1826} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1827} \quad \mathcal{T}_{1828}}{\langle S_2, \sigma_{1825} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1824} = \frac{\langle x := x + 1, \sigma_{1823} \rangle \longrightarrow \sigma_{1824} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1825} \quad \mathcal{T}_{1826}}{\langle S_2, \sigma_{1823} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1822} = \frac{\langle x := x + 1, \sigma_{1821} \rangle \longrightarrow \sigma_{1822} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1823} \quad \mathcal{T}_{1824}}{\langle S_2, \sigma_{1821} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1820} = \frac{\langle x := x + 1, \sigma_{1819} \rangle \longrightarrow \sigma_{1820} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1821} \quad \mathcal{T}_{1822}}{\langle S_2, \sigma_{1819} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1818} = \frac{\langle x := x + 1, \sigma_{1817} \rangle \longrightarrow \sigma_{1818} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1819} \quad \mathcal{T}_{1820}}{\langle S_2, \sigma_{1817} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1816} = \frac{\langle x := x + 1, \sigma_{1815} \rangle \longrightarrow \sigma_{1816} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1817} \quad \mathcal{T}_{1818}}{\langle S_2, \sigma_{1815} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1814} = \frac{\langle x := x + 1, \sigma_{1813} \rangle \longrightarrow \sigma_{1814} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1815} \quad \mathcal{T}_{1816}}{\langle S_2, \sigma_{1813} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1812} = \frac{\langle x := x + 1, \sigma_{1811} \rangle \longrightarrow \sigma_{1812} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1813} \quad \mathcal{T}_{1814}}{\langle S_2, \sigma_{1811} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1810} = \frac{\langle x := x + 1, \sigma_{1809} \rangle \longrightarrow \sigma_{1810} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1811} \quad \mathcal{T}_{1812}}{\langle S_2, \sigma_{1809} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1808} = \frac{\langle x := x + 1, \sigma_{1807} \rangle \longrightarrow \sigma_{1808} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1809} \quad \mathcal{T}_{1810}}{\langle S_2, \sigma_{1807} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1806} = \frac{\langle x := x + 1, \sigma_{1805} \rangle \longrightarrow \sigma_{1806} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1807} \quad \mathcal{T}_{1808}}{\langle S_2, \sigma_{1805} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1804} = \frac{\langle x := x + 1, \sigma_{1803} \rangle \longrightarrow \sigma_{1804} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1805} \quad \mathcal{T}_{1806}}{\langle S_2, \sigma_{1803} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1802} = \frac{\langle x := x + 1, \sigma_{1801} \rangle \longrightarrow \sigma_{1802} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1803} \quad \mathcal{T}_{1804}}{\langle S_2, \sigma_{1801} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1800} = \frac{\langle x := x + 1, \sigma_{1799} \rangle \longrightarrow \sigma_{1800} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1801} \quad \mathcal{T}_{1802}}{\langle S_2, \sigma_{1799} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1798} = \frac{\langle x := x + 1, \sigma_{1797} \rangle \longrightarrow \sigma_{1798} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1799} \quad \mathcal{T}_{1800}}{\langle S_2, \sigma_{1797} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1796} = \frac{\langle x := x + 1, \sigma_{1795} \rangle \longrightarrow \sigma_{1796} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1797} \quad \mathcal{T}_{1798}}{\langle S_2, \sigma_{1795} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1794} = \frac{\langle x := x + 1, \sigma_{1793} \rangle \longrightarrow \sigma_{1794} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1795} \quad \mathcal{T}_{1796}}{\langle S_2, \sigma_{1793} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1792} = \frac{\langle x := x + 1, \sigma_{1791} \rangle \longrightarrow \sigma_{1792} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1793} \quad \mathcal{T}_{1794}}{\langle S_2, \sigma_{1791} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1790} = \frac{\langle x := x + 1, \sigma_{1789} \rangle \longrightarrow \sigma_{1790} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1791} \quad \mathcal{T}_{1792}}{\langle S_2, \sigma_{1789} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1788} = \frac{\langle x := x + 1, \sigma_{1787} \rangle \longrightarrow \sigma_{1788} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1789} \quad \mathcal{T}_{1790}}{\langle S_2, \sigma_{1787} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1786} = \frac{\langle x := x + 1, \sigma_{1785} \rangle \longrightarrow \sigma_{1786} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1787} \quad \mathcal{T}_{1788}}{\langle S_2, \sigma_{1785} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1784} = \frac{\langle x := x + 1, \sigma_{1783} \rangle \longrightarrow \sigma_{1784} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1785} \quad \mathcal{T}_{1786}}{\langle S_2, \sigma_{1783} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1782} = \frac{\langle x := x + 1, \sigma_{1781} \rangle \longrightarrow \sigma_{1782} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1783} \quad \mathcal{T}_{1784}}{\langle S_2, \sigma_{1781} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1780} = \frac{\langle x := x + 1, \sigma_{1779} \rangle \longrightarrow \sigma_{1780} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1781} \quad \mathcal{T}_{1782}}{\langle S_2, \sigma_{1779} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1778} = \frac{\langle x := x + 1, \sigma_{1777} \rangle \longrightarrow \sigma_{1778} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1779} \quad \mathcal{T}_{1780}}{\langle S_2, \sigma_{1777} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1776} = \frac{\langle x := x + 1, \sigma_{1775} \rangle \longrightarrow \sigma_{1776} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1777} \quad \mathcal{T}_{1778}}{\langle S_2, \sigma_{1775} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1774} = \frac{\langle x := x + 1, \sigma_{1773} \rangle \longrightarrow \sigma_{1774} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1775} \quad \mathcal{T}_{1776}}{\langle S_2, \sigma_{1773} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1772} = \frac{\langle x := x + 1, \sigma_{1771} \rangle \longrightarrow \sigma_{1772} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1773} \quad \mathcal{T}_{1774}}{\langle S_2, \sigma_{1771} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1770} = \frac{\langle x := x + 1, \sigma_{1769} \rangle \longrightarrow \sigma_{1770} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1771} \quad \mathcal{T}_{1772}}{\langle S_2, \sigma_{1769} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1768} = \frac{\langle x := x + 1, \sigma_{1767} \rangle \longrightarrow \sigma_{1768} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1769} \quad \mathcal{T}_{1770}}{\langle S_2, \sigma_{1767} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1766} = \frac{\langle x := x + 1, \sigma_{1765} \rangle \longrightarrow \sigma_{1766} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1767} \quad \mathcal{T}_{1768}}{\langle S_2, \sigma_{1765} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1764} = \frac{\langle x := x + 1, \sigma_{1763} \rangle \longrightarrow \sigma_{1764} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1765} \quad \mathcal{T}_{1766}}{\langle S_2, \sigma_{1763} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1762} = \frac{\langle x := x + 1, \sigma_{1761} \rangle \longrightarrow \sigma_{1762} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1763} \quad \mathcal{T}_{1764}}{\langle S_2, \sigma_{1761} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1760} = \frac{\langle x := x + 1, \sigma_{1759} \rangle \longrightarrow \sigma_{1760} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1761} \quad \mathcal{T}_{1762}}{\langle S_2, \sigma_{1759} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1758} = \frac{\langle x := x + 1, \sigma_{1757} \rangle \longrightarrow \sigma_{1758} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1759} \quad \mathcal{T}_{1760}}{\langle S_2, \sigma_{1757} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1756} = \frac{\langle x := x + 1, \sigma_{1755} \rangle \longrightarrow \sigma_{1756} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1757} \quad \mathcal{T}_{1758}}{\langle S_2, \sigma_{1755} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1754} = \frac{\langle x := x + 1, \sigma_{1753} \rangle \longrightarrow \sigma_{1754} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1755} \quad \mathcal{T}_{1756}}{\langle S_2, \sigma_{1753} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1752} = \frac{\langle x := x + 1, \sigma_{1751} \rangle \longrightarrow \sigma_{1752} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1753} \quad \mathcal{T}_{1754}}{\langle S_2, \sigma_{1751} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1750} = \frac{\langle x := x + 1, \sigma_{1749} \rangle \longrightarrow \sigma_{1750} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1751} \quad \mathcal{T}_{1752}}{\langle S_2, \sigma_{1749} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1748} = \frac{\langle x := x + 1, \sigma_{1747} \rangle \longrightarrow \sigma_{1748} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1749} \quad \mathcal{T}_{1750}}{\langle S_2, \sigma_{1747} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1746} = \frac{\langle x := x + 1, \sigma_{1745} \rangle \longrightarrow \sigma_{1746} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1747} \quad \mathcal{T}_{1748}}{\langle S_2, \sigma_{1745} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1744} = \frac{\langle x := x + 1, \sigma_{1743} \rangle \longrightarrow \sigma_{1744} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1745} \quad \mathcal{T}_{1746}}{\langle S_2, \sigma_{1743} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1742} = \frac{\langle x := x + 1, \sigma_{1741} \rangle \longrightarrow \sigma_{1742} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1743} \quad \mathcal{T}_{1744}}{\langle S_2, \sigma_{1741} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1740} = \frac{\langle x := x + 1, \sigma_{1739} \rangle \longrightarrow \sigma_{1740} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1741} \quad \mathcal{T}_{1742}}{\langle S_2, \sigma_{1739} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1738} = \frac{\langle x := x + 1, \sigma_{1737} \rangle \longrightarrow \sigma_{1738} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1739} \quad \mathcal{T}_{1740}}{\langle S_2, \sigma_{1737} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1736} = \frac{\langle x := x + 1, \sigma_{1735} \rangle \longrightarrow \sigma_{1736} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1737} \quad \mathcal{T}_{1738}}{\langle S_2, \sigma_{1735} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1734} = \frac{\langle x := x + 1, \sigma_{1733} \rangle \longrightarrow \sigma_{1734} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1735} \quad \mathcal{T}_{1736}}{\langle S_2, \sigma_{1733} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1732} = \frac{\langle x := x + 1, \sigma_{1731} \rangle \longrightarrow \sigma_{1732} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1733} \quad \mathcal{T}_{1734}}{\langle S_2, \sigma_{1731} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1730} = \frac{\langle x := x + 1, \sigma_{1729} \rangle \longrightarrow \sigma_{1730} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1731} \quad \mathcal{T}_{1732}}{\langle S_2, \sigma_{1729} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1728} = \frac{\langle x := x + 1, \sigma_{1727} \rangle \longrightarrow \sigma_{1728} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1729} \quad \mathcal{T}_{1730}}{\langle S_2, \sigma_{1727} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1726} = \frac{\langle x := x + 1, \sigma_{1725} \rangle \longrightarrow \sigma_{1726} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1727} \quad \mathcal{T}_{1728}}{\langle S_2, \sigma_{1725} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1724} = \frac{\langle x := x + 1, \sigma_{1723} \rangle \longrightarrow \sigma_{1724} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1725} \quad \mathcal{T}_{1726}}{\langle S_2, \sigma_{1723} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1722} = \frac{\langle x := x + 1, \sigma_{1721} \rangle \longrightarrow \sigma_{1722} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1723} \quad \mathcal{T}_{1724}}{\langle S_2, \sigma_{1721} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1720} = \frac{\langle x := x + 1, \sigma_{1719} \rangle \longrightarrow \sigma_{1720} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1721} \quad \mathcal{T}_{1722}}{\langle S_2, \sigma_{1719} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1718} = \frac{\langle x := x + 1, \sigma_{1717} \rangle \longrightarrow \sigma_{1718} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1719} \quad \mathcal{T}_{1720}}{\langle S_2, \sigma_{1717} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1716} = \frac{\langle x := x + 1, \sigma_{1715} \rangle \longrightarrow \sigma_{1716} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1717} \quad \mathcal{T}_{1718}}{\langle S_2, \sigma_{1715} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1714} = \frac{\langle x := x + 1, \sigma_{1713} \rangle \longrightarrow \sigma_{1714} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1715} \quad \mathcal{T}_{1716}}{\langle S_2, \sigma_{1713} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1712} = \frac{\langle x := x + 1, \sigma_{1711} \rangle \longrightarrow \sigma_{1712} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1713} \quad \mathcal{T}_{1714}}{\langle S_2, \sigma_{1711} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1710} = \frac{\langle x := x + 1, \sigma_{1709} \rangle \longrightarrow \sigma_{1710} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1711} \quad \mathcal{T}_{1712}}{\langle S_2, \sigma_{1709} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1708} = \frac{\langle x := x + 1, \sigma_{1707} \rangle \longrightarrow \sigma_{1708} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1709} \quad \mathcal{T}_{1710}}{\langle S_2, \sigma_{1707} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1706} = \frac{\langle x := x + 1, \sigma_{1705} \rangle \longrightarrow \sigma_{1706} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1707} \quad \mathcal{T}_{1708}}{\langle S_2, \sigma_{1705} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1704} = \frac{\langle x := x + 1, \sigma_{1703} \rangle \longrightarrow \sigma_{1704} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1705} \quad \mathcal{T}_{1706}}{\langle S_2, \sigma_{1703} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1702} = \frac{\langle x := x + 1, \sigma_{1701} \rangle \longrightarrow \sigma_{1702} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1703} \quad \mathcal{T}_{1704}}{\langle S_2, \sigma_{1701} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1700} = \frac{\langle x := x + 1, \sigma_{1699} \rangle \longrightarrow \sigma_{1700} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1701} \quad \mathcal{T}_{1702}}{\langle S_2, \sigma_{1699} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1698} = \frac{\langle x := x + 1, \sigma_{1697} \rangle \longrightarrow \sigma_{1698} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1699} \quad \mathcal{T}_{1700}}{\langle S_2, \sigma_{1697} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1696} = \frac{\langle x := x + 1, \sigma_{1695} \rangle \longrightarrow \sigma_{1696} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1697} \quad \mathcal{T}_{1698}}{\langle S_2, \sigma_{1695} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1694} = \frac{\langle x := x + 1, \sigma_{1693} \rangle \longrightarrow \sigma_{1694} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1695} \quad \mathcal{T}_{1696}}{\langle S_2, \sigma_{1693} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1692} = \frac{\langle x := x + 1, \sigma_{1691} \rangle \longrightarrow \sigma_{1692} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1693} \quad \mathcal{T}_{1694}}{\langle S_2, \sigma_{1691} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1690} = \frac{\langle x := x + 1, \sigma_{1689} \rangle \longrightarrow \sigma_{1690} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1691} \quad \mathcal{T}_{1692}}{\langle S_2, \sigma_{1689} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1688} = \frac{\langle x := x + 1, \sigma_{1687} \rangle \longrightarrow \sigma_{1688} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1689} \quad \mathcal{T}_{1690}}{\langle S_2, \sigma_{1687} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1686} = \frac{\langle x := x + 1, \sigma_{1685} \rangle \longrightarrow \sigma_{1686} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1687} \quad \mathcal{T}_{1688}}{\langle S_2, \sigma_{1685} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1684} = \frac{\langle x := x + 1, \sigma_{1683} \rangle \longrightarrow \sigma_{1684} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1685} \quad \mathcal{T}_{1686}}{\langle S_2, \sigma_{1683} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1682} = \frac{\langle x := x + 1, \sigma_{1681} \rangle \longrightarrow \sigma_{1682} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1683} \quad \mathcal{T}_{1684}}{\langle S_2, \sigma_{1681} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1680} = \frac{\langle x := x + 1, \sigma_{1679} \rangle \longrightarrow \sigma_{1680} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1681} \quad \mathcal{T}_{1682}}{\langle S_2, \sigma_{1679} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1678} = \frac{\langle x := x + 1, \sigma_{1677} \rangle \longrightarrow \sigma_{1678} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1679} \quad \mathcal{T}_{1680}}{\langle S_2, \sigma_{1677} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1676} = \frac{\langle x := x + 1, \sigma_{1675} \rangle \longrightarrow \sigma_{1676} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1677} \quad \mathcal{T}_{1678}}{\langle S_2, \sigma_{1675} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1674} = \frac{\langle x := x + 1, \sigma_{1673} \rangle \longrightarrow \sigma_{1674} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1675} \quad \mathcal{T}_{1676}}{\langle S_2, \sigma_{1673} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1672} = \frac{\langle x := x + 1, \sigma_{1671} \rangle \longrightarrow \sigma_{1672} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1673} \quad \mathcal{T}_{1674}}{\langle S_2, \sigma_{1671} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1670} = \frac{\langle x := x + 1, \sigma_{1669} \rangle \longrightarrow \sigma_{1670} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1671} \quad \mathcal{T}_{1672}}{\langle S_2, \sigma_{1669} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1668} = \frac{\langle x := x + 1, \sigma_{1667} \rangle \longrightarrow \sigma_{1668} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1669} \quad \mathcal{T}_{1670}}{\langle S_2, \sigma_{1667} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1666} = \frac{\langle x := x + 1, \sigma_{1665} \rangle \longrightarrow \sigma_{1666} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1667} \quad \mathcal{T}_{1668}}{\langle S_2, \sigma_{1665} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1664} = \frac{\langle x := x + 1, \sigma_{1663} \rangle \longrightarrow \sigma_{1664} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1665} \quad \mathcal{T}_{1666}}{\langle S_2, \sigma_{1663} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1662} = \frac{\langle x := x + 1, \sigma_{1661} \rangle \longrightarrow \sigma_{1662} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1663} \quad \mathcal{T}_{1664}}{\langle S_2, \sigma_{1661} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1660} = \frac{\langle x := x + 1, \sigma_{1659} \rangle \longrightarrow \sigma_{1660} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1661} \quad \mathcal{T}_{1662}}{\langle S_2, \sigma_{1659} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1658} = \frac{\langle x := x + 1, \sigma_{1657} \rangle \longrightarrow \sigma_{1658} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1659} \quad \mathcal{T}_{1660}}{\langle S_2, \sigma_{1657} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1656} = \frac{\langle x := x + 1, \sigma_{1655} \rangle \longrightarrow \sigma_{1656} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1657} \quad \mathcal{T}_{1658}}{\langle S_2, \sigma_{1655} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1654} = \frac{\langle x := x + 1, \sigma_{1653} \rangle \longrightarrow \sigma_{1654} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1655} \quad \mathcal{T}_{1656}}{\langle S_2, \sigma_{1653} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1652} = \frac{\langle x := x + 1, \sigma_{1651} \rangle \longrightarrow \sigma_{1652} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1653} \quad \mathcal{T}_{1654}}{\langle S_2, \sigma_{1651} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1650} = \frac{\langle x := x + 1, \sigma_{1649} \rangle \longrightarrow \sigma_{1650} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1651} \quad \mathcal{T}_{1652}}{\langle S_2, \sigma_{1649} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1648} = \frac{\langle x := x + 1, \sigma_{1647} \rangle \longrightarrow \sigma_{1648} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1649} \quad \mathcal{T}_{1650}}{\langle S_2, \sigma_{1647} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1646} = \frac{\langle x := x + 1, \sigma_{1645} \rangle \longrightarrow \sigma_{1646} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1647} \quad \mathcal{T}_{1648}}{\langle S_2, \sigma_{1645} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1644} = \frac{\langle x := x + 1, \sigma_{1643} \rangle \longrightarrow \sigma_{1644} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1645} \quad \mathcal{T}_{1646}}{\langle S_2, \sigma_{1643} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1642} = \frac{\langle x := x + 1, \sigma_{1641} \rangle \longrightarrow \sigma_{1642} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1643} \quad \mathcal{T}_{1644}}{\langle S_2, \sigma_{1641} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1640} = \frac{\langle x := x + 1, \sigma_{1639} \rangle \longrightarrow \sigma_{1640} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1641} \quad \mathcal{T}_{1642}}{\langle S_2, \sigma_{1639} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1638} = \frac{\langle x := x + 1, \sigma_{1637} \rangle \longrightarrow \sigma_{1638} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1639} \quad \mathcal{T}_{1640}}{\langle S_2, \sigma_{1637} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1636} = \frac{\langle x := x + 1, \sigma_{1635} \rangle \longrightarrow \sigma_{1636} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1637} \quad \mathcal{T}_{1638}}{\langle S_2, \sigma_{1635} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1634} = \frac{\langle x := x + 1, \sigma_{1633} \rangle \longrightarrow \sigma_{1634} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1635} \quad \mathcal{T}_{1636}}{\langle S_2, \sigma_{1633} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1632} = \frac{\langle x := x + 1, \sigma_{1631} \rangle \longrightarrow \sigma_{1632} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1633} \quad \mathcal{T}_{1634}}{\langle S_2, \sigma_{1631} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1630} = \frac{\langle x := x + 1, \sigma_{1629} \rangle \longrightarrow \sigma_{1630} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1631} \quad \mathcal{T}_{1632}}{\langle S_2, \sigma_{1629} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1628} = \frac{\langle x := x + 1, \sigma_{1627} \rangle \longrightarrow \sigma_{1628} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1629} \quad \mathcal{T}_{1630}}{\langle S_2, \sigma_{1627} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1626} = \frac{\langle x := x + 1, \sigma_{1625} \rangle \longrightarrow \sigma_{1626} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1627} \quad \mathcal{T}_{1628}}{\langle S_2, \sigma_{1625} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1624} = \frac{\langle x := x + 1, \sigma_{1623} \rangle \longrightarrow \sigma_{1624} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1625} \quad \mathcal{T}_{1626}}{\langle S_2, \sigma_{1623} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1622} = \frac{\langle x := x + 1, \sigma_{1621} \rangle \longrightarrow \sigma_{1622} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1623} \quad \mathcal{T}_{1624}}{\langle S_2, \sigma_{1621} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1620} = \frac{\langle x := x + 1, \sigma_{1619} \rangle \longrightarrow \sigma_{1620} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1621} \quad \mathcal{T}_{1622}}{\langle S_2, \sigma_{1619} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1618} = \frac{\langle x := x + 1, \sigma_{1617} \rangle \longrightarrow \sigma_{1618} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1619} \quad \mathcal{T}_{1620}}{\langle S_2, \sigma_{1617} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1616} = \frac{\langle x := x + 1, \sigma_{1615} \rangle \longrightarrow \sigma_{1616} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1617} \quad \mathcal{T}_{1618}}{\langle S_2, \sigma_{1615} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1614} = \frac{\langle x := x + 1, \sigma_{1613} \rangle \longrightarrow \sigma_{1614} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1615} \quad \mathcal{T}_{1616}}{\langle S_2, \sigma_{1613} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1612} = \frac{\langle x := x + 1, \sigma_{1611} \rangle \longrightarrow \sigma_{1612} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1613} \quad \mathcal{T}_{1614}}{\langle S_2, \sigma_{1611} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1610} = \frac{\langle x := x + 1, \sigma_{1609} \rangle \longrightarrow \sigma_{1610} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1611} \quad \mathcal{T}_{1612}}{\langle S_2, \sigma_{1609} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1608} = \frac{\langle x := x + 1, \sigma_{1607} \rangle \longrightarrow \sigma_{1608} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1609} \quad \mathcal{T}_{1610}}{\langle S_2, \sigma_{1607} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1606} = \frac{\langle x := x + 1, \sigma_{1605} \rangle \longrightarrow \sigma_{1606} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1607} \quad \mathcal{T}_{1608}}{\langle S_2, \sigma_{1605} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1604} = \frac{\langle x := x + 1, \sigma_{1603} \rangle \longrightarrow \sigma_{1604} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1605} \quad \mathcal{T}_{1606}}{\langle S_2, \sigma_{1603} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1602} = \frac{\langle x := x + 1, \sigma_{1601} \rangle \longrightarrow \sigma_{1602} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1603} \quad \mathcal{T}_{1604}}{\langle S_2, \sigma_{1601} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1600} = \frac{\langle x := x + 1, \sigma_{1599} \rangle \longrightarrow \sigma_{1600} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1601} \quad \mathcal{T}_{1602}}{\langle S_2, \sigma_{1599} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1598} = \frac{\langle x := x + 1, \sigma_{1597} \rangle \longrightarrow \sigma_{1598} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1599} \quad \mathcal{T}_{1600}}{\langle S_2, \sigma_{1597} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1596} = \frac{\langle x := x + 1, \sigma_{1595} \rangle \longrightarrow \sigma_{1596} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1597} \quad \mathcal{T}_{1598}}{\langle S_2, \sigma_{1595} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1594} = \frac{\langle x := x + 1, \sigma_{1593} \rangle \longrightarrow \sigma_{1594} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1595} \quad \mathcal{T}_{1596}}{\langle S_2, \sigma_{1593} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1592} = \frac{\langle x := x + 1, \sigma_{1591} \rangle \longrightarrow \sigma_{1592} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1593} \quad \mathcal{T}_{1594}}{\langle S_2, \sigma_{1591} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1590} = \frac{\langle x := x + 1, \sigma_{1589} \rangle \longrightarrow \sigma_{1590} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1591} \quad \mathcal{T}_{1592}}{\langle S_2, \sigma_{1589} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1588} = \frac{\langle x := x + 1, \sigma_{1587} \rangle \longrightarrow \sigma_{1588} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1589} \quad \mathcal{T}_{1590}}{\langle S_2, \sigma_{1587} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1586} = \frac{\langle x := x + 1, \sigma_{1585} \rangle \longrightarrow \sigma_{1586} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1587} \quad \mathcal{T}_{1588}}{\langle S_2, \sigma_{1585} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1584} = \frac{\langle x := x + 1, \sigma_{1583} \rangle \longrightarrow \sigma_{1584} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1585} \quad \mathcal{T}_{1586}}{\langle S_2, \sigma_{1583} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1582} = \frac{\langle x := x + 1, \sigma_{1581} \rangle \longrightarrow \sigma_{1582} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1583} \quad \mathcal{T}_{1584}}{\langle S_2, \sigma_{1581} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1580} = \frac{\langle x := x + 1, \sigma_{1579} \rangle \longrightarrow \sigma_{1580} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1581} \quad \mathcal{T}_{1582}}{\langle S_2, \sigma_{1579} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1578} = \frac{\langle x := x + 1, \sigma_{1577} \rangle \longrightarrow \sigma_{1578} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1579} \quad \mathcal{T}_{1580}}{\langle S_2, \sigma_{1577} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1576} = \frac{\langle x := x + 1, \sigma_{1575} \rangle \longrightarrow \sigma_{1576} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1577} \quad \mathcal{T}_{1578}}{\langle S_2, \sigma_{1575} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1574} = \frac{\langle x := x + 1, \sigma_{1573} \rangle \longrightarrow \sigma_{1574} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1575} \quad \mathcal{T}_{1576}}{\langle S_2, \sigma_{1573} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1572} = \frac{\langle x := x + 1, \sigma_{1571} \rangle \longrightarrow \sigma_{1572} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1573} \quad \mathcal{T}_{1574}}{\langle S_2, \sigma_{1571} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1570} = \frac{\langle x := x + 1, \sigma_{1569} \rangle \longrightarrow \sigma_{1570} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1571} \quad \mathcal{T}_{1572}}{\langle S_2, \sigma_{1569} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1568} = \frac{\langle x := x + 1, \sigma_{1567} \rangle \longrightarrow \sigma_{1568} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1569} \quad \mathcal{T}_{1570}}{\langle S_2, \sigma_{1567} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1566} = \frac{\langle x := x + 1, \sigma_{1565} \rangle \longrightarrow \sigma_{1566} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1567} \quad \mathcal{T}_{1568}}{\langle S_2, \sigma_{1565} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1564} = \frac{\langle x := x + 1, \sigma_{1563} \rangle \longrightarrow \sigma_{1564} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1565} \quad \mathcal{T}_{1566}}{\langle S_2, \sigma_{1563} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1562} = \frac{\langle x := x + 1, \sigma_{1561} \rangle \longrightarrow \sigma_{1562} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1563} \quad \mathcal{T}_{1564}}{\langle S_2, \sigma_{1561} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1560} = \frac{\langle x := x + 1, \sigma_{1559} \rangle \longrightarrow \sigma_{1560} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1561} \quad \mathcal{T}_{1562}}{\langle S_2, \sigma_{1559} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1558} = \frac{\langle x := x + 1, \sigma_{1557} \rangle \longrightarrow \sigma_{1558} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1559} \quad \mathcal{T}_{1560}}{\langle S_2, \sigma_{1557} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1556} = \frac{\langle x := x + 1, \sigma_{1555} \rangle \longrightarrow \sigma_{1556} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1557} \quad \mathcal{T}_{1558}}{\langle S_2, \sigma_{1555} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1554} = \frac{\langle x := x + 1, \sigma_{1553} \rangle \longrightarrow \sigma_{1554} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1555} \quad \mathcal{T}_{1556}}{\langle S_2, \sigma_{1553} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1552} = \frac{\langle x := x + 1, \sigma_{1551} \rangle \longrightarrow \sigma_{1552} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1553} \quad \mathcal{T}_{1554}}{\langle S_2, \sigma_{1551} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1550} = \frac{\langle x := x + 1, \sigma_{1549} \rangle \longrightarrow \sigma_{1550} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1551} \quad \mathcal{T}_{1552}}{\langle S_2, \sigma_{1549} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1548} = \frac{\langle x := x + 1, \sigma_{1547} \rangle \longrightarrow \sigma_{1548} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1549} \quad \mathcal{T}_{1550}}{\langle S_2, \sigma_{1547} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1546} = \frac{\langle x := x + 1, \sigma_{1545} \rangle \longrightarrow \sigma_{1546} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1547} \quad \mathcal{T}_{1548}}{\langle S_2, \sigma_{1545} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1544} = \frac{\langle x := x + 1, \sigma_{1543} \rangle \longrightarrow \sigma_{1544} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1545} \quad \mathcal{T}_{1546}}{\langle S_2, \sigma_{1543} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1542} = \frac{\langle x := x + 1, \sigma_{1541} \rangle \longrightarrow \sigma_{1542} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1543} \quad \mathcal{T}_{1544}}{\langle S_2, \sigma_{1541} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1540} = \frac{\langle x := x + 1, \sigma_{1539} \rangle \longrightarrow \sigma_{1540} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1541} \quad \mathcal{T}_{1542}}{\langle S_2, \sigma_{1539} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1538} = \frac{\langle x := x + 1, \sigma_{1537} \rangle \longrightarrow \sigma_{1538} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1539} \quad \mathcal{T}_{1540}}{\langle S_2, \sigma_{1537} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1536} = \frac{\langle x := x + 1, \sigma_{1535} \rangle \longrightarrow \sigma_{1536} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1537} \quad \mathcal{T}_{1538}}{\langle S_2, \sigma_{1535} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1534} = \frac{\langle x := x + 1, \sigma_{1533} \rangle \longrightarrow \sigma_{1534} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1535} \quad \mathcal{T}_{1536}}{\langle S_2, \sigma_{1533} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1532} = \frac{\langle x := x + 1, \sigma_{1531} \rangle \longrightarrow \sigma_{1532} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1533} \quad \mathcal{T}_{1534}}{\langle S_2, \sigma_{1531} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1530} = \frac{\langle x := x + 1, \sigma_{1529} \rangle \longrightarrow \sigma_{1530} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1531} \quad \mathcal{T}_{1532}}{\langle S_2, \sigma_{1529} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1528} = \frac{\langle x := x + 1, \sigma_{1527} \rangle \longrightarrow \sigma_{1528} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1529} \quad \mathcal{T}_{1530}}{\langle S_2, \sigma_{1527} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1526} = \frac{\langle x := x + 1, \sigma_{1525} \rangle \longrightarrow \sigma_{1526} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1527} \quad \mathcal{T}_{1528}}{\langle S_2, \sigma_{1525} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1524} = \frac{\langle x := x + 1, \sigma_{1523} \rangle \longrightarrow \sigma_{1524} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1525} \quad \mathcal{T}_{1526}}{\langle S_2, \sigma_{1523} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1522} = \frac{\langle x := x + 1, \sigma_{1521} \rangle \longrightarrow \sigma_{1522} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1523} \quad \mathcal{T}_{1524}}{\langle S_2, \sigma_{1521} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1520} = \frac{\langle x := x + 1, \sigma_{1519} \rangle \longrightarrow \sigma_{1520} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1521} \quad \mathcal{T}_{1522}}{\langle S_2, \sigma_{1519} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1518} = \frac{\langle x := x + 1, \sigma_{1517} \rangle \longrightarrow \sigma_{1518} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1519} \quad \mathcal{T}_{1520}}{\langle S_2, \sigma_{1517} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1516} = \frac{\langle x := x + 1, \sigma_{1515} \rangle \longrightarrow \sigma_{1516} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1517} \quad \mathcal{T}_{1518}}{\langle S_2, \sigma_{1515} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1514} = \frac{\langle x := x + 1, \sigma_{1513} \rangle \longrightarrow \sigma_{1514} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1515} \quad \mathcal{T}_{1516}}{\langle S_2, \sigma_{1513} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1512} = \frac{\langle x := x + 1, \sigma_{1511} \rangle \longrightarrow \sigma_{1512} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1513} \quad \mathcal{T}_{1514}}{\langle S_2, \sigma_{1511} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1510} = \frac{\langle x := x + 1, \sigma_{1509} \rangle \longrightarrow \sigma_{1510} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1511} \quad \mathcal{T}_{1512}}{\langle S_2, \sigma_{1509} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1508} = \frac{\langle x := x + 1, \sigma_{1507} \rangle \longrightarrow \sigma_{1508} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1509} \quad \mathcal{T}_{1510}}{\langle S_2, \sigma_{1507} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1506} = \frac{\langle x := x + 1, \sigma_{1505} \rangle \longrightarrow \sigma_{1506} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1507} \quad \mathcal{T}_{1508}}{\langle S_2, \sigma_{1505} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1504} = \frac{\langle x := x + 1, \sigma_{1503} \rangle \longrightarrow \sigma_{1504} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1505} \quad \mathcal{T}_{1506}}{\langle S_2, \sigma_{1503} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1502} = \frac{\langle x := x + 1, \sigma_{1501} \rangle \longrightarrow \sigma_{1502} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1503} \quad \mathcal{T}_{1504}}{\langle S_2, \sigma_{1501} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1500} = \frac{\langle x := x + 1, \sigma_{1499} \rangle \longrightarrow \sigma_{1500} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1501} \quad \mathcal{T}_{1502}}{\langle S_2, \sigma_{1499} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1498} = \frac{\langle x := x + 1, \sigma_{1497} \rangle \longrightarrow \sigma_{1498} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1499} \quad \mathcal{T}_{1500}}{\langle S_2, \sigma_{1497} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1496} = \frac{\langle x := x + 1, \sigma_{1495} \rangle \longrightarrow \sigma_{1496} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1497} \quad \mathcal{T}_{1498}}{\langle S_2, \sigma_{1495} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1494} = \frac{\langle x := x + 1, \sigma_{1493} \rangle \longrightarrow \sigma_{1494} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1495} \quad \mathcal{T}_{1496}}{\langle S_2, \sigma_{1493} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1492} = \frac{\langle x := x + 1, \sigma_{1491} \rangle \longrightarrow \sigma_{1492} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1493} \quad \mathcal{T}_{1494}}{\langle S_2, \sigma_{1491} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1490} = \frac{\langle x := x + 1, \sigma_{1489} \rangle \longrightarrow \sigma_{1490} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1491} \quad \mathcal{T}_{1492}}{\langle S_2, \sigma_{1489} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1488} = \frac{\langle x := x + 1, \sigma_{1487} \rangle \longrightarrow \sigma_{1488} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1489} \quad \mathcal{T}_{1490}}{\langle S_2, \sigma_{1487} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1486} = \frac{\langle x := x + 1, \sigma_{1485} \rangle \longrightarrow \sigma_{1486} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1487} \quad \mathcal{T}_{1488}}{\langle S_2, \sigma_{1485} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1484} = \frac{\langle x := x + 1, \sigma_{1483} \rangle \longrightarrow \sigma_{1484} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1485} \quad \mathcal{T}_{1486}}{\langle S_2, \sigma_{1483} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1482} = \frac{\langle x := x + 1, \sigma_{1481} \rangle \longrightarrow \sigma_{1482} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1483} \quad \mathcal{T}_{1484}}{\langle S_2, \sigma_{1481} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1480} = \frac{\langle x := x + 1, \sigma_{1479} \rangle \longrightarrow \sigma_{1480} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1481} \quad \mathcal{T}_{1482}}{\langle S_2, \sigma_{1479} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1478} = \frac{\langle x := x + 1, \sigma_{1477} \rangle \longrightarrow \sigma_{1478} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1479} \quad \mathcal{T}_{1480}}{\langle S_2, \sigma_{1477} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1476} = \frac{\langle x := x + 1, \sigma_{1475} \rangle \longrightarrow \sigma_{1476} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1477} \quad \mathcal{T}_{1478}}{\langle S_2, \sigma_{1475} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1474} = \frac{\langle x := x + 1, \sigma_{1473} \rangle \longrightarrow \sigma_{1474} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1475} \quad \mathcal{T}_{1476}}{\langle S_2, \sigma_{1473} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1472} = \frac{\langle x := x + 1, \sigma_{1471} \rangle \longrightarrow \sigma_{1472} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1473} \quad \mathcal{T}_{1474}}{\langle S_2, \sigma_{1471} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1470} = \frac{\langle x := x + 1, \sigma_{1469} \rangle \longrightarrow \sigma_{1470} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1471} \quad \mathcal{T}_{1472}}{\langle S_2, \sigma_{1469} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1468} = \frac{\langle x := x + 1, \sigma_{1467} \rangle \longrightarrow \sigma_{1468} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1469} \quad \mathcal{T}_{1470}}{\langle S_2, \sigma_{1467} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1466} = \frac{\langle x := x + 1, \sigma_{1465} \rangle \longrightarrow \sigma_{1466} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1467} \quad \mathcal{T}_{1468}}{\langle S_2, \sigma_{1465} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1464} = \frac{\langle x := x + 1, \sigma_{1463} \rangle \longrightarrow \sigma_{1464} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1465} \quad \mathcal{T}_{1466}}{\langle S_2, \sigma_{1463} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1462} = \frac{\langle x := x + 1, \sigma_{1461} \rangle \longrightarrow \sigma_{1462} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1463} \quad \mathcal{T}_{1464}}{\langle S_2, \sigma_{1461} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1460} = \frac{\langle x := x + 1, \sigma_{1459} \rangle \longrightarrow \sigma_{1460} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1461} \quad \mathcal{T}_{1462}}{\langle S_2, \sigma_{1459} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1458} = \frac{\langle x := x + 1, \sigma_{1457} \rangle \longrightarrow \sigma_{1458} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1459} \quad \mathcal{T}_{1460}}{\langle S_2, \sigma_{1457} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1456} = \frac{\langle x := x + 1, \sigma_{1455} \rangle \longrightarrow \sigma_{1456} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1457} \quad \mathcal{T}_{1458}}{\langle S_2, \sigma_{1455} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1454} = \frac{\langle x := x + 1, \sigma_{1453} \rangle \longrightarrow \sigma_{1454} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1455} \quad \mathcal{T}_{1456}}{\langle S_2, \sigma_{1453} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1452} = \frac{\langle x := x + 1, \sigma_{1451} \rangle \longrightarrow \sigma_{1452} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1453} \quad \mathcal{T}_{1454}}{\langle S_2, \sigma_{1451} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1450} = \frac{\langle x := x + 1, \sigma_{1449} \rangle \longrightarrow \sigma_{1450} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1451} \quad \mathcal{T}_{1452}}{\langle S_2, \sigma_{1449} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1448} = \frac{\langle x := x + 1, \sigma_{1447} \rangle \longrightarrow \sigma_{1448} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1449} \quad \mathcal{T}_{1450}}{\langle S_2, \sigma_{1447} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1446} = \frac{\langle x := x + 1, \sigma_{1445} \rangle \longrightarrow \sigma_{1446} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1447} \quad \mathcal{T}_{1448}}{\langle S_2, \sigma_{1445} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1444} = \frac{\langle x := x + 1, \sigma_{1443} \rangle \longrightarrow \sigma_{1444} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1445} \quad \mathcal{T}_{1446}}{\langle S_2, \sigma_{1443} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1442} = \frac{\langle x := x + 1, \sigma_{1441} \rangle \longrightarrow \sigma_{1442} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1443} \quad \mathcal{T}_{1444}}{\langle S_2, \sigma_{1441} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1440} = \frac{\langle x := x + 1, \sigma_{1439} \rangle \longrightarrow \sigma_{1440} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1441} \quad \mathcal{T}_{1442}}{\langle S_2, \sigma_{1439} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1438} = \frac{\langle x := x + 1, \sigma_{1437} \rangle \longrightarrow \sigma_{1438} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1439} \quad \mathcal{T}_{1440}}{\langle S_2, \sigma_{1437} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1436} = \frac{\langle x := x + 1, \sigma_{1435} \rangle \longrightarrow \sigma_{1436} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1437} \quad \mathcal{T}_{1438}}{\langle S_2, \sigma_{1435} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1434} = \frac{\langle x := x + 1, \sigma_{1433} \rangle \longrightarrow \sigma_{1434} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1435} \quad \mathcal{T}_{1436}}{\langle S_2, \sigma_{1433} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1432} = \frac{\langle x := x + 1, \sigma_{1431} \rangle \longrightarrow \sigma_{1432} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1433} \quad \mathcal{T}_{1434}}{\langle S_2, \sigma_{1431} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1430} = \frac{\langle x := x + 1, \sigma_{1429} \rangle \longrightarrow \sigma_{1430} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1431} \quad \mathcal{T}_{1432}}{\langle S_2, \sigma_{1429} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1428} = \frac{\langle x := x + 1, \sigma_{1427} \rangle \longrightarrow \sigma_{1428} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1429} \quad \mathcal{T}_{1430}}{\langle S_2, \sigma_{1427} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1426} = \frac{\langle x := x + 1, \sigma_{1425} \rangle \longrightarrow \sigma_{1426} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1427} \quad \mathcal{T}_{1428}}{\langle S_2, \sigma_{1425} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1424} = \frac{\langle x := x + 1, \sigma_{1423} \rangle \longrightarrow \sigma_{1424} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1425} \quad \mathcal{T}_{1426}}{\langle S_2, \sigma_{1423} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1422} = \frac{\langle x := x + 1, \sigma_{1421} \rangle \longrightarrow \sigma_{1422} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1423} \quad \mathcal{T}_{1424}}{\langle S_2, \sigma_{1421} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1420} = \frac{\langle x := x + 1, \sigma_{1419} \rangle \longrightarrow \sigma_{1420} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1421} \quad \mathcal{T}_{1422}}{\langle S_2, \sigma_{1419} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1418} = \frac{\langle x := x + 1, \sigma_{1417} \rangle \longrightarrow \sigma_{1418} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1419} \quad \mathcal{T}_{1420}}{\langle S_2, \sigma_{1417} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1416} = \frac{\langle x := x + 1, \sigma_{1415} \rangle \longrightarrow \sigma_{1416} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1417} \quad \mathcal{T}_{1418}}{\langle S_2, \sigma_{1415} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1414} = \frac{\langle x := x + 1, \sigma_{1413} \rangle \longrightarrow \sigma_{1414} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1415} \quad \mathcal{T}_{1416}}{\langle S_2, \sigma_{1413} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1412} = \frac{\langle x := x + 1, \sigma_{1411} \rangle \longrightarrow \sigma_{1412} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1413} \quad \mathcal{T}_{1414}}{\langle S_2, \sigma_{1411} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1410} = \frac{\langle x := x + 1, \sigma_{1409} \rangle \longrightarrow \sigma_{1410} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1411} \quad \mathcal{T}_{1412}}{\langle S_2, \sigma_{1409} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1408} = \frac{\langle x := x + 1, \sigma_{1407} \rangle \longrightarrow \sigma_{1408} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1409} \quad \mathcal{T}_{1410}}{\langle S_2, \sigma_{1407} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1406} = \frac{\langle x := x + 1, \sigma_{1405} \rangle \longrightarrow \sigma_{1406} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1407} \quad \mathcal{T}_{1408}}{\langle S_2, \sigma_{1405} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1404} = \frac{\langle x := x + 1, \sigma_{1403} \rangle \longrightarrow \sigma_{1404} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1405} \quad \mathcal{T}_{1406}}{\langle S_2, \sigma_{1403} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1402} = \frac{\langle x := x + 1, \sigma_{1401} \rangle \longrightarrow \sigma_{1402} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1403} \quad \mathcal{T}_{1404}}{\langle S_2, \sigma_{1401} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1400} = \frac{\langle x := x + 1, \sigma_{1399} \rangle \longrightarrow \sigma_{1400} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1401} \quad \mathcal{T}_{1402}}{\langle S_2, \sigma_{1399} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1398} = \frac{\langle x := x + 1, \sigma_{1397} \rangle \longrightarrow \sigma_{1398} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1399} \quad \mathcal{T}_{1400}}{\langle S_2, \sigma_{1397} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1396} = \frac{\langle x := x + 1, \sigma_{1395} \rangle \longrightarrow \sigma_{1396} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1397} \quad \mathcal{T}_{1398}}{\langle S_2, \sigma_{1395} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1394} = \frac{\langle x := x + 1, \sigma_{1393} \rangle \longrightarrow \sigma_{1394} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1395} \quad \mathcal{T}_{1396}}{\langle S_2, \sigma_{1393} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1392} = \frac{\langle x := x + 1, \sigma_{1391} \rangle \longrightarrow \sigma_{1392} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1393} \quad \mathcal{T}_{1394}}{\langle S_2, \sigma_{1391} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1390} = \frac{\langle x := x + 1, \sigma_{1389} \rangle \longrightarrow \sigma_{1390} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1391} \quad \mathcal{T}_{1392}}{\langle S_2, \sigma_{1389} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1388} = \frac{\langle x := x + 1, \sigma_{1387} \rangle \longrightarrow \sigma_{1388} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1389} \quad \mathcal{T}_{1390}}{\langle S_2, \sigma_{1387} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1386} = \frac{\langle x := x + 1, \sigma_{1385} \rangle \longrightarrow \sigma_{1386} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1387} \quad \mathcal{T}_{1388}}{\langle S_2, \sigma_{1385} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1384} = \frac{\langle x := x + 1, \sigma_{1383} \rangle \longrightarrow \sigma_{1384} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1385} \quad \mathcal{T}_{1386}}{\langle S_2, \sigma_{1383} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1382} = \frac{\langle x := x + 1, \sigma_{1381} \rangle \longrightarrow \sigma_{1382} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1383} \quad \mathcal{T}_{1384}}{\langle S_2, \sigma_{1381} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1380} = \frac{\langle x := x + 1, \sigma_{1379} \rangle \longrightarrow \sigma_{1380} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1381} \quad \mathcal{T}_{1382}}{\langle S_2, \sigma_{1379} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1378} = \frac{\langle x := x + 1, \sigma_{1377} \rangle \longrightarrow \sigma_{1378} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1379} \quad \mathcal{T}_{1380}}{\langle S_2, \sigma_{1377} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1376} = \frac{\langle x := x + 1, \sigma_{1375} \rangle \longrightarrow \sigma_{1376} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1377} \quad \mathcal{T}_{1378}}{\langle S_2, \sigma_{1375} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1374} = \frac{\langle x := x + 1, \sigma_{1373} \rangle \longrightarrow \sigma_{1374} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1375} \quad \mathcal{T}_{1376}}{\langle S_2, \sigma_{1373} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1372} = \frac{\langle x := x + 1, \sigma_{1371} \rangle \longrightarrow \sigma_{1372} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1373} \quad \mathcal{T}_{1374}}{\langle S_2, \sigma_{1371} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1370} = \frac{\langle x := x + 1, \sigma_{1369} \rangle \longrightarrow \sigma_{1370} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1371} \quad \mathcal{T}_{1372}}{\langle S_2, \sigma_{1369} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1368} = \frac{\langle x := x + 1, \sigma_{1367} \rangle \longrightarrow \sigma_{1368} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1369} \quad \mathcal{T}_{1370}}{\langle S_2, \sigma_{1367} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1366} = \frac{\langle x := x + 1, \sigma_{1365} \rangle \longrightarrow \sigma_{1366} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1367} \quad \mathcal{T}_{1368}}{\langle S_2, \sigma_{1365} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1364} = \frac{\langle x := x + 1, \sigma_{1363} \rangle \longrightarrow \sigma_{1364} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1365} \quad \mathcal{T}_{1366}}{\langle S_2, \sigma_{1363} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1362} = \frac{\langle x := x + 1, \sigma_{1361} \rangle \longrightarrow \sigma_{1362} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1363} \quad \mathcal{T}_{1364}}{\langle S_2, \sigma_{1361} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1360} = \frac{\langle x := x + 1, \sigma_{1359} \rangle \longrightarrow \sigma_{1360} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{1361} \quad \mathcal{T}_{1362}}{\langle S_2, \sigma_{1359} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1358} = \frac{\langle x := x + 1, \sigma_{1357} \rangle \longrightarrow \sigma_{1358} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1359} \quad \mathcal{T}_{1360}}{\langle S_2, \sigma_{1357} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1356} = \frac{\langle x := x + 1, \sigma_{1355} \rangle \longrightarrow \sigma_{1356} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1357} \quad \mathcal{T}_{1358}}{\langle S_2, \sigma_{1355} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1354} = \frac{\langle x := x + 1, \sigma_{1353} \rangle \longrightarrow \sigma_{1354} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1355} \quad \mathcal{T}_{1356}}{\langle S_2, \sigma_{1353} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1352} = \frac{\langle x := x + 1, \sigma_{1351} \rangle \longrightarrow \sigma_{1352} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1353} \quad \mathcal{T}_{1354}}{\langle S_2, \sigma_{1351} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1350} = \frac{\langle x := x + 1, \sigma_{1349} \rangle \longrightarrow \sigma_{1350} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1351} \quad \mathcal{T}_{1352}}{\langle S_2, \sigma_{1349} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1348} = \frac{\langle x := x + 1, \sigma_{1347} \rangle \longrightarrow \sigma_{1348} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1349} \quad \mathcal{T}_{1350}}{\langle S_2, \sigma_{1347} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1346} = \frac{\langle x := x + 1, \sigma_{1345} \rangle \longrightarrow \sigma_{1346} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1347} \quad \mathcal{T}_{1348}}{\langle S_2, \sigma_{1345} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1344} = \frac{\langle x := x + 1, \sigma_{1343} \rangle \longrightarrow \sigma_{1344} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1345} \quad \mathcal{T}_{1346}}{\langle S_2, \sigma_{1343} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1342} = \frac{\langle x := x + 1, \sigma_{1341} \rangle \longrightarrow \sigma_{1342} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1343} \quad \mathcal{T}_{1344}}{\langle S_2, \sigma_{1341} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1340} = \frac{\langle x := x + 1, \sigma_{1339} \rangle \longrightarrow \sigma_{1340} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1341} \quad \mathcal{T}_{1342}}{\langle S_2, \sigma_{1339} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1338} = \frac{\langle x := x + 1, \sigma_{1337} \rangle \longrightarrow \sigma_{1338} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1339} \quad \mathcal{T}_{1340}}{\langle S_2, \sigma_{1337} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1336} = \frac{\langle x := x + 1, \sigma_{1335} \rangle \longrightarrow \sigma_{1336} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1337} \quad \mathcal{T}_{1338}}{\langle S_2, \sigma_{1335} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1334} = \frac{\langle x := x + 1, \sigma_{1333} \rangle \longrightarrow \sigma_{1334} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1335} \quad \mathcal{T}_{1336}}{\langle S_2, \sigma_{1333} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1332} = \frac{\langle x := x + 1, \sigma_{1331} \rangle \longrightarrow \sigma_{1332} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1333} \quad \mathcal{T}_{1334}}{\langle S_2, \sigma_{1331} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1330} = \frac{\langle x := x + 1, \sigma_{1329} \rangle \longrightarrow \sigma_{1330} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1331} \quad \mathcal{T}_{1332}}{\langle S_2, \sigma_{1329} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1328} = \frac{\langle x := x + 1, \sigma_{1327} \rangle \longrightarrow \sigma_{1328} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1329} \quad \mathcal{T}_{1330}}{\langle S_2, \sigma_{1327} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1326} = \frac{\langle x := x + 1, \sigma_{1325} \rangle \longrightarrow \sigma_{1326} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1327} \quad \mathcal{T}_{1328}}{\langle S_2, \sigma_{1325} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1324} = \frac{\langle x := x + 1, \sigma_{1323} \rangle \longrightarrow \sigma_{1324} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1325} \quad \mathcal{T}_{1326}}{\langle S_2, \sigma_{1323} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1322} = \frac{\langle x := x + 1, \sigma_{1321} \rangle \longrightarrow \sigma_{1322} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1323} \quad \mathcal{T}_{1324}}{\langle S_2, \sigma_{1321} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1320} = \frac{\langle x := x + 1, \sigma_{1319} \rangle \longrightarrow \sigma_{1320} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1321} \quad \mathcal{T}_{1322}}{\langle S_2, \sigma_{1319} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1318} = \frac{\langle x := x + 1, \sigma_{1317} \rangle \longrightarrow \sigma_{1318} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1319} \quad \mathcal{T}_{1320}}{\langle S_2, \sigma_{1317} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1316} = \frac{\langle x := x + 1, \sigma_{1315} \rangle \longrightarrow \sigma_{1316} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1317} \quad \mathcal{T}_{1318}}{\langle S_2, \sigma_{1315} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1314} = \frac{\langle x := x + 1, \sigma_{1313} \rangle \longrightarrow \sigma_{1314} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1315} \quad \mathcal{T}_{1316}}{\langle S_2, \sigma_{1313} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1312} = \frac{\langle x := x + 1, \sigma_{1311} \rangle \longrightarrow \sigma_{1312} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1313} \quad \mathcal{T}_{1314}}{\langle S_2, \sigma_{1311} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1310} = \frac{\langle x := x + 1, \sigma_{1309} \rangle \longrightarrow \sigma_{1310} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1311} \quad \mathcal{T}_{1312}}{\langle S_2, \sigma_{1309} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1308} = \frac{\langle x := x + 1, \sigma_{1307} \rangle \longrightarrow \sigma_{1308} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1309} \quad \mathcal{T}_{1310}}{\langle S_2, \sigma_{1307} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1306} = \frac{\langle x := x + 1, \sigma_{1305} \rangle \longrightarrow \sigma_{1306} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1307} \quad \mathcal{T}_{1308}}{\langle S_2, \sigma_{1305} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1304} = \frac{\langle x := x + 1, \sigma_{1303} \rangle \longrightarrow \sigma_{1304} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1305} \quad \mathcal{T}_{1306}}{\langle S_2, \sigma_{1303} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1302} = \frac{\langle x := x + 1, \sigma_{1301} \rangle \longrightarrow \sigma_{1302} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1303} \quad \mathcal{T}_{1304}}{\langle S_2, \sigma_{1301} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1300} = \frac{\langle x := x + 1, \sigma_{1299} \rangle \longrightarrow \sigma_{1300} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1301} \quad \mathcal{T}_{1302}}{\langle S_2, \sigma_{1299} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1298} = \frac{\langle x := x + 1, \sigma_{1297} \rangle \longrightarrow \sigma_{1298} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1299} \quad \mathcal{T}_{1300}}{\langle S_2, \sigma_{1297} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1296} = \frac{\langle x := x + 1, \sigma_{1295} \rangle \longrightarrow \sigma_{1296} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1297} \quad \mathcal{T}_{1298}}{\langle S_2, \sigma_{1295} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1294} = \frac{\langle x := x + 1, \sigma_{1293} \rangle \longrightarrow \sigma_{1294} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1295} \quad \mathcal{T}_{1296}}{\langle S_2, \sigma_{1293} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1292} = \frac{\langle x := x + 1, \sigma_{1291} \rangle \longrightarrow \sigma_{1292} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1293} \quad \mathcal{T}_{1294}}{\langle S_2, \sigma_{1291} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1290} = \frac{\langle x := x + 1, \sigma_{1289} \rangle \longrightarrow \sigma_{1290} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1291} \quad \mathcal{T}_{1292}}{\langle S_2, \sigma_{1289} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1288} = \frac{\langle x := x + 1, \sigma_{1287} \rangle \longrightarrow \sigma_{1288} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1289} \quad \mathcal{T}_{1290}}{\langle S_2, \sigma_{1287} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1286} = \frac{\langle x := x + 1, \sigma_{1285} \rangle \longrightarrow \sigma_{1286} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1287} \quad \mathcal{T}_{1288}}{\langle S_2, \sigma_{1285} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1284} = \frac{\langle x := x + 1, \sigma_{1283} \rangle \longrightarrow \sigma_{1284} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1285} \quad \mathcal{T}_{1286}}{\langle S_2, \sigma_{1283} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1282} = \frac{\langle x := x + 1, \sigma_{1281} \rangle \longrightarrow \sigma_{1282} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1283} \quad \mathcal{T}_{1284}}{\langle S_2, \sigma_{1281} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1280} = \frac{\langle x := x + 1, \sigma_{1279} \rangle \longrightarrow \sigma_{1280} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1281} \quad \mathcal{T}_{1282}}{\langle S_2, \sigma_{1279} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1278} = \frac{\langle x := x + 1, \sigma_{1277} \rangle \longrightarrow \sigma_{1278} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1279} \quad \mathcal{T}_{1280}}{\langle S_2, \sigma_{1277} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1276} = \frac{\langle x := x + 1, \sigma_{1275} \rangle \longrightarrow \sigma_{1276} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1277} \quad \mathcal{T}_{1278}}{\langle S_2, \sigma_{1275} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1274} = \frac{\langle x := x + 1, \sigma_{1273} \rangle \longrightarrow \sigma_{1274} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1275} \quad \mathcal{T}_{1276}}{\langle S_2, \sigma_{1273} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1272} = \frac{\langle x := x + 1, \sigma_{1271} \rangle \longrightarrow \sigma_{1272} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1273} \quad \mathcal{T}_{1274}}{\langle S_2, \sigma_{1271} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1270} = \frac{\langle x := x + 1, \sigma_{1269} \rangle \longrightarrow \sigma_{1270} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1271} \quad \mathcal{T}_{1272}}{\langle S_2, \sigma_{1269} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1268} = \frac{\langle x := x + 1, \sigma_{1267} \rangle \longrightarrow \sigma_{1268} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1269} \quad \mathcal{T}_{1270}}{\langle S_2, \sigma_{1267} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1266} = \frac{\langle x := x + 1, \sigma_{1265} \rangle \longrightarrow \sigma_{1266} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1267} \quad \mathcal{T}_{1268}}{\langle S_2, \sigma_{1265} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1264} = \frac{\langle x := x + 1, \sigma_{1263} \rangle \longrightarrow \sigma_{1264} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1265} \quad \mathcal{T}_{1266}}{\langle S_2, \sigma_{1263} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1262} = \frac{\langle x := x + 1, \sigma_{1261} \rangle \longrightarrow \sigma_{1262} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1263} \quad \mathcal{T}_{1264}}{\langle S_2, \sigma_{1261} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1260} = \frac{\langle x := x + 1, \sigma_{1259} \rangle \longrightarrow \sigma_{1260} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1261} \quad \mathcal{T}_{1262}}{\langle S_2, \sigma_{1259} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1258} = \frac{\langle x := x + 1, \sigma_{1257} \rangle \longrightarrow \sigma_{1258} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1259} \quad \mathcal{T}_{1260}}{\langle S_2, \sigma_{1257} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1256} = \frac{\langle x := x + 1, \sigma_{1255} \rangle \longrightarrow \sigma_{1256} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1257} \quad \mathcal{T}_{1258}}{\langle S_2, \sigma_{1255} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1254} = \frac{\langle x := x + 1, \sigma_{1253} \rangle \longrightarrow \sigma_{1254} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1255} \quad \mathcal{T}_{1256}}{\langle S_2, \sigma_{1253} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1252} = \frac{\langle x := x + 1, \sigma_{1251} \rangle \longrightarrow \sigma_{1252} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1253} \quad \mathcal{T}_{1254}}{\langle S_2, \sigma_{1251} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1250} = \frac{\langle x := x + 1, \sigma_{1249} \rangle \longrightarrow \sigma_{1250} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1251} \quad \mathcal{T}_{1252}}{\langle S_2, \sigma_{1249} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1248} = \frac{\langle x := x + 1, \sigma_{1247} \rangle \longrightarrow \sigma_{1248} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1249} \quad \mathcal{T}_{1250}}{\langle S_2, \sigma_{1247} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1246} = \frac{\langle x := x + 1, \sigma_{1245} \rangle \longrightarrow \sigma_{1246} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1247} \quad \mathcal{T}_{1248}}{\langle S_2, \sigma_{1245} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1244} = \frac{\langle x := x + 1, \sigma_{1243} \rangle \longrightarrow \sigma_{1244} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1245} \quad \mathcal{T}_{1246}}{\langle S_2, \sigma_{1243} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1242} = \frac{\langle x := x + 1, \sigma_{1241} \rangle \longrightarrow \sigma_{1242} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1243} \quad \mathcal{T}_{1244}}{\langle S_2, \sigma_{1241} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1240} = \frac{\langle x := x + 1, \sigma_{1239} \rangle \longrightarrow \sigma_{1240} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1241} \quad \mathcal{T}_{1242}}{\langle S_2, \sigma_{1239} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1238} = \frac{\langle x := x + 1, \sigma_{1237} \rangle \longrightarrow \sigma_{1238} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1239} \quad \mathcal{T}_{1240}}{\langle S_2, \sigma_{1237} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1236} = \frac{\langle x := x + 1, \sigma_{1235} \rangle \longrightarrow \sigma_{1236} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1237} \quad \mathcal{T}_{1238}}{\langle S_2, \sigma_{1235} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1234} = \frac{\langle x := x + 1, \sigma_{1233} \rangle \longrightarrow \sigma_{1234} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1235} \quad \mathcal{T}_{1236}}{\langle S_2, \sigma_{1233} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1232} = \frac{\langle x := x + 1, \sigma_{1231} \rangle \longrightarrow \sigma_{1232} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1233} \quad \mathcal{T}_{1234}}{\langle S_2, \sigma_{1231} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1230} = \frac{\langle x := x + 1, \sigma_{1229} \rangle \longrightarrow \sigma_{1230} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1231} \quad \mathcal{T}_{1232}}{\langle S_2, \sigma_{1229} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1228} = \frac{\langle x := x + 1, \sigma_{1227} \rangle \longrightarrow \sigma_{1228} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1229} \quad \mathcal{T}_{1230}}{\langle S_2, \sigma_{1227} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1226} = \frac{\langle x := x + 1, \sigma_{1225} \rangle \longrightarrow \sigma_{1226} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1227} \quad \mathcal{T}_{1228}}{\langle S_2, \sigma_{1225} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1224} = \frac{\langle x := x + 1, \sigma_{1223} \rangle \longrightarrow \sigma_{1224} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1225} \quad \mathcal{T}_{1226}}{\langle S_2, \sigma_{1223} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1222} = \frac{\langle x := x + 1, \sigma_{1221} \rangle \longrightarrow \sigma_{1222} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1223} \quad \mathcal{T}_{1224}}{\langle S_2, \sigma_{1221} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1220} = \frac{\langle x := x + 1, \sigma_{1219} \rangle \longrightarrow \sigma_{1220} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1221} \quad \mathcal{T}_{1222}}{\langle S_2, \sigma_{1219} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1218} = \frac{\langle x := x + 1, \sigma_{1217} \rangle \longrightarrow \sigma_{1218} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1219} \quad \mathcal{T}_{1220}}{\langle S_2, \sigma_{1217} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1216} = \frac{\langle x := x + 1, \sigma_{1215} \rangle \longrightarrow \sigma_{1216} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1217} \quad \mathcal{T}_{1218}}{\langle S_2, \sigma_{1215} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1214} = \frac{\langle x := x + 1, \sigma_{1213} \rangle \longrightarrow \sigma_{1214} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1215} \quad \mathcal{T}_{1216}}{\langle S_2, \sigma_{1213} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1212} = \frac{\langle x := x + 1, \sigma_{1211} \rangle \longrightarrow \sigma_{1212} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1213} \quad \mathcal{T}_{1214}}{\langle S_2, \sigma_{1211} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1210} = \frac{\langle x := x + 1, \sigma_{1209} \rangle \longrightarrow \sigma_{1210} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1211} \quad \mathcal{T}_{1212}}{\langle S_2, \sigma_{1209} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1208} = \frac{\langle x := x + 1, \sigma_{1207} \rangle \longrightarrow \sigma_{1208} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1209} \quad \mathcal{T}_{1210}}{\langle S_2, \sigma_{1207} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1206} = \frac{\langle x := x + 1, \sigma_{1205} \rangle \longrightarrow \sigma_{1206} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1207} \quad \mathcal{T}_{1208}}{\langle S_2, \sigma_{1205} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1204} = \frac{\langle x := x + 1, \sigma_{1203} \rangle \longrightarrow \sigma_{1204} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1205} \quad \mathcal{T}_{1206}}{\langle S_2, \sigma_{1203} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1202} = \frac{\langle x := x + 1, \sigma_{1201} \rangle \longrightarrow \sigma_{1202} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1203} \quad \mathcal{T}_{1204}}{\langle S_2, \sigma_{1201} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1200} = \frac{\langle x := x + 1, \sigma_{1199} \rangle \longrightarrow \sigma_{1200} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1201} \quad \mathcal{T}_{1202}}{\langle S_2, \sigma_{1199} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1198} = \frac{\langle x := x + 1, \sigma_{1197} \rangle \longrightarrow \sigma_{1198} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1199} \quad \mathcal{T}_{1200}}{\langle S_2, \sigma_{1197} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1196} = \frac{\langle x := x + 1, \sigma_{1195} \rangle \longrightarrow \sigma_{1196} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1197} \quad \mathcal{T}_{1198}}{\langle S_2, \sigma_{1195} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1194} = \frac{\langle x := x + 1, \sigma_{1193} \rangle \longrightarrow \sigma_{1194} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1195} \quad \mathcal{T}_{1196}}{\langle S_2, \sigma_{1193} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1192} = \frac{\langle x := x + 1, \sigma_{1191} \rangle \longrightarrow \sigma_{1192} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1193} \quad \mathcal{T}_{1194}}{\langle S_2, \sigma_{1191} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1190} = \frac{\langle x := x + 1, \sigma_{1189} \rangle \longrightarrow \sigma_{1190} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1191} \quad \mathcal{T}_{1192}}{\langle S_2, \sigma_{1189} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1188} = \frac{\langle x := x + 1, \sigma_{1187} \rangle \longrightarrow \sigma_{1188} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1189} \quad \mathcal{T}_{1190}}{\langle S_2, \sigma_{1187} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1186} = \frac{\langle x := x + 1, \sigma_{1185} \rangle \longrightarrow \sigma_{1186} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1187} \quad \mathcal{T}_{1188}}{\langle S_2, \sigma_{1185} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1184} = \frac{\langle x := x + 1, \sigma_{1183} \rangle \longrightarrow \sigma_{1184} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1185} \quad \mathcal{T}_{1186}}{\langle S_2, \sigma_{1183} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1182} = \frac{\langle x := x + 1, \sigma_{1181} \rangle \longrightarrow \sigma_{1182} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1183} \quad \mathcal{T}_{1184}}{\langle S_2, \sigma_{1181} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1180} = \frac{\langle x := x + 1, \sigma_{1179} \rangle \longrightarrow \sigma_{1180} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1181} \quad \mathcal{T}_{1182}}{\langle S_2, \sigma_{1179} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1178} = \frac{\langle x := x + 1, \sigma_{1177} \rangle \longrightarrow \sigma_{1178} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1179} \quad \mathcal{T}_{1180}}{\langle S_2, \sigma_{1177} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1176} = \frac{\langle x := x + 1, \sigma_{1175} \rangle \longrightarrow \sigma_{1176} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1177} \quad \mathcal{T}_{1178}}{\langle S_2, \sigma_{1175} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1174} = \frac{\langle x := x + 1, \sigma_{1173} \rangle \longrightarrow \sigma_{1174} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1175} \quad \mathcal{T}_{1176}}{\langle S_2, \sigma_{1173} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1172} = \frac{\langle x := x + 1, \sigma_{1171} \rangle \longrightarrow \sigma_{1172} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1173} \quad \mathcal{T}_{1174}}{\langle S_2, \sigma_{1171} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1170} = \frac{\langle x := x + 1, \sigma_{1169} \rangle \longrightarrow \sigma_{1170} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1171} \quad \mathcal{T}_{1172}}{\langle S_2, \sigma_{1169} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1168} = \frac{\langle x := x + 1, \sigma_{1167} \rangle \longrightarrow \sigma_{1168} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1169} \quad \mathcal{T}_{1170}}{\langle S_2, \sigma_{1167} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1166} = \frac{\langle x := x + 1, \sigma_{1165} \rangle \longrightarrow \sigma_{1166} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1167} \quad \mathcal{T}_{1168}}{\langle S_2, \sigma_{1165} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1164} = \frac{\langle x := x + 1, \sigma_{1163} \rangle \longrightarrow \sigma_{1164} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1165} \quad \mathcal{T}_{1166}}{\langle S_2, \sigma_{1163} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1162} = \frac{\langle x := x + 1, \sigma_{1161} \rangle \longrightarrow \sigma_{1162} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1163} \quad \mathcal{T}_{1164}}{\langle S_2, \sigma_{1161} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1160} = \frac{\langle x := x + 1, \sigma_{1159} \rangle \longrightarrow \sigma_{1160} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1161} \quad \mathcal{T}_{1162}}{\langle S_2, \sigma_{1159} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1158} = \frac{\langle x := x + 1, \sigma_{1157} \rangle \longrightarrow \sigma_{1158} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1159} \quad \mathcal{T}_{1160}}{\langle S_2, \sigma_{1157} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1156} = \frac{\langle x := x + 1, \sigma_{1155} \rangle \longrightarrow \sigma_{1156} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1157} \quad \mathcal{T}_{1158}}{\langle S_2, \sigma_{1155} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1154} = \frac{\langle x := x + 1, \sigma_{1153} \rangle \longrightarrow \sigma_{1154} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1155} \quad \mathcal{T}_{1156}}{\langle S_2, \sigma_{1153} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1152} = \frac{\langle x := x + 1, \sigma_{1151} \rangle \longrightarrow \sigma_{1152} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1153} \quad \mathcal{T}_{1154}}{\langle S_2, \sigma_{1151} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1150} = \frac{\langle x := x + 1, \sigma_{1149} \rangle \longrightarrow \sigma_{1150} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1151} \quad \mathcal{T}_{1152}}{\langle S_2, \sigma_{1149} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1148} = \frac{\langle x := x + 1, \sigma_{1147} \rangle \longrightarrow \sigma_{1148} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1149} \quad \mathcal{T}_{1150}}{\langle S_2, \sigma_{1147} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1146} = \frac{\langle x := x + 1, \sigma_{1145} \rangle \longrightarrow \sigma_{1146} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1147} \quad \mathcal{T}_{1148}}{\langle S_2, \sigma_{1145} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1144} = \frac{\langle x := x + 1, \sigma_{1143} \rangle \longrightarrow \sigma_{1144} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1145} \quad \mathcal{T}_{1146}}{\langle S_2, \sigma_{1143} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1142} = \frac{\langle x := x + 1, \sigma_{1141} \rangle \longrightarrow \sigma_{1142} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1143} \quad \mathcal{T}_{1144}}{\langle S_2, \sigma_{1141} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1140} = \frac{\langle x := x + 1, \sigma_{1139} \rangle \longrightarrow \sigma_{1140} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1141} \quad \mathcal{T}_{1142}}{\langle S_2, \sigma_{1139} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1138} = \frac{\langle x := x + 1, \sigma_{1137} \rangle \longrightarrow \sigma_{1138} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1139} \quad \mathcal{T}_{1140}}{\langle S_2, \sigma_{1137} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1136} = \frac{\langle x := x + 1, \sigma_{1135} \rangle \longrightarrow \sigma_{1136} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1137} \quad \mathcal{T}_{1138}}{\langle S_2, \sigma_{1135} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1134} = \frac{\langle x := x + 1, \sigma_{1133} \rangle \longrightarrow \sigma_{1134} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1135} \quad \mathcal{T}_{1136}}{\langle S_2, \sigma_{1133} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1132} = \frac{\langle x := x + 1, \sigma_{1131} \rangle \longrightarrow \sigma_{1132} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1133} \quad \mathcal{T}_{1134}}{\langle S_2, \sigma_{1131} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1130} = \frac{\langle x := x + 1, \sigma_{1129} \rangle \longrightarrow \sigma_{1130} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1131} \quad \mathcal{T}_{1132}}{\langle S_2, \sigma_{1129} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1128} = \frac{\langle x := x + 1, \sigma_{1127} \rangle \longrightarrow \sigma_{1128} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1129} \quad \mathcal{T}_{1130}}{\langle S_2, \sigma_{1127} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1126} = \frac{\langle x := x + 1, \sigma_{1125} \rangle \longrightarrow \sigma_{1126} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1127} \quad \mathcal{T}_{1128}}{\langle S_2, \sigma_{1125} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1124} = \frac{\langle x := x + 1, \sigma_{1123} \rangle \longrightarrow \sigma_{1124} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1125} \quad \mathcal{T}_{1126}}{\langle S_2, \sigma_{1123} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1122} = \frac{\langle x := x + 1, \sigma_{1121} \rangle \longrightarrow \sigma_{1122} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1123} \quad \mathcal{T}_{1124}}{\langle S_2, \sigma_{1121} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1120} = \frac{\langle x := x + 1, \sigma_{1119} \rangle \longrightarrow \sigma_{1120} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1121} \quad \mathcal{T}_{1122}}{\langle S_2, \sigma_{1119} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1118} = \frac{\langle x := x + 1, \sigma_{1117} \rangle \longrightarrow \sigma_{1118} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1119} \quad \mathcal{T}_{1120}}{\langle S_2, \sigma_{1117} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1116} = \frac{\langle x := x + 1, \sigma_{1115} \rangle \longrightarrow \sigma_{1116} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1117} \quad \mathcal{T}_{1118}}{\langle S_2, \sigma_{1115} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1114} = \frac{\langle x := x + 1, \sigma_{1113} \rangle \longrightarrow \sigma_{1114} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1115} \quad \mathcal{T}_{1116}}{\langle S_2, \sigma_{1113} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1112} = \frac{\langle x := x + 1, \sigma_{1111} \rangle \longrightarrow \sigma_{1112} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1113} \quad \mathcal{T}_{1114}}{\langle S_2, \sigma_{1111} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1110} = \frac{\langle x := x + 1, \sigma_{1109} \rangle \longrightarrow \sigma_{1110} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1111} \quad \mathcal{T}_{1112}}{\langle S_2, \sigma_{1109} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1108} = \frac{\langle x := x + 1, \sigma_{1107} \rangle \longrightarrow \sigma_{1108} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1109} \quad \mathcal{T}_{1110}}{\langle S_2, \sigma_{1107} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1106} = \frac{\langle x := x + 1, \sigma_{1105} \rangle \longrightarrow \sigma_{1106} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1107} \quad \mathcal{T}_{1108}}{\langle S_2, \sigma_{1105} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1104} = \frac{\langle x := x + 1, \sigma_{1103} \rangle \longrightarrow \sigma_{1104} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1105} \quad \mathcal{T}_{1106}}{\langle S_2, \sigma_{1103} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1102} = \frac{\langle x := x + 1, \sigma_{1101} \rangle \longrightarrow \sigma_{1102} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1103} \quad \mathcal{T}_{1104}}{\langle S_2, \sigma_{1101} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1100} = \frac{\langle x := x + 1, \sigma_{1099} \rangle \longrightarrow \sigma_{1100} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1101} \quad \mathcal{T}_{1102}}{\langle S_2, \sigma_{1099} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1098} = \frac{\langle x := x + 1, \sigma_{1097} \rangle \longrightarrow \sigma_{1098} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1099} \quad \mathcal{T}_{1100}}{\langle S_2, \sigma_{1097} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1096} = \frac{\langle x := x + 1, \sigma_{1095} \rangle \longrightarrow \sigma_{1096} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1097} \quad \mathcal{T}_{1098}}{\langle S_2, \sigma_{1095} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1094} = \frac{\langle x := x + 1, \sigma_{1093} \rangle \longrightarrow \sigma_{1094} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1095} \quad \mathcal{T}_{1096}}{\langle S_2, \sigma_{1093} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1092} = \frac{\langle x := x + 1, \sigma_{1091} \rangle \longrightarrow \sigma_{1092} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1093} \quad \mathcal{T}_{1094}}{\langle S_2, \sigma_{1091} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1090} = \frac{\langle x := x + 1, \sigma_{1089} \rangle \longrightarrow \sigma_{1090} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1091} \quad \mathcal{T}_{1092}}{\langle S_2, \sigma_{1089} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1088} = \frac{\langle x := x + 1, \sigma_{1087} \rangle \longrightarrow \sigma_{1088} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1089} \quad \mathcal{T}_{1090}}{\langle S_2, \sigma_{1087} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1086} = \frac{\langle x := x + 1, \sigma_{1085} \rangle \longrightarrow \sigma_{1086} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1087} \quad \mathcal{T}_{1088}}{\langle S_2, \sigma_{1085} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1084} = \frac{\langle x := x + 1, \sigma_{1083} \rangle \longrightarrow \sigma_{1084} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1085} \quad \mathcal{T}_{1086}}{\langle S_2, \sigma_{1083} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1082} = \frac{\langle x := x + 1, \sigma_{1081} \rangle \longrightarrow \sigma_{1082} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1083} \quad \mathcal{T}_{1084}}{\langle S_2, \sigma_{1081} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1080} = \frac{\langle x := x + 1, \sigma_{1079} \rangle \longrightarrow \sigma_{1080} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1081} \quad \mathcal{T}_{1082}}{\langle S_2, \sigma_{1079} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1078} = \frac{\langle x := x + 1, \sigma_{1077} \rangle \longrightarrow \sigma_{1078} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1079} \quad \mathcal{T}_{1080}}{\langle S_2, \sigma_{1077} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1076} = \frac{\langle x := x + 1, \sigma_{1075} \rangle \longrightarrow \sigma_{1076} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1077} \quad \mathcal{T}_{1078}}{\langle S_2, \sigma_{1075} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1074} = \frac{\langle x := x + 1, \sigma_{1073} \rangle \longrightarrow \sigma_{1074} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1075} \quad \mathcal{T}_{1076}}{\langle S_2, \sigma_{1073} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1072} = \frac{\langle x := x + 1, \sigma_{1071} \rangle \longrightarrow \sigma_{1072} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1073} \quad \mathcal{T}_{1074}}{\langle S_2, \sigma_{1071} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1070} = \frac{\langle x := x + 1, \sigma_{1069} \rangle \longrightarrow \sigma_{1070} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1071} \quad \mathcal{T}_{1072}}{\langle S_2, \sigma_{1069} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1068} = \frac{\langle x := x + 1, \sigma_{1067} \rangle \longrightarrow \sigma_{1068} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1069} \quad \mathcal{T}_{1070}}{\langle S_2, \sigma_{1067} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1066} = \frac{\langle x := x + 1, \sigma_{1065} \rangle \longrightarrow \sigma_{1066} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1067} \quad \mathcal{T}_{1068}}{\langle S_2, \sigma_{1065} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1064} = \frac{\langle x := x + 1, \sigma_{1063} \rangle \longrightarrow \sigma_{1064} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1065} \quad \mathcal{T}_{1066}}{\langle S_2, \sigma_{1063} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1062} = \frac{\langle x := x + 1, \sigma_{1061} \rangle \longrightarrow \sigma_{1062} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1063} \quad \mathcal{T}_{1064}}{\langle S_2, \sigma_{1061} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1060} = \frac{\langle x := x + 1, \sigma_{1059} \rangle \longrightarrow \sigma_{1060} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1061} \quad \mathcal{T}_{1062}}{\langle S_2, \sigma_{1059} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1058} = \frac{\langle x := x + 1, \sigma_{1057} \rangle \longrightarrow \sigma_{1058} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1059} \quad \mathcal{T}_{1060}}{\langle S_2, \sigma_{1057} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1056} = \frac{\langle x := x + 1, \sigma_{1055} \rangle \longrightarrow \sigma_{1056} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1057} \quad \mathcal{T}_{1058}}{\langle S_2, \sigma_{1055} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1054} = \frac{\langle x := x + 1, \sigma_{1053} \rangle \longrightarrow \sigma_{1054} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1055} \quad \mathcal{T}_{1056}}{\langle S_2, \sigma_{1053} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1052} = \frac{\langle x := x + 1, \sigma_{1051} \rangle \longrightarrow \sigma_{1052} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1053} \quad \mathcal{T}_{1054}}{\langle S_2, \sigma_{1051} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1050} = \frac{\langle x := x + 1, \sigma_{1049} \rangle \longrightarrow \sigma_{1050} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1051} \quad \mathcal{T}_{1052}}{\langle S_2, \sigma_{1049} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1048} = \frac{\langle x := x + 1, \sigma_{1047} \rangle \longrightarrow \sigma_{1048} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1049} \quad \mathcal{T}_{1050}}{\langle S_2, \sigma_{1047} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1046} = \frac{\langle x := x + 1, \sigma_{1045} \rangle \longrightarrow \sigma_{1046} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1047} \quad \mathcal{T}_{1048}}{\langle S_2, \sigma_{1045} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1044} = \frac{\langle x := x + 1, \sigma_{1043} \rangle \longrightarrow \sigma_{1044} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1045} \quad \mathcal{T}_{1046}}{\langle S_2, \sigma_{1043} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1042} = \frac{\langle x := x + 1, \sigma_{1041} \rangle \longrightarrow \sigma_{1042} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1043} \quad \mathcal{T}_{1044}}{\langle S_2, \sigma_{1041} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1040} = \frac{\langle x := x + 1, \sigma_{1039} \rangle \longrightarrow \sigma_{1040} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1041} \quad \mathcal{T}_{1042}}{\langle S_2, \sigma_{1039} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1038} = \frac{\langle x := x + 1, \sigma_{1037} \rangle \longrightarrow \sigma_{1038} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1039} \quad \mathcal{T}_{1040}}{\langle S_2, \sigma_{1037} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1036} = \frac{\langle x := x + 1, \sigma_{1035} \rangle \longrightarrow \sigma_{1036} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1037} \quad \mathcal{T}_{1038}}{\langle S_2, \sigma_{1035} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1034} = \frac{\langle x := x + 1, \sigma_{1033} \rangle \longrightarrow \sigma_{1034} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1035} \quad \mathcal{T}_{1036}}{\langle S_2, \sigma_{1033} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1032} = \frac{\langle x := x + 1, \sigma_{1031} \rangle \longrightarrow \sigma_{1032} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1033} \quad \mathcal{T}_{1034}}{\langle S_2, \sigma_{1031} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1030} = \frac{\langle x := x + 1, \sigma_{1029} \rangle \longrightarrow \sigma_{1030} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1031} \quad \mathcal{T}_{1032}}{\langle S_2, \sigma_{1029} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1028} = \frac{\langle x := x + 1, \sigma_{1027} \rangle \longrightarrow \sigma_{1028} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1029} \quad \mathcal{T}_{1030}}{\langle S_2, \sigma_{1027} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1026} = \frac{\langle x := x + 1, \sigma_{1025} \rangle \longrightarrow \sigma_{1026} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1027} \quad \mathcal{T}_{1028}}{\langle S_2, \sigma_{1025} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1024} = \frac{\langle x := x + 1, \sigma_{1023} \rangle \longrightarrow \sigma_{1024} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1025} \quad \mathcal{T}_{1026}}{\langle S_2, \sigma_{1023} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1022} = \frac{\langle x := x + 1, \sigma_{1021} \rangle \longrightarrow \sigma_{1022} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1023} \quad \mathcal{T}_{1024}}{\langle S_2, \sigma_{1021} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1020} = \frac{\langle x := x + 1, \sigma_{1019} \rangle \longrightarrow \sigma_{1020} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1021} \quad \mathcal{T}_{1022}}{\langle S_2, \sigma_{1019} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1018} = \frac{\langle x := x + 1, \sigma_{1017} \rangle \longrightarrow \sigma_{1018} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1019} \quad \mathcal{T}_{1020}}{\langle S_2, \sigma_{1017} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1016} = \frac{\langle x := x + 1, \sigma_{1015} \rangle \longrightarrow \sigma_{1016} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1017} \quad \mathcal{T}_{1018}}{\langle S_2, \sigma_{1015} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1014} = \frac{\langle x := x + 1, \sigma_{1013} \rangle \longrightarrow \sigma_{1014} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1015} \quad \mathcal{T}_{1016}}{\langle S_2, \sigma_{1013} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1012} = \frac{\langle x := x + 1, \sigma_{1011} \rangle \longrightarrow \sigma_{1012} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1013} \quad \mathcal{T}_{1014}}{\langle S_2, \sigma_{1011} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1010} = \frac{\langle x := x + 1, \sigma_{1009} \rangle \longrightarrow \sigma_{1010} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1011} \quad \mathcal{T}_{1012}}{\langle S_2, \sigma_{1009} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1008} = \frac{\langle x := x + 1, \sigma_{1007} \rangle \longrightarrow \sigma_{1008} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1009} \quad \mathcal{T}_{1010}}{\langle S_2, \sigma_{1007} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1006} = \frac{\langle x := x + 1, \sigma_{1005} \rangle \longrightarrow \sigma_{1006} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1007} \quad \mathcal{T}_{1008}}{\langle S_2, \sigma_{1005} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1004} = \frac{\langle x := x + 1, \sigma_{1003} \rangle \longrightarrow \sigma_{1004} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1005} \quad \mathcal{T}_{1006}}{\langle S_2, \sigma_{1003} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1002} = \frac{\langle x := x + 1, \sigma_{1001} \rangle \longrightarrow \sigma_{1002} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1003} \quad \mathcal{T}_{1004}}{\langle S_2, \sigma_{1001} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{1000} = \frac{\langle x := x + 1, \sigma_{999} \rangle \longrightarrow \sigma_{1000} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{1001} \quad \mathcal{T}_{1002}}{\langle S_2, \sigma_{999} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{998} = \frac{\langle x := x + 1, \sigma_{997} \rangle \longrightarrow \sigma_{998} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{999} \quad \mathcal{T}_{1000}}{\langle S_2, \sigma_{997} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{996} = \frac{\langle x := x + 1, \sigma_{995} \rangle \longrightarrow \sigma_{996} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{997} \quad \mathcal{T}_{998}}{\langle S_2, \sigma_{995} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{994} = \frac{\langle x := x + 1, \sigma_{993} \rangle \longrightarrow \sigma_{994} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{995} \quad \mathcal{T}_{996}}{\langle S_2, \sigma_{993} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{992} = \frac{\langle x := x + 1, \sigma_{991} \rangle \longrightarrow \sigma_{992} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{993} \quad \mathcal{T}_{994}}{\langle S_2, \sigma_{991} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{990} = \frac{\langle x := x + 1, \sigma_{989} \rangle \longrightarrow \sigma_{990} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{991} \quad \mathcal{T}_{992}}{\langle S_2, \sigma_{989} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{988} = \frac{\langle x := x + 1, \sigma_{987} \rangle \longrightarrow \sigma_{988} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{989} \quad \mathcal{T}_{990}}{\langle S_2, \sigma_{987} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{986} = \frac{\langle x := x + 1, \sigma_{985} \rangle \longrightarrow \sigma_{986} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{987} \quad \mathcal{T}_{988}}{\langle S_2, \sigma_{985} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{984} = \frac{\langle x := x + 1, \sigma_{983} \rangle \longrightarrow \sigma_{984} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{985} \quad \mathcal{T}_{986}}{\langle S_2, \sigma_{983} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{982} = \frac{\langle x := x + 1, \sigma_{981} \rangle \longrightarrow \sigma_{982} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{983} \quad \mathcal{T}_{984}}{\langle S_2, \sigma_{981} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{980} = \frac{\langle x := x + 1, \sigma_{979} \rangle \longrightarrow \sigma_{980} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{981} \quad \mathcal{T}_{982}}{\langle S_2, \sigma_{979} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{978} = \frac{\langle x := x + 1, \sigma_{977} \rangle \longrightarrow \sigma_{978} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{979} \quad \mathcal{T}_{980}}{\langle S_2, \sigma_{977} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{976} = \frac{\langle x := x + 1, \sigma_{975} \rangle \longrightarrow \sigma_{976} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{977} \quad \mathcal{T}_{978}}{\langle S_2, \sigma_{975} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{974} = \frac{\langle x := x + 1, \sigma_{973} \rangle \longrightarrow \sigma_{974} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{975} \quad \mathcal{T}_{976}}{\langle S_2, \sigma_{973} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{972} = \frac{\langle x := x + 1, \sigma_{971} \rangle \longrightarrow \sigma_{972} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{973} \quad \mathcal{T}_{974}}{\langle S_2, \sigma_{971} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{970} = \frac{\langle x := x + 1, \sigma_{969} \rangle \longrightarrow \sigma_{970} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{971} \quad \mathcal{T}_{972}}{\langle S_2, \sigma_{969} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{968} = \frac{\langle x := x + 1, \sigma_{967} \rangle \longrightarrow \sigma_{968} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{969} \quad \mathcal{T}_{970}}{\langle S_2, \sigma_{967} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{966} = \frac{\langle x := x + 1, \sigma_{965} \rangle \longrightarrow \sigma_{966} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{967} \quad \mathcal{T}_{968}}{\langle S_2, \sigma_{965} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{964} = \frac{\langle x := x + 1, \sigma_{963} \rangle \longrightarrow \sigma_{964} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{965} \quad \mathcal{T}_{966}}{\langle S_2, \sigma_{963} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{962} = \frac{\langle x := x + 1, \sigma_{961} \rangle \longrightarrow \sigma_{962} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{963} \quad \mathcal{T}_{964}}{\langle S_2, \sigma_{961} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{960} = \frac{\langle x := x + 1, \sigma_{959} \rangle \longrightarrow \sigma_{960} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{961} \quad \mathcal{T}_{962}}{\langle S_2, \sigma_{959} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{958} = \frac{\langle x := x + 1, \sigma_{957} \rangle \longrightarrow \sigma_{958} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{959} \quad \mathcal{T}_{960}}{\langle S_2, \sigma_{957} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{956} = \frac{\langle x := x + 1, \sigma_{955} \rangle \longrightarrow \sigma_{956} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{957} \quad \mathcal{T}_{958}}{\langle S_2, \sigma_{955} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{954} = \frac{\langle x := x + 1, \sigma_{953} \rangle \longrightarrow \sigma_{954} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{955} \quad \mathcal{T}_{956}}{\langle S_2, \sigma_{953} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{952} = \frac{\langle x := x + 1, \sigma_{951} \rangle \longrightarrow \sigma_{952} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{953} \quad \mathcal{T}_{954}}{\langle S_2, \sigma_{951} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{950} = \frac{\langle x := x + 1, \sigma_{949} \rangle \longrightarrow \sigma_{950} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{951} \quad \mathcal{T}_{952}}{\langle S_2, \sigma_{949} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{948} = \frac{\langle x := x + 1, \sigma_{947} \rangle \longrightarrow \sigma_{948} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{949} \quad \mathcal{T}_{950}}{\langle S_2, \sigma_{947} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{946} = \frac{\langle x := x + 1, \sigma_{945} \rangle \longrightarrow \sigma_{946} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{947} \quad \mathcal{T}_{948}}{\langle S_2, \sigma_{945} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{944} = \frac{\langle x := x + 1, \sigma_{943} \rangle \longrightarrow \sigma_{944} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{945} \quad \mathcal{T}_{946}}{\langle S_2, \sigma_{943} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{942} = \frac{\langle x := x + 1, \sigma_{941} \rangle \longrightarrow \sigma_{942} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{943} \quad \mathcal{T}_{944}}{\langle S_2, \sigma_{941} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{940} = \frac{\langle x := x + 1, \sigma_{939} \rangle \longrightarrow \sigma_{940} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{941} \quad \mathcal{T}_{942}}{\langle S_2, \sigma_{939} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{938} = \frac{\langle x := x + 1, \sigma_{937} \rangle \longrightarrow \sigma_{938} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{939} \quad \mathcal{T}_{940}}{\langle S_2, \sigma_{937} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{936} = \frac{\langle x := x + 1, \sigma_{935} \rangle \longrightarrow \sigma_{936} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{937} \quad \mathcal{T}_{938}}{\langle S_2, \sigma_{935} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{934} = \frac{\langle x := x + 1, \sigma_{933} \rangle \longrightarrow \sigma_{934} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{935} \quad \mathcal{T}_{936}}{\langle S_2, \sigma_{933} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{932} = \frac{\langle x := x + 1, \sigma_{931} \rangle \longrightarrow \sigma_{932} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{933} \quad \mathcal{T}_{934}}{\langle S_2, \sigma_{931} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{930} = \frac{\langle x := x + 1, \sigma_{929} \rangle \longrightarrow \sigma_{930} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{931} \quad \mathcal{T}_{932}}{\langle S_2, \sigma_{929} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{928} = \frac{\langle x := x + 1, \sigma_{927} \rangle \longrightarrow \sigma_{928} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{929} \quad \mathcal{T}_{930}}{\langle S_2, \sigma_{927} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{926} = \frac{\langle x := x + 1, \sigma_{925} \rangle \longrightarrow \sigma_{926} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{927} \quad \mathcal{T}_{928}}{\langle S_2, \sigma_{925} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{924} = \frac{\langle x := x + 1, \sigma_{923} \rangle \longrightarrow \sigma_{924} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{925} \quad \mathcal{T}_{926}}{\langle S_2, \sigma_{923} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{922} = \frac{\langle x := x + 1, \sigma_{921} \rangle \longrightarrow \sigma_{922} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{923} \quad \mathcal{T}_{924}}{\langle S_2, \sigma_{921} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{920} = \frac{\langle x := x + 1, \sigma_{919} \rangle \longrightarrow \sigma_{920} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{921} \quad \mathcal{T}_{922}}{\langle S_2, \sigma_{919} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{918} = \frac{\langle x := x + 1, \sigma_{917} \rangle \longrightarrow \sigma_{918} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{919} \quad \mathcal{T}_{920}}{\langle S_2, \sigma_{917} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{916} = \frac{\langle x := x + 1, \sigma_{915} \rangle \longrightarrow \sigma_{916} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{917} \quad \mathcal{T}_{918}}{\langle S_2, \sigma_{915} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{914} = \frac{\langle x := x + 1, \sigma_{913} \rangle \longrightarrow \sigma_{914} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{915} \quad \mathcal{T}_{916}}{\langle S_2, \sigma_{913} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{912} = \frac{\langle x := x + 1, \sigma_{911} \rangle \longrightarrow \sigma_{912} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{913} \quad \mathcal{T}_{914}}{\langle S_2, \sigma_{911} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{910} = \frac{\langle x := x + 1, \sigma_{909} \rangle \longrightarrow \sigma_{910} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{911} \quad \mathcal{T}_{912}}{\langle S_2, \sigma_{909} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{908} = \frac{\langle x := x + 1, \sigma_{907} \rangle \longrightarrow \sigma_{908} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{909} \quad \mathcal{T}_{910}}{\langle S_2, \sigma_{907} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{906} = \frac{\langle x := x + 1, \sigma_{905} \rangle \longrightarrow \sigma_{906} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{907} \quad \mathcal{T}_{908}}{\langle S_2, \sigma_{905} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{904} = \frac{\langle x := x + 1, \sigma_{903} \rangle \longrightarrow \sigma_{904} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{905} \quad \mathcal{T}_{906}}{\langle S_2, \sigma_{903} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{902} = \frac{\langle x := x + 1, \sigma_{901} \rangle \longrightarrow \sigma_{902} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{903} \quad \mathcal{T}_{904}}{\langle S_2, \sigma_{901} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{900} = \frac{\langle x := x + 1, \sigma_{899} \rangle \longrightarrow \sigma_{900} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{901} \quad \mathcal{T}_{902}}{\langle S_2, \sigma_{899} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{898} = \frac{\langle x := x + 1, \sigma_{897} \rangle \longrightarrow \sigma_{898} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{899} \quad \mathcal{T}_{900}}{\langle S_2, \sigma_{897} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{896} = \frac{\langle x := x + 1, \sigma_{895} \rangle \longrightarrow \sigma_{896} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{897} \quad \mathcal{T}_{898}}{\langle S_2, \sigma_{895} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{894} = \frac{\langle x := x + 1, \sigma_{893} \rangle \longrightarrow \sigma_{894} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{895} \quad \mathcal{T}_{896}}{\langle S_2, \sigma_{893} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{892} = \frac{\langle x := x + 1, \sigma_{891} \rangle \longrightarrow \sigma_{892} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{893} \quad \mathcal{T}_{894}}{\langle S_2, \sigma_{891} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{890} = \frac{\langle x := x + 1, \sigma_{889} \rangle \longrightarrow \sigma_{890} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{891} \quad \mathcal{T}_{892}}{\langle S_2, \sigma_{889} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{888} = \frac{\langle x := x + 1, \sigma_{887} \rangle \longrightarrow \sigma_{888} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{889} \quad \mathcal{T}_{890}}{\langle S_2, \sigma_{887} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{886} = \frac{\langle x := x + 1, \sigma_{885} \rangle \longrightarrow \sigma_{886} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{887} \quad \mathcal{T}_{888}}{\langle S_2, \sigma_{885} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{884} = \frac{\langle x := x + 1, \sigma_{883} \rangle \longrightarrow \sigma_{884} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{885} \quad \mathcal{T}_{886}}{\langle S_2, \sigma_{883} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{882} = \frac{\langle x := x + 1, \sigma_{881} \rangle \longrightarrow \sigma_{882} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{883} \quad \mathcal{T}_{884}}{\langle S_2, \sigma_{881} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{880} = \frac{\langle x := x + 1, \sigma_{879} \rangle \longrightarrow \sigma_{880} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{881} \quad \mathcal{T}_{882}}{\langle S_2, \sigma_{879} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{878} = \frac{\langle x := x + 1, \sigma_{877} \rangle \longrightarrow \sigma_{878} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{879} \quad \mathcal{T}_{880}}{\langle S_2, \sigma_{877} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{876} = \frac{\langle x := x + 1, \sigma_{875} \rangle \longrightarrow \sigma_{876} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{877} \quad \mathcal{T}_{878}}{\langle S_2, \sigma_{875} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{874} = \frac{\langle x := x + 1, \sigma_{873} \rangle \longrightarrow \sigma_{874} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{875} \quad \mathcal{T}_{876}}{\langle S_2, \sigma_{873} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{872} = \frac{\langle x := x + 1, \sigma_{871} \rangle \longrightarrow \sigma_{872} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{873} \quad \mathcal{T}_{874}}{\langle S_2, \sigma_{871} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{870} = \frac{\langle x := x + 1, \sigma_{869} \rangle \longrightarrow \sigma_{870} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{871} \quad \mathcal{T}_{872}}{\langle S_2, \sigma_{869} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{868} = \frac{\langle x := x + 1, \sigma_{867} \rangle \longrightarrow \sigma_{868} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{869} \quad \mathcal{T}_{870}}{\langle S_2, \sigma_{867} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{866} = \frac{\langle x := x + 1, \sigma_{865} \rangle \longrightarrow \sigma_{866} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{867} \quad \mathcal{T}_{868}}{\langle S_2, \sigma_{865} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{864} = \frac{\langle x := x + 1, \sigma_{863} \rangle \longrightarrow \sigma_{864} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{865} \quad \mathcal{T}_{866}}{\langle S_2, \sigma_{863} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{862} = \frac{\langle x := x + 1, \sigma_{861} \rangle \longrightarrow \sigma_{862} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{863} \quad \mathcal{T}_{864}}{\langle S_2, \sigma_{861} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{860} = \frac{\langle x := x + 1, \sigma_{859} \rangle \longrightarrow \sigma_{860} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{861} \quad \mathcal{T}_{862}}{\langle S_2, \sigma_{859} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{858} = \frac{\langle x := x + 1, \sigma_{857} \rangle \longrightarrow \sigma_{858} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{859} \quad \mathcal{T}_{860}}{\langle S_2, \sigma_{857} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{856} = \frac{\langle x := x + 1, \sigma_{855} \rangle \longrightarrow \sigma_{856} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{857} \quad \mathcal{T}_{858}}{\langle S_2, \sigma_{855} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{854} = \frac{\langle x := x + 1, \sigma_{853} \rangle \longrightarrow \sigma_{854} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{855} \quad \mathcal{T}_{856}}{\langle S_2, \sigma_{853} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{852} = \frac{\langle x := x + 1, \sigma_{851} \rangle \longrightarrow \sigma_{852} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{853} \quad \mathcal{T}_{854}}{\langle S_2, \sigma_{851} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{850} = \frac{\langle x := x + 1, \sigma_{849} \rangle \longrightarrow \sigma_{850} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{851} \quad \mathcal{T}_{852}}{\langle S_2, \sigma_{849} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{848} = \frac{\langle x := x + 1, \sigma_{847} \rangle \longrightarrow \sigma_{848} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{849} \quad \mathcal{T}_{850}}{\langle S_2, \sigma_{847} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{846} = \frac{\langle x := x + 1, \sigma_{845} \rangle \longrightarrow \sigma_{846} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{847} \quad \mathcal{T}_{848}}{\langle S_2, \sigma_{845} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{844} = \frac{\langle x := x + 1, \sigma_{843} \rangle \longrightarrow \sigma_{844} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{845} \quad \mathcal{T}_{846}}{\langle S_2, \sigma_{843} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{842} = \frac{\langle x := x + 1, \sigma_{841} \rangle \longrightarrow \sigma_{842} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{843} \quad \mathcal{T}_{844}}{\langle S_2, \sigma_{841} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{840} = \frac{\langle x := x + 1, \sigma_{839} \rangle \longrightarrow \sigma_{840} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{841} \quad \mathcal{T}_{842}}{\langle S_2, \sigma_{839} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{838} = \frac{\langle x := x + 1, \sigma_{837} \rangle \longrightarrow \sigma_{838} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{839} \quad \mathcal{T}_{840}}{\langle S_2, \sigma_{837} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{836} = \frac{\langle x := x + 1, \sigma_{835} \rangle \longrightarrow \sigma_{836} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{837} \quad \mathcal{T}_{838}}{\langle S_2, \sigma_{835} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{834} = \frac{\langle x := x + 1, \sigma_{833} \rangle \longrightarrow \sigma_{834} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{835} \quad \mathcal{T}_{836}}{\langle S_2, \sigma_{833} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{832} = \frac{\langle x := x + 1, \sigma_{831} \rangle \longrightarrow \sigma_{832} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{833} \quad \mathcal{T}_{834}}{\langle S_2, \sigma_{831} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{830} = \frac{\langle x := x + 1, \sigma_{829} \rangle \longrightarrow \sigma_{830} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{831} \quad \mathcal{T}_{832}}{\langle S_2, \sigma_{829} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{828} = \frac{\langle x := x + 1, \sigma_{827} \rangle \longrightarrow \sigma_{828} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{829} \quad \mathcal{T}_{830}}{\langle S_2, \sigma_{827} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{826} = \frac{\langle x := x + 1, \sigma_{825} \rangle \longrightarrow \sigma_{826} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{827} \quad \mathcal{T}_{828}}{\langle S_2, \sigma_{825} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{824} = \frac{\langle x := x + 1, \sigma_{823} \rangle \longrightarrow \sigma_{824} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{825} \quad \mathcal{T}_{826}}{\langle S_2, \sigma_{823} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{822} = \frac{\langle x := x + 1, \sigma_{821} \rangle \longrightarrow \sigma_{822} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{823} \quad \mathcal{T}_{824}}{\langle S_2, \sigma_{821} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{820} = \frac{\langle x := x + 1, \sigma_{819} \rangle \longrightarrow \sigma_{820} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{821} \quad \mathcal{T}_{822}}{\langle S_2, \sigma_{819} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{818} = \frac{\langle x := x + 1, \sigma_{817} \rangle \longrightarrow \sigma_{818} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{819} \quad \mathcal{T}_{820}}{\langle S_2, \sigma_{817} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{816} = \frac{\langle x := x + 1, \sigma_{815} \rangle \longrightarrow \sigma_{816} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{817} \quad \mathcal{T}_{818}}{\langle S_2, \sigma_{815} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{814} = \frac{\langle x := x + 1, \sigma_{813} \rangle \longrightarrow \sigma_{814} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{815} \quad \mathcal{T}_{816}}{\langle S_2, \sigma_{813} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{812} = \frac{\langle x := x + 1, \sigma_{811} \rangle \longrightarrow \sigma_{812} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{813} \quad \mathcal{T}_{814}}{\langle S_2, \sigma_{811} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{810} = \frac{\langle x := x + 1, \sigma_{809} \rangle \longrightarrow \sigma_{810} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{811} \quad \mathcal{T}_{812}}{\langle S_2, \sigma_{809} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{808} = \frac{\langle x := x + 1, \sigma_{807} \rangle \longrightarrow \sigma_{808} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{809} \quad \mathcal{T}_{810}}{\langle S_2, \sigma_{807} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{806} = \frac{\langle x := x + 1, \sigma_{805} \rangle \longrightarrow \sigma_{806} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{807} \quad \mathcal{T}_{808}}{\langle S_2, \sigma_{805} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{804} = \frac{\langle x := x + 1, \sigma_{803} \rangle \longrightarrow \sigma_{804} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{805} \quad \mathcal{T}_{806}}{\langle S_2, \sigma_{803} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{802} = \frac{\langle x := x + 1, \sigma_{801} \rangle \longrightarrow \sigma_{802} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{803} \quad \mathcal{T}_{804}}{\langle S_2, \sigma_{801} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{800} = \frac{\langle x := x + 1, \sigma_{799} \rangle \longrightarrow \sigma_{800} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{801} \quad \mathcal{T}_{802}}{\langle S_2, \sigma_{799} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{798} = \frac{\langle x := x + 1, \sigma_{797} \rangle \longrightarrow \sigma_{798} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{799} \quad \mathcal{T}_{800}}{\langle S_2, \sigma_{797} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{796} = \frac{\langle x := x + 1, \sigma_{795} \rangle \longrightarrow \sigma_{796} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{797} \quad \mathcal{T}_{798}}{\langle S_2, \sigma_{795} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{794} = \frac{\langle x := x + 1, \sigma_{793} \rangle \longrightarrow \sigma_{794} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{795} \quad \mathcal{T}_{796}}{\langle S_2, \sigma_{793} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{792} = \frac{\langle x := x + 1, \sigma_{791} \rangle \longrightarrow \sigma_{792} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{793} \quad \mathcal{T}_{794}}{\langle S_2, \sigma_{791} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{790} = \frac{\langle x := x + 1, \sigma_{789} \rangle \longrightarrow \sigma_{790} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{791} \quad \mathcal{T}_{792}}{\langle S_2, \sigma_{789} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{788} = \frac{\langle x := x + 1, \sigma_{787} \rangle \longrightarrow \sigma_{788} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{789} \quad \mathcal{T}_{790}}{\langle S_2, \sigma_{787} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{786} = \frac{\langle x := x + 1, \sigma_{785} \rangle \longrightarrow \sigma_{786} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{787} \quad \mathcal{T}_{788}}{\langle S_2, \sigma_{785} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{784} = \frac{\langle x := x + 1, \sigma_{783} \rangle \longrightarrow \sigma_{784} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{785} \quad \mathcal{T}_{786}}{\langle S_2, \sigma_{783} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{782} = \frac{\langle x := x + 1, \sigma_{781} \rangle \longrightarrow \sigma_{782} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{783} \quad \mathcal{T}_{784}}{\langle S_2, \sigma_{781} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{780} = \frac{\langle x := x + 1, \sigma_{779} \rangle \longrightarrow \sigma_{780} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{781} \quad \mathcal{T}_{782}}{\langle S_2, \sigma_{779} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{778} = \frac{\langle x := x + 1, \sigma_{777} \rangle \longrightarrow \sigma_{778} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{779} \quad \mathcal{T}_{780}}{\langle S_2, \sigma_{777} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{776} = \frac{\langle x := x + 1, \sigma_{775} \rangle \longrightarrow \sigma_{776} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{777} \quad \mathcal{T}_{778}}{\langle S_2, \sigma_{775} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{774} = \frac{\langle x := x + 1, \sigma_{773} \rangle \longrightarrow \sigma_{774} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{775} \quad \mathcal{T}_{776}}{\langle S_2, \sigma_{773} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{772} = \frac{\langle x := x + 1, \sigma_{771} \rangle \longrightarrow \sigma_{772} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{773} \quad \mathcal{T}_{774}}{\langle S_2, \sigma_{771} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{770} = \frac{\langle x := x + 1, \sigma_{769} \rangle \longrightarrow \sigma_{770} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{771} \quad \mathcal{T}_{772}}{\langle S_2, \sigma_{769} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{768} = \frac{\langle x := x + 1, \sigma_{767} \rangle \longrightarrow \sigma_{768} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{769} \quad \mathcal{T}_{770}}{\langle S_2, \sigma_{767} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{766} = \frac{\langle x := x + 1, \sigma_{765} \rangle \longrightarrow \sigma_{766} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{767} \quad \mathcal{T}_{768}}{\langle S_2, \sigma_{765} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{764} = \frac{\langle x := x + 1, \sigma_{763} \rangle \longrightarrow \sigma_{764} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{765} \quad \mathcal{T}_{766}}{\langle S_2, \sigma_{763} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{762} = \frac{\langle x := x + 1, \sigma_{761} \rangle \longrightarrow \sigma_{762} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{763} \quad \mathcal{T}_{764}}{\langle S_2, \sigma_{761} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{760} = \frac{\langle x := x + 1, \sigma_{759} \rangle \longrightarrow \sigma_{760} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{761} \quad \mathcal{T}_{762}}{\langle S_2, \sigma_{759} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{758} = \frac{\langle x := x + 1, \sigma_{757} \rangle \longrightarrow \sigma_{758} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{759} \quad \mathcal{T}_{760}}{\langle S_2, \sigma_{757} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{756} = \frac{\langle x := x + 1, \sigma_{755} \rangle \longrightarrow \sigma_{756} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{757} \quad \mathcal{T}_{758}}{\langle S_2, \sigma_{755} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{754} = \frac{\langle x := x + 1, \sigma_{753} \rangle \longrightarrow \sigma_{754} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{755} \quad \mathcal{T}_{756}}{\langle S_2, \sigma_{753} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{752} = \frac{\langle x := x + 1, \sigma_{751} \rangle \longrightarrow \sigma_{752} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{753} \quad \mathcal{T}_{754}}{\langle S_2, \sigma_{751} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{750} = \frac{\langle x := x + 1, \sigma_{749} \rangle \longrightarrow \sigma_{750} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{751} \quad \mathcal{T}_{752}}{\langle S_2, \sigma_{749} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{748} = \frac{\langle x := x + 1, \sigma_{747} \rangle \longrightarrow \sigma_{748} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{749} \quad \mathcal{T}_{750}}{\langle S_2, \sigma_{747} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{746} = \frac{\langle x := x + 1, \sigma_{745} \rangle \longrightarrow \sigma_{746} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{747} \quad \mathcal{T}_{748}}{\langle S_2, \sigma_{745} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{744} = \frac{\langle x := x + 1, \sigma_{743} \rangle \longrightarrow \sigma_{744} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{745} \quad \mathcal{T}_{746}}{\langle S_2, \sigma_{743} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{742} = \frac{\langle x := x + 1, \sigma_{741} \rangle \longrightarrow \sigma_{742} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{743} \quad \mathcal{T}_{744}}{\langle S_2, \sigma_{741} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{740} = \frac{\langle x := x + 1, \sigma_{739} \rangle \longrightarrow \sigma_{740} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{741} \quad \mathcal{T}_{742}}{\langle S_2, \sigma_{739} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{738} = \frac{\langle x := x + 1, \sigma_{737} \rangle \longrightarrow \sigma_{738} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{739} \quad \mathcal{T}_{740}}{\langle S_2, \sigma_{737} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{736} = \frac{\langle x := x + 1, \sigma_{735} \rangle \longrightarrow \sigma_{736} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{737} \quad \mathcal{T}_{738}}{\langle S_2, \sigma_{735} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{734} = \frac{\langle x := x + 1, \sigma_{733} \rangle \longrightarrow \sigma_{734} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{735} \quad \mathcal{T}_{736}}{\langle S_2, \sigma_{733} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{732} = \frac{\langle x := x + 1, \sigma_{731} \rangle \longrightarrow \sigma_{732} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{733} \quad \mathcal{T}_{734}}{\langle S_2, \sigma_{731} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{730} = \frac{\langle x := x + 1, \sigma_{729} \rangle \longrightarrow \sigma_{730} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{731} \quad \mathcal{T}_{732}}{\langle S_2, \sigma_{729} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{728} = \frac{\langle x := x + 1, \sigma_{727} \rangle \longrightarrow \sigma_{728} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{729} \quad \mathcal{T}_{730}}{\langle S_2, \sigma_{727} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{726} = \frac{\langle x := x + 1, \sigma_{725} \rangle \longrightarrow \sigma_{726} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{727} \quad \mathcal{T}_{728}}{\langle S_2, \sigma_{725} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{724} = \frac{\langle x := x + 1, \sigma_{723} \rangle \longrightarrow \sigma_{724} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{725} \quad \mathcal{T}_{726}}{\langle S_2, \sigma_{723} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{722} = \frac{\langle x := x + 1, \sigma_{721} \rangle \longrightarrow \sigma_{722} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{723} \quad \mathcal{T}_{724}}{\langle S_2, \sigma_{721} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{720} = \frac{\langle x := x + 1, \sigma_{719} \rangle \longrightarrow \sigma_{720} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{721} \quad \mathcal{T}_{722}}{\langle S_2, \sigma_{719} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{718} = \frac{\langle x := x + 1, \sigma_{717} \rangle \longrightarrow \sigma_{718} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{719} \quad \mathcal{T}_{720}}{\langle S_2, \sigma_{717} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{716} = \frac{\langle x := x + 1, \sigma_{715} \rangle \longrightarrow \sigma_{716} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{717} \quad \mathcal{T}_{718}}{\langle S_2, \sigma_{715} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{714} = \frac{\langle x := x + 1, \sigma_{713} \rangle \longrightarrow \sigma_{714} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{715} \quad \mathcal{T}_{716}}{\langle S_2, \sigma_{713} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{712} = \frac{\langle x := x + 1, \sigma_{711} \rangle \longrightarrow \sigma_{712} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{713} \quad \mathcal{T}_{714}}{\langle S_2, \sigma_{711} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{710} = \frac{\langle x := x + 1, \sigma_{709} \rangle \longrightarrow \sigma_{710} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{711} \quad \mathcal{T}_{712}}{\langle S_2, \sigma_{709} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{708} = \frac{\langle x := x + 1, \sigma_{707} \rangle \longrightarrow \sigma_{708} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{709} \quad \mathcal{T}_{710}}{\langle S_2, \sigma_{707} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{706} = \frac{\langle x := x + 1, \sigma_{705} \rangle \longrightarrow \sigma_{706} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{707} \quad \mathcal{T}_{708}}{\langle S_2, \sigma_{705} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{704} = \frac{\langle x := x + 1, \sigma_{703} \rangle \longrightarrow \sigma_{704} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{705} \quad \mathcal{T}_{706}}{\langle S_2, \sigma_{703} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{702} = \frac{\langle x := x + 1, \sigma_{701} \rangle \longrightarrow \sigma_{702} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{703} \quad \mathcal{T}_{704}}{\langle S_2, \sigma_{701} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{700} = \frac{\langle x := x + 1, \sigma_{699} \rangle \longrightarrow \sigma_{700} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{701} \quad \mathcal{T}_{702}}{\langle S_2, \sigma_{699} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{698} = \frac{\langle x := x + 1, \sigma_{697} \rangle \longrightarrow \sigma_{698} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{699} \quad \mathcal{T}_{700}}{\langle S_2, \sigma_{697} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{696} = \frac{\langle x := x + 1, \sigma_{695} \rangle \longrightarrow \sigma_{696} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{697} \quad \mathcal{T}_{698}}{\langle S_2, \sigma_{695} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{694} = \frac{\langle x := x + 1, \sigma_{693} \rangle \longrightarrow \sigma_{694} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{695} \quad \mathcal{T}_{696}}{\langle S_2, \sigma_{693} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{692} = \frac{\langle x := x + 1, \sigma_{691} \rangle \longrightarrow \sigma_{692} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{693} \quad \mathcal{T}_{694}}{\langle S_2, \sigma_{691} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{690} = \frac{\langle x := x + 1, \sigma_{689} \rangle \longrightarrow \sigma_{690} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{691} \quad \mathcal{T}_{692}}{\langle S_2, \sigma_{689} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{688} = \frac{\langle x := x + 1, \sigma_{687} \rangle \longrightarrow \sigma_{688} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{689} \quad \mathcal{T}_{690}}{\langle S_2, \sigma_{687} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{686} = \frac{\langle x := x + 1, \sigma_{685} \rangle \longrightarrow \sigma_{686} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{687} \quad \mathcal{T}_{688}}{\langle S_2, \sigma_{685} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{684} = \frac{\langle x := x + 1, \sigma_{683} \rangle \longrightarrow \sigma_{684} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{685} \quad \mathcal{T}_{686}}{\langle S_2, \sigma_{683} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{682} = \frac{\langle x := x + 1, \sigma_{681} \rangle \longrightarrow \sigma_{682} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{683} \quad \mathcal{T}_{684}}{\langle S_2, \sigma_{681} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{680} = \frac{\langle x := x + 1, \sigma_{679} \rangle \longrightarrow \sigma_{680} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{681} \quad \mathcal{T}_{682}}{\langle S_2, \sigma_{679} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{678} = \frac{\langle x := x + 1, \sigma_{677} \rangle \longrightarrow \sigma_{678} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{679} \quad \mathcal{T}_{680}}{\langle S_2, \sigma_{677} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{676} = \frac{\langle x := x + 1, \sigma_{675} \rangle \longrightarrow \sigma_{676} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{677} \quad \mathcal{T}_{678}}{\langle S_2, \sigma_{675} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{674} = \frac{\langle x := x + 1, \sigma_{673} \rangle \longrightarrow \sigma_{674} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{675} \quad \mathcal{T}_{676}}{\langle S_2, \sigma_{673} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{672} = \frac{\langle x := x + 1, \sigma_{671} \rangle \longrightarrow \sigma_{672} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{673} \quad \mathcal{T}_{674}}{\langle S_2, \sigma_{671} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{670} = \frac{\langle x := x + 1, \sigma_{669} \rangle \longrightarrow \sigma_{670} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{671} \quad \mathcal{T}_{672}}{\langle S_2, \sigma_{669} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{668} = \frac{\langle x := x + 1, \sigma_{667} \rangle \longrightarrow \sigma_{668} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{669} \quad \mathcal{T}_{670}}{\langle S_2, \sigma_{667} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{666} = \frac{\langle x := x + 1, \sigma_{665} \rangle \longrightarrow \sigma_{666} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{667} \quad \mathcal{T}_{668}}{\langle S_2, \sigma_{665} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{664} = \frac{\langle x := x + 1, \sigma_{663} \rangle \longrightarrow \sigma_{664} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{665} \quad \mathcal{T}_{666}}{\langle S_2, \sigma_{663} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{662} = \frac{\langle x := x + 1, \sigma_{661} \rangle \longrightarrow \sigma_{662} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{663} \quad \mathcal{T}_{664}}{\langle S_2, \sigma_{661} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{660} = \frac{\langle x := x + 1, \sigma_{659} \rangle \longrightarrow \sigma_{660} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{661} \quad \mathcal{T}_{662}}{\langle S_2, \sigma_{659} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{658} = \frac{\langle x := x + 1, \sigma_{657} \rangle \longrightarrow \sigma_{658} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{659} \quad \mathcal{T}_{660}}{\langle S_2, \sigma_{657} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{656} = \frac{\langle x := x + 1, \sigma_{655} \rangle \longrightarrow \sigma_{656} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{657} \quad \mathcal{T}_{658}}{\langle S_2, \sigma_{655} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{654} = \frac{\langle x := x + 1, \sigma_{653} \rangle \longrightarrow \sigma_{654} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{655} \quad \mathcal{T}_{656}}{\langle S_2, \sigma_{653} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{652} = \frac{\langle x := x + 1, \sigma_{651} \rangle \longrightarrow \sigma_{652} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{653} \quad \mathcal{T}_{654}}{\langle S_2, \sigma_{651} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{650} = \frac{\langle x := x + 1, \sigma_{649} \rangle \longrightarrow \sigma_{650} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{651} \quad \mathcal{T}_{652}}{\langle S_2, \sigma_{649} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{648} = \frac{\langle x := x + 1, \sigma_{647} \rangle \longrightarrow \sigma_{648} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{649} \quad \mathcal{T}_{650}}{\langle S_2, \sigma_{647} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{646} = \frac{\langle x := x + 1, \sigma_{645} \rangle \longrightarrow \sigma_{646} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{647} \quad \mathcal{T}_{648}}{\langle S_2, \sigma_{645} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{644} = \frac{\langle x := x + 1, \sigma_{643} \rangle \longrightarrow \sigma_{644} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{645} \quad \mathcal{T}_{646}}{\langle S_2, \sigma_{643} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{642} = \frac{\langle x := x + 1, \sigma_{641} \rangle \longrightarrow \sigma_{642} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{643} \quad \mathcal{T}_{644}}{\langle S_2, \sigma_{641} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{640} = \frac{\langle x := x + 1, \sigma_{639} \rangle \longrightarrow \sigma_{640} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{641} \quad \mathcal{T}_{642}}{\langle S_2, \sigma_{639} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{638} = \frac{\langle x := x + 1, \sigma_{637} \rangle \longrightarrow \sigma_{638} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{639} \quad \mathcal{T}_{640}}{\langle S_2, \sigma_{637} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{636} = \frac{\langle x := x + 1, \sigma_{635} \rangle \longrightarrow \sigma_{636} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{637} \quad \mathcal{T}_{638}}{\langle S_2, \sigma_{635} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{634} = \frac{\langle x := x + 1, \sigma_{633} \rangle \longrightarrow \sigma_{634} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{635} \quad \mathcal{T}_{636}}{\langle S_2, \sigma_{633} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{632} = \frac{\langle x := x + 1, \sigma_{631} \rangle \longrightarrow \sigma_{632} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{633} \quad \mathcal{T}_{634}}{\langle S_2, \sigma_{631} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{630} = \frac{\langle x := x + 1, \sigma_{629} \rangle \longrightarrow \sigma_{630} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{631} \quad \mathcal{T}_{632}}{\langle S_2, \sigma_{629} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{628} = \frac{\langle x := x + 1, \sigma_{627} \rangle \longrightarrow \sigma_{628} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{629} \quad \mathcal{T}_{630}}{\langle S_2, \sigma_{627} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{626} = \frac{\langle x := x + 1, \sigma_{625} \rangle \longrightarrow \sigma_{626} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{627} \quad \mathcal{T}_{628}}{\langle S_2, \sigma_{625} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{624} = \frac{\langle x := x + 1, \sigma_{623} \rangle \longrightarrow \sigma_{624} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{625} \quad \mathcal{T}_{626}}{\langle S_2, \sigma_{623} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{622} = \frac{\langle x := x + 1, \sigma_{621} \rangle \longrightarrow \sigma_{622} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{623} \quad \mathcal{T}_{624}}{\langle S_2, \sigma_{621} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{620} = \frac{\langle x := x + 1, \sigma_{619} \rangle \longrightarrow \sigma_{620} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{621} \quad \mathcal{T}_{622}}{\langle S_2, \sigma_{619} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{618} = \frac{\langle x := x + 1, \sigma_{617} \rangle \longrightarrow \sigma_{618} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{619} \quad \mathcal{T}_{620}}{\langle S_2, \sigma_{617} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{616} = \frac{\langle x := x + 1, \sigma_{615} \rangle \longrightarrow \sigma_{616} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{617} \quad \mathcal{T}_{618}}{\langle S_2, \sigma_{615} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{614} = \frac{\langle x := x + 1, \sigma_{613} \rangle \longrightarrow \sigma_{614} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{615} \quad \mathcal{T}_{616}}{\langle S_2, \sigma_{613} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{612} = \frac{\langle x := x + 1, \sigma_{611} \rangle \longrightarrow \sigma_{612} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{613} \quad \mathcal{T}_{614}}{\langle S_2, \sigma_{611} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{610} = \frac{\langle x := x + 1, \sigma_{609} \rangle \longrightarrow \sigma_{610} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{611} \quad \mathcal{T}_{612}}{\langle S_2, \sigma_{609} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{608} = \frac{\langle x := x + 1, \sigma_{607} \rangle \longrightarrow \sigma_{608} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{609} \quad \mathcal{T}_{610}}{\langle S_2, \sigma_{607} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{606} = \frac{\langle x := x + 1, \sigma_{605} \rangle \longrightarrow \sigma_{606} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{607} \quad \mathcal{T}_{608}}{\langle S_2, \sigma_{605} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{604} = \frac{\langle x := x + 1, \sigma_{603} \rangle \longrightarrow \sigma_{604} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{605} \quad \mathcal{T}_{606}}{\langle S_2, \sigma_{603} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{602} = \frac{\langle x := x + 1, \sigma_{601} \rangle \longrightarrow \sigma_{602} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{603} \quad \mathcal{T}_{604}}{\langle S_2, \sigma_{601} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{600} = \frac{\langle x := x + 1, \sigma_{599} \rangle \longrightarrow \sigma_{600} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{601} \quad \mathcal{T}_{602}}{\langle S_2, \sigma_{599} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{598} = \frac{\langle x := x + 1, \sigma_{597} \rangle \longrightarrow \sigma_{598} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{599} \quad \mathcal{T}_{600}}{\langle S_2, \sigma_{597} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{596} = \frac{\langle x := x + 1, \sigma_{595} \rangle \longrightarrow \sigma_{596} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{597} \quad \mathcal{T}_{598}}{\langle S_2, \sigma_{595} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{594} = \frac{\langle x := x + 1, \sigma_{593} \rangle \longrightarrow \sigma_{594} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{595} \quad \mathcal{T}_{596}}{\langle S_2, \sigma_{593} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{592} = \frac{\langle x := x + 1, \sigma_{591} \rangle \longrightarrow \sigma_{592} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{593} \quad \mathcal{T}_{594}}{\langle S_2, \sigma_{591} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{590} = \frac{\langle x := x + 1, \sigma_{589} \rangle \longrightarrow \sigma_{590} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{591} \quad \mathcal{T}_{592}}{\langle S_2, \sigma_{589} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{588} = \frac{\langle x := x + 1, \sigma_{587} \rangle \longrightarrow \sigma_{588} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{589} \quad \mathcal{T}_{590}}{\langle S_2, \sigma_{587} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{586} = \frac{\langle x := x + 1, \sigma_{585} \rangle \longrightarrow \sigma_{586} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{587} \quad \mathcal{T}_{588}}{\langle S_2, \sigma_{585} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{584} = \frac{\langle x := x + 1, \sigma_{583} \rangle \longrightarrow \sigma_{584} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{585} \quad \mathcal{T}_{586}}{\langle S_2, \sigma_{583} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{582} = \frac{\langle x := x + 1, \sigma_{581} \rangle \longrightarrow \sigma_{582} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{583} \quad \mathcal{T}_{584}}{\langle S_2, \sigma_{581} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{580} = \frac{\langle x := x + 1, \sigma_{579} \rangle \longrightarrow \sigma_{580} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{581} \quad \mathcal{T}_{582}}{\langle S_2, \sigma_{579} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{578} = \frac{\langle x := x + 1, \sigma_{577} \rangle \longrightarrow \sigma_{578} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{579} \quad \mathcal{T}_{580}}{\langle S_2, \sigma_{577} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{576} = \frac{\langle x := x + 1, \sigma_{575} \rangle \longrightarrow \sigma_{576} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{577} \quad \mathcal{T}_{578}}{\langle S_2, \sigma_{575} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{574} = \frac{\langle x := x + 1, \sigma_{573} \rangle \longrightarrow \sigma_{574} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{575} \quad \mathcal{T}_{576}}{\langle S_2, \sigma_{573} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{572} = \frac{\langle x := x + 1, \sigma_{571} \rangle \longrightarrow \sigma_{572} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{573} \quad \mathcal{T}_{574}}{\langle S_2, \sigma_{571} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{570} = \frac{\langle x := x + 1, \sigma_{569} \rangle \longrightarrow \sigma_{570} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{571} \quad \mathcal{T}_{572}}{\langle S_2, \sigma_{569} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{568} = \frac{\langle x := x + 1, \sigma_{567} \rangle \longrightarrow \sigma_{568} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{569} \quad \mathcal{T}_{570}}{\langle S_2, \sigma_{567} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{566} = \frac{\langle x := x + 1, \sigma_{565} \rangle \longrightarrow \sigma_{566} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{567} \quad \mathcal{T}_{568}}{\langle S_2, \sigma_{565} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{564} = \frac{\langle x := x + 1, \sigma_{563} \rangle \longrightarrow \sigma_{564} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{565} \quad \mathcal{T}_{566}}{\langle S_2, \sigma_{563} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{562} = \frac{\langle x := x + 1, \sigma_{561} \rangle \longrightarrow \sigma_{562} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{563} \quad \mathcal{T}_{564}}{\langle S_2, \sigma_{561} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{560} = \frac{\langle x := x + 1, \sigma_{559} \rangle \longrightarrow \sigma_{560} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{561} \quad \mathcal{T}_{562}}{\langle S_2, \sigma_{559} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{558} = \frac{\langle x := x + 1, \sigma_{557} \rangle \longrightarrow \sigma_{558} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{559} \quad \mathcal{T}_{560}}{\langle S_2, \sigma_{557} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{556} = \frac{\langle x := x + 1, \sigma_{555} \rangle \longrightarrow \sigma_{556} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{557} \quad \mathcal{T}_{558}}{\langle S_2, \sigma_{555} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{554} = \frac{\langle x := x + 1, \sigma_{553} \rangle \longrightarrow \sigma_{554} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{555} \quad \mathcal{T}_{556}}{\langle S_2, \sigma_{553} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{552} = \frac{\langle x := x + 1, \sigma_{551} \rangle \longrightarrow \sigma_{552} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{553} \quad \mathcal{T}_{554}}{\langle S_2, \sigma_{551} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{550} = \frac{\langle x := x + 1, \sigma_{549} \rangle \longrightarrow \sigma_{550} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{551} \quad \mathcal{T}_{552}}{\langle S_2, \sigma_{549} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{548} = \frac{\langle x := x + 1, \sigma_{547} \rangle \longrightarrow \sigma_{548} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{549} \quad \mathcal{T}_{550}}{\langle S_2, \sigma_{547} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{546} = \frac{\langle x := x + 1, \sigma_{545} \rangle \longrightarrow \sigma_{546} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{547} \quad \mathcal{T}_{548}}{\langle S_2, \sigma_{545} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{544} = \frac{\langle x := x + 1, \sigma_{543} \rangle \longrightarrow \sigma_{544} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{545} \quad \mathcal{T}_{546}}{\langle S_2, \sigma_{543} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{542} = \frac{\langle x := x + 1, \sigma_{541} \rangle \longrightarrow \sigma_{542} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{543} \quad \mathcal{T}_{544}}{\langle S_2, \sigma_{541} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{540} = \frac{\langle x := x + 1, \sigma_{539} \rangle \longrightarrow \sigma_{540} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{541} \quad \mathcal{T}_{542}}{\langle S_2, \sigma_{539} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{538} = \frac{\langle x := x + 1, \sigma_{537} \rangle \longrightarrow \sigma_{538} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{539} \quad \mathcal{T}_{540}}{\langle S_2, \sigma_{537} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{536} = \frac{\langle x := x + 1, \sigma_{535} \rangle \longrightarrow \sigma_{536} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{537} \quad \mathcal{T}_{538}}{\langle S_2, \sigma_{535} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{534} = \frac{\langle x := x + 1, \sigma_{533} \rangle \longrightarrow \sigma_{534} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{535} \quad \mathcal{T}_{536}}{\langle S_2, \sigma_{533} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{532} = \frac{\langle x := x + 1, \sigma_{531} \rangle \longrightarrow \sigma_{532} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{533} \quad \mathcal{T}_{534}}{\langle S_2, \sigma_{531} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{530} = \frac{\langle x := x + 1, \sigma_{529} \rangle \longrightarrow \sigma_{530} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{531} \quad \mathcal{T}_{532}}{\langle S_2, \sigma_{529} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{528} = \frac{\langle x := x + 1, \sigma_{527} \rangle \longrightarrow \sigma_{528} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{529} \quad \mathcal{T}_{530}}{\langle S_2, \sigma_{527} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{526} = \frac{\langle x := x + 1, \sigma_{525} \rangle \longrightarrow \sigma_{526} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{527} \quad \mathcal{T}_{528}}{\langle S_2, \sigma_{525} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{524} = \frac{\langle x := x + 1, \sigma_{523} \rangle \longrightarrow \sigma_{524} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{525} \quad \mathcal{T}_{526}}{\langle S_2, \sigma_{523} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{522} = \frac{\langle x := x + 1, \sigma_{521} \rangle \longrightarrow \sigma_{522} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{523} \quad \mathcal{T}_{524}}{\langle S_2, \sigma_{521} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{520} = \frac{\langle x := x + 1, \sigma_{519} \rangle \longrightarrow \sigma_{520} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{521} \quad \mathcal{T}_{522}}{\langle S_2, \sigma_{519} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{518} = \frac{\langle x := x + 1, \sigma_{517} \rangle \longrightarrow \sigma_{518} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{519} \quad \mathcal{T}_{520}}{\langle S_2, \sigma_{517} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{516} = \frac{\langle x := x + 1, \sigma_{515} \rangle \longrightarrow \sigma_{516} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{517} \quad \mathcal{T}_{518}}{\langle S_2, \sigma_{515} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{514} = \frac{\langle x := x + 1, \sigma_{513} \rangle \longrightarrow \sigma_{514} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{515} \quad \mathcal{T}_{516}}{\langle S_2, \sigma_{513} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{512} = \frac{\langle x := x + 1, \sigma_{511} \rangle \longrightarrow \sigma_{512} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{513} \quad \mathcal{T}_{514}}{\langle S_2, \sigma_{511} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{510} = \frac{\langle x := x + 1, \sigma_{509} \rangle \longrightarrow \sigma_{510} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{511} \quad \mathcal{T}_{512}}{\langle S_2, \sigma_{509} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{508} = \frac{\langle x := x + 1, \sigma_{507} \rangle \longrightarrow \sigma_{508} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{509} \quad \mathcal{T}_{510}}{\langle S_2, \sigma_{507} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{506} = \frac{\langle x := x + 1, \sigma_{505} \rangle \longrightarrow \sigma_{506} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{507} \quad \mathcal{T}_{508}}{\langle S_2, \sigma_{505} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{504} = \frac{\langle x := x + 1, \sigma_{503} \rangle \longrightarrow \sigma_{504} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{505} \quad \mathcal{T}_{506}}{\langle S_2, \sigma_{503} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{502} = \frac{\langle x := x + 1, \sigma_{501} \rangle \longrightarrow \sigma_{502} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{503} \quad \mathcal{T}_{504}}{\langle S_2, \sigma_{501} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{500} = \frac{\langle x := x + 1, \sigma_{499} \rangle \longrightarrow \sigma_{500} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{501} \quad \mathcal{T}_{502}}{\langle S_2, \sigma_{499} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{498} = \frac{\langle x := x + 1, \sigma_{497} \rangle \longrightarrow \sigma_{498} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{499} \quad \mathcal{T}_{500}}{\langle S_2, \sigma_{497} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{496} = \frac{\langle x := x + 1, \sigma_{495} \rangle \longrightarrow \sigma_{496} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{497} \quad \mathcal{T}_{498}}{\langle S_2, \sigma_{495} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{494} = \frac{\langle x := x + 1, \sigma_{493} \rangle \longrightarrow \sigma_{494} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{495} \quad \mathcal{T}_{496}}{\langle S_2, \sigma_{493} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{492} = \frac{\langle x := x + 1, \sigma_{491} \rangle \longrightarrow \sigma_{492} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{493} \quad \mathcal{T}_{494}}{\langle S_2, \sigma_{491} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{490} = \frac{\langle x := x + 1, \sigma_{489} \rangle \longrightarrow \sigma_{490} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{491} \quad \mathcal{T}_{492}}{\langle S_2, \sigma_{489} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{488} = \frac{\langle x := x + 1, \sigma_{487} \rangle \longrightarrow \sigma_{488} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{489} \quad \mathcal{T}_{490}}{\langle S_2, \sigma_{487} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{486} = \frac{\langle x := x + 1, \sigma_{485} \rangle \longrightarrow \sigma_{486} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{487} \quad \mathcal{T}_{488}}{\langle S_2, \sigma_{485} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{484} = \frac{\langle x := x + 1, \sigma_{483} \rangle \longrightarrow \sigma_{484} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{485} \quad \mathcal{T}_{486}}{\langle S_2, \sigma_{483} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{482} = \frac{\langle x := x + 1, \sigma_{481} \rangle \longrightarrow \sigma_{482} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{483} \quad \mathcal{T}_{484}}{\langle S_2, \sigma_{481} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{480} = \frac{\langle x := x + 1, \sigma_{479} \rangle \longrightarrow \sigma_{480} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{481} \quad \mathcal{T}_{482}}{\langle S_2, \sigma_{479} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{478} = \frac{\langle x := x + 1, \sigma_{477} \rangle \longrightarrow \sigma_{478} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{479} \quad \mathcal{T}_{480}}{\langle S_2, \sigma_{477} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{476} = \frac{\langle x := x + 1, \sigma_{475} \rangle \longrightarrow \sigma_{476} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{477} \quad \mathcal{T}_{478}}{\langle S_2, \sigma_{475} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{474} = \frac{\langle x := x + 1, \sigma_{473} \rangle \longrightarrow \sigma_{474} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{475} \quad \mathcal{T}_{476}}{\langle S_2, \sigma_{473} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{472} = \frac{\langle x := x + 1, \sigma_{471} \rangle \longrightarrow \sigma_{472} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{473} \quad \mathcal{T}_{474}}{\langle S_2, \sigma_{471} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{470} = \frac{\langle x := x + 1, \sigma_{469} \rangle \longrightarrow \sigma_{470} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{471} \quad \mathcal{T}_{472}}{\langle S_2, \sigma_{469} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{468} = \frac{\langle x := x + 1, \sigma_{467} \rangle \longrightarrow \sigma_{468} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{469} \quad \mathcal{T}_{470}}{\langle S_2, \sigma_{467} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{466} = \frac{\langle x := x + 1, \sigma_{465} \rangle \longrightarrow \sigma_{466} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{467} \quad \mathcal{T}_{468}}{\langle S_2, \sigma_{465} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{464} = \frac{\langle x := x + 1, \sigma_{463} \rangle \longrightarrow \sigma_{464} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{465} \quad \mathcal{T}_{466}}{\langle S_2, \sigma_{463} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{462} = \frac{\langle x := x + 1, \sigma_{461} \rangle \longrightarrow \sigma_{462} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{463} \quad \mathcal{T}_{464}}{\langle S_2, \sigma_{461} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{460} = \frac{\langle x := x + 1, \sigma_{459} \rangle \longrightarrow \sigma_{460} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{461} \quad \mathcal{T}_{462}}{\langle S_2, \sigma_{459} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{458} = \frac{\langle x := x + 1, \sigma_{457} \rangle \longrightarrow \sigma_{458} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{459} \quad \mathcal{T}_{460}}{\langle S_2, \sigma_{457} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{456} = \frac{\langle x := x + 1, \sigma_{455} \rangle \longrightarrow \sigma_{456} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{457} \quad \mathcal{T}_{458}}{\langle S_2, \sigma_{455} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{454} = \frac{\langle x := x + 1, \sigma_{453} \rangle \longrightarrow \sigma_{454} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{455} \quad \mathcal{T}_{456}}{\langle S_2, \sigma_{453} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{452} = \frac{\langle x := x + 1, \sigma_{451} \rangle \longrightarrow \sigma_{452} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{453} \quad \mathcal{T}_{454}}{\langle S_2, \sigma_{451} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{450} = \frac{\langle x := x + 1, \sigma_{449} \rangle \longrightarrow \sigma_{450} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{451} \quad \mathcal{T}_{452}}{\langle S_2, \sigma_{449} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{448} = \frac{\langle x := x + 1, \sigma_{447} \rangle \longrightarrow \sigma_{448} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{449} \quad \mathcal{T}_{450}}{\langle S_2, \sigma_{447} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{446} = \frac{\langle x := x + 1, \sigma_{445} \rangle \longrightarrow \sigma_{446} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{447} \quad \mathcal{T}_{448}}{\langle S_2, \sigma_{445} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{444} = \frac{\langle x := x + 1, \sigma_{443} \rangle \longrightarrow \sigma_{444} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{445} \quad \mathcal{T}_{446}}{\langle S_2, \sigma_{443} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{442} = \frac{\langle x := x + 1, \sigma_{441} \rangle \longrightarrow \sigma_{442} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{443} \quad \mathcal{T}_{444}}{\langle S_2, \sigma_{441} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{440} = \frac{\langle x := x + 1, \sigma_{439} \rangle \longrightarrow \sigma_{440} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{441} \quad \mathcal{T}_{442}}{\langle S_2, \sigma_{439} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{438} = \frac{\langle x := x + 1, \sigma_{437} \rangle \longrightarrow \sigma_{438} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{439} \quad \mathcal{T}_{440}}{\langle S_2, \sigma_{437} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{436} = \frac{\langle x := x + 1, \sigma_{435} \rangle \longrightarrow \sigma_{436} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{437} \quad \mathcal{T}_{438}}{\langle S_2, \sigma_{435} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{434} = \frac{\langle x := x + 1, \sigma_{433} \rangle \longrightarrow \sigma_{434} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{435} \quad \mathcal{T}_{436}}{\langle S_2, \sigma_{433} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{432} = \frac{\langle x := x + 1, \sigma_{431} \rangle \longrightarrow \sigma_{432} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{433} \quad \mathcal{T}_{434}}{\langle S_2, \sigma_{431} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{430} = \frac{\langle x := x + 1, \sigma_{429} \rangle \longrightarrow \sigma_{430} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{431} \quad \mathcal{T}_{432}}{\langle S_2, \sigma_{429} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{428} = \frac{\langle x := x + 1, \sigma_{427} \rangle \longrightarrow \sigma_{428} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{429} \quad \mathcal{T}_{430}}{\langle S_2, \sigma_{427} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{426} = \frac{\langle x := x + 1, \sigma_{425} \rangle \longrightarrow \sigma_{426} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{427} \quad \mathcal{T}_{428}}{\langle S_2, \sigma_{425} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{424} = \frac{\langle x := x + 1, \sigma_{423} \rangle \longrightarrow \sigma_{424} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{425} \quad \mathcal{T}_{426}}{\langle S_2, \sigma_{423} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{422} = \frac{\langle x := x + 1, \sigma_{421} \rangle \longrightarrow \sigma_{422} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{423} \quad \mathcal{T}_{424}}{\langle S_2, \sigma_{421} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{420} = \frac{\langle x := x + 1, \sigma_{419} \rangle \longrightarrow \sigma_{420} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{421} \quad \mathcal{T}_{422}}{\langle S_2, \sigma_{419} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{418} = \frac{\langle x := x + 1, \sigma_{417} \rangle \longrightarrow \sigma_{418} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{419} \quad \mathcal{T}_{420}}{\langle S_2, \sigma_{417} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{416} = \frac{\langle x := x + 1, \sigma_{415} \rangle \longrightarrow \sigma_{416} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{417} \quad \mathcal{T}_{418}}{\langle S_2, \sigma_{415} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{414} = \frac{\langle x := x + 1, \sigma_{413} \rangle \longrightarrow \sigma_{414} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{415} \quad \mathcal{T}_{416}}{\langle S_2, \sigma_{413} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{412} = \frac{\langle x := x + 1, \sigma_{411} \rangle \longrightarrow \sigma_{412} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{413} \quad \mathcal{T}_{414}}{\langle S_2, \sigma_{411} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{410} = \frac{\langle x := x + 1, \sigma_{409} \rangle \longrightarrow \sigma_{410} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{411} \quad \mathcal{T}_{412}}{\langle S_2, \sigma_{409} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{408} = \frac{\langle x := x + 1, \sigma_{407} \rangle \longrightarrow \sigma_{408} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{409} \quad \mathcal{T}_{410}}{\langle S_2, \sigma_{407} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{406} = \frac{\langle x := x + 1, \sigma_{405} \rangle \longrightarrow \sigma_{406} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{407} \quad \mathcal{T}_{408}}{\langle S_2, \sigma_{405} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{404} = \frac{\langle x := x + 1, \sigma_{403} \rangle \longrightarrow \sigma_{404} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{405} \quad \mathcal{T}_{406}}{\langle S_2, \sigma_{403} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{402} = \frac{\langle x := x + 1, \sigma_{401} \rangle \longrightarrow \sigma_{402} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{403} \quad \mathcal{T}_{404}}{\langle S_2, \sigma_{401} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{400} = \frac{\langle x := x + 1, \sigma_{399} \rangle \longrightarrow \sigma_{400} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{401} \quad \mathcal{T}_{402}}{\langle S_2, \sigma_{399} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{398} = \frac{\langle x := x + 1, \sigma_{397} \rangle \longrightarrow \sigma_{398} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{399} \quad \mathcal{T}_{400}}{\langle S_2, \sigma_{397} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{396} = \frac{\langle x := x + 1, \sigma_{395} \rangle \longrightarrow \sigma_{396} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{397} \quad \mathcal{T}_{398}}{\langle S_2, \sigma_{395} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{394} = \frac{\langle x := x + 1, \sigma_{393} \rangle \longrightarrow \sigma_{394} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{395} \quad \mathcal{T}_{396}}{\langle S_2, \sigma_{393} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{392} = \frac{\langle x := x + 1, \sigma_{391} \rangle \longrightarrow \sigma_{392} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{393} \quad \mathcal{T}_{394}}{\langle S_2, \sigma_{391} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{390} = \frac{\langle x := x + 1, \sigma_{389} \rangle \longrightarrow \sigma_{390} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{391} \quad \mathcal{T}_{392}}{\langle S_2, \sigma_{389} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{388} = \frac{\langle x := x + 1, \sigma_{387} \rangle \longrightarrow \sigma_{388} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{389} \quad \mathcal{T}_{390}}{\langle S_2, \sigma_{387} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{386} = \frac{\langle x := x + 1, \sigma_{385} \rangle \longrightarrow \sigma_{386} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{387} \quad \mathcal{T}_{388}}{\langle S_2, \sigma_{385} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{384} = \frac{\langle x := x + 1, \sigma_{383} \rangle \longrightarrow \sigma_{384} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{385} \quad \mathcal{T}_{386}}{\langle S_2, \sigma_{383} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{382} = \frac{\langle x := x + 1, \sigma_{381} \rangle \longrightarrow \sigma_{382} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{383} \quad \mathcal{T}_{384}}{\langle S_2, \sigma_{381} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{380} = \frac{\langle x := x + 1, \sigma_{379} \rangle \longrightarrow \sigma_{380} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{381} \quad \mathcal{T}_{382}}{\langle S_2, \sigma_{379} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{378} = \frac{\langle x := x + 1, \sigma_{377} \rangle \longrightarrow \sigma_{378} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{379} \quad \mathcal{T}_{380}}{\langle S_2, \sigma_{377} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{376} = \frac{\langle x := x + 1, \sigma_{375} \rangle \longrightarrow \sigma_{376} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{377} \quad \mathcal{T}_{378}}{\langle S_2, \sigma_{375} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{374} = \frac{\langle x := x + 1, \sigma_{373} \rangle \longrightarrow \sigma_{374} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{375} \quad \mathcal{T}_{376}}{\langle S_2, \sigma_{373} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{372} = \frac{\langle x := x + 1, \sigma_{371} \rangle \longrightarrow \sigma_{372} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{373} \quad \mathcal{T}_{374}}{\langle S_2, \sigma_{371} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{370} = \frac{\langle x := x + 1, \sigma_{369} \rangle \longrightarrow \sigma_{370} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{371} \quad \mathcal{T}_{372}}{\langle S_2, \sigma_{369} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{368} = \frac{\langle x := x + 1, \sigma_{367} \rangle \longrightarrow \sigma_{368} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{369} \quad \mathcal{T}_{370}}{\langle S_2, \sigma_{367} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{366} = \frac{\langle x := x + 1, \sigma_{365} \rangle \longrightarrow \sigma_{366} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{367} \quad \mathcal{T}_{368}}{\langle S_2, \sigma_{365} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{364} = \frac{\langle x := x + 1, \sigma_{363} \rangle \longrightarrow \sigma_{364} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{365} \quad \mathcal{T}_{366}}{\langle S_2, \sigma_{363} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{362} = \frac{\langle x := x + 1, \sigma_{361} \rangle \longrightarrow \sigma_{362} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{363} \quad \mathcal{T}_{364}}{\langle S_2, \sigma_{361} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{360} = \frac{\langle x := x + 1, \sigma_{359} \rangle \longrightarrow \sigma_{360} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{361} \quad \mathcal{T}_{362}}{\langle S_2, \sigma_{359} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{358} = \frac{\langle x := x + 1, \sigma_{357} \rangle \longrightarrow \sigma_{358} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{359} \quad \mathcal{T}_{360}}{\langle S_2, \sigma_{357} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{356} = \frac{\langle x := x + 1, \sigma_{355} \rangle \longrightarrow \sigma_{356} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{357} \quad \mathcal{T}_{358}}{\langle S_2, \sigma_{355} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{354} = \frac{\langle x := x + 1, \sigma_{353} \rangle \longrightarrow \sigma_{354} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{355} \quad \mathcal{T}_{356}}{\langle S_2, \sigma_{353} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{352} = \frac{\langle x := x + 1, \sigma_{351} \rangle \longrightarrow \sigma_{352} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{353} \quad \mathcal{T}_{354}}{\langle S_2, \sigma_{351} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{350} = \frac{\langle x := x + 1, \sigma_{349} \rangle \longrightarrow \sigma_{350} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{351} \quad \mathcal{T}_{352}}{\langle S_2, \sigma_{349} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{348} = \frac{\langle x := x + 1, \sigma_{347} \rangle \longrightarrow \sigma_{348} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{349} \quad \mathcal{T}_{350}}{\langle S_2, \sigma_{347} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{346} = \frac{\langle x := x + 1, \sigma_{345} \rangle \longrightarrow \sigma_{346} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{347} \quad \mathcal{T}_{348}}{\langle S_2, \sigma_{345} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{344} = \frac{\langle x := x + 1, \sigma_{343} \rangle \longrightarrow \sigma_{344} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{345} \quad \mathcal{T}_{346}}{\langle S_2, \sigma_{343} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{342} = \frac{\langle x := x + 1, \sigma_{341} \rangle \longrightarrow \sigma_{342} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{343} \quad \mathcal{T}_{344}}{\langle S_2, \sigma_{341} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{340} = \frac{\langle x := x + 1, \sigma_{339} \rangle \longrightarrow \sigma_{340} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{341} \quad \mathcal{T}_{342}}{\langle S_2, \sigma_{339} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{338} = \frac{\langle x := x + 1, \sigma_{337} \rangle \longrightarrow \sigma_{338} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{339} \quad \mathcal{T}_{340}}{\langle S_2, \sigma_{337} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{336} = \frac{\langle x := x + 1, \sigma_{335} \rangle \longrightarrow \sigma_{336} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{337} \quad \mathcal{T}_{338}}{\langle S_2, \sigma_{335} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{334} = \frac{\langle x := x + 1, \sigma_{333} \rangle \longrightarrow \sigma_{334} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{335} \quad \mathcal{T}_{336}}{\langle S_2, \sigma_{333} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{332} = \frac{\langle x := x + 1, \sigma_{331} \rangle \longrightarrow \sigma_{332} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{333} \quad \mathcal{T}_{334}}{\langle S_2, \sigma_{331} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{330} = \frac{\langle x := x + 1, \sigma_{329} \rangle \longrightarrow \sigma_{330} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{331} \quad \mathcal{T}_{332}}{\langle S_2, \sigma_{329} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{328} = \frac{\langle x := x + 1, \sigma_{327} \rangle \longrightarrow \sigma_{328} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{329} \quad \mathcal{T}_{330}}{\langle S_2, \sigma_{327} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{326} = \frac{\langle x := x + 1, \sigma_{325} \rangle \longrightarrow \sigma_{326} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{327} \quad \mathcal{T}_{328}}{\langle S_2, \sigma_{325} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{324} = \frac{\langle x := x + 1, \sigma_{323} \rangle \longrightarrow \sigma_{324} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{325} \quad \mathcal{T}_{326}}{\langle S_2, \sigma_{323} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{322} = \frac{\langle x := x + 1, \sigma_{321} \rangle \longrightarrow \sigma_{322} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{323} \quad \mathcal{T}_{324}}{\langle S_2, \sigma_{321} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{320} = \frac{\langle x := x + 1, \sigma_{319} \rangle \longrightarrow \sigma_{320} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{321} \quad \mathcal{T}_{322}}{\langle S_2, \sigma_{319} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{318} = \frac{\langle x := x + 1, \sigma_{317} \rangle \longrightarrow \sigma_{318} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{319} \quad \mathcal{T}_{320}}{\langle S_2, \sigma_{317} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{316} = \frac{\langle x := x + 1, \sigma_{315} \rangle \longrightarrow \sigma_{316} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{317} \quad \mathcal{T}_{318}}{\langle S_2, \sigma_{315} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{314} = \frac{\langle x := x + 1, \sigma_{313} \rangle \longrightarrow \sigma_{314} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{315} \quad \mathcal{T}_{316}}{\langle S_2, \sigma_{313} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{312} = \frac{\langle x := x + 1, \sigma_{311} \rangle \longrightarrow \sigma_{312} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{313} \quad \mathcal{T}_{314}}{\langle S_2, \sigma_{311} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{310} = \frac{\langle x := x + 1, \sigma_{309} \rangle \longrightarrow \sigma_{310} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{311} \quad \mathcal{T}_{312}}{\langle S_2, \sigma_{309} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{308} = \frac{\langle x := x + 1, \sigma_{307} \rangle \longrightarrow \sigma_{308} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{309} \quad \mathcal{T}_{310}}{\langle S_2, \sigma_{307} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{306} = \frac{\langle x := x + 1, \sigma_{305} \rangle \longrightarrow \sigma_{306} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{307} \quad \mathcal{T}_{308}}{\langle S_2, \sigma_{305} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{304} = \frac{\langle x := x + 1, \sigma_{303} \rangle \longrightarrow \sigma_{304} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{305} \quad \mathcal{T}_{306}}{\langle S_2, \sigma_{303} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{302} = \frac{\langle x := x + 1, \sigma_{301} \rangle \longrightarrow \sigma_{302} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{303} \quad \mathcal{T}_{304}}{\langle S_2, \sigma_{301} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{300} = \frac{\langle x := x + 1, \sigma_{299} \rangle \longrightarrow \sigma_{300} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{301} \quad \mathcal{T}_{302}}{\langle S_2, \sigma_{299} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{298} = \frac{\langle x := x + 1, \sigma_{297} \rangle \longrightarrow \sigma_{298} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{299} \quad \mathcal{T}_{300}}{\langle S_2, \sigma_{297} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{296} = \frac{\langle x := x + 1, \sigma_{295} \rangle \longrightarrow \sigma_{296} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{297} \quad \mathcal{T}_{298}}{\langle S_2, \sigma_{295} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{294} = \frac{\langle x := x + 1, \sigma_{293} \rangle \longrightarrow \sigma_{294} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{295} \quad \mathcal{T}_{296}}{\langle S_2, \sigma_{293} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{292} = \frac{\langle x := x + 1, \sigma_{291} \rangle \longrightarrow \sigma_{292} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{293} \quad \mathcal{T}_{294}}{\langle S_2, \sigma_{291} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{290} = \frac{\langle x := x + 1, \sigma_{289} \rangle \longrightarrow \sigma_{290} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{291} \quad \mathcal{T}_{292}}{\langle S_2, \sigma_{289} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{288} = \frac{\langle x := x + 1, \sigma_{287} \rangle \longrightarrow \sigma_{288} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{289} \quad \mathcal{T}_{290}}{\langle S_2, \sigma_{287} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{286} = \frac{\langle x := x + 1, \sigma_{285} \rangle \longrightarrow \sigma_{286} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{287} \quad \mathcal{T}_{288}}{\langle S_2, \sigma_{285} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{284} = \frac{\langle x := x + 1, \sigma_{283} \rangle \longrightarrow \sigma_{284} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{285} \quad \mathcal{T}_{286}}{\langle S_2, \sigma_{283} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{282} = \frac{\langle x := x + 1, \sigma_{281} \rangle \longrightarrow \sigma_{282} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{283} \quad \mathcal{T}_{284}}{\langle S_2, \sigma_{281} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{280} = \frac{\langle x := x + 1, \sigma_{279} \rangle \longrightarrow \sigma_{280} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{281} \quad \mathcal{T}_{282}}{\langle S_2, \sigma_{279} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{278} = \frac{\langle x := x + 1, \sigma_{277} \rangle \longrightarrow \sigma_{278} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{279} \quad \mathcal{T}_{280}}{\langle S_2, \sigma_{277} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{276} = \frac{\langle x := x + 1, \sigma_{275} \rangle \longrightarrow \sigma_{276} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{277} \quad \mathcal{T}_{278}}{\langle S_2, \sigma_{275} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{274} = \frac{\langle x := x + 1, \sigma_{273} \rangle \longrightarrow \sigma_{274} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{275} \quad \mathcal{T}_{276}}{\langle S_2, \sigma_{273} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{272} = \frac{\langle x := x + 1, \sigma_{271} \rangle \longrightarrow \sigma_{272} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{273} \quad \mathcal{T}_{274}}{\langle S_2, \sigma_{271} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{270} = \frac{\langle x := x + 1, \sigma_{269} \rangle \longrightarrow \sigma_{270} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{271} \quad \mathcal{T}_{272}}{\langle S_2, \sigma_{269} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{268} = \frac{\langle x := x + 1, \sigma_{267} \rangle \longrightarrow \sigma_{268} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{269} \quad \mathcal{T}_{270}}{\langle S_2, \sigma_{267} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{266} = \frac{\langle x := x + 1, \sigma_{265} \rangle \longrightarrow \sigma_{266} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{267} \quad \mathcal{T}_{268}}{\langle S_2, \sigma_{265} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{264} = \frac{\langle x := x + 1, \sigma_{263} \rangle \longrightarrow \sigma_{264} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{265} \quad \mathcal{T}_{266}}{\langle S_2, \sigma_{263} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{262} = \frac{\langle x := x + 1, \sigma_{261} \rangle \longrightarrow \sigma_{262} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{263} \quad \mathcal{T}_{264}}{\langle S_2, \sigma_{261} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{260} = \frac{\langle x := x + 1, \sigma_{259} \rangle \longrightarrow \sigma_{260} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{261} \quad \mathcal{T}_{262}}{\langle S_2, \sigma_{259} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{258} = \frac{\langle x := x + 1, \sigma_{257} \rangle \longrightarrow \sigma_{258} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{259} \quad \mathcal{T}_{260}}{\langle S_2, \sigma_{257} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{256} = \frac{\langle x := x + 1, \sigma_{255} \rangle \longrightarrow \sigma_{256} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{257} \quad \mathcal{T}_{258}}{\langle S_2, \sigma_{255} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{254} = \frac{\langle x := x + 1, \sigma_{253} \rangle \longrightarrow \sigma_{254} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{255} \quad \mathcal{T}_{256}}{\langle S_2, \sigma_{253} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{252} = \frac{\langle x := x + 1, \sigma_{251} \rangle \longrightarrow \sigma_{252} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{253} \quad \mathcal{T}_{254}}{\langle S_2, \sigma_{251} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{250} = \frac{\langle x := x + 1, \sigma_{249} \rangle \longrightarrow \sigma_{250} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{251} \quad \mathcal{T}_{252}}{\langle S_2, \sigma_{249} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{248} = \frac{\langle x := x + 1, \sigma_{247} \rangle \longrightarrow \sigma_{248} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{249} \quad \mathcal{T}_{250}}{\langle S_2, \sigma_{247} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{246} = \frac{\langle x := x + 1, \sigma_{245} \rangle \longrightarrow \sigma_{246} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{247} \quad \mathcal{T}_{248}}{\langle S_2, \sigma_{245} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{244} = \frac{\langle x := x + 1, \sigma_{243} \rangle \longrightarrow \sigma_{244} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{245} \quad \mathcal{T}_{246}}{\langle S_2, \sigma_{243} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{242} = \frac{\langle x := x + 1, \sigma_{241} \rangle \longrightarrow \sigma_{242} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{243} \quad \mathcal{T}_{244}}{\langle S_2, \sigma_{241} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{240} = \frac{\langle x := x + 1, \sigma_{239} \rangle \longrightarrow \sigma_{240} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{241} \quad \mathcal{T}_{242}}{\langle S_2, \sigma_{239} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{238} = \frac{\langle x := x + 1, \sigma_{237} \rangle \longrightarrow \sigma_{238} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{239} \quad \mathcal{T}_{240}}{\langle S_2, \sigma_{237} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{236} = \frac{\langle x := x + 1, \sigma_{235} \rangle \longrightarrow \sigma_{236} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{237} \quad \mathcal{T}_{238}}{\langle S_2, \sigma_{235} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{234} = \frac{\langle x := x + 1, \sigma_{233} \rangle \longrightarrow \sigma_{234} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{235} \quad \mathcal{T}_{236}}{\langle S_2, \sigma_{233} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{232} = \frac{\langle x := x + 1, \sigma_{231} \rangle \longrightarrow \sigma_{232} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{233} \quad \mathcal{T}_{234}}{\langle S_2, \sigma_{231} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{230} = \frac{\langle x := x + 1, \sigma_{229} \rangle \longrightarrow \sigma_{230} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{231} \quad \mathcal{T}_{232}}{\langle S_2, \sigma_{229} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{228} = \frac{\langle x := x + 1, \sigma_{227} \rangle \longrightarrow \sigma_{228} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{229} \quad \mathcal{T}_{230}}{\langle S_2, \sigma_{227} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{226} = \frac{\langle x := x + 1, \sigma_{225} \rangle \longrightarrow \sigma_{226} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{227} \quad \mathcal{T}_{228}}{\langle S_2, \sigma_{225} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{224} = \frac{\langle x := x + 1, \sigma_{223} \rangle \longrightarrow \sigma_{224} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{225} \quad \mathcal{T}_{226}}{\langle S_2, \sigma_{223} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{222} = \frac{\langle x := x + 1, \sigma_{221} \rangle \longrightarrow \sigma_{222} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{223} \quad \mathcal{T}_{224}}{\langle S_2, \sigma_{221} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{220} = \frac{\langle x := x + 1, \sigma_{219} \rangle \longrightarrow \sigma_{220} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{221} \quad \mathcal{T}_{222}}{\langle S_2, \sigma_{219} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{218} = \frac{\langle x := x + 1, \sigma_{217} \rangle \longrightarrow \sigma_{218} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{219} \quad \mathcal{T}_{220}}{\langle S_2, \sigma_{217} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{216} = \frac{\langle x := x + 1, \sigma_{215} \rangle \longrightarrow \sigma_{216} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{217} \quad \mathcal{T}_{218}}{\langle S_2, \sigma_{215} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{214} = \frac{\langle x := x + 1, \sigma_{213} \rangle \longrightarrow \sigma_{214} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{215} \quad \mathcal{T}_{216}}{\langle S_2, \sigma_{213} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{212} = \frac{\langle x := x + 1, \sigma_{211} \rangle \longrightarrow \sigma_{212} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{213} \quad \mathcal{T}_{214}}{\langle S_2, \sigma_{211} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{210} = \frac{\langle x := x + 1, \sigma_{209} \rangle \longrightarrow \sigma_{210} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{211} \quad \mathcal{T}_{212}}{\langle S_2, \sigma_{209} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{208} = \frac{\langle x := x + 1, \sigma_{207} \rangle \longrightarrow \sigma_{208} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{209} \quad \mathcal{T}_{210}}{\langle S_2, \sigma_{207} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{206} = \frac{\langle x := x + 1, \sigma_{205} \rangle \longrightarrow \sigma_{206} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{207} \quad \mathcal{T}_{208}}{\langle S_2, \sigma_{205} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{204} = \frac{\langle x := x + 1, \sigma_{203} \rangle \longrightarrow \sigma_{204} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{205} \quad \mathcal{T}_{206}}{\langle S_2, \sigma_{203} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{202} = \frac{\langle x := x + 1, \sigma_{201} \rangle \longrightarrow \sigma_{202} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{203} \quad \mathcal{T}_{204}}{\langle S_2, \sigma_{201} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{200} = \frac{\langle x := x + 1, \sigma_{199} \rangle \longrightarrow \sigma_{200} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{201} \quad \mathcal{T}_{202}}{\langle S_2, \sigma_{199} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{198} = \frac{\langle x := x + 1, \sigma_{197} \rangle \longrightarrow \sigma_{198} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{199} \quad \mathcal{T}_{200}}{\langle S_2, \sigma_{197} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{196} = \frac{\langle x := x + 1, \sigma_{195} \rangle \longrightarrow \sigma_{196} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{197} \quad \mathcal{T}_{198}}{\langle S_2, \sigma_{195} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{194} = \frac{\langle x := x + 1, \sigma_{193} \rangle \longrightarrow \sigma_{194} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{195} \quad \mathcal{T}_{196}}{\langle S_2, \sigma_{193} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{192} = \frac{\langle x := x + 1, \sigma_{191} \rangle \longrightarrow \sigma_{192} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{193} \quad \mathcal{T}_{194}}{\langle S_2, \sigma_{191} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{190} = \frac{\langle x := x + 1, \sigma_{189} \rangle \longrightarrow \sigma_{190} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{191} \quad \mathcal{T}_{192}}{\langle S_2, \sigma_{189} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{188} = \frac{\langle x := x + 1, \sigma_{187} \rangle \longrightarrow \sigma_{188} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{189} \quad \mathcal{T}_{190}}{\langle S_2, \sigma_{187} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{186} = \frac{\langle x := x + 1, \sigma_{185} \rangle \longrightarrow \sigma_{186} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{187} \quad \mathcal{T}_{188}}{\langle S_2, \sigma_{185} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{184} = \frac{\langle x := x + 1, \sigma_{183} \rangle \longrightarrow \sigma_{184} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{185} \quad \mathcal{T}_{186}}{\langle S_2, \sigma_{183} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{182} = \frac{\langle x := x + 1, \sigma_{181} \rangle \longrightarrow \sigma_{182} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{183} \quad \mathcal{T}_{184}}{\langle S_2, \sigma_{181} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{180} = \frac{\langle x := x + 1, \sigma_{179} \rangle \longrightarrow \sigma_{180} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{181} \quad \mathcal{T}_{182}}{\langle S_2, \sigma_{179} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{178} = \frac{\langle x := x + 1, \sigma_{177} \rangle \longrightarrow \sigma_{178} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{179} \quad \mathcal{T}_{180}}{\langle S_2, \sigma_{177} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{176} = \frac{\langle x := x + 1, \sigma_{175} \rangle \longrightarrow \sigma_{176} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{177} \quad \mathcal{T}_{178}}{\langle S_2, \sigma_{175} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{174} = \frac{\langle x := x + 1, \sigma_{173} \rangle \longrightarrow \sigma_{174} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{175} \quad \mathcal{T}_{176}}{\langle S_2, \sigma_{173} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{172} = \frac{\langle x := x + 1, \sigma_{171} \rangle \longrightarrow \sigma_{172} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{173} \quad \mathcal{T}_{174}}{\langle S_2, \sigma_{171} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{170} = \frac{\langle x := x + 1, \sigma_{169} \rangle \longrightarrow \sigma_{170} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{171} \quad \mathcal{T}_{172}}{\langle S_2, \sigma_{169} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{168} = \frac{\langle x := x + 1, \sigma_{167} \rangle \longrightarrow \sigma_{168} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{169} \quad \mathcal{T}_{170}}{\langle S_2, \sigma_{167} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{166} = \frac{\langle x := x + 1, \sigma_{165} \rangle \longrightarrow \sigma_{166} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{167} \quad \mathcal{T}_{168}}{\langle S_2, \sigma_{165} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{164} = \frac{\langle x := x + 1, \sigma_{163} \rangle \longrightarrow \sigma_{164} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{165} \quad \mathcal{T}_{166}}{\langle S_2, \sigma_{163} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{162} = \frac{\langle x := x + 1, \sigma_{161} \rangle \longrightarrow \sigma_{162} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{163} \quad \mathcal{T}_{164}}{\langle S_2, \sigma_{161} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{160} = \frac{\langle x := x + 1, \sigma_{159} \rangle \longrightarrow \sigma_{160} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{161} \quad \mathcal{T}_{162}}{\langle S_2, \sigma_{159} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{158} = \frac{\langle x := x + 1, \sigma_{157} \rangle \longrightarrow \sigma_{158} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{159} \quad \mathcal{T}_{160}}{\langle S_2, \sigma_{157} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{156} = \frac{\langle x := x + 1, \sigma_{155} \rangle \longrightarrow \sigma_{156} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{157} \quad \mathcal{T}_{158}}{\langle S_2, \sigma_{155} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{154} = \frac{\langle x := x + 1, \sigma_{153} \rangle \longrightarrow \sigma_{154} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{155} \quad \mathcal{T}_{156}}{\langle S_2, \sigma_{153} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{152} = \frac{\langle x := x + 1, \sigma_{151} \rangle \longrightarrow \sigma_{152} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{153} \quad \mathcal{T}_{154}}{\langle S_2, \sigma_{151} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{150} = \frac{\langle x := x + 1, \sigma_{149} \rangle \longrightarrow \sigma_{150} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{151} \quad \mathcal{T}_{152}}{\langle S_2, \sigma_{149} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{148} = \frac{\langle x := x + 1, \sigma_{147} \rangle \longrightarrow \sigma_{148} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{149} \quad \mathcal{T}_{150}}{\langle S_2, \sigma_{147} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{146} = \frac{\langle x := x + 1, \sigma_{145} \rangle \longrightarrow \sigma_{146} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{147} \quad \mathcal{T}_{148}}{\langle S_2, \sigma_{145} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{144} = \frac{\langle x := x + 1, \sigma_{143} \rangle \longrightarrow \sigma_{144} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{145} \quad \mathcal{T}_{146}}{\langle S_2, \sigma_{143} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{142} = \frac{\langle x := x + 1, \sigma_{141} \rangle \longrightarrow \sigma_{142} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{143} \quad \mathcal{T}_{144}}{\langle S_2, \sigma_{141} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{140} = \frac{\langle x := x + 1, \sigma_{139} \rangle \longrightarrow \sigma_{140} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{141} \quad \mathcal{T}_{142}}{\langle S_2, \sigma_{139} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{138} = \frac{\langle x := x + 1, \sigma_{137} \rangle \longrightarrow \sigma_{138} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{139} \quad \mathcal{T}_{140}}{\langle S_2, \sigma_{137} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{136} = \frac{\langle x := x + 1, \sigma_{135} \rangle \longrightarrow \sigma_{136} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{137} \quad \mathcal{T}_{138}}{\langle S_2, \sigma_{135} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{134} = \frac{\langle x := x + 1, \sigma_{133} \rangle \longrightarrow \sigma_{134} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{135} \quad \mathcal{T}_{136}}{\langle S_2, \sigma_{133} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{132} = \frac{\langle x := x + 1, \sigma_{131} \rangle \longrightarrow \sigma_{132} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{133} \quad \mathcal{T}_{134}}{\langle S_2, \sigma_{131} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{130} = \frac{\langle x := x + 1, \sigma_{129} \rangle \longrightarrow \sigma_{130} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{131} \quad \mathcal{T}_{132}}{\langle S_2, \sigma_{129} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{128} = \frac{\langle x := x + 1, \sigma_{127} \rangle \longrightarrow \sigma_{128} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{129} \quad \mathcal{T}_{130}}{\langle S_2, \sigma_{127} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{126} = \frac{\langle x := x + 1, \sigma_{125} \rangle \longrightarrow \sigma_{126} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{127} \quad \mathcal{T}_{128}}{\langle S_2, \sigma_{125} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{124} = \frac{\langle x := x + 1, \sigma_{123} \rangle \longrightarrow \sigma_{124} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{125} \quad \mathcal{T}_{126}}{\langle S_2, \sigma_{123} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{122} = \frac{\langle x := x + 1, \sigma_{121} \rangle \longrightarrow \sigma_{122} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{123} \quad \mathcal{T}_{124}}{\langle S_2, \sigma_{121} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{120} = \frac{\langle x := x + 1, \sigma_{119} \rangle \longrightarrow \sigma_{120} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{121} \quad \mathcal{T}_{122}}{\langle S_2, \sigma_{119} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{118} = \frac{\langle x := x + 1, \sigma_{117} \rangle \longrightarrow \sigma_{118} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{119} \quad \mathcal{T}_{120}}{\langle S_2, \sigma_{117} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{116} = \frac{\langle x := x + 1, \sigma_{115} \rangle \longrightarrow \sigma_{116} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{117} \quad \mathcal{T}_{118}}{\langle S_2, \sigma_{115} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{114} = \frac{\langle x := x + 1, \sigma_{113} \rangle \longrightarrow \sigma_{114} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{115} \quad \mathcal{T}_{116}}{\langle S_2, \sigma_{113} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{112} = \frac{\langle x := x + 1, \sigma_{111} \rangle \longrightarrow \sigma_{112} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{113} \quad \mathcal{T}_{114}}{\langle S_2, \sigma_{111} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{110} = \frac{\langle x := x + 1, \sigma_{109} \rangle \longrightarrow \sigma_{110} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{111} \quad \mathcal{T}_{112}}{\langle S_2, \sigma_{109} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{108} = \frac{\langle x := x + 1, \sigma_{107} \rangle \longrightarrow \sigma_{108} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{109} \quad \mathcal{T}_{110}}{\langle S_2, \sigma_{107} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{106} = \frac{\langle x := x + 1, \sigma_{105} \rangle \longrightarrow \sigma_{106} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{107} \quad \mathcal{T}_{108}}{\langle S_2, \sigma_{105} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{104} = \frac{\langle x := x + 1, \sigma_{103} \rangle \longrightarrow \sigma_{104} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{105} \quad \mathcal{T}_{106}}{\langle S_2, \sigma_{103} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{102} = \frac{\langle x := x + 1, \sigma_{101} \rangle \longrightarrow \sigma_{102} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{103} \quad \mathcal{T}_{104}}{\langle S_2, \sigma_{101} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{100} = \frac{\langle x := x + 1, \sigma_{99} \rangle \longrightarrow \sigma_{100} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{101} \quad \mathcal{T}_{102}}{\langle S_2, \sigma_{99} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{98} = \frac{\langle x := x + 1, \sigma_{97} \rangle \longrightarrow \sigma_{98} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{99} \quad \mathcal{T}_{100}}{\langle S_2, \sigma_{97} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{96} = \frac{\langle x := x + 1, \sigma_{95} \rangle \longrightarrow \sigma_{96} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{97} \quad \mathcal{T}_{98}}{\langle S_2, \sigma_{95} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{94} = \frac{\langle x := x + 1, \sigma_{93} \rangle \longrightarrow \sigma_{94} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{95} \quad \mathcal{T}_{96}}{\langle S_2, \sigma_{93} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{92} = \frac{\langle x := x + 1, \sigma_{91} \rangle \longrightarrow \sigma_{92} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{93} \quad \mathcal{T}_{94}}{\langle S_2, \sigma_{91} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{90} = \frac{\langle x := x + 1, \sigma_{89} \rangle \longrightarrow \sigma_{90} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{91} \quad \mathcal{T}_{92}}{\langle S_2, \sigma_{89} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{88} = \frac{\langle x := x + 1, \sigma_{87} \rangle \longrightarrow \sigma_{88} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{89} \quad \mathcal{T}_{90}}{\langle S_2, \sigma_{87} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{86} = \frac{\langle x := x + 1, \sigma_{85} \rangle \longrightarrow \sigma_{86} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{87} \quad \mathcal{T}_{88}}{\langle S_2, \sigma_{85} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{84} = \frac{\langle x := x + 1, \sigma_{83} \rangle \longrightarrow \sigma_{84} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{85} \quad \mathcal{T}_{86}}{\langle S_2, \sigma_{83} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{82} = \frac{\langle x := x + 1, \sigma_{81} \rangle \longrightarrow \sigma_{82} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{83} \quad \mathcal{T}_{84}}{\langle S_2, \sigma_{81} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{80} = \frac{\langle x := x + 1, \sigma_{79} \rangle \longrightarrow \sigma_{80} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{81} \quad \mathcal{T}_{82}}{\langle S_2, \sigma_{79} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{78} = \frac{\langle x := x + 1, \sigma_{77} \rangle \longrightarrow \sigma_{78} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{79} \quad \mathcal{T}_{80}}{\langle S_2, \sigma_{77} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{76} = \frac{\langle x := x + 1, \sigma_{75} \rangle \longrightarrow \sigma_{76} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{77} \quad \mathcal{T}_{78}}{\langle S_2, \sigma_{75} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{74} = \frac{\langle x := x + 1, \sigma_{73} \rangle \longrightarrow \sigma_{74} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{75} \quad \mathcal{T}_{76}}{\langle S_2, \sigma_{73} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{72} = \frac{\langle x := x + 1, \sigma_{71} \rangle \longrightarrow \sigma_{72} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{73} \quad \mathcal{T}_{74}}{\langle S_2, \sigma_{71} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{70} = \frac{\langle x := x + 1, \sigma_{69} \rangle \longrightarrow \sigma_{70} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{71} \quad \mathcal{T}_{72}}{\langle S_2, \sigma_{69} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{68} = \frac{\langle x := x + 1, \sigma_{67} \rangle \longrightarrow \sigma_{68} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{69} \quad \mathcal{T}_{70}}{\langle S_2, \sigma_{67} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{66} = \frac{\langle x := x + 1, \sigma_{65} \rangle \longrightarrow \sigma_{66} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{67} \quad \mathcal{T}_{68}}{\langle S_2, \sigma_{65} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{64} = \frac{\langle x := x + 1, \sigma_{63} \rangle \longrightarrow \sigma_{64} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{65} \quad \mathcal{T}_{66}}{\langle S_2, \sigma_{63} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{62} = \frac{\langle x := x + 1, \sigma_{61} \rangle \longrightarrow \sigma_{62} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{63} \quad \mathcal{T}_{64}}{\langle S_2, \sigma_{61} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{60} = \frac{\langle x := x + 1, \sigma_{59} \rangle \longrightarrow \sigma_{60} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{61} \quad \mathcal{T}_{62}}{\langle S_2, \sigma_{59} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{58} = \frac{\langle x := x + 1, \sigma_{57} \rangle \longrightarrow \sigma_{58} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{59} \quad \mathcal{T}_{60}}{\langle S_2, \sigma_{57} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{56} = \frac{\langle x := x + 1, \sigma_{55} \rangle \longrightarrow \sigma_{56} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{57} \quad \mathcal{T}_{58}}{\langle S_2, \sigma_{55} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{54} = \frac{\langle x := x + 1, \sigma_{53} \rangle \longrightarrow \sigma_{54} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{55} \quad \mathcal{T}_{56}}{\langle S_2, \sigma_{53} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{52} = \frac{\langle x := x + 1, \sigma_{51} \rangle \longrightarrow \sigma_{52} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{53} \quad \mathcal{T}_{54}}{\langle S_2, \sigma_{51} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{50} = \frac{\langle x := x + 1, \sigma_{49} \rangle \longrightarrow \sigma_{50} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{51} \quad \mathcal{T}_{52}}{\langle S_2, \sigma_{49} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{48} = \frac{\langle x := x + 1, \sigma_{47} \rangle \longrightarrow \sigma_{48} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{49} \quad \mathcal{T}_{50}}{\langle S_2, \sigma_{47} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{46} = \frac{\langle x := x + 1, \sigma_{45} \rangle \longrightarrow \sigma_{46} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{47} \quad \mathcal{T}_{48}}{\langle S_2, \sigma_{45} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{44} = \frac{\langle x := x + 1, \sigma_{43} \rangle \longrightarrow \sigma_{44} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{45} \quad \mathcal{T}_{46}}{\langle S_2, \sigma_{43} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{42} = \frac{\langle x := x + 1, \sigma_{41} \rangle \longrightarrow \sigma_{42} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{43} \quad \mathcal{T}_{44}}{\langle S_2, \sigma_{41} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{40} = \frac{\langle x := x + 1, \sigma_{39} \rangle \longrightarrow \sigma_{40} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{41} \quad \mathcal{T}_{42}}{\langle S_2, \sigma_{39} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{38} = \frac{\langle x := x + 1, \sigma_{37} \rangle \longrightarrow \sigma_{38} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{39} \quad \mathcal{T}_{40}}{\langle S_2, \sigma_{37} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{36} = \frac{\langle x := x + 1, \sigma_{35} \rangle \longrightarrow \sigma_{36} [\text{aSS}_{\text{ns}}] \quad \mathcal{T}_{37} \quad \mathcal{T}_{38}}{\langle S_2, \sigma_{35} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{34} = \frac{\langle x := x + 1, \sigma_{33} \rangle \longrightarrow \sigma_{34} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{35} \quad \mathcal{T}_{36}}{\langle S_2, \sigma_{33} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{32} = \frac{\langle x := x + 1, \sigma_{31} \rangle \longrightarrow \sigma_{32} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{33} \quad \mathcal{T}_{34}}{\langle S_2, \sigma_{31} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{30} = \frac{\langle x := x + 1, \sigma_{29} \rangle \longrightarrow \sigma_{30} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{31} \quad \mathcal{T}_{32}}{\langle S_2, \sigma_{29} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{28} = \frac{\langle x := x + 1, \sigma_{27} \rangle \longrightarrow \sigma_{28} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{29} \quad \mathcal{T}_{30}}{\langle S_2, \sigma_{27} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{26} = \frac{\langle x := x + 1, \sigma_{25} \rangle \longrightarrow \sigma_{26} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{27} \quad \mathcal{T}_{28}}{\langle S_2, \sigma_{25} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{24} = \frac{\langle x := x + 1, \sigma_{23} \rangle \longrightarrow \sigma_{24} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{25} \quad \mathcal{T}_{26}}{\langle S_2, \sigma_{23} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{22} = \frac{\langle x := x + 1, \sigma_{21} \rangle \longrightarrow \sigma_{22} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{23} \quad \mathcal{T}_{24}}{\langle S_2, \sigma_{21} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{20} = \frac{\langle x := x + 1, \sigma_{19} \rangle \longrightarrow \sigma_{20} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{21} \quad \mathcal{T}_{22}}{\langle S_2, \sigma_{19} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{18} = \frac{\langle x := x + 1, \sigma_{17} \rangle \longrightarrow \sigma_{18} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{19} \quad \mathcal{T}_{20}}{\langle S_2, \sigma_{17} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{16} = \frac{\langle x := x + 1, \sigma_{15} \rangle \longrightarrow \sigma_{16} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{17} \quad \mathcal{T}_{18}}{\langle S_2, \sigma_{15} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{14} = \frac{\langle x := x + 1, \sigma_{13} \rangle \longrightarrow \sigma_{14} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{15} \quad \mathcal{T}_{16}}{\langle S_2, \sigma_{13} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{12} = \frac{\langle x := x + 1, \sigma_{11} \rangle \longrightarrow \sigma_{12} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{13} \quad \mathcal{T}_{14}}{\langle S_2, \sigma_{11} \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{10} = \frac{\langle x := x + 1, \sigma_9 \rangle \longrightarrow \sigma_{10} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{11} \quad \mathcal{T}_{12}}{\langle S_2, \sigma_9 \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\begin{aligned}
\mathcal{T}_8 &= \frac{\langle x := x + 1, \sigma_7 \rangle \longrightarrow \sigma_8 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_9 \quad \mathcal{T}_{10}}{\langle S_2, \sigma_7 \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_6 &= \frac{\langle x := x + 1, \sigma_5 \rangle \longrightarrow \sigma_6 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_7 \quad \mathcal{T}_8}{\langle S_2, \sigma_5 \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_4 &= \frac{\langle x := x + 1, \sigma_3 \rangle \longrightarrow \sigma_4 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_5 \quad \mathcal{T}_6}{\langle S_2, \sigma_3 \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_2 &= \frac{\langle x := 0, \sigma_1 \rangle \longrightarrow \sigma_2 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_3 \quad \mathcal{T}_4}{\langle S_3, \sigma_1 \rangle \longrightarrow \perp} [\text{for} - \text{tt}_{\text{ns}}] \\
\mathcal{T}_1 &= \frac{\langle y := 0, \emptyset \rangle \longrightarrow \sigma_1 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_2}{\langle S_4, \emptyset \rangle \longrightarrow \perp} [\text{comp}_{\text{ns}}]
\end{aligned}$$

Sentencias Abreviadas

$S_1 = \text{assert } x \leq 999 \text{ before } y := y + 1$
 $S_2 = \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1$
 $S_3 = \text{for } x := 0 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1$
 $S_4 = (y := 0 ; \text{for } x := 0 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

Estados

$\sigma_1 = \{y \mapsto 0\}$	$\sigma_{10} = \{x \mapsto 4, y \mapsto 4\}$	$\sigma_{19} = \{x \mapsto 8, y \mapsto 9\}$
$\sigma_2 = \{x \mapsto 0, y \mapsto 0\}$	$\sigma_{11} = \{x \mapsto 4, y \mapsto 5\}$	$\sigma_{20} = \{x \mapsto 9, y \mapsto 9\}$
$\sigma_3 = \{x \mapsto 0, y \mapsto 1\}$	$\sigma_{12} = \{x \mapsto 5, y \mapsto 5\}$	$\sigma_{21} = \{x \mapsto 9, y \mapsto 10\}$
$\sigma_4 = \{x \mapsto 1, y \mapsto 1\}$	$\sigma_{13} = \{x \mapsto 5, y \mapsto 6\}$	$\sigma_{22} = \{x \mapsto 10, y \mapsto 10\}$
$\sigma_5 = \{x \mapsto 1, y \mapsto 2\}$	$\sigma_{14} = \{x \mapsto 6, y \mapsto 6\}$	$\sigma_{23} = \{x \mapsto 10, y \mapsto 11\}$
$\sigma_6 = \{x \mapsto 2, y \mapsto 2\}$	$\sigma_{15} = \{x \mapsto 6, y \mapsto 7\}$	$\sigma_{24} = \{x \mapsto 11, y \mapsto 11\}$
$\sigma_7 = \{x \mapsto 2, y \mapsto 3\}$	$\sigma_{16} = \{x \mapsto 7, y \mapsto 7\}$	$\sigma_{25} = \{x \mapsto 11, y \mapsto 12\}$
$\sigma_8 = \{x \mapsto 3, y \mapsto 3\}$	$\sigma_{17} = \{x \mapsto 7, y \mapsto 8\}$	$\sigma_{26} = \{x \mapsto 12, y \mapsto 12\}$
$\sigma_9 = \{x \mapsto 3, y \mapsto 4\}$	$\sigma_{18} = \{x \mapsto 8, y \mapsto 8\}$	$\sigma_{27} = \{x \mapsto 12, y \mapsto 13\}$

$$\begin{aligned}
\sigma_{100} &= \{x \mapsto 49, y \mapsto 49\} \\
\sigma_{101} &= \{x \mapsto 49, y \mapsto 50\} \\
\sigma_{102} &= \{x \mapsto 50, y \mapsto 50\} \\
\sigma_{103} &= \{x \mapsto 50, y \mapsto 51\} \\
\sigma_{104} &= \{x \mapsto 51, y \mapsto 51\} \\
\sigma_{105} &= \{x \mapsto 51, y \mapsto 52\} \\
\sigma_{106} &= \{x \mapsto 52, y \mapsto 52\} \\
\sigma_{107} &= \{x \mapsto 52, y \mapsto 53\} \\
\sigma_{108} &= \{x \mapsto 53, y \mapsto 53\} \\
\sigma_{109} &= \{x \mapsto 53, y \mapsto 54\} \\
\sigma_{110} &= \{x \mapsto 54, y \mapsto 54\} \\
\sigma_{111} &= \{x \mapsto 54, y \mapsto 55\} \\
\sigma_{112} &= \{x \mapsto 55, y \mapsto 55\} \\
\sigma_{113} &= \{x \mapsto 55, y \mapsto 56\} \\
\sigma_{114} &= \{x \mapsto 56, y \mapsto 56\} \\
\sigma_{115} &= \{x \mapsto 56, y \mapsto 57\} \\
\sigma_{116} &= \{x \mapsto 57, y \mapsto 57\} \\
\sigma_{117} &= \{x \mapsto 57, y \mapsto 58\} \\
\sigma_{118} &= \{x \mapsto 58, y \mapsto 58\} \\
\sigma_{119} &= \{x \mapsto 58, y \mapsto 59\} \\
\sigma_{120} &= \{x \mapsto 59, y \mapsto 59\} \\
\sigma_{121} &= \{x \mapsto 59, y \mapsto 60\} \\
\sigma_{122} &= \{x \mapsto 60, y \mapsto 60\} \\
\sigma_{123} &= \{x \mapsto 60, y \mapsto 61\} \\
\sigma_{124} &= \{x \mapsto 61, y \mapsto 61\} \\
\sigma_{125} &= \{x \mapsto 61, y \mapsto 62\} \\
\sigma_{126} &= \{x \mapsto 62, y \mapsto 62\} \\
\sigma_{127} &= \{x \mapsto 62, y \mapsto 63\} \\
\sigma_{128} &= \{x \mapsto 63, y \mapsto 63\} \\
\sigma_{129} &= \{x \mapsto 63, y \mapsto 64\} \\
\sigma_{130} &= \{x \mapsto 64, y \mapsto 64\} \\
\sigma_{131} &= \{x \mapsto 64, y \mapsto 65\} \\
\sigma_{132} &= \{x \mapsto 65, y \mapsto 65\} \\
\sigma_{133} &= \{x \mapsto 65, y \mapsto 66\} \\
\sigma_{134} &= \{x \mapsto 66, y \mapsto 66\} \\
\sigma_{135} &= \{x \mapsto 66, y \mapsto 67\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{208} &= \{x \mapsto 103, y \mapsto 103\} \\
\sigma_{209} &= \{x \mapsto 103, y \mapsto 104\} \\
\sigma_{210} &= \{x \mapsto 104, y \mapsto 104\} \\
\sigma_{211} &= \{x \mapsto 104, y \mapsto 105\} \\
\sigma_{212} &= \{x \mapsto 105, y \mapsto 105\} \\
\sigma_{213} &= \{x \mapsto 105, y \mapsto 106\} \\
\sigma_{214} &= \{x \mapsto 106, y \mapsto 106\} \\
\sigma_{215} &= \{x \mapsto 106, y \mapsto 107\} \\
\sigma_{216} &= \{x \mapsto 107, y \mapsto 107\} \\
\sigma_{217} &= \{x \mapsto 107, y \mapsto 108\} \\
\sigma_{218} &= \{x \mapsto 108, y \mapsto 108\} \\
\sigma_{219} &= \{x \mapsto 108, y \mapsto 109\} \\
\sigma_{220} &= \{x \mapsto 109, y \mapsto 109\} \\
\sigma_{221} &= \{x \mapsto 109, y \mapsto 110\} \\
\sigma_{222} &= \{x \mapsto 110, y \mapsto 110\} \\
\sigma_{223} &= \{x \mapsto 110, y \mapsto 111\} \\
\sigma_{224} &= \{x \mapsto 111, y \mapsto 111\} \\
\sigma_{225} &= \{x \mapsto 111, y \mapsto 112\} \\
\sigma_{226} &= \{x \mapsto 112, y \mapsto 112\} \\
\sigma_{227} &= \{x \mapsto 112, y \mapsto 113\} \\
\sigma_{228} &= \{x \mapsto 113, y \mapsto 113\} \\
\sigma_{229} &= \{x \mapsto 113, y \mapsto 114\} \\
\sigma_{230} &= \{x \mapsto 114, y \mapsto 114\} \\
\sigma_{231} &= \{x \mapsto 114, y \mapsto 115\} \\
\sigma_{232} &= \{x \mapsto 115, y \mapsto 115\} \\
\sigma_{233} &= \{x \mapsto 115, y \mapsto 116\} \\
\sigma_{234} &= \{x \mapsto 116, y \mapsto 116\} \\
\sigma_{235} &= \{x \mapsto 116, y \mapsto 117\} \\
\sigma_{236} &= \{x \mapsto 117, y \mapsto 117\} \\
\sigma_{237} &= \{x \mapsto 117, y \mapsto 118\} \\
\sigma_{238} &= \{x \mapsto 118, y \mapsto 118\} \\
\sigma_{239} &= \{x \mapsto 118, y \mapsto 119\} \\
\sigma_{240} &= \{x \mapsto 119, y \mapsto 119\} \\
\sigma_{241} &= \{x \mapsto 119, y \mapsto 120\} \\
\sigma_{242} &= \{x \mapsto 120, y \mapsto 120\} \\
\sigma_{243} &= \{x \mapsto 120, y \mapsto 121\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{316} &= \{x \mapsto 157, y \mapsto 157\} \\
\sigma_{317} &= \{x \mapsto 157, y \mapsto 158\} \\
\sigma_{318} &= \{x \mapsto 158, y \mapsto 158\} \\
\sigma_{319} &= \{x \mapsto 158, y \mapsto 159\} \\
\sigma_{320} &= \{x \mapsto 159, y \mapsto 159\} \\
\sigma_{321} &= \{x \mapsto 159, y \mapsto 160\} \\
\sigma_{322} &= \{x \mapsto 160, y \mapsto 160\} \\
\sigma_{323} &= \{x \mapsto 160, y \mapsto 161\} \\
\sigma_{324} &= \{x \mapsto 161, y \mapsto 161\} \\
\sigma_{325} &= \{x \mapsto 161, y \mapsto 162\} \\
\sigma_{326} &= \{x \mapsto 162, y \mapsto 162\} \\
\sigma_{327} &= \{x \mapsto 162, y \mapsto 163\} \\
\sigma_{328} &= \{x \mapsto 163, y \mapsto 163\} \\
\sigma_{329} &= \{x \mapsto 163, y \mapsto 164\} \\
\sigma_{330} &= \{x \mapsto 164, y \mapsto 164\} \\
\sigma_{331} &= \{x \mapsto 164, y \mapsto 165\} \\
\sigma_{332} &= \{x \mapsto 165, y \mapsto 165\} \\
\sigma_{333} &= \{x \mapsto 165, y \mapsto 166\} \\
\sigma_{334} &= \{x \mapsto 166, y \mapsto 166\} \\
\sigma_{335} &= \{x \mapsto 166, y \mapsto 167\} \\
\sigma_{336} &= \{x \mapsto 167, y \mapsto 167\} \\
\sigma_{337} &= \{x \mapsto 167, y \mapsto 168\} \\
\sigma_{338} &= \{x \mapsto 168, y \mapsto 168\} \\
\sigma_{339} &= \{x \mapsto 168, y \mapsto 169\} \\
\sigma_{340} &= \{x \mapsto 169, y \mapsto 169\} \\
\sigma_{341} &= \{x \mapsto 169, y \mapsto 170\} \\
\sigma_{342} &= \{x \mapsto 170, y \mapsto 170\} \\
\sigma_{343} &= \{x \mapsto 170, y \mapsto 171\} \\
\sigma_{344} &= \{x \mapsto 171, y \mapsto 171\} \\
\sigma_{345} &= \{x \mapsto 171, y \mapsto 172\} \\
\sigma_{346} &= \{x \mapsto 172, y \mapsto 172\} \\
\sigma_{347} &= \{x \mapsto 172, y \mapsto 173\} \\
\sigma_{348} &= \{x \mapsto 173, y \mapsto 173\} \\
\sigma_{349} &= \{x \mapsto 173, y \mapsto 174\} \\
\sigma_{350} &= \{x \mapsto 174, y \mapsto 174\} \\
\sigma_{351} &= \{x \mapsto 174, y \mapsto 175\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{424} &= \{x \mapsto 211, y \mapsto 211\} \\
\sigma_{425} &= \{x \mapsto 211, y \mapsto 212\} \\
\sigma_{426} &= \{x \mapsto 212, y \mapsto 212\} \\
\sigma_{427} &= \{x \mapsto 212, y \mapsto 213\} \\
\sigma_{428} &= \{x \mapsto 213, y \mapsto 213\} \\
\sigma_{429} &= \{x \mapsto 213, y \mapsto 214\} \\
\sigma_{430} &= \{x \mapsto 214, y \mapsto 214\} \\
\sigma_{431} &= \{x \mapsto 214, y \mapsto 215\} \\
\sigma_{432} &= \{x \mapsto 215, y \mapsto 215\} \\
\sigma_{433} &= \{x \mapsto 215, y \mapsto 216\} \\
\sigma_{434} &= \{x \mapsto 216, y \mapsto 216\} \\
\sigma_{435} &= \{x \mapsto 216, y \mapsto 217\} \\
\sigma_{436} &= \{x \mapsto 217, y \mapsto 217\} \\
\sigma_{437} &= \{x \mapsto 217, y \mapsto 218\} \\
\sigma_{438} &= \{x \mapsto 218, y \mapsto 218\} \\
\sigma_{439} &= \{x \mapsto 218, y \mapsto 219\} \\
\sigma_{440} &= \{x \mapsto 219, y \mapsto 219\} \\
\sigma_{441} &= \{x \mapsto 219, y \mapsto 220\} \\
\sigma_{442} &= \{x \mapsto 220, y \mapsto 220\} \\
\sigma_{443} &= \{x \mapsto 220, y \mapsto 221\} \\
\sigma_{444} &= \{x \mapsto 221, y \mapsto 221\} \\
\sigma_{445} &= \{x \mapsto 221, y \mapsto 222\} \\
\sigma_{446} &= \{x \mapsto 222, y \mapsto 222\} \\
\sigma_{447} &= \{x \mapsto 222, y \mapsto 223\} \\
\sigma_{448} &= \{x \mapsto 223, y \mapsto 223\} \\
\sigma_{449} &= \{x \mapsto 223, y \mapsto 224\} \\
\sigma_{450} &= \{x \mapsto 224, y \mapsto 224\} \\
\sigma_{451} &= \{x \mapsto 224, y \mapsto 225\} \\
\sigma_{452} &= \{x \mapsto 225, y \mapsto 225\} \\
\sigma_{453} &= \{x \mapsto 225, y \mapsto 226\} \\
\sigma_{454} &= \{x \mapsto 226, y \mapsto 226\} \\
\sigma_{455} &= \{x \mapsto 226, y \mapsto 227\} \\
\sigma_{456} &= \{x \mapsto 227, y \mapsto 227\} \\
\sigma_{457} &= \{x \mapsto 227, y \mapsto 228\} \\
\sigma_{458} &= \{x \mapsto 228, y \mapsto 228\} \\
\sigma_{459} &= \{x \mapsto 228, y \mapsto 229\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{532} &= \{x \mapsto 265, y \mapsto 265\} \\
\sigma_{533} &= \{x \mapsto 265, y \mapsto 266\} \\
\sigma_{534} &= \{x \mapsto 266, y \mapsto 266\} \\
\sigma_{535} &= \{x \mapsto 266, y \mapsto 267\} \\
\sigma_{536} &= \{x \mapsto 267, y \mapsto 267\} \\
\sigma_{537} &= \{x \mapsto 267, y \mapsto 268\} \\
\sigma_{538} &= \{x \mapsto 268, y \mapsto 268\} \\
\sigma_{539} &= \{x \mapsto 268, y \mapsto 269\} \\
\sigma_{540} &= \{x \mapsto 269, y \mapsto 269\} \\
\sigma_{541} &= \{x \mapsto 269, y \mapsto 270\} \\
\sigma_{542} &= \{x \mapsto 270, y \mapsto 270\} \\
\sigma_{543} &= \{x \mapsto 270, y \mapsto 271\} \\
\sigma_{544} &= \{x \mapsto 271, y \mapsto 271\} \\
\sigma_{545} &= \{x \mapsto 271, y \mapsto 272\} \\
\sigma_{546} &= \{x \mapsto 272, y \mapsto 272\} \\
\sigma_{547} &= \{x \mapsto 272, y \mapsto 273\} \\
\sigma_{548} &= \{x \mapsto 273, y \mapsto 273\} \\
\sigma_{549} &= \{x \mapsto 273, y \mapsto 274\} \\
\sigma_{550} &= \{x \mapsto 274, y \mapsto 274\} \\
\sigma_{551} &= \{x \mapsto 274, y \mapsto 275\} \\
\sigma_{552} &= \{x \mapsto 275, y \mapsto 275\} \\
\sigma_{553} &= \{x \mapsto 275, y \mapsto 276\} \\
\sigma_{554} &= \{x \mapsto 276, y \mapsto 276\} \\
\sigma_{555} &= \{x \mapsto 276, y \mapsto 277\} \\
\sigma_{556} &= \{x \mapsto 277, y \mapsto 277\} \\
\sigma_{557} &= \{x \mapsto 277, y \mapsto 278\} \\
\sigma_{558} &= \{x \mapsto 278, y \mapsto 278\} \\
\sigma_{559} &= \{x \mapsto 278, y \mapsto 279\} \\
\sigma_{560} &= \{x \mapsto 279, y \mapsto 279\} \\
\sigma_{561} &= \{x \mapsto 279, y \mapsto 280\} \\
\sigma_{562} &= \{x \mapsto 280, y \mapsto 280\} \\
\sigma_{563} &= \{x \mapsto 280, y \mapsto 281\} \\
\sigma_{564} &= \{x \mapsto 281, y \mapsto 281\} \\
\sigma_{565} &= \{x \mapsto 281, y \mapsto 282\} \\
\sigma_{566} &= \{x \mapsto 282, y \mapsto 282\} \\
\sigma_{567} &= \{x \mapsto 282, y \mapsto 283\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{640} &= \{x \mapsto 319, y \mapsto 319\} \\
\sigma_{641} &= \{x \mapsto 319, y \mapsto 320\} \\
\sigma_{642} &= \{x \mapsto 320, y \mapsto 320\} \\
\sigma_{643} &= \{x \mapsto 320, y \mapsto 321\} \\
\sigma_{644} &= \{x \mapsto 321, y \mapsto 321\} \\
\sigma_{645} &= \{x \mapsto 321, y \mapsto 322\} \\
\sigma_{646} &= \{x \mapsto 322, y \mapsto 322\} \\
\sigma_{647} &= \{x \mapsto 322, y \mapsto 323\} \\
\sigma_{648} &= \{x \mapsto 323, y \mapsto 323\} \\
\sigma_{649} &= \{x \mapsto 323, y \mapsto 324\} \\
\sigma_{650} &= \{x \mapsto 324, y \mapsto 324\} \\
\sigma_{651} &= \{x \mapsto 324, y \mapsto 325\} \\
\sigma_{652} &= \{x \mapsto 325, y \mapsto 325\} \\
\sigma_{653} &= \{x \mapsto 325, y \mapsto 326\} \\
\sigma_{654} &= \{x \mapsto 326, y \mapsto 326\} \\
\sigma_{655} &= \{x \mapsto 326, y \mapsto 327\} \\
\sigma_{656} &= \{x \mapsto 327, y \mapsto 327\} \\
\sigma_{657} &= \{x \mapsto 327, y \mapsto 328\} \\
\sigma_{658} &= \{x \mapsto 328, y \mapsto 328\} \\
\sigma_{659} &= \{x \mapsto 328, y \mapsto 329\} \\
\sigma_{660} &= \{x \mapsto 329, y \mapsto 329\} \\
\sigma_{661} &= \{x \mapsto 329, y \mapsto 330\} \\
\sigma_{662} &= \{x \mapsto 330, y \mapsto 330\} \\
\sigma_{663} &= \{x \mapsto 330, y \mapsto 331\} \\
\sigma_{664} &= \{x \mapsto 331, y \mapsto 331\} \\
\sigma_{665} &= \{x \mapsto 331, y \mapsto 332\} \\
\sigma_{666} &= \{x \mapsto 332, y \mapsto 332\} \\
\sigma_{667} &= \{x \mapsto 332, y \mapsto 333\} \\
\sigma_{668} &= \{x \mapsto 333, y \mapsto 333\} \\
\sigma_{669} &= \{x \mapsto 333, y \mapsto 334\} \\
\sigma_{670} &= \{x \mapsto 334, y \mapsto 334\} \\
\sigma_{671} &= \{x \mapsto 334, y \mapsto 335\} \\
\sigma_{672} &= \{x \mapsto 335, y \mapsto 335\} \\
\sigma_{673} &= \{x \mapsto 335, y \mapsto 336\} \\
\sigma_{674} &= \{x \mapsto 336, y \mapsto 336\} \\
\sigma_{675} &= \{x \mapsto 336, y \mapsto 337\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{748} &= \{x \mapsto 373, y \mapsto 373\} \\
\sigma_{749} &= \{x \mapsto 373, y \mapsto 374\} \\
\sigma_{750} &= \{x \mapsto 374, y \mapsto 374\} \\
\sigma_{751} &= \{x \mapsto 374, y \mapsto 375\} \\
\sigma_{752} &= \{x \mapsto 375, y \mapsto 375\} \\
\sigma_{753} &= \{x \mapsto 375, y \mapsto 376\} \\
\sigma_{754} &= \{x \mapsto 376, y \mapsto 376\} \\
\sigma_{755} &= \{x \mapsto 376, y \mapsto 377\} \\
\sigma_{756} &= \{x \mapsto 377, y \mapsto 377\} \\
\sigma_{757} &= \{x \mapsto 377, y \mapsto 378\} \\
\sigma_{758} &= \{x \mapsto 378, y \mapsto 378\} \\
\sigma_{759} &= \{x \mapsto 378, y \mapsto 379\} \\
\sigma_{760} &= \{x \mapsto 379, y \mapsto 379\} \\
\sigma_{761} &= \{x \mapsto 379, y \mapsto 380\} \\
\sigma_{762} &= \{x \mapsto 380, y \mapsto 380\} \\
\sigma_{763} &= \{x \mapsto 380, y \mapsto 381\} \\
\sigma_{764} &= \{x \mapsto 381, y \mapsto 381\} \\
\sigma_{765} &= \{x \mapsto 381, y \mapsto 382\} \\
\sigma_{766} &= \{x \mapsto 382, y \mapsto 382\} \\
\sigma_{767} &= \{x \mapsto 382, y \mapsto 383\} \\
\sigma_{768} &= \{x \mapsto 383, y \mapsto 383\} \\
\sigma_{769} &= \{x \mapsto 383, y \mapsto 384\} \\
\sigma_{770} &= \{x \mapsto 384, y \mapsto 384\} \\
\sigma_{771} &= \{x \mapsto 384, y \mapsto 385\} \\
\sigma_{772} &= \{x \mapsto 385, y \mapsto 385\} \\
\sigma_{773} &= \{x \mapsto 385, y \mapsto 386\} \\
\sigma_{774} &= \{x \mapsto 386, y \mapsto 386\} \\
\sigma_{775} &= \{x \mapsto 386, y \mapsto 387\} \\
\sigma_{776} &= \{x \mapsto 387, y \mapsto 387\} \\
\sigma_{777} &= \{x \mapsto 387, y \mapsto 388\} \\
\sigma_{778} &= \{x \mapsto 388, y \mapsto 388\} \\
\sigma_{779} &= \{x \mapsto 388, y \mapsto 389\} \\
\sigma_{780} &= \{x \mapsto 389, y \mapsto 389\} \\
\sigma_{781} &= \{x \mapsto 389, y \mapsto 390\} \\
\sigma_{782} &= \{x \mapsto 390, y \mapsto 390\} \\
\sigma_{783} &= \{x \mapsto 390, y \mapsto 391\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{856} &= \{x \mapsto 427, y \mapsto 427\} \\
\sigma_{857} &= \{x \mapsto 427, y \mapsto 428\} \\
\sigma_{858} &= \{x \mapsto 428, y \mapsto 428\} \\
\sigma_{859} &= \{x \mapsto 428, y \mapsto 429\} \\
\sigma_{860} &= \{x \mapsto 429, y \mapsto 429\} \\
\sigma_{861} &= \{x \mapsto 429, y \mapsto 430\} \\
\sigma_{862} &= \{x \mapsto 430, y \mapsto 430\} \\
\sigma_{863} &= \{x \mapsto 430, y \mapsto 431\} \\
\sigma_{864} &= \{x \mapsto 431, y \mapsto 431\} \\
\sigma_{865} &= \{x \mapsto 431, y \mapsto 432\} \\
\sigma_{866} &= \{x \mapsto 432, y \mapsto 432\} \\
\sigma_{867} &= \{x \mapsto 432, y \mapsto 433\} \\
\sigma_{868} &= \{x \mapsto 433, y \mapsto 433\} \\
\sigma_{869} &= \{x \mapsto 433, y \mapsto 434\} \\
\sigma_{870} &= \{x \mapsto 434, y \mapsto 434\} \\
\sigma_{871} &= \{x \mapsto 434, y \mapsto 435\} \\
\sigma_{872} &= \{x \mapsto 435, y \mapsto 435\} \\
\sigma_{873} &= \{x \mapsto 435, y \mapsto 436\} \\
\sigma_{874} &= \{x \mapsto 436, y \mapsto 436\} \\
\sigma_{875} &= \{x \mapsto 436, y \mapsto 437\} \\
\sigma_{876} &= \{x \mapsto 437, y \mapsto 437\} \\
\sigma_{877} &= \{x \mapsto 437, y \mapsto 438\} \\
\sigma_{878} &= \{x \mapsto 438, y \mapsto 438\} \\
\sigma_{879} &= \{x \mapsto 438, y \mapsto 439\} \\
\sigma_{880} &= \{x \mapsto 439, y \mapsto 439\} \\
\sigma_{881} &= \{x \mapsto 439, y \mapsto 440\} \\
\sigma_{882} &= \{x \mapsto 440, y \mapsto 440\} \\
\sigma_{883} &= \{x \mapsto 440, y \mapsto 441\} \\
\sigma_{884} &= \{x \mapsto 441, y \mapsto 441\} \\
\sigma_{885} &= \{x \mapsto 441, y \mapsto 442\} \\
\sigma_{886} &= \{x \mapsto 442, y \mapsto 442\} \\
\sigma_{887} &= \{x \mapsto 442, y \mapsto 443\} \\
\sigma_{888} &= \{x \mapsto 443, y \mapsto 443\} \\
\sigma_{889} &= \{x \mapsto 443, y \mapsto 444\} \\
\sigma_{890} &= \{x \mapsto 444, y \mapsto 444\} \\
\sigma_{891} &= \{x \mapsto 444, y \mapsto 445\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{964} &= \{x \mapsto 481, y \mapsto 481\} \\
\sigma_{965} &= \{x \mapsto 481, y \mapsto 482\} \\
\sigma_{966} &= \{x \mapsto 482, y \mapsto 482\} \\
\sigma_{967} &= \{x \mapsto 482, y \mapsto 483\} \\
\sigma_{968} &= \{x \mapsto 483, y \mapsto 483\} \\
\sigma_{969} &= \{x \mapsto 483, y \mapsto 484\} \\
\sigma_{970} &= \{x \mapsto 484, y \mapsto 484\} \\
\sigma_{971} &= \{x \mapsto 484, y \mapsto 485\} \\
\sigma_{972} &= \{x \mapsto 485, y \mapsto 485\} \\
\sigma_{973} &= \{x \mapsto 485, y \mapsto 486\} \\
\sigma_{974} &= \{x \mapsto 486, y \mapsto 486\} \\
\sigma_{975} &= \{x \mapsto 486, y \mapsto 487\} \\
\sigma_{976} &= \{x \mapsto 487, y \mapsto 487\} \\
\sigma_{977} &= \{x \mapsto 487, y \mapsto 488\} \\
\sigma_{978} &= \{x \mapsto 488, y \mapsto 488\} \\
\sigma_{979} &= \{x \mapsto 488, y \mapsto 489\} \\
\sigma_{980} &= \{x \mapsto 489, y \mapsto 489\} \\
\sigma_{981} &= \{x \mapsto 489, y \mapsto 490\} \\
\sigma_{982} &= \{x \mapsto 490, y \mapsto 490\} \\
\sigma_{983} &= \{x \mapsto 490, y \mapsto 491\} \\
\sigma_{984} &= \{x \mapsto 491, y \mapsto 491\} \\
\sigma_{985} &= \{x \mapsto 491, y \mapsto 492\} \\
\sigma_{986} &= \{x \mapsto 492, y \mapsto 492\} \\
\sigma_{987} &= \{x \mapsto 492, y \mapsto 493\} \\
\sigma_{988} &= \{x \mapsto 493, y \mapsto 493\} \\
\sigma_{989} &= \{x \mapsto 493, y \mapsto 494\} \\
\sigma_{990} &= \{x \mapsto 494, y \mapsto 494\} \\
\sigma_{991} &= \{x \mapsto 494, y \mapsto 495\} \\
\sigma_{992} &= \{x \mapsto 495, y \mapsto 495\} \\
\sigma_{993} &= \{x \mapsto 495, y \mapsto 496\} \\
\sigma_{994} &= \{x \mapsto 496, y \mapsto 496\} \\
\sigma_{995} &= \{x \mapsto 496, y \mapsto 497\} \\
\sigma_{996} &= \{x \mapsto 497, y \mapsto 497\} \\
\sigma_{997} &= \{x \mapsto 497, y \mapsto 498\} \\
\sigma_{998} &= \{x \mapsto 498, y \mapsto 498\} \\
\sigma_{999} &= \{x \mapsto 498, y \mapsto 499\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1072} &= \{x \mapsto 535, y \mapsto 535\} \\
\sigma_{1073} &= \{x \mapsto 535, y \mapsto 536\} \\
\sigma_{1074} &= \{x \mapsto 536, y \mapsto 536\} \\
\sigma_{1075} &= \{x \mapsto 536, y \mapsto 537\} \\
\sigma_{1076} &= \{x \mapsto 537, y \mapsto 537\} \\
\sigma_{1077} &= \{x \mapsto 537, y \mapsto 538\} \\
\sigma_{1078} &= \{x \mapsto 538, y \mapsto 538\} \\
\sigma_{1079} &= \{x \mapsto 538, y \mapsto 539\} \\
\sigma_{1080} &= \{x \mapsto 539, y \mapsto 539\} \\
\sigma_{1081} &= \{x \mapsto 539, y \mapsto 540\} \\
\sigma_{1082} &= \{x \mapsto 540, y \mapsto 540\} \\
\sigma_{1083} &= \{x \mapsto 540, y \mapsto 541\} \\
\sigma_{1084} &= \{x \mapsto 541, y \mapsto 541\} \\
\sigma_{1085} &= \{x \mapsto 541, y \mapsto 542\} \\
\sigma_{1086} &= \{x \mapsto 542, y \mapsto 542\} \\
\sigma_{1087} &= \{x \mapsto 542, y \mapsto 543\} \\
\sigma_{1088} &= \{x \mapsto 543, y \mapsto 543\} \\
\sigma_{1089} &= \{x \mapsto 543, y \mapsto 544\} \\
\sigma_{1090} &= \{x \mapsto 544, y \mapsto 544\} \\
\sigma_{1091} &= \{x \mapsto 544, y \mapsto 545\} \\
\sigma_{1092} &= \{x \mapsto 545, y \mapsto 545\} \\
\sigma_{1093} &= \{x \mapsto 545, y \mapsto 546\} \\
\sigma_{1094} &= \{x \mapsto 546, y \mapsto 546\} \\
\sigma_{1095} &= \{x \mapsto 546, y \mapsto 547\} \\
\sigma_{1096} &= \{x \mapsto 547, y \mapsto 547\} \\
\sigma_{1097} &= \{x \mapsto 547, y \mapsto 548\} \\
\sigma_{1098} &= \{x \mapsto 548, y \mapsto 548\} \\
\sigma_{1099} &= \{x \mapsto 548, y \mapsto 549\} \\
\sigma_{1100} &= \{x \mapsto 549, y \mapsto 549\} \\
\sigma_{1101} &= \{x \mapsto 549, y \mapsto 550\} \\
\sigma_{1102} &= \{x \mapsto 550, y \mapsto 550\} \\
\sigma_{1103} &= \{x \mapsto 550, y \mapsto 551\} \\
\sigma_{1104} &= \{x \mapsto 551, y \mapsto 551\} \\
\sigma_{1105} &= \{x \mapsto 551, y \mapsto 552\} \\
\sigma_{1106} &= \{x \mapsto 552, y \mapsto 552\} \\
\sigma_{1107} &= \{x \mapsto 552, y \mapsto 553\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1288} &= \{x \mapsto 643, y \mapsto 643\} \\
\sigma_{1289} &= \{x \mapsto 643, y \mapsto 644\} \\
\sigma_{1290} &= \{x \mapsto 644, y \mapsto 644\} \\
\sigma_{1291} &= \{x \mapsto 644, y \mapsto 645\} \\
\sigma_{1292} &= \{x \mapsto 645, y \mapsto 645\} \\
\sigma_{1293} &= \{x \mapsto 645, y \mapsto 646\} \\
\sigma_{1294} &= \{x \mapsto 646, y \mapsto 646\} \\
\sigma_{1295} &= \{x \mapsto 646, y \mapsto 647\} \\
\sigma_{1296} &= \{x \mapsto 647, y \mapsto 647\} \\
\sigma_{1297} &= \{x \mapsto 647, y \mapsto 648\} \\
\sigma_{1298} &= \{x \mapsto 648, y \mapsto 648\} \\
\sigma_{1299} &= \{x \mapsto 648, y \mapsto 649\} \\
\sigma_{1300} &= \{x \mapsto 649, y \mapsto 649\} \\
\sigma_{1301} &= \{x \mapsto 649, y \mapsto 650\} \\
\sigma_{1302} &= \{x \mapsto 650, y \mapsto 650\} \\
\sigma_{1303} &= \{x \mapsto 650, y \mapsto 651\} \\
\sigma_{1304} &= \{x \mapsto 651, y \mapsto 651\} \\
\sigma_{1305} &= \{x \mapsto 651, y \mapsto 652\} \\
\sigma_{1306} &= \{x \mapsto 652, y \mapsto 652\} \\
\sigma_{1307} &= \{x \mapsto 652, y \mapsto 653\} \\
\sigma_{1308} &= \{x \mapsto 653, y \mapsto 653\} \\
\sigma_{1309} &= \{x \mapsto 653, y \mapsto 654\} \\
\sigma_{1310} &= \{x \mapsto 654, y \mapsto 654\} \\
\sigma_{1311} &= \{x \mapsto 654, y \mapsto 655\} \\
\sigma_{1312} &= \{x \mapsto 655, y \mapsto 655\} \\
\sigma_{1313} &= \{x \mapsto 655, y \mapsto 656\} \\
\sigma_{1314} &= \{x \mapsto 656, y \mapsto 656\} \\
\sigma_{1315} &= \{x \mapsto 656, y \mapsto 657\} \\
\sigma_{1316} &= \{x \mapsto 657, y \mapsto 657\} \\
\sigma_{1317} &= \{x \mapsto 657, y \mapsto 658\} \\
\sigma_{1318} &= \{x \mapsto 658, y \mapsto 658\} \\
\sigma_{1319} &= \{x \mapsto 658, y \mapsto 659\} \\
\sigma_{1320} &= \{x \mapsto 659, y \mapsto 659\} \\
\sigma_{1321} &= \{x \mapsto 659, y \mapsto 660\} \\
\sigma_{1322} &= \{x \mapsto 660, y \mapsto 660\} \\
\sigma_{1323} &= \{x \mapsto 660, y \mapsto 661\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1972} &= \{x \mapsto 985, y \mapsto 985\} \\
\sigma_{1973} &= \{x \mapsto 985, y \mapsto 986\} \\
\sigma_{1974} &= \{x \mapsto 986, y \mapsto 986\} \\
\sigma_{1975} &= \{x \mapsto 986, y \mapsto 987\} \\
\sigma_{1976} &= \{x \mapsto 987, y \mapsto 987\} \\
\sigma_{1977} &= \{x \mapsto 987, y \mapsto 988\} \\
\sigma_{1978} &= \{x \mapsto 988, y \mapsto 988\} \\
\sigma_{1979} &= \{x \mapsto 988, y \mapsto 989\} \\
\sigma_{1980} &= \{x \mapsto 989, y \mapsto 989\} \\
\sigma_{1981} &= \{x \mapsto 989, y \mapsto 990\} \\
\sigma_{1982} &= \{x \mapsto 990, y \mapsto 990\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1983} &= \{x \mapsto 990, y \mapsto 991\} \\
\sigma_{1984} &= \{x \mapsto 991, y \mapsto 991\} \\
\sigma_{1985} &= \{x \mapsto 991, y \mapsto 992\} \\
\sigma_{1986} &= \{x \mapsto 992, y \mapsto 992\} \\
\sigma_{1987} &= \{x \mapsto 992, y \mapsto 993\} \\
\sigma_{1988} &= \{x \mapsto 993, y \mapsto 993\} \\
\sigma_{1989} &= \{x \mapsto 993, y \mapsto 994\} \\
\sigma_{1990} &= \{x \mapsto 994, y \mapsto 994\} \\
\sigma_{1991} &= \{x \mapsto 994, y \mapsto 995\} \\
\sigma_{1992} &= \{x \mapsto 995, y \mapsto 995\} \\
\sigma_{1993} &= \{x \mapsto 995, y \mapsto 996\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1994} &= \{x \mapsto 996, y \mapsto 996\} \\
\sigma_{1995} &= \{x \mapsto 996, y \mapsto 997\} \\
\sigma_{1996} &= \{x \mapsto 997, y \mapsto 997\} \\
\sigma_{1997} &= \{x \mapsto 997, y \mapsto 998\} \\
\sigma_{1998} &= \{x \mapsto 998, y \mapsto 998\} \\
\sigma_{1999} &= \{x \mapsto 998, y \mapsto 999\} \\
\sigma_{2000} &= \{x \mapsto 999, y \mapsto 999\} \\
\sigma_{2001} &= \{x \mapsto 999, y \mapsto 1000\} \\
\sigma_{2002} &= \{x \mapsto 1000, y \mapsto 1000\}
\end{aligned}$$

2. Semántica Operacional Estructurada (SOS - Small-Step)

$$\mathcal{T}_1 = \frac{\langle y := 0, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \emptyset \rangle \Rightarrow \langle S_2, \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \langle S_2, \sigma_1 \rangle \Rightarrow \langle S_3, \sigma_1 \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \langle S_3, \sigma_1 \rangle \Rightarrow \langle S_4, \sigma_1 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_5 = \frac{\langle x := 0, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_1 \rangle \Rightarrow \langle S_6, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_4 = \frac{\mathcal{T}_5}{\langle S_4, \sigma_1 \rangle \Rightarrow \langle S_7, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_6 = \frac{\langle S_6, \sigma_2 \rangle \Rightarrow \langle S_8, \sigma_2 \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_2 \rangle \Rightarrow \langle S_9, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_8 = \langle S_8, \sigma_2 \rangle \Rightarrow \langle y := y + 1, \sigma_2 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_7 = \frac{\mathcal{T}_8}{\langle S_9, \sigma_2 \rangle \Rightarrow \langle S_{10}, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_9 = \frac{\langle y := y + 1, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_2 \rangle \Rightarrow \langle S_{11}, \sigma_3 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{10} = \langle S_{11}, \sigma_3 \rangle \Rightarrow \langle S_{12}, \sigma_3 \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{11} = \langle S_{12}, \sigma_3 \rangle \Rightarrow \langle S_{13}, \sigma_3 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{13} = \frac{\langle x := x + 1, \sigma_3 \rangle \Rightarrow \sigma_4 [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_3 \rangle \Rightarrow \langle S_6, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{12} = \frac{\mathcal{T}_{13}}{\langle S_{13}, \sigma_3 \rangle \Rightarrow \langle S_7, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{14} = \frac{\langle S_6, \sigma_4 \rangle \Rightarrow \langle S_8, \sigma_4 \rangle [\text{assert}_{\text{sos}}]}{\langle S_{15}, \sigma_4 \rangle \Rightarrow \langle S_9, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{16} = \langle S_8, \sigma_4 \rangle \Rightarrow \langle y := y + 1, \sigma_4 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{15} = \frac{\mathcal{T}_{16}}{\langle S_9, \sigma_4 \rangle \Rightarrow \langle S_{10}, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{17} = \frac{\langle y := y + 1, \sigma_4 \rangle \Rightarrow \sigma_5 [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_4 \rangle \Rightarrow \langle S_{11}, \sigma_5 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{18} = \langle S_{11}, \sigma_5 \rangle \Rightarrow \langle S_{12}, \sigma_5 \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{19} = \langle S_{12}, \sigma_5 \rangle \Rightarrow \langle S_{13}, \sigma_5 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{21} = \frac{\langle x := x + 1, \sigma_5 \rangle \Rightarrow \sigma_6 [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_5 \rangle \Rightarrow \langle S_6, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{20} = \frac{\mathcal{T}_{21}}{\langle S_{13}, \sigma_5 \rangle \Rightarrow \langle S_7, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{22} = \frac{\langle S_6, \sigma_6 \rangle \Rightarrow \langle S_8, \sigma_6 \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_6 \rangle \Rightarrow \langle S_9, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{24} = \langle S_8, \sigma_6 \rangle \Rightarrow \langle y := y + 1, \sigma_6 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{23} = \frac{\mathcal{T}_{24}}{\langle S_9, \sigma_6 \rangle \Rightarrow \langle S_{10}, \sigma_6 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{25} = \frac{\langle y := y + 1, \sigma_6 \rangle \Rightarrow \sigma_7 [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_6 \rangle \Rightarrow \langle S_{11}, \sigma_7 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{26} = \langle S_{11}, \sigma_7 \rangle \Rightarrow \langle S_{12}, \sigma_7 \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{27} = \langle S_{12}, \sigma_7 \rangle \Rightarrow \langle S_{13}, \sigma_7 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{29} = \frac{\langle x := x + 1, \sigma_7 \rangle \Rightarrow \sigma_8 [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_7 \rangle \Rightarrow \langle S_6, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{28} = \frac{\mathcal{T}_{29}}{\langle S_{13}, \sigma_7 \rangle \Rightarrow \langle S_7, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{30} = \frac{\langle S_6, \sigma_8 \rangle \Rightarrow \langle S_8, \sigma_8 \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_8 \rangle \Rightarrow \langle S_9, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{32} = \langle S_8, \sigma_8 \rangle \Rightarrow \langle y := y + 1, \sigma_8 \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{31} = \frac{\mathcal{T}_{32}}{\langle S_9, \sigma_8 \rangle \Rightarrow \langle S_{10}, \sigma_8 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{33} = \frac{\langle y := y + 1, \sigma_8 \rangle \Rightarrow \sigma_9 \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_8 \rangle \Rightarrow \langle S_{11}, \sigma_9 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{34} = \langle S_{11}, \sigma_9 \rangle \Rightarrow \langle S_{12}, \sigma_9 \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{35} = \langle S_{12}, \sigma_9 \rangle \Rightarrow \langle S_{13}, \sigma_9 \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{37} = \frac{\langle x := x + 1, \sigma_9 \rangle \Rightarrow \sigma_{10} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_9 \rangle \Rightarrow \langle S_6, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{36} = \frac{\mathcal{T}_{37}}{\langle S_{13}, \sigma_9 \rangle \Rightarrow \langle S_7, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{38} = \frac{\langle S_6, \sigma_{10} \rangle \Rightarrow \langle S_8, \sigma_{10} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_{15}, \sigma_{10} \rangle \Rightarrow \langle S_9, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{40} = \langle S_8, \sigma_{10} \rangle \Rightarrow \langle y := y + 1, \sigma_{10} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{39} = \frac{\mathcal{T}_{40}}{\langle S_9, \sigma_{10} \rangle \Rightarrow \langle S_{10}, \sigma_{10} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{41} = \frac{\langle y := y + 1, \sigma_{10} \rangle \Rightarrow \sigma_{11} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{10} \rangle \Rightarrow \langle S_{11}, \sigma_{11} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{42} = \langle S_{11}, \sigma_{11} \rangle \Rightarrow \langle S_{12}, \sigma_{11} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{43} = \langle S_{12}, \sigma_{11} \rangle \Rightarrow \langle S_{13}, \sigma_{11} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{45} = \frac{\langle x := x + 1, \sigma_{11} \rangle \Rightarrow \sigma_{12} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{11} \rangle \Rightarrow \langle S_6, \sigma_{12} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{44} = \frac{\mathcal{T}_{45}}{\langle S_{13}, \sigma_{11} \rangle \Rightarrow \langle S_7, \sigma_{12} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{46} = \frac{\langle S_6, \sigma_{12} \rangle \Rightarrow \langle S_8, \sigma_{12} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{12} \rangle \Rightarrow \langle S_9, \sigma_{12} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{48} = \langle S_8, \sigma_{12} \rangle \Rightarrow \langle y := y + 1, \sigma_{12} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{47} = \frac{\mathcal{T}_{48}}{\langle S_9, \sigma_{12} \rangle \Rightarrow \langle S_{10}, \sigma_{12} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{49} = \frac{\langle y := y + 1, \sigma_{12} \rangle \Rightarrow \sigma_{13} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{12} \rangle \Rightarrow \langle S_{11}, \sigma_{13} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{50} = \langle S_{11}, \sigma_{13} \rangle \Rightarrow \langle S_{12}, \sigma_{13} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{51} = \langle S_{12}, \sigma_{13} \rangle \Rightarrow \langle S_{13}, \sigma_{13} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{53} = \frac{\langle x := x + 1, \sigma_{13} \rangle \Rightarrow \sigma_{14} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{13} \rangle \Rightarrow \langle S_6, \sigma_{14} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{52} = \frac{\mathcal{T}_{53}}{\langle S_{13}, \sigma_{13} \rangle \Rightarrow \langle S_7, \sigma_{14} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{54} = \frac{\langle S_6, \sigma_{14} \rangle \Rightarrow \langle S_8, \sigma_{14} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{14} \rangle \Rightarrow \langle S_9, \sigma_{14} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{56} = \langle S_8, \sigma_{14} \rangle \Rightarrow \langle y := y + 1, \sigma_{14} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{55} = \frac{\mathcal{T}_{56}}{\langle S_9, \sigma_{14} \rangle \Rightarrow \langle S_{10}, \sigma_{14} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{57} = \frac{\langle y := y + 1, \sigma_{14} \rangle \Rightarrow \sigma_{15} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{14} \rangle \Rightarrow \langle S_{11}, \sigma_{15} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{58} = \langle S_{11}, \sigma_{15} \rangle \Rightarrow \langle S_{12}, \sigma_{15} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{59} = \langle S_{12}, \sigma_{15} \rangle \Rightarrow \langle S_{13}, \sigma_{15} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{61} = \frac{\langle x := x + 1, \sigma_{15} \rangle \Rightarrow \sigma_{16} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{15} \rangle \Rightarrow \langle S_6, \sigma_{16} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{60} = \frac{\mathcal{T}_{61}}{\langle S_{13}, \sigma_{15} \rangle \Rightarrow \langle S_7, \sigma_{16} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{62} = \frac{\langle S_6, \sigma_{16} \rangle \Rightarrow \langle S_8, \sigma_{16} \rangle [\text{assert}_{\text{sos}}]}{\langle S_{15}, \sigma_{16} \rangle \Rightarrow \langle S_9, \sigma_{16} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{64} = \langle S_8, \sigma_{16} \rangle \Rightarrow \langle y := y + 1, \sigma_{16} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{63} = \frac{\mathcal{T}_{64}}{\langle S_9, \sigma_{16} \rangle \Rightarrow \langle S_{10}, \sigma_{16} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{65} = \frac{\langle y := y + 1, \sigma_{16} \rangle \Rightarrow \sigma_{17} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{16} \rangle \Rightarrow \langle S_{11}, \sigma_{17} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{66} = \langle S_{11}, \sigma_{17} \rangle \Rightarrow \langle S_{12}, \sigma_{17} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{67} = \langle S_{12}, \sigma_{17} \rangle \Rightarrow \langle S_{13}, \sigma_{17} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{69} = \frac{\langle x := x + 1, \sigma_{17} \rangle \Rightarrow \sigma_{18} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{17} \rangle \Rightarrow \langle S_6, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{68} = \frac{\mathcal{T}_{69}}{\langle S_{13}, \sigma_{17} \rangle \Rightarrow \langle S_7, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{70} = \frac{\langle S_6, \sigma_{18} \rangle \Rightarrow \langle S_8, \sigma_{18} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{18} \rangle \Rightarrow \langle S_9, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{72} = \langle S_8, \sigma_{18} \rangle \Rightarrow \langle y := y + 1, \sigma_{18} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{71} = \frac{\mathcal{T}_{72}}{\langle S_9, \sigma_{18} \rangle \Rightarrow \langle S_{10}, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{73} = \frac{\langle y := y + 1, \sigma_{18} \rangle \Rightarrow \sigma_{19} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{18} \rangle \Rightarrow \langle S_{11}, \sigma_{19} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{74} = \langle S_{11}, \sigma_{19} \rangle \Rightarrow \langle S_{12}, \sigma_{19} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{75} = \langle S_{12}, \sigma_{19} \rangle \Rightarrow \langle S_{13}, \sigma_{19} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{77} = \frac{\langle x := x + 1, \sigma_{19} \rangle \Rightarrow \sigma_{20} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{19} \rangle \Rightarrow \langle S_6, \sigma_{20} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{76} = \frac{\mathcal{T}_{77}}{\langle S_{13}, \sigma_{19} \rangle \Rightarrow \langle S_7, \sigma_{20} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{78} = \frac{\langle S_6, \sigma_{20} \rangle \Rightarrow \langle S_8, \sigma_{20} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{20} \rangle \Rightarrow \langle S_9, \sigma_{20} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{80} = \langle S_8, \sigma_{20} \rangle \Rightarrow \langle y := y + 1, \sigma_{20} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{79} = \frac{\mathcal{T}_{80}}{\langle S_9, \sigma_{20} \rangle \Rightarrow \langle S_{10}, \sigma_{20} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{81} = \frac{\langle y := y + 1, \sigma_{20} \rangle \Rightarrow \sigma_{21} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{20} \rangle \Rightarrow \langle S_{11}, \sigma_{21} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{82} = \langle S_{11}, \sigma_{21} \rangle \Rightarrow \langle S_{12}, \sigma_{21} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{83} = \langle S_{12}, \sigma_{21} \rangle \Rightarrow \langle S_{13}, \sigma_{21} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{85} = \frac{\langle x := x + 1, \sigma_{21} \rangle \Rightarrow \sigma_{22} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{21} \rangle \Rightarrow \langle S_6, \sigma_{22} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{84} = \frac{\mathcal{T}_{85}}{\langle S_{13}, \sigma_{21} \rangle \Rightarrow \langle S_7, \sigma_{22} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{86} = \frac{\langle S_6, \sigma_{22} \rangle \Rightarrow \langle S_8, \sigma_{22} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{22} \rangle \Rightarrow \langle S_9, \sigma_{22} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{88} = \langle S_8, \sigma_{22} \rangle \Rightarrow \langle y := y + 1, \sigma_{22} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{87} = \frac{\mathcal{T}_{88}}{\langle S_9, \sigma_{22} \rangle \Rightarrow \langle S_{10}, \sigma_{22} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{89} = \frac{\langle y := y + 1, \sigma_{22} \rangle \Rightarrow \sigma_{23} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{22} \rangle \Rightarrow \langle S_{11}, \sigma_{23} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{90} = \langle S_{11}, \sigma_{23} \rangle \Rightarrow \langle S_{12}, \sigma_{23} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{91} = \langle S_{12}, \sigma_{23} \rangle \Rightarrow \langle S_{16}, \sigma_{23} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{93} = \frac{\langle x := x + 1, \sigma_{23} \rangle \Rightarrow \sigma_{24} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{23} \rangle \Rightarrow \langle S_6, \sigma_{24} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{92} = \frac{\mathcal{T}_{93}}{\langle S_{13}, \sigma_{23} \rangle \Rightarrow \langle S_7, \sigma_{24} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{94} = \frac{\langle S_6, \sigma_{24} \rangle \Rightarrow \langle S_8, \sigma_{24} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{24} \rangle \Rightarrow \langle S_9, \sigma_{24} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{96} = \langle S_8, \sigma_{24} \rangle \Rightarrow \langle y := y + 1, \sigma_{24} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{95} = \frac{\mathcal{T}_{96}}{\langle S_9, \sigma_{24} \rangle \Rightarrow \langle S_{10}, \sigma_{24} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{97} = \frac{\langle y := y + 1, \sigma_{24} \rangle \Rightarrow \sigma_{25} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{24} \rangle \Rightarrow \langle S_{11}, \sigma_{25} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{98} = \langle S_{11}, \sigma_{25} \rangle \Rightarrow \langle S_{12}, \sigma_{25} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{99} = \langle S_{12}, \sigma_{25} \rangle \Rightarrow \langle S_{16}, \sigma_{25} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{101} = \frac{\langle x := x + 1, \sigma_{25} \rangle \Rightarrow \sigma_{26} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{25} \rangle \Rightarrow \langle S_6, \sigma_{26} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{100} = \frac{\mathcal{T}_{101}}{\langle S_{13}, \sigma_{25} \rangle \Rightarrow \langle S_7, \sigma_{26} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{102} = \frac{\langle S_6, \sigma_{26} \rangle \Rightarrow \langle S_8, \sigma_{26} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{26} \rangle \Rightarrow \langle S_9, \sigma_{26} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{104} = \langle S_8, \sigma_{26} \rangle \Rightarrow \langle y := y + 1, \sigma_{26} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{103} = \frac{\mathcal{T}_{104}}{\langle S_9, \sigma_{26} \rangle \Rightarrow \langle S_{10}, \sigma_{26} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{105} = \frac{\langle y := y + 1, \sigma_{26} \rangle \Rightarrow \sigma_{27} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{26} \rangle \Rightarrow \langle S_{11}, \sigma_{27} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{106} = \langle S_{11}, \sigma_{27} \rangle \Rightarrow \langle S_{12}, \sigma_{27} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{107} = \langle S_{12}, \sigma_{27} \rangle \Rightarrow \langle S_{16}, \sigma_{27} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{109} = \frac{\langle x := x + 1, \sigma_{27} \rangle \Rightarrow \sigma_{28} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{27} \rangle \Rightarrow \langle S_6, \sigma_{28} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{108} = \frac{\mathcal{T}_{109}}{\langle S_{13}, \sigma_{27} \rangle \Rightarrow \langle S_7, \sigma_{28} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{110} = \frac{\langle S_6, \sigma_{28} \rangle \Rightarrow \langle S_8, \sigma_{28} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{28} \rangle \Rightarrow \langle S_9, \sigma_{28} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{112} = \langle S_8, \sigma_{28} \rangle \Rightarrow \langle y := y + 1, \sigma_{28} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{111} = \frac{\mathcal{T}_{112}}{\langle S_9, \sigma_{28} \rangle \Rightarrow \langle S_{10}, \sigma_{28} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{113} = \frac{\langle y := y + 1, \sigma_{28} \rangle \Rightarrow \sigma_{29} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{28} \rangle \Rightarrow \langle S_{11}, \sigma_{29} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{114} = \langle S_{11}, \sigma_{29} \rangle \Rightarrow \langle S_{12}, \sigma_{29} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{115} = \langle S_{12}, \sigma_{29} \rangle \Rightarrow \langle S_{16}, \sigma_{29} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{117} = \frac{\langle x := x + 1, \sigma_{29} \rangle \Rightarrow \sigma_{30} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{29} \rangle \Rightarrow \langle S_6, \sigma_{30} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{116} = \frac{\mathcal{T}_{117}}{\langle S_{13}, \sigma_{29} \rangle \Rightarrow \langle S_7, \sigma_{30} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{118} = \frac{\langle S_6, \sigma_{30} \rangle \Rightarrow \langle S_8, \sigma_{30} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{30} \rangle \Rightarrow \langle S_9, \sigma_{30} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{120} = \langle S_8, \sigma_{30} \rangle \Rightarrow \langle y := y + 1, \sigma_{30} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{119} = \frac{\mathcal{T}_{120}}{\langle S_9, \sigma_{30} \rangle \Rightarrow \langle S_{10}, \sigma_{30} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{121} = \frac{\langle y := y + 1, \sigma_{30} \rangle \Rightarrow \sigma_{31} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{30} \rangle \Rightarrow \langle S_{11}, \sigma_{31} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{122} = \langle S_{11}, \sigma_{31} \rangle \Rightarrow \langle S_{12}, \sigma_{31} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{123} = \langle S_{12}, \sigma_{31} \rangle \Rightarrow \langle S_{16}, \sigma_{31} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{125} = \frac{\langle x := x + 1, \sigma_{31} \rangle \Rightarrow \sigma_{32} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{31} \rangle \Rightarrow \langle S_6, \sigma_{32} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{124} = \frac{\mathcal{T}_{125}}{\langle S_{13}, \sigma_{31} \rangle \Rightarrow \langle S_7, \sigma_{32} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{126} = \frac{\langle S_6, \sigma_{32} \rangle \Rightarrow \langle S_8, \sigma_{32} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{32} \rangle \Rightarrow \langle S_9, \sigma_{32} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{128} = \langle S_8, \sigma_{32} \rangle \Rightarrow \langle y := y + 1, \sigma_{32} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{127} = \frac{\mathcal{T}_{128}}{\langle S_9, \sigma_{32} \rangle \Rightarrow \langle S_{10}, \sigma_{32} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{129} = \frac{\langle y := y + 1, \sigma_{32} \rangle \Rightarrow \sigma_{33} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{32} \rangle \Rightarrow \langle S_{11}, \sigma_{33} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{130} = \langle S_{11}, \sigma_{33} \rangle \Rightarrow \langle S_{12}, \sigma_{33} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{131} = \langle S_{12}, \sigma_{33} \rangle \Rightarrow \langle S_{16}, \sigma_{33} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{133} = \frac{\langle x := x + 1, \sigma_{33} \rangle \Rightarrow \sigma_{34} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{33} \rangle \Rightarrow \langle S_6, \sigma_{34} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{132} = \frac{\mathcal{T}_{133}}{\langle S_{13}, \sigma_{33} \rangle \Rightarrow \langle S_7, \sigma_{34} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{134} = \frac{\langle S_6, \sigma_{34} \rangle \Rightarrow \langle S_8, \sigma_{34} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{34} \rangle \Rightarrow \langle S_9, \sigma_{34} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{136} = \langle S_8, \sigma_{34} \rangle \Rightarrow \langle y := y + 1, \sigma_{34} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{135} = \frac{\mathcal{T}_{136}}{\langle S_9, \sigma_{34} \rangle \Rightarrow \langle S_{10}, \sigma_{34} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{137} = \frac{\langle y := y + 1, \sigma_{34} \rangle \Rightarrow \sigma_{35} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{34} \rangle \Rightarrow \langle S_{11}, \sigma_{35} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{138} = \langle S_{11}, \sigma_{35} \rangle \Rightarrow \langle S_{12}, \sigma_{35} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{139} = \langle S_{12}, \sigma_{35} \rangle \Rightarrow \langle S_{16}, \sigma_{35} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{141} = \frac{\langle x := x + 1, \sigma_{35} \rangle \Rightarrow \sigma_{36} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{35} \rangle \Rightarrow \langle S_6, \sigma_{36} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{140} = \frac{\mathcal{T}_{141}}{\langle S_{13}, \sigma_{35} \rangle \Rightarrow \langle S_7, \sigma_{36} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{142} = \frac{\langle S_6, \sigma_{36} \rangle \Rightarrow \langle S_8, \sigma_{36} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{36} \rangle \Rightarrow \langle S_9, \sigma_{36} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{144} = \langle S_8, \sigma_{36} \rangle \Rightarrow \langle y := y + 1, \sigma_{36} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{143} = \frac{\mathcal{T}_{144}}{\langle S_9, \sigma_{36} \rangle \Rightarrow \langle S_{10}, \sigma_{36} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{145} = \frac{\langle y := y + 1, \sigma_{36} \rangle \Rightarrow \sigma_{37} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{36} \rangle \Rightarrow \langle S_{11}, \sigma_{37} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{146} = \langle S_{11}, \sigma_{37} \rangle \Rightarrow \langle S_{12}, \sigma_{37} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{147} = \langle S_{12}, \sigma_{37} \rangle \Rightarrow \langle S_{16}, \sigma_{37} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{149} = \frac{\langle x := x + 1, \sigma_{37} \rangle \Rightarrow \sigma_{38} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{37} \rangle \Rightarrow \langle S_6, \sigma_{38} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{148} = \frac{\mathcal{T}_{149}}{\langle S_{13}, \sigma_{37} \rangle \Rightarrow \langle S_7, \sigma_{38} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{150} = \frac{\langle S_6, \sigma_{38} \rangle \Rightarrow \langle S_8, \sigma_{38} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{38} \rangle \Rightarrow \langle S_9, \sigma_{38} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{152} = \langle S_8, \sigma_{38} \rangle \Rightarrow \langle y := y + 1, \sigma_{38} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{151} = \frac{\mathcal{T}_{152}}{\langle S_9, \sigma_{38} \rangle \Rightarrow \langle S_{10}, \sigma_{38} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{153} = \frac{\langle y := y + 1, \sigma_{38} \rangle \Rightarrow \sigma_{39} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{38} \rangle \Rightarrow \langle S_{11}, \sigma_{39} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{154} = \langle S_{11}, \sigma_{39} \rangle \Rightarrow \langle S_{12}, \sigma_{39} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{155} = \langle S_{12}, \sigma_{39} \rangle \Rightarrow \langle S_{16}, \sigma_{39} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{157} = \frac{\langle x := x + 1, \sigma_{39} \rangle \Rightarrow \sigma_{40} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{39} \rangle \Rightarrow \langle S_6, \sigma_{40} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{156} = \frac{\mathcal{T}_{157}}{\langle S_{13}, \sigma_{39} \rangle \Rightarrow \langle S_7, \sigma_{40} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{158} = \frac{\langle S_6, \sigma_{40} \rangle \Rightarrow \langle S_8, \sigma_{40} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{40} \rangle \Rightarrow \langle S_9, \sigma_{40} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{160} = \langle S_8, \sigma_{40} \rangle \Rightarrow \langle y := y + 1, \sigma_{40} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{159} = \frac{\mathcal{T}_{160}}{\langle S_9, \sigma_{40} \rangle \Rightarrow \langle S_{10}, \sigma_{40} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{161} = \frac{\langle y := y + 1, \sigma_{40} \rangle \Rightarrow \sigma_{41} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{40} \rangle \Rightarrow \langle S_{11}, \sigma_{41} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{162} = \langle S_{11}, \sigma_{41} \rangle \Rightarrow \langle S_{12}, \sigma_{41} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{163} = \langle S_{12}, \sigma_{41} \rangle \Rightarrow \langle S_{16}, \sigma_{41} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{165} = \frac{\langle x := x + 1, \sigma_{41} \rangle \Rightarrow \sigma_{42} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{41} \rangle \Rightarrow \langle S_6, \sigma_{42} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{164} = \frac{\mathcal{T}_{165}}{\langle S_{13}, \sigma_{41} \rangle \Rightarrow \langle S_7, \sigma_{42} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{166} = \frac{\langle S_6, \sigma_{42} \rangle \Rightarrow \langle S_8, \sigma_{42} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{42} \rangle \Rightarrow \langle S_9, \sigma_{42} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{168} = \langle S_8, \sigma_{42} \rangle \Rightarrow \langle y := y + 1, \sigma_{42} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{167} = \frac{\mathcal{T}_{168}}{\langle S_9, \sigma_{42} \rangle \Rightarrow \langle S_{10}, \sigma_{42} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{169} = \frac{\langle y := y + 1, \sigma_{42} \rangle \Rightarrow \sigma_{43} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{42} \rangle \Rightarrow \langle S_{11}, \sigma_{43} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{170} = \langle S_{11}, \sigma_{43} \rangle \Rightarrow \langle S_{12}, \sigma_{43} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{171} = \langle S_{12}, \sigma_{43} \rangle \Rightarrow \langle S_{16}, \sigma_{43} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{173} = \frac{\langle x := x + 1, \sigma_{43} \rangle \Rightarrow \sigma_{44} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{43} \rangle \Rightarrow \langle S_6, \sigma_{44} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{172} = \frac{\mathcal{T}_{173}}{\langle S_{13}, \sigma_{43} \rangle \Rightarrow \langle S_7, \sigma_{44} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{174} = \frac{\langle S_6, \sigma_{44} \rangle \Rightarrow \langle S_8, \sigma_{44} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{44} \rangle \Rightarrow \langle S_9, \sigma_{44} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{176} = \langle S_8, \sigma_{44} \rangle \Rightarrow \langle y := y + 1, \sigma_{44} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{175} = \frac{\mathcal{T}_{176}}{\langle S_9, \sigma_{44} \rangle \Rightarrow \langle S_{10}, \sigma_{44} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{177} = \frac{\langle y := y + 1, \sigma_{44} \rangle \Rightarrow \sigma_{45} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{44} \rangle \Rightarrow \langle S_{11}, \sigma_{45} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{178} = \langle S_{11}, \sigma_{45} \rangle \Rightarrow \langle S_{12}, \sigma_{45} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{179} = \langle S_{12}, \sigma_{45} \rangle \Rightarrow \langle S_{16}, \sigma_{45} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{181} = \frac{\langle x := x + 1, \sigma_{45} \rangle \Rightarrow \sigma_{46} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{45} \rangle \Rightarrow \langle S_6, \sigma_{46} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{180} = \frac{\mathcal{T}_{181}}{\langle S_{13}, \sigma_{45} \rangle \Rightarrow \langle S_7, \sigma_{46} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{182} = \frac{\langle S_6, \sigma_{46} \rangle \Rightarrow \langle S_8, \sigma_{46} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{46} \rangle \Rightarrow \langle S_9, \sigma_{46} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{184} = \langle S_8, \sigma_{46} \rangle \Rightarrow \langle y := y + 1, \sigma_{46} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{183} = \frac{\mathcal{T}_{184}}{\langle S_9, \sigma_{46} \rangle \Rightarrow \langle S_{10}, \sigma_{46} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{185} = \frac{\langle y := y + 1, \sigma_{46} \rangle \Rightarrow \sigma_{47} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{46} \rangle \Rightarrow \langle S_{11}, \sigma_{47} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{186} = \langle S_{11}, \sigma_{47} \rangle \Rightarrow \langle S_{12}, \sigma_{47} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{187} = \langle S_{12}, \sigma_{47} \rangle \Rightarrow \langle S_{16}, \sigma_{47} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{189} = \frac{\langle x := x + 1, \sigma_{47} \rangle \Rightarrow \sigma_{48} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{47} \rangle \Rightarrow \langle S_6, \sigma_{48} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{188} = \frac{\mathcal{T}_{189}}{\langle S_{13}, \sigma_{47} \rangle \Rightarrow \langle S_7, \sigma_{48} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{190} = \frac{\langle S_6, \sigma_{48} \rangle \Rightarrow \langle S_8, \sigma_{48} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{48} \rangle \Rightarrow \langle S_9, \sigma_{48} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{192} = \langle S_8, \sigma_{48} \rangle \Rightarrow \langle y := y + 1, \sigma_{48} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{191} = \frac{\mathcal{T}_{192}}{\langle S_9, \sigma_{48} \rangle \Rightarrow \langle S_{10}, \sigma_{48} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{193} = \frac{\langle y := y + 1, \sigma_{48} \rangle \Rightarrow \sigma_{49} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{48} \rangle \Rightarrow \langle S_{11}, \sigma_{49} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{194} = \langle S_{11}, \sigma_{49} \rangle \Rightarrow \langle S_{12}, \sigma_{49} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{195} = \langle S_{12}, \sigma_{49} \rangle \Rightarrow \langle S_{16}, \sigma_{49} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{197} = \frac{\langle x := x + 1, \sigma_{49} \rangle \Rightarrow \sigma_{50} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{49} \rangle \Rightarrow \langle S_6, \sigma_{50} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{196} = \frac{\mathcal{T}_{197}}{\langle S_{13}, \sigma_{49} \rangle \Rightarrow \langle S_7, \sigma_{50} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{198} = \frac{\langle S_6, \sigma_{50} \rangle \Rightarrow \langle S_8, \sigma_{50} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{50} \rangle \Rightarrow \langle S_9, \sigma_{50} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{200} = \langle S_8, \sigma_{50} \rangle \Rightarrow \langle y := y + 1, \sigma_{50} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{199} = \frac{\mathcal{T}_{200}}{\langle S_9, \sigma_{50} \rangle \Rightarrow \langle S_{10}, \sigma_{50} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{201} = \frac{\langle y := y + 1, \sigma_{50} \rangle \Rightarrow \sigma_{51} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{50} \rangle \Rightarrow \langle S_{11}, \sigma_{51} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{202} = \langle S_{11}, \sigma_{51} \rangle \Rightarrow \langle S_{12}, \sigma_{51} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{203} = \langle S_{12}, \sigma_{51} \rangle \Rightarrow \langle S_{16}, \sigma_{51} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{205} = \frac{\langle x := x + 1, \sigma_{51} \rangle \Rightarrow \sigma_{52} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{51} \rangle \Rightarrow \langle S_6, \sigma_{52} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{204} = \frac{\mathcal{T}_{205}}{\langle S_{13}, \sigma_{51} \rangle \Rightarrow \langle S_7, \sigma_{52} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{206} = \frac{\langle S_6, \sigma_{52} \rangle \Rightarrow \langle S_8, \sigma_{52} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{52} \rangle \Rightarrow \langle S_9, \sigma_{52} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{208} = \langle S_8, \sigma_{52} \rangle \Rightarrow \langle y := y + 1, \sigma_{52} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{207} = \frac{\mathcal{T}_{208}}{\langle S_9, \sigma_{52} \rangle \Rightarrow \langle S_{10}, \sigma_{52} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{209} = \frac{\langle y := y + 1, \sigma_{52} \rangle \Rightarrow \sigma_{53} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{52} \rangle \Rightarrow \langle S_{11}, \sigma_{53} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{210} = \langle S_{11}, \sigma_{53} \rangle \Rightarrow \langle S_{12}, \sigma_{53} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{211} = \langle S_{12}, \sigma_{53} \rangle \Rightarrow \langle S_{16}, \sigma_{53} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{213} = \frac{\langle x := x + 1, \sigma_{53} \rangle \Rightarrow \sigma_{54} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{53} \rangle \Rightarrow \langle S_6, \sigma_{54} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{212} = \frac{\mathcal{T}_{213}}{\langle S_{13}, \sigma_{53} \rangle \Rightarrow \langle S_7, \sigma_{54} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{214} = \frac{\langle S_6, \sigma_{54} \rangle \Rightarrow \langle S_8, \sigma_{54} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{54} \rangle \Rightarrow \langle S_9, \sigma_{54} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{216} = \langle S_8, \sigma_{54} \rangle \Rightarrow \langle y := y + 1, \sigma_{54} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{215} = \frac{\mathcal{T}_{216}}{\langle S_9, \sigma_{54} \rangle \Rightarrow \langle S_{10}, \sigma_{54} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{217} = \frac{\langle y := y + 1, \sigma_{54} \rangle \Rightarrow \sigma_{55} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{54} \rangle \Rightarrow \langle S_{11}, \sigma_{55} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{218} = \langle S_{11}, \sigma_{55} \rangle \Rightarrow \langle S_{12}, \sigma_{55} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{219} = \langle S_{12}, \sigma_{55} \rangle \Rightarrow \langle S_{16}, \sigma_{55} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{221} = \frac{\langle x := x + 1, \sigma_{55} \rangle \Rightarrow \sigma_{56} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{55} \rangle \Rightarrow \langle S_6, \sigma_{56} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{220} = \frac{\mathcal{T}_{221}}{\langle S_{13}, \sigma_{55} \rangle \Rightarrow \langle S_7, \sigma_{56} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{222} = \frac{\langle S_6, \sigma_{56} \rangle \Rightarrow \langle S_8, \sigma_{56} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{56} \rangle \Rightarrow \langle S_9, \sigma_{56} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{224} = \langle S_8, \sigma_{56} \rangle \Rightarrow \langle y := y + 1, \sigma_{56} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{223} = \frac{\mathcal{T}_{224}}{\langle S_9, \sigma_{56} \rangle \Rightarrow \langle S_{10}, \sigma_{56} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{225} = \frac{\langle y := y + 1, \sigma_{56} \rangle \Rightarrow \sigma_{57} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{56} \rangle \Rightarrow \langle S_{11}, \sigma_{57} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{226} = \langle S_{11}, \sigma_{57} \rangle \Rightarrow \langle S_{12}, \sigma_{57} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{227} = \langle S_{12}, \sigma_{57} \rangle \Rightarrow \langle S_{16}, \sigma_{57} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{229} = \frac{\langle x := x + 1, \sigma_{57} \rangle \Rightarrow \sigma_{58} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{57} \rangle \Rightarrow \langle S_6, \sigma_{58} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{228} = \frac{\mathcal{T}_{229}}{\langle S_{13}, \sigma_{57} \rangle \Rightarrow \langle S_7, \sigma_{58} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{230} = \frac{\langle S_6, \sigma_{58} \rangle \Rightarrow \langle S_8, \sigma_{58} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{58} \rangle \Rightarrow \langle S_9, \sigma_{58} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{232} = \langle S_8, \sigma_{58} \rangle \Rightarrow \langle y := y + 1, \sigma_{58} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{231} = \frac{\mathcal{T}_{232}}{\langle S_9, \sigma_{58} \rangle \Rightarrow \langle S_{10}, \sigma_{58} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{233} = \frac{\langle y := y + 1, \sigma_{58} \rangle \Rightarrow \sigma_{59} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{58} \rangle \Rightarrow \langle S_{11}, \sigma_{59} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{234} = \langle S_{11}, \sigma_{59} \rangle \Rightarrow \langle S_{12}, \sigma_{59} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{235} = \langle S_{12}, \sigma_{59} \rangle \Rightarrow \langle S_{16}, \sigma_{59} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{237} = \frac{\langle x := x + 1, \sigma_{59} \rangle \Rightarrow \sigma_{60} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{59} \rangle \Rightarrow \langle S_6, \sigma_{60} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{236} = \frac{\mathcal{T}_{237}}{\langle S_{13}, \sigma_{59} \rangle \Rightarrow \langle S_7, \sigma_{60} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{238} = \frac{\langle S_6, \sigma_{60} \rangle \Rightarrow \langle S_8, \sigma_{60} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{60} \rangle \Rightarrow \langle S_9, \sigma_{60} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{240} = \langle S_8, \sigma_{60} \rangle \Rightarrow \langle y := y + 1, \sigma_{60} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{239} = \frac{\mathcal{T}_{240}}{\langle S_9, \sigma_{60} \rangle \Rightarrow \langle S_{10}, \sigma_{60} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{241} = \frac{\langle y := y + 1, \sigma_{60} \rangle \Rightarrow \sigma_{61} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{60} \rangle \Rightarrow \langle S_{11}, \sigma_{61} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{242} = \langle S_{11}, \sigma_{61} \rangle \Rightarrow \langle S_{12}, \sigma_{61} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{243} = \langle S_{12}, \sigma_{61} \rangle \Rightarrow \langle S_{16}, \sigma_{61} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{245} = \frac{\langle x := x + 1, \sigma_{61} \rangle \Rightarrow \sigma_{62} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{61} \rangle \Rightarrow \langle S_6, \sigma_{62} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{244} = \frac{\mathcal{T}_{245}}{\langle S_{13}, \sigma_{61} \rangle \Rightarrow \langle S_7, \sigma_{62} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{246} = \frac{\langle S_6, \sigma_{62} \rangle \Rightarrow \langle S_8, \sigma_{62} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{62} \rangle \Rightarrow \langle S_9, \sigma_{62} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{248} = \langle S_8, \sigma_{62} \rangle \Rightarrow \langle y := y + 1, \sigma_{62} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{247} = \frac{\mathcal{T}_{248}}{\langle S_9, \sigma_{62} \rangle \Rightarrow \langle S_{10}, \sigma_{62} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{249} = \frac{\langle y := y + 1, \sigma_{62} \rangle \Rightarrow \sigma_{63} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{62} \rangle \Rightarrow \langle S_{11}, \sigma_{63} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{250} = \langle S_{11}, \sigma_{63} \rangle \Rightarrow \langle S_{12}, \sigma_{63} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{251} = \langle S_{12}, \sigma_{63} \rangle \Rightarrow \langle S_{16}, \sigma_{63} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{253} = \frac{\langle x := x + 1, \sigma_{63} \rangle \Rightarrow \sigma_{64} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{63} \rangle \Rightarrow \langle S_6, \sigma_{64} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{252} = \frac{\mathcal{T}_{253}}{\langle S_{13}, \sigma_{63} \rangle \Rightarrow \langle S_7, \sigma_{64} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{254} = \frac{\langle S_6, \sigma_{64} \rangle \Rightarrow \langle S_8, \sigma_{64} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{64} \rangle \Rightarrow \langle S_9, \sigma_{64} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{256} = \langle S_8, \sigma_{64} \rangle \Rightarrow \langle y := y + 1, \sigma_{64} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{255} = \frac{\mathcal{T}_{256}}{\langle S_9, \sigma_{64} \rangle \Rightarrow \langle S_{10}, \sigma_{64} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{257} = \frac{\langle y := y + 1, \sigma_{64} \rangle \Rightarrow \sigma_{65} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{64} \rangle \Rightarrow \langle S_{11}, \sigma_{65} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{258} = \langle S_{11}, \sigma_{65} \rangle \Rightarrow \langle S_{12}, \sigma_{65} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{259} = \langle S_{12}, \sigma_{65} \rangle \Rightarrow \langle S_{16}, \sigma_{65} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{261} = \frac{\langle x := x + 1, \sigma_{65} \rangle \Rightarrow \sigma_{66} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{65} \rangle \Rightarrow \langle S_6, \sigma_{66} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{260} = \frac{\mathcal{T}_{261}}{\langle S_{13}, \sigma_{65} \rangle \Rightarrow \langle S_7, \sigma_{66} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{262} = \frac{\langle S_6, \sigma_{66} \rangle \Rightarrow \langle S_8, \sigma_{66} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{66} \rangle \Rightarrow \langle S_9, \sigma_{66} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{264} = \langle S_8, \sigma_{66} \rangle \Rightarrow \langle y := y + 1, \sigma_{66} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{263} = \frac{\mathcal{T}_{264}}{\langle S_9, \sigma_{66} \rangle \Rightarrow \langle S_{10}, \sigma_{66} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{265} = \frac{\langle y := y + 1, \sigma_{66} \rangle \Rightarrow \sigma_{67} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{66} \rangle \Rightarrow \langle S_{11}, \sigma_{67} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{266} = \langle S_{11}, \sigma_{67} \rangle \Rightarrow \langle S_{12}, \sigma_{67} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{267} = \langle S_{12}, \sigma_{67} \rangle \Rightarrow \langle S_{16}, \sigma_{67} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{269} = \frac{\langle x := x + 1, \sigma_{67} \rangle \Rightarrow \sigma_{68} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{67} \rangle \Rightarrow \langle S_6, \sigma_{68} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{268} = \frac{\mathcal{T}_{269}}{\langle S_{13}, \sigma_{67} \rangle \Rightarrow \langle S_7, \sigma_{68} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{270} = \frac{\langle S_6, \sigma_{68} \rangle \Rightarrow \langle S_8, \sigma_{68} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{68} \rangle \Rightarrow \langle S_9, \sigma_{68} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{272} = \langle S_8, \sigma_{68} \rangle \Rightarrow \langle y := y + 1, \sigma_{68} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{271} = \frac{\mathcal{T}_{272}}{\langle S_9, \sigma_{68} \rangle \Rightarrow \langle S_{10}, \sigma_{68} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{273} = \frac{\langle y := y + 1, \sigma_{68} \rangle \Rightarrow \sigma_{69} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{68} \rangle \Rightarrow \langle S_{11}, \sigma_{69} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{274} = \langle S_{11}, \sigma_{69} \rangle \Rightarrow \langle S_{12}, \sigma_{69} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{275} = \langle S_{12}, \sigma_{69} \rangle \Rightarrow \langle S_{16}, \sigma_{69} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{277} = \frac{\langle x := x + 1, \sigma_{69} \rangle \Rightarrow \sigma_{70} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{69} \rangle \Rightarrow \langle S_6, \sigma_{70} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{276} = \frac{\mathcal{T}_{277}}{\langle S_{13}, \sigma_{69} \rangle \Rightarrow \langle S_7, \sigma_{70} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{278} = \frac{\langle S_6, \sigma_{70} \rangle \Rightarrow \langle S_8, \sigma_{70} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{70} \rangle \Rightarrow \langle S_9, \sigma_{70} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{280} = \langle S_8, \sigma_{70} \rangle \Rightarrow \langle y := y + 1, \sigma_{70} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{279} = \frac{\mathcal{T}_{280}}{\langle S_9, \sigma_{70} \rangle \Rightarrow \langle S_{10}, \sigma_{70} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{281} = \frac{\langle y := y + 1, \sigma_{70} \rangle \Rightarrow \sigma_{71} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{70} \rangle \Rightarrow \langle S_{11}, \sigma_{71} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{282} = \langle S_{11}, \sigma_{71} \rangle \Rightarrow \langle S_{12}, \sigma_{71} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{283} = \langle S_{12}, \sigma_{71} \rangle \Rightarrow \langle S_{16}, \sigma_{71} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{285} = \frac{\langle x := x + 1, \sigma_{71} \rangle \Rightarrow \sigma_{72} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{71} \rangle \Rightarrow \langle S_6, \sigma_{72} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{284} = \frac{\mathcal{T}_{285}}{\langle S_{13}, \sigma_{71} \rangle \Rightarrow \langle S_7, \sigma_{72} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{286} = \frac{\langle S_6, \sigma_{72} \rangle \Rightarrow \langle S_8, \sigma_{72} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{72} \rangle \Rightarrow \langle S_9, \sigma_{72} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{288} = \langle S_8, \sigma_{72} \rangle \Rightarrow \langle y := y + 1, \sigma_{72} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{287} = \frac{\mathcal{T}_{288}}{\langle S_9, \sigma_{72} \rangle \Rightarrow \langle S_{10}, \sigma_{72} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{289} = \frac{\langle y := y + 1, \sigma_{72} \rangle \Rightarrow \sigma_{73} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{72} \rangle \Rightarrow \langle S_{11}, \sigma_{73} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{290} = \langle S_{11}, \sigma_{73} \rangle \Rightarrow \langle S_{12}, \sigma_{73} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{291} = \langle S_{12}, \sigma_{73} \rangle \Rightarrow \langle S_{16}, \sigma_{73} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{293} = \frac{\langle x := x + 1, \sigma_{73} \rangle \Rightarrow \sigma_{74} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{73} \rangle \Rightarrow \langle S_6, \sigma_{74} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{292} = \frac{\mathcal{T}_{293}}{\langle S_{13}, \sigma_{73} \rangle \Rightarrow \langle S_7, \sigma_{74} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{294} = \frac{\langle S_6, \sigma_{74} \rangle \Rightarrow \langle S_8, \sigma_{74} \rangle \text{ [assert}_{\text{tt}_{\text{sos}}}]}{\langle S_7, \sigma_{74} \rangle \Rightarrow \langle S_9, \sigma_{74} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{296} = \langle S_8, \sigma_{74} \rangle \Rightarrow \langle y := y + 1, \sigma_{74} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{295} = \frac{\mathcal{T}_{296}}{\langle S_9, \sigma_{74} \rangle \Rightarrow \langle S_{10}, \sigma_{74} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{297} = \frac{\langle y := y + 1, \sigma_{74} \rangle \Rightarrow \sigma_{75} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{74} \rangle \Rightarrow \langle S_{11}, \sigma_{75} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{298} = \langle S_{11}, \sigma_{75} \rangle \Rightarrow \langle S_{12}, \sigma_{75} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{299} = \langle S_{12}, \sigma_{75} \rangle \Rightarrow \langle S_{16}, \sigma_{75} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{301} = \frac{\langle x := x + 1, \sigma_{75} \rangle \Rightarrow \sigma_{76} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{75} \rangle \Rightarrow \langle S_6, \sigma_{76} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{300} = \frac{\mathcal{T}_{301}}{\langle S_{13}, \sigma_{75} \rangle \Rightarrow \langle S_7, \sigma_{76} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{302} = \frac{\langle S_6, \sigma_{76} \rangle \Rightarrow \langle S_8, \sigma_{76} \rangle \text{ [assert}_{\text{tt}_{\text{sos}}}]}{\langle S_7, \sigma_{76} \rangle \Rightarrow \langle S_9, \sigma_{76} \rangle} [\text{comp}_{\text{tt}_{\text{sos}}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{304} = \langle S_8, \sigma_{76} \rangle \Rightarrow \langle y := y + 1, \sigma_{76} \rangle \text{ [if - tt}_{\text{tt}_{\text{sos}}}]$$

$$\mathcal{T}_{303} = \frac{\mathcal{T}_{304}}{\langle S_9, \sigma_{76} \rangle \Rightarrow \langle S_{10}, \sigma_{76} \rangle} [\text{comp}_{\text{tt}_{\text{sos}}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{305} = \frac{\langle y := y + 1, \sigma_{76} \rangle \Rightarrow \sigma_{77} \text{ [ass}_{\text{tt}_{\text{sos}}}]}{\langle S_{10}, \sigma_{76} \rangle \Rightarrow \langle S_{11}, \sigma_{77} \rangle} [\text{comp}_{\text{tt}_{\text{sos}}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{306} = \langle S_{11}, \sigma_{77} \rangle \Rightarrow \langle S_{12}, \sigma_{77} \rangle \text{ [for}_{\text{tt}_{\text{tt}_{\text{sos}}}}]$$

$$\Downarrow$$

$$\mathcal{T}_{307} = \langle S_{12}, \sigma_{77} \rangle \Rightarrow \langle S_{16}, \sigma_{77} \rangle \text{ [if - tt}_{\text{tt}_{\text{tt}_{\text{sos}}}}]$$

$$\Downarrow$$

$$\mathcal{T}_{309} = \frac{\langle x := x + 1, \sigma_{77} \rangle \Rightarrow \sigma_{78} \text{ [ass}_{\text{tt}_{\text{tt}_{\text{tt}_{\text{sos}}}}}]}{\langle S_{14}, \sigma_{77} \rangle \Rightarrow \langle S_6, \sigma_{78} \rangle} [\text{comp}_{\text{tt}_{\text{tt}_{\text{tt}_{\text{tt}_{\text{sos}}}}}}^2]$$

$$\mathcal{T}_{308} = \frac{\mathcal{T}_{309}}{\langle S_{13}, \sigma_{77} \rangle \Rightarrow \langle S_7, \sigma_{78} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{310} = \frac{\langle S_6, \sigma_{78} \rangle \Rightarrow \langle S_8, \sigma_{78} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{78} \rangle \Rightarrow \langle S_9, \sigma_{78} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{312} = \langle S_8, \sigma_{78} \rangle \Rightarrow \langle y := y + 1, \sigma_{78} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{311} = \frac{\mathcal{T}_{312}}{\langle S_9, \sigma_{78} \rangle \Rightarrow \langle S_{10}, \sigma_{78} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{313} = \frac{\langle y := y + 1, \sigma_{78} \rangle \Rightarrow \sigma_{79} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{78} \rangle \Rightarrow \langle S_{11}, \sigma_{79} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{314} = \langle S_{11}, \sigma_{79} \rangle \Rightarrow \langle S_{12}, \sigma_{79} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{315} = \langle S_{12}, \sigma_{79} \rangle \Rightarrow \langle S_{16}, \sigma_{79} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{317} = \frac{\langle x := x + 1, \sigma_{79} \rangle \Rightarrow \sigma_{80} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{79} \rangle \Rightarrow \langle S_6, \sigma_{80} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{316} = \frac{\mathcal{T}_{317}}{\langle S_{13}, \sigma_{79} \rangle \Rightarrow \langle S_7, \sigma_{80} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{318} = \frac{\langle S_6, \sigma_{80} \rangle \Rightarrow \langle S_8, \sigma_{80} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{80} \rangle \Rightarrow \langle S_9, \sigma_{80} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{320} = \langle S_8, \sigma_{80} \rangle \Rightarrow \langle y := y + 1, \sigma_{80} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{319} = \frac{\mathcal{T}_{320}}{\langle S_9, \sigma_{80} \rangle \Rightarrow \langle S_{10}, \sigma_{80} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{321} = \frac{\langle y := y + 1, \sigma_{80} \rangle \Rightarrow \sigma_{81} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{80} \rangle \Rightarrow \langle S_{11}, \sigma_{81} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{322} = \langle S_{11}, \sigma_{81} \rangle \Rightarrow \langle S_{12}, \sigma_{81} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{323} = \langle S_{12}, \sigma_{81} \rangle \Rightarrow \langle S_{16}, \sigma_{81} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{325} = \frac{\langle x := x + 1, \sigma_{81} \rangle \Rightarrow \sigma_{82} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{81} \rangle \Rightarrow \langle S_6, \sigma_{82} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{324} = \frac{\mathcal{T}_{325}}{\langle S_{13}, \sigma_{81} \rangle \Rightarrow \langle S_7, \sigma_{82} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{326} = \frac{\langle S_6, \sigma_{82} \rangle \Rightarrow \langle S_8, \sigma_{82} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{82} \rangle \Rightarrow \langle S_9, \sigma_{82} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{328} = \langle S_8, \sigma_{82} \rangle \Rightarrow \langle y := y + 1, \sigma_{82} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{327} = \frac{\mathcal{T}_{328}}{\langle S_9, \sigma_{82} \rangle \Rightarrow \langle S_{10}, \sigma_{82} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{329} = \frac{\langle y := y + 1, \sigma_{82} \rangle \Rightarrow \sigma_{83} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{82} \rangle \Rightarrow \langle S_{11}, \sigma_{83} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{330} = \langle S_{11}, \sigma_{83} \rangle \Rightarrow \langle S_{12}, \sigma_{83} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{331} = \langle S_{12}, \sigma_{83} \rangle \Rightarrow \langle S_{16}, \sigma_{83} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{333} = \frac{\langle x := x + 1, \sigma_{83} \rangle \Rightarrow \sigma_{84} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{83} \rangle \Rightarrow \langle S_6, \sigma_{84} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{332} = \frac{\mathcal{T}_{333}}{\langle S_{13}, \sigma_{83} \rangle \Rightarrow \langle S_7, \sigma_{84} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{334} = \frac{\langle S_6, \sigma_{84} \rangle \Rightarrow \langle S_8, \sigma_{84} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{84} \rangle \Rightarrow \langle S_9, \sigma_{84} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{336} = \langle S_8, \sigma_{84} \rangle \Rightarrow \langle y := y + 1, \sigma_{84} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{335} = \frac{\mathcal{T}_{336}}{\langle S_9, \sigma_{84} \rangle \Rightarrow \langle S_{10}, \sigma_{84} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{337} = \frac{\langle y := y + 1, \sigma_{84} \rangle \Rightarrow \sigma_{85} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{84} \rangle \Rightarrow \langle S_{11}, \sigma_{85} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{338} = \langle S_{11}, \sigma_{85} \rangle \Rightarrow \langle S_{12}, \sigma_{85} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{339} = \langle S_{12}, \sigma_{85} \rangle \Rightarrow \langle S_{16}, \sigma_{85} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{341} = \frac{\langle x := x + 1, \sigma_{85} \rangle \Rightarrow \sigma_{86} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{85} \rangle \Rightarrow \langle S_6, \sigma_{86} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{340} = \frac{\mathcal{T}_{341}}{\langle S_{13}, \sigma_{85} \rangle \Rightarrow \langle S_7, \sigma_{86} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{342} = \frac{\langle S_6, \sigma_{86} \rangle \Rightarrow \langle S_8, \sigma_{86} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{86} \rangle \Rightarrow \langle S_9, \sigma_{86} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{344} = \langle S_8, \sigma_{86} \rangle \Rightarrow \langle y := y + 1, \sigma_{86} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{343} = \frac{\mathcal{T}_{344}}{\langle S_9, \sigma_{86} \rangle \Rightarrow \langle S_{10}, \sigma_{86} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{345} = \frac{\langle y := y + 1, \sigma_{86} \rangle \Rightarrow \sigma_{87} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{86} \rangle \Rightarrow \langle S_{11}, \sigma_{87} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{346} = \langle S_{11}, \sigma_{87} \rangle \Rightarrow \langle S_{12}, \sigma_{87} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{347} = \langle S_{12}, \sigma_{87} \rangle \Rightarrow \langle S_{16}, \sigma_{87} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{349} = \frac{\langle x := x + 1, \sigma_{87} \rangle \Rightarrow \sigma_{88} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{87} \rangle \Rightarrow \langle S_6, \sigma_{88} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{348} = \frac{\mathcal{T}_{349}}{\langle S_{13}, \sigma_{87} \rangle \Rightarrow \langle S_7, \sigma_{88} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{350} = \frac{\langle S_6, \sigma_{88} \rangle \Rightarrow \langle S_8, \sigma_{88} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{88} \rangle \Rightarrow \langle S_9, \sigma_{88} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{352} = \langle S_8, \sigma_{88} \rangle \Rightarrow \langle y := y + 1, \sigma_{88} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{351} = \frac{\mathcal{T}_{352}}{\langle S_9, \sigma_{88} \rangle \Rightarrow \langle S_{10}, \sigma_{88} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{353} = \frac{\langle y := y + 1, \sigma_{88} \rangle \Rightarrow \sigma_{89} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{88} \rangle \Rightarrow \langle S_{11}, \sigma_{89} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{354} = \langle S_{11}, \sigma_{89} \rangle \Rightarrow \langle S_{12}, \sigma_{89} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{355} = \langle S_{12}, \sigma_{89} \rangle \Rightarrow \langle S_{16}, \sigma_{89} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{357} = \frac{\langle x := x + 1, \sigma_{89} \rangle \Rightarrow \sigma_{90} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{89} \rangle \Rightarrow \langle S_6, \sigma_{90} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{356} = \frac{\mathcal{T}_{357}}{\langle S_{13}, \sigma_{89} \rangle \Rightarrow \langle S_7, \sigma_{90} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{358} = \frac{\langle S_6, \sigma_{90} \rangle \Rightarrow \langle S_8, \sigma_{90} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{90} \rangle \Rightarrow \langle S_9, \sigma_{90} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{360} = \langle S_8, \sigma_{90} \rangle \Rightarrow \langle y := y + 1, \sigma_{90} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{359} = \frac{\mathcal{T}_{360}}{\langle S_9, \sigma_{90} \rangle \Rightarrow \langle S_{10}, \sigma_{90} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{361} = \frac{\langle y := y + 1, \sigma_{90} \rangle \Rightarrow \sigma_{91} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{90} \rangle \Rightarrow \langle S_{11}, \sigma_{91} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{362} = \langle S_{11}, \sigma_{91} \rangle \Rightarrow \langle S_{12}, \sigma_{91} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{363} = \langle S_{12}, \sigma_{91} \rangle \Rightarrow \langle S_{16}, \sigma_{91} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{365} = \frac{\langle x := x + 1, \sigma_{91} \rangle \Rightarrow \sigma_{92} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{91} \rangle \Rightarrow \langle S_6, \sigma_{92} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{364} = \frac{\mathcal{T}_{365}}{\langle S_{13}, \sigma_{91} \rangle \Rightarrow \langle S_7, \sigma_{92} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{366} = \frac{\langle S_6, \sigma_{92} \rangle \Rightarrow \langle S_8, \sigma_{92} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{92} \rangle \Rightarrow \langle S_9, \sigma_{92} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{368} = \langle S_8, \sigma_{92} \rangle \Rightarrow \langle y := y + 1, \sigma_{92} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{367} = \frac{\mathcal{T}_{368}}{\langle S_9, \sigma_{92} \rangle \Rightarrow \langle S_{10}, \sigma_{92} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{369} = \frac{\langle y := y + 1, \sigma_{92} \rangle \Rightarrow \sigma_{93} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{92} \rangle \Rightarrow \langle S_{11}, \sigma_{93} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{370} = \langle S_{11}, \sigma_{93} \rangle \Rightarrow \langle S_{12}, \sigma_{93} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{371} = \langle S_{12}, \sigma_{93} \rangle \Rightarrow \langle S_{16}, \sigma_{93} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{373} = \frac{\langle x := x + 1, \sigma_{93} \rangle \Rightarrow \sigma_{94} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{93} \rangle \Rightarrow \langle S_6, \sigma_{94} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{372} = \frac{\mathcal{T}_{373}}{\langle S_{13}, \sigma_{93} \rangle \Rightarrow \langle S_7, \sigma_{94} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{374} = \frac{\langle S_6, \sigma_{94} \rangle \Rightarrow \langle S_8, \sigma_{94} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{94} \rangle \Rightarrow \langle S_9, \sigma_{94} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{376} = \langle S_8, \sigma_{94} \rangle \Rightarrow \langle y := y + 1, \sigma_{94} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{375} = \frac{\mathcal{T}_{376}}{\langle S_9, \sigma_{94} \rangle \Rightarrow \langle S_{10}, \sigma_{94} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{377} = \frac{\langle y := y + 1, \sigma_{94} \rangle \Rightarrow \sigma_{95} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{94} \rangle \Rightarrow \langle S_{11}, \sigma_{95} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{378} = \langle S_{11}, \sigma_{95} \rangle \Rightarrow \langle S_{12}, \sigma_{95} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{379} = \langle S_{12}, \sigma_{95} \rangle \Rightarrow \langle S_{16}, \sigma_{95} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{381} = \frac{\langle x := x + 1, \sigma_{95} \rangle \Rightarrow \sigma_{96} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{95} \rangle \Rightarrow \langle S_6, \sigma_{96} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{380} = \frac{\mathcal{T}_{381}}{\langle S_{13}, \sigma_{95} \rangle \Rightarrow \langle S_7, \sigma_{96} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{382} = \frac{\langle S_6, \sigma_{96} \rangle \Rightarrow \langle S_8, \sigma_{96} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{96} \rangle \Rightarrow \langle S_9, \sigma_{96} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{384} = \langle S_8, \sigma_{96} \rangle \Rightarrow \langle y := y + 1, \sigma_{96} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{383} = \frac{\mathcal{T}_{384}}{\langle S_9, \sigma_{96} \rangle \Rightarrow \langle S_{10}, \sigma_{96} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{385} = \frac{\langle y := y + 1, \sigma_{96} \rangle \Rightarrow \sigma_{97} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{96} \rangle \Rightarrow \langle S_{11}, \sigma_{97} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{386} = \langle S_{11}, \sigma_{97} \rangle \Rightarrow \langle S_{12}, \sigma_{97} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{387} = \langle S_{12}, \sigma_{97} \rangle \Rightarrow \langle S_{16}, \sigma_{97} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{389} = \frac{\langle x := x + 1, \sigma_{97} \rangle \Rightarrow \sigma_{98} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{97} \rangle \Rightarrow \langle S_6, \sigma_{98} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{388} = \frac{\mathcal{T}_{389}}{\langle S_{13}, \sigma_{97} \rangle \Rightarrow \langle S_7, \sigma_{98} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{390} = \frac{\langle S_6, \sigma_{98} \rangle \Rightarrow \langle S_8, \sigma_{98} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{98} \rangle \Rightarrow \langle S_9, \sigma_{98} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{392} = \langle S_8, \sigma_{98} \rangle \Rightarrow \langle y := y + 1, \sigma_{98} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{391} = \frac{\mathcal{T}_{392}}{\langle S_9, \sigma_{98} \rangle \Rightarrow \langle S_{10}, \sigma_{98} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{393} = \frac{\langle y := y + 1, \sigma_{98} \rangle \Rightarrow \sigma_{99} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{98} \rangle \Rightarrow \langle S_{11}, \sigma_{99} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{394} = \langle S_{11}, \sigma_{99} \rangle \Rightarrow \langle S_{12}, \sigma_{99} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{395} = \langle S_{12}, \sigma_{99} \rangle \Rightarrow \langle S_{16}, \sigma_{99} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{397} = \frac{\langle x := x + 1, \sigma_{99} \rangle \Rightarrow \sigma_{100} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{99} \rangle \Rightarrow \langle S_6, \sigma_{100} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{396} = \frac{\mathcal{T}_{397}}{\langle S_{13}, \sigma_{99} \rangle \Rightarrow \langle S_7, \sigma_{100} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{398} = \frac{\langle S_6, \sigma_{100} \rangle \Rightarrow \langle S_8, \sigma_{100} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{100} \rangle \Rightarrow \langle S_9, \sigma_{100} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{400} = \langle S_8, \sigma_{100} \rangle \Rightarrow \langle y := y + 1, \sigma_{100} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{399} = \frac{\mathcal{T}_{400}}{\langle S_9, \sigma_{100} \rangle \Rightarrow \langle S_{10}, \sigma_{100} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{401} = \frac{\langle y := y + 1, \sigma_{100} \rangle \Rightarrow \sigma_{101} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{100} \rangle \Rightarrow \langle S_{11}, \sigma_{101} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{402} = \langle S_{11}, \sigma_{101} \rangle \Rightarrow \langle S_{12}, \sigma_{101} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{403} = \langle S_{12}, \sigma_{101} \rangle \Rightarrow \langle S_{16}, \sigma_{101} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{405} = \frac{\langle x := x + 1, \sigma_{101} \rangle \Rightarrow \sigma_{102} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{101} \rangle \Rightarrow \langle S_6, \sigma_{102} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{404} = \frac{\mathcal{T}_{405}}{\langle S_{13}, \sigma_{101} \rangle \Rightarrow \langle S_7, \sigma_{102} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{406} = \frac{\langle S_6, \sigma_{102} \rangle \Rightarrow \langle S_8, \sigma_{102} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{102} \rangle \Rightarrow \langle S_9, \sigma_{102} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{408} = \langle S_8, \sigma_{102} \rangle \Rightarrow \langle y := y + 1, \sigma_{102} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{407} = \frac{\mathcal{T}_{408}}{\langle S_9, \sigma_{102} \rangle \Rightarrow \langle S_{10}, \sigma_{102} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{409} = \frac{\langle y := y + 1, \sigma_{102} \rangle \Rightarrow \sigma_{103} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{102} \rangle \Rightarrow \langle S_{11}, \sigma_{103} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{410} = \langle S_{11}, \sigma_{103} \rangle \Rightarrow \langle S_{12}, \sigma_{103} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{411} = \langle S_{12}, \sigma_{103} \rangle \Rightarrow \langle S_{16}, \sigma_{103} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{413} = \frac{\langle x := x + 1, \sigma_{103} \rangle \Rightarrow \sigma_{104} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{103} \rangle \Rightarrow \langle S_6, \sigma_{104} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{412} = \frac{\mathcal{T}_{413}}{\langle S_{13}, \sigma_{103} \rangle \Rightarrow \langle S_7, \sigma_{104} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{414} = \frac{\langle S_6, \sigma_{104} \rangle \Rightarrow \langle S_8, \sigma_{104} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{104} \rangle \Rightarrow \langle S_9, \sigma_{104} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{416} = \langle S_8, \sigma_{104} \rangle \Rightarrow \langle y := y + 1, \sigma_{104} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{415} = \frac{\mathcal{T}_{416}}{\langle S_9, \sigma_{104} \rangle \Rightarrow \langle S_{10}, \sigma_{104} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{417} = \frac{\langle y := y + 1, \sigma_{104} \rangle \Rightarrow \sigma_{105} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{104} \rangle \Rightarrow \langle S_{11}, \sigma_{105} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{418} = \langle S_{11}, \sigma_{105} \rangle \Rightarrow \langle S_{12}, \sigma_{105} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{419} = \langle S_{12}, \sigma_{105} \rangle \Rightarrow \langle S_{16}, \sigma_{105} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{421} = \frac{\langle x := x + 1, \sigma_{105} \rangle \Rightarrow \sigma_{106} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{105} \rangle \Rightarrow \langle S_6, \sigma_{106} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{420} = \frac{\mathcal{T}_{421}}{\langle S_{13}, \sigma_{105} \rangle \Rightarrow \langle S_7, \sigma_{106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{422} = \frac{\langle S_6, \sigma_{106} \rangle \Rightarrow \langle S_8, \sigma_{106} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{106} \rangle \Rightarrow \langle S_9, \sigma_{106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{424} = \langle S_8, \sigma_{106} \rangle \Rightarrow \langle y := y + 1, \sigma_{106} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{423} = \frac{\mathcal{T}_{424}}{\langle S_9, \sigma_{106} \rangle \Rightarrow \langle S_{10}, \sigma_{106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{425} = \frac{\langle y := y + 1, \sigma_{106} \rangle \Rightarrow \sigma_{107} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{106} \rangle \Rightarrow \langle S_{11}, \sigma_{107} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{426} = \langle S_{11}, \sigma_{107} \rangle \Rightarrow \langle S_{12}, \sigma_{107} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{427} = \langle S_{12}, \sigma_{107} \rangle \Rightarrow \langle S_{16}, \sigma_{107} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{429} = \frac{\langle x := x + 1, \sigma_{107} \rangle \Rightarrow \sigma_{108} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{107} \rangle \Rightarrow \langle S_6, \sigma_{108} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{428} = \frac{\mathcal{T}_{429}}{\langle S_{13}, \sigma_{107} \rangle \Rightarrow \langle S_7, \sigma_{108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{430} = \frac{\langle S_6, \sigma_{108} \rangle \Rightarrow \langle S_8, \sigma_{108} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{108} \rangle \Rightarrow \langle S_9, \sigma_{108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{432} = \langle S_8, \sigma_{108} \rangle \Rightarrow \langle y := y + 1, \sigma_{108} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{431} = \frac{\mathcal{T}_{432}}{\langle S_9, \sigma_{108} \rangle \Rightarrow \langle S_{10}, \sigma_{108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{433} = \frac{\langle y := y + 1, \sigma_{108} \rangle \Rightarrow \sigma_{109} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{108} \rangle \Rightarrow \langle S_{11}, \sigma_{109} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{434} = \langle S_{11}, \sigma_{109} \rangle \Rightarrow \langle S_{12}, \sigma_{109} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{435} = \langle S_{12}, \sigma_{109} \rangle \Rightarrow \langle S_{16}, \sigma_{109} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{437} = \frac{\langle x := x + 1, \sigma_{109} \rangle \Rightarrow \sigma_{110} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{109} \rangle \Rightarrow \langle S_6, \sigma_{110} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{436} = \frac{\mathcal{T}_{437}}{\langle S_{13}, \sigma_{109} \rangle \Rightarrow \langle S_7, \sigma_{110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{438} = \frac{\langle S_6, \sigma_{110} \rangle \Rightarrow \langle S_8, \sigma_{110} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{110} \rangle \Rightarrow \langle S_9, \sigma_{110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{440} = \langle S_8, \sigma_{110} \rangle \Rightarrow \langle y := y + 1, \sigma_{110} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{439} = \frac{\mathcal{T}_{440}}{\langle S_9, \sigma_{110} \rangle \Rightarrow \langle S_{10}, \sigma_{110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{441} = \frac{\langle y := y + 1, \sigma_{110} \rangle \Rightarrow \sigma_{111} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{110} \rangle \Rightarrow \langle S_{11}, \sigma_{111} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{442} = \langle S_{11}, \sigma_{111} \rangle \Rightarrow \langle S_{12}, \sigma_{111} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{443} = \langle S_{12}, \sigma_{111} \rangle \Rightarrow \langle S_{16}, \sigma_{111} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{445} = \frac{\langle x := x + 1, \sigma_{111} \rangle \Rightarrow \sigma_{112} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{111} \rangle \Rightarrow \langle S_6, \sigma_{112} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{444} = \frac{\mathcal{T}_{445}}{\langle S_{13}, \sigma_{111} \rangle \Rightarrow \langle S_7, \sigma_{112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{446} = \frac{\langle S_6, \sigma_{112} \rangle \Rightarrow \langle S_8, \sigma_{112} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{112} \rangle \Rightarrow \langle S_9, \sigma_{112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{448} = \langle S_8, \sigma_{112} \rangle \Rightarrow \langle y := y + 1, \sigma_{112} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{447} = \frac{\mathcal{T}_{448}}{\langle S_9, \sigma_{112} \rangle \Rightarrow \langle S_{10}, \sigma_{112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{449} = \frac{\langle y := y + 1, \sigma_{112} \rangle \Rightarrow \sigma_{113} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{112} \rangle \Rightarrow \langle S_{11}, \sigma_{113} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{450} = \langle S_{11}, \sigma_{113} \rangle \Rightarrow \langle S_{12}, \sigma_{113} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{451} = \langle S_{12}, \sigma_{113} \rangle \Rightarrow \langle S_{16}, \sigma_{113} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{453} = \frac{\langle x := x + 1, \sigma_{113} \rangle \Rightarrow \sigma_{114} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{113} \rangle \Rightarrow \langle S_6, \sigma_{114} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{452} = \frac{\mathcal{T}_{453}}{\langle S_{13}, \sigma_{113} \rangle \Rightarrow \langle S_7, \sigma_{114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{454} = \frac{\langle S_6, \sigma_{114} \rangle \Rightarrow \langle S_8, \sigma_{114} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{114} \rangle \Rightarrow \langle S_9, \sigma_{114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{456} = \langle S_8, \sigma_{114} \rangle \Rightarrow \langle y := y + 1, \sigma_{114} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{455} = \frac{\mathcal{T}_{456}}{\langle S_9, \sigma_{114} \rangle \Rightarrow \langle S_{10}, \sigma_{114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{457} = \frac{\langle y := y + 1, \sigma_{114} \rangle \Rightarrow \sigma_{115} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{114} \rangle \Rightarrow \langle S_{11}, \sigma_{115} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{458} = \langle S_{11}, \sigma_{115} \rangle \Rightarrow \langle S_{12}, \sigma_{115} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{459} = \langle S_{12}, \sigma_{115} \rangle \Rightarrow \langle S_{16}, \sigma_{115} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{461} = \frac{\langle x := x + 1, \sigma_{115} \rangle \Rightarrow \sigma_{116} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{115} \rangle \Rightarrow \langle S_6, \sigma_{116} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{460} = \frac{\mathcal{T}_{461}}{\langle S_{13}, \sigma_{115} \rangle \Rightarrow \langle S_7, \sigma_{116} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{462} = \frac{\langle S_6, \sigma_{116} \rangle \Rightarrow \langle S_8, \sigma_{116} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{116} \rangle \Rightarrow \langle S_9, \sigma_{116} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{464} = \langle S_8, \sigma_{116} \rangle \Rightarrow \langle y := y + 1, \sigma_{116} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{463} = \frac{\mathcal{T}_{464}}{\langle S_9, \sigma_{116} \rangle \Rightarrow \langle S_{10}, \sigma_{116} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{465} = \frac{\langle y := y + 1, \sigma_{116} \rangle \Rightarrow \sigma_{117} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{116} \rangle \Rightarrow \langle S_{11}, \sigma_{117} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{466} = \langle S_{11}, \sigma_{117} \rangle \Rightarrow \langle S_{12}, \sigma_{117} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{467} = \langle S_{12}, \sigma_{117} \rangle \Rightarrow \langle S_{16}, \sigma_{117} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{469} = \frac{\langle x := x + 1, \sigma_{117} \rangle \Rightarrow \sigma_{118} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{117} \rangle \Rightarrow \langle S_6, \sigma_{118} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{468} = \frac{\mathcal{T}_{469}}{\langle S_{13}, \sigma_{117} \rangle \Rightarrow \langle S_7, \sigma_{118} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{470} = \frac{\langle S_6, \sigma_{118} \rangle \Rightarrow \langle S_8, \sigma_{118} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{118} \rangle \Rightarrow \langle S_9, \sigma_{118} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{472} = \langle S_8, \sigma_{118} \rangle \Rightarrow \langle y := y + 1, \sigma_{118} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{471} = \frac{\mathcal{T}_{472}}{\langle S_9, \sigma_{118} \rangle \Rightarrow \langle S_{10}, \sigma_{118} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{473} = \frac{\langle y := y + 1, \sigma_{118} \rangle \Rightarrow \sigma_{119} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{118} \rangle \Rightarrow \langle S_{11}, \sigma_{119} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{474} = \langle S_{11}, \sigma_{119} \rangle \Rightarrow \langle S_{12}, \sigma_{119} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{475} = \langle S_{12}, \sigma_{119} \rangle \Rightarrow \langle S_{16}, \sigma_{119} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{477} = \frac{\langle x := x + 1, \sigma_{119} \rangle \Rightarrow \sigma_{120} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{119} \rangle \Rightarrow \langle S_6, \sigma_{120} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{476} = \frac{\mathcal{T}_{477}}{\langle S_{13}, \sigma_{119} \rangle \Rightarrow \langle S_7, \sigma_{120} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{478} = \frac{\langle S_6, \sigma_{120} \rangle \Rightarrow \langle S_8, \sigma_{120} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{120} \rangle \Rightarrow \langle S_9, \sigma_{120} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{480} = \langle S_8, \sigma_{120} \rangle \Rightarrow \langle y := y + 1, \sigma_{120} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{479} = \frac{\mathcal{T}_{480}}{\langle S_9, \sigma_{120} \rangle \Rightarrow \langle S_{10}, \sigma_{120} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{481} = \frac{\langle y := y + 1, \sigma_{120} \rangle \Rightarrow \sigma_{121} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{120} \rangle \Rightarrow \langle S_{11}, \sigma_{121} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{482} = \langle S_{11}, \sigma_{121} \rangle \Rightarrow \langle S_{12}, \sigma_{121} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{483} = \langle S_{12}, \sigma_{121} \rangle \Rightarrow \langle S_{16}, \sigma_{121} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{485} = \frac{\langle x := x + 1, \sigma_{121} \rangle \Rightarrow \sigma_{122} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{121} \rangle \Rightarrow \langle S_6, \sigma_{122} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{484} = \frac{\mathcal{T}_{485}}{\langle S_{13}, \sigma_{121} \rangle \Rightarrow \langle S_7, \sigma_{122} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{486} = \frac{\langle S_6, \sigma_{122} \rangle \Rightarrow \langle S_8, \sigma_{122} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{122} \rangle \Rightarrow \langle S_9, \sigma_{122} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{488} = \langle S_8, \sigma_{122} \rangle \Rightarrow \langle y := y + 1, \sigma_{122} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{487} = \frac{\mathcal{T}_{488}}{\langle S_9, \sigma_{122} \rangle \Rightarrow \langle S_{10}, \sigma_{122} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$\Downarrow$

$$\mathcal{T}_{489} = \frac{\langle y := y + 1, \sigma_{122} \rangle \Rightarrow \sigma_{123} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{122} \rangle \Rightarrow \langle S_{11}, \sigma_{123} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$\Downarrow$

$$\mathcal{T}_{490} = \langle S_{11}, \sigma_{123} \rangle \Rightarrow \langle S_{12}, \sigma_{123} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{491} = \langle S_{12}, \sigma_{123} \rangle \Rightarrow \langle S_{16}, \sigma_{123} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{493} = \frac{\langle x := x + 1, \sigma_{123} \rangle \Rightarrow \sigma_{124} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{123} \rangle \Rightarrow \langle S_6, \sigma_{124} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{492} = \frac{\mathcal{T}_{493}}{\langle S_{13}, \sigma_{123} \rangle \Rightarrow \langle S_7, \sigma_{124} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$\Downarrow$

$$\mathcal{T}_{494} = \frac{\langle S_6, \sigma_{124} \rangle \Rightarrow \langle S_8, \sigma_{124} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{124} \rangle \Rightarrow \langle S_9, \sigma_{124} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{496} = \langle S_8, \sigma_{124} \rangle \Rightarrow \langle y := y + 1, \sigma_{124} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{495} = \frac{\mathcal{T}_{496}}{\langle S_9, \sigma_{124} \rangle \Rightarrow \langle S_{10}, \sigma_{124} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{497} = \frac{\langle y := y + 1, \sigma_{124} \rangle \Rightarrow \sigma_{125} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{124} \rangle \Rightarrow \langle S_{11}, \sigma_{125} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{498} = \langle S_{11}, \sigma_{125} \rangle \Rightarrow \langle S_{12}, \sigma_{125} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{499} = \langle S_{12}, \sigma_{125} \rangle \Rightarrow \langle S_{16}, \sigma_{125} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{501} = \frac{\langle x := x + 1, \sigma_{125} \rangle \Rightarrow \sigma_{126} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{125} \rangle \Rightarrow \langle S_6, \sigma_{126} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{500} = \frac{\mathcal{T}_{501}}{\langle S_{13}, \sigma_{125} \rangle \Rightarrow \langle S_7, \sigma_{126} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{502} = \frac{\langle S_6, \sigma_{126} \rangle \Rightarrow \langle S_8, \sigma_{126} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{126} \rangle \Rightarrow \langle S_9, \sigma_{126} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{504} = \langle S_8, \sigma_{126} \rangle \Rightarrow \langle y := y + 1, \sigma_{126} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{503} = \frac{\mathcal{T}_{504}}{\langle S_9, \sigma_{126} \rangle \Rightarrow \langle S_{10}, \sigma_{126} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{505} = \frac{\langle y := y + 1, \sigma_{126} \rangle \Rightarrow \sigma_{127} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{126} \rangle \Rightarrow \langle S_{11}, \sigma_{127} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{506} = \langle S_{11}, \sigma_{127} \rangle \Rightarrow \langle S_{12}, \sigma_{127} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{507} = \langle S_{12}, \sigma_{127} \rangle \Rightarrow \langle S_{16}, \sigma_{127} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{509} = \frac{\langle x := x + 1, \sigma_{127} \rangle \Rightarrow \sigma_{128} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{127} \rangle \Rightarrow \langle S_6, \sigma_{128} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{508} = \frac{\mathcal{T}_{509}}{\langle S_{13}, \sigma_{127} \rangle \Rightarrow \langle S_7, \sigma_{128} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{510} = \frac{\langle S_6, \sigma_{128} \rangle \Rightarrow \langle S_8, \sigma_{128} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{128} \rangle \Rightarrow \langle S_9, \sigma_{128} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{512} = \langle S_8, \sigma_{128} \rangle \Rightarrow \langle y := y + 1, \sigma_{128} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{511} = \frac{\mathcal{T}_{512}}{\langle S_9, \sigma_{128} \rangle \Rightarrow \langle S_{10}, \sigma_{128} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{513} = \frac{\langle y := y + 1, \sigma_{128} \rangle \Rightarrow \sigma_{129} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{128} \rangle \Rightarrow \langle S_{11}, \sigma_{129} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{514} = \langle S_{11}, \sigma_{129} \rangle \Rightarrow \langle S_{12}, \sigma_{129} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{515} = \langle S_{12}, \sigma_{129} \rangle \Rightarrow \langle S_{16}, \sigma_{129} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{517} = \frac{\langle x := x + 1, \sigma_{129} \rangle \Rightarrow \sigma_{130} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{129} \rangle \Rightarrow \langle S_6, \sigma_{130} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{516} = \frac{\mathcal{T}_{517}}{\langle S_{13}, \sigma_{129} \rangle \Rightarrow \langle S_7, \sigma_{130} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{518} = \frac{\langle S_6, \sigma_{130} \rangle \Rightarrow \langle S_8, \sigma_{130} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{130} \rangle \Rightarrow \langle S_9, \sigma_{130} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{520} = \langle S_8, \sigma_{130} \rangle \Rightarrow \langle y := y + 1, \sigma_{130} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{519} = \frac{\mathcal{T}_{520}}{\langle S_9, \sigma_{130} \rangle \Rightarrow \langle S_{10}, \sigma_{130} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{521} = \frac{\langle y := y + 1, \sigma_{130} \rangle \Rightarrow \sigma_{131} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{130} \rangle \Rightarrow \langle S_{11}, \sigma_{131} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{522} = \langle S_{11}, \sigma_{131} \rangle \Rightarrow \langle S_{12}, \sigma_{131} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{523} = \langle S_{12}, \sigma_{131} \rangle \Rightarrow \langle S_{16}, \sigma_{131} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{525} = \frac{\langle x := x + 1, \sigma_{131} \rangle \Rightarrow \sigma_{132} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{131} \rangle \Rightarrow \langle S_6, \sigma_{132} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{524} = \frac{\mathcal{T}_{525}}{\langle S_{13}, \sigma_{131} \rangle \Rightarrow \langle S_7, \sigma_{132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{526} = \frac{\langle S_6, \sigma_{132} \rangle \Rightarrow \langle S_8, \sigma_{132} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{132} \rangle \Rightarrow \langle S_9, \sigma_{132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{528} = \langle S_8, \sigma_{132} \rangle \Rightarrow \langle y := y + 1, \sigma_{132} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{527} = \frac{\mathcal{T}_{528}}{\langle S_9, \sigma_{132} \rangle \Rightarrow \langle S_{10}, \sigma_{132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{529} = \frac{\langle y := y + 1, \sigma_{132} \rangle \Rightarrow \sigma_{133} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{132} \rangle \Rightarrow \langle S_{11}, \sigma_{133} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{530} = \langle S_{11}, \sigma_{133} \rangle \Rightarrow \langle S_{12}, \sigma_{133} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{531} = \langle S_{12}, \sigma_{133} \rangle \Rightarrow \langle S_{16}, \sigma_{133} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{533} = \frac{\langle x := x + 1, \sigma_{133} \rangle \Rightarrow \sigma_{134} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{133} \rangle \Rightarrow \langle S_6, \sigma_{134} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{532} = \frac{\mathcal{T}_{533}}{\langle S_{13}, \sigma_{133} \rangle \Rightarrow \langle S_7, \sigma_{134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{534} = \frac{\langle S_6, \sigma_{134} \rangle \Rightarrow \langle S_8, \sigma_{134} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{134} \rangle \Rightarrow \langle S_9, \sigma_{134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{536} = \langle S_8, \sigma_{134} \rangle \Rightarrow \langle y := y + 1, \sigma_{134} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{535} = \frac{\mathcal{T}_{536}}{\langle S_9, \sigma_{134} \rangle \Rightarrow \langle S_{10}, \sigma_{134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{537} = \frac{\langle y := y + 1, \sigma_{134} \rangle \Rightarrow \sigma_{135} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{134} \rangle \Rightarrow \langle S_{11}, \sigma_{135} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{538} = \langle S_{11}, \sigma_{135} \rangle \Rightarrow \langle S_{12}, \sigma_{135} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{539} = \langle S_{12}, \sigma_{135} \rangle \Rightarrow \langle S_{16}, \sigma_{135} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{541} = \frac{\langle x := x + 1, \sigma_{135} \rangle \Rightarrow \sigma_{136} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{135} \rangle \Rightarrow \langle S_6, \sigma_{136} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{540} = \frac{\mathcal{T}_{541}}{\langle S_{13}, \sigma_{135} \rangle \Rightarrow \langle S_7, \sigma_{136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{542} = \frac{\langle S_6, \sigma_{136} \rangle \Rightarrow \langle S_8, \sigma_{136} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{136} \rangle \Rightarrow \langle S_9, \sigma_{136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{544} = \langle S_8, \sigma_{136} \rangle \Rightarrow \langle y := y + 1, \sigma_{136} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{543} = \frac{\mathcal{T}_{544}}{\langle S_9, \sigma_{136} \rangle \Rightarrow \langle S_{10}, \sigma_{136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{545} = \frac{\langle y := y + 1, \sigma_{136} \rangle \Rightarrow \sigma_{137} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{136} \rangle \Rightarrow \langle S_{11}, \sigma_{137} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{546} = \langle S_{11}, \sigma_{137} \rangle \Rightarrow \langle S_{12}, \sigma_{137} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{547} = \langle S_{12}, \sigma_{137} \rangle \Rightarrow \langle S_{16}, \sigma_{137} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{549} = \frac{\langle x := x + 1, \sigma_{137} \rangle \Rightarrow \sigma_{138} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{137} \rangle \Rightarrow \langle S_6, \sigma_{138} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{548} = \frac{\mathcal{T}_{549}}{\langle S_{13}, \sigma_{137} \rangle \Rightarrow \langle S_7, \sigma_{138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{550} = \frac{\langle S_6, \sigma_{138} \rangle \Rightarrow \langle S_8, \sigma_{138} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{138} \rangle \Rightarrow \langle S_9, \sigma_{138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{552} = \langle S_8, \sigma_{138} \rangle \Rightarrow \langle y := y + 1, \sigma_{138} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{551} = \frac{\mathcal{T}_{552}}{\langle S_9, \sigma_{138} \rangle \Rightarrow \langle S_{10}, \sigma_{138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{553} = \frac{\langle y := y + 1, \sigma_{138} \rangle \Rightarrow \sigma_{139} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{138} \rangle \Rightarrow \langle S_{11}, \sigma_{139} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{554} = \langle S_{11}, \sigma_{139} \rangle \Rightarrow \langle S_{12}, \sigma_{139} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{555} = \langle S_{12}, \sigma_{139} \rangle \Rightarrow \langle S_{16}, \sigma_{139} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{557} = \frac{\langle x := x + 1, \sigma_{139} \rangle \Rightarrow \sigma_{140} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{139} \rangle \Rightarrow \langle S_6, \sigma_{140} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{556} = \frac{\mathcal{T}_{557}}{\langle S_{13}, \sigma_{139} \rangle \Rightarrow \langle S_7, \sigma_{140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{558} = \frac{\langle S_6, \sigma_{140} \rangle \Rightarrow \langle S_8, \sigma_{140} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{140} \rangle \Rightarrow \langle S_9, \sigma_{140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{560} = \langle S_8, \sigma_{140} \rangle \Rightarrow \langle y := y + 1, \sigma_{140} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{559} = \frac{\mathcal{T}_{560}}{\langle S_9, \sigma_{140} \rangle \Rightarrow \langle S_{10}, \sigma_{140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{561} = \frac{\langle y := y + 1, \sigma_{140} \rangle \Rightarrow \sigma_{141} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{140} \rangle \Rightarrow \langle S_{11}, \sigma_{141} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{562} = \langle S_{11}, \sigma_{141} \rangle \Rightarrow \langle S_{12}, \sigma_{141} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{563} = \langle S_{12}, \sigma_{141} \rangle \Rightarrow \langle S_{16}, \sigma_{141} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{565} = \frac{\langle x := x + 1, \sigma_{141} \rangle \Rightarrow \sigma_{142} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{141} \rangle \Rightarrow \langle S_6, \sigma_{142} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{564} = \frac{\mathcal{T}_{565}}{\langle S_{13}, \sigma_{141} \rangle \Rightarrow \langle S_7, \sigma_{142} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{566} = \frac{\langle S_6, \sigma_{142} \rangle \Rightarrow \langle S_8, \sigma_{142} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{142} \rangle \Rightarrow \langle S_9, \sigma_{142} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{568} = \langle S_8, \sigma_{142} \rangle \Rightarrow \langle y := y + 1, \sigma_{142} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{567} = \frac{\mathcal{T}_{568}}{\langle S_9, \sigma_{142} \rangle \Rightarrow \langle S_{10}, \sigma_{142} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{569} = \frac{\langle y := y + 1, \sigma_{142} \rangle \Rightarrow \sigma_{143} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{142} \rangle \Rightarrow \langle S_{11}, \sigma_{143} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{570} = \langle S_{11}, \sigma_{143} \rangle \Rightarrow \langle S_{12}, \sigma_{143} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{571} = \langle S_{12}, \sigma_{143} \rangle \Rightarrow \langle S_{16}, \sigma_{143} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{573} = \frac{\langle x := x + 1, \sigma_{143} \rangle \Rightarrow \sigma_{144} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{143} \rangle \Rightarrow \langle S_6, \sigma_{144} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{572} = \frac{\mathcal{T}_{573}}{\langle S_{13}, \sigma_{143} \rangle \Rightarrow \langle S_7, \sigma_{144} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{574} = \frac{\langle S_6, \sigma_{144} \rangle \Rightarrow \langle S_8, \sigma_{144} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{144} \rangle \Rightarrow \langle S_9, \sigma_{144} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{576} = \langle S_8, \sigma_{144} \rangle \Rightarrow \langle y := y + 1, \sigma_{144} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{575} = \frac{\mathcal{T}_{576}}{\langle S_9, \sigma_{144} \rangle \Rightarrow \langle S_{10}, \sigma_{144} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{577} = \frac{\langle y := y + 1, \sigma_{144} \rangle \Rightarrow \sigma_{145} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{144} \rangle \Rightarrow \langle S_{11}, \sigma_{145} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{578} = \langle S_{11}, \sigma_{145} \rangle \Rightarrow \langle S_{12}, \sigma_{145} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{579} = \langle S_{12}, \sigma_{145} \rangle \Rightarrow \langle S_{16}, \sigma_{145} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{581} = \frac{\langle x := x + 1, \sigma_{145} \rangle \Rightarrow \sigma_{146} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{145} \rangle \Rightarrow \langle S_6, \sigma_{146} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{580} = \frac{\mathcal{T}_{581}}{\langle S_{13}, \sigma_{145} \rangle \Rightarrow \langle S_7, \sigma_{146} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{582} = \frac{\langle S_6, \sigma_{146} \rangle \Rightarrow \langle S_8, \sigma_{146} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{146} \rangle \Rightarrow \langle S_9, \sigma_{146} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{584} = \langle S_8, \sigma_{146} \rangle \Rightarrow \langle y := y + 1, \sigma_{146} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{583} = \frac{\mathcal{T}_{584}}{\langle S_9, \sigma_{146} \rangle \Rightarrow \langle S_{10}, \sigma_{146} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{585} = \frac{\langle y := y + 1, \sigma_{146} \rangle \Rightarrow \sigma_{147} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{146} \rangle \Rightarrow \langle S_{11}, \sigma_{147} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{586} = \langle S_{11}, \sigma_{147} \rangle \Rightarrow \langle S_{12}, \sigma_{147} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{587} = \langle S_{12}, \sigma_{147} \rangle \Rightarrow \langle S_{16}, \sigma_{147} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{589} = \frac{\langle x := x + 1, \sigma_{147} \rangle \Rightarrow \sigma_{148} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{147} \rangle \Rightarrow \langle S_6, \sigma_{148} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{588} = \frac{\mathcal{T}_{589}}{\langle S_{13}, \sigma_{147} \rangle \Rightarrow \langle S_7, \sigma_{148} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{590} = \frac{\langle S_6, \sigma_{148} \rangle \Rightarrow \langle S_8, \sigma_{148} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{148} \rangle \Rightarrow \langle S_9, \sigma_{148} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{592} = \langle S_8, \sigma_{148} \rangle \Rightarrow \langle y := y + 1, \sigma_{148} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{591} = \frac{\mathcal{T}_{592}}{\langle S_9, \sigma_{148} \rangle \Rightarrow \langle S_{10}, \sigma_{148} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{593} = \frac{\langle y := y + 1, \sigma_{148} \rangle \Rightarrow \sigma_{149} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{148} \rangle \Rightarrow \langle S_{11}, \sigma_{149} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{594} = \langle S_{11}, \sigma_{149} \rangle \Rightarrow \langle S_{12}, \sigma_{149} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{595} = \langle S_{12}, \sigma_{149} \rangle \Rightarrow \langle S_{16}, \sigma_{149} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{597} = \frac{\langle x := x + 1, \sigma_{149} \rangle \Rightarrow \sigma_{150} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{149} \rangle \Rightarrow \langle S_6, \sigma_{150} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{596} = \frac{\mathcal{T}_{597}}{\langle S_{13}, \sigma_{149} \rangle \Rightarrow \langle S_7, \sigma_{150} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{598} = \frac{\langle S_6, \sigma_{150} \rangle \Rightarrow \langle S_8, \sigma_{150} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{150} \rangle \Rightarrow \langle S_9, \sigma_{150} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{600} = \langle S_8, \sigma_{150} \rangle \Rightarrow \langle y := y + 1, \sigma_{150} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{599} = \frac{\mathcal{T}_{600}}{\langle S_9, \sigma_{150} \rangle \Rightarrow \langle S_{10}, \sigma_{150} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{601} = \frac{\langle y := y + 1, \sigma_{150} \rangle \Rightarrow \sigma_{151} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{150} \rangle \Rightarrow \langle S_{11}, \sigma_{151} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{602} = \langle S_{11}, \sigma_{151} \rangle \Rightarrow \langle S_{12}, \sigma_{151} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{603} = \langle S_{12}, \sigma_{151} \rangle \Rightarrow \langle S_{16}, \sigma_{151} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{605} = \frac{\langle x := x + 1, \sigma_{151} \rangle \Rightarrow \sigma_{152} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{151} \rangle \Rightarrow \langle S_6, \sigma_{152} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{604} = \frac{\mathcal{T}_{605}}{\langle S_{13}, \sigma_{151} \rangle \Rightarrow \langle S_7, \sigma_{152} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{606} = \frac{\langle S_6, \sigma_{152} \rangle \Rightarrow \langle S_8, \sigma_{152} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{152} \rangle \Rightarrow \langle S_9, \sigma_{152} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{608} = \langle S_8, \sigma_{152} \rangle \Rightarrow \langle y := y + 1, \sigma_{152} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{607} = \frac{\mathcal{T}_{608}}{\langle S_9, \sigma_{152} \rangle \Rightarrow \langle S_{10}, \sigma_{152} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{609} = \frac{\langle y := y + 1, \sigma_{152} \rangle \Rightarrow \sigma_{153} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{152} \rangle \Rightarrow \langle S_{11}, \sigma_{153} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{610} = \langle S_{11}, \sigma_{153} \rangle \Rightarrow \langle S_{12}, \sigma_{153} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{611} = \langle S_{12}, \sigma_{153} \rangle \Rightarrow \langle S_{16}, \sigma_{153} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{613} = \frac{\langle x := x + 1, \sigma_{153} \rangle \Rightarrow \sigma_{154} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{153} \rangle \Rightarrow \langle S_6, \sigma_{154} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{612} = \frac{\mathcal{T}_{613}}{\langle S_{13}, \sigma_{153} \rangle \Rightarrow \langle S_7, \sigma_{154} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{614} = \frac{\langle S_6, \sigma_{154} \rangle \Rightarrow \langle S_8, \sigma_{154} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{154} \rangle \Rightarrow \langle S_9, \sigma_{154} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{616} = \langle S_8, \sigma_{154} \rangle \Rightarrow \langle y := y + 1, \sigma_{154} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{615} = \frac{\mathcal{T}_{616}}{\langle S_9, \sigma_{154} \rangle \Rightarrow \langle S_{10}, \sigma_{154} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{617} = \frac{\langle y := y + 1, \sigma_{154} \rangle \Rightarrow \sigma_{155} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{154} \rangle \Rightarrow \langle S_{11}, \sigma_{155} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{618} = \langle S_{11}, \sigma_{155} \rangle \Rightarrow \langle S_{12}, \sigma_{155} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{619} = \langle S_{12}, \sigma_{155} \rangle \Rightarrow \langle S_{16}, \sigma_{155} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{621} = \frac{\langle x := x + 1, \sigma_{155} \rangle \Rightarrow \sigma_{156} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{155} \rangle \Rightarrow \langle S_6, \sigma_{156} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{620} = \frac{\mathcal{T}_{621}}{\langle S_{13}, \sigma_{155} \rangle \Rightarrow \langle S_7, \sigma_{156} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{622} = \frac{\langle S_6, \sigma_{156} \rangle \Rightarrow \langle S_8, \sigma_{156} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{156} \rangle \Rightarrow \langle S_9, \sigma_{156} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{624} = \langle S_8, \sigma_{156} \rangle \Rightarrow \langle y := y + 1, \sigma_{156} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{623} = \frac{\mathcal{T}_{624}}{\langle S_9, \sigma_{156} \rangle \Rightarrow \langle S_{10}, \sigma_{156} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{625} = \frac{\langle y := y + 1, \sigma_{156} \rangle \Rightarrow \sigma_{157} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{156} \rangle \Rightarrow \langle S_{11}, \sigma_{157} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{626} = \langle S_{11}, \sigma_{157} \rangle \Rightarrow \langle S_{12}, \sigma_{157} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{627} = \langle S_{12}, \sigma_{157} \rangle \Rightarrow \langle S_{16}, \sigma_{157} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{629} = \frac{\langle x := x + 1, \sigma_{157} \rangle \Rightarrow \sigma_{158} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{157} \rangle \Rightarrow \langle S_6, \sigma_{158} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{628} = \frac{\mathcal{T}_{629}}{\langle S_{13}, \sigma_{157} \rangle \Rightarrow \langle S_7, \sigma_{158} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{630} = \frac{\langle S_6, \sigma_{158} \rangle \Rightarrow \langle S_8, \sigma_{158} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{158} \rangle \Rightarrow \langle S_9, \sigma_{158} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{632} = \langle S_8, \sigma_{158} \rangle \Rightarrow \langle y := y + 1, \sigma_{158} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{631} = \frac{\mathcal{T}_{632}}{\langle S_9, \sigma_{158} \rangle \Rightarrow \langle S_{10}, \sigma_{158} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{633} = \frac{\langle y := y + 1, \sigma_{158} \rangle \Rightarrow \sigma_{159} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{158} \rangle \Rightarrow \langle S_{11}, \sigma_{159} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{634} = \langle S_{11}, \sigma_{159} \rangle \Rightarrow \langle S_{12}, \sigma_{159} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{635} = \langle S_{12}, \sigma_{159} \rangle \Rightarrow \langle S_{16}, \sigma_{159} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{637} = \frac{\langle x := x + 1, \sigma_{159} \rangle \Rightarrow \sigma_{160} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{159} \rangle \Rightarrow \langle S_6, \sigma_{160} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{636} = \frac{\mathcal{T}_{637}}{\langle S_{13}, \sigma_{159} \rangle \Rightarrow \langle S_7, \sigma_{160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{638} = \frac{\langle S_6, \sigma_{160} \rangle \Rightarrow \langle S_8, \sigma_{160} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{160} \rangle \Rightarrow \langle S_9, \sigma_{160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{640} = \langle S_8, \sigma_{160} \rangle \Rightarrow \langle y := y + 1, \sigma_{160} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{639} = \frac{\mathcal{T}_{640}}{\langle S_9, \sigma_{160} \rangle \Rightarrow \langle S_{10}, \sigma_{160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{641} = \frac{\langle y := y + 1, \sigma_{160} \rangle \Rightarrow \sigma_{161} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{160} \rangle \Rightarrow \langle S_{11}, \sigma_{161} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{642} = \langle S_{11}, \sigma_{161} \rangle \Rightarrow \langle S_{12}, \sigma_{161} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{643} = \langle S_{12}, \sigma_{161} \rangle \Rightarrow \langle S_{16}, \sigma_{161} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{645} = \frac{\langle x := x + 1, \sigma_{161} \rangle \Rightarrow \sigma_{162} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{161} \rangle \Rightarrow \langle S_6, \sigma_{162} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{644} = \frac{\mathcal{T}_{645}}{\langle S_{13}, \sigma_{161} \rangle \Rightarrow \langle S_7, \sigma_{162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{646} = \frac{\langle S_6, \sigma_{162} \rangle \Rightarrow \langle S_8, \sigma_{162} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{162} \rangle \Rightarrow \langle S_9, \sigma_{162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{648} = \langle S_8, \sigma_{162} \rangle \Rightarrow \langle y := y + 1, \sigma_{162} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{647} = \frac{\mathcal{T}_{648}}{\langle S_9, \sigma_{162} \rangle \Rightarrow \langle S_{10}, \sigma_{162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{649} = \frac{\langle y := y + 1, \sigma_{162} \rangle \Rightarrow \sigma_{163} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{162} \rangle \Rightarrow \langle S_{11}, \sigma_{163} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{650} = \langle S_{11}, \sigma_{163} \rangle \Rightarrow \langle S_{12}, \sigma_{163} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{651} = \langle S_{12}, \sigma_{163} \rangle \Rightarrow \langle S_{16}, \sigma_{163} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{653} = \frac{\langle x := x + 1, \sigma_{163} \rangle \Rightarrow \sigma_{164} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{163} \rangle \Rightarrow \langle S_6, \sigma_{164} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{652} = \frac{\mathcal{T}_{653}}{\langle S_{13}, \sigma_{163} \rangle \Rightarrow \langle S_7, \sigma_{164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{654} = \frac{\langle S_6, \sigma_{164} \rangle \Rightarrow \langle S_8, \sigma_{164} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{164} \rangle \Rightarrow \langle S_9, \sigma_{164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{656} = \langle S_8, \sigma_{164} \rangle \Rightarrow \langle y := y + 1, \sigma_{164} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{655} = \frac{\mathcal{T}_{656}}{\langle S_9, \sigma_{164} \rangle \Rightarrow \langle S_{10}, \sigma_{164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{657} = \frac{\langle y := y + 1, \sigma_{164} \rangle \Rightarrow \sigma_{165} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{164} \rangle \Rightarrow \langle S_{11}, \sigma_{165} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{658} = \langle S_{11}, \sigma_{165} \rangle \Rightarrow \langle S_{12}, \sigma_{165} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{659} = \langle S_{12}, \sigma_{165} \rangle \Rightarrow \langle S_{16}, \sigma_{165} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{661} = \frac{\langle x := x + 1, \sigma_{165} \rangle \Rightarrow \sigma_{166} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{165} \rangle \Rightarrow \langle S_6, \sigma_{166} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{660} = \frac{\mathcal{T}_{661}}{\langle S_{13}, \sigma_{165} \rangle \Rightarrow \langle S_7, \sigma_{166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{662} = \frac{\langle S_6, \sigma_{166} \rangle \Rightarrow \langle S_8, \sigma_{166} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{166} \rangle \Rightarrow \langle S_9, \sigma_{166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{664} = \langle S_8, \sigma_{166} \rangle \Rightarrow \langle y := y + 1, \sigma_{166} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{663} = \frac{\mathcal{T}_{664}}{\langle S_9, \sigma_{166} \rangle \Rightarrow \langle S_{10}, \sigma_{166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{665} = \frac{\langle y := y + 1, \sigma_{166} \rangle \Rightarrow \sigma_{167} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{166} \rangle \Rightarrow \langle S_{11}, \sigma_{167} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{666} = \langle S_{11}, \sigma_{167} \rangle \Rightarrow \langle S_{12}, \sigma_{167} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{667} = \langle S_{12}, \sigma_{167} \rangle \Rightarrow \langle S_{16}, \sigma_{167} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{669} = \frac{\langle x := x + 1, \sigma_{167} \rangle \Rightarrow \sigma_{168} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{167} \rangle \Rightarrow \langle S_6, \sigma_{168} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{668} = \frac{\mathcal{T}_{669}}{\langle S_{13}, \sigma_{167} \rangle \Rightarrow \langle S_7, \sigma_{168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{670} = \frac{\langle S_6, \sigma_{168} \rangle \Rightarrow \langle S_8, \sigma_{168} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{168} \rangle \Rightarrow \langle S_9, \sigma_{168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{672} = \langle S_8, \sigma_{168} \rangle \Rightarrow \langle y := y + 1, \sigma_{168} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{671} = \frac{\mathcal{T}_{672}}{\langle S_9, \sigma_{168} \rangle \Rightarrow \langle S_{10}, \sigma_{168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{673} = \frac{\langle y := y + 1, \sigma_{168} \rangle \Rightarrow \sigma_{169} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{168} \rangle \Rightarrow \langle S_{11}, \sigma_{169} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{674} = \langle S_{11}, \sigma_{169} \rangle \Rightarrow \langle S_{12}, \sigma_{169} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{675} = \langle S_{12}, \sigma_{169} \rangle \Rightarrow \langle S_{16}, \sigma_{169} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{677} = \frac{\langle x := x + 1, \sigma_{169} \rangle \Rightarrow \sigma_{170} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{169} \rangle \Rightarrow \langle S_6, \sigma_{170} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{676} = \frac{\mathcal{T}_{677}}{\langle S_{13}, \sigma_{169} \rangle \Rightarrow \langle S_7, \sigma_{170} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{678} = \frac{\langle S_6, \sigma_{170} \rangle \Rightarrow \langle S_8, \sigma_{170} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{170} \rangle \Rightarrow \langle S_9, \sigma_{170} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{680} = \langle S_8, \sigma_{170} \rangle \Rightarrow \langle y := y + 1, \sigma_{170} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{679} = \frac{\mathcal{T}_{680}}{\langle S_9, \sigma_{170} \rangle \Rightarrow \langle S_{10}, \sigma_{170} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{681} = \frac{\langle y := y + 1, \sigma_{170} \rangle \Rightarrow \sigma_{171} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{170} \rangle \Rightarrow \langle S_{11}, \sigma_{171} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{682} = \langle S_{11}, \sigma_{171} \rangle \Rightarrow \langle S_{12}, \sigma_{171} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{683} = \langle S_{12}, \sigma_{171} \rangle \Rightarrow \langle S_{16}, \sigma_{171} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{685} = \frac{\langle x := x + 1, \sigma_{171} \rangle \Rightarrow \sigma_{172} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{171} \rangle \Rightarrow \langle S_6, \sigma_{172} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{684} = \frac{\mathcal{T}_{685}}{\langle S_{13}, \sigma_{171} \rangle \Rightarrow \langle S_7, \sigma_{172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{686} = \frac{\langle S_6, \sigma_{172} \rangle \Rightarrow \langle S_8, \sigma_{172} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{172} \rangle \Rightarrow \langle S_9, \sigma_{172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{688} = \langle S_8, \sigma_{172} \rangle \Rightarrow \langle y := y + 1, \sigma_{172} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{687} = \frac{\mathcal{T}_{688}}{\langle S_9, \sigma_{172} \rangle \Rightarrow \langle S_{10}, \sigma_{172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{689} = \frac{\langle y := y + 1, \sigma_{172} \rangle \Rightarrow \sigma_{173} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{172} \rangle \Rightarrow \langle S_{11}, \sigma_{173} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{690} = \langle S_{11}, \sigma_{173} \rangle \Rightarrow \langle S_{12}, \sigma_{173} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{691} = \langle S_{12}, \sigma_{173} \rangle \Rightarrow \langle S_{16}, \sigma_{173} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{693} = \frac{\langle x := x + 1, \sigma_{173} \rangle \Rightarrow \sigma_{174} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{173} \rangle \Rightarrow \langle S_6, \sigma_{174} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{692} = \frac{\mathcal{T}_{693}}{\langle S_{13}, \sigma_{173} \rangle \Rightarrow \langle S_7, \sigma_{174} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{694} = \frac{\langle S_6, \sigma_{174} \rangle \Rightarrow \langle S_8, \sigma_{174} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{174} \rangle \Rightarrow \langle S_9, \sigma_{174} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{696} = \langle S_8, \sigma_{174} \rangle \Rightarrow \langle y := y + 1, \sigma_{174} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{695} = \frac{\mathcal{T}_{696}}{\langle S_9, \sigma_{174} \rangle \Rightarrow \langle S_{10}, \sigma_{174} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{697} = \frac{\langle y := y + 1, \sigma_{174} \rangle \Rightarrow \sigma_{175} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{174} \rangle \Rightarrow \langle S_{11}, \sigma_{175} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{698} = \langle S_{11}, \sigma_{175} \rangle \Rightarrow \langle S_{12}, \sigma_{175} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{699} = \langle S_{12}, \sigma_{175} \rangle \Rightarrow \langle S_{16}, \sigma_{175} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{701} = \frac{\langle x := x + 1, \sigma_{175} \rangle \Rightarrow \sigma_{176} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{175} \rangle \Rightarrow \langle S_6, \sigma_{176} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{700} = \frac{\mathcal{T}_{701}}{\langle S_{13}, \sigma_{175} \rangle \Rightarrow \langle S_7, \sigma_{176} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{702} = \frac{\langle S_6, \sigma_{176} \rangle \Rightarrow \langle S_8, \sigma_{176} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{176} \rangle \Rightarrow \langle S_9, \sigma_{176} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{704} = \langle S_8, \sigma_{176} \rangle \Rightarrow \langle y := y + 1, \sigma_{176} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{703} = \frac{\mathcal{T}_{704}}{\langle S_9, \sigma_{176} \rangle \Rightarrow \langle S_{10}, \sigma_{176} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{705} = \frac{\langle y := y + 1, \sigma_{176} \rangle \Rightarrow \sigma_{177} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{176} \rangle \Rightarrow \langle S_{11}, \sigma_{177} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{706} = \langle S_{11}, \sigma_{177} \rangle \Rightarrow \langle S_{12}, \sigma_{177} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{707} = \langle S_{12}, \sigma_{177} \rangle \Rightarrow \langle S_{16}, \sigma_{177} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{709} = \frac{\langle x := x + 1, \sigma_{177} \rangle \Rightarrow \sigma_{178} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{177} \rangle \Rightarrow \langle S_6, \sigma_{178} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{708} = \frac{\mathcal{T}_{709}}{\langle S_{13}, \sigma_{177} \rangle \Rightarrow \langle S_7, \sigma_{178} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{710} = \frac{\langle S_6, \sigma_{178} \rangle \Rightarrow \langle S_8, \sigma_{178} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{178} \rangle \Rightarrow \langle S_9, \sigma_{178} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{712} = \langle S_8, \sigma_{178} \rangle \Rightarrow \langle y := y + 1, \sigma_{178} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{711} = \frac{\mathcal{T}_{712}}{\langle S_9, \sigma_{178} \rangle \Rightarrow \langle S_{10}, \sigma_{178} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{713} = \frac{\langle y := y + 1, \sigma_{178} \rangle \Rightarrow \sigma_{179} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{178} \rangle \Rightarrow \langle S_{11}, \sigma_{179} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{714} = \langle S_{11}, \sigma_{179} \rangle \Rightarrow \langle S_{12}, \sigma_{179} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{715} = \langle S_{12}, \sigma_{179} \rangle \Rightarrow \langle S_{16}, \sigma_{179} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{717} = \frac{\langle x := x + 1, \sigma_{179} \rangle \Rightarrow \sigma_{180} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{179} \rangle \Rightarrow \langle S_6, \sigma_{180} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{716} = \frac{\mathcal{T}_{717}}{\langle S_{13}, \sigma_{179} \rangle \Rightarrow \langle S_7, \sigma_{180} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{718} = \frac{\langle S_6, \sigma_{180} \rangle \Rightarrow \langle S_8, \sigma_{180} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{180} \rangle \Rightarrow \langle S_9, \sigma_{180} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{720} = \langle S_8, \sigma_{180} \rangle \Rightarrow \langle y := y + 1, \sigma_{180} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{719} = \frac{\mathcal{T}_{720}}{\langle S_9, \sigma_{180} \rangle \Rightarrow \langle S_{10}, \sigma_{180} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{721} = \frac{\langle y := y + 1, \sigma_{180} \rangle \Rightarrow \sigma_{181} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{180} \rangle \Rightarrow \langle S_{11}, \sigma_{181} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{722} = \langle S_{11}, \sigma_{181} \rangle \Rightarrow \langle S_{12}, \sigma_{181} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{723} = \langle S_{12}, \sigma_{181} \rangle \Rightarrow \langle S_{16}, \sigma_{181} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{725} = \frac{\langle x := x + 1, \sigma_{181} \rangle \Rightarrow \sigma_{182} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{181} \rangle \Rightarrow \langle S_6, \sigma_{182} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{724} = \frac{\mathcal{T}_{725}}{\langle S_{13}, \sigma_{181} \rangle \Rightarrow \langle S_7, \sigma_{182} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{726} = \frac{\langle S_6, \sigma_{182} \rangle \Rightarrow \langle S_8, \sigma_{182} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{182} \rangle \Rightarrow \langle S_9, \sigma_{182} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{728} = \langle S_8, \sigma_{182} \rangle \Rightarrow \langle y := y + 1, \sigma_{182} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{727} = \frac{\mathcal{T}_{728}}{\langle S_9, \sigma_{182} \rangle \Rightarrow \langle S_{10}, \sigma_{182} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{729} = \frac{\langle y := y + 1, \sigma_{182} \rangle \Rightarrow \sigma_{183} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{182} \rangle \Rightarrow \langle S_{11}, \sigma_{183} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{730} = \langle S_{11}, \sigma_{183} \rangle \Rightarrow \langle S_{12}, \sigma_{183} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{731} = \langle S_{12}, \sigma_{183} \rangle \Rightarrow \langle S_{16}, \sigma_{183} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{733} = \frac{\langle x := x + 1, \sigma_{183} \rangle \Rightarrow \sigma_{184} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{183} \rangle \Rightarrow \langle S_6, \sigma_{184} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{732} = \frac{\mathcal{T}_{733}}{\langle S_{13}, \sigma_{183} \rangle \Rightarrow \langle S_7, \sigma_{184} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{734} = \frac{\langle S_6, \sigma_{184} \rangle \Rightarrow \langle S_8, \sigma_{184} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{184} \rangle \Rightarrow \langle S_9, \sigma_{184} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{736} = \langle S_8, \sigma_{184} \rangle \Rightarrow \langle y := y + 1, \sigma_{184} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{735} = \frac{\mathcal{T}_{736}}{\langle S_9, \sigma_{184} \rangle \Rightarrow \langle S_{10}, \sigma_{184} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{737} = \frac{\langle y := y + 1, \sigma_{184} \rangle \Rightarrow \sigma_{185} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{184} \rangle \Rightarrow \langle S_{11}, \sigma_{185} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{738} = \langle S_{11}, \sigma_{185} \rangle \Rightarrow \langle S_{12}, \sigma_{185} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{739} = \langle S_{12}, \sigma_{185} \rangle \Rightarrow \langle S_{16}, \sigma_{185} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{741} = \frac{\langle x := x + 1, \sigma_{185} \rangle \Rightarrow \sigma_{186} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{185} \rangle \Rightarrow \langle S_6, \sigma_{186} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{740} = \frac{\mathcal{T}_{741}}{\langle S_{13}, \sigma_{185} \rangle \Rightarrow \langle S_7, \sigma_{186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{742} = \frac{\langle S_6, \sigma_{186} \rangle \Rightarrow \langle S_8, \sigma_{186} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{186} \rangle \Rightarrow \langle S_9, \sigma_{186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{744} = \langle S_8, \sigma_{186} \rangle \Rightarrow \langle y := y + 1, \sigma_{186} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{743} = \frac{\mathcal{T}_{744}}{\langle S_9, \sigma_{186} \rangle \Rightarrow \langle S_{10}, \sigma_{186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{745} = \frac{\langle y := y + 1, \sigma_{186} \rangle \Rightarrow \sigma_{187} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{186} \rangle \Rightarrow \langle S_{11}, \sigma_{187} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{746} = \langle S_{11}, \sigma_{187} \rangle \Rightarrow \langle S_{12}, \sigma_{187} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{747} = \langle S_{12}, \sigma_{187} \rangle \Rightarrow \langle S_{16}, \sigma_{187} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{749} = \frac{\langle x := x + 1, \sigma_{187} \rangle \Rightarrow \sigma_{188} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{187} \rangle \Rightarrow \langle S_6, \sigma_{188} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{748} = \frac{\mathcal{T}_{749}}{\langle S_{13}, \sigma_{187} \rangle \Rightarrow \langle S_7, \sigma_{188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{750} = \frac{\langle S_6, \sigma_{188} \rangle \Rightarrow \langle S_8, \sigma_{188} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{188} \rangle \Rightarrow \langle S_9, \sigma_{188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{752} = \langle S_8, \sigma_{188} \rangle \Rightarrow \langle y := y + 1, \sigma_{188} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{751} = \frac{\mathcal{T}_{752}}{\langle S_9, \sigma_{188} \rangle \Rightarrow \langle S_{10}, \sigma_{188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{753} = \frac{\langle y := y + 1, \sigma_{188} \rangle \Rightarrow \sigma_{189} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{188} \rangle \Rightarrow \langle S_{11}, \sigma_{189} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{754} = \langle S_{11}, \sigma_{189} \rangle \Rightarrow \langle S_{12}, \sigma_{189} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{755} = \langle S_{12}, \sigma_{189} \rangle \Rightarrow \langle S_{16}, \sigma_{189} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{757} = \frac{\langle x := x + 1, \sigma_{189} \rangle \Rightarrow \sigma_{190} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{189} \rangle \Rightarrow \langle S_6, \sigma_{190} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{756} = \frac{\mathcal{T}_{757}}{\langle S_{13}, \sigma_{189} \rangle \Rightarrow \langle S_7, \sigma_{190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{758} = \frac{\langle S_6, \sigma_{190} \rangle \Rightarrow \langle S_8, \sigma_{190} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{190} \rangle \Rightarrow \langle S_9, \sigma_{190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{760} = \langle S_8, \sigma_{190} \rangle \Rightarrow \langle y := y + 1, \sigma_{190} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{759} = \frac{\mathcal{T}_{760}}{\langle S_9, \sigma_{190} \rangle \Rightarrow \langle S_{10}, \sigma_{190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{761} = \frac{\langle y := y + 1, \sigma_{190} \rangle \Rightarrow \sigma_{191} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{190} \rangle \Rightarrow \langle S_{11}, \sigma_{191} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{762} = \langle S_{11}, \sigma_{191} \rangle \Rightarrow \langle S_{12}, \sigma_{191} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{763} = \langle S_{12}, \sigma_{191} \rangle \Rightarrow \langle S_{16}, \sigma_{191} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{765} = \frac{\langle x := x + 1, \sigma_{191} \rangle \Rightarrow \sigma_{192} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{191} \rangle \Rightarrow \langle S_6, \sigma_{192} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{764} = \frac{\mathcal{T}_{765}}{\langle S_{13}, \sigma_{191} \rangle \Rightarrow \langle S_7, \sigma_{192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{766} = \frac{\langle S_6, \sigma_{192} \rangle \Rightarrow \langle S_8, \sigma_{192} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{192} \rangle \Rightarrow \langle S_9, \sigma_{192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{768} = \langle S_8, \sigma_{192} \rangle \Rightarrow \langle y := y + 1, \sigma_{192} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{767} = \frac{\mathcal{T}_{768}}{\langle S_9, \sigma_{192} \rangle \Rightarrow \langle S_{10}, \sigma_{192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{769} = \frac{\langle y := y + 1, \sigma_{192} \rangle \Rightarrow \sigma_{193} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{192} \rangle \Rightarrow \langle S_{11}, \sigma_{193} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{770} = \langle S_{11}, \sigma_{193} \rangle \Rightarrow \langle S_{12}, \sigma_{193} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{771} = \langle S_{12}, \sigma_{193} \rangle \Rightarrow \langle S_{16}, \sigma_{193} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{773} = \frac{\langle x := x + 1, \sigma_{193} \rangle \Rightarrow \sigma_{194} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{193} \rangle \Rightarrow \langle S_6, \sigma_{194} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{772} = \frac{\mathcal{T}_{773}}{\langle S_{13}, \sigma_{193} \rangle \Rightarrow \langle S_7, \sigma_{194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{774} = \frac{\langle S_6, \sigma_{194} \rangle \Rightarrow \langle S_8, \sigma_{194} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{194} \rangle \Rightarrow \langle S_9, \sigma_{194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{776} = \langle S_8, \sigma_{194} \rangle \Rightarrow \langle y := y + 1, \sigma_{194} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{775} = \frac{\mathcal{T}_{776}}{\langle S_9, \sigma_{194} \rangle \Rightarrow \langle S_{10}, \sigma_{194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{777} = \frac{\langle y := y + 1, \sigma_{194} \rangle \Rightarrow \sigma_{195} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{194} \rangle \Rightarrow \langle S_{11}, \sigma_{195} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{778} = \langle S_{11}, \sigma_{195} \rangle \Rightarrow \langle S_{12}, \sigma_{195} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{779} = \langle S_{12}, \sigma_{195} \rangle \Rightarrow \langle S_{16}, \sigma_{195} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{781} = \frac{\langle x := x + 1, \sigma_{195} \rangle \Rightarrow \sigma_{196} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{195} \rangle \Rightarrow \langle S_6, \sigma_{196} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{780} = \frac{\mathcal{T}_{781}}{\langle S_{13}, \sigma_{195} \rangle \Rightarrow \langle S_7, \sigma_{196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{782} = \frac{\langle S_6, \sigma_{196} \rangle \Rightarrow \langle S_8, \sigma_{196} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{196} \rangle \Rightarrow \langle S_9, \sigma_{196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{784} = \langle S_8, \sigma_{196} \rangle \Rightarrow \langle y := y + 1, \sigma_{196} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{783} = \frac{\mathcal{T}_{784}}{\langle S_9, \sigma_{196} \rangle \Rightarrow \langle S_{10}, \sigma_{196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{785} = \frac{\langle y := y + 1, \sigma_{196} \rangle \Rightarrow \sigma_{197} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{196} \rangle \Rightarrow \langle S_{11}, \sigma_{197} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{786} = \langle S_{11}, \sigma_{197} \rangle \Rightarrow \langle S_{12}, \sigma_{197} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{787} = \langle S_{12}, \sigma_{197} \rangle \Rightarrow \langle S_{16}, \sigma_{197} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{789} = \frac{\langle x := x + 1, \sigma_{197} \rangle \Rightarrow \sigma_{198} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{197} \rangle \Rightarrow \langle S_6, \sigma_{198} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{788} = \frac{\mathcal{T}_{789}}{\langle S_{13}, \sigma_{197} \rangle \Rightarrow \langle S_7, \sigma_{198} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{790} = \frac{\langle S_6, \sigma_{198} \rangle \Rightarrow \langle S_8, \sigma_{198} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{198} \rangle \Rightarrow \langle S_9, \sigma_{198} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{792} = \langle S_8, \sigma_{198} \rangle \Rightarrow \langle y := y + 1, \sigma_{198} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{791} = \frac{\mathcal{T}_{792}}{\langle S_9, \sigma_{198} \rangle \Rightarrow \langle S_{10}, \sigma_{198} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{793} = \frac{\langle y := y + 1, \sigma_{198} \rangle \Rightarrow \sigma_{199} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{198} \rangle \Rightarrow \langle S_{11}, \sigma_{199} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{794} = \langle S_{11}, \sigma_{199} \rangle \Rightarrow \langle S_{12}, \sigma_{199} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{795} = \langle S_{12}, \sigma_{199} \rangle \Rightarrow \langle S_{16}, \sigma_{199} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{797} = \frac{\langle x := x + 1, \sigma_{199} \rangle \Rightarrow \sigma_{200} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{199} \rangle \Rightarrow \langle S_6, \sigma_{200} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{796} = \frac{\mathcal{T}_{797}}{\langle S_{13}, \sigma_{199} \rangle \Rightarrow \langle S_7, \sigma_{200} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{798} = \frac{\langle S_6, \sigma_{200} \rangle \Rightarrow \langle S_8, \sigma_{200} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{200} \rangle \Rightarrow \langle S_9, \sigma_{200} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{800} = \langle S_8, \sigma_{200} \rangle \Rightarrow \langle y := y + 1, \sigma_{200} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{799} = \frac{\mathcal{T}_{800}}{\langle S_9, \sigma_{200} \rangle \Rightarrow \langle S_{10}, \sigma_{200} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{801} = \frac{\langle y := y + 1, \sigma_{200} \rangle \Rightarrow \sigma_{201} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{200} \rangle \Rightarrow \langle S_{11}, \sigma_{201} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{802} = \langle S_{11}, \sigma_{201} \rangle \Rightarrow \langle S_{12}, \sigma_{201} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{803} = \langle S_{12}, \sigma_{201} \rangle \Rightarrow \langle S_{16}, \sigma_{201} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{805} = \frac{\langle x := x + 1, \sigma_{201} \rangle \Rightarrow \sigma_{202} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{201} \rangle \Rightarrow \langle S_6, \sigma_{202} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{804} = \frac{\mathcal{T}_{805}}{\langle S_{13}, \sigma_{201} \rangle \Rightarrow \langle S_7, \sigma_{202} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{806} = \frac{\langle S_6, \sigma_{202} \rangle \Rightarrow \langle S_8, \sigma_{202} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{202} \rangle \Rightarrow \langle S_9, \sigma_{202} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{808} = \langle S_8, \sigma_{202} \rangle \Rightarrow \langle y := y + 1, \sigma_{202} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{807} = \frac{\mathcal{T}_{808}}{\langle S_9, \sigma_{202} \rangle \Rightarrow \langle S_{10}, \sigma_{202} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{809} = \frac{\langle y := y + 1, \sigma_{202} \rangle \Rightarrow \sigma_{203} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{202} \rangle \Rightarrow \langle S_{11}, \sigma_{203} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{810} = \langle S_{11}, \sigma_{203} \rangle \Rightarrow \langle S_{12}, \sigma_{203} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{811} = \langle S_{12}, \sigma_{203} \rangle \Rightarrow \langle S_{13}, \sigma_{203} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{813} = \frac{\langle x := x + 1, \sigma_{203} \rangle \Rightarrow \sigma_{204} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{203} \rangle \Rightarrow \langle S_6, \sigma_{204} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{812} = \frac{\mathcal{T}_{813}}{\langle S_{13}, \sigma_{203} \rangle \Rightarrow \langle S_{15}, \sigma_{204} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{814} = \frac{\langle S_6, \sigma_{204} \rangle \Rightarrow \langle S_8, \sigma_{204} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{204} \rangle \Rightarrow \langle S_9, \sigma_{204} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{816} = \langle S_8, \sigma_{204} \rangle \Rightarrow \langle y := y + 1, \sigma_{204} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{815} = \frac{\mathcal{T}_{816}}{\langle S_9, \sigma_{204} \rangle \Rightarrow \langle S_{10}, \sigma_{204} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{817} = \frac{\langle y := y + 1, \sigma_{204} \rangle \Rightarrow \sigma_{205} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{204} \rangle \Rightarrow \langle S_{11}, \sigma_{205} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{818} = \langle S_{11}, \sigma_{205} \rangle \Rightarrow \langle S_{12}, \sigma_{205} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{819} = \langle S_{12}, \sigma_{205} \rangle \Rightarrow \langle S_{13}, \sigma_{205} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{821} = \frac{\langle x := x + 1, \sigma_{205} \rangle \Rightarrow \sigma_{206} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{205} \rangle \Rightarrow \langle S_6, \sigma_{206} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{820} = \frac{\mathcal{T}_{821}}{\langle S_{13}, \sigma_{205} \rangle \Rightarrow \langle S_{15}, \sigma_{206} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{822} = \frac{\langle S_6, \sigma_{206} \rangle \Rightarrow \langle S_8, \sigma_{206} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{206} \rangle \Rightarrow \langle S_9, \sigma_{206} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{824} = \langle S_8, \sigma_{206} \rangle \Rightarrow \langle y := y + 1, \sigma_{206} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{823} = \frac{\mathcal{T}_{824}}{\langle S_9, \sigma_{206} \rangle \Rightarrow \langle S_{10}, \sigma_{206} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{825} = \frac{\langle y := y + 1, \sigma_{206} \rangle \Rightarrow \sigma_{207} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{206} \rangle \Rightarrow \langle S_{11}, \sigma_{207} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{826} = \langle S_{11}, \sigma_{207} \rangle \Rightarrow \langle S_{12}, \sigma_{207} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{827} = \langle S_{12}, \sigma_{207} \rangle \Rightarrow \langle S_{13}, \sigma_{207} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{829} = \frac{\langle x := x + 1, \sigma_{207} \rangle \Rightarrow \sigma_{208} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{207} \rangle \Rightarrow \langle S_6, \sigma_{208} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{828} = \frac{\mathcal{T}_{829}}{\langle S_{13}, \sigma_{207} \rangle \Rightarrow \langle S_{15}, \sigma_{208} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{830} = \frac{\langle S_6, \sigma_{208} \rangle \Rightarrow \langle S_8, \sigma_{208} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{208} \rangle \Rightarrow \langle S_9, \sigma_{208} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{832} = \langle S_8, \sigma_{208} \rangle \Rightarrow \langle y := y + 1, \sigma_{208} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{831} = \frac{\mathcal{T}_{832}}{\langle S_9, \sigma_{208} \rangle \Rightarrow \langle S_{10}, \sigma_{208} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{833} = \frac{\langle y := y + 1, \sigma_{208} \rangle \Rightarrow \sigma_{209} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{208} \rangle \Rightarrow \langle S_{11}, \sigma_{209} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{834} = \langle S_{11}, \sigma_{209} \rangle \Rightarrow \langle S_{12}, \sigma_{209} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{835} = \langle S_{12}, \sigma_{209} \rangle \Rightarrow \langle S_{13}, \sigma_{209} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{837} = \frac{\langle x := x + 1, \sigma_{209} \rangle \Rightarrow \sigma_{210} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{209} \rangle \Rightarrow \langle S_6, \sigma_{210} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{836} = \frac{\mathcal{T}_{837}}{\langle S_{13}, \sigma_{209} \rangle \Rightarrow \langle S_{15}, \sigma_{210} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{838} = \frac{\langle S_6, \sigma_{210} \rangle \Rightarrow \langle S_8, \sigma_{210} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{210} \rangle \Rightarrow \langle S_9, \sigma_{210} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{840} = \langle S_8, \sigma_{210} \rangle \Rightarrow \langle y := y + 1, \sigma_{210} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{839} = \frac{\mathcal{T}_{840}}{\langle S_9, \sigma_{210} \rangle \Rightarrow \langle S_{10}, \sigma_{210} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{841} = \frac{\langle y := y + 1, \sigma_{210} \rangle \Rightarrow \sigma_{211} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{210} \rangle \Rightarrow \langle S_{11}, \sigma_{211} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{842} = \langle S_{11}, \sigma_{211} \rangle \Rightarrow \langle S_{12}, \sigma_{211} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{843} = \langle S_{12}, \sigma_{211} \rangle \Rightarrow \langle S_{13}, \sigma_{211} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{845} = \frac{\langle x := x + 1, \sigma_{211} \rangle \Rightarrow \sigma_{212} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{211} \rangle \Rightarrow \langle S_6, \sigma_{212} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{844} = \frac{\mathcal{T}_{845}}{\langle S_{13}, \sigma_{211} \rangle \Rightarrow \langle S_{15}, \sigma_{212} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{846} = \frac{\langle S_6, \sigma_{212} \rangle \Rightarrow \langle S_8, \sigma_{212} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{212} \rangle \Rightarrow \langle S_9, \sigma_{212} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{848} = \langle S_8, \sigma_{212} \rangle \Rightarrow \langle y := y + 1, \sigma_{212} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{847} = \frac{\mathcal{T}_{848}}{\langle S_9, \sigma_{212} \rangle \Rightarrow \langle S_{10}, \sigma_{212} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{849} = \frac{\langle y := y + 1, \sigma_{212} \rangle \Rightarrow \sigma_{213} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{212} \rangle \Rightarrow \langle S_{11}, \sigma_{213} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{850} = \langle S_{11}, \sigma_{213} \rangle \Rightarrow \langle S_{12}, \sigma_{213} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{851} = \langle S_{12}, \sigma_{213} \rangle \Rightarrow \langle S_{13}, \sigma_{213} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{853} = \frac{\langle x := x + 1, \sigma_{213} \rangle \Rightarrow \sigma_{214} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{213} \rangle \Rightarrow \langle S_6, \sigma_{214} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{852} = \frac{\mathcal{T}_{853}}{\langle S_{13}, \sigma_{213} \rangle \Rightarrow \langle S_{15}, \sigma_{214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{854} = \frac{\langle S_6, \sigma_{214} \rangle \Rightarrow \langle S_8, \sigma_{214} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{214} \rangle \Rightarrow \langle S_9, \sigma_{214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{856} = \langle S_8, \sigma_{214} \rangle \Rightarrow \langle y := y + 1, \sigma_{214} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{855} = \frac{\mathcal{T}_{856}}{\langle S_9, \sigma_{214} \rangle \Rightarrow \langle S_{10}, \sigma_{214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{857} = \frac{\langle y := y + 1, \sigma_{214} \rangle \Rightarrow \sigma_{215} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{214} \rangle \Rightarrow \langle S_{11}, \sigma_{215} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{858} = \langle S_{11}, \sigma_{215} \rangle \Rightarrow \langle S_{12}, \sigma_{215} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{859} = \langle S_{12}, \sigma_{215} \rangle \Rightarrow \langle S_{13}, \sigma_{215} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{861} = \frac{\langle x := x + 1, \sigma_{215} \rangle \Rightarrow \sigma_{216} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{215} \rangle \Rightarrow \langle S_6, \sigma_{216} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{860} = \frac{\mathcal{T}_{861}}{\langle S_{13}, \sigma_{215} \rangle \Rightarrow \langle S_{15}, \sigma_{216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{862} = \frac{\langle S_6, \sigma_{216} \rangle \Rightarrow \langle S_8, \sigma_{216} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{216} \rangle \Rightarrow \langle S_9, \sigma_{216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{864} = \langle S_8, \sigma_{216} \rangle \Rightarrow \langle y := y + 1, \sigma_{216} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{863} = \frac{\mathcal{T}_{864}}{\langle S_9, \sigma_{216} \rangle \Rightarrow \langle S_{10}, \sigma_{216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{865} = \frac{\langle y := y + 1, \sigma_{216} \rangle \Rightarrow \sigma_{217} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{216} \rangle \Rightarrow \langle S_{11}, \sigma_{217} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{866} = \langle S_{11}, \sigma_{217} \rangle \Rightarrow \langle S_{12}, \sigma_{217} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{867} = \langle S_{12}, \sigma_{217} \rangle \Rightarrow \langle S_{13}, \sigma_{217} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{869} = \frac{\langle x := x + 1, \sigma_{217} \rangle \Rightarrow \sigma_{218} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{217} \rangle \Rightarrow \langle S_6, \sigma_{218} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{868} = \frac{\mathcal{T}_{869}}{\langle S_{13}, \sigma_{217} \rangle \Rightarrow \langle S_{15}, \sigma_{218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{870} = \frac{\langle S_6, \sigma_{218} \rangle \Rightarrow \langle S_8, \sigma_{218} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{218} \rangle \Rightarrow \langle S_9, \sigma_{218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{872} = \langle S_8, \sigma_{218} \rangle \Rightarrow \langle y := y + 1, \sigma_{218} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{871} = \frac{\mathcal{T}_{872}}{\langle S_9, \sigma_{218} \rangle \Rightarrow \langle S_{10}, \sigma_{218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{873} = \frac{\langle y := y + 1, \sigma_{218} \rangle \Rightarrow \sigma_{219} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{218} \rangle \Rightarrow \langle S_{11}, \sigma_{219} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{874} = \langle S_{11}, \sigma_{219} \rangle \Rightarrow \langle S_{12}, \sigma_{219} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{875} = \langle S_{12}, \sigma_{219} \rangle \Rightarrow \langle S_{13}, \sigma_{219} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{877} = \frac{\langle x := x + 1, \sigma_{219} \rangle \Rightarrow \sigma_{220} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{219} \rangle \Rightarrow \langle S_6, \sigma_{220} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{876} = \frac{\mathcal{T}_{877}}{\langle S_{13}, \sigma_{219} \rangle \Rightarrow \langle S_{15}, \sigma_{220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{878} = \frac{\langle S_6, \sigma_{220} \rangle \Rightarrow \langle S_8, \sigma_{220} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{220} \rangle \Rightarrow \langle S_9, \sigma_{220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{880} = \langle S_8, \sigma_{220} \rangle \Rightarrow \langle y := y + 1, \sigma_{220} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{879} = \frac{\mathcal{T}_{880}}{\langle S_9, \sigma_{220} \rangle \Rightarrow \langle S_{10}, \sigma_{220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{881} = \frac{\langle y := y + 1, \sigma_{220} \rangle \Rightarrow \sigma_{221} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{220} \rangle \Rightarrow \langle S_{11}, \sigma_{221} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{882} = \langle S_{11}, \sigma_{221} \rangle \Rightarrow \langle S_{12}, \sigma_{221} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{883} = \langle S_{12}, \sigma_{221} \rangle \Rightarrow \langle S_{13}, \sigma_{221} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{885} = \frac{\langle x := x + 1, \sigma_{221} \rangle \Rightarrow \sigma_{222} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{221} \rangle \Rightarrow \langle S_6, \sigma_{222} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{884} = \frac{\mathcal{T}_{885}}{\langle S_{13}, \sigma_{221} \rangle \Rightarrow \langle S_{15}, \sigma_{222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{886} = \frac{\langle S_6, \sigma_{222} \rangle \Rightarrow \langle S_8, \sigma_{222} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{222} \rangle \Rightarrow \langle S_9, \sigma_{222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{888} = \langle S_8, \sigma_{222} \rangle \Rightarrow \langle y := y + 1, \sigma_{222} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{887} = \frac{\mathcal{T}_{888}}{\langle S_9, \sigma_{222} \rangle \Rightarrow \langle S_{10}, \sigma_{222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{889} = \frac{\langle y := y + 1, \sigma_{222} \rangle \Rightarrow \sigma_{223} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{222} \rangle \Rightarrow \langle S_{11}, \sigma_{223} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{890} = \langle S_{11}, \sigma_{223} \rangle \Rightarrow \langle S_{12}, \sigma_{223} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{891} = \langle S_{12}, \sigma_{223} \rangle \Rightarrow \langle S_{13}, \sigma_{223} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{893} = \frac{\langle x := x + 1, \sigma_{223} \rangle \Rightarrow \sigma_{224} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{223} \rangle \Rightarrow \langle S_6, \sigma_{224} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{892} = \frac{\mathcal{T}_{893}}{\langle S_{13}, \sigma_{223} \rangle \Rightarrow \langle S_{15}, \sigma_{224} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{894} = \frac{\langle S_6, \sigma_{224} \rangle \Rightarrow \langle S_8, \sigma_{224} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{224} \rangle \Rightarrow \langle S_9, \sigma_{224} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{896} = \langle S_8, \sigma_{224} \rangle \Rightarrow \langle y := y + 1, \sigma_{224} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{895} = \frac{\mathcal{T}_{896}}{\langle S_9, \sigma_{224} \rangle \Rightarrow \langle S_{10}, \sigma_{224} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{897} = \frac{\langle y := y + 1, \sigma_{224} \rangle \Rightarrow \sigma_{225} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{224} \rangle \Rightarrow \langle S_{11}, \sigma_{225} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{898} = \langle S_{11}, \sigma_{225} \rangle \Rightarrow \langle S_{12}, \sigma_{225} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{899} = \langle S_{12}, \sigma_{225} \rangle \Rightarrow \langle S_{13}, \sigma_{225} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{901} = \frac{\langle x := x + 1, \sigma_{225} \rangle \Rightarrow \sigma_{226} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{225} \rangle \Rightarrow \langle S_6, \sigma_{226} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{900} = \frac{\mathcal{T}_{901}}{\langle S_{13}, \sigma_{225} \rangle \Rightarrow \langle S_{15}, \sigma_{226} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{902} = \frac{\langle S_6, \sigma_{226} \rangle \Rightarrow \langle S_8, \sigma_{226} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{226} \rangle \Rightarrow \langle S_9, \sigma_{226} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{904} = \langle S_8, \sigma_{226} \rangle \Rightarrow \langle y := y + 1, \sigma_{226} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{903} = \frac{\mathcal{T}_{904}}{\langle S_9, \sigma_{226} \rangle \Rightarrow \langle S_{10}, \sigma_{226} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{905} = \frac{\langle y := y + 1, \sigma_{226} \rangle \Rightarrow \sigma_{227} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{226} \rangle \Rightarrow \langle S_{11}, \sigma_{227} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{906} = \langle S_{11}, \sigma_{227} \rangle \Rightarrow \langle S_{12}, \sigma_{227} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{907} = \langle S_{12}, \sigma_{227} \rangle \Rightarrow \langle S_{13}, \sigma_{227} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{909} = \frac{\langle x := x + 1, \sigma_{227} \rangle \Rightarrow \sigma_{228} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{227} \rangle \Rightarrow \langle S_6, \sigma_{228} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{908} = \frac{\mathcal{T}_{909}}{\langle S_{13}, \sigma_{227} \rangle \Rightarrow \langle S_{15}, \sigma_{228} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{910} = \frac{\langle S_6, \sigma_{228} \rangle \Rightarrow \langle S_8, \sigma_{228} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{228} \rangle \Rightarrow \langle S_9, \sigma_{228} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{912} = \langle S_8, \sigma_{228} \rangle \Rightarrow \langle y := y + 1, \sigma_{228} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{911} = \frac{\mathcal{T}_{912}}{\langle S_9, \sigma_{228} \rangle \Rightarrow \langle S_{10}, \sigma_{228} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{913} = \frac{\langle y := y + 1, \sigma_{228} \rangle \Rightarrow \sigma_{229} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{228} \rangle \Rightarrow \langle S_{11}, \sigma_{229} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{914} = \langle S_{11}, \sigma_{229} \rangle \Rightarrow \langle S_{12}, \sigma_{229} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{915} = \langle S_{12}, \sigma_{229} \rangle \Rightarrow \langle S_{13}, \sigma_{229} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{917} = \frac{\langle x := x + 1, \sigma_{229} \rangle \Rightarrow \sigma_{230} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{229} \rangle \Rightarrow \langle S_6, \sigma_{230} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{916} = \frac{\mathcal{T}_{917}}{\langle S_{13}, \sigma_{229} \rangle \Rightarrow \langle S_{15}, \sigma_{230} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{918} = \frac{\langle S_6, \sigma_{230} \rangle \Rightarrow \langle S_8, \sigma_{230} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{230} \rangle \Rightarrow \langle S_9, \sigma_{230} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{920} = \langle S_8, \sigma_{230} \rangle \Rightarrow \langle y := y + 1, \sigma_{230} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{919} = \frac{\mathcal{T}_{920}}{\langle S_9, \sigma_{230} \rangle \Rightarrow \langle S_{10}, \sigma_{230} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{921} = \frac{\langle y := y + 1, \sigma_{230} \rangle \Rightarrow \sigma_{231} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{230} \rangle \Rightarrow \langle S_{11}, \sigma_{231} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{922} = \langle S_{11}, \sigma_{231} \rangle \Rightarrow \langle S_{12}, \sigma_{231} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{923} = \langle S_{12}, \sigma_{231} \rangle \Rightarrow \langle S_{13}, \sigma_{231} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{925} = \frac{\langle x := x + 1, \sigma_{231} \rangle \Rightarrow \sigma_{232} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{231} \rangle \Rightarrow \langle S_6, \sigma_{232} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{924} = \frac{\mathcal{T}_{925}}{\langle S_{13}, \sigma_{231} \rangle \Rightarrow \langle S_{15}, \sigma_{232} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{926} = \frac{\langle S_6, \sigma_{232} \rangle \Rightarrow \langle S_8, \sigma_{232} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{232} \rangle \Rightarrow \langle S_9, \sigma_{232} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{928} = \langle S_8, \sigma_{232} \rangle \Rightarrow \langle y := y + 1, \sigma_{232} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{927} = \frac{\mathcal{T}_{928}}{\langle S_9, \sigma_{232} \rangle \Rightarrow \langle S_{10}, \sigma_{232} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{929} = \frac{\langle y := y + 1, \sigma_{232} \rangle \Rightarrow \sigma_{233} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{232} \rangle \Rightarrow \langle S_{11}, \sigma_{233} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{930} = \langle S_{11}, \sigma_{233} \rangle \Rightarrow \langle S_{12}, \sigma_{233} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{931} = \langle S_{12}, \sigma_{233} \rangle \Rightarrow \langle S_{13}, \sigma_{233} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{933} = \frac{\langle x := x + 1, \sigma_{233} \rangle \Rightarrow \sigma_{234} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{233} \rangle \Rightarrow \langle S_6, \sigma_{234} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{932} = \frac{\mathcal{T}_{933}}{\langle S_{13}, \sigma_{233} \rangle \Rightarrow \langle S_{15}, \sigma_{234} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{934} = \frac{\langle S_6, \sigma_{234} \rangle \Rightarrow \langle S_8, \sigma_{234} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{234} \rangle \Rightarrow \langle S_9, \sigma_{234} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{936} = \langle S_8, \sigma_{234} \rangle \Rightarrow \langle y := y + 1, \sigma_{234} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{935} = \frac{\mathcal{T}_{936}}{\langle S_9, \sigma_{234} \rangle \Rightarrow \langle S_{10}, \sigma_{234} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{937} = \frac{\langle y := y + 1, \sigma_{234} \rangle \Rightarrow \sigma_{235} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{234} \rangle \Rightarrow \langle S_{11}, \sigma_{235} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{938} = \langle S_{11}, \sigma_{235} \rangle \Rightarrow \langle S_{12}, \sigma_{235} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{939} = \langle S_{12}, \sigma_{235} \rangle \Rightarrow \langle S_{13}, \sigma_{235} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{941} = \frac{\langle x := x + 1, \sigma_{235} \rangle \Rightarrow \sigma_{236} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{235} \rangle \Rightarrow \langle S_6, \sigma_{236} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{940} = \frac{\mathcal{T}_{941}}{\langle S_{13}, \sigma_{235} \rangle \Rightarrow \langle S_{15}, \sigma_{236} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{942} = \frac{\langle S_6, \sigma_{236} \rangle \Rightarrow \langle S_8, \sigma_{236} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{236} \rangle \Rightarrow \langle S_9, \sigma_{236} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{944} = \langle S_8, \sigma_{236} \rangle \Rightarrow \langle y := y + 1, \sigma_{236} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{943} = \frac{\mathcal{T}_{944}}{\langle S_9, \sigma_{236} \rangle \Rightarrow \langle S_{10}, \sigma_{236} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{945} = \frac{\langle y := y + 1, \sigma_{236} \rangle \Rightarrow \sigma_{237} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{236} \rangle \Rightarrow \langle S_{11}, \sigma_{237} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{946} = \langle S_{11}, \sigma_{237} \rangle \Rightarrow \langle S_{12}, \sigma_{237} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{947} = \langle S_{12}, \sigma_{237} \rangle \Rightarrow \langle S_{13}, \sigma_{237} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{949} = \frac{\langle x := x + 1, \sigma_{237} \rangle \Rightarrow \sigma_{238} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{237} \rangle \Rightarrow \langle S_6, \sigma_{238} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{948} = \frac{\mathcal{T}_{949}}{\langle S_{13}, \sigma_{237} \rangle \Rightarrow \langle S_{15}, \sigma_{238} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{950} = \frac{\langle S_6, \sigma_{238} \rangle \Rightarrow \langle S_8, \sigma_{238} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{238} \rangle \Rightarrow \langle S_9, \sigma_{238} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{952} = \langle S_8, \sigma_{238} \rangle \Rightarrow \langle y := y + 1, \sigma_{238} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{951} = \frac{\mathcal{T}_{952}}{\langle S_9, \sigma_{238} \rangle \Rightarrow \langle S_{10}, \sigma_{238} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{953} = \frac{\langle y := y + 1, \sigma_{238} \rangle \Rightarrow \sigma_{239} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{238} \rangle \Rightarrow \langle S_{11}, \sigma_{239} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{954} = \langle S_{11}, \sigma_{239} \rangle \Rightarrow \langle S_{12}, \sigma_{239} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{955} = \langle S_{12}, \sigma_{239} \rangle \Rightarrow \langle S_{13}, \sigma_{239} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{957} = \frac{\langle x := x + 1, \sigma_{239} \rangle \Rightarrow \sigma_{240} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{239} \rangle \Rightarrow \langle S_6, \sigma_{240} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{956} = \frac{\mathcal{T}_{957}}{\langle S_{13}, \sigma_{239} \rangle \Rightarrow \langle S_{15}, \sigma_{240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{958} = \frac{\langle S_6, \sigma_{240} \rangle \Rightarrow \langle S_8, \sigma_{240} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{240} \rangle \Rightarrow \langle S_9, \sigma_{240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{960} = \langle S_8, \sigma_{240} \rangle \Rightarrow \langle y := y + 1, \sigma_{240} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{959} = \frac{\mathcal{T}_{960}}{\langle S_9, \sigma_{240} \rangle \Rightarrow \langle S_{10}, \sigma_{240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{961} = \frac{\langle y := y + 1, \sigma_{240} \rangle \Rightarrow \sigma_{241} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{240} \rangle \Rightarrow \langle S_{11}, \sigma_{241} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{962} = \langle S_{11}, \sigma_{241} \rangle \Rightarrow \langle S_{12}, \sigma_{241} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{963} = \langle S_{12}, \sigma_{241} \rangle \Rightarrow \langle S_{13}, \sigma_{241} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{965} = \frac{\langle x := x + 1, \sigma_{241} \rangle \Rightarrow \sigma_{242} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{241} \rangle \Rightarrow \langle S_6, \sigma_{242} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{964} = \frac{\mathcal{T}_{965}}{\langle S_{13}, \sigma_{241} \rangle \Rightarrow \langle S_{15}, \sigma_{242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{966} = \frac{\langle S_6, \sigma_{242} \rangle \Rightarrow \langle S_8, \sigma_{242} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{242} \rangle \Rightarrow \langle S_9, \sigma_{242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{968} = \langle S_8, \sigma_{242} \rangle \Rightarrow \langle y := y + 1, \sigma_{242} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{967} = \frac{\mathcal{T}_{968}}{\langle S_9, \sigma_{242} \rangle \Rightarrow \langle S_{10}, \sigma_{242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{969} = \frac{\langle y := y + 1, \sigma_{242} \rangle \Rightarrow \sigma_{243} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{242} \rangle \Rightarrow \langle S_{11}, \sigma_{243} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{970} = \langle S_{11}, \sigma_{243} \rangle \Rightarrow \langle S_{12}, \sigma_{243} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{971} = \langle S_{12}, \sigma_{243} \rangle \Rightarrow \langle S_{13}, \sigma_{243} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{973} = \frac{\langle x := x + 1, \sigma_{243} \rangle \Rightarrow \sigma_{244} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{243} \rangle \Rightarrow \langle S_6, \sigma_{244} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{972} = \frac{\mathcal{T}_{973}}{\langle S_{13}, \sigma_{243} \rangle \Rightarrow \langle S_{15}, \sigma_{244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{974} = \frac{\langle S_6, \sigma_{244} \rangle \Rightarrow \langle S_8, \sigma_{244} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{244} \rangle \Rightarrow \langle S_9, \sigma_{244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{976} = \langle S_8, \sigma_{244} \rangle \Rightarrow \langle y := y + 1, \sigma_{244} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{975} = \frac{\mathcal{T}_{976}}{\langle S_9, \sigma_{244} \rangle \Rightarrow \langle S_{10}, \sigma_{244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{977} = \frac{\langle y := y + 1, \sigma_{244} \rangle \Rightarrow \sigma_{245} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{244} \rangle \Rightarrow \langle S_{11}, \sigma_{245} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{978} = \langle S_{11}, \sigma_{245} \rangle \Rightarrow \langle S_{12}, \sigma_{245} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{979} = \langle S_{12}, \sigma_{245} \rangle \Rightarrow \langle S_{13}, \sigma_{245} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{981} = \frac{\langle x := x + 1, \sigma_{245} \rangle \Rightarrow \sigma_{246} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{245} \rangle \Rightarrow \langle S_6, \sigma_{246} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{980} = \frac{\mathcal{T}_{981}}{\langle S_{13}, \sigma_{245} \rangle \Rightarrow \langle S_{15}, \sigma_{246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{982} = \frac{\langle S_6, \sigma_{246} \rangle \Rightarrow \langle S_8, \sigma_{246} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{246} \rangle \Rightarrow \langle S_9, \sigma_{246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{984} = \langle S_8, \sigma_{246} \rangle \Rightarrow \langle y := y + 1, \sigma_{246} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{983} = \frac{\mathcal{T}_{984}}{\langle S_9, \sigma_{246} \rangle \Rightarrow \langle S_{10}, \sigma_{246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{985} = \frac{\langle y := y + 1, \sigma_{246} \rangle \Rightarrow \sigma_{247} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{246} \rangle \Rightarrow \langle S_{11}, \sigma_{247} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{986} = \langle S_{11}, \sigma_{247} \rangle \Rightarrow \langle S_{12}, \sigma_{247} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{987} = \langle S_{12}, \sigma_{247} \rangle \Rightarrow \langle S_{13}, \sigma_{247} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{989} = \frac{\langle x := x + 1, \sigma_{247} \rangle \Rightarrow \sigma_{248} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{247} \rangle \Rightarrow \langle S_6, \sigma_{248} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{988} = \frac{\mathcal{T}_{989}}{\langle S_{13}, \sigma_{247} \rangle \Rightarrow \langle S_{15}, \sigma_{248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{990} = \frac{\langle S_6, \sigma_{248} \rangle \Rightarrow \langle S_8, \sigma_{248} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{248} \rangle \Rightarrow \langle S_9, \sigma_{248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{992} = \langle S_8, \sigma_{248} \rangle \Rightarrow \langle y := y + 1, \sigma_{248} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{991} = \frac{\mathcal{T}_{992}}{\langle S_9, \sigma_{248} \rangle \Rightarrow \langle S_{10}, \sigma_{248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{993} = \frac{\langle y := y + 1, \sigma_{248} \rangle \Rightarrow \sigma_{249} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{248} \rangle \Rightarrow \langle S_{11}, \sigma_{249} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{994} = \langle S_{11}, \sigma_{249} \rangle \Rightarrow \langle S_{12}, \sigma_{249} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{995} = \langle S_{12}, \sigma_{249} \rangle \Rightarrow \langle S_{13}, \sigma_{249} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{997} = \frac{\langle x := x + 1, \sigma_{249} \rangle \Rightarrow \sigma_{250} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{249} \rangle \Rightarrow \langle S_6, \sigma_{250} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{996} = \frac{\mathcal{T}_{997}}{\langle S_{13}, \sigma_{249} \rangle \Rightarrow \langle S_{15}, \sigma_{250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{998} = \frac{\langle S_6, \sigma_{250} \rangle \Rightarrow \langle S_8, \sigma_{250} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{250} \rangle \Rightarrow \langle S_9, \sigma_{250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1000} = \langle S_8, \sigma_{250} \rangle \Rightarrow \langle y := y + 1, \sigma_{250} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{999} = \frac{\mathcal{T}_{1000}}{\langle S_9, \sigma_{250} \rangle \Rightarrow \langle S_{10}, \sigma_{250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1001} = \frac{\langle y := y + 1, \sigma_{250} \rangle \Rightarrow \sigma_{251} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{250} \rangle \Rightarrow \langle S_{11}, \sigma_{251} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1002} = \langle S_{11}, \sigma_{251} \rangle \Rightarrow \langle S_{12}, \sigma_{251} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1003} = \langle S_{12}, \sigma_{251} \rangle \Rightarrow \langle S_{13}, \sigma_{251} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1005} = \frac{\langle x := x + 1, \sigma_{251} \rangle \Rightarrow \sigma_{252} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{251} \rangle \Rightarrow \langle S_6, \sigma_{252} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1004} = \frac{\mathcal{T}_{1005}}{\langle S_{13}, \sigma_{251} \rangle \Rightarrow \langle S_{15}, \sigma_{252} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1006} = \frac{\langle S_6, \sigma_{252} \rangle \Rightarrow \langle S_8, \sigma_{252} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{252} \rangle \Rightarrow \langle S_9, \sigma_{252} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1008} = \langle S_8, \sigma_{252} \rangle \Rightarrow \langle y := y + 1, \sigma_{252} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1007} = \frac{\mathcal{T}_{1008}}{\langle S_9, \sigma_{252} \rangle \Rightarrow \langle S_{10}, \sigma_{252} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1009} = \frac{\langle y := y + 1, \sigma_{252} \rangle \Rightarrow \sigma_{253} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{252} \rangle \Rightarrow \langle S_{11}, \sigma_{253} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1010} = \langle S_{11}, \sigma_{253} \rangle \Rightarrow \langle S_{12}, \sigma_{253} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1011} = \langle S_{12}, \sigma_{253} \rangle \Rightarrow \langle S_{13}, \sigma_{253} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1013} = \frac{\langle x := x + 1, \sigma_{253} \rangle \Rightarrow \sigma_{254} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{253} \rangle \Rightarrow \langle S_6, \sigma_{254} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1012} = \frac{\mathcal{T}_{1013}}{\langle S_{13}, \sigma_{253} \rangle \Rightarrow \langle S_{15}, \sigma_{254} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1014} = \frac{\langle S_6, \sigma_{254} \rangle \Rightarrow \langle S_8, \sigma_{254} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{254} \rangle \Rightarrow \langle S_9, \sigma_{254} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1016} = \langle S_8, \sigma_{254} \rangle \Rightarrow \langle y := y + 1, \sigma_{254} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1015} = \frac{\mathcal{T}_{1016}}{\langle S_9, \sigma_{254} \rangle \Rightarrow \langle S_{10}, \sigma_{254} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1017} = \frac{\langle y := y + 1, \sigma_{254} \rangle \Rightarrow \sigma_{255} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{254} \rangle \Rightarrow \langle S_{11}, \sigma_{255} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1018} = \langle S_{11}, \sigma_{255} \rangle \Rightarrow \langle S_{12}, \sigma_{255} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1019} = \langle S_{12}, \sigma_{255} \rangle \Rightarrow \langle S_{13}, \sigma_{255} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1021} = \frac{\langle x := x + 1, \sigma_{255} \rangle \Rightarrow \sigma_{256} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{255} \rangle \Rightarrow \langle S_6, \sigma_{256} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1020} = \frac{\mathcal{T}_{1021}}{\langle S_{13}, \sigma_{255} \rangle \Rightarrow \langle S_{15}, \sigma_{256} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1022} = \frac{\langle S_6, \sigma_{256} \rangle \Rightarrow \langle S_8, \sigma_{256} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{256} \rangle \Rightarrow \langle S_9, \sigma_{256} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1024} = \langle S_8, \sigma_{256} \rangle \Rightarrow \langle y := y + 1, \sigma_{256} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1023} = \frac{\mathcal{T}_{1024}}{\langle S_9, \sigma_{256} \rangle \Rightarrow \langle S_{10}, \sigma_{256} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1025} = \frac{\langle y := y + 1, \sigma_{256} \rangle \Rightarrow \sigma_{257} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{256} \rangle \Rightarrow \langle S_{11}, \sigma_{257} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1026} = \langle S_{11}, \sigma_{257} \rangle \Rightarrow \langle S_{12}, \sigma_{257} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1027} = \langle S_{12}, \sigma_{257} \rangle \Rightarrow \langle S_{13}, \sigma_{257} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1029} = \frac{\langle x := x + 1, \sigma_{257} \rangle \Rightarrow \sigma_{258} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{257} \rangle \Rightarrow \langle S_6, \sigma_{258} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1028} = \frac{\mathcal{T}_{1029}}{\langle S_{13}, \sigma_{257} \rangle \Rightarrow \langle S_{15}, \sigma_{258} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1030} = \frac{\langle S_6, \sigma_{258} \rangle \Rightarrow \langle S_8, \sigma_{258} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{258} \rangle \Rightarrow \langle S_9, \sigma_{258} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1032} = \langle S_8, \sigma_{258} \rangle \Rightarrow \langle y := y + 1, \sigma_{258} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1031} = \frac{\mathcal{T}_{1032}}{\langle S_9, \sigma_{258} \rangle \Rightarrow \langle S_{10}, \sigma_{258} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1033} = \frac{\langle y := y + 1, \sigma_{258} \rangle \Rightarrow \sigma_{259} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{258} \rangle \Rightarrow \langle S_{11}, \sigma_{259} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{1034} = \langle S_{11}, \sigma_{259} \rangle \Rightarrow \langle S_{12}, \sigma_{259} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1035} = \langle S_{12}, \sigma_{259} \rangle \Rightarrow \langle S_{13}, \sigma_{259} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1037} = \frac{\langle x := x + 1, \sigma_{259} \rangle \Rightarrow \sigma_{260} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{259} \rangle \Rightarrow \langle S_6, \sigma_{260} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1036} = \frac{\mathcal{T}_{1037}}{\langle S_{13}, \sigma_{259} \rangle \Rightarrow \langle S_{15}, \sigma_{260} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1038} = \frac{\langle S_6, \sigma_{260} \rangle \Rightarrow \langle S_8, \sigma_{260} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{260} \rangle \Rightarrow \langle S_9, \sigma_{260} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1040} = \langle S_8, \sigma_{260} \rangle \Rightarrow \langle y := y + 1, \sigma_{260} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1039} = \frac{\mathcal{T}_{1040}}{\langle S_9, \sigma_{260} \rangle \Rightarrow \langle S_{10}, \sigma_{260} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1041} = \frac{\langle y := y + 1, \sigma_{260} \rangle \Rightarrow \sigma_{261} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{260} \rangle \Rightarrow \langle S_{11}, \sigma_{261} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1042} = \langle S_{11}, \sigma_{261} \rangle \Rightarrow \langle S_{12}, \sigma_{261} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1043} = \langle S_{12}, \sigma_{261} \rangle \Rightarrow \langle S_{13}, \sigma_{261} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1045} = \frac{\langle x := x + 1, \sigma_{261} \rangle \Rightarrow \sigma_{262} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{261} \rangle \Rightarrow \langle S_6, \sigma_{262} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1044} = \frac{\mathcal{T}_{1045}}{\langle S_{13}, \sigma_{261} \rangle \Rightarrow \langle S_{15}, \sigma_{262} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1046} = \frac{\langle S_6, \sigma_{262} \rangle \Rightarrow \langle S_8, \sigma_{262} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{262} \rangle \Rightarrow \langle S_9, \sigma_{262} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1048} = \langle S_8, \sigma_{262} \rangle \Rightarrow \langle y := y + 1, \sigma_{262} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1047} = \frac{\mathcal{T}_{1048}}{\langle S_9, \sigma_{262} \rangle \Rightarrow \langle S_{10}, \sigma_{262} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1049} = \frac{\langle y := y + 1, \sigma_{262} \rangle \Rightarrow \sigma_{263} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{262} \rangle \Rightarrow \langle S_{11}, \sigma_{263} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1050} = \langle S_{11}, \sigma_{263} \rangle \Rightarrow \langle S_{12}, \sigma_{263} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1051} = \langle S_{12}, \sigma_{263} \rangle \Rightarrow \langle S_{13}, \sigma_{263} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1053} = \frac{\langle x := x + 1, \sigma_{263} \rangle \Rightarrow \sigma_{264} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{263} \rangle \Rightarrow \langle S_6, \sigma_{264} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1052} = \frac{\mathcal{T}_{1053}}{\langle S_{13}, \sigma_{263} \rangle \Rightarrow \langle S_{15}, \sigma_{264} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1054} = \frac{\langle S_6, \sigma_{264} \rangle \Rightarrow \langle S_8, \sigma_{264} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{264} \rangle \Rightarrow \langle S_9, \sigma_{264} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1056} = \langle S_8, \sigma_{264} \rangle \Rightarrow \langle y := y + 1, \sigma_{264} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1055} = \frac{\mathcal{T}_{1056}}{\langle S_9, \sigma_{264} \rangle \Rightarrow \langle S_{10}, \sigma_{264} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1057} = \frac{\langle y := y + 1, \sigma_{264} \rangle \Rightarrow \sigma_{265} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{264} \rangle \Rightarrow \langle S_{11}, \sigma_{265} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1058} = \langle S_{11}, \sigma_{265} \rangle \Rightarrow \langle S_{12}, \sigma_{265} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1059} = \langle S_{12}, \sigma_{265} \rangle \Rightarrow \langle S_{13}, \sigma_{265} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1061} = \frac{\langle x := x + 1, \sigma_{265} \rangle \Rightarrow \sigma_{266} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{265} \rangle \Rightarrow \langle S_6, \sigma_{266} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1060} = \frac{\mathcal{T}_{1061}}{\langle S_{13}, \sigma_{265} \rangle \Rightarrow \langle S_{15}, \sigma_{266} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1062} = \frac{\langle S_6, \sigma_{266} \rangle \Rightarrow \langle S_8, \sigma_{266} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{266} \rangle \Rightarrow \langle S_9, \sigma_{266} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1064} = \langle S_8, \sigma_{266} \rangle \Rightarrow \langle y := y + 1, \sigma_{266} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1063} = \frac{\mathcal{T}_{1064}}{\langle S_9, \sigma_{266} \rangle \Rightarrow \langle S_{10}, \sigma_{266} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1065} = \frac{\langle y := y + 1, \sigma_{266} \rangle \Rightarrow \sigma_{267} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{266} \rangle \Rightarrow \langle S_{11}, \sigma_{267} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1066} = \langle S_{11}, \sigma_{267} \rangle \Rightarrow \langle S_{12}, \sigma_{267} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1067} = \langle S_{12}, \sigma_{267} \rangle \Rightarrow \langle S_{13}, \sigma_{267} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1069} = \frac{\langle x := x + 1, \sigma_{267} \rangle \Rightarrow \sigma_{268} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{267} \rangle \Rightarrow \langle S_6, \sigma_{268} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1068} = \frac{\mathcal{T}_{1069}}{\langle S_{13}, \sigma_{267} \rangle \Rightarrow \langle S_{15}, \sigma_{268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1070} = \frac{\langle S_6, \sigma_{268} \rangle \Rightarrow \langle S_8, \sigma_{268} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{268} \rangle \Rightarrow \langle S_9, \sigma_{268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1072} = \langle S_8, \sigma_{268} \rangle \Rightarrow \langle y := y + 1, \sigma_{268} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1071} = \frac{\mathcal{T}_{1072}}{\langle S_9, \sigma_{268} \rangle \Rightarrow \langle S_{10}, \sigma_{268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1073} = \frac{\langle y := y + 1, \sigma_{268} \rangle \Rightarrow \sigma_{269} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{268} \rangle \Rightarrow \langle S_{11}, \sigma_{269} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1074} = \langle S_{11}, \sigma_{269} \rangle \Rightarrow \langle S_{12}, \sigma_{269} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1075} = \langle S_{12}, \sigma_{269} \rangle \Rightarrow \langle S_{13}, \sigma_{269} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1077} = \frac{\langle x := x + 1, \sigma_{269} \rangle \Rightarrow \sigma_{270} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{269} \rangle \Rightarrow \langle S_6, \sigma_{270} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1076} = \frac{\mathcal{T}_{1077}}{\langle S_{13}, \sigma_{269} \rangle \Rightarrow \langle S_{15}, \sigma_{270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1078} = \frac{\langle S_6, \sigma_{270} \rangle \Rightarrow \langle S_8, \sigma_{270} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{270} \rangle \Rightarrow \langle S_9, \sigma_{270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1080} = \langle S_8, \sigma_{270} \rangle \Rightarrow \langle y := y + 1, \sigma_{270} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1079} = \frac{\mathcal{T}_{1080}}{\langle S_9, \sigma_{270} \rangle \Rightarrow \langle S_{10}, \sigma_{270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1081} = \frac{\langle y := y + 1, \sigma_{270} \rangle \Rightarrow \sigma_{271} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{270} \rangle \Rightarrow \langle S_{11}, \sigma_{271} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1082} = \langle S_{11}, \sigma_{271} \rangle \Rightarrow \langle S_{12}, \sigma_{271} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1083} = \langle S_{12}, \sigma_{271} \rangle \Rightarrow \langle S_{13}, \sigma_{271} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1085} = \frac{\langle x := x + 1, \sigma_{271} \rangle \Rightarrow \sigma_{272} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{271} \rangle \Rightarrow \langle S_6, \sigma_{272} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1084} = \frac{\mathcal{T}_{1085}}{\langle S_{13}, \sigma_{271} \rangle \Rightarrow \langle S_{15}, \sigma_{272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1086} = \frac{\langle S_6, \sigma_{272} \rangle \Rightarrow \langle S_8, \sigma_{272} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{272} \rangle \Rightarrow \langle S_9, \sigma_{272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1088} = \langle S_8, \sigma_{272} \rangle \Rightarrow \langle y := y + 1, \sigma_{272} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1087} = \frac{\mathcal{T}_{1088}}{\langle S_9, \sigma_{272} \rangle \Rightarrow \langle S_{10}, \sigma_{272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1089} = \frac{\langle y := y + 1, \sigma_{272} \rangle \Rightarrow \sigma_{273} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{272} \rangle \Rightarrow \langle S_{11}, \sigma_{273} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1090} = \langle S_{11}, \sigma_{273} \rangle \Rightarrow \langle S_{12}, \sigma_{273} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1091} = \langle S_{12}, \sigma_{273} \rangle \Rightarrow \langle S_{13}, \sigma_{273} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1093} = \frac{\langle x := x + 1, \sigma_{273} \rangle \Rightarrow \sigma_{274} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{273} \rangle \Rightarrow \langle S_6, \sigma_{274} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1092} = \frac{\mathcal{T}_{1093}}{\langle S_{13}, \sigma_{273} \rangle \Rightarrow \langle S_{15}, \sigma_{274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1094} = \frac{\langle S_6, \sigma_{274} \rangle \Rightarrow \langle S_8, \sigma_{274} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{274} \rangle \Rightarrow \langle S_9, \sigma_{274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1096} = \langle S_8, \sigma_{274} \rangle \Rightarrow \langle y := y + 1, \sigma_{274} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1095} = \frac{\mathcal{T}_{1096}}{\langle S_9, \sigma_{274} \rangle \Rightarrow \langle S_{10}, \sigma_{274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1097} = \frac{\langle y := y + 1, \sigma_{274} \rangle \Rightarrow \sigma_{275} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{274} \rangle \Rightarrow \langle S_{11}, \sigma_{275} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1098} = \langle S_{11}, \sigma_{275} \rangle \Rightarrow \langle S_{12}, \sigma_{275} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1099} = \langle S_{12}, \sigma_{275} \rangle \Rightarrow \langle S_{13}, \sigma_{275} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1101} = \frac{\langle x := x + 1, \sigma_{275} \rangle \Rightarrow \sigma_{276} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{275} \rangle \Rightarrow \langle S_6, \sigma_{276} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1100} = \frac{\mathcal{T}_{1101}}{\langle S_{13}, \sigma_{275} \rangle \Rightarrow \langle S_{15}, \sigma_{276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1102} = \frac{\langle S_6, \sigma_{276} \rangle \Rightarrow \langle S_8, \sigma_{276} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{276} \rangle \Rightarrow \langle S_9, \sigma_{276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1104} = \langle S_8, \sigma_{276} \rangle \Rightarrow \langle y := y + 1, \sigma_{276} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1103} = \frac{\mathcal{T}_{1104}}{\langle S_9, \sigma_{276} \rangle \Rightarrow \langle S_{10}, \sigma_{276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1105} = \frac{\langle y := y + 1, \sigma_{276} \rangle \Rightarrow \sigma_{277} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{276} \rangle \Rightarrow \langle S_{11}, \sigma_{277} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1106} = \langle S_{11}, \sigma_{277} \rangle \Rightarrow \langle S_{12}, \sigma_{277} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1107} = \langle S_{12}, \sigma_{277} \rangle \Rightarrow \langle S_{13}, \sigma_{277} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1109} = \frac{\langle x := x + 1, \sigma_{277} \rangle \Rightarrow \sigma_{278} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{277} \rangle \Rightarrow \langle S_6, \sigma_{278} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1108} = \frac{\mathcal{T}_{1109}}{\langle S_{13}, \sigma_{277} \rangle \Rightarrow \langle S_{15}, \sigma_{278} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1110} = \frac{\langle S_6, \sigma_{278} \rangle \Rightarrow \langle S_8, \sigma_{278} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{278} \rangle \Rightarrow \langle S_9, \sigma_{278} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1112} = \langle S_8, \sigma_{278} \rangle \Rightarrow \langle y := y + 1, \sigma_{278} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1111} = \frac{\mathcal{T}_{1112}}{\langle S_9, \sigma_{278} \rangle \Rightarrow \langle S_{10}, \sigma_{278} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1113} = \frac{\langle y := y + 1, \sigma_{278} \rangle \Rightarrow \sigma_{279} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{278} \rangle \Rightarrow \langle S_{11}, \sigma_{279} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1114} = \langle S_{11}, \sigma_{279} \rangle \Rightarrow \langle S_{12}, \sigma_{279} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1115} = \langle S_{12}, \sigma_{279} \rangle \Rightarrow \langle S_{13}, \sigma_{279} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1117} = \frac{\langle x := x + 1, \sigma_{279} \rangle \Rightarrow \sigma_{280} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{279} \rangle \Rightarrow \langle S_6, \sigma_{280} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1116} = \frac{\mathcal{T}_{1117}}{\langle S_{13}, \sigma_{279} \rangle \Rightarrow \langle S_{15}, \sigma_{280} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1118} = \frac{\langle S_6, \sigma_{280} \rangle \Rightarrow \langle S_8, \sigma_{280} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{280} \rangle \Rightarrow \langle S_9, \sigma_{280} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1120} = \langle S_8, \sigma_{280} \rangle \Rightarrow \langle y := y + 1, \sigma_{280} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1119} = \frac{\mathcal{T}_{1120}}{\langle S_9, \sigma_{280} \rangle \Rightarrow \langle S_{10}, \sigma_{280} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1121} = \frac{\langle y := y + 1, \sigma_{280} \rangle \Rightarrow \sigma_{281} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{280} \rangle \Rightarrow \langle S_{11}, \sigma_{281} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1122} = \langle S_{11}, \sigma_{281} \rangle \Rightarrow \langle S_{12}, \sigma_{281} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1123} = \langle S_{12}, \sigma_{281} \rangle \Rightarrow \langle S_{13}, \sigma_{281} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1125} = \frac{\langle x := x + 1, \sigma_{281} \rangle \Rightarrow \sigma_{282} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{281} \rangle \Rightarrow \langle S_6, \sigma_{282} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1124} = \frac{\mathcal{T}_{1125}}{\langle S_{13}, \sigma_{281} \rangle \Rightarrow \langle S_{15}, \sigma_{282} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1126} = \frac{\langle S_6, \sigma_{282} \rangle \Rightarrow \langle S_8, \sigma_{282} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{282} \rangle \Rightarrow \langle S_9, \sigma_{282} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1128} = \langle S_8, \sigma_{282} \rangle \Rightarrow \langle y := y + 1, \sigma_{282} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1127} = \frac{\mathcal{T}_{1128}}{\langle S_9, \sigma_{282} \rangle \Rightarrow \langle S_{10}, \sigma_{282} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1129} = \frac{\langle y := y + 1, \sigma_{282} \rangle \Rightarrow \sigma_{283} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{282} \rangle \Rightarrow \langle S_{11}, \sigma_{283} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1130} = \langle S_{11}, \sigma_{283} \rangle \Rightarrow \langle S_{12}, \sigma_{283} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1131} = \langle S_{12}, \sigma_{283} \rangle \Rightarrow \langle S_{13}, \sigma_{283} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1133} = \frac{\langle x := x + 1, \sigma_{283} \rangle \Rightarrow \sigma_{284} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{283} \rangle \Rightarrow \langle S_6, \sigma_{284} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1132} = \frac{\mathcal{T}_{1133}}{\langle S_{13}, \sigma_{283} \rangle \Rightarrow \langle S_{15}, \sigma_{284} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1134} = \frac{\langle S_6, \sigma_{284} \rangle \Rightarrow \langle S_8, \sigma_{284} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{284} \rangle \Rightarrow \langle S_9, \sigma_{284} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1136} = \langle S_8, \sigma_{284} \rangle \Rightarrow \langle y := y + 1, \sigma_{284} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1135} = \frac{\mathcal{T}_{1136}}{\langle S_9, \sigma_{284} \rangle \Rightarrow \langle S_{10}, \sigma_{284} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1137} = \frac{\langle y := y + 1, \sigma_{284} \rangle \Rightarrow \sigma_{285} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{284} \rangle \Rightarrow \langle S_{11}, \sigma_{285} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1138} = \langle S_{11}, \sigma_{285} \rangle \Rightarrow \langle S_{12}, \sigma_{285} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1139} = \langle S_{12}, \sigma_{285} \rangle \Rightarrow \langle S_{13}, \sigma_{285} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1141} = \frac{\langle x := x + 1, \sigma_{285} \rangle \Rightarrow \sigma_{286} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{285} \rangle \Rightarrow \langle S_6, \sigma_{286} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1140} = \frac{\mathcal{T}_{1141}}{\langle S_{13}, \sigma_{285} \rangle \Rightarrow \langle S_{15}, \sigma_{286} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1142} = \frac{\langle S_6, \sigma_{286} \rangle \Rightarrow \langle S_8, \sigma_{286} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{286} \rangle \Rightarrow \langle S_9, \sigma_{286} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1144} = \langle S_8, \sigma_{286} \rangle \Rightarrow \langle y := y + 1, \sigma_{286} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1143} = \frac{\mathcal{T}_{1144}}{\langle S_9, \sigma_{286} \rangle \Rightarrow \langle S_{10}, \sigma_{286} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1145} = \frac{\langle y := y + 1, \sigma_{286} \rangle \Rightarrow \sigma_{287} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{286} \rangle \Rightarrow \langle S_{11}, \sigma_{287} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1146} = \langle S_{11}, \sigma_{287} \rangle \Rightarrow \langle S_{12}, \sigma_{287} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1147} = \langle S_{12}, \sigma_{287} \rangle \Rightarrow \langle S_{13}, \sigma_{287} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1149} = \frac{\langle x := x + 1, \sigma_{287} \rangle \Rightarrow \sigma_{288} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{287} \rangle \Rightarrow \langle S_6, \sigma_{288} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1148} = \frac{\mathcal{T}_{1149}}{\langle S_{13}, \sigma_{287} \rangle \Rightarrow \langle S_{15}, \sigma_{288} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1150} = \frac{\langle S_6, \sigma_{288} \rangle \Rightarrow \langle S_8, \sigma_{288} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{288} \rangle \Rightarrow \langle S_9, \sigma_{288} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1152} = \langle S_8, \sigma_{288} \rangle \Rightarrow \langle y := y + 1, \sigma_{288} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1151} = \frac{\mathcal{T}_{1152}}{\langle S_9, \sigma_{288} \rangle \Rightarrow \langle S_{10}, \sigma_{288} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1153} = \frac{\langle y := y + 1, \sigma_{288} \rangle \Rightarrow \sigma_{289} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{288} \rangle \Rightarrow \langle S_{11}, \sigma_{289} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1154} = \langle S_{11}, \sigma_{289} \rangle \Rightarrow \langle S_{12}, \sigma_{289} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1155} = \langle S_{12}, \sigma_{289} \rangle \Rightarrow \langle S_{13}, \sigma_{289} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1157} = \frac{\langle x := x + 1, \sigma_{289} \rangle \Rightarrow \sigma_{290} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{289} \rangle \Rightarrow \langle S_6, \sigma_{290} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1156} = \frac{\mathcal{T}_{1157}}{\langle S_{13}, \sigma_{289} \rangle \Rightarrow \langle S_{15}, \sigma_{290} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1158} = \frac{\langle S_6, \sigma_{290} \rangle \Rightarrow \langle S_8, \sigma_{290} \rangle \text{ [assert}_{\text{tt}_{\text{sos}}}]}{\langle S_7, \sigma_{290} \rangle \Rightarrow \langle S_9, \sigma_{290} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1160} = \langle S_8, \sigma_{290} \rangle \Rightarrow \langle y := y + 1, \sigma_{290} \rangle \text{ [if - tt}_{\text{tt}_{\text{sos}}}]$$

$$\mathcal{T}_{1159} = \frac{\mathcal{T}_{1160}}{\langle S_9, \sigma_{290} \rangle \Rightarrow \langle S_{10}, \sigma_{290} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1161} = \frac{\langle y := y + 1, \sigma_{290} \rangle \Rightarrow \sigma_{291} \text{ [ass}_{\text{ss}_{\text{sos}}}]}{\langle S_{10}, \sigma_{290} \rangle \Rightarrow \langle S_{11}, \sigma_{291} \rangle} \text{ [comp}_{\text{ss}_{\text{ss}_{\text{ss}}}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1162} = \langle S_{11}, \sigma_{291} \rangle \Rightarrow \langle S_{12}, \sigma_{291} \rangle \text{ [for}_{\text{ss}_{\text{ss}}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1163} = \langle S_{12}, \sigma_{291} \rangle \Rightarrow \langle S_{13}, \sigma_{291} \rangle \text{ [if - tt}_{\text{ss}_{\text{ss}}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1165} = \frac{\langle x := x + 1, \sigma_{291} \rangle \Rightarrow \sigma_{292} \text{ [ass}_{\text{ss}_{\text{ss}}}]}{\langle S_{14}, \sigma_{291} \rangle \Rightarrow \langle S_6, \sigma_{292} \rangle} \text{ [comp}_{\text{ss}_{\text{ss}}}}^2]$$

$$\mathcal{T}_{1164} = \frac{\mathcal{T}_{1165}}{\langle S_{13}, \sigma_{291} \rangle \Rightarrow \langle S_{15}, \sigma_{292} \rangle} \text{ [comp}_{\text{ss}_{\text{ss}}}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1166} = \frac{\langle S_6, \sigma_{292} \rangle \Rightarrow \langle S_8, \sigma_{292} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{292} \rangle \Rightarrow \langle S_9, \sigma_{292} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1168} = \langle S_8, \sigma_{292} \rangle \Rightarrow \langle y := y + 1, \sigma_{292} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1167} = \frac{\mathcal{T}_{1168}}{\langle S_9, \sigma_{292} \rangle \Rightarrow \langle S_{10}, \sigma_{292} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1169} = \frac{\langle y := y + 1, \sigma_{292} \rangle \Rightarrow \sigma_{293} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{292} \rangle \Rightarrow \langle S_{11}, \sigma_{293} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1170} = \langle S_{11}, \sigma_{293} \rangle \Rightarrow \langle S_{12}, \sigma_{293} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1171} = \langle S_{12}, \sigma_{293} \rangle \Rightarrow \langle S_{13}, \sigma_{293} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1173} = \frac{\langle x := x + 1, \sigma_{293} \rangle \Rightarrow \sigma_{294} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{293} \rangle \Rightarrow \langle S_6, \sigma_{294} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1172} = \frac{\mathcal{T}_{1173}}{\langle S_{13}, \sigma_{293} \rangle \Rightarrow \langle S_{15}, \sigma_{294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1174} = \frac{\langle S_6, \sigma_{294} \rangle \Rightarrow \langle S_8, \sigma_{294} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{294} \rangle \Rightarrow \langle S_9, \sigma_{294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1176} = \langle S_8, \sigma_{294} \rangle \Rightarrow \langle y := y + 1, \sigma_{294} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1175} = \frac{\mathcal{T}_{1176}}{\langle S_9, \sigma_{294} \rangle \Rightarrow \langle S_{10}, \sigma_{294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1177} = \frac{\langle y := y + 1, \sigma_{294} \rangle \Rightarrow \sigma_{295} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{294} \rangle \Rightarrow \langle S_{11}, \sigma_{295} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1178} = \langle S_{11}, \sigma_{295} \rangle \Rightarrow \langle S_{12}, \sigma_{295} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1179} = \langle S_{12}, \sigma_{295} \rangle \Rightarrow \langle S_{13}, \sigma_{295} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1181} = \frac{\langle x := x + 1, \sigma_{295} \rangle \Rightarrow \sigma_{296} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{295} \rangle \Rightarrow \langle S_6, \sigma_{296} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1180} = \frac{\mathcal{T}_{1181}}{\langle S_{13}, \sigma_{295} \rangle \Rightarrow \langle S_{15}, \sigma_{296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1182} = \frac{\langle S_6, \sigma_{296} \rangle \Rightarrow \langle S_8, \sigma_{296} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{296} \rangle \Rightarrow \langle S_9, \sigma_{296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1184} = \langle S_8, \sigma_{296} \rangle \Rightarrow \langle y := y + 1, \sigma_{296} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1183} = \frac{\mathcal{T}_{1184}}{\langle S_9, \sigma_{296} \rangle \Rightarrow \langle S_{10}, \sigma_{296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1185} = \frac{\langle y := y + 1, \sigma_{296} \rangle \Rightarrow \sigma_{297} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{296} \rangle \Rightarrow \langle S_{11}, \sigma_{297} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1186} = \langle S_{11}, \sigma_{297} \rangle \Rightarrow \langle S_{12}, \sigma_{297} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1187} = \langle S_{12}, \sigma_{297} \rangle \Rightarrow \langle S_{13}, \sigma_{297} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1189} = \frac{\langle x := x + 1, \sigma_{297} \rangle \Rightarrow \sigma_{298} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{297} \rangle \Rightarrow \langle S_6, \sigma_{298} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1188} = \frac{\mathcal{T}_{1189}}{\langle S_{13}, \sigma_{297} \rangle \Rightarrow \langle S_{15}, \sigma_{298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1190} = \frac{\langle S_6, \sigma_{298} \rangle \Rightarrow \langle S_8, \sigma_{298} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{298} \rangle \Rightarrow \langle S_9, \sigma_{298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1192} = \langle S_8, \sigma_{298} \rangle \Rightarrow \langle y := y + 1, \sigma_{298} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1191} = \frac{\mathcal{T}_{1192}}{\langle S_9, \sigma_{298} \rangle \Rightarrow \langle S_{10}, \sigma_{298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1193} = \frac{\langle y := y + 1, \sigma_{298} \rangle \Rightarrow \sigma_{299} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{298} \rangle \Rightarrow \langle S_{11}, \sigma_{299} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1194} = \langle S_{11}, \sigma_{299} \rangle \Rightarrow \langle S_{12}, \sigma_{299} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1195} = \langle S_{12}, \sigma_{299} \rangle \Rightarrow \langle S_{13}, \sigma_{299} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1197} = \frac{\langle x := x + 1, \sigma_{299} \rangle \Rightarrow \sigma_{300} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{299} \rangle \Rightarrow \langle S_6, \sigma_{300} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1196} = \frac{\mathcal{T}_{1197}}{\langle S_{13}, \sigma_{299} \rangle \Rightarrow \langle S_{15}, \sigma_{300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1198} = \frac{\langle S_6, \sigma_{300} \rangle \Rightarrow \langle S_8, \sigma_{300} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{300} \rangle \Rightarrow \langle S_9, \sigma_{300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1200} = \langle S_8, \sigma_{300} \rangle \Rightarrow \langle y := y + 1, \sigma_{300} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1199} = \frac{\mathcal{T}_{1200}}{\langle S_9, \sigma_{300} \rangle \Rightarrow \langle S_{10}, \sigma_{300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1201} = \frac{\langle y := y + 1, \sigma_{300} \rangle \Rightarrow \sigma_{301} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{300} \rangle \Rightarrow \langle S_{11}, \sigma_{301} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1202} = \langle S_{11}, \sigma_{301} \rangle \Rightarrow \langle S_{12}, \sigma_{301} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1203} = \langle S_{12}, \sigma_{301} \rangle \Rightarrow \langle S_{13}, \sigma_{301} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1205} = \frac{\langle x := x + 1, \sigma_{301} \rangle \Rightarrow \sigma_{302} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{301} \rangle \Rightarrow \langle S_6, \sigma_{302} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1204} = \frac{\mathcal{T}_{1205}}{\langle S_{13}, \sigma_{301} \rangle \Rightarrow \langle S_{15}, \sigma_{302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1206} = \frac{\langle S_6, \sigma_{302} \rangle \Rightarrow \langle S_8, \sigma_{302} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{302} \rangle \Rightarrow \langle S_9, \sigma_{302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1208} = \langle S_8, \sigma_{302} \rangle \Rightarrow \langle y := y + 1, \sigma_{302} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1207} = \frac{\mathcal{T}_{1208}}{\langle S_9, \sigma_{302} \rangle \Rightarrow \langle S_{10}, \sigma_{302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1209} = \frac{\langle y := y + 1, \sigma_{302} \rangle \Rightarrow \sigma_{303} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{302} \rangle \Rightarrow \langle S_{11}, \sigma_{303} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1210} = \langle S_{11}, \sigma_{303} \rangle \Rightarrow \langle S_{12}, \sigma_{303} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1211} = \langle S_{12}, \sigma_{303} \rangle \Rightarrow \langle S_{13}, \sigma_{303} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1213} = \frac{\langle x := x + 1, \sigma_{303} \rangle \Rightarrow \sigma_{304} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{303} \rangle \Rightarrow \langle S_6, \sigma_{304} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1212} = \frac{\mathcal{T}_{1213}}{\langle S_{13}, \sigma_{303} \rangle \Rightarrow \langle S_{15}, \sigma_{304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1214} = \frac{\langle S_6, \sigma_{304} \rangle \Rightarrow \langle S_8, \sigma_{304} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{304} \rangle \Rightarrow \langle S_9, \sigma_{304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1216} = \langle S_8, \sigma_{304} \rangle \Rightarrow \langle y := y + 1, \sigma_{304} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1215} = \frac{\mathcal{T}_{1216}}{\langle S_9, \sigma_{304} \rangle \Rightarrow \langle S_{10}, \sigma_{304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1217} = \frac{\langle y := y + 1, \sigma_{304} \rangle \Rightarrow \sigma_{305} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{304} \rangle \Rightarrow \langle S_{11}, \sigma_{305} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1218} = \langle S_{11}, \sigma_{305} \rangle \Rightarrow \langle S_{12}, \sigma_{305} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1219} = \langle S_{12}, \sigma_{305} \rangle \Rightarrow \langle S_{13}, \sigma_{305} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1221} = \frac{\langle x := x + 1, \sigma_{305} \rangle \Rightarrow \sigma_{306} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{305} \rangle \Rightarrow \langle S_6, \sigma_{306} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1220} = \frac{\mathcal{T}_{1221}}{\langle S_{13}, \sigma_{305} \rangle \Rightarrow \langle S_{15}, \sigma_{306} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1222} = \frac{\langle S_6, \sigma_{306} \rangle \Rightarrow \langle S_8, \sigma_{306} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{306} \rangle \Rightarrow \langle S_9, \sigma_{306} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1224} = \langle S_8, \sigma_{306} \rangle \Rightarrow \langle y := y + 1, \sigma_{306} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1223} = \frac{\mathcal{T}_{1224}}{\langle S_9, \sigma_{306} \rangle \Rightarrow \langle S_{10}, \sigma_{306} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1225} = \frac{\langle y := y + 1, \sigma_{306} \rangle \Rightarrow \sigma_{307} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{306} \rangle \Rightarrow \langle S_{11}, \sigma_{307} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1226} = \langle S_{11}, \sigma_{307} \rangle \Rightarrow \langle S_{12}, \sigma_{307} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1227} = \langle S_{12}, \sigma_{307} \rangle \Rightarrow \langle S_{13}, \sigma_{307} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1229} = \frac{\langle x := x + 1, \sigma_{307} \rangle \Rightarrow \sigma_{308} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{307} \rangle \Rightarrow \langle S_6, \sigma_{308} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1228} = \frac{\mathcal{T}_{1229}}{\langle S_{13}, \sigma_{307} \rangle \Rightarrow \langle S_{15}, \sigma_{308} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1230} = \frac{\langle S_6, \sigma_{308} \rangle \Rightarrow \langle S_8, \sigma_{308} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{308} \rangle \Rightarrow \langle S_9, \sigma_{308} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1232} = \langle S_8, \sigma_{308} \rangle \Rightarrow \langle y := y + 1, \sigma_{308} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1231} = \frac{\mathcal{T}_{1232}}{\langle S_9, \sigma_{308} \rangle \Rightarrow \langle S_{10}, \sigma_{308} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1233} = \frac{\langle y := y + 1, \sigma_{308} \rangle \Rightarrow \sigma_{309} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{308} \rangle \Rightarrow \langle S_{11}, \sigma_{309} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1234} = \langle S_{11}, \sigma_{309} \rangle \Rightarrow \langle S_{12}, \sigma_{309} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1235} = \langle S_{12}, \sigma_{309} \rangle \Rightarrow \langle S_{13}, \sigma_{309} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1237} = \frac{\langle x := x + 1, \sigma_{309} \rangle \Rightarrow \sigma_{310} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{309} \rangle \Rightarrow \langle S_6, \sigma_{310} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1236} = \frac{\mathcal{T}_{1237}}{\langle S_{13}, \sigma_{309} \rangle \Rightarrow \langle S_{15}, \sigma_{310} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1238} = \frac{\langle S_6, \sigma_{310} \rangle \Rightarrow \langle S_8, \sigma_{310} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{310} \rangle \Rightarrow \langle S_9, \sigma_{310} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1240} = \langle S_8, \sigma_{310} \rangle \Rightarrow \langle y := y + 1, \sigma_{310} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1239} = \frac{\mathcal{T}_{1240}}{\langle S_9, \sigma_{310} \rangle \Rightarrow \langle S_{10}, \sigma_{310} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1241} = \frac{\langle y := y + 1, \sigma_{310} \rangle \Rightarrow \sigma_{311} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{310} \rangle \Rightarrow \langle S_{11}, \sigma_{311} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1242} = \langle S_{11}, \sigma_{311} \rangle \Rightarrow \langle S_{12}, \sigma_{311} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1243} = \langle S_{12}, \sigma_{311} \rangle \Rightarrow \langle S_{13}, \sigma_{311} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1245} = \frac{\langle x := x + 1, \sigma_{311} \rangle \Rightarrow \sigma_{312} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{311} \rangle \Rightarrow \langle S_6, \sigma_{312} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1244} = \frac{\mathcal{T}_{1245}}{\langle S_{13}, \sigma_{311} \rangle \Rightarrow \langle S_{15}, \sigma_{312} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1246} = \frac{\langle S_6, \sigma_{312} \rangle \Rightarrow \langle S_8, \sigma_{312} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{312} \rangle \Rightarrow \langle S_9, \sigma_{312} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1248} = \langle S_8, \sigma_{312} \rangle \Rightarrow \langle y := y + 1, \sigma_{312} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1247} = \frac{\mathcal{T}_{1248}}{\langle S_9, \sigma_{312} \rangle \Rightarrow \langle S_{10}, \sigma_{312} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1249} = \frac{\langle y := y + 1, \sigma_{312} \rangle \Rightarrow \sigma_{313} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{312} \rangle \Rightarrow \langle S_{11}, \sigma_{313} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1250} = \langle S_{11}, \sigma_{313} \rangle \Rightarrow \langle S_{12}, \sigma_{313} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1251} = \langle S_{12}, \sigma_{313} \rangle \Rightarrow \langle S_{13}, \sigma_{313} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1253} = \frac{\langle x := x + 1, \sigma_{313} \rangle \Rightarrow \sigma_{314} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{313} \rangle \Rightarrow \langle S_6, \sigma_{314} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1252} = \frac{\mathcal{T}_{1253}}{\langle S_{13}, \sigma_{313} \rangle \Rightarrow \langle S_{15}, \sigma_{314} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1254} = \frac{\langle S_6, \sigma_{314} \rangle \Rightarrow \langle S_8, \sigma_{314} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{314} \rangle \Rightarrow \langle S_9, \sigma_{314} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1256} = \langle S_8, \sigma_{314} \rangle \Rightarrow \langle y := y + 1, \sigma_{314} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1255} = \frac{\mathcal{T}_{1256}}{\langle S_9, \sigma_{314} \rangle \Rightarrow \langle S_{10}, \sigma_{314} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1257} = \frac{\langle y := y + 1, \sigma_{314} \rangle \Rightarrow \sigma_{315} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{314} \rangle \Rightarrow \langle S_{11}, \sigma_{315} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1258} = \langle S_{11}, \sigma_{315} \rangle \Rightarrow \langle S_{12}, \sigma_{315} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1259} = \langle S_{12}, \sigma_{315} \rangle \Rightarrow \langle S_{13}, \sigma_{315} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1261} = \frac{\langle x := x + 1, \sigma_{315} \rangle \Rightarrow \sigma_{316} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{315} \rangle \Rightarrow \langle S_6, \sigma_{316} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1260} = \frac{\mathcal{T}_{1261}}{\langle S_{13}, \sigma_{315} \rangle \Rightarrow \langle S_{15}, \sigma_{316} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1262} = \frac{\langle S_6, \sigma_{316} \rangle \Rightarrow \langle S_8, \sigma_{316} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{316} \rangle \Rightarrow \langle S_9, \sigma_{316} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1264} = \langle S_8, \sigma_{316} \rangle \Rightarrow \langle y := y + 1, \sigma_{316} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1263} = \frac{\mathcal{T}_{1264}}{\langle S_9, \sigma_{316} \rangle \Rightarrow \langle S_{10}, \sigma_{316} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1265} = \frac{\langle y := y + 1, \sigma_{316} \rangle \Rightarrow \sigma_{317} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{316} \rangle \Rightarrow \langle S_{11}, \sigma_{317} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1266} = \langle S_{11}, \sigma_{317} \rangle \Rightarrow \langle S_{12}, \sigma_{317} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1267} = \langle S_{12}, \sigma_{317} \rangle \Rightarrow \langle S_{13}, \sigma_{317} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1269} = \frac{\langle x := x + 1, \sigma_{317} \rangle \Rightarrow \sigma_{318} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{317} \rangle \Rightarrow \langle S_6, \sigma_{318} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1268} = \frac{\mathcal{T}_{1269}}{\langle S_{13}, \sigma_{317} \rangle \Rightarrow \langle S_{15}, \sigma_{318} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1270} = \frac{\langle S_6, \sigma_{318} \rangle \Rightarrow \langle S_8, \sigma_{318} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{318} \rangle \Rightarrow \langle S_9, \sigma_{318} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1272} = \langle S_8, \sigma_{318} \rangle \Rightarrow \langle y := y + 1, \sigma_{318} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1271} = \frac{\mathcal{T}_{1272}}{\langle S_9, \sigma_{318} \rangle \Rightarrow \langle S_{10}, \sigma_{318} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1273} = \frac{\langle y := y + 1, \sigma_{318} \rangle \Rightarrow \sigma_{319} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{318} \rangle \Rightarrow \langle S_{11}, \sigma_{319} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1274} = \langle S_{11}, \sigma_{319} \rangle \Rightarrow \langle S_{12}, \sigma_{319} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1275} = \langle S_{12}, \sigma_{319} \rangle \Rightarrow \langle S_{13}, \sigma_{319} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1277} = \frac{\langle x := x + 1, \sigma_{319} \rangle \Rightarrow \sigma_{320} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{319} \rangle \Rightarrow \langle S_6, \sigma_{320} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1276} = \frac{\mathcal{T}_{1277}}{\langle S_{13}, \sigma_{319} \rangle \Rightarrow \langle S_{15}, \sigma_{320} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1278} = \frac{\langle S_6, \sigma_{320} \rangle \Rightarrow \langle S_8, \sigma_{320} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{320} \rangle \Rightarrow \langle S_9, \sigma_{320} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1280} = \langle S_8, \sigma_{320} \rangle \Rightarrow \langle y := y + 1, \sigma_{320} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1279} = \frac{\mathcal{T}_{1280}}{\langle S_9, \sigma_{320} \rangle \Rightarrow \langle S_{10}, \sigma_{320} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1281} = \frac{\langle y := y + 1, \sigma_{320} \rangle \Rightarrow \sigma_{321} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{320} \rangle \Rightarrow \langle S_{11}, \sigma_{321} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1282} = \langle S_{11}, \sigma_{321} \rangle \Rightarrow \langle S_{12}, \sigma_{321} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1283} = \langle S_{12}, \sigma_{321} \rangle \Rightarrow \langle S_{13}, \sigma_{321} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1285} = \frac{\langle x := x + 1, \sigma_{321} \rangle \Rightarrow \sigma_{322} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{321} \rangle \Rightarrow \langle S_6, \sigma_{322} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1284} = \frac{\mathcal{T}_{1285}}{\langle S_{13}, \sigma_{321} \rangle \Rightarrow \langle S_{15}, \sigma_{322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1286} = \frac{\langle S_6, \sigma_{322} \rangle \Rightarrow \langle S_8, \sigma_{322} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{322} \rangle \Rightarrow \langle S_9, \sigma_{322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1288} = \langle S_8, \sigma_{322} \rangle \Rightarrow \langle y := y + 1, \sigma_{322} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1287} = \frac{\mathcal{T}_{1288}}{\langle S_9, \sigma_{322} \rangle \Rightarrow \langle S_{10}, \sigma_{322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1289} = \frac{\langle y := y + 1, \sigma_{322} \rangle \Rightarrow \sigma_{323} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{322} \rangle \Rightarrow \langle S_{11}, \sigma_{323} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1290} = \langle S_{11}, \sigma_{323} \rangle \Rightarrow \langle S_{12}, \sigma_{323} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1291} = \langle S_{12}, \sigma_{323} \rangle \Rightarrow \langle S_{13}, \sigma_{323} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1293} = \frac{\langle x := x + 1, \sigma_{323} \rangle \Rightarrow \sigma_{324} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{323} \rangle \Rightarrow \langle S_6, \sigma_{324} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1292} = \frac{\mathcal{T}_{1293}}{\langle S_{13}, \sigma_{323} \rangle \Rightarrow \langle S_{15}, \sigma_{324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1294} = \frac{\langle S_6, \sigma_{324} \rangle \Rightarrow \langle S_8, \sigma_{324} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{324} \rangle \Rightarrow \langle S_9, \sigma_{324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1296} = \langle S_8, \sigma_{324} \rangle \Rightarrow \langle y := y + 1, \sigma_{324} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1295} = \frac{\mathcal{T}_{1296}}{\langle S_9, \sigma_{324} \rangle \Rightarrow \langle S_{10}, \sigma_{324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1297} = \frac{\langle y := y + 1, \sigma_{324} \rangle \Rightarrow \sigma_{325} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{324} \rangle \Rightarrow \langle S_{11}, \sigma_{325} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1298} = \langle S_{11}, \sigma_{325} \rangle \Rightarrow \langle S_{12}, \sigma_{325} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1299} = \langle S_{12}, \sigma_{325} \rangle \Rightarrow \langle S_{13}, \sigma_{325} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1301} = \frac{\langle x := x + 1, \sigma_{325} \rangle \Rightarrow \sigma_{326} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{325} \rangle \Rightarrow \langle S_6, \sigma_{326} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1300} = \frac{\mathcal{T}_{1301}}{\langle S_{13}, \sigma_{325} \rangle \Rightarrow \langle S_{15}, \sigma_{326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1302} = \frac{\langle S_6, \sigma_{326} \rangle \Rightarrow \langle S_8, \sigma_{326} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{326} \rangle \Rightarrow \langle S_9, \sigma_{326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1304} = \langle S_8, \sigma_{326} \rangle \Rightarrow \langle y := y + 1, \sigma_{326} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1303} = \frac{\mathcal{T}_{1304}}{\langle S_9, \sigma_{326} \rangle \Rightarrow \langle S_{10}, \sigma_{326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1305} = \frac{\langle y := y + 1, \sigma_{326} \rangle \Rightarrow \sigma_{327} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{326} \rangle \Rightarrow \langle S_{11}, \sigma_{327} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1306} = \langle S_{11}, \sigma_{327} \rangle \Rightarrow \langle S_{12}, \sigma_{327} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1307} = \langle S_{12}, \sigma_{327} \rangle \Rightarrow \langle S_{13}, \sigma_{327} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1309} = \frac{\langle x := x + 1, \sigma_{327} \rangle \Rightarrow \sigma_{328} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{327} \rangle \Rightarrow \langle S_6, \sigma_{328} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1308} = \frac{\mathcal{T}_{1309}}{\langle S_{13}, \sigma_{327} \rangle \Rightarrow \langle S_{15}, \sigma_{328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1310} = \frac{\langle S_6, \sigma_{328} \rangle \Rightarrow \langle S_8, \sigma_{328} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{328} \rangle \Rightarrow \langle S_9, \sigma_{328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1312} = \langle S_8, \sigma_{328} \rangle \Rightarrow \langle y := y + 1, \sigma_{328} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1311} = \frac{\mathcal{T}_{1312}}{\langle S_9, \sigma_{328} \rangle \Rightarrow \langle S_{10}, \sigma_{328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1313} = \frac{\langle y := y + 1, \sigma_{328} \rangle \Rightarrow \sigma_{329} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{328} \rangle \Rightarrow \langle S_{11}, \sigma_{329} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1314} = \langle S_{11}, \sigma_{329} \rangle \Rightarrow \langle S_{12}, \sigma_{329} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1315} = \langle S_{12}, \sigma_{329} \rangle \Rightarrow \langle S_{13}, \sigma_{329} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1317} = \frac{\langle x := x + 1, \sigma_{329} \rangle \Rightarrow \sigma_{330} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{329} \rangle \Rightarrow \langle S_6, \sigma_{330} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1316} = \frac{\mathcal{T}_{1317}}{\langle S_{13}, \sigma_{329} \rangle \Rightarrow \langle S_{15}, \sigma_{330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1318} = \frac{\langle S_6, \sigma_{330} \rangle \Rightarrow \langle S_8, \sigma_{330} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{330} \rangle \Rightarrow \langle S_9, \sigma_{330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1320} = \langle S_8, \sigma_{330} \rangle \Rightarrow \langle y := y + 1, \sigma_{330} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1319} = \frac{\mathcal{T}_{1320}}{\langle S_9, \sigma_{330} \rangle \Rightarrow \langle S_{10}, \sigma_{330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1321} = \frac{\langle y := y + 1, \sigma_{330} \rangle \Rightarrow \sigma_{331} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{330} \rangle \Rightarrow \langle S_{11}, \sigma_{331} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1322} = \langle S_{11}, \sigma_{331} \rangle \Rightarrow \langle S_{12}, \sigma_{331} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1323} = \langle S_{12}, \sigma_{331} \rangle \Rightarrow \langle S_{13}, \sigma_{331} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1325} = \frac{\langle x := x + 1, \sigma_{331} \rangle \Rightarrow \sigma_{332} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{331} \rangle \Rightarrow \langle S_6, \sigma_{332} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1324} = \frac{\mathcal{T}_{1325}}{\langle S_{13}, \sigma_{331} \rangle \Rightarrow \langle S_{15}, \sigma_{332} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1326} = \frac{\langle S_6, \sigma_{332} \rangle \Rightarrow \langle S_8, \sigma_{332} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{332} \rangle \Rightarrow \langle S_9, \sigma_{332} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1328} = \langle S_8, \sigma_{332} \rangle \Rightarrow \langle y := y + 1, \sigma_{332} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1327} = \frac{\mathcal{T}_{1328}}{\langle S_9, \sigma_{332} \rangle \Rightarrow \langle S_{10}, \sigma_{332} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1329} = \frac{\langle y := y + 1, \sigma_{332} \rangle \Rightarrow \sigma_{333} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{332} \rangle \Rightarrow \langle S_{11}, \sigma_{333} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1330} = \langle S_{11}, \sigma_{333} \rangle \Rightarrow \langle S_{12}, \sigma_{333} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1331} = \langle S_{12}, \sigma_{333} \rangle \Rightarrow \langle S_{13}, \sigma_{333} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1333} = \frac{\langle x := x + 1, \sigma_{333} \rangle \Rightarrow \sigma_{334} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{333} \rangle \Rightarrow \langle S_6, \sigma_{334} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1332} = \frac{\mathcal{T}_{1333}}{\langle S_{13}, \sigma_{333} \rangle \Rightarrow \langle S_{15}, \sigma_{334} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1334} = \frac{\langle S_6, \sigma_{334} \rangle \Rightarrow \langle S_8, \sigma_{334} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{334} \rangle \Rightarrow \langle S_9, \sigma_{334} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1336} = \langle S_8, \sigma_{334} \rangle \Rightarrow \langle y := y + 1, \sigma_{334} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1335} = \frac{\mathcal{T}_{1336}}{\langle S_9, \sigma_{334} \rangle \Rightarrow \langle S_{10}, \sigma_{334} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1337} = \frac{\langle y := y + 1, \sigma_{334} \rangle \Rightarrow \sigma_{335} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{334} \rangle \Rightarrow \langle S_{11}, \sigma_{335} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1338} = \langle S_{11}, \sigma_{335} \rangle \Rightarrow \langle S_{12}, \sigma_{335} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1339} = \langle S_{12}, \sigma_{335} \rangle \Rightarrow \langle S_{13}, \sigma_{335} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1341} = \frac{\langle x := x + 1, \sigma_{335} \rangle \Rightarrow \sigma_{336} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{335} \rangle \Rightarrow \langle S_6, \sigma_{336} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1340} = \frac{\mathcal{T}_{1341}}{\langle S_{13}, \sigma_{335} \rangle \Rightarrow \langle S_{15}, \sigma_{336} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1342} = \frac{\langle S_6, \sigma_{336} \rangle \Rightarrow \langle S_8, \sigma_{336} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{336} \rangle \Rightarrow \langle S_9, \sigma_{336} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1344} = \langle S_8, \sigma_{336} \rangle \Rightarrow \langle y := y + 1, \sigma_{336} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1343} = \frac{\mathcal{T}_{1344}}{\langle S_9, \sigma_{336} \rangle \Rightarrow \langle S_{10}, \sigma_{336} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1345} = \frac{\langle y := y + 1, \sigma_{336} \rangle \Rightarrow \sigma_{337} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{336} \rangle \Rightarrow \langle S_{11}, \sigma_{337} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1346} = \langle S_{11}, \sigma_{337} \rangle \Rightarrow \langle S_{12}, \sigma_{337} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1347} = \langle S_{12}, \sigma_{337} \rangle \Rightarrow \langle S_{13}, \sigma_{337} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1349} = \frac{\langle x := x + 1, \sigma_{337} \rangle \Rightarrow \sigma_{338} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{337} \rangle \Rightarrow \langle S_6, \sigma_{338} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1348} = \frac{\mathcal{T}_{1349}}{\langle S_{13}, \sigma_{337} \rangle \Rightarrow \langle S_{15}, \sigma_{338} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1350} = \frac{\langle S_6, \sigma_{338} \rangle \Rightarrow \langle S_8, \sigma_{338} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{338} \rangle \Rightarrow \langle S_9, \sigma_{338} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1352} = \langle S_8, \sigma_{338} \rangle \Rightarrow \langle y := y + 1, \sigma_{338} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1351} = \frac{\mathcal{T}_{1352}}{\langle S_9, \sigma_{338} \rangle \Rightarrow \langle S_{10}, \sigma_{338} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1353} = \frac{\langle y := y + 1, \sigma_{338} \rangle \Rightarrow \sigma_{339} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{338} \rangle \Rightarrow \langle S_{11}, \sigma_{339} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1354} = \langle S_{11}, \sigma_{339} \rangle \Rightarrow \langle S_{12}, \sigma_{339} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1355} = \langle S_{12}, \sigma_{339} \rangle \Rightarrow \langle S_{13}, \sigma_{339} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1357} = \frac{\langle x := x + 1, \sigma_{339} \rangle \Rightarrow \sigma_{340} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{339} \rangle \Rightarrow \langle S_6, \sigma_{340} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1356} = \frac{\mathcal{T}_{1357}}{\langle S_{13}, \sigma_{339} \rangle \Rightarrow \langle S_{15}, \sigma_{340} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1358} = \frac{\langle S_6, \sigma_{340} \rangle \Rightarrow \langle S_8, \sigma_{340} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{340} \rangle \Rightarrow \langle S_9, \sigma_{340} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1360} = \langle S_8, \sigma_{340} \rangle \Rightarrow \langle y := y + 1, \sigma_{340} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1359} = \frac{\mathcal{T}_{1360}}{\langle S_9, \sigma_{340} \rangle \Rightarrow \langle S_{10}, \sigma_{340} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1361} = \frac{\langle y := y + 1, \sigma_{340} \rangle \Rightarrow \sigma_{341} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{340} \rangle \Rightarrow \langle S_{11}, \sigma_{341} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1362} = \langle S_{11}, \sigma_{341} \rangle \Rightarrow \langle S_{12}, \sigma_{341} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1363} = \langle S_{12}, \sigma_{341} \rangle \Rightarrow \langle S_{13}, \sigma_{341} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1365} = \frac{\langle x := x + 1, \sigma_{341} \rangle \Rightarrow \sigma_{342} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{341} \rangle \Rightarrow \langle S_6, \sigma_{342} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1364} = \frac{\mathcal{T}_{1365}}{\langle S_{13}, \sigma_{341} \rangle \Rightarrow \langle S_{15}, \sigma_{342} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1366} = \frac{\langle S_6, \sigma_{342} \rangle \Rightarrow \langle S_8, \sigma_{342} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{342} \rangle \Rightarrow \langle S_9, \sigma_{342} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1368} = \langle S_8, \sigma_{342} \rangle \Rightarrow \langle y := y + 1, \sigma_{342} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1367} = \frac{\mathcal{T}_{1368}}{\langle S_9, \sigma_{342} \rangle \Rightarrow \langle S_{10}, \sigma_{342} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1369} = \frac{\langle y := y + 1, \sigma_{342} \rangle \Rightarrow \sigma_{343} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{342} \rangle \Rightarrow \langle S_{11}, \sigma_{343} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1370} = \langle S_{11}, \sigma_{343} \rangle \Rightarrow \langle S_{12}, \sigma_{343} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1371} = \langle S_{12}, \sigma_{343} \rangle \Rightarrow \langle S_{13}, \sigma_{343} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1373} = \frac{\langle x := x + 1, \sigma_{343} \rangle \Rightarrow \sigma_{344} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{343} \rangle \Rightarrow \langle S_6, \sigma_{344} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1372} = \frac{\mathcal{T}_{1373}}{\langle S_{13}, \sigma_{343} \rangle \Rightarrow \langle S_{15}, \sigma_{344} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{1374} = \frac{\langle S_6, \sigma_{344} \rangle \Rightarrow \langle S_8, \sigma_{344} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{344} \rangle \Rightarrow \langle S_9, \sigma_{344} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{1376} = \langle S_8, \sigma_{344} \rangle \Rightarrow \langle y := y + 1, \sigma_{344} \rangle \text{ [if } -\text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1375} = \frac{\mathcal{T}_{1376}}{\langle S_9, \sigma_{344} \rangle \Rightarrow \langle S_{10}, \sigma_{344} \rangle} [\text{comp}_{\text{sol}}^1]$$

$$\mathcal{T}_{1377} = \frac{\langle y := y + 1, \sigma_{344} \rangle \Rightarrow \sigma_{345} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{344} \rangle \Rightarrow \langle S_{11}, \sigma_{345} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1378} = \langle S_{11}, \sigma_{345} \rangle \Rightarrow \langle S_{12}, \sigma_{345} \rangle \text{ [for}_{\text{sos}}]$$

$$\mathcal{T}_{1379} = \langle S_{12}, \sigma_{345} \rangle \Rightarrow \langle S_{13}, \sigma_{345} \rangle \text{ [if } -\text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1381} = \frac{\langle x := x + 1, \sigma_{345} \rangle \Rightarrow \sigma_{346} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{345} \rangle \Rightarrow \langle S_6, \sigma_{346} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1380} = \frac{\mathcal{T}_{1381}}{\langle S_{13}, \sigma_{345} \rangle \Rightarrow \langle S_{15}, \sigma_{346} \rangle} [\text{comp}_{\text{sol}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1382} = \frac{\langle S_6, \sigma_{346} \rangle \Rightarrow \langle S_8, \sigma_{346} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{346} \rangle \Rightarrow \langle S_9, \sigma_{346} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1384} = \langle S_8, \sigma_{346} \rangle \Rightarrow \langle y := y + 1, \sigma_{346} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1383} = \frac{\mathcal{T}_{1384}}{\langle S_9, \sigma_{346} \rangle \Rightarrow \langle S_{10}, \sigma_{346} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1385} = \frac{\langle y := y + 1, \sigma_{346} \rangle \Rightarrow \sigma_{347} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{346} \rangle \Rightarrow \langle S_{11}, \sigma_{347} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1386} = \langle S_{11}, \sigma_{347} \rangle \Rightarrow \langle S_{12}, \sigma_{347} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1387} = \langle S_{12}, \sigma_{347} \rangle \Rightarrow \langle S_{13}, \sigma_{347} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1389} = \frac{\langle x := x + 1, \sigma_{347} \rangle \Rightarrow \sigma_{348} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{347} \rangle \Rightarrow \langle S_6, \sigma_{348} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1388} = \frac{\mathcal{T}_{1389}}{\langle S_{13}, \sigma_{347} \rangle \Rightarrow \langle S_{15}, \sigma_{348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1390} = \frac{\langle S_6, \sigma_{348} \rangle \Rightarrow \langle S_8, \sigma_{348} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{348} \rangle \Rightarrow \langle S_9, \sigma_{348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1392} = \langle S_8, \sigma_{348} \rangle \Rightarrow \langle y := y + 1, \sigma_{348} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1391} = \frac{\mathcal{T}_{1392}}{\langle S_9, \sigma_{348} \rangle \Rightarrow \langle S_{10}, \sigma_{348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1393} = \frac{\langle y := y + 1, \sigma_{348} \rangle \Rightarrow \sigma_{349} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{348} \rangle \Rightarrow \langle S_{11}, \sigma_{349} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1394} = \langle S_{11}, \sigma_{349} \rangle \Rightarrow \langle S_{12}, \sigma_{349} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1395} = \langle S_{12}, \sigma_{349} \rangle \Rightarrow \langle S_{13}, \sigma_{349} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1397} = \frac{\langle x := x + 1, \sigma_{349} \rangle \Rightarrow \sigma_{350} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{349} \rangle \Rightarrow \langle S_6, \sigma_{350} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1396} = \frac{\mathcal{T}_{1397}}{\langle S_{13}, \sigma_{349} \rangle \Rightarrow \langle S_{15}, \sigma_{350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1398} = \frac{\langle S_6, \sigma_{350} \rangle \Rightarrow \langle S_8, \sigma_{350} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{350} \rangle \Rightarrow \langle S_9, \sigma_{350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1400} = \langle S_8, \sigma_{350} \rangle \Rightarrow \langle y := y + 1, \sigma_{350} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1399} = \frac{\mathcal{T}_{1400}}{\langle S_9, \sigma_{350} \rangle \Rightarrow \langle S_{10}, \sigma_{350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1401} = \frac{\langle y := y + 1, \sigma_{350} \rangle \Rightarrow \sigma_{351} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{350} \rangle \Rightarrow \langle S_{11}, \sigma_{351} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1402} = \langle S_{11}, \sigma_{351} \rangle \Rightarrow \langle S_{12}, \sigma_{351} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1403} = \langle S_{12}, \sigma_{351} \rangle \Rightarrow \langle S_{13}, \sigma_{351} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1405} = \frac{\langle x := x + 1, \sigma_{351} \rangle \Rightarrow \sigma_{352} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{351} \rangle \Rightarrow \langle S_6, \sigma_{352} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1404} = \frac{\mathcal{T}_{1405}}{\langle S_{13}, \sigma_{351} \rangle \Rightarrow \langle S_{15}, \sigma_{352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1406} = \frac{\langle S_6, \sigma_{352} \rangle \Rightarrow \langle S_8, \sigma_{352} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{352} \rangle \Rightarrow \langle S_9, \sigma_{352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1408} = \langle S_8, \sigma_{352} \rangle \Rightarrow \langle y := y + 1, \sigma_{352} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1407} = \frac{\mathcal{T}_{1408}}{\langle S_9, \sigma_{352} \rangle \Rightarrow \langle S_{10}, \sigma_{352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1409} = \frac{\langle y := y + 1, \sigma_{352} \rangle \Rightarrow \sigma_{353} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{352} \rangle \Rightarrow \langle S_{11}, \sigma_{353} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1410} = \langle S_{11}, \sigma_{353} \rangle \Rightarrow \langle S_{12}, \sigma_{353} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1411} = \langle S_{12}, \sigma_{353} \rangle \Rightarrow \langle S_{13}, \sigma_{353} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1413} = \frac{\langle x := x + 1, \sigma_{353} \rangle \Rightarrow \sigma_{354} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{353} \rangle \Rightarrow \langle S_6, \sigma_{354} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1412} = \frac{\mathcal{T}_{1413}}{\langle S_{13}, \sigma_{353} \rangle \Rightarrow \langle S_{15}, \sigma_{354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1414} = \frac{\langle S_6, \sigma_{354} \rangle \Rightarrow \langle S_8, \sigma_{354} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{354} \rangle \Rightarrow \langle S_9, \sigma_{354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1416} = \langle S_8, \sigma_{354} \rangle \Rightarrow \langle y := y + 1, \sigma_{354} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1415} = \frac{\mathcal{T}_{1416}}{\langle S_9, \sigma_{354} \rangle \Rightarrow \langle S_{10}, \sigma_{354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1417} = \frac{\langle y := y + 1, \sigma_{354} \rangle \Rightarrow \sigma_{355} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{354} \rangle \Rightarrow \langle S_{11}, \sigma_{355} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1418} = \langle S_{11}, \sigma_{355} \rangle \Rightarrow \langle S_{12}, \sigma_{355} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1419} = \langle S_{12}, \sigma_{355} \rangle \Rightarrow \langle S_{13}, \sigma_{355} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1421} = \frac{\langle x := x + 1, \sigma_{355} \rangle \Rightarrow \sigma_{356} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{355} \rangle \Rightarrow \langle S_6, \sigma_{356} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1420} = \frac{\mathcal{T}_{1421}}{\langle S_{13}, \sigma_{355} \rangle \Rightarrow \langle S_{15}, \sigma_{356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1422} = \frac{\langle S_6, \sigma_{356} \rangle \Rightarrow \langle S_8, \sigma_{356} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{356} \rangle \Rightarrow \langle S_9, \sigma_{356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1424} = \langle S_8, \sigma_{356} \rangle \Rightarrow \langle y := y + 1, \sigma_{356} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1423} = \frac{\mathcal{T}_{1424}}{\langle S_9, \sigma_{356} \rangle \Rightarrow \langle S_{10}, \sigma_{356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1425} = \frac{\langle y := y + 1, \sigma_{356} \rangle \Rightarrow \sigma_{357} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{356} \rangle \Rightarrow \langle S_{11}, \sigma_{357} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1426} = \langle S_{11}, \sigma_{357} \rangle \Rightarrow \langle S_{12}, \sigma_{357} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1427} = \langle S_{12}, \sigma_{357} \rangle \Rightarrow \langle S_{13}, \sigma_{357} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1429} = \frac{\langle x := x + 1, \sigma_{357} \rangle \Rightarrow \sigma_{358} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{357} \rangle \Rightarrow \langle S_6, \sigma_{358} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1428} = \frac{\mathcal{T}_{1429}}{\langle S_{13}, \sigma_{357} \rangle \Rightarrow \langle S_{15}, \sigma_{358} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1430} = \frac{\langle S_6, \sigma_{358} \rangle \Rightarrow \langle S_8, \sigma_{358} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{358} \rangle \Rightarrow \langle S_9, \sigma_{358} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1432} = \langle S_8, \sigma_{358} \rangle \Rightarrow \langle y := y + 1, \sigma_{358} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1431} = \frac{\mathcal{T}_{1432}}{\langle S_9, \sigma_{358} \rangle \Rightarrow \langle S_{10}, \sigma_{358} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1433} = \frac{\langle y := y + 1, \sigma_{358} \rangle \Rightarrow \sigma_{359} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{358} \rangle \Rightarrow \langle S_{11}, \sigma_{359} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{1434} = \langle S_{11}, \sigma_{359} \rangle \Rightarrow \langle S_{12}, \sigma_{359} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1435} = \langle S_{12}, \sigma_{359} \rangle \Rightarrow \langle S_{13}, \sigma_{359} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1437} = \frac{\langle x := x + 1, \sigma_{359} \rangle \Rightarrow \sigma_{360} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{359} \rangle \Rightarrow \langle S_6, \sigma_{360} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1436} = \frac{\mathcal{T}_{1437}}{\langle S_{13}, \sigma_{359} \rangle \Rightarrow \langle S_{15}, \sigma_{360} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1438} = \frac{\langle S_6, \sigma_{360} \rangle \Rightarrow \langle S_8, \sigma_{360} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{360} \rangle \Rightarrow \langle S_9, \sigma_{360} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1440} = \langle S_8, \sigma_{360} \rangle \Rightarrow \langle y := y + 1, \sigma_{360} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1439} = \frac{\mathcal{T}_{1440}}{\langle S_9, \sigma_{360} \rangle \Rightarrow \langle S_{10}, \sigma_{360} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1441} = \frac{\langle y := y + 1, \sigma_{360} \rangle \Rightarrow \sigma_{361} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{360} \rangle \Rightarrow \langle S_{11}, \sigma_{361} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1442} = \langle S_{11}, \sigma_{361} \rangle \Rightarrow \langle S_{12}, \sigma_{361} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1443} = \langle S_{12}, \sigma_{361} \rangle \Rightarrow \langle S_{13}, \sigma_{361} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1445} = \frac{\langle x := x + 1, \sigma_{361} \rangle \Rightarrow \sigma_{362} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{361} \rangle \Rightarrow \langle S_6, \sigma_{362} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1444} = \frac{\mathcal{T}_{1445}}{\langle S_{13}, \sigma_{361} \rangle \Rightarrow \langle S_{15}, \sigma_{362} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1446} = \frac{\langle S_6, \sigma_{362} \rangle \Rightarrow \langle S_8, \sigma_{362} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{362} \rangle \Rightarrow \langle S_9, \sigma_{362} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1448} = \langle S_8, \sigma_{362} \rangle \Rightarrow \langle y := y + 1, \sigma_{362} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1447} = \frac{\mathcal{T}_{1448}}{\langle S_9, \sigma_{362} \rangle \Rightarrow \langle S_{10}, \sigma_{362} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1449} = \frac{\langle y := y + 1, \sigma_{362} \rangle \Rightarrow \sigma_{363} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{362} \rangle \Rightarrow \langle S_{11}, \sigma_{363} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1450} = \langle S_{11}, \sigma_{363} \rangle \Rightarrow \langle S_{12}, \sigma_{363} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1451} = \langle S_{12}, \sigma_{363} \rangle \Rightarrow \langle S_{13}, \sigma_{363} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1453} = \frac{\langle x := x + 1, \sigma_{363} \rangle \Rightarrow \sigma_{364} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{363} \rangle \Rightarrow \langle S_6, \sigma_{364} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1452} = \frac{\mathcal{T}_{1453}}{\langle S_{13}, \sigma_{363} \rangle \Rightarrow \langle S_{15}, \sigma_{364} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1454} = \frac{\langle S_6, \sigma_{364} \rangle \Rightarrow \langle S_8, \sigma_{364} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{364} \rangle \Rightarrow \langle S_9, \sigma_{364} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1456} = \langle S_8, \sigma_{364} \rangle \Rightarrow \langle y := y + 1, \sigma_{364} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1455} = \frac{\mathcal{T}_{1456}}{\langle S_9, \sigma_{364} \rangle \Rightarrow \langle S_{10}, \sigma_{364} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1457} = \frac{\langle y := y + 1, \sigma_{364} \rangle \Rightarrow \sigma_{365} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{364} \rangle \Rightarrow \langle S_{11}, \sigma_{365} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1458} = \langle S_{11}, \sigma_{365} \rangle \Rightarrow \langle S_{12}, \sigma_{365} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1459} = \langle S_{12}, \sigma_{365} \rangle \Rightarrow \langle S_{13}, \sigma_{365} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1461} = \frac{\langle x := x + 1, \sigma_{365} \rangle \Rightarrow \sigma_{366} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{365} \rangle \Rightarrow \langle S_6, \sigma_{366} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1460} = \frac{\mathcal{T}_{1461}}{\langle S_{13}, \sigma_{365} \rangle \Rightarrow \langle S_{15}, \sigma_{366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1462} = \frac{\langle S_6, \sigma_{366} \rangle \Rightarrow \langle S_8, \sigma_{366} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{366} \rangle \Rightarrow \langle S_9, \sigma_{366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1464} = \langle S_8, \sigma_{366} \rangle \Rightarrow \langle y := y + 1, \sigma_{366} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1463} = \frac{\mathcal{T}_{1464}}{\langle S_9, \sigma_{366} \rangle \Rightarrow \langle S_{10}, \sigma_{366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1465} = \frac{\langle y := y + 1, \sigma_{366} \rangle \Rightarrow \sigma_{367} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{366} \rangle \Rightarrow \langle S_{11}, \sigma_{367} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1466} = \langle S_{11}, \sigma_{367} \rangle \Rightarrow \langle S_{12}, \sigma_{367} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1467} = \langle S_{12}, \sigma_{367} \rangle \Rightarrow \langle S_{13}, \sigma_{367} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1469} = \frac{\langle x := x + 1, \sigma_{367} \rangle \Rightarrow \sigma_{368} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{367} \rangle \Rightarrow \langle S_6, \sigma_{368} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1468} = \frac{\mathcal{T}_{1469}}{\langle S_{13}, \sigma_{367} \rangle \Rightarrow \langle S_{15}, \sigma_{368} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1470} = \frac{\langle S_6, \sigma_{368} \rangle \Rightarrow \langle S_8, \sigma_{368} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{368} \rangle \Rightarrow \langle S_9, \sigma_{368} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1472} = \langle S_8, \sigma_{368} \rangle \Rightarrow \langle y := y + 1, \sigma_{368} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1471} = \frac{\mathcal{T}_{1472}}{\langle S_9, \sigma_{368} \rangle \Rightarrow \langle S_{10}, \sigma_{368} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1473} = \frac{\langle y := y + 1, \sigma_{368} \rangle \Rightarrow \sigma_{369} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{368} \rangle \Rightarrow \langle S_{11}, \sigma_{369} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1474} = \langle S_{11}, \sigma_{369} \rangle \Rightarrow \langle S_{12}, \sigma_{369} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1475} = \langle S_{12}, \sigma_{369} \rangle \Rightarrow \langle S_{13}, \sigma_{369} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1477} = \frac{\langle x := x + 1, \sigma_{369} \rangle \Rightarrow \sigma_{370} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{369} \rangle \Rightarrow \langle S_6, \sigma_{370} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1476} = \frac{\mathcal{T}_{1477}}{\langle S_{13}, \sigma_{369} \rangle \Rightarrow \langle S_{15}, \sigma_{370} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1478} = \frac{\langle S_6, \sigma_{370} \rangle \Rightarrow \langle S_8, \sigma_{370} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{370} \rangle \Rightarrow \langle S_9, \sigma_{370} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1480} = \langle S_8, \sigma_{370} \rangle \Rightarrow \langle y := y + 1, \sigma_{370} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1479} = \frac{\mathcal{T}_{1480}}{\langle S_9, \sigma_{370} \rangle \Rightarrow \langle S_{10}, \sigma_{370} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1481} = \frac{\langle y := y + 1, \sigma_{370} \rangle \Rightarrow \sigma_{371} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{370} \rangle \Rightarrow \langle S_{11}, \sigma_{371} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1482} = \langle S_{11}, \sigma_{371} \rangle \Rightarrow \langle S_{12}, \sigma_{371} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1483} = \langle S_{12}, \sigma_{371} \rangle \Rightarrow \langle S_{13}, \sigma_{371} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1485} = \frac{\langle x := x + 1, \sigma_{371} \rangle \Rightarrow \sigma_{372} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{371} \rangle \Rightarrow \langle S_6, \sigma_{372} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1484} = \frac{\mathcal{T}_{1485}}{\langle S_{13}, \sigma_{371} \rangle \Rightarrow \langle S_{15}, \sigma_{372} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1486} = \frac{\langle S_6, \sigma_{372} \rangle \Rightarrow \langle S_8, \sigma_{372} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{372} \rangle \Rightarrow \langle S_9, \sigma_{372} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1488} = \langle S_8, \sigma_{372} \rangle \Rightarrow \langle y := y + 1, \sigma_{372} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1487} = \frac{\mathcal{T}_{1488}}{\langle S_9, \sigma_{372} \rangle \Rightarrow \langle S_{10}, \sigma_{372} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1489} = \frac{\langle y := y + 1, \sigma_{372} \rangle \Rightarrow \sigma_{373} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{372} \rangle \Rightarrow \langle S_{11}, \sigma_{373} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1490} = \langle S_{11}, \sigma_{373} \rangle \Rightarrow \langle S_{12}, \sigma_{373} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1491} = \langle S_{12}, \sigma_{373} \rangle \Rightarrow \langle S_{13}, \sigma_{373} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1493} = \frac{\langle x := x + 1, \sigma_{373} \rangle \Rightarrow \sigma_{374} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{373} \rangle \Rightarrow \langle S_6, \sigma_{374} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1492} = \frac{\mathcal{T}_{1493}}{\langle S_{13}, \sigma_{373} \rangle \Rightarrow \langle S_{15}, \sigma_{374} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1494} = \frac{\langle S_6, \sigma_{374} \rangle \Rightarrow \langle S_8, \sigma_{374} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{374} \rangle \Rightarrow \langle S_9, \sigma_{374} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1496} = \langle S_8, \sigma_{374} \rangle \Rightarrow \langle y := y + 1, \sigma_{374} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1495} = \frac{\mathcal{T}_{1496}}{\langle S_9, \sigma_{374} \rangle \Rightarrow \langle S_{10}, \sigma_{374} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1497} = \frac{\langle y := y + 1, \sigma_{374} \rangle \Rightarrow \sigma_{375} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{374} \rangle \Rightarrow \langle S_{11}, \sigma_{375} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1498} = \langle S_{11}, \sigma_{375} \rangle \Rightarrow \langle S_{12}, \sigma_{375} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1499} = \langle S_{12}, \sigma_{375} \rangle \Rightarrow \langle S_{13}, \sigma_{375} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1501} = \frac{\langle x := x + 1, \sigma_{375} \rangle \Rightarrow \sigma_{376} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{375} \rangle \Rightarrow \langle S_6, \sigma_{376} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1500} = \frac{\mathcal{T}_{1501}}{\langle S_{13}, \sigma_{375} \rangle \Rightarrow \langle S_{15}, \sigma_{376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1502} = \frac{\langle S_6, \sigma_{376} \rangle \Rightarrow \langle S_8, \sigma_{376} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{376} \rangle \Rightarrow \langle S_9, \sigma_{376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1504} = \langle S_8, \sigma_{376} \rangle \Rightarrow \langle y := y + 1, \sigma_{376} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1503} = \frac{\mathcal{T}_{1504}}{\langle S_9, \sigma_{376} \rangle \Rightarrow \langle S_{10}, \sigma_{376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1505} = \frac{\langle y := y + 1, \sigma_{376} \rangle \Rightarrow \sigma_{377} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{376} \rangle \Rightarrow \langle S_{11}, \sigma_{377} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1506} = \langle S_{11}, \sigma_{377} \rangle \Rightarrow \langle S_{12}, \sigma_{377} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1507} = \langle S_{12}, \sigma_{377} \rangle \Rightarrow \langle S_{13}, \sigma_{377} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1509} = \frac{\langle x := x + 1, \sigma_{377} \rangle \Rightarrow \sigma_{378} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{377} \rangle \Rightarrow \langle S_6, \sigma_{378} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1508} = \frac{\mathcal{T}_{1509}}{\langle S_{13}, \sigma_{377} \rangle \Rightarrow \langle S_{15}, \sigma_{378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1510} = \frac{\langle S_6, \sigma_{378} \rangle \Rightarrow \langle S_8, \sigma_{378} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{378} \rangle \Rightarrow \langle S_9, \sigma_{378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1512} = \langle S_8, \sigma_{378} \rangle \Rightarrow \langle y := y + 1, \sigma_{378} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1511} = \frac{\mathcal{T}_{1512}}{\langle S_9, \sigma_{378} \rangle \Rightarrow \langle S_{10}, \sigma_{378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1513} = \frac{\langle y := y + 1, \sigma_{378} \rangle \Rightarrow \sigma_{379} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{378} \rangle \Rightarrow \langle S_{11}, \sigma_{379} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1514} = \langle S_{11}, \sigma_{379} \rangle \Rightarrow \langle S_{12}, \sigma_{379} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1515} = \langle S_{12}, \sigma_{379} \rangle \Rightarrow \langle S_{13}, \sigma_{379} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1517} = \frac{\langle x := x + 1, \sigma_{379} \rangle \Rightarrow \sigma_{380} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{379} \rangle \Rightarrow \langle S_6, \sigma_{380} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1516} = \frac{\mathcal{T}_{1517}}{\langle S_{13}, \sigma_{379} \rangle \Rightarrow \langle S_{15}, \sigma_{380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1518} = \frac{\langle S_6, \sigma_{380} \rangle \Rightarrow \langle S_8, \sigma_{380} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{380} \rangle \Rightarrow \langle S_9, \sigma_{380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1520} = \langle S_8, \sigma_{380} \rangle \Rightarrow \langle y := y + 1, \sigma_{380} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1519} = \frac{\mathcal{T}_{1520}}{\langle S_9, \sigma_{380} \rangle \Rightarrow \langle S_{10}, \sigma_{380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1521} = \frac{\langle y := y + 1, \sigma_{380} \rangle \Rightarrow \sigma_{381} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{380} \rangle \Rightarrow \langle S_{11}, \sigma_{381} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1522} = \langle S_{11}, \sigma_{381} \rangle \Rightarrow \langle S_{12}, \sigma_{381} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1523} = \langle S_{12}, \sigma_{381} \rangle \Rightarrow \langle S_{13}, \sigma_{381} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1525} = \frac{\langle x := x + 1, \sigma_{381} \rangle \Rightarrow \sigma_{382} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{381} \rangle \Rightarrow \langle S_6, \sigma_{382} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1524} = \frac{\mathcal{T}_{1525}}{\langle S_{13}, \sigma_{381} \rangle \Rightarrow \langle S_{15}, \sigma_{382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1526} = \frac{\langle S_6, \sigma_{382} \rangle \Rightarrow \langle S_8, \sigma_{382} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{382} \rangle \Rightarrow \langle S_9, \sigma_{382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1528} = \langle S_8, \sigma_{382} \rangle \Rightarrow \langle y := y + 1, \sigma_{382} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1527} = \frac{\mathcal{T}_{1528}}{\langle S_9, \sigma_{382} \rangle \Rightarrow \langle S_{10}, \sigma_{382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1529} = \frac{\langle y := y + 1, \sigma_{382} \rangle \Rightarrow \sigma_{383} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{382} \rangle \Rightarrow \langle S_{11}, \sigma_{383} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1530} = \langle S_{11}, \sigma_{383} \rangle \Rightarrow \langle S_{12}, \sigma_{383} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1531} = \langle S_{12}, \sigma_{383} \rangle \Rightarrow \langle S_{13}, \sigma_{383} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1533} = \frac{\langle x := x + 1, \sigma_{383} \rangle \Rightarrow \sigma_{384} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{383} \rangle \Rightarrow \langle S_6, \sigma_{384} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1532} = \frac{\mathcal{T}_{1533}}{\langle S_{13}, \sigma_{383} \rangle \Rightarrow \langle S_{15}, \sigma_{384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1534} = \frac{\langle S_6, \sigma_{384} \rangle \Rightarrow \langle S_8, \sigma_{384} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{384} \rangle \Rightarrow \langle S_9, \sigma_{384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1536} = \langle S_8, \sigma_{384} \rangle \Rightarrow \langle y := y + 1, \sigma_{384} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1535} = \frac{\mathcal{T}_{1536}}{\langle S_9, \sigma_{384} \rangle \Rightarrow \langle S_{10}, \sigma_{384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1537} = \frac{\langle y := y + 1, \sigma_{384} \rangle \Rightarrow \sigma_{385} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{384} \rangle \Rightarrow \langle S_{11}, \sigma_{385} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1538} = \langle S_{11}, \sigma_{385} \rangle \Rightarrow \langle S_{12}, \sigma_{385} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1539} = \langle S_{12}, \sigma_{385} \rangle \Rightarrow \langle S_{13}, \sigma_{385} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1541} = \frac{\langle x := x + 1, \sigma_{385} \rangle \Rightarrow \sigma_{386} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{385} \rangle \Rightarrow \langle S_6, \sigma_{386} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1540} = \frac{\mathcal{T}_{1541}}{\langle S_{13}, \sigma_{385} \rangle \Rightarrow \langle S_{15}, \sigma_{386} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1542} = \frac{\langle S_6, \sigma_{386} \rangle \Rightarrow \langle S_8, \sigma_{386} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{386} \rangle \Rightarrow \langle S_9, \sigma_{386} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1544} = \langle S_8, \sigma_{386} \rangle \Rightarrow \langle y := y + 1, \sigma_{386} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1543} = \frac{\mathcal{T}_{1544}}{\langle S_9, \sigma_{386} \rangle \Rightarrow \langle S_{10}, \sigma_{386} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1545} = \frac{\langle y := y + 1, \sigma_{386} \rangle \Rightarrow \sigma_{387} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{386} \rangle \Rightarrow \langle S_{11}, \sigma_{387} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1546} = \langle S_{11}, \sigma_{387} \rangle \Rightarrow \langle S_{12}, \sigma_{387} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1547} = \langle S_{12}, \sigma_{387} \rangle \Rightarrow \langle S_{13}, \sigma_{387} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1549} = \frac{\langle x := x + 1, \sigma_{387} \rangle \Rightarrow \sigma_{388} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{387} \rangle \Rightarrow \langle S_6, \sigma_{388} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1548} = \frac{\mathcal{T}_{1549}}{\langle S_{13}, \sigma_{387} \rangle \Rightarrow \langle S_{15}, \sigma_{388} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1550} = \frac{\langle S_6, \sigma_{388} \rangle \Rightarrow \langle S_8, \sigma_{388} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{388} \rangle \Rightarrow \langle S_9, \sigma_{388} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1552} = \langle S_8, \sigma_{388} \rangle \Rightarrow \langle y := y + 1, \sigma_{388} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1551} = \frac{\mathcal{T}_{1552}}{\langle S_9, \sigma_{388} \rangle \Rightarrow \langle S_{10}, \sigma_{388} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1553} = \frac{\langle y := y + 1, \sigma_{388} \rangle \Rightarrow \sigma_{389} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{388} \rangle \Rightarrow \langle S_{11}, \sigma_{389} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1554} = \langle S_{11}, \sigma_{389} \rangle \Rightarrow \langle S_{12}, \sigma_{389} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1555} = \langle S_{12}, \sigma_{389} \rangle \Rightarrow \langle S_{13}, \sigma_{389} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1557} = \frac{\langle x := x + 1, \sigma_{389} \rangle \Rightarrow \sigma_{390} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{389} \rangle \Rightarrow \langle S_6, \sigma_{390} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1556} = \frac{\mathcal{T}_{1557}}{\langle S_{13}, \sigma_{389} \rangle \Rightarrow \langle S_{15}, \sigma_{390} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1558} = \frac{\langle S_6, \sigma_{390} \rangle \Rightarrow \langle S_8, \sigma_{390} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{390} \rangle \Rightarrow \langle S_9, \sigma_{390} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1560} = \langle S_8, \sigma_{390} \rangle \Rightarrow \langle y := y + 1, \sigma_{390} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1559} = \frac{\mathcal{T}_{1560}}{\langle S_9, \sigma_{390} \rangle \Rightarrow \langle S_{10}, \sigma_{390} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1561} = \frac{\langle y := y + 1, \sigma_{390} \rangle \Rightarrow \sigma_{391} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{390} \rangle \Rightarrow \langle S_{11}, \sigma_{391} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1562} = \langle S_{11}, \sigma_{391} \rangle \Rightarrow \langle S_{12}, \sigma_{391} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1563} = \langle S_{12}, \sigma_{391} \rangle \Rightarrow \langle S_{13}, \sigma_{391} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1565} = \frac{\langle x := x + 1, \sigma_{391} \rangle \Rightarrow \sigma_{392} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{391} \rangle \Rightarrow \langle S_6, \sigma_{392} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1564} = \frac{\mathcal{T}_{1565}}{\langle S_{13}, \sigma_{391} \rangle \Rightarrow \langle S_{15}, \sigma_{392} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1566} = \frac{\langle S_6, \sigma_{392} \rangle \Rightarrow \langle S_8, \sigma_{392} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{392} \rangle \Rightarrow \langle S_9, \sigma_{392} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1568} = \langle S_8, \sigma_{392} \rangle \Rightarrow \langle y := y + 1, \sigma_{392} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1567} = \frac{\mathcal{T}_{1568}}{\langle S_9, \sigma_{392} \rangle \Rightarrow \langle S_{10}, \sigma_{392} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1569} = \frac{\langle y := y + 1, \sigma_{392} \rangle \Rightarrow \sigma_{393} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{392} \rangle \Rightarrow \langle S_{11}, \sigma_{393} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1570} = \langle S_{11}, \sigma_{393} \rangle \Rightarrow \langle S_{12}, \sigma_{393} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1571} = \langle S_{12}, \sigma_{393} \rangle \Rightarrow \langle S_{13}, \sigma_{393} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1573} = \frac{\langle x := x + 1, \sigma_{393} \rangle \Rightarrow \sigma_{394} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{393} \rangle \Rightarrow \langle S_6, \sigma_{394} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1572} = \frac{\mathcal{T}_{1573}}{\langle S_{13}, \sigma_{393} \rangle \Rightarrow \langle S_{15}, \sigma_{394} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1574} = \frac{\langle S_6, \sigma_{394} \rangle \Rightarrow \langle S_8, \sigma_{394} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{394} \rangle \Rightarrow \langle S_9, \sigma_{394} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1576} = \langle S_8, \sigma_{394} \rangle \Rightarrow \langle y := y + 1, \sigma_{394} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1575} = \frac{\mathcal{T}_{1576}}{\langle S_9, \sigma_{394} \rangle \Rightarrow \langle S_{10}, \sigma_{394} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1577} = \frac{\langle y := y + 1, \sigma_{394} \rangle \Rightarrow \sigma_{395} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{394} \rangle \Rightarrow \langle S_{11}, \sigma_{395} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1578} = \langle S_{11}, \sigma_{395} \rangle \Rightarrow \langle S_{12}, \sigma_{395} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1579} = \langle S_{12}, \sigma_{395} \rangle \Rightarrow \langle S_{13}, \sigma_{395} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1581} = \frac{\langle x := x + 1, \sigma_{395} \rangle \Rightarrow \sigma_{396} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{395} \rangle \Rightarrow \langle S_6, \sigma_{396} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1580} = \frac{\mathcal{T}_{1581}}{\langle S_{13}, \sigma_{395} \rangle \Rightarrow \langle S_{15}, \sigma_{396} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1582} = \frac{\langle S_6, \sigma_{396} \rangle \Rightarrow \langle S_8, \sigma_{396} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{396} \rangle \Rightarrow \langle S_9, \sigma_{396} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1584} = \langle S_8, \sigma_{396} \rangle \Rightarrow \langle y := y + 1, \sigma_{396} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1583} = \frac{\mathcal{T}_{1584}}{\langle S_9, \sigma_{396} \rangle \Rightarrow \langle S_{10}, \sigma_{396} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1585} = \frac{\langle y := y + 1, \sigma_{396} \rangle \Rightarrow \sigma_{397} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{396} \rangle \Rightarrow \langle S_{11}, \sigma_{397} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1586} = \langle S_{11}, \sigma_{397} \rangle \Rightarrow \langle S_{12}, \sigma_{397} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1587} = \langle S_{12}, \sigma_{397} \rangle \Rightarrow \langle S_{13}, \sigma_{397} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1589} = \frac{\langle x := x + 1, \sigma_{397} \rangle \Rightarrow \sigma_{398} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{397} \rangle \Rightarrow \langle S_6, \sigma_{398} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1588} = \frac{\mathcal{T}_{1589}}{\langle S_{13}, \sigma_{397} \rangle \Rightarrow \langle S_{15}, \sigma_{398} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1590} = \frac{\langle S_6, \sigma_{398} \rangle \Rightarrow \langle S_8, \sigma_{398} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{398} \rangle \Rightarrow \langle S_9, \sigma_{398} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1592} = \langle S_8, \sigma_{398} \rangle \Rightarrow \langle y := y + 1, \sigma_{398} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1591} = \frac{\mathcal{T}_{1592}}{\langle S_9, \sigma_{398} \rangle \Rightarrow \langle S_{10}, \sigma_{398} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1593} = \frac{\langle y := y + 1, \sigma_{398} \rangle \Rightarrow \sigma_{399} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{398} \rangle \Rightarrow \langle S_{11}, \sigma_{399} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1594} = \langle S_{11}, \sigma_{399} \rangle \Rightarrow \langle S_{12}, \sigma_{399} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1595} = \langle S_{12}, \sigma_{399} \rangle \Rightarrow \langle S_{13}, \sigma_{399} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1597} = \frac{\langle x := x + 1, \sigma_{399} \rangle \Rightarrow \sigma_{400} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{399} \rangle \Rightarrow \langle S_6, \sigma_{400} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1596} = \frac{\mathcal{T}_{1597}}{\langle S_{13}, \sigma_{399} \rangle \Rightarrow \langle S_{15}, \sigma_{400} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1598} = \frac{\langle S_6, \sigma_{400} \rangle \Rightarrow \langle S_8, \sigma_{400} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{400} \rangle \Rightarrow \langle S_9, \sigma_{400} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1600} = \langle S_8, \sigma_{400} \rangle \Rightarrow \langle y := y + 1, \sigma_{400} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1599} = \frac{\mathcal{T}_{1600}}{\langle S_9, \sigma_{400} \rangle \Rightarrow \langle S_{10}, \sigma_{400} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1601} = \frac{\langle y := y + 1, \sigma_{400} \rangle \Rightarrow \sigma_{401} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{400} \rangle \Rightarrow \langle S_{11}, \sigma_{401} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{1602} = \langle S_{11}, \sigma_{401} \rangle \Rightarrow \langle S_{12}, \sigma_{401} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1603} = \langle S_{12}, \sigma_{401} \rangle \Rightarrow \langle S_{13}, \sigma_{401} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1605} = \frac{\langle x := x + 1, \sigma_{401} \rangle \Rightarrow \sigma_{402} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{401} \rangle \Rightarrow \langle S_6, \sigma_{402} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1604} = \frac{\mathcal{T}_{1605}}{\langle S_{13}, \sigma_{401} \rangle \Rightarrow \langle S_{15}, \sigma_{402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1606} = \frac{\langle S_6, \sigma_{402} \rangle \Rightarrow \langle S_8, \sigma_{402} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{402} \rangle \Rightarrow \langle S_9, \sigma_{402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1608} = \langle S_8, \sigma_{402} \rangle \Rightarrow \langle y := y + 1, \sigma_{402} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1607} = \frac{\mathcal{T}_{1608}}{\langle S_9, \sigma_{402} \rangle \Rightarrow \langle S_{10}, \sigma_{402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1609} = \frac{\langle y := y + 1, \sigma_{402} \rangle \Rightarrow \sigma_{403} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{402} \rangle \Rightarrow \langle S_{11}, \sigma_{403} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1610} = \langle S_{11}, \sigma_{403} \rangle \Rightarrow \langle S_{12}, \sigma_{403} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1611} = \langle S_{12}, \sigma_{403} \rangle \Rightarrow \langle S_{13}, \sigma_{403} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1613} = \frac{\langle x := x + 1, \sigma_{403} \rangle \Rightarrow \sigma_{404} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{403} \rangle \Rightarrow \langle S_6, \sigma_{404} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1612} = \frac{\mathcal{T}_{1613}}{\langle S_{13}, \sigma_{403} \rangle \Rightarrow \langle S_{15}, \sigma_{404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1614} = \frac{\langle S_6, \sigma_{404} \rangle \Rightarrow \langle S_8, \sigma_{404} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{404} \rangle \Rightarrow \langle S_9, \sigma_{404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1616} = \langle S_8, \sigma_{404} \rangle \Rightarrow \langle y := y + 1, \sigma_{404} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1615} = \frac{\mathcal{T}_{1616}}{\langle S_9, \sigma_{404} \rangle \Rightarrow \langle S_{10}, \sigma_{404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1617} = \frac{\langle y := y + 1, \sigma_{404} \rangle \Rightarrow \sigma_{405} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{404} \rangle \Rightarrow \langle S_{11}, \sigma_{405} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1618} = \langle S_{11}, \sigma_{405} \rangle \Rightarrow \langle S_{12}, \sigma_{405} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1619} = \langle S_{12}, \sigma_{405} \rangle \Rightarrow \langle S_{13}, \sigma_{405} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1621} = \frac{\langle x := x + 1, \sigma_{405} \rangle \Rightarrow \sigma_{406} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{405} \rangle \Rightarrow \langle S_6, \sigma_{406} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1620} = \frac{\mathcal{T}_{1621}}{\langle S_{13}, \sigma_{405} \rangle \Rightarrow \langle S_{15}, \sigma_{406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1622} = \frac{\langle S_6, \sigma_{406} \rangle \Rightarrow \langle S_8, \sigma_{406} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{406} \rangle \Rightarrow \langle S_9, \sigma_{406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1624} = \langle S_8, \sigma_{406} \rangle \Rightarrow \langle y := y + 1, \sigma_{406} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1623} = \frac{\mathcal{T}_{1624}}{\langle S_9, \sigma_{406} \rangle \Rightarrow \langle S_{10}, \sigma_{406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1625} = \frac{\langle y := y + 1, \sigma_{406} \rangle \Rightarrow \sigma_{407} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{406} \rangle \Rightarrow \langle S_{11}, \sigma_{407} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1626} = \langle S_{11}, \sigma_{407} \rangle \Rightarrow \langle S_{12}, \sigma_{407} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1627} = \langle S_{12}, \sigma_{407} \rangle \Rightarrow \langle S_{13}, \sigma_{407} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1629} = \frac{\langle x := x + 1, \sigma_{407} \rangle \Rightarrow \sigma_{408} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{407} \rangle \Rightarrow \langle S_6, \sigma_{408} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1628} = \frac{\mathcal{T}_{1629}}{\langle S_{13}, \sigma_{407} \rangle \Rightarrow \langle S_{15}, \sigma_{408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1630} = \frac{\langle S_6, \sigma_{408} \rangle \Rightarrow \langle S_8, \sigma_{408} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{408} \rangle \Rightarrow \langle S_9, \sigma_{408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1632} = \langle S_8, \sigma_{408} \rangle \Rightarrow \langle y := y + 1, \sigma_{408} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1631} = \frac{\mathcal{T}_{1632}}{\langle S_9, \sigma_{408} \rangle \Rightarrow \langle S_{10}, \sigma_{408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1633} = \frac{\langle y := y + 1, \sigma_{408} \rangle \Rightarrow \sigma_{409} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{408} \rangle \Rightarrow \langle S_{11}, \sigma_{409} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1634} = \langle S_{11}, \sigma_{409} \rangle \Rightarrow \langle S_{12}, \sigma_{409} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1635} = \langle S_{12}, \sigma_{409} \rangle \Rightarrow \langle S_{13}, \sigma_{409} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1637} = \frac{\langle x := x + 1, \sigma_{409} \rangle \Rightarrow \sigma_{410} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{409} \rangle \Rightarrow \langle S_6, \sigma_{410} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1636} = \frac{\mathcal{T}_{1637}}{\langle S_{13}, \sigma_{409} \rangle \Rightarrow \langle S_{15}, \sigma_{410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1638} = \frac{\langle S_6, \sigma_{410} \rangle \Rightarrow \langle S_8, \sigma_{410} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{410} \rangle \Rightarrow \langle S_9, \sigma_{410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1640} = \langle S_8, \sigma_{410} \rangle \Rightarrow \langle y := y + 1, \sigma_{410} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1639} = \frac{\mathcal{T}_{1640}}{\langle S_9, \sigma_{410} \rangle \Rightarrow \langle S_{10}, \sigma_{410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1641} = \frac{\langle y := y + 1, \sigma_{410} \rangle \Rightarrow \sigma_{411} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{410} \rangle \Rightarrow \langle S_{11}, \sigma_{411} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1642} = \langle S_{11}, \sigma_{411} \rangle \Rightarrow \langle S_{12}, \sigma_{411} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1643} = \langle S_{12}, \sigma_{411} \rangle \Rightarrow \langle S_{13}, \sigma_{411} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1645} = \frac{\langle x := x + 1, \sigma_{411} \rangle \Rightarrow \sigma_{412} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{411} \rangle \Rightarrow \langle S_6, \sigma_{412} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1644} = \frac{\mathcal{T}_{1645}}{\langle S_{13}, \sigma_{411} \rangle \Rightarrow \langle S_{15}, \sigma_{412} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1646} = \frac{\langle S_6, \sigma_{412} \rangle \Rightarrow \langle S_8, \sigma_{412} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{412} \rangle \Rightarrow \langle S_9, \sigma_{412} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1648} = \langle S_8, \sigma_{412} \rangle \Rightarrow \langle y := y + 1, \sigma_{412} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1647} = \frac{\mathcal{T}_{1648}}{\langle S_9, \sigma_{412} \rangle \Rightarrow \langle S_{10}, \sigma_{412} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1649} = \frac{\langle y := y + 1, \sigma_{412} \rangle \Rightarrow \sigma_{413} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{412} \rangle \Rightarrow \langle S_{11}, \sigma_{413} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{1650} = \langle S_{11}, \sigma_{413} \rangle \Rightarrow \langle S_{12}, \sigma_{413} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1651} = \langle S_{12}, \sigma_{413} \rangle \Rightarrow \langle S_{13}, \sigma_{413} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1653} = \frac{\langle x := x + 1, \sigma_{413} \rangle \Rightarrow \sigma_{414} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{413} \rangle \Rightarrow \langle S_6, \sigma_{414} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1652} = \frac{\mathcal{T}_{1653}}{\langle S_{13}, \sigma_{413} \rangle \Rightarrow \langle S_{15}, \sigma_{414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1654} = \frac{\langle S_6, \sigma_{414} \rangle \Rightarrow \langle S_8, \sigma_{414} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{414} \rangle \Rightarrow \langle S_9, \sigma_{414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1656} = \langle S_8, \sigma_{414} \rangle \Rightarrow \langle y := y + 1, \sigma_{414} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1655} = \frac{\mathcal{T}_{1656}}{\langle S_9, \sigma_{414} \rangle \Rightarrow \langle S_{10}, \sigma_{414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1657} = \frac{\langle y := y + 1, \sigma_{414} \rangle \Rightarrow \sigma_{415} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{414} \rangle \Rightarrow \langle S_{11}, \sigma_{415} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1658} = \langle S_{11}, \sigma_{415} \rangle \Rightarrow \langle S_{12}, \sigma_{415} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1659} = \langle S_{12}, \sigma_{415} \rangle \Rightarrow \langle S_{13}, \sigma_{415} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1661} = \frac{\langle x := x + 1, \sigma_{415} \rangle \Rightarrow \sigma_{416} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{415} \rangle \Rightarrow \langle S_6, \sigma_{416} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1660} = \frac{\mathcal{T}_{1661}}{\langle S_{13}, \sigma_{415} \rangle \Rightarrow \langle S_{15}, \sigma_{416} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1662} = \frac{\langle S_6, \sigma_{416} \rangle \Rightarrow \langle S_8, \sigma_{416} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{416} \rangle \Rightarrow \langle S_9, \sigma_{416} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1664} = \langle S_8, \sigma_{416} \rangle \Rightarrow \langle y := y + 1, \sigma_{416} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1663} = \frac{\mathcal{T}_{1664}}{\langle S_9, \sigma_{416} \rangle \Rightarrow \langle S_{10}, \sigma_{416} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1665} = \frac{\langle y := y + 1, \sigma_{416} \rangle \Rightarrow \sigma_{417} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{416} \rangle \Rightarrow \langle S_{11}, \sigma_{417} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1666} = \langle S_{11}, \sigma_{417} \rangle \Rightarrow \langle S_{12}, \sigma_{417} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{1667} = \langle S_{12}, \sigma_{417} \rangle \Rightarrow \langle S_{13}, \sigma_{417} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1669} = \frac{\langle x := x + 1, \sigma_{417} \rangle \Rightarrow \sigma_{418} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{417} \rangle \Rightarrow \langle S_6, \sigma_{418} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1668} = \frac{\mathcal{T}_{1669}}{\langle S_{13}, \sigma_{417} \rangle \Rightarrow \langle S_{15}, \sigma_{418} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1670} = \frac{\langle S_6, \sigma_{418} \rangle \Rightarrow \langle S_8, \sigma_{418} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{418} \rangle \Rightarrow \langle S_9, \sigma_{418} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1672} = \langle S_8, \sigma_{418} \rangle \Rightarrow \langle y := y + 1, \sigma_{418} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1671} = \frac{\mathcal{T}_{1672}}{\langle S_9, \sigma_{418} \rangle \Rightarrow \langle S_{10}, \sigma_{418} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1673} = \frac{\langle y := y + 1, \sigma_{418} \rangle \Rightarrow \sigma_{419} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{418} \rangle \Rightarrow \langle S_{11}, \sigma_{419} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1674} = \langle S_{11}, \sigma_{419} \rangle \Rightarrow \langle S_{12}, \sigma_{419} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{1675} = \langle S_{12}, \sigma_{419} \rangle \Rightarrow \langle S_{13}, \sigma_{419} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{1677} = \frac{\langle x := x + 1, \sigma_{419} \rangle \Rightarrow \sigma_{420} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{419} \rangle \Rightarrow \langle S_6, \sigma_{420} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1676} = \frac{\mathcal{T}_{1677}}{\langle S_{13}, \sigma_{419} \rangle \Rightarrow \langle S_{15}, \sigma_{420} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1678} = \frac{\langle S_6, \sigma_{420} \rangle \Rightarrow \langle S_8, \sigma_{420} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{420} \rangle \Rightarrow \langle S_9, \sigma_{420} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1680} = \langle S_8, \sigma_{420} \rangle \Rightarrow \langle y := y + 1, \sigma_{420} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1679} = \frac{\mathcal{T}_{1680}}{\langle S_9, \sigma_{420} \rangle \Rightarrow \langle S_{10}, \sigma_{420} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{1681} = \frac{\langle y := y + 1, \sigma_{420} \rangle \Rightarrow \sigma_{421} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{420} \rangle \Rightarrow \langle S_{11}, \sigma_{421} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{1682} = \langle S_{11}, \sigma_{421} \rangle \Rightarrow \langle S_{12}, \sigma_{421} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1683} = \langle S_{12}, \sigma_{421} \rangle \Rightarrow \langle S_{13}, \sigma_{421} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1685} = \frac{\langle x := x + 1, \sigma_{421} \rangle \Rightarrow \sigma_{422} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{421} \rangle \Rightarrow \langle S_6, \sigma_{422} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1684} = \frac{\mathcal{T}_{1685}}{\langle S_{13}, \sigma_{421} \rangle \Rightarrow \langle S_{15}, \sigma_{422} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1686} = \frac{\langle S_6, \sigma_{422} \rangle \Rightarrow \langle S_8, \sigma_{422} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{422} \rangle \Rightarrow \langle S_9, \sigma_{422} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1688} = \langle S_8, \sigma_{422} \rangle \Rightarrow \langle y := y + 1, \sigma_{422} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1687} = \frac{\mathcal{T}_{1688}}{\langle S_9, \sigma_{422} \rangle \Rightarrow \langle S_{10}, \sigma_{422} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1689} = \frac{\langle y := y + 1, \sigma_{422} \rangle \Rightarrow \sigma_{423} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{422} \rangle \Rightarrow \langle S_{11}, \sigma_{423} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1690} = \langle S_{11}, \sigma_{423} \rangle \Rightarrow \langle S_{12}, \sigma_{423} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1691} = \langle S_{12}, \sigma_{423} \rangle \Rightarrow \langle S_{13}, \sigma_{423} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1693} = \frac{\langle x := x + 1, \sigma_{423} \rangle \Rightarrow \sigma_{424} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{423} \rangle \Rightarrow \langle S_6, \sigma_{424} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1692} = \frac{\mathcal{T}_{1693}}{\langle S_{13}, \sigma_{423} \rangle \Rightarrow \langle S_{15}, \sigma_{424} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1694} = \frac{\langle S_6, \sigma_{424} \rangle \Rightarrow \langle S_8, \sigma_{424} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{424} \rangle \Rightarrow \langle S_9, \sigma_{424} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1696} = \langle S_8, \sigma_{424} \rangle \Rightarrow \langle y := y + 1, \sigma_{424} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1695} = \frac{\mathcal{T}_{1696}}{\langle S_9, \sigma_{424} \rangle \Rightarrow \langle S_{10}, \sigma_{424} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1697} = \frac{\langle y := y + 1, \sigma_{424} \rangle \Rightarrow \sigma_{425} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{424} \rangle \Rightarrow \langle S_{11}, \sigma_{425} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1698} = \langle S_{11}, \sigma_{425} \rangle \Rightarrow \langle S_{12}, \sigma_{425} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1699} = \langle S_{12}, \sigma_{425} \rangle \Rightarrow \langle S_{13}, \sigma_{425} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1701} = \frac{\langle x := x + 1, \sigma_{425} \rangle \Rightarrow \sigma_{426} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{425} \rangle \Rightarrow \langle S_6, \sigma_{426} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1700} = \frac{\mathcal{T}_{1701}}{\langle S_{13}, \sigma_{425} \rangle \Rightarrow \langle S_{15}, \sigma_{426} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1702} = \frac{\langle S_6, \sigma_{426} \rangle \Rightarrow \langle S_8, \sigma_{426} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{426} \rangle \Rightarrow \langle S_9, \sigma_{426} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1704} = \langle S_8, \sigma_{426} \rangle \Rightarrow \langle y := y + 1, \sigma_{426} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1703} = \frac{\mathcal{T}_{1704}}{\langle S_9, \sigma_{426} \rangle \Rightarrow \langle S_{10}, \sigma_{426} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1705} = \frac{\langle y := y + 1, \sigma_{426} \rangle \Rightarrow \sigma_{427} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{426} \rangle \Rightarrow \langle S_{11}, \sigma_{427} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1706} = \langle S_{11}, \sigma_{427} \rangle \Rightarrow \langle S_{12}, \sigma_{427} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1707} = \langle S_{12}, \sigma_{427} \rangle \Rightarrow \langle S_{13}, \sigma_{427} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1709} = \frac{\langle x := x + 1, \sigma_{427} \rangle \Rightarrow \sigma_{428} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{427} \rangle \Rightarrow \langle S_6, \sigma_{428} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1708} = \frac{\mathcal{T}_{1709}}{\langle S_{13}, \sigma_{427} \rangle \Rightarrow \langle S_{15}, \sigma_{428} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1710} = \frac{\langle S_6, \sigma_{428} \rangle \Rightarrow \langle S_8, \sigma_{428} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{428} \rangle \Rightarrow \langle S_9, \sigma_{428} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1712} = \langle S_8, \sigma_{428} \rangle \Rightarrow \langle y := y + 1, \sigma_{428} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1711} = \frac{\mathcal{T}_{1712}}{\langle S_9, \sigma_{428} \rangle \Rightarrow \langle S_{10}, \sigma_{428} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1713} = \frac{\langle y := y + 1, \sigma_{428} \rangle \Rightarrow \sigma_{429} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{428} \rangle \Rightarrow \langle S_{11}, \sigma_{429} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1714} = \langle S_{11}, \sigma_{429} \rangle \Rightarrow \langle S_{12}, \sigma_{429} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1715} = \langle S_{12}, \sigma_{429} \rangle \Rightarrow \langle S_{13}, \sigma_{429} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1717} = \frac{\langle x := x + 1, \sigma_{429} \rangle \Rightarrow \sigma_{430} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{429} \rangle \Rightarrow \langle S_6, \sigma_{430} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1716} = \frac{\mathcal{T}_{1717}}{\langle S_{13}, \sigma_{429} \rangle \Rightarrow \langle S_{15}, \sigma_{430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1718} = \frac{\langle S_6, \sigma_{430} \rangle \Rightarrow \langle S_8, \sigma_{430} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{430} \rangle \Rightarrow \langle S_9, \sigma_{430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1720} = \langle S_8, \sigma_{430} \rangle \Rightarrow \langle y := y + 1, \sigma_{430} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1719} = \frac{\mathcal{T}_{1720}}{\langle S_9, \sigma_{430} \rangle \Rightarrow \langle S_{10}, \sigma_{430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1721} = \frac{\langle y := y + 1, \sigma_{430} \rangle \Rightarrow \sigma_{431} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{430} \rangle \Rightarrow \langle S_{11}, \sigma_{431} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1722} = \langle S_{11}, \sigma_{431} \rangle \Rightarrow \langle S_{12}, \sigma_{431} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1723} = \langle S_{12}, \sigma_{431} \rangle \Rightarrow \langle S_{13}, \sigma_{431} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1725} = \frac{\langle x := x + 1, \sigma_{431} \rangle \Rightarrow \sigma_{432} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{431} \rangle \Rightarrow \langle S_6, \sigma_{432} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1724} = \frac{\mathcal{T}_{1725}}{\langle S_{13}, \sigma_{431} \rangle \Rightarrow \langle S_{15}, \sigma_{432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1726} = \frac{\langle S_6, \sigma_{432} \rangle \Rightarrow \langle S_8, \sigma_{432} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{432} \rangle \Rightarrow \langle S_9, \sigma_{432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1728} = \langle S_8, \sigma_{432} \rangle \Rightarrow \langle y := y + 1, \sigma_{432} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1727} = \frac{\mathcal{T}_{1728}}{\langle S_9, \sigma_{432} \rangle \Rightarrow \langle S_{10}, \sigma_{432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1729} = \frac{\langle y := y + 1, \sigma_{432} \rangle \Rightarrow \sigma_{433} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{432} \rangle \Rightarrow \langle S_{11}, \sigma_{433} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1730} = \langle S_{11}, \sigma_{433} \rangle \Rightarrow \langle S_{12}, \sigma_{433} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1731} = \langle S_{12}, \sigma_{433} \rangle \Rightarrow \langle S_{13}, \sigma_{433} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1733} = \frac{\langle x := x + 1, \sigma_{433} \rangle \Rightarrow \sigma_{434} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{433} \rangle \Rightarrow \langle S_6, \sigma_{434} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1732} = \frac{\mathcal{T}_{1733}}{\langle S_{13}, \sigma_{433} \rangle \Rightarrow \langle S_{15}, \sigma_{434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1734} = \frac{\langle S_6, \sigma_{434} \rangle \Rightarrow \langle S_8, \sigma_{434} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{434} \rangle \Rightarrow \langle S_9, \sigma_{434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1736} = \langle S_8, \sigma_{434} \rangle \Rightarrow \langle y := y + 1, \sigma_{434} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1735} = \frac{\mathcal{T}_{1736}}{\langle S_9, \sigma_{434} \rangle \Rightarrow \langle S_{10}, \sigma_{434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1737} = \frac{\langle y := y + 1, \sigma_{434} \rangle \Rightarrow \sigma_{435} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{434} \rangle \Rightarrow \langle S_{11}, \sigma_{435} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1738} = \langle S_{11}, \sigma_{435} \rangle \Rightarrow \langle S_{12}, \sigma_{435} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1739} = \langle S_{12}, \sigma_{435} \rangle \Rightarrow \langle S_{13}, \sigma_{435} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1741} = \frac{\langle x := x + 1, \sigma_{435} \rangle \Rightarrow \sigma_{436} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{435} \rangle \Rightarrow \langle S_6, \sigma_{436} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1740} = \frac{\mathcal{T}_{1741}}{\langle S_{13}, \sigma_{435} \rangle \Rightarrow \langle S_{15}, \sigma_{436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1742} = \frac{\langle S_6, \sigma_{436} \rangle \Rightarrow \langle S_8, \sigma_{436} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{436} \rangle \Rightarrow \langle S_9, \sigma_{436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1744} = \langle S_8, \sigma_{436} \rangle \Rightarrow \langle y := y + 1, \sigma_{436} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1743} = \frac{\mathcal{T}_{1744}}{\langle S_9, \sigma_{436} \rangle \Rightarrow \langle S_{10}, \sigma_{436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1745} = \frac{\langle y := y + 1, \sigma_{436} \rangle \Rightarrow \sigma_{437} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{436} \rangle \Rightarrow \langle S_{11}, \sigma_{437} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1746} = \langle S_{11}, \sigma_{437} \rangle \Rightarrow \langle S_{12}, \sigma_{437} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1747} = \langle S_{12}, \sigma_{437} \rangle \Rightarrow \langle S_{13}, \sigma_{437} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1749} = \frac{\langle x := x + 1, \sigma_{437} \rangle \Rightarrow \sigma_{438} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{437} \rangle \Rightarrow \langle S_6, \sigma_{438} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1748} = \frac{\mathcal{T}_{1749}}{\langle S_{13}, \sigma_{437} \rangle \Rightarrow \langle S_{15}, \sigma_{438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1750} = \frac{\langle S_6, \sigma_{438} \rangle \Rightarrow \langle S_8, \sigma_{438} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{438} \rangle \Rightarrow \langle S_9, \sigma_{438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1752} = \langle S_8, \sigma_{438} \rangle \Rightarrow \langle y := y + 1, \sigma_{438} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1751} = \frac{\mathcal{T}_{1752}}{\langle S_9, \sigma_{438} \rangle \Rightarrow \langle S_{10}, \sigma_{438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1753} = \frac{\langle y := y + 1, \sigma_{438} \rangle \Rightarrow \sigma_{439} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{438} \rangle \Rightarrow \langle S_{11}, \sigma_{439} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1754} = \langle S_{11}, \sigma_{439} \rangle \Rightarrow \langle S_{12}, \sigma_{439} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1755} = \langle S_{12}, \sigma_{439} \rangle \Rightarrow \langle S_{13}, \sigma_{439} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1757} = \frac{\langle x := x + 1, \sigma_{439} \rangle \Rightarrow \sigma_{440} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{439} \rangle \Rightarrow \langle S_6, \sigma_{440} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1756} = \frac{\mathcal{T}_{1757}}{\langle S_{13}, \sigma_{439} \rangle \Rightarrow \langle S_{15}, \sigma_{440} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1758} = \frac{\langle S_6, \sigma_{440} \rangle \Rightarrow \langle S_8, \sigma_{440} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{440} \rangle \Rightarrow \langle S_9, \sigma_{440} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1760} = \langle S_8, \sigma_{440} \rangle \Rightarrow \langle y := y + 1, \sigma_{440} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1759} = \frac{\mathcal{T}_{1760}}{\langle S_9, \sigma_{440} \rangle \Rightarrow \langle S_{10}, \sigma_{440} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1761} = \frac{\langle y := y + 1, \sigma_{440} \rangle \Rightarrow \sigma_{441} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{440} \rangle \Rightarrow \langle S_{11}, \sigma_{441} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1762} = \langle S_{11}, \sigma_{441} \rangle \Rightarrow \langle S_{12}, \sigma_{441} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1763} = \langle S_{12}, \sigma_{441} \rangle \Rightarrow \langle S_{13}, \sigma_{441} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1765} = \frac{\langle x := x + 1, \sigma_{441} \rangle \Rightarrow \sigma_{442} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{441} \rangle \Rightarrow \langle S_6, \sigma_{442} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1764} = \frac{\mathcal{T}_{1765}}{\langle S_{13}, \sigma_{441} \rangle \Rightarrow \langle S_{15}, \sigma_{442} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1766} = \frac{\langle S_6, \sigma_{442} \rangle \Rightarrow \langle S_8, \sigma_{442} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{442} \rangle \Rightarrow \langle S_9, \sigma_{442} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1768} = \langle S_8, \sigma_{442} \rangle \Rightarrow \langle y := y + 1, \sigma_{442} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1767} = \frac{\mathcal{T}_{1768}}{\langle S_9, \sigma_{442} \rangle \Rightarrow \langle S_{10}, \sigma_{442} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1769} = \frac{\langle y := y + 1, \sigma_{442} \rangle \Rightarrow \sigma_{443} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{442} \rangle \Rightarrow \langle S_{11}, \sigma_{443} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1770} = \langle S_{11}, \sigma_{443} \rangle \Rightarrow \langle S_{12}, \sigma_{443} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1771} = \langle S_{12}, \sigma_{443} \rangle \Rightarrow \langle S_{13}, \sigma_{443} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1773} = \frac{\langle x := x + 1, \sigma_{443} \rangle \Rightarrow \sigma_{444} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{443} \rangle \Rightarrow \langle S_6, \sigma_{444} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1772} = \frac{\mathcal{T}_{1773}}{\langle S_{13}, \sigma_{443} \rangle \Rightarrow \langle S_{15}, \sigma_{444} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1774} = \frac{\langle S_6, \sigma_{444} \rangle \Rightarrow \langle S_8, \sigma_{444} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{444} \rangle \Rightarrow \langle S_9, \sigma_{444} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1776} = \langle S_8, \sigma_{444} \rangle \Rightarrow \langle y := y + 1, \sigma_{444} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1775} = \frac{\mathcal{T}_{1776}}{\langle S_9, \sigma_{444} \rangle \Rightarrow \langle S_{10}, \sigma_{444} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1777} = \frac{\langle y := y + 1, \sigma_{444} \rangle \Rightarrow \sigma_{445} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{444} \rangle \Rightarrow \langle S_{11}, \sigma_{445} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1778} = \langle S_{11}, \sigma_{445} \rangle \Rightarrow \langle S_{12}, \sigma_{445} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1779} = \langle S_{12}, \sigma_{445} \rangle \Rightarrow \langle S_{13}, \sigma_{445} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1781} = \frac{\langle x := x + 1, \sigma_{445} \rangle \Rightarrow \sigma_{446} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{445} \rangle \Rightarrow \langle S_6, \sigma_{446} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1780} = \frac{\mathcal{T}_{1781}}{\langle S_{13}, \sigma_{445} \rangle \Rightarrow \langle S_{15}, \sigma_{446} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1782} = \frac{\langle S_6, \sigma_{446} \rangle \Rightarrow \langle S_8, \sigma_{446} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{446} \rangle \Rightarrow \langle S_9, \sigma_{446} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1784} = \langle S_8, \sigma_{446} \rangle \Rightarrow \langle y := y + 1, \sigma_{446} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1783} = \frac{\mathcal{T}_{1784}}{\langle S_9, \sigma_{446} \rangle \Rightarrow \langle S_{10}, \sigma_{446} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1785} = \frac{\langle y := y + 1, \sigma_{446} \rangle \Rightarrow \sigma_{447} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{446} \rangle \Rightarrow \langle S_{11}, \sigma_{447} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1786} = \langle S_{11}, \sigma_{447} \rangle \Rightarrow \langle S_{12}, \sigma_{447} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1787} = \langle S_{12}, \sigma_{447} \rangle \Rightarrow \langle S_{13}, \sigma_{447} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1789} = \frac{\langle x := x + 1, \sigma_{447} \rangle \Rightarrow \sigma_{448} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{447} \rangle \Rightarrow \langle S_6, \sigma_{448} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1788} = \frac{\mathcal{T}_{1789}}{\langle S_{13}, \sigma_{447} \rangle \Rightarrow \langle S_{15}, \sigma_{448} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1790} = \frac{\langle S_6, \sigma_{448} \rangle \Rightarrow \langle S_8, \sigma_{448} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{448} \rangle \Rightarrow \langle S_9, \sigma_{448} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1792} = \langle S_8, \sigma_{448} \rangle \Rightarrow \langle y := y + 1, \sigma_{448} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1791} = \frac{\mathcal{T}_{1792}}{\langle S_9, \sigma_{448} \rangle \Rightarrow \langle S_{10}, \sigma_{448} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1793} = \frac{\langle y := y + 1, \sigma_{448} \rangle \Rightarrow \sigma_{449} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{448} \rangle \Rightarrow \langle S_{11}, \sigma_{449} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1794} = \langle S_{11}, \sigma_{449} \rangle \Rightarrow \langle S_{12}, \sigma_{449} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1795} = \langle S_{12}, \sigma_{449} \rangle \Rightarrow \langle S_{13}, \sigma_{449} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1797} = \frac{\langle x := x + 1, \sigma_{449} \rangle \Rightarrow \sigma_{450} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{449} \rangle \Rightarrow \langle S_6, \sigma_{450} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1796} = \frac{\mathcal{T}_{1797}}{\langle S_{13}, \sigma_{449} \rangle \Rightarrow \langle S_{15}, \sigma_{450} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1798} = \frac{\langle S_6, \sigma_{450} \rangle \Rightarrow \langle S_8, \sigma_{450} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{450} \rangle \Rightarrow \langle S_9, \sigma_{450} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1800} = \langle S_8, \sigma_{450} \rangle \Rightarrow \langle y := y + 1, \sigma_{450} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1799} = \frac{\mathcal{T}_{1800}}{\langle S_9, \sigma_{450} \rangle \Rightarrow \langle S_{10}, \sigma_{450} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1801} = \frac{\langle y := y + 1, \sigma_{450} \rangle \Rightarrow \sigma_{451} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{450} \rangle \Rightarrow \langle S_{11}, \sigma_{451} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1802} = \langle S_{11}, \sigma_{451} \rangle \Rightarrow \langle S_{12}, \sigma_{451} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1803} = \langle S_{12}, \sigma_{451} \rangle \Rightarrow \langle S_{13}, \sigma_{451} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1805} = \frac{\langle x := x + 1, \sigma_{451} \rangle \Rightarrow \sigma_{452} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{451} \rangle \Rightarrow \langle S_6, \sigma_{452} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1804} = \frac{\mathcal{T}_{1805}}{\langle S_{13}, \sigma_{451} \rangle \Rightarrow \langle S_{15}, \sigma_{452} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1806} = \frac{\langle S_6, \sigma_{452} \rangle \Rightarrow \langle S_8, \sigma_{452} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{452} \rangle \Rightarrow \langle S_9, \sigma_{452} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1808} = \langle S_8, \sigma_{452} \rangle \Rightarrow \langle y := y + 1, \sigma_{452} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1807} = \frac{\mathcal{T}_{1808}}{\langle S_9, \sigma_{452} \rangle \Rightarrow \langle S_{10}, \sigma_{452} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1809} = \frac{\langle y := y + 1, \sigma_{452} \rangle \Rightarrow \sigma_{453} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{452} \rangle \Rightarrow \langle S_{11}, \sigma_{453} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1810} = \langle S_{11}, \sigma_{453} \rangle \Rightarrow \langle S_{12}, \sigma_{453} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1811} = \langle S_{12}, \sigma_{453} \rangle \Rightarrow \langle S_{13}, \sigma_{453} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1813} = \frac{\langle x := x + 1, \sigma_{453} \rangle \Rightarrow \sigma_{454} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{453} \rangle \Rightarrow \langle S_6, \sigma_{454} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1812} = \frac{\mathcal{T}_{1813}}{\langle S_{13}, \sigma_{453} \rangle \Rightarrow \langle S_{15}, \sigma_{454} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1814} = \frac{\langle S_6, \sigma_{454} \rangle \Rightarrow \langle S_8, \sigma_{454} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{454} \rangle \Rightarrow \langle S_9, \sigma_{454} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1816} = \langle S_8, \sigma_{454} \rangle \Rightarrow \langle y := y + 1, \sigma_{454} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1815} = \frac{\mathcal{T}_{1816}}{\langle S_9, \sigma_{454} \rangle \Rightarrow \langle S_{10}, \sigma_{454} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1817} = \frac{\langle y := y + 1, \sigma_{454} \rangle \Rightarrow \sigma_{455} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{454} \rangle \Rightarrow \langle S_{11}, \sigma_{455} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1818} = \langle S_{11}, \sigma_{455} \rangle \Rightarrow \langle S_{12}, \sigma_{455} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1819} = \langle S_{12}, \sigma_{455} \rangle \Rightarrow \langle S_{13}, \sigma_{455} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1821} = \frac{\langle x := x + 1, \sigma_{455} \rangle \Rightarrow \sigma_{456} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{455} \rangle \Rightarrow \langle S_6, \sigma_{456} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1820} = \frac{\mathcal{T}_{1821}}{\langle S_{13}, \sigma_{455} \rangle \Rightarrow \langle S_{15}, \sigma_{456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1822} = \frac{\langle S_6, \sigma_{456} \rangle \Rightarrow \langle S_8, \sigma_{456} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{456} \rangle \Rightarrow \langle S_9, \sigma_{456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1824} = \langle S_8, \sigma_{456} \rangle \Rightarrow \langle y := y + 1, \sigma_{456} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1823} = \frac{\mathcal{T}_{1824}}{\langle S_9, \sigma_{456} \rangle \Rightarrow \langle S_{10}, \sigma_{456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1825} = \frac{\langle y := y + 1, \sigma_{456} \rangle \Rightarrow \sigma_{457} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{456} \rangle \Rightarrow \langle S_{11}, \sigma_{457} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1826} = \langle S_{11}, \sigma_{457} \rangle \Rightarrow \langle S_{12}, \sigma_{457} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1827} = \langle S_{12}, \sigma_{457} \rangle \Rightarrow \langle S_{13}, \sigma_{457} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1829} = \frac{\langle x := x + 1, \sigma_{457} \rangle \Rightarrow \sigma_{458} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{457} \rangle \Rightarrow \langle S_6, \sigma_{458} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1828} = \frac{\mathcal{T}_{1829}}{\langle S_{13}, \sigma_{457} \rangle \Rightarrow \langle S_{15}, \sigma_{458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1830} = \frac{\langle S_6, \sigma_{458} \rangle \Rightarrow \langle S_8, \sigma_{458} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{458} \rangle \Rightarrow \langle S_9, \sigma_{458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1832} = \langle S_8, \sigma_{458} \rangle \Rightarrow \langle y := y + 1, \sigma_{458} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1831} = \frac{\mathcal{T}_{1832}}{\langle S_9, \sigma_{458} \rangle \Rightarrow \langle S_{10}, \sigma_{458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1833} = \frac{\langle y := y + 1, \sigma_{458} \rangle \Rightarrow \sigma_{459} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{458} \rangle \Rightarrow \langle S_{11}, \sigma_{459} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1834} = \langle S_{11}, \sigma_{459} \rangle \Rightarrow \langle S_{12}, \sigma_{459} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1835} = \langle S_{12}, \sigma_{459} \rangle \Rightarrow \langle S_{13}, \sigma_{459} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1837} = \frac{\langle x := x + 1, \sigma_{459} \rangle \Rightarrow \sigma_{460} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{459} \rangle \Rightarrow \langle S_6, \sigma_{460} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1836} = \frac{\mathcal{T}_{1837}}{\langle S_{13}, \sigma_{459} \rangle \Rightarrow \langle S_{15}, \sigma_{460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1838} = \frac{\langle S_6, \sigma_{460} \rangle \Rightarrow \langle S_8, \sigma_{460} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{460} \rangle \Rightarrow \langle S_9, \sigma_{460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1840} = \langle S_8, \sigma_{460} \rangle \Rightarrow \langle y := y + 1, \sigma_{460} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1839} = \frac{\mathcal{T}_{1840}}{\langle S_9, \sigma_{460} \rangle \Rightarrow \langle S_{10}, \sigma_{460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1841} = \frac{\langle y := y + 1, \sigma_{460} \rangle \Rightarrow \sigma_{461} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{460} \rangle \Rightarrow \langle S_{11}, \sigma_{461} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1842} = \langle S_{11}, \sigma_{461} \rangle \Rightarrow \langle S_{12}, \sigma_{461} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1843} = \langle S_{12}, \sigma_{461} \rangle \Rightarrow \langle S_{13}, \sigma_{461} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1845} = \frac{\langle x := x + 1, \sigma_{461} \rangle \Rightarrow \sigma_{462} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{461} \rangle \Rightarrow \langle S_6, \sigma_{462} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1844} = \frac{\mathcal{T}_{1845}}{\langle S_{13}, \sigma_{461} \rangle \Rightarrow \langle S_{15}, \sigma_{462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1846} = \frac{\langle S_6, \sigma_{462} \rangle \Rightarrow \langle S_8, \sigma_{462} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{462} \rangle \Rightarrow \langle S_9, \sigma_{462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1848} = \langle S_8, \sigma_{462} \rangle \Rightarrow \langle y := y + 1, \sigma_{462} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1847} = \frac{\mathcal{T}_{1848}}{\langle S_9, \sigma_{462} \rangle \Rightarrow \langle S_{10}, \sigma_{462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1849} = \frac{\langle y := y + 1, \sigma_{462} \rangle \Rightarrow \sigma_{463} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{462} \rangle \Rightarrow \langle S_{11}, \sigma_{463} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1850} = \langle S_{11}, \sigma_{463} \rangle \Rightarrow \langle S_{12}, \sigma_{463} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1851} = \langle S_{12}, \sigma_{463} \rangle \Rightarrow \langle S_{13}, \sigma_{463} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1853} = \frac{\langle x := x + 1, \sigma_{463} \rangle \Rightarrow \sigma_{464} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{463} \rangle \Rightarrow \langle S_6, \sigma_{464} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1852} = \frac{\mathcal{T}_{1853}}{\langle S_{13}, \sigma_{463} \rangle \Rightarrow \langle S_{15}, \sigma_{464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1854} = \frac{\langle S_6, \sigma_{464} \rangle \Rightarrow \langle S_8, \sigma_{464} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{464} \rangle \Rightarrow \langle S_9, \sigma_{464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1856} = \langle S_8, \sigma_{464} \rangle \Rightarrow \langle y := y + 1, \sigma_{464} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1855} = \frac{\mathcal{T}_{1856}}{\langle S_9, \sigma_{464} \rangle \Rightarrow \langle S_{10}, \sigma_{464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1857} = \frac{\langle y := y + 1, \sigma_{464} \rangle \Rightarrow \sigma_{465} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{464} \rangle \Rightarrow \langle S_{11}, \sigma_{465} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1858} = \langle S_{11}, \sigma_{465} \rangle \Rightarrow \langle S_{12}, \sigma_{465} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1859} = \langle S_{12}, \sigma_{465} \rangle \Rightarrow \langle S_{13}, \sigma_{465} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1861} = \frac{\langle x := x + 1, \sigma_{465} \rangle \Rightarrow \sigma_{466} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{465} \rangle \Rightarrow \langle S_6, \sigma_{466} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1860} = \frac{\mathcal{T}_{1861}}{\langle S_{13}, \sigma_{465} \rangle \Rightarrow \langle S_{15}, \sigma_{466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1862} = \frac{\langle S_6, \sigma_{466} \rangle \Rightarrow \langle S_8, \sigma_{466} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{466} \rangle \Rightarrow \langle S_9, \sigma_{466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1864} = \langle S_8, \sigma_{466} \rangle \Rightarrow \langle y := y + 1, \sigma_{466} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1863} = \frac{\mathcal{T}_{1864}}{\langle S_9, \sigma_{466} \rangle \Rightarrow \langle S_{10}, \sigma_{466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1865} = \frac{\langle y := y + 1, \sigma_{466} \rangle \Rightarrow \sigma_{467} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{466} \rangle \Rightarrow \langle S_{11}, \sigma_{467} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1866} = \langle S_{11}, \sigma_{467} \rangle \Rightarrow \langle S_{12}, \sigma_{467} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1867} = \langle S_{12}, \sigma_{467} \rangle \Rightarrow \langle S_{13}, \sigma_{467} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1869} = \frac{\langle x := x + 1, \sigma_{467} \rangle \Rightarrow \sigma_{468} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{467} \rangle \Rightarrow \langle S_6, \sigma_{468} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1868} = \frac{\mathcal{T}_{1869}}{\langle S_{13}, \sigma_{467} \rangle \Rightarrow \langle S_{15}, \sigma_{468} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1870} = \frac{\langle S_6, \sigma_{468} \rangle \Rightarrow \langle S_8, \sigma_{468} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{468} \rangle \Rightarrow \langle S_9, \sigma_{468} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1872} = \langle S_8, \sigma_{468} \rangle \Rightarrow \langle y := y + 1, \sigma_{468} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1871} = \frac{\mathcal{T}_{1872}}{\langle S_9, \sigma_{468} \rangle \Rightarrow \langle S_{10}, \sigma_{468} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1873} = \frac{\langle y := y + 1, \sigma_{468} \rangle \Rightarrow \sigma_{469} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{468} \rangle \Rightarrow \langle S_{11}, \sigma_{469} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1874} = \langle S_{11}, \sigma_{469} \rangle \Rightarrow \langle S_{12}, \sigma_{469} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1875} = \langle S_{12}, \sigma_{469} \rangle \Rightarrow \langle S_{13}, \sigma_{469} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1877} = \frac{\langle x := x + 1, \sigma_{469} \rangle \Rightarrow \sigma_{470} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{469} \rangle \Rightarrow \langle S_6, \sigma_{470} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1876} = \frac{\mathcal{T}_{1877}}{\langle S_{13}, \sigma_{469} \rangle \Rightarrow \langle S_{15}, \sigma_{470} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1878} = \frac{\langle S_6, \sigma_{470} \rangle \Rightarrow \langle S_8, \sigma_{470} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{470} \rangle \Rightarrow \langle S_9, \sigma_{470} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1880} = \langle S_8, \sigma_{470} \rangle \Rightarrow \langle y := y + 1, \sigma_{470} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1879} = \frac{\mathcal{T}_{1880}}{\langle S_9, \sigma_{470} \rangle \Rightarrow \langle S_{10}, \sigma_{470} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1881} = \frac{\langle y := y + 1, \sigma_{470} \rangle \Rightarrow \sigma_{471} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{470} \rangle \Rightarrow \langle S_{11}, \sigma_{471} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1882} = \langle S_{11}, \sigma_{471} \rangle \Rightarrow \langle S_{12}, \sigma_{471} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1883} = \langle S_{12}, \sigma_{471} \rangle \Rightarrow \langle S_{13}, \sigma_{471} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1885} = \frac{\langle x := x + 1, \sigma_{471} \rangle \Rightarrow \sigma_{472} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{471} \rangle \Rightarrow \langle S_6, \sigma_{472} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{1884} = \frac{\mathcal{T}_{1885}}{\langle S_{13}, \sigma_{471} \rangle \Rightarrow \langle S_{15}, \sigma_{472} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1886} = \frac{\langle S_6, \sigma_{472} \rangle \Rightarrow \langle S_8, \sigma_{472} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{472} \rangle \Rightarrow \langle S_9, \sigma_{472} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1888} = \langle S_8, \sigma_{472} \rangle \Rightarrow \langle y := y + 1, \sigma_{472} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{1887} = \frac{\mathcal{T}_{1888}}{\langle S_9, \sigma_{472} \rangle \Rightarrow \langle S_{10}, \sigma_{472} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1889} = \frac{\langle y := y + 1, \sigma_{472} \rangle \Rightarrow \sigma_{473} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{472} \rangle \Rightarrow \langle S_{11}, \sigma_{473} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{1890} = \langle S_{11}, \sigma_{473} \rangle \Rightarrow \langle S_{12}, \sigma_{473} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{1891} = \langle S_{12}, \sigma_{473} \rangle \Rightarrow \langle S_{13}, \sigma_{473} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1893} = \frac{\langle x := x + 1, \sigma_{473} \rangle \Rightarrow \sigma_{474} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{473} \rangle \Rightarrow \langle S_6, \sigma_{474} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1892} = \frac{\mathcal{T}_{1893}}{\langle S_{13}, \sigma_{473} \rangle \Rightarrow \langle S_{15}, \sigma_{474} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1894} = \frac{\langle S_6, \sigma_{474} \rangle \Rightarrow \langle S_8, \sigma_{474} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{474} \rangle \Rightarrow \langle S_9, \sigma_{474} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1896} = \langle S_8, \sigma_{474} \rangle \Rightarrow \langle y := y + 1, \sigma_{474} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1895} = \frac{\mathcal{T}_{1896}}{\langle S_9, \sigma_{474} \rangle \Rightarrow \langle S_{10}, \sigma_{474} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1897} = \frac{\langle y := y + 1, \sigma_{474} \rangle \Rightarrow \sigma_{475} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{474} \rangle \Rightarrow \langle S_{11}, \sigma_{475} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1898} = \langle S_{11}, \sigma_{475} \rangle \Rightarrow \langle S_{12}, \sigma_{475} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1899} = \langle S_{12}, \sigma_{475} \rangle \Rightarrow \langle S_{13}, \sigma_{475} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1901} = \frac{\langle x := x + 1, \sigma_{475} \rangle \Rightarrow \sigma_{476} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{475} \rangle \Rightarrow \langle S_6, \sigma_{476} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1900} = \frac{\mathcal{T}_{1901}}{\langle S_{13}, \sigma_{475} \rangle \Rightarrow \langle S_{15}, \sigma_{476} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1902} = \frac{\langle S_6, \sigma_{476} \rangle \Rightarrow \langle S_8, \sigma_{476} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{476} \rangle \Rightarrow \langle S_9, \sigma_{476} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1904} = \langle S_8, \sigma_{476} \rangle \Rightarrow \langle y := y + 1, \sigma_{476} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1903} = \frac{\mathcal{T}_{1904}}{\langle S_9, \sigma_{476} \rangle \Rightarrow \langle S_{10}, \sigma_{476} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1905} = \frac{\langle y := y + 1, \sigma_{476} \rangle \Rightarrow \sigma_{477} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{476} \rangle \Rightarrow \langle S_{11}, \sigma_{477} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1906} = \langle S_{11}, \sigma_{477} \rangle \Rightarrow \langle S_{12}, \sigma_{477} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1907} = \langle S_{12}, \sigma_{477} \rangle \Rightarrow \langle S_{13}, \sigma_{477} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1909} = \frac{\langle x := x + 1, \sigma_{477} \rangle \Rightarrow \sigma_{478} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{477} \rangle \Rightarrow \langle S_6, \sigma_{478} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1908} = \frac{\mathcal{T}_{1909}}{\langle S_{13}, \sigma_{477} \rangle \Rightarrow \langle S_{15}, \sigma_{478} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1910} = \frac{\langle S_6, \sigma_{478} \rangle \Rightarrow \langle S_8, \sigma_{478} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{478} \rangle \Rightarrow \langle S_9, \sigma_{478} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1912} = \langle S_8, \sigma_{478} \rangle \Rightarrow \langle y := y + 1, \sigma_{478} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1911} = \frac{\mathcal{T}_{1912}}{\langle S_9, \sigma_{478} \rangle \Rightarrow \langle S_{10}, \sigma_{478} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1913} = \frac{\langle y := y + 1, \sigma_{478} \rangle \Rightarrow \sigma_{479} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{478} \rangle \Rightarrow \langle S_{11}, \sigma_{479} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1914} = \langle S_{11}, \sigma_{479} \rangle \Rightarrow \langle S_{12}, \sigma_{479} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1915} = \langle S_{12}, \sigma_{479} \rangle \Rightarrow \langle S_{13}, \sigma_{479} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1917} = \frac{\langle x := x + 1, \sigma_{479} \rangle \Rightarrow \sigma_{480} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{479} \rangle \Rightarrow \langle S_6, \sigma_{480} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1916} = \frac{\mathcal{T}_{1917}}{\langle S_{13}, \sigma_{479} \rangle \Rightarrow \langle S_{15}, \sigma_{480} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1918} = \frac{\langle S_6, \sigma_{480} \rangle \Rightarrow \langle S_8, \sigma_{480} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{480} \rangle \Rightarrow \langle S_9, \sigma_{480} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1920} = \langle S_8, \sigma_{480} \rangle \Rightarrow \langle y := y + 1, \sigma_{480} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1919} = \frac{\mathcal{T}_{1920}}{\langle S_9, \sigma_{480} \rangle \Rightarrow \langle S_{10}, \sigma_{480} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1921} = \frac{\langle y := y + 1, \sigma_{480} \rangle \Rightarrow \sigma_{481} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{480} \rangle \Rightarrow \langle S_{11}, \sigma_{481} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1922} = \langle S_{11}, \sigma_{481} \rangle \Rightarrow \langle S_{12}, \sigma_{481} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1923} = \langle S_{12}, \sigma_{481} \rangle \Rightarrow \langle S_{13}, \sigma_{481} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1925} = \frac{\langle x := x + 1, \sigma_{481} \rangle \Rightarrow \sigma_{482} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{481} \rangle \Rightarrow \langle S_6, \sigma_{482} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1924} = \frac{\mathcal{T}_{1925}}{\langle S_{13}, \sigma_{481} \rangle \Rightarrow \langle S_{15}, \sigma_{482} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1926} = \frac{\langle S_6, \sigma_{482} \rangle \Rightarrow \langle S_8, \sigma_{482} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{482} \rangle \Rightarrow \langle S_9, \sigma_{482} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1928} = \langle S_8, \sigma_{482} \rangle \Rightarrow \langle y := y + 1, \sigma_{482} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{1927} = \frac{\mathcal{T}_{1928}}{\langle S_9, \sigma_{482} \rangle \Rightarrow \langle S_{10}, \sigma_{482} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1929} = \frac{\langle y := y + 1, \sigma_{482} \rangle \Rightarrow \sigma_{483} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{482} \rangle \Rightarrow \langle S_{11}, \sigma_{483} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{1930} = \langle S_{11}, \sigma_{483} \rangle \Rightarrow \langle S_{12}, \sigma_{483} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1931} = \langle S_{12}, \sigma_{483} \rangle \Rightarrow \langle S_{13}, \sigma_{483} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{1933} = \frac{\langle x := x + 1, \sigma_{483} \rangle \Rightarrow \sigma_{484} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{483} \rangle \Rightarrow \langle S_6, \sigma_{484} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1932} = \frac{\mathcal{T}_{1933}}{\langle S_{13}, \sigma_{483} \rangle \Rightarrow \langle S_{15}, \sigma_{484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1934} = \frac{\langle S_6, \sigma_{484} \rangle \Rightarrow \langle S_8, \sigma_{484} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{484} \rangle \Rightarrow \langle S_9, \sigma_{484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1936} = \langle S_8, \sigma_{484} \rangle \Rightarrow \langle y := y + 1, \sigma_{484} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1935} = \frac{\mathcal{T}_{1936}}{\langle S_9, \sigma_{484} \rangle \Rightarrow \langle S_{10}, \sigma_{484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{1937} = \frac{\langle y := y + 1, \sigma_{484} \rangle \Rightarrow \sigma_{485} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{484} \rangle \Rightarrow \langle S_{11}, \sigma_{485} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1938} = \langle S_{11}, \sigma_{485} \rangle \Rightarrow \langle S_{12}, \sigma_{485} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1939} = \langle S_{12}, \sigma_{485} \rangle \Rightarrow \langle S_{13}, \sigma_{485} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1941} = \frac{\langle x := x + 1, \sigma_{485} \rangle \Rightarrow \sigma_{486} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{485} \rangle \Rightarrow \langle S_6, \sigma_{486} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1940} = \frac{\mathcal{T}_{1941}}{\langle S_{13}, \sigma_{485} \rangle \Rightarrow \langle S_{15}, \sigma_{486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1942} = \frac{\langle S_6, \sigma_{486} \rangle \Rightarrow \langle S_8, \sigma_{486} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{486} \rangle \Rightarrow \langle S_9, \sigma_{486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1944} = \langle S_8, \sigma_{486} \rangle \Rightarrow \langle y := y + 1, \sigma_{486} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1943} = \frac{\mathcal{T}_{1944}}{\langle S_9, \sigma_{486} \rangle \Rightarrow \langle S_{10}, \sigma_{486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1945} = \frac{\langle y := y + 1, \sigma_{486} \rangle \Rightarrow \sigma_{487} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{486} \rangle \Rightarrow \langle S_{11}, \sigma_{487} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1946} = \langle S_{11}, \sigma_{487} \rangle \Rightarrow \langle S_{12}, \sigma_{487} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1947} = \langle S_{12}, \sigma_{487} \rangle \Rightarrow \langle S_{13}, \sigma_{487} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1949} = \frac{\langle x := x + 1, \sigma_{487} \rangle \Rightarrow \sigma_{488} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{487} \rangle \Rightarrow \langle S_6, \sigma_{488} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1948} = \frac{\mathcal{T}_{1949}}{\langle S_{13}, \sigma_{487} \rangle \Rightarrow \langle S_{15}, \sigma_{488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1950} = \frac{\langle S_6, \sigma_{488} \rangle \Rightarrow \langle S_8, \sigma_{488} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{488} \rangle \Rightarrow \langle S_9, \sigma_{488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1952} = \langle S_8, \sigma_{488} \rangle \Rightarrow \langle y := y + 1, \sigma_{488} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1951} = \frac{\mathcal{T}_{1952}}{\langle S_9, \sigma_{488} \rangle \Rightarrow \langle S_{10}, \sigma_{488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1953} = \frac{\langle y := y + 1, \sigma_{488} \rangle \Rightarrow \sigma_{489} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{488} \rangle \Rightarrow \langle S_{11}, \sigma_{489} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1954} = \langle S_{11}, \sigma_{489} \rangle \Rightarrow \langle S_{12}, \sigma_{489} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1955} = \langle S_{12}, \sigma_{489} \rangle \Rightarrow \langle S_{13}, \sigma_{489} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1957} = \frac{\langle x := x + 1, \sigma_{489} \rangle \Rightarrow \sigma_{490} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{489} \rangle \Rightarrow \langle S_6, \sigma_{490} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1956} = \frac{\mathcal{T}_{1957}}{\langle S_{13}, \sigma_{489} \rangle \Rightarrow \langle S_{15}, \sigma_{490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1958} = \frac{\langle S_6, \sigma_{490} \rangle \Rightarrow \langle S_8, \sigma_{490} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{490} \rangle \Rightarrow \langle S_9, \sigma_{490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1960} = \langle S_8, \sigma_{490} \rangle \Rightarrow \langle y := y + 1, \sigma_{490} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1959} = \frac{\mathcal{T}_{1960}}{\langle S_9, \sigma_{490} \rangle \Rightarrow \langle S_{10}, \sigma_{490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1961} = \frac{\langle y := y + 1, \sigma_{490} \rangle \Rightarrow \sigma_{491} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{490} \rangle \Rightarrow \langle S_{11}, \sigma_{491} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1962} = \langle S_{11}, \sigma_{491} \rangle \Rightarrow \langle S_{12}, \sigma_{491} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1963} = \langle S_{12}, \sigma_{491} \rangle \Rightarrow \langle S_{13}, \sigma_{491} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1965} = \frac{\langle x := x + 1, \sigma_{491} \rangle \Rightarrow \sigma_{492} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{491} \rangle \Rightarrow \langle S_6, \sigma_{492} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1964} = \frac{\mathcal{T}_{1965}}{\langle S_{13}, \sigma_{491} \rangle \Rightarrow \langle S_{15}, \sigma_{492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1966} = \frac{\langle S_6, \sigma_{492} \rangle \Rightarrow \langle S_8, \sigma_{492} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{492} \rangle \Rightarrow \langle S_9, \sigma_{492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1968} = \langle S_8, \sigma_{492} \rangle \Rightarrow \langle y := y + 1, \sigma_{492} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1967} = \frac{\mathcal{T}_{1968}}{\langle S_9, \sigma_{492} \rangle \Rightarrow \langle S_{10}, \sigma_{492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1969} = \frac{\langle y := y + 1, \sigma_{492} \rangle \Rightarrow \sigma_{493} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{492} \rangle \Rightarrow \langle S_{11}, \sigma_{493} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1970} = \langle S_{11}, \sigma_{493} \rangle \Rightarrow \langle S_{12}, \sigma_{493} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1971} = \langle S_{12}, \sigma_{493} \rangle \Rightarrow \langle S_{13}, \sigma_{493} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1973} = \frac{\langle x := x + 1, \sigma_{493} \rangle \Rightarrow \sigma_{494} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{493} \rangle \Rightarrow \langle S_6, \sigma_{494} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1972} = \frac{\mathcal{T}_{1973}}{\langle S_{13}, \sigma_{493} \rangle \Rightarrow \langle S_{15}, \sigma_{494} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1974} = \frac{\langle S_6, \sigma_{494} \rangle \Rightarrow \langle S_8, \sigma_{494} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{494} \rangle \Rightarrow \langle S_9, \sigma_{494} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1976} = \langle S_8, \sigma_{494} \rangle \Rightarrow \langle y := y + 1, \sigma_{494} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1975} = \frac{\mathcal{T}_{1976}}{\langle S_9, \sigma_{494} \rangle \Rightarrow \langle S_{10}, \sigma_{494} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1977} = \frac{\langle y := y + 1, \sigma_{494} \rangle \Rightarrow \sigma_{495} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{494} \rangle \Rightarrow \langle S_{11}, \sigma_{495} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1978} = \langle S_{11}, \sigma_{495} \rangle \Rightarrow \langle S_{12}, \sigma_{495} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1979} = \langle S_{12}, \sigma_{495} \rangle \Rightarrow \langle S_{13}, \sigma_{495} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1981} = \frac{\langle x := x + 1, \sigma_{495} \rangle \Rightarrow \sigma_{496} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{495} \rangle \Rightarrow \langle S_6, \sigma_{496} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1980} = \frac{\mathcal{T}_{1981}}{\langle S_{13}, \sigma_{495} \rangle \Rightarrow \langle S_{15}, \sigma_{496} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1982} = \frac{\langle S_6, \sigma_{496} \rangle \Rightarrow \langle S_8, \sigma_{496} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{496} \rangle \Rightarrow \langle S_9, \sigma_{496} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1984} = \langle S_8, \sigma_{496} \rangle \Rightarrow \langle y := y + 1, \sigma_{496} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1983} = \frac{\mathcal{T}_{1984}}{\langle S_9, \sigma_{496} \rangle \Rightarrow \langle S_{10}, \sigma_{496} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1985} = \frac{\langle y := y + 1, \sigma_{496} \rangle \Rightarrow \sigma_{497} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{496} \rangle \Rightarrow \langle S_{11}, \sigma_{497} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1986} = \langle S_{11}, \sigma_{497} \rangle \Rightarrow \langle S_{12}, \sigma_{497} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1987} = \langle S_{12}, \sigma_{497} \rangle \Rightarrow \langle S_{13}, \sigma_{497} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1989} = \frac{\langle x := x + 1, \sigma_{497} \rangle \Rightarrow \sigma_{498} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{497} \rangle \Rightarrow \langle S_6, \sigma_{498} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1988} = \frac{\mathcal{T}_{1989}}{\langle S_{13}, \sigma_{497} \rangle \Rightarrow \langle S_{15}, \sigma_{498} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1990} = \frac{\langle S_6, \sigma_{498} \rangle \Rightarrow \langle S_8, \sigma_{498} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{498} \rangle \Rightarrow \langle S_9, \sigma_{498} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1992} = \langle S_8, \sigma_{498} \rangle \Rightarrow \langle y := y + 1, \sigma_{498} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1991} = \frac{\mathcal{T}_{1992}}{\langle S_9, \sigma_{498} \rangle \Rightarrow \langle S_{10}, \sigma_{498} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1993} = \frac{\langle y := y + 1, \sigma_{498} \rangle \Rightarrow \sigma_{499} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{498} \rangle \Rightarrow \langle S_{11}, \sigma_{499} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{1994} = \langle S_{11}, \sigma_{499} \rangle \Rightarrow \langle S_{12}, \sigma_{499} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1995} = \langle S_{12}, \sigma_{499} \rangle \Rightarrow \langle S_{13}, \sigma_{499} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{1997} = \frac{\langle x := x + 1, \sigma_{499} \rangle \Rightarrow \sigma_{500} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{499} \rangle \Rightarrow \langle S_6, \sigma_{500} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{1996} = \frac{\mathcal{T}_{1997}}{\langle S_{13}, \sigma_{499} \rangle \Rightarrow \langle S_{15}, \sigma_{500} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{1998} = \frac{\langle S_6, \sigma_{500} \rangle \Rightarrow \langle S_8, \sigma_{500} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{500} \rangle \Rightarrow \langle S_9, \sigma_{500} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2000} = \langle S_8, \sigma_{500} \rangle \Rightarrow \langle y := y + 1, \sigma_{500} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{1999} = \frac{\mathcal{T}_{2000}}{\langle S_9, \sigma_{500} \rangle \Rightarrow \langle S_{10}, \sigma_{500} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2001} = \frac{\langle y := y + 1, \sigma_{500} \rangle \Rightarrow \sigma_{501} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{500} \rangle \Rightarrow \langle S_{11}, \sigma_{501} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2002} = \langle S_{11}, \sigma_{501} \rangle \Rightarrow \langle S_{12}, \sigma_{501} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2003} = \langle S_{12}, \sigma_{501} \rangle \Rightarrow \langle S_{13}, \sigma_{501} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2005} = \frac{\langle x := x + 1, \sigma_{501} \rangle \Rightarrow \sigma_{502} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{501} \rangle \Rightarrow \langle S_6, \sigma_{502} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2004} = \frac{\mathcal{T}_{2005}}{\langle S_{13}, \sigma_{501} \rangle \Rightarrow \langle S_{15}, \sigma_{502} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2006} = \frac{\langle S_6, \sigma_{502} \rangle \Rightarrow \langle S_8, \sigma_{502} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{502} \rangle \Rightarrow \langle S_9, \sigma_{502} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2008} = \langle S_8, \sigma_{502} \rangle \Rightarrow \langle y := y + 1, \sigma_{502} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2007} = \frac{\mathcal{T}_{2008}}{\langle S_9, \sigma_{502} \rangle \Rightarrow \langle S_{10}, \sigma_{502} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2009} = \frac{\langle y := y + 1, \sigma_{502} \rangle \Rightarrow \sigma_{503} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{502} \rangle \Rightarrow \langle S_{11}, \sigma_{503} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2010} = \langle S_{11}, \sigma_{503} \rangle \Rightarrow \langle S_{12}, \sigma_{503} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2011} = \langle S_{12}, \sigma_{503} \rangle \Rightarrow \langle S_{13}, \sigma_{503} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2013} = \frac{\langle x := x + 1, \sigma_{503} \rangle \Rightarrow \sigma_{504} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{503} \rangle \Rightarrow \langle S_6, \sigma_{504} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2012} = \frac{\mathcal{T}_{2013}}{\langle S_{13}, \sigma_{503} \rangle \Rightarrow \langle S_{15}, \sigma_{504} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2014} = \frac{\langle S_6, \sigma_{504} \rangle \Rightarrow \langle S_8, \sigma_{504} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{504} \rangle \Rightarrow \langle S_9, \sigma_{504} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2016} = \langle S_8, \sigma_{504} \rangle \Rightarrow \langle y := y + 1, \sigma_{504} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2015} = \frac{\mathcal{T}_{2016}}{\langle S_9, \sigma_{504} \rangle \Rightarrow \langle S_{10}, \sigma_{504} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2017} = \frac{\langle y := y + 1, \sigma_{504} \rangle \Rightarrow \sigma_{505} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{504} \rangle \Rightarrow \langle S_{11}, \sigma_{505} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2018} = \langle S_{11}, \sigma_{505} \rangle \Rightarrow \langle S_{12}, \sigma_{505} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2019} = \langle S_{12}, \sigma_{505} \rangle \Rightarrow \langle S_{13}, \sigma_{505} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2021} = \frac{\langle x := x + 1, \sigma_{505} \rangle \Rightarrow \sigma_{506} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{505} \rangle \Rightarrow \langle S_6, \sigma_{506} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2020} = \frac{\mathcal{T}_{2021}}{\langle S_{13}, \sigma_{505} \rangle \Rightarrow \langle S_{15}, \sigma_{506} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2022} = \frac{\langle S_6, \sigma_{506} \rangle \Rightarrow \langle S_8, \sigma_{506} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{506} \rangle \Rightarrow \langle S_9, \sigma_{506} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2024} = \langle S_8, \sigma_{506} \rangle \Rightarrow \langle y := y + 1, \sigma_{506} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2023} = \frac{\mathcal{T}_{2024}}{\langle S_9, \sigma_{506} \rangle \Rightarrow \langle S_{10}, \sigma_{506} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2025} = \frac{\langle y := y + 1, \sigma_{506} \rangle \Rightarrow \sigma_{507} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{506} \rangle \Rightarrow \langle S_{11}, \sigma_{507} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{2026} = \langle S_{11}, \sigma_{507} \rangle \Rightarrow \langle S_{12}, \sigma_{507} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2027} = \langle S_{12}, \sigma_{507} \rangle \Rightarrow \langle S_{13}, \sigma_{507} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2029} = \frac{\langle x := x + 1, \sigma_{507} \rangle \Rightarrow \sigma_{508} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{507} \rangle \Rightarrow \langle S_6, \sigma_{508} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2028} = \frac{\mathcal{T}_{2029}}{\langle S_{13}, \sigma_{507} \rangle \Rightarrow \langle S_{15}, \sigma_{508} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2030} = \frac{\langle S_6, \sigma_{508} \rangle \Rightarrow \langle S_8, \sigma_{508} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{508} \rangle \Rightarrow \langle S_9, \sigma_{508} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2032} = \langle S_8, \sigma_{508} \rangle \Rightarrow \langle y := y + 1, \sigma_{508} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2031} = \frac{\mathcal{T}_{2032}}{\langle S_9, \sigma_{508} \rangle \Rightarrow \langle S_{10}, \sigma_{508} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2033} = \frac{\langle y := y + 1, \sigma_{508} \rangle \Rightarrow \sigma_{509} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{508} \rangle \Rightarrow \langle S_{11}, \sigma_{509} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2034} = \langle S_{11}, \sigma_{509} \rangle \Rightarrow \langle S_{12}, \sigma_{509} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2035} = \langle S_{12}, \sigma_{509} \rangle \Rightarrow \langle S_{13}, \sigma_{509} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2037} = \frac{\langle x := x + 1, \sigma_{509} \rangle \Rightarrow \sigma_{510} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{509} \rangle \Rightarrow \langle S_6, \sigma_{510} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2036} = \frac{\mathcal{T}_{2037}}{\langle S_{13}, \sigma_{509} \rangle \Rightarrow \langle S_{15}, \sigma_{510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2038} = \frac{\langle S_6, \sigma_{510} \rangle \Rightarrow \langle S_8, \sigma_{510} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{510} \rangle \Rightarrow \langle S_9, \sigma_{510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2040} = \langle S_8, \sigma_{510} \rangle \Rightarrow \langle y := y + 1, \sigma_{510} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2039} = \frac{\mathcal{T}_{2040}}{\langle S_9, \sigma_{510} \rangle \Rightarrow \langle S_{10}, \sigma_{510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2041} = \frac{\langle y := y + 1, \sigma_{510} \rangle \Rightarrow \sigma_{511} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{510} \rangle \Rightarrow \langle S_{11}, \sigma_{511} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2042} = \langle S_{11}, \sigma_{511} \rangle \Rightarrow \langle S_{12}, \sigma_{511} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2043} = \langle S_{12}, \sigma_{511} \rangle \Rightarrow \langle S_{13}, \sigma_{511} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2045} = \frac{\langle x := x + 1, \sigma_{511} \rangle \Rightarrow \sigma_{512} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{511} \rangle \Rightarrow \langle S_6, \sigma_{512} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2044} = \frac{\mathcal{T}_{2045}}{\langle S_{13}, \sigma_{511} \rangle \Rightarrow \langle S_{15}, \sigma_{512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2046} = \frac{\langle S_6, \sigma_{512} \rangle \Rightarrow \langle S_8, \sigma_{512} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{512} \rangle \Rightarrow \langle S_9, \sigma_{512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2048} = \langle S_8, \sigma_{512} \rangle \Rightarrow \langle y := y + 1, \sigma_{512} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2047} = \frac{\mathcal{T}_{2048}}{\langle S_9, \sigma_{512} \rangle \Rightarrow \langle S_{10}, \sigma_{512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2049} = \frac{\langle y := y + 1, \sigma_{512} \rangle \Rightarrow \sigma_{513} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{512} \rangle \Rightarrow \langle S_{11}, \sigma_{513} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2050} = \langle S_{11}, \sigma_{513} \rangle \Rightarrow \langle S_{12}, \sigma_{513} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2051} = \langle S_{12}, \sigma_{513} \rangle \Rightarrow \langle S_{13}, \sigma_{513} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2053} = \frac{\langle x := x + 1, \sigma_{513} \rangle \Rightarrow \sigma_{514} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{513} \rangle \Rightarrow \langle S_6, \sigma_{514} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2052} = \frac{\mathcal{T}_{2053}}{\langle S_{13}, \sigma_{513} \rangle \Rightarrow \langle S_{15}, \sigma_{514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2054} = \frac{\langle S_6, \sigma_{514} \rangle \Rightarrow \langle S_8, \sigma_{514} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{514} \rangle \Rightarrow \langle S_9, \sigma_{514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2056} = \langle S_8, \sigma_{514} \rangle \Rightarrow \langle y := y + 1, \sigma_{514} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2055} = \frac{\mathcal{T}_{2056}}{\langle S_9, \sigma_{514} \rangle \Rightarrow \langle S_{10}, \sigma_{514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2057} = \frac{\langle y := y + 1, \sigma_{514} \rangle \Rightarrow \sigma_{515} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{514} \rangle \Rightarrow \langle S_{11}, \sigma_{515} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2058} = \langle S_{11}, \sigma_{515} \rangle \Rightarrow \langle S_{12}, \sigma_{515} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2059} = \langle S_{12}, \sigma_{515} \rangle \Rightarrow \langle S_{13}, \sigma_{515} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2061} = \frac{\langle x := x + 1, \sigma_{515} \rangle \Rightarrow \sigma_{516} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{515} \rangle \Rightarrow \langle S_6, \sigma_{516} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2060} = \frac{\mathcal{T}_{2061}}{\langle S_{13}, \sigma_{515} \rangle \Rightarrow \langle S_{15}, \sigma_{516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2062} = \frac{\langle S_6, \sigma_{516} \rangle \Rightarrow \langle S_8, \sigma_{516} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{516} \rangle \Rightarrow \langle S_9, \sigma_{516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2064} = \langle S_8, \sigma_{516} \rangle \Rightarrow \langle y := y + 1, \sigma_{516} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2063} = \frac{\mathcal{T}_{2064}}{\langle S_9, \sigma_{516} \rangle \Rightarrow \langle S_{10}, \sigma_{516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2065} = \frac{\langle y := y + 1, \sigma_{516} \rangle \Rightarrow \sigma_{517} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{516} \rangle \Rightarrow \langle S_{11}, \sigma_{517} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2066} = \langle S_{11}, \sigma_{517} \rangle \Rightarrow \langle S_{12}, \sigma_{517} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2067} = \langle S_{12}, \sigma_{517} \rangle \Rightarrow \langle S_{13}, \sigma_{517} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2069} = \frac{\langle x := x + 1, \sigma_{517} \rangle \Rightarrow \sigma_{518} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{517} \rangle \Rightarrow \langle S_6, \sigma_{518} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2068} = \frac{\mathcal{T}_{2069}}{\langle S_{13}, \sigma_{517} \rangle \Rightarrow \langle S_{15}, \sigma_{518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2070} = \frac{\langle S_6, \sigma_{518} \rangle \Rightarrow \langle S_8, \sigma_{518} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{518} \rangle \Rightarrow \langle S_9, \sigma_{518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2072} = \langle S_8, \sigma_{518} \rangle \Rightarrow \langle y := y + 1, \sigma_{518} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2071} = \frac{\mathcal{T}_{2072}}{\langle S_9, \sigma_{518} \rangle \Rightarrow \langle S_{10}, \sigma_{518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2073} = \frac{\langle y := y + 1, \sigma_{518} \rangle \Rightarrow \sigma_{519} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{518} \rangle \Rightarrow \langle S_{11}, \sigma_{519} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2074} = \langle S_{11}, \sigma_{519} \rangle \Rightarrow \langle S_{12}, \sigma_{519} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2075} = \langle S_{12}, \sigma_{519} \rangle \Rightarrow \langle S_{13}, \sigma_{519} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2077} = \frac{\langle x := x + 1, \sigma_{519} \rangle \Rightarrow \sigma_{520} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{519} \rangle \Rightarrow \langle S_6, \sigma_{520} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2076} = \frac{\mathcal{T}_{2077}}{\langle S_{13}, \sigma_{519} \rangle \Rightarrow \langle S_{15}, \sigma_{520} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2078} = \frac{\langle S_6, \sigma_{520} \rangle \Rightarrow \langle S_8, \sigma_{520} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{520} \rangle \Rightarrow \langle S_9, \sigma_{520} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2080} = \langle S_8, \sigma_{520} \rangle \Rightarrow \langle y := y + 1, \sigma_{520} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2079} = \frac{\mathcal{T}_{2080}}{\langle S_9, \sigma_{520} \rangle \Rightarrow \langle S_{10}, \sigma_{520} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2081} = \frac{\langle y := y + 1, \sigma_{520} \rangle \Rightarrow \sigma_{521} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{520} \rangle \Rightarrow \langle S_{11}, \sigma_{521} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2082} = \langle S_{11}, \sigma_{521} \rangle \Rightarrow \langle S_{12}, \sigma_{521} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2083} = \langle S_{12}, \sigma_{521} \rangle \Rightarrow \langle S_{13}, \sigma_{521} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2085} = \frac{\langle x := x + 1, \sigma_{521} \rangle \Rightarrow \sigma_{522} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{521} \rangle \Rightarrow \langle S_6, \sigma_{522} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2084} = \frac{\mathcal{T}_{2085}}{\langle S_{13}, \sigma_{521} \rangle \Rightarrow \langle S_{15}, \sigma_{522} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2086} = \frac{\langle S_6, \sigma_{522} \rangle \Rightarrow \langle S_8, \sigma_{522} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{522} \rangle \Rightarrow \langle S_9, \sigma_{522} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2088} = \langle S_8, \sigma_{522} \rangle \Rightarrow \langle y := y + 1, \sigma_{522} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2087} = \frac{\mathcal{T}_{2088}}{\langle S_9, \sigma_{522} \rangle \Rightarrow \langle S_{10}, \sigma_{522} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2089} = \frac{\langle y := y + 1, \sigma_{522} \rangle \Rightarrow \sigma_{523} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{522} \rangle \Rightarrow \langle S_{11}, \sigma_{523} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2090} = \langle S_{11}, \sigma_{523} \rangle \Rightarrow \langle S_{12}, \sigma_{523} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2091} = \langle S_{12}, \sigma_{523} \rangle \Rightarrow \langle S_{13}, \sigma_{523} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2093} = \frac{\langle x := x + 1, \sigma_{523} \rangle \Rightarrow \sigma_{524} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{523} \rangle \Rightarrow \langle S_6, \sigma_{524} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2092} = \frac{\mathcal{T}_{2093}}{\langle S_{13}, \sigma_{523} \rangle \Rightarrow \langle S_{15}, \sigma_{524} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2094} = \frac{\langle S_6, \sigma_{524} \rangle \Rightarrow \langle S_8, \sigma_{524} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{524} \rangle \Rightarrow \langle S_9, \sigma_{524} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2096} = \langle S_8, \sigma_{524} \rangle \Rightarrow \langle y := y + 1, \sigma_{524} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2095} = \frac{\mathcal{T}_{2096}}{\langle S_9, \sigma_{524} \rangle \Rightarrow \langle S_{10}, \sigma_{524} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2097} = \frac{\langle y := y + 1, \sigma_{524} \rangle \Rightarrow \sigma_{525} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{524} \rangle \Rightarrow \langle S_{11}, \sigma_{525} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2098} = \langle S_{11}, \sigma_{525} \rangle \Rightarrow \langle S_{12}, \sigma_{525} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2099} = \langle S_{12}, \sigma_{525} \rangle \Rightarrow \langle S_{13}, \sigma_{525} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2101} = \frac{\langle x := x + 1, \sigma_{525} \rangle \Rightarrow \sigma_{526} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{525} \rangle \Rightarrow \langle S_6, \sigma_{526} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2100} = \frac{\mathcal{T}_{2101}}{\langle S_{13}, \sigma_{525} \rangle \Rightarrow \langle S_{15}, \sigma_{526} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2102} = \frac{\langle S_6, \sigma_{526} \rangle \Rightarrow \langle S_8, \sigma_{526} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{526} \rangle \Rightarrow \langle S_9, \sigma_{526} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2104} = \langle S_8, \sigma_{526} \rangle \Rightarrow \langle y := y + 1, \sigma_{526} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2103} = \frac{\mathcal{T}_{2104}}{\langle S_9, \sigma_{526} \rangle \Rightarrow \langle S_{10}, \sigma_{526} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2105} = \frac{\langle y := y + 1, \sigma_{526} \rangle \Rightarrow \sigma_{527} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{526} \rangle \Rightarrow \langle S_{11}, \sigma_{527} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2106} = \langle S_{11}, \sigma_{527} \rangle \Rightarrow \langle S_{12}, \sigma_{527} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2107} = \langle S_{12}, \sigma_{527} \rangle \Rightarrow \langle S_{13}, \sigma_{527} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2109} = \frac{\langle x := x + 1, \sigma_{527} \rangle \Rightarrow \sigma_{528} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{527} \rangle \Rightarrow \langle S_6, \sigma_{528} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2108} = \frac{\mathcal{T}_{2109}}{\langle S_{13}, \sigma_{527} \rangle \Rightarrow \langle S_{15}, \sigma_{528} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2110} = \frac{\langle S_6, \sigma_{528} \rangle \Rightarrow \langle S_8, \sigma_{528} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{528} \rangle \Rightarrow \langle S_9, \sigma_{528} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2112} = \langle S_8, \sigma_{528} \rangle \Rightarrow \langle y := y + 1, \sigma_{528} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2111} = \frac{\mathcal{T}_{2112}}{\langle S_9, \sigma_{528} \rangle \Rightarrow \langle S_{10}, \sigma_{528} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2113} = \frac{\langle y := y + 1, \sigma_{528} \rangle \Rightarrow \sigma_{529} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{528} \rangle \Rightarrow \langle S_{11}, \sigma_{529} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{2114} = \langle S_{11}, \sigma_{529} \rangle \Rightarrow \langle S_{12}, \sigma_{529} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2115} = \langle S_{12}, \sigma_{529} \rangle \Rightarrow \langle S_{13}, \sigma_{529} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2117} = \frac{\langle x := x + 1, \sigma_{529} \rangle \Rightarrow \sigma_{530} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{529} \rangle \Rightarrow \langle S_6, \sigma_{530} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2116} = \frac{\mathcal{T}_{2117}}{\langle S_{13}, \sigma_{529} \rangle \Rightarrow \langle S_{15}, \sigma_{530} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2118} = \frac{\langle S_6, \sigma_{530} \rangle \Rightarrow \langle S_8, \sigma_{530} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{530} \rangle \Rightarrow \langle S_9, \sigma_{530} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2120} = \langle S_8, \sigma_{530} \rangle \Rightarrow \langle y := y + 1, \sigma_{530} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2119} = \frac{\mathcal{T}_{2120}}{\langle S_9, \sigma_{530} \rangle \Rightarrow \langle S_{10}, \sigma_{530} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2121} = \frac{\langle y := y + 1, \sigma_{530} \rangle \Rightarrow \sigma_{531} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{530} \rangle \Rightarrow \langle S_{11}, \sigma_{531} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2122} = \langle S_{11}, \sigma_{531} \rangle \Rightarrow \langle S_{12}, \sigma_{531} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2123} = \langle S_{12}, \sigma_{531} \rangle \Rightarrow \langle S_{13}, \sigma_{531} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2125} = \frac{\langle x := x + 1, \sigma_{531} \rangle \Rightarrow \sigma_{532} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{531} \rangle \Rightarrow \langle S_6, \sigma_{532} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2124} = \frac{\mathcal{T}_{2125}}{\langle S_{13}, \sigma_{531} \rangle \Rightarrow \langle S_{15}, \sigma_{532} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2126} = \frac{\langle S_6, \sigma_{532} \rangle \Rightarrow \langle S_8, \sigma_{532} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{532} \rangle \Rightarrow \langle S_9, \sigma_{532} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2128} = \langle S_8, \sigma_{532} \rangle \Rightarrow \langle y := y + 1, \sigma_{532} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2127} = \frac{\mathcal{T}_{2128}}{\langle S_9, \sigma_{532} \rangle \Rightarrow \langle S_{10}, \sigma_{532} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2129} = \frac{\langle y := y + 1, \sigma_{532} \rangle \Rightarrow \sigma_{533} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{532} \rangle \Rightarrow \langle S_{11}, \sigma_{533} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2130} = \langle S_{11}, \sigma_{533} \rangle \Rightarrow \langle S_{12}, \sigma_{533} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2131} = \langle S_{12}, \sigma_{533} \rangle \Rightarrow \langle S_{13}, \sigma_{533} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2133} = \frac{\langle x := x + 1, \sigma_{533} \rangle \Rightarrow \sigma_{534} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{533} \rangle \Rightarrow \langle S_6, \sigma_{534} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2132} = \frac{\mathcal{T}_{2133}}{\langle S_{13}, \sigma_{533} \rangle \Rightarrow \langle S_{15}, \sigma_{534} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2134} = \frac{\langle S_6, \sigma_{534} \rangle \Rightarrow \langle S_8, \sigma_{534} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{534} \rangle \Rightarrow \langle S_9, \sigma_{534} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2136} = \langle S_8, \sigma_{534} \rangle \Rightarrow \langle y := y + 1, \sigma_{534} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2135} = \frac{\mathcal{T}_{2136}}{\langle S_9, \sigma_{534} \rangle \Rightarrow \langle S_{10}, \sigma_{534} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2137} = \frac{\langle y := y + 1, \sigma_{534} \rangle \Rightarrow \sigma_{535} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{534} \rangle \Rightarrow \langle S_{11}, \sigma_{535} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2138} = \langle S_{11}, \sigma_{535} \rangle \Rightarrow \langle S_{12}, \sigma_{535} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2139} = \langle S_{12}, \sigma_{535} \rangle \Rightarrow \langle S_{13}, \sigma_{535} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2141} = \frac{\langle x := x + 1, \sigma_{535} \rangle \Rightarrow \sigma_{536} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{535} \rangle \Rightarrow \langle S_6, \sigma_{536} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2140} = \frac{\mathcal{T}_{2141}}{\langle S_{13}, \sigma_{535} \rangle \Rightarrow \langle S_{15}, \sigma_{536} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2142} = \frac{\langle S_6, \sigma_{536} \rangle \Rightarrow \langle S_8, \sigma_{536} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{536} \rangle \Rightarrow \langle S_9, \sigma_{536} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2144} = \langle S_8, \sigma_{536} \rangle \Rightarrow \langle y := y + 1, \sigma_{536} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2143} = \frac{\mathcal{T}_{2144}}{\langle S_9, \sigma_{536} \rangle \Rightarrow \langle S_{10}, \sigma_{536} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2145} = \frac{\langle y := y + 1, \sigma_{536} \rangle \Rightarrow \sigma_{537} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{536} \rangle \Rightarrow \langle S_{11}, \sigma_{537} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2146} = \langle S_{11}, \sigma_{537} \rangle \Rightarrow \langle S_{12}, \sigma_{537} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2147} = \langle S_{12}, \sigma_{537} \rangle \Rightarrow \langle S_{13}, \sigma_{537} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2149} = \frac{\langle x := x + 1, \sigma_{537} \rangle \Rightarrow \sigma_{538} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{537} \rangle \Rightarrow \langle S_6, \sigma_{538} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2148} = \frac{\mathcal{T}_{2149}}{\langle S_{13}, \sigma_{537} \rangle \Rightarrow \langle S_{15}, \sigma_{538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2150} = \frac{\langle S_6, \sigma_{538} \rangle \Rightarrow \langle S_8, \sigma_{538} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{538} \rangle \Rightarrow \langle S_9, \sigma_{538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2152} = \langle S_8, \sigma_{538} \rangle \Rightarrow \langle y := y + 1, \sigma_{538} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2151} = \frac{\mathcal{T}_{2152}}{\langle S_9, \sigma_{538} \rangle \Rightarrow \langle S_{10}, \sigma_{538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2153} = \frac{\langle y := y + 1, \sigma_{538} \rangle \Rightarrow \sigma_{539} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{538} \rangle \Rightarrow \langle S_{11}, \sigma_{539} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2154} = \langle S_{11}, \sigma_{539} \rangle \Rightarrow \langle S_{12}, \sigma_{539} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2155} = \langle S_{12}, \sigma_{539} \rangle \Rightarrow \langle S_{13}, \sigma_{539} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2157} = \frac{\langle x := x + 1, \sigma_{539} \rangle \Rightarrow \sigma_{540} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{539} \rangle \Rightarrow \langle S_6, \sigma_{540} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2156} = \frac{\mathcal{T}_{2157}}{\langle S_{13}, \sigma_{539} \rangle \Rightarrow \langle S_{15}, \sigma_{540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2158} = \frac{\langle S_6, \sigma_{540} \rangle \Rightarrow \langle S_8, \sigma_{540} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{540} \rangle \Rightarrow \langle S_9, \sigma_{540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2160} = \langle S_8, \sigma_{540} \rangle \Rightarrow \langle y := y + 1, \sigma_{540} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2159} = \frac{\mathcal{T}_{2160}}{\langle S_9, \sigma_{540} \rangle \Rightarrow \langle S_{10}, \sigma_{540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2161} = \frac{\langle y := y + 1, \sigma_{540} \rangle \Rightarrow \sigma_{541} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{540} \rangle \Rightarrow \langle S_{11}, \sigma_{541} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2162} = \langle S_{11}, \sigma_{541} \rangle \Rightarrow \langle S_{12}, \sigma_{541} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2163} = \langle S_{12}, \sigma_{541} \rangle \Rightarrow \langle S_{13}, \sigma_{541} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2165} = \frac{\langle x := x + 1, \sigma_{541} \rangle \Rightarrow \sigma_{542} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{541} \rangle \Rightarrow \langle S_6, \sigma_{542} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2164} = \frac{\mathcal{T}_{2165}}{\langle S_{13}, \sigma_{541} \rangle \Rightarrow \langle S_{15}, \sigma_{542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2166} = \frac{\langle S_6, \sigma_{542} \rangle \Rightarrow \langle S_8, \sigma_{542} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{542} \rangle \Rightarrow \langle S_9, \sigma_{542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2168} = \langle S_8, \sigma_{542} \rangle \Rightarrow \langle y := y + 1, \sigma_{542} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2167} = \frac{\mathcal{T}_{2168}}{\langle S_9, \sigma_{542} \rangle \Rightarrow \langle S_{10}, \sigma_{542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2169} = \frac{\langle y := y + 1, \sigma_{542} \rangle \Rightarrow \sigma_{543} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{542} \rangle \Rightarrow \langle S_{11}, \sigma_{543} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2170} = \langle S_{11}, \sigma_{543} \rangle \Rightarrow \langle S_{12}, \sigma_{543} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2171} = \langle S_{12}, \sigma_{543} \rangle \Rightarrow \langle S_{13}, \sigma_{543} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2173} = \frac{\langle x := x + 1, \sigma_{543} \rangle \Rightarrow \sigma_{544} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{543} \rangle \Rightarrow \langle S_6, \sigma_{544} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2172} = \frac{\mathcal{T}_{2173}}{\langle S_{13}, \sigma_{543} \rangle \Rightarrow \langle S_{15}, \sigma_{544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2174} = \frac{\langle S_6, \sigma_{544} \rangle \Rightarrow \langle S_8, \sigma_{544} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{544} \rangle \Rightarrow \langle S_9, \sigma_{544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2176} = \langle S_8, \sigma_{544} \rangle \Rightarrow \langle y := y + 1, \sigma_{544} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2175} = \frac{\mathcal{T}_{2176}}{\langle S_9, \sigma_{544} \rangle \Rightarrow \langle S_{10}, \sigma_{544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2177} = \frac{\langle y := y + 1, \sigma_{544} \rangle \Rightarrow \sigma_{545} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{544} \rangle \Rightarrow \langle S_{11}, \sigma_{545} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2178} = \langle S_{11}, \sigma_{545} \rangle \Rightarrow \langle S_{12}, \sigma_{545} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2179} = \langle S_{12}, \sigma_{545} \rangle \Rightarrow \langle S_{13}, \sigma_{545} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2181} = \frac{\langle x := x + 1, \sigma_{545} \rangle \Rightarrow \sigma_{546} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{545} \rangle \Rightarrow \langle S_6, \sigma_{546} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2180} = \frac{\mathcal{T}_{2181}}{\langle S_{13}, \sigma_{545} \rangle \Rightarrow \langle S_{15}, \sigma_{546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2182} = \frac{\langle S_6, \sigma_{546} \rangle \Rightarrow \langle S_8, \sigma_{546} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{546} \rangle \Rightarrow \langle S_9, \sigma_{546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2184} = \langle S_8, \sigma_{546} \rangle \Rightarrow \langle y := y + 1, \sigma_{546} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2183} = \frac{\mathcal{T}_{2184}}{\langle S_9, \sigma_{546} \rangle \Rightarrow \langle S_{10}, \sigma_{546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2185} = \frac{\langle y := y + 1, \sigma_{546} \rangle \Rightarrow \sigma_{547} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{546} \rangle \Rightarrow \langle S_{11}, \sigma_{547} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2186} = \langle S_{11}, \sigma_{547} \rangle \Rightarrow \langle S_{12}, \sigma_{547} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2187} = \langle S_{12}, \sigma_{547} \rangle \Rightarrow \langle S_{13}, \sigma_{547} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2189} = \frac{\langle x := x + 1, \sigma_{547} \rangle \Rightarrow \sigma_{548} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{547} \rangle \Rightarrow \langle S_6, \sigma_{548} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2188} = \frac{\mathcal{T}_{2189}}{\langle S_{13}, \sigma_{547} \rangle \Rightarrow \langle S_{15}, \sigma_{548} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2190} = \frac{\langle S_6, \sigma_{548} \rangle \Rightarrow \langle S_8, \sigma_{548} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{548} \rangle \Rightarrow \langle S_9, \sigma_{548} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2192} = \langle S_8, \sigma_{548} \rangle \Rightarrow \langle y := y + 1, \sigma_{548} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2191} = \frac{\mathcal{T}_{2192}}{\langle S_9, \sigma_{548} \rangle \Rightarrow \langle S_{10}, \sigma_{548} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2193} = \frac{\langle y := y + 1, \sigma_{548} \rangle \Rightarrow \sigma_{549} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{548} \rangle \Rightarrow \langle S_{11}, \sigma_{549} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2194} = \langle S_{11}, \sigma_{549} \rangle \Rightarrow \langle S_{12}, \sigma_{549} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2195} = \langle S_{12}, \sigma_{549} \rangle \Rightarrow \langle S_{13}, \sigma_{549} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2197} = \frac{\langle x := x + 1, \sigma_{549} \rangle \Rightarrow \sigma_{550} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{549} \rangle \Rightarrow \langle S_6, \sigma_{550} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2196} = \frac{\mathcal{T}_{2197}}{\langle S_{13}, \sigma_{549} \rangle \Rightarrow \langle S_{15}, \sigma_{550} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2198} = \frac{\langle S_6, \sigma_{550} \rangle \Rightarrow \langle S_8, \sigma_{550} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{550} \rangle \Rightarrow \langle S_9, \sigma_{550} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2200} = \langle S_8, \sigma_{550} \rangle \Rightarrow \langle y := y + 1, \sigma_{550} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2199} = \frac{\mathcal{T}_{2200}}{\langle S_9, \sigma_{550} \rangle \Rightarrow \langle S_{10}, \sigma_{550} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2201} = \frac{\langle y := y + 1, \sigma_{550} \rangle \Rightarrow \sigma_{551} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{550} \rangle \Rightarrow \langle S_{11}, \sigma_{551} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2202} = \langle S_{11}, \sigma_{551} \rangle \Rightarrow \langle S_{12}, \sigma_{551} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2203} = \langle S_{12}, \sigma_{551} \rangle \Rightarrow \langle S_{13}, \sigma_{551} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2205} = \frac{\langle x := x + 1, \sigma_{551} \rangle \Rightarrow \sigma_{552} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{551} \rangle \Rightarrow \langle S_6, \sigma_{552} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2204} = \frac{\mathcal{T}_{2205}}{\langle S_{13}, \sigma_{551} \rangle \Rightarrow \langle S_{15}, \sigma_{552} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2206} = \frac{\langle S_6, \sigma_{552} \rangle \Rightarrow \langle S_8, \sigma_{552} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{552} \rangle \Rightarrow \langle S_9, \sigma_{552} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2208} = \langle S_8, \sigma_{552} \rangle \Rightarrow \langle y := y + 1, \sigma_{552} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2207} = \frac{\mathcal{T}_{2208}}{\langle S_9, \sigma_{552} \rangle \Rightarrow \langle S_{10}, \sigma_{552} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2209} = \frac{\langle y := y + 1, \sigma_{552} \rangle \Rightarrow \sigma_{553} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{552} \rangle \Rightarrow \langle S_{11}, \sigma_{553} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2210} = \langle S_{11}, \sigma_{553} \rangle \Rightarrow \langle S_{12}, \sigma_{553} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2211} = \langle S_{12}, \sigma_{553} \rangle \Rightarrow \langle S_{13}, \sigma_{553} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2213} = \frac{\langle x := x + 1, \sigma_{553} \rangle \Rightarrow \sigma_{554} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{553} \rangle \Rightarrow \langle S_6, \sigma_{554} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2212} = \frac{\mathcal{T}_{2213}}{\langle S_{13}, \sigma_{553} \rangle \Rightarrow \langle S_{15}, \sigma_{554} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2214} = \frac{\langle S_6, \sigma_{554} \rangle \Rightarrow \langle S_8, \sigma_{554} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{554} \rangle \Rightarrow \langle S_9, \sigma_{554} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2216} = \langle S_8, \sigma_{554} \rangle \Rightarrow \langle y := y + 1, \sigma_{554} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2215} = \frac{\mathcal{T}_{2216}}{\langle S_9, \sigma_{554} \rangle \Rightarrow \langle S_{10}, \sigma_{554} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2217} = \frac{\langle y := y + 1, \sigma_{554} \rangle \Rightarrow \sigma_{555} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{554} \rangle \Rightarrow \langle S_{11}, \sigma_{555} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2218} = \langle S_{11}, \sigma_{555} \rangle \Rightarrow \langle S_{12}, \sigma_{555} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2219} = \langle S_{12}, \sigma_{555} \rangle \Rightarrow \langle S_{13}, \sigma_{555} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2221} = \frac{\langle x := x + 1, \sigma_{555} \rangle \Rightarrow \sigma_{556} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{555} \rangle \Rightarrow \langle S_6, \sigma_{556} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2220} = \frac{\mathcal{T}_{2221}}{\langle S_{13}, \sigma_{555} \rangle \Rightarrow \langle S_{15}, \sigma_{556} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2222} = \frac{\langle S_6, \sigma_{556} \rangle \Rightarrow \langle S_8, \sigma_{556} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{556} \rangle \Rightarrow \langle S_9, \sigma_{556} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2224} = \langle S_8, \sigma_{556} \rangle \Rightarrow \langle y := y + 1, \sigma_{556} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2223} = \frac{\mathcal{T}_{2224}}{\langle S_9, \sigma_{556} \rangle \Rightarrow \langle S_{10}, \sigma_{556} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2225} = \frac{\langle y := y + 1, \sigma_{556} \rangle \Rightarrow \sigma_{557} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{556} \rangle \Rightarrow \langle S_{11}, \sigma_{557} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2226} = \langle S_{11}, \sigma_{557} \rangle \Rightarrow \langle S_{12}, \sigma_{557} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2227} = \langle S_{12}, \sigma_{557} \rangle \Rightarrow \langle S_{13}, \sigma_{557} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2229} = \frac{\langle x := x + 1, \sigma_{557} \rangle \Rightarrow \sigma_{558} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{557} \rangle \Rightarrow \langle S_6, \sigma_{558} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2228} = \frac{\mathcal{T}_{2229}}{\langle S_{13}, \sigma_{557} \rangle \Rightarrow \langle S_{15}, \sigma_{558} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2230} = \frac{\langle S_6, \sigma_{558} \rangle \Rightarrow \langle S_8, \sigma_{558} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{558} \rangle \Rightarrow \langle S_9, \sigma_{558} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2232} = \langle S_8, \sigma_{558} \rangle \Rightarrow \langle y := y + 1, \sigma_{558} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2231} = \frac{\mathcal{T}_{2232}}{\langle S_9, \sigma_{558} \rangle \Rightarrow \langle S_{10}, \sigma_{558} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2233} = \frac{\langle y := y + 1, \sigma_{558} \rangle \Rightarrow \sigma_{559} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{558} \rangle \Rightarrow \langle S_{11}, \sigma_{559} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2234} = \langle S_{11}, \sigma_{559} \rangle \Rightarrow \langle S_{12}, \sigma_{559} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2235} = \langle S_{12}, \sigma_{559} \rangle \Rightarrow \langle S_{13}, \sigma_{559} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2237} = \frac{\langle x := x + 1, \sigma_{559} \rangle \Rightarrow \sigma_{560} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{559} \rangle \Rightarrow \langle S_6, \sigma_{560} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2236} = \frac{\mathcal{T}_{2237}}{\langle S_{13}, \sigma_{559} \rangle \Rightarrow \langle S_{15}, \sigma_{560} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{2238} = \frac{\langle S_6, \sigma_{560} \rangle \Rightarrow \langle S_8, \sigma_{560} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{560} \rangle \Rightarrow \langle S_9, \sigma_{560} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{2240} = \langle S_8, \sigma_{560} \rangle \Rightarrow \langle y := y + 1, \sigma_{560} \rangle \text{ [if } -\text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2239} = \frac{\mathcal{T}_{2240}}{\langle S_9, \sigma_{560} \rangle \Rightarrow \langle S_{10}, \sigma_{560} \rangle} [\text{comp}_{\text{sol}}^1]$$

$$\mathcal{T}_{2241} = \frac{\langle y := y + 1, \sigma_{560} \rangle \Rightarrow \sigma_{561} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{560} \rangle \Rightarrow \langle S_{11}, \sigma_{561} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2242} = \langle S_{11}, \sigma_{561} \rangle \Rightarrow \langle S_{12}, \sigma_{561} \rangle \text{ [for}_{\text{sos}}]$$

$$\mathcal{T}_{2243} = \langle S_{12}, \sigma_{561} \rangle \Rightarrow \langle S_{13}, \sigma_{561} \rangle \text{ [if } -\text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2245} = \frac{\langle x := x + 1, \sigma_{561} \rangle \Rightarrow \sigma_{562} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{561} \rangle \Rightarrow \langle S_6, \sigma_{562} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2244} = \frac{\mathcal{T}_{2245}}{\langle S_{13}, \sigma_{561} \rangle \Rightarrow \langle S_{15}, \sigma_{562} \rangle} [\text{comp}_{\text{sol}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2246} = \frac{\langle S_6, \sigma_{562} \rangle \Rightarrow \langle S_8, \sigma_{562} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{562} \rangle \Rightarrow \langle S_9, \sigma_{562} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2248} = \langle S_8, \sigma_{562} \rangle \Rightarrow \langle y := y + 1, \sigma_{562} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2247} = \frac{\mathcal{T}_{2248}}{\langle S_9, \sigma_{562} \rangle \Rightarrow \langle S_{10}, \sigma_{562} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2249} = \frac{\langle y := y + 1, \sigma_{562} \rangle \Rightarrow \sigma_{563} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{562} \rangle \Rightarrow \langle S_{11}, \sigma_{563} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2250} = \langle S_{11}, \sigma_{563} \rangle \Rightarrow \langle S_{12}, \sigma_{563} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2251} = \langle S_{12}, \sigma_{563} \rangle \Rightarrow \langle S_{13}, \sigma_{563} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2253} = \frac{\langle x := x + 1, \sigma_{563} \rangle \Rightarrow \sigma_{564} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{563} \rangle \Rightarrow \langle S_6, \sigma_{564} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2252} = \frac{\mathcal{T}_{2253}}{\langle S_{13}, \sigma_{563} \rangle \Rightarrow \langle S_{15}, \sigma_{564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2254} = \frac{\langle S_6, \sigma_{564} \rangle \Rightarrow \langle S_8, \sigma_{564} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{564} \rangle \Rightarrow \langle S_9, \sigma_{564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2256} = \langle S_8, \sigma_{564} \rangle \Rightarrow \langle y := y + 1, \sigma_{564} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2255} = \frac{\mathcal{T}_{2256}}{\langle S_9, \sigma_{564} \rangle \Rightarrow \langle S_{10}, \sigma_{564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2257} = \frac{\langle y := y + 1, \sigma_{564} \rangle \Rightarrow \sigma_{565} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{564} \rangle \Rightarrow \langle S_{11}, \sigma_{565} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2258} = \langle S_{11}, \sigma_{565} \rangle \Rightarrow \langle S_{12}, \sigma_{565} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2259} = \langle S_{12}, \sigma_{565} \rangle \Rightarrow \langle S_{13}, \sigma_{565} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2261} = \frac{\langle x := x + 1, \sigma_{565} \rangle \Rightarrow \sigma_{566} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{565} \rangle \Rightarrow \langle S_6, \sigma_{566} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2260} = \frac{\mathcal{T}_{2261}}{\langle S_{13}, \sigma_{565} \rangle \Rightarrow \langle S_{15}, \sigma_{566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2262} = \frac{\langle S_6, \sigma_{566} \rangle \Rightarrow \langle S_8, \sigma_{566} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{566} \rangle \Rightarrow \langle S_9, \sigma_{566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2264} = \langle S_8, \sigma_{566} \rangle \Rightarrow \langle y := y + 1, \sigma_{566} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2263} = \frac{\mathcal{T}_{2264}}{\langle S_9, \sigma_{566} \rangle \Rightarrow \langle S_{10}, \sigma_{566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2265} = \frac{\langle y := y + 1, \sigma_{566} \rangle \Rightarrow \sigma_{567} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{566} \rangle \Rightarrow \langle S_{11}, \sigma_{567} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2266} = \langle S_{11}, \sigma_{567} \rangle \Rightarrow \langle S_{12}, \sigma_{567} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2267} = \langle S_{12}, \sigma_{567} \rangle \Rightarrow \langle S_{13}, \sigma_{567} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2269} = \frac{\langle x := x + 1, \sigma_{567} \rangle \Rightarrow \sigma_{568} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{567} \rangle \Rightarrow \langle S_6, \sigma_{568} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2268} = \frac{\mathcal{T}_{2269}}{\langle S_{13}, \sigma_{567} \rangle \Rightarrow \langle S_{15}, \sigma_{568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2270} = \frac{\langle S_6, \sigma_{568} \rangle \Rightarrow \langle S_8, \sigma_{568} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{568} \rangle \Rightarrow \langle S_9, \sigma_{568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2272} = \langle S_8, \sigma_{568} \rangle \Rightarrow \langle y := y + 1, \sigma_{568} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2271} = \frac{\mathcal{T}_{2272}}{\langle S_9, \sigma_{568} \rangle \Rightarrow \langle S_{10}, \sigma_{568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2273} = \frac{\langle y := y + 1, \sigma_{568} \rangle \Rightarrow \sigma_{569} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{568} \rangle \Rightarrow \langle S_{11}, \sigma_{569} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2274} = \langle S_{11}, \sigma_{569} \rangle \Rightarrow \langle S_{12}, \sigma_{569} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2275} = \langle S_{12}, \sigma_{569} \rangle \Rightarrow \langle S_{13}, \sigma_{569} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2277} = \frac{\langle x := x + 1, \sigma_{569} \rangle \Rightarrow \sigma_{570} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{569} \rangle \Rightarrow \langle S_6, \sigma_{570} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2276} = \frac{\mathcal{T}_{2277}}{\langle S_{13}, \sigma_{569} \rangle \Rightarrow \langle S_{15}, \sigma_{570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2278} = \frac{\langle S_6, \sigma_{570} \rangle \Rightarrow \langle S_8, \sigma_{570} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{570} \rangle \Rightarrow \langle S_9, \sigma_{570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2280} = \langle S_8, \sigma_{570} \rangle \Rightarrow \langle y := y + 1, \sigma_{570} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2279} = \frac{\mathcal{T}_{2280}}{\langle S_9, \sigma_{570} \rangle \Rightarrow \langle S_{10}, \sigma_{570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2281} = \frac{\langle y := y + 1, \sigma_{570} \rangle \Rightarrow \sigma_{571} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{570} \rangle \Rightarrow \langle S_{11}, \sigma_{571} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2282} = \langle S_{11}, \sigma_{571} \rangle \Rightarrow \langle S_{12}, \sigma_{571} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2283} = \langle S_{12}, \sigma_{571} \rangle \Rightarrow \langle S_{13}, \sigma_{571} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2285} = \frac{\langle x := x + 1, \sigma_{571} \rangle \Rightarrow \sigma_{572} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{571} \rangle \Rightarrow \langle S_6, \sigma_{572} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2284} = \frac{\mathcal{T}_{2285}}{\langle S_{13}, \sigma_{571} \rangle \Rightarrow \langle S_{15}, \sigma_{572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2286} = \frac{\langle S_6, \sigma_{572} \rangle \Rightarrow \langle S_8, \sigma_{572} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{572} \rangle \Rightarrow \langle S_9, \sigma_{572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2288} = \langle S_8, \sigma_{572} \rangle \Rightarrow \langle y := y + 1, \sigma_{572} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2287} = \frac{\mathcal{T}_{2288}}{\langle S_9, \sigma_{572} \rangle \Rightarrow \langle S_{10}, \sigma_{572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2289} = \frac{\langle y := y + 1, \sigma_{572} \rangle \Rightarrow \sigma_{573} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{572} \rangle \Rightarrow \langle S_{11}, \sigma_{573} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2290} = \langle S_{11}, \sigma_{573} \rangle \Rightarrow \langle S_{12}, \sigma_{573} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2291} = \langle S_{12}, \sigma_{573} \rangle \Rightarrow \langle S_{13}, \sigma_{573} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2293} = \frac{\langle x := x + 1, \sigma_{573} \rangle \Rightarrow \sigma_{574} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{573} \rangle \Rightarrow \langle S_6, \sigma_{574} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2292} = \frac{\mathcal{T}_{2293}}{\langle S_{13}, \sigma_{573} \rangle \Rightarrow \langle S_{15}, \sigma_{574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2294} = \frac{\langle S_6, \sigma_{574} \rangle \Rightarrow \langle S_8, \sigma_{574} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{574} \rangle \Rightarrow \langle S_9, \sigma_{574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2296} = \langle S_8, \sigma_{574} \rangle \Rightarrow \langle y := y + 1, \sigma_{574} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2295} = \frac{\mathcal{T}_{2296}}{\langle S_9, \sigma_{574} \rangle \Rightarrow \langle S_{10}, \sigma_{574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2297} = \frac{\langle y := y + 1, \sigma_{574} \rangle \Rightarrow \sigma_{575} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{574} \rangle \Rightarrow \langle S_{11}, \sigma_{575} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2298} = \langle S_{11}, \sigma_{575} \rangle \Rightarrow \langle S_{12}, \sigma_{575} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2299} = \langle S_{12}, \sigma_{575} \rangle \Rightarrow \langle S_{13}, \sigma_{575} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2301} = \frac{\langle x := x + 1, \sigma_{575} \rangle \Rightarrow \sigma_{576} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{575} \rangle \Rightarrow \langle S_6, \sigma_{576} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2300} = \frac{\mathcal{T}_{2301}}{\langle S_{13}, \sigma_{575} \rangle \Rightarrow \langle S_{15}, \sigma_{576} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2302} = \frac{\langle S_6, \sigma_{576} \rangle \Rightarrow \langle S_8, \sigma_{576} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{576} \rangle \Rightarrow \langle S_9, \sigma_{576} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2304} = \langle S_8, \sigma_{576} \rangle \Rightarrow \langle y := y + 1, \sigma_{576} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2303} = \frac{\mathcal{T}_{2304}}{\langle S_9, \sigma_{576} \rangle \Rightarrow \langle S_{10}, \sigma_{576} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2305} = \frac{\langle y := y + 1, \sigma_{576} \rangle \Rightarrow \sigma_{577} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{576} \rangle \Rightarrow \langle S_{11}, \sigma_{577} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2306} = \langle S_{11}, \sigma_{577} \rangle \Rightarrow \langle S_{12}, \sigma_{577} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2307} = \langle S_{12}, \sigma_{577} \rangle \Rightarrow \langle S_{13}, \sigma_{577} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2309} = \frac{\langle x := x + 1, \sigma_{577} \rangle \Rightarrow \sigma_{578} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{577} \rangle \Rightarrow \langle S_6, \sigma_{578} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2308} = \frac{\mathcal{T}_{2309}}{\langle S_{13}, \sigma_{577} \rangle \Rightarrow \langle S_{15}, \sigma_{578} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2310} = \frac{\langle S_6, \sigma_{578} \rangle \Rightarrow \langle S_8, \sigma_{578} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{578} \rangle \Rightarrow \langle S_9, \sigma_{578} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2312} = \langle S_8, \sigma_{578} \rangle \Rightarrow \langle y := y + 1, \sigma_{578} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2311} = \frac{\mathcal{T}_{2312}}{\langle S_9, \sigma_{578} \rangle \Rightarrow \langle S_{10}, \sigma_{578} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2313} = \frac{\langle y := y + 1, \sigma_{578} \rangle \Rightarrow \sigma_{579} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{578} \rangle \Rightarrow \langle S_{11}, \sigma_{579} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2314} = \langle S_{11}, \sigma_{579} \rangle \Rightarrow \langle S_{12}, \sigma_{579} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2315} = \langle S_{12}, \sigma_{579} \rangle \Rightarrow \langle S_{13}, \sigma_{579} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2317} = \frac{\langle x := x + 1, \sigma_{579} \rangle \Rightarrow \sigma_{580} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{579} \rangle \Rightarrow \langle S_6, \sigma_{580} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2316} = \frac{\mathcal{T}_{2317}}{\langle S_{13}, \sigma_{579} \rangle \Rightarrow \langle S_{15}, \sigma_{580} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2318} = \frac{\langle S_6, \sigma_{580} \rangle \Rightarrow \langle S_8, \sigma_{580} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{580} \rangle \Rightarrow \langle S_9, \sigma_{580} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2320} = \langle S_8, \sigma_{580} \rangle \Rightarrow \langle y := y + 1, \sigma_{580} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2319} = \frac{\mathcal{T}_{2320}}{\langle S_9, \sigma_{580} \rangle \Rightarrow \langle S_{10}, \sigma_{580} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2321} = \frac{\langle y := y + 1, \sigma_{580} \rangle \Rightarrow \sigma_{581} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{580} \rangle \Rightarrow \langle S_{11}, \sigma_{581} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2322} = \langle S_{11}, \sigma_{581} \rangle \Rightarrow \langle S_{12}, \sigma_{581} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2323} = \langle S_{12}, \sigma_{581} \rangle \Rightarrow \langle S_{13}, \sigma_{581} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2325} = \frac{\langle x := x + 1, \sigma_{581} \rangle \Rightarrow \sigma_{582} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{581} \rangle \Rightarrow \langle S_6, \sigma_{582} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2324} = \frac{\mathcal{T}_{2325}}{\langle S_{13}, \sigma_{581} \rangle \Rightarrow \langle S_{15}, \sigma_{582} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2326} = \frac{\langle S_6, \sigma_{582} \rangle \Rightarrow \langle S_8, \sigma_{582} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{582} \rangle \Rightarrow \langle S_9, \sigma_{582} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2328} = \langle S_8, \sigma_{582} \rangle \Rightarrow \langle y := y + 1, \sigma_{582} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2327} = \frac{\mathcal{T}_{2328}}{\langle S_9, \sigma_{582} \rangle \Rightarrow \langle S_{10}, \sigma_{582} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2329} = \frac{\langle y := y + 1, \sigma_{582} \rangle \Rightarrow \sigma_{583} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{582} \rangle \Rightarrow \langle S_{11}, \sigma_{583} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2330} = \langle S_{11}, \sigma_{583} \rangle \Rightarrow \langle S_{12}, \sigma_{583} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2331} = \langle S_{12}, \sigma_{583} \rangle \Rightarrow \langle S_{13}, \sigma_{583} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2333} = \frac{\langle x := x + 1, \sigma_{583} \rangle \Rightarrow \sigma_{584} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{583} \rangle \Rightarrow \langle S_6, \sigma_{584} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2332} = \frac{\mathcal{T}_{2333}}{\langle S_{13}, \sigma_{583} \rangle \Rightarrow \langle S_{15}, \sigma_{584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2334} = \frac{\langle S_6, \sigma_{584} \rangle \Rightarrow \langle S_8, \sigma_{584} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{584} \rangle \Rightarrow \langle S_9, \sigma_{584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2336} = \langle S_8, \sigma_{584} \rangle \Rightarrow \langle y := y + 1, \sigma_{584} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2335} = \frac{\mathcal{T}_{2336}}{\langle S_9, \sigma_{584} \rangle \Rightarrow \langle S_{10}, \sigma_{584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2337} = \frac{\langle y := y + 1, \sigma_{584} \rangle \Rightarrow \sigma_{585} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{584} \rangle \Rightarrow \langle S_{11}, \sigma_{585} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2338} = \langle S_{11}, \sigma_{585} \rangle \Rightarrow \langle S_{12}, \sigma_{585} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2339} = \langle S_{12}, \sigma_{585} \rangle \Rightarrow \langle S_{13}, \sigma_{585} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2341} = \frac{\langle x := x + 1, \sigma_{585} \rangle \Rightarrow \sigma_{586} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{585} \rangle \Rightarrow \langle S_6, \sigma_{586} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2340} = \frac{\mathcal{T}_{2341}}{\langle S_{13}, \sigma_{585} \rangle \Rightarrow \langle S_{15}, \sigma_{586} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2342} = \frac{\langle S_6, \sigma_{586} \rangle \Rightarrow \langle S_8, \sigma_{586} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{586} \rangle \Rightarrow \langle S_9, \sigma_{586} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2344} = \langle S_8, \sigma_{586} \rangle \Rightarrow \langle y := y + 1, \sigma_{586} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2343} = \frac{\mathcal{T}_{2344}}{\langle S_9, \sigma_{586} \rangle \Rightarrow \langle S_{10}, \sigma_{586} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2345} = \frac{\langle y := y + 1, \sigma_{586} \rangle \Rightarrow \sigma_{587} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{586} \rangle \Rightarrow \langle S_{11}, \sigma_{587} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2346} = \langle S_{11}, \sigma_{587} \rangle \Rightarrow \langle S_{12}, \sigma_{587} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2347} = \langle S_{12}, \sigma_{587} \rangle \Rightarrow \langle S_{13}, \sigma_{587} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2349} = \frac{\langle x := x + 1, \sigma_{587} \rangle \Rightarrow \sigma_{588} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{587} \rangle \Rightarrow \langle S_6, \sigma_{588} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2348} = \frac{\mathcal{T}_{2349}}{\langle S_{13}, \sigma_{587} \rangle \Rightarrow \langle S_{15}, \sigma_{588} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2350} = \frac{\langle S_6, \sigma_{588} \rangle \Rightarrow \langle S_8, \sigma_{588} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{588} \rangle \Rightarrow \langle S_9, \sigma_{588} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2352} = \langle S_8, \sigma_{588} \rangle \Rightarrow \langle y := y + 1, \sigma_{588} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2351} = \frac{\mathcal{T}_{2352}}{\langle S_9, \sigma_{588} \rangle \Rightarrow \langle S_{10}, \sigma_{588} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2353} = \frac{\langle y := y + 1, \sigma_{588} \rangle \Rightarrow \sigma_{589} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{588} \rangle \Rightarrow \langle S_{11}, \sigma_{589} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2354} = \langle S_{11}, \sigma_{589} \rangle \Rightarrow \langle S_{12}, \sigma_{589} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2355} = \langle S_{12}, \sigma_{589} \rangle \Rightarrow \langle S_{13}, \sigma_{589} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2357} = \frac{\langle x := x + 1, \sigma_{589} \rangle \Rightarrow \sigma_{590} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{589} \rangle \Rightarrow \langle S_6, \sigma_{590} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2356} = \frac{\mathcal{T}_{2357}}{\langle S_{13}, \sigma_{589} \rangle \Rightarrow \langle S_{15}, \sigma_{590} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2358} = \frac{\langle S_6, \sigma_{590} \rangle \Rightarrow \langle S_8, \sigma_{590} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{590} \rangle \Rightarrow \langle S_9, \sigma_{590} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2360} = \langle S_8, \sigma_{590} \rangle \Rightarrow \langle y := y + 1, \sigma_{590} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2359} = \frac{\mathcal{T}_{2360}}{\langle S_9, \sigma_{590} \rangle \Rightarrow \langle S_{10}, \sigma_{590} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2361} = \frac{\langle y := y + 1, \sigma_{590} \rangle \Rightarrow \sigma_{591} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{590} \rangle \Rightarrow \langle S_{11}, \sigma_{591} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2362} = \langle S_{11}, \sigma_{591} \rangle \Rightarrow \langle S_{12}, \sigma_{591} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2363} = \langle S_{12}, \sigma_{591} \rangle \Rightarrow \langle S_{13}, \sigma_{591} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2365} = \frac{\langle x := x + 1, \sigma_{591} \rangle \Rightarrow \sigma_{592} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{591} \rangle \Rightarrow \langle S_6, \sigma_{592} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2364} = \frac{\mathcal{T}_{2365}}{\langle S_{13}, \sigma_{591} \rangle \Rightarrow \langle S_{15}, \sigma_{592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2366} = \frac{\langle S_6, \sigma_{592} \rangle \Rightarrow \langle S_8, \sigma_{592} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{592} \rangle \Rightarrow \langle S_9, \sigma_{592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2368} = \langle S_8, \sigma_{592} \rangle \Rightarrow \langle y := y + 1, \sigma_{592} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2367} = \frac{\mathcal{T}_{2368}}{\langle S_9, \sigma_{592} \rangle \Rightarrow \langle S_{10}, \sigma_{592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2369} = \frac{\langle y := y + 1, \sigma_{592} \rangle \Rightarrow \sigma_{593} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{592} \rangle \Rightarrow \langle S_{11}, \sigma_{593} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2370} = \langle S_{11}, \sigma_{593} \rangle \Rightarrow \langle S_{12}, \sigma_{593} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2371} = \langle S_{12}, \sigma_{593} \rangle \Rightarrow \langle S_{13}, \sigma_{593} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2373} = \frac{\langle x := x + 1, \sigma_{593} \rangle \Rightarrow \sigma_{594} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{593} \rangle \Rightarrow \langle S_6, \sigma_{594} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2372} = \frac{\mathcal{T}_{2373}}{\langle S_{13}, \sigma_{593} \rangle \Rightarrow \langle S_{15}, \sigma_{594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2374} = \frac{\langle S_6, \sigma_{594} \rangle \Rightarrow \langle S_8, \sigma_{594} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{594} \rangle \Rightarrow \langle S_9, \sigma_{594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2376} = \langle S_8, \sigma_{594} \rangle \Rightarrow \langle y := y + 1, \sigma_{594} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2375} = \frac{\mathcal{T}_{2376}}{\langle S_9, \sigma_{594} \rangle \Rightarrow \langle S_{10}, \sigma_{594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2377} = \frac{\langle y := y + 1, \sigma_{594} \rangle \Rightarrow \sigma_{595} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{594} \rangle \Rightarrow \langle S_{11}, \sigma_{595} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2378} = \langle S_{11}, \sigma_{595} \rangle \Rightarrow \langle S_{12}, \sigma_{595} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2379} = \langle S_{12}, \sigma_{595} \rangle \Rightarrow \langle S_{13}, \sigma_{595} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2381} = \frac{\langle x := x + 1, \sigma_{595} \rangle \Rightarrow \sigma_{596} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{595} \rangle \Rightarrow \langle S_6, \sigma_{596} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2380} = \frac{\mathcal{T}_{2381}}{\langle S_{13}, \sigma_{595} \rangle \Rightarrow \langle S_{15}, \sigma_{596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2382} = \frac{\langle S_6, \sigma_{596} \rangle \Rightarrow \langle S_8, \sigma_{596} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{596} \rangle \Rightarrow \langle S_9, \sigma_{596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2384} = \langle S_8, \sigma_{596} \rangle \Rightarrow \langle y := y + 1, \sigma_{596} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2383} = \frac{\mathcal{T}_{2384}}{\langle S_9, \sigma_{596} \rangle \Rightarrow \langle S_{10}, \sigma_{596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2385} = \frac{\langle y := y + 1, \sigma_{596} \rangle \Rightarrow \sigma_{597} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{596} \rangle \Rightarrow \langle S_{11}, \sigma_{597} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2386} = \langle S_{11}, \sigma_{597} \rangle \Rightarrow \langle S_{12}, \sigma_{597} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2387} = \langle S_{12}, \sigma_{597} \rangle \Rightarrow \langle S_{13}, \sigma_{597} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2389} = \frac{\langle x := x + 1, \sigma_{597} \rangle \Rightarrow \sigma_{598} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{597} \rangle \Rightarrow \langle S_6, \sigma_{598} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2388} = \frac{\mathcal{T}_{2389}}{\langle S_{13}, \sigma_{597} \rangle \Rightarrow \langle S_{15}, \sigma_{598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2390} = \frac{\langle S_6, \sigma_{598} \rangle \Rightarrow \langle S_8, \sigma_{598} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{598} \rangle \Rightarrow \langle S_9, \sigma_{598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2392} = \langle S_8, \sigma_{598} \rangle \Rightarrow \langle y := y + 1, \sigma_{598} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2391} = \frac{\mathcal{T}_{2392}}{\langle S_9, \sigma_{598} \rangle \Rightarrow \langle S_{10}, \sigma_{598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2393} = \frac{\langle y := y + 1, \sigma_{598} \rangle \Rightarrow \sigma_{599} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{598} \rangle \Rightarrow \langle S_{11}, \sigma_{599} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2394} = \langle S_{11}, \sigma_{599} \rangle \Rightarrow \langle S_{12}, \sigma_{599} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2395} = \langle S_{12}, \sigma_{599} \rangle \Rightarrow \langle S_{13}, \sigma_{599} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2397} = \frac{\langle x := x + 1, \sigma_{599} \rangle \Rightarrow \sigma_{600} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{599} \rangle \Rightarrow \langle S_6, \sigma_{600} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2396} = \frac{\mathcal{T}_{2397}}{\langle S_{13}, \sigma_{599} \rangle \Rightarrow \langle S_{15}, \sigma_{600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2398} = \frac{\langle S_6, \sigma_{600} \rangle \Rightarrow \langle S_8, \sigma_{600} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{600} \rangle \Rightarrow \langle S_9, \sigma_{600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2400} = \langle S_8, \sigma_{600} \rangle \Rightarrow \langle y := y + 1, \sigma_{600} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2399} = \frac{\mathcal{T}_{2400}}{\langle S_9, \sigma_{600} \rangle \Rightarrow \langle S_{10}, \sigma_{600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2401} = \frac{\langle y := y + 1, \sigma_{600} \rangle \Rightarrow \sigma_{601} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{600} \rangle \Rightarrow \langle S_{11}, \sigma_{601} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2402} = \langle S_{11}, \sigma_{601} \rangle \Rightarrow \langle S_{12}, \sigma_{601} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2403} = \langle S_{12}, \sigma_{601} \rangle \Rightarrow \langle S_{13}, \sigma_{601} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2405} = \frac{\langle x := x + 1, \sigma_{601} \rangle \Rightarrow \sigma_{602} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{601} \rangle \Rightarrow \langle S_6, \sigma_{602} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2404} = \frac{\mathcal{T}_{2405}}{\langle S_{13}, \sigma_{601} \rangle \Rightarrow \langle S_{15}, \sigma_{602} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2406} = \frac{\langle S_6, \sigma_{602} \rangle \Rightarrow \langle S_8, \sigma_{602} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{602} \rangle \Rightarrow \langle S_9, \sigma_{602} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2408} = \langle S_8, \sigma_{602} \rangle \Rightarrow \langle y := y + 1, \sigma_{602} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2407} = \frac{\mathcal{T}_{2408}}{\langle S_9, \sigma_{602} \rangle \Rightarrow \langle S_{10}, \sigma_{602} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2409} = \frac{\langle y := y + 1, \sigma_{602} \rangle \Rightarrow \sigma_{603} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{602} \rangle \Rightarrow \langle S_{11}, \sigma_{603} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2410} = \langle S_{11}, \sigma_{603} \rangle \Rightarrow \langle S_{12}, \sigma_{603} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2411} = \langle S_{12}, \sigma_{603} \rangle \Rightarrow \langle S_{13}, \sigma_{603} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2413} = \frac{\langle x := x + 1, \sigma_{603} \rangle \Rightarrow \sigma_{604} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{603} \rangle \Rightarrow \langle S_6, \sigma_{604} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2412} = \frac{\mathcal{T}_{2413}}{\langle S_{13}, \sigma_{603} \rangle \Rightarrow \langle S_{15}, \sigma_{604} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2414} = \frac{\langle S_6, \sigma_{604} \rangle \Rightarrow \langle S_8, \sigma_{604} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{604} \rangle \Rightarrow \langle S_9, \sigma_{604} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2416} = \langle S_8, \sigma_{604} \rangle \Rightarrow \langle y := y + 1, \sigma_{604} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2415} = \frac{\mathcal{T}_{2416}}{\langle S_9, \sigma_{604} \rangle \Rightarrow \langle S_{10}, \sigma_{604} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2417} = \frac{\langle y := y + 1, \sigma_{604} \rangle \Rightarrow \sigma_{605} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{604} \rangle \Rightarrow \langle S_{11}, \sigma_{605} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2418} = \langle S_{11}, \sigma_{605} \rangle \Rightarrow \langle S_{12}, \sigma_{605} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2419} = \langle S_{12}, \sigma_{605} \rangle \Rightarrow \langle S_{13}, \sigma_{605} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2421} = \frac{\langle x := x + 1, \sigma_{605} \rangle \Rightarrow \sigma_{606} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{605} \rangle \Rightarrow \langle S_6, \sigma_{606} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2420} = \frac{\mathcal{T}_{2421}}{\langle S_{13}, \sigma_{605} \rangle \Rightarrow \langle S_{15}, \sigma_{606} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2422} = \frac{\langle S_6, \sigma_{606} \rangle \Rightarrow \langle S_8, \sigma_{606} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{606} \rangle \Rightarrow \langle S_9, \sigma_{606} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2424} = \langle S_8, \sigma_{606} \rangle \Rightarrow \langle y := y + 1, \sigma_{606} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2423} = \frac{\mathcal{T}_{2424}}{\langle S_9, \sigma_{606} \rangle \Rightarrow \langle S_{10}, \sigma_{606} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2425} = \frac{\langle y := y + 1, \sigma_{606} \rangle \Rightarrow \sigma_{607} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{606} \rangle \Rightarrow \langle S_{11}, \sigma_{607} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2426} = \langle S_{11}, \sigma_{607} \rangle \Rightarrow \langle S_{12}, \sigma_{607} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2427} = \langle S_{12}, \sigma_{607} \rangle \Rightarrow \langle S_{13}, \sigma_{607} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2429} = \frac{\langle x := x + 1, \sigma_{607} \rangle \Rightarrow \sigma_{608} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{607} \rangle \Rightarrow \langle S_6, \sigma_{608} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2428} = \frac{\mathcal{T}_{2429}}{\langle S_{13}, \sigma_{607} \rangle \Rightarrow \langle S_{15}, \sigma_{608} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2430} = \frac{\langle S_6, \sigma_{608} \rangle \Rightarrow \langle S_8, \sigma_{608} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{608} \rangle \Rightarrow \langle S_9, \sigma_{608} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2432} = \langle S_8, \sigma_{608} \rangle \Rightarrow \langle y := y + 1, \sigma_{608} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2431} = \frac{\mathcal{T}_{2432}}{\langle S_9, \sigma_{608} \rangle \Rightarrow \langle S_{10}, \sigma_{608} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2433} = \frac{\langle y := y + 1, \sigma_{608} \rangle \Rightarrow \sigma_{609} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{608} \rangle \Rightarrow \langle S_{11}, \sigma_{609} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2434} = \langle S_{11}, \sigma_{609} \rangle \Rightarrow \langle S_{12}, \sigma_{609} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2435} = \langle S_{12}, \sigma_{609} \rangle \Rightarrow \langle S_{13}, \sigma_{609} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2437} = \frac{\langle x := x + 1, \sigma_{609} \rangle \Rightarrow \sigma_{610} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{609} \rangle \Rightarrow \langle S_6, \sigma_{610} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2436} = \frac{\mathcal{T}_{2437}}{\langle S_{13}, \sigma_{609} \rangle \Rightarrow \langle S_{15}, \sigma_{610} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2438} = \frac{\langle S_6, \sigma_{610} \rangle \Rightarrow \langle S_8, \sigma_{610} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{610} \rangle \Rightarrow \langle S_9, \sigma_{610} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2440} = \langle S_8, \sigma_{610} \rangle \Rightarrow \langle y := y + 1, \sigma_{610} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2439} = \frac{\mathcal{T}_{2440}}{\langle S_9, \sigma_{610} \rangle \Rightarrow \langle S_{10}, \sigma_{610} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2441} = \frac{\langle y := y + 1, \sigma_{610} \rangle \Rightarrow \sigma_{611} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{610} \rangle \Rightarrow \langle S_{11}, \sigma_{611} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2442} = \langle S_{11}, \sigma_{611} \rangle \Rightarrow \langle S_{12}, \sigma_{611} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2443} = \langle S_{12}, \sigma_{611} \rangle \Rightarrow \langle S_{13}, \sigma_{611} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2445} = \frac{\langle x := x + 1, \sigma_{611} \rangle \Rightarrow \sigma_{612} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{611} \rangle \Rightarrow \langle S_6, \sigma_{612} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2444} = \frac{\mathcal{T}_{2445}}{\langle S_{13}, \sigma_{611} \rangle \Rightarrow \langle S_{15}, \sigma_{612} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2446} = \frac{\langle S_6, \sigma_{612} \rangle \Rightarrow \langle S_8, \sigma_{612} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{612} \rangle \Rightarrow \langle S_9, \sigma_{612} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2448} = \langle S_8, \sigma_{612} \rangle \Rightarrow \langle y := y + 1, \sigma_{612} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2447} = \frac{\mathcal{T}_{2448}}{\langle S_9, \sigma_{612} \rangle \Rightarrow \langle S_{10}, \sigma_{612} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2449} = \frac{\langle y := y + 1, \sigma_{612} \rangle \Rightarrow \sigma_{613} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{612} \rangle \Rightarrow \langle S_{11}, \sigma_{613} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2450} = \langle S_{11}, \sigma_{613} \rangle \Rightarrow \langle S_{12}, \sigma_{613} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2451} = \langle S_{12}, \sigma_{613} \rangle \Rightarrow \langle S_{13}, \sigma_{613} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2453} = \frac{\langle x := x + 1, \sigma_{613} \rangle \Rightarrow \sigma_{614} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{613} \rangle \Rightarrow \langle S_6, \sigma_{614} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2452} = \frac{\mathcal{T}_{2453}}{\langle S_{13}, \sigma_{613} \rangle \Rightarrow \langle S_{15}, \sigma_{614} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2454} = \frac{\langle S_6, \sigma_{614} \rangle \Rightarrow \langle S_8, \sigma_{614} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{614} \rangle \Rightarrow \langle S_9, \sigma_{614} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2456} = \langle S_8, \sigma_{614} \rangle \Rightarrow \langle y := y + 1, \sigma_{614} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2455} = \frac{\mathcal{T}_{2456}}{\langle S_9, \sigma_{614} \rangle \Rightarrow \langle S_{10}, \sigma_{614} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2457} = \frac{\langle y := y + 1, \sigma_{614} \rangle \Rightarrow \sigma_{615} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{614} \rangle \Rightarrow \langle S_{11}, \sigma_{615} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2458} = \langle S_{11}, \sigma_{615} \rangle \Rightarrow \langle S_{12}, \sigma_{615} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2459} = \langle S_{12}, \sigma_{615} \rangle \Rightarrow \langle S_{13}, \sigma_{615} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2461} = \frac{\langle x := x + 1, \sigma_{615} \rangle \Rightarrow \sigma_{616} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{615} \rangle \Rightarrow \langle S_6, \sigma_{616} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2460} = \frac{\mathcal{T}_{2461}}{\langle S_{13}, \sigma_{615} \rangle \Rightarrow \langle S_{15}, \sigma_{616} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2462} = \frac{\langle S_6, \sigma_{616} \rangle \Rightarrow \langle S_8, \sigma_{616} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{616} \rangle \Rightarrow \langle S_9, \sigma_{616} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2464} = \langle S_8, \sigma_{616} \rangle \Rightarrow \langle y := y + 1, \sigma_{616} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2463} = \frac{\mathcal{T}_{2464}}{\langle S_9, \sigma_{616} \rangle \Rightarrow \langle S_{10}, \sigma_{616} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2465} = \frac{\langle y := y + 1, \sigma_{616} \rangle \Rightarrow \sigma_{617} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{616} \rangle \Rightarrow \langle S_{11}, \sigma_{617} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{2466} = \langle S_{11}, \sigma_{617} \rangle \Rightarrow \langle S_{12}, \sigma_{617} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2467} = \langle S_{12}, \sigma_{617} \rangle \Rightarrow \langle S_{13}, \sigma_{617} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2469} = \frac{\langle x := x + 1, \sigma_{617} \rangle \Rightarrow \sigma_{618} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{617} \rangle \Rightarrow \langle S_6, \sigma_{618} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2468} = \frac{\mathcal{T}_{2469}}{\langle S_{13}, \sigma_{617} \rangle \Rightarrow \langle S_{15}, \sigma_{618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2470} = \frac{\langle S_6, \sigma_{618} \rangle \Rightarrow \langle S_8, \sigma_{618} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{618} \rangle \Rightarrow \langle S_9, \sigma_{618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2472} = \langle S_8, \sigma_{618} \rangle \Rightarrow \langle y := y + 1, \sigma_{618} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2471} = \frac{\mathcal{T}_{2472}}{\langle S_9, \sigma_{618} \rangle \Rightarrow \langle S_{10}, \sigma_{618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2473} = \frac{\langle y := y + 1, \sigma_{618} \rangle \Rightarrow \sigma_{619} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{618} \rangle \Rightarrow \langle S_{11}, \sigma_{619} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2474} = \langle S_{11}, \sigma_{619} \rangle \Rightarrow \langle S_{12}, \sigma_{619} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2475} = \langle S_{12}, \sigma_{619} \rangle \Rightarrow \langle S_{13}, \sigma_{619} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2477} = \frac{\langle x := x + 1, \sigma_{619} \rangle \Rightarrow \sigma_{620} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{619} \rangle \Rightarrow \langle S_6, \sigma_{620} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2476} = \frac{\mathcal{T}_{2477}}{\langle S_{13}, \sigma_{619} \rangle \Rightarrow \langle S_{15}, \sigma_{620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2478} = \frac{\langle S_6, \sigma_{620} \rangle \Rightarrow \langle S_8, \sigma_{620} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{620} \rangle \Rightarrow \langle S_9, \sigma_{620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2480} = \langle S_8, \sigma_{620} \rangle \Rightarrow \langle y := y + 1, \sigma_{620} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2479} = \frac{\mathcal{T}_{2480}}{\langle S_9, \sigma_{620} \rangle \Rightarrow \langle S_{10}, \sigma_{620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2481} = \frac{\langle y := y + 1, \sigma_{620} \rangle \Rightarrow \sigma_{621} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{620} \rangle \Rightarrow \langle S_{11}, \sigma_{621} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2482} = \langle S_{11}, \sigma_{621} \rangle \Rightarrow \langle S_{12}, \sigma_{621} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2483} = \langle S_{12}, \sigma_{621} \rangle \Rightarrow \langle S_{13}, \sigma_{621} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2485} = \frac{\langle x := x + 1, \sigma_{621} \rangle \Rightarrow \sigma_{622} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{621} \rangle \Rightarrow \langle S_6, \sigma_{622} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2484} = \frac{\mathcal{T}_{2485}}{\langle S_{13}, \sigma_{621} \rangle \Rightarrow \langle S_{15}, \sigma_{622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2486} = \frac{\langle S_6, \sigma_{622} \rangle \Rightarrow \langle S_8, \sigma_{622} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{622} \rangle \Rightarrow \langle S_9, \sigma_{622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2488} = \langle S_8, \sigma_{622} \rangle \Rightarrow \langle y := y + 1, \sigma_{622} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2487} = \frac{\mathcal{T}_{2488}}{\langle S_9, \sigma_{622} \rangle \Rightarrow \langle S_{10}, \sigma_{622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2489} = \frac{\langle y := y + 1, \sigma_{622} \rangle \Rightarrow \sigma_{623} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{622} \rangle \Rightarrow \langle S_{11}, \sigma_{623} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2490} = \langle S_{11}, \sigma_{623} \rangle \Rightarrow \langle S_{12}, \sigma_{623} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2491} = \langle S_{12}, \sigma_{623} \rangle \Rightarrow \langle S_{13}, \sigma_{623} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2493} = \frac{\langle x := x + 1, \sigma_{623} \rangle \Rightarrow \sigma_{624} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{623} \rangle \Rightarrow \langle S_6, \sigma_{624} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2492} = \frac{\mathcal{T}_{2493}}{\langle S_{13}, \sigma_{623} \rangle \Rightarrow \langle S_{15}, \sigma_{624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2494} = \frac{\langle S_6, \sigma_{624} \rangle \Rightarrow \langle S_8, \sigma_{624} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{624} \rangle \Rightarrow \langle S_9, \sigma_{624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2496} = \langle S_8, \sigma_{624} \rangle \Rightarrow \langle y := y + 1, \sigma_{624} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2495} = \frac{\mathcal{T}_{2496}}{\langle S_9, \sigma_{624} \rangle \Rightarrow \langle S_{10}, \sigma_{624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2497} = \frac{\langle y := y + 1, \sigma_{624} \rangle \Rightarrow \sigma_{625} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{624} \rangle \Rightarrow \langle S_{11}, \sigma_{625} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2498} = \langle S_{11}, \sigma_{625} \rangle \Rightarrow \langle S_{12}, \sigma_{625} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2499} = \langle S_{12}, \sigma_{625} \rangle \Rightarrow \langle S_{13}, \sigma_{625} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2501} = \frac{\langle x := x + 1, \sigma_{625} \rangle \Rightarrow \sigma_{626} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{625} \rangle \Rightarrow \langle S_6, \sigma_{626} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2500} = \frac{\mathcal{T}_{2501}}{\langle S_{13}, \sigma_{625} \rangle \Rightarrow \langle S_{15}, \sigma_{626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2502} = \frac{\langle S_6, \sigma_{626} \rangle \Rightarrow \langle S_8, \sigma_{626} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{626} \rangle \Rightarrow \langle S_9, \sigma_{626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2504} = \langle S_8, \sigma_{626} \rangle \Rightarrow \langle y := y + 1, \sigma_{626} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2503} = \frac{\mathcal{T}_{2504}}{\langle S_9, \sigma_{626} \rangle \Rightarrow \langle S_{10}, \sigma_{626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2505} = \frac{\langle y := y + 1, \sigma_{626} \rangle \Rightarrow \sigma_{627} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{626} \rangle \Rightarrow \langle S_{11}, \sigma_{627} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2506} = \langle S_{11}, \sigma_{627} \rangle \Rightarrow \langle S_{12}, \sigma_{627} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2507} = \langle S_{12}, \sigma_{627} \rangle \Rightarrow \langle S_{13}, \sigma_{627} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2509} = \frac{\langle x := x + 1, \sigma_{627} \rangle \Rightarrow \sigma_{628} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{627} \rangle \Rightarrow \langle S_6, \sigma_{628} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2508} = \frac{\mathcal{T}_{2509}}{\langle S_{13}, \sigma_{627} \rangle \Rightarrow \langle S_{15}, \sigma_{628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2510} = \frac{\langle S_6, \sigma_{628} \rangle \Rightarrow \langle S_8, \sigma_{628} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{628} \rangle \Rightarrow \langle S_9, \sigma_{628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2512} = \langle S_8, \sigma_{628} \rangle \Rightarrow \langle y := y + 1, \sigma_{628} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2511} = \frac{\mathcal{T}_{2512}}{\langle S_9, \sigma_{628} \rangle \Rightarrow \langle S_{10}, \sigma_{628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2513} = \frac{\langle y := y + 1, \sigma_{628} \rangle \Rightarrow \sigma_{629} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{628} \rangle \Rightarrow \langle S_{11}, \sigma_{629} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2514} = \langle S_{11}, \sigma_{629} \rangle \Rightarrow \langle S_{12}, \sigma_{629} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2515} = \langle S_{12}, \sigma_{629} \rangle \Rightarrow \langle S_{13}, \sigma_{629} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2517} = \frac{\langle x := x + 1, \sigma_{629} \rangle \Rightarrow \sigma_{630} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{629} \rangle \Rightarrow \langle S_6, \sigma_{630} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2516} = \frac{\mathcal{T}_{2517}}{\langle S_{13}, \sigma_{629} \rangle \Rightarrow \langle S_{15}, \sigma_{630} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2518} = \frac{\langle S_6, \sigma_{630} \rangle \Rightarrow \langle S_8, \sigma_{630} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{630} \rangle \Rightarrow \langle S_9, \sigma_{630} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2520} = \langle S_8, \sigma_{630} \rangle \Rightarrow \langle y := y + 1, \sigma_{630} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2519} = \frac{\mathcal{T}_{2520}}{\langle S_9, \sigma_{630} \rangle \Rightarrow \langle S_{10}, \sigma_{630} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2521} = \frac{\langle y := y + 1, \sigma_{630} \rangle \Rightarrow \sigma_{631} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{630} \rangle \Rightarrow \langle S_{11}, \sigma_{631} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2522} = \langle S_{11}, \sigma_{631} \rangle \Rightarrow \langle S_{12}, \sigma_{631} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2523} = \langle S_{12}, \sigma_{631} \rangle \Rightarrow \langle S_{13}, \sigma_{631} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2525} = \frac{\langle x := x + 1, \sigma_{631} \rangle \Rightarrow \sigma_{632} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{631} \rangle \Rightarrow \langle S_6, \sigma_{632} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2524} = \frac{\mathcal{T}_{2525}}{\langle S_{13}, \sigma_{631} \rangle \Rightarrow \langle S_{15}, \sigma_{632} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2526} = \frac{\langle S_6, \sigma_{632} \rangle \Rightarrow \langle S_8, \sigma_{632} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{632} \rangle \Rightarrow \langle S_9, \sigma_{632} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2528} = \langle S_8, \sigma_{632} \rangle \Rightarrow \langle y := y + 1, \sigma_{632} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2527} = \frac{\mathcal{T}_{2528}}{\langle S_9, \sigma_{632} \rangle \Rightarrow \langle S_{10}, \sigma_{632} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2529} = \frac{\langle y := y + 1, \sigma_{632} \rangle \Rightarrow \sigma_{633} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{632} \rangle \Rightarrow \langle S_{11}, \sigma_{633} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2530} = \langle S_{11}, \sigma_{633} \rangle \Rightarrow \langle S_{12}, \sigma_{633} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2531} = \langle S_{12}, \sigma_{633} \rangle \Rightarrow \langle S_{13}, \sigma_{633} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2533} = \frac{\langle x := x + 1, \sigma_{633} \rangle \Rightarrow \sigma_{634} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{633} \rangle \Rightarrow \langle S_6, \sigma_{634} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2532} = \frac{\mathcal{T}_{2533}}{\langle S_{13}, \sigma_{633} \rangle \Rightarrow \langle S_{15}, \sigma_{634} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2534} = \frac{\langle S_6, \sigma_{634} \rangle \Rightarrow \langle S_8, \sigma_{634} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{634} \rangle \Rightarrow \langle S_9, \sigma_{634} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2536} = \langle S_8, \sigma_{634} \rangle \Rightarrow \langle y := y + 1, \sigma_{634} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2535} = \frac{\mathcal{T}_{2536}}{\langle S_9, \sigma_{634} \rangle \Rightarrow \langle S_{10}, \sigma_{634} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2537} = \frac{\langle y := y + 1, \sigma_{634} \rangle \Rightarrow \sigma_{635} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{634} \rangle \Rightarrow \langle S_{11}, \sigma_{635} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2538} = \langle S_{11}, \sigma_{635} \rangle \Rightarrow \langle S_{12}, \sigma_{635} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2539} = \langle S_{12}, \sigma_{635} \rangle \Rightarrow \langle S_{13}, \sigma_{635} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2541} = \frac{\langle x := x + 1, \sigma_{635} \rangle \Rightarrow \sigma_{636} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{635} \rangle \Rightarrow \langle S_6, \sigma_{636} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2540} = \frac{\mathcal{T}_{2541}}{\langle S_{13}, \sigma_{635} \rangle \Rightarrow \langle S_{15}, \sigma_{636} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2542} = \frac{\langle S_6, \sigma_{636} \rangle \Rightarrow \langle S_8, \sigma_{636} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{636} \rangle \Rightarrow \langle S_9, \sigma_{636} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2544} = \langle S_8, \sigma_{636} \rangle \Rightarrow \langle y := y + 1, \sigma_{636} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2543} = \frac{\mathcal{T}_{2544}}{\langle S_9, \sigma_{636} \rangle \Rightarrow \langle S_{10}, \sigma_{636} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2545} = \frac{\langle y := y + 1, \sigma_{636} \rangle \Rightarrow \sigma_{637} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{636} \rangle \Rightarrow \langle S_{11}, \sigma_{637} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2546} = \langle S_{11}, \sigma_{637} \rangle \Rightarrow \langle S_{12}, \sigma_{637} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2547} = \langle S_{12}, \sigma_{637} \rangle \Rightarrow \langle S_{13}, \sigma_{637} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2549} = \frac{\langle x := x + 1, \sigma_{637} \rangle \Rightarrow \sigma_{638} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{637} \rangle \Rightarrow \langle S_6, \sigma_{638} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2548} = \frac{\mathcal{T}_{2549}}{\langle S_{13}, \sigma_{637} \rangle \Rightarrow \langle S_{15}, \sigma_{638} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2550} = \frac{\langle S_6, \sigma_{638} \rangle \Rightarrow \langle S_8, \sigma_{638} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{638} \rangle \Rightarrow \langle S_9, \sigma_{638} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2552} = \langle S_8, \sigma_{638} \rangle \Rightarrow \langle y := y + 1, \sigma_{638} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2551} = \frac{\mathcal{T}_{2552}}{\langle S_9, \sigma_{638} \rangle \Rightarrow \langle S_{10}, \sigma_{638} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2553} = \frac{\langle y := y + 1, \sigma_{638} \rangle \Rightarrow \sigma_{639} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{638} \rangle \Rightarrow \langle S_{11}, \sigma_{639} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2554} = \langle S_{11}, \sigma_{639} \rangle \Rightarrow \langle S_{12}, \sigma_{639} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2555} = \langle S_{12}, \sigma_{639} \rangle \Rightarrow \langle S_{13}, \sigma_{639} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2557} = \frac{\langle x := x + 1, \sigma_{639} \rangle \Rightarrow \sigma_{640} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{639} \rangle \Rightarrow \langle S_6, \sigma_{640} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2556} = \frac{\mathcal{T}_{2557}}{\langle S_{13}, \sigma_{639} \rangle \Rightarrow \langle S_{15}, \sigma_{640} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2558} = \frac{\langle S_6, \sigma_{640} \rangle \Rightarrow \langle S_8, \sigma_{640} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{640} \rangle \Rightarrow \langle S_9, \sigma_{640} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2560} = \langle S_8, \sigma_{640} \rangle \Rightarrow \langle y := y + 1, \sigma_{640} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2559} = \frac{\mathcal{T}_{2560}}{\langle S_9, \sigma_{640} \rangle \Rightarrow \langle S_{10}, \sigma_{640} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2561} = \frac{\langle y := y + 1, \sigma_{640} \rangle \Rightarrow \sigma_{641} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{640} \rangle \Rightarrow \langle S_{11}, \sigma_{641} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2562} = \langle S_{11}, \sigma_{641} \rangle \Rightarrow \langle S_{12}, \sigma_{641} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2563} = \langle S_{12}, \sigma_{641} \rangle \Rightarrow \langle S_{13}, \sigma_{641} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2565} = \frac{\langle x := x + 1, \sigma_{641} \rangle \Rightarrow \sigma_{642} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{641} \rangle \Rightarrow \langle S_6, \sigma_{642} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2564} = \frac{\mathcal{T}_{2565}}{\langle S_{13}, \sigma_{641} \rangle \Rightarrow \langle S_{15}, \sigma_{642} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2566} = \frac{\langle S_6, \sigma_{642} \rangle \Rightarrow \langle S_8, \sigma_{642} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{642} \rangle \Rightarrow \langle S_9, \sigma_{642} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2568} = \langle S_8, \sigma_{642} \rangle \Rightarrow \langle y := y + 1, \sigma_{642} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2567} = \frac{\mathcal{T}_{2568}}{\langle S_9, \sigma_{642} \rangle \Rightarrow \langle S_{10}, \sigma_{642} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2569} = \frac{\langle y := y + 1, \sigma_{642} \rangle \Rightarrow \sigma_{643} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{642} \rangle \Rightarrow \langle S_{11}, \sigma_{643} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2570} = \langle S_{11}, \sigma_{643} \rangle \Rightarrow \langle S_{12}, \sigma_{643} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2571} = \langle S_{12}, \sigma_{643} \rangle \Rightarrow \langle S_{13}, \sigma_{643} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2573} = \frac{\langle x := x + 1, \sigma_{643} \rangle \Rightarrow \sigma_{644} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{643} \rangle \Rightarrow \langle S_6, \sigma_{644} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2572} = \frac{\mathcal{T}_{2573}}{\langle S_{13}, \sigma_{643} \rangle \Rightarrow \langle S_{15}, \sigma_{644} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2574} = \frac{\langle S_6, \sigma_{644} \rangle \Rightarrow \langle S_8, \sigma_{644} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{644} \rangle \Rightarrow \langle S_9, \sigma_{644} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2576} = \langle S_8, \sigma_{644} \rangle \Rightarrow \langle y := y + 1, \sigma_{644} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2575} = \frac{\mathcal{T}_{2576}}{\langle S_9, \sigma_{644} \rangle \Rightarrow \langle S_{10}, \sigma_{644} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2577} = \frac{\langle y := y + 1, \sigma_{644} \rangle \Rightarrow \sigma_{645} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{644} \rangle \Rightarrow \langle S_{11}, \sigma_{645} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2578} = \langle S_{11}, \sigma_{645} \rangle \Rightarrow \langle S_{12}, \sigma_{645} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2579} = \langle S_{12}, \sigma_{645} \rangle \Rightarrow \langle S_{13}, \sigma_{645} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2581} = \frac{\langle x := x + 1, \sigma_{645} \rangle \Rightarrow \sigma_{646} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{645} \rangle \Rightarrow \langle S_6, \sigma_{646} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2580} = \frac{\mathcal{T}_{2581}}{\langle S_{13}, \sigma_{645} \rangle \Rightarrow \langle S_{15}, \sigma_{646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2582} = \frac{\langle S_6, \sigma_{646} \rangle \Rightarrow \langle S_8, \sigma_{646} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{646} \rangle \Rightarrow \langle S_9, \sigma_{646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2584} = \langle S_8, \sigma_{646} \rangle \Rightarrow \langle y := y + 1, \sigma_{646} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2583} = \frac{\mathcal{T}_{2584}}{\langle S_9, \sigma_{646} \rangle \Rightarrow \langle S_{10}, \sigma_{646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2585} = \frac{\langle y := y + 1, \sigma_{646} \rangle \Rightarrow \sigma_{647} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{646} \rangle \Rightarrow \langle S_{11}, \sigma_{647} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2586} = \langle S_{11}, \sigma_{647} \rangle \Rightarrow \langle S_{12}, \sigma_{647} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2587} = \langle S_{12}, \sigma_{647} \rangle \Rightarrow \langle S_{13}, \sigma_{647} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2589} = \frac{\langle x := x + 1, \sigma_{647} \rangle \Rightarrow \sigma_{648} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{647} \rangle \Rightarrow \langle S_6, \sigma_{648} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2588} = \frac{\mathcal{T}_{2589}}{\langle S_{13}, \sigma_{647} \rangle \Rightarrow \langle S_{15}, \sigma_{648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2590} = \frac{\langle S_6, \sigma_{648} \rangle \Rightarrow \langle S_8, \sigma_{648} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{648} \rangle \Rightarrow \langle S_9, \sigma_{648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2592} = \langle S_8, \sigma_{648} \rangle \Rightarrow \langle y := y + 1, \sigma_{648} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2591} = \frac{\mathcal{T}_{2592}}{\langle S_9, \sigma_{648} \rangle \Rightarrow \langle S_{10}, \sigma_{648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2593} = \frac{\langle y := y + 1, \sigma_{648} \rangle \Rightarrow \sigma_{649} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{648} \rangle \Rightarrow \langle S_{11}, \sigma_{649} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2594} = \langle S_{11}, \sigma_{649} \rangle \Rightarrow \langle S_{12}, \sigma_{649} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2595} = \langle S_{12}, \sigma_{649} \rangle \Rightarrow \langle S_{13}, \sigma_{649} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2597} = \frac{\langle x := x + 1, \sigma_{649} \rangle \Rightarrow \sigma_{650} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{649} \rangle \Rightarrow \langle S_6, \sigma_{650} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2596} = \frac{\mathcal{T}_{2597}}{\langle S_{13}, \sigma_{649} \rangle \Rightarrow \langle S_{15}, \sigma_{650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2598} = \frac{\langle S_6, \sigma_{650} \rangle \Rightarrow \langle S_8, \sigma_{650} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{650} \rangle \Rightarrow \langle S_9, \sigma_{650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2600} = \langle S_8, \sigma_{650} \rangle \Rightarrow \langle y := y + 1, \sigma_{650} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2599} = \frac{\mathcal{T}_{2600}}{\langle S_9, \sigma_{650} \rangle \Rightarrow \langle S_{10}, \sigma_{650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2601} = \frac{\langle y := y + 1, \sigma_{650} \rangle \Rightarrow \sigma_{651} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{650} \rangle \Rightarrow \langle S_{11}, \sigma_{651} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2602} = \langle S_{11}, \sigma_{651} \rangle \Rightarrow \langle S_{12}, \sigma_{651} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2603} = \langle S_{12}, \sigma_{651} \rangle \Rightarrow \langle S_{13}, \sigma_{651} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2605} = \frac{\langle x := x + 1, \sigma_{651} \rangle \Rightarrow \sigma_{652} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{651} \rangle \Rightarrow \langle S_6, \sigma_{652} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2604} = \frac{\mathcal{T}_{2605}}{\langle S_{13}, \sigma_{651} \rangle \Rightarrow \langle S_{15}, \sigma_{652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2606} = \frac{\langle S_6, \sigma_{652} \rangle \Rightarrow \langle S_8, \sigma_{652} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{652} \rangle \Rightarrow \langle S_9, \sigma_{652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2608} = \langle S_8, \sigma_{652} \rangle \Rightarrow \langle y := y + 1, \sigma_{652} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2607} = \frac{\mathcal{T}_{2608}}{\langle S_9, \sigma_{652} \rangle \Rightarrow \langle S_{10}, \sigma_{652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2609} = \frac{\langle y := y + 1, \sigma_{652} \rangle \Rightarrow \sigma_{653} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{652} \rangle \Rightarrow \langle S_{11}, \sigma_{653} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2610} = \langle S_{11}, \sigma_{653} \rangle \Rightarrow \langle S_{12}, \sigma_{653} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2611} = \langle S_{12}, \sigma_{653} \rangle \Rightarrow \langle S_{13}, \sigma_{653} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2613} = \frac{\langle x := x + 1, \sigma_{653} \rangle \Rightarrow \sigma_{654} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{653} \rangle \Rightarrow \langle S_6, \sigma_{654} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2612} = \frac{\mathcal{T}_{2613}}{\langle S_{13}, \sigma_{653} \rangle \Rightarrow \langle S_{15}, \sigma_{654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2614} = \frac{\langle S_6, \sigma_{654} \rangle \Rightarrow \langle S_8, \sigma_{654} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{654} \rangle \Rightarrow \langle S_9, \sigma_{654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2616} = \langle S_8, \sigma_{654} \rangle \Rightarrow \langle y := y + 1, \sigma_{654} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2615} = \frac{\mathcal{T}_{2616}}{\langle S_9, \sigma_{654} \rangle \Rightarrow \langle S_{10}, \sigma_{654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2617} = \frac{\langle y := y + 1, \sigma_{654} \rangle \Rightarrow \sigma_{655} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{654} \rangle \Rightarrow \langle S_{11}, \sigma_{655} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2618} = \langle S_{11}, \sigma_{655} \rangle \Rightarrow \langle S_{12}, \sigma_{655} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2619} = \langle S_{12}, \sigma_{655} \rangle \Rightarrow \langle S_{13}, \sigma_{655} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2621} = \frac{\langle x := x + 1, \sigma_{655} \rangle \Rightarrow \sigma_{656} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{655} \rangle \Rightarrow \langle S_6, \sigma_{656} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2620} = \frac{\mathcal{T}_{2621}}{\langle S_{13}, \sigma_{655} \rangle \Rightarrow \langle S_{15}, \sigma_{656} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2622} = \frac{\langle S_6, \sigma_{656} \rangle \Rightarrow \langle S_8, \sigma_{656} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{656} \rangle \Rightarrow \langle S_9, \sigma_{656} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2624} = \langle S_8, \sigma_{656} \rangle \Rightarrow \langle y := y + 1, \sigma_{656} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2623} = \frac{\mathcal{T}_{2624}}{\langle S_9, \sigma_{656} \rangle \Rightarrow \langle S_{10}, \sigma_{656} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2625} = \frac{\langle y := y + 1, \sigma_{656} \rangle \Rightarrow \sigma_{657} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{656} \rangle \Rightarrow \langle S_{11}, \sigma_{657} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2626} = \langle S_{11}, \sigma_{657} \rangle \Rightarrow \langle S_{12}, \sigma_{657} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2627} = \langle S_{12}, \sigma_{657} \rangle \Rightarrow \langle S_{13}, \sigma_{657} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2629} = \frac{\langle x := x + 1, \sigma_{657} \rangle \Rightarrow \sigma_{658} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{657} \rangle \Rightarrow \langle S_6, \sigma_{658} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2628} = \frac{\mathcal{T}_{2629}}{\langle S_{13}, \sigma_{657} \rangle \Rightarrow \langle S_{15}, \sigma_{658} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2630} = \frac{\langle S_6, \sigma_{658} \rangle \Rightarrow \langle S_8, \sigma_{658} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{658} \rangle \Rightarrow \langle S_9, \sigma_{658} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2632} = \langle S_8, \sigma_{658} \rangle \Rightarrow \langle y := y + 1, \sigma_{658} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2631} = \frac{\mathcal{T}_{2632}}{\langle S_9, \sigma_{658} \rangle \Rightarrow \langle S_{10}, \sigma_{658} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2633} = \frac{\langle y := y + 1, \sigma_{658} \rangle \Rightarrow \sigma_{659} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{658} \rangle \Rightarrow \langle S_{11}, \sigma_{659} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2634} = \langle S_{11}, \sigma_{659} \rangle \Rightarrow \langle S_{12}, \sigma_{659} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2635} = \langle S_{12}, \sigma_{659} \rangle \Rightarrow \langle S_{13}, \sigma_{659} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2637} = \frac{\langle x := x + 1, \sigma_{659} \rangle \Rightarrow \sigma_{660} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{659} \rangle \Rightarrow \langle S_6, \sigma_{660} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2636} = \frac{\mathcal{T}_{2637}}{\langle S_{13}, \sigma_{659} \rangle \Rightarrow \langle S_{15}, \sigma_{660} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2638} = \frac{\langle S_6, \sigma_{660} \rangle \Rightarrow \langle S_8, \sigma_{660} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{660} \rangle \Rightarrow \langle S_9, \sigma_{660} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2640} = \langle S_8, \sigma_{660} \rangle \Rightarrow \langle y := y + 1, \sigma_{660} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2639} = \frac{\mathcal{T}_{2640}}{\langle S_9, \sigma_{660} \rangle \Rightarrow \langle S_{10}, \sigma_{660} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2641} = \frac{\langle y := y + 1, \sigma_{660} \rangle \Rightarrow \sigma_{661} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{660} \rangle \Rightarrow \langle S_{11}, \sigma_{661} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2642} = \langle S_{11}, \sigma_{661} \rangle \Rightarrow \langle S_{12}, \sigma_{661} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2643} = \langle S_{12}, \sigma_{661} \rangle \Rightarrow \langle S_{13}, \sigma_{661} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2645} = \frac{\langle x := x + 1, \sigma_{661} \rangle \Rightarrow \sigma_{662} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{661} \rangle \Rightarrow \langle S_6, \sigma_{662} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2644} = \frac{\mathcal{T}_{2645}}{\langle S_{13}, \sigma_{661} \rangle \Rightarrow \langle S_{15}, \sigma_{662} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2646} = \frac{\langle S_6, \sigma_{662} \rangle \Rightarrow \langle S_8, \sigma_{662} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{662} \rangle \Rightarrow \langle S_9, \sigma_{662} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2648} = \langle S_8, \sigma_{662} \rangle \Rightarrow \langle y := y + 1, \sigma_{662} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2647} = \frac{\mathcal{T}_{2648}}{\langle S_9, \sigma_{662} \rangle \Rightarrow \langle S_{10}, \sigma_{662} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2649} = \frac{\langle y := y + 1, \sigma_{662} \rangle \Rightarrow \sigma_{663} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{662} \rangle \Rightarrow \langle S_{11}, \sigma_{663} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2650} = \langle S_{11}, \sigma_{663} \rangle \Rightarrow \langle S_{12}, \sigma_{663} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2651} = \langle S_{12}, \sigma_{663} \rangle \Rightarrow \langle S_{13}, \sigma_{663} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2653} = \frac{\langle x := x + 1, \sigma_{663} \rangle \Rightarrow \sigma_{664} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{663} \rangle \Rightarrow \langle S_6, \sigma_{664} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2652} = \frac{\mathcal{T}_{2653}}{\langle S_{13}, \sigma_{663} \rangle \Rightarrow \langle S_{15}, \sigma_{664} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2654} = \frac{\langle S_6, \sigma_{664} \rangle \Rightarrow \langle S_8, \sigma_{664} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{664} \rangle \Rightarrow \langle S_9, \sigma_{664} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2656} = \langle S_8, \sigma_{664} \rangle \Rightarrow \langle y := y + 1, \sigma_{664} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2655} = \frac{\mathcal{T}_{2656}}{\langle S_9, \sigma_{664} \rangle \Rightarrow \langle S_{10}, \sigma_{664} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2657} = \frac{\langle y := y + 1, \sigma_{664} \rangle \Rightarrow \sigma_{665} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{664} \rangle \Rightarrow \langle S_{11}, \sigma_{665} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2658} = \langle S_{11}, \sigma_{665} \rangle \Rightarrow \langle S_{12}, \sigma_{665} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2659} = \langle S_{12}, \sigma_{665} \rangle \Rightarrow \langle S_{13}, \sigma_{665} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2661} = \frac{\langle x := x + 1, \sigma_{665} \rangle \Rightarrow \sigma_{666} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{665} \rangle \Rightarrow \langle S_6, \sigma_{666} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2660} = \frac{\mathcal{T}_{2661}}{\langle S_{13}, \sigma_{665} \rangle \Rightarrow \langle S_{15}, \sigma_{666} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2662} = \frac{\langle S_6, \sigma_{666} \rangle \Rightarrow \langle S_8, \sigma_{666} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{666} \rangle \Rightarrow \langle S_9, \sigma_{666} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2664} = \langle S_8, \sigma_{666} \rangle \Rightarrow \langle y := y + 1, \sigma_{666} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2663} = \frac{\mathcal{T}_{2664}}{\langle S_9, \sigma_{666} \rangle \Rightarrow \langle S_{10}, \sigma_{666} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2665} = \frac{\langle y := y + 1, \sigma_{666} \rangle \Rightarrow \sigma_{667} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{666} \rangle \Rightarrow \langle S_{11}, \sigma_{667} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2666} = \langle S_{11}, \sigma_{667} \rangle \Rightarrow \langle S_{12}, \sigma_{667} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2667} = \langle S_{12}, \sigma_{667} \rangle \Rightarrow \langle S_{13}, \sigma_{667} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2669} = \frac{\langle x := x + 1, \sigma_{667} \rangle \Rightarrow \sigma_{668} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{667} \rangle \Rightarrow \langle S_6, \sigma_{668} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2668} = \frac{\mathcal{T}_{2669}}{\langle S_{13}, \sigma_{667} \rangle \Rightarrow \langle S_{15}, \sigma_{668} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2670} = \frac{\langle S_6, \sigma_{668} \rangle \Rightarrow \langle S_8, \sigma_{668} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{668} \rangle \Rightarrow \langle S_9, \sigma_{668} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2672} = \langle S_8, \sigma_{668} \rangle \Rightarrow \langle y := y + 1, \sigma_{668} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2671} = \frac{\mathcal{T}_{2672}}{\langle S_9, \sigma_{668} \rangle \Rightarrow \langle S_{10}, \sigma_{668} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2673} = \frac{\langle y := y + 1, \sigma_{668} \rangle \Rightarrow \sigma_{669} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{668} \rangle \Rightarrow \langle S_{11}, \sigma_{669} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{2674} = \langle S_{11}, \sigma_{669} \rangle \Rightarrow \langle S_{12}, \sigma_{669} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2675} = \langle S_{12}, \sigma_{669} \rangle \Rightarrow \langle S_{13}, \sigma_{669} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2677} = \frac{\langle x := x + 1, \sigma_{669} \rangle \Rightarrow \sigma_{670} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{669} \rangle \Rightarrow \langle S_6, \sigma_{670} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2676} = \frac{\mathcal{T}_{2677}}{\langle S_{13}, \sigma_{669} \rangle \Rightarrow \langle S_{15}, \sigma_{670} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2678} = \frac{\langle S_6, \sigma_{670} \rangle \Rightarrow \langle S_8, \sigma_{670} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{670} \rangle \Rightarrow \langle S_9, \sigma_{670} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2680} = \langle S_8, \sigma_{670} \rangle \Rightarrow \langle y := y + 1, \sigma_{670} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2679} = \frac{\mathcal{T}_{2680}}{\langle S_9, \sigma_{670} \rangle \Rightarrow \langle S_{10}, \sigma_{670} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2681} = \frac{\langle y := y + 1, \sigma_{670} \rangle \Rightarrow \sigma_{671} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{670} \rangle \Rightarrow \langle S_{11}, \sigma_{671} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2682} = \langle S_{11}, \sigma_{671} \rangle \Rightarrow \langle S_{12}, \sigma_{671} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2683} = \langle S_{12}, \sigma_{671} \rangle \Rightarrow \langle S_{13}, \sigma_{671} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2685} = \frac{\langle x := x + 1, \sigma_{671} \rangle \Rightarrow \sigma_{672} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{671} \rangle \Rightarrow \langle S_6, \sigma_{672} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2684} = \frac{\mathcal{T}_{2685}}{\langle S_{13}, \sigma_{671} \rangle \Rightarrow \langle S_{15}, \sigma_{672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2686} = \frac{\langle S_6, \sigma_{672} \rangle \Rightarrow \langle S_8, \sigma_{672} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{672} \rangle \Rightarrow \langle S_9, \sigma_{672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2688} = \langle S_8, \sigma_{672} \rangle \Rightarrow \langle y := y + 1, \sigma_{672} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2687} = \frac{\mathcal{T}_{2688}}{\langle S_9, \sigma_{672} \rangle \Rightarrow \langle S_{10}, \sigma_{672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2689} = \frac{\langle y := y + 1, \sigma_{672} \rangle \Rightarrow \sigma_{673} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{672} \rangle \Rightarrow \langle S_{11}, \sigma_{673} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2690} = \langle S_{11}, \sigma_{673} \rangle \Rightarrow \langle S_{12}, \sigma_{673} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2691} = \langle S_{12}, \sigma_{673} \rangle \Rightarrow \langle S_{13}, \sigma_{673} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2693} = \frac{\langle x := x + 1, \sigma_{673} \rangle \Rightarrow \sigma_{674} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{673} \rangle \Rightarrow \langle S_6, \sigma_{674} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2692} = \frac{\mathcal{T}_{2693}}{\langle S_{13}, \sigma_{673} \rangle \Rightarrow \langle S_{15}, \sigma_{674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2694} = \frac{\langle S_6, \sigma_{674} \rangle \Rightarrow \langle S_8, \sigma_{674} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{674} \rangle \Rightarrow \langle S_9, \sigma_{674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2696} = \langle S_8, \sigma_{674} \rangle \Rightarrow \langle y := y + 1, \sigma_{674} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2695} = \frac{\mathcal{T}_{2696}}{\langle S_9, \sigma_{674} \rangle \Rightarrow \langle S_{10}, \sigma_{674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2697} = \frac{\langle y := y + 1, \sigma_{674} \rangle \Rightarrow \sigma_{675} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{674} \rangle \Rightarrow \langle S_{11}, \sigma_{675} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2698} = \langle S_{11}, \sigma_{675} \rangle \Rightarrow \langle S_{12}, \sigma_{675} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2699} = \langle S_{12}, \sigma_{675} \rangle \Rightarrow \langle S_{13}, \sigma_{675} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2701} = \frac{\langle x := x + 1, \sigma_{675} \rangle \Rightarrow \sigma_{676} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{675} \rangle \Rightarrow \langle S_6, \sigma_{676} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2700} = \frac{\mathcal{T}_{2701}}{\langle S_{13}, \sigma_{675} \rangle \Rightarrow \langle S_{15}, \sigma_{676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2702} = \frac{\langle S_6, \sigma_{676} \rangle \Rightarrow \langle S_8, \sigma_{676} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{676} \rangle \Rightarrow \langle S_9, \sigma_{676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2704} = \langle S_8, \sigma_{676} \rangle \Rightarrow \langle y := y + 1, \sigma_{676} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2703} = \frac{\mathcal{T}_{2704}}{\langle S_9, \sigma_{676} \rangle \Rightarrow \langle S_{10}, \sigma_{676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2705} = \frac{\langle y := y + 1, \sigma_{676} \rangle \Rightarrow \sigma_{677} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{676} \rangle \Rightarrow \langle S_{11}, \sigma_{677} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2706} = \langle S_{11}, \sigma_{677} \rangle \Rightarrow \langle S_{12}, \sigma_{677} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2707} = \langle S_{12}, \sigma_{677} \rangle \Rightarrow \langle S_{13}, \sigma_{677} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2709} = \frac{\langle x := x + 1, \sigma_{677} \rangle \Rightarrow \sigma_{678} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{677} \rangle \Rightarrow \langle S_6, \sigma_{678} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2708} = \frac{\mathcal{T}_{2709}}{\langle S_{13}, \sigma_{677} \rangle \Rightarrow \langle S_{15}, \sigma_{678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2710} = \frac{\langle S_6, \sigma_{678} \rangle \Rightarrow \langle S_8, \sigma_{678} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{678} \rangle \Rightarrow \langle S_9, \sigma_{678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2712} = \langle S_8, \sigma_{678} \rangle \Rightarrow \langle y := y + 1, \sigma_{678} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2711} = \frac{\mathcal{T}_{2712}}{\langle S_9, \sigma_{678} \rangle \Rightarrow \langle S_{10}, \sigma_{678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2713} = \frac{\langle y := y + 1, \sigma_{678} \rangle \Rightarrow \sigma_{679} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{678} \rangle \Rightarrow \langle S_{11}, \sigma_{679} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2714} = \langle S_{11}, \sigma_{679} \rangle \Rightarrow \langle S_{12}, \sigma_{679} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2715} = \langle S_{12}, \sigma_{679} \rangle \Rightarrow \langle S_{13}, \sigma_{679} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2717} = \frac{\langle x := x + 1, \sigma_{679} \rangle \Rightarrow \sigma_{680} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{679} \rangle \Rightarrow \langle S_6, \sigma_{680} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2716} = \frac{\mathcal{T}_{2717}}{\langle S_{13}, \sigma_{679} \rangle \Rightarrow \langle S_{15}, \sigma_{680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2718} = \frac{\langle S_6, \sigma_{680} \rangle \Rightarrow \langle S_8, \sigma_{680} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{680} \rangle \Rightarrow \langle S_9, \sigma_{680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2720} = \langle S_8, \sigma_{680} \rangle \Rightarrow \langle y := y + 1, \sigma_{680} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2719} = \frac{\mathcal{T}_{2720}}{\langle S_9, \sigma_{680} \rangle \Rightarrow \langle S_{10}, \sigma_{680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2721} = \frac{\langle y := y + 1, \sigma_{680} \rangle \Rightarrow \sigma_{681} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{680} \rangle \Rightarrow \langle S_{11}, \sigma_{681} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2722} = \langle S_{11}, \sigma_{681} \rangle \Rightarrow \langle S_{12}, \sigma_{681} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2723} = \langle S_{12}, \sigma_{681} \rangle \Rightarrow \langle S_{13}, \sigma_{681} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2725} = \frac{\langle x := x + 1, \sigma_{681} \rangle \Rightarrow \sigma_{682} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{681} \rangle \Rightarrow \langle S_6, \sigma_{682} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2724} = \frac{\mathcal{T}_{2725}}{\langle S_{13}, \sigma_{681} \rangle \Rightarrow \langle S_{15}, \sigma_{682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2726} = \frac{\langle S_6, \sigma_{682} \rangle \Rightarrow \langle S_8, \sigma_{682} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{682} \rangle \Rightarrow \langle S_9, \sigma_{682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2728} = \langle S_8, \sigma_{682} \rangle \Rightarrow \langle y := y + 1, \sigma_{682} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2727} = \frac{\mathcal{T}_{2728}}{\langle S_9, \sigma_{682} \rangle \Rightarrow \langle S_{10}, \sigma_{682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2729} = \frac{\langle y := y + 1, \sigma_{682} \rangle \Rightarrow \sigma_{683} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{682} \rangle \Rightarrow \langle S_{11}, \sigma_{683} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2730} = \langle S_{11}, \sigma_{683} \rangle \Rightarrow \langle S_{12}, \sigma_{683} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2731} = \langle S_{12}, \sigma_{683} \rangle \Rightarrow \langle S_{13}, \sigma_{683} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2733} = \frac{\langle x := x + 1, \sigma_{683} \rangle \Rightarrow \sigma_{684} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{683} \rangle \Rightarrow \langle S_6, \sigma_{684} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2732} = \frac{\mathcal{T}_{2733}}{\langle S_{13}, \sigma_{683} \rangle \Rightarrow \langle S_{15}, \sigma_{684} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2734} = \frac{\langle S_6, \sigma_{684} \rangle \Rightarrow \langle S_8, \sigma_{684} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{684} \rangle \Rightarrow \langle S_9, \sigma_{684} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2736} = \langle S_8, \sigma_{684} \rangle \Rightarrow \langle y := y + 1, \sigma_{684} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2735} = \frac{\mathcal{T}_{2736}}{\langle S_9, \sigma_{684} \rangle \Rightarrow \langle S_{10}, \sigma_{684} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2737} = \frac{\langle y := y + 1, \sigma_{684} \rangle \Rightarrow \sigma_{685} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{684} \rangle \Rightarrow \langle S_{11}, \sigma_{685} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2738} = \langle S_{11}, \sigma_{685} \rangle \Rightarrow \langle S_{12}, \sigma_{685} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2739} = \langle S_{12}, \sigma_{685} \rangle \Rightarrow \langle S_{13}, \sigma_{685} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2741} = \frac{\langle x := x + 1, \sigma_{685} \rangle \Rightarrow \sigma_{686} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{685} \rangle \Rightarrow \langle S_6, \sigma_{686} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2740} = \frac{\mathcal{T}_{2741}}{\langle S_{13}, \sigma_{685} \rangle \Rightarrow \langle S_{15}, \sigma_{686} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2742} = \frac{\langle S_6, \sigma_{686} \rangle \Rightarrow \langle S_8, \sigma_{686} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{686} \rangle \Rightarrow \langle S_9, \sigma_{686} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2744} = \langle S_8, \sigma_{686} \rangle \Rightarrow \langle y := y + 1, \sigma_{686} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2743} = \frac{\mathcal{T}_{2744}}{\langle S_9, \sigma_{686} \rangle \Rightarrow \langle S_{10}, \sigma_{686} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2745} = \frac{\langle y := y + 1, \sigma_{686} \rangle \Rightarrow \sigma_{687} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{686} \rangle \Rightarrow \langle S_{11}, \sigma_{687} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2746} = \langle S_{11}, \sigma_{687} \rangle \Rightarrow \langle S_{12}, \sigma_{687} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2747} = \langle S_{12}, \sigma_{687} \rangle \Rightarrow \langle S_{13}, \sigma_{687} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2749} = \frac{\langle x := x + 1, \sigma_{687} \rangle \Rightarrow \sigma_{688} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{687} \rangle \Rightarrow \langle S_6, \sigma_{688} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2748} = \frac{\mathcal{T}_{2749}}{\langle S_{13}, \sigma_{687} \rangle \Rightarrow \langle S_{15}, \sigma_{688} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2750} = \frac{\langle S_6, \sigma_{688} \rangle \Rightarrow \langle S_8, \sigma_{688} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{688} \rangle \Rightarrow \langle S_9, \sigma_{688} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2752} = \langle S_8, \sigma_{688} \rangle \Rightarrow \langle y := y + 1, \sigma_{688} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2751} = \frac{\mathcal{T}_{2752}}{\langle S_9, \sigma_{688} \rangle \Rightarrow \langle S_{10}, \sigma_{688} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2753} = \frac{\langle y := y + 1, \sigma_{688} \rangle \Rightarrow \sigma_{689} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{688} \rangle \Rightarrow \langle S_{11}, \sigma_{689} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2754} = \langle S_{11}, \sigma_{689} \rangle \Rightarrow \langle S_{12}, \sigma_{689} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2755} = \langle S_{12}, \sigma_{689} \rangle \Rightarrow \langle S_{13}, \sigma_{689} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2757} = \frac{\langle x := x + 1, \sigma_{689} \rangle \Rightarrow \sigma_{690} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{689} \rangle \Rightarrow \langle S_6, \sigma_{690} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2756} = \frac{\mathcal{T}_{2757}}{\langle S_{13}, \sigma_{689} \rangle \Rightarrow \langle S_{15}, \sigma_{690} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2758} = \frac{\langle S_6, \sigma_{690} \rangle \Rightarrow \langle S_8, \sigma_{690} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{690} \rangle \Rightarrow \langle S_9, \sigma_{690} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2760} = \langle S_8, \sigma_{690} \rangle \Rightarrow \langle y := y + 1, \sigma_{690} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2759} = \frac{\mathcal{T}_{2760}}{\langle S_9, \sigma_{690} \rangle \Rightarrow \langle S_{10}, \sigma_{690} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2761} = \frac{\langle y := y + 1, \sigma_{690} \rangle \Rightarrow \sigma_{691} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{690} \rangle \Rightarrow \langle S_{11}, \sigma_{691} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2762} = \langle S_{11}, \sigma_{691} \rangle \Rightarrow \langle S_{12}, \sigma_{691} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2763} = \langle S_{12}, \sigma_{691} \rangle \Rightarrow \langle S_{13}, \sigma_{691} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2765} = \frac{\langle x := x + 1, \sigma_{691} \rangle \Rightarrow \sigma_{692} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{691} \rangle \Rightarrow \langle S_6, \sigma_{692} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2764} = \frac{\mathcal{T}_{2765}}{\langle S_{13}, \sigma_{691} \rangle \Rightarrow \langle S_{15}, \sigma_{692} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2766} = \frac{\langle S_6, \sigma_{692} \rangle \Rightarrow \langle S_8, \sigma_{692} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{692} \rangle \Rightarrow \langle S_9, \sigma_{692} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2768} = \langle S_8, \sigma_{692} \rangle \Rightarrow \langle y := y + 1, \sigma_{692} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2767} = \frac{\mathcal{T}_{2768}}{\langle S_9, \sigma_{692} \rangle \Rightarrow \langle S_{10}, \sigma_{692} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2769} = \frac{\langle y := y + 1, \sigma_{692} \rangle \Rightarrow \sigma_{693} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{692} \rangle \Rightarrow \langle S_{11}, \sigma_{693} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2770} = \langle S_{11}, \sigma_{693} \rangle \Rightarrow \langle S_{12}, \sigma_{693} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2771} = \langle S_{12}, \sigma_{693} \rangle \Rightarrow \langle S_{13}, \sigma_{693} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2773} = \frac{\langle x := x + 1, \sigma_{693} \rangle \Rightarrow \sigma_{694} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{693} \rangle \Rightarrow \langle S_6, \sigma_{694} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2772} = \frac{\mathcal{T}_{2773}}{\langle S_{13}, \sigma_{693} \rangle \Rightarrow \langle S_{15}, \sigma_{694} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2774} = \frac{\langle S_6, \sigma_{694} \rangle \Rightarrow \langle S_8, \sigma_{694} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{694} \rangle \Rightarrow \langle S_9, \sigma_{694} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2776} = \langle S_8, \sigma_{694} \rangle \Rightarrow \langle y := y + 1, \sigma_{694} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2775} = \frac{\mathcal{T}_{2776}}{\langle S_9, \sigma_{694} \rangle \Rightarrow \langle S_{10}, \sigma_{694} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2777} = \frac{\langle y := y + 1, \sigma_{694} \rangle \Rightarrow \sigma_{695} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{694} \rangle \Rightarrow \langle S_{11}, \sigma_{695} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2778} = \langle S_{11}, \sigma_{695} \rangle \Rightarrow \langle S_{12}, \sigma_{695} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2779} = \langle S_{12}, \sigma_{695} \rangle \Rightarrow \langle S_{13}, \sigma_{695} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2781} = \frac{\langle x := x + 1, \sigma_{695} \rangle \Rightarrow \sigma_{696} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{695} \rangle \Rightarrow \langle S_6, \sigma_{696} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2780} = \frac{\mathcal{T}_{2781}}{\langle S_{13}, \sigma_{695} \rangle \Rightarrow \langle S_{15}, \sigma_{696} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2782} = \frac{\langle S_6, \sigma_{696} \rangle \Rightarrow \langle S_8, \sigma_{696} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{696} \rangle \Rightarrow \langle S_9, \sigma_{696} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2784} = \langle S_8, \sigma_{696} \rangle \Rightarrow \langle y := y + 1, \sigma_{696} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2783} = \frac{\mathcal{T}_{2784}}{\langle S_9, \sigma_{696} \rangle \Rightarrow \langle S_{10}, \sigma_{696} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2785} = \frac{\langle y := y + 1, \sigma_{696} \rangle \Rightarrow \sigma_{697} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{696} \rangle \Rightarrow \langle S_{11}, \sigma_{697} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2786} = \langle S_{11}, \sigma_{697} \rangle \Rightarrow \langle S_{12}, \sigma_{697} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2787} = \langle S_{12}, \sigma_{697} \rangle \Rightarrow \langle S_{13}, \sigma_{697} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2789} = \frac{\langle x := x + 1, \sigma_{697} \rangle \Rightarrow \sigma_{698} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{697} \rangle \Rightarrow \langle S_6, \sigma_{698} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2788} = \frac{\mathcal{T}_{2789}}{\langle S_{13}, \sigma_{697} \rangle \Rightarrow \langle S_{15}, \sigma_{698} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2790} = \frac{\langle S_6, \sigma_{698} \rangle \Rightarrow \langle S_8, \sigma_{698} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{698} \rangle \Rightarrow \langle S_9, \sigma_{698} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2792} = \langle S_8, \sigma_{698} \rangle \Rightarrow \langle y := y + 1, \sigma_{698} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2791} = \frac{\mathcal{T}_{2792}}{\langle S_9, \sigma_{698} \rangle \Rightarrow \langle S_{10}, \sigma_{698} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2793} = \frac{\langle y := y + 1, \sigma_{698} \rangle \Rightarrow \sigma_{699} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{698} \rangle \Rightarrow \langle S_{11}, \sigma_{699} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2794} = \langle S_{11}, \sigma_{699} \rangle \Rightarrow \langle S_{12}, \sigma_{699} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2795} = \langle S_{12}, \sigma_{699} \rangle \Rightarrow \langle S_{13}, \sigma_{699} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2797} = \frac{\langle x := x + 1, \sigma_{699} \rangle \Rightarrow \sigma_{700} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{699} \rangle \Rightarrow \langle S_6, \sigma_{700} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2796} = \frac{\mathcal{T}_{2797}}{\langle S_{13}, \sigma_{699} \rangle \Rightarrow \langle S_{15}, \sigma_{700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2798} = \frac{\langle S_6, \sigma_{700} \rangle \Rightarrow \langle S_8, \sigma_{700} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{700} \rangle \Rightarrow \langle S_9, \sigma_{700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2800} = \langle S_8, \sigma_{700} \rangle \Rightarrow \langle y := y + 1, \sigma_{700} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2799} = \frac{\mathcal{T}_{2800}}{\langle S_9, \sigma_{700} \rangle \Rightarrow \langle S_{10}, \sigma_{700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2801} = \frac{\langle y := y + 1, \sigma_{700} \rangle \Rightarrow \sigma_{701} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{700} \rangle \Rightarrow \langle S_{11}, \sigma_{701} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2802} = \langle S_{11}, \sigma_{701} \rangle \Rightarrow \langle S_{12}, \sigma_{701} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2803} = \langle S_{12}, \sigma_{701} \rangle \Rightarrow \langle S_{13}, \sigma_{701} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2805} = \frac{\langle x := x + 1, \sigma_{701} \rangle \Rightarrow \sigma_{702} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{701} \rangle \Rightarrow \langle S_6, \sigma_{702} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2804} = \frac{\mathcal{T}_{2805}}{\langle S_{13}, \sigma_{701} \rangle \Rightarrow \langle S_{15}, \sigma_{702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2806} = \frac{\langle S_6, \sigma_{702} \rangle \Rightarrow \langle S_8, \sigma_{702} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{702} \rangle \Rightarrow \langle S_9, \sigma_{702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2808} = \langle S_8, \sigma_{702} \rangle \Rightarrow \langle y := y + 1, \sigma_{702} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2807} = \frac{\mathcal{T}_{2808}}{\langle S_9, \sigma_{702} \rangle \Rightarrow \langle S_{10}, \sigma_{702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2809} = \frac{\langle y := y + 1, \sigma_{702} \rangle \Rightarrow \sigma_{703} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{702} \rangle \Rightarrow \langle S_{11}, \sigma_{703} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2810} = \langle S_{11}, \sigma_{703} \rangle \Rightarrow \langle S_{12}, \sigma_{703} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2811} = \langle S_{12}, \sigma_{703} \rangle \Rightarrow \langle S_{13}, \sigma_{703} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2813} = \frac{\langle x := x + 1, \sigma_{703} \rangle \Rightarrow \sigma_{704} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{703} \rangle \Rightarrow \langle S_6, \sigma_{704} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2812} = \frac{\mathcal{T}_{2813}}{\langle S_{13}, \sigma_{703} \rangle \Rightarrow \langle S_{15}, \sigma_{704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2814} = \frac{\langle S_6, \sigma_{704} \rangle \Rightarrow \langle S_8, \sigma_{704} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{704} \rangle \Rightarrow \langle S_9, \sigma_{704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2816} = \langle S_8, \sigma_{704} \rangle \Rightarrow \langle y := y + 1, \sigma_{704} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2815} = \frac{\mathcal{T}_{2816}}{\langle S_9, \sigma_{704} \rangle \Rightarrow \langle S_{10}, \sigma_{704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2817} = \frac{\langle y := y + 1, \sigma_{704} \rangle \Rightarrow \sigma_{705} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{704} \rangle \Rightarrow \langle S_{11}, \sigma_{705} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2818} = \langle S_{11}, \sigma_{705} \rangle \Rightarrow \langle S_{12}, \sigma_{705} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2819} = \langle S_{12}, \sigma_{705} \rangle \Rightarrow \langle S_{13}, \sigma_{705} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2821} = \frac{\langle x := x + 1, \sigma_{705} \rangle \Rightarrow \sigma_{706} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{705} \rangle \Rightarrow \langle S_6, \sigma_{706} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2820} = \frac{\mathcal{T}_{2821}}{\langle S_{13}, \sigma_{705} \rangle \Rightarrow \langle S_{15}, \sigma_{706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2822} = \frac{\langle S_6, \sigma_{706} \rangle \Rightarrow \langle S_8, \sigma_{706} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{706} \rangle \Rightarrow \langle S_9, \sigma_{706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2824} = \langle S_8, \sigma_{706} \rangle \Rightarrow \langle y := y + 1, \sigma_{706} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2823} = \frac{\mathcal{T}_{2824}}{\langle S_9, \sigma_{706} \rangle \Rightarrow \langle S_{10}, \sigma_{706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2825} = \frac{\langle y := y + 1, \sigma_{706} \rangle \Rightarrow \sigma_{707} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{706} \rangle \Rightarrow \langle S_{11}, \sigma_{707} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2826} = \langle S_{11}, \sigma_{707} \rangle \Rightarrow \langle S_{12}, \sigma_{707} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2827} = \langle S_{12}, \sigma_{707} \rangle \Rightarrow \langle S_{13}, \sigma_{707} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2829} = \frac{\langle x := x + 1, \sigma_{707} \rangle \Rightarrow \sigma_{708} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{707} \rangle \Rightarrow \langle S_6, \sigma_{708} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2828} = \frac{\mathcal{T}_{2829}}{\langle S_{13}, \sigma_{707} \rangle \Rightarrow \langle S_{15}, \sigma_{708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2830} = \frac{\langle S_6, \sigma_{708} \rangle \Rightarrow \langle S_8, \sigma_{708} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{708} \rangle \Rightarrow \langle S_9, \sigma_{708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2832} = \langle S_8, \sigma_{708} \rangle \Rightarrow \langle y := y + 1, \sigma_{708} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2831} = \frac{\mathcal{T}_{2832}}{\langle S_9, \sigma_{708} \rangle \Rightarrow \langle S_{10}, \sigma_{708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2833} = \frac{\langle y := y + 1, \sigma_{708} \rangle \Rightarrow \sigma_{709} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{708} \rangle \Rightarrow \langle S_{11}, \sigma_{709} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2834} = \langle S_{11}, \sigma_{709} \rangle \Rightarrow \langle S_{12}, \sigma_{709} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2835} = \langle S_{12}, \sigma_{709} \rangle \Rightarrow \langle S_{13}, \sigma_{709} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2837} = \frac{\langle x := x + 1, \sigma_{709} \rangle \Rightarrow \sigma_{710} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{709} \rangle \Rightarrow \langle S_6, \sigma_{710} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2836} = \frac{\mathcal{T}_{2837}}{\langle S_{13}, \sigma_{709} \rangle \Rightarrow \langle S_{15}, \sigma_{710} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2838} = \frac{\langle S_6, \sigma_{710} \rangle \Rightarrow \langle S_8, \sigma_{710} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{710} \rangle \Rightarrow \langle S_9, \sigma_{710} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2840} = \langle S_8, \sigma_{710} \rangle \Rightarrow \langle y := y + 1, \sigma_{710} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2839} = \frac{\mathcal{T}_{2840}}{\langle S_9, \sigma_{710} \rangle \Rightarrow \langle S_{10}, \sigma_{710} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2841} = \frac{\langle y := y + 1, \sigma_{710} \rangle \Rightarrow \sigma_{711} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{710} \rangle \Rightarrow \langle S_{11}, \sigma_{711} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2842} = \langle S_{11}, \sigma_{711} \rangle \Rightarrow \langle S_{12}, \sigma_{711} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2843} = \langle S_{12}, \sigma_{711} \rangle \Rightarrow \langle S_{13}, \sigma_{711} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2845} = \frac{\langle x := x + 1, \sigma_{711} \rangle \Rightarrow \sigma_{712} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{711} \rangle \Rightarrow \langle S_6, \sigma_{712} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2844} = \frac{\mathcal{T}_{2845}}{\langle S_{13}, \sigma_{711} \rangle \Rightarrow \langle S_{15}, \sigma_{712} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2846} = \frac{\langle S_6, \sigma_{712} \rangle \Rightarrow \langle S_8, \sigma_{712} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{712} \rangle \Rightarrow \langle S_9, \sigma_{712} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2848} = \langle S_8, \sigma_{712} \rangle \Rightarrow \langle y := y + 1, \sigma_{712} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2847} = \frac{\mathcal{T}_{2848}}{\langle S_9, \sigma_{712} \rangle \Rightarrow \langle S_{10}, \sigma_{712} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2849} = \frac{\langle y := y + 1, \sigma_{712} \rangle \Rightarrow \sigma_{713} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{712} \rangle \Rightarrow \langle S_{11}, \sigma_{713} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2850} = \langle S_{11}, \sigma_{713} \rangle \Rightarrow \langle S_{12}, \sigma_{713} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2851} = \langle S_{12}, \sigma_{713} \rangle \Rightarrow \langle S_{13}, \sigma_{713} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2853} = \frac{\langle x := x + 1, \sigma_{713} \rangle \Rightarrow \sigma_{714} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{713} \rangle \Rightarrow \langle S_6, \sigma_{714} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2852} = \frac{\mathcal{T}_{2853}}{\langle S_{13}, \sigma_{713} \rangle \Rightarrow \langle S_{15}, \sigma_{714} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2854} = \frac{\langle S_6, \sigma_{714} \rangle \Rightarrow \langle S_8, \sigma_{714} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{714} \rangle \Rightarrow \langle S_9, \sigma_{714} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2856} = \langle S_8, \sigma_{714} \rangle \Rightarrow \langle y := y + 1, \sigma_{714} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2855} = \frac{\mathcal{T}_{2856}}{\langle S_9, \sigma_{714} \rangle \Rightarrow \langle S_{10}, \sigma_{714} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2857} = \frac{\langle y := y + 1, \sigma_{714} \rangle \Rightarrow \sigma_{715} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{714} \rangle \Rightarrow \langle S_{11}, \sigma_{715} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2858} = \langle S_{11}, \sigma_{715} \rangle \Rightarrow \langle S_{12}, \sigma_{715} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2859} = \langle S_{12}, \sigma_{715} \rangle \Rightarrow \langle S_{13}, \sigma_{715} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2861} = \frac{\langle x := x + 1, \sigma_{715} \rangle \Rightarrow \sigma_{716} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{715} \rangle \Rightarrow \langle S_6, \sigma_{716} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2860} = \frac{\mathcal{T}_{2861}}{\langle S_{13}, \sigma_{715} \rangle \Rightarrow \langle S_{15}, \sigma_{716} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2862} = \frac{\langle S_6, \sigma_{716} \rangle \Rightarrow \langle S_8, \sigma_{716} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{716} \rangle \Rightarrow \langle S_9, \sigma_{716} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2864} = \langle S_8, \sigma_{716} \rangle \Rightarrow \langle y := y + 1, \sigma_{716} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2863} = \frac{\mathcal{T}_{2864}}{\langle S_9, \sigma_{716} \rangle \Rightarrow \langle S_{10}, \sigma_{716} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2865} = \frac{\langle y := y + 1, \sigma_{716} \rangle \Rightarrow \sigma_{717} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{716} \rangle \Rightarrow \langle S_{11}, \sigma_{717} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2866} = \langle S_{11}, \sigma_{717} \rangle \Rightarrow \langle S_{12}, \sigma_{717} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2867} = \langle S_{12}, \sigma_{717} \rangle \Rightarrow \langle S_{13}, \sigma_{717} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2869} = \frac{\langle x := x + 1, \sigma_{717} \rangle \Rightarrow \sigma_{718} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{717} \rangle \Rightarrow \langle S_6, \sigma_{718} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2868} = \frac{\mathcal{T}_{2869}}{\langle S_{13}, \sigma_{717} \rangle \Rightarrow \langle S_{15}, \sigma_{718} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2870} = \frac{\langle S_6, \sigma_{718} \rangle \Rightarrow \langle S_8, \sigma_{718} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{718} \rangle \Rightarrow \langle S_9, \sigma_{718} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2872} = \langle S_8, \sigma_{718} \rangle \Rightarrow \langle y := y + 1, \sigma_{718} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2871} = \frac{\mathcal{T}_{2872}}{\langle S_9, \sigma_{718} \rangle \Rightarrow \langle S_{10}, \sigma_{718} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2873} = \frac{\langle y := y + 1, \sigma_{718} \rangle \Rightarrow \sigma_{719} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{718} \rangle \Rightarrow \langle S_{11}, \sigma_{719} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2874} = \langle S_{11}, \sigma_{719} \rangle \Rightarrow \langle S_{12}, \sigma_{719} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2875} = \langle S_{12}, \sigma_{719} \rangle \Rightarrow \langle S_{13}, \sigma_{719} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2877} = \frac{\langle x := x + 1, \sigma_{719} \rangle \Rightarrow \sigma_{720} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{719} \rangle \Rightarrow \langle S_6, \sigma_{720} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2876} = \frac{\mathcal{T}_{2877}}{\langle S_{13}, \sigma_{719} \rangle \Rightarrow \langle S_{15}, \sigma_{720} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2878} = \frac{\langle S_6, \sigma_{720} \rangle \Rightarrow \langle S_8, \sigma_{720} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{720} \rangle \Rightarrow \langle S_9, \sigma_{720} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2880} = \langle S_8, \sigma_{720} \rangle \Rightarrow \langle y := y + 1, \sigma_{720} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2879} = \frac{\mathcal{T}_{2880}}{\langle S_9, \sigma_{720} \rangle \Rightarrow \langle S_{10}, \sigma_{720} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2881} = \frac{\langle y := y + 1, \sigma_{720} \rangle \Rightarrow \sigma_{721} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{720} \rangle \Rightarrow \langle S_{11}, \sigma_{721} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2882} = \langle S_{11}, \sigma_{721} \rangle \Rightarrow \langle S_{12}, \sigma_{721} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2883} = \langle S_{12}, \sigma_{721} \rangle \Rightarrow \langle S_{13}, \sigma_{721} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2885} = \frac{\langle x := x + 1, \sigma_{721} \rangle \Rightarrow \sigma_{722} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{721} \rangle \Rightarrow \langle S_6, \sigma_{722} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2884} = \frac{\mathcal{T}_{2885}}{\langle S_{13}, \sigma_{721} \rangle \Rightarrow \langle S_{15}, \sigma_{722} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2886} = \frac{\langle S_6, \sigma_{722} \rangle \Rightarrow \langle S_8, \sigma_{722} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{722} \rangle \Rightarrow \langle S_9, \sigma_{722} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2888} = \langle S_8, \sigma_{722} \rangle \Rightarrow \langle y := y + 1, \sigma_{722} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2887} = \frac{\mathcal{T}_{2888}}{\langle S_9, \sigma_{722} \rangle \Rightarrow \langle S_{10}, \sigma_{722} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2889} = \frac{\langle y := y + 1, \sigma_{722} \rangle \Rightarrow \sigma_{723} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{722} \rangle \Rightarrow \langle S_{11}, \sigma_{723} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{2890} = \langle S_{11}, \sigma_{723} \rangle \Rightarrow \langle S_{12}, \sigma_{723} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2891} = \langle S_{12}, \sigma_{723} \rangle \Rightarrow \langle S_{13}, \sigma_{723} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2893} = \frac{\langle x := x + 1, \sigma_{723} \rangle \Rightarrow \sigma_{724} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{723} \rangle \Rightarrow \langle S_6, \sigma_{724} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2892} = \frac{\mathcal{T}_{2893}}{\langle S_{13}, \sigma_{723} \rangle \Rightarrow \langle S_{15}, \sigma_{724} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2894} = \frac{\langle S_6, \sigma_{724} \rangle \Rightarrow \langle S_8, \sigma_{724} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{724} \rangle \Rightarrow \langle S_9, \sigma_{724} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2896} = \langle S_8, \sigma_{724} \rangle \Rightarrow \langle y := y + 1, \sigma_{724} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2895} = \frac{\mathcal{T}_{2896}}{\langle S_9, \sigma_{724} \rangle \Rightarrow \langle S_{10}, \sigma_{724} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2897} = \frac{\langle y := y + 1, \sigma_{724} \rangle \Rightarrow \sigma_{725} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{724} \rangle \Rightarrow \langle S_{11}, \sigma_{725} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2898} = \langle S_{11}, \sigma_{725} \rangle \Rightarrow \langle S_{12}, \sigma_{725} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2899} = \langle S_{12}, \sigma_{725} \rangle \Rightarrow \langle S_{13}, \sigma_{725} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2901} = \frac{\langle x := x + 1, \sigma_{725} \rangle \Rightarrow \sigma_{726} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{725} \rangle \Rightarrow \langle S_6, \sigma_{726} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2900} = \frac{\mathcal{T}_{2901}}{\langle S_{13}, \sigma_{725} \rangle \Rightarrow \langle S_{15}, \sigma_{726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2902} = \frac{\langle S_6, \sigma_{726} \rangle \Rightarrow \langle S_8, \sigma_{726} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{726} \rangle \Rightarrow \langle S_9, \sigma_{726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2904} = \langle S_8, \sigma_{726} \rangle \Rightarrow \langle y := y + 1, \sigma_{726} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2903} = \frac{\mathcal{T}_{2904}}{\langle S_9, \sigma_{726} \rangle \Rightarrow \langle S_{10}, \sigma_{726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2905} = \frac{\langle y := y + 1, \sigma_{726} \rangle \Rightarrow \sigma_{727} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{726} \rangle \Rightarrow \langle S_{11}, \sigma_{727} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2906} = \langle S_{11}, \sigma_{727} \rangle \Rightarrow \langle S_{12}, \sigma_{727} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2907} = \langle S_{12}, \sigma_{727} \rangle \Rightarrow \langle S_{13}, \sigma_{727} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2909} = \frac{\langle x := x + 1, \sigma_{727} \rangle \Rightarrow \sigma_{728} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{727} \rangle \Rightarrow \langle S_6, \sigma_{728} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2908} = \frac{\mathcal{T}_{2909}}{\langle S_{13}, \sigma_{727} \rangle \Rightarrow \langle S_{15}, \sigma_{728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2910} = \frac{\langle S_6, \sigma_{728} \rangle \Rightarrow \langle S_8, \sigma_{728} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{728} \rangle \Rightarrow \langle S_9, \sigma_{728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2912} = \langle S_8, \sigma_{728} \rangle \Rightarrow \langle y := y + 1, \sigma_{728} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2911} = \frac{\mathcal{T}_{2912}}{\langle S_9, \sigma_{728} \rangle \Rightarrow \langle S_{10}, \sigma_{728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2913} = \frac{\langle y := y + 1, \sigma_{728} \rangle \Rightarrow \sigma_{729} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{728} \rangle \Rightarrow \langle S_{11}, \sigma_{729} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2914} = \langle S_{11}, \sigma_{729} \rangle \Rightarrow \langle S_{12}, \sigma_{729} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2915} = \langle S_{12}, \sigma_{729} \rangle \Rightarrow \langle S_{13}, \sigma_{729} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2917} = \frac{\langle x := x + 1, \sigma_{729} \rangle \Rightarrow \sigma_{730} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{729} \rangle \Rightarrow \langle S_6, \sigma_{730} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2916} = \frac{\mathcal{T}_{2917}}{\langle S_{13}, \sigma_{729} \rangle \Rightarrow \langle S_{15}, \sigma_{730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2918} = \frac{\langle S_6, \sigma_{730} \rangle \Rightarrow \langle S_8, \sigma_{730} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{730} \rangle \Rightarrow \langle S_9, \sigma_{730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2920} = \langle S_8, \sigma_{730} \rangle \Rightarrow \langle y := y + 1, \sigma_{730} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2919} = \frac{\mathcal{T}_{2920}}{\langle S_9, \sigma_{730} \rangle \Rightarrow \langle S_{10}, \sigma_{730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2921} = \frac{\langle y := y + 1, \sigma_{730} \rangle \Rightarrow \sigma_{731} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{730} \rangle \Rightarrow \langle S_{11}, \sigma_{731} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2922} = \langle S_{11}, \sigma_{731} \rangle \Rightarrow \langle S_{12}, \sigma_{731} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2923} = \langle S_{12}, \sigma_{731} \rangle \Rightarrow \langle S_{13}, \sigma_{731} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2925} = \frac{\langle x := x + 1, \sigma_{731} \rangle \Rightarrow \sigma_{732} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{731} \rangle \Rightarrow \langle S_6, \sigma_{732} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2924} = \frac{\mathcal{T}_{2925}}{\langle S_{13}, \sigma_{731} \rangle \Rightarrow \langle S_{15}, \sigma_{732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2926} = \frac{\langle S_6, \sigma_{732} \rangle \Rightarrow \langle S_8, \sigma_{732} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{732} \rangle \Rightarrow \langle S_9, \sigma_{732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2928} = \langle S_8, \sigma_{732} \rangle \Rightarrow \langle y := y + 1, \sigma_{732} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2927} = \frac{\mathcal{T}_{2928}}{\langle S_9, \sigma_{732} \rangle \Rightarrow \langle S_{10}, \sigma_{732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{2929} = \frac{\langle y := y + 1, \sigma_{732} \rangle \Rightarrow \sigma_{733} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{732} \rangle \Rightarrow \langle S_{11}, \sigma_{733} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2930} = \langle S_{11}, \sigma_{733} \rangle \Rightarrow \langle S_{12}, \sigma_{733} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{2931} = \langle S_{12}, \sigma_{733} \rangle \Rightarrow \langle S_{13}, \sigma_{733} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2933} = \frac{\langle x := x + 1, \sigma_{733} \rangle \Rightarrow \sigma_{734} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{733} \rangle \Rightarrow \langle S_6, \sigma_{734} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2932} = \frac{\mathcal{T}_{2933}}{\langle S_{13}, \sigma_{733} \rangle \Rightarrow \langle S_{15}, \sigma_{734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2934} = \frac{\langle S_6, \sigma_{734} \rangle \Rightarrow \langle S_8, \sigma_{734} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{734} \rangle \Rightarrow \langle S_9, \sigma_{734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2936} = \langle S_8, \sigma_{734} \rangle \Rightarrow \langle y := y + 1, \sigma_{734} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2935} = \frac{\mathcal{T}_{2936}}{\langle S_9, \sigma_{734} \rangle \Rightarrow \langle S_{10}, \sigma_{734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2937} = \frac{\langle y := y + 1, \sigma_{734} \rangle \Rightarrow \sigma_{735} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{734} \rangle \Rightarrow \langle S_{11}, \sigma_{735} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2938} = \langle S_{11}, \sigma_{735} \rangle \Rightarrow \langle S_{12}, \sigma_{735} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2939} = \langle S_{12}, \sigma_{735} \rangle \Rightarrow \langle S_{13}, \sigma_{735} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2941} = \frac{\langle x := x + 1, \sigma_{735} \rangle \Rightarrow \sigma_{736} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{735} \rangle \Rightarrow \langle S_6, \sigma_{736} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2940} = \frac{\mathcal{T}_{2941}}{\langle S_{13}, \sigma_{735} \rangle \Rightarrow \langle S_{15}, \sigma_{736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2942} = \frac{\langle S_6, \sigma_{736} \rangle \Rightarrow \langle S_8, \sigma_{736} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{736} \rangle \Rightarrow \langle S_9, \sigma_{736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2944} = \langle S_8, \sigma_{736} \rangle \Rightarrow \langle y := y + 1, \sigma_{736} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2943} = \frac{\mathcal{T}_{2944}}{\langle S_9, \sigma_{736} \rangle \Rightarrow \langle S_{10}, \sigma_{736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2945} = \frac{\langle y := y + 1, \sigma_{736} \rangle \Rightarrow \sigma_{737} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{736} \rangle \Rightarrow \langle S_{11}, \sigma_{737} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2946} = \langle S_{11}, \sigma_{737} \rangle \Rightarrow \langle S_{12}, \sigma_{737} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2947} = \langle S_{12}, \sigma_{737} \rangle \Rightarrow \langle S_{13}, \sigma_{737} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2949} = \frac{\langle x := x + 1, \sigma_{737} \rangle \Rightarrow \sigma_{738} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{737} \rangle \Rightarrow \langle S_6, \sigma_{738} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2948} = \frac{\mathcal{T}_{2949}}{\langle S_{13}, \sigma_{737} \rangle \Rightarrow \langle S_{15}, \sigma_{738} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2950} = \frac{\langle S_6, \sigma_{738} \rangle \Rightarrow \langle S_8, \sigma_{738} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{738} \rangle \Rightarrow \langle S_9, \sigma_{738} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2952} = \langle S_8, \sigma_{738} \rangle \Rightarrow \langle y := y + 1, \sigma_{738} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2951} = \frac{\mathcal{T}_{2952}}{\langle S_9, \sigma_{738} \rangle \Rightarrow \langle S_{10}, \sigma_{738} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2953} = \frac{\langle y := y + 1, \sigma_{738} \rangle \Rightarrow \sigma_{739} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{738} \rangle \Rightarrow \langle S_{11}, \sigma_{739} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2954} = \langle S_{11}, \sigma_{739} \rangle \Rightarrow \langle S_{12}, \sigma_{739} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2955} = \langle S_{12}, \sigma_{739} \rangle \Rightarrow \langle S_{13}, \sigma_{739} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2957} = \frac{\langle x := x + 1, \sigma_{739} \rangle \Rightarrow \sigma_{740} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{739} \rangle \Rightarrow \langle S_6, \sigma_{740} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2956} = \frac{\mathcal{T}_{2957}}{\langle S_{13}, \sigma_{739} \rangle \Rightarrow \langle S_{15}, \sigma_{740} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2958} = \frac{\langle S_6, \sigma_{740} \rangle \Rightarrow \langle S_8, \sigma_{740} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{740} \rangle \Rightarrow \langle S_9, \sigma_{740} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2960} = \langle S_8, \sigma_{740} \rangle \Rightarrow \langle y := y + 1, \sigma_{740} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2959} = \frac{\mathcal{T}_{2960}}{\langle S_9, \sigma_{740} \rangle \Rightarrow \langle S_{10}, \sigma_{740} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2961} = \frac{\langle y := y + 1, \sigma_{740} \rangle \Rightarrow \sigma_{741} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{740} \rangle \Rightarrow \langle S_{11}, \sigma_{741} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2962} = \langle S_{11}, \sigma_{741} \rangle \Rightarrow \langle S_{12}, \sigma_{741} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2963} = \langle S_{12}, \sigma_{741} \rangle \Rightarrow \langle S_{13}, \sigma_{741} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2965} = \frac{\langle x := x + 1, \sigma_{741} \rangle \Rightarrow \sigma_{742} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{741} \rangle \Rightarrow \langle S_6, \sigma_{742} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{2964} = \frac{\mathcal{T}_{2965}}{\langle S_{13}, \sigma_{741} \rangle \Rightarrow \langle S_{15}, \sigma_{742} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2966} = \frac{\langle S_6, \sigma_{742} \rangle \Rightarrow \langle S_8, \sigma_{742} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{742} \rangle \Rightarrow \langle S_9, \sigma_{742} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2968} = \langle S_8, \sigma_{742} \rangle \Rightarrow \langle y := y + 1, \sigma_{742} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{2967} = \frac{\mathcal{T}_{2968}}{\langle S_9, \sigma_{742} \rangle \Rightarrow \langle S_{10}, \sigma_{742} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2969} = \frac{\langle y := y + 1, \sigma_{742} \rangle \Rightarrow \sigma_{743} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{742} \rangle \Rightarrow \langle S_{11}, \sigma_{743} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2970} = \langle S_{11}, \sigma_{743} \rangle \Rightarrow \langle S_{12}, \sigma_{743} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{2971} = \langle S_{12}, \sigma_{743} \rangle \Rightarrow \langle S_{13}, \sigma_{743} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2973} = \frac{\langle x := x + 1, \sigma_{743} \rangle \Rightarrow \sigma_{744} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{743} \rangle \Rightarrow \langle S_6, \sigma_{744} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2972} = \frac{\mathcal{T}_{2973}}{\langle S_{13}, \sigma_{743} \rangle \Rightarrow \langle S_{15}, \sigma_{744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2974} = \frac{\langle S_6, \sigma_{744} \rangle \Rightarrow \langle S_8, \sigma_{744} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{744} \rangle \Rightarrow \langle S_9, \sigma_{744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2976} = \langle S_8, \sigma_{744} \rangle \Rightarrow \langle y := y + 1, \sigma_{744} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{2975} = \frac{\mathcal{T}_{2976}}{\langle S_9, \sigma_{744} \rangle \Rightarrow \langle S_{10}, \sigma_{744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2977} = \frac{\langle y := y + 1, \sigma_{744} \rangle \Rightarrow \sigma_{745} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{744} \rangle \Rightarrow \langle S_{11}, \sigma_{745} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2978} = \langle S_{11}, \sigma_{745} \rangle \Rightarrow \langle S_{12}, \sigma_{745} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2979} = \langle S_{12}, \sigma_{745} \rangle \Rightarrow \langle S_{13}, \sigma_{745} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2981} = \frac{\langle x := x + 1, \sigma_{745} \rangle \Rightarrow \sigma_{746} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{745} \rangle \Rightarrow \langle S_6, \sigma_{746} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2980} = \frac{\mathcal{T}_{2981}}{\langle S_{13}, \sigma_{745} \rangle \Rightarrow \langle S_{15}, \sigma_{746} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2982} = \frac{\langle S_6, \sigma_{746} \rangle \Rightarrow \langle S_8, \sigma_{746} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{746} \rangle \Rightarrow \langle S_9, \sigma_{746} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2984} = \langle S_8, \sigma_{746} \rangle \Rightarrow \langle y := y + 1, \sigma_{746} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2983} = \frac{\mathcal{T}_{2984}}{\langle S_9, \sigma_{746} \rangle \Rightarrow \langle S_{10}, \sigma_{746} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2985} = \frac{\langle y := y + 1, \sigma_{746} \rangle \Rightarrow \sigma_{747} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{746} \rangle \Rightarrow \langle S_{11}, \sigma_{747} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{2986} = \langle S_{11}, \sigma_{747} \rangle \Rightarrow \langle S_{12}, \sigma_{747} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2987} = \langle S_{12}, \sigma_{747} \rangle \Rightarrow \langle S_{13}, \sigma_{747} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{2989} = \frac{\langle x := x + 1, \sigma_{747} \rangle \Rightarrow \sigma_{748} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{747} \rangle \Rightarrow \langle S_6, \sigma_{748} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2988} = \frac{\mathcal{T}_{2989}}{\langle S_{13}, \sigma_{747} \rangle \Rightarrow \langle S_{15}, \sigma_{748} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2990} = \frac{\langle S_6, \sigma_{748} \rangle \Rightarrow \langle S_8, \sigma_{748} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{748} \rangle \Rightarrow \langle S_9, \sigma_{748} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2992} = \langle S_8, \sigma_{748} \rangle \Rightarrow \langle y := y + 1, \sigma_{748} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2991} = \frac{\mathcal{T}_{2992}}{\langle S_9, \sigma_{748} \rangle \Rightarrow \langle S_{10}, \sigma_{748} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{2993} = \frac{\langle y := y + 1, \sigma_{748} \rangle \Rightarrow \sigma_{749} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{748} \rangle \Rightarrow \langle S_{11}, \sigma_{749} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{2994} = \langle S_{11}, \sigma_{749} \rangle \Rightarrow \langle S_{12}, \sigma_{749} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2995} = \langle S_{12}, \sigma_{749} \rangle \Rightarrow \langle S_{13}, \sigma_{749} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{2997} = \frac{\langle x := x + 1, \sigma_{749} \rangle \Rightarrow \sigma_{750} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{749} \rangle \Rightarrow \langle S_6, \sigma_{750} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{2996} = \frac{\mathcal{T}_{2997}}{\langle S_{13}, \sigma_{749} \rangle \Rightarrow \langle S_{15}, \sigma_{750} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{2998} = \frac{\langle S_6, \sigma_{750} \rangle \Rightarrow \langle S_8, \sigma_{750} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{750} \rangle \Rightarrow \langle S_9, \sigma_{750} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3000} = \langle S_8, \sigma_{750} \rangle \Rightarrow \langle y := y + 1, \sigma_{750} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{2999} = \frac{\mathcal{T}_{3000}}{\langle S_9, \sigma_{750} \rangle \Rightarrow \langle S_{10}, \sigma_{750} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3001} = \frac{\langle y := y + 1, \sigma_{750} \rangle \Rightarrow \sigma_{751} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{750} \rangle \Rightarrow \langle S_{11}, \sigma_{751} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3002} = \langle S_{11}, \sigma_{751} \rangle \Rightarrow \langle S_{12}, \sigma_{751} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3003} = \langle S_{12}, \sigma_{751} \rangle \Rightarrow \langle S_{13}, \sigma_{751} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3005} = \frac{\langle x := x + 1, \sigma_{751} \rangle \Rightarrow \sigma_{752} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{751} \rangle \Rightarrow \langle S_6, \sigma_{752} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3004} = \frac{\mathcal{T}_{3005}}{\langle S_{13}, \sigma_{751} \rangle \Rightarrow \langle S_{15}, \sigma_{752} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3006} = \frac{\langle S_6, \sigma_{752} \rangle \Rightarrow \langle S_8, \sigma_{752} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{752} \rangle \Rightarrow \langle S_9, \sigma_{752} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3008} = \langle S_8, \sigma_{752} \rangle \Rightarrow \langle y := y + 1, \sigma_{752} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3007} = \frac{\mathcal{T}_{3008}}{\langle S_9, \sigma_{752} \rangle \Rightarrow \langle S_{10}, \sigma_{752} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3009} = \frac{\langle y := y + 1, \sigma_{752} \rangle \Rightarrow \sigma_{753} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{752} \rangle \Rightarrow \langle S_{11}, \sigma_{753} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3010} = \langle S_{11}, \sigma_{753} \rangle \Rightarrow \langle S_{12}, \sigma_{753} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3011} = \langle S_{12}, \sigma_{753} \rangle \Rightarrow \langle S_{13}, \sigma_{753} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3013} = \frac{\langle x := x + 1, \sigma_{753} \rangle \Rightarrow \sigma_{754} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{753} \rangle \Rightarrow \langle S_6, \sigma_{754} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3012} = \frac{\mathcal{T}_{3013}}{\langle S_{13}, \sigma_{753} \rangle \Rightarrow \langle S_{15}, \sigma_{754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3014} = \frac{\langle S_6, \sigma_{754} \rangle \Rightarrow \langle S_8, \sigma_{754} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{754} \rangle \Rightarrow \langle S_9, \sigma_{754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3016} = \langle S_8, \sigma_{754} \rangle \Rightarrow \langle y := y + 1, \sigma_{754} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3015} = \frac{\mathcal{T}_{3016}}{\langle S_9, \sigma_{754} \rangle \Rightarrow \langle S_{10}, \sigma_{754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3017} = \frac{\langle y := y + 1, \sigma_{754} \rangle \Rightarrow \sigma_{755} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{754} \rangle \Rightarrow \langle S_{11}, \sigma_{755} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3018} = \langle S_{11}, \sigma_{755} \rangle \Rightarrow \langle S_{12}, \sigma_{755} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3019} = \langle S_{12}, \sigma_{755} \rangle \Rightarrow \langle S_{13}, \sigma_{755} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3021} = \frac{\langle x := x + 1, \sigma_{755} \rangle \Rightarrow \sigma_{756} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{755} \rangle \Rightarrow \langle S_6, \sigma_{756} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3020} = \frac{\mathcal{T}_{3021}}{\langle S_{13}, \sigma_{755} \rangle \Rightarrow \langle S_{15}, \sigma_{756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3022} = \frac{\langle S_6, \sigma_{756} \rangle \Rightarrow \langle S_8, \sigma_{756} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{756} \rangle \Rightarrow \langle S_9, \sigma_{756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3024} = \langle S_8, \sigma_{756} \rangle \Rightarrow \langle y := y + 1, \sigma_{756} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3023} = \frac{\mathcal{T}_{3024}}{\langle S_9, \sigma_{756} \rangle \Rightarrow \langle S_{10}, \sigma_{756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3025} = \frac{\langle y := y + 1, \sigma_{756} \rangle \Rightarrow \sigma_{757} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{756} \rangle \Rightarrow \langle S_{11}, \sigma_{757} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3026} = \langle S_{11}, \sigma_{757} \rangle \Rightarrow \langle S_{12}, \sigma_{757} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3027} = \langle S_{12}, \sigma_{757} \rangle \Rightarrow \langle S_{13}, \sigma_{757} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3029} = \frac{\langle x := x + 1, \sigma_{757} \rangle \Rightarrow \sigma_{758} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{757} \rangle \Rightarrow \langle S_6, \sigma_{758} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3028} = \frac{\mathcal{T}_{3029}}{\langle S_{13}, \sigma_{757} \rangle \Rightarrow \langle S_{15}, \sigma_{758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3030} = \frac{\langle S_6, \sigma_{758} \rangle \Rightarrow \langle S_8, \sigma_{758} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{758} \rangle \Rightarrow \langle S_9, \sigma_{758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3032} = \langle S_8, \sigma_{758} \rangle \Rightarrow \langle y := y + 1, \sigma_{758} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3031} = \frac{\mathcal{T}_{3032}}{\langle S_9, \sigma_{758} \rangle \Rightarrow \langle S_{10}, \sigma_{758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3033} = \frac{\langle y := y + 1, \sigma_{758} \rangle \Rightarrow \sigma_{759} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{758} \rangle \Rightarrow \langle S_{11}, \sigma_{759} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3034} = \langle S_{11}, \sigma_{759} \rangle \Rightarrow \langle S_{12}, \sigma_{759} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3035} = \langle S_{12}, \sigma_{759} \rangle \Rightarrow \langle S_{13}, \sigma_{759} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3037} = \frac{\langle x := x + 1, \sigma_{759} \rangle \Rightarrow \sigma_{760} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{759} \rangle \Rightarrow \langle S_6, \sigma_{760} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3036} = \frac{\mathcal{T}_{3037}}{\langle S_{13}, \sigma_{759} \rangle \Rightarrow \langle S_{15}, \sigma_{760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3038} = \frac{\langle S_6, \sigma_{760} \rangle \Rightarrow \langle S_8, \sigma_{760} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{760} \rangle \Rightarrow \langle S_9, \sigma_{760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3040} = \langle S_8, \sigma_{760} \rangle \Rightarrow \langle y := y + 1, \sigma_{760} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3039} = \frac{\mathcal{T}_{3040}}{\langle S_9, \sigma_{760} \rangle \Rightarrow \langle S_{10}, \sigma_{760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3041} = \frac{\langle y := y + 1, \sigma_{760} \rangle \Rightarrow \sigma_{761} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{760} \rangle \Rightarrow \langle S_{11}, \sigma_{761} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3042} = \langle S_{11}, \sigma_{761} \rangle \Rightarrow \langle S_{12}, \sigma_{761} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3043} = \langle S_{12}, \sigma_{761} \rangle \Rightarrow \langle S_{13}, \sigma_{761} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3045} = \frac{\langle x := x + 1, \sigma_{761} \rangle \Rightarrow \sigma_{762} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{761} \rangle \Rightarrow \langle S_6, \sigma_{762} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3044} = \frac{\mathcal{T}_{3045}}{\langle S_{13}, \sigma_{761} \rangle \Rightarrow \langle S_{15}, \sigma_{762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3046} = \frac{\langle S_6, \sigma_{762} \rangle \Rightarrow \langle S_8, \sigma_{762} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{762} \rangle \Rightarrow \langle S_9, \sigma_{762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3048} = \langle S_8, \sigma_{762} \rangle \Rightarrow \langle y := y + 1, \sigma_{762} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3047} = \frac{\mathcal{T}_{3048}}{\langle S_9, \sigma_{762} \rangle \Rightarrow \langle S_{10}, \sigma_{762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3049} = \frac{\langle y := y + 1, \sigma_{762} \rangle \Rightarrow \sigma_{763} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{762} \rangle \Rightarrow \langle S_{11}, \sigma_{763} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3050} = \langle S_{11}, \sigma_{763} \rangle \Rightarrow \langle S_{12}, \sigma_{763} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3051} = \langle S_{12}, \sigma_{763} \rangle \Rightarrow \langle S_{13}, \sigma_{763} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3053} = \frac{\langle x := x + 1, \sigma_{763} \rangle \Rightarrow \sigma_{764} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{763} \rangle \Rightarrow \langle S_6, \sigma_{764} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3052} = \frac{\mathcal{T}_{3053}}{\langle S_{13}, \sigma_{763} \rangle \Rightarrow \langle S_{15}, \sigma_{764} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3054} = \frac{\langle S_6, \sigma_{764} \rangle \Rightarrow \langle S_8, \sigma_{764} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{764} \rangle \Rightarrow \langle S_9, \sigma_{764} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3056} = \langle S_8, \sigma_{764} \rangle \Rightarrow \langle y := y + 1, \sigma_{764} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3055} = \frac{\mathcal{T}_{3056}}{\langle S_9, \sigma_{764} \rangle \Rightarrow \langle S_{10}, \sigma_{764} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3057} = \frac{\langle y := y + 1, \sigma_{764} \rangle \Rightarrow \sigma_{765} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{764} \rangle \Rightarrow \langle S_{11}, \sigma_{765} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3058} = \langle S_{11}, \sigma_{765} \rangle \Rightarrow \langle S_{12}, \sigma_{765} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3059} = \langle S_{12}, \sigma_{765} \rangle \Rightarrow \langle S_{13}, \sigma_{765} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3061} = \frac{\langle x := x + 1, \sigma_{765} \rangle \Rightarrow \sigma_{766} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{765} \rangle \Rightarrow \langle S_6, \sigma_{766} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3060} = \frac{\mathcal{T}_{3061}}{\langle S_{13}, \sigma_{765} \rangle \Rightarrow \langle S_{15}, \sigma_{766} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3062} = \frac{\langle S_6, \sigma_{766} \rangle \Rightarrow \langle S_8, \sigma_{766} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{766} \rangle \Rightarrow \langle S_9, \sigma_{766} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3064} = \langle S_8, \sigma_{766} \rangle \Rightarrow \langle y := y + 1, \sigma_{766} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3063} = \frac{\mathcal{T}_{3064}}{\langle S_9, \sigma_{766} \rangle \Rightarrow \langle S_{10}, \sigma_{766} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3065} = \frac{\langle y := y + 1, \sigma_{766} \rangle \Rightarrow \sigma_{767} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{766} \rangle \Rightarrow \langle S_{11}, \sigma_{767} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3066} = \langle S_{11}, \sigma_{767} \rangle \Rightarrow \langle S_{12}, \sigma_{767} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3067} = \langle S_{12}, \sigma_{767} \rangle \Rightarrow \langle S_{13}, \sigma_{767} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3069} = \frac{\langle x := x + 1, \sigma_{767} \rangle \Rightarrow \sigma_{768} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{767} \rangle \Rightarrow \langle S_6, \sigma_{768} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3068} = \frac{\mathcal{T}_{3069}}{\langle S_{13}, \sigma_{767} \rangle \Rightarrow \langle S_{15}, \sigma_{768} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3070} = \frac{\langle S_6, \sigma_{768} \rangle \Rightarrow \langle S_8, \sigma_{768} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{768} \rangle \Rightarrow \langle S_9, \sigma_{768} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3072} = \langle S_8, \sigma_{768} \rangle \Rightarrow \langle y := y + 1, \sigma_{768} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3071} = \frac{\mathcal{T}_{3072}}{\langle S_9, \sigma_{768} \rangle \Rightarrow \langle S_{10}, \sigma_{768} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3073} = \frac{\langle y := y + 1, \sigma_{768} \rangle \Rightarrow \sigma_{769} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{768} \rangle \Rightarrow \langle S_{11}, \sigma_{769} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3074} = \langle S_{11}, \sigma_{769} \rangle \Rightarrow \langle S_{12}, \sigma_{769} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3075} = \langle S_{12}, \sigma_{769} \rangle \Rightarrow \langle S_{13}, \sigma_{769} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3077} = \frac{\langle x := x + 1, \sigma_{769} \rangle \Rightarrow \sigma_{770} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{769} \rangle \Rightarrow \langle S_6, \sigma_{770} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3076} = \frac{\mathcal{T}_{3077}}{\langle S_{13}, \sigma_{769} \rangle \Rightarrow \langle S_{15}, \sigma_{770} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3078} = \frac{\langle S_6, \sigma_{770} \rangle \Rightarrow \langle S_8, \sigma_{770} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{770} \rangle \Rightarrow \langle S_9, \sigma_{770} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3080} = \langle S_8, \sigma_{770} \rangle \Rightarrow \langle y := y + 1, \sigma_{770} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3079} = \frac{\mathcal{T}_{3080}}{\langle S_9, \sigma_{770} \rangle \Rightarrow \langle S_{10}, \sigma_{770} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3081} = \frac{\langle y := y + 1, \sigma_{770} \rangle \Rightarrow \sigma_{771} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{770} \rangle \Rightarrow \langle S_{11}, \sigma_{771} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3082} = \langle S_{11}, \sigma_{771} \rangle \Rightarrow \langle S_{12}, \sigma_{771} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3083} = \langle S_{12}, \sigma_{771} \rangle \Rightarrow \langle S_{13}, \sigma_{771} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3085} = \frac{\langle x := x + 1, \sigma_{771} \rangle \Rightarrow \sigma_{772} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{771} \rangle \Rightarrow \langle S_6, \sigma_{772} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3084} = \frac{\mathcal{T}_{3085}}{\langle S_{13}, \sigma_{771} \rangle \Rightarrow \langle S_{15}, \sigma_{772} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3086} = \frac{\langle S_6, \sigma_{772} \rangle \Rightarrow \langle S_8, \sigma_{772} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{772} \rangle \Rightarrow \langle S_9, \sigma_{772} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3088} = \langle S_8, \sigma_{772} \rangle \Rightarrow \langle y := y + 1, \sigma_{772} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3087} = \frac{\mathcal{T}_{3088}}{\langle S_9, \sigma_{772} \rangle \Rightarrow \langle S_{10}, \sigma_{772} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3089} = \frac{\langle y := y + 1, \sigma_{772} \rangle \Rightarrow \sigma_{773} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{772} \rangle \Rightarrow \langle S_{11}, \sigma_{773} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3090} = \langle S_{11}, \sigma_{773} \rangle \Rightarrow \langle S_{12}, \sigma_{773} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3091} = \langle S_{12}, \sigma_{773} \rangle \Rightarrow \langle S_{13}, \sigma_{773} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3093} = \frac{\langle x := x + 1, \sigma_{773} \rangle \Rightarrow \sigma_{774} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{773} \rangle \Rightarrow \langle S_6, \sigma_{774} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3092} = \frac{\mathcal{T}_{3093}}{\langle S_{13}, \sigma_{773} \rangle \Rightarrow \langle S_{15}, \sigma_{774} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3094} = \frac{\langle S_6, \sigma_{774} \rangle \Rightarrow \langle S_8, \sigma_{774} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{774} \rangle \Rightarrow \langle S_9, \sigma_{774} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3096} = \langle S_8, \sigma_{774} \rangle \Rightarrow \langle y := y + 1, \sigma_{774} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3095} = \frac{\mathcal{T}_{3096}}{\langle S_9, \sigma_{774} \rangle \Rightarrow \langle S_{10}, \sigma_{774} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3097} = \frac{\langle y := y + 1, \sigma_{774} \rangle \Rightarrow \sigma_{775} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{774} \rangle \Rightarrow \langle S_{11}, \sigma_{775} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3098} = \langle S_{11}, \sigma_{775} \rangle \Rightarrow \langle S_{12}, \sigma_{775} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3099} = \langle S_{12}, \sigma_{775} \rangle \Rightarrow \langle S_{13}, \sigma_{775} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3101} = \frac{\langle x := x + 1, \sigma_{775} \rangle \Rightarrow \sigma_{776} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{775} \rangle \Rightarrow \langle S_6, \sigma_{776} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3100} = \frac{\mathcal{T}_{3101}}{\langle S_{13}, \sigma_{775} \rangle \Rightarrow \langle S_{15}, \sigma_{776} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3102} = \frac{\langle S_6, \sigma_{776} \rangle \Rightarrow \langle S_8, \sigma_{776} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{776} \rangle \Rightarrow \langle S_9, \sigma_{776} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3104} = \langle S_8, \sigma_{776} \rangle \Rightarrow \langle y := y + 1, \sigma_{776} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3103} = \frac{\mathcal{T}_{3104}}{\langle S_9, \sigma_{776} \rangle \Rightarrow \langle S_{10}, \sigma_{776} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3105} = \frac{\langle y := y + 1, \sigma_{776} \rangle \Rightarrow \sigma_{777} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{776} \rangle \Rightarrow \langle S_{11}, \sigma_{777} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3106} = \langle S_{11}, \sigma_{777} \rangle \Rightarrow \langle S_{12}, \sigma_{777} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3107} = \langle S_{12}, \sigma_{777} \rangle \Rightarrow \langle S_{13}, \sigma_{777} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3109} = \frac{\langle x := x + 1, \sigma_{777} \rangle \Rightarrow \sigma_{778} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{777} \rangle \Rightarrow \langle S_6, \sigma_{778} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3108} = \frac{\mathcal{T}_{3109}}{\langle S_{13}, \sigma_{777} \rangle \Rightarrow \langle S_{15}, \sigma_{778} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3110} = \frac{\langle S_6, \sigma_{778} \rangle \Rightarrow \langle S_8, \sigma_{778} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{778} \rangle \Rightarrow \langle S_9, \sigma_{778} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3112} = \langle S_8, \sigma_{778} \rangle \Rightarrow \langle y := y + 1, \sigma_{778} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3111} = \frac{\mathcal{T}_{3112}}{\langle S_9, \sigma_{778} \rangle \Rightarrow \langle S_{10}, \sigma_{778} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3113} = \frac{\langle y := y + 1, \sigma_{778} \rangle \Rightarrow \sigma_{779} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{778} \rangle \Rightarrow \langle S_{11}, \sigma_{779} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3114} = \langle S_{11}, \sigma_{779} \rangle \Rightarrow \langle S_{12}, \sigma_{779} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3115} = \langle S_{12}, \sigma_{779} \rangle \Rightarrow \langle S_{13}, \sigma_{779} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3117} = \frac{\langle x := x + 1, \sigma_{779} \rangle \Rightarrow \sigma_{780} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{779} \rangle \Rightarrow \langle S_6, \sigma_{780} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3116} = \frac{\mathcal{T}_{3117}}{\langle S_{13}, \sigma_{779} \rangle \Rightarrow \langle S_{15}, \sigma_{780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3118} = \frac{\langle S_6, \sigma_{780} \rangle \Rightarrow \langle S_8, \sigma_{780} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{780} \rangle \Rightarrow \langle S_9, \sigma_{780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3120} = \langle S_8, \sigma_{780} \rangle \Rightarrow \langle y := y + 1, \sigma_{780} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3119} = \frac{\mathcal{T}_{3120}}{\langle S_9, \sigma_{780} \rangle \Rightarrow \langle S_{10}, \sigma_{780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3121} = \frac{\langle y := y + 1, \sigma_{780} \rangle \Rightarrow \sigma_{781} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{780} \rangle \Rightarrow \langle S_{11}, \sigma_{781} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3122} = \langle S_{11}, \sigma_{781} \rangle \Rightarrow \langle S_{12}, \sigma_{781} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3123} = \langle S_{12}, \sigma_{781} \rangle \Rightarrow \langle S_{13}, \sigma_{781} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3125} = \frac{\langle x := x + 1, \sigma_{781} \rangle \Rightarrow \sigma_{782} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{781} \rangle \Rightarrow \langle S_6, \sigma_{782} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3124} = \frac{\mathcal{T}_{3125}}{\langle S_{13}, \sigma_{781} \rangle \Rightarrow \langle S_{15}, \sigma_{782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3126} = \frac{\langle S_6, \sigma_{782} \rangle \Rightarrow \langle S_8, \sigma_{782} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{782} \rangle \Rightarrow \langle S_9, \sigma_{782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3128} = \langle S_8, \sigma_{782} \rangle \Rightarrow \langle y := y + 1, \sigma_{782} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3127} = \frac{\mathcal{T}_{3128}}{\langle S_9, \sigma_{782} \rangle \Rightarrow \langle S_{10}, \sigma_{782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3129} = \frac{\langle y := y + 1, \sigma_{782} \rangle \Rightarrow \sigma_{783} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{782} \rangle \Rightarrow \langle S_{11}, \sigma_{783} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3130} = \langle S_{11}, \sigma_{783} \rangle \Rightarrow \langle S_{12}, \sigma_{783} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3131} = \langle S_{12}, \sigma_{783} \rangle \Rightarrow \langle S_{13}, \sigma_{783} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3133} = \frac{\langle x := x + 1, \sigma_{783} \rangle \Rightarrow \sigma_{784} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{783} \rangle \Rightarrow \langle S_6, \sigma_{784} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3132} = \frac{\mathcal{T}_{3133}}{\langle S_{13}, \sigma_{783} \rangle \Rightarrow \langle S_{15}, \sigma_{784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3134} = \frac{\langle S_6, \sigma_{784} \rangle \Rightarrow \langle S_8, \sigma_{784} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{784} \rangle \Rightarrow \langle S_9, \sigma_{784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3136} = \langle S_8, \sigma_{784} \rangle \Rightarrow \langle y := y + 1, \sigma_{784} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3135} = \frac{\mathcal{T}_{3136}}{\langle S_9, \sigma_{784} \rangle \Rightarrow \langle S_{10}, \sigma_{784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3137} = \frac{\langle y := y + 1, \sigma_{784} \rangle \Rightarrow \sigma_{785} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{784} \rangle \Rightarrow \langle S_{11}, \sigma_{785} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3138} = \langle S_{11}, \sigma_{785} \rangle \Rightarrow \langle S_{12}, \sigma_{785} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3139} = \langle S_{12}, \sigma_{785} \rangle \Rightarrow \langle S_{13}, \sigma_{785} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3141} = \frac{\langle x := x + 1, \sigma_{785} \rangle \Rightarrow \sigma_{786} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{785} \rangle \Rightarrow \langle S_6, \sigma_{786} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3140} = \frac{\mathcal{T}_{3141}}{\langle S_{13}, \sigma_{785} \rangle \Rightarrow \langle S_{15}, \sigma_{786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3142} = \frac{\langle S_6, \sigma_{786} \rangle \Rightarrow \langle S_8, \sigma_{786} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{786} \rangle \Rightarrow \langle S_9, \sigma_{786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3144} = \langle S_8, \sigma_{786} \rangle \Rightarrow \langle y := y + 1, \sigma_{786} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3143} = \frac{\mathcal{T}_{3144}}{\langle S_9, \sigma_{786} \rangle \Rightarrow \langle S_{10}, \sigma_{786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3145} = \frac{\langle y := y + 1, \sigma_{786} \rangle \Rightarrow \sigma_{787} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{786} \rangle \Rightarrow \langle S_{11}, \sigma_{787} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3146} = \langle S_{11}, \sigma_{787} \rangle \Rightarrow \langle S_{12}, \sigma_{787} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3147} = \langle S_{12}, \sigma_{787} \rangle \Rightarrow \langle S_{13}, \sigma_{787} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3149} = \frac{\langle x := x + 1, \sigma_{787} \rangle \Rightarrow \sigma_{788} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{787} \rangle \Rightarrow \langle S_6, \sigma_{788} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3148} = \frac{\mathcal{T}_{3149}}{\langle S_{13}, \sigma_{787} \rangle \Rightarrow \langle S_{15}, \sigma_{788} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3150} = \frac{\langle S_6, \sigma_{788} \rangle \Rightarrow \langle S_8, \sigma_{788} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{788} \rangle \Rightarrow \langle S_9, \sigma_{788} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3152} = \langle S_8, \sigma_{788} \rangle \Rightarrow \langle y := y + 1, \sigma_{788} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3151} = \frac{\mathcal{T}_{3152}}{\langle S_9, \sigma_{788} \rangle \Rightarrow \langle S_{10}, \sigma_{788} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3153} = \frac{\langle y := y + 1, \sigma_{788} \rangle \Rightarrow \sigma_{789} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{788} \rangle \Rightarrow \langle S_{11}, \sigma_{789} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3154} = \langle S_{11}, \sigma_{789} \rangle \Rightarrow \langle S_{12}, \sigma_{789} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3155} = \langle S_{12}, \sigma_{789} \rangle \Rightarrow \langle S_{13}, \sigma_{789} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3157} = \frac{\langle x := x + 1, \sigma_{789} \rangle \Rightarrow \sigma_{790} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{789} \rangle \Rightarrow \langle S_6, \sigma_{790} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3156} = \frac{\mathcal{T}_{3157}}{\langle S_{13}, \sigma_{789} \rangle \Rightarrow \langle S_{15}, \sigma_{790} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3158} = \frac{\langle S_6, \sigma_{790} \rangle \Rightarrow \langle S_8, \sigma_{790} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{790} \rangle \Rightarrow \langle S_9, \sigma_{790} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3160} = \langle S_8, \sigma_{790} \rangle \Rightarrow \langle y := y + 1, \sigma_{790} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3159} = \frac{\mathcal{T}_{3160}}{\langle S_9, \sigma_{790} \rangle \Rightarrow \langle S_{10}, \sigma_{790} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3161} = \frac{\langle y := y + 1, \sigma_{790} \rangle \Rightarrow \sigma_{791} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{790} \rangle \Rightarrow \langle S_{11}, \sigma_{791} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3162} = \langle S_{11}, \sigma_{791} \rangle \Rightarrow \langle S_{12}, \sigma_{791} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3163} = \langle S_{12}, \sigma_{791} \rangle \Rightarrow \langle S_{13}, \sigma_{791} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3165} = \frac{\langle x := x + 1, \sigma_{791} \rangle \Rightarrow \sigma_{792} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{791} \rangle \Rightarrow \langle S_6, \sigma_{792} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3164} = \frac{\mathcal{T}_{3165}}{\langle S_{13}, \sigma_{791} \rangle \Rightarrow \langle S_{15}, \sigma_{792} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3166} = \frac{\langle S_6, \sigma_{792} \rangle \Rightarrow \langle S_8, \sigma_{792} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{792} \rangle \Rightarrow \langle S_9, \sigma_{792} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3168} = \langle S_8, \sigma_{792} \rangle \Rightarrow \langle y := y + 1, \sigma_{792} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3167} = \frac{\mathcal{T}_{3168}}{\langle S_9, \sigma_{792} \rangle \Rightarrow \langle S_{10}, \sigma_{792} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3169} = \frac{\langle y := y + 1, \sigma_{792} \rangle \Rightarrow \sigma_{793} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{792} \rangle \Rightarrow \langle S_{11}, \sigma_{793} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3170} = \langle S_{11}, \sigma_{793} \rangle \Rightarrow \langle S_{12}, \sigma_{793} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3171} = \langle S_{12}, \sigma_{793} \rangle \Rightarrow \langle S_{13}, \sigma_{793} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3173} = \frac{\langle x := x + 1, \sigma_{793} \rangle \Rightarrow \sigma_{794} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{793} \rangle \Rightarrow \langle S_6, \sigma_{794} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3172} = \frac{\mathcal{T}_{3173}}{\langle S_{13}, \sigma_{793} \rangle \Rightarrow \langle S_{15}, \sigma_{794} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3174} = \frac{\langle S_6, \sigma_{794} \rangle \Rightarrow \langle S_8, \sigma_{794} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{794} \rangle \Rightarrow \langle S_9, \sigma_{794} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3176} = \langle S_8, \sigma_{794} \rangle \Rightarrow \langle y := y + 1, \sigma_{794} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3175} = \frac{\mathcal{T}_{3176}}{\langle S_9, \sigma_{794} \rangle \Rightarrow \langle S_{10}, \sigma_{794} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3177} = \frac{\langle y := y + 1, \sigma_{794} \rangle \Rightarrow \sigma_{795} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{794} \rangle \Rightarrow \langle S_{11}, \sigma_{795} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3178} = \langle S_{11}, \sigma_{795} \rangle \Rightarrow \langle S_{12}, \sigma_{795} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3179} = \langle S_{12}, \sigma_{795} \rangle \Rightarrow \langle S_{13}, \sigma_{795} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3181} = \frac{\langle x := x + 1, \sigma_{795} \rangle \Rightarrow \sigma_{796} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{795} \rangle \Rightarrow \langle S_6, \sigma_{796} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3180} = \frac{\mathcal{T}_{3181}}{\langle S_{13}, \sigma_{795} \rangle \Rightarrow \langle S_{15}, \sigma_{796} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3182} = \frac{\langle S_6, \sigma_{796} \rangle \Rightarrow \langle S_8, \sigma_{796} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{796} \rangle \Rightarrow \langle S_9, \sigma_{796} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3184} = \langle S_8, \sigma_{796} \rangle \Rightarrow \langle y := y + 1, \sigma_{796} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3183} = \frac{\mathcal{T}_{3184}}{\langle S_9, \sigma_{796} \rangle \Rightarrow \langle S_{10}, \sigma_{796} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3185} = \frac{\langle y := y + 1, \sigma_{796} \rangle \Rightarrow \sigma_{797} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{796} \rangle \Rightarrow \langle S_{11}, \sigma_{797} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3186} = \langle S_{11}, \sigma_{797} \rangle \Rightarrow \langle S_{12}, \sigma_{797} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3187} = \langle S_{12}, \sigma_{797} \rangle \Rightarrow \langle S_{13}, \sigma_{797} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3189} = \frac{\langle x := x + 1, \sigma_{797} \rangle \Rightarrow \sigma_{798} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{797} \rangle \Rightarrow \langle S_6, \sigma_{798} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3188} = \frac{\mathcal{T}_{3189}}{\langle S_{13}, \sigma_{797} \rangle \Rightarrow \langle S_{15}, \sigma_{798} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3190} = \frac{\langle S_6, \sigma_{798} \rangle \Rightarrow \langle S_8, \sigma_{798} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{798} \rangle \Rightarrow \langle S_9, \sigma_{798} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3192} = \langle S_8, \sigma_{798} \rangle \Rightarrow \langle y := y + 1, \sigma_{798} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3191} = \frac{\mathcal{T}_{3192}}{\langle S_9, \sigma_{798} \rangle \Rightarrow \langle S_{10}, \sigma_{798} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3193} = \frac{\langle y := y + 1, \sigma_{798} \rangle \Rightarrow \sigma_{799} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{798} \rangle \Rightarrow \langle S_{11}, \sigma_{799} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{3194} = \langle S_{11}, \sigma_{799} \rangle \Rightarrow \langle S_{12}, \sigma_{799} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3195} = \langle S_{12}, \sigma_{799} \rangle \Rightarrow \langle S_{13}, \sigma_{799} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3197} = \frac{\langle x := x + 1, \sigma_{799} \rangle \Rightarrow \sigma_{800} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{799} \rangle \Rightarrow \langle S_6, \sigma_{800} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3196} = \frac{\mathcal{T}_{3197}}{\langle S_{13}, \sigma_{799} \rangle \Rightarrow \langle S_{15}, \sigma_{800} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3198} = \frac{\langle S_6, \sigma_{800} \rangle \Rightarrow \langle S_8, \sigma_{800} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{800} \rangle \Rightarrow \langle S_9, \sigma_{800} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3200} = \langle S_8, \sigma_{800} \rangle \Rightarrow \langle y := y + 1, \sigma_{800} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3199} = \frac{\mathcal{T}_{3200}}{\langle S_9, \sigma_{800} \rangle \Rightarrow \langle S_{10}, \sigma_{800} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3201} = \frac{\langle y := y + 1, \sigma_{800} \rangle \Rightarrow \sigma_{801} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{800} \rangle \Rightarrow \langle S_{11}, \sigma_{801} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3202} = \langle S_{11}, \sigma_{801} \rangle \Rightarrow \langle S_{12}, \sigma_{801} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3203} = \langle S_{12}, \sigma_{801} \rangle \Rightarrow \langle S_{13}, \sigma_{801} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3205} = \frac{\langle x := x + 1, \sigma_{801} \rangle \Rightarrow \sigma_{802} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{801} \rangle \Rightarrow \langle S_6, \sigma_{802} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3204} = \frac{\mathcal{T}_{3205}}{\langle S_{13}, \sigma_{801} \rangle \Rightarrow \langle S_{15}, \sigma_{802} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3206} = \frac{\langle S_6, \sigma_{802} \rangle \Rightarrow \langle S_8, \sigma_{802} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{802} \rangle \Rightarrow \langle S_9, \sigma_{802} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3208} = \langle S_8, \sigma_{802} \rangle \Rightarrow \langle y := y + 1, \sigma_{802} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3207} = \frac{\mathcal{T}_{3208}}{\langle S_9, \sigma_{802} \rangle \Rightarrow \langle S_{10}, \sigma_{802} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3209} = \frac{\langle y := y + 1, \sigma_{802} \rangle \Rightarrow \sigma_{803} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{802} \rangle \Rightarrow \langle S_{11}, \sigma_{803} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3210} = \langle S_{11}, \sigma_{803} \rangle \Rightarrow \langle S_{12}, \sigma_{803} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3211} = \langle S_{12}, \sigma_{803} \rangle \Rightarrow \langle S_{13}, \sigma_{803} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3213} = \frac{\langle x := x + 1, \sigma_{803} \rangle \Rightarrow \sigma_{804} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{803} \rangle \Rightarrow \langle S_6, \sigma_{804} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3212} = \frac{\mathcal{T}_{3213}}{\langle S_{13}, \sigma_{803} \rangle \Rightarrow \langle S_{15}, \sigma_{804} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3214} = \frac{\langle S_6, \sigma_{804} \rangle \Rightarrow \langle S_8, \sigma_{804} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{804} \rangle \Rightarrow \langle S_9, \sigma_{804} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3216} = \langle S_8, \sigma_{804} \rangle \Rightarrow \langle y := y + 1, \sigma_{804} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3215} = \frac{\mathcal{T}_{3216}}{\langle S_9, \sigma_{804} \rangle \Rightarrow \langle S_{10}, \sigma_{804} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3217} = \frac{\langle y := y + 1, \sigma_{804} \rangle \Rightarrow \sigma_{805} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{804} \rangle \Rightarrow \langle S_{11}, \sigma_{805} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3218} = \langle S_{11}, \sigma_{805} \rangle \Rightarrow \langle S_{12}, \sigma_{805} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3219} = \langle S_{12}, \sigma_{805} \rangle \Rightarrow \langle S_{13}, \sigma_{805} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3221} = \frac{\langle x := x + 1, \sigma_{805} \rangle \Rightarrow \sigma_{806} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{805} \rangle \Rightarrow \langle S_6, \sigma_{806} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3220} = \frac{\mathcal{T}_{3221}}{\langle S_{13}, \sigma_{805} \rangle \Rightarrow \langle S_{15}, \sigma_{806} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3222} = \frac{\langle S_6, \sigma_{806} \rangle \Rightarrow \langle S_8, \sigma_{806} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{806} \rangle \Rightarrow \langle S_9, \sigma_{806} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3224} = \langle S_8, \sigma_{806} \rangle \Rightarrow \langle y := y + 1, \sigma_{806} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3223} = \frac{\mathcal{T}_{3224}}{\langle S_9, \sigma_{806} \rangle \Rightarrow \langle S_{10}, \sigma_{806} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3225} = \frac{\langle y := y + 1, \sigma_{806} \rangle \Rightarrow \sigma_{807} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{806} \rangle \Rightarrow \langle S_{11}, \sigma_{807} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3226} = \langle S_{11}, \sigma_{807} \rangle \Rightarrow \langle S_{12}, \sigma_{807} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3227} = \langle S_{12}, \sigma_{807} \rangle \Rightarrow \langle S_{13}, \sigma_{807} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3229} = \frac{\langle x := x + 1, \sigma_{807} \rangle \Rightarrow \sigma_{808} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{807} \rangle \Rightarrow \langle S_6, \sigma_{808} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3228} = \frac{\mathcal{T}_{3229}}{\langle S_{13}, \sigma_{807} \rangle \Rightarrow \langle S_{15}, \sigma_{808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3230} = \frac{\langle S_6, \sigma_{808} \rangle \Rightarrow \langle S_8, \sigma_{808} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{808} \rangle \Rightarrow \langle S_9, \sigma_{808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3232} = \langle S_8, \sigma_{808} \rangle \Rightarrow \langle y := y + 1, \sigma_{808} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3231} = \frac{\mathcal{T}_{3232}}{\langle S_9, \sigma_{808} \rangle \Rightarrow \langle S_{10}, \sigma_{808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3233} = \frac{\langle y := y + 1, \sigma_{808} \rangle \Rightarrow \sigma_{809} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{808} \rangle \Rightarrow \langle S_{11}, \sigma_{809} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3234} = \langle S_{11}, \sigma_{809} \rangle \Rightarrow \langle S_{12}, \sigma_{809} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3235} = \langle S_{12}, \sigma_{809} \rangle \Rightarrow \langle S_{13}, \sigma_{809} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3237} = \frac{\langle x := x + 1, \sigma_{809} \rangle \Rightarrow \sigma_{810} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{809} \rangle \Rightarrow \langle S_6, \sigma_{810} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3236} = \frac{\mathcal{T}_{3237}}{\langle S_{13}, \sigma_{809} \rangle \Rightarrow \langle S_{15}, \sigma_{810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3238} = \frac{\langle S_6, \sigma_{810} \rangle \Rightarrow \langle S_8, \sigma_{810} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{810} \rangle \Rightarrow \langle S_9, \sigma_{810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3240} = \langle S_8, \sigma_{810} \rangle \Rightarrow \langle y := y + 1, \sigma_{810} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3239} = \frac{\mathcal{T}_{3240}}{\langle S_9, \sigma_{810} \rangle \Rightarrow \langle S_{10}, \sigma_{810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3241} = \frac{\langle y := y + 1, \sigma_{810} \rangle \Rightarrow \sigma_{811} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{810} \rangle \Rightarrow \langle S_{11}, \sigma_{811} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3242} = \langle S_{11}, \sigma_{811} \rangle \Rightarrow \langle S_{12}, \sigma_{811} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3243} = \langle S_{12}, \sigma_{811} \rangle \Rightarrow \langle S_{13}, \sigma_{811} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3245} = \frac{\langle x := x + 1, \sigma_{811} \rangle \Rightarrow \sigma_{812} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{811} \rangle \Rightarrow \langle S_6, \sigma_{812} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3244} = \frac{\mathcal{T}_{3245}}{\langle S_{13}, \sigma_{811} \rangle \Rightarrow \langle S_{15}, \sigma_{812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3246} = \frac{\langle S_6, \sigma_{812} \rangle \Rightarrow \langle S_8, \sigma_{812} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{812} \rangle \Rightarrow \langle S_9, \sigma_{812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3248} = \langle S_8, \sigma_{812} \rangle \Rightarrow \langle y := y + 1, \sigma_{812} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3247} = \frac{\mathcal{T}_{3248}}{\langle S_9, \sigma_{812} \rangle \Rightarrow \langle S_{10}, \sigma_{812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3249} = \frac{\langle y := y + 1, \sigma_{812} \rangle \Rightarrow \sigma_{813} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{812} \rangle \Rightarrow \langle S_{11}, \sigma_{813} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3250} = \langle S_{11}, \sigma_{813} \rangle \Rightarrow \langle S_{12}, \sigma_{813} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3251} = \langle S_{12}, \sigma_{813} \rangle \Rightarrow \langle S_{13}, \sigma_{813} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3253} = \frac{\langle x := x + 1, \sigma_{813} \rangle \Rightarrow \sigma_{814} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{813} \rangle \Rightarrow \langle S_6, \sigma_{814} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3252} = \frac{\mathcal{T}_{3253}}{\langle S_{13}, \sigma_{813} \rangle \Rightarrow \langle S_{15}, \sigma_{814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3254} = \frac{\langle S_6, \sigma_{814} \rangle \Rightarrow \langle S_8, \sigma_{814} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{814} \rangle \Rightarrow \langle S_9, \sigma_{814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3256} = \langle S_8, \sigma_{814} \rangle \Rightarrow \langle y := y + 1, \sigma_{814} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3255} = \frac{\mathcal{T}_{3256}}{\langle S_9, \sigma_{814} \rangle \Rightarrow \langle S_{10}, \sigma_{814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3257} = \frac{\langle y := y + 1, \sigma_{814} \rangle \Rightarrow \sigma_{815} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{814} \rangle \Rightarrow \langle S_{11}, \sigma_{815} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3258} = \langle S_{11}, \sigma_{815} \rangle \Rightarrow \langle S_{12}, \sigma_{815} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3259} = \langle S_{12}, \sigma_{815} \rangle \Rightarrow \langle S_{13}, \sigma_{815} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3261} = \frac{\langle x := x + 1, \sigma_{815} \rangle \Rightarrow \sigma_{816} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{815} \rangle \Rightarrow \langle S_6, \sigma_{816} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3260} = \frac{\mathcal{T}_{3261}}{\langle S_{13}, \sigma_{815} \rangle \Rightarrow \langle S_{15}, \sigma_{816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3262} = \frac{\langle S_6, \sigma_{816} \rangle \Rightarrow \langle S_8, \sigma_{816} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{816} \rangle \Rightarrow \langle S_9, \sigma_{816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3264} = \langle S_8, \sigma_{816} \rangle \Rightarrow \langle y := y + 1, \sigma_{816} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3263} = \frac{\mathcal{T}_{3264}}{\langle S_9, \sigma_{816} \rangle \Rightarrow \langle S_{10}, \sigma_{816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3265} = \frac{\langle y := y + 1, \sigma_{816} \rangle \Rightarrow \sigma_{817} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{816} \rangle \Rightarrow \langle S_{11}, \sigma_{817} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3266} = \langle S_{11}, \sigma_{817} \rangle \Rightarrow \langle S_{12}, \sigma_{817} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3267} = \langle S_{12}, \sigma_{817} \rangle \Rightarrow \langle S_{13}, \sigma_{817} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3269} = \frac{\langle x := x + 1, \sigma_{817} \rangle \Rightarrow \sigma_{818} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{817} \rangle \Rightarrow \langle S_6, \sigma_{818} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3268} = \frac{\mathcal{T}_{3269}}{\langle S_{13}, \sigma_{817} \rangle \Rightarrow \langle S_{15}, \sigma_{818} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3270} = \frac{\langle S_6, \sigma_{818} \rangle \Rightarrow \langle S_8, \sigma_{818} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{818} \rangle \Rightarrow \langle S_9, \sigma_{818} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3272} = \langle S_8, \sigma_{818} \rangle \Rightarrow \langle y := y + 1, \sigma_{818} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3271} = \frac{\mathcal{T}_{3272}}{\langle S_9, \sigma_{818} \rangle \Rightarrow \langle S_{10}, \sigma_{818} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3273} = \frac{\langle y := y + 1, \sigma_{818} \rangle \Rightarrow \sigma_{819} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{818} \rangle \Rightarrow \langle S_{11}, \sigma_{819} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3274} = \langle S_{11}, \sigma_{819} \rangle \Rightarrow \langle S_{12}, \sigma_{819} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3275} = \langle S_{12}, \sigma_{819} \rangle \Rightarrow \langle S_{13}, \sigma_{819} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3277} = \frac{\langle x := x + 1, \sigma_{819} \rangle \Rightarrow \sigma_{820} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{819} \rangle \Rightarrow \langle S_6, \sigma_{820} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3276} = \frac{\mathcal{T}_{3277}}{\langle S_{13}, \sigma_{819} \rangle \Rightarrow \langle S_{15}, \sigma_{820} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3278} = \frac{\langle S_6, \sigma_{820} \rangle \Rightarrow \langle S_8, \sigma_{820} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{820} \rangle \Rightarrow \langle S_9, \sigma_{820} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3280} = \langle S_8, \sigma_{820} \rangle \Rightarrow \langle y := y + 1, \sigma_{820} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3279} = \frac{\mathcal{T}_{3280}}{\langle S_9, \sigma_{820} \rangle \Rightarrow \langle S_{10}, \sigma_{820} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3281} = \frac{\langle y := y + 1, \sigma_{820} \rangle \Rightarrow \sigma_{821} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{820} \rangle \Rightarrow \langle S_{11}, \sigma_{821} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3282} = \langle S_{11}, \sigma_{821} \rangle \Rightarrow \langle S_{12}, \sigma_{821} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3283} = \langle S_{12}, \sigma_{821} \rangle \Rightarrow \langle S_{13}, \sigma_{821} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3285} = \frac{\langle x := x + 1, \sigma_{821} \rangle \Rightarrow \sigma_{822} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{821} \rangle \Rightarrow \langle S_6, \sigma_{822} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3284} = \frac{\mathcal{T}_{3285}}{\langle S_{13}, \sigma_{821} \rangle \Rightarrow \langle S_{15}, \sigma_{822} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3286} = \frac{\langle S_6, \sigma_{822} \rangle \Rightarrow \langle S_8, \sigma_{822} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{822} \rangle \Rightarrow \langle S_9, \sigma_{822} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3288} = \langle S_8, \sigma_{822} \rangle \Rightarrow \langle y := y + 1, \sigma_{822} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3287} = \frac{\mathcal{T}_{3288}}{\langle S_9, \sigma_{822} \rangle \Rightarrow \langle S_{10}, \sigma_{822} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3289} = \frac{\langle y := y + 1, \sigma_{822} \rangle \Rightarrow \sigma_{823} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{822} \rangle \Rightarrow \langle S_{11}, \sigma_{823} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3290} = \langle S_{11}, \sigma_{823} \rangle \Rightarrow \langle S_{12}, \sigma_{823} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3291} = \langle S_{12}, \sigma_{823} \rangle \Rightarrow \langle S_{13}, \sigma_{823} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3293} = \frac{\langle x := x + 1, \sigma_{823} \rangle \Rightarrow \sigma_{824} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{823} \rangle \Rightarrow \langle S_6, \sigma_{824} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3292} = \frac{\mathcal{T}_{3293}}{\langle S_{13}, \sigma_{823} \rangle \Rightarrow \langle S_{15}, \sigma_{824} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3294} = \frac{\langle S_6, \sigma_{824} \rangle \Rightarrow \langle S_8, \sigma_{824} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{824} \rangle \Rightarrow \langle S_9, \sigma_{824} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3296} = \langle S_8, \sigma_{824} \rangle \Rightarrow \langle y := y + 1, \sigma_{824} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3295} = \frac{\mathcal{T}_{3296}}{\langle S_9, \sigma_{824} \rangle \Rightarrow \langle S_{10}, \sigma_{824} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3297} = \frac{\langle y := y + 1, \sigma_{824} \rangle \Rightarrow \sigma_{825} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{824} \rangle \Rightarrow \langle S_{11}, \sigma_{825} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3298} = \langle S_{11}, \sigma_{825} \rangle \Rightarrow \langle S_{12}, \sigma_{825} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3299} = \langle S_{12}, \sigma_{825} \rangle \Rightarrow \langle S_{13}, \sigma_{825} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3301} = \frac{\langle x := x + 1, \sigma_{825} \rangle \Rightarrow \sigma_{826} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{825} \rangle \Rightarrow \langle S_6, \sigma_{826} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3300} = \frac{\mathcal{T}_{3301}}{\langle S_{13}, \sigma_{825} \rangle \Rightarrow \langle S_{15}, \sigma_{826} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3302} = \frac{\langle S_6, \sigma_{826} \rangle \Rightarrow \langle S_8, \sigma_{826} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{826} \rangle \Rightarrow \langle S_9, \sigma_{826} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3304} = \langle S_8, \sigma_{826} \rangle \Rightarrow \langle y := y + 1, \sigma_{826} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3303} = \frac{\mathcal{T}_{3304}}{\langle S_9, \sigma_{826} \rangle \Rightarrow \langle S_{10}, \sigma_{826} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3305} = \frac{\langle y := y + 1, \sigma_{826} \rangle \Rightarrow \sigma_{827} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{826} \rangle \Rightarrow \langle S_{11}, \sigma_{827} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3306} = \langle S_{11}, \sigma_{827} \rangle \Rightarrow \langle S_{12}, \sigma_{827} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3307} = \langle S_{12}, \sigma_{827} \rangle \Rightarrow \langle S_{13}, \sigma_{827} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3309} = \frac{\langle x := x + 1, \sigma_{827} \rangle \Rightarrow \sigma_{828} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{827} \rangle \Rightarrow \langle S_6, \sigma_{828} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3308} = \frac{\mathcal{T}_{3309}}{\langle S_{13}, \sigma_{827} \rangle \Rightarrow \langle S_{15}, \sigma_{828} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3310} = \frac{\langle S_6, \sigma_{828} \rangle \Rightarrow \langle S_8, \sigma_{828} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{828} \rangle \Rightarrow \langle S_9, \sigma_{828} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3312} = \langle S_8, \sigma_{828} \rangle \Rightarrow \langle y := y + 1, \sigma_{828} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3311} = \frac{\mathcal{T}_{3312}}{\langle S_9, \sigma_{828} \rangle \Rightarrow \langle S_{10}, \sigma_{828} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3313} = \frac{\langle y := y + 1, \sigma_{828} \rangle \Rightarrow \sigma_{829} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{828} \rangle \Rightarrow \langle S_{11}, \sigma_{829} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3314} = \langle S_{11}, \sigma_{829} \rangle \Rightarrow \langle S_{12}, \sigma_{829} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3315} = \langle S_{12}, \sigma_{829} \rangle \Rightarrow \langle S_{13}, \sigma_{829} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3317} = \frac{\langle x := x + 1, \sigma_{829} \rangle \Rightarrow \sigma_{830} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{829} \rangle \Rightarrow \langle S_6, \sigma_{830} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3316} = \frac{\mathcal{T}_{3317}}{\langle S_{13}, \sigma_{829} \rangle \Rightarrow \langle S_{15}, \sigma_{830} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3318} = \frac{\langle S_6, \sigma_{830} \rangle \Rightarrow \langle S_8, \sigma_{830} \rangle [\text{assert}_{\text{tos}}]}{\langle S_7, \sigma_{830} \rangle \Rightarrow \langle S_9, \sigma_{830} \rangle} [\text{comp}_{\text{tos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3320} = \langle S_8, \sigma_{830} \rangle \Rightarrow \langle y := y + 1, \sigma_{830} \rangle [\text{if} - \text{tt}_{\text{tos}}]$$

$$\mathcal{T}_{3319} = \frac{\mathcal{T}_{3320}}{\langle S_9, \sigma_{830} \rangle \Rightarrow \langle S_{10}, \sigma_{830} \rangle} [\text{comp}_{\text{tos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3321} = \frac{\langle y := y + 1, \sigma_{830} \rangle \Rightarrow \sigma_{831} [\text{ass}_{\text{tos}}]}{\langle S_{10}, \sigma_{830} \rangle \Rightarrow \langle S_{11}, \sigma_{831} \rangle} [\text{comp}_{\text{tos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3322} = \langle S_{11}, \sigma_{831} \rangle \Rightarrow \langle S_{12}, \sigma_{831} \rangle [\text{for}_{\text{tos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3323} = \langle S_{12}, \sigma_{831} \rangle \Rightarrow \langle S_{13}, \sigma_{831} \rangle [\text{if} - \text{tt}_{\text{tos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3325} = \frac{\langle x := x + 1, \sigma_{831} \rangle \Rightarrow \sigma_{832} [\text{ass}_{\text{tos}}]}{\langle S_{14}, \sigma_{831} \rangle \Rightarrow \langle S_6, \sigma_{832} \rangle} [\text{comp}_{\text{tos}}^2]$$

$$\mathcal{T}_{3324} = \frac{\mathcal{T}_{3325}}{\langle S_{13}, \sigma_{831} \rangle \Rightarrow \langle S_{15}, \sigma_{832} \rangle} [\text{comp}_{\text{tos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3326} = \frac{\langle S_6, \sigma_{832} \rangle \Rightarrow \langle S_8, \sigma_{832} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{832} \rangle \Rightarrow \langle S_9, \sigma_{832} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3328} = \langle S_8, \sigma_{832} \rangle \Rightarrow \langle y := y + 1, \sigma_{832} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3327} = \frac{\mathcal{T}_{3328}}{\langle S_9, \sigma_{832} \rangle \Rightarrow \langle S_{10}, \sigma_{832} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3329} = \frac{\langle y := y + 1, \sigma_{832} \rangle \Rightarrow \sigma_{833} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{832} \rangle \Rightarrow \langle S_{11}, \sigma_{833} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3330} = \langle S_{11}, \sigma_{833} \rangle \Rightarrow \langle S_{12}, \sigma_{833} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3331} = \langle S_{12}, \sigma_{833} \rangle \Rightarrow \langle S_{13}, \sigma_{833} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3333} = \frac{\langle x := x + 1, \sigma_{833} \rangle \Rightarrow \sigma_{834} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{833} \rangle \Rightarrow \langle S_6, \sigma_{834} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3332} = \frac{\mathcal{T}_{3333}}{\langle S_{13}, \sigma_{833} \rangle \Rightarrow \langle S_{15}, \sigma_{834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3334} = \frac{\langle S_6, \sigma_{834} \rangle \Rightarrow \langle S_8, \sigma_{834} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{834} \rangle \Rightarrow \langle S_9, \sigma_{834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3336} = \langle S_8, \sigma_{834} \rangle \Rightarrow \langle y := y + 1, \sigma_{834} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3335} = \frac{\mathcal{T}_{3336}}{\langle S_9, \sigma_{834} \rangle \Rightarrow \langle S_{10}, \sigma_{834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3337} = \frac{\langle y := y + 1, \sigma_{834} \rangle \Rightarrow \sigma_{835} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{834} \rangle \Rightarrow \langle S_{11}, \sigma_{835} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3338} = \langle S_{11}, \sigma_{835} \rangle \Rightarrow \langle S_{12}, \sigma_{835} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3339} = \langle S_{12}, \sigma_{835} \rangle \Rightarrow \langle S_{13}, \sigma_{835} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3341} = \frac{\langle x := x + 1, \sigma_{835} \rangle \Rightarrow \sigma_{836} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{835} \rangle \Rightarrow \langle S_6, \sigma_{836} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3340} = \frac{\mathcal{T}_{3341}}{\langle S_{13}, \sigma_{835} \rangle \Rightarrow \langle S_{15}, \sigma_{836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3342} = \frac{\langle S_6, \sigma_{836} \rangle \Rightarrow \langle S_8, \sigma_{836} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{836} \rangle \Rightarrow \langle S_9, \sigma_{836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3344} = \langle S_8, \sigma_{836} \rangle \Rightarrow \langle y := y + 1, \sigma_{836} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3343} = \frac{\mathcal{T}_{3344}}{\langle S_9, \sigma_{836} \rangle \Rightarrow \langle S_{10}, \sigma_{836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3345} = \frac{\langle y := y + 1, \sigma_{836} \rangle \Rightarrow \sigma_{837} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{836} \rangle \Rightarrow \langle S_{11}, \sigma_{837} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3346} = \langle S_{11}, \sigma_{837} \rangle \Rightarrow \langle S_{12}, \sigma_{837} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3347} = \langle S_{12}, \sigma_{837} \rangle \Rightarrow \langle S_{13}, \sigma_{837} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3349} = \frac{\langle x := x + 1, \sigma_{837} \rangle \Rightarrow \sigma_{838} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{837} \rangle \Rightarrow \langle S_6, \sigma_{838} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3348} = \frac{\mathcal{T}_{3349}}{\langle S_{13}, \sigma_{837} \rangle \Rightarrow \langle S_{15}, \sigma_{838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3350} = \frac{\langle S_6, \sigma_{838} \rangle \Rightarrow \langle S_8, \sigma_{838} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{838} \rangle \Rightarrow \langle S_9, \sigma_{838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3352} = \langle S_8, \sigma_{838} \rangle \Rightarrow \langle y := y + 1, \sigma_{838} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3351} = \frac{\mathcal{T}_{3352}}{\langle S_9, \sigma_{838} \rangle \Rightarrow \langle S_{10}, \sigma_{838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3353} = \frac{\langle y := y + 1, \sigma_{838} \rangle \Rightarrow \sigma_{839} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{838} \rangle \Rightarrow \langle S_{11}, \sigma_{839} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3354} = \langle S_{11}, \sigma_{839} \rangle \Rightarrow \langle S_{12}, \sigma_{839} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3355} = \langle S_{12}, \sigma_{839} \rangle \Rightarrow \langle S_{13}, \sigma_{839} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3357} = \frac{\langle x := x + 1, \sigma_{839} \rangle \Rightarrow \sigma_{840} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{839} \rangle \Rightarrow \langle S_6, \sigma_{840} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3356} = \frac{\mathcal{T}_{3357}}{\langle S_{13}, \sigma_{839} \rangle \Rightarrow \langle S_{15}, \sigma_{840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3358} = \frac{\langle S_6, \sigma_{840} \rangle \Rightarrow \langle S_8, \sigma_{840} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{840} \rangle \Rightarrow \langle S_9, \sigma_{840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3360} = \langle S_8, \sigma_{840} \rangle \Rightarrow \langle y := y + 1, \sigma_{840} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3359} = \frac{\mathcal{T}_{3360}}{\langle S_9, \sigma_{840} \rangle \Rightarrow \langle S_{10}, \sigma_{840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3361} = \frac{\langle y := y + 1, \sigma_{840} \rangle \Rightarrow \sigma_{841} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{840} \rangle \Rightarrow \langle S_{11}, \sigma_{841} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3362} = \langle S_{11}, \sigma_{841} \rangle \Rightarrow \langle S_{12}, \sigma_{841} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3363} = \langle S_{12}, \sigma_{841} \rangle \Rightarrow \langle S_{13}, \sigma_{841} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3365} = \frac{\langle x := x + 1, \sigma_{841} \rangle \Rightarrow \sigma_{842} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{841} \rangle \Rightarrow \langle S_6, \sigma_{842} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3364} = \frac{\mathcal{T}_{3365}}{\langle S_{13}, \sigma_{841} \rangle \Rightarrow \langle S_{15}, \sigma_{842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3366} = \frac{\langle S_6, \sigma_{842} \rangle \Rightarrow \langle S_8, \sigma_{842} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{842} \rangle \Rightarrow \langle S_9, \sigma_{842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3368} = \langle S_8, \sigma_{842} \rangle \Rightarrow \langle y := y + 1, \sigma_{842} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3367} = \frac{\mathcal{T}_{3368}}{\langle S_9, \sigma_{842} \rangle \Rightarrow \langle S_{10}, \sigma_{842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3369} = \frac{\langle y := y + 1, \sigma_{842} \rangle \Rightarrow \sigma_{843} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{842} \rangle \Rightarrow \langle S_{11}, \sigma_{843} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3370} = \langle S_{11}, \sigma_{843} \rangle \Rightarrow \langle S_{12}, \sigma_{843} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3371} = \langle S_{12}, \sigma_{843} \rangle \Rightarrow \langle S_{13}, \sigma_{843} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3373} = \frac{\langle x := x + 1, \sigma_{843} \rangle \Rightarrow \sigma_{844} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{843} \rangle \Rightarrow \langle S_6, \sigma_{844} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3372} = \frac{\mathcal{T}_{3373}}{\langle S_{13}, \sigma_{843} \rangle \Rightarrow \langle S_{15}, \sigma_{844} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3374} = \frac{\langle S_6, \sigma_{844} \rangle \Rightarrow \langle S_8, \sigma_{844} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{844} \rangle \Rightarrow \langle S_9, \sigma_{844} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3376} = \langle S_8, \sigma_{844} \rangle \Rightarrow \langle y := y + 1, \sigma_{844} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3375} = \frac{\mathcal{T}_{3376}}{\langle S_9, \sigma_{844} \rangle \Rightarrow \langle S_{10}, \sigma_{844} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3377} = \frac{\langle y := y + 1, \sigma_{844} \rangle \Rightarrow \sigma_{845} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{844} \rangle \Rightarrow \langle S_{11}, \sigma_{845} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3378} = \langle S_{11}, \sigma_{845} \rangle \Rightarrow \langle S_{12}, \sigma_{845} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3379} = \langle S_{12}, \sigma_{845} \rangle \Rightarrow \langle S_{13}, \sigma_{845} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3381} = \frac{\langle x := x + 1, \sigma_{845} \rangle \Rightarrow \sigma_{846} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{845} \rangle \Rightarrow \langle S_6, \sigma_{846} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3380} = \frac{\mathcal{T}_{3381}}{\langle S_{13}, \sigma_{845} \rangle \Rightarrow \langle S_{15}, \sigma_{846} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3382} = \frac{\langle S_6, \sigma_{846} \rangle \Rightarrow \langle S_8, \sigma_{846} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{846} \rangle \Rightarrow \langle S_9, \sigma_{846} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3384} = \langle S_8, \sigma_{846} \rangle \Rightarrow \langle y := y + 1, \sigma_{846} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3383} = \frac{\mathcal{T}_{3384}}{\langle S_9, \sigma_{846} \rangle \Rightarrow \langle S_{10}, \sigma_{846} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3385} = \frac{\langle y := y + 1, \sigma_{846} \rangle \Rightarrow \sigma_{847} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{846} \rangle \Rightarrow \langle S_{11}, \sigma_{847} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3386} = \langle S_{11}, \sigma_{847} \rangle \Rightarrow \langle S_{12}, \sigma_{847} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3387} = \langle S_{12}, \sigma_{847} \rangle \Rightarrow \langle S_{13}, \sigma_{847} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3389} = \frac{\langle x := x + 1, \sigma_{847} \rangle \Rightarrow \sigma_{848} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{847} \rangle \Rightarrow \langle S_6, \sigma_{848} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3388} = \frac{\mathcal{T}_{3389}}{\langle S_{13}, \sigma_{847} \rangle \Rightarrow \langle S_{15}, \sigma_{848} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3390} = \frac{\langle S_6, \sigma_{848} \rangle \Rightarrow \langle S_8, \sigma_{848} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{848} \rangle \Rightarrow \langle S_9, \sigma_{848} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3392} = \langle S_8, \sigma_{848} \rangle \Rightarrow \langle y := y + 1, \sigma_{848} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3391} = \frac{\mathcal{T}_{3392}}{\langle S_9, \sigma_{848} \rangle \Rightarrow \langle S_{10}, \sigma_{848} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3393} = \frac{\langle y := y + 1, \sigma_{848} \rangle \Rightarrow \sigma_{849} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{848} \rangle \Rightarrow \langle S_{11}, \sigma_{849} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3394} = \langle S_{11}, \sigma_{849} \rangle \Rightarrow \langle S_{12}, \sigma_{849} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3395} = \langle S_{12}, \sigma_{849} \rangle \Rightarrow \langle S_{13}, \sigma_{849} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3397} = \frac{\langle x := x + 1, \sigma_{849} \rangle \Rightarrow \sigma_{850} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{849} \rangle \Rightarrow \langle S_6, \sigma_{850} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3396} = \frac{\mathcal{T}_{3397}}{\langle S_{13}, \sigma_{849} \rangle \Rightarrow \langle S_{15}, \sigma_{850} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3398} = \frac{\langle S_6, \sigma_{850} \rangle \Rightarrow \langle S_8, \sigma_{850} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{850} \rangle \Rightarrow \langle S_9, \sigma_{850} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3400} = \langle S_8, \sigma_{850} \rangle \Rightarrow \langle y := y + 1, \sigma_{850} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3399} = \frac{\mathcal{T}_{3400}}{\langle S_9, \sigma_{850} \rangle \Rightarrow \langle S_{10}, \sigma_{850} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3401} = \frac{\langle y := y + 1, \sigma_{850} \rangle \Rightarrow \sigma_{851} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{850} \rangle \Rightarrow \langle S_{11}, \sigma_{851} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3402} = \langle S_{11}, \sigma_{851} \rangle \Rightarrow \langle S_{12}, \sigma_{851} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3403} = \langle S_{12}, \sigma_{851} \rangle \Rightarrow \langle S_{13}, \sigma_{851} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3405} = \frac{\langle x := x + 1, \sigma_{851} \rangle \Rightarrow \sigma_{852} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{851} \rangle \Rightarrow \langle S_6, \sigma_{852} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3404} = \frac{\mathcal{T}_{3405}}{\langle S_{13}, \sigma_{851} \rangle \Rightarrow \langle S_{15}, \sigma_{852} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3406} = \frac{\langle S_6, \sigma_{852} \rangle \Rightarrow \langle S_8, \sigma_{852} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{852} \rangle \Rightarrow \langle S_9, \sigma_{852} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3408} = \langle S_8, \sigma_{852} \rangle \Rightarrow \langle y := y + 1, \sigma_{852} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3407} = \frac{\mathcal{T}_{3408}}{\langle S_9, \sigma_{852} \rangle \Rightarrow \langle S_{10}, \sigma_{852} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3409} = \frac{\langle y := y + 1, \sigma_{852} \rangle \Rightarrow \sigma_{853} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{852} \rangle \Rightarrow \langle S_{11}, \sigma_{853} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3410} = \langle S_{11}, \sigma_{853} \rangle \Rightarrow \langle S_{12}, \sigma_{853} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3411} = \langle S_{12}, \sigma_{853} \rangle \Rightarrow \langle S_{13}, \sigma_{853} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3413} = \frac{\langle x := x + 1, \sigma_{853} \rangle \Rightarrow \sigma_{854} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{853} \rangle \Rightarrow \langle S_6, \sigma_{854} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3412} = \frac{\mathcal{T}_{3413}}{\langle S_{13}, \sigma_{853} \rangle \Rightarrow \langle S_{15}, \sigma_{854} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3414} = \frac{\langle S_6, \sigma_{854} \rangle \Rightarrow \langle S_8, \sigma_{854} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{854} \rangle \Rightarrow \langle S_9, \sigma_{854} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3416} = \langle S_8, \sigma_{854} \rangle \Rightarrow \langle y := y + 1, \sigma_{854} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3415} = \frac{\mathcal{T}_{3416}}{\langle S_9, \sigma_{854} \rangle \Rightarrow \langle S_{10}, \sigma_{854} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3417} = \frac{\langle y := y + 1, \sigma_{854} \rangle \Rightarrow \sigma_{855} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{854} \rangle \Rightarrow \langle S_{11}, \sigma_{855} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3418} = \langle S_{11}, \sigma_{855} \rangle \Rightarrow \langle S_{12}, \sigma_{855} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3419} = \langle S_{12}, \sigma_{855} \rangle \Rightarrow \langle S_{13}, \sigma_{855} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3421} = \frac{\langle x := x + 1, \sigma_{855} \rangle \Rightarrow \sigma_{856} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{855} \rangle \Rightarrow \langle S_6, \sigma_{856} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3420} = \frac{\mathcal{T}_{3421}}{\langle S_{13}, \sigma_{855} \rangle \Rightarrow \langle S_{15}, \sigma_{856} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3422} = \frac{\langle S_6, \sigma_{856} \rangle \Rightarrow \langle S_8, \sigma_{856} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{856} \rangle \Rightarrow \langle S_9, \sigma_{856} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3424} = \langle S_8, \sigma_{856} \rangle \Rightarrow \langle y := y + 1, \sigma_{856} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3423} = \frac{\mathcal{T}_{3424}}{\langle S_9, \sigma_{856} \rangle \Rightarrow \langle S_{10}, \sigma_{856} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3425} = \frac{\langle y := y + 1, \sigma_{856} \rangle \Rightarrow \sigma_{857} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{856} \rangle \Rightarrow \langle S_{11}, \sigma_{857} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3426} = \langle S_{11}, \sigma_{857} \rangle \Rightarrow \langle S_{12}, \sigma_{857} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3427} = \langle S_{12}, \sigma_{857} \rangle \Rightarrow \langle S_{13}, \sigma_{857} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3429} = \frac{\langle x := x + 1, \sigma_{857} \rangle \Rightarrow \sigma_{858} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{857} \rangle \Rightarrow \langle S_6, \sigma_{858} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3428} = \frac{\mathcal{T}_{3429}}{\langle S_{13}, \sigma_{857} \rangle \Rightarrow \langle S_{15}, \sigma_{858} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3430} = \frac{\langle S_6, \sigma_{858} \rangle \Rightarrow \langle S_8, \sigma_{858} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{858} \rangle \Rightarrow \langle S_9, \sigma_{858} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3432} = \langle S_8, \sigma_{858} \rangle \Rightarrow \langle y := y + 1, \sigma_{858} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3431} = \frac{\mathcal{T}_{3432}}{\langle S_9, \sigma_{858} \rangle \Rightarrow \langle S_{10}, \sigma_{858} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3433} = \frac{\langle y := y + 1, \sigma_{858} \rangle \Rightarrow \sigma_{859} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{858} \rangle \Rightarrow \langle S_{11}, \sigma_{859} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3434} = \langle S_{11}, \sigma_{859} \rangle \Rightarrow \langle S_{12}, \sigma_{859} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3435} = \langle S_{12}, \sigma_{859} \rangle \Rightarrow \langle S_{13}, \sigma_{859} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3437} = \frac{\langle x := x + 1, \sigma_{859} \rangle \Rightarrow \sigma_{860} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{859} \rangle \Rightarrow \langle S_6, \sigma_{860} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3436} = \frac{\mathcal{T}_{3437}}{\langle S_{13}, \sigma_{859} \rangle \Rightarrow \langle S_{15}, \sigma_{860} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3438} = \frac{\langle S_6, \sigma_{860} \rangle \Rightarrow \langle S_8, \sigma_{860} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{860} \rangle \Rightarrow \langle S_9, \sigma_{860} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3440} = \langle S_8, \sigma_{860} \rangle \Rightarrow \langle y := y + 1, \sigma_{860} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3439} = \frac{\mathcal{T}_{3440}}{\langle S_9, \sigma_{860} \rangle \Rightarrow \langle S_{10}, \sigma_{860} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3441} = \frac{\langle y := y + 1, \sigma_{860} \rangle \Rightarrow \sigma_{861} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{860} \rangle \Rightarrow \langle S_{11}, \sigma_{861} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3442} = \langle S_{11}, \sigma_{861} \rangle \Rightarrow \langle S_{12}, \sigma_{861} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3443} = \langle S_{12}, \sigma_{861} \rangle \Rightarrow \langle S_{13}, \sigma_{861} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3445} = \frac{\langle x := x + 1, \sigma_{861} \rangle \Rightarrow \sigma_{862} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{861} \rangle \Rightarrow \langle S_6, \sigma_{862} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3444} = \frac{\mathcal{T}_{3445}}{\langle S_{13}, \sigma_{861} \rangle \Rightarrow \langle S_{15}, \sigma_{862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3446} = \frac{\langle S_6, \sigma_{862} \rangle \Rightarrow \langle S_8, \sigma_{862} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{862} \rangle \Rightarrow \langle S_9, \sigma_{862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3448} = \langle S_8, \sigma_{862} \rangle \Rightarrow \langle y := y + 1, \sigma_{862} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3447} = \frac{\mathcal{T}_{3448}}{\langle S_9, \sigma_{862} \rangle \Rightarrow \langle S_{10}, \sigma_{862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3449} = \frac{\langle y := y + 1, \sigma_{862} \rangle \Rightarrow \sigma_{863} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{862} \rangle \Rightarrow \langle S_{11}, \sigma_{863} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3450} = \langle S_{11}, \sigma_{863} \rangle \Rightarrow \langle S_{12}, \sigma_{863} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3451} = \langle S_{12}, \sigma_{863} \rangle \Rightarrow \langle S_{13}, \sigma_{863} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3453} = \frac{\langle x := x + 1, \sigma_{863} \rangle \Rightarrow \sigma_{864} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{863} \rangle \Rightarrow \langle S_6, \sigma_{864} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3452} = \frac{\mathcal{T}_{3453}}{\langle S_{13}, \sigma_{863} \rangle \Rightarrow \langle S_{15}, \sigma_{864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3454} = \frac{\langle S_6, \sigma_{864} \rangle \Rightarrow \langle S_8, \sigma_{864} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{864} \rangle \Rightarrow \langle S_9, \sigma_{864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3456} = \langle S_8, \sigma_{864} \rangle \Rightarrow \langle y := y + 1, \sigma_{864} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3455} = \frac{\mathcal{T}_{3456}}{\langle S_9, \sigma_{864} \rangle \Rightarrow \langle S_{10}, \sigma_{864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3457} = \frac{\langle y := y + 1, \sigma_{864} \rangle \Rightarrow \sigma_{865} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{864} \rangle \Rightarrow \langle S_{11}, \sigma_{865} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3458} = \langle S_{11}, \sigma_{865} \rangle \Rightarrow \langle S_{12}, \sigma_{865} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3459} = \langle S_{12}, \sigma_{865} \rangle \Rightarrow \langle S_{13}, \sigma_{865} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3461} = \frac{\langle x := x + 1, \sigma_{865} \rangle \Rightarrow \sigma_{866} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{865} \rangle \Rightarrow \langle S_6, \sigma_{866} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3460} = \frac{\mathcal{T}_{3461}}{\langle S_{13}, \sigma_{865} \rangle \Rightarrow \langle S_{15}, \sigma_{866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3462} = \frac{\langle S_6, \sigma_{866} \rangle \Rightarrow \langle S_8, \sigma_{866} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{866} \rangle \Rightarrow \langle S_9, \sigma_{866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3464} = \langle S_8, \sigma_{866} \rangle \Rightarrow \langle y := y + 1, \sigma_{866} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3463} = \frac{\mathcal{T}_{3464}}{\langle S_9, \sigma_{866} \rangle \Rightarrow \langle S_{10}, \sigma_{866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3465} = \frac{\langle y := y + 1, \sigma_{866} \rangle \Rightarrow \sigma_{867} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{866} \rangle \Rightarrow \langle S_{11}, \sigma_{867} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3466} = \langle S_{11}, \sigma_{867} \rangle \Rightarrow \langle S_{12}, \sigma_{867} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3467} = \langle S_{12}, \sigma_{867} \rangle \Rightarrow \langle S_{13}, \sigma_{867} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3469} = \frac{\langle x := x + 1, \sigma_{867} \rangle \Rightarrow \sigma_{868} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{867} \rangle \Rightarrow \langle S_6, \sigma_{868} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3468} = \frac{\mathcal{T}_{3469}}{\langle S_{13}, \sigma_{867} \rangle \Rightarrow \langle S_{15}, \sigma_{868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3470} = \frac{\langle S_6, \sigma_{868} \rangle \Rightarrow \langle S_8, \sigma_{868} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{868} \rangle \Rightarrow \langle S_9, \sigma_{868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3472} = \langle S_8, \sigma_{868} \rangle \Rightarrow \langle y := y + 1, \sigma_{868} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3471} = \frac{\mathcal{T}_{3472}}{\langle S_9, \sigma_{868} \rangle \Rightarrow \langle S_{10}, \sigma_{868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3473} = \frac{\langle y := y + 1, \sigma_{868} \rangle \Rightarrow \sigma_{869} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{868} \rangle \Rightarrow \langle S_{11}, \sigma_{869} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3474} = \langle S_{11}, \sigma_{869} \rangle \Rightarrow \langle S_{12}, \sigma_{869} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3475} = \langle S_{12}, \sigma_{869} \rangle \Rightarrow \langle S_{13}, \sigma_{869} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3477} = \frac{\langle x := x + 1, \sigma_{869} \rangle \Rightarrow \sigma_{870} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{869} \rangle \Rightarrow \langle S_6, \sigma_{870} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3476} = \frac{\mathcal{T}_{3477}}{\langle S_{13}, \sigma_{869} \rangle \Rightarrow \langle S_{15}, \sigma_{870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3478} = \frac{\langle S_6, \sigma_{870} \rangle \Rightarrow \langle S_8, \sigma_{870} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{870} \rangle \Rightarrow \langle S_9, \sigma_{870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3480} = \langle S_8, \sigma_{870} \rangle \Rightarrow \langle y := y + 1, \sigma_{870} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3479} = \frac{\mathcal{T}_{3480}}{\langle S_9, \sigma_{870} \rangle \Rightarrow \langle S_{10}, \sigma_{870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3481} = \frac{\langle y := y + 1, \sigma_{870} \rangle \Rightarrow \sigma_{871} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{870} \rangle \Rightarrow \langle S_{11}, \sigma_{871} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3482} = \langle S_{11}, \sigma_{871} \rangle \Rightarrow \langle S_{12}, \sigma_{871} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3483} = \langle S_{12}, \sigma_{871} \rangle \Rightarrow \langle S_{13}, \sigma_{871} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3485} = \frac{\langle x := x + 1, \sigma_{871} \rangle \Rightarrow \sigma_{872} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{871} \rangle \Rightarrow \langle S_6, \sigma_{872} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3484} = \frac{\mathcal{T}_{3485}}{\langle S_{13}, \sigma_{871} \rangle \Rightarrow \langle S_{15}, \sigma_{872} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3486} = \frac{\langle S_6, \sigma_{872} \rangle \Rightarrow \langle S_8, \sigma_{872} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{872} \rangle \Rightarrow \langle S_9, \sigma_{872} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3488} = \langle S_8, \sigma_{872} \rangle \Rightarrow \langle y := y + 1, \sigma_{872} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3487} = \frac{\mathcal{T}_{3488}}{\langle S_9, \sigma_{872} \rangle \Rightarrow \langle S_{10}, \sigma_{872} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3489} = \frac{\langle y := y + 1, \sigma_{872} \rangle \Rightarrow \sigma_{873} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{872} \rangle \Rightarrow \langle S_{11}, \sigma_{873} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3490} = \langle S_{11}, \sigma_{873} \rangle \Rightarrow \langle S_{12}, \sigma_{873} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3491} = \langle S_{12}, \sigma_{873} \rangle \Rightarrow \langle S_{13}, \sigma_{873} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3493} = \frac{\langle x := x + 1, \sigma_{873} \rangle \Rightarrow \sigma_{874} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{873} \rangle \Rightarrow \langle S_6, \sigma_{874} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3492} = \frac{\mathcal{T}_{3493}}{\langle S_{13}, \sigma_{873} \rangle \Rightarrow \langle S_{15}, \sigma_{874} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3494} = \frac{\langle S_6, \sigma_{874} \rangle \Rightarrow \langle S_8, \sigma_{874} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{874} \rangle \Rightarrow \langle S_9, \sigma_{874} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3496} = \langle S_8, \sigma_{874} \rangle \Rightarrow \langle y := y + 1, \sigma_{874} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3495} = \frac{\mathcal{T}_{3496}}{\langle S_9, \sigma_{874} \rangle \Rightarrow \langle S_{10}, \sigma_{874} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3497} = \frac{\langle y := y + 1, \sigma_{874} \rangle \Rightarrow \sigma_{875} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{874} \rangle \Rightarrow \langle S_{11}, \sigma_{875} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3498} = \langle S_{11}, \sigma_{875} \rangle \Rightarrow \langle S_{12}, \sigma_{875} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3499} = \langle S_{12}, \sigma_{875} \rangle \Rightarrow \langle S_{13}, \sigma_{875} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3501} = \frac{\langle x := x + 1, \sigma_{875} \rangle \Rightarrow \sigma_{876} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{875} \rangle \Rightarrow \langle S_6, \sigma_{876} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3500} = \frac{\mathcal{T}_{3501}}{\langle S_{13}, \sigma_{875} \rangle \Rightarrow \langle S_{15}, \sigma_{876} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3502} = \frac{\langle S_6, \sigma_{876} \rangle \Rightarrow \langle S_8, \sigma_{876} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{876} \rangle \Rightarrow \langle S_9, \sigma_{876} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3504} = \langle S_8, \sigma_{876} \rangle \Rightarrow \langle y := y + 1, \sigma_{876} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3503} = \frac{\mathcal{T}_{3504}}{\langle S_9, \sigma_{876} \rangle \Rightarrow \langle S_{10}, \sigma_{876} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3505} = \frac{\langle y := y + 1, \sigma_{876} \rangle \Rightarrow \sigma_{877} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{876} \rangle \Rightarrow \langle S_{11}, \sigma_{877} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3506} = \langle S_{11}, \sigma_{877} \rangle \Rightarrow \langle S_{12}, \sigma_{877} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3507} = \langle S_{12}, \sigma_{877} \rangle \Rightarrow \langle S_{13}, \sigma_{877} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3509} = \frac{\langle x := x + 1, \sigma_{877} \rangle \Rightarrow \sigma_{878} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{877} \rangle \Rightarrow \langle S_6, \sigma_{878} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3508} = \frac{\mathcal{T}_{3509}}{\langle S_{13}, \sigma_{877} \rangle \Rightarrow \langle S_{15}, \sigma_{878} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3510} = \frac{\langle S_6, \sigma_{878} \rangle \Rightarrow \langle S_8, \sigma_{878} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{878} \rangle \Rightarrow \langle S_9, \sigma_{878} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3512} = \langle S_8, \sigma_{878} \rangle \Rightarrow \langle y := y + 1, \sigma_{878} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3511} = \frac{\mathcal{T}_{3512}}{\langle S_9, \sigma_{878} \rangle \Rightarrow \langle S_{10}, \sigma_{878} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3513} = \frac{\langle y := y + 1, \sigma_{878} \rangle \Rightarrow \sigma_{879} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{878} \rangle \Rightarrow \langle S_{11}, \sigma_{879} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3514} = \langle S_{11}, \sigma_{879} \rangle \Rightarrow \langle S_{12}, \sigma_{879} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3515} = \langle S_{12}, \sigma_{879} \rangle \Rightarrow \langle S_{13}, \sigma_{879} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3517} = \frac{\langle x := x + 1, \sigma_{879} \rangle \Rightarrow \sigma_{880} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{879} \rangle \Rightarrow \langle S_6, \sigma_{880} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3516} = \frac{\mathcal{T}_{3517}}{\langle S_{13}, \sigma_{879} \rangle \Rightarrow \langle S_{15}, \sigma_{880} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3518} = \frac{\langle S_6, \sigma_{880} \rangle \Rightarrow \langle S_8, \sigma_{880} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{880} \rangle \Rightarrow \langle S_9, \sigma_{880} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3520} = \langle S_8, \sigma_{880} \rangle \Rightarrow \langle y := y + 1, \sigma_{880} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3519} = \frac{\mathcal{T}_{3520}}{\langle S_9, \sigma_{880} \rangle \Rightarrow \langle S_{10}, \sigma_{880} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3521} = \frac{\langle y := y + 1, \sigma_{880} \rangle \Rightarrow \sigma_{881} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{880} \rangle \Rightarrow \langle S_{11}, \sigma_{881} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3522} = \langle S_{11}, \sigma_{881} \rangle \Rightarrow \langle S_{12}, \sigma_{881} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3523} = \langle S_{12}, \sigma_{881} \rangle \Rightarrow \langle S_{13}, \sigma_{881} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3525} = \frac{\langle x := x + 1, \sigma_{881} \rangle \Rightarrow \sigma_{882} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{881} \rangle \Rightarrow \langle S_6, \sigma_{882} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3524} = \frac{\mathcal{T}_{3525}}{\langle S_{13}, \sigma_{881} \rangle \Rightarrow \langle S_{15}, \sigma_{882} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3526} = \frac{\langle S_6, \sigma_{882} \rangle \Rightarrow \langle S_8, \sigma_{882} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{882} \rangle \Rightarrow \langle S_9, \sigma_{882} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3528} = \langle S_8, \sigma_{882} \rangle \Rightarrow \langle y := y + 1, \sigma_{882} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3527} = \frac{\mathcal{T}_{3528}}{\langle S_9, \sigma_{882} \rangle \Rightarrow \langle S_{10}, \sigma_{882} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3529} = \frac{\langle y := y + 1, \sigma_{882} \rangle \Rightarrow \sigma_{883} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{882} \rangle \Rightarrow \langle S_{11}, \sigma_{883} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3530} = \langle S_{11}, \sigma_{883} \rangle \Rightarrow \langle S_{12}, \sigma_{883} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3531} = \langle S_{12}, \sigma_{883} \rangle \Rightarrow \langle S_{13}, \sigma_{883} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3533} = \frac{\langle x := x + 1, \sigma_{883} \rangle \Rightarrow \sigma_{884} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{883} \rangle \Rightarrow \langle S_6, \sigma_{884} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3532} = \frac{\mathcal{T}_{3533}}{\langle S_{13}, \sigma_{883} \rangle \Rightarrow \langle S_{15}, \sigma_{884} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3534} = \frac{\langle S_6, \sigma_{884} \rangle \Rightarrow \langle S_8, \sigma_{884} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{884} \rangle \Rightarrow \langle S_9, \sigma_{884} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3536} = \langle S_8, \sigma_{884} \rangle \Rightarrow \langle y := y + 1, \sigma_{884} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3535} = \frac{\mathcal{T}_{3536}}{\langle S_9, \sigma_{884} \rangle \Rightarrow \langle S_{10}, \sigma_{884} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3537} = \frac{\langle y := y + 1, \sigma_{884} \rangle \Rightarrow \sigma_{885} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{884} \rangle \Rightarrow \langle S_{11}, \sigma_{885} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3538} = \langle S_{11}, \sigma_{885} \rangle \Rightarrow \langle S_{12}, \sigma_{885} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3539} = \langle S_{12}, \sigma_{885} \rangle \Rightarrow \langle S_{13}, \sigma_{885} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3541} = \frac{\langle x := x + 1, \sigma_{885} \rangle \Rightarrow \sigma_{886} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{885} \rangle \Rightarrow \langle S_6, \sigma_{886} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3540} = \frac{\mathcal{T}_{3541}}{\langle S_{13}, \sigma_{885} \rangle \Rightarrow \langle S_{15}, \sigma_{886} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3542} = \frac{\langle S_6, \sigma_{886} \rangle \Rightarrow \langle S_8, \sigma_{886} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{886} \rangle \Rightarrow \langle S_9, \sigma_{886} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3544} = \langle S_8, \sigma_{886} \rangle \Rightarrow \langle y := y + 1, \sigma_{886} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3543} = \frac{\mathcal{T}_{3544}}{\langle S_9, \sigma_{886} \rangle \Rightarrow \langle S_{10}, \sigma_{886} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3545} = \frac{\langle y := y + 1, \sigma_{886} \rangle \Rightarrow \sigma_{887} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{886} \rangle \Rightarrow \langle S_{11}, \sigma_{887} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{3546} = \langle S_{11}, \sigma_{887} \rangle \Rightarrow \langle S_{12}, \sigma_{887} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3547} = \langle S_{12}, \sigma_{887} \rangle \Rightarrow \langle S_{13}, \sigma_{887} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3549} = \frac{\langle x := x + 1, \sigma_{887} \rangle \Rightarrow \sigma_{888} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{887} \rangle \Rightarrow \langle S_6, \sigma_{888} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3548} = \frac{\mathcal{T}_{3549}}{\langle S_{13}, \sigma_{887} \rangle \Rightarrow \langle S_{15}, \sigma_{888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3550} = \frac{\langle S_6, \sigma_{888} \rangle \Rightarrow \langle S_8, \sigma_{888} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{888} \rangle \Rightarrow \langle S_9, \sigma_{888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3552} = \langle S_8, \sigma_{888} \rangle \Rightarrow \langle y := y + 1, \sigma_{888} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3551} = \frac{\mathcal{T}_{3552}}{\langle S_9, \sigma_{888} \rangle \Rightarrow \langle S_{10}, \sigma_{888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3553} = \frac{\langle y := y + 1, \sigma_{888} \rangle \Rightarrow \sigma_{889} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{888} \rangle \Rightarrow \langle S_{11}, \sigma_{889} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3554} = \langle S_{11}, \sigma_{889} \rangle \Rightarrow \langle S_{12}, \sigma_{889} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3555} = \langle S_{12}, \sigma_{889} \rangle \Rightarrow \langle S_{13}, \sigma_{889} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3557} = \frac{\langle x := x + 1, \sigma_{889} \rangle \Rightarrow \sigma_{890} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{889} \rangle \Rightarrow \langle S_6, \sigma_{890} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3556} = \frac{\mathcal{T}_{3557}}{\langle S_{13}, \sigma_{889} \rangle \Rightarrow \langle S_{15}, \sigma_{890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3558} = \frac{\langle S_6, \sigma_{890} \rangle \Rightarrow \langle S_8, \sigma_{890} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{890} \rangle \Rightarrow \langle S_9, \sigma_{890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3560} = \langle S_8, \sigma_{890} \rangle \Rightarrow \langle y := y + 1, \sigma_{890} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3559} = \frac{\mathcal{T}_{3560}}{\langle S_9, \sigma_{890} \rangle \Rightarrow \langle S_{10}, \sigma_{890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3561} = \frac{\langle y := y + 1, \sigma_{890} \rangle \Rightarrow \sigma_{891} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{890} \rangle \Rightarrow \langle S_{11}, \sigma_{891} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3562} = \langle S_{11}, \sigma_{891} \rangle \Rightarrow \langle S_{12}, \sigma_{891} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3563} = \langle S_{12}, \sigma_{891} \rangle \Rightarrow \langle S_{13}, \sigma_{891} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3565} = \frac{\langle x := x + 1, \sigma_{891} \rangle \Rightarrow \sigma_{892} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{891} \rangle \Rightarrow \langle S_6, \sigma_{892} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3564} = \frac{\mathcal{T}_{3565}}{\langle S_{13}, \sigma_{891} \rangle \Rightarrow \langle S_{15}, \sigma_{892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3566} = \frac{\langle S_6, \sigma_{892} \rangle \Rightarrow \langle S_8, \sigma_{892} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{892} \rangle \Rightarrow \langle S_9, \sigma_{892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3568} = \langle S_8, \sigma_{892} \rangle \Rightarrow \langle y := y + 1, \sigma_{892} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3567} = \frac{\mathcal{T}_{3568}}{\langle S_9, \sigma_{892} \rangle \Rightarrow \langle S_{10}, \sigma_{892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3569} = \frac{\langle y := y + 1, \sigma_{892} \rangle \Rightarrow \sigma_{893} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{892} \rangle \Rightarrow \langle S_{11}, \sigma_{893} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3570} = \langle S_{11}, \sigma_{893} \rangle \Rightarrow \langle S_{12}, \sigma_{893} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3571} = \langle S_{12}, \sigma_{893} \rangle \Rightarrow \langle S_{13}, \sigma_{893} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3573} = \frac{\langle x := x + 1, \sigma_{893} \rangle \Rightarrow \sigma_{894} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{893} \rangle \Rightarrow \langle S_6, \sigma_{894} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3572} = \frac{\mathcal{T}_{3573}}{\langle S_{13}, \sigma_{893} \rangle \Rightarrow \langle S_{15}, \sigma_{894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3574} = \frac{\langle S_6, \sigma_{894} \rangle \Rightarrow \langle S_8, \sigma_{894} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{894} \rangle \Rightarrow \langle S_9, \sigma_{894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3576} = \langle S_8, \sigma_{894} \rangle \Rightarrow \langle y := y + 1, \sigma_{894} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3575} = \frac{\mathcal{T}_{3576}}{\langle S_9, \sigma_{894} \rangle \Rightarrow \langle S_{10}, \sigma_{894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3577} = \frac{\langle y := y + 1, \sigma_{894} \rangle \Rightarrow \sigma_{895} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{894} \rangle \Rightarrow \langle S_{11}, \sigma_{895} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3578} = \langle S_{11}, \sigma_{895} \rangle \Rightarrow \langle S_{12}, \sigma_{895} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3579} = \langle S_{12}, \sigma_{895} \rangle \Rightarrow \langle S_{13}, \sigma_{895} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3581} = \frac{\langle x := x + 1, \sigma_{895} \rangle \Rightarrow \sigma_{896} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{895} \rangle \Rightarrow \langle S_6, \sigma_{896} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3580} = \frac{\mathcal{T}_{3581}}{\langle S_{13}, \sigma_{895} \rangle \Rightarrow \langle S_{15}, \sigma_{896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3582} = \frac{\langle S_6, \sigma_{896} \rangle \Rightarrow \langle S_8, \sigma_{896} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{896} \rangle \Rightarrow \langle S_9, \sigma_{896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3584} = \langle S_8, \sigma_{896} \rangle \Rightarrow \langle y := y + 1, \sigma_{896} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3583} = \frac{\mathcal{T}_{3584}}{\langle S_9, \sigma_{896} \rangle \Rightarrow \langle S_{10}, \sigma_{896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3585} = \frac{\langle y := y + 1, \sigma_{896} \rangle \Rightarrow \sigma_{897} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{896} \rangle \Rightarrow \langle S_{11}, \sigma_{897} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3586} = \langle S_{11}, \sigma_{897} \rangle \Rightarrow \langle S_{12}, \sigma_{897} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3587} = \langle S_{12}, \sigma_{897} \rangle \Rightarrow \langle S_{13}, \sigma_{897} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3589} = \frac{\langle x := x + 1, \sigma_{897} \rangle \Rightarrow \sigma_{898} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{897} \rangle \Rightarrow \langle S_6, \sigma_{898} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3588} = \frac{\mathcal{T}_{3589}}{\langle S_{13}, \sigma_{897} \rangle \Rightarrow \langle S_{15}, \sigma_{898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3590} = \frac{\langle S_6, \sigma_{898} \rangle \Rightarrow \langle S_8, \sigma_{898} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{898} \rangle \Rightarrow \langle S_9, \sigma_{898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3592} = \langle S_8, \sigma_{898} \rangle \Rightarrow \langle y := y + 1, \sigma_{898} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3591} = \frac{\mathcal{T}_{3592}}{\langle S_9, \sigma_{898} \rangle \Rightarrow \langle S_{10}, \sigma_{898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3593} = \frac{\langle y := y + 1, \sigma_{898} \rangle \Rightarrow \sigma_{899} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{898} \rangle \Rightarrow \langle S_{11}, \sigma_{899} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3594} = \langle S_{11}, \sigma_{899} \rangle \Rightarrow \langle S_{12}, \sigma_{899} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3595} = \langle S_{12}, \sigma_{899} \rangle \Rightarrow \langle S_{13}, \sigma_{899} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3597} = \frac{\langle x := x + 1, \sigma_{899} \rangle \Rightarrow \sigma_{900} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{899} \rangle \Rightarrow \langle S_6, \sigma_{900} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3596} = \frac{\mathcal{T}_{3597}}{\langle S_{13}, \sigma_{899} \rangle \Rightarrow \langle S_{15}, \sigma_{900} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3598} = \frac{\langle S_6, \sigma_{900} \rangle \Rightarrow \langle S_8, \sigma_{900} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{900} \rangle \Rightarrow \langle S_9, \sigma_{900} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3600} = \langle S_8, \sigma_{900} \rangle \Rightarrow \langle y := y + 1, \sigma_{900} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3599} = \frac{\mathcal{T}_{3600}}{\langle S_9, \sigma_{900} \rangle \Rightarrow \langle S_{10}, \sigma_{900} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3601} = \frac{\langle y := y + 1, \sigma_{900} \rangle \Rightarrow \sigma_{901} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{900} \rangle \Rightarrow \langle S_{11}, \sigma_{901} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3602} = \langle S_{11}, \sigma_{901} \rangle \Rightarrow \langle S_{12}, \sigma_{901} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3603} = \langle S_{12}, \sigma_{901} \rangle \Rightarrow \langle S_{13}, \sigma_{901} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3605} = \frac{\langle x := x + 1, \sigma_{901} \rangle \Rightarrow \sigma_{902} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{901} \rangle \Rightarrow \langle S_6, \sigma_{902} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3604} = \frac{\mathcal{T}_{3605}}{\langle S_{13}, \sigma_{901} \rangle \Rightarrow \langle S_{15}, \sigma_{902} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3606} = \frac{\langle S_6, \sigma_{902} \rangle \Rightarrow \langle S_8, \sigma_{902} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{902} \rangle \Rightarrow \langle S_9, \sigma_{902} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3608} = \langle S_8, \sigma_{902} \rangle \Rightarrow \langle y := y + 1, \sigma_{902} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3607} = \frac{\mathcal{T}_{3608}}{\langle S_9, \sigma_{902} \rangle \Rightarrow \langle S_{10}, \sigma_{902} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3609} = \frac{\langle y := y + 1, \sigma_{902} \rangle \Rightarrow \sigma_{903} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{902} \rangle \Rightarrow \langle S_{11}, \sigma_{903} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3610} = \langle S_{11}, \sigma_{903} \rangle \Rightarrow \langle S_{12}, \sigma_{903} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3611} = \langle S_{12}, \sigma_{903} \rangle \Rightarrow \langle S_{13}, \sigma_{903} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3613} = \frac{\langle x := x + 1, \sigma_{903} \rangle \Rightarrow \sigma_{904} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{903} \rangle \Rightarrow \langle S_6, \sigma_{904} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3612} = \frac{\mathcal{T}_{3613}}{\langle S_{13}, \sigma_{903} \rangle \Rightarrow \langle S_{15}, \sigma_{904} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3614} = \frac{\langle S_6, \sigma_{904} \rangle \Rightarrow \langle S_8, \sigma_{904} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{904} \rangle \Rightarrow \langle S_9, \sigma_{904} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3616} = \langle S_8, \sigma_{904} \rangle \Rightarrow \langle y := y + 1, \sigma_{904} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3615} = \frac{\mathcal{T}_{3616}}{\langle S_9, \sigma_{904} \rangle \Rightarrow \langle S_{10}, \sigma_{904} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3617} = \frac{\langle y := y + 1, \sigma_{904} \rangle \Rightarrow \sigma_{905} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{904} \rangle \Rightarrow \langle S_{11}, \sigma_{905} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3618} = \langle S_{11}, \sigma_{905} \rangle \Rightarrow \langle S_{12}, \sigma_{905} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3619} = \langle S_{12}, \sigma_{905} \rangle \Rightarrow \langle S_{13}, \sigma_{905} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3621} = \frac{\langle x := x + 1, \sigma_{905} \rangle \Rightarrow \sigma_{906} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{905} \rangle \Rightarrow \langle S_6, \sigma_{906} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3620} = \frac{\mathcal{T}_{3621}}{\langle S_{13}, \sigma_{905} \rangle \Rightarrow \langle S_{15}, \sigma_{906} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3622} = \frac{\langle S_6, \sigma_{906} \rangle \Rightarrow \langle S_8, \sigma_{906} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{906} \rangle \Rightarrow \langle S_9, \sigma_{906} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3624} = \langle S_8, \sigma_{906} \rangle \Rightarrow \langle y := y + 1, \sigma_{906} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3623} = \frac{\mathcal{T}_{3624}}{\langle S_9, \sigma_{906} \rangle \Rightarrow \langle S_{10}, \sigma_{906} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3625} = \frac{\langle y := y + 1, \sigma_{906} \rangle \Rightarrow \sigma_{907} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{906} \rangle \Rightarrow \langle S_{11}, \sigma_{907} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3626} = \langle S_{11}, \sigma_{907} \rangle \Rightarrow \langle S_{12}, \sigma_{907} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3627} = \langle S_{12}, \sigma_{907} \rangle \Rightarrow \langle S_{13}, \sigma_{907} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3629} = \frac{\langle x := x + 1, \sigma_{907} \rangle \Rightarrow \sigma_{908} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{907} \rangle \Rightarrow \langle S_6, \sigma_{908} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3628} = \frac{\mathcal{T}_{3629}}{\langle S_{13}, \sigma_{907} \rangle \Rightarrow \langle S_{15}, \sigma_{908} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3630} = \frac{\langle S_6, \sigma_{908} \rangle \Rightarrow \langle S_8, \sigma_{908} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{908} \rangle \Rightarrow \langle S_9, \sigma_{908} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3632} = \langle S_8, \sigma_{908} \rangle \Rightarrow \langle y := y + 1, \sigma_{908} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3631} = \frac{\mathcal{T}_{3632}}{\langle S_9, \sigma_{908} \rangle \Rightarrow \langle S_{10}, \sigma_{908} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3633} = \frac{\langle y := y + 1, \sigma_{908} \rangle \Rightarrow \sigma_{909} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{908} \rangle \Rightarrow \langle S_{11}, \sigma_{909} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3634} = \langle S_{11}, \sigma_{909} \rangle \Rightarrow \langle S_{12}, \sigma_{909} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3635} = \langle S_{12}, \sigma_{909} \rangle \Rightarrow \langle S_{13}, \sigma_{909} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3637} = \frac{\langle x := x + 1, \sigma_{909} \rangle \Rightarrow \sigma_{910} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{909} \rangle \Rightarrow \langle S_6, \sigma_{910} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3636} = \frac{\mathcal{T}_{3637}}{\langle S_{13}, \sigma_{909} \rangle \Rightarrow \langle S_{15}, \sigma_{910} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3638} = \frac{\langle S_6, \sigma_{910} \rangle \Rightarrow \langle S_8, \sigma_{910} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{910} \rangle \Rightarrow \langle S_9, \sigma_{910} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3640} = \langle S_8, \sigma_{910} \rangle \Rightarrow \langle y := y + 1, \sigma_{910} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3639} = \frac{\mathcal{T}_{3640}}{\langle S_9, \sigma_{910} \rangle \Rightarrow \langle S_{10}, \sigma_{910} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3641} = \frac{\langle y := y + 1, \sigma_{910} \rangle \Rightarrow \sigma_{911} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{910} \rangle \Rightarrow \langle S_{11}, \sigma_{911} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$\Downarrow$

$$\mathcal{T}_{3642} = \langle S_{11}, \sigma_{911} \rangle \Rightarrow \langle S_{12}, \sigma_{911} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3643} = \langle S_{12}, \sigma_{911} \rangle \Rightarrow \langle S_{13}, \sigma_{911} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3645} = \frac{\langle x := x + 1, \sigma_{911} \rangle \Rightarrow \sigma_{912} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{911} \rangle \Rightarrow \langle S_6, \sigma_{912} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3644} = \frac{\mathcal{T}_{3645}}{\langle S_{13}, \sigma_{911} \rangle \Rightarrow \langle S_{15}, \sigma_{912} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3646} = \frac{\langle S_6, \sigma_{912} \rangle \Rightarrow \langle S_8, \sigma_{912} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{912} \rangle \Rightarrow \langle S_9, \sigma_{912} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3648} = \langle S_8, \sigma_{912} \rangle \Rightarrow \langle y := y + 1, \sigma_{912} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3647} = \frac{\mathcal{T}_{3648}}{\langle S_9, \sigma_{912} \rangle \Rightarrow \langle S_{10}, \sigma_{912} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3649} = \frac{\langle y := y + 1, \sigma_{912} \rangle \Rightarrow \sigma_{913} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{912} \rangle \Rightarrow \langle S_{11}, \sigma_{913} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3650} = \langle S_{11}, \sigma_{913} \rangle \Rightarrow \langle S_{12}, \sigma_{913} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3651} = \langle S_{12}, \sigma_{913} \rangle \Rightarrow \langle S_{13}, \sigma_{913} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3653} = \frac{\langle x := x + 1, \sigma_{913} \rangle \Rightarrow \sigma_{914} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{913} \rangle \Rightarrow \langle S_6, \sigma_{914} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3652} = \frac{\mathcal{T}_{3653}}{\langle S_{13}, \sigma_{913} \rangle \Rightarrow \langle S_{15}, \sigma_{914} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3654} = \frac{\langle S_6, \sigma_{914} \rangle \Rightarrow \langle S_8, \sigma_{914} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{914} \rangle \Rightarrow \langle S_9, \sigma_{914} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3656} = \langle S_8, \sigma_{914} \rangle \Rightarrow \langle y := y + 1, \sigma_{914} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3655} = \frac{\mathcal{T}_{3656}}{\langle S_9, \sigma_{914} \rangle \Rightarrow \langle S_{10}, \sigma_{914} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3657} = \frac{\langle y := y + 1, \sigma_{914} \rangle \Rightarrow \sigma_{915} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{914} \rangle \Rightarrow \langle S_{11}, \sigma_{915} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3658} = \langle S_{11}, \sigma_{915} \rangle \Rightarrow \langle S_{12}, \sigma_{915} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3659} = \langle S_{12}, \sigma_{915} \rangle \Rightarrow \langle S_{13}, \sigma_{915} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3661} = \frac{\langle x := x + 1, \sigma_{915} \rangle \Rightarrow \sigma_{916} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{915} \rangle \Rightarrow \langle S_6, \sigma_{916} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3660} = \frac{\mathcal{T}_{3661}}{\langle S_{13}, \sigma_{915} \rangle \Rightarrow \langle S_{15}, \sigma_{916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3662} = \frac{\langle S_6, \sigma_{916} \rangle \Rightarrow \langle S_8, \sigma_{916} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{916} \rangle \Rightarrow \langle S_9, \sigma_{916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3664} = \langle S_8, \sigma_{916} \rangle \Rightarrow \langle y := y + 1, \sigma_{916} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3663} = \frac{\mathcal{T}_{3664}}{\langle S_9, \sigma_{916} \rangle \Rightarrow \langle S_{10}, \sigma_{916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3665} = \frac{\langle y := y + 1, \sigma_{916} \rangle \Rightarrow \sigma_{917} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{916} \rangle \Rightarrow \langle S_{11}, \sigma_{917} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3666} = \langle S_{11}, \sigma_{917} \rangle \Rightarrow \langle S_{12}, \sigma_{917} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3667} = \langle S_{12}, \sigma_{917} \rangle \Rightarrow \langle S_{13}, \sigma_{917} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3669} = \frac{\langle x := x + 1, \sigma_{917} \rangle \Rightarrow \sigma_{918} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{917} \rangle \Rightarrow \langle S_6, \sigma_{918} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3668} = \frac{\mathcal{T}_{3669}}{\langle S_{13}, \sigma_{917} \rangle \Rightarrow \langle S_{15}, \sigma_{918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3670} = \frac{\langle S_6, \sigma_{918} \rangle \Rightarrow \langle S_8, \sigma_{918} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{918} \rangle \Rightarrow \langle S_9, \sigma_{918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3672} = \langle S_8, \sigma_{918} \rangle \Rightarrow \langle y := y + 1, \sigma_{918} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3671} = \frac{\mathcal{T}_{3672}}{\langle S_9, \sigma_{918} \rangle \Rightarrow \langle S_{10}, \sigma_{918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3673} = \frac{\langle y := y + 1, \sigma_{918} \rangle \Rightarrow \sigma_{919} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{918} \rangle \Rightarrow \langle S_{11}, \sigma_{919} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{3674} = \langle S_{11}, \sigma_{919} \rangle \Rightarrow \langle S_{12}, \sigma_{919} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3675} = \langle S_{12}, \sigma_{919} \rangle \Rightarrow \langle S_{13}, \sigma_{919} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3677} = \frac{\langle x := x + 1, \sigma_{919} \rangle \Rightarrow \sigma_{920} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{919} \rangle \Rightarrow \langle S_6, \sigma_{920} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3676} = \frac{\mathcal{T}_{3677}}{\langle S_{13}, \sigma_{919} \rangle \Rightarrow \langle S_{15}, \sigma_{920} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3678} = \frac{\langle S_6, \sigma_{920} \rangle \Rightarrow \langle S_8, \sigma_{920} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{920} \rangle \Rightarrow \langle S_9, \sigma_{920} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3680} = \langle S_8, \sigma_{920} \rangle \Rightarrow \langle y := y + 1, \sigma_{920} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3679} = \frac{\mathcal{T}_{3680}}{\langle S_9, \sigma_{920} \rangle \Rightarrow \langle S_{10}, \sigma_{920} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3681} = \frac{\langle y := y + 1, \sigma_{920} \rangle \Rightarrow \sigma_{921} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{920} \rangle \Rightarrow \langle S_{11}, \sigma_{921} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3682} = \langle S_{11}, \sigma_{921} \rangle \Rightarrow \langle S_{12}, \sigma_{921} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3683} = \langle S_{12}, \sigma_{921} \rangle \Rightarrow \langle S_{13}, \sigma_{921} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3685} = \frac{\langle x := x + 1, \sigma_{921} \rangle \Rightarrow \sigma_{922} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{921} \rangle \Rightarrow \langle S_6, \sigma_{922} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3684} = \frac{\mathcal{T}_{3685}}{\langle S_{13}, \sigma_{921} \rangle \Rightarrow \langle S_{15}, \sigma_{922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3686} = \frac{\langle S_6, \sigma_{922} \rangle \Rightarrow \langle S_8, \sigma_{922} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{922} \rangle \Rightarrow \langle S_9, \sigma_{922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3688} = \langle S_8, \sigma_{922} \rangle \Rightarrow \langle y := y + 1, \sigma_{922} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3687} = \frac{\mathcal{T}_{3688}}{\langle S_9, \sigma_{922} \rangle \Rightarrow \langle S_{10}, \sigma_{922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3689} = \frac{\langle y := y + 1, \sigma_{922} \rangle \Rightarrow \sigma_{923} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{922} \rangle \Rightarrow \langle S_{11}, \sigma_{923} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3690} = \langle S_{11}, \sigma_{923} \rangle \Rightarrow \langle S_{12}, \sigma_{923} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3691} = \langle S_{12}, \sigma_{923} \rangle \Rightarrow \langle S_{13}, \sigma_{923} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3693} = \frac{\langle x := x + 1, \sigma_{923} \rangle \Rightarrow \sigma_{924} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{923} \rangle \Rightarrow \langle S_6, \sigma_{924} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3692} = \frac{\mathcal{T}_{3693}}{\langle S_{13}, \sigma_{923} \rangle \Rightarrow \langle S_{15}, \sigma_{924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3694} = \frac{\langle S_6, \sigma_{924} \rangle \Rightarrow \langle S_8, \sigma_{924} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{924} \rangle \Rightarrow \langle S_9, \sigma_{924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3696} = \langle S_8, \sigma_{924} \rangle \Rightarrow \langle y := y + 1, \sigma_{924} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3695} = \frac{\mathcal{T}_{3696}}{\langle S_9, \sigma_{924} \rangle \Rightarrow \langle S_{10}, \sigma_{924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3697} = \frac{\langle y := y + 1, \sigma_{924} \rangle \Rightarrow \sigma_{925} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{924} \rangle \Rightarrow \langle S_{11}, \sigma_{925} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3698} = \langle S_{11}, \sigma_{925} \rangle \Rightarrow \langle S_{12}, \sigma_{925} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3699} = \langle S_{12}, \sigma_{925} \rangle \Rightarrow \langle S_{13}, \sigma_{925} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3701} = \frac{\langle x := x + 1, \sigma_{925} \rangle \Rightarrow \sigma_{926} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{925} \rangle \Rightarrow \langle S_6, \sigma_{926} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3700} = \frac{\mathcal{T}_{3701}}{\langle S_{13}, \sigma_{925} \rangle \Rightarrow \langle S_{15}, \sigma_{926} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3702} = \frac{\langle S_6, \sigma_{926} \rangle \Rightarrow \langle S_8, \sigma_{926} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{926} \rangle \Rightarrow \langle S_9, \sigma_{926} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3704} = \langle S_8, \sigma_{926} \rangle \Rightarrow \langle y := y + 1, \sigma_{926} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3703} = \frac{\mathcal{T}_{3704}}{\langle S_9, \sigma_{926} \rangle \Rightarrow \langle S_{10}, \sigma_{926} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3705} = \frac{\langle y := y + 1, \sigma_{926} \rangle \Rightarrow \sigma_{927} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{926} \rangle \Rightarrow \langle S_{11}, \sigma_{927} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3706} = \langle S_{11}, \sigma_{927} \rangle \Rightarrow \langle S_{12}, \sigma_{927} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3707} = \langle S_{12}, \sigma_{927} \rangle \Rightarrow \langle S_{13}, \sigma_{927} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3709} = \frac{\langle x := x + 1, \sigma_{927} \rangle \Rightarrow \sigma_{928} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{927} \rangle \Rightarrow \langle S_6, \sigma_{928} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3708} = \frac{\mathcal{T}_{3709}}{\langle S_{13}, \sigma_{927} \rangle \Rightarrow \langle S_{15}, \sigma_{928} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3710} = \frac{\langle S_6, \sigma_{928} \rangle \Rightarrow \langle S_8, \sigma_{928} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{928} \rangle \Rightarrow \langle S_9, \sigma_{928} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3712} = \langle S_8, \sigma_{928} \rangle \Rightarrow \langle y := y + 1, \sigma_{928} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3711} = \frac{\mathcal{T}_{3712}}{\langle S_9, \sigma_{928} \rangle \Rightarrow \langle S_{10}, \sigma_{928} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3713} = \frac{\langle y := y + 1, \sigma_{928} \rangle \Rightarrow \sigma_{929} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{928} \rangle \Rightarrow \langle S_{11}, \sigma_{929} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3714} = \langle S_{11}, \sigma_{929} \rangle \Rightarrow \langle S_{12}, \sigma_{929} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3715} = \langle S_{12}, \sigma_{929} \rangle \Rightarrow \langle S_{13}, \sigma_{929} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3717} = \frac{\langle x := x + 1, \sigma_{929} \rangle \Rightarrow \sigma_{930} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{929} \rangle \Rightarrow \langle S_6, \sigma_{930} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3716} = \frac{\mathcal{T}_{3717}}{\langle S_{13}, \sigma_{929} \rangle \Rightarrow \langle S_{15}, \sigma_{930} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3718} = \frac{\langle S_6, \sigma_{930} \rangle \Rightarrow \langle S_8, \sigma_{930} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{930} \rangle \Rightarrow \langle S_9, \sigma_{930} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3720} = \langle S_8, \sigma_{930} \rangle \Rightarrow \langle y := y + 1, \sigma_{930} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3719} = \frac{\mathcal{T}_{3720}}{\langle S_9, \sigma_{930} \rangle \Rightarrow \langle S_{10}, \sigma_{930} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3721} = \frac{\langle y := y + 1, \sigma_{930} \rangle \Rightarrow \sigma_{931} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{930} \rangle \Rightarrow \langle S_{11}, \sigma_{931} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3722} = \langle S_{11}, \sigma_{931} \rangle \Rightarrow \langle S_{12}, \sigma_{931} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3723} = \langle S_{12}, \sigma_{931} \rangle \Rightarrow \langle S_{13}, \sigma_{931} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3725} = \frac{\langle x := x + 1, \sigma_{931} \rangle \Rightarrow \sigma_{932} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{931} \rangle \Rightarrow \langle S_6, \sigma_{932} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3724} = \frac{\mathcal{T}_{3725}}{\langle S_{13}, \sigma_{931} \rangle \Rightarrow \langle S_{15}, \sigma_{932} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3726} = \frac{\langle S_6, \sigma_{932} \rangle \Rightarrow \langle S_8, \sigma_{932} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{932} \rangle \Rightarrow \langle S_9, \sigma_{932} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3728} = \langle S_8, \sigma_{932} \rangle \Rightarrow \langle y := y + 1, \sigma_{932} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3727} = \frac{\mathcal{T}_{3728}}{\langle S_9, \sigma_{932} \rangle \Rightarrow \langle S_{10}, \sigma_{932} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3729} = \frac{\langle y := y + 1, \sigma_{932} \rangle \Rightarrow \sigma_{933} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{932} \rangle \Rightarrow \langle S_{11}, \sigma_{933} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3730} = \langle S_{11}, \sigma_{933} \rangle \Rightarrow \langle S_{12}, \sigma_{933} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3731} = \langle S_{12}, \sigma_{933} \rangle \Rightarrow \langle S_{13}, \sigma_{933} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3733} = \frac{\langle x := x + 1, \sigma_{933} \rangle \Rightarrow \sigma_{934} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{933} \rangle \Rightarrow \langle S_6, \sigma_{934} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3732} = \frac{\mathcal{T}_{3733}}{\langle S_{13}, \sigma_{933} \rangle \Rightarrow \langle S_{15}, \sigma_{934} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3734} = \frac{\langle S_6, \sigma_{934} \rangle \Rightarrow \langle S_8, \sigma_{934} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{934} \rangle \Rightarrow \langle S_9, \sigma_{934} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3736} = \langle S_8, \sigma_{934} \rangle \Rightarrow \langle y := y + 1, \sigma_{934} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3735} = \frac{\mathcal{T}_{3736}}{\langle S_9, \sigma_{934} \rangle \Rightarrow \langle S_{10}, \sigma_{934} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3737} = \frac{\langle y := y + 1, \sigma_{934} \rangle \Rightarrow \sigma_{935} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{934} \rangle \Rightarrow \langle S_{11}, \sigma_{935} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3738} = \langle S_{11}, \sigma_{935} \rangle \Rightarrow \langle S_{12}, \sigma_{935} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3739} = \langle S_{12}, \sigma_{935} \rangle \Rightarrow \langle S_{13}, \sigma_{935} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3741} = \frac{\langle x := x + 1, \sigma_{935} \rangle \Rightarrow \sigma_{936} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{935} \rangle \Rightarrow \langle S_6, \sigma_{936} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3740} = \frac{\mathcal{T}_{3741}}{\langle S_{13}, \sigma_{935} \rangle \Rightarrow \langle S_{15}, \sigma_{936} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3742} = \frac{\langle S_6, \sigma_{936} \rangle \Rightarrow \langle S_8, \sigma_{936} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{936} \rangle \Rightarrow \langle S_9, \sigma_{936} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3744} = \langle S_8, \sigma_{936} \rangle \Rightarrow \langle y := y + 1, \sigma_{936} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3743} = \frac{\mathcal{T}_{3744}}{\langle S_9, \sigma_{936} \rangle \Rightarrow \langle S_{10}, \sigma_{936} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3745} = \frac{\langle y := y + 1, \sigma_{936} \rangle \Rightarrow \sigma_{937} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{936} \rangle \Rightarrow \langle S_{11}, \sigma_{937} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3746} = \langle S_{11}, \sigma_{937} \rangle \Rightarrow \langle S_{12}, \sigma_{937} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3747} = \langle S_{12}, \sigma_{937} \rangle \Rightarrow \langle S_{13}, \sigma_{937} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3749} = \frac{\langle x := x + 1, \sigma_{937} \rangle \Rightarrow \sigma_{938} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{937} \rangle \Rightarrow \langle S_6, \sigma_{938} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3748} = \frac{\mathcal{T}_{3749}}{\langle S_{13}, \sigma_{937} \rangle \Rightarrow \langle S_{15}, \sigma_{938} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3750} = \frac{\langle S_6, \sigma_{938} \rangle \Rightarrow \langle S_8, \sigma_{938} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{938} \rangle \Rightarrow \langle S_9, \sigma_{938} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3752} = \langle S_8, \sigma_{938} \rangle \Rightarrow \langle y := y + 1, \sigma_{938} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3751} = \frac{\mathcal{T}_{3752}}{\langle S_9, \sigma_{938} \rangle \Rightarrow \langle S_{10}, \sigma_{938} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3753} = \frac{\langle y := y + 1, \sigma_{938} \rangle \Rightarrow \sigma_{939} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{938} \rangle \Rightarrow \langle S_{11}, \sigma_{939} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{3754} = \langle S_{11}, \sigma_{939} \rangle \Rightarrow \langle S_{12}, \sigma_{939} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3755} = \langle S_{12}, \sigma_{939} \rangle \Rightarrow \langle S_{13}, \sigma_{939} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3757} = \frac{\langle x := x + 1, \sigma_{939} \rangle \Rightarrow \sigma_{940} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{939} \rangle \Rightarrow \langle S_6, \sigma_{940} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3756} = \frac{\mathcal{T}_{3757}}{\langle S_{13}, \sigma_{939} \rangle \Rightarrow \langle S_{15}, \sigma_{940} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3758} = \frac{\langle S_6, \sigma_{940} \rangle \Rightarrow \langle S_8, \sigma_{940} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{940} \rangle \Rightarrow \langle S_9, \sigma_{940} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3760} = \langle S_8, \sigma_{940} \rangle \Rightarrow \langle y := y + 1, \sigma_{940} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3759} = \frac{\mathcal{T}_{3760}}{\langle S_9, \sigma_{940} \rangle \Rightarrow \langle S_{10}, \sigma_{940} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{3761} = \frac{\langle y := y + 1, \sigma_{940} \rangle \Rightarrow \sigma_{941} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{940} \rangle \Rightarrow \langle S_{11}, \sigma_{941} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{3762} = \langle S_{11}, \sigma_{941} \rangle \Rightarrow \langle S_{12}, \sigma_{941} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3763} = \langle S_{12}, \sigma_{941} \rangle \Rightarrow \langle S_{13}, \sigma_{941} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{3765} = \frac{\langle x := x + 1, \sigma_{941} \rangle \Rightarrow \sigma_{942} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{941} \rangle \Rightarrow \langle S_6, \sigma_{942} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3764} = \frac{\mathcal{T}_{3765}}{\langle S_{13}, \sigma_{941} \rangle \Rightarrow \langle S_{15}, \sigma_{942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3766} = \frac{\langle S_6, \sigma_{942} \rangle \Rightarrow \langle S_8, \sigma_{942} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{942} \rangle \Rightarrow \langle S_9, \sigma_{942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3768} = \langle S_8, \sigma_{942} \rangle \Rightarrow \langle y := y + 1, \sigma_{942} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3767} = \frac{\mathcal{T}_{3768}}{\langle S_9, \sigma_{942} \rangle \Rightarrow \langle S_{10}, \sigma_{942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3769} = \frac{\langle y := y + 1, \sigma_{942} \rangle \Rightarrow \sigma_{943} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{942} \rangle \Rightarrow \langle S_{11}, \sigma_{943} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3770} = \langle S_{11}, \sigma_{943} \rangle \Rightarrow \langle S_{12}, \sigma_{943} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3771} = \langle S_{12}, \sigma_{943} \rangle \Rightarrow \langle S_{13}, \sigma_{943} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3773} = \frac{\langle x := x + 1, \sigma_{943} \rangle \Rightarrow \sigma_{944} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{943} \rangle \Rightarrow \langle S_6, \sigma_{944} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3772} = \frac{\mathcal{T}_{3773}}{\langle S_{13}, \sigma_{943} \rangle \Rightarrow \langle S_{15}, \sigma_{944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3774} = \frac{\langle S_6, \sigma_{944} \rangle \Rightarrow \langle S_8, \sigma_{944} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{944} \rangle \Rightarrow \langle S_9, \sigma_{944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3776} = \langle S_8, \sigma_{944} \rangle \Rightarrow \langle y := y + 1, \sigma_{944} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3775} = \frac{\mathcal{T}_{3776}}{\langle S_9, \sigma_{944} \rangle \Rightarrow \langle S_{10}, \sigma_{944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3777} = \frac{\langle y := y + 1, \sigma_{944} \rangle \Rightarrow \sigma_{945} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{944} \rangle \Rightarrow \langle S_{11}, \sigma_{945} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3778} = \langle S_{11}, \sigma_{945} \rangle \Rightarrow \langle S_{12}, \sigma_{945} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3779} = \langle S_{12}, \sigma_{945} \rangle \Rightarrow \langle S_{13}, \sigma_{945} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3781} = \frac{\langle x := x + 1, \sigma_{945} \rangle \Rightarrow \sigma_{946} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{945} \rangle \Rightarrow \langle S_6, \sigma_{946} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3780} = \frac{\mathcal{T}_{3781}}{\langle S_{13}, \sigma_{945} \rangle \Rightarrow \langle S_{15}, \sigma_{946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3782} = \frac{\langle S_6, \sigma_{946} \rangle \Rightarrow \langle S_8, \sigma_{946} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{946} \rangle \Rightarrow \langle S_9, \sigma_{946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3784} = \langle S_8, \sigma_{946} \rangle \Rightarrow \langle y := y + 1, \sigma_{946} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3783} = \frac{\mathcal{T}_{3784}}{\langle S_9, \sigma_{946} \rangle \Rightarrow \langle S_{10}, \sigma_{946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3785} = \frac{\langle y := y + 1, \sigma_{946} \rangle \Rightarrow \sigma_{947} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{946} \rangle \Rightarrow \langle S_{11}, \sigma_{947} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3786} = \langle S_{11}, \sigma_{947} \rangle \Rightarrow \langle S_{12}, \sigma_{947} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3787} = \langle S_{12}, \sigma_{947} \rangle \Rightarrow \langle S_{13}, \sigma_{947} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3789} = \frac{\langle x := x + 1, \sigma_{947} \rangle \Rightarrow \sigma_{948} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{947} \rangle \Rightarrow \langle S_6, \sigma_{948} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3788} = \frac{\mathcal{T}_{3789}}{\langle S_{13}, \sigma_{947} \rangle \Rightarrow \langle S_{15}, \sigma_{948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3790} = \frac{\langle S_6, \sigma_{948} \rangle \Rightarrow \langle S_8, \sigma_{948} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{948} \rangle \Rightarrow \langle S_9, \sigma_{948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3792} = \langle S_8, \sigma_{948} \rangle \Rightarrow \langle y := y + 1, \sigma_{948} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3791} = \frac{\mathcal{T}_{3792}}{\langle S_9, \sigma_{948} \rangle \Rightarrow \langle S_{10}, \sigma_{948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3793} = \frac{\langle y := y + 1, \sigma_{948} \rangle \Rightarrow \sigma_{949} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{948} \rangle \Rightarrow \langle S_{11}, \sigma_{949} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3794} = \langle S_{11}, \sigma_{949} \rangle \Rightarrow \langle S_{12}, \sigma_{949} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3795} = \langle S_{12}, \sigma_{949} \rangle \Rightarrow \langle S_{13}, \sigma_{949} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3797} = \frac{\langle x := x + 1, \sigma_{949} \rangle \Rightarrow \sigma_{950} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{949} \rangle \Rightarrow \langle S_6, \sigma_{950} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3796} = \frac{\mathcal{T}_{3797}}{\langle S_{13}, \sigma_{949} \rangle \Rightarrow \langle S_{15}, \sigma_{950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3798} = \frac{\langle S_6, \sigma_{950} \rangle \Rightarrow \langle S_8, \sigma_{950} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{950} \rangle \Rightarrow \langle S_9, \sigma_{950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3800} = \langle S_8, \sigma_{950} \rangle \Rightarrow \langle y := y + 1, \sigma_{950} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3799} = \frac{\mathcal{T}_{3800}}{\langle S_9, \sigma_{950} \rangle \Rightarrow \langle S_{10}, \sigma_{950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3801} = \frac{\langle y := y + 1, \sigma_{950} \rangle \Rightarrow \sigma_{951} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{950} \rangle \Rightarrow \langle S_{11}, \sigma_{951} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3802} = \langle S_{11}, \sigma_{951} \rangle \Rightarrow \langle S_{12}, \sigma_{951} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3803} = \langle S_{12}, \sigma_{951} \rangle \Rightarrow \langle S_{13}, \sigma_{951} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3805} = \frac{\langle x := x + 1, \sigma_{951} \rangle \Rightarrow \sigma_{952} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{951} \rangle \Rightarrow \langle S_6, \sigma_{952} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3804} = \frac{\mathcal{T}_{3805}}{\langle S_{13}, \sigma_{951} \rangle \Rightarrow \langle S_{15}, \sigma_{952} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3806} = \frac{\langle S_6, \sigma_{952} \rangle \Rightarrow \langle S_8, \sigma_{952} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{952} \rangle \Rightarrow \langle S_9, \sigma_{952} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3808} = \langle S_8, \sigma_{952} \rangle \Rightarrow \langle y := y + 1, \sigma_{952} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3807} = \frac{\mathcal{T}_{3808}}{\langle S_9, \sigma_{952} \rangle \Rightarrow \langle S_{10}, \sigma_{952} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3809} = \frac{\langle y := y + 1, \sigma_{952} \rangle \Rightarrow \sigma_{953} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{952} \rangle \Rightarrow \langle S_{11}, \sigma_{953} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3810} = \langle S_{11}, \sigma_{953} \rangle \Rightarrow \langle S_{12}, \sigma_{953} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3811} = \langle S_{12}, \sigma_{953} \rangle \Rightarrow \langle S_{13}, \sigma_{953} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3813} = \frac{\langle x := x + 1, \sigma_{953} \rangle \Rightarrow \sigma_{954} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{953} \rangle \Rightarrow \langle S_6, \sigma_{954} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3812} = \frac{\mathcal{T}_{3813}}{\langle S_{13}, \sigma_{953} \rangle \Rightarrow \langle S_{15}, \sigma_{954} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3814} = \frac{\langle S_6, \sigma_{954} \rangle \Rightarrow \langle S_8, \sigma_{954} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{954} \rangle \Rightarrow \langle S_9, \sigma_{954} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3816} = \langle S_8, \sigma_{954} \rangle \Rightarrow \langle y := y + 1, \sigma_{954} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3815} = \frac{\mathcal{T}_{3816}}{\langle S_9, \sigma_{954} \rangle \Rightarrow \langle S_{10}, \sigma_{954} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3817} = \frac{\langle y := y + 1, \sigma_{954} \rangle \Rightarrow \sigma_{955} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{954} \rangle \Rightarrow \langle S_{11}, \sigma_{955} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3818} = \langle S_{11}, \sigma_{955} \rangle \Rightarrow \langle S_{12}, \sigma_{955} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3819} = \langle S_{12}, \sigma_{955} \rangle \Rightarrow \langle S_{13}, \sigma_{955} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3821} = \frac{\langle x := x + 1, \sigma_{955} \rangle \Rightarrow \sigma_{956} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{955} \rangle \Rightarrow \langle S_6, \sigma_{956} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3820} = \frac{\mathcal{T}_{3821}}{\langle S_{13}, \sigma_{955} \rangle \Rightarrow \langle S_{15}, \sigma_{956} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3822} = \frac{\langle S_6, \sigma_{956} \rangle \Rightarrow \langle S_8, \sigma_{956} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{956} \rangle \Rightarrow \langle S_9, \sigma_{956} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3824} = \langle S_8, \sigma_{956} \rangle \Rightarrow \langle y := y + 1, \sigma_{956} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3823} = \frac{\mathcal{T}_{3824}}{\langle S_9, \sigma_{956} \rangle \Rightarrow \langle S_{10}, \sigma_{956} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3825} = \frac{\langle y := y + 1, \sigma_{956} \rangle \Rightarrow \sigma_{957} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{956} \rangle \Rightarrow \langle S_{11}, \sigma_{957} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3826} = \langle S_{11}, \sigma_{957} \rangle \Rightarrow \langle S_{12}, \sigma_{957} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$\Downarrow$

$$\mathcal{T}_{3827} = \langle S_{12}, \sigma_{957} \rangle \Rightarrow \langle S_{13}, \sigma_{957} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{3829} = \frac{\langle x := x + 1, \sigma_{957} \rangle \Rightarrow \sigma_{958} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{957} \rangle \Rightarrow \langle S_6, \sigma_{958} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{3828} = \frac{\mathcal{T}_{3829}}{\langle S_{13}, \sigma_{957} \rangle \Rightarrow \langle S_{15}, \sigma_{958} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{3830} = \frac{\langle S_6, \sigma_{958} \rangle \Rightarrow \langle S_8, \sigma_{958} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{958} \rangle \Rightarrow \langle S_9, \sigma_{958} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{3832} = \langle S_8, \sigma_{958} \rangle \Rightarrow \langle y := y + 1, \sigma_{958} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{3831} = \frac{\mathcal{T}_{3832}}{\langle S_9, \sigma_{958} \rangle \Rightarrow \langle S_{10}, \sigma_{958} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{3833} = \frac{\langle y := y + 1, \sigma_{958} \rangle \Rightarrow \sigma_{959} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{958} \rangle \Rightarrow \langle S_{11}, \sigma_{959} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

 $\Downarrow$

$$\mathcal{T}_{3834} = \langle S_{11}, \sigma_{959} \rangle \Rightarrow \langle S_{12}, \sigma_{959} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{3835} = \langle S_{12}, \sigma_{959} \rangle \Rightarrow \langle S_{13}, \sigma_{959} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3837} = \frac{\langle x := x + 1, \sigma_{959} \rangle \Rightarrow \sigma_{960} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{959} \rangle \Rightarrow \langle S_6, \sigma_{960} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3836} = \frac{\mathcal{T}_{3837}}{\langle S_{13}, \sigma_{959} \rangle \Rightarrow \langle S_{15}, \sigma_{960} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3838} = \frac{\langle S_6, \sigma_{960} \rangle \Rightarrow \langle S_8, \sigma_{960} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{960} \rangle \Rightarrow \langle S_9, \sigma_{960} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3840} = \langle S_8, \sigma_{960} \rangle \Rightarrow \langle y := y + 1, \sigma_{960} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3839} = \frac{\mathcal{T}_{3840}}{\langle S_9, \sigma_{960} \rangle \Rightarrow \langle S_{10}, \sigma_{960} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3841} = \frac{\langle y := y + 1, \sigma_{960} \rangle \Rightarrow \sigma_{961} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{960} \rangle \Rightarrow \langle S_{11}, \sigma_{961} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3842} = \langle S_{11}, \sigma_{961} \rangle \Rightarrow \langle S_{12}, \sigma_{961} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3843} = \langle S_{12}, \sigma_{961} \rangle \Rightarrow \langle S_{13}, \sigma_{961} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3845} = \frac{\langle x := x + 1, \sigma_{961} \rangle \Rightarrow \sigma_{962} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{961} \rangle \Rightarrow \langle S_6, \sigma_{962} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3844} = \frac{\mathcal{T}_{3845}}{\langle S_{13}, \sigma_{961} \rangle \Rightarrow \langle S_{15}, \sigma_{962} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3846} = \frac{\langle S_6, \sigma_{962} \rangle \Rightarrow \langle S_8, \sigma_{962} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{962} \rangle \Rightarrow \langle S_9, \sigma_{962} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3848} = \langle S_8, \sigma_{962} \rangle \Rightarrow \langle y := y + 1, \sigma_{962} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3847} = \frac{\mathcal{T}_{3848}}{\langle S_9, \sigma_{962} \rangle \Rightarrow \langle S_{10}, \sigma_{962} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3849} = \frac{\langle y := y + 1, \sigma_{962} \rangle \Rightarrow \sigma_{963} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{962} \rangle \Rightarrow \langle S_{11}, \sigma_{963} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3850} = \langle S_{11}, \sigma_{963} \rangle \Rightarrow \langle S_{12}, \sigma_{963} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3851} = \langle S_{12}, \sigma_{963} \rangle \Rightarrow \langle S_{13}, \sigma_{963} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3853} = \frac{\langle x := x + 1, \sigma_{963} \rangle \Rightarrow \sigma_{964} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{963} \rangle \Rightarrow \langle S_6, \sigma_{964} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3852} = \frac{\mathcal{T}_{3853}}{\langle S_{13}, \sigma_{963} \rangle \Rightarrow \langle S_{15}, \sigma_{964} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3854} = \frac{\langle S_6, \sigma_{964} \rangle \Rightarrow \langle S_8, \sigma_{964} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{964} \rangle \Rightarrow \langle S_9, \sigma_{964} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3856} = \langle S_8, \sigma_{964} \rangle \Rightarrow \langle y := y + 1, \sigma_{964} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3855} = \frac{\mathcal{T}_{3856}}{\langle S_9, \sigma_{964} \rangle \Rightarrow \langle S_{10}, \sigma_{964} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3857} = \frac{\langle y := y + 1, \sigma_{964} \rangle \Rightarrow \sigma_{965} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{964} \rangle \Rightarrow \langle S_{11}, \sigma_{965} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3858} = \langle S_{11}, \sigma_{965} \rangle \Rightarrow \langle S_{12}, \sigma_{965} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3859} = \langle S_{12}, \sigma_{965} \rangle \Rightarrow \langle S_{13}, \sigma_{965} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3861} = \frac{\langle x := x + 1, \sigma_{965} \rangle \Rightarrow \sigma_{966} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{965} \rangle \Rightarrow \langle S_6, \sigma_{966} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3860} = \frac{\mathcal{T}_{3861}}{\langle S_{13}, \sigma_{965} \rangle \Rightarrow \langle S_{15}, \sigma_{966} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3862} = \frac{\langle S_6, \sigma_{966} \rangle \Rightarrow \langle S_8, \sigma_{966} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{966} \rangle \Rightarrow \langle S_9, \sigma_{966} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3864} = \langle S_8, \sigma_{966} \rangle \Rightarrow \langle y := y + 1, \sigma_{966} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3863} = \frac{\mathcal{T}_{3864}}{\langle S_9, \sigma_{966} \rangle \Rightarrow \langle S_{10}, \sigma_{966} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3865} = \frac{\langle y := y + 1, \sigma_{966} \rangle \Rightarrow \sigma_{967} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{966} \rangle \Rightarrow \langle S_{11}, \sigma_{967} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3866} = \langle S_{11}, \sigma_{967} \rangle \Rightarrow \langle S_{12}, \sigma_{967} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3867} = \langle S_{12}, \sigma_{967} \rangle \Rightarrow \langle S_{13}, \sigma_{967} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3869} = \frac{\langle x := x + 1, \sigma_{967} \rangle \Rightarrow \sigma_{968} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{967} \rangle \Rightarrow \langle S_6, \sigma_{968} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3868} = \frac{\mathcal{T}_{3869}}{\langle S_{13}, \sigma_{967} \rangle \Rightarrow \langle S_{15}, \sigma_{968} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3870} = \frac{\langle S_6, \sigma_{968} \rangle \Rightarrow \langle S_8, \sigma_{968} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{968} \rangle \Rightarrow \langle S_9, \sigma_{968} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3872} = \langle S_8, \sigma_{968} \rangle \Rightarrow \langle y := y + 1, \sigma_{968} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3871} = \frac{\mathcal{T}_{3872}}{\langle S_9, \sigma_{968} \rangle \Rightarrow \langle S_{10}, \sigma_{968} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3873} = \frac{\langle y := y + 1, \sigma_{968} \rangle \Rightarrow \sigma_{969} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{968} \rangle \Rightarrow \langle S_{11}, \sigma_{969} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3874} = \langle S_{11}, \sigma_{969} \rangle \Rightarrow \langle S_{12}, \sigma_{969} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3875} = \langle S_{12}, \sigma_{969} \rangle \Rightarrow \langle S_{13}, \sigma_{969} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3877} = \frac{\langle x := x + 1, \sigma_{969} \rangle \Rightarrow \sigma_{970} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{969} \rangle \Rightarrow \langle S_6, \sigma_{970} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3876} = \frac{\mathcal{T}_{3877}}{\langle S_{13}, \sigma_{969} \rangle \Rightarrow \langle S_{15}, \sigma_{970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3878} = \frac{\langle S_6, \sigma_{970} \rangle \Rightarrow \langle S_8, \sigma_{970} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{970} \rangle \Rightarrow \langle S_9, \sigma_{970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3880} = \langle S_8, \sigma_{970} \rangle \Rightarrow \langle y := y + 1, \sigma_{970} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3879} = \frac{\mathcal{T}_{3880}}{\langle S_9, \sigma_{970} \rangle \Rightarrow \langle S_{10}, \sigma_{970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3881} = \frac{\langle y := y + 1, \sigma_{970} \rangle \Rightarrow \sigma_{971} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{970} \rangle \Rightarrow \langle S_{11}, \sigma_{971} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3882} = \langle S_{11}, \sigma_{971} \rangle \Rightarrow \langle S_{12}, \sigma_{971} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3883} = \langle S_{12}, \sigma_{971} \rangle \Rightarrow \langle S_{13}, \sigma_{971} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3885} = \frac{\langle x := x + 1, \sigma_{971} \rangle \Rightarrow \sigma_{972} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{971} \rangle \Rightarrow \langle S_6, \sigma_{972} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3884} = \frac{\mathcal{T}_{3885}}{\langle S_{13}, \sigma_{971} \rangle \Rightarrow \langle S_{15}, \sigma_{972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3886} = \frac{\langle S_6, \sigma_{972} \rangle \Rightarrow \langle S_8, \sigma_{972} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{972} \rangle \Rightarrow \langle S_9, \sigma_{972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3888} = \langle S_8, \sigma_{972} \rangle \Rightarrow \langle y := y + 1, \sigma_{972} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3887} = \frac{\mathcal{T}_{3888}}{\langle S_9, \sigma_{972} \rangle \Rightarrow \langle S_{10}, \sigma_{972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3889} = \frac{\langle y := y + 1, \sigma_{972} \rangle \Rightarrow \sigma_{973} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{972} \rangle \Rightarrow \langle S_{11}, \sigma_{973} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3890} = \langle S_{11}, \sigma_{973} \rangle \Rightarrow \langle S_{12}, \sigma_{973} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3891} = \langle S_{12}, \sigma_{973} \rangle \Rightarrow \langle S_{13}, \sigma_{973} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3893} = \frac{\langle x := x + 1, \sigma_{973} \rangle \Rightarrow \sigma_{974} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{973} \rangle \Rightarrow \langle S_6, \sigma_{974} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3892} = \frac{\mathcal{T}_{3893}}{\langle S_{13}, \sigma_{973} \rangle \Rightarrow \langle S_{15}, \sigma_{974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3894} = \frac{\langle S_6, \sigma_{974} \rangle \Rightarrow \langle S_8, \sigma_{974} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{974} \rangle \Rightarrow \langle S_9, \sigma_{974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3896} = \langle S_8, \sigma_{974} \rangle \Rightarrow \langle y := y + 1, \sigma_{974} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3895} = \frac{\mathcal{T}_{3896}}{\langle S_9, \sigma_{974} \rangle \Rightarrow \langle S_{10}, \sigma_{974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3897} = \frac{\langle y := y + 1, \sigma_{974} \rangle \Rightarrow \sigma_{975} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{974} \rangle \Rightarrow \langle S_{11}, \sigma_{975} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3898} = \langle S_{11}, \sigma_{975} \rangle \Rightarrow \langle S_{12}, \sigma_{975} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3899} = \langle S_{12}, \sigma_{975} \rangle \Rightarrow \langle S_{13}, \sigma_{975} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3901} = \frac{\langle x := x + 1, \sigma_{975} \rangle \Rightarrow \sigma_{976} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{975} \rangle \Rightarrow \langle S_6, \sigma_{976} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3900} = \frac{\mathcal{T}_{3901}}{\langle S_{13}, \sigma_{975} \rangle \Rightarrow \langle S_{15}, \sigma_{976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3902} = \frac{\langle S_6, \sigma_{976} \rangle \Rightarrow \langle S_8, \sigma_{976} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{976} \rangle \Rightarrow \langle S_9, \sigma_{976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3904} = \langle S_8, \sigma_{976} \rangle \Rightarrow \langle y := y + 1, \sigma_{976} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3903} = \frac{\mathcal{T}_{3904}}{\langle S_9, \sigma_{976} \rangle \Rightarrow \langle S_{10}, \sigma_{976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3905} = \frac{\langle y := y + 1, \sigma_{976} \rangle \Rightarrow \sigma_{977} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{976} \rangle \Rightarrow \langle S_{11}, \sigma_{977} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3906} = \langle S_{11}, \sigma_{977} \rangle \Rightarrow \langle S_{12}, \sigma_{977} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3907} = \langle S_{12}, \sigma_{977} \rangle \Rightarrow \langle S_{13}, \sigma_{977} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3909} = \frac{\langle x := x + 1, \sigma_{977} \rangle \Rightarrow \sigma_{978} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{977} \rangle \Rightarrow \langle S_6, \sigma_{978} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3908} = \frac{\mathcal{T}_{3909}}{\langle S_{13}, \sigma_{977} \rangle \Rightarrow \langle S_{15}, \sigma_{978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3910} = \frac{\langle S_6, \sigma_{978} \rangle \Rightarrow \langle S_8, \sigma_{978} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{978} \rangle \Rightarrow \langle S_9, \sigma_{978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3912} = \langle S_8, \sigma_{978} \rangle \Rightarrow \langle y := y + 1, \sigma_{978} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3911} = \frac{\mathcal{T}_{3912}}{\langle S_9, \sigma_{978} \rangle \Rightarrow \langle S_{10}, \sigma_{978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3913} = \frac{\langle y := y + 1, \sigma_{978} \rangle \Rightarrow \sigma_{979} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{978} \rangle \Rightarrow \langle S_{11}, \sigma_{979} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3914} = \langle S_{11}, \sigma_{979} \rangle \Rightarrow \langle S_{12}, \sigma_{979} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3915} = \langle S_{12}, \sigma_{979} \rangle \Rightarrow \langle S_{13}, \sigma_{979} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3917} = \frac{\langle x := x + 1, \sigma_{979} \rangle \Rightarrow \sigma_{980} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{979} \rangle \Rightarrow \langle S_6, \sigma_{980} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3916} = \frac{\mathcal{T}_{3917}}{\langle S_{13}, \sigma_{979} \rangle \Rightarrow \langle S_{15}, \sigma_{980} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3918} = \frac{\langle S_6, \sigma_{980} \rangle \Rightarrow \langle S_8, \sigma_{980} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{980} \rangle \Rightarrow \langle S_9, \sigma_{980} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3920} = \langle S_8, \sigma_{980} \rangle \Rightarrow \langle y := y + 1, \sigma_{980} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3919} = \frac{\mathcal{T}_{3920}}{\langle S_9, \sigma_{980} \rangle \Rightarrow \langle S_{10}, \sigma_{980} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3921} = \frac{\langle y := y + 1, \sigma_{980} \rangle \Rightarrow \sigma_{981} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{980} \rangle \Rightarrow \langle S_{11}, \sigma_{981} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3922} = \langle S_{11}, \sigma_{981} \rangle \Rightarrow \langle S_{12}, \sigma_{981} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3923} = \langle S_{12}, \sigma_{981} \rangle \Rightarrow \langle S_{13}, \sigma_{981} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3925} = \frac{\langle x := x + 1, \sigma_{981} \rangle \Rightarrow \sigma_{982} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{981} \rangle \Rightarrow \langle S_6, \sigma_{982} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3924} = \frac{\mathcal{T}_{3925}}{\langle S_{13}, \sigma_{981} \rangle \Rightarrow \langle S_{15}, \sigma_{982} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3926} = \frac{\langle S_6, \sigma_{982} \rangle \Rightarrow \langle S_8, \sigma_{982} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{982} \rangle \Rightarrow \langle S_9, \sigma_{982} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3928} = \langle S_8, \sigma_{982} \rangle \Rightarrow \langle y := y + 1, \sigma_{982} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3927} = \frac{\mathcal{T}_{3928}}{\langle S_9, \sigma_{982} \rangle \Rightarrow \langle S_{10}, \sigma_{982} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3929} = \frac{\langle y := y + 1, \sigma_{982} \rangle \Rightarrow \sigma_{983} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{982} \rangle \Rightarrow \langle S_{11}, \sigma_{983} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3930} = \langle S_{11}, \sigma_{983} \rangle \Rightarrow \langle S_{12}, \sigma_{983} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3931} = \langle S_{12}, \sigma_{983} \rangle \Rightarrow \langle S_{13}, \sigma_{983} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3933} = \frac{\langle x := x + 1, \sigma_{983} \rangle \Rightarrow \sigma_{984} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{983} \rangle \Rightarrow \langle S_6, \sigma_{984} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3932} = \frac{\mathcal{T}_{3933}}{\langle S_{13}, \sigma_{983} \rangle \Rightarrow \langle S_{15}, \sigma_{984} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3934} = \frac{\langle S_6, \sigma_{984} \rangle \Rightarrow \langle S_8, \sigma_{984} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{984} \rangle \Rightarrow \langle S_9, \sigma_{984} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3936} = \langle S_8, \sigma_{984} \rangle \Rightarrow \langle y := y + 1, \sigma_{984} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3935} = \frac{\mathcal{T}_{3936}}{\langle S_9, \sigma_{984} \rangle \Rightarrow \langle S_{10}, \sigma_{984} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3937} = \frac{\langle y := y + 1, \sigma_{984} \rangle \Rightarrow \sigma_{985} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{984} \rangle \Rightarrow \langle S_{11}, \sigma_{985} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3938} = \langle S_{11}, \sigma_{985} \rangle \Rightarrow \langle S_{12}, \sigma_{985} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3939} = \langle S_{12}, \sigma_{985} \rangle \Rightarrow \langle S_{13}, \sigma_{985} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3941} = \frac{\langle x := x + 1, \sigma_{985} \rangle \Rightarrow \sigma_{986} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{985} \rangle \Rightarrow \langle S_6, \sigma_{986} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3940} = \frac{\mathcal{T}_{3941}}{\langle S_{13}, \sigma_{985} \rangle \Rightarrow \langle S_{15}, \sigma_{986} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3942} = \frac{\langle S_6, \sigma_{986} \rangle \Rightarrow \langle S_8, \sigma_{986} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{986} \rangle \Rightarrow \langle S_9, \sigma_{986} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3944} = \langle S_8, \sigma_{986} \rangle \Rightarrow \langle y := y + 1, \sigma_{986} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3943} = \frac{\mathcal{T}_{3944}}{\langle S_9, \sigma_{986} \rangle \Rightarrow \langle S_{10}, \sigma_{986} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3945} = \frac{\langle y := y + 1, \sigma_{986} \rangle \Rightarrow \sigma_{987} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{986} \rangle \Rightarrow \langle S_{11}, \sigma_{987} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3946} = \langle S_{11}, \sigma_{987} \rangle \Rightarrow \langle S_{12}, \sigma_{987} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3947} = \langle S_{12}, \sigma_{987} \rangle \Rightarrow \langle S_{13}, \sigma_{987} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3949} = \frac{\langle x := x + 1, \sigma_{987} \rangle \Rightarrow \sigma_{988} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{987} \rangle \Rightarrow \langle S_6, \sigma_{988} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3948} = \frac{\mathcal{T}_{3949}}{\langle S_{13}, \sigma_{987} \rangle \Rightarrow \langle S_{15}, \sigma_{988} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3950} = \frac{\langle S_6, \sigma_{988} \rangle \Rightarrow \langle S_8, \sigma_{988} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{988} \rangle \Rightarrow \langle S_9, \sigma_{988} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3952} = \langle S_8, \sigma_{988} \rangle \Rightarrow \langle y := y + 1, \sigma_{988} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3951} = \frac{\mathcal{T}_{3952}}{\langle S_9, \sigma_{988} \rangle \Rightarrow \langle S_{10}, \sigma_{988} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3953} = \frac{\langle y := y + 1, \sigma_{988} \rangle \Rightarrow \sigma_{989} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{988} \rangle \Rightarrow \langle S_{11}, \sigma_{989} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3954} = \langle S_{11}, \sigma_{989} \rangle \Rightarrow \langle S_{12}, \sigma_{989} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3955} = \langle S_{12}, \sigma_{989} \rangle \Rightarrow \langle S_{13}, \sigma_{989} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3957} = \frac{\langle x := x + 1, \sigma_{989} \rangle \Rightarrow \sigma_{990} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{989} \rangle \Rightarrow \langle S_6, \sigma_{990} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3956} = \frac{\mathcal{T}_{3957}}{\langle S_{13}, \sigma_{989} \rangle \Rightarrow \langle S_{15}, \sigma_{990} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3958} = \frac{\langle S_6, \sigma_{990} \rangle \Rightarrow \langle S_8, \sigma_{990} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{990} \rangle \Rightarrow \langle S_9, \sigma_{990} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3960} = \langle S_8, \sigma_{990} \rangle \Rightarrow \langle y := y + 1, \sigma_{990} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3959} = \frac{\mathcal{T}_{3960}}{\langle S_9, \sigma_{990} \rangle \Rightarrow \langle S_{10}, \sigma_{990} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3961} = \frac{\langle y := y + 1, \sigma_{990} \rangle \Rightarrow \sigma_{991} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{990} \rangle \Rightarrow \langle S_{11}, \sigma_{991} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{3962} = \langle S_{11}, \sigma_{991} \rangle \Rightarrow \langle S_{12}, \sigma_{991} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3963} = \langle S_{12}, \sigma_{991} \rangle \Rightarrow \langle S_{13}, \sigma_{991} \rangle \text{ [if - tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{3965} = \frac{\langle x := x + 1, \sigma_{991} \rangle \Rightarrow \sigma_{992} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{991} \rangle \Rightarrow \langle S_6, \sigma_{992} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3964} = \frac{\mathcal{T}_{3965}}{\langle S_{13}, \sigma_{991} \rangle \Rightarrow \langle S_{15}, \sigma_{992} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3966} = \frac{\langle S_6, \sigma_{992} \rangle \Rightarrow \langle S_8, \sigma_{992} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{992} \rangle \Rightarrow \langle S_9, \sigma_{992} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3968} = \langle S_8, \sigma_{992} \rangle \Rightarrow \langle y := y + 1, \sigma_{992} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3967} = \frac{\mathcal{T}_{3968}}{\langle S_9, \sigma_{992} \rangle \Rightarrow \langle S_{10}, \sigma_{992} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3969} = \frac{\langle y := y + 1, \sigma_{992} \rangle \Rightarrow \sigma_{993} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{992} \rangle \Rightarrow \langle S_{11}, \sigma_{993} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3970} = \langle S_{11}, \sigma_{993} \rangle \Rightarrow \langle S_{12}, \sigma_{993} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3971} = \langle S_{12}, \sigma_{993} \rangle \Rightarrow \langle S_{13}, \sigma_{993} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3973} = \frac{\langle x := x + 1, \sigma_{993} \rangle \Rightarrow \sigma_{994} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{993} \rangle \Rightarrow \langle S_6, \sigma_{994} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3972} = \frac{\mathcal{T}_{3973}}{\langle S_{13}, \sigma_{993} \rangle \Rightarrow \langle S_{15}, \sigma_{994} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3974} = \frac{\langle S_6, \sigma_{994} \rangle \Rightarrow \langle S_8, \sigma_{994} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{994} \rangle \Rightarrow \langle S_9, \sigma_{994} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3976} = \langle S_8, \sigma_{994} \rangle \Rightarrow \langle y := y + 1, \sigma_{994} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{3975} = \frac{\mathcal{T}_{3976}}{\langle S_9, \sigma_{994} \rangle \Rightarrow \langle S_{10}, \sigma_{994} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3977} = \frac{\langle y := y + 1, \sigma_{994} \rangle \Rightarrow \sigma_{995} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{994} \rangle \Rightarrow \langle S_{11}, \sigma_{995} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3978} = \langle S_{11}, \sigma_{995} \rangle \Rightarrow \langle S_{12}, \sigma_{995} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3979} = \langle S_{12}, \sigma_{995} \rangle \Rightarrow \langle S_{13}, \sigma_{995} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3981} = \frac{\langle x := x + 1, \sigma_{995} \rangle \Rightarrow \sigma_{996} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{995} \rangle \Rightarrow \langle S_6, \sigma_{996} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3980} = \frac{\mathcal{T}_{3981}}{\langle S_{13}, \sigma_{995} \rangle \Rightarrow \langle S_{15}, \sigma_{996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3982} = \frac{\langle S_6, \sigma_{996} \rangle \Rightarrow \langle S_8, \sigma_{996} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{996} \rangle \Rightarrow \langle S_9, \sigma_{996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3984} = \langle S_8, \sigma_{996} \rangle \Rightarrow \langle y := y + 1, \sigma_{996} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3983} = \frac{\mathcal{T}_{3984}}{\langle S_9, \sigma_{996} \rangle \Rightarrow \langle S_{10}, \sigma_{996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3985} = \frac{\langle y := y + 1, \sigma_{996} \rangle \Rightarrow \sigma_{997} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{996} \rangle \Rightarrow \langle S_{11}, \sigma_{997} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3986} = \langle S_{11}, \sigma_{997} \rangle \Rightarrow \langle S_{12}, \sigma_{997} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3987} = \langle S_{12}, \sigma_{997} \rangle \Rightarrow \langle S_{13}, \sigma_{997} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3989} = \frac{\langle x := x + 1, \sigma_{997} \rangle \Rightarrow \sigma_{998} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{997} \rangle \Rightarrow \langle S_6, \sigma_{998} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3988} = \frac{\mathcal{T}_{3989}}{\langle S_{13}, \sigma_{997} \rangle \Rightarrow \langle S_{15}, \sigma_{998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3990} = \frac{\langle S_6, \sigma_{998} \rangle \Rightarrow \langle S_8, \sigma_{998} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{998} \rangle \Rightarrow \langle S_9, \sigma_{998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3992} = \langle S_8, \sigma_{998} \rangle \Rightarrow \langle y := y + 1, \sigma_{998} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3991} = \frac{\mathcal{T}_{3992}}{\langle S_9, \sigma_{998} \rangle \Rightarrow \langle S_{10}, \sigma_{998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{3993} = \frac{\langle y := y + 1, \sigma_{998} \rangle \Rightarrow \sigma_{999} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{998} \rangle \Rightarrow \langle S_{11}, \sigma_{999} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{3994} = \langle S_{11}, \sigma_{999} \rangle \Rightarrow \langle S_{12}, \sigma_{999} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3995} = \langle S_{12}, \sigma_{999} \rangle \Rightarrow \langle S_{13}, \sigma_{999} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{3997} = \frac{\langle x := x + 1, \sigma_{999} \rangle \Rightarrow \sigma_{1000} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{999} \rangle \Rightarrow \langle S_6, \sigma_{1000} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{3996} = \frac{\mathcal{T}_{3997}}{\langle S_{13}, \sigma_{999} \rangle \Rightarrow \langle S_{15}, \sigma_{1000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{3998} = \frac{\langle S_6, \sigma_{1000} \rangle \Rightarrow \langle S_8, \sigma_{1000} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1000} \rangle \Rightarrow \langle S_9, \sigma_{1000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4000} = \langle S_8, \sigma_{1000} \rangle \Rightarrow \langle y := y + 1, \sigma_{1000} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{3999} = \frac{\mathcal{T}_{4000}}{\langle S_9, \sigma_{1000} \rangle \Rightarrow \langle S_{10}, \sigma_{1000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4001} = \frac{\langle y := y + 1, \sigma_{1000} \rangle \Rightarrow \sigma_{1001} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1000} \rangle \Rightarrow \langle S_{11}, \sigma_{1001} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4002} = \langle S_{11}, \sigma_{1001} \rangle \Rightarrow \langle S_{12}, \sigma_{1001} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4003} = \langle S_{12}, \sigma_{1001} \rangle \Rightarrow \langle S_{13}, \sigma_{1001} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4005} = \frac{\langle x := x + 1, \sigma_{1001} \rangle \Rightarrow \sigma_{1002} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1001} \rangle \Rightarrow \langle S_6, \sigma_{1002} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4004} = \frac{\mathcal{T}_{4005}}{\langle S_{13}, \sigma_{1001} \rangle \Rightarrow \langle S_{15}, \sigma_{1002} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4006} = \frac{\langle S_6, \sigma_{1002} \rangle \Rightarrow \langle S_8, \sigma_{1002} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1002} \rangle \Rightarrow \langle S_9, \sigma_{1002} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4008} = \langle S_8, \sigma_{1002} \rangle \Rightarrow \langle y := y + 1, \sigma_{1002} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4007} = \frac{\mathcal{T}_{4008}}{\langle S_9, \sigma_{1002} \rangle \Rightarrow \langle S_{10}, \sigma_{1002} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4009} = \frac{\langle y := y + 1, \sigma_{1002} \rangle \Rightarrow \sigma_{1003} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1002} \rangle \Rightarrow \langle S_{11}, \sigma_{1003} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4010} = \langle S_{11}, \sigma_{1003} \rangle \Rightarrow \langle S_{12}, \sigma_{1003} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4011} = \langle S_{12}, \sigma_{1003} \rangle \Rightarrow \langle S_{13}, \sigma_{1003} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4013} = \frac{\langle x := x + 1, \sigma_{1003} \rangle \Rightarrow \sigma_{1004} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1003} \rangle \Rightarrow \langle S_6, \sigma_{1004} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4012} = \frac{\mathcal{T}_{4013}}{\langle S_{13}, \sigma_{1003} \rangle \Rightarrow \langle S_{15}, \sigma_{1004} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4014} = \frac{\langle S_6, \sigma_{1004} \rangle \Rightarrow \langle S_8, \sigma_{1004} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1004} \rangle \Rightarrow \langle S_9, \sigma_{1004} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4016} = \langle S_8, \sigma_{1004} \rangle \Rightarrow \langle y := y + 1, \sigma_{1004} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4015} = \frac{\mathcal{T}_{4016}}{\langle S_9, \sigma_{1004} \rangle \Rightarrow \langle S_{10}, \sigma_{1004} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4017} = \frac{\langle y := y + 1, \sigma_{1004} \rangle \Rightarrow \sigma_{1005} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1004} \rangle \Rightarrow \langle S_{11}, \sigma_{1005} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4018} = \langle S_{11}, \sigma_{1005} \rangle \Rightarrow \langle S_{12}, \sigma_{1005} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4019} = \langle S_{12}, \sigma_{1005} \rangle \Rightarrow \langle S_{13}, \sigma_{1005} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4021} = \frac{\langle x := x + 1, \sigma_{1005} \rangle \Rightarrow \sigma_{1006} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1005} \rangle \Rightarrow \langle S_6, \sigma_{1006} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4020} = \frac{\mathcal{T}_{4021}}{\langle S_{13}, \sigma_{1005} \rangle \Rightarrow \langle S_{15}, \sigma_{1006} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4022} = \frac{\langle S_6, \sigma_{1006} \rangle \Rightarrow \langle S_8, \sigma_{1006} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1006} \rangle \Rightarrow \langle S_9, \sigma_{1006} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4024} = \langle S_8, \sigma_{1006} \rangle \Rightarrow \langle y := y + 1, \sigma_{1006} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4023} = \frac{\mathcal{T}_{4024}}{\langle S_9, \sigma_{1006} \rangle \Rightarrow \langle S_{10}, \sigma_{1006} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4025} = \frac{\langle y := y + 1, \sigma_{1006} \rangle \Rightarrow \sigma_{1007} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1006} \rangle \Rightarrow \langle S_{11}, \sigma_{1007} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4026} = \langle S_{11}, \sigma_{1007} \rangle \Rightarrow \langle S_{12}, \sigma_{1007} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4027} = \langle S_{12}, \sigma_{1007} \rangle \Rightarrow \langle S_{13}, \sigma_{1007} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4029} = \frac{\langle x := x + 1, \sigma_{1007} \rangle \Rightarrow \sigma_{1008} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1007} \rangle \Rightarrow \langle S_6, \sigma_{1008} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4028} = \frac{\mathcal{T}_{4029}}{\langle S_{13}, \sigma_{1007} \rangle \Rightarrow \langle S_{15}, \sigma_{1008} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4030} = \frac{\langle S_6, \sigma_{1008} \rangle \Rightarrow \langle S_8, \sigma_{1008} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1008} \rangle \Rightarrow \langle S_9, \sigma_{1008} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4032} = \langle S_8, \sigma_{1008} \rangle \Rightarrow \langle y := y + 1, \sigma_{1008} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4031} = \frac{\mathcal{T}_{4032}}{\langle S_9, \sigma_{1008} \rangle \Rightarrow \langle S_{10}, \sigma_{1008} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4033} = \frac{\langle y := y + 1, \sigma_{1008} \rangle \Rightarrow \sigma_{1009} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1008} \rangle \Rightarrow \langle S_{11}, \sigma_{1009} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4034} = \langle S_{11}, \sigma_{1009} \rangle \Rightarrow \langle S_{12}, \sigma_{1009} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4035} = \langle S_{12}, \sigma_{1009} \rangle \Rightarrow \langle S_{13}, \sigma_{1009} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4037} = \frac{\langle x := x + 1, \sigma_{1009} \rangle \Rightarrow \sigma_{1010} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1009} \rangle \Rightarrow \langle S_6, \sigma_{1010} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4036} = \frac{\mathcal{T}_{4037}}{\langle S_{13}, \sigma_{1009} \rangle \Rightarrow \langle S_{15}, \sigma_{1010} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4038} = \frac{\langle S_6, \sigma_{1010} \rangle \Rightarrow \langle S_8, \sigma_{1010} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1010} \rangle \Rightarrow \langle S_9, \sigma_{1010} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4040} = \langle S_8, \sigma_{1010} \rangle \Rightarrow \langle y := y + 1, \sigma_{1010} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4039} = \frac{\mathcal{T}_{4040}}{\langle S_9, \sigma_{1010} \rangle \Rightarrow \langle S_{10}, \sigma_{1010} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4041} = \frac{\langle y := y + 1, \sigma_{1010} \rangle \Rightarrow \sigma_{1011} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1010} \rangle \Rightarrow \langle S_{11}, \sigma_{1011} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4042} = \langle S_{11}, \sigma_{1011} \rangle \Rightarrow \langle S_{12}, \sigma_{1011} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4043} = \langle S_{12}, \sigma_{1011} \rangle \Rightarrow \langle S_{13}, \sigma_{1011} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4045} = \frac{\langle x := x + 1, \sigma_{1011} \rangle \Rightarrow \sigma_{1012} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1011} \rangle \Rightarrow \langle S_6, \sigma_{1012} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4044} = \frac{\mathcal{T}_{4045}}{\langle S_{13}, \sigma_{1011} \rangle \Rightarrow \langle S_{15}, \sigma_{1012} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4046} = \frac{\langle S_6, \sigma_{1012} \rangle \Rightarrow \langle S_8, \sigma_{1012} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1012} \rangle \Rightarrow \langle S_9, \sigma_{1012} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4048} = \langle S_8, \sigma_{1012} \rangle \Rightarrow \langle y := y + 1, \sigma_{1012} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4047} = \frac{\mathcal{T}_{4048}}{\langle S_9, \sigma_{1012} \rangle \Rightarrow \langle S_{10}, \sigma_{1012} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4049} = \frac{\langle y := y + 1, \sigma_{1012} \rangle \Rightarrow \sigma_{1013} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1012} \rangle \Rightarrow \langle S_{11}, \sigma_{1013} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4050} = \langle S_{11}, \sigma_{1013} \rangle \Rightarrow \langle S_{12}, \sigma_{1013} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4051} = \langle S_{12}, \sigma_{1013} \rangle \Rightarrow \langle S_{13}, \sigma_{1013} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4053} = \frac{\langle x := x + 1, \sigma_{1013} \rangle \Rightarrow \sigma_{1014} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1013} \rangle \Rightarrow \langle S_6, \sigma_{1014} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4052} = \frac{\mathcal{T}_{4053}}{\langle S_{13}, \sigma_{1013} \rangle \Rightarrow \langle S_{15}, \sigma_{1014} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4054} = \frac{\langle S_6, \sigma_{1014} \rangle \Rightarrow \langle S_8, \sigma_{1014} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1014} \rangle \Rightarrow \langle S_9, \sigma_{1014} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4056} = \langle S_8, \sigma_{1014} \rangle \Rightarrow \langle y := y + 1, \sigma_{1014} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4055} = \frac{\mathcal{T}_{4056}}{\langle S_9, \sigma_{1014} \rangle \Rightarrow \langle S_{10}, \sigma_{1014} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4057} = \frac{\langle y := y + 1, \sigma_{1014} \rangle \Rightarrow \sigma_{1015} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1014} \rangle \Rightarrow \langle S_{11}, \sigma_{1015} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4058} = \langle S_{11}, \sigma_{1015} \rangle \Rightarrow \langle S_{12}, \sigma_{1015} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4059} = \langle S_{12}, \sigma_{1015} \rangle \Rightarrow \langle S_{13}, \sigma_{1015} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4061} = \frac{\langle x := x + 1, \sigma_{1015} \rangle \Rightarrow \sigma_{1016} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1015} \rangle \Rightarrow \langle S_6, \sigma_{1016} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4060} = \frac{\mathcal{T}_{4061}}{\langle S_{13}, \sigma_{1015} \rangle \Rightarrow \langle S_{15}, \sigma_{1016} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4062} = \frac{\langle S_6, \sigma_{1016} \rangle \Rightarrow \langle S_8, \sigma_{1016} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1016} \rangle \Rightarrow \langle S_9, \sigma_{1016} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4064} = \langle S_8, \sigma_{1016} \rangle \Rightarrow \langle y := y + 1, \sigma_{1016} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4063} = \frac{\mathcal{T}_{4064}}{\langle S_9, \sigma_{1016} \rangle \Rightarrow \langle S_{10}, \sigma_{1016} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4065} = \frac{\langle y := y + 1, \sigma_{1016} \rangle \Rightarrow \sigma_{1017} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1016} \rangle \Rightarrow \langle S_{11}, \sigma_{1017} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4066} = \langle S_{11}, \sigma_{1017} \rangle \Rightarrow \langle S_{12}, \sigma_{1017} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4067} = \langle S_{12}, \sigma_{1017} \rangle \Rightarrow \langle S_{13}, \sigma_{1017} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4069} = \frac{\langle x := x + 1, \sigma_{1017} \rangle \Rightarrow \sigma_{1018} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1017} \rangle \Rightarrow \langle S_6, \sigma_{1018} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4068} = \frac{\mathcal{T}_{4069}}{\langle S_{13}, \sigma_{1017} \rangle \Rightarrow \langle S_{15}, \sigma_{1018} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4070} = \frac{\langle S_6, \sigma_{1018} \rangle \Rightarrow \langle S_8, \sigma_{1018} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1018} \rangle \Rightarrow \langle S_9, \sigma_{1018} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4072} = \langle S_8, \sigma_{1018} \rangle \Rightarrow \langle y := y + 1, \sigma_{1018} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4071} = \frac{\mathcal{T}_{4072}}{\langle S_9, \sigma_{1018} \rangle \Rightarrow \langle S_{10}, \sigma_{1018} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4073} = \frac{\langle y := y + 1, \sigma_{1018} \rangle \Rightarrow \sigma_{1019} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1018} \rangle \Rightarrow \langle S_{11}, \sigma_{1019} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4074} = \langle S_{11}, \sigma_{1019} \rangle \Rightarrow \langle S_{12}, \sigma_{1019} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4075} = \langle S_{12}, \sigma_{1019} \rangle \Rightarrow \langle S_{13}, \sigma_{1019} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4077} = \frac{\langle x := x + 1, \sigma_{1019} \rangle \Rightarrow \sigma_{1020} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1019} \rangle \Rightarrow \langle S_6, \sigma_{1020} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4076} = \frac{\mathcal{T}_{4077}}{\langle S_{13}, \sigma_{1019} \rangle \Rightarrow \langle S_{15}, \sigma_{1020} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4078} = \frac{\langle S_6, \sigma_{1020} \rangle \Rightarrow \langle S_8, \sigma_{1020} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1020} \rangle \Rightarrow \langle S_9, \sigma_{1020} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4080} = \langle S_8, \sigma_{1020} \rangle \Rightarrow \langle y := y + 1, \sigma_{1020} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4079} = \frac{\mathcal{T}_{4080}}{\langle S_9, \sigma_{1020} \rangle \Rightarrow \langle S_{10}, \sigma_{1020} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4081} = \frac{\langle y := y + 1, \sigma_{1020} \rangle \Rightarrow \sigma_{1021} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1020} \rangle \Rightarrow \langle S_{11}, \sigma_{1021} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4082} = \langle S_{11}, \sigma_{1021} \rangle \Rightarrow \langle S_{12}, \sigma_{1021} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4083} = \langle S_{12}, \sigma_{1021} \rangle \Rightarrow \langle S_{13}, \sigma_{1021} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4085} = \frac{\langle x := x + 1, \sigma_{1021} \rangle \Rightarrow \sigma_{1022} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1021} \rangle \Rightarrow \langle S_6, \sigma_{1022} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4084} = \frac{\mathcal{T}_{4085}}{\langle S_{13}, \sigma_{1021} \rangle \Rightarrow \langle S_{15}, \sigma_{1022} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4086} = \frac{\langle S_6, \sigma_{1022} \rangle \Rightarrow \langle S_8, \sigma_{1022} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1022} \rangle \Rightarrow \langle S_9, \sigma_{1022} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4088} = \langle S_8, \sigma_{1022} \rangle \Rightarrow \langle y := y + 1, \sigma_{1022} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4087} = \frac{\mathcal{T}_{4088}}{\langle S_9, \sigma_{1022} \rangle \Rightarrow \langle S_{10}, \sigma_{1022} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4089} = \frac{\langle y := y + 1, \sigma_{1022} \rangle \Rightarrow \sigma_{1023} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1022} \rangle \Rightarrow \langle S_{11}, \sigma_{1023} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4090} = \langle S_{11}, \sigma_{1023} \rangle \Rightarrow \langle S_{12}, \sigma_{1023} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4091} = \langle S_{12}, \sigma_{1023} \rangle \Rightarrow \langle S_{13}, \sigma_{1023} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4093} = \frac{\langle x := x + 1, \sigma_{1023} \rangle \Rightarrow \sigma_{1024} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1023} \rangle \Rightarrow \langle S_6, \sigma_{1024} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4092} = \frac{\mathcal{T}_{4093}}{\langle S_{13}, \sigma_{1023} \rangle \Rightarrow \langle S_{15}, \sigma_{1024} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4094} = \frac{\langle S_6, \sigma_{1024} \rangle \Rightarrow \langle S_8, \sigma_{1024} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1024} \rangle \Rightarrow \langle S_9, \sigma_{1024} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4096} = \langle S_8, \sigma_{1024} \rangle \Rightarrow \langle y := y + 1, \sigma_{1024} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4095} = \frac{\mathcal{T}_{4096}}{\langle S_9, \sigma_{1024} \rangle \Rightarrow \langle S_{10}, \sigma_{1024} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4097} = \frac{\langle y := y + 1, \sigma_{1024} \rangle \Rightarrow \sigma_{1025} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1024} \rangle \Rightarrow \langle S_{11}, \sigma_{1025} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4098} = \langle S_{11}, \sigma_{1025} \rangle \Rightarrow \langle S_{12}, \sigma_{1025} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4099} = \langle S_{12}, \sigma_{1025} \rangle \Rightarrow \langle S_{13}, \sigma_{1025} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4101} = \frac{\langle x := x + 1, \sigma_{1025} \rangle \Rightarrow \sigma_{1026} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1025} \rangle \Rightarrow \langle S_6, \sigma_{1026} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4100} = \frac{\mathcal{T}_{4101}}{\langle S_{13}, \sigma_{1025} \rangle \Rightarrow \langle S_{15}, \sigma_{1026} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4102} = \frac{\langle S_6, \sigma_{1026} \rangle \Rightarrow \langle S_8, \sigma_{1026} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1026} \rangle \Rightarrow \langle S_9, \sigma_{1026} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4104} = \langle S_8, \sigma_{1026} \rangle \Rightarrow \langle y := y + 1, \sigma_{1026} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4103} = \frac{\mathcal{T}_{4104}}{\langle S_9, \sigma_{1026} \rangle \Rightarrow \langle S_{10}, \sigma_{1026} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4105} = \frac{\langle y := y + 1, \sigma_{1026} \rangle \Rightarrow \sigma_{1027} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1026} \rangle \Rightarrow \langle S_{11}, \sigma_{1027} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4106} = \langle S_{11}, \sigma_{1027} \rangle \Rightarrow \langle S_{12}, \sigma_{1027} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4107} = \langle S_{12}, \sigma_{1027} \rangle \Rightarrow \langle S_{13}, \sigma_{1027} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4109} = \frac{\langle x := x + 1, \sigma_{1027} \rangle \Rightarrow \sigma_{1028} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1027} \rangle \Rightarrow \langle S_6, \sigma_{1028} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4108} = \frac{\mathcal{T}_{4109}}{\langle S_{13}, \sigma_{1027} \rangle \Rightarrow \langle S_{15}, \sigma_{1028} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4110} = \frac{\langle S_6, \sigma_{1028} \rangle \Rightarrow \langle S_8, \sigma_{1028} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1028} \rangle \Rightarrow \langle S_9, \sigma_{1028} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4112} = \langle S_8, \sigma_{1028} \rangle \Rightarrow \langle y := y + 1, \sigma_{1028} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4111} = \frac{\mathcal{T}_{4112}}{\langle S_9, \sigma_{1028} \rangle \Rightarrow \langle S_{10}, \sigma_{1028} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4113} = \frac{\langle y := y + 1, \sigma_{1028} \rangle \Rightarrow \sigma_{1029} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1028} \rangle \Rightarrow \langle S_{11}, \sigma_{1029} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4114} = \langle S_{11}, \sigma_{1029} \rangle \Rightarrow \langle S_{12}, \sigma_{1029} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4115} = \langle S_{12}, \sigma_{1029} \rangle \Rightarrow \langle S_{13}, \sigma_{1029} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4117} = \frac{\langle x := x + 1, \sigma_{1029} \rangle \Rightarrow \sigma_{1030} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1029} \rangle \Rightarrow \langle S_6, \sigma_{1030} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4116} = \frac{\mathcal{T}_{4117}}{\langle S_{13}, \sigma_{1029} \rangle \Rightarrow \langle S_{15}, \sigma_{1030} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4118} = \frac{\langle S_6, \sigma_{1030} \rangle \Rightarrow \langle S_8, \sigma_{1030} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1030} \rangle \Rightarrow \langle S_9, \sigma_{1030} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4120} = \langle S_8, \sigma_{1030} \rangle \Rightarrow \langle y := y + 1, \sigma_{1030} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4119} = \frac{\mathcal{T}_{4120}}{\langle S_9, \sigma_{1030} \rangle \Rightarrow \langle S_{10}, \sigma_{1030} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4121} = \frac{\langle y := y + 1, \sigma_{1030} \rangle \Rightarrow \sigma_{1031} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1030} \rangle \Rightarrow \langle S_{11}, \sigma_{1031} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4122} = \langle S_{11}, \sigma_{1031} \rangle \Rightarrow \langle S_{12}, \sigma_{1031} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4123} = \langle S_{12}, \sigma_{1031} \rangle \Rightarrow \langle S_{13}, \sigma_{1031} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4125} = \frac{\langle x := x + 1, \sigma_{1031} \rangle \Rightarrow \sigma_{1032} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1031} \rangle \Rightarrow \langle S_6, \sigma_{1032} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4124} = \frac{\mathcal{T}_{4125}}{\langle S_{13}, \sigma_{1031} \rangle \Rightarrow \langle S_{15}, \sigma_{1032} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4126} = \frac{\langle S_6, \sigma_{1032} \rangle \Rightarrow \langle S_8, \sigma_{1032} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1032} \rangle \Rightarrow \langle S_9, \sigma_{1032} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4128} = \langle S_8, \sigma_{1032} \rangle \Rightarrow \langle y := y + 1, \sigma_{1032} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4127} = \frac{\mathcal{T}_{4128}}{\langle S_9, \sigma_{1032} \rangle \Rightarrow \langle S_{10}, \sigma_{1032} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4129} = \frac{\langle y := y + 1, \sigma_{1032} \rangle \Rightarrow \sigma_{1033} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1032} \rangle \Rightarrow \langle S_{11}, \sigma_{1033} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4130} = \langle S_{11}, \sigma_{1033} \rangle \Rightarrow \langle S_{12}, \sigma_{1033} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4131} = \langle S_{12}, \sigma_{1033} \rangle \Rightarrow \langle S_{13}, \sigma_{1033} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4133} = \frac{\langle x := x + 1, \sigma_{1033} \rangle \Rightarrow \sigma_{1034} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1033} \rangle \Rightarrow \langle S_6, \sigma_{1034} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4132} = \frac{\mathcal{T}_{4133}}{\langle S_{13}, \sigma_{1033} \rangle \Rightarrow \langle S_{15}, \sigma_{1034} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4134} = \frac{\langle S_6, \sigma_{1034} \rangle \Rightarrow \langle S_8, \sigma_{1034} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1034} \rangle \Rightarrow \langle S_9, \sigma_{1034} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4136} = \langle S_8, \sigma_{1034} \rangle \Rightarrow \langle y := y + 1, \sigma_{1034} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4135} = \frac{\mathcal{T}_{4136}}{\langle S_9, \sigma_{1034} \rangle \Rightarrow \langle S_{10}, \sigma_{1034} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4137} = \frac{\langle y := y + 1, \sigma_{1034} \rangle \Rightarrow \sigma_{1035} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1034} \rangle \Rightarrow \langle S_{11}, \sigma_{1035} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4138} = \langle S_{11}, \sigma_{1035} \rangle \Rightarrow \langle S_{12}, \sigma_{1035} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4139} = \langle S_{12}, \sigma_{1035} \rangle \Rightarrow \langle S_{13}, \sigma_{1035} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4141} = \frac{\langle x := x + 1, \sigma_{1035} \rangle \Rightarrow \sigma_{1036} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1035} \rangle \Rightarrow \langle S_6, \sigma_{1036} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4140} = \frac{\mathcal{T}_{4141}}{\langle S_{13}, \sigma_{1035} \rangle \Rightarrow \langle S_{15}, \sigma_{1036} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4142} = \frac{\langle S_6, \sigma_{1036} \rangle \Rightarrow \langle S_8, \sigma_{1036} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1036} \rangle \Rightarrow \langle S_9, \sigma_{1036} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4144} = \langle S_8, \sigma_{1036} \rangle \Rightarrow \langle y := y + 1, \sigma_{1036} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4143} = \frac{\mathcal{T}_{4144}}{\langle S_9, \sigma_{1036} \rangle \Rightarrow \langle S_{10}, \sigma_{1036} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4145} = \frac{\langle y := y + 1, \sigma_{1036} \rangle \Rightarrow \sigma_{1037} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1036} \rangle \Rightarrow \langle S_{11}, \sigma_{1037} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4146} = \langle S_{11}, \sigma_{1037} \rangle \Rightarrow \langle S_{12}, \sigma_{1037} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4147} = \langle S_{12}, \sigma_{1037} \rangle \Rightarrow \langle S_{13}, \sigma_{1037} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4149} = \frac{\langle x := x + 1, \sigma_{1037} \rangle \Rightarrow \sigma_{1038} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1037} \rangle \Rightarrow \langle S_6, \sigma_{1038} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4148} = \frac{\mathcal{T}_{4149}}{\langle S_{13}, \sigma_{1037} \rangle \Rightarrow \langle S_{15}, \sigma_{1038} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4150} = \frac{\langle S_6, \sigma_{1038} \rangle \Rightarrow \langle S_8, \sigma_{1038} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1038} \rangle \Rightarrow \langle S_9, \sigma_{1038} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4152} = \langle S_8, \sigma_{1038} \rangle \Rightarrow \langle y := y + 1, \sigma_{1038} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4151} = \frac{\mathcal{T}_{4152}}{\langle S_9, \sigma_{1038} \rangle \Rightarrow \langle S_{10}, \sigma_{1038} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4153} = \frac{\langle y := y + 1, \sigma_{1038} \rangle \Rightarrow \sigma_{1039} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1038} \rangle \Rightarrow \langle S_{11}, \sigma_{1039} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4154} = \langle S_{11}, \sigma_{1039} \rangle \Rightarrow \langle S_{12}, \sigma_{1039} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4155} = \langle S_{12}, \sigma_{1039} \rangle \Rightarrow \langle S_{13}, \sigma_{1039} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4157} = \frac{\langle x := x + 1, \sigma_{1039} \rangle \Rightarrow \sigma_{1040} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1039} \rangle \Rightarrow \langle S_6, \sigma_{1040} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4156} = \frac{\mathcal{T}_{4157}}{\langle S_{13}, \sigma_{1039} \rangle \Rightarrow \langle S_{15}, \sigma_{1040} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4158} = \frac{\langle S_6, \sigma_{1040} \rangle \Rightarrow \langle S_8, \sigma_{1040} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1040} \rangle \Rightarrow \langle S_9, \sigma_{1040} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4160} = \langle S_8, \sigma_{1040} \rangle \Rightarrow \langle y := y + 1, \sigma_{1040} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4159} = \frac{\mathcal{T}_{4160}}{\langle S_9, \sigma_{1040} \rangle \Rightarrow \langle S_{10}, \sigma_{1040} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4161} = \frac{\langle y := y + 1, \sigma_{1040} \rangle \Rightarrow \sigma_{1041} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1040} \rangle \Rightarrow \langle S_{11}, \sigma_{1041} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4162} = \langle S_{11}, \sigma_{1041} \rangle \Rightarrow \langle S_{12}, \sigma_{1041} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4163} = \langle S_{12}, \sigma_{1041} \rangle \Rightarrow \langle S_{13}, \sigma_{1041} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4165} = \frac{\langle x := x + 1, \sigma_{1041} \rangle \Rightarrow \sigma_{1042} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1041} \rangle \Rightarrow \langle S_6, \sigma_{1042} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4164} = \frac{\mathcal{T}_{4165}}{\langle S_{13}, \sigma_{1041} \rangle \Rightarrow \langle S_{15}, \sigma_{1042} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4166} = \frac{\langle S_6, \sigma_{1042} \rangle \Rightarrow \langle S_8, \sigma_{1042} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1042} \rangle \Rightarrow \langle S_9, \sigma_{1042} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4168} = \langle S_8, \sigma_{1042} \rangle \Rightarrow \langle y := y + 1, \sigma_{1042} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4167} = \frac{\mathcal{T}_{4168}}{\langle S_9, \sigma_{1042} \rangle \Rightarrow \langle S_{10}, \sigma_{1042} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4169} = \frac{\langle y := y + 1, \sigma_{1042} \rangle \Rightarrow \sigma_{1043} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1042} \rangle \Rightarrow \langle S_{11}, \sigma_{1043} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4170} = \langle S_{11}, \sigma_{1043} \rangle \Rightarrow \langle S_{12}, \sigma_{1043} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4171} = \langle S_{12}, \sigma_{1043} \rangle \Rightarrow \langle S_{13}, \sigma_{1043} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4173} = \frac{\langle x := x + 1, \sigma_{1043} \rangle \Rightarrow \sigma_{1044} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1043} \rangle \Rightarrow \langle S_6, \sigma_{1044} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4172} = \frac{\mathcal{T}_{4173}}{\langle S_{13}, \sigma_{1043} \rangle \Rightarrow \langle S_{15}, \sigma_{1044} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4174} = \frac{\langle S_6, \sigma_{1044} \rangle \Rightarrow \langle S_8, \sigma_{1044} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1044} \rangle \Rightarrow \langle S_9, \sigma_{1044} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4176} = \langle S_8, \sigma_{1044} \rangle \Rightarrow \langle y := y + 1, \sigma_{1044} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4175} = \frac{\mathcal{T}_{4176}}{\langle S_9, \sigma_{1044} \rangle \Rightarrow \langle S_{10}, \sigma_{1044} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4177} = \frac{\langle y := y + 1, \sigma_{1044} \rangle \Rightarrow \sigma_{1045} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1044} \rangle \Rightarrow \langle S_{11}, \sigma_{1045} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4178} = \langle S_{11}, \sigma_{1045} \rangle \Rightarrow \langle S_{12}, \sigma_{1045} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4179} = \langle S_{12}, \sigma_{1045} \rangle \Rightarrow \langle S_{13}, \sigma_{1045} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4181} = \frac{\langle x := x + 1, \sigma_{1045} \rangle \Rightarrow \sigma_{1046} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1045} \rangle \Rightarrow \langle S_6, \sigma_{1046} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4180} = \frac{\mathcal{T}_{4181}}{\langle S_{13}, \sigma_{1045} \rangle \Rightarrow \langle S_{15}, \sigma_{1046} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4182} = \frac{\langle S_6, \sigma_{1046} \rangle \Rightarrow \langle S_8, \sigma_{1046} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1046} \rangle \Rightarrow \langle S_9, \sigma_{1046} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4184} = \langle S_8, \sigma_{1046} \rangle \Rightarrow \langle y := y + 1, \sigma_{1046} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4183} = \frac{\mathcal{T}_{4184}}{\langle S_9, \sigma_{1046} \rangle \Rightarrow \langle S_{10}, \sigma_{1046} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4185} = \frac{\langle y := y + 1, \sigma_{1046} \rangle \Rightarrow \sigma_{1047} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1046} \rangle \Rightarrow \langle S_{11}, \sigma_{1047} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4186} = \langle S_{11}, \sigma_{1047} \rangle \Rightarrow \langle S_{12}, \sigma_{1047} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4187} = \langle S_{12}, \sigma_{1047} \rangle \Rightarrow \langle S_{13}, \sigma_{1047} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4189} = \frac{\langle x := x + 1, \sigma_{1047} \rangle \Rightarrow \sigma_{1048} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1047} \rangle \Rightarrow \langle S_6, \sigma_{1048} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4188} = \frac{\mathcal{T}_{4189}}{\langle S_{13}, \sigma_{1047} \rangle \Rightarrow \langle S_{15}, \sigma_{1048} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4190} = \frac{\langle S_6, \sigma_{1048} \rangle \Rightarrow \langle S_8, \sigma_{1048} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1048} \rangle \Rightarrow \langle S_9, \sigma_{1048} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{4192} = \langle S_8, \sigma_{1048} \rangle \Rightarrow \langle y := y + 1, \sigma_{1048} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4191} = \frac{\mathcal{T}_{4192}}{\langle S_9, \sigma_{1048} \rangle \Rightarrow \langle S_{10}, \sigma_{1048} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{4193} = \frac{\langle y := y + 1, \sigma_{1048} \rangle \Rightarrow \sigma_{1049} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1048} \rangle \Rightarrow \langle S_{11}, \sigma_{1049} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{4194} = \langle S_{11}, \sigma_{1049} \rangle \Rightarrow \langle S_{12}, \sigma_{1049} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{4195} = \langle S_{12}, \sigma_{1049} \rangle \Rightarrow \langle S_{13}, \sigma_{1049} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{4197} = \frac{\langle x := x + 1, \sigma_{1049} \rangle \Rightarrow \sigma_{1050} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1049} \rangle \Rightarrow \langle S_6, \sigma_{1050} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4196} = \frac{\mathcal{T}_{4197}}{\langle S_{13}, \sigma_{1049} \rangle \Rightarrow \langle S_{15}, \sigma_{1050} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4198} = \frac{\langle S_6, \sigma_{1050} \rangle \Rightarrow \langle S_8, \sigma_{1050} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1050} \rangle \Rightarrow \langle S_9, \sigma_{1050} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4200} = \langle S_8, \sigma_{1050} \rangle \Rightarrow \langle y := y + 1, \sigma_{1050} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4199} = \frac{\mathcal{T}_{4200}}{\langle S_9, \sigma_{1050} \rangle \Rightarrow \langle S_{10}, \sigma_{1050} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4201} = \frac{\langle y := y + 1, \sigma_{1050} \rangle \Rightarrow \sigma_{1051} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1050} \rangle \Rightarrow \langle S_{11}, \sigma_{1051} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4202} = \langle S_{11}, \sigma_{1051} \rangle \Rightarrow \langle S_{12}, \sigma_{1051} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4203} = \langle S_{12}, \sigma_{1051} \rangle \Rightarrow \langle S_{13}, \sigma_{1051} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4205} = \frac{\langle x := x + 1, \sigma_{1051} \rangle \Rightarrow \sigma_{1052} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1051} \rangle \Rightarrow \langle S_6, \sigma_{1052} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4204} = \frac{\mathcal{T}_{4205}}{\langle S_{13}, \sigma_{1051} \rangle \Rightarrow \langle S_{15}, \sigma_{1052} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4206} = \frac{\langle S_6, \sigma_{1052} \rangle \Rightarrow \langle S_8, \sigma_{1052} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1052} \rangle \Rightarrow \langle S_9, \sigma_{1052} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4208} = \langle S_8, \sigma_{1052} \rangle \Rightarrow \langle y := y + 1, \sigma_{1052} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4207} = \frac{\mathcal{T}_{4208}}{\langle S_9, \sigma_{1052} \rangle \Rightarrow \langle S_{10}, \sigma_{1052} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4209} = \frac{\langle y := y + 1, \sigma_{1052} \rangle \Rightarrow \sigma_{1053} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1052} \rangle \Rightarrow \langle S_{11}, \sigma_{1053} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4210} = \langle S_{11}, \sigma_{1053} \rangle \Rightarrow \langle S_{12}, \sigma_{1053} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4211} = \langle S_{12}, \sigma_{1053} \rangle \Rightarrow \langle S_{13}, \sigma_{1053} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4213} = \frac{\langle x := x + 1, \sigma_{1053} \rangle \Rightarrow \sigma_{1054} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1053} \rangle \Rightarrow \langle S_6, \sigma_{1054} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4212} = \frac{\mathcal{T}_{4213}}{\langle S_{13}, \sigma_{1053} \rangle \Rightarrow \langle S_{15}, \sigma_{1054} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4214} = \frac{\langle S_6, \sigma_{1054} \rangle \Rightarrow \langle S_8, \sigma_{1054} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1054} \rangle \Rightarrow \langle S_9, \sigma_{1054} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4216} = \langle S_8, \sigma_{1054} \rangle \Rightarrow \langle y := y + 1, \sigma_{1054} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4215} = \frac{\mathcal{T}_{4216}}{\langle S_9, \sigma_{1054} \rangle \Rightarrow \langle S_{10}, \sigma_{1054} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4217} = \frac{\langle y := y + 1, \sigma_{1054} \rangle \Rightarrow \sigma_{1055} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1054} \rangle \Rightarrow \langle S_{11}, \sigma_{1055} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4218} = \langle S_{11}, \sigma_{1055} \rangle \Rightarrow \langle S_{12}, \sigma_{1055} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4219} = \langle S_{12}, \sigma_{1055} \rangle \Rightarrow \langle S_{13}, \sigma_{1055} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4221} = \frac{\langle x := x + 1, \sigma_{1055} \rangle \Rightarrow \sigma_{1056} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1055} \rangle \Rightarrow \langle S_6, \sigma_{1056} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4220} = \frac{\mathcal{T}_{4221}}{\langle S_{13}, \sigma_{1055} \rangle \Rightarrow \langle S_{15}, \sigma_{1056} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4222} = \frac{\langle S_6, \sigma_{1056} \rangle \Rightarrow \langle S_8, \sigma_{1056} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1056} \rangle \Rightarrow \langle S_9, \sigma_{1056} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4224} = \langle S_8, \sigma_{1056} \rangle \Rightarrow \langle y := y + 1, \sigma_{1056} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4223} = \frac{\mathcal{T}_{4224}}{\langle S_9, \sigma_{1056} \rangle \Rightarrow \langle S_{10}, \sigma_{1056} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4225} = \frac{\langle y := y + 1, \sigma_{1056} \rangle \Rightarrow \sigma_{1057} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1056} \rangle \Rightarrow \langle S_{11}, \sigma_{1057} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4226} = \langle S_{11}, \sigma_{1057} \rangle \Rightarrow \langle S_{12}, \sigma_{1057} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4227} = \langle S_{12}, \sigma_{1057} \rangle \Rightarrow \langle S_{13}, \sigma_{1057} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4229} = \frac{\langle x := x + 1, \sigma_{1057} \rangle \Rightarrow \sigma_{1058} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1057} \rangle \Rightarrow \langle S_6, \sigma_{1058} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4228} = \frac{\mathcal{T}_{4229}}{\langle S_{13}, \sigma_{1057} \rangle \Rightarrow \langle S_{15}, \sigma_{1058} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4230} = \frac{\langle S_6, \sigma_{1058} \rangle \Rightarrow \langle S_8, \sigma_{1058} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1058} \rangle \Rightarrow \langle S_9, \sigma_{1058} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4232} = \langle S_8, \sigma_{1058} \rangle \Rightarrow \langle y := y + 1, \sigma_{1058} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4231} = \frac{\mathcal{T}_{4232}}{\langle S_9, \sigma_{1058} \rangle \Rightarrow \langle S_{10}, \sigma_{1058} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4233} = \frac{\langle y := y + 1, \sigma_{1058} \rangle \Rightarrow \sigma_{1059} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1058} \rangle \Rightarrow \langle S_{11}, \sigma_{1059} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4234} = \langle S_{11}, \sigma_{1059} \rangle \Rightarrow \langle S_{12}, \sigma_{1059} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4235} = \langle S_{12}, \sigma_{1059} \rangle \Rightarrow \langle S_{13}, \sigma_{1059} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4237} = \frac{\langle x := x + 1, \sigma_{1059} \rangle \Rightarrow \sigma_{1060} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1059} \rangle \Rightarrow \langle S_6, \sigma_{1060} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4236} = \frac{\mathcal{T}_{4237}}{\langle S_{13}, \sigma_{1059} \rangle \Rightarrow \langle S_{15}, \sigma_{1060} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4238} = \frac{\langle S_6, \sigma_{1060} \rangle \Rightarrow \langle S_8, \sigma_{1060} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1060} \rangle \Rightarrow \langle S_9, \sigma_{1060} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4240} = \langle S_8, \sigma_{1060} \rangle \Rightarrow \langle y := y + 1, \sigma_{1060} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4239} = \frac{\mathcal{T}_{4240}}{\langle S_9, \sigma_{1060} \rangle \Rightarrow \langle S_{10}, \sigma_{1060} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4241} = \frac{\langle y := y + 1, \sigma_{1060} \rangle \Rightarrow \sigma_{1061} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1060} \rangle \Rightarrow \langle S_{11}, \sigma_{1061} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4242} = \langle S_{11}, \sigma_{1061} \rangle \Rightarrow \langle S_{12}, \sigma_{1061} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4243} = \langle S_{12}, \sigma_{1061} \rangle \Rightarrow \langle S_{13}, \sigma_{1061} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4245} = \frac{\langle x := x + 1, \sigma_{1061} \rangle \Rightarrow \sigma_{1062} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1061} \rangle \Rightarrow \langle S_6, \sigma_{1062} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4244} = \frac{\mathcal{T}_{4245}}{\langle S_{13}, \sigma_{1061} \rangle \Rightarrow \langle S_{15}, \sigma_{1062} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4246} = \frac{\langle S_6, \sigma_{1062} \rangle \Rightarrow \langle S_8, \sigma_{1062} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1062} \rangle \Rightarrow \langle S_9, \sigma_{1062} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4248} = \langle S_8, \sigma_{1062} \rangle \Rightarrow \langle y := y + 1, \sigma_{1062} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4247} = \frac{\mathcal{T}_{4248}}{\langle S_9, \sigma_{1062} \rangle \Rightarrow \langle S_{10}, \sigma_{1062} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4249} = \frac{\langle y := y + 1, \sigma_{1062} \rangle \Rightarrow \sigma_{1063} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1062} \rangle \Rightarrow \langle S_{11}, \sigma_{1063} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4250} = \langle S_{11}, \sigma_{1063} \rangle \Rightarrow \langle S_{12}, \sigma_{1063} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4251} = \langle S_{12}, \sigma_{1063} \rangle \Rightarrow \langle S_{13}, \sigma_{1063} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4253} = \frac{\langle x := x + 1, \sigma_{1063} \rangle \Rightarrow \sigma_{1064} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1063} \rangle \Rightarrow \langle S_6, \sigma_{1064} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4252} = \frac{\mathcal{T}_{4253}}{\langle S_{13}, \sigma_{1063} \rangle \Rightarrow \langle S_{15}, \sigma_{1064} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4254} = \frac{\langle S_6, \sigma_{1064} \rangle \Rightarrow \langle S_8, \sigma_{1064} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1064} \rangle \Rightarrow \langle S_9, \sigma_{1064} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4256} = \langle S_8, \sigma_{1064} \rangle \Rightarrow \langle y := y + 1, \sigma_{1064} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4255} = \frac{\mathcal{T}_{4256}}{\langle S_9, \sigma_{1064} \rangle \Rightarrow \langle S_{10}, \sigma_{1064} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4257} = \frac{\langle y := y + 1, \sigma_{1064} \rangle \Rightarrow \sigma_{1065} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1064} \rangle \Rightarrow \langle S_{11}, \sigma_{1065} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4258} = \langle S_{11}, \sigma_{1065} \rangle \Rightarrow \langle S_{12}, \sigma_{1065} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4259} = \langle S_{12}, \sigma_{1065} \rangle \Rightarrow \langle S_{13}, \sigma_{1065} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4261} = \frac{\langle x := x + 1, \sigma_{1065} \rangle \Rightarrow \sigma_{1066} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1065} \rangle \Rightarrow \langle S_6, \sigma_{1066} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4260} = \frac{\mathcal{T}_{4261}}{\langle S_{13}, \sigma_{1065} \rangle \Rightarrow \langle S_{15}, \sigma_{1066} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4262} = \frac{\langle S_6, \sigma_{1066} \rangle \Rightarrow \langle S_8, \sigma_{1066} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1066} \rangle \Rightarrow \langle S_9, \sigma_{1066} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4264} = \langle S_8, \sigma_{1066} \rangle \Rightarrow \langle y := y + 1, \sigma_{1066} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4263} = \frac{\mathcal{T}_{4264}}{\langle S_9, \sigma_{1066} \rangle \Rightarrow \langle S_{10}, \sigma_{1066} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4265} = \frac{\langle y := y + 1, \sigma_{1066} \rangle \Rightarrow \sigma_{1067} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1066} \rangle \Rightarrow \langle S_{11}, \sigma_{1067} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4266} = \langle S_{11}, \sigma_{1067} \rangle \Rightarrow \langle S_{12}, \sigma_{1067} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4267} = \langle S_{12}, \sigma_{1067} \rangle \Rightarrow \langle S_{13}, \sigma_{1067} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4269} = \frac{\langle x := x + 1, \sigma_{1067} \rangle \Rightarrow \sigma_{1068} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1067} \rangle \Rightarrow \langle S_6, \sigma_{1068} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4268} = \frac{\mathcal{T}_{4269}}{\langle S_{13}, \sigma_{1067} \rangle \Rightarrow \langle S_{15}, \sigma_{1068} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4270} = \frac{\langle S_6, \sigma_{1068} \rangle \Rightarrow \langle S_8, \sigma_{1068} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1068} \rangle \Rightarrow \langle S_9, \sigma_{1068} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4272} = \langle S_8, \sigma_{1068} \rangle \Rightarrow \langle y := y + 1, \sigma_{1068} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4271} = \frac{\mathcal{T}_{4272}}{\langle S_9, \sigma_{1068} \rangle \Rightarrow \langle S_{10}, \sigma_{1068} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4273} = \frac{\langle y := y + 1, \sigma_{1068} \rangle \Rightarrow \sigma_{1069} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1068} \rangle \Rightarrow \langle S_{11}, \sigma_{1069} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4274} = \langle S_{11}, \sigma_{1069} \rangle \Rightarrow \langle S_{12}, \sigma_{1069} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4275} = \langle S_{12}, \sigma_{1069} \rangle \Rightarrow \langle S_{13}, \sigma_{1069} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4277} = \frac{\langle x := x + 1, \sigma_{1069} \rangle \Rightarrow \sigma_{1070} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1069} \rangle \Rightarrow \langle S_6, \sigma_{1070} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4276} = \frac{\mathcal{T}_{4277}}{\langle S_{13}, \sigma_{1069} \rangle \Rightarrow \langle S_{15}, \sigma_{1070} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4278} = \frac{\langle S_6, \sigma_{1070} \rangle \Rightarrow \langle S_8, \sigma_{1070} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1070} \rangle \Rightarrow \langle S_9, \sigma_{1070} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4280} = \langle S_8, \sigma_{1070} \rangle \Rightarrow \langle y := y + 1, \sigma_{1070} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4279} = \frac{\mathcal{T}_{4280}}{\langle S_9, \sigma_{1070} \rangle \Rightarrow \langle S_{10}, \sigma_{1070} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4281} = \frac{\langle y := y + 1, \sigma_{1070} \rangle \Rightarrow \sigma_{1071} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1070} \rangle \Rightarrow \langle S_{11}, \sigma_{1071} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4282} = \langle S_{11}, \sigma_{1071} \rangle \Rightarrow \langle S_{12}, \sigma_{1071} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4283} = \langle S_{12}, \sigma_{1071} \rangle \Rightarrow \langle S_{13}, \sigma_{1071} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4285} = \frac{\langle x := x + 1, \sigma_{1071} \rangle \Rightarrow \sigma_{1072} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1071} \rangle \Rightarrow \langle S_6, \sigma_{1072} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4284} = \frac{\mathcal{T}_{4285}}{\langle S_{13}, \sigma_{1071} \rangle \Rightarrow \langle S_{15}, \sigma_{1072} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4286} = \frac{\langle S_6, \sigma_{1072} \rangle \Rightarrow \langle S_8, \sigma_{1072} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1072} \rangle \Rightarrow \langle S_9, \sigma_{1072} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4288} = \langle S_8, \sigma_{1072} \rangle \Rightarrow \langle y := y + 1, \sigma_{1072} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4287} = \frac{\mathcal{T}_{4288}}{\langle S_9, \sigma_{1072} \rangle \Rightarrow \langle S_{10}, \sigma_{1072} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4289} = \frac{\langle y := y + 1, \sigma_{1072} \rangle \Rightarrow \sigma_{1073} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1072} \rangle \Rightarrow \langle S_{11}, \sigma_{1073} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4290} = \langle S_{11}, \sigma_{1073} \rangle \Rightarrow \langle S_{12}, \sigma_{1073} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{4291} = \langle S_{12}, \sigma_{1073} \rangle \Rightarrow \langle S_{13}, \sigma_{1073} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{4293} = \frac{\langle x := x + 1, \sigma_{1073} \rangle \Rightarrow \sigma_{1074} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1073} \rangle \Rightarrow \langle S_6, \sigma_{1074} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4292} = \frac{\mathcal{T}_{4293}}{\langle S_{13}, \sigma_{1073} \rangle \Rightarrow \langle S_{15}, \sigma_{1074} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{4294} = \frac{\langle S_6, \sigma_{1074} \rangle \Rightarrow \langle S_8, \sigma_{1074} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1074} \rangle \Rightarrow \langle S_9, \sigma_{1074} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{4296} = \langle S_8, \sigma_{1074} \rangle \Rightarrow \langle y := y + 1, \sigma_{1074} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4295} = \frac{\mathcal{T}_{4296}}{\langle S_9, \sigma_{1074} \rangle \Rightarrow \langle S_{10}, \sigma_{1074} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{4297} = \frac{\langle y := y + 1, \sigma_{1074} \rangle \Rightarrow \sigma_{1075} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1074} \rangle \Rightarrow \langle S_{11}, \sigma_{1075} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4298} = \langle S_{11}, \sigma_{1075} \rangle \Rightarrow \langle S_{12}, \sigma_{1075} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4299} = \langle S_{12}, \sigma_{1075} \rangle \Rightarrow \langle S_{13}, \sigma_{1075} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4301} = \frac{\langle x := x + 1, \sigma_{1075} \rangle \Rightarrow \sigma_{1076} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1075} \rangle \Rightarrow \langle S_6, \sigma_{1076} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4300} = \frac{\mathcal{T}_{4301}}{\langle S_{13}, \sigma_{1075} \rangle \Rightarrow \langle S_{15}, \sigma_{1076} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4302} = \frac{\langle S_6, \sigma_{1076} \rangle \Rightarrow \langle S_8, \sigma_{1076} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1076} \rangle \Rightarrow \langle S_9, \sigma_{1076} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4304} = \langle S_8, \sigma_{1076} \rangle \Rightarrow \langle y := y + 1, \sigma_{1076} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4303} = \frac{\mathcal{T}_{4304}}{\langle S_9, \sigma_{1076} \rangle \Rightarrow \langle S_{10}, \sigma_{1076} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4305} = \frac{\langle y := y + 1, \sigma_{1076} \rangle \Rightarrow \sigma_{1077} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1076} \rangle \Rightarrow \langle S_{11}, \sigma_{1077} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4306} = \langle S_{11}, \sigma_{1077} \rangle \Rightarrow \langle S_{12}, \sigma_{1077} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4307} = \langle S_{12}, \sigma_{1077} \rangle \Rightarrow \langle S_{13}, \sigma_{1077} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4309} = \frac{\langle x := x + 1, \sigma_{1077} \rangle \Rightarrow \sigma_{1078} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1077} \rangle \Rightarrow \langle S_6, \sigma_{1078} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4308} = \frac{\mathcal{T}_{4309}}{\langle S_{13}, \sigma_{1077} \rangle \Rightarrow \langle S_{15}, \sigma_{1078} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4310} = \frac{\langle S_6, \sigma_{1078} \rangle \Rightarrow \langle S_8, \sigma_{1078} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1078} \rangle \Rightarrow \langle S_9, \sigma_{1078} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4312} = \langle S_8, \sigma_{1078} \rangle \Rightarrow \langle y := y + 1, \sigma_{1078} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4311} = \frac{\mathcal{T}_{4312}}{\langle S_9, \sigma_{1078} \rangle \Rightarrow \langle S_{10}, \sigma_{1078} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4313} = \frac{\langle y := y + 1, \sigma_{1078} \rangle \Rightarrow \sigma_{1079} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1078} \rangle \Rightarrow \langle S_{11}, \sigma_{1079} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4314} = \langle S_{11}, \sigma_{1079} \rangle \Rightarrow \langle S_{12}, \sigma_{1079} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4315} = \langle S_{12}, \sigma_{1079} \rangle \Rightarrow \langle S_{13}, \sigma_{1079} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4317} = \frac{\langle x := x + 1, \sigma_{1079} \rangle \Rightarrow \sigma_{1080} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1079} \rangle \Rightarrow \langle S_6, \sigma_{1080} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4316} = \frac{\mathcal{T}_{4317}}{\langle S_{13}, \sigma_{1079} \rangle \Rightarrow \langle S_{15}, \sigma_{1080} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4318} = \frac{\langle S_6, \sigma_{1080} \rangle \Rightarrow \langle S_8, \sigma_{1080} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1080} \rangle \Rightarrow \langle S_9, \sigma_{1080} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4320} = \langle S_8, \sigma_{1080} \rangle \Rightarrow \langle y := y + 1, \sigma_{1080} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4319} = \frac{\mathcal{T}_{4320}}{\langle S_9, \sigma_{1080} \rangle \Rightarrow \langle S_{10}, \sigma_{1080} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4321} = \frac{\langle y := y + 1, \sigma_{1080} \rangle \Rightarrow \sigma_{1081} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1080} \rangle \Rightarrow \langle S_{11}, \sigma_{1081} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4322} = \langle S_{11}, \sigma_{1081} \rangle \Rightarrow \langle S_{12}, \sigma_{1081} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4323} = \langle S_{12}, \sigma_{1081} \rangle \Rightarrow \langle S_{13}, \sigma_{1081} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4325} = \frac{\langle x := x + 1, \sigma_{1081} \rangle \Rightarrow \sigma_{1082} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1081} \rangle \Rightarrow \langle S_6, \sigma_{1082} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4324} = \frac{\mathcal{T}_{4325}}{\langle S_{13}, \sigma_{1081} \rangle \Rightarrow \langle S_{15}, \sigma_{1082} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4326} = \frac{\langle S_6, \sigma_{1082} \rangle \Rightarrow \langle S_8, \sigma_{1082} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1082} \rangle \Rightarrow \langle S_9, \sigma_{1082} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4328} = \langle S_8, \sigma_{1082} \rangle \Rightarrow \langle y := y + 1, \sigma_{1082} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4327} = \frac{\mathcal{T}_{4328}}{\langle S_9, \sigma_{1082} \rangle \Rightarrow \langle S_{10}, \sigma_{1082} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4329} = \frac{\langle y := y + 1, \sigma_{1082} \rangle \Rightarrow \sigma_{1083} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1082} \rangle \Rightarrow \langle S_{11}, \sigma_{1083} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4330} = \langle S_{11}, \sigma_{1083} \rangle \Rightarrow \langle S_{12}, \sigma_{1083} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4331} = \langle S_{12}, \sigma_{1083} \rangle \Rightarrow \langle S_{13}, \sigma_{1083} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4333} = \frac{\langle x := x + 1, \sigma_{1083} \rangle \Rightarrow \sigma_{1084} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1083} \rangle \Rightarrow \langle S_6, \sigma_{1084} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4332} = \frac{\mathcal{T}_{4333}}{\langle S_{13}, \sigma_{1083} \rangle \Rightarrow \langle S_{15}, \sigma_{1084} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4334} = \frac{\langle S_6, \sigma_{1084} \rangle \Rightarrow \langle S_8, \sigma_{1084} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1084} \rangle \Rightarrow \langle S_9, \sigma_{1084} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4336} = \langle S_8, \sigma_{1084} \rangle \Rightarrow \langle y := y + 1, \sigma_{1084} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4335} = \frac{\mathcal{T}_{4336}}{\langle S_9, \sigma_{1084} \rangle \Rightarrow \langle S_{10}, \sigma_{1084} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4337} = \frac{\langle y := y + 1, \sigma_{1084} \rangle \Rightarrow \sigma_{1085} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1084} \rangle \Rightarrow \langle S_{11}, \sigma_{1085} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4338} = \langle S_{11}, \sigma_{1085} \rangle \Rightarrow \langle S_{12}, \sigma_{1085} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4339} = \langle S_{12}, \sigma_{1085} \rangle \Rightarrow \langle S_{13}, \sigma_{1085} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4341} = \frac{\langle x := x + 1, \sigma_{1085} \rangle \Rightarrow \sigma_{1086} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1085} \rangle \Rightarrow \langle S_6, \sigma_{1086} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4340} = \frac{\mathcal{T}_{4341}}{\langle S_{13}, \sigma_{1085} \rangle \Rightarrow \langle S_{15}, \sigma_{1086} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4342} = \frac{\langle S_6, \sigma_{1086} \rangle \Rightarrow \langle S_8, \sigma_{1086} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1086} \rangle \Rightarrow \langle S_9, \sigma_{1086} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4344} = \langle S_8, \sigma_{1086} \rangle \Rightarrow \langle y := y + 1, \sigma_{1086} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4343} = \frac{\mathcal{T}_{4344}}{\langle S_9, \sigma_{1086} \rangle \Rightarrow \langle S_{10}, \sigma_{1086} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4345} = \frac{\langle y := y + 1, \sigma_{1086} \rangle \Rightarrow \sigma_{1087} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1086} \rangle \Rightarrow \langle S_{11}, \sigma_{1087} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4346} = \langle S_{11}, \sigma_{1087} \rangle \Rightarrow \langle S_{12}, \sigma_{1087} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4347} = \langle S_{12}, \sigma_{1087} \rangle \Rightarrow \langle S_{13}, \sigma_{1087} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4349} = \frac{\langle x := x + 1, \sigma_{1087} \rangle \Rightarrow \sigma_{1088} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1087} \rangle \Rightarrow \langle S_6, \sigma_{1088} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4348} = \frac{\mathcal{T}_{4349}}{\langle S_{13}, \sigma_{1087} \rangle \Rightarrow \langle S_{15}, \sigma_{1088} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4350} = \frac{\langle S_6, \sigma_{1088} \rangle \Rightarrow \langle S_8, \sigma_{1088} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1088} \rangle \Rightarrow \langle S_9, \sigma_{1088} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4352} = \langle S_8, \sigma_{1088} \rangle \Rightarrow \langle y := y + 1, \sigma_{1088} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4351} = \frac{\mathcal{T}_{4352}}{\langle S_9, \sigma_{1088} \rangle \Rightarrow \langle S_{10}, \sigma_{1088} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4353} = \frac{\langle y := y + 1, \sigma_{1088} \rangle \Rightarrow \sigma_{1089} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1088} \rangle \Rightarrow \langle S_{11}, \sigma_{1089} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4354} = \langle S_{11}, \sigma_{1089} \rangle \Rightarrow \langle S_{12}, \sigma_{1089} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4355} = \langle S_{12}, \sigma_{1089} \rangle \Rightarrow \langle S_{13}, \sigma_{1089} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4357} = \frac{\langle x := x + 1, \sigma_{1089} \rangle \Rightarrow \sigma_{1090} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1089} \rangle \Rightarrow \langle S_6, \sigma_{1090} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4356} = \frac{\mathcal{T}_{4357}}{\langle S_{13}, \sigma_{1089} \rangle \Rightarrow \langle S_{15}, \sigma_{1090} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4358} = \frac{\langle S_6, \sigma_{1090} \rangle \Rightarrow \langle S_8, \sigma_{1090} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1090} \rangle \Rightarrow \langle S_9, \sigma_{1090} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4360} = \langle S_8, \sigma_{1090} \rangle \Rightarrow \langle y := y + 1, \sigma_{1090} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4359} = \frac{\mathcal{T}_{4360}}{\langle S_9, \sigma_{1090} \rangle \Rightarrow \langle S_{10}, \sigma_{1090} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4361} = \frac{\langle y := y + 1, \sigma_{1090} \rangle \Rightarrow \sigma_{1091} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1090} \rangle \Rightarrow \langle S_{11}, \sigma_{1091} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4362} = \langle S_{11}, \sigma_{1091} \rangle \Rightarrow \langle S_{12}, \sigma_{1091} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4363} = \langle S_{12}, \sigma_{1091} \rangle \Rightarrow \langle S_{13}, \sigma_{1091} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4365} = \frac{\langle x := x + 1, \sigma_{1091} \rangle \Rightarrow \sigma_{1092} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1091} \rangle \Rightarrow \langle S_6, \sigma_{1092} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4364} = \frac{\mathcal{T}_{4365}}{\langle S_{13}, \sigma_{1091} \rangle \Rightarrow \langle S_{15}, \sigma_{1092} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4366} = \frac{\langle S_6, \sigma_{1092} \rangle \Rightarrow \langle S_8, \sigma_{1092} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1092} \rangle \Rightarrow \langle S_9, \sigma_{1092} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4368} = \langle S_8, \sigma_{1092} \rangle \Rightarrow \langle y := y + 1, \sigma_{1092} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4367} = \frac{\mathcal{T}_{4368}}{\langle S_9, \sigma_{1092} \rangle \Rightarrow \langle S_{10}, \sigma_{1092} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4369} = \frac{\langle y := y + 1, \sigma_{1092} \rangle \Rightarrow \sigma_{1093} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1092} \rangle \Rightarrow \langle S_{11}, \sigma_{1093} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4370} = \langle S_{11}, \sigma_{1093} \rangle \Rightarrow \langle S_{12}, \sigma_{1093} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4371} = \langle S_{12}, \sigma_{1093} \rangle \Rightarrow \langle S_{13}, \sigma_{1093} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4373} = \frac{\langle x := x + 1, \sigma_{1093} \rangle \Rightarrow \sigma_{1094} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1093} \rangle \Rightarrow \langle S_6, \sigma_{1094} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4372} = \frac{\mathcal{T}_{4373}}{\langle S_{13}, \sigma_{1093} \rangle \Rightarrow \langle S_{15}, \sigma_{1094} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4374} = \frac{\langle S_6, \sigma_{1094} \rangle \Rightarrow \langle S_8, \sigma_{1094} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1094} \rangle \Rightarrow \langle S_9, \sigma_{1094} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4376} = \langle S_8, \sigma_{1094} \rangle \Rightarrow \langle y := y + 1, \sigma_{1094} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4375} = \frac{\mathcal{T}_{4376}}{\langle S_9, \sigma_{1094} \rangle \Rightarrow \langle S_{10}, \sigma_{1094} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4377} = \frac{\langle y := y + 1, \sigma_{1094} \rangle \Rightarrow \sigma_{1095} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1094} \rangle \Rightarrow \langle S_{11}, \sigma_{1095} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4378} = \langle S_{11}, \sigma_{1095} \rangle \Rightarrow \langle S_{12}, \sigma_{1095} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4379} = \langle S_{12}, \sigma_{1095} \rangle \Rightarrow \langle S_{13}, \sigma_{1095} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4381} = \frac{\langle x := x + 1, \sigma_{1095} \rangle \Rightarrow \sigma_{1096} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1095} \rangle \Rightarrow \langle S_6, \sigma_{1096} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4380} = \frac{\mathcal{T}_{4381}}{\langle S_{13}, \sigma_{1095} \rangle \Rightarrow \langle S_{15}, \sigma_{1096} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4382} = \frac{\langle S_6, \sigma_{1096} \rangle \Rightarrow \langle S_8, \sigma_{1096} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1096} \rangle \Rightarrow \langle S_9, \sigma_{1096} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4384} = \langle S_8, \sigma_{1096} \rangle \Rightarrow \langle y := y + 1, \sigma_{1096} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4383} = \frac{\mathcal{T}_{4384}}{\langle S_9, \sigma_{1096} \rangle \Rightarrow \langle S_{10}, \sigma_{1096} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4385} = \frac{\langle y := y + 1, \sigma_{1096} \rangle \Rightarrow \sigma_{1097} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1096} \rangle \Rightarrow \langle S_{11}, \sigma_{1097} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4386} = \langle S_{11}, \sigma_{1097} \rangle \Rightarrow \langle S_{12}, \sigma_{1097} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4387} = \langle S_{12}, \sigma_{1097} \rangle \Rightarrow \langle S_{13}, \sigma_{1097} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4389} = \frac{\langle x := x + 1, \sigma_{1097} \rangle \Rightarrow \sigma_{1098} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1097} \rangle \Rightarrow \langle S_6, \sigma_{1098} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4388} = \frac{\mathcal{T}_{4389}}{\langle S_{13}, \sigma_{1097} \rangle \Rightarrow \langle S_{15}, \sigma_{1098} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4390} = \frac{\langle S_6, \sigma_{1098} \rangle \Rightarrow \langle S_8, \sigma_{1098} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1098} \rangle \Rightarrow \langle S_9, \sigma_{1098} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4392} = \langle S_8, \sigma_{1098} \rangle \Rightarrow \langle y := y + 1, \sigma_{1098} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4391} = \frac{\mathcal{T}_{4392}}{\langle S_9, \sigma_{1098} \rangle \Rightarrow \langle S_{10}, \sigma_{1098} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4393} = \frac{\langle y := y + 1, \sigma_{1098} \rangle \Rightarrow \sigma_{1099} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1098} \rangle \Rightarrow \langle S_{11}, \sigma_{1099} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4394} = \langle S_{11}, \sigma_{1099} \rangle \Rightarrow \langle S_{12}, \sigma_{1099} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4395} = \langle S_{12}, \sigma_{1099} \rangle \Rightarrow \langle S_{13}, \sigma_{1099} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4397} = \frac{\langle x := x + 1, \sigma_{1099} \rangle \Rightarrow \sigma_{1100} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1099} \rangle \Rightarrow \langle S_6, \sigma_{1100} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4396} = \frac{\mathcal{T}_{4397}}{\langle S_{13}, \sigma_{1099} \rangle \Rightarrow \langle S_{15}, \sigma_{1100} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4398} = \frac{\langle S_6, \sigma_{1100} \rangle \Rightarrow \langle S_8, \sigma_{1100} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1100} \rangle \Rightarrow \langle S_9, \sigma_{1100} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4400} = \langle S_8, \sigma_{1100} \rangle \Rightarrow \langle y := y + 1, \sigma_{1100} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4399} = \frac{\mathcal{T}_{4400}}{\langle S_9, \sigma_{1100} \rangle \Rightarrow \langle S_{10}, \sigma_{1100} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4401} = \frac{\langle y := y + 1, \sigma_{1100} \rangle \Rightarrow \sigma_{1101} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1100} \rangle \Rightarrow \langle S_{11}, \sigma_{1101} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4402} = \langle S_{11}, \sigma_{1101} \rangle \Rightarrow \langle S_{12}, \sigma_{1101} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4403} = \langle S_{12}, \sigma_{1101} \rangle \Rightarrow \langle S_{13}, \sigma_{1101} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4405} = \frac{\langle x := x + 1, \sigma_{1101} \rangle \Rightarrow \sigma_{1102} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1101} \rangle \Rightarrow \langle S_6, \sigma_{1102} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4404} = \frac{\mathcal{T}_{4405}}{\langle S_{13}, \sigma_{1101} \rangle \Rightarrow \langle S_{15}, \sigma_{1102} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4406} = \frac{\langle S_6, \sigma_{1102} \rangle \Rightarrow \langle S_8, \sigma_{1102} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1102} \rangle \Rightarrow \langle S_9, \sigma_{1102} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4408} = \langle S_8, \sigma_{1102} \rangle \Rightarrow \langle y := y + 1, \sigma_{1102} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4407} = \frac{\mathcal{T}_{4408}}{\langle S_9, \sigma_{1102} \rangle \Rightarrow \langle S_{10}, \sigma_{1102} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4409} = \frac{\langle y := y + 1, \sigma_{1102} \rangle \Rightarrow \sigma_{1103} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1102} \rangle \Rightarrow \langle S_{11}, \sigma_{1103} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4410} = \langle S_{11}, \sigma_{1103} \rangle \Rightarrow \langle S_{12}, \sigma_{1103} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4411} = \langle S_{12}, \sigma_{1103} \rangle \Rightarrow \langle S_{13}, \sigma_{1103} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4413} = \frac{\langle x := x + 1, \sigma_{1103} \rangle \Rightarrow \sigma_{1104} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1103} \rangle \Rightarrow \langle S_6, \sigma_{1104} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4412} = \frac{\mathcal{T}_{4413}}{\langle S_{13}, \sigma_{1103} \rangle \Rightarrow \langle S_{15}, \sigma_{1104} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4414} = \frac{\langle S_6, \sigma_{1104} \rangle \Rightarrow \langle S_8, \sigma_{1104} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1104} \rangle \Rightarrow \langle S_9, \sigma_{1104} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4416} = \langle S_8, \sigma_{1104} \rangle \Rightarrow \langle y := y + 1, \sigma_{1104} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4415} = \frac{\mathcal{T}_{4416}}{\langle S_9, \sigma_{1104} \rangle \Rightarrow \langle S_{10}, \sigma_{1104} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4417} = \frac{\langle y := y + 1, \sigma_{1104} \rangle \Rightarrow \sigma_{1105} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1104} \rangle \Rightarrow \langle S_{11}, \sigma_{1105} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4418} = \langle S_{11}, \sigma_{1105} \rangle \Rightarrow \langle S_{12}, \sigma_{1105} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4419} = \langle S_{12}, \sigma_{1105} \rangle \Rightarrow \langle S_{13}, \sigma_{1105} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4421} = \frac{\langle x := x + 1, \sigma_{1105} \rangle \Rightarrow \sigma_{1106} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1105} \rangle \Rightarrow \langle S_6, \sigma_{1106} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4420} = \frac{\mathcal{T}_{4421}}{\langle S_{13}, \sigma_{1105} \rangle \Rightarrow \langle S_{15}, \sigma_{1106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4422} = \frac{\langle S_6, \sigma_{1106} \rangle \Rightarrow \langle S_8, \sigma_{1106} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1106} \rangle \Rightarrow \langle S_9, \sigma_{1106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4424} = \langle S_8, \sigma_{1106} \rangle \Rightarrow \langle y := y + 1, \sigma_{1106} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4423} = \frac{\mathcal{T}_{4424}}{\langle S_9, \sigma_{1106} \rangle \Rightarrow \langle S_{10}, \sigma_{1106} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4425} = \frac{\langle y := y + 1, \sigma_{1106} \rangle \Rightarrow \sigma_{1107} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1106} \rangle \Rightarrow \langle S_{11}, \sigma_{1107} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4426} = \langle S_{11}, \sigma_{1107} \rangle \Rightarrow \langle S_{12}, \sigma_{1107} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4427} = \langle S_{12}, \sigma_{1107} \rangle \Rightarrow \langle S_{13}, \sigma_{1107} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4429} = \frac{\langle x := x + 1, \sigma_{1107} \rangle \Rightarrow \sigma_{1108} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1107} \rangle \Rightarrow \langle S_6, \sigma_{1108} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4428} = \frac{\mathcal{T}_{4429}}{\langle S_{13}, \sigma_{1107} \rangle \Rightarrow \langle S_{15}, \sigma_{1108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4430} = \frac{\langle S_6, \sigma_{1108} \rangle \Rightarrow \langle S_8, \sigma_{1108} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1108} \rangle \Rightarrow \langle S_9, \sigma_{1108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4432} = \langle S_8, \sigma_{1108} \rangle \Rightarrow \langle y := y + 1, \sigma_{1108} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4431} = \frac{\mathcal{T}_{4432}}{\langle S_9, \sigma_{1108} \rangle \Rightarrow \langle S_{10}, \sigma_{1108} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4433} = \frac{\langle y := y + 1, \sigma_{1108} \rangle \Rightarrow \sigma_{1109} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1108} \rangle \Rightarrow \langle S_{11}, \sigma_{1109} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4434} = \langle S_{11}, \sigma_{1109} \rangle \Rightarrow \langle S_{12}, \sigma_{1109} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4435} = \langle S_{12}, \sigma_{1109} \rangle \Rightarrow \langle S_{13}, \sigma_{1109} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4437} = \frac{\langle x := x + 1, \sigma_{1109} \rangle \Rightarrow \sigma_{1110} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1109} \rangle \Rightarrow \langle S_6, \sigma_{1110} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4436} = \frac{\mathcal{T}_{4437}}{\langle S_{13}, \sigma_{1109} \rangle \Rightarrow \langle S_{15}, \sigma_{1110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4438} = \frac{\langle S_6, \sigma_{1110} \rangle \Rightarrow \langle S_8, \sigma_{1110} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1110} \rangle \Rightarrow \langle S_9, \sigma_{1110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4440} = \langle S_8, \sigma_{1110} \rangle \Rightarrow \langle y := y + 1, \sigma_{1110} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4439} = \frac{\mathcal{T}_{4440}}{\langle S_9, \sigma_{1110} \rangle \Rightarrow \langle S_{10}, \sigma_{1110} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4441} = \frac{\langle y := y + 1, \sigma_{1110} \rangle \Rightarrow \sigma_{1111} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1110} \rangle \Rightarrow \langle S_{11}, \sigma_{1111} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4442} = \langle S_{11}, \sigma_{1111} \rangle \Rightarrow \langle S_{12}, \sigma_{1111} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4443} = \langle S_{12}, \sigma_{1111} \rangle \Rightarrow \langle S_{13}, \sigma_{1111} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4445} = \frac{\langle x := x + 1, \sigma_{1111} \rangle \Rightarrow \sigma_{1112} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1111} \rangle \Rightarrow \langle S_6, \sigma_{1112} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4444} = \frac{\mathcal{T}_{4445}}{\langle S_{13}, \sigma_{1111} \rangle \Rightarrow \langle S_{15}, \sigma_{1112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4446} = \frac{\langle S_6, \sigma_{1112} \rangle \Rightarrow \langle S_8, \sigma_{1112} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1112} \rangle \Rightarrow \langle S_9, \sigma_{1112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4448} = \langle S_8, \sigma_{1112} \rangle \Rightarrow \langle y := y + 1, \sigma_{1112} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4447} = \frac{\mathcal{T}_{4448}}{\langle S_9, \sigma_{1112} \rangle \Rightarrow \langle S_{10}, \sigma_{1112} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4449} = \frac{\langle y := y + 1, \sigma_{1112} \rangle \Rightarrow \sigma_{1113} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1112} \rangle \Rightarrow \langle S_{11}, \sigma_{1113} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4450} = \langle S_{11}, \sigma_{1113} \rangle \Rightarrow \langle S_{12}, \sigma_{1113} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4451} = \langle S_{12}, \sigma_{1113} \rangle \Rightarrow \langle S_{13}, \sigma_{1113} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4453} = \frac{\langle x := x + 1, \sigma_{1113} \rangle \Rightarrow \sigma_{1114} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1113} \rangle \Rightarrow \langle S_6, \sigma_{1114} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4452} = \frac{\mathcal{T}_{4453}}{\langle S_{13}, \sigma_{1113} \rangle \Rightarrow \langle S_{15}, \sigma_{1114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4454} = \frac{\langle S_6, \sigma_{1114} \rangle \Rightarrow \langle S_8, \sigma_{1114} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1114} \rangle \Rightarrow \langle S_9, \sigma_{1114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4456} = \langle S_8, \sigma_{1114} \rangle \Rightarrow \langle y := y + 1, \sigma_{1114} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4455} = \frac{\mathcal{T}_{4456}}{\langle S_9, \sigma_{1114} \rangle \Rightarrow \langle S_{10}, \sigma_{1114} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4457} = \frac{\langle y := y + 1, \sigma_{1114} \rangle \Rightarrow \sigma_{1115} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1114} \rangle \Rightarrow \langle S_{11}, \sigma_{1115} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4458} = \langle S_{11}, \sigma_{1115} \rangle \Rightarrow \langle S_{12}, \sigma_{1115} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4459} = \langle S_{12}, \sigma_{1115} \rangle \Rightarrow \langle S_{13}, \sigma_{1115} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4461} = \frac{\langle x := x + 1, \sigma_{1115} \rangle \Rightarrow \sigma_{1116} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1115} \rangle \Rightarrow \langle S_6, \sigma_{1116} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4460} = \frac{\mathcal{T}_{4461}}{\langle S_{13}, \sigma_{1115} \rangle \Rightarrow \langle S_{15}, \sigma_{1116} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4462} = \frac{\langle S_6, \sigma_{1116} \rangle \Rightarrow \langle S_8, \sigma_{1116} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1116} \rangle \Rightarrow \langle S_9, \sigma_{1116} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4464} = \langle S_8, \sigma_{1116} \rangle \Rightarrow \langle y := y + 1, \sigma_{1116} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4463} = \frac{\mathcal{T}_{4464}}{\langle S_9, \sigma_{1116} \rangle \Rightarrow \langle S_{10}, \sigma_{1116} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4465} = \frac{\langle y := y + 1, \sigma_{1116} \rangle \Rightarrow \sigma_{1117} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1116} \rangle \Rightarrow \langle S_{11}, \sigma_{1117} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4466} = \langle S_{11}, \sigma_{1117} \rangle \Rightarrow \langle S_{12}, \sigma_{1117} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4467} = \langle S_{12}, \sigma_{1117} \rangle \Rightarrow \langle S_{13}, \sigma_{1117} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4469} = \frac{\langle x := x + 1, \sigma_{1117} \rangle \Rightarrow \sigma_{1118} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1117} \rangle \Rightarrow \langle S_6, \sigma_{1118} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4468} = \frac{\mathcal{T}_{4469}}{\langle S_{13}, \sigma_{1117} \rangle \Rightarrow \langle S_{15}, \sigma_{1118} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4470} = \frac{\langle S_6, \sigma_{1118} \rangle \Rightarrow \langle S_8, \sigma_{1118} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1118} \rangle \Rightarrow \langle S_9, \sigma_{1118} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4472} = \langle S_8, \sigma_{1118} \rangle \Rightarrow \langle y := y + 1, \sigma_{1118} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4471} = \frac{\mathcal{T}_{4472}}{\langle S_9, \sigma_{1118} \rangle \Rightarrow \langle S_{10}, \sigma_{1118} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4473} = \frac{\langle y := y + 1, \sigma_{1118} \rangle \Rightarrow \sigma_{1119} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1118} \rangle \Rightarrow \langle S_{11}, \sigma_{1119} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4474} = \langle S_{11}, \sigma_{1119} \rangle \Rightarrow \langle S_{12}, \sigma_{1119} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4475} = \langle S_{12}, \sigma_{1119} \rangle \Rightarrow \langle S_{13}, \sigma_{1119} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4477} = \frac{\langle x := x + 1, \sigma_{1119} \rangle \Rightarrow \sigma_{1120} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1119} \rangle \Rightarrow \langle S_6, \sigma_{1120} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4476} = \frac{\mathcal{T}_{4477}}{\langle S_{13}, \sigma_{1119} \rangle \Rightarrow \langle S_{15}, \sigma_{1120} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4478} = \frac{\langle S_6, \sigma_{1120} \rangle \Rightarrow \langle S_8, \sigma_{1120} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1120} \rangle \Rightarrow \langle S_9, \sigma_{1120} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4480} = \langle S_8, \sigma_{1120} \rangle \Rightarrow \langle y := y + 1, \sigma_{1120} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4479} = \frac{\mathcal{T}_{4480}}{\langle S_9, \sigma_{1120} \rangle \Rightarrow \langle S_{10}, \sigma_{1120} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4481} = \frac{\langle y := y + 1, \sigma_{1120} \rangle \Rightarrow \sigma_{1121} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1120} \rangle \Rightarrow \langle S_{11}, \sigma_{1121} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4482} = \langle S_{11}, \sigma_{1121} \rangle \Rightarrow \langle S_{12}, \sigma_{1121} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4483} = \langle S_{12}, \sigma_{1121} \rangle \Rightarrow \langle S_{13}, \sigma_{1121} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4485} = \frac{\langle x := x + 1, \sigma_{1121} \rangle \Rightarrow \sigma_{1122} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1121} \rangle \Rightarrow \langle S_6, \sigma_{1122} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4484} = \frac{\mathcal{T}_{4485}}{\langle S_{13}, \sigma_{1121} \rangle \Rightarrow \langle S_{15}, \sigma_{1122} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4486} = \frac{\langle S_6, \sigma_{1122} \rangle \Rightarrow \langle S_8, \sigma_{1122} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1122} \rangle \Rightarrow \langle S_9, \sigma_{1122} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4488} = \langle S_8, \sigma_{1122} \rangle \Rightarrow \langle y := y + 1, \sigma_{1122} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4487} = \frac{\mathcal{T}_{4488}}{\langle S_9, \sigma_{1122} \rangle \Rightarrow \langle S_{10}, \sigma_{1122} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4489} = \frac{\langle y := y + 1, \sigma_{1122} \rangle \Rightarrow \sigma_{1123} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1122} \rangle \Rightarrow \langle S_{11}, \sigma_{1123} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4490} = \langle S_{11}, \sigma_{1123} \rangle \Rightarrow \langle S_{12}, \sigma_{1123} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4491} = \langle S_{12}, \sigma_{1123} \rangle \Rightarrow \langle S_{13}, \sigma_{1123} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4493} = \frac{\langle x := x + 1, \sigma_{1123} \rangle \Rightarrow \sigma_{1124} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1123} \rangle \Rightarrow \langle S_6, \sigma_{1124} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4492} = \frac{\mathcal{T}_{4493}}{\langle S_{13}, \sigma_{1123} \rangle \Rightarrow \langle S_{15}, \sigma_{1124} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4494} = \frac{\langle S_6, \sigma_{1124} \rangle \Rightarrow \langle S_8, \sigma_{1124} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1124} \rangle \Rightarrow \langle S_9, \sigma_{1124} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4496} = \langle S_8, \sigma_{1124} \rangle \Rightarrow \langle y := y + 1, \sigma_{1124} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4495} = \frac{\mathcal{T}_{4496}}{\langle S_9, \sigma_{1124} \rangle \Rightarrow \langle S_{10}, \sigma_{1124} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4497} = \frac{\langle y := y + 1, \sigma_{1124} \rangle \Rightarrow \sigma_{1125} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1124} \rangle \Rightarrow \langle S_{11}, \sigma_{1125} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4498} = \langle S_{11}, \sigma_{1125} \rangle \Rightarrow \langle S_{12}, \sigma_{1125} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4499} = \langle S_{12}, \sigma_{1125} \rangle \Rightarrow \langle S_{13}, \sigma_{1125} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4501} = \frac{\langle x := x + 1, \sigma_{1125} \rangle \Rightarrow \sigma_{1126} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1125} \rangle \Rightarrow \langle S_6, \sigma_{1126} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4500} = \frac{\mathcal{T}_{4501}}{\langle S_{13}, \sigma_{1125} \rangle \Rightarrow \langle S_{15}, \sigma_{1126} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4502} = \frac{\langle S_6, \sigma_{1126} \rangle \Rightarrow \langle S_8, \sigma_{1126} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1126} \rangle \Rightarrow \langle S_9, \sigma_{1126} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4504} = \langle S_8, \sigma_{1126} \rangle \Rightarrow \langle y := y + 1, \sigma_{1126} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4503} = \frac{\mathcal{T}_{4504}}{\langle S_9, \sigma_{1126} \rangle \Rightarrow \langle S_{10}, \sigma_{1126} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4505} = \frac{\langle y := y + 1, \sigma_{1126} \rangle \Rightarrow \sigma_{1127} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1126} \rangle \Rightarrow \langle S_{11}, \sigma_{1127} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4506} = \langle S_{11}, \sigma_{1127} \rangle \Rightarrow \langle S_{12}, \sigma_{1127} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4507} = \langle S_{12}, \sigma_{1127} \rangle \Rightarrow \langle S_{13}, \sigma_{1127} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4509} = \frac{\langle x := x + 1, \sigma_{1127} \rangle \Rightarrow \sigma_{1128} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1127} \rangle \Rightarrow \langle S_6, \sigma_{1128} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4508} = \frac{\mathcal{T}_{4509}}{\langle S_{13}, \sigma_{1127} \rangle \Rightarrow \langle S_{15}, \sigma_{1128} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4510} = \frac{\langle S_6, \sigma_{1128} \rangle \Rightarrow \langle S_8, \sigma_{1128} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1128} \rangle \Rightarrow \langle S_9, \sigma_{1128} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4512} = \langle S_8, \sigma_{1128} \rangle \Rightarrow \langle y := y + 1, \sigma_{1128} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4511} = \frac{\mathcal{T}_{4512}}{\langle S_9, \sigma_{1128} \rangle \Rightarrow \langle S_{10}, \sigma_{1128} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4513} = \frac{\langle y := y + 1, \sigma_{1128} \rangle \Rightarrow \sigma_{1129} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1128} \rangle \Rightarrow \langle S_{11}, \sigma_{1129} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4514} = \langle S_{11}, \sigma_{1129} \rangle \Rightarrow \langle S_{12}, \sigma_{1129} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4515} = \langle S_{12}, \sigma_{1129} \rangle \Rightarrow \langle S_{13}, \sigma_{1129} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4517} = \frac{\langle x := x + 1, \sigma_{1129} \rangle \Rightarrow \sigma_{1130} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1129} \rangle \Rightarrow \langle S_6, \sigma_{1130} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4516} = \frac{\mathcal{T}_{4517}}{\langle S_{13}, \sigma_{1129} \rangle \Rightarrow \langle S_{15}, \sigma_{1130} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4518} = \frac{\langle S_6, \sigma_{1130} \rangle \Rightarrow \langle S_8, \sigma_{1130} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1130} \rangle \Rightarrow \langle S_9, \sigma_{1130} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4520} = \langle S_8, \sigma_{1130} \rangle \Rightarrow \langle y := y + 1, \sigma_{1130} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4519} = \frac{\mathcal{T}_{4520}}{\langle S_9, \sigma_{1130} \rangle \Rightarrow \langle S_{10}, \sigma_{1130} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4521} = \frac{\langle y := y + 1, \sigma_{1130} \rangle \Rightarrow \sigma_{1131} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1130} \rangle \Rightarrow \langle S_{11}, \sigma_{1131} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4522} = \langle S_{11}, \sigma_{1131} \rangle \Rightarrow \langle S_{12}, \sigma_{1131} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4523} = \langle S_{12}, \sigma_{1131} \rangle \Rightarrow \langle S_{13}, \sigma_{1131} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4525} = \frac{\langle x := x + 1, \sigma_{1131} \rangle \Rightarrow \sigma_{1132} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1131} \rangle \Rightarrow \langle S_6, \sigma_{1132} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4524} = \frac{\mathcal{T}_{4525}}{\langle S_{13}, \sigma_{1131} \rangle \Rightarrow \langle S_{15}, \sigma_{1132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4526} = \frac{\langle S_6, \sigma_{1132} \rangle \Rightarrow \langle S_8, \sigma_{1132} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1132} \rangle \Rightarrow \langle S_9, \sigma_{1132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4528} = \langle S_8, \sigma_{1132} \rangle \Rightarrow \langle y := y + 1, \sigma_{1132} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4527} = \frac{\mathcal{T}_{4528}}{\langle S_9, \sigma_{1132} \rangle \Rightarrow \langle S_{10}, \sigma_{1132} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4529} = \frac{\langle y := y + 1, \sigma_{1132} \rangle \Rightarrow \sigma_{1133} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1132} \rangle \Rightarrow \langle S_{11}, \sigma_{1133} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4530} = \langle S_{11}, \sigma_{1133} \rangle \Rightarrow \langle S_{12}, \sigma_{1133} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4531} = \langle S_{12}, \sigma_{1133} \rangle \Rightarrow \langle S_{13}, \sigma_{1133} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4533} = \frac{\langle x := x + 1, \sigma_{1133} \rangle \Rightarrow \sigma_{1134} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1133} \rangle \Rightarrow \langle S_6, \sigma_{1134} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4532} = \frac{\mathcal{T}_{4533}}{\langle S_{13}, \sigma_{1133} \rangle \Rightarrow \langle S_{15}, \sigma_{1134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4534} = \frac{\langle S_6, \sigma_{1134} \rangle \Rightarrow \langle S_8, \sigma_{1134} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1134} \rangle \Rightarrow \langle S_9, \sigma_{1134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4536} = \langle S_8, \sigma_{1134} \rangle \Rightarrow \langle y := y + 1, \sigma_{1134} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4535} = \frac{\mathcal{T}_{4536}}{\langle S_9, \sigma_{1134} \rangle \Rightarrow \langle S_{10}, \sigma_{1134} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4537} = \frac{\langle y := y + 1, \sigma_{1134} \rangle \Rightarrow \sigma_{1135} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1134} \rangle \Rightarrow \langle S_{11}, \sigma_{1135} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4538} = \langle S_{11}, \sigma_{1135} \rangle \Rightarrow \langle S_{12}, \sigma_{1135} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4539} = \langle S_{12}, \sigma_{1135} \rangle \Rightarrow \langle S_{13}, \sigma_{1135} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4541} = \frac{\langle x := x + 1, \sigma_{1135} \rangle \Rightarrow \sigma_{1136} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1135} \rangle \Rightarrow \langle S_6, \sigma_{1136} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4540} = \frac{\mathcal{T}_{4541}}{\langle S_{13}, \sigma_{1135} \rangle \Rightarrow \langle S_{15}, \sigma_{1136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4542} = \frac{\langle S_6, \sigma_{1136} \rangle \Rightarrow \langle S_8, \sigma_{1136} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1136} \rangle \Rightarrow \langle S_9, \sigma_{1136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4544} = \langle S_8, \sigma_{1136} \rangle \Rightarrow \langle y := y + 1, \sigma_{1136} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4543} = \frac{\mathcal{T}_{4544}}{\langle S_9, \sigma_{1136} \rangle \Rightarrow \langle S_{10}, \sigma_{1136} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4545} = \frac{\langle y := y + 1, \sigma_{1136} \rangle \Rightarrow \sigma_{1137} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1136} \rangle \Rightarrow \langle S_{11}, \sigma_{1137} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4546} = \langle S_{11}, \sigma_{1137} \rangle \Rightarrow \langle S_{12}, \sigma_{1137} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4547} = \langle S_{12}, \sigma_{1137} \rangle \Rightarrow \langle S_{13}, \sigma_{1137} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4549} = \frac{\langle x := x + 1, \sigma_{1137} \rangle \Rightarrow \sigma_{1138} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1137} \rangle \Rightarrow \langle S_6, \sigma_{1138} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4548} = \frac{\mathcal{T}_{4549}}{\langle S_{13}, \sigma_{1137} \rangle \Rightarrow \langle S_{15}, \sigma_{1138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4550} = \frac{\langle S_6, \sigma_{1138} \rangle \Rightarrow \langle S_8, \sigma_{1138} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1138} \rangle \Rightarrow \langle S_9, \sigma_{1138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4552} = \langle S_8, \sigma_{1138} \rangle \Rightarrow \langle y := y + 1, \sigma_{1138} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4551} = \frac{\mathcal{T}_{4552}}{\langle S_9, \sigma_{1138} \rangle \Rightarrow \langle S_{10}, \sigma_{1138} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4553} = \frac{\langle y := y + 1, \sigma_{1138} \rangle \Rightarrow \sigma_{1139} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1138} \rangle \Rightarrow \langle S_{11}, \sigma_{1139} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4554} = \langle S_{11}, \sigma_{1139} \rangle \Rightarrow \langle S_{12}, \sigma_{1139} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4555} = \langle S_{12}, \sigma_{1139} \rangle \Rightarrow \langle S_{13}, \sigma_{1139} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4557} = \frac{\langle x := x + 1, \sigma_{1139} \rangle \Rightarrow \sigma_{1140} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1139} \rangle \Rightarrow \langle S_6, \sigma_{1140} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4556} = \frac{\mathcal{T}_{4557}}{\langle S_{13}, \sigma_{1139} \rangle \Rightarrow \langle S_{15}, \sigma_{1140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4558} = \frac{\langle S_6, \sigma_{1140} \rangle \Rightarrow \langle S_8, \sigma_{1140} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1140} \rangle \Rightarrow \langle S_9, \sigma_{1140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4560} = \langle S_8, \sigma_{1140} \rangle \Rightarrow \langle y := y + 1, \sigma_{1140} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4559} = \frac{\mathcal{T}_{4560}}{\langle S_9, \sigma_{1140} \rangle \Rightarrow \langle S_{10}, \sigma_{1140} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4561} = \frac{\langle y := y + 1, \sigma_{1140} \rangle \Rightarrow \sigma_{1141} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1140} \rangle \Rightarrow \langle S_{11}, \sigma_{1141} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4562} = \langle S_{11}, \sigma_{1141} \rangle \Rightarrow \langle S_{12}, \sigma_{1141} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4563} = \langle S_{12}, \sigma_{1141} \rangle \Rightarrow \langle S_{13}, \sigma_{1141} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4565} = \frac{\langle x := x + 1, \sigma_{1141} \rangle \Rightarrow \sigma_{1142} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1141} \rangle \Rightarrow \langle S_6, \sigma_{1142} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4564} = \frac{\mathcal{T}_{4565}}{\langle S_{13}, \sigma_{1141} \rangle \Rightarrow \langle S_{15}, \sigma_{1142} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4566} = \frac{\langle S_6, \sigma_{1142} \rangle \Rightarrow \langle S_8, \sigma_{1142} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1142} \rangle \Rightarrow \langle S_9, \sigma_{1142} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4568} = \langle S_8, \sigma_{1142} \rangle \Rightarrow \langle y := y + 1, \sigma_{1142} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4567} = \frac{\mathcal{T}_{4568}}{\langle S_9, \sigma_{1142} \rangle \Rightarrow \langle S_{10}, \sigma_{1142} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4569} = \frac{\langle y := y + 1, \sigma_{1142} \rangle \Rightarrow \sigma_{1143} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1142} \rangle \Rightarrow \langle S_{11}, \sigma_{1143} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4570} = \langle S_{11}, \sigma_{1143} \rangle \Rightarrow \langle S_{12}, \sigma_{1143} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4571} = \langle S_{12}, \sigma_{1143} \rangle \Rightarrow \langle S_{13}, \sigma_{1143} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4573} = \frac{\langle x := x + 1, \sigma_{1143} \rangle \Rightarrow \sigma_{1144} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1143} \rangle \Rightarrow \langle S_6, \sigma_{1144} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4572} = \frac{\mathcal{T}_{4573}}{\langle S_{13}, \sigma_{1143} \rangle \Rightarrow \langle S_{15}, \sigma_{1144} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4574} = \frac{\langle S_6, \sigma_{1144} \rangle \Rightarrow \langle S_8, \sigma_{1144} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1144} \rangle \Rightarrow \langle S_9, \sigma_{1144} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4576} = \langle S_8, \sigma_{1144} \rangle \Rightarrow \langle y := y + 1, \sigma_{1144} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4575} = \frac{\mathcal{T}_{4576}}{\langle S_9, \sigma_{1144} \rangle \Rightarrow \langle S_{10}, \sigma_{1144} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4577} = \frac{\langle y := y + 1, \sigma_{1144} \rangle \Rightarrow \sigma_{1145} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1144} \rangle \Rightarrow \langle S_{11}, \sigma_{1145} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4578} = \langle S_{11}, \sigma_{1145} \rangle \Rightarrow \langle S_{12}, \sigma_{1145} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4579} = \langle S_{12}, \sigma_{1145} \rangle \Rightarrow \langle S_{13}, \sigma_{1145} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4581} = \frac{\langle x := x + 1, \sigma_{1145} \rangle \Rightarrow \sigma_{1146} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1145} \rangle \Rightarrow \langle S_6, \sigma_{1146} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4580} = \frac{\mathcal{T}_{4581}}{\langle S_{13}, \sigma_{1145} \rangle \Rightarrow \langle S_{15}, \sigma_{1146} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4582} = \frac{\langle S_6, \sigma_{1146} \rangle \Rightarrow \langle S_8, \sigma_{1146} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1146} \rangle \Rightarrow \langle S_9, \sigma_{1146} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4584} = \langle S_8, \sigma_{1146} \rangle \Rightarrow \langle y := y + 1, \sigma_{1146} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4583} = \frac{\mathcal{T}_{4584}}{\langle S_9, \sigma_{1146} \rangle \Rightarrow \langle S_{10}, \sigma_{1146} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4585} = \frac{\langle y := y + 1, \sigma_{1146} \rangle \Rightarrow \sigma_{1147} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1146} \rangle \Rightarrow \langle S_{11}, \sigma_{1147} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4586} = \langle S_{11}, \sigma_{1147} \rangle \Rightarrow \langle S_{12}, \sigma_{1147} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4587} = \langle S_{12}, \sigma_{1147} \rangle \Rightarrow \langle S_{13}, \sigma_{1147} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4589} = \frac{\langle x := x + 1, \sigma_{1147} \rangle \Rightarrow \sigma_{1148} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1147} \rangle \Rightarrow \langle S_6, \sigma_{1148} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4588} = \frac{\mathcal{T}_{4589}}{\langle S_{13}, \sigma_{1147} \rangle \Rightarrow \langle S_{15}, \sigma_{1148} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4590} = \frac{\langle S_6, \sigma_{1148} \rangle \Rightarrow \langle S_8, \sigma_{1148} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1148} \rangle \Rightarrow \langle S_9, \sigma_{1148} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4592} = \langle S_8, \sigma_{1148} \rangle \Rightarrow \langle y := y + 1, \sigma_{1148} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4591} = \frac{\mathcal{T}_{4592}}{\langle S_9, \sigma_{1148} \rangle \Rightarrow \langle S_{10}, \sigma_{1148} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4593} = \frac{\langle y := y + 1, \sigma_{1148} \rangle \Rightarrow \sigma_{1149} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1148} \rangle \Rightarrow \langle S_{11}, \sigma_{1149} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4594} = \langle S_{11}, \sigma_{1149} \rangle \Rightarrow \langle S_{12}, \sigma_{1149} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4595} = \langle S_{12}, \sigma_{1149} \rangle \Rightarrow \langle S_{13}, \sigma_{1149} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4597} = \frac{\langle x := x + 1, \sigma_{1149} \rangle \Rightarrow \sigma_{1150} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1149} \rangle \Rightarrow \langle S_6, \sigma_{1150} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4596} = \frac{\mathcal{T}_{4597}}{\langle S_{13}, \sigma_{1149} \rangle \Rightarrow \langle S_{15}, \sigma_{1150} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4598} = \frac{\langle S_6, \sigma_{1150} \rangle \Rightarrow \langle S_8, \sigma_{1150} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1150} \rangle \Rightarrow \langle S_9, \sigma_{1150} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4600} = \langle S_8, \sigma_{1150} \rangle \Rightarrow \langle y := y + 1, \sigma_{1150} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4599} = \frac{\mathcal{T}_{4600}}{\langle S_9, \sigma_{1150} \rangle \Rightarrow \langle S_{10}, \sigma_{1150} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4601} = \frac{\langle y := y + 1, \sigma_{1150} \rangle \Rightarrow \sigma_{1151} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1150} \rangle \Rightarrow \langle S_{11}, \sigma_{1151} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4602} = \langle S_{11}, \sigma_{1151} \rangle \Rightarrow \langle S_{12}, \sigma_{1151} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4603} = \langle S_{12}, \sigma_{1151} \rangle \Rightarrow \langle S_{13}, \sigma_{1151} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4605} = \frac{\langle x := x + 1, \sigma_{1151} \rangle \Rightarrow \sigma_{1152} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1151} \rangle \Rightarrow \langle S_6, \sigma_{1152} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4604} = \frac{\mathcal{T}_{4605}}{\langle S_{13}, \sigma_{1151} \rangle \Rightarrow \langle S_{15}, \sigma_{1152} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4606} = \frac{\langle S_6, \sigma_{1152} \rangle \Rightarrow \langle S_8, \sigma_{1152} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1152} \rangle \Rightarrow \langle S_9, \sigma_{1152} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4608} = \langle S_8, \sigma_{1152} \rangle \Rightarrow \langle y := y + 1, \sigma_{1152} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4607} = \frac{\mathcal{T}_{4608}}{\langle S_9, \sigma_{1152} \rangle \Rightarrow \langle S_{10}, \sigma_{1152} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4609} = \frac{\langle y := y + 1, \sigma_{1152} \rangle \Rightarrow \sigma_{1153} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1152} \rangle \Rightarrow \langle S_{11}, \sigma_{1153} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4610} = \langle S_{11}, \sigma_{1153} \rangle \Rightarrow \langle S_{12}, \sigma_{1153} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4611} = \langle S_{12}, \sigma_{1153} \rangle \Rightarrow \langle S_{13}, \sigma_{1153} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4613} = \frac{\langle x := x + 1, \sigma_{1153} \rangle \Rightarrow \sigma_{1154} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1153} \rangle \Rightarrow \langle S_6, \sigma_{1154} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4612} = \frac{\mathcal{T}_{4613}}{\langle S_{13}, \sigma_{1153} \rangle \Rightarrow \langle S_{15}, \sigma_{1154} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4614} = \frac{\langle S_6, \sigma_{1154} \rangle \Rightarrow \langle S_8, \sigma_{1154} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1154} \rangle \Rightarrow \langle S_9, \sigma_{1154} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4616} = \langle S_8, \sigma_{1154} \rangle \Rightarrow \langle y := y + 1, \sigma_{1154} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4615} = \frac{\mathcal{T}_{4616}}{\langle S_9, \sigma_{1154} \rangle \Rightarrow \langle S_{10}, \sigma_{1154} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4617} = \frac{\langle y := y + 1, \sigma_{1154} \rangle \Rightarrow \sigma_{1155} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1154} \rangle \Rightarrow \langle S_{11}, \sigma_{1155} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4618} = \langle S_{11}, \sigma_{1155} \rangle \Rightarrow \langle S_{12}, \sigma_{1155} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4619} = \langle S_{12}, \sigma_{1155} \rangle \Rightarrow \langle S_{13}, \sigma_{1155} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4621} = \frac{\langle x := x + 1, \sigma_{1155} \rangle \Rightarrow \sigma_{1156} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1155} \rangle \Rightarrow \langle S_6, \sigma_{1156} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4620} = \frac{\mathcal{T}_{4621}}{\langle S_{13}, \sigma_{1155} \rangle \Rightarrow \langle S_{15}, \sigma_{1156} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4622} = \frac{\langle S_6, \sigma_{1156} \rangle \Rightarrow \langle S_8, \sigma_{1156} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1156} \rangle \Rightarrow \langle S_9, \sigma_{1156} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4624} = \langle S_8, \sigma_{1156} \rangle \Rightarrow \langle y := y + 1, \sigma_{1156} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4623} = \frac{\mathcal{T}_{4624}}{\langle S_9, \sigma_{1156} \rangle \Rightarrow \langle S_{10}, \sigma_{1156} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4625} = \frac{\langle y := y + 1, \sigma_{1156} \rangle \Rightarrow \sigma_{1157} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1156} \rangle \Rightarrow \langle S_{11}, \sigma_{1157} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4626} = \langle S_{11}, \sigma_{1157} \rangle \Rightarrow \langle S_{12}, \sigma_{1157} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4627} = \langle S_{12}, \sigma_{1157} \rangle \Rightarrow \langle S_{13}, \sigma_{1157} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4629} = \frac{\langle x := x + 1, \sigma_{1157} \rangle \Rightarrow \sigma_{1158} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1157} \rangle \Rightarrow \langle S_6, \sigma_{1158} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4628} = \frac{\mathcal{T}_{4629}}{\langle S_{13}, \sigma_{1157} \rangle \Rightarrow \langle S_{15}, \sigma_{1158} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4630} = \frac{\langle S_6, \sigma_{1158} \rangle \Rightarrow \langle S_8, \sigma_{1158} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1158} \rangle \Rightarrow \langle S_9, \sigma_{1158} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4632} = \langle S_8, \sigma_{1158} \rangle \Rightarrow \langle y := y + 1, \sigma_{1158} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4631} = \frac{\mathcal{T}_{4632}}{\langle S_9, \sigma_{1158} \rangle \Rightarrow \langle S_{10}, \sigma_{1158} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4633} = \frac{\langle y := y + 1, \sigma_{1158} \rangle \Rightarrow \sigma_{1159} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1158} \rangle \Rightarrow \langle S_{11}, \sigma_{1159} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4634} = \langle S_{11}, \sigma_{1159} \rangle \Rightarrow \langle S_{12}, \sigma_{1159} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4635} = \langle S_{12}, \sigma_{1159} \rangle \Rightarrow \langle S_{13}, \sigma_{1159} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4637} = \frac{\langle x := x + 1, \sigma_{1159} \rangle \Rightarrow \sigma_{1160} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1159} \rangle \Rightarrow \langle S_6, \sigma_{1160} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4636} = \frac{\mathcal{T}_{4637}}{\langle S_{13}, \sigma_{1159} \rangle \Rightarrow \langle S_{15}, \sigma_{1160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4638} = \frac{\langle S_6, \sigma_{1160} \rangle \Rightarrow \langle S_8, \sigma_{1160} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1160} \rangle \Rightarrow \langle S_9, \sigma_{1160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4640} = \langle S_8, \sigma_{1160} \rangle \Rightarrow \langle y := y + 1, \sigma_{1160} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4639} = \frac{\mathcal{T}_{4640}}{\langle S_9, \sigma_{1160} \rangle \Rightarrow \langle S_{10}, \sigma_{1160} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4641} = \frac{\langle y := y + 1, \sigma_{1160} \rangle \Rightarrow \sigma_{1161} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1160} \rangle \Rightarrow \langle S_{11}, \sigma_{1161} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4642} = \langle S_{11}, \sigma_{1161} \rangle \Rightarrow \langle S_{12}, \sigma_{1161} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4643} = \langle S_{12}, \sigma_{1161} \rangle \Rightarrow \langle S_{13}, \sigma_{1161} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4645} = \frac{\langle x := x + 1, \sigma_{1161} \rangle \Rightarrow \sigma_{1162} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1161} \rangle \Rightarrow \langle S_6, \sigma_{1162} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4644} = \frac{\mathcal{T}_{4645}}{\langle S_{13}, \sigma_{1161} \rangle \Rightarrow \langle S_{15}, \sigma_{1162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4646} = \frac{\langle S_6, \sigma_{1162} \rangle \Rightarrow \langle S_8, \sigma_{1162} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1162} \rangle \Rightarrow \langle S_9, \sigma_{1162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4648} = \langle S_8, \sigma_{1162} \rangle \Rightarrow \langle y := y + 1, \sigma_{1162} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4647} = \frac{\mathcal{T}_{4648}}{\langle S_9, \sigma_{1162} \rangle \Rightarrow \langle S_{10}, \sigma_{1162} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4649} = \frac{\langle y := y + 1, \sigma_{1162} \rangle \Rightarrow \sigma_{1163} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1162} \rangle \Rightarrow \langle S_{11}, \sigma_{1163} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4650} = \langle S_{11}, \sigma_{1163} \rangle \Rightarrow \langle S_{12}, \sigma_{1163} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4651} = \langle S_{12}, \sigma_{1163} \rangle \Rightarrow \langle S_{13}, \sigma_{1163} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4653} = \frac{\langle x := x + 1, \sigma_{1163} \rangle \Rightarrow \sigma_{1164} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1163} \rangle \Rightarrow \langle S_6, \sigma_{1164} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4652} = \frac{\mathcal{T}_{4653}}{\langle S_{13}, \sigma_{1163} \rangle \Rightarrow \langle S_{15}, \sigma_{1164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4654} = \frac{\langle S_6, \sigma_{1164} \rangle \Rightarrow \langle S_8, \sigma_{1164} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1164} \rangle \Rightarrow \langle S_9, \sigma_{1164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4656} = \langle S_8, \sigma_{1164} \rangle \Rightarrow \langle y := y + 1, \sigma_{1164} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4655} = \frac{\mathcal{T}_{4656}}{\langle S_9, \sigma_{1164} \rangle \Rightarrow \langle S_{10}, \sigma_{1164} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4657} = \frac{\langle y := y + 1, \sigma_{1164} \rangle \Rightarrow \sigma_{1165} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1164} \rangle \Rightarrow \langle S_{11}, \sigma_{1165} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4658} = \langle S_{11}, \sigma_{1165} \rangle \Rightarrow \langle S_{12}, \sigma_{1165} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4659} = \langle S_{12}, \sigma_{1165} \rangle \Rightarrow \langle S_{13}, \sigma_{1165} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4661} = \frac{\langle x := x + 1, \sigma_{1165} \rangle \Rightarrow \sigma_{1166} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1165} \rangle \Rightarrow \langle S_6, \sigma_{1166} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4660} = \frac{\mathcal{T}_{4661}}{\langle S_{13}, \sigma_{1165} \rangle \Rightarrow \langle S_{15}, \sigma_{1166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4662} = \frac{\langle S_6, \sigma_{1166} \rangle \Rightarrow \langle S_8, \sigma_{1166} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1166} \rangle \Rightarrow \langle S_9, \sigma_{1166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4664} = \langle S_8, \sigma_{1166} \rangle \Rightarrow \langle y := y + 1, \sigma_{1166} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4663} = \frac{\mathcal{T}_{4664}}{\langle S_9, \sigma_{1166} \rangle \Rightarrow \langle S_{10}, \sigma_{1166} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4665} = \frac{\langle y := y + 1, \sigma_{1166} \rangle \Rightarrow \sigma_{1167} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1166} \rangle \Rightarrow \langle S_{11}, \sigma_{1167} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4666} = \langle S_{11}, \sigma_{1167} \rangle \Rightarrow \langle S_{12}, \sigma_{1167} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4667} = \langle S_{12}, \sigma_{1167} \rangle \Rightarrow \langle S_{13}, \sigma_{1167} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4669} = \frac{\langle x := x + 1, \sigma_{1167} \rangle \Rightarrow \sigma_{1168} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1167} \rangle \Rightarrow \langle S_6, \sigma_{1168} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4668} = \frac{\mathcal{T}_{4669}}{\langle S_{13}, \sigma_{1167} \rangle \Rightarrow \langle S_{15}, \sigma_{1168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4670} = \frac{\langle S_6, \sigma_{1168} \rangle \Rightarrow \langle S_8, \sigma_{1168} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1168} \rangle \Rightarrow \langle S_9, \sigma_{1168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4672} = \langle S_8, \sigma_{1168} \rangle \Rightarrow \langle y := y + 1, \sigma_{1168} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4671} = \frac{\mathcal{T}_{4672}}{\langle S_9, \sigma_{1168} \rangle \Rightarrow \langle S_{10}, \sigma_{1168} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4673} = \frac{\langle y := y + 1, \sigma_{1168} \rangle \Rightarrow \sigma_{1169} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1168} \rangle \Rightarrow \langle S_{11}, \sigma_{1169} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4674} = \langle S_{11}, \sigma_{1169} \rangle \Rightarrow \langle S_{12}, \sigma_{1169} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4675} = \langle S_{12}, \sigma_{1169} \rangle \Rightarrow \langle S_{13}, \sigma_{1169} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4677} = \frac{\langle x := x + 1, \sigma_{1169} \rangle \Rightarrow \sigma_{1170} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1169} \rangle \Rightarrow \langle S_6, \sigma_{1170} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4676} = \frac{\mathcal{T}_{4677}}{\langle S_{13}, \sigma_{1169} \rangle \Rightarrow \langle S_{15}, \sigma_{1170} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4678} = \frac{\langle S_6, \sigma_{1170} \rangle \Rightarrow \langle S_8, \sigma_{1170} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1170} \rangle \Rightarrow \langle S_9, \sigma_{1170} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4680} = \langle S_8, \sigma_{1170} \rangle \Rightarrow \langle y := y + 1, \sigma_{1170} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4679} = \frac{\mathcal{T}_{4680}}{\langle S_9, \sigma_{1170} \rangle \Rightarrow \langle S_{10}, \sigma_{1170} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4681} = \frac{\langle y := y + 1, \sigma_{1170} \rangle \Rightarrow \sigma_{1171} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1170} \rangle \Rightarrow \langle S_{11}, \sigma_{1171} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4682} = \langle S_{11}, \sigma_{1171} \rangle \Rightarrow \langle S_{12}, \sigma_{1171} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4683} = \langle S_{12}, \sigma_{1171} \rangle \Rightarrow \langle S_{13}, \sigma_{1171} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4685} = \frac{\langle x := x + 1, \sigma_{1171} \rangle \Rightarrow \sigma_{1172} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1171} \rangle \Rightarrow \langle S_6, \sigma_{1172} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4684} = \frac{\mathcal{T}_{4685}}{\langle S_{13}, \sigma_{1171} \rangle \Rightarrow \langle S_{15}, \sigma_{1172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4686} = \frac{\langle S_6, \sigma_{1172} \rangle \Rightarrow \langle S_8, \sigma_{1172} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1172} \rangle \Rightarrow \langle S_9, \sigma_{1172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4688} = \langle S_8, \sigma_{1172} \rangle \Rightarrow \langle y := y + 1, \sigma_{1172} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4687} = \frac{\mathcal{T}_{4688}}{\langle S_9, \sigma_{1172} \rangle \Rightarrow \langle S_{10}, \sigma_{1172} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4689} = \frac{\langle y := y + 1, \sigma_{1172} \rangle \Rightarrow \sigma_{1173} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1172} \rangle \Rightarrow \langle S_{11}, \sigma_{1173} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4690} = \langle S_{11}, \sigma_{1173} \rangle \Rightarrow \langle S_{12}, \sigma_{1173} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4691} = \langle S_{12}, \sigma_{1173} \rangle \Rightarrow \langle S_{13}, \sigma_{1173} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4693} = \frac{\langle x := x + 1, \sigma_{1173} \rangle \Rightarrow \sigma_{1174} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1173} \rangle \Rightarrow \langle S_6, \sigma_{1174} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4692} = \frac{\mathcal{T}_{4693}}{\langle S_{13}, \sigma_{1173} \rangle \Rightarrow \langle S_{15}, \sigma_{1174} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4694} = \frac{\langle S_6, \sigma_{1174} \rangle \Rightarrow \langle S_8, \sigma_{1174} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1174} \rangle \Rightarrow \langle S_9, \sigma_{1174} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4696} = \langle S_8, \sigma_{1174} \rangle \Rightarrow \langle y := y + 1, \sigma_{1174} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4695} = \frac{\mathcal{T}_{4696}}{\langle S_9, \sigma_{1174} \rangle \Rightarrow \langle S_{10}, \sigma_{1174} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4697} = \frac{\langle y := y + 1, \sigma_{1174} \rangle \Rightarrow \sigma_{1175} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1174} \rangle \Rightarrow \langle S_{11}, \sigma_{1175} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4698} = \langle S_{11}, \sigma_{1175} \rangle \Rightarrow \langle S_{12}, \sigma_{1175} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4699} = \langle S_{12}, \sigma_{1175} \rangle \Rightarrow \langle S_{13}, \sigma_{1175} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4701} = \frac{\langle x := x + 1, \sigma_{1175} \rangle \Rightarrow \sigma_{1176} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1175} \rangle \Rightarrow \langle S_6, \sigma_{1176} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4700} = \frac{\mathcal{T}_{4701}}{\langle S_{13}, \sigma_{1175} \rangle \Rightarrow \langle S_{15}, \sigma_{1176} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4702} = \frac{\langle S_6, \sigma_{1176} \rangle \Rightarrow \langle S_8, \sigma_{1176} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1176} \rangle \Rightarrow \langle S_9, \sigma_{1176} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4704} = \langle S_8, \sigma_{1176} \rangle \Rightarrow \langle y := y + 1, \sigma_{1176} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4703} = \frac{\mathcal{T}_{4704}}{\langle S_9, \sigma_{1176} \rangle \Rightarrow \langle S_{10}, \sigma_{1176} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4705} = \frac{\langle y := y + 1, \sigma_{1176} \rangle \Rightarrow \sigma_{1177} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1176} \rangle \Rightarrow \langle S_{11}, \sigma_{1177} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4706} = \langle S_{11}, \sigma_{1177} \rangle \Rightarrow \langle S_{12}, \sigma_{1177} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4707} = \langle S_{12}, \sigma_{1177} \rangle \Rightarrow \langle S_{13}, \sigma_{1177} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4709} = \frac{\langle x := x + 1, \sigma_{1177} \rangle \Rightarrow \sigma_{1178} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1177} \rangle \Rightarrow \langle S_6, \sigma_{1178} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4708} = \frac{\mathcal{T}_{4709}}{\langle S_{13}, \sigma_{1177} \rangle \Rightarrow \langle S_{15}, \sigma_{1178} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4710} = \frac{\langle S_6, \sigma_{1178} \rangle \Rightarrow \langle S_8, \sigma_{1178} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1178} \rangle \Rightarrow \langle S_9, \sigma_{1178} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4712} = \langle S_8, \sigma_{1178} \rangle \Rightarrow \langle y := y + 1, \sigma_{1178} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4711} = \frac{\mathcal{T}_{4712}}{\langle S_9, \sigma_{1178} \rangle \Rightarrow \langle S_{10}, \sigma_{1178} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4713} = \frac{\langle y := y + 1, \sigma_{1178} \rangle \Rightarrow \sigma_{1179} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1178} \rangle \Rightarrow \langle S_{11}, \sigma_{1179} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4714} = \langle S_{11}, \sigma_{1179} \rangle \Rightarrow \langle S_{12}, \sigma_{1179} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4715} = \langle S_{12}, \sigma_{1179} \rangle \Rightarrow \langle S_{13}, \sigma_{1179} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4717} = \frac{\langle x := x + 1, \sigma_{1179} \rangle \Rightarrow \sigma_{1180} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1179} \rangle \Rightarrow \langle S_6, \sigma_{1180} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4716} = \frac{\mathcal{T}_{4717}}{\langle S_{13}, \sigma_{1179} \rangle \Rightarrow \langle S_{15}, \sigma_{1180} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4718} = \frac{\langle S_6, \sigma_{1180} \rangle \Rightarrow \langle S_8, \sigma_{1180} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1180} \rangle \Rightarrow \langle S_9, \sigma_{1180} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4720} = \langle S_8, \sigma_{1180} \rangle \Rightarrow \langle y := y + 1, \sigma_{1180} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4719} = \frac{\mathcal{T}_{4720}}{\langle S_9, \sigma_{1180} \rangle \Rightarrow \langle S_{10}, \sigma_{1180} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4721} = \frac{\langle y := y + 1, \sigma_{1180} \rangle \Rightarrow \sigma_{1181} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1180} \rangle \Rightarrow \langle S_{11}, \sigma_{1181} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4722} = \langle S_{11}, \sigma_{1181} \rangle \Rightarrow \langle S_{12}, \sigma_{1181} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4723} = \langle S_{12}, \sigma_{1181} \rangle \Rightarrow \langle S_{13}, \sigma_{1181} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4725} = \frac{\langle x := x + 1, \sigma_{1181} \rangle \Rightarrow \sigma_{1182} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1181} \rangle \Rightarrow \langle S_6, \sigma_{1182} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4724} = \frac{\mathcal{T}_{4725}}{\langle S_{13}, \sigma_{1181} \rangle \Rightarrow \langle S_{15}, \sigma_{1182} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4726} = \frac{\langle S_6, \sigma_{1182} \rangle \Rightarrow \langle S_8, \sigma_{1182} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1182} \rangle \Rightarrow \langle S_9, \sigma_{1182} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4728} = \langle S_8, \sigma_{1182} \rangle \Rightarrow \langle y := y + 1, \sigma_{1182} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4727} = \frac{\mathcal{T}_{4728}}{\langle S_9, \sigma_{1182} \rangle \Rightarrow \langle S_{10}, \sigma_{1182} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4729} = \frac{\langle y := y + 1, \sigma_{1182} \rangle \Rightarrow \sigma_{1183} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1182} \rangle \Rightarrow \langle S_{11}, \sigma_{1183} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4730} = \langle S_{11}, \sigma_{1183} \rangle \Rightarrow \langle S_{12}, \sigma_{1183} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4731} = \langle S_{12}, \sigma_{1183} \rangle \Rightarrow \langle S_{13}, \sigma_{1183} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4733} = \frac{\langle x := x + 1, \sigma_{1183} \rangle \Rightarrow \sigma_{1184} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1183} \rangle \Rightarrow \langle S_6, \sigma_{1184} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4732} = \frac{\mathcal{T}_{4733}}{\langle S_{13}, \sigma_{1183} \rangle \Rightarrow \langle S_{15}, \sigma_{1184} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4734} = \frac{\langle S_6, \sigma_{1184} \rangle \Rightarrow \langle S_8, \sigma_{1184} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1184} \rangle \Rightarrow \langle S_9, \sigma_{1184} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4736} = \langle S_8, \sigma_{1184} \rangle \Rightarrow \langle y := y + 1, \sigma_{1184} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4735} = \frac{\mathcal{T}_{4736}}{\langle S_9, \sigma_{1184} \rangle \Rightarrow \langle S_{10}, \sigma_{1184} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4737} = \frac{\langle y := y + 1, \sigma_{1184} \rangle \Rightarrow \sigma_{1185} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1184} \rangle \Rightarrow \langle S_{11}, \sigma_{1185} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4738} = \langle S_{11}, \sigma_{1185} \rangle \Rightarrow \langle S_{12}, \sigma_{1185} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4739} = \langle S_{12}, \sigma_{1185} \rangle \Rightarrow \langle S_{13}, \sigma_{1185} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4741} = \frac{\langle x := x + 1, \sigma_{1185} \rangle \Rightarrow \sigma_{1186} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1185} \rangle \Rightarrow \langle S_6, \sigma_{1186} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4740} = \frac{\mathcal{T}_{4741}}{\langle S_{13}, \sigma_{1185} \rangle \Rightarrow \langle S_{15}, \sigma_{1186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4742} = \frac{\langle S_6, \sigma_{1186} \rangle \Rightarrow \langle S_8, \sigma_{1186} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1186} \rangle \Rightarrow \langle S_9, \sigma_{1186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4744} = \langle S_8, \sigma_{1186} \rangle \Rightarrow \langle y := y + 1, \sigma_{1186} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4743} = \frac{\mathcal{T}_{4744}}{\langle S_9, \sigma_{1186} \rangle \Rightarrow \langle S_{10}, \sigma_{1186} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4745} = \frac{\langle y := y + 1, \sigma_{1186} \rangle \Rightarrow \sigma_{1187} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1186} \rangle \Rightarrow \langle S_{11}, \sigma_{1187} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4746} = \langle S_{11}, \sigma_{1187} \rangle \Rightarrow \langle S_{12}, \sigma_{1187} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4747} = \langle S_{12}, \sigma_{1187} \rangle \Rightarrow \langle S_{13}, \sigma_{1187} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4749} = \frac{\langle x := x + 1, \sigma_{1187} \rangle \Rightarrow \sigma_{1188} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1187} \rangle \Rightarrow \langle S_6, \sigma_{1188} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4748} = \frac{\mathcal{T}_{4749}}{\langle S_{13}, \sigma_{1187} \rangle \Rightarrow \langle S_{15}, \sigma_{1188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4750} = \frac{\langle S_6, \sigma_{1188} \rangle \Rightarrow \langle S_8, \sigma_{1188} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1188} \rangle \Rightarrow \langle S_9, \sigma_{1188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4752} = \langle S_8, \sigma_{1188} \rangle \Rightarrow \langle y := y + 1, \sigma_{1188} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4751} = \frac{\mathcal{T}_{4752}}{\langle S_9, \sigma_{1188} \rangle \Rightarrow \langle S_{10}, \sigma_{1188} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4753} = \frac{\langle y := y + 1, \sigma_{1188} \rangle \Rightarrow \sigma_{1189} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1188} \rangle \Rightarrow \langle S_{11}, \sigma_{1189} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4754} = \langle S_{11}, \sigma_{1189} \rangle \Rightarrow \langle S_{12}, \sigma_{1189} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4755} = \langle S_{12}, \sigma_{1189} \rangle \Rightarrow \langle S_{13}, \sigma_{1189} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4757} = \frac{\langle x := x + 1, \sigma_{1189} \rangle \Rightarrow \sigma_{1190} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1189} \rangle \Rightarrow \langle S_6, \sigma_{1190} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4756} = \frac{\mathcal{T}_{4757}}{\langle S_{13}, \sigma_{1189} \rangle \Rightarrow \langle S_{15}, \sigma_{1190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4758} = \frac{\langle S_6, \sigma_{1190} \rangle \Rightarrow \langle S_8, \sigma_{1190} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1190} \rangle \Rightarrow \langle S_9, \sigma_{1190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4760} = \langle S_8, \sigma_{1190} \rangle \Rightarrow \langle y := y + 1, \sigma_{1190} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4759} = \frac{\mathcal{T}_{4760}}{\langle S_9, \sigma_{1190} \rangle \Rightarrow \langle S_{10}, \sigma_{1190} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4761} = \frac{\langle y := y + 1, \sigma_{1190} \rangle \Rightarrow \sigma_{1191} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1190} \rangle \Rightarrow \langle S_{11}, \sigma_{1191} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4762} = \langle S_{11}, \sigma_{1191} \rangle \Rightarrow \langle S_{12}, \sigma_{1191} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4763} = \langle S_{12}, \sigma_{1191} \rangle \Rightarrow \langle S_{13}, \sigma_{1191} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4765} = \frac{\langle x := x + 1, \sigma_{1191} \rangle \Rightarrow \sigma_{1192} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1191} \rangle \Rightarrow \langle S_6, \sigma_{1192} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4764} = \frac{\mathcal{T}_{4765}}{\langle S_{13}, \sigma_{1191} \rangle \Rightarrow \langle S_{15}, \sigma_{1192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4766} = \frac{\langle S_6, \sigma_{1192} \rangle \Rightarrow \langle S_8, \sigma_{1192} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1192} \rangle \Rightarrow \langle S_9, \sigma_{1192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4768} = \langle S_8, \sigma_{1192} \rangle \Rightarrow \langle y := y + 1, \sigma_{1192} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4767} = \frac{\mathcal{T}_{4768}}{\langle S_9, \sigma_{1192} \rangle \Rightarrow \langle S_{10}, \sigma_{1192} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4769} = \frac{\langle y := y + 1, \sigma_{1192} \rangle \Rightarrow \sigma_{1193} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1192} \rangle \Rightarrow \langle S_{11}, \sigma_{1193} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4770} = \langle S_{11}, \sigma_{1193} \rangle \Rightarrow \langle S_{12}, \sigma_{1193} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4771} = \langle S_{12}, \sigma_{1193} \rangle \Rightarrow \langle S_{13}, \sigma_{1193} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4773} = \frac{\langle x := x + 1, \sigma_{1193} \rangle \Rightarrow \sigma_{1194} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1193} \rangle \Rightarrow \langle S_6, \sigma_{1194} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4772} = \frac{\mathcal{T}_{4773}}{\langle S_{13}, \sigma_{1193} \rangle \Rightarrow \langle S_{15}, \sigma_{1194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4774} = \frac{\langle S_6, \sigma_{1194} \rangle \Rightarrow \langle S_8, \sigma_{1194} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1194} \rangle \Rightarrow \langle S_9, \sigma_{1194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4776} = \langle S_8, \sigma_{1194} \rangle \Rightarrow \langle y := y + 1, \sigma_{1194} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4775} = \frac{\mathcal{T}_{4776}}{\langle S_9, \sigma_{1194} \rangle \Rightarrow \langle S_{10}, \sigma_{1194} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4777} = \frac{\langle y := y + 1, \sigma_{1194} \rangle \Rightarrow \sigma_{1195} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1194} \rangle \Rightarrow \langle S_{11}, \sigma_{1195} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4778} = \langle S_{11}, \sigma_{1195} \rangle \Rightarrow \langle S_{12}, \sigma_{1195} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4779} = \langle S_{12}, \sigma_{1195} \rangle \Rightarrow \langle S_{13}, \sigma_{1195} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4781} = \frac{\langle x := x + 1, \sigma_{1195} \rangle \Rightarrow \sigma_{1196} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1195} \rangle \Rightarrow \langle S_6, \sigma_{1196} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4780} = \frac{\mathcal{T}_{4781}}{\langle S_{13}, \sigma_{1195} \rangle \Rightarrow \langle S_{15}, \sigma_{1196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4782} = \frac{\langle S_6, \sigma_{1196} \rangle \Rightarrow \langle S_8, \sigma_{1196} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1196} \rangle \Rightarrow \langle S_9, \sigma_{1196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4784} = \langle S_8, \sigma_{1196} \rangle \Rightarrow \langle y := y + 1, \sigma_{1196} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4783} = \frac{\mathcal{T}_{4784}}{\langle S_9, \sigma_{1196} \rangle \Rightarrow \langle S_{10}, \sigma_{1196} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4785} = \frac{\langle y := y + 1, \sigma_{1196} \rangle \Rightarrow \sigma_{1197} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1196} \rangle \Rightarrow \langle S_{11}, \sigma_{1197} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4786} = \langle S_{11}, \sigma_{1197} \rangle \Rightarrow \langle S_{12}, \sigma_{1197} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4787} = \langle S_{12}, \sigma_{1197} \rangle \Rightarrow \langle S_{13}, \sigma_{1197} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4789} = \frac{\langle x := x + 1, \sigma_{1197} \rangle \Rightarrow \sigma_{1198} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1197} \rangle \Rightarrow \langle S_6, \sigma_{1198} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4788} = \frac{\mathcal{T}_{4789}}{\langle S_{13}, \sigma_{1197} \rangle \Rightarrow \langle S_{15}, \sigma_{1198} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4790} = \frac{\langle S_6, \sigma_{1198} \rangle \Rightarrow \langle S_8, \sigma_{1198} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1198} \rangle \Rightarrow \langle S_9, \sigma_{1198} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4792} = \langle S_8, \sigma_{1198} \rangle \Rightarrow \langle y := y + 1, \sigma_{1198} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4791} = \frac{\mathcal{T}_{4792}}{\langle S_9, \sigma_{1198} \rangle \Rightarrow \langle S_{10}, \sigma_{1198} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4793} = \frac{\langle y := y + 1, \sigma_{1198} \rangle \Rightarrow \sigma_{1199} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1198} \rangle \Rightarrow \langle S_{11}, \sigma_{1199} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4794} = \langle S_{11}, \sigma_{1199} \rangle \Rightarrow \langle S_{12}, \sigma_{1199} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4795} = \langle S_{12}, \sigma_{1199} \rangle \Rightarrow \langle S_{13}, \sigma_{1199} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4797} = \frac{\langle x := x + 1, \sigma_{1199} \rangle \Rightarrow \sigma_{1200} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1199} \rangle \Rightarrow \langle S_6, \sigma_{1200} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4796} = \frac{\mathcal{T}_{4797}}{\langle S_{13}, \sigma_{1199} \rangle \Rightarrow \langle S_{15}, \sigma_{1200} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4798} = \frac{\langle S_6, \sigma_{1200} \rangle \Rightarrow \langle S_8, \sigma_{1200} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1200} \rangle \Rightarrow \langle S_9, \sigma_{1200} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4800} = \langle S_8, \sigma_{1200} \rangle \Rightarrow \langle y := y + 1, \sigma_{1200} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4799} = \frac{\mathcal{T}_{4800}}{\langle S_9, \sigma_{1200} \rangle \Rightarrow \langle S_{10}, \sigma_{1200} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4801} = \frac{\langle y := y + 1, \sigma_{1200} \rangle \Rightarrow \sigma_{1201} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1200} \rangle \Rightarrow \langle S_{11}, \sigma_{1201} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4802} = \langle S_{11}, \sigma_{1201} \rangle \Rightarrow \langle S_{12}, \sigma_{1201} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4803} = \langle S_{12}, \sigma_{1201} \rangle \Rightarrow \langle S_{13}, \sigma_{1201} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4805} = \frac{\langle x := x + 1, \sigma_{1201} \rangle \Rightarrow \sigma_{1202} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1201} \rangle \Rightarrow \langle S_6, \sigma_{1202} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4804} = \frac{\mathcal{T}_{4805}}{\langle S_{13}, \sigma_{1201} \rangle \Rightarrow \langle S_{15}, \sigma_{1202} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4806} = \frac{\langle S_6, \sigma_{1202} \rangle \Rightarrow \langle S_8, \sigma_{1202} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1202} \rangle \Rightarrow \langle S_9, \sigma_{1202} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4808} = \langle S_8, \sigma_{1202} \rangle \Rightarrow \langle y := y + 1, \sigma_{1202} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4807} = \frac{\mathcal{T}_{4808}}{\langle S_9, \sigma_{1202} \rangle \Rightarrow \langle S_{10}, \sigma_{1202} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4809} = \frac{\langle y := y + 1, \sigma_{1202} \rangle \Rightarrow \sigma_{1203} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1202} \rangle \Rightarrow \langle S_{11}, \sigma_{1203} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4810} = \langle S_{11}, \sigma_{1203} \rangle \Rightarrow \langle S_{12}, \sigma_{1203} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4811} = \langle S_{12}, \sigma_{1203} \rangle \Rightarrow \langle S_{13}, \sigma_{1203} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4813} = \frac{\langle x := x + 1, \sigma_{1203} \rangle \Rightarrow \sigma_{1204} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1203} \rangle \Rightarrow \langle S_6, \sigma_{1204} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4812} = \frac{\mathcal{T}_{4813}}{\langle S_{13}, \sigma_{1203} \rangle \Rightarrow \langle S_{15}, \sigma_{1204} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4814} = \frac{\langle S_6, \sigma_{1204} \rangle \Rightarrow \langle S_8, \sigma_{1204} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1204} \rangle \Rightarrow \langle S_9, \sigma_{1204} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4816} = \langle S_8, \sigma_{1204} \rangle \Rightarrow \langle y := y + 1, \sigma_{1204} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4815} = \frac{\mathcal{T}_{4816}}{\langle S_9, \sigma_{1204} \rangle \Rightarrow \langle S_{10}, \sigma_{1204} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4817} = \frac{\langle y := y + 1, \sigma_{1204} \rangle \Rightarrow \sigma_{1205} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1204} \rangle \Rightarrow \langle S_{11}, \sigma_{1205} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4818} = \langle S_{11}, \sigma_{1205} \rangle \Rightarrow \langle S_{12}, \sigma_{1205} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4819} = \langle S_{12}, \sigma_{1205} \rangle \Rightarrow \langle S_{13}, \sigma_{1205} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4821} = \frac{\langle x := x + 1, \sigma_{1205} \rangle \Rightarrow \sigma_{1206} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1205} \rangle \Rightarrow \langle S_6, \sigma_{1206} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4820} = \frac{\mathcal{T}_{4821}}{\langle S_{13}, \sigma_{1205} \rangle \Rightarrow \langle S_{15}, \sigma_{1206} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4822} = \frac{\langle S_6, \sigma_{1206} \rangle \Rightarrow \langle S_8, \sigma_{1206} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1206} \rangle \Rightarrow \langle S_9, \sigma_{1206} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4824} = \langle S_8, \sigma_{1206} \rangle \Rightarrow \langle y := y + 1, \sigma_{1206} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4823} = \frac{\mathcal{T}_{4824}}{\langle S_9, \sigma_{1206} \rangle \Rightarrow \langle S_{10}, \sigma_{1206} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4825} = \frac{\langle y := y + 1, \sigma_{1206} \rangle \Rightarrow \sigma_{1207} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1206} \rangle \Rightarrow \langle S_{11}, \sigma_{1207} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4826} = \langle S_{11}, \sigma_{1207} \rangle \Rightarrow \langle S_{12}, \sigma_{1207} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4827} = \langle S_{12}, \sigma_{1207} \rangle \Rightarrow \langle S_{13}, \sigma_{1207} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4829} = \frac{\langle x := x + 1, \sigma_{1207} \rangle \Rightarrow \sigma_{1208} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1207} \rangle \Rightarrow \langle S_6, \sigma_{1208} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4828} = \frac{\mathcal{T}_{4829}}{\langle S_{13}, \sigma_{1207} \rangle \Rightarrow \langle S_{15}, \sigma_{1208} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4830} = \frac{\langle S_6, \sigma_{1208} \rangle \Rightarrow \langle S_8, \sigma_{1208} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1208} \rangle \Rightarrow \langle S_9, \sigma_{1208} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4832} = \langle S_8, \sigma_{1208} \rangle \Rightarrow \langle y := y + 1, \sigma_{1208} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4831} = \frac{\mathcal{T}_{4832}}{\langle S_9, \sigma_{1208} \rangle \Rightarrow \langle S_{10}, \sigma_{1208} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4833} = \frac{\langle y := y + 1, \sigma_{1208} \rangle \Rightarrow \sigma_{1209} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1208} \rangle \Rightarrow \langle S_{11}, \sigma_{1209} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4834} = \langle S_{11}, \sigma_{1209} \rangle \Rightarrow \langle S_{12}, \sigma_{1209} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4835} = \langle S_{12}, \sigma_{1209} \rangle \Rightarrow \langle S_{13}, \sigma_{1209} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4837} = \frac{\langle x := x + 1, \sigma_{1209} \rangle \Rightarrow \sigma_{1210} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1209} \rangle \Rightarrow \langle S_6, \sigma_{1210} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4836} = \frac{\mathcal{T}_{4837}}{\langle S_{13}, \sigma_{1209} \rangle \Rightarrow \langle S_{15}, \sigma_{1210} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4838} = \frac{\langle S_6, \sigma_{1210} \rangle \Rightarrow \langle S_8, \sigma_{1210} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1210} \rangle \Rightarrow \langle S_9, \sigma_{1210} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4840} = \langle S_8, \sigma_{1210} \rangle \Rightarrow \langle y := y + 1, \sigma_{1210} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4839} = \frac{\mathcal{T}_{4840}}{\langle S_9, \sigma_{1210} \rangle \Rightarrow \langle S_{10}, \sigma_{1210} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4841} = \frac{\langle y := y + 1, \sigma_{1210} \rangle \Rightarrow \sigma_{1211} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1210} \rangle \Rightarrow \langle S_{11}, \sigma_{1211} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4842} = \langle S_{11}, \sigma_{1211} \rangle \Rightarrow \langle S_{12}, \sigma_{1211} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4843} = \langle S_{12}, \sigma_{1211} \rangle \Rightarrow \langle S_{13}, \sigma_{1211} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4845} = \frac{\langle x := x + 1, \sigma_{1211} \rangle \Rightarrow \sigma_{1212} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1211} \rangle \Rightarrow \langle S_6, \sigma_{1212} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4844} = \frac{\mathcal{T}_{4845}}{\langle S_{13}, \sigma_{1211} \rangle \Rightarrow \langle S_{15}, \sigma_{1212} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4846} = \frac{\langle S_6, \sigma_{1212} \rangle \Rightarrow \langle S_8, \sigma_{1212} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1212} \rangle \Rightarrow \langle S_9, \sigma_{1212} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4848} = \langle S_8, \sigma_{1212} \rangle \Rightarrow \langle y := y + 1, \sigma_{1212} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4847} = \frac{\mathcal{T}_{4848}}{\langle S_9, \sigma_{1212} \rangle \Rightarrow \langle S_{10}, \sigma_{1212} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4849} = \frac{\langle y := y + 1, \sigma_{1212} \rangle \Rightarrow \sigma_{1213} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1212} \rangle \Rightarrow \langle S_{11}, \sigma_{1213} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4850} = \langle S_{11}, \sigma_{1213} \rangle \Rightarrow \langle S_{12}, \sigma_{1213} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4851} = \langle S_{12}, \sigma_{1213} \rangle \Rightarrow \langle S_{13}, \sigma_{1213} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4853} = \frac{\langle x := x + 1, \sigma_{1213} \rangle \Rightarrow \sigma_{1214} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1213} \rangle \Rightarrow \langle S_6, \sigma_{1214} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4852} = \frac{\mathcal{T}_{4853}}{\langle S_{13}, \sigma_{1213} \rangle \Rightarrow \langle S_{15}, \sigma_{1214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4854} = \frac{\langle S_6, \sigma_{1214} \rangle \Rightarrow \langle S_8, \sigma_{1214} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1214} \rangle \Rightarrow \langle S_9, \sigma_{1214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4856} = \langle S_8, \sigma_{1214} \rangle \Rightarrow \langle y := y + 1, \sigma_{1214} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4855} = \frac{\mathcal{T}_{4856}}{\langle S_9, \sigma_{1214} \rangle \Rightarrow \langle S_{10}, \sigma_{1214} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4857} = \frac{\langle y := y + 1, \sigma_{1214} \rangle \Rightarrow \sigma_{1215} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1214} \rangle \Rightarrow \langle S_{11}, \sigma_{1215} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4858} = \langle S_{11}, \sigma_{1215} \rangle \Rightarrow \langle S_{12}, \sigma_{1215} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4859} = \langle S_{12}, \sigma_{1215} \rangle \Rightarrow \langle S_{13}, \sigma_{1215} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4861} = \frac{\langle x := x + 1, \sigma_{1215} \rangle \Rightarrow \sigma_{1216} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1215} \rangle \Rightarrow \langle S_6, \sigma_{1216} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4860} = \frac{\mathcal{T}_{4861}}{\langle S_{13}, \sigma_{1215} \rangle \Rightarrow \langle S_{15}, \sigma_{1216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4862} = \frac{\langle S_6, \sigma_{1216} \rangle \Rightarrow \langle S_8, \sigma_{1216} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1216} \rangle \Rightarrow \langle S_9, \sigma_{1216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4864} = \langle S_8, \sigma_{1216} \rangle \Rightarrow \langle y := y + 1, \sigma_{1216} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4863} = \frac{\mathcal{T}_{4864}}{\langle S_9, \sigma_{1216} \rangle \Rightarrow \langle S_{10}, \sigma_{1216} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4865} = \frac{\langle y := y + 1, \sigma_{1216} \rangle \Rightarrow \sigma_{1217} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1216} \rangle \Rightarrow \langle S_{11}, \sigma_{1217} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4866} = \langle S_{11}, \sigma_{1217} \rangle \Rightarrow \langle S_{12}, \sigma_{1217} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4867} = \langle S_{12}, \sigma_{1217} \rangle \Rightarrow \langle S_{13}, \sigma_{1217} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4869} = \frac{\langle x := x + 1, \sigma_{1217} \rangle \Rightarrow \sigma_{1218} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1217} \rangle \Rightarrow \langle S_6, \sigma_{1218} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4868} = \frac{\mathcal{T}_{4869}}{\langle S_{13}, \sigma_{1217} \rangle \Rightarrow \langle S_{15}, \sigma_{1218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4870} = \frac{\langle S_6, \sigma_{1218} \rangle \Rightarrow \langle S_8, \sigma_{1218} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1218} \rangle \Rightarrow \langle S_9, \sigma_{1218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4872} = \langle S_8, \sigma_{1218} \rangle \Rightarrow \langle y := y + 1, \sigma_{1218} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4871} = \frac{\mathcal{T}_{4872}}{\langle S_9, \sigma_{1218} \rangle \Rightarrow \langle S_{10}, \sigma_{1218} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4873} = \frac{\langle y := y + 1, \sigma_{1218} \rangle \Rightarrow \sigma_{1219} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1218} \rangle \Rightarrow \langle S_{11}, \sigma_{1219} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4874} = \langle S_{11}, \sigma_{1219} \rangle \Rightarrow \langle S_{12}, \sigma_{1219} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4875} = \langle S_{12}, \sigma_{1219} \rangle \Rightarrow \langle S_{13}, \sigma_{1219} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4877} = \frac{\langle x := x + 1, \sigma_{1219} \rangle \Rightarrow \sigma_{1220} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1219} \rangle \Rightarrow \langle S_6, \sigma_{1220} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4876} = \frac{\mathcal{T}_{4877}}{\langle S_{13}, \sigma_{1219} \rangle \Rightarrow \langle S_{15}, \sigma_{1220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4878} = \frac{\langle S_6, \sigma_{1220} \rangle \Rightarrow \langle S_8, \sigma_{1220} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1220} \rangle \Rightarrow \langle S_9, \sigma_{1220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4880} = \langle S_8, \sigma_{1220} \rangle \Rightarrow \langle y := y + 1, \sigma_{1220} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4879} = \frac{\mathcal{T}_{4880}}{\langle S_9, \sigma_{1220} \rangle \Rightarrow \langle S_{10}, \sigma_{1220} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4881} = \frac{\langle y := y + 1, \sigma_{1220} \rangle \Rightarrow \sigma_{1221} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1220} \rangle \Rightarrow \langle S_{11}, \sigma_{1221} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4882} = \langle S_{11}, \sigma_{1221} \rangle \Rightarrow \langle S_{12}, \sigma_{1221} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4883} = \langle S_{12}, \sigma_{1221} \rangle \Rightarrow \langle S_{13}, \sigma_{1221} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4885} = \frac{\langle x := x + 1, \sigma_{1221} \rangle \Rightarrow \sigma_{1222} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1221} \rangle \Rightarrow \langle S_6, \sigma_{1222} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4884} = \frac{\mathcal{T}_{4885}}{\langle S_{13}, \sigma_{1221} \rangle \Rightarrow \langle S_{15}, \sigma_{1222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4886} = \frac{\langle S_6, \sigma_{1222} \rangle \Rightarrow \langle S_8, \sigma_{1222} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1222} \rangle \Rightarrow \langle S_9, \sigma_{1222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4888} = \langle S_8, \sigma_{1222} \rangle \Rightarrow \langle y := y + 1, \sigma_{1222} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4887} = \frac{\mathcal{T}_{4888}}{\langle S_9, \sigma_{1222} \rangle \Rightarrow \langle S_{10}, \sigma_{1222} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4889} = \frac{\langle y := y + 1, \sigma_{1222} \rangle \Rightarrow \sigma_{1223} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1222} \rangle \Rightarrow \langle S_{11}, \sigma_{1223} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4890} = \langle S_{11}, \sigma_{1223} \rangle \Rightarrow \langle S_{12}, \sigma_{1223} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4891} = \langle S_{12}, \sigma_{1223} \rangle \Rightarrow \langle S_{13}, \sigma_{1223} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4893} = \frac{\langle x := x + 1, \sigma_{1223} \rangle \Rightarrow \sigma_{1224} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1223} \rangle \Rightarrow \langle S_6, \sigma_{1224} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4892} = \frac{\mathcal{T}_{4893}}{\langle S_{13}, \sigma_{1223} \rangle \Rightarrow \langle S_{15}, \sigma_{1224} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4894} = \frac{\langle S_6, \sigma_{1224} \rangle \Rightarrow \langle S_8, \sigma_{1224} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1224} \rangle \Rightarrow \langle S_9, \sigma_{1224} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4896} = \langle S_8, \sigma_{1224} \rangle \Rightarrow \langle y := y + 1, \sigma_{1224} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4895} = \frac{\mathcal{T}_{4896}}{\langle S_9, \sigma_{1224} \rangle \Rightarrow \langle S_{10}, \sigma_{1224} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4897} = \frac{\langle y := y + 1, \sigma_{1224} \rangle \Rightarrow \sigma_{1225} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1224} \rangle \Rightarrow \langle S_{11}, \sigma_{1225} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4898} = \langle S_{11}, \sigma_{1225} \rangle \Rightarrow \langle S_{12}, \sigma_{1225} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4899} = \langle S_{12}, \sigma_{1225} \rangle \Rightarrow \langle S_{13}, \sigma_{1225} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4901} = \frac{\langle x := x + 1, \sigma_{1225} \rangle \Rightarrow \sigma_{1226} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1225} \rangle \Rightarrow \langle S_6, \sigma_{1226} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4900} = \frac{\mathcal{T}_{4901}}{\langle S_{13}, \sigma_{1225} \rangle \Rightarrow \langle S_{15}, \sigma_{1226} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4902} = \frac{\langle S_6, \sigma_{1226} \rangle \Rightarrow \langle S_8, \sigma_{1226} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1226} \rangle \Rightarrow \langle S_9, \sigma_{1226} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4904} = \langle S_8, \sigma_{1226} \rangle \Rightarrow \langle y := y + 1, \sigma_{1226} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4903} = \frac{\mathcal{T}_{4904}}{\langle S_9, \sigma_{1226} \rangle \Rightarrow \langle S_{10}, \sigma_{1226} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4905} = \frac{\langle y := y + 1, \sigma_{1226} \rangle \Rightarrow \sigma_{1227} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1226} \rangle \Rightarrow \langle S_{11}, \sigma_{1227} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4906} = \langle S_{11}, \sigma_{1227} \rangle \Rightarrow \langle S_{12}, \sigma_{1227} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4907} = \langle S_{12}, \sigma_{1227} \rangle \Rightarrow \langle S_{13}, \sigma_{1227} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4909} = \frac{\langle x := x + 1, \sigma_{1227} \rangle \Rightarrow \sigma_{1228} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1227} \rangle \Rightarrow \langle S_6, \sigma_{1228} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4908} = \frac{\mathcal{T}_{4909}}{\langle S_{13}, \sigma_{1227} \rangle \Rightarrow \langle S_{15}, \sigma_{1228} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4910} = \frac{\langle S_6, \sigma_{1228} \rangle \Rightarrow \langle S_8, \sigma_{1228} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1228} \rangle \Rightarrow \langle S_9, \sigma_{1228} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4912} = \langle S_8, \sigma_{1228} \rangle \Rightarrow \langle y := y + 1, \sigma_{1228} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4911} = \frac{\mathcal{T}_{4912}}{\langle S_9, \sigma_{1228} \rangle \Rightarrow \langle S_{10}, \sigma_{1228} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4913} = \frac{\langle y := y + 1, \sigma_{1228} \rangle \Rightarrow \sigma_{1229} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1228} \rangle \Rightarrow \langle S_{11}, \sigma_{1229} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4914} = \langle S_{11}, \sigma_{1229} \rangle \Rightarrow \langle S_{12}, \sigma_{1229} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4915} = \langle S_{12}, \sigma_{1229} \rangle \Rightarrow \langle S_{13}, \sigma_{1229} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4917} = \frac{\langle x := x + 1, \sigma_{1229} \rangle \Rightarrow \sigma_{1230} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1229} \rangle \Rightarrow \langle S_6, \sigma_{1230} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{4916} = \frac{\mathcal{T}_{4917}}{\langle S_{13}, \sigma_{1229} \rangle \Rightarrow \langle S_{15}, \sigma_{1230} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4918} = \frac{\langle S_6, \sigma_{1230} \rangle \Rightarrow \langle S_8, \sigma_{1230} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1230} \rangle \Rightarrow \langle S_9, \sigma_{1230} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4920} = \langle S_8, \sigma_{1230} \rangle \Rightarrow \langle y := y + 1, \sigma_{1230} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{4919} = \frac{\mathcal{T}_{4920}}{\langle S_9, \sigma_{1230} \rangle \Rightarrow \langle S_{10}, \sigma_{1230} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4921} = \frac{\langle y := y + 1, \sigma_{1230} \rangle \Rightarrow \sigma_{1231} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1230} \rangle \Rightarrow \langle S_{11}, \sigma_{1231} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{4922} = \langle S_{11}, \sigma_{1231} \rangle \Rightarrow \langle S_{12}, \sigma_{1231} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4923} = \langle S_{12}, \sigma_{1231} \rangle \Rightarrow \langle S_{13}, \sigma_{1231} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4925} = \frac{\langle x := x + 1, \sigma_{1231} \rangle \Rightarrow \sigma_{1232} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1231} \rangle \Rightarrow \langle S_6, \sigma_{1232} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4924} = \frac{\mathcal{T}_{4925}}{\langle S_{13}, \sigma_{1231} \rangle \Rightarrow \langle S_{15}, \sigma_{1232} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4926} = \frac{\langle S_6, \sigma_{1232} \rangle \Rightarrow \langle S_8, \sigma_{1232} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1232} \rangle \Rightarrow \langle S_9, \sigma_{1232} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4928} = \langle S_8, \sigma_{1232} \rangle \Rightarrow \langle y := y + 1, \sigma_{1232} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4927} = \frac{\mathcal{T}_{4928}}{\langle S_9, \sigma_{1232} \rangle \Rightarrow \langle S_{10}, \sigma_{1232} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4929} = \frac{\langle y := y + 1, \sigma_{1232} \rangle \Rightarrow \sigma_{1233} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1232} \rangle \Rightarrow \langle S_{11}, \sigma_{1233} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4930} = \langle S_{11}, \sigma_{1233} \rangle \Rightarrow \langle S_{12}, \sigma_{1233} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4931} = \langle S_{12}, \sigma_{1233} \rangle \Rightarrow \langle S_{13}, \sigma_{1233} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4933} = \frac{\langle x := x + 1, \sigma_{1233} \rangle \Rightarrow \sigma_{1234} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1233} \rangle \Rightarrow \langle S_6, \sigma_{1234} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4932} = \frac{\mathcal{T}_{4933}}{\langle S_{13}, \sigma_{1233} \rangle \Rightarrow \langle S_{15}, \sigma_{1234} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4934} = \frac{\langle S_6, \sigma_{1234} \rangle \Rightarrow \langle S_8, \sigma_{1234} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1234} \rangle \Rightarrow \langle S_9, \sigma_{1234} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4936} = \langle S_8, \sigma_{1234} \rangle \Rightarrow \langle y := y + 1, \sigma_{1234} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4935} = \frac{\mathcal{T}_{4936}}{\langle S_9, \sigma_{1234} \rangle \Rightarrow \langle S_{10}, \sigma_{1234} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4937} = \frac{\langle y := y + 1, \sigma_{1234} \rangle \Rightarrow \sigma_{1235} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1234} \rangle \Rightarrow \langle S_{11}, \sigma_{1235} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4938} = \langle S_{11}, \sigma_{1235} \rangle \Rightarrow \langle S_{12}, \sigma_{1235} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4939} = \langle S_{12}, \sigma_{1235} \rangle \Rightarrow \langle S_{13}, \sigma_{1235} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4941} = \frac{\langle x := x + 1, \sigma_{1235} \rangle \Rightarrow \sigma_{1236} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1235} \rangle \Rightarrow \langle S_6, \sigma_{1236} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4940} = \frac{\mathcal{T}_{4941}}{\langle S_{13}, \sigma_{1235} \rangle \Rightarrow \langle S_{15}, \sigma_{1236} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4942} = \frac{\langle S_6, \sigma_{1236} \rangle \Rightarrow \langle S_8, \sigma_{1236} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1236} \rangle \Rightarrow \langle S_9, \sigma_{1236} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4944} = \langle S_8, \sigma_{1236} \rangle \Rightarrow \langle y := y + 1, \sigma_{1236} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4943} = \frac{\mathcal{T}_{4944}}{\langle S_9, \sigma_{1236} \rangle \Rightarrow \langle S_{10}, \sigma_{1236} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4945} = \frac{\langle y := y + 1, \sigma_{1236} \rangle \Rightarrow \sigma_{1237} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1236} \rangle \Rightarrow \langle S_{11}, \sigma_{1237} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4946} = \langle S_{11}, \sigma_{1237} \rangle \Rightarrow \langle S_{12}, \sigma_{1237} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4947} = \langle S_{12}, \sigma_{1237} \rangle \Rightarrow \langle S_{13}, \sigma_{1237} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4949} = \frac{\langle x := x + 1, \sigma_{1237} \rangle \Rightarrow \sigma_{1238} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1237} \rangle \Rightarrow \langle S_6, \sigma_{1238} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4948} = \frac{\mathcal{T}_{4949}}{\langle S_{13}, \sigma_{1237} \rangle \Rightarrow \langle S_{15}, \sigma_{1238} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4950} = \frac{\langle S_6, \sigma_{1238} \rangle \Rightarrow \langle S_8, \sigma_{1238} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1238} \rangle \Rightarrow \langle S_9, \sigma_{1238} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4952} = \langle S_8, \sigma_{1238} \rangle \Rightarrow \langle y := y + 1, \sigma_{1238} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{4951} = \frac{\mathcal{T}_{4952}}{\langle S_9, \sigma_{1238} \rangle \Rightarrow \langle S_{10}, \sigma_{1238} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4953} = \frac{\langle y := y + 1, \sigma_{1238} \rangle \Rightarrow \sigma_{1239} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1238} \rangle \Rightarrow \langle S_{11}, \sigma_{1239} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4954} = \langle S_{11}, \sigma_{1239} \rangle \Rightarrow \langle S_{12}, \sigma_{1239} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4955} = \langle S_{12}, \sigma_{1239} \rangle \Rightarrow \langle S_{13}, \sigma_{1239} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4957} = \frac{\langle x := x + 1, \sigma_{1239} \rangle \Rightarrow \sigma_{1240} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1239} \rangle \Rightarrow \langle S_6, \sigma_{1240} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4956} = \frac{\mathcal{T}_{4957}}{\langle S_{13}, \sigma_{1239} \rangle \Rightarrow \langle S_{15}, \sigma_{1240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4958} = \frac{\langle S_6, \sigma_{1240} \rangle \Rightarrow \langle S_8, \sigma_{1240} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1240} \rangle \Rightarrow \langle S_9, \sigma_{1240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4960} = \langle S_8, \sigma_{1240} \rangle \Rightarrow \langle y := y + 1, \sigma_{1240} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4959} = \frac{\mathcal{T}_{4960}}{\langle S_9, \sigma_{1240} \rangle \Rightarrow \langle S_{10}, \sigma_{1240} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4961} = \frac{\langle y := y + 1, \sigma_{1240} \rangle \Rightarrow \sigma_{1241} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1240} \rangle \Rightarrow \langle S_{11}, \sigma_{1241} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{4962} = \langle S_{11}, \sigma_{1241} \rangle \Rightarrow \langle S_{12}, \sigma_{1241} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4963} = \langle S_{12}, \sigma_{1241} \rangle \Rightarrow \langle S_{13}, \sigma_{1241} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{4965} = \frac{\langle x := x + 1, \sigma_{1241} \rangle \Rightarrow \sigma_{1242} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1241} \rangle \Rightarrow \langle S_6, \sigma_{1242} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4964} = \frac{\mathcal{T}_{4965}}{\langle S_{13}, \sigma_{1241} \rangle \Rightarrow \langle S_{15}, \sigma_{1242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4966} = \frac{\langle S_6, \sigma_{1242} \rangle \Rightarrow \langle S_8, \sigma_{1242} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1242} \rangle \Rightarrow \langle S_9, \sigma_{1242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{4968} = \langle S_8, \sigma_{1242} \rangle \Rightarrow \langle y := y + 1, \sigma_{1242} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4967} = \frac{\mathcal{T}_{4968}}{\langle S_9, \sigma_{1242} \rangle \Rightarrow \langle S_{10}, \sigma_{1242} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4969} = \frac{\langle y := y + 1, \sigma_{1242} \rangle \Rightarrow \sigma_{1243} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1242} \rangle \Rightarrow \langle S_{11}, \sigma_{1243} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4970} = \langle S_{11}, \sigma_{1243} \rangle \Rightarrow \langle S_{12}, \sigma_{1243} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4971} = \langle S_{12}, \sigma_{1243} \rangle \Rightarrow \langle S_{13}, \sigma_{1243} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4973} = \frac{\langle x := x + 1, \sigma_{1243} \rangle \Rightarrow \sigma_{1244} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1243} \rangle \Rightarrow \langle S_6, \sigma_{1244} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4972} = \frac{\mathcal{T}_{4973}}{\langle S_{13}, \sigma_{1243} \rangle \Rightarrow \langle S_{15}, \sigma_{1244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4974} = \frac{\langle S_6, \sigma_{1244} \rangle \Rightarrow \langle S_8, \sigma_{1244} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1244} \rangle \Rightarrow \langle S_9, \sigma_{1244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4976} = \langle S_8, \sigma_{1244} \rangle \Rightarrow \langle y := y + 1, \sigma_{1244} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4975} = \frac{\mathcal{T}_{4976}}{\langle S_9, \sigma_{1244} \rangle \Rightarrow \langle S_{10}, \sigma_{1244} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4977} = \frac{\langle y := y + 1, \sigma_{1244} \rangle \Rightarrow \sigma_{1245} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1244} \rangle \Rightarrow \langle S_{11}, \sigma_{1245} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4978} = \langle S_{11}, \sigma_{1245} \rangle \Rightarrow \langle S_{12}, \sigma_{1245} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4979} = \langle S_{12}, \sigma_{1245} \rangle \Rightarrow \langle S_{13}, \sigma_{1245} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4981} = \frac{\langle x := x + 1, \sigma_{1245} \rangle \Rightarrow \sigma_{1246} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1245} \rangle \Rightarrow \langle S_6, \sigma_{1246} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4980} = \frac{\mathcal{T}_{4981}}{\langle S_{13}, \sigma_{1245} \rangle \Rightarrow \langle S_{15}, \sigma_{1246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4982} = \frac{\langle S_6, \sigma_{1246} \rangle \Rightarrow \langle S_8, \sigma_{1246} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1246} \rangle \Rightarrow \langle S_9, \sigma_{1246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4984} = \langle S_8, \sigma_{1246} \rangle \Rightarrow \langle y := y + 1, \sigma_{1246} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4983} = \frac{\mathcal{T}_{4984}}{\langle S_9, \sigma_{1246} \rangle \Rightarrow \langle S_{10}, \sigma_{1246} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4985} = \frac{\langle y := y + 1, \sigma_{1246} \rangle \Rightarrow \sigma_{1247} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1246} \rangle \Rightarrow \langle S_{11}, \sigma_{1247} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4986} = \langle S_{11}, \sigma_{1247} \rangle \Rightarrow \langle S_{12}, \sigma_{1247} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4987} = \langle S_{12}, \sigma_{1247} \rangle \Rightarrow \langle S_{13}, \sigma_{1247} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4989} = \frac{\langle x := x + 1, \sigma_{1247} \rangle \Rightarrow \sigma_{1248} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1247} \rangle \Rightarrow \langle S_6, \sigma_{1248} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4988} = \frac{\mathcal{T}_{4989}}{\langle S_{13}, \sigma_{1247} \rangle \Rightarrow \langle S_{15}, \sigma_{1248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4990} = \frac{\langle S_6, \sigma_{1248} \rangle \Rightarrow \langle S_8, \sigma_{1248} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1248} \rangle \Rightarrow \langle S_9, \sigma_{1248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4992} = \langle S_8, \sigma_{1248} \rangle \Rightarrow \langle y := y + 1, \sigma_{1248} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4991} = \frac{\mathcal{T}_{4992}}{\langle S_9, \sigma_{1248} \rangle \Rightarrow \langle S_{10}, \sigma_{1248} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4993} = \frac{\langle y := y + 1, \sigma_{1248} \rangle \Rightarrow \sigma_{1249} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1248} \rangle \Rightarrow \langle S_{11}, \sigma_{1249} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{4994} = \langle S_{11}, \sigma_{1249} \rangle \Rightarrow \langle S_{12}, \sigma_{1249} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4995} = \langle S_{12}, \sigma_{1249} \rangle \Rightarrow \langle S_{13}, \sigma_{1249} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{4997} = \frac{\langle x := x + 1, \sigma_{1249} \rangle \Rightarrow \sigma_{1250} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1249} \rangle \Rightarrow \langle S_6, \sigma_{1250} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{4996} = \frac{\mathcal{T}_{4997}}{\langle S_{13}, \sigma_{1249} \rangle \Rightarrow \langle S_{15}, \sigma_{1250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{4998} = \frac{\langle S_6, \sigma_{1250} \rangle \Rightarrow \langle S_8, \sigma_{1250} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1250} \rangle \Rightarrow \langle S_9, \sigma_{1250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5000} = \langle S_8, \sigma_{1250} \rangle \Rightarrow \langle y := y + 1, \sigma_{1250} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{4999} = \frac{\mathcal{T}_{5000}}{\langle S_9, \sigma_{1250} \rangle \Rightarrow \langle S_{10}, \sigma_{1250} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5001} = \frac{\langle y := y + 1, \sigma_{1250} \rangle \Rightarrow \sigma_{1251} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1250} \rangle \Rightarrow \langle S_{11}, \sigma_{1251} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5002} = \langle S_{11}, \sigma_{1251} \rangle \Rightarrow \langle S_{12}, \sigma_{1251} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5003} = \langle S_{12}, \sigma_{1251} \rangle \Rightarrow \langle S_{13}, \sigma_{1251} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5005} = \frac{\langle x := x + 1, \sigma_{1251} \rangle \Rightarrow \sigma_{1252} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1251} \rangle \Rightarrow \langle S_6, \sigma_{1252} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5004} = \frac{\mathcal{T}_{5005}}{\langle S_{13}, \sigma_{1251} \rangle \Rightarrow \langle S_{15}, \sigma_{1252} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5006} = \frac{\langle S_6, \sigma_{1252} \rangle \Rightarrow \langle S_8, \sigma_{1252} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1252} \rangle \Rightarrow \langle S_9, \sigma_{1252} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5008} = \langle S_8, \sigma_{1252} \rangle \Rightarrow \langle y := y + 1, \sigma_{1252} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5007} = \frac{\mathcal{T}_{5008}}{\langle S_9, \sigma_{1252} \rangle \Rightarrow \langle S_{10}, \sigma_{1252} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5009} = \frac{\langle y := y + 1, \sigma_{1252} \rangle \Rightarrow \sigma_{1253} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1252} \rangle \Rightarrow \langle S_{11}, \sigma_{1253} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5010} = \langle S_{11}, \sigma_{1253} \rangle \Rightarrow \langle S_{12}, \sigma_{1253} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5011} = \langle S_{12}, \sigma_{1253} \rangle \Rightarrow \langle S_{13}, \sigma_{1253} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5013} = \frac{\langle x := x + 1, \sigma_{1253} \rangle \Rightarrow \sigma_{1254} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1253} \rangle \Rightarrow \langle S_6, \sigma_{1254} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5012} = \frac{\mathcal{T}_{5013}}{\langle S_{13}, \sigma_{1253} \rangle \Rightarrow \langle S_{15}, \sigma_{1254} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5014} = \frac{\langle S_6, \sigma_{1254} \rangle \Rightarrow \langle S_8, \sigma_{1254} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1254} \rangle \Rightarrow \langle S_9, \sigma_{1254} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5016} = \langle S_8, \sigma_{1254} \rangle \Rightarrow \langle y := y + 1, \sigma_{1254} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5015} = \frac{\mathcal{T}_{5016}}{\langle S_9, \sigma_{1254} \rangle \Rightarrow \langle S_{10}, \sigma_{1254} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5017} = \frac{\langle y := y + 1, \sigma_{1254} \rangle \Rightarrow \sigma_{1255} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1254} \rangle \Rightarrow \langle S_{11}, \sigma_{1255} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5018} = \langle S_{11}, \sigma_{1255} \rangle \Rightarrow \langle S_{12}, \sigma_{1255} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5019} = \langle S_{12}, \sigma_{1255} \rangle \Rightarrow \langle S_{13}, \sigma_{1255} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5021} = \frac{\langle x := x + 1, \sigma_{1255} \rangle \Rightarrow \sigma_{1256} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1255} \rangle \Rightarrow \langle S_6, \sigma_{1256} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5020} = \frac{\mathcal{T}_{5021}}{\langle S_{13}, \sigma_{1255} \rangle \Rightarrow \langle S_{15}, \sigma_{1256} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5022} = \frac{\langle S_6, \sigma_{1256} \rangle \Rightarrow \langle S_8, \sigma_{1256} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1256} \rangle \Rightarrow \langle S_9, \sigma_{1256} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5024} = \langle S_8, \sigma_{1256} \rangle \Rightarrow \langle y := y + 1, \sigma_{1256} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5023} = \frac{\mathcal{T}_{5024}}{\langle S_9, \sigma_{1256} \rangle \Rightarrow \langle S_{10}, \sigma_{1256} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5025} = \frac{\langle y := y + 1, \sigma_{1256} \rangle \Rightarrow \sigma_{1257} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1256} \rangle \Rightarrow \langle S_{11}, \sigma_{1257} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5026} = \langle S_{11}, \sigma_{1257} \rangle \Rightarrow \langle S_{12}, \sigma_{1257} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5027} = \langle S_{12}, \sigma_{1257} \rangle \Rightarrow \langle S_{13}, \sigma_{1257} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5029} = \frac{\langle x := x + 1, \sigma_{1257} \rangle \Rightarrow \sigma_{1258} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1257} \rangle \Rightarrow \langle S_6, \sigma_{1258} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5028} = \frac{\mathcal{T}_{5029}}{\langle S_{13}, \sigma_{1257} \rangle \Rightarrow \langle S_{15}, \sigma_{1258} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5030} = \frac{\langle S_6, \sigma_{1258} \rangle \Rightarrow \langle S_8, \sigma_{1258} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1258} \rangle \Rightarrow \langle S_9, \sigma_{1258} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5032} = \langle S_8, \sigma_{1258} \rangle \Rightarrow \langle y := y + 1, \sigma_{1258} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5031} = \frac{\mathcal{T}_{5032}}{\langle S_9, \sigma_{1258} \rangle \Rightarrow \langle S_{10}, \sigma_{1258} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5033} = \frac{\langle y := y + 1, \sigma_{1258} \rangle \Rightarrow \sigma_{1259} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1258} \rangle \Rightarrow \langle S_{11}, \sigma_{1259} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5034} = \langle S_{11}, \sigma_{1259} \rangle \Rightarrow \langle S_{12}, \sigma_{1259} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5035} = \langle S_{12}, \sigma_{1259} \rangle \Rightarrow \langle S_{13}, \sigma_{1259} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5037} = \frac{\langle x := x + 1, \sigma_{1259} \rangle \Rightarrow \sigma_{1260} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1259} \rangle \Rightarrow \langle S_6, \sigma_{1260} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5036} = \frac{\mathcal{T}_{5037}}{\langle S_{13}, \sigma_{1259} \rangle \Rightarrow \langle S_{15}, \sigma_{1260} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5038} = \frac{\langle S_6, \sigma_{1260} \rangle \Rightarrow \langle S_8, \sigma_{1260} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1260} \rangle \Rightarrow \langle S_9, \sigma_{1260} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5040} = \langle S_8, \sigma_{1260} \rangle \Rightarrow \langle y := y + 1, \sigma_{1260} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5039} = \frac{\mathcal{T}_{5040}}{\langle S_9, \sigma_{1260} \rangle \Rightarrow \langle S_{10}, \sigma_{1260} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5041} = \frac{\langle y := y + 1, \sigma_{1260} \rangle \Rightarrow \sigma_{1261} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1260} \rangle \Rightarrow \langle S_{11}, \sigma_{1261} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5042} = \langle S_{11}, \sigma_{1261} \rangle \Rightarrow \langle S_{12}, \sigma_{1261} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5043} = \langle S_{12}, \sigma_{1261} \rangle \Rightarrow \langle S_{13}, \sigma_{1261} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5045} = \frac{\langle x := x + 1, \sigma_{1261} \rangle \Rightarrow \sigma_{1262} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1261} \rangle \Rightarrow \langle S_6, \sigma_{1262} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5044} = \frac{\mathcal{T}_{5045}}{\langle S_{13}, \sigma_{1261} \rangle \Rightarrow \langle S_{15}, \sigma_{1262} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5046} = \frac{\langle S_6, \sigma_{1262} \rangle \Rightarrow \langle S_8, \sigma_{1262} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1262} \rangle \Rightarrow \langle S_9, \sigma_{1262} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5048} = \langle S_8, \sigma_{1262} \rangle \Rightarrow \langle y := y + 1, \sigma_{1262} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5047} = \frac{\mathcal{T}_{5048}}{\langle S_9, \sigma_{1262} \rangle \Rightarrow \langle S_{10}, \sigma_{1262} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5049} = \frac{\langle y := y + 1, \sigma_{1262} \rangle \Rightarrow \sigma_{1263} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1262} \rangle \Rightarrow \langle S_{11}, \sigma_{1263} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5050} = \langle S_{11}, \sigma_{1263} \rangle \Rightarrow \langle S_{12}, \sigma_{1263} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5051} = \langle S_{12}, \sigma_{1263} \rangle \Rightarrow \langle S_{13}, \sigma_{1263} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5053} = \frac{\langle x := x + 1, \sigma_{1263} \rangle \Rightarrow \sigma_{1264} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1263} \rangle \Rightarrow \langle S_6, \sigma_{1264} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5052} = \frac{\mathcal{T}_{5053}}{\langle S_{13}, \sigma_{1263} \rangle \Rightarrow \langle S_{15}, \sigma_{1264} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5054} = \frac{\langle S_6, \sigma_{1264} \rangle \Rightarrow \langle S_8, \sigma_{1264} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1264} \rangle \Rightarrow \langle S_9, \sigma_{1264} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5056} = \langle S_8, \sigma_{1264} \rangle \Rightarrow \langle y := y + 1, \sigma_{1264} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5055} = \frac{\mathcal{T}_{5056}}{\langle S_9, \sigma_{1264} \rangle \Rightarrow \langle S_{10}, \sigma_{1264} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5057} = \frac{\langle y := y + 1, \sigma_{1264} \rangle \Rightarrow \sigma_{1265} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1264} \rangle \Rightarrow \langle S_{11}, \sigma_{1265} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5058} = \langle S_{11}, \sigma_{1265} \rangle \Rightarrow \langle S_{12}, \sigma_{1265} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5059} = \langle S_{12}, \sigma_{1265} \rangle \Rightarrow \langle S_{13}, \sigma_{1265} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5061} = \frac{\langle x := x + 1, \sigma_{1265} \rangle \Rightarrow \sigma_{1266} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1265} \rangle \Rightarrow \langle S_6, \sigma_{1266} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5060} = \frac{\mathcal{T}_{5061}}{\langle S_{13}, \sigma_{1265} \rangle \Rightarrow \langle S_{15}, \sigma_{1266} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5062} = \frac{\langle S_6, \sigma_{1266} \rangle \Rightarrow \langle S_8, \sigma_{1266} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1266} \rangle \Rightarrow \langle S_9, \sigma_{1266} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5064} = \langle S_8, \sigma_{1266} \rangle \Rightarrow \langle y := y + 1, \sigma_{1266} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5063} = \frac{\mathcal{T}_{5064}}{\langle S_9, \sigma_{1266} \rangle \Rightarrow \langle S_{10}, \sigma_{1266} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5065} = \frac{\langle y := y + 1, \sigma_{1266} \rangle \Rightarrow \sigma_{1267} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1266} \rangle \Rightarrow \langle S_{11}, \sigma_{1267} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5066} = \langle S_{11}, \sigma_{1267} \rangle \Rightarrow \langle S_{12}, \sigma_{1267} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5067} = \langle S_{12}, \sigma_{1267} \rangle \Rightarrow \langle S_{13}, \sigma_{1267} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5069} = \frac{\langle x := x + 1, \sigma_{1267} \rangle \Rightarrow \sigma_{1268} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1267} \rangle \Rightarrow \langle S_6, \sigma_{1268} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5068} = \frac{\mathcal{T}_{5069}}{\langle S_{13}, \sigma_{1267} \rangle \Rightarrow \langle S_{15}, \sigma_{1268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5070} = \frac{\langle S_6, \sigma_{1268} \rangle \Rightarrow \langle S_8, \sigma_{1268} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1268} \rangle \Rightarrow \langle S_9, \sigma_{1268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5072} = \langle S_8, \sigma_{1268} \rangle \Rightarrow \langle y := y + 1, \sigma_{1268} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5071} = \frac{\mathcal{T}_{5072}}{\langle S_9, \sigma_{1268} \rangle \Rightarrow \langle S_{10}, \sigma_{1268} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5073} = \frac{\langle y := y + 1, \sigma_{1268} \rangle \Rightarrow \sigma_{1269} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1268} \rangle \Rightarrow \langle S_{11}, \sigma_{1269} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5074} = \langle S_{11}, \sigma_{1269} \rangle \Rightarrow \langle S_{12}, \sigma_{1269} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5075} = \langle S_{12}, \sigma_{1269} \rangle \Rightarrow \langle S_{13}, \sigma_{1269} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5077} = \frac{\langle x := x + 1, \sigma_{1269} \rangle \Rightarrow \sigma_{1270} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1269} \rangle \Rightarrow \langle S_6, \sigma_{1270} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5076} = \frac{\mathcal{T}_{5077}}{\langle S_{13}, \sigma_{1269} \rangle \Rightarrow \langle S_{15}, \sigma_{1270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5078} = \frac{\langle S_6, \sigma_{1270} \rangle \Rightarrow \langle S_8, \sigma_{1270} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1270} \rangle \Rightarrow \langle S_9, \sigma_{1270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5080} = \langle S_8, \sigma_{1270} \rangle \Rightarrow \langle y := y + 1, \sigma_{1270} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5079} = \frac{\mathcal{T}_{5080}}{\langle S_9, \sigma_{1270} \rangle \Rightarrow \langle S_{10}, \sigma_{1270} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5081} = \frac{\langle y := y + 1, \sigma_{1270} \rangle \Rightarrow \sigma_{1271} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1270} \rangle \Rightarrow \langle S_{11}, \sigma_{1271} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5082} = \langle S_{11}, \sigma_{1271} \rangle \Rightarrow \langle S_{12}, \sigma_{1271} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5083} = \langle S_{12}, \sigma_{1271} \rangle \Rightarrow \langle S_{13}, \sigma_{1271} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5085} = \frac{\langle x := x + 1, \sigma_{1271} \rangle \Rightarrow \sigma_{1272} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1271} \rangle \Rightarrow \langle S_6, \sigma_{1272} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5084} = \frac{\mathcal{T}_{5085}}{\langle S_{13}, \sigma_{1271} \rangle \Rightarrow \langle S_{15}, \sigma_{1272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5086} = \frac{\langle S_6, \sigma_{1272} \rangle \Rightarrow \langle S_8, \sigma_{1272} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1272} \rangle \Rightarrow \langle S_9, \sigma_{1272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5088} = \langle S_8, \sigma_{1272} \rangle \Rightarrow \langle y := y + 1, \sigma_{1272} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5087} = \frac{\mathcal{T}_{5088}}{\langle S_9, \sigma_{1272} \rangle \Rightarrow \langle S_{10}, \sigma_{1272} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5089} = \frac{\langle y := y + 1, \sigma_{1272} \rangle \Rightarrow \sigma_{1273} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1272} \rangle \Rightarrow \langle S_{11}, \sigma_{1273} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5090} = \langle S_{11}, \sigma_{1273} \rangle \Rightarrow \langle S_{12}, \sigma_{1273} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5091} = \langle S_{12}, \sigma_{1273} \rangle \Rightarrow \langle S_{13}, \sigma_{1273} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5093} = \frac{\langle x := x + 1, \sigma_{1273} \rangle \Rightarrow \sigma_{1274} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1273} \rangle \Rightarrow \langle S_6, \sigma_{1274} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5092} = \frac{\mathcal{T}_{5093}}{\langle S_{13}, \sigma_{1273} \rangle \Rightarrow \langle S_{15}, \sigma_{1274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5094} = \frac{\langle S_6, \sigma_{1274} \rangle \Rightarrow \langle S_8, \sigma_{1274} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1274} \rangle \Rightarrow \langle S_9, \sigma_{1274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5096} = \langle S_8, \sigma_{1274} \rangle \Rightarrow \langle y := y + 1, \sigma_{1274} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5095} = \frac{\mathcal{T}_{5096}}{\langle S_9, \sigma_{1274} \rangle \Rightarrow \langle S_{10}, \sigma_{1274} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5097} = \frac{\langle y := y + 1, \sigma_{1274} \rangle \Rightarrow \sigma_{1275} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1274} \rangle \Rightarrow \langle S_{11}, \sigma_{1275} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5098} = \langle S_{11}, \sigma_{1275} \rangle \Rightarrow \langle S_{12}, \sigma_{1275} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5099} = \langle S_{12}, \sigma_{1275} \rangle \Rightarrow \langle S_{13}, \sigma_{1275} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5101} = \frac{\langle x := x + 1, \sigma_{1275} \rangle \Rightarrow \sigma_{1276} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1275} \rangle \Rightarrow \langle S_6, \sigma_{1276} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5100} = \frac{\mathcal{T}_{5101}}{\langle S_{13}, \sigma_{1275} \rangle \Rightarrow \langle S_{15}, \sigma_{1276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5102} = \frac{\langle S_6, \sigma_{1276} \rangle \Rightarrow \langle S_8, \sigma_{1276} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1276} \rangle \Rightarrow \langle S_9, \sigma_{1276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5104} = \langle S_8, \sigma_{1276} \rangle \Rightarrow \langle y := y + 1, \sigma_{1276} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5103} = \frac{\mathcal{T}_{5104}}{\langle S_9, \sigma_{1276} \rangle \Rightarrow \langle S_{10}, \sigma_{1276} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5105} = \frac{\langle y := y + 1, \sigma_{1276} \rangle \Rightarrow \sigma_{1277} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1276} \rangle \Rightarrow \langle S_{11}, \sigma_{1277} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5106} = \langle S_{11}, \sigma_{1277} \rangle \Rightarrow \langle S_{12}, \sigma_{1277} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5107} = \langle S_{12}, \sigma_{1277} \rangle \Rightarrow \langle S_{13}, \sigma_{1277} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5109} = \frac{\langle x := x + 1, \sigma_{1277} \rangle \Rightarrow \sigma_{1278} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1277} \rangle \Rightarrow \langle S_6, \sigma_{1278} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5108} = \frac{\mathcal{T}_{5109}}{\langle S_{13}, \sigma_{1277} \rangle \Rightarrow \langle S_{15}, \sigma_{1278} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5110} = \frac{\langle S_6, \sigma_{1278} \rangle \Rightarrow \langle S_8, \sigma_{1278} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1278} \rangle \Rightarrow \langle S_9, \sigma_{1278} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5112} = \langle S_8, \sigma_{1278} \rangle \Rightarrow \langle y := y + 1, \sigma_{1278} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5111} = \frac{\mathcal{T}_{5112}}{\langle S_9, \sigma_{1278} \rangle \Rightarrow \langle S_{10}, \sigma_{1278} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5113} = \frac{\langle y := y + 1, \sigma_{1278} \rangle \Rightarrow \sigma_{1279} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1278} \rangle \Rightarrow \langle S_{11}, \sigma_{1279} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5114} = \langle S_{11}, \sigma_{1279} \rangle \Rightarrow \langle S_{12}, \sigma_{1279} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5115} = \langle S_{12}, \sigma_{1279} \rangle \Rightarrow \langle S_{13}, \sigma_{1279} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5117} = \frac{\langle x := x + 1, \sigma_{1279} \rangle \Rightarrow \sigma_{1280} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1279} \rangle \Rightarrow \langle S_6, \sigma_{1280} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5116} = \frac{\mathcal{T}_{5117}}{\langle S_{13}, \sigma_{1279} \rangle \Rightarrow \langle S_{15}, \sigma_{1280} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5118} = \frac{\langle S_6, \sigma_{1280} \rangle \Rightarrow \langle S_8, \sigma_{1280} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1280} \rangle \Rightarrow \langle S_9, \sigma_{1280} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5120} = \langle S_8, \sigma_{1280} \rangle \Rightarrow \langle y := y + 1, \sigma_{1280} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5119} = \frac{\mathcal{T}_{5120}}{\langle S_9, \sigma_{1280} \rangle \Rightarrow \langle S_{10}, \sigma_{1280} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5121} = \frac{\langle y := y + 1, \sigma_{1280} \rangle \Rightarrow \sigma_{1281} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1280} \rangle \Rightarrow \langle S_{11}, \sigma_{1281} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5122} = \langle S_{11}, \sigma_{1281} \rangle \Rightarrow \langle S_{12}, \sigma_{1281} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5123} = \langle S_{12}, \sigma_{1281} \rangle \Rightarrow \langle S_{13}, \sigma_{1281} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5125} = \frac{\langle x := x + 1, \sigma_{1281} \rangle \Rightarrow \sigma_{1282} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1281} \rangle \Rightarrow \langle S_6, \sigma_{1282} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5124} = \frac{\mathcal{T}_{5125}}{\langle S_{13}, \sigma_{1281} \rangle \Rightarrow \langle S_{15}, \sigma_{1282} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5126} = \frac{\langle S_6, \sigma_{1282} \rangle \Rightarrow \langle S_8, \sigma_{1282} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1282} \rangle \Rightarrow \langle S_9, \sigma_{1282} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5128} = \langle S_8, \sigma_{1282} \rangle \Rightarrow \langle y := y + 1, \sigma_{1282} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5127} = \frac{\mathcal{T}_{5128}}{\langle S_9, \sigma_{1282} \rangle \Rightarrow \langle S_{10}, \sigma_{1282} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5129} = \frac{\langle y := y + 1, \sigma_{1282} \rangle \Rightarrow \sigma_{1283} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1282} \rangle \Rightarrow \langle S_{11}, \sigma_{1283} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5130} = \langle S_{11}, \sigma_{1283} \rangle \Rightarrow \langle S_{12}, \sigma_{1283} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5131} = \langle S_{12}, \sigma_{1283} \rangle \Rightarrow \langle S_{13}, \sigma_{1283} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5133} = \frac{\langle x := x + 1, \sigma_{1283} \rangle \Rightarrow \sigma_{1284} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1283} \rangle \Rightarrow \langle S_6, \sigma_{1284} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5132} = \frac{\mathcal{T}_{5133}}{\langle S_{13}, \sigma_{1283} \rangle \Rightarrow \langle S_{15}, \sigma_{1284} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5134} = \frac{\langle S_6, \sigma_{1284} \rangle \Rightarrow \langle S_8, \sigma_{1284} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1284} \rangle \Rightarrow \langle S_9, \sigma_{1284} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5136} = \langle S_8, \sigma_{1284} \rangle \Rightarrow \langle y := y + 1, \sigma_{1284} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5135} = \frac{\mathcal{T}_{5136}}{\langle S_9, \sigma_{1284} \rangle \Rightarrow \langle S_{10}, \sigma_{1284} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5137} = \frac{\langle y := y + 1, \sigma_{1284} \rangle \Rightarrow \sigma_{1285} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1284} \rangle \Rightarrow \langle S_{11}, \sigma_{1285} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5138} = \langle S_{11}, \sigma_{1285} \rangle \Rightarrow \langle S_{12}, \sigma_{1285} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5139} = \langle S_{12}, \sigma_{1285} \rangle \Rightarrow \langle S_{13}, \sigma_{1285} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5141} = \frac{\langle x := x + 1, \sigma_{1285} \rangle \Rightarrow \sigma_{1286} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1285} \rangle \Rightarrow \langle S_6, \sigma_{1286} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5140} = \frac{\mathcal{T}_{5141}}{\langle S_{13}, \sigma_{1285} \rangle \Rightarrow \langle S_{15}, \sigma_{1286} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5142} = \frac{\langle S_6, \sigma_{1286} \rangle \Rightarrow \langle S_8, \sigma_{1286} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1286} \rangle \Rightarrow \langle S_9, \sigma_{1286} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5144} = \langle S_8, \sigma_{1286} \rangle \Rightarrow \langle y := y + 1, \sigma_{1286} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5143} = \frac{\mathcal{T}_{5144}}{\langle S_9, \sigma_{1286} \rangle \Rightarrow \langle S_{10}, \sigma_{1286} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5145} = \frac{\langle y := y + 1, \sigma_{1286} \rangle \Rightarrow \sigma_{1287} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1286} \rangle \Rightarrow \langle S_{11}, \sigma_{1287} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5146} = \langle S_{11}, \sigma_{1287} \rangle \Rightarrow \langle S_{12}, \sigma_{1287} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5147} = \langle S_{12}, \sigma_{1287} \rangle \Rightarrow \langle S_{13}, \sigma_{1287} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5149} = \frac{\langle x := x + 1, \sigma_{1287} \rangle \Rightarrow \sigma_{1288} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1287} \rangle \Rightarrow \langle S_6, \sigma_{1288} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5148} = \frac{\mathcal{T}_{5149}}{\langle S_{13}, \sigma_{1287} \rangle \Rightarrow \langle S_{15}, \sigma_{1288} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5150} = \frac{\langle S_6, \sigma_{1288} \rangle \Rightarrow \langle S_8, \sigma_{1288} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1288} \rangle \Rightarrow \langle S_9, \sigma_{1288} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5152} = \langle S_8, \sigma_{1288} \rangle \Rightarrow \langle y := y + 1, \sigma_{1288} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5151} = \frac{\mathcal{T}_{5152}}{\langle S_9, \sigma_{1288} \rangle \Rightarrow \langle S_{10}, \sigma_{1288} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5153} = \frac{\langle y := y + 1, \sigma_{1288} \rangle \Rightarrow \sigma_{1289} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1288} \rangle \Rightarrow \langle S_{11}, \sigma_{1289} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5154} = \langle S_{11}, \sigma_{1289} \rangle \Rightarrow \langle S_{12}, \sigma_{1289} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5155} = \langle S_{12}, \sigma_{1289} \rangle \Rightarrow \langle S_{13}, \sigma_{1289} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5157} = \frac{\langle x := x + 1, \sigma_{1289} \rangle \Rightarrow \sigma_{1290} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1289} \rangle \Rightarrow \langle S_6, \sigma_{1290} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5156} = \frac{\mathcal{T}_{5157}}{\langle S_{13}, \sigma_{1289} \rangle \Rightarrow \langle S_{15}, \sigma_{1290} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5158} = \frac{\langle S_6, \sigma_{1290} \rangle \Rightarrow \langle S_8, \sigma_{1290} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1290} \rangle \Rightarrow \langle S_9, \sigma_{1290} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5160} = \langle S_8, \sigma_{1290} \rangle \Rightarrow \langle y := y + 1, \sigma_{1290} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5159} = \frac{\mathcal{T}_{5160}}{\langle S_9, \sigma_{1290} \rangle \Rightarrow \langle S_{10}, \sigma_{1290} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5161} = \frac{\langle y := y + 1, \sigma_{1290} \rangle \Rightarrow \sigma_{1291} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1290} \rangle \Rightarrow \langle S_{11}, \sigma_{1291} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5162} = \langle S_{11}, \sigma_{1291} \rangle \Rightarrow \langle S_{12}, \sigma_{1291} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5163} = \langle S_{12}, \sigma_{1291} \rangle \Rightarrow \langle S_{13}, \sigma_{1291} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5165} = \frac{\langle x := x + 1, \sigma_{1291} \rangle \Rightarrow \sigma_{1292} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1291} \rangle \Rightarrow \langle S_6, \sigma_{1292} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5164} = \frac{\mathcal{T}_{5165}}{\langle S_{13}, \sigma_{1291} \rangle \Rightarrow \langle S_{15}, \sigma_{1292} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5166} = \frac{\langle S_6, \sigma_{1292} \rangle \Rightarrow \langle S_8, \sigma_{1292} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1292} \rangle \Rightarrow \langle S_9, \sigma_{1292} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5168} = \langle S_8, \sigma_{1292} \rangle \Rightarrow \langle y := y + 1, \sigma_{1292} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5167} = \frac{\mathcal{T}_{5168}}{\langle S_9, \sigma_{1292} \rangle \Rightarrow \langle S_{10}, \sigma_{1292} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5169} = \frac{\langle y := y + 1, \sigma_{1292} \rangle \Rightarrow \sigma_{1293} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1292} \rangle \Rightarrow \langle S_{11}, \sigma_{1293} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5170} = \langle S_{11}, \sigma_{1293} \rangle \Rightarrow \langle S_{12}, \sigma_{1293} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5171} = \langle S_{12}, \sigma_{1293} \rangle \Rightarrow \langle S_{13}, \sigma_{1293} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5173} = \frac{\langle x := x + 1, \sigma_{1293} \rangle \Rightarrow \sigma_{1294} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1293} \rangle \Rightarrow \langle S_6, \sigma_{1294} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5172} = \frac{\mathcal{T}_{5173}}{\langle S_{13}, \sigma_{1293} \rangle \Rightarrow \langle S_{15}, \sigma_{1294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5174} = \frac{\langle S_6, \sigma_{1294} \rangle \Rightarrow \langle S_8, \sigma_{1294} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1294} \rangle \Rightarrow \langle S_9, \sigma_{1294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5176} = \langle S_8, \sigma_{1294} \rangle \Rightarrow \langle y := y + 1, \sigma_{1294} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5175} = \frac{\mathcal{T}_{5176}}{\langle S_9, \sigma_{1294} \rangle \Rightarrow \langle S_{10}, \sigma_{1294} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5177} = \frac{\langle y := y + 1, \sigma_{1294} \rangle \Rightarrow \sigma_{1295} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1294} \rangle \Rightarrow \langle S_{11}, \sigma_{1295} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5178} = \langle S_{11}, \sigma_{1295} \rangle \Rightarrow \langle S_{12}, \sigma_{1295} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5179} = \langle S_{12}, \sigma_{1295} \rangle \Rightarrow \langle S_{13}, \sigma_{1295} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5181} = \frac{\langle x := x + 1, \sigma_{1295} \rangle \Rightarrow \sigma_{1296} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1295} \rangle \Rightarrow \langle S_6, \sigma_{1296} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5180} = \frac{\mathcal{T}_{5181}}{\langle S_{13}, \sigma_{1295} \rangle \Rightarrow \langle S_{15}, \sigma_{1296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5182} = \frac{\langle S_6, \sigma_{1296} \rangle \Rightarrow \langle S_8, \sigma_{1296} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1296} \rangle \Rightarrow \langle S_9, \sigma_{1296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5184} = \langle S_8, \sigma_{1296} \rangle \Rightarrow \langle y := y + 1, \sigma_{1296} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5183} = \frac{\mathcal{T}_{5184}}{\langle S_9, \sigma_{1296} \rangle \Rightarrow \langle S_{10}, \sigma_{1296} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5185} = \frac{\langle y := y + 1, \sigma_{1296} \rangle \Rightarrow \sigma_{1297} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1296} \rangle \Rightarrow \langle S_{11}, \sigma_{1297} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5186} = \langle S_{11}, \sigma_{1297} \rangle \Rightarrow \langle S_{12}, \sigma_{1297} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5187} = \langle S_{12}, \sigma_{1297} \rangle \Rightarrow \langle S_{13}, \sigma_{1297} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5189} = \frac{\langle x := x + 1, \sigma_{1297} \rangle \Rightarrow \sigma_{1298} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1297} \rangle \Rightarrow \langle S_6, \sigma_{1298} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5188} = \frac{\mathcal{T}_{5189}}{\langle S_{13}, \sigma_{1297} \rangle \Rightarrow \langle S_{15}, \sigma_{1298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5190} = \frac{\langle S_6, \sigma_{1298} \rangle \Rightarrow \langle S_8, \sigma_{1298} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1298} \rangle \Rightarrow \langle S_9, \sigma_{1298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5192} = \langle S_8, \sigma_{1298} \rangle \Rightarrow \langle y := y + 1, \sigma_{1298} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5191} = \frac{\mathcal{T}_{5192}}{\langle S_9, \sigma_{1298} \rangle \Rightarrow \langle S_{10}, \sigma_{1298} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5193} = \frac{\langle y := y + 1, \sigma_{1298} \rangle \Rightarrow \sigma_{1299} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1298} \rangle \Rightarrow \langle S_{11}, \sigma_{1299} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5194} = \langle S_{11}, \sigma_{1299} \rangle \Rightarrow \langle S_{12}, \sigma_{1299} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5195} = \langle S_{12}, \sigma_{1299} \rangle \Rightarrow \langle S_{13}, \sigma_{1299} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5197} = \frac{\langle x := x + 1, \sigma_{1299} \rangle \Rightarrow \sigma_{1300} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1299} \rangle \Rightarrow \langle S_6, \sigma_{1300} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5196} = \frac{\mathcal{T}_{5197}}{\langle S_{13}, \sigma_{1299} \rangle \Rightarrow \langle S_{15}, \sigma_{1300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5198} = \frac{\langle S_6, \sigma_{1300} \rangle \Rightarrow \langle S_8, \sigma_{1300} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1300} \rangle \Rightarrow \langle S_9, \sigma_{1300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5200} = \langle S_8, \sigma_{1300} \rangle \Rightarrow \langle y := y + 1, \sigma_{1300} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5199} = \frac{\mathcal{T}_{5200}}{\langle S_9, \sigma_{1300} \rangle \Rightarrow \langle S_{10}, \sigma_{1300} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5201} = \frac{\langle y := y + 1, \sigma_{1300} \rangle \Rightarrow \sigma_{1301} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1300} \rangle \Rightarrow \langle S_{11}, \sigma_{1301} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5202} = \langle S_{11}, \sigma_{1301} \rangle \Rightarrow \langle S_{12}, \sigma_{1301} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5203} = \langle S_{12}, \sigma_{1301} \rangle \Rightarrow \langle S_{13}, \sigma_{1301} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5205} = \frac{\langle x := x + 1, \sigma_{1301} \rangle \Rightarrow \sigma_{1302} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1301} \rangle \Rightarrow \langle S_6, \sigma_{1302} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5204} = \frac{\mathcal{T}_{5205}}{\langle S_{13}, \sigma_{1301} \rangle \Rightarrow \langle S_{15}, \sigma_{1302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5206} = \frac{\langle S_6, \sigma_{1302} \rangle \Rightarrow \langle S_8, \sigma_{1302} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1302} \rangle \Rightarrow \langle S_9, \sigma_{1302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5208} = \langle S_8, \sigma_{1302} \rangle \Rightarrow \langle y := y + 1, \sigma_{1302} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5207} = \frac{\mathcal{T}_{5208}}{\langle S_9, \sigma_{1302} \rangle \Rightarrow \langle S_{10}, \sigma_{1302} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5209} = \frac{\langle y := y + 1, \sigma_{1302} \rangle \Rightarrow \sigma_{1303} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1302} \rangle \Rightarrow \langle S_{11}, \sigma_{1303} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5210} = \langle S_{11}, \sigma_{1303} \rangle \Rightarrow \langle S_{12}, \sigma_{1303} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5211} = \langle S_{12}, \sigma_{1303} \rangle \Rightarrow \langle S_{13}, \sigma_{1303} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5213} = \frac{\langle x := x + 1, \sigma_{1303} \rangle \Rightarrow \sigma_{1304} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1303} \rangle \Rightarrow \langle S_6, \sigma_{1304} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5212} = \frac{\mathcal{T}_{5213}}{\langle S_{13}, \sigma_{1303} \rangle \Rightarrow \langle S_{15}, \sigma_{1304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5214} = \frac{\langle S_6, \sigma_{1304} \rangle \Rightarrow \langle S_8, \sigma_{1304} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1304} \rangle \Rightarrow \langle S_9, \sigma_{1304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5216} = \langle S_8, \sigma_{1304} \rangle \Rightarrow \langle y := y + 1, \sigma_{1304} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5215} = \frac{\mathcal{T}_{5216}}{\langle S_9, \sigma_{1304} \rangle \Rightarrow \langle S_{10}, \sigma_{1304} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5217} = \frac{\langle y := y + 1, \sigma_{1304} \rangle \Rightarrow \sigma_{1305} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1304} \rangle \Rightarrow \langle S_{11}, \sigma_{1305} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5218} = \langle S_{11}, \sigma_{1305} \rangle \Rightarrow \langle S_{12}, \sigma_{1305} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5219} = \langle S_{12}, \sigma_{1305} \rangle \Rightarrow \langle S_{13}, \sigma_{1305} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5221} = \frac{\langle x := x + 1, \sigma_{1305} \rangle \Rightarrow \sigma_{1306} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1305} \rangle \Rightarrow \langle S_6, \sigma_{1306} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5220} = \frac{\mathcal{T}_{5221}}{\langle S_{13}, \sigma_{1305} \rangle \Rightarrow \langle S_{15}, \sigma_{1306} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5222} = \frac{\langle S_6, \sigma_{1306} \rangle \Rightarrow \langle S_8, \sigma_{1306} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1306} \rangle \Rightarrow \langle S_9, \sigma_{1306} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5224} = \langle S_8, \sigma_{1306} \rangle \Rightarrow \langle y := y + 1, \sigma_{1306} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5223} = \frac{\mathcal{T}_{5224}}{\langle S_9, \sigma_{1306} \rangle \Rightarrow \langle S_{10}, \sigma_{1306} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5225} = \frac{\langle y := y + 1, \sigma_{1306} \rangle \Rightarrow \sigma_{1307} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1306} \rangle \Rightarrow \langle S_{11}, \sigma_{1307} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5226} = \langle S_{11}, \sigma_{1307} \rangle \Rightarrow \langle S_{12}, \sigma_{1307} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5227} = \langle S_{12}, \sigma_{1307} \rangle \Rightarrow \langle S_{13}, \sigma_{1307} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5229} = \frac{\langle x := x + 1, \sigma_{1307} \rangle \Rightarrow \sigma_{1308} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1307} \rangle \Rightarrow \langle S_6, \sigma_{1308} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5228} = \frac{\mathcal{T}_{5229}}{\langle S_{13}, \sigma_{1307} \rangle \Rightarrow \langle S_{15}, \sigma_{1308} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5230} = \frac{\langle S_6, \sigma_{1308} \rangle \Rightarrow \langle S_8, \sigma_{1308} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1308} \rangle \Rightarrow \langle S_9, \sigma_{1308} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5232} = \langle S_8, \sigma_{1308} \rangle \Rightarrow \langle y := y + 1, \sigma_{1308} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5231} = \frac{\mathcal{T}_{5232}}{\langle S_9, \sigma_{1308} \rangle \Rightarrow \langle S_{10}, \sigma_{1308} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5233} = \frac{\langle y := y + 1, \sigma_{1308} \rangle \Rightarrow \sigma_{1309} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1308} \rangle \Rightarrow \langle S_{11}, \sigma_{1309} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5234} = \langle S_{11}, \sigma_{1309} \rangle \Rightarrow \langle S_{12}, \sigma_{1309} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5235} = \langle S_{12}, \sigma_{1309} \rangle \Rightarrow \langle S_{13}, \sigma_{1309} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5237} = \frac{\langle x := x + 1, \sigma_{1309} \rangle \Rightarrow \sigma_{1310} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1309} \rangle \Rightarrow \langle S_6, \sigma_{1310} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5236} = \frac{\mathcal{T}_{5237}}{\langle S_{13}, \sigma_{1309} \rangle \Rightarrow \langle S_{15}, \sigma_{1310} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5238} = \frac{\langle S_6, \sigma_{1310} \rangle \Rightarrow \langle S_8, \sigma_{1310} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1310} \rangle \Rightarrow \langle S_9, \sigma_{1310} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5240} = \langle S_8, \sigma_{1310} \rangle \Rightarrow \langle y := y + 1, \sigma_{1310} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5239} = \frac{\mathcal{T}_{5240}}{\langle S_9, \sigma_{1310} \rangle \Rightarrow \langle S_{10}, \sigma_{1310} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5241} = \frac{\langle y := y + 1, \sigma_{1310} \rangle \Rightarrow \sigma_{1311} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1310} \rangle \Rightarrow \langle S_{11}, \sigma_{1311} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5242} = \langle S_{11}, \sigma_{1311} \rangle \Rightarrow \langle S_{12}, \sigma_{1311} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5243} = \langle S_{12}, \sigma_{1311} \rangle \Rightarrow \langle S_{13}, \sigma_{1311} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5245} = \frac{\langle x := x + 1, \sigma_{1311} \rangle \Rightarrow \sigma_{1312} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1311} \rangle \Rightarrow \langle S_6, \sigma_{1312} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5244} = \frac{\mathcal{T}_{5245}}{\langle S_{13}, \sigma_{1311} \rangle \Rightarrow \langle S_{15}, \sigma_{1312} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5246} = \frac{\langle S_6, \sigma_{1312} \rangle \Rightarrow \langle S_8, \sigma_{1312} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1312} \rangle \Rightarrow \langle S_9, \sigma_{1312} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5248} = \langle S_8, \sigma_{1312} \rangle \Rightarrow \langle y := y + 1, \sigma_{1312} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5247} = \frac{\mathcal{T}_{5248}}{\langle S_9, \sigma_{1312} \rangle \Rightarrow \langle S_{10}, \sigma_{1312} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5249} = \frac{\langle y := y + 1, \sigma_{1312} \rangle \Rightarrow \sigma_{1313} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1312} \rangle \Rightarrow \langle S_{11}, \sigma_{1313} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5250} = \langle S_{11}, \sigma_{1313} \rangle \Rightarrow \langle S_{12}, \sigma_{1313} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5251} = \langle S_{12}, \sigma_{1313} \rangle \Rightarrow \langle S_{13}, \sigma_{1313} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5253} = \frac{\langle x := x + 1, \sigma_{1313} \rangle \Rightarrow \sigma_{1314} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1313} \rangle \Rightarrow \langle S_6, \sigma_{1314} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5252} = \frac{\mathcal{T}_{5253}}{\langle S_{13}, \sigma_{1313} \rangle \Rightarrow \langle S_{15}, \sigma_{1314} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5254} = \frac{\langle S_6, \sigma_{1314} \rangle \Rightarrow \langle S_8, \sigma_{1314} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1314} \rangle \Rightarrow \langle S_9, \sigma_{1314} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5256} = \langle S_8, \sigma_{1314} \rangle \Rightarrow \langle y := y + 1, \sigma_{1314} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5255} = \frac{\mathcal{T}_{5256}}{\langle S_9, \sigma_{1314} \rangle \Rightarrow \langle S_{10}, \sigma_{1314} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5257} = \frac{\langle y := y + 1, \sigma_{1314} \rangle \Rightarrow \sigma_{1315} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1314} \rangle \Rightarrow \langle S_{11}, \sigma_{1315} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5258} = \langle S_{11}, \sigma_{1315} \rangle \Rightarrow \langle S_{12}, \sigma_{1315} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5259} = \langle S_{12}, \sigma_{1315} \rangle \Rightarrow \langle S_{13}, \sigma_{1315} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5261} = \frac{\langle x := x + 1, \sigma_{1315} \rangle \Rightarrow \sigma_{1316} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1315} \rangle \Rightarrow \langle S_6, \sigma_{1316} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5260} = \frac{\mathcal{T}_{5261}}{\langle S_{13}, \sigma_{1315} \rangle \Rightarrow \langle S_{15}, \sigma_{1316} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5262} = \frac{\langle S_6, \sigma_{1316} \rangle \Rightarrow \langle S_8, \sigma_{1316} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1316} \rangle \Rightarrow \langle S_9, \sigma_{1316} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5264} = \langle S_8, \sigma_{1316} \rangle \Rightarrow \langle y := y + 1, \sigma_{1316} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5263} = \frac{\mathcal{T}_{5264}}{\langle S_9, \sigma_{1316} \rangle \Rightarrow \langle S_{10}, \sigma_{1316} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5265} = \frac{\langle y := y + 1, \sigma_{1316} \rangle \Rightarrow \sigma_{1317} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1316} \rangle \Rightarrow \langle S_{11}, \sigma_{1317} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5266} = \langle S_{11}, \sigma_{1317} \rangle \Rightarrow \langle S_{12}, \sigma_{1317} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5267} = \langle S_{12}, \sigma_{1317} \rangle \Rightarrow \langle S_{13}, \sigma_{1317} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5269} = \frac{\langle x := x + 1, \sigma_{1317} \rangle \Rightarrow \sigma_{1318} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1317} \rangle \Rightarrow \langle S_6, \sigma_{1318} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5268} = \frac{\mathcal{T}_{5269}}{\langle S_{13}, \sigma_{1317} \rangle \Rightarrow \langle S_{15}, \sigma_{1318} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5270} = \frac{\langle S_6, \sigma_{1318} \rangle \Rightarrow \langle S_8, \sigma_{1318} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1318} \rangle \Rightarrow \langle S_9, \sigma_{1318} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5272} = \langle S_8, \sigma_{1318} \rangle \Rightarrow \langle y := y + 1, \sigma_{1318} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5271} = \frac{\mathcal{T}_{5272}}{\langle S_9, \sigma_{1318} \rangle \Rightarrow \langle S_{10}, \sigma_{1318} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5273} = \frac{\langle y := y + 1, \sigma_{1318} \rangle \Rightarrow \sigma_{1319} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1318} \rangle \Rightarrow \langle S_{11}, \sigma_{1319} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5274} = \langle S_{11}, \sigma_{1319} \rangle \Rightarrow \langle S_{12}, \sigma_{1319} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5275} = \langle S_{12}, \sigma_{1319} \rangle \Rightarrow \langle S_{13}, \sigma_{1319} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5277} = \frac{\langle x := x + 1, \sigma_{1319} \rangle \Rightarrow \sigma_{1320} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1319} \rangle \Rightarrow \langle S_6, \sigma_{1320} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5276} = \frac{\mathcal{T}_{5277}}{\langle S_{13}, \sigma_{1319} \rangle \Rightarrow \langle S_{15}, \sigma_{1320} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5278} = \frac{\langle S_6, \sigma_{1320} \rangle \Rightarrow \langle S_8, \sigma_{1320} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1320} \rangle \Rightarrow \langle S_9, \sigma_{1320} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5280} = \langle S_8, \sigma_{1320} \rangle \Rightarrow \langle y := y + 1, \sigma_{1320} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5279} = \frac{\mathcal{T}_{5280}}{\langle S_9, \sigma_{1320} \rangle \Rightarrow \langle S_{10}, \sigma_{1320} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5281} = \frac{\langle y := y + 1, \sigma_{1320} \rangle \Rightarrow \sigma_{1321} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1320} \rangle \Rightarrow \langle S_{11}, \sigma_{1321} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5282} = \langle S_{11}, \sigma_{1321} \rangle \Rightarrow \langle S_{12}, \sigma_{1321} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5283} = \langle S_{12}, \sigma_{1321} \rangle \Rightarrow \langle S_{13}, \sigma_{1321} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5285} = \frac{\langle x := x + 1, \sigma_{1321} \rangle \Rightarrow \sigma_{1322} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1321} \rangle \Rightarrow \langle S_6, \sigma_{1322} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5284} = \frac{\mathcal{T}_{5285}}{\langle S_{13}, \sigma_{1321} \rangle \Rightarrow \langle S_{15}, \sigma_{1322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5286} = \frac{\langle S_6, \sigma_{1322} \rangle \Rightarrow \langle S_8, \sigma_{1322} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1322} \rangle \Rightarrow \langle S_9, \sigma_{1322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5288} = \langle S_8, \sigma_{1322} \rangle \Rightarrow \langle y := y + 1, \sigma_{1322} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5287} = \frac{\mathcal{T}_{5288}}{\langle S_9, \sigma_{1322} \rangle \Rightarrow \langle S_{10}, \sigma_{1322} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5289} = \frac{\langle y := y + 1, \sigma_{1322} \rangle \Rightarrow \sigma_{1323} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1322} \rangle \Rightarrow \langle S_{11}, \sigma_{1323} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5290} = \langle S_{11}, \sigma_{1323} \rangle \Rightarrow \langle S_{12}, \sigma_{1323} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5291} = \langle S_{12}, \sigma_{1323} \rangle \Rightarrow \langle S_{13}, \sigma_{1323} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5293} = \frac{\langle x := x + 1, \sigma_{1323} \rangle \Rightarrow \sigma_{1324} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1323} \rangle \Rightarrow \langle S_6, \sigma_{1324} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5292} = \frac{\mathcal{T}_{5293}}{\langle S_{13}, \sigma_{1323} \rangle \Rightarrow \langle S_{15}, \sigma_{1324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5294} = \frac{\langle S_6, \sigma_{1324} \rangle \Rightarrow \langle S_8, \sigma_{1324} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1324} \rangle \Rightarrow \langle S_9, \sigma_{1324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5296} = \langle S_8, \sigma_{1324} \rangle \Rightarrow \langle y := y + 1, \sigma_{1324} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5295} = \frac{\mathcal{T}_{5296}}{\langle S_9, \sigma_{1324} \rangle \Rightarrow \langle S_{10}, \sigma_{1324} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5297} = \frac{\langle y := y + 1, \sigma_{1324} \rangle \Rightarrow \sigma_{1325} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1324} \rangle \Rightarrow \langle S_{11}, \sigma_{1325} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5298} = \langle S_{11}, \sigma_{1325} \rangle \Rightarrow \langle S_{12}, \sigma_{1325} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5299} = \langle S_{12}, \sigma_{1325} \rangle \Rightarrow \langle S_{13}, \sigma_{1325} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5301} = \frac{\langle x := x + 1, \sigma_{1325} \rangle \Rightarrow \sigma_{1326} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1325} \rangle \Rightarrow \langle S_6, \sigma_{1326} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5300} = \frac{\mathcal{T}_{5301}}{\langle S_{13}, \sigma_{1325} \rangle \Rightarrow \langle S_{15}, \sigma_{1326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5302} = \frac{\langle S_6, \sigma_{1326} \rangle \Rightarrow \langle S_8, \sigma_{1326} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1326} \rangle \Rightarrow \langle S_9, \sigma_{1326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5304} = \langle S_8, \sigma_{1326} \rangle \Rightarrow \langle y := y + 1, \sigma_{1326} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5303} = \frac{\mathcal{T}_{5304}}{\langle S_9, \sigma_{1326} \rangle \Rightarrow \langle S_{10}, \sigma_{1326} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5305} = \frac{\langle y := y + 1, \sigma_{1326} \rangle \Rightarrow \sigma_{1327} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1326} \rangle \Rightarrow \langle S_{11}, \sigma_{1327} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5306} = \langle S_{11}, \sigma_{1327} \rangle \Rightarrow \langle S_{12}, \sigma_{1327} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5307} = \langle S_{12}, \sigma_{1327} \rangle \Rightarrow \langle S_{13}, \sigma_{1327} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5309} = \frac{\langle x := x + 1, \sigma_{1327} \rangle \Rightarrow \sigma_{1328} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1327} \rangle \Rightarrow \langle S_6, \sigma_{1328} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5308} = \frac{\mathcal{T}_{5309}}{\langle S_{13}, \sigma_{1327} \rangle \Rightarrow \langle S_{15}, \sigma_{1328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5310} = \frac{\langle S_6, \sigma_{1328} \rangle \Rightarrow \langle S_8, \sigma_{1328} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1328} \rangle \Rightarrow \langle S_9, \sigma_{1328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5312} = \langle S_8, \sigma_{1328} \rangle \Rightarrow \langle y := y + 1, \sigma_{1328} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5311} = \frac{\mathcal{T}_{5312}}{\langle S_9, \sigma_{1328} \rangle \Rightarrow \langle S_{10}, \sigma_{1328} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5313} = \frac{\langle y := y + 1, \sigma_{1328} \rangle \Rightarrow \sigma_{1329} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1328} \rangle \Rightarrow \langle S_{11}, \sigma_{1329} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5314} = \langle S_{11}, \sigma_{1329} \rangle \Rightarrow \langle S_{12}, \sigma_{1329} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5315} = \langle S_{12}, \sigma_{1329} \rangle \Rightarrow \langle S_{13}, \sigma_{1329} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5317} = \frac{\langle x := x + 1, \sigma_{1329} \rangle \Rightarrow \sigma_{1330} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1329} \rangle \Rightarrow \langle S_6, \sigma_{1330} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5316} = \frac{\mathcal{T}_{5317}}{\langle S_{13}, \sigma_{1329} \rangle \Rightarrow \langle S_{15}, \sigma_{1330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5318} = \frac{\langle S_6, \sigma_{1330} \rangle \Rightarrow \langle S_8, \sigma_{1330} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1330} \rangle \Rightarrow \langle S_9, \sigma_{1330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5320} = \langle S_8, \sigma_{1330} \rangle \Rightarrow \langle y := y + 1, \sigma_{1330} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5319} = \frac{\mathcal{T}_{5320}}{\langle S_9, \sigma_{1330} \rangle \Rightarrow \langle S_{10}, \sigma_{1330} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5321} = \frac{\langle y := y + 1, \sigma_{1330} \rangle \Rightarrow \sigma_{1331} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1330} \rangle \Rightarrow \langle S_{11}, \sigma_{1331} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5322} = \langle S_{11}, \sigma_{1331} \rangle \Rightarrow \langle S_{12}, \sigma_{1331} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5323} = \langle S_{12}, \sigma_{1331} \rangle \Rightarrow \langle S_{13}, \sigma_{1331} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5325} = \frac{\langle x := x + 1, \sigma_{1331} \rangle \Rightarrow \sigma_{1332} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1331} \rangle \Rightarrow \langle S_6, \sigma_{1332} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5324} = \frac{\mathcal{T}_{5325}}{\langle S_{13}, \sigma_{1331} \rangle \Rightarrow \langle S_{15}, \sigma_{1332} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5326} = \frac{\langle S_6, \sigma_{1332} \rangle \Rightarrow \langle S_8, \sigma_{1332} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1332} \rangle \Rightarrow \langle S_9, \sigma_{1332} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5328} = \langle S_8, \sigma_{1332} \rangle \Rightarrow \langle y := y + 1, \sigma_{1332} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5327} = \frac{\mathcal{T}_{5328}}{\langle S_9, \sigma_{1332} \rangle \Rightarrow \langle S_{10}, \sigma_{1332} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5329} = \frac{\langle y := y + 1, \sigma_{1332} \rangle \Rightarrow \sigma_{1333} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1332} \rangle \Rightarrow \langle S_{11}, \sigma_{1333} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5330} = \langle S_{11}, \sigma_{1333} \rangle \Rightarrow \langle S_{12}, \sigma_{1333} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5331} = \langle S_{12}, \sigma_{1333} \rangle \Rightarrow \langle S_{13}, \sigma_{1333} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5333} = \frac{\langle x := x + 1, \sigma_{1333} \rangle \Rightarrow \sigma_{1334} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1333} \rangle \Rightarrow \langle S_6, \sigma_{1334} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5332} = \frac{\mathcal{T}_{5333}}{\langle S_{13}, \sigma_{1333} \rangle \Rightarrow \langle S_{15}, \sigma_{1334} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5334} = \frac{\langle S_6, \sigma_{1334} \rangle \Rightarrow \langle S_8, \sigma_{1334} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1334} \rangle \Rightarrow \langle S_9, \sigma_{1334} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5336} = \langle S_8, \sigma_{1334} \rangle \Rightarrow \langle y := y + 1, \sigma_{1334} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5335} = \frac{\mathcal{T}_{5336}}{\langle S_9, \sigma_{1334} \rangle \Rightarrow \langle S_{10}, \sigma_{1334} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5337} = \frac{\langle y := y + 1, \sigma_{1334} \rangle \Rightarrow \sigma_{1335} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1334} \rangle \Rightarrow \langle S_{11}, \sigma_{1335} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5338} = \langle S_{11}, \sigma_{1335} \rangle \Rightarrow \langle S_{12}, \sigma_{1335} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5339} = \langle S_{12}, \sigma_{1335} \rangle \Rightarrow \langle S_{13}, \sigma_{1335} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5341} = \frac{\langle x := x + 1, \sigma_{1335} \rangle \Rightarrow \sigma_{1336} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1335} \rangle \Rightarrow \langle S_6, \sigma_{1336} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5340} = \frac{\mathcal{T}_{5341}}{\langle S_{13}, \sigma_{1335} \rangle \Rightarrow \langle S_{15}, \sigma_{1336} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5342} = \frac{\langle S_6, \sigma_{1336} \rangle \Rightarrow \langle S_8, \sigma_{1336} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1336} \rangle \Rightarrow \langle S_9, \sigma_{1336} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5344} = \langle S_8, \sigma_{1336} \rangle \Rightarrow \langle y := y + 1, \sigma_{1336} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5343} = \frac{\mathcal{T}_{5344}}{\langle S_9, \sigma_{1336} \rangle \Rightarrow \langle S_{10}, \sigma_{1336} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5345} = \frac{\langle y := y + 1, \sigma_{1336} \rangle \Rightarrow \sigma_{1337} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1336} \rangle \Rightarrow \langle S_{11}, \sigma_{1337} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5346} = \langle S_{11}, \sigma_{1337} \rangle \Rightarrow \langle S_{12}, \sigma_{1337} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5347} = \langle S_{12}, \sigma_{1337} \rangle \Rightarrow \langle S_{13}, \sigma_{1337} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5349} = \frac{\langle x := x + 1, \sigma_{1337} \rangle \Rightarrow \sigma_{1338} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1337} \rangle \Rightarrow \langle S_6, \sigma_{1338} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5348} = \frac{\mathcal{T}_{5349}}{\langle S_{13}, \sigma_{1337} \rangle \Rightarrow \langle S_{15}, \sigma_{1338} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5350} = \frac{\langle S_6, \sigma_{1338} \rangle \Rightarrow \langle S_8, \sigma_{1338} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1338} \rangle \Rightarrow \langle S_9, \sigma_{1338} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5352} = \langle S_8, \sigma_{1338} \rangle \Rightarrow \langle y := y + 1, \sigma_{1338} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5351} = \frac{\mathcal{T}_{5352}}{\langle S_9, \sigma_{1338} \rangle \Rightarrow \langle S_{10}, \sigma_{1338} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5353} = \frac{\langle y := y + 1, \sigma_{1338} \rangle \Rightarrow \sigma_{1339} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1338} \rangle \Rightarrow \langle S_{11}, \sigma_{1339} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5354} = \langle S_{11}, \sigma_{1339} \rangle \Rightarrow \langle S_{12}, \sigma_{1339} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5355} = \langle S_{12}, \sigma_{1339} \rangle \Rightarrow \langle S_{13}, \sigma_{1339} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5357} = \frac{\langle x := x + 1, \sigma_{1339} \rangle \Rightarrow \sigma_{1340} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1339} \rangle \Rightarrow \langle S_6, \sigma_{1340} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5356} = \frac{\mathcal{T}_{5357}}{\langle S_{13}, \sigma_{1339} \rangle \Rightarrow \langle S_{15}, \sigma_{1340} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5358} = \frac{\langle S_6, \sigma_{1340} \rangle \Rightarrow \langle S_8, \sigma_{1340} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1340} \rangle \Rightarrow \langle S_9, \sigma_{1340} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5360} = \langle S_8, \sigma_{1340} \rangle \Rightarrow \langle y := y + 1, \sigma_{1340} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5359} = \frac{\mathcal{T}_{5360}}{\langle S_9, \sigma_{1340} \rangle \Rightarrow \langle S_{10}, \sigma_{1340} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5361} = \frac{\langle y := y + 1, \sigma_{1340} \rangle \Rightarrow \sigma_{1341} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1340} \rangle \Rightarrow \langle S_{11}, \sigma_{1341} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5362} = \langle S_{11}, \sigma_{1341} \rangle \Rightarrow \langle S_{12}, \sigma_{1341} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5363} = \langle S_{12}, \sigma_{1341} \rangle \Rightarrow \langle S_{13}, \sigma_{1341} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5365} = \frac{\langle x := x + 1, \sigma_{1341} \rangle \Rightarrow \sigma_{1342} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1341} \rangle \Rightarrow \langle S_6, \sigma_{1342} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5364} = \frac{\mathcal{T}_{5365}}{\langle S_{13}, \sigma_{1341} \rangle \Rightarrow \langle S_{15}, \sigma_{1342} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5366} = \frac{\langle S_6, \sigma_{1342} \rangle \Rightarrow \langle S_8, \sigma_{1342} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1342} \rangle \Rightarrow \langle S_9, \sigma_{1342} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5368} = \langle S_8, \sigma_{1342} \rangle \Rightarrow \langle y := y + 1, \sigma_{1342} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5367} = \frac{\mathcal{T}_{5368}}{\langle S_9, \sigma_{1342} \rangle \Rightarrow \langle S_{10}, \sigma_{1342} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5369} = \frac{\langle y := y + 1, \sigma_{1342} \rangle \Rightarrow \sigma_{1343} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1342} \rangle \Rightarrow \langle S_{11}, \sigma_{1343} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5370} = \langle S_{11}, \sigma_{1343} \rangle \Rightarrow \langle S_{12}, \sigma_{1343} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5371} = \langle S_{12}, \sigma_{1343} \rangle \Rightarrow \langle S_{13}, \sigma_{1343} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5373} = \frac{\langle x := x + 1, \sigma_{1343} \rangle \Rightarrow \sigma_{1344} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1343} \rangle \Rightarrow \langle S_6, \sigma_{1344} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5372} = \frac{\mathcal{T}_{5373}}{\langle S_{13}, \sigma_{1343} \rangle \Rightarrow \langle S_{15}, \sigma_{1344} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5374} = \frac{\langle S_6, \sigma_{1344} \rangle \Rightarrow \langle S_8, \sigma_{1344} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1344} \rangle \Rightarrow \langle S_9, \sigma_{1344} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5376} = \langle S_8, \sigma_{1344} \rangle \Rightarrow \langle y := y + 1, \sigma_{1344} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5375} = \frac{\mathcal{T}_{5376}}{\langle S_9, \sigma_{1344} \rangle \Rightarrow \langle S_{10}, \sigma_{1344} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5377} = \frac{\langle y := y + 1, \sigma_{1344} \rangle \Rightarrow \sigma_{1345} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1344} \rangle \Rightarrow \langle S_{11}, \sigma_{1345} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5378} = \langle S_{11}, \sigma_{1345} \rangle \Rightarrow \langle S_{12}, \sigma_{1345} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5379} = \langle S_{12}, \sigma_{1345} \rangle \Rightarrow \langle S_{13}, \sigma_{1345} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5381} = \frac{\langle x := x + 1, \sigma_{1345} \rangle \Rightarrow \sigma_{1346} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1345} \rangle \Rightarrow \langle S_6, \sigma_{1346} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5380} = \frac{\mathcal{T}_{5381}}{\langle S_{13}, \sigma_{1345} \rangle \Rightarrow \langle S_{15}, \sigma_{1346} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5382} = \frac{\langle S_6, \sigma_{1346} \rangle \Rightarrow \langle S_8, \sigma_{1346} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1346} \rangle \Rightarrow \langle S_9, \sigma_{1346} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5384} = \langle S_8, \sigma_{1346} \rangle \Rightarrow \langle y := y + 1, \sigma_{1346} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5383} = \frac{\mathcal{T}_{5384}}{\langle S_9, \sigma_{1346} \rangle \Rightarrow \langle S_{10}, \sigma_{1346} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5385} = \frac{\langle y := y + 1, \sigma_{1346} \rangle \Rightarrow \sigma_{1347} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1346} \rangle \Rightarrow \langle S_{11}, \sigma_{1347} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5386} = \langle S_{11}, \sigma_{1347} \rangle \Rightarrow \langle S_{12}, \sigma_{1347} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5387} = \langle S_{12}, \sigma_{1347} \rangle \Rightarrow \langle S_{13}, \sigma_{1347} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5389} = \frac{\langle x := x + 1, \sigma_{1347} \rangle \Rightarrow \sigma_{1348} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1347} \rangle \Rightarrow \langle S_6, \sigma_{1348} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5388} = \frac{\mathcal{T}_{5389}}{\langle S_{13}, \sigma_{1347} \rangle \Rightarrow \langle S_{15}, \sigma_{1348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5390} = \frac{\langle S_6, \sigma_{1348} \rangle \Rightarrow \langle S_8, \sigma_{1348} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1348} \rangle \Rightarrow \langle S_9, \sigma_{1348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5392} = \langle S_8, \sigma_{1348} \rangle \Rightarrow \langle y := y + 1, \sigma_{1348} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5391} = \frac{\mathcal{T}_{5392}}{\langle S_9, \sigma_{1348} \rangle \Rightarrow \langle S_{10}, \sigma_{1348} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5393} = \frac{\langle y := y + 1, \sigma_{1348} \rangle \Rightarrow \sigma_{1349} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1348} \rangle \Rightarrow \langle S_{11}, \sigma_{1349} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5394} = \langle S_{11}, \sigma_{1349} \rangle \Rightarrow \langle S_{12}, \sigma_{1349} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5395} = \langle S_{12}, \sigma_{1349} \rangle \Rightarrow \langle S_{13}, \sigma_{1349} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5397} = \frac{\langle x := x + 1, \sigma_{1349} \rangle \Rightarrow \sigma_{1350} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1349} \rangle \Rightarrow \langle S_6, \sigma_{1350} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5396} = \frac{\mathcal{T}_{5397}}{\langle S_{13}, \sigma_{1349} \rangle \Rightarrow \langle S_{15}, \sigma_{1350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5398} = \frac{\langle S_6, \sigma_{1350} \rangle \Rightarrow \langle S_8, \sigma_{1350} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1350} \rangle \Rightarrow \langle S_9, \sigma_{1350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5400} = \langle S_8, \sigma_{1350} \rangle \Rightarrow \langle y := y + 1, \sigma_{1350} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5399} = \frac{\mathcal{T}_{5400}}{\langle S_9, \sigma_{1350} \rangle \Rightarrow \langle S_{10}, \sigma_{1350} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5401} = \frac{\langle y := y + 1, \sigma_{1350} \rangle \Rightarrow \sigma_{1351} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1350} \rangle \Rightarrow \langle S_{11}, \sigma_{1351} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5402} = \langle S_{11}, \sigma_{1351} \rangle \Rightarrow \langle S_{12}, \sigma_{1351} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5403} = \langle S_{12}, \sigma_{1351} \rangle \Rightarrow \langle S_{13}, \sigma_{1351} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5405} = \frac{\langle x := x + 1, \sigma_{1351} \rangle \Rightarrow \sigma_{1352} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1351} \rangle \Rightarrow \langle S_6, \sigma_{1352} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5404} = \frac{\mathcal{T}_{5405}}{\langle S_{13}, \sigma_{1351} \rangle \Rightarrow \langle S_{15}, \sigma_{1352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5406} = \frac{\langle S_6, \sigma_{1352} \rangle \Rightarrow \langle S_8, \sigma_{1352} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1352} \rangle \Rightarrow \langle S_9, \sigma_{1352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5408} = \langle S_8, \sigma_{1352} \rangle \Rightarrow \langle y := y + 1, \sigma_{1352} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5407} = \frac{\mathcal{T}_{5408}}{\langle S_9, \sigma_{1352} \rangle \Rightarrow \langle S_{10}, \sigma_{1352} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5409} = \frac{\langle y := y + 1, \sigma_{1352} \rangle \Rightarrow \sigma_{1353} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1352} \rangle \Rightarrow \langle S_{11}, \sigma_{1353} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5410} = \langle S_{11}, \sigma_{1353} \rangle \Rightarrow \langle S_{12}, \sigma_{1353} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5411} = \langle S_{12}, \sigma_{1353} \rangle \Rightarrow \langle S_{13}, \sigma_{1353} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5413} = \frac{\langle x := x + 1, \sigma_{1353} \rangle \Rightarrow \sigma_{1354} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1353} \rangle \Rightarrow \langle S_6, \sigma_{1354} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5412} = \frac{\mathcal{T}_{5413}}{\langle S_{13}, \sigma_{1353} \rangle \Rightarrow \langle S_{15}, \sigma_{1354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5414} = \frac{\langle S_6, \sigma_{1354} \rangle \Rightarrow \langle S_8, \sigma_{1354} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1354} \rangle \Rightarrow \langle S_9, \sigma_{1354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5416} = \langle S_8, \sigma_{1354} \rangle \Rightarrow \langle y := y + 1, \sigma_{1354} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5415} = \frac{\mathcal{T}_{5416}}{\langle S_9, \sigma_{1354} \rangle \Rightarrow \langle S_{10}, \sigma_{1354} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5417} = \frac{\langle y := y + 1, \sigma_{1354} \rangle \Rightarrow \sigma_{1355} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1354} \rangle \Rightarrow \langle S_{11}, \sigma_{1355} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5418} = \langle S_{11}, \sigma_{1355} \rangle \Rightarrow \langle S_{12}, \sigma_{1355} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5419} = \langle S_{12}, \sigma_{1355} \rangle \Rightarrow \langle S_{13}, \sigma_{1355} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5421} = \frac{\langle x := x + 1, \sigma_{1355} \rangle \Rightarrow \sigma_{1356} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1355} \rangle \Rightarrow \langle S_6, \sigma_{1356} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5420} = \frac{\mathcal{T}_{5421}}{\langle S_{13}, \sigma_{1355} \rangle \Rightarrow \langle S_{15}, \sigma_{1356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5422} = \frac{\langle S_6, \sigma_{1356} \rangle \Rightarrow \langle S_8, \sigma_{1356} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1356} \rangle \Rightarrow \langle S_9, \sigma_{1356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5424} = \langle S_8, \sigma_{1356} \rangle \Rightarrow \langle y := y + 1, \sigma_{1356} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5423} = \frac{\mathcal{T}_{5424}}{\langle S_9, \sigma_{1356} \rangle \Rightarrow \langle S_{10}, \sigma_{1356} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5425} = \frac{\langle y := y + 1, \sigma_{1356} \rangle \Rightarrow \sigma_{1357} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1356} \rangle \Rightarrow \langle S_{11}, \sigma_{1357} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5426} = \langle S_{11}, \sigma_{1357} \rangle \Rightarrow \langle S_{12}, \sigma_{1357} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5427} = \langle S_{12}, \sigma_{1357} \rangle \Rightarrow \langle S_{13}, \sigma_{1357} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5429} = \frac{\langle x := x + 1, \sigma_{1357} \rangle \Rightarrow \sigma_{1358} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1357} \rangle \Rightarrow \langle S_6, \sigma_{1358} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5428} = \frac{\mathcal{T}_{5429}}{\langle S_{13}, \sigma_{1357} \rangle \Rightarrow \langle S_{15}, \sigma_{1358} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5430} = \frac{\langle S_6, \sigma_{1358} \rangle \Rightarrow \langle S_8, \sigma_{1358} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1358} \rangle \Rightarrow \langle S_9, \sigma_{1358} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5432} = \langle S_8, \sigma_{1358} \rangle \Rightarrow \langle y := y + 1, \sigma_{1358} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5431} = \frac{\mathcal{T}_{5432}}{\langle S_9, \sigma_{1358} \rangle \Rightarrow \langle S_{10}, \sigma_{1358} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5433} = \frac{\langle y := y + 1, \sigma_{1358} \rangle \Rightarrow \sigma_{1359} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1358} \rangle \Rightarrow \langle S_{11}, \sigma_{1359} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5434} = \langle S_{11}, \sigma_{1359} \rangle \Rightarrow \langle S_{12}, \sigma_{1359} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5435} = \langle S_{12}, \sigma_{1359} \rangle \Rightarrow \langle S_{13}, \sigma_{1359} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5437} = \frac{\langle x := x + 1, \sigma_{1359} \rangle \Rightarrow \sigma_{1360} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1359} \rangle \Rightarrow \langle S_6, \sigma_{1360} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5436} = \frac{\mathcal{T}_{5437}}{\langle S_{13}, \sigma_{1359} \rangle \Rightarrow \langle S_{15}, \sigma_{1360} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5438} = \frac{\langle S_6, \sigma_{1360} \rangle \Rightarrow \langle S_8, \sigma_{1360} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1360} \rangle \Rightarrow \langle S_9, \sigma_{1360} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5440} = \langle S_8, \sigma_{1360} \rangle \Rightarrow \langle y := y + 1, \sigma_{1360} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5439} = \frac{\mathcal{T}_{5440}}{\langle S_9, \sigma_{1360} \rangle \Rightarrow \langle S_{10}, \sigma_{1360} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5441} = \frac{\langle y := y + 1, \sigma_{1360} \rangle \Rightarrow \sigma_{1361} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1360} \rangle \Rightarrow \langle S_{11}, \sigma_{1361} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5442} = \langle S_{11}, \sigma_{1361} \rangle \Rightarrow \langle S_{12}, \sigma_{1361} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5443} = \langle S_{12}, \sigma_{1361} \rangle \Rightarrow \langle S_{13}, \sigma_{1361} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5445} = \frac{\langle x := x + 1, \sigma_{1361} \rangle \Rightarrow \sigma_{1362} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1361} \rangle \Rightarrow \langle S_6, \sigma_{1362} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5444} = \frac{\mathcal{T}_{5445}}{\langle S_{13}, \sigma_{1361} \rangle \Rightarrow \langle S_{15}, \sigma_{1362} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5446} = \frac{\langle S_6, \sigma_{1362} \rangle \Rightarrow \langle S_8, \sigma_{1362} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1362} \rangle \Rightarrow \langle S_9, \sigma_{1362} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5448} = \langle S_8, \sigma_{1362} \rangle \Rightarrow \langle y := y + 1, \sigma_{1362} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5447} = \frac{\mathcal{T}_{5448}}{\langle S_9, \sigma_{1362} \rangle \Rightarrow \langle S_{10}, \sigma_{1362} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5449} = \frac{\langle y := y + 1, \sigma_{1362} \rangle \Rightarrow \sigma_{1363} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1362} \rangle \Rightarrow \langle S_{11}, \sigma_{1363} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5450} = \langle S_{11}, \sigma_{1363} \rangle \Rightarrow \langle S_{12}, \sigma_{1363} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5451} = \langle S_{12}, \sigma_{1363} \rangle \Rightarrow \langle S_{13}, \sigma_{1363} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5453} = \frac{\langle x := x + 1, \sigma_{1363} \rangle \Rightarrow \sigma_{1364} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1363} \rangle \Rightarrow \langle S_6, \sigma_{1364} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5452} = \frac{\mathcal{T}_{5453}}{\langle S_{13}, \sigma_{1363} \rangle \Rightarrow \langle S_{15}, \sigma_{1364} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5454} = \frac{\langle S_6, \sigma_{1364} \rangle \Rightarrow \langle S_8, \sigma_{1364} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1364} \rangle \Rightarrow \langle S_9, \sigma_{1364} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5456} = \langle S_8, \sigma_{1364} \rangle \Rightarrow \langle y := y + 1, \sigma_{1364} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5455} = \frac{\mathcal{T}_{5456}}{\langle S_9, \sigma_{1364} \rangle \Rightarrow \langle S_{10}, \sigma_{1364} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5457} = \frac{\langle y := y + 1, \sigma_{1364} \rangle \Rightarrow \sigma_{1365} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1364} \rangle \Rightarrow \langle S_{11}, \sigma_{1365} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5458} = \langle S_{11}, \sigma_{1365} \rangle \Rightarrow \langle S_{12}, \sigma_{1365} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5459} = \langle S_{12}, \sigma_{1365} \rangle \Rightarrow \langle S_{13}, \sigma_{1365} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5461} = \frac{\langle x := x + 1, \sigma_{1365} \rangle \Rightarrow \sigma_{1366} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1365} \rangle \Rightarrow \langle S_6, \sigma_{1366} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5460} = \frac{\mathcal{T}_{5461}}{\langle S_{13}, \sigma_{1365} \rangle \Rightarrow \langle S_{15}, \sigma_{1366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5462} = \frac{\langle S_6, \sigma_{1366} \rangle \Rightarrow \langle S_8, \sigma_{1366} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1366} \rangle \Rightarrow \langle S_9, \sigma_{1366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5464} = \langle S_8, \sigma_{1366} \rangle \Rightarrow \langle y := y + 1, \sigma_{1366} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5463} = \frac{\mathcal{T}_{5464}}{\langle S_9, \sigma_{1366} \rangle \Rightarrow \langle S_{10}, \sigma_{1366} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5465} = \frac{\langle y := y + 1, \sigma_{1366} \rangle \Rightarrow \sigma_{1367} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1366} \rangle \Rightarrow \langle S_{11}, \sigma_{1367} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5466} = \langle S_{11}, \sigma_{1367} \rangle \Rightarrow \langle S_{12}, \sigma_{1367} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5467} = \langle S_{12}, \sigma_{1367} \rangle \Rightarrow \langle S_{13}, \sigma_{1367} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5469} = \frac{\langle x := x + 1, \sigma_{1367} \rangle \Rightarrow \sigma_{1368} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1367} \rangle \Rightarrow \langle S_6, \sigma_{1368} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5468} = \frac{\mathcal{T}_{5469}}{\langle S_{13}, \sigma_{1367} \rangle \Rightarrow \langle S_{15}, \sigma_{1368} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5470} = \frac{\langle S_6, \sigma_{1368} \rangle \Rightarrow \langle S_8, \sigma_{1368} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1368} \rangle \Rightarrow \langle S_9, \sigma_{1368} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5472} = \langle S_8, \sigma_{1368} \rangle \Rightarrow \langle y := y + 1, \sigma_{1368} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5471} = \frac{\mathcal{T}_{5472}}{\langle S_9, \sigma_{1368} \rangle \Rightarrow \langle S_{10}, \sigma_{1368} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5473} = \frac{\langle y := y + 1, \sigma_{1368} \rangle \Rightarrow \sigma_{1369} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1368} \rangle \Rightarrow \langle S_{11}, \sigma_{1369} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5474} = \langle S_{11}, \sigma_{1369} \rangle \Rightarrow \langle S_{12}, \sigma_{1369} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5475} = \langle S_{12}, \sigma_{1369} \rangle \Rightarrow \langle S_{13}, \sigma_{1369} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5477} = \frac{\langle x := x + 1, \sigma_{1369} \rangle \Rightarrow \sigma_{1370} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1369} \rangle \Rightarrow \langle S_6, \sigma_{1370} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5476} = \frac{\mathcal{T}_{5477}}{\langle S_{13}, \sigma_{1369} \rangle \Rightarrow \langle S_{15}, \sigma_{1370} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5478} = \frac{\langle S_6, \sigma_{1370} \rangle \Rightarrow \langle S_8, \sigma_{1370} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1370} \rangle \Rightarrow \langle S_9, \sigma_{1370} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5480} = \langle S_8, \sigma_{1370} \rangle \Rightarrow \langle y := y + 1, \sigma_{1370} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5479} = \frac{\mathcal{T}_{5480}}{\langle S_9, \sigma_{1370} \rangle \Rightarrow \langle S_{10}, \sigma_{1370} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5481} = \frac{\langle y := y + 1, \sigma_{1370} \rangle \Rightarrow \sigma_{1371} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1370} \rangle \Rightarrow \langle S_{11}, \sigma_{1371} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5482} = \langle S_{11}, \sigma_{1371} \rangle \Rightarrow \langle S_{12}, \sigma_{1371} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5483} = \langle S_{12}, \sigma_{1371} \rangle \Rightarrow \langle S_{13}, \sigma_{1371} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5485} = \frac{\langle x := x + 1, \sigma_{1371} \rangle \Rightarrow \sigma_{1372} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1371} \rangle \Rightarrow \langle S_6, \sigma_{1372} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5484} = \frac{\mathcal{T}_{5485}}{\langle S_{13}, \sigma_{1371} \rangle \Rightarrow \langle S_{15}, \sigma_{1372} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5486} = \frac{\langle S_6, \sigma_{1372} \rangle \Rightarrow \langle S_8, \sigma_{1372} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1372} \rangle \Rightarrow \langle S_9, \sigma_{1372} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5488} = \langle S_8, \sigma_{1372} \rangle \Rightarrow \langle y := y + 1, \sigma_{1372} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5487} = \frac{\mathcal{T}_{5488}}{\langle S_9, \sigma_{1372} \rangle \Rightarrow \langle S_{10}, \sigma_{1372} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5489} = \frac{\langle y := y + 1, \sigma_{1372} \rangle \Rightarrow \sigma_{1373} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1372} \rangle \Rightarrow \langle S_{11}, \sigma_{1373} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5490} = \langle S_{11}, \sigma_{1373} \rangle \Rightarrow \langle S_{12}, \sigma_{1373} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5491} = \langle S_{12}, \sigma_{1373} \rangle \Rightarrow \langle S_{13}, \sigma_{1373} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5493} = \frac{\langle x := x + 1, \sigma_{1373} \rangle \Rightarrow \sigma_{1374} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1373} \rangle \Rightarrow \langle S_6, \sigma_{1374} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5492} = \frac{\mathcal{T}_{5493}}{\langle S_{13}, \sigma_{1373} \rangle \Rightarrow \langle S_{15}, \sigma_{1374} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5494} = \frac{\langle S_6, \sigma_{1374} \rangle \Rightarrow \langle S_8, \sigma_{1374} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1374} \rangle \Rightarrow \langle S_9, \sigma_{1374} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5496} = \langle S_8, \sigma_{1374} \rangle \Rightarrow \langle y := y + 1, \sigma_{1374} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5495} = \frac{\mathcal{T}_{5496}}{\langle S_9, \sigma_{1374} \rangle \Rightarrow \langle S_{10}, \sigma_{1374} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5497} = \frac{\langle y := y + 1, \sigma_{1374} \rangle \Rightarrow \sigma_{1375} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1374} \rangle \Rightarrow \langle S_{11}, \sigma_{1375} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5498} = \langle S_{11}, \sigma_{1375} \rangle \Rightarrow \langle S_{12}, \sigma_{1375} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5499} = \langle S_{12}, \sigma_{1375} \rangle \Rightarrow \langle S_{13}, \sigma_{1375} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5501} = \frac{\langle x := x + 1, \sigma_{1375} \rangle \Rightarrow \sigma_{1376} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1375} \rangle \Rightarrow \langle S_6, \sigma_{1376} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5500} = \frac{\mathcal{T}_{5501}}{\langle S_{13}, \sigma_{1375} \rangle \Rightarrow \langle S_{15}, \sigma_{1376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5502} = \frac{\langle S_6, \sigma_{1376} \rangle \Rightarrow \langle S_8, \sigma_{1376} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1376} \rangle \Rightarrow \langle S_9, \sigma_{1376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5504} = \langle S_8, \sigma_{1376} \rangle \Rightarrow \langle y := y + 1, \sigma_{1376} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5503} = \frac{\mathcal{T}_{5504}}{\langle S_9, \sigma_{1376} \rangle \Rightarrow \langle S_{10}, \sigma_{1376} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5505} = \frac{\langle y := y + 1, \sigma_{1376} \rangle \Rightarrow \sigma_{1377} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1376} \rangle \Rightarrow \langle S_{11}, \sigma_{1377} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5506} = \langle S_{11}, \sigma_{1377} \rangle \Rightarrow \langle S_{12}, \sigma_{1377} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5507} = \langle S_{12}, \sigma_{1377} \rangle \Rightarrow \langle S_{13}, \sigma_{1377} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5509} = \frac{\langle x := x + 1, \sigma_{1377} \rangle \Rightarrow \sigma_{1378} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1377} \rangle \Rightarrow \langle S_6, \sigma_{1378} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5508} = \frac{\mathcal{T}_{5509}}{\langle S_{13}, \sigma_{1377} \rangle \Rightarrow \langle S_{15}, \sigma_{1378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5510} = \frac{\langle S_6, \sigma_{1378} \rangle \Rightarrow \langle S_8, \sigma_{1378} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1378} \rangle \Rightarrow \langle S_9, \sigma_{1378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5512} = \langle S_8, \sigma_{1378} \rangle \Rightarrow \langle y := y + 1, \sigma_{1378} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5511} = \frac{\mathcal{T}_{5512}}{\langle S_9, \sigma_{1378} \rangle \Rightarrow \langle S_{10}, \sigma_{1378} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5513} = \frac{\langle y := y + 1, \sigma_{1378} \rangle \Rightarrow \sigma_{1379} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1378} \rangle \Rightarrow \langle S_{11}, \sigma_{1379} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5514} = \langle S_{11}, \sigma_{1379} \rangle \Rightarrow \langle S_{12}, \sigma_{1379} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5515} = \langle S_{12}, \sigma_{1379} \rangle \Rightarrow \langle S_{13}, \sigma_{1379} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5517} = \frac{\langle x := x + 1, \sigma_{1379} \rangle \Rightarrow \sigma_{1380} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1379} \rangle \Rightarrow \langle S_6, \sigma_{1380} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5516} = \frac{\mathcal{T}_{5517}}{\langle S_{13}, \sigma_{1379} \rangle \Rightarrow \langle S_{15}, \sigma_{1380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5518} = \frac{\langle S_6, \sigma_{1380} \rangle \Rightarrow \langle S_8, \sigma_{1380} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1380} \rangle \Rightarrow \langle S_9, \sigma_{1380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5520} = \langle S_8, \sigma_{1380} \rangle \Rightarrow \langle y := y + 1, \sigma_{1380} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5519} = \frac{\mathcal{T}_{5520}}{\langle S_9, \sigma_{1380} \rangle \Rightarrow \langle S_{10}, \sigma_{1380} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5521} = \frac{\langle y := y + 1, \sigma_{1380} \rangle \Rightarrow \sigma_{1381} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1380} \rangle \Rightarrow \langle S_{11}, \sigma_{1381} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5522} = \langle S_{11}, \sigma_{1381} \rangle \Rightarrow \langle S_{12}, \sigma_{1381} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5523} = \langle S_{12}, \sigma_{1381} \rangle \Rightarrow \langle S_{13}, \sigma_{1381} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5525} = \frac{\langle x := x + 1, \sigma_{1381} \rangle \Rightarrow \sigma_{1382} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1381} \rangle \Rightarrow \langle S_6, \sigma_{1382} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5524} = \frac{\mathcal{T}_{5525}}{\langle S_{13}, \sigma_{1381} \rangle \Rightarrow \langle S_{15}, \sigma_{1382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5526} = \frac{\langle S_6, \sigma_{1382} \rangle \Rightarrow \langle S_8, \sigma_{1382} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1382} \rangle \Rightarrow \langle S_9, \sigma_{1382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5528} = \langle S_8, \sigma_{1382} \rangle \Rightarrow \langle y := y + 1, \sigma_{1382} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5527} = \frac{\mathcal{T}_{5528}}{\langle S_9, \sigma_{1382} \rangle \Rightarrow \langle S_{10}, \sigma_{1382} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5529} = \frac{\langle y := y + 1, \sigma_{1382} \rangle \Rightarrow \sigma_{1383} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1382} \rangle \Rightarrow \langle S_{11}, \sigma_{1383} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5530} = \langle S_{11}, \sigma_{1383} \rangle \Rightarrow \langle S_{12}, \sigma_{1383} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5531} = \langle S_{12}, \sigma_{1383} \rangle \Rightarrow \langle S_{13}, \sigma_{1383} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5533} = \frac{\langle x := x + 1, \sigma_{1383} \rangle \Rightarrow \sigma_{1384} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1383} \rangle \Rightarrow \langle S_6, \sigma_{1384} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5532} = \frac{\mathcal{T}_{5533}}{\langle S_{13}, \sigma_{1383} \rangle \Rightarrow \langle S_{15}, \sigma_{1384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5534} = \frac{\langle S_6, \sigma_{1384} \rangle \Rightarrow \langle S_8, \sigma_{1384} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1384} \rangle \Rightarrow \langle S_9, \sigma_{1384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5536} = \langle S_8, \sigma_{1384} \rangle \Rightarrow \langle y := y + 1, \sigma_{1384} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5535} = \frac{\mathcal{T}_{5536}}{\langle S_9, \sigma_{1384} \rangle \Rightarrow \langle S_{10}, \sigma_{1384} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5537} = \frac{\langle y := y + 1, \sigma_{1384} \rangle \Rightarrow \sigma_{1385} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1384} \rangle \Rightarrow \langle S_{11}, \sigma_{1385} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5538} = \langle S_{11}, \sigma_{1385} \rangle \Rightarrow \langle S_{12}, \sigma_{1385} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5539} = \langle S_{12}, \sigma_{1385} \rangle \Rightarrow \langle S_{13}, \sigma_{1385} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5541} = \frac{\langle x := x + 1, \sigma_{1385} \rangle \Rightarrow \sigma_{1386} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1385} \rangle \Rightarrow \langle S_6, \sigma_{1386} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5540} = \frac{\mathcal{T}_{5541}}{\langle S_{13}, \sigma_{1385} \rangle \Rightarrow \langle S_{15}, \sigma_{1386} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5542} = \frac{\langle S_6, \sigma_{1386} \rangle \Rightarrow \langle S_8, \sigma_{1386} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1386} \rangle \Rightarrow \langle S_9, \sigma_{1386} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5544} = \langle S_8, \sigma_{1386} \rangle \Rightarrow \langle y := y + 1, \sigma_{1386} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5543} = \frac{\mathcal{T}_{5544}}{\langle S_9, \sigma_{1386} \rangle \Rightarrow \langle S_{10}, \sigma_{1386} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5545} = \frac{\langle y := y + 1, \sigma_{1386} \rangle \Rightarrow \sigma_{1387} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1386} \rangle \Rightarrow \langle S_{11}, \sigma_{1387} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5546} = \langle S_{11}, \sigma_{1387} \rangle \Rightarrow \langle S_{12}, \sigma_{1387} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5547} = \langle S_{12}, \sigma_{1387} \rangle \Rightarrow \langle S_{13}, \sigma_{1387} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5549} = \frac{\langle x := x + 1, \sigma_{1387} \rangle \Rightarrow \sigma_{1388} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1387} \rangle \Rightarrow \langle S_6, \sigma_{1388} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5548} = \frac{\mathcal{T}_{5549}}{\langle S_{13}, \sigma_{1387} \rangle \Rightarrow \langle S_{15}, \sigma_{1388} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5550} = \frac{\langle S_6, \sigma_{1388} \rangle \Rightarrow \langle S_8, \sigma_{1388} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1388} \rangle \Rightarrow \langle S_9, \sigma_{1388} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5552} = \langle S_8, \sigma_{1388} \rangle \Rightarrow \langle y := y + 1, \sigma_{1388} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5551} = \frac{\mathcal{T}_{5552}}{\langle S_9, \sigma_{1388} \rangle \Rightarrow \langle S_{10}, \sigma_{1388} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5553} = \frac{\langle y := y + 1, \sigma_{1388} \rangle \Rightarrow \sigma_{1389} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1388} \rangle \Rightarrow \langle S_{11}, \sigma_{1389} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5554} = \langle S_{11}, \sigma_{1389} \rangle \Rightarrow \langle S_{12}, \sigma_{1389} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5555} = \langle S_{12}, \sigma_{1389} \rangle \Rightarrow \langle S_{13}, \sigma_{1389} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5557} = \frac{\langle x := x + 1, \sigma_{1389} \rangle \Rightarrow \sigma_{1390} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1389} \rangle \Rightarrow \langle S_6, \sigma_{1390} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5556} = \frac{\mathcal{T}_{5557}}{\langle S_{13}, \sigma_{1389} \rangle \Rightarrow \langle S_{15}, \sigma_{1390} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5558} = \frac{\langle S_6, \sigma_{1390} \rangle \Rightarrow \langle S_8, \sigma_{1390} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1390} \rangle \Rightarrow \langle S_9, \sigma_{1390} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5560} = \langle S_8, \sigma_{1390} \rangle \Rightarrow \langle y := y + 1, \sigma_{1390} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5559} = \frac{\mathcal{T}_{5560}}{\langle S_9, \sigma_{1390} \rangle \Rightarrow \langle S_{10}, \sigma_{1390} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5561} = \frac{\langle y := y + 1, \sigma_{1390} \rangle \Rightarrow \sigma_{1391} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1390} \rangle \Rightarrow \langle S_{11}, \sigma_{1391} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5562} = \langle S_{11}, \sigma_{1391} \rangle \Rightarrow \langle S_{12}, \sigma_{1391} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5563} = \langle S_{12}, \sigma_{1391} \rangle \Rightarrow \langle S_{13}, \sigma_{1391} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5565} = \frac{\langle x := x + 1, \sigma_{1391} \rangle \Rightarrow \sigma_{1392} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1391} \rangle \Rightarrow \langle S_6, \sigma_{1392} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5564} = \frac{\mathcal{T}_{5565}}{\langle S_{13}, \sigma_{1391} \rangle \Rightarrow \langle S_{15}, \sigma_{1392} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5566} = \frac{\langle S_6, \sigma_{1392} \rangle \Rightarrow \langle S_8, \sigma_{1392} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1392} \rangle \Rightarrow \langle S_9, \sigma_{1392} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5568} = \langle S_8, \sigma_{1392} \rangle \Rightarrow \langle y := y + 1, \sigma_{1392} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5567} = \frac{\mathcal{T}_{5568}}{\langle S_9, \sigma_{1392} \rangle \Rightarrow \langle S_{10}, \sigma_{1392} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5569} = \frac{\langle y := y + 1, \sigma_{1392} \rangle \Rightarrow \sigma_{1393} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1392} \rangle \Rightarrow \langle S_{11}, \sigma_{1393} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5570} = \langle S_{11}, \sigma_{1393} \rangle \Rightarrow \langle S_{12}, \sigma_{1393} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5571} = \langle S_{12}, \sigma_{1393} \rangle \Rightarrow \langle S_{13}, \sigma_{1393} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5573} = \frac{\langle x := x + 1, \sigma_{1393} \rangle \Rightarrow \sigma_{1394} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1393} \rangle \Rightarrow \langle S_6, \sigma_{1394} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5572} = \frac{\mathcal{T}_{5573}}{\langle S_{13}, \sigma_{1393} \rangle \Rightarrow \langle S_{15}, \sigma_{1394} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5574} = \frac{\langle S_6, \sigma_{1394} \rangle \Rightarrow \langle S_8, \sigma_{1394} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1394} \rangle \Rightarrow \langle S_9, \sigma_{1394} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5576} = \langle S_8, \sigma_{1394} \rangle \Rightarrow \langle y := y + 1, \sigma_{1394} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5575} = \frac{\mathcal{T}_{5576}}{\langle S_9, \sigma_{1394} \rangle \Rightarrow \langle S_{10}, \sigma_{1394} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5577} = \frac{\langle y := y + 1, \sigma_{1394} \rangle \Rightarrow \sigma_{1395} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1394} \rangle \Rightarrow \langle S_{11}, \sigma_{1395} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5578} = \langle S_{11}, \sigma_{1395} \rangle \Rightarrow \langle S_{12}, \sigma_{1395} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5579} = \langle S_{12}, \sigma_{1395} \rangle \Rightarrow \langle S_{13}, \sigma_{1395} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5581} = \frac{\langle x := x + 1, \sigma_{1395} \rangle \Rightarrow \sigma_{1396} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1395} \rangle \Rightarrow \langle S_6, \sigma_{1396} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5580} = \frac{\mathcal{T}_{5581}}{\langle S_{13}, \sigma_{1395} \rangle \Rightarrow \langle S_{15}, \sigma_{1396} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5582} = \frac{\langle S_6, \sigma_{1396} \rangle \Rightarrow \langle S_8, \sigma_{1396} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1396} \rangle \Rightarrow \langle S_9, \sigma_{1396} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5584} = \langle S_8, \sigma_{1396} \rangle \Rightarrow \langle y := y + 1, \sigma_{1396} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5583} = \frac{\mathcal{T}_{5584}}{\langle S_9, \sigma_{1396} \rangle \Rightarrow \langle S_{10}, \sigma_{1396} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5585} = \frac{\langle y := y + 1, \sigma_{1396} \rangle \Rightarrow \sigma_{1397} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1396} \rangle \Rightarrow \langle S_{11}, \sigma_{1397} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5586} = \langle S_{11}, \sigma_{1397} \rangle \Rightarrow \langle S_{12}, \sigma_{1397} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5587} = \langle S_{12}, \sigma_{1397} \rangle \Rightarrow \langle S_{13}, \sigma_{1397} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5589} = \frac{\langle x := x + 1, \sigma_{1397} \rangle \Rightarrow \sigma_{1398} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1397} \rangle \Rightarrow \langle S_6, \sigma_{1398} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5588} = \frac{\mathcal{T}_{5589}}{\langle S_{13}, \sigma_{1397} \rangle \Rightarrow \langle S_{15}, \sigma_{1398} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5590} = \frac{\langle S_6, \sigma_{1398} \rangle \Rightarrow \langle S_8, \sigma_{1398} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1398} \rangle \Rightarrow \langle S_9, \sigma_{1398} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5592} = \langle S_8, \sigma_{1398} \rangle \Rightarrow \langle y := y + 1, \sigma_{1398} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5591} = \frac{\mathcal{T}_{5592}}{\langle S_9, \sigma_{1398} \rangle \Rightarrow \langle S_{10}, \sigma_{1398} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5593} = \frac{\langle y := y + 1, \sigma_{1398} \rangle \Rightarrow \sigma_{1399} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1398} \rangle \Rightarrow \langle S_{11}, \sigma_{1399} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5594} = \langle S_{11}, \sigma_{1399} \rangle \Rightarrow \langle S_{12}, \sigma_{1399} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5595} = \langle S_{12}, \sigma_{1399} \rangle \Rightarrow \langle S_{13}, \sigma_{1399} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5597} = \frac{\langle x := x + 1, \sigma_{1399} \rangle \Rightarrow \sigma_{1400} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1399} \rangle \Rightarrow \langle S_6, \sigma_{1400} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5596} = \frac{\mathcal{T}_{5597}}{\langle S_{13}, \sigma_{1399} \rangle \Rightarrow \langle S_{15}, \sigma_{1400} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5598} = \frac{\langle S_6, \sigma_{1400} \rangle \Rightarrow \langle S_8, \sigma_{1400} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1400} \rangle \Rightarrow \langle S_9, \sigma_{1400} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5600} = \langle S_8, \sigma_{1400} \rangle \Rightarrow \langle y := y + 1, \sigma_{1400} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5599} = \frac{\mathcal{T}_{5600}}{\langle S_9, \sigma_{1400} \rangle \Rightarrow \langle S_{10}, \sigma_{1400} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5601} = \frac{\langle y := y + 1, \sigma_{1400} \rangle \Rightarrow \sigma_{1401} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1400} \rangle \Rightarrow \langle S_{11}, \sigma_{1401} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5602} = \langle S_{11}, \sigma_{1401} \rangle \Rightarrow \langle S_{12}, \sigma_{1401} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5603} = \langle S_{12}, \sigma_{1401} \rangle \Rightarrow \langle S_{13}, \sigma_{1401} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5605} = \frac{\langle x := x + 1, \sigma_{1401} \rangle \Rightarrow \sigma_{1402} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1401} \rangle \Rightarrow \langle S_6, \sigma_{1402} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5604} = \frac{\mathcal{T}_{5605}}{\langle S_{13}, \sigma_{1401} \rangle \Rightarrow \langle S_{15}, \sigma_{1402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5606} = \frac{\langle S_6, \sigma_{1402} \rangle \Rightarrow \langle S_8, \sigma_{1402} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1402} \rangle \Rightarrow \langle S_9, \sigma_{1402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5608} = \langle S_8, \sigma_{1402} \rangle \Rightarrow \langle y := y + 1, \sigma_{1402} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5607} = \frac{\mathcal{T}_{5608}}{\langle S_9, \sigma_{1402} \rangle \Rightarrow \langle S_{10}, \sigma_{1402} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5609} = \frac{\langle y := y + 1, \sigma_{1402} \rangle \Rightarrow \sigma_{1403} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1402} \rangle \Rightarrow \langle S_{11}, \sigma_{1403} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5610} = \langle S_{11}, \sigma_{1403} \rangle \Rightarrow \langle S_{12}, \sigma_{1403} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5611} = \langle S_{12}, \sigma_{1403} \rangle \Rightarrow \langle S_{13}, \sigma_{1403} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5613} = \frac{\langle x := x + 1, \sigma_{1403} \rangle \Rightarrow \sigma_{1404} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1403} \rangle \Rightarrow \langle S_6, \sigma_{1404} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5612} = \frac{\mathcal{T}_{5613}}{\langle S_{13}, \sigma_{1403} \rangle \Rightarrow \langle S_{15}, \sigma_{1404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5614} = \frac{\langle S_6, \sigma_{1404} \rangle \Rightarrow \langle S_8, \sigma_{1404} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1404} \rangle \Rightarrow \langle S_9, \sigma_{1404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5616} = \langle S_8, \sigma_{1404} \rangle \Rightarrow \langle y := y + 1, \sigma_{1404} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5615} = \frac{\mathcal{T}_{5616}}{\langle S_9, \sigma_{1404} \rangle \Rightarrow \langle S_{10}, \sigma_{1404} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5617} = \frac{\langle y := y + 1, \sigma_{1404} \rangle \Rightarrow \sigma_{1405} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1404} \rangle \Rightarrow \langle S_{11}, \sigma_{1405} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5618} = \langle S_{11}, \sigma_{1405} \rangle \Rightarrow \langle S_{12}, \sigma_{1405} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5619} = \langle S_{12}, \sigma_{1405} \rangle \Rightarrow \langle S_{13}, \sigma_{1405} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5621} = \frac{\langle x := x + 1, \sigma_{1405} \rangle \Rightarrow \sigma_{1406} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1405} \rangle \Rightarrow \langle S_6, \sigma_{1406} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5620} = \frac{\mathcal{T}_{5621}}{\langle S_{13}, \sigma_{1405} \rangle \Rightarrow \langle S_{15}, \sigma_{1406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5622} = \frac{\langle S_6, \sigma_{1406} \rangle \Rightarrow \langle S_8, \sigma_{1406} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1406} \rangle \Rightarrow \langle S_9, \sigma_{1406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5624} = \langle S_8, \sigma_{1406} \rangle \Rightarrow \langle y := y + 1, \sigma_{1406} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5623} = \frac{\mathcal{T}_{5624}}{\langle S_9, \sigma_{1406} \rangle \Rightarrow \langle S_{10}, \sigma_{1406} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5625} = \frac{\langle y := y + 1, \sigma_{1406} \rangle \Rightarrow \sigma_{1407} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1406} \rangle \Rightarrow \langle S_{11}, \sigma_{1407} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5626} = \langle S_{11}, \sigma_{1407} \rangle \Rightarrow \langle S_{12}, \sigma_{1407} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5627} = \langle S_{12}, \sigma_{1407} \rangle \Rightarrow \langle S_{13}, \sigma_{1407} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5629} = \frac{\langle x := x + 1, \sigma_{1407} \rangle \Rightarrow \sigma_{1408} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1407} \rangle \Rightarrow \langle S_6, \sigma_{1408} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5628} = \frac{\mathcal{T}_{5629}}{\langle S_{13}, \sigma_{1407} \rangle \Rightarrow \langle S_{15}, \sigma_{1408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5630} = \frac{\langle S_6, \sigma_{1408} \rangle \Rightarrow \langle S_8, \sigma_{1408} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1408} \rangle \Rightarrow \langle S_9, \sigma_{1408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5632} = \langle S_8, \sigma_{1408} \rangle \Rightarrow \langle y := y + 1, \sigma_{1408} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5631} = \frac{\mathcal{T}_{5632}}{\langle S_9, \sigma_{1408} \rangle \Rightarrow \langle S_{10}, \sigma_{1408} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5633} = \frac{\langle y := y + 1, \sigma_{1408} \rangle \Rightarrow \sigma_{1409} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1408} \rangle \Rightarrow \langle S_{11}, \sigma_{1409} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5634} = \langle S_{11}, \sigma_{1409} \rangle \Rightarrow \langle S_{12}, \sigma_{1409} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5635} = \langle S_{12}, \sigma_{1409} \rangle \Rightarrow \langle S_{13}, \sigma_{1409} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5637} = \frac{\langle x := x + 1, \sigma_{1409} \rangle \Rightarrow \sigma_{1410} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1409} \rangle \Rightarrow \langle S_6, \sigma_{1410} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5636} = \frac{\mathcal{T}_{5637}}{\langle S_{13}, \sigma_{1409} \rangle \Rightarrow \langle S_{15}, \sigma_{1410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5638} = \frac{\langle S_6, \sigma_{1410} \rangle \Rightarrow \langle S_8, \sigma_{1410} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1410} \rangle \Rightarrow \langle S_9, \sigma_{1410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5640} = \langle S_8, \sigma_{1410} \rangle \Rightarrow \langle y := y + 1, \sigma_{1410} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5639} = \frac{\mathcal{T}_{5640}}{\langle S_9, \sigma_{1410} \rangle \Rightarrow \langle S_{10}, \sigma_{1410} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5641} = \frac{\langle y := y + 1, \sigma_{1410} \rangle \Rightarrow \sigma_{1411} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1410} \rangle \Rightarrow \langle S_{11}, \sigma_{1411} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5642} = \langle S_{11}, \sigma_{1411} \rangle \Rightarrow \langle S_{12}, \sigma_{1411} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5643} = \langle S_{12}, \sigma_{1411} \rangle \Rightarrow \langle S_{13}, \sigma_{1411} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5645} = \frac{\langle x := x + 1, \sigma_{1411} \rangle \Rightarrow \sigma_{1412} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1411} \rangle \Rightarrow \langle S_6, \sigma_{1412} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5644} = \frac{\mathcal{T}_{5645}}{\langle S_{13}, \sigma_{1411} \rangle \Rightarrow \langle S_{15}, \sigma_{1412} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5646} = \frac{\langle S_6, \sigma_{1412} \rangle \Rightarrow \langle S_8, \sigma_{1412} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1412} \rangle \Rightarrow \langle S_9, \sigma_{1412} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5648} = \langle S_8, \sigma_{1412} \rangle \Rightarrow \langle y := y + 1, \sigma_{1412} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5647} = \frac{\mathcal{T}_{5648}}{\langle S_9, \sigma_{1412} \rangle \Rightarrow \langle S_{10}, \sigma_{1412} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5649} = \frac{\langle y := y + 1, \sigma_{1412} \rangle \Rightarrow \sigma_{1413} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1412} \rangle \Rightarrow \langle S_{11}, \sigma_{1413} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5650} = \langle S_{11}, \sigma_{1413} \rangle \Rightarrow \langle S_{12}, \sigma_{1413} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5651} = \langle S_{12}, \sigma_{1413} \rangle \Rightarrow \langle S_{13}, \sigma_{1413} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5653} = \frac{\langle x := x + 1, \sigma_{1413} \rangle \Rightarrow \sigma_{1414} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1413} \rangle \Rightarrow \langle S_6, \sigma_{1414} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5652} = \frac{\mathcal{T}_{5653}}{\langle S_{13}, \sigma_{1413} \rangle \Rightarrow \langle S_{15}, \sigma_{1414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5654} = \frac{\langle S_6, \sigma_{1414} \rangle \Rightarrow \langle S_8, \sigma_{1414} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1414} \rangle \Rightarrow \langle S_9, \sigma_{1414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5656} = \langle S_8, \sigma_{1414} \rangle \Rightarrow \langle y := y + 1, \sigma_{1414} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5655} = \frac{\mathcal{T}_{5656}}{\langle S_9, \sigma_{1414} \rangle \Rightarrow \langle S_{10}, \sigma_{1414} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5657} = \frac{\langle y := y + 1, \sigma_{1414} \rangle \Rightarrow \sigma_{1415} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1414} \rangle \Rightarrow \langle S_{11}, \sigma_{1415} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5658} = \langle S_{11}, \sigma_{1415} \rangle \Rightarrow \langle S_{12}, \sigma_{1415} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5659} = \langle S_{12}, \sigma_{1415} \rangle \Rightarrow \langle S_{13}, \sigma_{1415} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5661} = \frac{\langle x := x + 1, \sigma_{1415} \rangle \Rightarrow \sigma_{1416} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1415} \rangle \Rightarrow \langle S_6, \sigma_{1416} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5660} = \frac{\mathcal{T}_{5661}}{\langle S_{13}, \sigma_{1415} \rangle \Rightarrow \langle S_{15}, \sigma_{1416} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5662} = \frac{\langle S_6, \sigma_{1416} \rangle \Rightarrow \langle S_8, \sigma_{1416} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1416} \rangle \Rightarrow \langle S_9, \sigma_{1416} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5664} = \langle S_8, \sigma_{1416} \rangle \Rightarrow \langle y := y + 1, \sigma_{1416} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5663} = \frac{\mathcal{T}_{5664}}{\langle S_9, \sigma_{1416} \rangle \Rightarrow \langle S_{10}, \sigma_{1416} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5665} = \frac{\langle y := y + 1, \sigma_{1416} \rangle \Rightarrow \sigma_{1417} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1416} \rangle \Rightarrow \langle S_{11}, \sigma_{1417} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5666} = \langle S_{11}, \sigma_{1417} \rangle \Rightarrow \langle S_{12}, \sigma_{1417} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5667} = \langle S_{12}, \sigma_{1417} \rangle \Rightarrow \langle S_{13}, \sigma_{1417} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5669} = \frac{\langle x := x + 1, \sigma_{1417} \rangle \Rightarrow \sigma_{1418} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1417} \rangle \Rightarrow \langle S_6, \sigma_{1418} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5668} = \frac{\mathcal{T}_{5669}}{\langle S_{13}, \sigma_{1417} \rangle \Rightarrow \langle S_{15}, \sigma_{1418} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5670} = \frac{\langle S_6, \sigma_{1418} \rangle \Rightarrow \langle S_8, \sigma_{1418} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1418} \rangle \Rightarrow \langle S_9, \sigma_{1418} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5672} = \langle S_8, \sigma_{1418} \rangle \Rightarrow \langle y := y + 1, \sigma_{1418} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5671} = \frac{\mathcal{T}_{5672}}{\langle S_9, \sigma_{1418} \rangle \Rightarrow \langle S_{10}, \sigma_{1418} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5673} = \frac{\langle y := y + 1, \sigma_{1418} \rangle \Rightarrow \sigma_{1419} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1418} \rangle \Rightarrow \langle S_{11}, \sigma_{1419} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5674} = \langle S_{11}, \sigma_{1419} \rangle \Rightarrow \langle S_{12}, \sigma_{1419} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5675} = \langle S_{12}, \sigma_{1419} \rangle \Rightarrow \langle S_{13}, \sigma_{1419} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5677} = \frac{\langle x := x + 1, \sigma_{1419} \rangle \Rightarrow \sigma_{1420} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1419} \rangle \Rightarrow \langle S_6, \sigma_{1420} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5676} = \frac{\mathcal{T}_{5677}}{\langle S_{13}, \sigma_{1419} \rangle \Rightarrow \langle S_{15}, \sigma_{1420} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5678} = \frac{\langle S_6, \sigma_{1420} \rangle \Rightarrow \langle S_8, \sigma_{1420} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1420} \rangle \Rightarrow \langle S_9, \sigma_{1420} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5680} = \langle S_8, \sigma_{1420} \rangle \Rightarrow \langle y := y + 1, \sigma_{1420} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5679} = \frac{\mathcal{T}_{5680}}{\langle S_9, \sigma_{1420} \rangle \Rightarrow \langle S_{10}, \sigma_{1420} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5681} = \frac{\langle y := y + 1, \sigma_{1420} \rangle \Rightarrow \sigma_{1421} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1420} \rangle \Rightarrow \langle S_{11}, \sigma_{1421} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5682} = \langle S_{11}, \sigma_{1421} \rangle \Rightarrow \langle S_{12}, \sigma_{1421} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5683} = \langle S_{12}, \sigma_{1421} \rangle \Rightarrow \langle S_{13}, \sigma_{1421} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5685} = \frac{\langle x := x + 1, \sigma_{1421} \rangle \Rightarrow \sigma_{1422} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1421} \rangle \Rightarrow \langle S_6, \sigma_{1422} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5684} = \frac{\mathcal{T}_{5685}}{\langle S_{13}, \sigma_{1421} \rangle \Rightarrow \langle S_{15}, \sigma_{1422} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5686} = \frac{\langle S_6, \sigma_{1422} \rangle \Rightarrow \langle S_8, \sigma_{1422} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1422} \rangle \Rightarrow \langle S_9, \sigma_{1422} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5688} = \langle S_8, \sigma_{1422} \rangle \Rightarrow \langle y := y + 1, \sigma_{1422} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5687} = \frac{\mathcal{T}_{5688}}{\langle S_9, \sigma_{1422} \rangle \Rightarrow \langle S_{10}, \sigma_{1422} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5689} = \frac{\langle y := y + 1, \sigma_{1422} \rangle \Rightarrow \sigma_{1423} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1422} \rangle \Rightarrow \langle S_{11}, \sigma_{1423} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5690} = \langle S_{11}, \sigma_{1423} \rangle \Rightarrow \langle S_{12}, \sigma_{1423} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5691} = \langle S_{12}, \sigma_{1423} \rangle \Rightarrow \langle S_{13}, \sigma_{1423} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5693} = \frac{\langle x := x + 1, \sigma_{1423} \rangle \Rightarrow \sigma_{1424} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1423} \rangle \Rightarrow \langle S_6, \sigma_{1424} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5692} = \frac{\mathcal{T}_{5693}}{\langle S_{13}, \sigma_{1423} \rangle \Rightarrow \langle S_{15}, \sigma_{1424} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5694} = \frac{\langle S_6, \sigma_{1424} \rangle \Rightarrow \langle S_8, \sigma_{1424} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1424} \rangle \Rightarrow \langle S_9, \sigma_{1424} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5696} = \langle S_8, \sigma_{1424} \rangle \Rightarrow \langle y := y + 1, \sigma_{1424} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5695} = \frac{\mathcal{T}_{5696}}{\langle S_9, \sigma_{1424} \rangle \Rightarrow \langle S_{10}, \sigma_{1424} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5697} = \frac{\langle y := y + 1, \sigma_{1424} \rangle \Rightarrow \sigma_{1425} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1424} \rangle \Rightarrow \langle S_{11}, \sigma_{1425} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5698} = \langle S_{11}, \sigma_{1425} \rangle \Rightarrow \langle S_{12}, \sigma_{1425} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5699} = \langle S_{12}, \sigma_{1425} \rangle \Rightarrow \langle S_{13}, \sigma_{1425} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5701} = \frac{\langle x := x + 1, \sigma_{1425} \rangle \Rightarrow \sigma_{1426} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1425} \rangle \Rightarrow \langle S_6, \sigma_{1426} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5700} = \frac{\mathcal{T}_{5701}}{\langle S_{13}, \sigma_{1425} \rangle \Rightarrow \langle S_{15}, \sigma_{1426} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5702} = \frac{\langle S_6, \sigma_{1426} \rangle \Rightarrow \langle S_8, \sigma_{1426} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1426} \rangle \Rightarrow \langle S_9, \sigma_{1426} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5704} = \langle S_8, \sigma_{1426} \rangle \Rightarrow \langle y := y + 1, \sigma_{1426} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5703} = \frac{\mathcal{T}_{5704}}{\langle S_9, \sigma_{1426} \rangle \Rightarrow \langle S_{10}, \sigma_{1426} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5705} = \frac{\langle y := y + 1, \sigma_{1426} \rangle \Rightarrow \sigma_{1427} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1426} \rangle \Rightarrow \langle S_{11}, \sigma_{1427} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5706} = \langle S_{11}, \sigma_{1427} \rangle \Rightarrow \langle S_{12}, \sigma_{1427} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5707} = \langle S_{12}, \sigma_{1427} \rangle \Rightarrow \langle S_{13}, \sigma_{1427} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5709} = \frac{\langle x := x + 1, \sigma_{1427} \rangle \Rightarrow \sigma_{1428} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1427} \rangle \Rightarrow \langle S_6, \sigma_{1428} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5708} = \frac{\mathcal{T}_{5709}}{\langle S_{13}, \sigma_{1427} \rangle \Rightarrow \langle S_{15}, \sigma_{1428} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5710} = \frac{\langle S_6, \sigma_{1428} \rangle \Rightarrow \langle S_8, \sigma_{1428} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1428} \rangle \Rightarrow \langle S_9, \sigma_{1428} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5712} = \langle S_8, \sigma_{1428} \rangle \Rightarrow \langle y := y + 1, \sigma_{1428} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5711} = \frac{\mathcal{T}_{5712}}{\langle S_9, \sigma_{1428} \rangle \Rightarrow \langle S_{10}, \sigma_{1428} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5713} = \frac{\langle y := y + 1, \sigma_{1428} \rangle \Rightarrow \sigma_{1429} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1428} \rangle \Rightarrow \langle S_{11}, \sigma_{1429} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5714} = \langle S_{11}, \sigma_{1429} \rangle \Rightarrow \langle S_{12}, \sigma_{1429} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5715} = \langle S_{12}, \sigma_{1429} \rangle \Rightarrow \langle S_{13}, \sigma_{1429} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5717} = \frac{\langle x := x + 1, \sigma_{1429} \rangle \Rightarrow \sigma_{1430} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1429} \rangle \Rightarrow \langle S_6, \sigma_{1430} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5716} = \frac{\mathcal{T}_{5717}}{\langle S_{13}, \sigma_{1429} \rangle \Rightarrow \langle S_{15}, \sigma_{1430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5718} = \frac{\langle S_6, \sigma_{1430} \rangle \Rightarrow \langle S_8, \sigma_{1430} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1430} \rangle \Rightarrow \langle S_9, \sigma_{1430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5720} = \langle S_8, \sigma_{1430} \rangle \Rightarrow \langle y := y + 1, \sigma_{1430} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5719} = \frac{\mathcal{T}_{5720}}{\langle S_9, \sigma_{1430} \rangle \Rightarrow \langle S_{10}, \sigma_{1430} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5721} = \frac{\langle y := y + 1, \sigma_{1430} \rangle \Rightarrow \sigma_{1431} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1430} \rangle \Rightarrow \langle S_{11}, \sigma_{1431} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5722} = \langle S_{11}, \sigma_{1431} \rangle \Rightarrow \langle S_{12}, \sigma_{1431} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5723} = \langle S_{12}, \sigma_{1431} \rangle \Rightarrow \langle S_{13}, \sigma_{1431} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5725} = \frac{\langle x := x + 1, \sigma_{1431} \rangle \Rightarrow \sigma_{1432} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1431} \rangle \Rightarrow \langle S_6, \sigma_{1432} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5724} = \frac{\mathcal{T}_{5725}}{\langle S_{13}, \sigma_{1431} \rangle \Rightarrow \langle S_{15}, \sigma_{1432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5726} = \frac{\langle S_6, \sigma_{1432} \rangle \Rightarrow \langle S_8, \sigma_{1432} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1432} \rangle \Rightarrow \langle S_9, \sigma_{1432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5728} = \langle S_8, \sigma_{1432} \rangle \Rightarrow \langle y := y + 1, \sigma_{1432} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5727} = \frac{\mathcal{T}_{5728}}{\langle S_9, \sigma_{1432} \rangle \Rightarrow \langle S_{10}, \sigma_{1432} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5729} = \frac{\langle y := y + 1, \sigma_{1432} \rangle \Rightarrow \sigma_{1433} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1432} \rangle \Rightarrow \langle S_{11}, \sigma_{1433} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5730} = \langle S_{11}, \sigma_{1433} \rangle \Rightarrow \langle S_{12}, \sigma_{1433} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5731} = \langle S_{12}, \sigma_{1433} \rangle \Rightarrow \langle S_{13}, \sigma_{1433} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5733} = \frac{\langle x := x + 1, \sigma_{1433} \rangle \Rightarrow \sigma_{1434} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1433} \rangle \Rightarrow \langle S_6, \sigma_{1434} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5732} = \frac{\mathcal{T}_{5733}}{\langle S_{13}, \sigma_{1433} \rangle \Rightarrow \langle S_{15}, \sigma_{1434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5734} = \frac{\langle S_6, \sigma_{1434} \rangle \Rightarrow \langle S_8, \sigma_{1434} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1434} \rangle \Rightarrow \langle S_9, \sigma_{1434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5736} = \langle S_8, \sigma_{1434} \rangle \Rightarrow \langle y := y + 1, \sigma_{1434} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5735} = \frac{\mathcal{T}_{5736}}{\langle S_9, \sigma_{1434} \rangle \Rightarrow \langle S_{10}, \sigma_{1434} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5737} = \frac{\langle y := y + 1, \sigma_{1434} \rangle \Rightarrow \sigma_{1435} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1434} \rangle \Rightarrow \langle S_{11}, \sigma_{1435} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5738} = \langle S_{11}, \sigma_{1435} \rangle \Rightarrow \langle S_{12}, \sigma_{1435} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5739} = \langle S_{12}, \sigma_{1435} \rangle \Rightarrow \langle S_{13}, \sigma_{1435} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5741} = \frac{\langle x := x + 1, \sigma_{1435} \rangle \Rightarrow \sigma_{1436} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1435} \rangle \Rightarrow \langle S_6, \sigma_{1436} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5740} = \frac{\mathcal{T}_{5741}}{\langle S_{13}, \sigma_{1435} \rangle \Rightarrow \langle S_{15}, \sigma_{1436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5742} = \frac{\langle S_6, \sigma_{1436} \rangle \Rightarrow \langle S_8, \sigma_{1436} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1436} \rangle \Rightarrow \langle S_9, \sigma_{1436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5744} = \langle S_8, \sigma_{1436} \rangle \Rightarrow \langle y := y + 1, \sigma_{1436} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5743} = \frac{\mathcal{T}_{5744}}{\langle S_9, \sigma_{1436} \rangle \Rightarrow \langle S_{10}, \sigma_{1436} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5745} = \frac{\langle y := y + 1, \sigma_{1436} \rangle \Rightarrow \sigma_{1437} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1436} \rangle \Rightarrow \langle S_{11}, \sigma_{1437} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5746} = \langle S_{11}, \sigma_{1437} \rangle \Rightarrow \langle S_{12}, \sigma_{1437} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5747} = \langle S_{12}, \sigma_{1437} \rangle \Rightarrow \langle S_{13}, \sigma_{1437} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5749} = \frac{\langle x := x + 1, \sigma_{1437} \rangle \Rightarrow \sigma_{1438} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1437} \rangle \Rightarrow \langle S_6, \sigma_{1438} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5748} = \frac{\mathcal{T}_{5749}}{\langle S_{13}, \sigma_{1437} \rangle \Rightarrow \langle S_{15}, \sigma_{1438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5750} = \frac{\langle S_6, \sigma_{1438} \rangle \Rightarrow \langle S_8, \sigma_{1438} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1438} \rangle \Rightarrow \langle S_9, \sigma_{1438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5752} = \langle S_8, \sigma_{1438} \rangle \Rightarrow \langle y := y + 1, \sigma_{1438} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5751} = \frac{\mathcal{T}_{5752}}{\langle S_9, \sigma_{1438} \rangle \Rightarrow \langle S_{10}, \sigma_{1438} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5753} = \frac{\langle y := y + 1, \sigma_{1438} \rangle \Rightarrow \sigma_{1439} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1438} \rangle \Rightarrow \langle S_{11}, \sigma_{1439} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5754} = \langle S_{11}, \sigma_{1439} \rangle \Rightarrow \langle S_{12}, \sigma_{1439} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5755} = \langle S_{12}, \sigma_{1439} \rangle \Rightarrow \langle S_{13}, \sigma_{1439} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5757} = \frac{\langle x := x + 1, \sigma_{1439} \rangle \Rightarrow \sigma_{1440} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1439} \rangle \Rightarrow \langle S_6, \sigma_{1440} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5756} = \frac{\mathcal{T}_{5757}}{\langle S_{13}, \sigma_{1439} \rangle \Rightarrow \langle S_{15}, \sigma_{1440} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5758} = \frac{\langle S_6, \sigma_{1440} \rangle \Rightarrow \langle S_8, \sigma_{1440} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1440} \rangle \Rightarrow \langle S_9, \sigma_{1440} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5760} = \langle S_8, \sigma_{1440} \rangle \Rightarrow \langle y := y + 1, \sigma_{1440} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5759} = \frac{\mathcal{T}_{5760}}{\langle S_9, \sigma_{1440} \rangle \Rightarrow \langle S_{10}, \sigma_{1440} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5761} = \frac{\langle y := y + 1, \sigma_{1440} \rangle \Rightarrow \sigma_{1441} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1440} \rangle \Rightarrow \langle S_{11}, \sigma_{1441} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5762} = \langle S_{11}, \sigma_{1441} \rangle \Rightarrow \langle S_{12}, \sigma_{1441} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5763} = \langle S_{12}, \sigma_{1441} \rangle \Rightarrow \langle S_{13}, \sigma_{1441} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5765} = \frac{\langle x := x + 1, \sigma_{1441} \rangle \Rightarrow \sigma_{1442} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1441} \rangle \Rightarrow \langle S_6, \sigma_{1442} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5764} = \frac{\mathcal{T}_{5765}}{\langle S_{13}, \sigma_{1441} \rangle \Rightarrow \langle S_{15}, \sigma_{1442} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5766} = \frac{\langle S_6, \sigma_{1442} \rangle \Rightarrow \langle S_8, \sigma_{1442} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1442} \rangle \Rightarrow \langle S_9, \sigma_{1442} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5768} = \langle S_8, \sigma_{1442} \rangle \Rightarrow \langle y := y + 1, \sigma_{1442} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5767} = \frac{\mathcal{T}_{5768}}{\langle S_9, \sigma_{1442} \rangle \Rightarrow \langle S_{10}, \sigma_{1442} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5769} = \frac{\langle y := y + 1, \sigma_{1442} \rangle \Rightarrow \sigma_{1443} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1442} \rangle \Rightarrow \langle S_{11}, \sigma_{1443} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5770} = \langle S_{11}, \sigma_{1443} \rangle \Rightarrow \langle S_{12}, \sigma_{1443} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5771} = \langle S_{12}, \sigma_{1443} \rangle \Rightarrow \langle S_{13}, \sigma_{1443} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5773} = \frac{\langle x := x + 1, \sigma_{1443} \rangle \Rightarrow \sigma_{1444} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1443} \rangle \Rightarrow \langle S_6, \sigma_{1444} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5772} = \frac{\mathcal{T}_{5773}}{\langle S_{13}, \sigma_{1443} \rangle \Rightarrow \langle S_{15}, \sigma_{1444} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5774} = \frac{\langle S_6, \sigma_{1444} \rangle \Rightarrow \langle S_8, \sigma_{1444} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1444} \rangle \Rightarrow \langle S_9, \sigma_{1444} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5776} = \langle S_8, \sigma_{1444} \rangle \Rightarrow \langle y := y + 1, \sigma_{1444} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5775} = \frac{\mathcal{T}_{5776}}{\langle S_9, \sigma_{1444} \rangle \Rightarrow \langle S_{10}, \sigma_{1444} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5777} = \frac{\langle y := y + 1, \sigma_{1444} \rangle \Rightarrow \sigma_{1445} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1444} \rangle \Rightarrow \langle S_{11}, \sigma_{1445} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5778} = \langle S_{11}, \sigma_{1445} \rangle \Rightarrow \langle S_{12}, \sigma_{1445} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5779} = \langle S_{12}, \sigma_{1445} \rangle \Rightarrow \langle S_{13}, \sigma_{1445} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5781} = \frac{\langle x := x + 1, \sigma_{1445} \rangle \Rightarrow \sigma_{1446} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1445} \rangle \Rightarrow \langle S_6, \sigma_{1446} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5780} = \frac{\mathcal{T}_{5781}}{\langle S_{13}, \sigma_{1445} \rangle \Rightarrow \langle S_{15}, \sigma_{1446} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5782} = \frac{\langle S_6, \sigma_{1446} \rangle \Rightarrow \langle S_8, \sigma_{1446} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1446} \rangle \Rightarrow \langle S_9, \sigma_{1446} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5784} = \langle S_8, \sigma_{1446} \rangle \Rightarrow \langle y := y + 1, \sigma_{1446} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5783} = \frac{\mathcal{T}_{5784}}{\langle S_9, \sigma_{1446} \rangle \Rightarrow \langle S_{10}, \sigma_{1446} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5785} = \frac{\langle y := y + 1, \sigma_{1446} \rangle \Rightarrow \sigma_{1447} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1446} \rangle \Rightarrow \langle S_{11}, \sigma_{1447} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5786} = \langle S_{11}, \sigma_{1447} \rangle \Rightarrow \langle S_{12}, \sigma_{1447} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5787} = \langle S_{12}, \sigma_{1447} \rangle \Rightarrow \langle S_{13}, \sigma_{1447} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5789} = \frac{\langle x := x + 1, \sigma_{1447} \rangle \Rightarrow \sigma_{1448} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1447} \rangle \Rightarrow \langle S_6, \sigma_{1448} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5788} = \frac{\mathcal{T}_{5789}}{\langle S_{13}, \sigma_{1447} \rangle \Rightarrow \langle S_{15}, \sigma_{1448} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5790} = \frac{\langle S_6, \sigma_{1448} \rangle \Rightarrow \langle S_8, \sigma_{1448} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1448} \rangle \Rightarrow \langle S_9, \sigma_{1448} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5792} = \langle S_8, \sigma_{1448} \rangle \Rightarrow \langle y := y + 1, \sigma_{1448} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5791} = \frac{\mathcal{T}_{5792}}{\langle S_9, \sigma_{1448} \rangle \Rightarrow \langle S_{10}, \sigma_{1448} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5793} = \frac{\langle y := y + 1, \sigma_{1448} \rangle \Rightarrow \sigma_{1449} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1448} \rangle \Rightarrow \langle S_{11}, \sigma_{1449} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5794} = \langle S_{11}, \sigma_{1449} \rangle \Rightarrow \langle S_{12}, \sigma_{1449} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5795} = \langle S_{12}, \sigma_{1449} \rangle \Rightarrow \langle S_{13}, \sigma_{1449} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5797} = \frac{\langle x := x + 1, \sigma_{1449} \rangle \Rightarrow \sigma_{1450} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1449} \rangle \Rightarrow \langle S_6, \sigma_{1450} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5796} = \frac{\mathcal{T}_{5797}}{\langle S_{13}, \sigma_{1449} \rangle \Rightarrow \langle S_{15}, \sigma_{1450} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5798} = \frac{\langle S_6, \sigma_{1450} \rangle \Rightarrow \langle S_8, \sigma_{1450} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1450} \rangle \Rightarrow \langle S_9, \sigma_{1450} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5800} = \langle S_8, \sigma_{1450} \rangle \Rightarrow \langle y := y + 1, \sigma_{1450} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5799} = \frac{\mathcal{T}_{5800}}{\langle S_9, \sigma_{1450} \rangle \Rightarrow \langle S_{10}, \sigma_{1450} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5801} = \frac{\langle y := y + 1, \sigma_{1450} \rangle \Rightarrow \sigma_{1451} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1450} \rangle \Rightarrow \langle S_{11}, \sigma_{1451} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5802} = \langle S_{11}, \sigma_{1451} \rangle \Rightarrow \langle S_{12}, \sigma_{1451} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{5803} = \langle S_{12}, \sigma_{1451} \rangle \Rightarrow \langle S_{13}, \sigma_{1451} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{5805} = \frac{\langle x := x + 1, \sigma_{1451} \rangle \Rightarrow \sigma_{1452} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1451} \rangle \Rightarrow \langle S_6, \sigma_{1452} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5804} = \frac{\mathcal{T}_{5805}}{\langle S_{13}, \sigma_{1451} \rangle \Rightarrow \langle S_{15}, \sigma_{1452} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{5806} = \frac{\langle S_6, \sigma_{1452} \rangle \Rightarrow \langle S_8, \sigma_{1452} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1452} \rangle \Rightarrow \langle S_9, \sigma_{1452} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{5808} = \langle S_8, \sigma_{1452} \rangle \Rightarrow \langle y := y + 1, \sigma_{1452} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5807} = \frac{\mathcal{T}_{5808}}{\langle S_9, \sigma_{1452} \rangle \Rightarrow \langle S_{10}, \sigma_{1452} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{5809} = \frac{\langle y := y + 1, \sigma_{1452} \rangle \Rightarrow \sigma_{1453} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1452} \rangle \Rightarrow \langle S_{11}, \sigma_{1453} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5810} = \langle S_{11}, \sigma_{1453} \rangle \Rightarrow \langle S_{12}, \sigma_{1453} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5811} = \langle S_{12}, \sigma_{1453} \rangle \Rightarrow \langle S_{13}, \sigma_{1453} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5813} = \frac{\langle x := x + 1, \sigma_{1453} \rangle \Rightarrow \sigma_{1454} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1453} \rangle \Rightarrow \langle S_6, \sigma_{1454} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5812} = \frac{\mathcal{T}_{5813}}{\langle S_{13}, \sigma_{1453} \rangle \Rightarrow \langle S_{15}, \sigma_{1454} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5814} = \frac{\langle S_6, \sigma_{1454} \rangle \Rightarrow \langle S_8, \sigma_{1454} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1454} \rangle \Rightarrow \langle S_9, \sigma_{1454} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5816} = \langle S_8, \sigma_{1454} \rangle \Rightarrow \langle y := y + 1, \sigma_{1454} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{5815} = \frac{\mathcal{T}_{5816}}{\langle S_9, \sigma_{1454} \rangle \Rightarrow \langle S_{10}, \sigma_{1454} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5817} = \frac{\langle y := y + 1, \sigma_{1454} \rangle \Rightarrow \sigma_{1455} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1454} \rangle \Rightarrow \langle S_{11}, \sigma_{1455} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5818} = \langle S_{11}, \sigma_{1455} \rangle \Rightarrow \langle S_{12}, \sigma_{1455} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5819} = \langle S_{12}, \sigma_{1455} \rangle \Rightarrow \langle S_{13}, \sigma_{1455} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5821} = \frac{\langle x := x + 1, \sigma_{1455} \rangle \Rightarrow \sigma_{1456} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1455} \rangle \Rightarrow \langle S_6, \sigma_{1456} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5820} = \frac{\mathcal{T}_{5821}}{\langle S_{13}, \sigma_{1455} \rangle \Rightarrow \langle S_{15}, \sigma_{1456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5822} = \frac{\langle S_6, \sigma_{1456} \rangle \Rightarrow \langle S_8, \sigma_{1456} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1456} \rangle \Rightarrow \langle S_9, \sigma_{1456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5824} = \langle S_8, \sigma_{1456} \rangle \Rightarrow \langle y := y + 1, \sigma_{1456} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5823} = \frac{\mathcal{T}_{5824}}{\langle S_9, \sigma_{1456} \rangle \Rightarrow \langle S_{10}, \sigma_{1456} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5825} = \frac{\langle y := y + 1, \sigma_{1456} \rangle \Rightarrow \sigma_{1457} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1456} \rangle \Rightarrow \langle S_{11}, \sigma_{1457} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5826} = \langle S_{11}, \sigma_{1457} \rangle \Rightarrow \langle S_{12}, \sigma_{1457} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5827} = \langle S_{12}, \sigma_{1457} \rangle \Rightarrow \langle S_{13}, \sigma_{1457} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5829} = \frac{\langle x := x + 1, \sigma_{1457} \rangle \Rightarrow \sigma_{1458} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1457} \rangle \Rightarrow \langle S_6, \sigma_{1458} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5828} = \frac{\mathcal{T}_{5829}}{\langle S_{13}, \sigma_{1457} \rangle \Rightarrow \langle S_{15}, \sigma_{1458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5830} = \frac{\langle S_6, \sigma_{1458} \rangle \Rightarrow \langle S_8, \sigma_{1458} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1458} \rangle \Rightarrow \langle S_9, \sigma_{1458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5832} = \langle S_8, \sigma_{1458} \rangle \Rightarrow \langle y := y + 1, \sigma_{1458} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5831} = \frac{\mathcal{T}_{5832}}{\langle S_9, \sigma_{1458} \rangle \Rightarrow \langle S_{10}, \sigma_{1458} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5833} = \frac{\langle y := y + 1, \sigma_{1458} \rangle \Rightarrow \sigma_{1459} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1458} \rangle \Rightarrow \langle S_{11}, \sigma_{1459} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5834} = \langle S_{11}, \sigma_{1459} \rangle \Rightarrow \langle S_{12}, \sigma_{1459} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5835} = \langle S_{12}, \sigma_{1459} \rangle \Rightarrow \langle S_{13}, \sigma_{1459} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5837} = \frac{\langle x := x + 1, \sigma_{1459} \rangle \Rightarrow \sigma_{1460} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1459} \rangle \Rightarrow \langle S_6, \sigma_{1460} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5836} = \frac{\mathcal{T}_{5837}}{\langle S_{13}, \sigma_{1459} \rangle \Rightarrow \langle S_{15}, \sigma_{1460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5838} = \frac{\langle S_6, \sigma_{1460} \rangle \Rightarrow \langle S_8, \sigma_{1460} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1460} \rangle \Rightarrow \langle S_9, \sigma_{1460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5840} = \langle S_8, \sigma_{1460} \rangle \Rightarrow \langle y := y + 1, \sigma_{1460} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5839} = \frac{\mathcal{T}_{5840}}{\langle S_9, \sigma_{1460} \rangle \Rightarrow \langle S_{10}, \sigma_{1460} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5841} = \frac{\langle y := y + 1, \sigma_{1460} \rangle \Rightarrow \sigma_{1461} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1460} \rangle \Rightarrow \langle S_{11}, \sigma_{1461} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5842} = \langle S_{11}, \sigma_{1461} \rangle \Rightarrow \langle S_{12}, \sigma_{1461} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5843} = \langle S_{12}, \sigma_{1461} \rangle \Rightarrow \langle S_{13}, \sigma_{1461} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5845} = \frac{\langle x := x + 1, \sigma_{1461} \rangle \Rightarrow \sigma_{1462} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1461} \rangle \Rightarrow \langle S_6, \sigma_{1462} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5844} = \frac{\mathcal{T}_{5845}}{\langle S_{13}, \sigma_{1461} \rangle \Rightarrow \langle S_{15}, \sigma_{1462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5846} = \frac{\langle S_6, \sigma_{1462} \rangle \Rightarrow \langle S_8, \sigma_{1462} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1462} \rangle \Rightarrow \langle S_9, \sigma_{1462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5848} = \langle S_8, \sigma_{1462} \rangle \Rightarrow \langle y := y + 1, \sigma_{1462} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5847} = \frac{\mathcal{T}_{5848}}{\langle S_9, \sigma_{1462} \rangle \Rightarrow \langle S_{10}, \sigma_{1462} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5849} = \frac{\langle y := y + 1, \sigma_{1462} \rangle \Rightarrow \sigma_{1463} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1462} \rangle \Rightarrow \langle S_{11}, \sigma_{1463} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5850} = \langle S_{11}, \sigma_{1463} \rangle \Rightarrow \langle S_{12}, \sigma_{1463} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5851} = \langle S_{12}, \sigma_{1463} \rangle \Rightarrow \langle S_{13}, \sigma_{1463} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5853} = \frac{\langle x := x + 1, \sigma_{1463} \rangle \Rightarrow \sigma_{1464} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1463} \rangle \Rightarrow \langle S_6, \sigma_{1464} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5852} = \frac{\mathcal{T}_{5853}}{\langle S_{13}, \sigma_{1463} \rangle \Rightarrow \langle S_{15}, \sigma_{1464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5854} = \frac{\langle S_6, \sigma_{1464} \rangle \Rightarrow \langle S_8, \sigma_{1464} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1464} \rangle \Rightarrow \langle S_9, \sigma_{1464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5856} = \langle S_8, \sigma_{1464} \rangle \Rightarrow \langle y := y + 1, \sigma_{1464} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5855} = \frac{\mathcal{T}_{5856}}{\langle S_9, \sigma_{1464} \rangle \Rightarrow \langle S_{10}, \sigma_{1464} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5857} = \frac{\langle y := y + 1, \sigma_{1464} \rangle \Rightarrow \sigma_{1465} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1464} \rangle \Rightarrow \langle S_{11}, \sigma_{1465} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5858} = \langle S_{11}, \sigma_{1465} \rangle \Rightarrow \langle S_{12}, \sigma_{1465} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5859} = \langle S_{12}, \sigma_{1465} \rangle \Rightarrow \langle S_{13}, \sigma_{1465} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5861} = \frac{\langle x := x + 1, \sigma_{1465} \rangle \Rightarrow \sigma_{1466} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1465} \rangle \Rightarrow \langle S_6, \sigma_{1466} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5860} = \frac{\mathcal{T}_{5861}}{\langle S_{13}, \sigma_{1465} \rangle \Rightarrow \langle S_{15}, \sigma_{1466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5862} = \frac{\langle S_6, \sigma_{1466} \rangle \Rightarrow \langle S_8, \sigma_{1466} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1466} \rangle \Rightarrow \langle S_9, \sigma_{1466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5864} = \langle S_8, \sigma_{1466} \rangle \Rightarrow \langle y := y + 1, \sigma_{1466} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5863} = \frac{\mathcal{T}_{5864}}{\langle S_9, \sigma_{1466} \rangle \Rightarrow \langle S_{10}, \sigma_{1466} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5865} = \frac{\langle y := y + 1, \sigma_{1466} \rangle \Rightarrow \sigma_{1467} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1466} \rangle \Rightarrow \langle S_{11}, \sigma_{1467} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5866} = \langle S_{11}, \sigma_{1467} \rangle \Rightarrow \langle S_{12}, \sigma_{1467} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5867} = \langle S_{12}, \sigma_{1467} \rangle \Rightarrow \langle S_{13}, \sigma_{1467} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5869} = \frac{\langle x := x + 1, \sigma_{1467} \rangle \Rightarrow \sigma_{1468} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1467} \rangle \Rightarrow \langle S_6, \sigma_{1468} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5868} = \frac{\mathcal{T}_{5869}}{\langle S_{13}, \sigma_{1467} \rangle \Rightarrow \langle S_{15}, \sigma_{1468} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5870} = \frac{\langle S_6, \sigma_{1468} \rangle \Rightarrow \langle S_8, \sigma_{1468} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1468} \rangle \Rightarrow \langle S_9, \sigma_{1468} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5872} = \langle S_8, \sigma_{1468} \rangle \Rightarrow \langle y := y + 1, \sigma_{1468} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5871} = \frac{\mathcal{T}_{5872}}{\langle S_9, \sigma_{1468} \rangle \Rightarrow \langle S_{10}, \sigma_{1468} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5873} = \frac{\langle y := y + 1, \sigma_{1468} \rangle \Rightarrow \sigma_{1469} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1468} \rangle \Rightarrow \langle S_{11}, \sigma_{1469} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5874} = \langle S_{11}, \sigma_{1469} \rangle \Rightarrow \langle S_{12}, \sigma_{1469} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5875} = \langle S_{12}, \sigma_{1469} \rangle \Rightarrow \langle S_{13}, \sigma_{1469} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5877} = \frac{\langle x := x + 1, \sigma_{1469} \rangle \Rightarrow \sigma_{1470} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1469} \rangle \Rightarrow \langle S_6, \sigma_{1470} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5876} = \frac{\mathcal{T}_{5877}}{\langle S_{13}, \sigma_{1469} \rangle \Rightarrow \langle S_{15}, \sigma_{1470} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5878} = \frac{\langle S_6, \sigma_{1470} \rangle \Rightarrow \langle S_8, \sigma_{1470} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1470} \rangle \Rightarrow \langle S_9, \sigma_{1470} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5880} = \langle S_8, \sigma_{1470} \rangle \Rightarrow \langle y := y + 1, \sigma_{1470} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5879} = \frac{\mathcal{T}_{5880}}{\langle S_9, \sigma_{1470} \rangle \Rightarrow \langle S_{10}, \sigma_{1470} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5881} = \frac{\langle y := y + 1, \sigma_{1470} \rangle \Rightarrow \sigma_{1471} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1470} \rangle \Rightarrow \langle S_{11}, \sigma_{1471} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5882} = \langle S_{11}, \sigma_{1471} \rangle \Rightarrow \langle S_{12}, \sigma_{1471} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5883} = \langle S_{12}, \sigma_{1471} \rangle \Rightarrow \langle S_{13}, \sigma_{1471} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5885} = \frac{\langle x := x + 1, \sigma_{1471} \rangle \Rightarrow \sigma_{1472} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1471} \rangle \Rightarrow \langle S_6, \sigma_{1472} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5884} = \frac{\mathcal{T}_{5885}}{\langle S_{13}, \sigma_{1471} \rangle \Rightarrow \langle S_{15}, \sigma_{1472} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5886} = \frac{\langle S_6, \sigma_{1472} \rangle \Rightarrow \langle S_8, \sigma_{1472} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1472} \rangle \Rightarrow \langle S_9, \sigma_{1472} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5888} = \langle S_8, \sigma_{1472} \rangle \Rightarrow \langle y := y + 1, \sigma_{1472} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5887} = \frac{\mathcal{T}_{5888}}{\langle S_9, \sigma_{1472} \rangle \Rightarrow \langle S_{10}, \sigma_{1472} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5889} = \frac{\langle y := y + 1, \sigma_{1472} \rangle \Rightarrow \sigma_{1473} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1472} \rangle \Rightarrow \langle S_{11}, \sigma_{1473} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5890} = \langle S_{11}, \sigma_{1473} \rangle \Rightarrow \langle S_{12}, \sigma_{1473} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5891} = \langle S_{12}, \sigma_{1473} \rangle \Rightarrow \langle S_{13}, \sigma_{1473} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5893} = \frac{\langle x := x + 1, \sigma_{1473} \rangle \Rightarrow \sigma_{1474} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1473} \rangle \Rightarrow \langle S_6, \sigma_{1474} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5892} = \frac{\mathcal{T}_{5893}}{\langle S_{13}, \sigma_{1473} \rangle \Rightarrow \langle S_{15}, \sigma_{1474} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5894} = \frac{\langle S_6, \sigma_{1474} \rangle \Rightarrow \langle S_8, \sigma_{1474} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1474} \rangle \Rightarrow \langle S_9, \sigma_{1474} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5896} = \langle S_8, \sigma_{1474} \rangle \Rightarrow \langle y := y + 1, \sigma_{1474} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5895} = \frac{\mathcal{T}_{5896}}{\langle S_9, \sigma_{1474} \rangle \Rightarrow \langle S_{10}, \sigma_{1474} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5897} = \frac{\langle y := y + 1, \sigma_{1474} \rangle \Rightarrow \sigma_{1475} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1474} \rangle \Rightarrow \langle S_{11}, \sigma_{1475} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5898} = \langle S_{11}, \sigma_{1475} \rangle \Rightarrow \langle S_{12}, \sigma_{1475} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5899} = \langle S_{12}, \sigma_{1475} \rangle \Rightarrow \langle S_{13}, \sigma_{1475} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5901} = \frac{\langle x := x + 1, \sigma_{1475} \rangle \Rightarrow \sigma_{1476} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1475} \rangle \Rightarrow \langle S_6, \sigma_{1476} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5900} = \frac{\mathcal{T}_{5901}}{\langle S_{13}, \sigma_{1475} \rangle \Rightarrow \langle S_{15}, \sigma_{1476} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5902} = \frac{\langle S_6, \sigma_{1476} \rangle \Rightarrow \langle S_8, \sigma_{1476} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1476} \rangle \Rightarrow \langle S_9, \sigma_{1476} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5904} = \langle S_8, \sigma_{1476} \rangle \Rightarrow \langle y := y + 1, \sigma_{1476} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5903} = \frac{\mathcal{T}_{5904}}{\langle S_9, \sigma_{1476} \rangle \Rightarrow \langle S_{10}, \sigma_{1476} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5905} = \frac{\langle y := y + 1, \sigma_{1476} \rangle \Rightarrow \sigma_{1477} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1476} \rangle \Rightarrow \langle S_{11}, \sigma_{1477} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5906} = \langle S_{11}, \sigma_{1477} \rangle \Rightarrow \langle S_{12}, \sigma_{1477} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5907} = \langle S_{12}, \sigma_{1477} \rangle \Rightarrow \langle S_{13}, \sigma_{1477} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5909} = \frac{\langle x := x + 1, \sigma_{1477} \rangle \Rightarrow \sigma_{1478} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1477} \rangle \Rightarrow \langle S_6, \sigma_{1478} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5908} = \frac{\mathcal{T}_{5909}}{\langle S_{13}, \sigma_{1477} \rangle \Rightarrow \langle S_{15}, \sigma_{1478} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5910} = \frac{\langle S_6, \sigma_{1478} \rangle \Rightarrow \langle S_8, \sigma_{1478} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1478} \rangle \Rightarrow \langle S_9, \sigma_{1478} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5912} = \langle S_8, \sigma_{1478} \rangle \Rightarrow \langle y := y + 1, \sigma_{1478} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5911} = \frac{\mathcal{T}_{5912}}{\langle S_9, \sigma_{1478} \rangle \Rightarrow \langle S_{10}, \sigma_{1478} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5913} = \frac{\langle y := y + 1, \sigma_{1478} \rangle \Rightarrow \sigma_{1479} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1478} \rangle \Rightarrow \langle S_{11}, \sigma_{1479} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5914} = \langle S_{11}, \sigma_{1479} \rangle \Rightarrow \langle S_{12}, \sigma_{1479} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5915} = \langle S_{12}, \sigma_{1479} \rangle \Rightarrow \langle S_{13}, \sigma_{1479} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5917} = \frac{\langle x := x + 1, \sigma_{1479} \rangle \Rightarrow \sigma_{1480} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1479} \rangle \Rightarrow \langle S_6, \sigma_{1480} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5916} = \frac{\mathcal{T}_{5917}}{\langle S_{13}, \sigma_{1479} \rangle \Rightarrow \langle S_{15}, \sigma_{1480} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5918} = \frac{\langle S_6, \sigma_{1480} \rangle \Rightarrow \langle S_8, \sigma_{1480} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1480} \rangle \Rightarrow \langle S_9, \sigma_{1480} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5920} = \langle S_8, \sigma_{1480} \rangle \Rightarrow \langle y := y + 1, \sigma_{1480} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5919} = \frac{\mathcal{T}_{5920}}{\langle S_9, \sigma_{1480} \rangle \Rightarrow \langle S_{10}, \sigma_{1480} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5921} = \frac{\langle y := y + 1, \sigma_{1480} \rangle \Rightarrow \sigma_{1481} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1480} \rangle \Rightarrow \langle S_{11}, \sigma_{1481} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5922} = \langle S_{11}, \sigma_{1481} \rangle \Rightarrow \langle S_{12}, \sigma_{1481} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5923} = \langle S_{12}, \sigma_{1481} \rangle \Rightarrow \langle S_{13}, \sigma_{1481} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5925} = \frac{\langle x := x + 1, \sigma_{1481} \rangle \Rightarrow \sigma_{1482} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1481} \rangle \Rightarrow \langle S_6, \sigma_{1482} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5924} = \frac{\mathcal{T}_{5925}}{\langle S_{13}, \sigma_{1481} \rangle \Rightarrow \langle S_{15}, \sigma_{1482} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5926} = \frac{\langle S_6, \sigma_{1482} \rangle \Rightarrow \langle S_8, \sigma_{1482} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1482} \rangle \Rightarrow \langle S_9, \sigma_{1482} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5928} = \langle S_8, \sigma_{1482} \rangle \Rightarrow \langle y := y + 1, \sigma_{1482} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5927} = \frac{\mathcal{T}_{5928}}{\langle S_9, \sigma_{1482} \rangle \Rightarrow \langle S_{10}, \sigma_{1482} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5929} = \frac{\langle y := y + 1, \sigma_{1482} \rangle \Rightarrow \sigma_{1483} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1482} \rangle \Rightarrow \langle S_{11}, \sigma_{1483} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5930} = \langle S_{11}, \sigma_{1483} \rangle \Rightarrow \langle S_{12}, \sigma_{1483} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5931} = \langle S_{12}, \sigma_{1483} \rangle \Rightarrow \langle S_{13}, \sigma_{1483} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5933} = \frac{\langle x := x + 1, \sigma_{1483} \rangle \Rightarrow \sigma_{1484} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1483} \rangle \Rightarrow \langle S_6, \sigma_{1484} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5932} = \frac{\mathcal{T}_{5933}}{\langle S_{13}, \sigma_{1483} \rangle \Rightarrow \langle S_{15}, \sigma_{1484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5934} = \frac{\langle S_6, \sigma_{1484} \rangle \Rightarrow \langle S_8, \sigma_{1484} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1484} \rangle \Rightarrow \langle S_9, \sigma_{1484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5936} = \langle S_8, \sigma_{1484} \rangle \Rightarrow \langle y := y + 1, \sigma_{1484} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5935} = \frac{\mathcal{T}_{5936}}{\langle S_9, \sigma_{1484} \rangle \Rightarrow \langle S_{10}, \sigma_{1484} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5937} = \frac{\langle y := y + 1, \sigma_{1484} \rangle \Rightarrow \sigma_{1485} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1484} \rangle \Rightarrow \langle S_{11}, \sigma_{1485} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5938} = \langle S_{11}, \sigma_{1485} \rangle \Rightarrow \langle S_{12}, \sigma_{1485} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5939} = \langle S_{12}, \sigma_{1485} \rangle \Rightarrow \langle S_{13}, \sigma_{1485} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5941} = \frac{\langle x := x + 1, \sigma_{1485} \rangle \Rightarrow \sigma_{1486} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1485} \rangle \Rightarrow \langle S_6, \sigma_{1486} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5940} = \frac{\mathcal{T}_{5941}}{\langle S_{13}, \sigma_{1485} \rangle \Rightarrow \langle S_{15}, \sigma_{1486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5942} = \frac{\langle S_6, \sigma_{1486} \rangle \Rightarrow \langle S_8, \sigma_{1486} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1486} \rangle \Rightarrow \langle S_9, \sigma_{1486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5944} = \langle S_8, \sigma_{1486} \rangle \Rightarrow \langle y := y + 1, \sigma_{1486} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5943} = \frac{\mathcal{T}_{5944}}{\langle S_9, \sigma_{1486} \rangle \Rightarrow \langle S_{10}, \sigma_{1486} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5945} = \frac{\langle y := y + 1, \sigma_{1486} \rangle \Rightarrow \sigma_{1487} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1486} \rangle \Rightarrow \langle S_{11}, \sigma_{1487} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5946} = \langle S_{11}, \sigma_{1487} \rangle \Rightarrow \langle S_{12}, \sigma_{1487} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5947} = \langle S_{12}, \sigma_{1487} \rangle \Rightarrow \langle S_{13}, \sigma_{1487} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5949} = \frac{\langle x := x + 1, \sigma_{1487} \rangle \Rightarrow \sigma_{1488} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1487} \rangle \Rightarrow \langle S_6, \sigma_{1488} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5948} = \frac{\mathcal{T}_{5949}}{\langle S_{13}, \sigma_{1487} \rangle \Rightarrow \langle S_{15}, \sigma_{1488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5950} = \frac{\langle S_6, \sigma_{1488} \rangle \Rightarrow \langle S_8, \sigma_{1488} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1488} \rangle \Rightarrow \langle S_9, \sigma_{1488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5952} = \langle S_8, \sigma_{1488} \rangle \Rightarrow \langle y := y + 1, \sigma_{1488} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5951} = \frac{\mathcal{T}_{5952}}{\langle S_9, \sigma_{1488} \rangle \Rightarrow \langle S_{10}, \sigma_{1488} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{5953} = \frac{\langle y := y + 1, \sigma_{1488} \rangle \Rightarrow \sigma_{1489} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1488} \rangle \Rightarrow \langle S_{11}, \sigma_{1489} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{5954} = \langle S_{11}, \sigma_{1489} \rangle \Rightarrow \langle S_{12}, \sigma_{1489} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{5955} = \langle S_{12}, \sigma_{1489} \rangle \Rightarrow \langle S_{13}, \sigma_{1489} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5957} = \frac{\langle x := x + 1, \sigma_{1489} \rangle \Rightarrow \sigma_{1490} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1489} \rangle \Rightarrow \langle S_6, \sigma_{1490} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5956} = \frac{\mathcal{T}_{5957}}{\langle S_{13}, \sigma_{1489} \rangle \Rightarrow \langle S_{15}, \sigma_{1490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5958} = \frac{\langle S_6, \sigma_{1490} \rangle \Rightarrow \langle S_8, \sigma_{1490} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1490} \rangle \Rightarrow \langle S_9, \sigma_{1490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5960} = \langle S_8, \sigma_{1490} \rangle \Rightarrow \langle y := y + 1, \sigma_{1490} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5959} = \frac{\mathcal{T}_{5960}}{\langle S_9, \sigma_{1490} \rangle \Rightarrow \langle S_{10}, \sigma_{1490} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5961} = \frac{\langle y := y + 1, \sigma_{1490} \rangle \Rightarrow \sigma_{1491} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1490} \rangle \Rightarrow \langle S_{11}, \sigma_{1491} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5962} = \langle S_{11}, \sigma_{1491} \rangle \Rightarrow \langle S_{12}, \sigma_{1491} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5963} = \langle S_{12}, \sigma_{1491} \rangle \Rightarrow \langle S_{13}, \sigma_{1491} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5965} = \frac{\langle x := x + 1, \sigma_{1491} \rangle \Rightarrow \sigma_{1492} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1491} \rangle \Rightarrow \langle S_6, \sigma_{1492} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5964} = \frac{\mathcal{T}_{5965}}{\langle S_{13}, \sigma_{1491} \rangle \Rightarrow \langle S_{15}, \sigma_{1492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5966} = \frac{\langle S_6, \sigma_{1492} \rangle \Rightarrow \langle S_8, \sigma_{1492} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1492} \rangle \Rightarrow \langle S_9, \sigma_{1492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5968} = \langle S_8, \sigma_{1492} \rangle \Rightarrow \langle y := y + 1, \sigma_{1492} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5967} = \frac{\mathcal{T}_{5968}}{\langle S_9, \sigma_{1492} \rangle \Rightarrow \langle S_{10}, \sigma_{1492} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5969} = \frac{\langle y := y + 1, \sigma_{1492} \rangle \Rightarrow \sigma_{1493} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1492} \rangle \Rightarrow \langle S_{11}, \sigma_{1493} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5970} = \langle S_{11}, \sigma_{1493} \rangle \Rightarrow \langle S_{12}, \sigma_{1493} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5971} = \langle S_{12}, \sigma_{1493} \rangle \Rightarrow \langle S_{13}, \sigma_{1493} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5973} = \frac{\langle x := x + 1, \sigma_{1493} \rangle \Rightarrow \sigma_{1494} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1493} \rangle \Rightarrow \langle S_6, \sigma_{1494} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5972} = \frac{\mathcal{T}_{5973}}{\langle S_{13}, \sigma_{1493} \rangle \Rightarrow \langle S_{15}, \sigma_{1494} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5974} = \frac{\langle S_6, \sigma_{1494} \rangle \Rightarrow \langle S_8, \sigma_{1494} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1494} \rangle \Rightarrow \langle S_9, \sigma_{1494} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5976} = \langle S_8, \sigma_{1494} \rangle \Rightarrow \langle y := y + 1, \sigma_{1494} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5975} = \frac{\mathcal{T}_{5976}}{\langle S_9, \sigma_{1494} \rangle \Rightarrow \langle S_{10}, \sigma_{1494} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5977} = \frac{\langle y := y + 1, \sigma_{1494} \rangle \Rightarrow \sigma_{1495} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1494} \rangle \Rightarrow \langle S_{11}, \sigma_{1495} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5978} = \langle S_{11}, \sigma_{1495} \rangle \Rightarrow \langle S_{12}, \sigma_{1495} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5979} = \langle S_{12}, \sigma_{1495} \rangle \Rightarrow \langle S_{13}, \sigma_{1495} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5981} = \frac{\langle x := x + 1, \sigma_{1495} \rangle \Rightarrow \sigma_{1496} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1495} \rangle \Rightarrow \langle S_6, \sigma_{1496} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{5980} = \frac{\mathcal{T}_{5981}}{\langle S_{13}, \sigma_{1495} \rangle \Rightarrow \langle S_{15}, \sigma_{1496} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5982} = \frac{\langle S_6, \sigma_{1496} \rangle \Rightarrow \langle S_8, \sigma_{1496} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1496} \rangle \Rightarrow \langle S_9, \sigma_{1496} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5984} = \langle S_8, \sigma_{1496} \rangle \Rightarrow \langle y := y + 1, \sigma_{1496} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{5983} = \frac{\mathcal{T}_{5984}}{\langle S_9, \sigma_{1496} \rangle \Rightarrow \langle S_{10}, \sigma_{1496} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{5985} = \frac{\langle y := y + 1, \sigma_{1496} \rangle \Rightarrow \sigma_{1497} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1496} \rangle \Rightarrow \langle S_{11}, \sigma_{1497} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{5986} = \langle S_{11}, \sigma_{1497} \rangle \Rightarrow \langle S_{12}, \sigma_{1497} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{5987} = \langle S_{12}, \sigma_{1497} \rangle \Rightarrow \langle S_{13}, \sigma_{1497} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5989} = \frac{\langle x := x + 1, \sigma_{1497} \rangle \Rightarrow \sigma_{1498} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1497} \rangle \Rightarrow \langle S_6, \sigma_{1498} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5988} = \frac{\mathcal{T}_{5989}}{\langle S_{13}, \sigma_{1497} \rangle \Rightarrow \langle S_{15}, \sigma_{1498} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5990} = \frac{\langle S_6, \sigma_{1498} \rangle \Rightarrow \langle S_8, \sigma_{1498} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1498} \rangle \Rightarrow \langle S_9, \sigma_{1498} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5992} = \langle S_8, \sigma_{1498} \rangle \Rightarrow \langle y := y + 1, \sigma_{1498} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5991} = \frac{\mathcal{T}_{5992}}{\langle S_9, \sigma_{1498} \rangle \Rightarrow \langle S_{10}, \sigma_{1498} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5993} = \frac{\langle y := y + 1, \sigma_{1498} \rangle \Rightarrow \sigma_{1499} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1498} \rangle \Rightarrow \langle S_{11}, \sigma_{1499} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5994} = \langle S_{11}, \sigma_{1499} \rangle \Rightarrow \langle S_{12}, \sigma_{1499} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5995} = \langle S_{12}, \sigma_{1499} \rangle \Rightarrow \langle S_{13}, \sigma_{1499} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5997} = \frac{\langle x := x + 1, \sigma_{1499} \rangle \Rightarrow \sigma_{1500} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1499} \rangle \Rightarrow \langle S_6, \sigma_{1500} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{5996} = \frac{\mathcal{T}_{5997}}{\langle S_{13}, \sigma_{1499} \rangle \Rightarrow \langle S_{15}, \sigma_{1500} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{5998} = \frac{\langle S_6, \sigma_{1500} \rangle \Rightarrow \langle S_8, \sigma_{1500} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1500} \rangle \Rightarrow \langle S_9, \sigma_{1500} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6000} = \langle S_8, \sigma_{1500} \rangle \Rightarrow \langle y := y + 1, \sigma_{1500} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{5999} = \frac{\mathcal{T}_{6000}}{\langle S_9, \sigma_{1500} \rangle \Rightarrow \langle S_{10}, \sigma_{1500} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6001} = \frac{\langle y := y + 1, \sigma_{1500} \rangle \Rightarrow \sigma_{1501} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1500} \rangle \Rightarrow \langle S_{11}, \sigma_{1501} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6002} = \langle S_{11}, \sigma_{1501} \rangle \Rightarrow \langle S_{12}, \sigma_{1501} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6003} = \langle S_{12}, \sigma_{1501} \rangle \Rightarrow \langle S_{13}, \sigma_{1501} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6005} = \frac{\langle x := x + 1, \sigma_{1501} \rangle \Rightarrow \sigma_{1502} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1501} \rangle \Rightarrow \langle S_6, \sigma_{1502} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6004} = \frac{\mathcal{T}_{6005}}{\langle S_{13}, \sigma_{1501} \rangle \Rightarrow \langle S_{15}, \sigma_{1502} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6006} = \frac{\langle S_6, \sigma_{1502} \rangle \Rightarrow \langle S_8, \sigma_{1502} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1502} \rangle \Rightarrow \langle S_9, \sigma_{1502} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6008} = \langle S_8, \sigma_{1502} \rangle \Rightarrow \langle y := y + 1, \sigma_{1502} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6007} = \frac{\mathcal{T}_{6008}}{\langle S_9, \sigma_{1502} \rangle \Rightarrow \langle S_{10}, \sigma_{1502} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6009} = \frac{\langle y := y + 1, \sigma_{1502} \rangle \Rightarrow \sigma_{1503} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1502} \rangle \Rightarrow \langle S_{11}, \sigma_{1503} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6010} = \langle S_{11}, \sigma_{1503} \rangle \Rightarrow \langle S_{12}, \sigma_{1503} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6011} = \langle S_{12}, \sigma_{1503} \rangle \Rightarrow \langle S_{13}, \sigma_{1503} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6013} = \frac{\langle x := x + 1, \sigma_{1503} \rangle \Rightarrow \sigma_{1504} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1503} \rangle \Rightarrow \langle S_6, \sigma_{1504} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6012} = \frac{\mathcal{T}_{6013}}{\langle S_{13}, \sigma_{1503} \rangle \Rightarrow \langle S_{15}, \sigma_{1504} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6014} = \frac{\langle S_6, \sigma_{1504} \rangle \Rightarrow \langle S_8, \sigma_{1504} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1504} \rangle \Rightarrow \langle S_9, \sigma_{1504} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6016} = \langle S_8, \sigma_{1504} \rangle \Rightarrow \langle y := y + 1, \sigma_{1504} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6015} = \frac{\mathcal{T}_{6016}}{\langle S_9, \sigma_{1504} \rangle \Rightarrow \langle S_{10}, \sigma_{1504} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6017} = \frac{\langle y := y + 1, \sigma_{1504} \rangle \Rightarrow \sigma_{1505} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1504} \rangle \Rightarrow \langle S_{11}, \sigma_{1505} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6018} = \langle S_{11}, \sigma_{1505} \rangle \Rightarrow \langle S_{12}, \sigma_{1505} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6019} = \langle S_{12}, \sigma_{1505} \rangle \Rightarrow \langle S_{13}, \sigma_{1505} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6021} = \frac{\langle x := x + 1, \sigma_{1505} \rangle \Rightarrow \sigma_{1506} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1505} \rangle \Rightarrow \langle S_6, \sigma_{1506} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6020} = \frac{\mathcal{T}_{6021}}{\langle S_{13}, \sigma_{1505} \rangle \Rightarrow \langle S_{15}, \sigma_{1506} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6022} = \frac{\langle S_6, \sigma_{1506} \rangle \Rightarrow \langle S_8, \sigma_{1506} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1506} \rangle \Rightarrow \langle S_9, \sigma_{1506} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6024} = \langle S_8, \sigma_{1506} \rangle \Rightarrow \langle y := y + 1, \sigma_{1506} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6023} = \frac{\mathcal{T}_{6024}}{\langle S_9, \sigma_{1506} \rangle \Rightarrow \langle S_{10}, \sigma_{1506} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6025} = \frac{\langle y := y + 1, \sigma_{1506} \rangle \Rightarrow \sigma_{1507} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1506} \rangle \Rightarrow \langle S_{11}, \sigma_{1507} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6026} = \langle S_{11}, \sigma_{1507} \rangle \Rightarrow \langle S_{12}, \sigma_{1507} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6027} = \langle S_{12}, \sigma_{1507} \rangle \Rightarrow \langle S_{13}, \sigma_{1507} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6029} = \frac{\langle x := x + 1, \sigma_{1507} \rangle \Rightarrow \sigma_{1508} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1507} \rangle \Rightarrow \langle S_6, \sigma_{1508} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6028} = \frac{\mathcal{T}_{6029}}{\langle S_{13}, \sigma_{1507} \rangle \Rightarrow \langle S_{15}, \sigma_{1508} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6030} = \frac{\langle S_6, \sigma_{1508} \rangle \Rightarrow \langle S_8, \sigma_{1508} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1508} \rangle \Rightarrow \langle S_9, \sigma_{1508} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6032} = \langle S_8, \sigma_{1508} \rangle \Rightarrow \langle y := y + 1, \sigma_{1508} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6031} = \frac{\mathcal{T}_{6032}}{\langle S_9, \sigma_{1508} \rangle \Rightarrow \langle S_{10}, \sigma_{1508} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6033} = \frac{\langle y := y + 1, \sigma_{1508} \rangle \Rightarrow \sigma_{1509} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1508} \rangle \Rightarrow \langle S_{11}, \sigma_{1509} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6034} = \langle S_{11}, \sigma_{1509} \rangle \Rightarrow \langle S_{12}, \sigma_{1509} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6035} = \langle S_{12}, \sigma_{1509} \rangle \Rightarrow \langle S_{13}, \sigma_{1509} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6037} = \frac{\langle x := x + 1, \sigma_{1509} \rangle \Rightarrow \sigma_{1510} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1509} \rangle \Rightarrow \langle S_6, \sigma_{1510} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6036} = \frac{\mathcal{T}_{6037}}{\langle S_{13}, \sigma_{1509} \rangle \Rightarrow \langle S_{15}, \sigma_{1510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6038} = \frac{\langle S_6, \sigma_{1510} \rangle \Rightarrow \langle S_8, \sigma_{1510} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1510} \rangle \Rightarrow \langle S_9, \sigma_{1510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6040} = \langle S_8, \sigma_{1510} \rangle \Rightarrow \langle y := y + 1, \sigma_{1510} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6039} = \frac{\mathcal{T}_{6040}}{\langle S_9, \sigma_{1510} \rangle \Rightarrow \langle S_{10}, \sigma_{1510} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6041} = \frac{\langle y := y + 1, \sigma_{1510} \rangle \Rightarrow \sigma_{1511} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1510} \rangle \Rightarrow \langle S_{11}, \sigma_{1511} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6042} = \langle S_{11}, \sigma_{1511} \rangle \Rightarrow \langle S_{12}, \sigma_{1511} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6043} = \langle S_{12}, \sigma_{1511} \rangle \Rightarrow \langle S_{13}, \sigma_{1511} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6045} = \frac{\langle x := x + 1, \sigma_{1511} \rangle \Rightarrow \sigma_{1512} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1511} \rangle \Rightarrow \langle S_6, \sigma_{1512} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6044} = \frac{\mathcal{T}_{6045}}{\langle S_{13}, \sigma_{1511} \rangle \Rightarrow \langle S_{15}, \sigma_{1512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6046} = \frac{\langle S_6, \sigma_{1512} \rangle \Rightarrow \langle S_8, \sigma_{1512} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1512} \rangle \Rightarrow \langle S_9, \sigma_{1512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6048} = \langle S_8, \sigma_{1512} \rangle \Rightarrow \langle y := y + 1, \sigma_{1512} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6047} = \frac{\mathcal{T}_{6048}}{\langle S_9, \sigma_{1512} \rangle \Rightarrow \langle S_{10}, \sigma_{1512} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6049} = \frac{\langle y := y + 1, \sigma_{1512} \rangle \Rightarrow \sigma_{1513} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1512} \rangle \Rightarrow \langle S_{11}, \sigma_{1513} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6050} = \langle S_{11}, \sigma_{1513} \rangle \Rightarrow \langle S_{12}, \sigma_{1513} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6051} = \langle S_{12}, \sigma_{1513} \rangle \Rightarrow \langle S_{13}, \sigma_{1513} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6053} = \frac{\langle x := x + 1, \sigma_{1513} \rangle \Rightarrow \sigma_{1514} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1513} \rangle \Rightarrow \langle S_6, \sigma_{1514} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6052} = \frac{\mathcal{T}_{6053}}{\langle S_{13}, \sigma_{1513} \rangle \Rightarrow \langle S_{15}, \sigma_{1514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6054} = \frac{\langle S_6, \sigma_{1514} \rangle \Rightarrow \langle S_8, \sigma_{1514} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1514} \rangle \Rightarrow \langle S_9, \sigma_{1514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6056} = \langle S_8, \sigma_{1514} \rangle \Rightarrow \langle y := y + 1, \sigma_{1514} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6055} = \frac{\mathcal{T}_{6056}}{\langle S_9, \sigma_{1514} \rangle \Rightarrow \langle S_{10}, \sigma_{1514} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6057} = \frac{\langle y := y + 1, \sigma_{1514} \rangle \Rightarrow \sigma_{1515} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1514} \rangle \Rightarrow \langle S_{11}, \sigma_{1515} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6058} = \langle S_{11}, \sigma_{1515} \rangle \Rightarrow \langle S_{12}, \sigma_{1515} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6059} = \langle S_{12}, \sigma_{1515} \rangle \Rightarrow \langle S_{13}, \sigma_{1515} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6061} = \frac{\langle x := x + 1, \sigma_{1515} \rangle \Rightarrow \sigma_{1516} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1515} \rangle \Rightarrow \langle S_6, \sigma_{1516} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6060} = \frac{\mathcal{T}_{6061}}{\langle S_{13}, \sigma_{1515} \rangle \Rightarrow \langle S_{15}, \sigma_{1516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6062} = \frac{\langle S_6, \sigma_{1516} \rangle \Rightarrow \langle S_8, \sigma_{1516} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1516} \rangle \Rightarrow \langle S_9, \sigma_{1516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6064} = \langle S_8, \sigma_{1516} \rangle \Rightarrow \langle y := y + 1, \sigma_{1516} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6063} = \frac{\mathcal{T}_{6064}}{\langle S_9, \sigma_{1516} \rangle \Rightarrow \langle S_{10}, \sigma_{1516} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6065} = \frac{\langle y := y + 1, \sigma_{1516} \rangle \Rightarrow \sigma_{1517} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1516} \rangle \Rightarrow \langle S_{11}, \sigma_{1517} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6066} = \langle S_{11}, \sigma_{1517} \rangle \Rightarrow \langle S_{12}, \sigma_{1517} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6067} = \langle S_{12}, \sigma_{1517} \rangle \Rightarrow \langle S_{13}, \sigma_{1517} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6069} = \frac{\langle x := x + 1, \sigma_{1517} \rangle \Rightarrow \sigma_{1518} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1517} \rangle \Rightarrow \langle S_6, \sigma_{1518} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6068} = \frac{\mathcal{T}_{6069}}{\langle S_{13}, \sigma_{1517} \rangle \Rightarrow \langle S_{15}, \sigma_{1518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6070} = \frac{\langle S_6, \sigma_{1518} \rangle \Rightarrow \langle S_8, \sigma_{1518} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1518} \rangle \Rightarrow \langle S_9, \sigma_{1518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6072} = \langle S_8, \sigma_{1518} \rangle \Rightarrow \langle y := y + 1, \sigma_{1518} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6071} = \frac{\mathcal{T}_{6072}}{\langle S_9, \sigma_{1518} \rangle \Rightarrow \langle S_{10}, \sigma_{1518} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6073} = \frac{\langle y := y + 1, \sigma_{1518} \rangle \Rightarrow \sigma_{1519} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1518} \rangle \Rightarrow \langle S_{11}, \sigma_{1519} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6074} = \langle S_{11}, \sigma_{1519} \rangle \Rightarrow \langle S_{12}, \sigma_{1519} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6075} = \langle S_{12}, \sigma_{1519} \rangle \Rightarrow \langle S_{13}, \sigma_{1519} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6077} = \frac{\langle x := x + 1, \sigma_{1519} \rangle \Rightarrow \sigma_{1520} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1519} \rangle \Rightarrow \langle S_6, \sigma_{1520} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6076} = \frac{\mathcal{T}_{6077}}{\langle S_{13}, \sigma_{1519} \rangle \Rightarrow \langle S_{15}, \sigma_{1520} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6078} = \frac{\langle S_6, \sigma_{1520} \rangle \Rightarrow \langle S_8, \sigma_{1520} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1520} \rangle \Rightarrow \langle S_9, \sigma_{1520} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6080} = \langle S_8, \sigma_{1520} \rangle \Rightarrow \langle y := y + 1, \sigma_{1520} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6079} = \frac{\mathcal{T}_{6080}}{\langle S_9, \sigma_{1520} \rangle \Rightarrow \langle S_{10}, \sigma_{1520} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6081} = \frac{\langle y := y + 1, \sigma_{1520} \rangle \Rightarrow \sigma_{1521} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1520} \rangle \Rightarrow \langle S_{11}, \sigma_{1521} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6082} = \langle S_{11}, \sigma_{1521} \rangle \Rightarrow \langle S_{12}, \sigma_{1521} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6083} = \langle S_{12}, \sigma_{1521} \rangle \Rightarrow \langle S_{13}, \sigma_{1521} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6085} = \frac{\langle x := x + 1, \sigma_{1521} \rangle \Rightarrow \sigma_{1522} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1521} \rangle \Rightarrow \langle S_6, \sigma_{1522} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6084} = \frac{\mathcal{T}_{6085}}{\langle S_{13}, \sigma_{1521} \rangle \Rightarrow \langle S_{15}, \sigma_{1522} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6086} = \frac{\langle S_6, \sigma_{1522} \rangle \Rightarrow \langle S_8, \sigma_{1522} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1522} \rangle \Rightarrow \langle S_9, \sigma_{1522} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6088} = \langle S_8, \sigma_{1522} \rangle \Rightarrow \langle y := y + 1, \sigma_{1522} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6087} = \frac{\mathcal{T}_{6088}}{\langle S_9, \sigma_{1522} \rangle \Rightarrow \langle S_{10}, \sigma_{1522} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6089} = \frac{\langle y := y + 1, \sigma_{1522} \rangle \Rightarrow \sigma_{1523} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1522} \rangle \Rightarrow \langle S_{11}, \sigma_{1523} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6090} = \langle S_{11}, \sigma_{1523} \rangle \Rightarrow \langle S_{12}, \sigma_{1523} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6091} = \langle S_{12}, \sigma_{1523} \rangle \Rightarrow \langle S_{13}, \sigma_{1523} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6093} = \frac{\langle x := x + 1, \sigma_{1523} \rangle \Rightarrow \sigma_{1524} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1523} \rangle \Rightarrow \langle S_6, \sigma_{1524} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6092} = \frac{\mathcal{T}_{6093}}{\langle S_{13}, \sigma_{1523} \rangle \Rightarrow \langle S_{15}, \sigma_{1524} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6094} = \frac{\langle S_6, \sigma_{1524} \rangle \Rightarrow \langle S_8, \sigma_{1524} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1524} \rangle \Rightarrow \langle S_9, \sigma_{1524} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6096} = \langle S_8, \sigma_{1524} \rangle \Rightarrow \langle y := y + 1, \sigma_{1524} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6095} = \frac{\mathcal{T}_{6096}}{\langle S_9, \sigma_{1524} \rangle \Rightarrow \langle S_{10}, \sigma_{1524} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6097} = \frac{\langle y := y + 1, \sigma_{1524} \rangle \Rightarrow \sigma_{1525} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1524} \rangle \Rightarrow \langle S_{11}, \sigma_{1525} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6098} = \langle S_{11}, \sigma_{1525} \rangle \Rightarrow \langle S_{12}, \sigma_{1525} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6099} = \langle S_{12}, \sigma_{1525} \rangle \Rightarrow \langle S_{13}, \sigma_{1525} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6101} = \frac{\langle x := x + 1, \sigma_{1525} \rangle \Rightarrow \sigma_{1526} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1525} \rangle \Rightarrow \langle S_6, \sigma_{1526} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6100} = \frac{\mathcal{T}_{6101}}{\langle S_{13}, \sigma_{1525} \rangle \Rightarrow \langle S_{15}, \sigma_{1526} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6102} = \frac{\langle S_6, \sigma_{1526} \rangle \Rightarrow \langle S_8, \sigma_{1526} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1526} \rangle \Rightarrow \langle S_9, \sigma_{1526} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6104} = \langle S_8, \sigma_{1526} \rangle \Rightarrow \langle y := y + 1, \sigma_{1526} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6103} = \frac{\mathcal{T}_{6104}}{\langle S_9, \sigma_{1526} \rangle \Rightarrow \langle S_{10}, \sigma_{1526} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6105} = \frac{\langle y := y + 1, \sigma_{1526} \rangle \Rightarrow \sigma_{1527} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1526} \rangle \Rightarrow \langle S_{11}, \sigma_{1527} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6106} = \langle S_{11}, \sigma_{1527} \rangle \Rightarrow \langle S_{12}, \sigma_{1527} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6107} = \langle S_{12}, \sigma_{1527} \rangle \Rightarrow \langle S_{13}, \sigma_{1527} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6109} = \frac{\langle x := x + 1, \sigma_{1527} \rangle \Rightarrow \sigma_{1528} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1527} \rangle \Rightarrow \langle S_6, \sigma_{1528} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6108} = \frac{\mathcal{T}_{6109}}{\langle S_{13}, \sigma_{1527} \rangle \Rightarrow \langle S_{15}, \sigma_{1528} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6110} = \frac{\langle S_6, \sigma_{1528} \rangle \Rightarrow \langle S_8, \sigma_{1528} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1528} \rangle \Rightarrow \langle S_9, \sigma_{1528} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6112} = \langle S_8, \sigma_{1528} \rangle \Rightarrow \langle y := y + 1, \sigma_{1528} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6111} = \frac{\mathcal{T}_{6112}}{\langle S_9, \sigma_{1528} \rangle \Rightarrow \langle S_{10}, \sigma_{1528} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6113} = \frac{\langle y := y + 1, \sigma_{1528} \rangle \Rightarrow \sigma_{1529} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1528} \rangle \Rightarrow \langle S_{11}, \sigma_{1529} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6114} = \langle S_{11}, \sigma_{1529} \rangle \Rightarrow \langle S_{12}, \sigma_{1529} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6115} = \langle S_{12}, \sigma_{1529} \rangle \Rightarrow \langle S_{13}, \sigma_{1529} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6117} = \frac{\langle x := x + 1, \sigma_{1529} \rangle \Rightarrow \sigma_{1530} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1529} \rangle \Rightarrow \langle S_6, \sigma_{1530} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6116} = \frac{\mathcal{T}_{6117}}{\langle S_{13}, \sigma_{1529} \rangle \Rightarrow \langle S_{15}, \sigma_{1530} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6118} = \frac{\langle S_6, \sigma_{1530} \rangle \Rightarrow \langle S_8, \sigma_{1530} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1530} \rangle \Rightarrow \langle S_9, \sigma_{1530} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6120} = \langle S_8, \sigma_{1530} \rangle \Rightarrow \langle y := y + 1, \sigma_{1530} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6119} = \frac{\mathcal{T}_{6120}}{\langle S_9, \sigma_{1530} \rangle \Rightarrow \langle S_{10}, \sigma_{1530} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6121} = \frac{\langle y := y + 1, \sigma_{1530} \rangle \Rightarrow \sigma_{1531} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1530} \rangle \Rightarrow \langle S_{11}, \sigma_{1531} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6122} = \langle S_{11}, \sigma_{1531} \rangle \Rightarrow \langle S_{12}, \sigma_{1531} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6123} = \langle S_{12}, \sigma_{1531} \rangle \Rightarrow \langle S_{13}, \sigma_{1531} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6125} = \frac{\langle x := x + 1, \sigma_{1531} \rangle \Rightarrow \sigma_{1532} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1531} \rangle \Rightarrow \langle S_6, \sigma_{1532} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6124} = \frac{\mathcal{T}_{6125}}{\langle S_{13}, \sigma_{1531} \rangle \Rightarrow \langle S_{15}, \sigma_{1532} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6126} = \frac{\langle S_6, \sigma_{1532} \rangle \Rightarrow \langle S_8, \sigma_{1532} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1532} \rangle \Rightarrow \langle S_9, \sigma_{1532} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6128} = \langle S_8, \sigma_{1532} \rangle \Rightarrow \langle y := y + 1, \sigma_{1532} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6127} = \frac{\mathcal{T}_{6128}}{\langle S_9, \sigma_{1532} \rangle \Rightarrow \langle S_{10}, \sigma_{1532} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6129} = \frac{\langle y := y + 1, \sigma_{1532} \rangle \Rightarrow \sigma_{1533} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1532} \rangle \Rightarrow \langle S_{11}, \sigma_{1533} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6130} = \langle S_{11}, \sigma_{1533} \rangle \Rightarrow \langle S_{12}, \sigma_{1533} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6131} = \langle S_{12}, \sigma_{1533} \rangle \Rightarrow \langle S_{13}, \sigma_{1533} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6133} = \frac{\langle x := x + 1, \sigma_{1533} \rangle \Rightarrow \sigma_{1534} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1533} \rangle \Rightarrow \langle S_6, \sigma_{1534} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6132} = \frac{\mathcal{T}_{6133}}{\langle S_{13}, \sigma_{1533} \rangle \Rightarrow \langle S_{15}, \sigma_{1534} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6134} = \frac{\langle S_6, \sigma_{1534} \rangle \Rightarrow \langle S_8, \sigma_{1534} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1534} \rangle \Rightarrow \langle S_9, \sigma_{1534} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6136} = \langle S_8, \sigma_{1534} \rangle \Rightarrow \langle y := y + 1, \sigma_{1534} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6135} = \frac{\mathcal{T}_{6136}}{\langle S_9, \sigma_{1534} \rangle \Rightarrow \langle S_{10}, \sigma_{1534} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6137} = \frac{\langle y := y + 1, \sigma_{1534} \rangle \Rightarrow \sigma_{1535} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1534} \rangle \Rightarrow \langle S_{11}, \sigma_{1535} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6138} = \langle S_{11}, \sigma_{1535} \rangle \Rightarrow \langle S_{12}, \sigma_{1535} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6139} = \langle S_{12}, \sigma_{1535} \rangle \Rightarrow \langle S_{13}, \sigma_{1535} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6141} = \frac{\langle x := x + 1, \sigma_{1535} \rangle \Rightarrow \sigma_{1536} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1535} \rangle \Rightarrow \langle S_6, \sigma_{1536} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6140} = \frac{\mathcal{T}_{6141}}{\langle S_{13}, \sigma_{1535} \rangle \Rightarrow \langle S_{15}, \sigma_{1536} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6142} = \frac{\langle S_6, \sigma_{1536} \rangle \Rightarrow \langle S_8, \sigma_{1536} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1536} \rangle \Rightarrow \langle S_9, \sigma_{1536} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6144} = \langle S_8, \sigma_{1536} \rangle \Rightarrow \langle y := y + 1, \sigma_{1536} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6143} = \frac{\mathcal{T}_{6144}}{\langle S_9, \sigma_{1536} \rangle \Rightarrow \langle S_{10}, \sigma_{1536} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6145} = \frac{\langle y := y + 1, \sigma_{1536} \rangle \Rightarrow \sigma_{1537} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1536} \rangle \Rightarrow \langle S_{11}, \sigma_{1537} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6146} = \langle S_{11}, \sigma_{1537} \rangle \Rightarrow \langle S_{12}, \sigma_{1537} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6147} = \langle S_{12}, \sigma_{1537} \rangle \Rightarrow \langle S_{13}, \sigma_{1537} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6149} = \frac{\langle x := x + 1, \sigma_{1537} \rangle \Rightarrow \sigma_{1538} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1537} \rangle \Rightarrow \langle S_6, \sigma_{1538} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6148} = \frac{\mathcal{T}_{6149}}{\langle S_{13}, \sigma_{1537} \rangle \Rightarrow \langle S_{15}, \sigma_{1538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6150} = \frac{\langle S_6, \sigma_{1538} \rangle \Rightarrow \langle S_8, \sigma_{1538} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1538} \rangle \Rightarrow \langle S_9, \sigma_{1538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6152} = \langle S_8, \sigma_{1538} \rangle \Rightarrow \langle y := y + 1, \sigma_{1538} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6151} = \frac{\mathcal{T}_{6152}}{\langle S_9, \sigma_{1538} \rangle \Rightarrow \langle S_{10}, \sigma_{1538} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6153} = \frac{\langle y := y + 1, \sigma_{1538} \rangle \Rightarrow \sigma_{1539} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1538} \rangle \Rightarrow \langle S_{11}, \sigma_{1539} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6154} = \langle S_{11}, \sigma_{1539} \rangle \Rightarrow \langle S_{12}, \sigma_{1539} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6155} = \langle S_{12}, \sigma_{1539} \rangle \Rightarrow \langle S_{13}, \sigma_{1539} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6157} = \frac{\langle x := x + 1, \sigma_{1539} \rangle \Rightarrow \sigma_{1540} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1539} \rangle \Rightarrow \langle S_6, \sigma_{1540} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6156} = \frac{\mathcal{T}_{6157}}{\langle S_{13}, \sigma_{1539} \rangle \Rightarrow \langle S_{15}, \sigma_{1540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6158} = \frac{\langle S_6, \sigma_{1540} \rangle \Rightarrow \langle S_8, \sigma_{1540} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1540} \rangle \Rightarrow \langle S_9, \sigma_{1540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6160} = \langle S_8, \sigma_{1540} \rangle \Rightarrow \langle y := y + 1, \sigma_{1540} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6159} = \frac{\mathcal{T}_{6160}}{\langle S_9, \sigma_{1540} \rangle \Rightarrow \langle S_{10}, \sigma_{1540} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6161} = \frac{\langle y := y + 1, \sigma_{1540} \rangle \Rightarrow \sigma_{1541} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1540} \rangle \Rightarrow \langle S_{11}, \sigma_{1541} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6162} = \langle S_{11}, \sigma_{1541} \rangle \Rightarrow \langle S_{12}, \sigma_{1541} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6163} = \langle S_{12}, \sigma_{1541} \rangle \Rightarrow \langle S_{13}, \sigma_{1541} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6165} = \frac{\langle x := x + 1, \sigma_{1541} \rangle \Rightarrow \sigma_{1542} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1541} \rangle \Rightarrow \langle S_6, \sigma_{1542} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6164} = \frac{\mathcal{T}_{6165}}{\langle S_{13}, \sigma_{1541} \rangle \Rightarrow \langle S_{15}, \sigma_{1542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6166} = \frac{\langle S_6, \sigma_{1542} \rangle \Rightarrow \langle S_8, \sigma_{1542} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1542} \rangle \Rightarrow \langle S_9, \sigma_{1542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6168} = \langle S_8, \sigma_{1542} \rangle \Rightarrow \langle y := y + 1, \sigma_{1542} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6167} = \frac{\mathcal{T}_{6168}}{\langle S_9, \sigma_{1542} \rangle \Rightarrow \langle S_{10}, \sigma_{1542} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6169} = \frac{\langle y := y + 1, \sigma_{1542} \rangle \Rightarrow \sigma_{1543} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1542} \rangle \Rightarrow \langle S_{11}, \sigma_{1543} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6170} = \langle S_{11}, \sigma_{1543} \rangle \Rightarrow \langle S_{12}, \sigma_{1543} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6171} = \langle S_{12}, \sigma_{1543} \rangle \Rightarrow \langle S_{13}, \sigma_{1543} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6173} = \frac{\langle x := x + 1, \sigma_{1543} \rangle \Rightarrow \sigma_{1544} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1543} \rangle \Rightarrow \langle S_6, \sigma_{1544} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6172} = \frac{\mathcal{T}_{6173}}{\langle S_{13}, \sigma_{1543} \rangle \Rightarrow \langle S_{15}, \sigma_{1544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6174} = \frac{\langle S_6, \sigma_{1544} \rangle \Rightarrow \langle S_8, \sigma_{1544} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1544} \rangle \Rightarrow \langle S_9, \sigma_{1544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6176} = \langle S_8, \sigma_{1544} \rangle \Rightarrow \langle y := y + 1, \sigma_{1544} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6175} = \frac{\mathcal{T}_{6176}}{\langle S_9, \sigma_{1544} \rangle \Rightarrow \langle S_{10}, \sigma_{1544} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6177} = \frac{\langle y := y + 1, \sigma_{1544} \rangle \Rightarrow \sigma_{1545} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1544} \rangle \Rightarrow \langle S_{11}, \sigma_{1545} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6178} = \langle S_{11}, \sigma_{1545} \rangle \Rightarrow \langle S_{12}, \sigma_{1545} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6179} = \langle S_{12}, \sigma_{1545} \rangle \Rightarrow \langle S_{13}, \sigma_{1545} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6181} = \frac{\langle x := x + 1, \sigma_{1545} \rangle \Rightarrow \sigma_{1546} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1545} \rangle \Rightarrow \langle S_6, \sigma_{1546} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6180} = \frac{\mathcal{T}_{6181}}{\langle S_{13}, \sigma_{1545} \rangle \Rightarrow \langle S_{15}, \sigma_{1546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6182} = \frac{\langle S_6, \sigma_{1546} \rangle \Rightarrow \langle S_8, \sigma_{1546} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1546} \rangle \Rightarrow \langle S_9, \sigma_{1546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6184} = \langle S_8, \sigma_{1546} \rangle \Rightarrow \langle y := y + 1, \sigma_{1546} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6183} = \frac{\mathcal{T}_{6184}}{\langle S_9, \sigma_{1546} \rangle \Rightarrow \langle S_{10}, \sigma_{1546} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6185} = \frac{\langle y := y + 1, \sigma_{1546} \rangle \Rightarrow \sigma_{1547} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1546} \rangle \Rightarrow \langle S_{11}, \sigma_{1547} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6186} = \langle S_{11}, \sigma_{1547} \rangle \Rightarrow \langle S_{12}, \sigma_{1547} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6187} = \langle S_{12}, \sigma_{1547} \rangle \Rightarrow \langle S_{13}, \sigma_{1547} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6189} = \frac{\langle x := x + 1, \sigma_{1547} \rangle \Rightarrow \sigma_{1548} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1547} \rangle \Rightarrow \langle S_6, \sigma_{1548} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6188} = \frac{\mathcal{T}_{6189}}{\langle S_{13}, \sigma_{1547} \rangle \Rightarrow \langle S_{15}, \sigma_{1548} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6190} = \frac{\langle S_6, \sigma_{1548} \rangle \Rightarrow \langle S_8, \sigma_{1548} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1548} \rangle \Rightarrow \langle S_9, \sigma_{1548} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6192} = \langle S_8, \sigma_{1548} \rangle \Rightarrow \langle y := y + 1, \sigma_{1548} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6191} = \frac{\mathcal{T}_{6192}}{\langle S_9, \sigma_{1548} \rangle \Rightarrow \langle S_{10}, \sigma_{1548} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6193} = \frac{\langle y := y + 1, \sigma_{1548} \rangle \Rightarrow \sigma_{1549} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1548} \rangle \Rightarrow \langle S_{11}, \sigma_{1549} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6194} = \langle S_{11}, \sigma_{1549} \rangle \Rightarrow \langle S_{12}, \sigma_{1549} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6195} = \langle S_{12}, \sigma_{1549} \rangle \Rightarrow \langle S_{13}, \sigma_{1549} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6197} = \frac{\langle x := x + 1, \sigma_{1549} \rangle \Rightarrow \sigma_{1550} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1549} \rangle \Rightarrow \langle S_6, \sigma_{1550} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6196} = \frac{\mathcal{T}_{6197}}{\langle S_{13}, \sigma_{1549} \rangle \Rightarrow \langle S_{15}, \sigma_{1550} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6198} = \frac{\langle S_6, \sigma_{1550} \rangle \Rightarrow \langle S_8, \sigma_{1550} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1550} \rangle \Rightarrow \langle S_9, \sigma_{1550} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6200} = \langle S_8, \sigma_{1550} \rangle \Rightarrow \langle y := y + 1, \sigma_{1550} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6199} = \frac{\mathcal{T}_{6200}}{\langle S_9, \sigma_{1550} \rangle \Rightarrow \langle S_{10}, \sigma_{1550} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6201} = \frac{\langle y := y + 1, \sigma_{1550} \rangle \Rightarrow \sigma_{1551} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1550} \rangle \Rightarrow \langle S_{11}, \sigma_{1551} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6202} = \langle S_{11}, \sigma_{1551} \rangle \Rightarrow \langle S_{12}, \sigma_{1551} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6203} = \langle S_{12}, \sigma_{1551} \rangle \Rightarrow \langle S_{13}, \sigma_{1551} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6205} = \frac{\langle x := x + 1, \sigma_{1551} \rangle \Rightarrow \sigma_{1552} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1551} \rangle \Rightarrow \langle S_6, \sigma_{1552} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6204} = \frac{\mathcal{T}_{6205}}{\langle S_{13}, \sigma_{1551} \rangle \Rightarrow \langle S_{15}, \sigma_{1552} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6206} = \frac{\langle S_6, \sigma_{1552} \rangle \Rightarrow \langle S_8, \sigma_{1552} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1552} \rangle \Rightarrow \langle S_9, \sigma_{1552} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6208} = \langle S_8, \sigma_{1552} \rangle \Rightarrow \langle y := y + 1, \sigma_{1552} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6207} = \frac{\mathcal{T}_{6208}}{\langle S_9, \sigma_{1552} \rangle \Rightarrow \langle S_{10}, \sigma_{1552} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6209} = \frac{\langle y := y + 1, \sigma_{1552} \rangle \Rightarrow \sigma_{1553} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1552} \rangle \Rightarrow \langle S_{11}, \sigma_{1553} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6210} = \langle S_{11}, \sigma_{1553} \rangle \Rightarrow \langle S_{12}, \sigma_{1553} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6211} = \langle S_{12}, \sigma_{1553} \rangle \Rightarrow \langle S_{13}, \sigma_{1553} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6213} = \frac{\langle x := x + 1, \sigma_{1553} \rangle \Rightarrow \sigma_{1554} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1553} \rangle \Rightarrow \langle S_6, \sigma_{1554} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6212} = \frac{\mathcal{T}_{6213}}{\langle S_{13}, \sigma_{1553} \rangle \Rightarrow \langle S_{15}, \sigma_{1554} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6214} = \frac{\langle S_6, \sigma_{1554} \rangle \Rightarrow \langle S_8, \sigma_{1554} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1554} \rangle \Rightarrow \langle S_9, \sigma_{1554} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6216} = \langle S_8, \sigma_{1554} \rangle \Rightarrow \langle y := y + 1, \sigma_{1554} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6215} = \frac{\mathcal{T}_{6216}}{\langle S_9, \sigma_{1554} \rangle \Rightarrow \langle S_{10}, \sigma_{1554} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6217} = \frac{\langle y := y + 1, \sigma_{1554} \rangle \Rightarrow \sigma_{1555} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1554} \rangle \Rightarrow \langle S_{11}, \sigma_{1555} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6218} = \langle S_{11}, \sigma_{1555} \rangle \Rightarrow \langle S_{12}, \sigma_{1555} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6219} = \langle S_{12}, \sigma_{1555} \rangle \Rightarrow \langle S_{13}, \sigma_{1555} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6221} = \frac{\langle x := x + 1, \sigma_{1555} \rangle \Rightarrow \sigma_{1556} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1555} \rangle \Rightarrow \langle S_6, \sigma_{1556} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6220} = \frac{\mathcal{T}_{6221}}{\langle S_{13}, \sigma_{1555} \rangle \Rightarrow \langle S_{15}, \sigma_{1556} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6222} = \frac{\langle S_6, \sigma_{1556} \rangle \Rightarrow \langle S_8, \sigma_{1556} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1556} \rangle \Rightarrow \langle S_9, \sigma_{1556} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6224} = \langle S_8, \sigma_{1556} \rangle \Rightarrow \langle y := y + 1, \sigma_{1556} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6223} = \frac{\mathcal{T}_{6224}}{\langle S_9, \sigma_{1556} \rangle \Rightarrow \langle S_{10}, \sigma_{1556} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6225} = \frac{\langle y := y + 1, \sigma_{1556} \rangle \Rightarrow \sigma_{1557} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1556} \rangle \Rightarrow \langle S_{11}, \sigma_{1557} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6226} = \langle S_{11}, \sigma_{1557} \rangle \Rightarrow \langle S_{12}, \sigma_{1557} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6227} = \langle S_{12}, \sigma_{1557} \rangle \Rightarrow \langle S_{13}, \sigma_{1557} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6229} = \frac{\langle x := x + 1, \sigma_{1557} \rangle \Rightarrow \sigma_{1558} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1557} \rangle \Rightarrow \langle S_6, \sigma_{1558} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6228} = \frac{\mathcal{T}_{6229}}{\langle S_{13}, \sigma_{1557} \rangle \Rightarrow \langle S_{15}, \sigma_{1558} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6230} = \frac{\langle S_6, \sigma_{1558} \rangle \Rightarrow \langle S_8, \sigma_{1558} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1558} \rangle \Rightarrow \langle S_9, \sigma_{1558} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6232} = \langle S_8, \sigma_{1558} \rangle \Rightarrow \langle y := y + 1, \sigma_{1558} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6231} = \frac{\mathcal{T}_{6232}}{\langle S_9, \sigma_{1558} \rangle \Rightarrow \langle S_{10}, \sigma_{1558} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6233} = \frac{\langle y := y + 1, \sigma_{1558} \rangle \Rightarrow \sigma_{1559} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1558} \rangle \Rightarrow \langle S_{11}, \sigma_{1559} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6234} = \langle S_{11}, \sigma_{1559} \rangle \Rightarrow \langle S_{12}, \sigma_{1559} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6235} = \langle S_{12}, \sigma_{1559} \rangle \Rightarrow \langle S_{13}, \sigma_{1559} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6237} = \frac{\langle x := x + 1, \sigma_{1559} \rangle \Rightarrow \sigma_{1560} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1559} \rangle \Rightarrow \langle S_6, \sigma_{1560} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6236} = \frac{\mathcal{T}_{6237}}{\langle S_{13}, \sigma_{1559} \rangle \Rightarrow \langle S_{15}, \sigma_{1560} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6238} = \frac{\langle S_6, \sigma_{1560} \rangle \Rightarrow \langle S_8, \sigma_{1560} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1560} \rangle \Rightarrow \langle S_9, \sigma_{1560} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6240} = \langle S_8, \sigma_{1560} \rangle \Rightarrow \langle y := y + 1, \sigma_{1560} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6239} = \frac{\mathcal{T}_{6240}}{\langle S_9, \sigma_{1560} \rangle \Rightarrow \langle S_{10}, \sigma_{1560} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6241} = \frac{\langle y := y + 1, \sigma_{1560} \rangle \Rightarrow \sigma_{1561} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1560} \rangle \Rightarrow \langle S_{11}, \sigma_{1561} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6242} = \langle S_{11}, \sigma_{1561} \rangle \Rightarrow \langle S_{12}, \sigma_{1561} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6243} = \langle S_{12}, \sigma_{1561} \rangle \Rightarrow \langle S_{13}, \sigma_{1561} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6245} = \frac{\langle x := x + 1, \sigma_{1561} \rangle \Rightarrow \sigma_{1562} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1561} \rangle \Rightarrow \langle S_6, \sigma_{1562} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6244} = \frac{\mathcal{T}_{6245}}{\langle S_{13}, \sigma_{1561} \rangle \Rightarrow \langle S_{15}, \sigma_{1562} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6246} = \frac{\langle S_6, \sigma_{1562} \rangle \Rightarrow \langle S_8, \sigma_{1562} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1562} \rangle \Rightarrow \langle S_9, \sigma_{1562} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6248} = \langle S_8, \sigma_{1562} \rangle \Rightarrow \langle y := y + 1, \sigma_{1562} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6247} = \frac{\mathcal{T}_{6248}}{\langle S_9, \sigma_{1562} \rangle \Rightarrow \langle S_{10}, \sigma_{1562} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6249} = \frac{\langle y := y + 1, \sigma_{1562} \rangle \Rightarrow \sigma_{1563} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1562} \rangle \Rightarrow \langle S_{11}, \sigma_{1563} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6250} = \langle S_{11}, \sigma_{1563} \rangle \Rightarrow \langle S_{12}, \sigma_{1563} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6251} = \langle S_{12}, \sigma_{1563} \rangle \Rightarrow \langle S_{13}, \sigma_{1563} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6253} = \frac{\langle x := x + 1, \sigma_{1563} \rangle \Rightarrow \sigma_{1564} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1563} \rangle \Rightarrow \langle S_6, \sigma_{1564} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6252} = \frac{\mathcal{T}_{6253}}{\langle S_{13}, \sigma_{1563} \rangle \Rightarrow \langle S_{15}, \sigma_{1564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6254} = \frac{\langle S_6, \sigma_{1564} \rangle \Rightarrow \langle S_8, \sigma_{1564} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1564} \rangle \Rightarrow \langle S_9, \sigma_{1564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6256} = \langle S_8, \sigma_{1564} \rangle \Rightarrow \langle y := y + 1, \sigma_{1564} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6255} = \frac{\mathcal{T}_{6256}}{\langle S_9, \sigma_{1564} \rangle \Rightarrow \langle S_{10}, \sigma_{1564} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6257} = \frac{\langle y := y + 1, \sigma_{1564} \rangle \Rightarrow \sigma_{1565} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1564} \rangle \Rightarrow \langle S_{11}, \sigma_{1565} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6258} = \langle S_{11}, \sigma_{1565} \rangle \Rightarrow \langle S_{12}, \sigma_{1565} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6259} = \langle S_{12}, \sigma_{1565} \rangle \Rightarrow \langle S_{13}, \sigma_{1565} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6261} = \frac{\langle x := x + 1, \sigma_{1565} \rangle \Rightarrow \sigma_{1566} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1565} \rangle \Rightarrow \langle S_6, \sigma_{1566} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6260} = \frac{\mathcal{T}_{6261}}{\langle S_{13}, \sigma_{1565} \rangle \Rightarrow \langle S_{15}, \sigma_{1566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6262} = \frac{\langle S_6, \sigma_{1566} \rangle \Rightarrow \langle S_8, \sigma_{1566} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1566} \rangle \Rightarrow \langle S_9, \sigma_{1566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6264} = \langle S_8, \sigma_{1566} \rangle \Rightarrow \langle y := y + 1, \sigma_{1566} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6263} = \frac{\mathcal{T}_{6264}}{\langle S_9, \sigma_{1566} \rangle \Rightarrow \langle S_{10}, \sigma_{1566} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6265} = \frac{\langle y := y + 1, \sigma_{1566} \rangle \Rightarrow \sigma_{1567} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1566} \rangle \Rightarrow \langle S_{11}, \sigma_{1567} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6266} = \langle S_{11}, \sigma_{1567} \rangle \Rightarrow \langle S_{12}, \sigma_{1567} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6267} = \langle S_{12}, \sigma_{1567} \rangle \Rightarrow \langle S_{13}, \sigma_{1567} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6269} = \frac{\langle x := x + 1, \sigma_{1567} \rangle \Rightarrow \sigma_{1568} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1567} \rangle \Rightarrow \langle S_6, \sigma_{1568} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6268} = \frac{\mathcal{T}_{6269}}{\langle S_{13}, \sigma_{1567} \rangle \Rightarrow \langle S_{15}, \sigma_{1568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6270} = \frac{\langle S_6, \sigma_{1568} \rangle \Rightarrow \langle S_8, \sigma_{1568} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1568} \rangle \Rightarrow \langle S_9, \sigma_{1568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6272} = \langle S_8, \sigma_{1568} \rangle \Rightarrow \langle y := y + 1, \sigma_{1568} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6271} = \frac{\mathcal{T}_{6272}}{\langle S_9, \sigma_{1568} \rangle \Rightarrow \langle S_{10}, \sigma_{1568} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6273} = \frac{\langle y := y + 1, \sigma_{1568} \rangle \Rightarrow \sigma_{1569} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1568} \rangle \Rightarrow \langle S_{11}, \sigma_{1569} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6274} = \langle S_{11}, \sigma_{1569} \rangle \Rightarrow \langle S_{12}, \sigma_{1569} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6275} = \langle S_{12}, \sigma_{1569} \rangle \Rightarrow \langle S_{13}, \sigma_{1569} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6277} = \frac{\langle x := x + 1, \sigma_{1569} \rangle \Rightarrow \sigma_{1570} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1569} \rangle \Rightarrow \langle S_6, \sigma_{1570} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6276} = \frac{\mathcal{T}_{6277}}{\langle S_{13}, \sigma_{1569} \rangle \Rightarrow \langle S_{15}, \sigma_{1570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6278} = \frac{\langle S_6, \sigma_{1570} \rangle \Rightarrow \langle S_8, \sigma_{1570} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1570} \rangle \Rightarrow \langle S_9, \sigma_{1570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6280} = \langle S_8, \sigma_{1570} \rangle \Rightarrow \langle y := y + 1, \sigma_{1570} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6279} = \frac{\mathcal{T}_{6280}}{\langle S_9, \sigma_{1570} \rangle \Rightarrow \langle S_{10}, \sigma_{1570} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6281} = \frac{\langle y := y + 1, \sigma_{1570} \rangle \Rightarrow \sigma_{1571} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1570} \rangle \Rightarrow \langle S_{11}, \sigma_{1571} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6282} = \langle S_{11}, \sigma_{1571} \rangle \Rightarrow \langle S_{12}, \sigma_{1571} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6283} = \langle S_{12}, \sigma_{1571} \rangle \Rightarrow \langle S_{13}, \sigma_{1571} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6285} = \frac{\langle x := x + 1, \sigma_{1571} \rangle \Rightarrow \sigma_{1572} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1571} \rangle \Rightarrow \langle S_6, \sigma_{1572} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6284} = \frac{\mathcal{T}_{6285}}{\langle S_{13}, \sigma_{1571} \rangle \Rightarrow \langle S_{15}, \sigma_{1572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6286} = \frac{\langle S_6, \sigma_{1572} \rangle \Rightarrow \langle S_8, \sigma_{1572} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1572} \rangle \Rightarrow \langle S_9, \sigma_{1572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6288} = \langle S_8, \sigma_{1572} \rangle \Rightarrow \langle y := y + 1, \sigma_{1572} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6287} = \frac{\mathcal{T}_{6288}}{\langle S_9, \sigma_{1572} \rangle \Rightarrow \langle S_{10}, \sigma_{1572} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6289} = \frac{\langle y := y + 1, \sigma_{1572} \rangle \Rightarrow \sigma_{1573} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1572} \rangle \Rightarrow \langle S_{11}, \sigma_{1573} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6290} = \langle S_{11}, \sigma_{1573} \rangle \Rightarrow \langle S_{12}, \sigma_{1573} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6291} = \langle S_{12}, \sigma_{1573} \rangle \Rightarrow \langle S_{13}, \sigma_{1573} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6293} = \frac{\langle x := x + 1, \sigma_{1573} \rangle \Rightarrow \sigma_{1574} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1573} \rangle \Rightarrow \langle S_6, \sigma_{1574} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6292} = \frac{\mathcal{T}_{6293}}{\langle S_{13}, \sigma_{1573} \rangle \Rightarrow \langle S_{15}, \sigma_{1574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6294} = \frac{\langle S_6, \sigma_{1574} \rangle \Rightarrow \langle S_8, \sigma_{1574} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1574} \rangle \Rightarrow \langle S_9, \sigma_{1574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6296} = \langle S_8, \sigma_{1574} \rangle \Rightarrow \langle y := y + 1, \sigma_{1574} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6295} = \frac{\mathcal{T}_{6296}}{\langle S_9, \sigma_{1574} \rangle \Rightarrow \langle S_{10}, \sigma_{1574} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6297} = \frac{\langle y := y + 1, \sigma_{1574} \rangle \Rightarrow \sigma_{1575} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1574} \rangle \Rightarrow \langle S_{11}, \sigma_{1575} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6298} = \langle S_{11}, \sigma_{1575} \rangle \Rightarrow \langle S_{12}, \sigma_{1575} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6299} = \langle S_{12}, \sigma_{1575} \rangle \Rightarrow \langle S_{13}, \sigma_{1575} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6301} = \frac{\langle x := x + 1, \sigma_{1575} \rangle \Rightarrow \sigma_{1576} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1575} \rangle \Rightarrow \langle S_6, \sigma_{1576} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6300} = \frac{\mathcal{T}_{6301}}{\langle S_{13}, \sigma_{1575} \rangle \Rightarrow \langle S_{15}, \sigma_{1576} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6302} = \frac{\langle S_6, \sigma_{1576} \rangle \Rightarrow \langle S_8, \sigma_{1576} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1576} \rangle \Rightarrow \langle S_9, \sigma_{1576} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6304} = \langle S_8, \sigma_{1576} \rangle \Rightarrow \langle y := y + 1, \sigma_{1576} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6303} = \frac{\mathcal{T}_{6304}}{\langle S_9, \sigma_{1576} \rangle \Rightarrow \langle S_{10}, \sigma_{1576} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6305} = \frac{\langle y := y + 1, \sigma_{1576} \rangle \Rightarrow \sigma_{1577} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1576} \rangle \Rightarrow \langle S_{11}, \sigma_{1577} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6306} = \langle S_{11}, \sigma_{1577} \rangle \Rightarrow \langle S_{12}, \sigma_{1577} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6307} = \langle S_{12}, \sigma_{1577} \rangle \Rightarrow \langle S_{13}, \sigma_{1577} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6309} = \frac{\langle x := x + 1, \sigma_{1577} \rangle \Rightarrow \sigma_{1578} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1577} \rangle \Rightarrow \langle S_6, \sigma_{1578} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6308} = \frac{\mathcal{T}_{6309}}{\langle S_{13}, \sigma_{1577} \rangle \Rightarrow \langle S_{15}, \sigma_{1578} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6310} = \frac{\langle S_6, \sigma_{1578} \rangle \Rightarrow \langle S_8, \sigma_{1578} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1578} \rangle \Rightarrow \langle S_9, \sigma_{1578} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6312} = \langle S_8, \sigma_{1578} \rangle \Rightarrow \langle y := y + 1, \sigma_{1578} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6311} = \frac{\mathcal{T}_{6312}}{\langle S_9, \sigma_{1578} \rangle \Rightarrow \langle S_{10}, \sigma_{1578} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6313} = \frac{\langle y := y + 1, \sigma_{1578} \rangle \Rightarrow \sigma_{1579} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1578} \rangle \Rightarrow \langle S_{11}, \sigma_{1579} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6314} = \langle S_{11}, \sigma_{1579} \rangle \Rightarrow \langle S_{12}, \sigma_{1579} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6315} = \langle S_{12}, \sigma_{1579} \rangle \Rightarrow \langle S_{13}, \sigma_{1579} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6317} = \frac{\langle x := x + 1, \sigma_{1579} \rangle \Rightarrow \sigma_{1580} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1579} \rangle \Rightarrow \langle S_6, \sigma_{1580} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6316} = \frac{\mathcal{T}_{6317}}{\langle S_{13}, \sigma_{1579} \rangle \Rightarrow \langle S_{15}, \sigma_{1580} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6318} = \frac{\langle S_6, \sigma_{1580} \rangle \Rightarrow \langle S_8, \sigma_{1580} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1580} \rangle \Rightarrow \langle S_9, \sigma_{1580} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6320} = \langle S_8, \sigma_{1580} \rangle \Rightarrow \langle y := y + 1, \sigma_{1580} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6319} = \frac{\mathcal{T}_{6320}}{\langle S_9, \sigma_{1580} \rangle \Rightarrow \langle S_{10}, \sigma_{1580} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6321} = \frac{\langle y := y + 1, \sigma_{1580} \rangle \Rightarrow \sigma_{1581} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1580} \rangle \Rightarrow \langle S_{11}, \sigma_{1581} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6322} = \langle S_{11}, \sigma_{1581} \rangle \Rightarrow \langle S_{12}, \sigma_{1581} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6323} = \langle S_{12}, \sigma_{1581} \rangle \Rightarrow \langle S_{13}, \sigma_{1581} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6325} = \frac{\langle x := x + 1, \sigma_{1581} \rangle \Rightarrow \sigma_{1582} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1581} \rangle \Rightarrow \langle S_6, \sigma_{1582} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6324} = \frac{\mathcal{T}_{6325}}{\langle S_{13}, \sigma_{1581} \rangle \Rightarrow \langle S_{15}, \sigma_{1582} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6326} = \frac{\langle S_6, \sigma_{1582} \rangle \Rightarrow \langle S_8, \sigma_{1582} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1582} \rangle \Rightarrow \langle S_9, \sigma_{1582} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6328} = \langle S_8, \sigma_{1582} \rangle \Rightarrow \langle y := y + 1, \sigma_{1582} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6327} = \frac{\mathcal{T}_{6328}}{\langle S_9, \sigma_{1582} \rangle \Rightarrow \langle S_{10}, \sigma_{1582} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6329} = \frac{\langle y := y + 1, \sigma_{1582} \rangle \Rightarrow \sigma_{1583} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1582} \rangle \Rightarrow \langle S_{11}, \sigma_{1583} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6330} = \langle S_{11}, \sigma_{1583} \rangle \Rightarrow \langle S_{12}, \sigma_{1583} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6331} = \langle S_{12}, \sigma_{1583} \rangle \Rightarrow \langle S_{13}, \sigma_{1583} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6333} = \frac{\langle x := x + 1, \sigma_{1583} \rangle \Rightarrow \sigma_{1584} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1583} \rangle \Rightarrow \langle S_6, \sigma_{1584} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6332} = \frac{\mathcal{T}_{6333}}{\langle S_{13}, \sigma_{1583} \rangle \Rightarrow \langle S_{15}, \sigma_{1584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6334} = \frac{\langle S_6, \sigma_{1584} \rangle \Rightarrow \langle S_8, \sigma_{1584} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1584} \rangle \Rightarrow \langle S_9, \sigma_{1584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6336} = \langle S_8, \sigma_{1584} \rangle \Rightarrow \langle y := y + 1, \sigma_{1584} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6335} = \frac{\mathcal{T}_{6336}}{\langle S_9, \sigma_{1584} \rangle \Rightarrow \langle S_{10}, \sigma_{1584} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6337} = \frac{\langle y := y + 1, \sigma_{1584} \rangle \Rightarrow \sigma_{1585} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1584} \rangle \Rightarrow \langle S_{11}, \sigma_{1585} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6338} = \langle S_{11}, \sigma_{1585} \rangle \Rightarrow \langle S_{12}, \sigma_{1585} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6339} = \langle S_{12}, \sigma_{1585} \rangle \Rightarrow \langle S_{13}, \sigma_{1585} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6341} = \frac{\langle x := x + 1, \sigma_{1585} \rangle \Rightarrow \sigma_{1586} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1585} \rangle \Rightarrow \langle S_6, \sigma_{1586} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6340} = \frac{\mathcal{T}_{6341}}{\langle S_{13}, \sigma_{1585} \rangle \Rightarrow \langle S_{15}, \sigma_{1586} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6342} = \frac{\langle S_6, \sigma_{1586} \rangle \Rightarrow \langle S_8, \sigma_{1586} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1586} \rangle \Rightarrow \langle S_9, \sigma_{1586} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6344} = \langle S_8, \sigma_{1586} \rangle \Rightarrow \langle y := y + 1, \sigma_{1586} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6343} = \frac{\mathcal{T}_{6344}}{\langle S_9, \sigma_{1586} \rangle \Rightarrow \langle S_{10}, \sigma_{1586} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6345} = \frac{\langle y := y + 1, \sigma_{1586} \rangle \Rightarrow \sigma_{1587} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1586} \rangle \Rightarrow \langle S_{11}, \sigma_{1587} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6346} = \langle S_{11}, \sigma_{1587} \rangle \Rightarrow \langle S_{12}, \sigma_{1587} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6347} = \langle S_{12}, \sigma_{1587} \rangle \Rightarrow \langle S_{13}, \sigma_{1587} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6349} = \frac{\langle x := x + 1, \sigma_{1587} \rangle \Rightarrow \sigma_{1588} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1587} \rangle \Rightarrow \langle S_6, \sigma_{1588} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6348} = \frac{\mathcal{T}_{6349}}{\langle S_{13}, \sigma_{1587} \rangle \Rightarrow \langle S_{15}, \sigma_{1588} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6350} = \frac{\langle S_6, \sigma_{1588} \rangle \Rightarrow \langle S_8, \sigma_{1588} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1588} \rangle \Rightarrow \langle S_9, \sigma_{1588} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6352} = \langle S_8, \sigma_{1588} \rangle \Rightarrow \langle y := y + 1, \sigma_{1588} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6351} = \frac{\mathcal{T}_{6352}}{\langle S_9, \sigma_{1588} \rangle \Rightarrow \langle S_{10}, \sigma_{1588} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6353} = \frac{\langle y := y + 1, \sigma_{1588} \rangle \Rightarrow \sigma_{1589} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1588} \rangle \Rightarrow \langle S_{11}, \sigma_{1589} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6354} = \langle S_{11}, \sigma_{1589} \rangle \Rightarrow \langle S_{12}, \sigma_{1589} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6355} = \langle S_{12}, \sigma_{1589} \rangle \Rightarrow \langle S_{13}, \sigma_{1589} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6357} = \frac{\langle x := x + 1, \sigma_{1589} \rangle \Rightarrow \sigma_{1590} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1589} \rangle \Rightarrow \langle S_6, \sigma_{1590} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6356} = \frac{\mathcal{T}_{6357}}{\langle S_{13}, \sigma_{1589} \rangle \Rightarrow \langle S_{15}, \sigma_{1590} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6358} = \frac{\langle S_6, \sigma_{1590} \rangle \Rightarrow \langle S_8, \sigma_{1590} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1590} \rangle \Rightarrow \langle S_9, \sigma_{1590} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6360} = \langle S_8, \sigma_{1590} \rangle \Rightarrow \langle y := y + 1, \sigma_{1590} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6359} = \frac{\mathcal{T}_{6360}}{\langle S_9, \sigma_{1590} \rangle \Rightarrow \langle S_{10}, \sigma_{1590} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6361} = \frac{\langle y := y + 1, \sigma_{1590} \rangle \Rightarrow \sigma_{1591} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1590} \rangle \Rightarrow \langle S_{11}, \sigma_{1591} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6362} = \langle S_{11}, \sigma_{1591} \rangle \Rightarrow \langle S_{12}, \sigma_{1591} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6363} = \langle S_{12}, \sigma_{1591} \rangle \Rightarrow \langle S_{13}, \sigma_{1591} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6365} = \frac{\langle x := x + 1, \sigma_{1591} \rangle \Rightarrow \sigma_{1592} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1591} \rangle \Rightarrow \langle S_6, \sigma_{1592} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6364} = \frac{\mathcal{T}_{6365}}{\langle S_{13}, \sigma_{1591} \rangle \Rightarrow \langle S_{15}, \sigma_{1592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6366} = \frac{\langle S_6, \sigma_{1592} \rangle \Rightarrow \langle S_8, \sigma_{1592} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1592} \rangle \Rightarrow \langle S_9, \sigma_{1592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6368} = \langle S_8, \sigma_{1592} \rangle \Rightarrow \langle y := y + 1, \sigma_{1592} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6367} = \frac{\mathcal{T}_{6368}}{\langle S_9, \sigma_{1592} \rangle \Rightarrow \langle S_{10}, \sigma_{1592} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6369} = \frac{\langle y := y + 1, \sigma_{1592} \rangle \Rightarrow \sigma_{1593} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1592} \rangle \Rightarrow \langle S_{11}, \sigma_{1593} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6370} = \langle S_{11}, \sigma_{1593} \rangle \Rightarrow \langle S_{12}, \sigma_{1593} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6371} = \langle S_{12}, \sigma_{1593} \rangle \Rightarrow \langle S_{13}, \sigma_{1593} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6373} = \frac{\langle x := x + 1, \sigma_{1593} \rangle \Rightarrow \sigma_{1594} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1593} \rangle \Rightarrow \langle S_6, \sigma_{1594} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6372} = \frac{\mathcal{T}_{6373}}{\langle S_{13}, \sigma_{1593} \rangle \Rightarrow \langle S_{15}, \sigma_{1594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6374} = \frac{\langle S_6, \sigma_{1594} \rangle \Rightarrow \langle S_8, \sigma_{1594} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1594} \rangle \Rightarrow \langle S_9, \sigma_{1594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6376} = \langle S_8, \sigma_{1594} \rangle \Rightarrow \langle y := y + 1, \sigma_{1594} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6375} = \frac{\mathcal{T}_{6376}}{\langle S_9, \sigma_{1594} \rangle \Rightarrow \langle S_{10}, \sigma_{1594} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6377} = \frac{\langle y := y + 1, \sigma_{1594} \rangle \Rightarrow \sigma_{1595} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1594} \rangle \Rightarrow \langle S_{11}, \sigma_{1595} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6378} = \langle S_{11}, \sigma_{1595} \rangle \Rightarrow \langle S_{12}, \sigma_{1595} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6379} = \langle S_{12}, \sigma_{1595} \rangle \Rightarrow \langle S_{13}, \sigma_{1595} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6381} = \frac{\langle x := x + 1, \sigma_{1595} \rangle \Rightarrow \sigma_{1596} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1595} \rangle \Rightarrow \langle S_6, \sigma_{1596} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6380} = \frac{\mathcal{T}_{6381}}{\langle S_{13}, \sigma_{1595} \rangle \Rightarrow \langle S_{15}, \sigma_{1596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6382} = \frac{\langle S_6, \sigma_{1596} \rangle \Rightarrow \langle S_8, \sigma_{1596} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1596} \rangle \Rightarrow \langle S_9, \sigma_{1596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6384} = \langle S_8, \sigma_{1596} \rangle \Rightarrow \langle y := y + 1, \sigma_{1596} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6383} = \frac{\mathcal{T}_{6384}}{\langle S_9, \sigma_{1596} \rangle \Rightarrow \langle S_{10}, \sigma_{1596} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6385} = \frac{\langle y := y + 1, \sigma_{1596} \rangle \Rightarrow \sigma_{1597} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1596} \rangle \Rightarrow \langle S_{11}, \sigma_{1597} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6386} = \langle S_{11}, \sigma_{1597} \rangle \Rightarrow \langle S_{12}, \sigma_{1597} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6387} = \langle S_{12}, \sigma_{1597} \rangle \Rightarrow \langle S_{13}, \sigma_{1597} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6389} = \frac{\langle x := x + 1, \sigma_{1597} \rangle \Rightarrow \sigma_{1598} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1597} \rangle \Rightarrow \langle S_6, \sigma_{1598} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6388} = \frac{\mathcal{T}_{6389}}{\langle S_{13}, \sigma_{1597} \rangle \Rightarrow \langle S_{15}, \sigma_{1598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6390} = \frac{\langle S_6, \sigma_{1598} \rangle \Rightarrow \langle S_8, \sigma_{1598} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1598} \rangle \Rightarrow \langle S_9, \sigma_{1598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6392} = \langle S_8, \sigma_{1598} \rangle \Rightarrow \langle y := y + 1, \sigma_{1598} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6391} = \frac{\mathcal{T}_{6392}}{\langle S_9, \sigma_{1598} \rangle \Rightarrow \langle S_{10}, \sigma_{1598} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6393} = \frac{\langle y := y + 1, \sigma_{1598} \rangle \Rightarrow \sigma_{1599} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1598} \rangle \Rightarrow \langle S_{11}, \sigma_{1599} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6394} = \langle S_{11}, \sigma_{1599} \rangle \Rightarrow \langle S_{12}, \sigma_{1599} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6395} = \langle S_{12}, \sigma_{1599} \rangle \Rightarrow \langle S_{13}, \sigma_{1599} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6397} = \frac{\langle x := x + 1, \sigma_{1599} \rangle \Rightarrow \sigma_{1600} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1599} \rangle \Rightarrow \langle S_6, \sigma_{1600} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6396} = \frac{\mathcal{T}_{6397}}{\langle S_{13}, \sigma_{1599} \rangle \Rightarrow \langle S_{15}, \sigma_{1600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6398} = \frac{\langle S_6, \sigma_{1600} \rangle \Rightarrow \langle S_8, \sigma_{1600} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1600} \rangle \Rightarrow \langle S_9, \sigma_{1600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6400} = \langle S_8, \sigma_{1600} \rangle \Rightarrow \langle y := y + 1, \sigma_{1600} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6399} = \frac{\mathcal{T}_{6400}}{\langle S_9, \sigma_{1600} \rangle \Rightarrow \langle S_{10}, \sigma_{1600} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6401} = \frac{\langle y := y + 1, \sigma_{1600} \rangle \Rightarrow \sigma_{1601} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1600} \rangle \Rightarrow \langle S_{11}, \sigma_{1601} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6402} = \langle S_{11}, \sigma_{1601} \rangle \Rightarrow \langle S_{12}, \sigma_{1601} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6403} = \langle S_{12}, \sigma_{1601} \rangle \Rightarrow \langle S_{13}, \sigma_{1601} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6405} = \frac{\langle x := x + 1, \sigma_{1601} \rangle \Rightarrow \sigma_{1602} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1601} \rangle \Rightarrow \langle S_6, \sigma_{1602} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6404} = \frac{\mathcal{T}_{6405}}{\langle S_{13}, \sigma_{1601} \rangle \Rightarrow \langle S_{15}, \sigma_{1602} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6406} = \frac{\langle S_6, \sigma_{1602} \rangle \Rightarrow \langle S_8, \sigma_{1602} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1602} \rangle \Rightarrow \langle S_9, \sigma_{1602} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6408} = \langle S_8, \sigma_{1602} \rangle \Rightarrow \langle y := y + 1, \sigma_{1602} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6407} = \frac{\mathcal{T}_{6408}}{\langle S_9, \sigma_{1602} \rangle \Rightarrow \langle S_{10}, \sigma_{1602} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6409} = \frac{\langle y := y + 1, \sigma_{1602} \rangle \Rightarrow \sigma_{1603} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1602} \rangle \Rightarrow \langle S_{11}, \sigma_{1603} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6410} = \langle S_{11}, \sigma_{1603} \rangle \Rightarrow \langle S_{12}, \sigma_{1603} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6411} = \langle S_{12}, \sigma_{1603} \rangle \Rightarrow \langle S_{13}, \sigma_{1603} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6413} = \frac{\langle x := x + 1, \sigma_{1603} \rangle \Rightarrow \sigma_{1604} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1603} \rangle \Rightarrow \langle S_6, \sigma_{1604} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6412} = \frac{\mathcal{T}_{6413}}{\langle S_{13}, \sigma_{1603} \rangle \Rightarrow \langle S_{15}, \sigma_{1604} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6414} = \frac{\langle S_6, \sigma_{1604} \rangle \Rightarrow \langle S_8, \sigma_{1604} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1604} \rangle \Rightarrow \langle S_9, \sigma_{1604} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6416} = \langle S_8, \sigma_{1604} \rangle \Rightarrow \langle y := y + 1, \sigma_{1604} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6415} = \frac{\mathcal{T}_{6416}}{\langle S_9, \sigma_{1604} \rangle \Rightarrow \langle S_{10}, \sigma_{1604} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6417} = \frac{\langle y := y + 1, \sigma_{1604} \rangle \Rightarrow \sigma_{1605} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1604} \rangle \Rightarrow \langle S_{11}, \sigma_{1605} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6418} = \langle S_{11}, \sigma_{1605} \rangle \Rightarrow \langle S_{12}, \sigma_{1605} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6419} = \langle S_{12}, \sigma_{1605} \rangle \Rightarrow \langle S_{13}, \sigma_{1605} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6421} = \frac{\langle x := x + 1, \sigma_{1605} \rangle \Rightarrow \sigma_{1606} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1605} \rangle \Rightarrow \langle S_6, \sigma_{1606} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6420} = \frac{\mathcal{T}_{6421}}{\langle S_{13}, \sigma_{1605} \rangle \Rightarrow \langle S_{15}, \sigma_{1606} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6422} = \frac{\langle S_6, \sigma_{1606} \rangle \Rightarrow \langle S_8, \sigma_{1606} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1606} \rangle \Rightarrow \langle S_9, \sigma_{1606} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6424} = \langle S_8, \sigma_{1606} \rangle \Rightarrow \langle y := y + 1, \sigma_{1606} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6423} = \frac{\mathcal{T}_{6424}}{\langle S_9, \sigma_{1606} \rangle \Rightarrow \langle S_{10}, \sigma_{1606} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6425} = \frac{\langle y := y + 1, \sigma_{1606} \rangle \Rightarrow \sigma_{1607} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1606} \rangle \Rightarrow \langle S_{11}, \sigma_{1607} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6426} = \langle S_{11}, \sigma_{1607} \rangle \Rightarrow \langle S_{12}, \sigma_{1607} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6427} = \langle S_{12}, \sigma_{1607} \rangle \Rightarrow \langle S_{13}, \sigma_{1607} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6429} = \frac{\langle x := x + 1, \sigma_{1607} \rangle \Rightarrow \sigma_{1608} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1607} \rangle \Rightarrow \langle S_6, \sigma_{1608} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6428} = \frac{\mathcal{T}_{6429}}{\langle S_{13}, \sigma_{1607} \rangle \Rightarrow \langle S_{15}, \sigma_{1608} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6430} = \frac{\langle S_6, \sigma_{1608} \rangle \Rightarrow \langle S_8, \sigma_{1608} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1608} \rangle \Rightarrow \langle S_9, \sigma_{1608} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6432} = \langle S_8, \sigma_{1608} \rangle \Rightarrow \langle y := y + 1, \sigma_{1608} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6431} = \frac{\mathcal{T}_{6432}}{\langle S_9, \sigma_{1608} \rangle \Rightarrow \langle S_{10}, \sigma_{1608} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6433} = \frac{\langle y := y + 1, \sigma_{1608} \rangle \Rightarrow \sigma_{1609} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1608} \rangle \Rightarrow \langle S_{11}, \sigma_{1609} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6434} = \langle S_{11}, \sigma_{1609} \rangle \Rightarrow \langle S_{12}, \sigma_{1609} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6435} = \langle S_{12}, \sigma_{1609} \rangle \Rightarrow \langle S_{13}, \sigma_{1609} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6437} = \frac{\langle x := x + 1, \sigma_{1609} \rangle \Rightarrow \sigma_{1610} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1609} \rangle \Rightarrow \langle S_6, \sigma_{1610} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6436} = \frac{\mathcal{T}_{6437}}{\langle S_{13}, \sigma_{1609} \rangle \Rightarrow \langle S_{15}, \sigma_{1610} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6438} = \frac{\langle S_6, \sigma_{1610} \rangle \Rightarrow \langle S_8, \sigma_{1610} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1610} \rangle \Rightarrow \langle S_9, \sigma_{1610} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6440} = \langle S_8, \sigma_{1610} \rangle \Rightarrow \langle y := y + 1, \sigma_{1610} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6439} = \frac{\mathcal{T}_{6440}}{\langle S_9, \sigma_{1610} \rangle \Rightarrow \langle S_{10}, \sigma_{1610} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6441} = \frac{\langle y := y + 1, \sigma_{1610} \rangle \Rightarrow \sigma_{1611} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1610} \rangle \Rightarrow \langle S_{11}, \sigma_{1611} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6442} = \langle S_{11}, \sigma_{1611} \rangle \Rightarrow \langle S_{12}, \sigma_{1611} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6443} = \langle S_{12}, \sigma_{1611} \rangle \Rightarrow \langle S_{13}, \sigma_{1611} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6445} = \frac{\langle x := x + 1, \sigma_{1611} \rangle \Rightarrow \sigma_{1612} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1611} \rangle \Rightarrow \langle S_6, \sigma_{1612} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6444} = \frac{\mathcal{T}_{6445}}{\langle S_{13}, \sigma_{1611} \rangle \Rightarrow \langle S_{15}, \sigma_{1612} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6446} = \frac{\langle S_6, \sigma_{1612} \rangle \Rightarrow \langle S_8, \sigma_{1612} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1612} \rangle \Rightarrow \langle S_9, \sigma_{1612} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6448} = \langle S_8, \sigma_{1612} \rangle \Rightarrow \langle y := y + 1, \sigma_{1612} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6447} = \frac{\mathcal{T}_{6448}}{\langle S_9, \sigma_{1612} \rangle \Rightarrow \langle S_{10}, \sigma_{1612} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6449} = \frac{\langle y := y + 1, \sigma_{1612} \rangle \Rightarrow \sigma_{1613} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1612} \rangle \Rightarrow \langle S_{11}, \sigma_{1613} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6450} = \langle S_{11}, \sigma_{1613} \rangle \Rightarrow \langle S_{12}, \sigma_{1613} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6451} = \langle S_{12}, \sigma_{1613} \rangle \Rightarrow \langle S_{13}, \sigma_{1613} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6453} = \frac{\langle x := x + 1, \sigma_{1613} \rangle \Rightarrow \sigma_{1614} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1613} \rangle \Rightarrow \langle S_6, \sigma_{1614} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6452} = \frac{\mathcal{T}_{6453}}{\langle S_{13}, \sigma_{1613} \rangle \Rightarrow \langle S_{15}, \sigma_{1614} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6454} = \frac{\langle S_6, \sigma_{1614} \rangle \Rightarrow \langle S_8, \sigma_{1614} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1614} \rangle \Rightarrow \langle S_9, \sigma_{1614} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6456} = \langle S_8, \sigma_{1614} \rangle \Rightarrow \langle y := y + 1, \sigma_{1614} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6455} = \frac{\mathcal{T}_{6456}}{\langle S_9, \sigma_{1614} \rangle \Rightarrow \langle S_{10}, \sigma_{1614} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6457} = \frac{\langle y := y + 1, \sigma_{1614} \rangle \Rightarrow \sigma_{1615} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1614} \rangle \Rightarrow \langle S_{11}, \sigma_{1615} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6458} = \langle S_{11}, \sigma_{1615} \rangle \Rightarrow \langle S_{12}, \sigma_{1615} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6459} = \langle S_{12}, \sigma_{1615} \rangle \Rightarrow \langle S_{13}, \sigma_{1615} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6461} = \frac{\langle x := x + 1, \sigma_{1615} \rangle \Rightarrow \sigma_{1616} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1615} \rangle \Rightarrow \langle S_6, \sigma_{1616} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6460} = \frac{\mathcal{T}_{6461}}{\langle S_{13}, \sigma_{1615} \rangle \Rightarrow \langle S_{15}, \sigma_{1616} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6462} = \frac{\langle S_6, \sigma_{1616} \rangle \Rightarrow \langle S_8, \sigma_{1616} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1616} \rangle \Rightarrow \langle S_9, \sigma_{1616} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6464} = \langle S_8, \sigma_{1616} \rangle \Rightarrow \langle y := y + 1, \sigma_{1616} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6463} = \frac{\mathcal{T}_{6464}}{\langle S_9, \sigma_{1616} \rangle \Rightarrow \langle S_{10}, \sigma_{1616} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6465} = \frac{\langle y := y + 1, \sigma_{1616} \rangle \Rightarrow \sigma_{1617} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1616} \rangle \Rightarrow \langle S_{11}, \sigma_{1617} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6466} = \langle S_{11}, \sigma_{1617} \rangle \Rightarrow \langle S_{12}, \sigma_{1617} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6467} = \langle S_{12}, \sigma_{1617} \rangle \Rightarrow \langle S_{13}, \sigma_{1617} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6469} = \frac{\langle x := x + 1, \sigma_{1617} \rangle \Rightarrow \sigma_{1618} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1617} \rangle \Rightarrow \langle S_6, \sigma_{1618} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6468} = \frac{\mathcal{T}_{6469}}{\langle S_{13}, \sigma_{1617} \rangle \Rightarrow \langle S_{15}, \sigma_{1618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6470} = \frac{\langle S_6, \sigma_{1618} \rangle \Rightarrow \langle S_8, \sigma_{1618} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1618} \rangle \Rightarrow \langle S_9, \sigma_{1618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6472} = \langle S_8, \sigma_{1618} \rangle \Rightarrow \langle y := y + 1, \sigma_{1618} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6471} = \frac{\mathcal{T}_{6472}}{\langle S_9, \sigma_{1618} \rangle \Rightarrow \langle S_{10}, \sigma_{1618} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6473} = \frac{\langle y := y + 1, \sigma_{1618} \rangle \Rightarrow \sigma_{1619} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1618} \rangle \Rightarrow \langle S_{11}, \sigma_{1619} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6474} = \langle S_{11}, \sigma_{1619} \rangle \Rightarrow \langle S_{12}, \sigma_{1619} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6475} = \langle S_{12}, \sigma_{1619} \rangle \Rightarrow \langle S_{13}, \sigma_{1619} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6477} = \frac{\langle x := x + 1, \sigma_{1619} \rangle \Rightarrow \sigma_{1620} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1619} \rangle \Rightarrow \langle S_6, \sigma_{1620} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6476} = \frac{\mathcal{T}_{6477}}{\langle S_{13}, \sigma_{1619} \rangle \Rightarrow \langle S_{15}, \sigma_{1620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6478} = \frac{\langle S_6, \sigma_{1620} \rangle \Rightarrow \langle S_8, \sigma_{1620} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1620} \rangle \Rightarrow \langle S_9, \sigma_{1620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6480} = \langle S_8, \sigma_{1620} \rangle \Rightarrow \langle y := y + 1, \sigma_{1620} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6479} = \frac{\mathcal{T}_{6480}}{\langle S_9, \sigma_{1620} \rangle \Rightarrow \langle S_{10}, \sigma_{1620} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6481} = \frac{\langle y := y + 1, \sigma_{1620} \rangle \Rightarrow \sigma_{1621} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1620} \rangle \Rightarrow \langle S_{11}, \sigma_{1621} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6482} = \langle S_{11}, \sigma_{1621} \rangle \Rightarrow \langle S_{12}, \sigma_{1621} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6483} = \langle S_{12}, \sigma_{1621} \rangle \Rightarrow \langle S_{13}, \sigma_{1621} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6485} = \frac{\langle x := x + 1, \sigma_{1621} \rangle \Rightarrow \sigma_{1622} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1621} \rangle \Rightarrow \langle S_6, \sigma_{1622} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6484} = \frac{\mathcal{T}_{6485}}{\langle S_{13}, \sigma_{1621} \rangle \Rightarrow \langle S_{15}, \sigma_{1622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6486} = \frac{\langle S_6, \sigma_{1622} \rangle \Rightarrow \langle S_8, \sigma_{1622} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1622} \rangle \Rightarrow \langle S_9, \sigma_{1622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6488} = \langle S_8, \sigma_{1622} \rangle \Rightarrow \langle y := y + 1, \sigma_{1622} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6487} = \frac{\mathcal{T}_{6488}}{\langle S_9, \sigma_{1622} \rangle \Rightarrow \langle S_{10}, \sigma_{1622} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6489} = \frac{\langle y := y + 1, \sigma_{1622} \rangle \Rightarrow \sigma_{1623} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1622} \rangle \Rightarrow \langle S_{11}, \sigma_{1623} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6490} = \langle S_{11}, \sigma_{1623} \rangle \Rightarrow \langle S_{12}, \sigma_{1623} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6491} = \langle S_{12}, \sigma_{1623} \rangle \Rightarrow \langle S_{13}, \sigma_{1623} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6493} = \frac{\langle x := x + 1, \sigma_{1623} \rangle \Rightarrow \sigma_{1624} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1623} \rangle \Rightarrow \langle S_6, \sigma_{1624} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6492} = \frac{\mathcal{T}_{6493}}{\langle S_{13}, \sigma_{1623} \rangle \Rightarrow \langle S_{15}, \sigma_{1624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6494} = \frac{\langle S_6, \sigma_{1624} \rangle \Rightarrow \langle S_8, \sigma_{1624} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1624} \rangle \Rightarrow \langle S_9, \sigma_{1624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6496} = \langle S_8, \sigma_{1624} \rangle \Rightarrow \langle y := y + 1, \sigma_{1624} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6495} = \frac{\mathcal{T}_{6496}}{\langle S_9, \sigma_{1624} \rangle \Rightarrow \langle S_{10}, \sigma_{1624} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6497} = \frac{\langle y := y + 1, \sigma_{1624} \rangle \Rightarrow \sigma_{1625} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1624} \rangle \Rightarrow \langle S_{11}, \sigma_{1625} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6498} = \langle S_{11}, \sigma_{1625} \rangle \Rightarrow \langle S_{12}, \sigma_{1625} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6499} = \langle S_{12}, \sigma_{1625} \rangle \Rightarrow \langle S_{13}, \sigma_{1625} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6501} = \frac{\langle x := x + 1, \sigma_{1625} \rangle \Rightarrow \sigma_{1626} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1625} \rangle \Rightarrow \langle S_6, \sigma_{1626} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6500} = \frac{\mathcal{T}_{6501}}{\langle S_{13}, \sigma_{1625} \rangle \Rightarrow \langle S_{15}, \sigma_{1626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6502} = \frac{\langle S_6, \sigma_{1626} \rangle \Rightarrow \langle S_8, \sigma_{1626} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1626} \rangle \Rightarrow \langle S_9, \sigma_{1626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6504} = \langle S_8, \sigma_{1626} \rangle \Rightarrow \langle y := y + 1, \sigma_{1626} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6503} = \frac{\mathcal{T}_{6504}}{\langle S_9, \sigma_{1626} \rangle \Rightarrow \langle S_{10}, \sigma_{1626} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6505} = \frac{\langle y := y + 1, \sigma_{1626} \rangle \Rightarrow \sigma_{1627} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1626} \rangle \Rightarrow \langle S_{11}, \sigma_{1627} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6506} = \langle S_{11}, \sigma_{1627} \rangle \Rightarrow \langle S_{12}, \sigma_{1627} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6507} = \langle S_{12}, \sigma_{1627} \rangle \Rightarrow \langle S_{13}, \sigma_{1627} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6509} = \frac{\langle x := x + 1, \sigma_{1627} \rangle \Rightarrow \sigma_{1628} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1627} \rangle \Rightarrow \langle S_6, \sigma_{1628} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6508} = \frac{\mathcal{T}_{6509}}{\langle S_{13}, \sigma_{1627} \rangle \Rightarrow \langle S_{15}, \sigma_{1628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6510} = \frac{\langle S_6, \sigma_{1628} \rangle \Rightarrow \langle S_8, \sigma_{1628} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1628} \rangle \Rightarrow \langle S_9, \sigma_{1628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6512} = \langle S_8, \sigma_{1628} \rangle \Rightarrow \langle y := y + 1, \sigma_{1628} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6511} = \frac{\mathcal{T}_{6512}}{\langle S_9, \sigma_{1628} \rangle \Rightarrow \langle S_{10}, \sigma_{1628} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6513} = \frac{\langle y := y + 1, \sigma_{1628} \rangle \Rightarrow \sigma_{1629} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1628} \rangle \Rightarrow \langle S_{11}, \sigma_{1629} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6514} = \langle S_{11}, \sigma_{1629} \rangle \Rightarrow \langle S_{12}, \sigma_{1629} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6515} = \langle S_{12}, \sigma_{1629} \rangle \Rightarrow \langle S_{13}, \sigma_{1629} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6517} = \frac{\langle x := x + 1, \sigma_{1629} \rangle \Rightarrow \sigma_{1630} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1629} \rangle \Rightarrow \langle S_6, \sigma_{1630} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6516} = \frac{\mathcal{T}_{6517}}{\langle S_{13}, \sigma_{1629} \rangle \Rightarrow \langle S_{15}, \sigma_{1630} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6518} = \frac{\langle S_6, \sigma_{1630} \rangle \Rightarrow \langle S_8, \sigma_{1630} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1630} \rangle \Rightarrow \langle S_9, \sigma_{1630} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6520} = \langle S_8, \sigma_{1630} \rangle \Rightarrow \langle y := y + 1, \sigma_{1630} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6519} = \frac{\mathcal{T}_{6520}}{\langle S_9, \sigma_{1630} \rangle \Rightarrow \langle S_{10}, \sigma_{1630} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6521} = \frac{\langle y := y + 1, \sigma_{1630} \rangle \Rightarrow \sigma_{1631} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1630} \rangle \Rightarrow \langle S_{11}, \sigma_{1631} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6522} = \langle S_{11}, \sigma_{1631} \rangle \Rightarrow \langle S_{12}, \sigma_{1631} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6523} = \langle S_{12}, \sigma_{1631} \rangle \Rightarrow \langle S_{13}, \sigma_{1631} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6525} = \frac{\langle x := x + 1, \sigma_{1631} \rangle \Rightarrow \sigma_{1632} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1631} \rangle \Rightarrow \langle S_6, \sigma_{1632} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6524} = \frac{\mathcal{T}_{6525}}{\langle S_{13}, \sigma_{1631} \rangle \Rightarrow \langle S_{15}, \sigma_{1632} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6526} = \frac{\langle S_6, \sigma_{1632} \rangle \Rightarrow \langle S_8, \sigma_{1632} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1632} \rangle \Rightarrow \langle S_9, \sigma_{1632} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6528} = \langle S_8, \sigma_{1632} \rangle \Rightarrow \langle y := y + 1, \sigma_{1632} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6527} = \frac{\mathcal{T}_{6528}}{\langle S_9, \sigma_{1632} \rangle \Rightarrow \langle S_{10}, \sigma_{1632} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6529} = \frac{\langle y := y + 1, \sigma_{1632} \rangle \Rightarrow \sigma_{1633} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1632} \rangle \Rightarrow \langle S_{11}, \sigma_{1633} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6530} = \langle S_{11}, \sigma_{1633} \rangle \Rightarrow \langle S_{12}, \sigma_{1633} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6531} = \langle S_{12}, \sigma_{1633} \rangle \Rightarrow \langle S_{13}, \sigma_{1633} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6533} = \frac{\langle x := x + 1, \sigma_{1633} \rangle \Rightarrow \sigma_{1634} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1633} \rangle \Rightarrow \langle S_6, \sigma_{1634} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6532} = \frac{\mathcal{T}_{6533}}{\langle S_{13}, \sigma_{1633} \rangle \Rightarrow \langle S_{15}, \sigma_{1634} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6534} = \frac{\langle S_6, \sigma_{1634} \rangle \Rightarrow \langle S_8, \sigma_{1634} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1634} \rangle \Rightarrow \langle S_9, \sigma_{1634} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6536} = \langle S_8, \sigma_{1634} \rangle \Rightarrow \langle y := y + 1, \sigma_{1634} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6535} = \frac{\mathcal{T}_{6536}}{\langle S_9, \sigma_{1634} \rangle \Rightarrow \langle S_{10}, \sigma_{1634} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6537} = \frac{\langle y := y + 1, \sigma_{1634} \rangle \Rightarrow \sigma_{1635} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1634} \rangle \Rightarrow \langle S_{11}, \sigma_{1635} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6538} = \langle S_{11}, \sigma_{1635} \rangle \Rightarrow \langle S_{12}, \sigma_{1635} \rangle \text{ [for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6539} = \langle S_{12}, \sigma_{1635} \rangle \Rightarrow \langle S_{13}, \sigma_{1635} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6541} = \frac{\langle x := x + 1, \sigma_{1635} \rangle \Rightarrow \sigma_{1636} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1635} \rangle \Rightarrow \langle S_6, \sigma_{1636} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6540} = \frac{\mathcal{T}_{6541}}{\langle S_{13}, \sigma_{1635} \rangle \Rightarrow \langle S_{15}, \sigma_{1636} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6542} = \frac{\langle S_6, \sigma_{1636} \rangle \Rightarrow \langle S_8, \sigma_{1636} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1636} \rangle \Rightarrow \langle S_9, \sigma_{1636} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6544} = \langle S_8, \sigma_{1636} \rangle \Rightarrow \langle y := y + 1, \sigma_{1636} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6543} = \frac{\mathcal{T}_{6544}}{\langle S_9, \sigma_{1636} \rangle \Rightarrow \langle S_{10}, \sigma_{1636} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6545} = \frac{\langle y := y + 1, \sigma_{1636} \rangle \Rightarrow \sigma_{1637} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1636} \rangle \Rightarrow \langle S_{11}, \sigma_{1637} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6546} = \langle S_{11}, \sigma_{1637} \rangle \Rightarrow \langle S_{12}, \sigma_{1637} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6547} = \langle S_{12}, \sigma_{1637} \rangle \Rightarrow \langle S_{13}, \sigma_{1637} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6549} = \frac{\langle x := x + 1, \sigma_{1637} \rangle \Rightarrow \sigma_{1638} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1637} \rangle \Rightarrow \langle S_6, \sigma_{1638} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6548} = \frac{\mathcal{T}_{6549}}{\langle S_{13}, \sigma_{1637} \rangle \Rightarrow \langle S_{15}, \sigma_{1638} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6550} = \frac{\langle S_6, \sigma_{1638} \rangle \Rightarrow \langle S_8, \sigma_{1638} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1638} \rangle \Rightarrow \langle S_9, \sigma_{1638} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6552} = \langle S_8, \sigma_{1638} \rangle \Rightarrow \langle y := y + 1, \sigma_{1638} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6551} = \frac{\mathcal{T}_{6552}}{\langle S_9, \sigma_{1638} \rangle \Rightarrow \langle S_{10}, \sigma_{1638} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6553} = \frac{\langle y := y + 1, \sigma_{1638} \rangle \Rightarrow \sigma_{1639} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1638} \rangle \Rightarrow \langle S_{11}, \sigma_{1639} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6554} = \langle S_{11}, \sigma_{1639} \rangle \Rightarrow \langle S_{12}, \sigma_{1639} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6555} = \langle S_{12}, \sigma_{1639} \rangle \Rightarrow \langle S_{13}, \sigma_{1639} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6557} = \frac{\langle x := x + 1, \sigma_{1639} \rangle \Rightarrow \sigma_{1640} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1639} \rangle \Rightarrow \langle S_6, \sigma_{1640} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6556} = \frac{\mathcal{T}_{6557}}{\langle S_{13}, \sigma_{1639} \rangle \Rightarrow \langle S_{15}, \sigma_{1640} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6558} = \frac{\langle S_6, \sigma_{1640} \rangle \Rightarrow \langle S_8, \sigma_{1640} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1640} \rangle \Rightarrow \langle S_9, \sigma_{1640} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6560} = \langle S_8, \sigma_{1640} \rangle \Rightarrow \langle y := y + 1, \sigma_{1640} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6559} = \frac{\mathcal{T}_{6560}}{\langle S_9, \sigma_{1640} \rangle \Rightarrow \langle S_{10}, \sigma_{1640} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6561} = \frac{\langle y := y + 1, \sigma_{1640} \rangle \Rightarrow \sigma_{1641} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1640} \rangle \Rightarrow \langle S_{11}, \sigma_{1641} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6562} = \langle S_{11}, \sigma_{1641} \rangle \Rightarrow \langle S_{12}, \sigma_{1641} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6563} = \langle S_{12}, \sigma_{1641} \rangle \Rightarrow \langle S_{13}, \sigma_{1641} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6565} = \frac{\langle x := x + 1, \sigma_{1641} \rangle \Rightarrow \sigma_{1642} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1641} \rangle \Rightarrow \langle S_6, \sigma_{1642} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6564} = \frac{\mathcal{T}_{6565}}{\langle S_{13}, \sigma_{1641} \rangle \Rightarrow \langle S_{15}, \sigma_{1642} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6566} = \frac{\langle S_6, \sigma_{1642} \rangle \Rightarrow \langle S_8, \sigma_{1642} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1642} \rangle \Rightarrow \langle S_9, \sigma_{1642} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{6568} = \langle S_8, \sigma_{1642} \rangle \Rightarrow \langle y := y + 1, \sigma_{1642} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6567} = \frac{\mathcal{T}_{6568}}{\langle S_9, \sigma_{1642} \rangle \Rightarrow \langle S_{10}, \sigma_{1642} \rangle} [\text{comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{6569} = \frac{\langle y := y + 1, \sigma_{1642} \rangle \Rightarrow \sigma_{1643} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1642} \rangle \Rightarrow \langle S_{11}, \sigma_{1643} \rangle} [\text{comp}_{\text{sos}}^2]$$

 $\Downarrow$

$$\mathcal{T}_{6570} = \langle S_{11}, \sigma_{1643} \rangle \Rightarrow \langle S_{12}, \sigma_{1643} \rangle [\text{for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{6571} = \langle S_{12}, \sigma_{1643} \rangle \Rightarrow \langle S_{13}, \sigma_{1643} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{6573} = \frac{\langle x := x + 1, \sigma_{1643} \rangle \Rightarrow \sigma_{1644} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1643} \rangle \Rightarrow \langle S_6, \sigma_{1644} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6572} = \frac{\mathcal{T}_{6573}}{\langle S_{13}, \sigma_{1643} \rangle \Rightarrow \langle S_{15}, \sigma_{1644} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6574} = \frac{\langle S_6, \sigma_{1644} \rangle \Rightarrow \langle S_8, \sigma_{1644} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1644} \rangle \Rightarrow \langle S_9, \sigma_{1644} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6576} = \langle S_8, \sigma_{1644} \rangle \Rightarrow \langle y := y + 1, \sigma_{1644} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6575} = \frac{\mathcal{T}_{6576}}{\langle S_9, \sigma_{1644} \rangle \Rightarrow \langle S_{10}, \sigma_{1644} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6577} = \frac{\langle y := y + 1, \sigma_{1644} \rangle \Rightarrow \sigma_{1645} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1644} \rangle \Rightarrow \langle S_{11}, \sigma_{1645} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6578} = \langle S_{11}, \sigma_{1645} \rangle \Rightarrow \langle S_{12}, \sigma_{1645} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6579} = \langle S_{12}, \sigma_{1645} \rangle \Rightarrow \langle S_{13}, \sigma_{1645} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6581} = \frac{\langle x := x + 1, \sigma_{1645} \rangle \Rightarrow \sigma_{1646} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1645} \rangle \Rightarrow \langle S_6, \sigma_{1646} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6580} = \frac{\mathcal{T}_{6581}}{\langle S_{13}, \sigma_{1645} \rangle \Rightarrow \langle S_{15}, \sigma_{1646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6582} = \frac{\langle S_6, \sigma_{1646} \rangle \Rightarrow \langle S_8, \sigma_{1646} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1646} \rangle \Rightarrow \langle S_9, \sigma_{1646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6584} = \langle S_8, \sigma_{1646} \rangle \Rightarrow \langle y := y + 1, \sigma_{1646} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6583} = \frac{\mathcal{T}_{6584}}{\langle S_9, \sigma_{1646} \rangle \Rightarrow \langle S_{10}, \sigma_{1646} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6585} = \frac{\langle y := y + 1, \sigma_{1646} \rangle \Rightarrow \sigma_{1647} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1646} \rangle \Rightarrow \langle S_{11}, \sigma_{1647} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6586} = \langle S_{11}, \sigma_{1647} \rangle \Rightarrow \langle S_{12}, \sigma_{1647} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6587} = \langle S_{12}, \sigma_{1647} \rangle \Rightarrow \langle S_{13}, \sigma_{1647} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6589} = \frac{\langle x := x + 1, \sigma_{1647} \rangle \Rightarrow \sigma_{1648} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1647} \rangle \Rightarrow \langle S_6, \sigma_{1648} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6588} = \frac{\mathcal{T}_{6589}}{\langle S_{13}, \sigma_{1647} \rangle \Rightarrow \langle S_{15}, \sigma_{1648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6590} = \frac{\langle S_6, \sigma_{1648} \rangle \Rightarrow \langle S_8, \sigma_{1648} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1648} \rangle \Rightarrow \langle S_9, \sigma_{1648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6592} = \langle S_8, \sigma_{1648} \rangle \Rightarrow \langle y := y + 1, \sigma_{1648} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6591} = \frac{\mathcal{T}_{6592}}{\langle S_9, \sigma_{1648} \rangle \Rightarrow \langle S_{10}, \sigma_{1648} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6593} = \frac{\langle y := y + 1, \sigma_{1648} \rangle \Rightarrow \sigma_{1649} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1648} \rangle \Rightarrow \langle S_{11}, \sigma_{1649} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6594} = \langle S_{11}, \sigma_{1649} \rangle \Rightarrow \langle S_{12}, \sigma_{1649} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6595} = \langle S_{12}, \sigma_{1649} \rangle \Rightarrow \langle S_{13}, \sigma_{1649} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6597} = \frac{\langle x := x + 1, \sigma_{1649} \rangle \Rightarrow \sigma_{1650} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1649} \rangle \Rightarrow \langle S_6, \sigma_{1650} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6596} = \frac{\mathcal{T}_{6597}}{\langle S_{13}, \sigma_{1649} \rangle \Rightarrow \langle S_{15}, \sigma_{1650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6598} = \frac{\langle S_6, \sigma_{1650} \rangle \Rightarrow \langle S_8, \sigma_{1650} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1650} \rangle \Rightarrow \langle S_9, \sigma_{1650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6600} = \langle S_8, \sigma_{1650} \rangle \Rightarrow \langle y := y + 1, \sigma_{1650} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6599} = \frac{\mathcal{T}_{6600}}{\langle S_9, \sigma_{1650} \rangle \Rightarrow \langle S_{10}, \sigma_{1650} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6601} = \frac{\langle y := y + 1, \sigma_{1650} \rangle \Rightarrow \sigma_{1651} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1650} \rangle \Rightarrow \langle S_{11}, \sigma_{1651} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6602} = \langle S_{11}, \sigma_{1651} \rangle \Rightarrow \langle S_{12}, \sigma_{1651} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6603} = \langle S_{12}, \sigma_{1651} \rangle \Rightarrow \langle S_{13}, \sigma_{1651} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6605} = \frac{\langle x := x + 1, \sigma_{1651} \rangle \Rightarrow \sigma_{1652} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1651} \rangle \Rightarrow \langle S_6, \sigma_{1652} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6604} = \frac{\mathcal{T}_{6605}}{\langle S_{13}, \sigma_{1651} \rangle \Rightarrow \langle S_{15}, \sigma_{1652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6606} = \frac{\langle S_6, \sigma_{1652} \rangle \Rightarrow \langle S_8, \sigma_{1652} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1652} \rangle \Rightarrow \langle S_9, \sigma_{1652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6608} = \langle S_8, \sigma_{1652} \rangle \Rightarrow \langle y := y + 1, \sigma_{1652} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6607} = \frac{\mathcal{T}_{6608}}{\langle S_9, \sigma_{1652} \rangle \Rightarrow \langle S_{10}, \sigma_{1652} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6609} = \frac{\langle y := y + 1, \sigma_{1652} \rangle \Rightarrow \sigma_{1653} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1652} \rangle \Rightarrow \langle S_{11}, \sigma_{1653} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6610} = \langle S_{11}, \sigma_{1653} \rangle \Rightarrow \langle S_{12}, \sigma_{1653} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6611} = \langle S_{12}, \sigma_{1653} \rangle \Rightarrow \langle S_{13}, \sigma_{1653} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6613} = \frac{\langle x := x + 1, \sigma_{1653} \rangle \Rightarrow \sigma_{1654} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1653} \rangle \Rightarrow \langle S_6, \sigma_{1654} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6612} = \frac{\mathcal{T}_{6613}}{\langle S_{13}, \sigma_{1653} \rangle \Rightarrow \langle S_{15}, \sigma_{1654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6614} = \frac{\langle S_6, \sigma_{1654} \rangle \Rightarrow \langle S_8, \sigma_{1654} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1654} \rangle \Rightarrow \langle S_9, \sigma_{1654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6616} = \langle S_8, \sigma_{1654} \rangle \Rightarrow \langle y := y + 1, \sigma_{1654} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6615} = \frac{\mathcal{T}_{6616}}{\langle S_9, \sigma_{1654} \rangle \Rightarrow \langle S_{10}, \sigma_{1654} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6617} = \frac{\langle y := y + 1, \sigma_{1654} \rangle \Rightarrow \sigma_{1655} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1654} \rangle \Rightarrow \langle S_{11}, \sigma_{1655} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6618} = \langle S_{11}, \sigma_{1655} \rangle \Rightarrow \langle S_{12}, \sigma_{1655} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6619} = \langle S_{12}, \sigma_{1655} \rangle \Rightarrow \langle S_{13}, \sigma_{1655} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6621} = \frac{\langle x := x + 1, \sigma_{1655} \rangle \Rightarrow \sigma_{1656} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1655} \rangle \Rightarrow \langle S_6, \sigma_{1656} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6620} = \frac{\mathcal{T}_{6621}}{\langle S_{13}, \sigma_{1655} \rangle \Rightarrow \langle S_{15}, \sigma_{1656} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6622} = \frac{\langle S_6, \sigma_{1656} \rangle \Rightarrow \langle S_8, \sigma_{1656} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1656} \rangle \Rightarrow \langle S_9, \sigma_{1656} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6624} = \langle S_8, \sigma_{1656} \rangle \Rightarrow \langle y := y + 1, \sigma_{1656} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6623} = \frac{\mathcal{T}_{6624}}{\langle S_9, \sigma_{1656} \rangle \Rightarrow \langle S_{10}, \sigma_{1656} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6625} = \frac{\langle y := y + 1, \sigma_{1656} \rangle \Rightarrow \sigma_{1657} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1656} \rangle \Rightarrow \langle S_{11}, \sigma_{1657} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6626} = \langle S_{11}, \sigma_{1657} \rangle \Rightarrow \langle S_{12}, \sigma_{1657} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6627} = \langle S_{12}, \sigma_{1657} \rangle \Rightarrow \langle S_{13}, \sigma_{1657} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6629} = \frac{\langle x := x + 1, \sigma_{1657} \rangle \Rightarrow \sigma_{1658} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1657} \rangle \Rightarrow \langle S_6, \sigma_{1658} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6628} = \frac{\mathcal{T}_{6629}}{\langle S_{13}, \sigma_{1657} \rangle \Rightarrow \langle S_{15}, \sigma_{1658} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6630} = \frac{\langle S_6, \sigma_{1658} \rangle \Rightarrow \langle S_8, \sigma_{1658} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1658} \rangle \Rightarrow \langle S_9, \sigma_{1658} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6632} = \langle S_8, \sigma_{1658} \rangle \Rightarrow \langle y := y + 1, \sigma_{1658} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6631} = \frac{\mathcal{T}_{6632}}{\langle S_9, \sigma_{1658} \rangle \Rightarrow \langle S_{10}, \sigma_{1658} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6633} = \frac{\langle y := y + 1, \sigma_{1658} \rangle \Rightarrow \sigma_{1659} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1658} \rangle \Rightarrow \langle S_{11}, \sigma_{1659} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6634} = \langle S_{11}, \sigma_{1659} \rangle \Rightarrow \langle S_{12}, \sigma_{1659} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6635} = \langle S_{12}, \sigma_{1659} \rangle \Rightarrow \langle S_{13}, \sigma_{1659} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6637} = \frac{\langle x := x + 1, \sigma_{1659} \rangle \Rightarrow \sigma_{1660} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1659} \rangle \Rightarrow \langle S_6, \sigma_{1660} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6636} = \frac{\mathcal{T}_{6637}}{\langle S_{13}, \sigma_{1659} \rangle \Rightarrow \langle S_{15}, \sigma_{1660} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6638} = \frac{\langle S_6, \sigma_{1660} \rangle \Rightarrow \langle S_8, \sigma_{1660} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1660} \rangle \Rightarrow \langle S_9, \sigma_{1660} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6640} = \langle S_8, \sigma_{1660} \rangle \Rightarrow \langle y := y + 1, \sigma_{1660} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6639} = \frac{\mathcal{T}_{6640}}{\langle S_9, \sigma_{1660} \rangle \Rightarrow \langle S_{10}, \sigma_{1660} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6641} = \frac{\langle y := y + 1, \sigma_{1660} \rangle \Rightarrow \sigma_{1661} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1660} \rangle \Rightarrow \langle S_{11}, \sigma_{1661} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6642} = \langle S_{11}, \sigma_{1661} \rangle \Rightarrow \langle S_{12}, \sigma_{1661} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6643} = \langle S_{12}, \sigma_{1661} \rangle \Rightarrow \langle S_{13}, \sigma_{1661} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6645} = \frac{\langle x := x + 1, \sigma_{1661} \rangle \Rightarrow \sigma_{1662} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1661} \rangle \Rightarrow \langle S_6, \sigma_{1662} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6644} = \frac{\mathcal{T}_{6645}}{\langle S_{13}, \sigma_{1661} \rangle \Rightarrow \langle S_{15}, \sigma_{1662} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6646} = \frac{\langle S_6, \sigma_{1662} \rangle \Rightarrow \langle S_8, \sigma_{1662} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1662} \rangle \Rightarrow \langle S_9, \sigma_{1662} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6648} = \langle S_8, \sigma_{1662} \rangle \Rightarrow \langle y := y + 1, \sigma_{1662} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6647} = \frac{\mathcal{T}_{6648}}{\langle S_9, \sigma_{1662} \rangle \Rightarrow \langle S_{10}, \sigma_{1662} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6649} = \frac{\langle y := y + 1, \sigma_{1662} \rangle \Rightarrow \sigma_{1663} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1662} \rangle \Rightarrow \langle S_{11}, \sigma_{1663} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6650} = \langle S_{11}, \sigma_{1663} \rangle \Rightarrow \langle S_{12}, \sigma_{1663} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6651} = \langle S_{12}, \sigma_{1663} \rangle \Rightarrow \langle S_{13}, \sigma_{1663} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6653} = \frac{\langle x := x + 1, \sigma_{1663} \rangle \Rightarrow \sigma_{1664} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1663} \rangle \Rightarrow \langle S_6, \sigma_{1664} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6652} = \frac{\mathcal{T}_{6653}}{\langle S_{13}, \sigma_{1663} \rangle \Rightarrow \langle S_{15}, \sigma_{1664} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6654} = \frac{\langle S_6, \sigma_{1664} \rangle \Rightarrow \langle S_8, \sigma_{1664} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1664} \rangle \Rightarrow \langle S_9, \sigma_{1664} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6656} = \langle S_8, \sigma_{1664} \rangle \Rightarrow \langle y := y + 1, \sigma_{1664} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6655} = \frac{\mathcal{T}_{6656}}{\langle S_9, \sigma_{1664} \rangle \Rightarrow \langle S_{10}, \sigma_{1664} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6657} = \frac{\langle y := y + 1, \sigma_{1664} \rangle \Rightarrow \sigma_{1665} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1664} \rangle \Rightarrow \langle S_{11}, \sigma_{1665} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6658} = \langle S_{11}, \sigma_{1665} \rangle \Rightarrow \langle S_{12}, \sigma_{1665} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6659} = \langle S_{12}, \sigma_{1665} \rangle \Rightarrow \langle S_{13}, \sigma_{1665} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6661} = \frac{\langle x := x + 1, \sigma_{1665} \rangle \Rightarrow \sigma_{1666} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1665} \rangle \Rightarrow \langle S_6, \sigma_{1666} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6660} = \frac{\mathcal{T}_{6661}}{\langle S_{13}, \sigma_{1665} \rangle \Rightarrow \langle S_{15}, \sigma_{1666} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6662} = \frac{\langle S_6, \sigma_{1666} \rangle \Rightarrow \langle S_8, \sigma_{1666} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1666} \rangle \Rightarrow \langle S_9, \sigma_{1666} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6664} = \langle S_8, \sigma_{1666} \rangle \Rightarrow \langle y := y + 1, \sigma_{1666} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6663} = \frac{\mathcal{T}_{6664}}{\langle S_9, \sigma_{1666} \rangle \Rightarrow \langle S_{10}, \sigma_{1666} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6665} = \frac{\langle y := y + 1, \sigma_{1666} \rangle \Rightarrow \sigma_{1667} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1666} \rangle \Rightarrow \langle S_{11}, \sigma_{1667} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6666} = \langle S_{11}, \sigma_{1667} \rangle \Rightarrow \langle S_{12}, \sigma_{1667} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6667} = \langle S_{12}, \sigma_{1667} \rangle \Rightarrow \langle S_{13}, \sigma_{1667} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6669} = \frac{\langle x := x + 1, \sigma_{1667} \rangle \Rightarrow \sigma_{1668} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1667} \rangle \Rightarrow \langle S_6, \sigma_{1668} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6668} = \frac{\mathcal{T}_{6669}}{\langle S_{13}, \sigma_{1667} \rangle \Rightarrow \langle S_{15}, \sigma_{1668} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6670} = \frac{\langle S_6, \sigma_{1668} \rangle \Rightarrow \langle S_8, \sigma_{1668} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1668} \rangle \Rightarrow \langle S_9, \sigma_{1668} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6672} = \langle S_8, \sigma_{1668} \rangle \Rightarrow \langle y := y + 1, \sigma_{1668} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6671} = \frac{\mathcal{T}_{6672}}{\langle S_9, \sigma_{1668} \rangle \Rightarrow \langle S_{10}, \sigma_{1668} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6673} = \frac{\langle y := y + 1, \sigma_{1668} \rangle \Rightarrow \sigma_{1669} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1668} \rangle \Rightarrow \langle S_{11}, \sigma_{1669} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6674} = \langle S_{11}, \sigma_{1669} \rangle \Rightarrow \langle S_{12}, \sigma_{1669} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6675} = \langle S_{12}, \sigma_{1669} \rangle \Rightarrow \langle S_{13}, \sigma_{1669} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6677} = \frac{\langle x := x + 1, \sigma_{1669} \rangle \Rightarrow \sigma_{1670} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1669} \rangle \Rightarrow \langle S_6, \sigma_{1670} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6676} = \frac{\mathcal{T}_{6677}}{\langle S_{13}, \sigma_{1669} \rangle \Rightarrow \langle S_{15}, \sigma_{1670} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6678} = \frac{\langle S_6, \sigma_{1670} \rangle \Rightarrow \langle S_8, \sigma_{1670} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1670} \rangle \Rightarrow \langle S_9, \sigma_{1670} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6680} = \langle S_8, \sigma_{1670} \rangle \Rightarrow \langle y := y + 1, \sigma_{1670} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6679} = \frac{\mathcal{T}_{6680}}{\langle S_9, \sigma_{1670} \rangle \Rightarrow \langle S_{10}, \sigma_{1670} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6681} = \frac{\langle y := y + 1, \sigma_{1670} \rangle \Rightarrow \sigma_{1671} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1670} \rangle \Rightarrow \langle S_{11}, \sigma_{1671} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6682} = \langle S_{11}, \sigma_{1671} \rangle \Rightarrow \langle S_{12}, \sigma_{1671} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6683} = \langle S_{12}, \sigma_{1671} \rangle \Rightarrow \langle S_{13}, \sigma_{1671} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6685} = \frac{\langle x := x + 1, \sigma_{1671} \rangle \Rightarrow \sigma_{1672} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1671} \rangle \Rightarrow \langle S_6, \sigma_{1672} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6684} = \frac{\mathcal{T}_{6685}}{\langle S_{13}, \sigma_{1671} \rangle \Rightarrow \langle S_{15}, \sigma_{1672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6686} = \frac{\langle S_6, \sigma_{1672} \rangle \Rightarrow \langle S_8, \sigma_{1672} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1672} \rangle \Rightarrow \langle S_9, \sigma_{1672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6688} = \langle S_8, \sigma_{1672} \rangle \Rightarrow \langle y := y + 1, \sigma_{1672} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6687} = \frac{\mathcal{T}_{6688}}{\langle S_9, \sigma_{1672} \rangle \Rightarrow \langle S_{10}, \sigma_{1672} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6689} = \frac{\langle y := y + 1, \sigma_{1672} \rangle \Rightarrow \sigma_{1673} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1672} \rangle \Rightarrow \langle S_{11}, \sigma_{1673} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6690} = \langle S_{11}, \sigma_{1673} \rangle \Rightarrow \langle S_{12}, \sigma_{1673} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6691} = \langle S_{12}, \sigma_{1673} \rangle \Rightarrow \langle S_{13}, \sigma_{1673} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6693} = \frac{\langle x := x + 1, \sigma_{1673} \rangle \Rightarrow \sigma_{1674} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1673} \rangle \Rightarrow \langle S_6, \sigma_{1674} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6692} = \frac{\mathcal{T}_{6693}}{\langle S_{13}, \sigma_{1673} \rangle \Rightarrow \langle S_{15}, \sigma_{1674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6694} = \frac{\langle S_6, \sigma_{1674} \rangle \Rightarrow \langle S_8, \sigma_{1674} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1674} \rangle \Rightarrow \langle S_9, \sigma_{1674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6696} = \langle S_8, \sigma_{1674} \rangle \Rightarrow \langle y := y + 1, \sigma_{1674} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6695} = \frac{\mathcal{T}_{6696}}{\langle S_9, \sigma_{1674} \rangle \Rightarrow \langle S_{10}, \sigma_{1674} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6697} = \frac{\langle y := y + 1, \sigma_{1674} \rangle \Rightarrow \sigma_{1675} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1674} \rangle \Rightarrow \langle S_{11}, \sigma_{1675} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6698} = \langle S_{11}, \sigma_{1675} \rangle \Rightarrow \langle S_{12}, \sigma_{1675} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6699} = \langle S_{12}, \sigma_{1675} \rangle \Rightarrow \langle S_{13}, \sigma_{1675} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6701} = \frac{\langle x := x + 1, \sigma_{1675} \rangle \Rightarrow \sigma_{1676} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1675} \rangle \Rightarrow \langle S_6, \sigma_{1676} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6700} = \frac{\mathcal{T}_{6701}}{\langle S_{13}, \sigma_{1675} \rangle \Rightarrow \langle S_{15}, \sigma_{1676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6702} = \frac{\langle S_6, \sigma_{1676} \rangle \Rightarrow \langle S_8, \sigma_{1676} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1676} \rangle \Rightarrow \langle S_9, \sigma_{1676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6704} = \langle S_8, \sigma_{1676} \rangle \Rightarrow \langle y := y + 1, \sigma_{1676} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6703} = \frac{\mathcal{T}_{6704}}{\langle S_9, \sigma_{1676} \rangle \Rightarrow \langle S_{10}, \sigma_{1676} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6705} = \frac{\langle y := y + 1, \sigma_{1676} \rangle \Rightarrow \sigma_{1677} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1676} \rangle \Rightarrow \langle S_{11}, \sigma_{1677} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6706} = \langle S_{11}, \sigma_{1677} \rangle \Rightarrow \langle S_{12}, \sigma_{1677} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6707} = \langle S_{12}, \sigma_{1677} \rangle \Rightarrow \langle S_{13}, \sigma_{1677} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6709} = \frac{\langle x := x + 1, \sigma_{1677} \rangle \Rightarrow \sigma_{1678} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1677} \rangle \Rightarrow \langle S_6, \sigma_{1678} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6708} = \frac{\mathcal{T}_{6709}}{\langle S_{13}, \sigma_{1677} \rangle \Rightarrow \langle S_{15}, \sigma_{1678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6710} = \frac{\langle S_6, \sigma_{1678} \rangle \Rightarrow \langle S_8, \sigma_{1678} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1678} \rangle \Rightarrow \langle S_9, \sigma_{1678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6712} = \langle S_8, \sigma_{1678} \rangle \Rightarrow \langle y := y + 1, \sigma_{1678} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6711} = \frac{\mathcal{T}_{6712}}{\langle S_9, \sigma_{1678} \rangle \Rightarrow \langle S_{10}, \sigma_{1678} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6713} = \frac{\langle y := y + 1, \sigma_{1678} \rangle \Rightarrow \sigma_{1679} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1678} \rangle \Rightarrow \langle S_{11}, \sigma_{1679} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6714} = \langle S_{11}, \sigma_{1679} \rangle \Rightarrow \langle S_{12}, \sigma_{1679} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6715} = \langle S_{12}, \sigma_{1679} \rangle \Rightarrow \langle S_{13}, \sigma_{1679} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6717} = \frac{\langle x := x + 1, \sigma_{1679} \rangle \Rightarrow \sigma_{1680} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1679} \rangle \Rightarrow \langle S_6, \sigma_{1680} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6716} = \frac{\mathcal{T}_{6717}}{\langle S_{13}, \sigma_{1679} \rangle \Rightarrow \langle S_{15}, \sigma_{1680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6718} = \frac{\langle S_6, \sigma_{1680} \rangle \Rightarrow \langle S_8, \sigma_{1680} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1680} \rangle \Rightarrow \langle S_9, \sigma_{1680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6720} = \langle S_8, \sigma_{1680} \rangle \Rightarrow \langle y := y + 1, \sigma_{1680} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6719} = \frac{\mathcal{T}_{6720}}{\langle S_9, \sigma_{1680} \rangle \Rightarrow \langle S_{10}, \sigma_{1680} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6721} = \frac{\langle y := y + 1, \sigma_{1680} \rangle \Rightarrow \sigma_{1681} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1680} \rangle \Rightarrow \langle S_{11}, \sigma_{1681} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6722} = \langle S_{11}, \sigma_{1681} \rangle \Rightarrow \langle S_{12}, \sigma_{1681} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6723} = \langle S_{12}, \sigma_{1681} \rangle \Rightarrow \langle S_{13}, \sigma_{1681} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6725} = \frac{\langle x := x + 1, \sigma_{1681} \rangle \Rightarrow \sigma_{1682} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1681} \rangle \Rightarrow \langle S_6, \sigma_{1682} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6724} = \frac{\mathcal{T}_{6725}}{\langle S_{13}, \sigma_{1681} \rangle \Rightarrow \langle S_{15}, \sigma_{1682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6726} = \frac{\langle S_6, \sigma_{1682} \rangle \Rightarrow \langle S_8, \sigma_{1682} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1682} \rangle \Rightarrow \langle S_9, \sigma_{1682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6728} = \langle S_8, \sigma_{1682} \rangle \Rightarrow \langle y := y + 1, \sigma_{1682} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6727} = \frac{\mathcal{T}_{6728}}{\langle S_9, \sigma_{1682} \rangle \Rightarrow \langle S_{10}, \sigma_{1682} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6729} = \frac{\langle y := y + 1, \sigma_{1682} \rangle \Rightarrow \sigma_{1683} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1682} \rangle \Rightarrow \langle S_{11}, \sigma_{1683} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6730} = \langle S_{11}, \sigma_{1683} \rangle \Rightarrow \langle S_{12}, \sigma_{1683} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6731} = \langle S_{12}, \sigma_{1683} \rangle \Rightarrow \langle S_{13}, \sigma_{1683} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6733} = \frac{\langle x := x + 1, \sigma_{1683} \rangle \Rightarrow \sigma_{1684} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1683} \rangle \Rightarrow \langle S_6, \sigma_{1684} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6732} = \frac{\mathcal{T}_{6733}}{\langle S_{13}, \sigma_{1683} \rangle \Rightarrow \langle S_{15}, \sigma_{1684} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6734} = \frac{\langle S_6, \sigma_{1684} \rangle \Rightarrow \langle S_8, \sigma_{1684} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1684} \rangle \Rightarrow \langle S_9, \sigma_{1684} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6736} = \langle S_8, \sigma_{1684} \rangle \Rightarrow \langle y := y + 1, \sigma_{1684} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6735} = \frac{\mathcal{T}_{6736}}{\langle S_9, \sigma_{1684} \rangle \Rightarrow \langle S_{10}, \sigma_{1684} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6737} = \frac{\langle y := y + 1, \sigma_{1684} \rangle \Rightarrow \sigma_{1685} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1684} \rangle \Rightarrow \langle S_{11}, \sigma_{1685} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6738} = \langle S_{11}, \sigma_{1685} \rangle \Rightarrow \langle S_{12}, \sigma_{1685} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6739} = \langle S_{12}, \sigma_{1685} \rangle \Rightarrow \langle S_{13}, \sigma_{1685} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6741} = \frac{\langle x := x + 1, \sigma_{1685} \rangle \Rightarrow \sigma_{1686} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1685} \rangle \Rightarrow \langle S_6, \sigma_{1686} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6740} = \frac{\mathcal{T}_{6741}}{\langle S_{13}, \sigma_{1685} \rangle \Rightarrow \langle S_{15}, \sigma_{1686} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6742} = \frac{\langle S_6, \sigma_{1686} \rangle \Rightarrow \langle S_8, \sigma_{1686} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1686} \rangle \Rightarrow \langle S_9, \sigma_{1686} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6744} = \langle S_8, \sigma_{1686} \rangle \Rightarrow \langle y := y + 1, \sigma_{1686} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6743} = \frac{\mathcal{T}_{6744}}{\langle S_9, \sigma_{1686} \rangle \Rightarrow \langle S_{10}, \sigma_{1686} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6745} = \frac{\langle y := y + 1, \sigma_{1686} \rangle \Rightarrow \sigma_{1687} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1686} \rangle \Rightarrow \langle S_{11}, \sigma_{1687} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6746} = \langle S_{11}, \sigma_{1687} \rangle \Rightarrow \langle S_{12}, \sigma_{1687} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6747} = \langle S_{12}, \sigma_{1687} \rangle \Rightarrow \langle S_{13}, \sigma_{1687} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6749} = \frac{\langle x := x + 1, \sigma_{1687} \rangle \Rightarrow \sigma_{1688} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1687} \rangle \Rightarrow \langle S_6, \sigma_{1688} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6748} = \frac{\mathcal{T}_{6749}}{\langle S_{13}, \sigma_{1687} \rangle \Rightarrow \langle S_{15}, \sigma_{1688} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6750} = \frac{\langle S_6, \sigma_{1688} \rangle \Rightarrow \langle S_8, \sigma_{1688} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1688} \rangle \Rightarrow \langle S_9, \sigma_{1688} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6752} = \langle S_8, \sigma_{1688} \rangle \Rightarrow \langle y := y + 1, \sigma_{1688} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6751} = \frac{\mathcal{T}_{6752}}{\langle S_9, \sigma_{1688} \rangle \Rightarrow \langle S_{10}, \sigma_{1688} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6753} = \frac{\langle y := y + 1, \sigma_{1688} \rangle \Rightarrow \sigma_{1689} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1688} \rangle \Rightarrow \langle S_{11}, \sigma_{1689} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6754} = \langle S_{11}, \sigma_{1689} \rangle \Rightarrow \langle S_{12}, \sigma_{1689} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6755} = \langle S_{12}, \sigma_{1689} \rangle \Rightarrow \langle S_{13}, \sigma_{1689} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6757} = \frac{\langle x := x + 1, \sigma_{1689} \rangle \Rightarrow \sigma_{1690} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1689} \rangle \Rightarrow \langle S_6, \sigma_{1690} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6756} = \frac{\mathcal{T}_{6757}}{\langle S_{13}, \sigma_{1689} \rangle \Rightarrow \langle S_{15}, \sigma_{1690} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6758} = \frac{\langle S_6, \sigma_{1690} \rangle \Rightarrow \langle S_8, \sigma_{1690} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1690} \rangle \Rightarrow \langle S_9, \sigma_{1690} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6760} = \langle S_8, \sigma_{1690} \rangle \Rightarrow \langle y := y + 1, \sigma_{1690} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6759} = \frac{\mathcal{T}_{6760}}{\langle S_9, \sigma_{1690} \rangle \Rightarrow \langle S_{10}, \sigma_{1690} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6761} = \frac{\langle y := y + 1, \sigma_{1690} \rangle \Rightarrow \sigma_{1691} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1690} \rangle \Rightarrow \langle S_{11}, \sigma_{1691} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6762} = \langle S_{11}, \sigma_{1691} \rangle \Rightarrow \langle S_{12}, \sigma_{1691} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6763} = \langle S_{12}, \sigma_{1691} \rangle \Rightarrow \langle S_{13}, \sigma_{1691} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6765} = \frac{\langle x := x + 1, \sigma_{1691} \rangle \Rightarrow \sigma_{1692} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1691} \rangle \Rightarrow \langle S_6, \sigma_{1692} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6764} = \frac{\mathcal{T}_{6765}}{\langle S_{13}, \sigma_{1691} \rangle \Rightarrow \langle S_{15}, \sigma_{1692} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6766} = \frac{\langle S_6, \sigma_{1692} \rangle \Rightarrow \langle S_8, \sigma_{1692} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1692} \rangle \Rightarrow \langle S_9, \sigma_{1692} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6768} = \langle S_8, \sigma_{1692} \rangle \Rightarrow \langle y := y + 1, \sigma_{1692} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6767} = \frac{\mathcal{T}_{6768}}{\langle S_9, \sigma_{1692} \rangle \Rightarrow \langle S_{10}, \sigma_{1692} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6769} = \frac{\langle y := y + 1, \sigma_{1692} \rangle \Rightarrow \sigma_{1693} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1692} \rangle \Rightarrow \langle S_{11}, \sigma_{1693} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6770} = \langle S_{11}, \sigma_{1693} \rangle \Rightarrow \langle S_{12}, \sigma_{1693} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6771} = \langle S_{12}, \sigma_{1693} \rangle \Rightarrow \langle S_{13}, \sigma_{1693} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6773} = \frac{\langle x := x + 1, \sigma_{1693} \rangle \Rightarrow \sigma_{1694} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1693} \rangle \Rightarrow \langle S_6, \sigma_{1694} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6772} = \frac{\mathcal{T}_{6773}}{\langle S_{13}, \sigma_{1693} \rangle \Rightarrow \langle S_{15}, \sigma_{1694} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6774} = \frac{\langle S_6, \sigma_{1694} \rangle \Rightarrow \langle S_8, \sigma_{1694} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1694} \rangle \Rightarrow \langle S_9, \sigma_{1694} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6776} = \langle S_8, \sigma_{1694} \rangle \Rightarrow \langle y := y + 1, \sigma_{1694} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6775} = \frac{\mathcal{T}_{6776}}{\langle S_9, \sigma_{1694} \rangle \Rightarrow \langle S_{10}, \sigma_{1694} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6777} = \frac{\langle y := y + 1, \sigma_{1694} \rangle \Rightarrow \sigma_{1695} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1694} \rangle \Rightarrow \langle S_{11}, \sigma_{1695} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6778} = \langle S_{11}, \sigma_{1695} \rangle \Rightarrow \langle S_{12}, \sigma_{1695} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6779} = \langle S_{12}, \sigma_{1695} \rangle \Rightarrow \langle S_{13}, \sigma_{1695} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6781} = \frac{\langle x := x + 1, \sigma_{1695} \rangle \Rightarrow \sigma_{1696} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1695} \rangle \Rightarrow \langle S_6, \sigma_{1696} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6780} = \frac{\mathcal{T}_{6781}}{\langle S_{13}, \sigma_{1695} \rangle \Rightarrow \langle S_{15}, \sigma_{1696} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6782} = \frac{\langle S_6, \sigma_{1696} \rangle \Rightarrow \langle S_8, \sigma_{1696} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1696} \rangle \Rightarrow \langle S_9, \sigma_{1696} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6784} = \langle S_8, \sigma_{1696} \rangle \Rightarrow \langle y := y + 1, \sigma_{1696} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6783} = \frac{\mathcal{T}_{6784}}{\langle S_9, \sigma_{1696} \rangle \Rightarrow \langle S_{10}, \sigma_{1696} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6785} = \frac{\langle y := y + 1, \sigma_{1696} \rangle \Rightarrow \sigma_{1697} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1696} \rangle \Rightarrow \langle S_{11}, \sigma_{1697} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6786} = \langle S_{11}, \sigma_{1697} \rangle \Rightarrow \langle S_{12}, \sigma_{1697} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6787} = \langle S_{12}, \sigma_{1697} \rangle \Rightarrow \langle S_{13}, \sigma_{1697} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6789} = \frac{\langle x := x + 1, \sigma_{1697} \rangle \Rightarrow \sigma_{1698} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1697} \rangle \Rightarrow \langle S_6, \sigma_{1698} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6788} = \frac{\mathcal{T}_{6789}}{\langle S_{13}, \sigma_{1697} \rangle \Rightarrow \langle S_{15}, \sigma_{1698} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6790} = \frac{\langle S_6, \sigma_{1698} \rangle \Rightarrow \langle S_8, \sigma_{1698} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1698} \rangle \Rightarrow \langle S_9, \sigma_{1698} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6792} = \langle S_8, \sigma_{1698} \rangle \Rightarrow \langle y := y + 1, \sigma_{1698} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6791} = \frac{\mathcal{T}_{6792}}{\langle S_9, \sigma_{1698} \rangle \Rightarrow \langle S_{10}, \sigma_{1698} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6793} = \frac{\langle y := y + 1, \sigma_{1698} \rangle \Rightarrow \sigma_{1699} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1698} \rangle \Rightarrow \langle S_{11}, \sigma_{1699} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6794} = \langle S_{11}, \sigma_{1699} \rangle \Rightarrow \langle S_{12}, \sigma_{1699} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6795} = \langle S_{12}, \sigma_{1699} \rangle \Rightarrow \langle S_{13}, \sigma_{1699} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6797} = \frac{\langle x := x + 1, \sigma_{1699} \rangle \Rightarrow \sigma_{1700} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1699} \rangle \Rightarrow \langle S_6, \sigma_{1700} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6796} = \frac{\mathcal{T}_{6797}}{\langle S_{13}, \sigma_{1699} \rangle \Rightarrow \langle S_{15}, \sigma_{1700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6798} = \frac{\langle S_6, \sigma_{1700} \rangle \Rightarrow \langle S_8, \sigma_{1700} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1700} \rangle \Rightarrow \langle S_9, \sigma_{1700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6800} = \langle S_8, \sigma_{1700} \rangle \Rightarrow \langle y := y + 1, \sigma_{1700} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6799} = \frac{\mathcal{T}_{6800}}{\langle S_9, \sigma_{1700} \rangle \Rightarrow \langle S_{10}, \sigma_{1700} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6801} = \frac{\langle y := y + 1, \sigma_{1700} \rangle \Rightarrow \sigma_{1701} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1700} \rangle \Rightarrow \langle S_{11}, \sigma_{1701} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6802} = \langle S_{11}, \sigma_{1701} \rangle \Rightarrow \langle S_{12}, \sigma_{1701} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6803} = \langle S_{12}, \sigma_{1701} \rangle \Rightarrow \langle S_{13}, \sigma_{1701} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6805} = \frac{\langle x := x + 1, \sigma_{1701} \rangle \Rightarrow \sigma_{1702} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1701} \rangle \Rightarrow \langle S_6, \sigma_{1702} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6804} = \frac{\mathcal{T}_{6805}}{\langle S_{13}, \sigma_{1701} \rangle \Rightarrow \langle S_{15}, \sigma_{1702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6806} = \frac{\langle S_6, \sigma_{1702} \rangle \Rightarrow \langle S_8, \sigma_{1702} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1702} \rangle \Rightarrow \langle S_9, \sigma_{1702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6808} = \langle S_8, \sigma_{1702} \rangle \Rightarrow \langle y := y + 1, \sigma_{1702} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6807} = \frac{\mathcal{T}_{6808}}{\langle S_9, \sigma_{1702} \rangle \Rightarrow \langle S_{10}, \sigma_{1702} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6809} = \frac{\langle y := y + 1, \sigma_{1702} \rangle \Rightarrow \sigma_{1703} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1702} \rangle \Rightarrow \langle S_{11}, \sigma_{1703} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6810} = \langle S_{11}, \sigma_{1703} \rangle \Rightarrow \langle S_{12}, \sigma_{1703} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6811} = \langle S_{12}, \sigma_{1703} \rangle \Rightarrow \langle S_{13}, \sigma_{1703} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6813} = \frac{\langle x := x + 1, \sigma_{1703} \rangle \Rightarrow \sigma_{1704} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1703} \rangle \Rightarrow \langle S_6, \sigma_{1704} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6812} = \frac{\mathcal{T}_{6813}}{\langle S_{13}, \sigma_{1703} \rangle \Rightarrow \langle S_{15}, \sigma_{1704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6814} = \frac{\langle S_6, \sigma_{1704} \rangle \Rightarrow \langle S_8, \sigma_{1704} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1704} \rangle \Rightarrow \langle S_9, \sigma_{1704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6816} = \langle S_8, \sigma_{1704} \rangle \Rightarrow \langle y := y + 1, \sigma_{1704} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6815} = \frac{\mathcal{T}_{6816}}{\langle S_9, \sigma_{1704} \rangle \Rightarrow \langle S_{10}, \sigma_{1704} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6817} = \frac{\langle y := y + 1, \sigma_{1704} \rangle \Rightarrow \sigma_{1705} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1704} \rangle \Rightarrow \langle S_{11}, \sigma_{1705} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6818} = \langle S_{11}, \sigma_{1705} \rangle \Rightarrow \langle S_{12}, \sigma_{1705} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6819} = \langle S_{12}, \sigma_{1705} \rangle \Rightarrow \langle S_{13}, \sigma_{1705} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6821} = \frac{\langle x := x + 1, \sigma_{1705} \rangle \Rightarrow \sigma_{1706} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1705} \rangle \Rightarrow \langle S_6, \sigma_{1706} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6820} = \frac{\mathcal{T}_{6821}}{\langle S_{13}, \sigma_{1705} \rangle \Rightarrow \langle S_{15}, \sigma_{1706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6822} = \frac{\langle S_6, \sigma_{1706} \rangle \Rightarrow \langle S_8, \sigma_{1706} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1706} \rangle \Rightarrow \langle S_9, \sigma_{1706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6824} = \langle S_8, \sigma_{1706} \rangle \Rightarrow \langle y := y + 1, \sigma_{1706} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6823} = \frac{\mathcal{T}_{6824}}{\langle S_9, \sigma_{1706} \rangle \Rightarrow \langle S_{10}, \sigma_{1706} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6825} = \frac{\langle y := y + 1, \sigma_{1706} \rangle \Rightarrow \sigma_{1707} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1706} \rangle \Rightarrow \langle S_{11}, \sigma_{1707} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6826} = \langle S_{11}, \sigma_{1707} \rangle \Rightarrow \langle S_{12}, \sigma_{1707} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6827} = \langle S_{12}, \sigma_{1707} \rangle \Rightarrow \langle S_{13}, \sigma_{1707} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6829} = \frac{\langle x := x + 1, \sigma_{1707} \rangle \Rightarrow \sigma_{1708} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1707} \rangle \Rightarrow \langle S_6, \sigma_{1708} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6828} = \frac{\mathcal{T}_{6829}}{\langle S_{13}, \sigma_{1707} \rangle \Rightarrow \langle S_{15}, \sigma_{1708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6830} = \frac{\langle S_6, \sigma_{1708} \rangle \Rightarrow \langle S_8, \sigma_{1708} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1708} \rangle \Rightarrow \langle S_9, \sigma_{1708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6832} = \langle S_8, \sigma_{1708} \rangle \Rightarrow \langle y := y + 1, \sigma_{1708} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6831} = \frac{\mathcal{T}_{6832}}{\langle S_9, \sigma_{1708} \rangle \Rightarrow \langle S_{10}, \sigma_{1708} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6833} = \frac{\langle y := y + 1, \sigma_{1708} \rangle \Rightarrow \sigma_{1709} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1708} \rangle \Rightarrow \langle S_{11}, \sigma_{1709} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6834} = \langle S_{11}, \sigma_{1709} \rangle \Rightarrow \langle S_{12}, \sigma_{1709} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6835} = \langle S_{12}, \sigma_{1709} \rangle \Rightarrow \langle S_{13}, \sigma_{1709} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6837} = \frac{\langle x := x + 1, \sigma_{1709} \rangle \Rightarrow \sigma_{1710} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1709} \rangle \Rightarrow \langle S_6, \sigma_{1710} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6836} = \frac{\mathcal{T}_{6837}}{\langle S_{13}, \sigma_{1709} \rangle \Rightarrow \langle S_{15}, \sigma_{1710} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6838} = \frac{\langle S_6, \sigma_{1710} \rangle \Rightarrow \langle S_8, \sigma_{1710} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1710} \rangle \Rightarrow \langle S_9, \sigma_{1710} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6840} = \langle S_8, \sigma_{1710} \rangle \Rightarrow \langle y := y + 1, \sigma_{1710} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6839} = \frac{\mathcal{T}_{6840}}{\langle S_9, \sigma_{1710} \rangle \Rightarrow \langle S_{10}, \sigma_{1710} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6841} = \frac{\langle y := y + 1, \sigma_{1710} \rangle \Rightarrow \sigma_{1711} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1710} \rangle \Rightarrow \langle S_{11}, \sigma_{1711} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6842} = \langle S_{11}, \sigma_{1711} \rangle \Rightarrow \langle S_{12}, \sigma_{1711} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6843} = \langle S_{12}, \sigma_{1711} \rangle \Rightarrow \langle S_{13}, \sigma_{1711} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6845} = \frac{\langle x := x + 1, \sigma_{1711} \rangle \Rightarrow \sigma_{1712} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1711} \rangle \Rightarrow \langle S_6, \sigma_{1712} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{6844} = \frac{\mathcal{T}_{6845}}{\langle S_{13}, \sigma_{1711} \rangle \Rightarrow \langle S_{15}, \sigma_{1712} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6846} = \frac{\langle S_6, \sigma_{1712} \rangle \Rightarrow \langle S_8, \sigma_{1712} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1712} \rangle \Rightarrow \langle S_9, \sigma_{1712} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6848} = \langle S_8, \sigma_{1712} \rangle \Rightarrow \langle y := y + 1, \sigma_{1712} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{6847} = \frac{\mathcal{T}_{6848}}{\langle S_9, \sigma_{1712} \rangle \Rightarrow \langle S_{10}, \sigma_{1712} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6849} = \frac{\langle y := y + 1, \sigma_{1712} \rangle \Rightarrow \sigma_{1713} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1712} \rangle \Rightarrow \langle S_{11}, \sigma_{1713} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6850} = \langle S_{11}, \sigma_{1713} \rangle \Rightarrow \langle S_{12}, \sigma_{1713} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{6851} = \langle S_{12}, \sigma_{1713} \rangle \Rightarrow \langle S_{13}, \sigma_{1713} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6853} = \frac{\langle x := x + 1, \sigma_{1713} \rangle \Rightarrow \sigma_{1714} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1713} \rangle \Rightarrow \langle S_6, \sigma_{1714} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6852} = \frac{\mathcal{T}_{6853}}{\langle S_{13}, \sigma_{1713} \rangle \Rightarrow \langle S_{15}, \sigma_{1714} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6854} = \frac{\langle S_6, \sigma_{1714} \rangle \Rightarrow \langle S_8, \sigma_{1714} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1714} \rangle \Rightarrow \langle S_9, \sigma_{1714} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6856} = \langle S_8, \sigma_{1714} \rangle \Rightarrow \langle y := y + 1, \sigma_{1714} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6855} = \frac{\mathcal{T}_{6856}}{\langle S_9, \sigma_{1714} \rangle \Rightarrow \langle S_{10}, \sigma_{1714} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6857} = \frac{\langle y := y + 1, \sigma_{1714} \rangle \Rightarrow \sigma_{1715} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1714} \rangle \Rightarrow \langle S_{11}, \sigma_{1715} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6858} = \langle S_{11}, \sigma_{1715} \rangle \Rightarrow \langle S_{12}, \sigma_{1715} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6859} = \langle S_{12}, \sigma_{1715} \rangle \Rightarrow \langle S_{13}, \sigma_{1715} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6861} = \frac{\langle x := x + 1, \sigma_{1715} \rangle \Rightarrow \sigma_{1716} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1715} \rangle \Rightarrow \langle S_6, \sigma_{1716} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6860} = \frac{\mathcal{T}_{6861}}{\langle S_{13}, \sigma_{1715} \rangle \Rightarrow \langle S_{15}, \sigma_{1716} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6862} = \frac{\langle S_6, \sigma_{1716} \rangle \Rightarrow \langle S_8, \sigma_{1716} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1716} \rangle \Rightarrow \langle S_9, \sigma_{1716} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6864} = \langle S_8, \sigma_{1716} \rangle \Rightarrow \langle y := y + 1, \sigma_{1716} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6863} = \frac{\mathcal{T}_{6864}}{\langle S_9, \sigma_{1716} \rangle \Rightarrow \langle S_{10}, \sigma_{1716} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6865} = \frac{\langle y := y + 1, \sigma_{1716} \rangle \Rightarrow \sigma_{1717} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1716} \rangle \Rightarrow \langle S_{11}, \sigma_{1717} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6866} = \langle S_{11}, \sigma_{1717} \rangle \Rightarrow \langle S_{12}, \sigma_{1717} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6867} = \langle S_{12}, \sigma_{1717} \rangle \Rightarrow \langle S_{13}, \sigma_{1717} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6869} = \frac{\langle x := x + 1, \sigma_{1717} \rangle \Rightarrow \sigma_{1718} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1717} \rangle \Rightarrow \langle S_6, \sigma_{1718} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6868} = \frac{\mathcal{T}_{6869}}{\langle S_{13}, \sigma_{1717} \rangle \Rightarrow \langle S_{15}, \sigma_{1718} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6870} = \frac{\langle S_6, \sigma_{1718} \rangle \Rightarrow \langle S_8, \sigma_{1718} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1718} \rangle \Rightarrow \langle S_9, \sigma_{1718} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6872} = \langle S_8, \sigma_{1718} \rangle \Rightarrow \langle y := y + 1, \sigma_{1718} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6871} = \frac{\mathcal{T}_{6872}}{\langle S_9, \sigma_{1718} \rangle \Rightarrow \langle S_{10}, \sigma_{1718} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6873} = \frac{\langle y := y + 1, \sigma_{1718} \rangle \Rightarrow \sigma_{1719} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1718} \rangle \Rightarrow \langle S_{11}, \sigma_{1719} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6874} = \langle S_{11}, \sigma_{1719} \rangle \Rightarrow \langle S_{12}, \sigma_{1719} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6875} = \langle S_{12}, \sigma_{1719} \rangle \Rightarrow \langle S_{13}, \sigma_{1719} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6877} = \frac{\langle x := x + 1, \sigma_{1719} \rangle \Rightarrow \sigma_{1720} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1719} \rangle \Rightarrow \langle S_6, \sigma_{1720} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6876} = \frac{\mathcal{T}_{6877}}{\langle S_{13}, \sigma_{1719} \rangle \Rightarrow \langle S_{15}, \sigma_{1720} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6878} = \frac{\langle S_6, \sigma_{1720} \rangle \Rightarrow \langle S_8, \sigma_{1720} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1720} \rangle \Rightarrow \langle S_9, \sigma_{1720} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6880} = \langle S_8, \sigma_{1720} \rangle \Rightarrow \langle y := y + 1, \sigma_{1720} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6879} = \frac{\mathcal{T}_{6880}}{\langle S_9, \sigma_{1720} \rangle \Rightarrow \langle S_{10}, \sigma_{1720} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6881} = \frac{\langle y := y + 1, \sigma_{1720} \rangle \Rightarrow \sigma_{1721} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1720} \rangle \Rightarrow \langle S_{11}, \sigma_{1721} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6882} = \langle S_{11}, \sigma_{1721} \rangle \Rightarrow \langle S_{12}, \sigma_{1721} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6883} = \langle S_{12}, \sigma_{1721} \rangle \Rightarrow \langle S_{13}, \sigma_{1721} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6885} = \frac{\langle x := x + 1, \sigma_{1721} \rangle \Rightarrow \sigma_{1722} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1721} \rangle \Rightarrow \langle S_6, \sigma_{1722} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6884} = \frac{\mathcal{T}_{6885}}{\langle S_{13}, \sigma_{1721} \rangle \Rightarrow \langle S_{15}, \sigma_{1722} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6886} = \frac{\langle S_6, \sigma_{1722} \rangle \Rightarrow \langle S_8, \sigma_{1722} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1722} \rangle \Rightarrow \langle S_9, \sigma_{1722} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6888} = \langle S_8, \sigma_{1722} \rangle \Rightarrow \langle y := y + 1, \sigma_{1722} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6887} = \frac{\mathcal{T}_{6888}}{\langle S_9, \sigma_{1722} \rangle \Rightarrow \langle S_{10}, \sigma_{1722} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6889} = \frac{\langle y := y + 1, \sigma_{1722} \rangle \Rightarrow \sigma_{1723} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1722} \rangle \Rightarrow \langle S_{11}, \sigma_{1723} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6890} = \langle S_{11}, \sigma_{1723} \rangle \Rightarrow \langle S_{12}, \sigma_{1723} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6891} = \langle S_{12}, \sigma_{1723} \rangle \Rightarrow \langle S_{13}, \sigma_{1723} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6893} = \frac{\langle x := x + 1, \sigma_{1723} \rangle \Rightarrow \sigma_{1724} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1723} \rangle \Rightarrow \langle S_6, \sigma_{1724} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6892} = \frac{\mathcal{T}_{6893}}{\langle S_{13}, \sigma_{1723} \rangle \Rightarrow \langle S_{15}, \sigma_{1724} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6894} = \frac{\langle S_6, \sigma_{1724} \rangle \Rightarrow \langle S_8, \sigma_{1724} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1724} \rangle \Rightarrow \langle S_9, \sigma_{1724} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6896} = \langle S_8, \sigma_{1724} \rangle \Rightarrow \langle y := y + 1, \sigma_{1724} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6895} = \frac{\mathcal{T}_{6896}}{\langle S_9, \sigma_{1724} \rangle \Rightarrow \langle S_{10}, \sigma_{1724} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6897} = \frac{\langle y := y + 1, \sigma_{1724} \rangle \Rightarrow \sigma_{1725} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1724} \rangle \Rightarrow \langle S_{11}, \sigma_{1725} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6898} = \langle S_{11}, \sigma_{1725} \rangle \Rightarrow \langle S_{12}, \sigma_{1725} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6899} = \langle S_{12}, \sigma_{1725} \rangle \Rightarrow \langle S_{13}, \sigma_{1725} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6901} = \frac{\langle x := x + 1, \sigma_{1725} \rangle \Rightarrow \sigma_{1726} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1725} \rangle \Rightarrow \langle S_6, \sigma_{1726} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6900} = \frac{\mathcal{T}_{6901}}{\langle S_{13}, \sigma_{1725} \rangle \Rightarrow \langle S_{15}, \sigma_{1726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6902} = \frac{\langle S_6, \sigma_{1726} \rangle \Rightarrow \langle S_8, \sigma_{1726} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1726} \rangle \Rightarrow \langle S_9, \sigma_{1726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6904} = \langle S_8, \sigma_{1726} \rangle \Rightarrow \langle y := y + 1, \sigma_{1726} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6903} = \frac{\mathcal{T}_{6904}}{\langle S_9, \sigma_{1726} \rangle \Rightarrow \langle S_{10}, \sigma_{1726} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6905} = \frac{\langle y := y + 1, \sigma_{1726} \rangle \Rightarrow \sigma_{1727} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1726} \rangle \Rightarrow \langle S_{11}, \sigma_{1727} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6906} = \langle S_{11}, \sigma_{1727} \rangle \Rightarrow \langle S_{12}, \sigma_{1727} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6907} = \langle S_{12}, \sigma_{1727} \rangle \Rightarrow \langle S_{13}, \sigma_{1727} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6909} = \frac{\langle x := x + 1, \sigma_{1727} \rangle \Rightarrow \sigma_{1728} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1727} \rangle \Rightarrow \langle S_6, \sigma_{1728} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6908} = \frac{\mathcal{T}_{6909}}{\langle S_{13}, \sigma_{1727} \rangle \Rightarrow \langle S_{15}, \sigma_{1728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6910} = \frac{\langle S_6, \sigma_{1728} \rangle \Rightarrow \langle S_8, \sigma_{1728} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1728} \rangle \Rightarrow \langle S_9, \sigma_{1728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6912} = \langle S_8, \sigma_{1728} \rangle \Rightarrow \langle y := y + 1, \sigma_{1728} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6911} = \frac{\mathcal{T}_{6912}}{\langle S_9, \sigma_{1728} \rangle \Rightarrow \langle S_{10}, \sigma_{1728} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6913} = \frac{\langle y := y + 1, \sigma_{1728} \rangle \Rightarrow \sigma_{1729} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1728} \rangle \Rightarrow \langle S_{11}, \sigma_{1729} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6914} = \langle S_{11}, \sigma_{1729} \rangle \Rightarrow \langle S_{12}, \sigma_{1729} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6915} = \langle S_{12}, \sigma_{1729} \rangle \Rightarrow \langle S_{13}, \sigma_{1729} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6917} = \frac{\langle x := x + 1, \sigma_{1729} \rangle \Rightarrow \sigma_{1730} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1729} \rangle \Rightarrow \langle S_6, \sigma_{1730} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6916} = \frac{\mathcal{T}_{6917}}{\langle S_{13}, \sigma_{1729} \rangle \Rightarrow \langle S_{15}, \sigma_{1730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6918} = \frac{\langle S_6, \sigma_{1730} \rangle \Rightarrow \langle S_8, \sigma_{1730} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1730} \rangle \Rightarrow \langle S_9, \sigma_{1730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6920} = \langle S_8, \sigma_{1730} \rangle \Rightarrow \langle y := y + 1, \sigma_{1730} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6919} = \frac{\mathcal{T}_{6920}}{\langle S_9, \sigma_{1730} \rangle \Rightarrow \langle S_{10}, \sigma_{1730} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6921} = \frac{\langle y := y + 1, \sigma_{1730} \rangle \Rightarrow \sigma_{1731} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1730} \rangle \Rightarrow \langle S_{11}, \sigma_{1731} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6922} = \langle S_{11}, \sigma_{1731} \rangle \Rightarrow \langle S_{12}, \sigma_{1731} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6923} = \langle S_{12}, \sigma_{1731} \rangle \Rightarrow \langle S_{13}, \sigma_{1731} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6925} = \frac{\langle x := x + 1, \sigma_{1731} \rangle \Rightarrow \sigma_{1732} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1731} \rangle \Rightarrow \langle S_6, \sigma_{1732} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6924} = \frac{\mathcal{T}_{6925}}{\langle S_{13}, \sigma_{1731} \rangle \Rightarrow \langle S_{15}, \sigma_{1732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6926} = \frac{\langle S_6, \sigma_{1732} \rangle \Rightarrow \langle S_8, \sigma_{1732} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1732} \rangle \Rightarrow \langle S_9, \sigma_{1732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6928} = \langle S_8, \sigma_{1732} \rangle \Rightarrow \langle y := y + 1, \sigma_{1732} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6927} = \frac{\mathcal{T}_{6928}}{\langle S_9, \sigma_{1732} \rangle \Rightarrow \langle S_{10}, \sigma_{1732} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6929} = \frac{\langle y := y + 1, \sigma_{1732} \rangle \Rightarrow \sigma_{1733} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1732} \rangle \Rightarrow \langle S_{11}, \sigma_{1733} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6930} = \langle S_{11}, \sigma_{1733} \rangle \Rightarrow \langle S_{12}, \sigma_{1733} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6931} = \langle S_{12}, \sigma_{1733} \rangle \Rightarrow \langle S_{13}, \sigma_{1733} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6933} = \frac{\langle x := x + 1, \sigma_{1733} \rangle \Rightarrow \sigma_{1734} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1733} \rangle \Rightarrow \langle S_6, \sigma_{1734} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6932} = \frac{\mathcal{T}_{6933}}{\langle S_{13}, \sigma_{1733} \rangle \Rightarrow \langle S_{15}, \sigma_{1734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6934} = \frac{\langle S_6, \sigma_{1734} \rangle \Rightarrow \langle S_8, \sigma_{1734} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1734} \rangle \Rightarrow \langle S_9, \sigma_{1734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6936} = \langle S_8, \sigma_{1734} \rangle \Rightarrow \langle y := y + 1, \sigma_{1734} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6935} = \frac{\mathcal{T}_{6936}}{\langle S_9, \sigma_{1734} \rangle \Rightarrow \langle S_{10}, \sigma_{1734} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6937} = \frac{\langle y := y + 1, \sigma_{1734} \rangle \Rightarrow \sigma_{1735} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1734} \rangle \Rightarrow \langle S_{11}, \sigma_{1735} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6938} = \langle S_{11}, \sigma_{1735} \rangle \Rightarrow \langle S_{12}, \sigma_{1735} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6939} = \langle S_{12}, \sigma_{1735} \rangle \Rightarrow \langle S_{13}, \sigma_{1735} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6941} = \frac{\langle x := x + 1, \sigma_{1735} \rangle \Rightarrow \sigma_{1736} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1735} \rangle \Rightarrow \langle S_6, \sigma_{1736} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6940} = \frac{\mathcal{T}_{6941}}{\langle S_{13}, \sigma_{1735} \rangle \Rightarrow \langle S_{15}, \sigma_{1736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6942} = \frac{\langle S_6, \sigma_{1736} \rangle \Rightarrow \langle S_8, \sigma_{1736} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1736} \rangle \Rightarrow \langle S_9, \sigma_{1736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6944} = \langle S_8, \sigma_{1736} \rangle \Rightarrow \langle y := y + 1, \sigma_{1736} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{6943} = \frac{\mathcal{T}_{6944}}{\langle S_9, \sigma_{1736} \rangle \Rightarrow \langle S_{10}, \sigma_{1736} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6945} = \frac{\langle y := y + 1, \sigma_{1736} \rangle \Rightarrow \sigma_{1737} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1736} \rangle \Rightarrow \langle S_{11}, \sigma_{1737} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6946} = \langle S_{11}, \sigma_{1737} \rangle \Rightarrow \langle S_{12}, \sigma_{1737} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6947} = \langle S_{12}, \sigma_{1737} \rangle \Rightarrow \langle S_{13}, \sigma_{1737} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6949} = \frac{\langle x := x + 1, \sigma_{1737} \rangle \Rightarrow \sigma_{1738} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1737} \rangle \Rightarrow \langle S_6, \sigma_{1738} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6948} = \frac{\mathcal{T}_{6949}}{\langle S_{13}, \sigma_{1737} \rangle \Rightarrow \langle S_{15}, \sigma_{1738} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6950} = \frac{\langle S_6, \sigma_{1738} \rangle \Rightarrow \langle S_8, \sigma_{1738} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1738} \rangle \Rightarrow \langle S_9, \sigma_{1738} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6952} = \langle S_8, \sigma_{1738} \rangle \Rightarrow \langle y := y + 1, \sigma_{1738} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6951} = \frac{\mathcal{T}_{6952}}{\langle S_9, \sigma_{1738} \rangle \Rightarrow \langle S_{10}, \sigma_{1738} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6953} = \frac{\langle y := y + 1, \sigma_{1738} \rangle \Rightarrow \sigma_{1739} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1738} \rangle \Rightarrow \langle S_{11}, \sigma_{1739} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6954} = \langle S_{11}, \sigma_{1739} \rangle \Rightarrow \langle S_{12}, \sigma_{1739} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6955} = \langle S_{12}, \sigma_{1739} \rangle \Rightarrow \langle S_{13}, \sigma_{1739} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6957} = \frac{\langle x := x + 1, \sigma_{1739} \rangle \Rightarrow \sigma_{1740} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1739} \rangle \Rightarrow \langle S_6, \sigma_{1740} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6956} = \frac{\mathcal{T}_{6957}}{\langle S_{13}, \sigma_{1739} \rangle \Rightarrow \langle S_{15}, \sigma_{1740} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6958} = \frac{\langle S_6, \sigma_{1740} \rangle \Rightarrow \langle S_8, \sigma_{1740} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1740} \rangle \Rightarrow \langle S_9, \sigma_{1740} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6960} = \langle S_8, \sigma_{1740} \rangle \Rightarrow \langle y := y + 1, \sigma_{1740} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6959} = \frac{\mathcal{T}_{6960}}{\langle S_9, \sigma_{1740} \rangle \Rightarrow \langle S_{10}, \sigma_{1740} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6961} = \frac{\langle y := y + 1, \sigma_{1740} \rangle \Rightarrow \sigma_{1741} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1740} \rangle \Rightarrow \langle S_{11}, \sigma_{1741} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6962} = \langle S_{11}, \sigma_{1741} \rangle \Rightarrow \langle S_{12}, \sigma_{1741} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6963} = \langle S_{12}, \sigma_{1741} \rangle \Rightarrow \langle S_{13}, \sigma_{1741} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6965} = \frac{\langle x := x + 1, \sigma_{1741} \rangle \Rightarrow \sigma_{1742} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1741} \rangle \Rightarrow \langle S_6, \sigma_{1742} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6964} = \frac{\mathcal{T}_{6965}}{\langle S_{13}, \sigma_{1741} \rangle \Rightarrow \langle S_{15}, \sigma_{1742} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6966} = \frac{\langle S_6, \sigma_{1742} \rangle \Rightarrow \langle S_8, \sigma_{1742} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1742} \rangle \Rightarrow \langle S_9, \sigma_{1742} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6968} = \langle S_8, \sigma_{1742} \rangle \Rightarrow \langle y := y + 1, \sigma_{1742} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6967} = \frac{\mathcal{T}_{6968}}{\langle S_9, \sigma_{1742} \rangle \Rightarrow \langle S_{10}, \sigma_{1742} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6969} = \frac{\langle y := y + 1, \sigma_{1742} \rangle \Rightarrow \sigma_{1743} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1742} \rangle \Rightarrow \langle S_{11}, \sigma_{1743} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6970} = \langle S_{11}, \sigma_{1743} \rangle \Rightarrow \langle S_{12}, \sigma_{1743} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6971} = \langle S_{12}, \sigma_{1743} \rangle \Rightarrow \langle S_{13}, \sigma_{1743} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6973} = \frac{\langle x := x + 1, \sigma_{1743} \rangle \Rightarrow \sigma_{1744} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1743} \rangle \Rightarrow \langle S_6, \sigma_{1744} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6972} = \frac{\mathcal{T}_{6973}}{\langle S_{13}, \sigma_{1743} \rangle \Rightarrow \langle S_{15}, \sigma_{1744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6974} = \frac{\langle S_6, \sigma_{1744} \rangle \Rightarrow \langle S_8, \sigma_{1744} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1744} \rangle \Rightarrow \langle S_9, \sigma_{1744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6976} = \langle S_8, \sigma_{1744} \rangle \Rightarrow \langle y := y + 1, \sigma_{1744} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6975} = \frac{\mathcal{T}_{6976}}{\langle S_9, \sigma_{1744} \rangle \Rightarrow \langle S_{10}, \sigma_{1744} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6977} = \frac{\langle y := y + 1, \sigma_{1744} \rangle \Rightarrow \sigma_{1745} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1744} \rangle \Rightarrow \langle S_{11}, \sigma_{1745} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6978} = \langle S_{11}, \sigma_{1745} \rangle \Rightarrow \langle S_{12}, \sigma_{1745} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6979} = \langle S_{12}, \sigma_{1745} \rangle \Rightarrow \langle S_{13}, \sigma_{1745} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6981} = \frac{\langle x := x + 1, \sigma_{1745} \rangle \Rightarrow \sigma_{1746} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1745} \rangle \Rightarrow \langle S_6, \sigma_{1746} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6980} = \frac{\mathcal{T}_{6981}}{\langle S_{13}, \sigma_{1745} \rangle \Rightarrow \langle S_{15}, \sigma_{1746} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6982} = \frac{\langle S_6, \sigma_{1746} \rangle \Rightarrow \langle S_8, \sigma_{1746} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1746} \rangle \Rightarrow \langle S_9, \sigma_{1746} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6984} = \langle S_8, \sigma_{1746} \rangle \Rightarrow \langle y := y + 1, \sigma_{1746} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6983} = \frac{\mathcal{T}_{6984}}{\langle S_9, \sigma_{1746} \rangle \Rightarrow \langle S_{10}, \sigma_{1746} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{6985} = \frac{\langle y := y + 1, \sigma_{1746} \rangle \Rightarrow \sigma_{1747} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1746} \rangle \Rightarrow \langle S_{11}, \sigma_{1747} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{6986} = \langle S_{11}, \sigma_{1747} \rangle \Rightarrow \langle S_{12}, \sigma_{1747} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6987} = \langle S_{12}, \sigma_{1747} \rangle \Rightarrow \langle S_{13}, \sigma_{1747} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{6989} = \frac{\langle x := x + 1, \sigma_{1747} \rangle \Rightarrow \sigma_{1748} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1747} \rangle \Rightarrow \langle S_6, \sigma_{1748} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6988} = \frac{\mathcal{T}_{6989}}{\langle S_{13}, \sigma_{1747} \rangle \Rightarrow \langle S_{15}, \sigma_{1748} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6990} = \frac{\langle S_6, \sigma_{1748} \rangle \Rightarrow \langle S_8, \sigma_{1748} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1748} \rangle \Rightarrow \langle S_9, \sigma_{1748} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6992} = \langle S_8, \sigma_{1748} \rangle \Rightarrow \langle y := y + 1, \sigma_{1748} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6991} = \frac{\mathcal{T}_{6992}}{\langle S_9, \sigma_{1748} \rangle \Rightarrow \langle S_{10}, \sigma_{1748} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6993} = \frac{\langle y := y + 1, \sigma_{1748} \rangle \Rightarrow \sigma_{1749} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1748} \rangle \Rightarrow \langle S_{11}, \sigma_{1749} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{6994} = \langle S_{11}, \sigma_{1749} \rangle \Rightarrow \langle S_{12}, \sigma_{1749} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6995} = \langle S_{12}, \sigma_{1749} \rangle \Rightarrow \langle S_{13}, \sigma_{1749} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{6997} = \frac{\langle x := x + 1, \sigma_{1749} \rangle \Rightarrow \sigma_{1750} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1749} \rangle \Rightarrow \langle S_6, \sigma_{1750} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{6996} = \frac{\mathcal{T}_{6997}}{\langle S_{13}, \sigma_{1749} \rangle \Rightarrow \langle S_{15}, \sigma_{1750} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{6998} = \frac{\langle S_6, \sigma_{1750} \rangle \Rightarrow \langle S_8, \sigma_{1750} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1750} \rangle \Rightarrow \langle S_9, \sigma_{1750} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7000} = \langle S_8, \sigma_{1750} \rangle \Rightarrow \langle y := y + 1, \sigma_{1750} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{6999} = \frac{\mathcal{T}_{7000}}{\langle S_9, \sigma_{1750} \rangle \Rightarrow \langle S_{10}, \sigma_{1750} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7001} = \frac{\langle y := y + 1, \sigma_{1750} \rangle \Rightarrow \sigma_{1751} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1750} \rangle \Rightarrow \langle S_{11}, \sigma_{1751} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7002} = \langle S_{11}, \sigma_{1751} \rangle \Rightarrow \langle S_{12}, \sigma_{1751} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7003} = \langle S_{12}, \sigma_{1751} \rangle \Rightarrow \langle S_{13}, \sigma_{1751} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7005} = \frac{\langle x := x + 1, \sigma_{1751} \rangle \Rightarrow \sigma_{1752} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1751} \rangle \Rightarrow \langle S_6, \sigma_{1752} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7004} = \frac{\mathcal{T}_{7005}}{\langle S_{13}, \sigma_{1751} \rangle \Rightarrow \langle S_{15}, \sigma_{1752} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7006} = \frac{\langle S_6, \sigma_{1752} \rangle \Rightarrow \langle S_8, \sigma_{1752} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1752} \rangle \Rightarrow \langle S_9, \sigma_{1752} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7008} = \langle S_8, \sigma_{1752} \rangle \Rightarrow \langle y := y + 1, \sigma_{1752} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7007} = \frac{\mathcal{T}_{7008}}{\langle S_9, \sigma_{1752} \rangle \Rightarrow \langle S_{10}, \sigma_{1752} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7009} = \frac{\langle y := y + 1, \sigma_{1752} \rangle \Rightarrow \sigma_{1753} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1752} \rangle \Rightarrow \langle S_{11}, \sigma_{1753} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7010} = \langle S_{11}, \sigma_{1753} \rangle \Rightarrow \langle S_{12}, \sigma_{1753} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7011} = \langle S_{12}, \sigma_{1753} \rangle \Rightarrow \langle S_{13}, \sigma_{1753} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7013} = \frac{\langle x := x + 1, \sigma_{1753} \rangle \Rightarrow \sigma_{1754} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1753} \rangle \Rightarrow \langle S_6, \sigma_{1754} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7012} = \frac{\mathcal{T}_{7013}}{\langle S_{13}, \sigma_{1753} \rangle \Rightarrow \langle S_{15}, \sigma_{1754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7014} = \frac{\langle S_6, \sigma_{1754} \rangle \Rightarrow \langle S_8, \sigma_{1754} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1754} \rangle \Rightarrow \langle S_9, \sigma_{1754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7016} = \langle S_8, \sigma_{1754} \rangle \Rightarrow \langle y := y + 1, \sigma_{1754} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7015} = \frac{\mathcal{T}_{7016}}{\langle S_9, \sigma_{1754} \rangle \Rightarrow \langle S_{10}, \sigma_{1754} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7017} = \frac{\langle y := y + 1, \sigma_{1754} \rangle \Rightarrow \sigma_{1755} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1754} \rangle \Rightarrow \langle S_{11}, \sigma_{1755} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7018} = \langle S_{11}, \sigma_{1755} \rangle \Rightarrow \langle S_{12}, \sigma_{1755} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7019} = \langle S_{12}, \sigma_{1755} \rangle \Rightarrow \langle S_{13}, \sigma_{1755} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7021} = \frac{\langle x := x + 1, \sigma_{1755} \rangle \Rightarrow \sigma_{1756} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1755} \rangle \Rightarrow \langle S_6, \sigma_{1756} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7020} = \frac{\mathcal{T}_{7021}}{\langle S_{13}, \sigma_{1755} \rangle \Rightarrow \langle S_{15}, \sigma_{1756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7022} = \frac{\langle S_6, \sigma_{1756} \rangle \Rightarrow \langle S_8, \sigma_{1756} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1756} \rangle \Rightarrow \langle S_9, \sigma_{1756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7024} = \langle S_8, \sigma_{1756} \rangle \Rightarrow \langle y := y + 1, \sigma_{1756} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7023} = \frac{\mathcal{T}_{7024}}{\langle S_9, \sigma_{1756} \rangle \Rightarrow \langle S_{10}, \sigma_{1756} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7025} = \frac{\langle y := y + 1, \sigma_{1756} \rangle \Rightarrow \sigma_{1757} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1756} \rangle \Rightarrow \langle S_{11}, \sigma_{1757} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7026} = \langle S_{11}, \sigma_{1757} \rangle \Rightarrow \langle S_{12}, \sigma_{1757} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7027} = \langle S_{12}, \sigma_{1757} \rangle \Rightarrow \langle S_{13}, \sigma_{1757} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7029} = \frac{\langle x := x + 1, \sigma_{1757} \rangle \Rightarrow \sigma_{1758} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1757} \rangle \Rightarrow \langle S_6, \sigma_{1758} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7028} = \frac{\mathcal{T}_{7029}}{\langle S_{13}, \sigma_{1757} \rangle \Rightarrow \langle S_{15}, \sigma_{1758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7030} = \frac{\langle S_6, \sigma_{1758} \rangle \Rightarrow \langle S_8, \sigma_{1758} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1758} \rangle \Rightarrow \langle S_9, \sigma_{1758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7032} = \langle S_8, \sigma_{1758} \rangle \Rightarrow \langle y := y + 1, \sigma_{1758} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7031} = \frac{\mathcal{T}_{7032}}{\langle S_9, \sigma_{1758} \rangle \Rightarrow \langle S_{10}, \sigma_{1758} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7033} = \frac{\langle y := y + 1, \sigma_{1758} \rangle \Rightarrow \sigma_{1759} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1758} \rangle \Rightarrow \langle S_{11}, \sigma_{1759} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7034} = \langle S_{11}, \sigma_{1759} \rangle \Rightarrow \langle S_{12}, \sigma_{1759} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7035} = \langle S_{12}, \sigma_{1759} \rangle \Rightarrow \langle S_{13}, \sigma_{1759} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7037} = \frac{\langle x := x + 1, \sigma_{1759} \rangle \Rightarrow \sigma_{1760} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1759} \rangle \Rightarrow \langle S_6, \sigma_{1760} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7036} = \frac{\mathcal{T}_{7037}}{\langle S_{13}, \sigma_{1759} \rangle \Rightarrow \langle S_{15}, \sigma_{1760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7038} = \frac{\langle S_6, \sigma_{1760} \rangle \Rightarrow \langle S_8, \sigma_{1760} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1760} \rangle \Rightarrow \langle S_9, \sigma_{1760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7040} = \langle S_8, \sigma_{1760} \rangle \Rightarrow \langle y := y + 1, \sigma_{1760} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7039} = \frac{\mathcal{T}_{7040}}{\langle S_9, \sigma_{1760} \rangle \Rightarrow \langle S_{10}, \sigma_{1760} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7041} = \frac{\langle y := y + 1, \sigma_{1760} \rangle \Rightarrow \sigma_{1761} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1760} \rangle \Rightarrow \langle S_{11}, \sigma_{1761} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7042} = \langle S_{11}, \sigma_{1761} \rangle \Rightarrow \langle S_{12}, \sigma_{1761} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7043} = \langle S_{12}, \sigma_{1761} \rangle \Rightarrow \langle S_{13}, \sigma_{1761} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7045} = \frac{\langle x := x + 1, \sigma_{1761} \rangle \Rightarrow \sigma_{1762} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1761} \rangle \Rightarrow \langle S_6, \sigma_{1762} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7044} = \frac{\mathcal{T}_{7045}}{\langle S_{13}, \sigma_{1761} \rangle \Rightarrow \langle S_{15}, \sigma_{1762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7046} = \frac{\langle S_6, \sigma_{1762} \rangle \Rightarrow \langle S_8, \sigma_{1762} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1762} \rangle \Rightarrow \langle S_9, \sigma_{1762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7048} = \langle S_8, \sigma_{1762} \rangle \Rightarrow \langle y := y + 1, \sigma_{1762} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7047} = \frac{\mathcal{T}_{7048}}{\langle S_9, \sigma_{1762} \rangle \Rightarrow \langle S_{10}, \sigma_{1762} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7049} = \frac{\langle y := y + 1, \sigma_{1762} \rangle \Rightarrow \sigma_{1763} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1762} \rangle \Rightarrow \langle S_{11}, \sigma_{1763} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7050} = \langle S_{11}, \sigma_{1763} \rangle \Rightarrow \langle S_{12}, \sigma_{1763} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7051} = \langle S_{12}, \sigma_{1763} \rangle \Rightarrow \langle S_{13}, \sigma_{1763} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7053} = \frac{\langle x := x + 1, \sigma_{1763} \rangle \Rightarrow \sigma_{1764} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1763} \rangle \Rightarrow \langle S_6, \sigma_{1764} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7052} = \frac{\mathcal{T}_{7053}}{\langle S_{13}, \sigma_{1763} \rangle \Rightarrow \langle S_{15}, \sigma_{1764} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7054} = \frac{\langle S_6, \sigma_{1764} \rangle \Rightarrow \langle S_8, \sigma_{1764} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1764} \rangle \Rightarrow \langle S_9, \sigma_{1764} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7056} = \langle S_8, \sigma_{1764} \rangle \Rightarrow \langle y := y + 1, \sigma_{1764} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7055} = \frac{\mathcal{T}_{7056}}{\langle S_9, \sigma_{1764} \rangle \Rightarrow \langle S_{10}, \sigma_{1764} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7057} = \frac{\langle y := y + 1, \sigma_{1764} \rangle \Rightarrow \sigma_{1765} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1764} \rangle \Rightarrow \langle S_{11}, \sigma_{1765} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7058} = \langle S_{11}, \sigma_{1765} \rangle \Rightarrow \langle S_{12}, \sigma_{1765} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7059} = \langle S_{12}, \sigma_{1765} \rangle \Rightarrow \langle S_{13}, \sigma_{1765} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7061} = \frac{\langle x := x + 1, \sigma_{1765} \rangle \Rightarrow \sigma_{1766} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1765} \rangle \Rightarrow \langle S_6, \sigma_{1766} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7060} = \frac{\mathcal{T}_{7061}}{\langle S_{13}, \sigma_{1765} \rangle \Rightarrow \langle S_{15}, \sigma_{1766} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7062} = \frac{\langle S_6, \sigma_{1766} \rangle \Rightarrow \langle S_8, \sigma_{1766} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1766} \rangle \Rightarrow \langle S_9, \sigma_{1766} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7064} = \langle S_8, \sigma_{1766} \rangle \Rightarrow \langle y := y + 1, \sigma_{1766} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7063} = \frac{\mathcal{T}_{7064}}{\langle S_9, \sigma_{1766} \rangle \Rightarrow \langle S_{10}, \sigma_{1766} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7065} = \frac{\langle y := y + 1, \sigma_{1766} \rangle \Rightarrow \sigma_{1767} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1766} \rangle \Rightarrow \langle S_{11}, \sigma_{1767} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7066} = \langle S_{11}, \sigma_{1767} \rangle \Rightarrow \langle S_{12}, \sigma_{1767} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7067} = \langle S_{12}, \sigma_{1767} \rangle \Rightarrow \langle S_{13}, \sigma_{1767} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7069} = \frac{\langle x := x + 1, \sigma_{1767} \rangle \Rightarrow \sigma_{1768} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1767} \rangle \Rightarrow \langle S_6, \sigma_{1768} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7068} = \frac{\mathcal{T}_{7069}}{\langle S_{13}, \sigma_{1767} \rangle \Rightarrow \langle S_{15}, \sigma_{1768} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7070} = \frac{\langle S_6, \sigma_{1768} \rangle \Rightarrow \langle S_8, \sigma_{1768} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1768} \rangle \Rightarrow \langle S_9, \sigma_{1768} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7072} = \langle S_8, \sigma_{1768} \rangle \Rightarrow \langle y := y + 1, \sigma_{1768} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7071} = \frac{\mathcal{T}_{7072}}{\langle S_9, \sigma_{1768} \rangle \Rightarrow \langle S_{10}, \sigma_{1768} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7073} = \frac{\langle y := y + 1, \sigma_{1768} \rangle \Rightarrow \sigma_{1769} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1768} \rangle \Rightarrow \langle S_{11}, \sigma_{1769} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7074} = \langle S_{11}, \sigma_{1769} \rangle \Rightarrow \langle S_{12}, \sigma_{1769} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7075} = \langle S_{12}, \sigma_{1769} \rangle \Rightarrow \langle S_{13}, \sigma_{1769} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7077} = \frac{\langle x := x + 1, \sigma_{1769} \rangle \Rightarrow \sigma_{1770} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1769} \rangle \Rightarrow \langle S_6, \sigma_{1770} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7076} = \frac{\mathcal{T}_{7077}}{\langle S_{13}, \sigma_{1769} \rangle \Rightarrow \langle S_{15}, \sigma_{1770} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7078} = \frac{\langle S_6, \sigma_{1770} \rangle \Rightarrow \langle S_8, \sigma_{1770} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1770} \rangle \Rightarrow \langle S_9, \sigma_{1770} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7080} = \langle S_8, \sigma_{1770} \rangle \Rightarrow \langle y := y + 1, \sigma_{1770} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7079} = \frac{\mathcal{T}_{7080}}{\langle S_9, \sigma_{1770} \rangle \Rightarrow \langle S_{10}, \sigma_{1770} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7081} = \frac{\langle y := y + 1, \sigma_{1770} \rangle \Rightarrow \sigma_{1771} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1770} \rangle \Rightarrow \langle S_{11}, \sigma_{1771} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7082} = \langle S_{11}, \sigma_{1771} \rangle \Rightarrow \langle S_{12}, \sigma_{1771} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7083} = \langle S_{12}, \sigma_{1771} \rangle \Rightarrow \langle S_{13}, \sigma_{1771} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7085} = \frac{\langle x := x + 1, \sigma_{1771} \rangle \Rightarrow \sigma_{1772} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1771} \rangle \Rightarrow \langle S_6, \sigma_{1772} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7084} = \frac{\mathcal{T}_{7085}}{\langle S_{13}, \sigma_{1771} \rangle \Rightarrow \langle S_{15}, \sigma_{1772} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7086} = \frac{\langle S_6, \sigma_{1772} \rangle \Rightarrow \langle S_8, \sigma_{1772} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1772} \rangle \Rightarrow \langle S_9, \sigma_{1772} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7088} = \langle S_8, \sigma_{1772} \rangle \Rightarrow \langle y := y + 1, \sigma_{1772} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7087} = \frac{\mathcal{T}_{7088}}{\langle S_9, \sigma_{1772} \rangle \Rightarrow \langle S_{10}, \sigma_{1772} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7089} = \frac{\langle y := y + 1, \sigma_{1772} \rangle \Rightarrow \sigma_{1773} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1772} \rangle \Rightarrow \langle S_{11}, \sigma_{1773} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7090} = \langle S_{11}, \sigma_{1773} \rangle \Rightarrow \langle S_{12}, \sigma_{1773} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7091} = \langle S_{12}, \sigma_{1773} \rangle \Rightarrow \langle S_{13}, \sigma_{1773} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7093} = \frac{\langle x := x + 1, \sigma_{1773} \rangle \Rightarrow \sigma_{1774} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1773} \rangle \Rightarrow \langle S_6, \sigma_{1774} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7092} = \frac{\mathcal{T}_{7093}}{\langle S_{13}, \sigma_{1773} \rangle \Rightarrow \langle S_{15}, \sigma_{1774} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7094} = \frac{\langle S_6, \sigma_{1774} \rangle \Rightarrow \langle S_8, \sigma_{1774} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1774} \rangle \Rightarrow \langle S_9, \sigma_{1774} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7096} = \langle S_8, \sigma_{1774} \rangle \Rightarrow \langle y := y + 1, \sigma_{1774} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7095} = \frac{\mathcal{T}_{7096}}{\langle S_9, \sigma_{1774} \rangle \Rightarrow \langle S_{10}, \sigma_{1774} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7097} = \frac{\langle y := y + 1, \sigma_{1774} \rangle \Rightarrow \sigma_{1775} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1774} \rangle \Rightarrow \langle S_{11}, \sigma_{1775} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7098} = \langle S_{11}, \sigma_{1775} \rangle \Rightarrow \langle S_{12}, \sigma_{1775} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7099} = \langle S_{12}, \sigma_{1775} \rangle \Rightarrow \langle S_{13}, \sigma_{1775} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7101} = \frac{\langle x := x + 1, \sigma_{1775} \rangle \Rightarrow \sigma_{1776} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1775} \rangle \Rightarrow \langle S_6, \sigma_{1776} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7100} = \frac{\mathcal{T}_{7101}}{\langle S_{13}, \sigma_{1775} \rangle \Rightarrow \langle S_{15}, \sigma_{1776} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7102} = \frac{\langle S_6, \sigma_{1776} \rangle \Rightarrow \langle S_8, \sigma_{1776} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1776} \rangle \Rightarrow \langle S_9, \sigma_{1776} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7104} = \langle S_8, \sigma_{1776} \rangle \Rightarrow \langle y := y + 1, \sigma_{1776} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7103} = \frac{\mathcal{T}_{7104}}{\langle S_9, \sigma_{1776} \rangle \Rightarrow \langle S_{10}, \sigma_{1776} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7105} = \frac{\langle y := y + 1, \sigma_{1776} \rangle \Rightarrow \sigma_{1777} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1776} \rangle \Rightarrow \langle S_{11}, \sigma_{1777} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7106} = \langle S_{11}, \sigma_{1777} \rangle \Rightarrow \langle S_{12}, \sigma_{1777} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7107} = \langle S_{12}, \sigma_{1777} \rangle \Rightarrow \langle S_{13}, \sigma_{1777} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7109} = \frac{\langle x := x + 1, \sigma_{1777} \rangle \Rightarrow \sigma_{1778} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1777} \rangle \Rightarrow \langle S_6, \sigma_{1778} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7108} = \frac{\mathcal{T}_{7109}}{\langle S_{13}, \sigma_{1777} \rangle \Rightarrow \langle S_{15}, \sigma_{1778} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7110} = \frac{\langle S_6, \sigma_{1778} \rangle \Rightarrow \langle S_8, \sigma_{1778} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1778} \rangle \Rightarrow \langle S_9, \sigma_{1778} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7112} = \langle S_8, \sigma_{1778} \rangle \Rightarrow \langle y := y + 1, \sigma_{1778} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7111} = \frac{\mathcal{T}_{7112}}{\langle S_9, \sigma_{1778} \rangle \Rightarrow \langle S_{10}, \sigma_{1778} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7113} = \frac{\langle y := y + 1, \sigma_{1778} \rangle \Rightarrow \sigma_{1779} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1778} \rangle \Rightarrow \langle S_{11}, \sigma_{1779} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7114} = \langle S_{11}, \sigma_{1779} \rangle \Rightarrow \langle S_{12}, \sigma_{1779} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7115} = \langle S_{12}, \sigma_{1779} \rangle \Rightarrow \langle S_{13}, \sigma_{1779} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7117} = \frac{\langle x := x + 1, \sigma_{1779} \rangle \Rightarrow \sigma_{1780} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1779} \rangle \Rightarrow \langle S_6, \sigma_{1780} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7116} = \frac{\mathcal{T}_{7117}}{\langle S_{13}, \sigma_{1779} \rangle \Rightarrow \langle S_{15}, \sigma_{1780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7118} = \frac{\langle S_6, \sigma_{1780} \rangle \Rightarrow \langle S_8, \sigma_{1780} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1780} \rangle \Rightarrow \langle S_9, \sigma_{1780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7120} = \langle S_8, \sigma_{1780} \rangle \Rightarrow \langle y := y + 1, \sigma_{1780} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7119} = \frac{\mathcal{T}_{7120}}{\langle S_9, \sigma_{1780} \rangle \Rightarrow \langle S_{10}, \sigma_{1780} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7121} = \frac{\langle y := y + 1, \sigma_{1780} \rangle \Rightarrow \sigma_{1781} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1780} \rangle \Rightarrow \langle S_{11}, \sigma_{1781} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7122} = \langle S_{11}, \sigma_{1781} \rangle \Rightarrow \langle S_{12}, \sigma_{1781} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7123} = \langle S_{12}, \sigma_{1781} \rangle \Rightarrow \langle S_{13}, \sigma_{1781} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7125} = \frac{\langle x := x + 1, \sigma_{1781} \rangle \Rightarrow \sigma_{1782} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1781} \rangle \Rightarrow \langle S_6, \sigma_{1782} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7124} = \frac{\mathcal{T}_{7125}}{\langle S_{13}, \sigma_{1781} \rangle \Rightarrow \langle S_{15}, \sigma_{1782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7126} = \frac{\langle S_6, \sigma_{1782} \rangle \Rightarrow \langle S_8, \sigma_{1782} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1782} \rangle \Rightarrow \langle S_9, \sigma_{1782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7128} = \langle S_8, \sigma_{1782} \rangle \Rightarrow \langle y := y + 1, \sigma_{1782} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7127} = \frac{\mathcal{T}_{7128}}{\langle S_9, \sigma_{1782} \rangle \Rightarrow \langle S_{10}, \sigma_{1782} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7129} = \frac{\langle y := y + 1, \sigma_{1782} \rangle \Rightarrow \sigma_{1783} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1782} \rangle \Rightarrow \langle S_{11}, \sigma_{1783} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7130} = \langle S_{11}, \sigma_{1783} \rangle \Rightarrow \langle S_{12}, \sigma_{1783} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7131} = \langle S_{12}, \sigma_{1783} \rangle \Rightarrow \langle S_{13}, \sigma_{1783} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7133} = \frac{\langle x := x + 1, \sigma_{1783} \rangle \Rightarrow \sigma_{1784} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1783} \rangle \Rightarrow \langle S_6, \sigma_{1784} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7132} = \frac{\mathcal{T}_{7133}}{\langle S_{13}, \sigma_{1783} \rangle \Rightarrow \langle S_{15}, \sigma_{1784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7134} = \frac{\langle S_6, \sigma_{1784} \rangle \Rightarrow \langle S_8, \sigma_{1784} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1784} \rangle \Rightarrow \langle S_9, \sigma_{1784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7136} = \langle S_8, \sigma_{1784} \rangle \Rightarrow \langle y := y + 1, \sigma_{1784} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7135} = \frac{\mathcal{T}_{7136}}{\langle S_9, \sigma_{1784} \rangle \Rightarrow \langle S_{10}, \sigma_{1784} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7137} = \frac{\langle y := y + 1, \sigma_{1784} \rangle \Rightarrow \sigma_{1785} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1784} \rangle \Rightarrow \langle S_{11}, \sigma_{1785} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7138} = \langle S_{11}, \sigma_{1785} \rangle \Rightarrow \langle S_{12}, \sigma_{1785} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7139} = \langle S_{12}, \sigma_{1785} \rangle \Rightarrow \langle S_{13}, \sigma_{1785} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7141} = \frac{\langle x := x + 1, \sigma_{1785} \rangle \Rightarrow \sigma_{1786} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1785} \rangle \Rightarrow \langle S_6, \sigma_{1786} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7140} = \frac{\mathcal{T}_{7141}}{\langle S_{13}, \sigma_{1785} \rangle \Rightarrow \langle S_{15}, \sigma_{1786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7142} = \frac{\langle S_6, \sigma_{1786} \rangle \Rightarrow \langle S_8, \sigma_{1786} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1786} \rangle \Rightarrow \langle S_9, \sigma_{1786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7144} = \langle S_8, \sigma_{1786} \rangle \Rightarrow \langle y := y + 1, \sigma_{1786} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7143} = \frac{\mathcal{T}_{7144}}{\langle S_9, \sigma_{1786} \rangle \Rightarrow \langle S_{10}, \sigma_{1786} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7145} = \frac{\langle y := y + 1, \sigma_{1786} \rangle \Rightarrow \sigma_{1787} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1786} \rangle \Rightarrow \langle S_{11}, \sigma_{1787} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7146} = \langle S_{11}, \sigma_{1787} \rangle \Rightarrow \langle S_{12}, \sigma_{1787} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7147} = \langle S_{12}, \sigma_{1787} \rangle \Rightarrow \langle S_{13}, \sigma_{1787} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7149} = \frac{\langle x := x + 1, \sigma_{1787} \rangle \Rightarrow \sigma_{1788} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1787} \rangle \Rightarrow \langle S_6, \sigma_{1788} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7148} = \frac{\mathcal{T}_{7149}}{\langle S_{13}, \sigma_{1787} \rangle \Rightarrow \langle S_{15}, \sigma_{1788} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7150} = \frac{\langle S_6, \sigma_{1788} \rangle \Rightarrow \langle S_8, \sigma_{1788} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1788} \rangle \Rightarrow \langle S_9, \sigma_{1788} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7152} = \langle S_8, \sigma_{1788} \rangle \Rightarrow \langle y := y + 1, \sigma_{1788} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7151} = \frac{\mathcal{T}_{7152}}{\langle S_9, \sigma_{1788} \rangle \Rightarrow \langle S_{10}, \sigma_{1788} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7153} = \frac{\langle y := y + 1, \sigma_{1788} \rangle \Rightarrow \sigma_{1789} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1788} \rangle \Rightarrow \langle S_{11}, \sigma_{1789} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7154} = \langle S_{11}, \sigma_{1789} \rangle \Rightarrow \langle S_{12}, \sigma_{1789} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7155} = \langle S_{12}, \sigma_{1789} \rangle \Rightarrow \langle S_{13}, \sigma_{1789} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7157} = \frac{\langle x := x + 1, \sigma_{1789} \rangle \Rightarrow \sigma_{1790} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1789} \rangle \Rightarrow \langle S_6, \sigma_{1790} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7156} = \frac{\mathcal{T}_{7157}}{\langle S_{13}, \sigma_{1789} \rangle \Rightarrow \langle S_{15}, \sigma_{1790} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7158} = \frac{\langle S_6, \sigma_{1790} \rangle \Rightarrow \langle S_8, \sigma_{1790} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1790} \rangle \Rightarrow \langle S_9, \sigma_{1790} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7160} = \langle S_8, \sigma_{1790} \rangle \Rightarrow \langle y := y + 1, \sigma_{1790} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7159} = \frac{\mathcal{T}_{7160}}{\langle S_9, \sigma_{1790} \rangle \Rightarrow \langle S_{10}, \sigma_{1790} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7161} = \frac{\langle y := y + 1, \sigma_{1790} \rangle \Rightarrow \sigma_{1791} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1790} \rangle \Rightarrow \langle S_{11}, \sigma_{1791} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7162} = \langle S_{11}, \sigma_{1791} \rangle \Rightarrow \langle S_{12}, \sigma_{1791} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7163} = \langle S_{12}, \sigma_{1791} \rangle \Rightarrow \langle S_{13}, \sigma_{1791} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7165} = \frac{\langle x := x + 1, \sigma_{1791} \rangle \Rightarrow \sigma_{1792} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1791} \rangle \Rightarrow \langle S_6, \sigma_{1792} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7164} = \frac{\mathcal{T}_{7165}}{\langle S_{13}, \sigma_{1791} \rangle \Rightarrow \langle S_{15}, \sigma_{1792} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7166} = \frac{\langle S_6, \sigma_{1792} \rangle \Rightarrow \langle S_8, \sigma_{1792} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1792} \rangle \Rightarrow \langle S_9, \sigma_{1792} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7168} = \langle S_8, \sigma_{1792} \rangle \Rightarrow \langle y := y + 1, \sigma_{1792} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7167} = \frac{\mathcal{T}_{7168}}{\langle S_9, \sigma_{1792} \rangle \Rightarrow \langle S_{10}, \sigma_{1792} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7169} = \frac{\langle y := y + 1, \sigma_{1792} \rangle \Rightarrow \sigma_{1793} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1792} \rangle \Rightarrow \langle S_{11}, \sigma_{1793} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7170} = \langle S_{11}, \sigma_{1793} \rangle \Rightarrow \langle S_{12}, \sigma_{1793} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7171} = \langle S_{12}, \sigma_{1793} \rangle \Rightarrow \langle S_{13}, \sigma_{1793} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7173} = \frac{\langle x := x + 1, \sigma_{1793} \rangle \Rightarrow \sigma_{1794} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1793} \rangle \Rightarrow \langle S_6, \sigma_{1794} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7172} = \frac{\mathcal{T}_{7173}}{\langle S_{13}, \sigma_{1793} \rangle \Rightarrow \langle S_{15}, \sigma_{1794} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7174} = \frac{\langle S_6, \sigma_{1794} \rangle \Rightarrow \langle S_8, \sigma_{1794} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1794} \rangle \Rightarrow \langle S_9, \sigma_{1794} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7176} = \langle S_8, \sigma_{1794} \rangle \Rightarrow \langle y := y + 1, \sigma_{1794} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7175} = \frac{\mathcal{T}_{7176}}{\langle S_9, \sigma_{1794} \rangle \Rightarrow \langle S_{10}, \sigma_{1794} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7177} = \frac{\langle y := y + 1, \sigma_{1794} \rangle \Rightarrow \sigma_{1795} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1794} \rangle \Rightarrow \langle S_{11}, \sigma_{1795} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7178} = \langle S_{11}, \sigma_{1795} \rangle \Rightarrow \langle S_{12}, \sigma_{1795} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7179} = \langle S_{12}, \sigma_{1795} \rangle \Rightarrow \langle S_{13}, \sigma_{1795} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7181} = \frac{\langle x := x + 1, \sigma_{1795} \rangle \Rightarrow \sigma_{1796} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1795} \rangle \Rightarrow \langle S_6, \sigma_{1796} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7180} = \frac{\mathcal{T}_{7181}}{\langle S_{13}, \sigma_{1795} \rangle \Rightarrow \langle S_{15}, \sigma_{1796} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7182} = \frac{\langle S_6, \sigma_{1796} \rangle \Rightarrow \langle S_8, \sigma_{1796} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1796} \rangle \Rightarrow \langle S_9, \sigma_{1796} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7184} = \langle S_8, \sigma_{1796} \rangle \Rightarrow \langle y := y + 1, \sigma_{1796} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7183} = \frac{\mathcal{T}_{7184}}{\langle S_9, \sigma_{1796} \rangle \Rightarrow \langle S_{10}, \sigma_{1796} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7185} = \frac{\langle y := y + 1, \sigma_{1796} \rangle \Rightarrow \sigma_{1797} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1796} \rangle \Rightarrow \langle S_{11}, \sigma_{1797} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7186} = \langle S_{11}, \sigma_{1797} \rangle \Rightarrow \langle S_{12}, \sigma_{1797} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7187} = \langle S_{12}, \sigma_{1797} \rangle \Rightarrow \langle S_{13}, \sigma_{1797} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7189} = \frac{\langle x := x + 1, \sigma_{1797} \rangle \Rightarrow \sigma_{1798} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1797} \rangle \Rightarrow \langle S_6, \sigma_{1798} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7188} = \frac{\mathcal{T}_{7189}}{\langle S_{13}, \sigma_{1797} \rangle \Rightarrow \langle S_{15}, \sigma_{1798} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7190} = \frac{\langle S_6, \sigma_{1798} \rangle \Rightarrow \langle S_8, \sigma_{1798} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1798} \rangle \Rightarrow \langle S_9, \sigma_{1798} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7192} = \langle S_8, \sigma_{1798} \rangle \Rightarrow \langle y := y + 1, \sigma_{1798} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7191} = \frac{\mathcal{T}_{7192}}{\langle S_9, \sigma_{1798} \rangle \Rightarrow \langle S_{10}, \sigma_{1798} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7193} = \frac{\langle y := y + 1, \sigma_{1798} \rangle \Rightarrow \sigma_{1799} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1798} \rangle \Rightarrow \langle S_{11}, \sigma_{1799} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7194} = \langle S_{11}, \sigma_{1799} \rangle \Rightarrow \langle S_{12}, \sigma_{1799} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7195} = \langle S_{12}, \sigma_{1799} \rangle \Rightarrow \langle S_{13}, \sigma_{1799} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7197} = \frac{\langle x := x + 1, \sigma_{1799} \rangle \Rightarrow \sigma_{1800} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1799} \rangle \Rightarrow \langle S_6, \sigma_{1800} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7196} = \frac{\mathcal{T}_{7197}}{\langle S_{13}, \sigma_{1799} \rangle \Rightarrow \langle S_{15}, \sigma_{1800} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7198} = \frac{\langle S_6, \sigma_{1800} \rangle \Rightarrow \langle S_8, \sigma_{1800} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1800} \rangle \Rightarrow \langle S_9, \sigma_{1800} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7200} = \langle S_8, \sigma_{1800} \rangle \Rightarrow \langle y := y + 1, \sigma_{1800} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7199} = \frac{\mathcal{T}_{7200}}{\langle S_9, \sigma_{1800} \rangle \Rightarrow \langle S_{10}, \sigma_{1800} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7201} = \frac{\langle y := y + 1, \sigma_{1800} \rangle \Rightarrow \sigma_{1801} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1800} \rangle \Rightarrow \langle S_{11}, \sigma_{1801} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7202} = \langle S_{11}, \sigma_{1801} \rangle \Rightarrow \langle S_{12}, \sigma_{1801} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7203} = \langle S_{12}, \sigma_{1801} \rangle \Rightarrow \langle S_{13}, \sigma_{1801} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7205} = \frac{\langle x := x + 1, \sigma_{1801} \rangle \Rightarrow \sigma_{1802} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1801} \rangle \Rightarrow \langle S_6, \sigma_{1802} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7204} = \frac{\mathcal{T}_{7205}}{\langle S_{13}, \sigma_{1801} \rangle \Rightarrow \langle S_{15}, \sigma_{1802} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7206} = \frac{\langle S_6, \sigma_{1802} \rangle \Rightarrow \langle S_8, \sigma_{1802} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1802} \rangle \Rightarrow \langle S_9, \sigma_{1802} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7208} = \langle S_8, \sigma_{1802} \rangle \Rightarrow \langle y := y + 1, \sigma_{1802} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7207} = \frac{\mathcal{T}_{7208}}{\langle S_9, \sigma_{1802} \rangle \Rightarrow \langle S_{10}, \sigma_{1802} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7209} = \frac{\langle y := y + 1, \sigma_{1802} \rangle \Rightarrow \sigma_{1803} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1802} \rangle \Rightarrow \langle S_{11}, \sigma_{1803} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7210} = \langle S_{11}, \sigma_{1803} \rangle \Rightarrow \langle S_{12}, \sigma_{1803} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7211} = \langle S_{12}, \sigma_{1803} \rangle \Rightarrow \langle S_{13}, \sigma_{1803} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7213} = \frac{\langle x := x + 1, \sigma_{1803} \rangle \Rightarrow \sigma_{1804} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1803} \rangle \Rightarrow \langle S_6, \sigma_{1804} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7212} = \frac{\mathcal{T}_{7213}}{\langle S_{13}, \sigma_{1803} \rangle \Rightarrow \langle S_{15}, \sigma_{1804} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7214} = \frac{\langle S_6, \sigma_{1804} \rangle \Rightarrow \langle S_8, \sigma_{1804} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1804} \rangle \Rightarrow \langle S_9, \sigma_{1804} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7216} = \langle S_8, \sigma_{1804} \rangle \Rightarrow \langle y := y + 1, \sigma_{1804} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7215} = \frac{\mathcal{T}_{7216}}{\langle S_9, \sigma_{1804} \rangle \Rightarrow \langle S_{10}, \sigma_{1804} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7217} = \frac{\langle y := y + 1, \sigma_{1804} \rangle \Rightarrow \sigma_{1805} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1804} \rangle \Rightarrow \langle S_{11}, \sigma_{1805} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7218} = \langle S_{11}, \sigma_{1805} \rangle \Rightarrow \langle S_{12}, \sigma_{1805} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7219} = \langle S_{12}, \sigma_{1805} \rangle \Rightarrow \langle S_{13}, \sigma_{1805} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7221} = \frac{\langle x := x + 1, \sigma_{1805} \rangle \Rightarrow \sigma_{1806} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1805} \rangle \Rightarrow \langle S_6, \sigma_{1806} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7220} = \frac{\mathcal{T}_{7221}}{\langle S_{13}, \sigma_{1805} \rangle \Rightarrow \langle S_{15}, \sigma_{1806} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7222} = \frac{\langle S_6, \sigma_{1806} \rangle \Rightarrow \langle S_8, \sigma_{1806} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1806} \rangle \Rightarrow \langle S_9, \sigma_{1806} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7224} = \langle S_8, \sigma_{1806} \rangle \Rightarrow \langle y := y + 1, \sigma_{1806} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7223} = \frac{\mathcal{T}_{7224}}{\langle S_9, \sigma_{1806} \rangle \Rightarrow \langle S_{10}, \sigma_{1806} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7225} = \frac{\langle y := y + 1, \sigma_{1806} \rangle \Rightarrow \sigma_{1807} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1806} \rangle \Rightarrow \langle S_{11}, \sigma_{1807} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7226} = \langle S_{11}, \sigma_{1807} \rangle \Rightarrow \langle S_{12}, \sigma_{1807} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7227} = \langle S_{12}, \sigma_{1807} \rangle \Rightarrow \langle S_{13}, \sigma_{1807} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7229} = \frac{\langle x := x + 1, \sigma_{1807} \rangle \Rightarrow \sigma_{1808} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1807} \rangle \Rightarrow \langle S_6, \sigma_{1808} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7228} = \frac{\mathcal{T}_{7229}}{\langle S_{13}, \sigma_{1807} \rangle \Rightarrow \langle S_{15}, \sigma_{1808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7230} = \frac{\langle S_6, \sigma_{1808} \rangle \Rightarrow \langle S_8, \sigma_{1808} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1808} \rangle \Rightarrow \langle S_9, \sigma_{1808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7232} = \langle S_8, \sigma_{1808} \rangle \Rightarrow \langle y := y + 1, \sigma_{1808} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7231} = \frac{\mathcal{T}_{7232}}{\langle S_9, \sigma_{1808} \rangle \Rightarrow \langle S_{10}, \sigma_{1808} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7233} = \frac{\langle y := y + 1, \sigma_{1808} \rangle \Rightarrow \sigma_{1809} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1808} \rangle \Rightarrow \langle S_{11}, \sigma_{1809} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7234} = \langle S_{11}, \sigma_{1809} \rangle \Rightarrow \langle S_{12}, \sigma_{1809} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7235} = \langle S_{12}, \sigma_{1809} \rangle \Rightarrow \langle S_{13}, \sigma_{1809} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7237} = \frac{\langle x := x + 1, \sigma_{1809} \rangle \Rightarrow \sigma_{1810} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1809} \rangle \Rightarrow \langle S_6, \sigma_{1810} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7236} = \frac{\mathcal{T}_{7237}}{\langle S_{13}, \sigma_{1809} \rangle \Rightarrow \langle S_{15}, \sigma_{1810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7238} = \frac{\langle S_6, \sigma_{1810} \rangle \Rightarrow \langle S_8, \sigma_{1810} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1810} \rangle \Rightarrow \langle S_9, \sigma_{1810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7240} = \langle S_8, \sigma_{1810} \rangle \Rightarrow \langle y := y + 1, \sigma_{1810} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7239} = \frac{\mathcal{T}_{7240}}{\langle S_9, \sigma_{1810} \rangle \Rightarrow \langle S_{10}, \sigma_{1810} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7241} = \frac{\langle y := y + 1, \sigma_{1810} \rangle \Rightarrow \sigma_{1811} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1810} \rangle \Rightarrow \langle S_{11}, \sigma_{1811} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7242} = \langle S_{11}, \sigma_{1811} \rangle \Rightarrow \langle S_{12}, \sigma_{1811} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7243} = \langle S_{12}, \sigma_{1811} \rangle \Rightarrow \langle S_{13}, \sigma_{1811} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7245} = \frac{\langle x := x + 1, \sigma_{1811} \rangle \Rightarrow \sigma_{1812} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1811} \rangle \Rightarrow \langle S_6, \sigma_{1812} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7244} = \frac{\mathcal{T}_{7245}}{\langle S_{13}, \sigma_{1811} \rangle \Rightarrow \langle S_{15}, \sigma_{1812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7246} = \frac{\langle S_6, \sigma_{1812} \rangle \Rightarrow \langle S_8, \sigma_{1812} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1812} \rangle \Rightarrow \langle S_9, \sigma_{1812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7248} = \langle S_8, \sigma_{1812} \rangle \Rightarrow \langle y := y + 1, \sigma_{1812} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7247} = \frac{\mathcal{T}_{7248}}{\langle S_9, \sigma_{1812} \rangle \Rightarrow \langle S_{10}, \sigma_{1812} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7249} = \frac{\langle y := y + 1, \sigma_{1812} \rangle \Rightarrow \sigma_{1813} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1812} \rangle \Rightarrow \langle S_{11}, \sigma_{1813} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7250} = \langle S_{11}, \sigma_{1813} \rangle \Rightarrow \langle S_{12}, \sigma_{1813} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7251} = \langle S_{12}, \sigma_{1813} \rangle \Rightarrow \langle S_{13}, \sigma_{1813} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7253} = \frac{\langle x := x + 1, \sigma_{1813} \rangle \Rightarrow \sigma_{1814} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1813} \rangle \Rightarrow \langle S_6, \sigma_{1814} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7252} = \frac{\mathcal{T}_{7253}}{\langle S_{13}, \sigma_{1813} \rangle \Rightarrow \langle S_{15}, \sigma_{1814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7254} = \frac{\langle S_6, \sigma_{1814} \rangle \Rightarrow \langle S_8, \sigma_{1814} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1814} \rangle \Rightarrow \langle S_9, \sigma_{1814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7256} = \langle S_8, \sigma_{1814} \rangle \Rightarrow \langle y := y + 1, \sigma_{1814} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7255} = \frac{\mathcal{T}_{7256}}{\langle S_9, \sigma_{1814} \rangle \Rightarrow \langle S_{10}, \sigma_{1814} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7257} = \frac{\langle y := y + 1, \sigma_{1814} \rangle \Rightarrow \sigma_{1815} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1814} \rangle \Rightarrow \langle S_{11}, \sigma_{1815} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7258} = \langle S_{11}, \sigma_{1815} \rangle \Rightarrow \langle S_{12}, \sigma_{1815} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7259} = \langle S_{12}, \sigma_{1815} \rangle \Rightarrow \langle S_{13}, \sigma_{1815} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7261} = \frac{\langle x := x + 1, \sigma_{1815} \rangle \Rightarrow \sigma_{1816} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1815} \rangle \Rightarrow \langle S_6, \sigma_{1816} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7260} = \frac{\mathcal{T}_{7261}}{\langle S_{13}, \sigma_{1815} \rangle \Rightarrow \langle S_{15}, \sigma_{1816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7262} = \frac{\langle S_6, \sigma_{1816} \rangle \Rightarrow \langle S_8, \sigma_{1816} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1816} \rangle \Rightarrow \langle S_9, \sigma_{1816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7264} = \langle S_8, \sigma_{1816} \rangle \Rightarrow \langle y := y + 1, \sigma_{1816} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7263} = \frac{\mathcal{T}_{7264}}{\langle S_9, \sigma_{1816} \rangle \Rightarrow \langle S_{10}, \sigma_{1816} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7265} = \frac{\langle y := y + 1, \sigma_{1816} \rangle \Rightarrow \sigma_{1817} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1816} \rangle \Rightarrow \langle S_{11}, \sigma_{1817} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7266} = \langle S_{11}, \sigma_{1817} \rangle \Rightarrow \langle S_{12}, \sigma_{1817} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7267} = \langle S_{12}, \sigma_{1817} \rangle \Rightarrow \langle S_{13}, \sigma_{1817} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7269} = \frac{\langle x := x + 1, \sigma_{1817} \rangle \Rightarrow \sigma_{1818} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1817} \rangle \Rightarrow \langle S_6, \sigma_{1818} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7268} = \frac{\mathcal{T}_{7269}}{\langle S_{13}, \sigma_{1817} \rangle \Rightarrow \langle S_{15}, \sigma_{1818} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7270} = \frac{\langle S_6, \sigma_{1818} \rangle \Rightarrow \langle S_8, \sigma_{1818} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1818} \rangle \Rightarrow \langle S_9, \sigma_{1818} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7272} = \langle S_8, \sigma_{1818} \rangle \Rightarrow \langle y := y + 1, \sigma_{1818} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7271} = \frac{\mathcal{T}_{7272}}{\langle S_9, \sigma_{1818} \rangle \Rightarrow \langle S_{10}, \sigma_{1818} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7273} = \frac{\langle y := y + 1, \sigma_{1818} \rangle \Rightarrow \sigma_{1819} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1818} \rangle \Rightarrow \langle S_{11}, \sigma_{1819} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7274} = \langle S_{11}, \sigma_{1819} \rangle \Rightarrow \langle S_{12}, \sigma_{1819} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7275} = \langle S_{12}, \sigma_{1819} \rangle \Rightarrow \langle S_{13}, \sigma_{1819} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7277} = \frac{\langle x := x + 1, \sigma_{1819} \rangle \Rightarrow \sigma_{1820} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1819} \rangle \Rightarrow \langle S_6, \sigma_{1820} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7276} = \frac{\mathcal{T}_{7277}}{\langle S_{13}, \sigma_{1819} \rangle \Rightarrow \langle S_{15}, \sigma_{1820} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7278} = \frac{\langle S_6, \sigma_{1820} \rangle \Rightarrow \langle S_8, \sigma_{1820} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1820} \rangle \Rightarrow \langle S_9, \sigma_{1820} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7280} = \langle S_8, \sigma_{1820} \rangle \Rightarrow \langle y := y + 1, \sigma_{1820} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7279} = \frac{\mathcal{T}_{7280}}{\langle S_9, \sigma_{1820} \rangle \Rightarrow \langle S_{10}, \sigma_{1820} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7281} = \frac{\langle y := y + 1, \sigma_{1820} \rangle \Rightarrow \sigma_{1821} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1820} \rangle \Rightarrow \langle S_{11}, \sigma_{1821} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7282} = \langle S_{11}, \sigma_{1821} \rangle \Rightarrow \langle S_{12}, \sigma_{1821} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7283} = \langle S_{12}, \sigma_{1821} \rangle \Rightarrow \langle S_{13}, \sigma_{1821} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7285} = \frac{\langle x := x + 1, \sigma_{1821} \rangle \Rightarrow \sigma_{1822} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1821} \rangle \Rightarrow \langle S_6, \sigma_{1822} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7284} = \frac{\mathcal{T}_{7285}}{\langle S_{13}, \sigma_{1821} \rangle \Rightarrow \langle S_{15}, \sigma_{1822} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7286} = \frac{\langle S_6, \sigma_{1822} \rangle \Rightarrow \langle S_8, \sigma_{1822} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1822} \rangle \Rightarrow \langle S_9, \sigma_{1822} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7288} = \langle S_8, \sigma_{1822} \rangle \Rightarrow \langle y := y + 1, \sigma_{1822} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7287} = \frac{\mathcal{T}_{7288}}{\langle S_9, \sigma_{1822} \rangle \Rightarrow \langle S_{10}, \sigma_{1822} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7289} = \frac{\langle y := y + 1, \sigma_{1822} \rangle \Rightarrow \sigma_{1823} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1822} \rangle \Rightarrow \langle S_{11}, \sigma_{1823} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7290} = \langle S_{11}, \sigma_{1823} \rangle \Rightarrow \langle S_{12}, \sigma_{1823} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7291} = \langle S_{12}, \sigma_{1823} \rangle \Rightarrow \langle S_{13}, \sigma_{1823} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7293} = \frac{\langle x := x + 1, \sigma_{1823} \rangle \Rightarrow \sigma_{1824} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1823} \rangle \Rightarrow \langle S_6, \sigma_{1824} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7292} = \frac{\mathcal{T}_{7293}}{\langle S_{13}, \sigma_{1823} \rangle \Rightarrow \langle S_{15}, \sigma_{1824} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7294} = \frac{\langle S_6, \sigma_{1824} \rangle \Rightarrow \langle S_8, \sigma_{1824} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1824} \rangle \Rightarrow \langle S_9, \sigma_{1824} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7296} = \langle S_8, \sigma_{1824} \rangle \Rightarrow \langle y := y + 1, \sigma_{1824} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7295} = \frac{\mathcal{T}_{7296}}{\langle S_9, \sigma_{1824} \rangle \Rightarrow \langle S_{10}, \sigma_{1824} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7297} = \frac{\langle y := y + 1, \sigma_{1824} \rangle \Rightarrow \sigma_{1825} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1824} \rangle \Rightarrow \langle S_{11}, \sigma_{1825} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7298} = \langle S_{11}, \sigma_{1825} \rangle \Rightarrow \langle S_{12}, \sigma_{1825} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7299} = \langle S_{12}, \sigma_{1825} \rangle \Rightarrow \langle S_{13}, \sigma_{1825} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7301} = \frac{\langle x := x + 1, \sigma_{1825} \rangle \Rightarrow \sigma_{1826} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1825} \rangle \Rightarrow \langle S_6, \sigma_{1826} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7300} = \frac{\mathcal{T}_{7301}}{\langle S_{13}, \sigma_{1825} \rangle \Rightarrow \langle S_{15}, \sigma_{1826} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7302} = \frac{\langle S_6, \sigma_{1826} \rangle \Rightarrow \langle S_8, \sigma_{1826} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1826} \rangle \Rightarrow \langle S_9, \sigma_{1826} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7304} = \langle S_8, \sigma_{1826} \rangle \Rightarrow \langle y := y + 1, \sigma_{1826} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7303} = \frac{\mathcal{T}_{7304}}{\langle S_9, \sigma_{1826} \rangle \Rightarrow \langle S_{10}, \sigma_{1826} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7305} = \frac{\langle y := y + 1, \sigma_{1826} \rangle \Rightarrow \sigma_{1827} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1826} \rangle \Rightarrow \langle S_{11}, \sigma_{1827} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7306} = \langle S_{11}, \sigma_{1827} \rangle \Rightarrow \langle S_{12}, \sigma_{1827} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7307} = \langle S_{12}, \sigma_{1827} \rangle \Rightarrow \langle S_{13}, \sigma_{1827} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7309} = \frac{\langle x := x + 1, \sigma_{1827} \rangle \Rightarrow \sigma_{1828} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1827} \rangle \Rightarrow \langle S_6, \sigma_{1828} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7308} = \frac{\mathcal{T}_{7309}}{\langle S_{13}, \sigma_{1827} \rangle \Rightarrow \langle S_{15}, \sigma_{1828} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7310} = \frac{\langle S_6, \sigma_{1828} \rangle \Rightarrow \langle S_8, \sigma_{1828} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1828} \rangle \Rightarrow \langle S_9, \sigma_{1828} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7312} = \langle S_8, \sigma_{1828} \rangle \Rightarrow \langle y := y + 1, \sigma_{1828} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7311} = \frac{\mathcal{T}_{7312}}{\langle S_9, \sigma_{1828} \rangle \Rightarrow \langle S_{10}, \sigma_{1828} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7313} = \frac{\langle y := y + 1, \sigma_{1828} \rangle \Rightarrow \sigma_{1829} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1828} \rangle \Rightarrow \langle S_{11}, \sigma_{1829} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7314} = \langle S_{11}, \sigma_{1829} \rangle \Rightarrow \langle S_{12}, \sigma_{1829} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{7315} = \langle S_{12}, \sigma_{1829} \rangle \Rightarrow \langle S_{13}, \sigma_{1829} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{7317} = \frac{\langle x := x + 1, \sigma_{1829} \rangle \Rightarrow \sigma_{1830} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1829} \rangle \Rightarrow \langle S_6, \sigma_{1830} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7316} = \frac{\mathcal{T}_{7317}}{\langle S_{13}, \sigma_{1829} \rangle \Rightarrow \langle S_{15}, \sigma_{1830} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7318} = \frac{\langle S_6, \sigma_{1830} \rangle \Rightarrow \langle S_8, \sigma_{1830} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1830} \rangle \Rightarrow \langle S_9, \sigma_{1830} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7320} = \langle S_8, \sigma_{1830} \rangle \Rightarrow \langle y := y + 1, \sigma_{1830} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7319} = \frac{\mathcal{T}_{7320}}{\langle S_9, \sigma_{1830} \rangle \Rightarrow \langle S_{10}, \sigma_{1830} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7321} = \frac{\langle y := y + 1, \sigma_{1830} \rangle \Rightarrow \sigma_{1831} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1830} \rangle \Rightarrow \langle S_{11}, \sigma_{1831} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7322} = \langle S_{11}, \sigma_{1831} \rangle \Rightarrow \langle S_{12}, \sigma_{1831} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7323} = \langle S_{12}, \sigma_{1831} \rangle \Rightarrow \langle S_{13}, \sigma_{1831} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7325} = \frac{\langle x := x + 1, \sigma_{1831} \rangle \Rightarrow \sigma_{1832} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1831} \rangle \Rightarrow \langle S_6, \sigma_{1832} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7324} = \frac{\mathcal{T}_{7325}}{\langle S_{13}, \sigma_{1831} \rangle \Rightarrow \langle S_{15}, \sigma_{1832} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7326} = \frac{\langle S_6, \sigma_{1832} \rangle \Rightarrow \langle S_8, \sigma_{1832} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1832} \rangle \Rightarrow \langle S_9, \sigma_{1832} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7328} = \langle S_8, \sigma_{1832} \rangle \Rightarrow \langle y := y + 1, \sigma_{1832} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7327} = \frac{\mathcal{T}_{7328}}{\langle S_9, \sigma_{1832} \rangle \Rightarrow \langle S_{10}, \sigma_{1832} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7329} = \frac{\langle y := y + 1, \sigma_{1832} \rangle \Rightarrow \sigma_{1833} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1832} \rangle \Rightarrow \langle S_{11}, \sigma_{1833} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7330} = \langle S_{11}, \sigma_{1833} \rangle \Rightarrow \langle S_{12}, \sigma_{1833} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7331} = \langle S_{12}, \sigma_{1833} \rangle \Rightarrow \langle S_{13}, \sigma_{1833} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7333} = \frac{\langle x := x + 1, \sigma_{1833} \rangle \Rightarrow \sigma_{1834} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1833} \rangle \Rightarrow \langle S_6, \sigma_{1834} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7332} = \frac{\mathcal{T}_{7333}}{\langle S_{13}, \sigma_{1833} \rangle \Rightarrow \langle S_{15}, \sigma_{1834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7334} = \frac{\langle S_6, \sigma_{1834} \rangle \Rightarrow \langle S_8, \sigma_{1834} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1834} \rangle \Rightarrow \langle S_9, \sigma_{1834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7336} = \langle S_8, \sigma_{1834} \rangle \Rightarrow \langle y := y + 1, \sigma_{1834} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7335} = \frac{\mathcal{T}_{7336}}{\langle S_9, \sigma_{1834} \rangle \Rightarrow \langle S_{10}, \sigma_{1834} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7337} = \frac{\langle y := y + 1, \sigma_{1834} \rangle \Rightarrow \sigma_{1835} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1834} \rangle \Rightarrow \langle S_{11}, \sigma_{1835} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7338} = \langle S_{11}, \sigma_{1835} \rangle \Rightarrow \langle S_{12}, \sigma_{1835} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7339} = \langle S_{12}, \sigma_{1835} \rangle \Rightarrow \langle S_{13}, \sigma_{1835} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7341} = \frac{\langle x := x + 1, \sigma_{1835} \rangle \Rightarrow \sigma_{1836} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1835} \rangle \Rightarrow \langle S_6, \sigma_{1836} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7340} = \frac{\mathcal{T}_{7341}}{\langle S_{13}, \sigma_{1835} \rangle \Rightarrow \langle S_{15}, \sigma_{1836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7342} = \frac{\langle S_6, \sigma_{1836} \rangle \Rightarrow \langle S_8, \sigma_{1836} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1836} \rangle \Rightarrow \langle S_9, \sigma_{1836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7344} = \langle S_8, \sigma_{1836} \rangle \Rightarrow \langle y := y + 1, \sigma_{1836} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7343} = \frac{\mathcal{T}_{7344}}{\langle S_9, \sigma_{1836} \rangle \Rightarrow \langle S_{10}, \sigma_{1836} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7345} = \frac{\langle y := y + 1, \sigma_{1836} \rangle \Rightarrow \sigma_{1837} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1836} \rangle \Rightarrow \langle S_{11}, \sigma_{1837} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7346} = \langle S_{11}, \sigma_{1837} \rangle \Rightarrow \langle S_{12}, \sigma_{1837} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7347} = \langle S_{12}, \sigma_{1837} \rangle \Rightarrow \langle S_{13}, \sigma_{1837} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7349} = \frac{\langle x := x + 1, \sigma_{1837} \rangle \Rightarrow \sigma_{1838} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1837} \rangle \Rightarrow \langle S_6, \sigma_{1838} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7348} = \frac{\mathcal{T}_{7349}}{\langle S_{13}, \sigma_{1837} \rangle \Rightarrow \langle S_{15}, \sigma_{1838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7350} = \frac{\langle S_6, \sigma_{1838} \rangle \Rightarrow \langle S_8, \sigma_{1838} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1838} \rangle \Rightarrow \langle S_9, \sigma_{1838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7352} = \langle S_8, \sigma_{1838} \rangle \Rightarrow \langle y := y + 1, \sigma_{1838} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7351} = \frac{\mathcal{T}_{7352}}{\langle S_9, \sigma_{1838} \rangle \Rightarrow \langle S_{10}, \sigma_{1838} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7353} = \frac{\langle y := y + 1, \sigma_{1838} \rangle \Rightarrow \sigma_{1839} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1838} \rangle \Rightarrow \langle S_{11}, \sigma_{1839} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7354} = \langle S_{11}, \sigma_{1839} \rangle \Rightarrow \langle S_{12}, \sigma_{1839} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7355} = \langle S_{12}, \sigma_{1839} \rangle \Rightarrow \langle S_{13}, \sigma_{1839} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7357} = \frac{\langle x := x + 1, \sigma_{1839} \rangle \Rightarrow \sigma_{1840} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1839} \rangle \Rightarrow \langle S_6, \sigma_{1840} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7356} = \frac{\mathcal{T}_{7357}}{\langle S_{13}, \sigma_{1839} \rangle \Rightarrow \langle S_{15}, \sigma_{1840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7358} = \frac{\langle S_6, \sigma_{1840} \rangle \Rightarrow \langle S_8, \sigma_{1840} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1840} \rangle \Rightarrow \langle S_9, \sigma_{1840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7360} = \langle S_8, \sigma_{1840} \rangle \Rightarrow \langle y := y + 1, \sigma_{1840} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7359} = \frac{\mathcal{T}_{7360}}{\langle S_9, \sigma_{1840} \rangle \Rightarrow \langle S_{10}, \sigma_{1840} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7361} = \frac{\langle y := y + 1, \sigma_{1840} \rangle \Rightarrow \sigma_{1841} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1840} \rangle \Rightarrow \langle S_{11}, \sigma_{1841} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7362} = \langle S_{11}, \sigma_{1841} \rangle \Rightarrow \langle S_{12}, \sigma_{1841} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7363} = \langle S_{12}, \sigma_{1841} \rangle \Rightarrow \langle S_{13}, \sigma_{1841} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7365} = \frac{\langle x := x + 1, \sigma_{1841} \rangle \Rightarrow \sigma_{1842} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1841} \rangle \Rightarrow \langle S_6, \sigma_{1842} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7364} = \frac{\mathcal{T}_{7365}}{\langle S_{13}, \sigma_{1841} \rangle \Rightarrow \langle S_{15}, \sigma_{1842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7366} = \frac{\langle S_6, \sigma_{1842} \rangle \Rightarrow \langle S_8, \sigma_{1842} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1842} \rangle \Rightarrow \langle S_9, \sigma_{1842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7368} = \langle S_8, \sigma_{1842} \rangle \Rightarrow \langle y := y + 1, \sigma_{1842} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7367} = \frac{\mathcal{T}_{7368}}{\langle S_9, \sigma_{1842} \rangle \Rightarrow \langle S_{10}, \sigma_{1842} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7369} = \frac{\langle y := y + 1, \sigma_{1842} \rangle \Rightarrow \sigma_{1843} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1842} \rangle \Rightarrow \langle S_{11}, \sigma_{1843} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7370} = \langle S_{11}, \sigma_{1843} \rangle \Rightarrow \langle S_{12}, \sigma_{1843} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7371} = \langle S_{12}, \sigma_{1843} \rangle \Rightarrow \langle S_{13}, \sigma_{1843} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7373} = \frac{\langle x := x + 1, \sigma_{1843} \rangle \Rightarrow \sigma_{1844} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1843} \rangle \Rightarrow \langle S_6, \sigma_{1844} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7372} = \frac{\mathcal{T}_{7373}}{\langle S_{13}, \sigma_{1843} \rangle \Rightarrow \langle S_{15}, \sigma_{1844} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7374} = \frac{\langle S_6, \sigma_{1844} \rangle \Rightarrow \langle S_8, \sigma_{1844} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1844} \rangle \Rightarrow \langle S_9, \sigma_{1844} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7376} = \langle S_8, \sigma_{1844} \rangle \Rightarrow \langle y := y + 1, \sigma_{1844} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7375} = \frac{\mathcal{T}_{7376}}{\langle S_9, \sigma_{1844} \rangle \Rightarrow \langle S_{10}, \sigma_{1844} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7377} = \frac{\langle y := y + 1, \sigma_{1844} \rangle \Rightarrow \sigma_{1845} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1844} \rangle \Rightarrow \langle S_{11}, \sigma_{1845} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7378} = \langle S_{11}, \sigma_{1845} \rangle \Rightarrow \langle S_{12}, \sigma_{1845} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7379} = \langle S_{12}, \sigma_{1845} \rangle \Rightarrow \langle S_{13}, \sigma_{1845} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7381} = \frac{\langle x := x + 1, \sigma_{1845} \rangle \Rightarrow \sigma_{1846} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1845} \rangle \Rightarrow \langle S_6, \sigma_{1846} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7380} = \frac{\mathcal{T}_{7381}}{\langle S_{13}, \sigma_{1845} \rangle \Rightarrow \langle S_{15}, \sigma_{1846} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7382} = \frac{\langle S_6, \sigma_{1846} \rangle \Rightarrow \langle S_8, \sigma_{1846} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1846} \rangle \Rightarrow \langle S_9, \sigma_{1846} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7384} = \langle S_8, \sigma_{1846} \rangle \Rightarrow \langle y := y + 1, \sigma_{1846} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7383} = \frac{\mathcal{T}_{7384}}{\langle S_9, \sigma_{1846} \rangle \Rightarrow \langle S_{10}, \sigma_{1846} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7385} = \frac{\langle y := y + 1, \sigma_{1846} \rangle \Rightarrow \sigma_{1847} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1846} \rangle \Rightarrow \langle S_{11}, \sigma_{1847} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7386} = \langle S_{11}, \sigma_{1847} \rangle \Rightarrow \langle S_{12}, \sigma_{1847} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7387} = \langle S_{12}, \sigma_{1847} \rangle \Rightarrow \langle S_{13}, \sigma_{1847} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7389} = \frac{\langle x := x + 1, \sigma_{1847} \rangle \Rightarrow \sigma_{1848} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1847} \rangle \Rightarrow \langle S_6, \sigma_{1848} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7388} = \frac{\mathcal{T}_{7389}}{\langle S_{13}, \sigma_{1847} \rangle \Rightarrow \langle S_{15}, \sigma_{1848} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7390} = \frac{\langle S_6, \sigma_{1848} \rangle \Rightarrow \langle S_8, \sigma_{1848} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1848} \rangle \Rightarrow \langle S_9, \sigma_{1848} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7392} = \langle S_8, \sigma_{1848} \rangle \Rightarrow \langle y := y + 1, \sigma_{1848} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7391} = \frac{\mathcal{T}_{7392}}{\langle S_9, \sigma_{1848} \rangle \Rightarrow \langle S_{10}, \sigma_{1848} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7393} = \frac{\langle y := y + 1, \sigma_{1848} \rangle \Rightarrow \sigma_{1849} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1848} \rangle \Rightarrow \langle S_{11}, \sigma_{1849} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7394} = \langle S_{11}, \sigma_{1849} \rangle \Rightarrow \langle S_{12}, \sigma_{1849} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7395} = \langle S_{12}, \sigma_{1849} \rangle \Rightarrow \langle S_{13}, \sigma_{1849} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7397} = \frac{\langle x := x + 1, \sigma_{1849} \rangle \Rightarrow \sigma_{1850} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1849} \rangle \Rightarrow \langle S_6, \sigma_{1850} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7396} = \frac{\mathcal{T}_{7397}}{\langle S_{13}, \sigma_{1849} \rangle \Rightarrow \langle S_{15}, \sigma_{1850} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7398} = \frac{\langle S_6, \sigma_{1850} \rangle \Rightarrow \langle S_8, \sigma_{1850} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1850} \rangle \Rightarrow \langle S_9, \sigma_{1850} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7400} = \langle S_8, \sigma_{1850} \rangle \Rightarrow \langle y := y + 1, \sigma_{1850} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7399} = \frac{\mathcal{T}_{7400}}{\langle S_9, \sigma_{1850} \rangle \Rightarrow \langle S_{10}, \sigma_{1850} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7401} = \frac{\langle y := y + 1, \sigma_{1850} \rangle \Rightarrow \sigma_{1851} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1850} \rangle \Rightarrow \langle S_{11}, \sigma_{1851} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7402} = \langle S_{11}, \sigma_{1851} \rangle \Rightarrow \langle S_{12}, \sigma_{1851} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7403} = \langle S_{12}, \sigma_{1851} \rangle \Rightarrow \langle S_{13}, \sigma_{1851} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7405} = \frac{\langle x := x + 1, \sigma_{1851} \rangle \Rightarrow \sigma_{1852} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1851} \rangle \Rightarrow \langle S_6, \sigma_{1852} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7404} = \frac{\mathcal{T}_{7405}}{\langle S_{13}, \sigma_{1851} \rangle \Rightarrow \langle S_{15}, \sigma_{1852} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7406} = \frac{\langle S_6, \sigma_{1852} \rangle \Rightarrow \langle S_8, \sigma_{1852} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1852} \rangle \Rightarrow \langle S_9, \sigma_{1852} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7408} = \langle S_8, \sigma_{1852} \rangle \Rightarrow \langle y := y + 1, \sigma_{1852} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7407} = \frac{\mathcal{T}_{7408}}{\langle S_9, \sigma_{1852} \rangle \Rightarrow \langle S_{10}, \sigma_{1852} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7409} = \frac{\langle y := y + 1, \sigma_{1852} \rangle \Rightarrow \sigma_{1853} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1852} \rangle \Rightarrow \langle S_{11}, \sigma_{1853} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7410} = \langle S_{11}, \sigma_{1853} \rangle \Rightarrow \langle S_{12}, \sigma_{1853} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7411} = \langle S_{12}, \sigma_{1853} \rangle \Rightarrow \langle S_{13}, \sigma_{1853} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7413} = \frac{\langle x := x + 1, \sigma_{1853} \rangle \Rightarrow \sigma_{1854} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1853} \rangle \Rightarrow \langle S_6, \sigma_{1854} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7412} = \frac{\mathcal{T}_{7413}}{\langle S_{13}, \sigma_{1853} \rangle \Rightarrow \langle S_{15}, \sigma_{1854} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7414} = \frac{\langle S_6, \sigma_{1854} \rangle \Rightarrow \langle S_8, \sigma_{1854} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1854} \rangle \Rightarrow \langle S_9, \sigma_{1854} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7416} = \langle S_8, \sigma_{1854} \rangle \Rightarrow \langle y := y + 1, \sigma_{1854} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7415} = \frac{\mathcal{T}_{7416}}{\langle S_9, \sigma_{1854} \rangle \Rightarrow \langle S_{10}, \sigma_{1854} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7417} = \frac{\langle y := y + 1, \sigma_{1854} \rangle \Rightarrow \sigma_{1855} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1854} \rangle \Rightarrow \langle S_{11}, \sigma_{1855} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7418} = \langle S_{11}, \sigma_{1855} \rangle \Rightarrow \langle S_{12}, \sigma_{1855} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7419} = \langle S_{12}, \sigma_{1855} \rangle \Rightarrow \langle S_{13}, \sigma_{1855} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7421} = \frac{\langle x := x + 1, \sigma_{1855} \rangle \Rightarrow \sigma_{1856} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1855} \rangle \Rightarrow \langle S_6, \sigma_{1856} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7420} = \frac{\mathcal{T}_{7421}}{\langle S_{13}, \sigma_{1855} \rangle \Rightarrow \langle S_{15}, \sigma_{1856} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7422} = \frac{\langle S_6, \sigma_{1856} \rangle \Rightarrow \langle S_8, \sigma_{1856} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1856} \rangle \Rightarrow \langle S_9, \sigma_{1856} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7424} = \langle S_8, \sigma_{1856} \rangle \Rightarrow \langle y := y + 1, \sigma_{1856} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7423} = \frac{\mathcal{T}_{7424}}{\langle S_9, \sigma_{1856} \rangle \Rightarrow \langle S_{10}, \sigma_{1856} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7425} = \frac{\langle y := y + 1, \sigma_{1856} \rangle \Rightarrow \sigma_{1857} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1856} \rangle \Rightarrow \langle S_{11}, \sigma_{1857} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7426} = \langle S_{11}, \sigma_{1857} \rangle \Rightarrow \langle S_{12}, \sigma_{1857} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7427} = \langle S_{12}, \sigma_{1857} \rangle \Rightarrow \langle S_{13}, \sigma_{1857} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7429} = \frac{\langle x := x + 1, \sigma_{1857} \rangle \Rightarrow \sigma_{1858} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1857} \rangle \Rightarrow \langle S_6, \sigma_{1858} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7428} = \frac{\mathcal{T}_{7429}}{\langle S_{13}, \sigma_{1857} \rangle \Rightarrow \langle S_{15}, \sigma_{1858} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7430} = \frac{\langle S_6, \sigma_{1858} \rangle \Rightarrow \langle S_8, \sigma_{1858} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1858} \rangle \Rightarrow \langle S_9, \sigma_{1858} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7432} = \langle S_8, \sigma_{1858} \rangle \Rightarrow \langle y := y + 1, \sigma_{1858} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7431} = \frac{\mathcal{T}_{7432}}{\langle S_9, \sigma_{1858} \rangle \Rightarrow \langle S_{10}, \sigma_{1858} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7433} = \frac{\langle y := y + 1, \sigma_{1858} \rangle \Rightarrow \sigma_{1859} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1858} \rangle \Rightarrow \langle S_{11}, \sigma_{1859} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7434} = \langle S_{11}, \sigma_{1859} \rangle \Rightarrow \langle S_{12}, \sigma_{1859} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7435} = \langle S_{12}, \sigma_{1859} \rangle \Rightarrow \langle S_{13}, \sigma_{1859} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7437} = \frac{\langle x := x + 1, \sigma_{1859} \rangle \Rightarrow \sigma_{1860} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1859} \rangle \Rightarrow \langle S_6, \sigma_{1860} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7436} = \frac{\mathcal{T}_{7437}}{\langle S_{13}, \sigma_{1859} \rangle \Rightarrow \langle S_{15}, \sigma_{1860} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7438} = \frac{\langle S_6, \sigma_{1860} \rangle \Rightarrow \langle S_8, \sigma_{1860} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1860} \rangle \Rightarrow \langle S_9, \sigma_{1860} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7440} = \langle S_8, \sigma_{1860} \rangle \Rightarrow \langle y := y + 1, \sigma_{1860} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7439} = \frac{\mathcal{T}_{7440}}{\langle S_9, \sigma_{1860} \rangle \Rightarrow \langle S_{10}, \sigma_{1860} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7441} = \frac{\langle y := y + 1, \sigma_{1860} \rangle \Rightarrow \sigma_{1861} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1860} \rangle \Rightarrow \langle S_{11}, \sigma_{1861} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7442} = \langle S_{11}, \sigma_{1861} \rangle \Rightarrow \langle S_{12}, \sigma_{1861} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7443} = \langle S_{12}, \sigma_{1861} \rangle \Rightarrow \langle S_{13}, \sigma_{1861} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7445} = \frac{\langle x := x + 1, \sigma_{1861} \rangle \Rightarrow \sigma_{1862} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1861} \rangle \Rightarrow \langle S_6, \sigma_{1862} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7444} = \frac{\mathcal{T}_{7445}}{\langle S_{13}, \sigma_{1861} \rangle \Rightarrow \langle S_{15}, \sigma_{1862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7446} = \frac{\langle S_6, \sigma_{1862} \rangle \Rightarrow \langle S_8, \sigma_{1862} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1862} \rangle \Rightarrow \langle S_9, \sigma_{1862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7448} = \langle S_8, \sigma_{1862} \rangle \Rightarrow \langle y := y + 1, \sigma_{1862} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7447} = \frac{\mathcal{T}_{7448}}{\langle S_9, \sigma_{1862} \rangle \Rightarrow \langle S_{10}, \sigma_{1862} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7449} = \frac{\langle y := y + 1, \sigma_{1862} \rangle \Rightarrow \sigma_{1863} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1862} \rangle \Rightarrow \langle S_{11}, \sigma_{1863} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7450} = \langle S_{11}, \sigma_{1863} \rangle \Rightarrow \langle S_{12}, \sigma_{1863} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7451} = \langle S_{12}, \sigma_{1863} \rangle \Rightarrow \langle S_{13}, \sigma_{1863} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7453} = \frac{\langle x := x + 1, \sigma_{1863} \rangle \Rightarrow \sigma_{1864} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1863} \rangle \Rightarrow \langle S_6, \sigma_{1864} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7452} = \frac{\mathcal{T}_{7453}}{\langle S_{13}, \sigma_{1863} \rangle \Rightarrow \langle S_{15}, \sigma_{1864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7454} = \frac{\langle S_6, \sigma_{1864} \rangle \Rightarrow \langle S_8, \sigma_{1864} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1864} \rangle \Rightarrow \langle S_9, \sigma_{1864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7456} = \langle S_8, \sigma_{1864} \rangle \Rightarrow \langle y := y + 1, \sigma_{1864} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7455} = \frac{\mathcal{T}_{7456}}{\langle S_9, \sigma_{1864} \rangle \Rightarrow \langle S_{10}, \sigma_{1864} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7457} = \frac{\langle y := y + 1, \sigma_{1864} \rangle \Rightarrow \sigma_{1865} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1864} \rangle \Rightarrow \langle S_{11}, \sigma_{1865} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7458} = \langle S_{11}, \sigma_{1865} \rangle \Rightarrow \langle S_{12}, \sigma_{1865} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7459} = \langle S_{12}, \sigma_{1865} \rangle \Rightarrow \langle S_{13}, \sigma_{1865} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7461} = \frac{\langle x := x + 1, \sigma_{1865} \rangle \Rightarrow \sigma_{1866} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1865} \rangle \Rightarrow \langle S_6, \sigma_{1866} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7460} = \frac{\mathcal{T}_{7461}}{\langle S_{13}, \sigma_{1865} \rangle \Rightarrow \langle S_{15}, \sigma_{1866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7462} = \frac{\langle S_6, \sigma_{1866} \rangle \Rightarrow \langle S_8, \sigma_{1866} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1866} \rangle \Rightarrow \langle S_9, \sigma_{1866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7464} = \langle S_8, \sigma_{1866} \rangle \Rightarrow \langle y := y + 1, \sigma_{1866} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7463} = \frac{\mathcal{T}_{7464}}{\langle S_9, \sigma_{1866} \rangle \Rightarrow \langle S_{10}, \sigma_{1866} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7465} = \frac{\langle y := y + 1, \sigma_{1866} \rangle \Rightarrow \sigma_{1867} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1866} \rangle \Rightarrow \langle S_{11}, \sigma_{1867} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7466} = \langle S_{11}, \sigma_{1867} \rangle \Rightarrow \langle S_{12}, \sigma_{1867} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7467} = \langle S_{12}, \sigma_{1867} \rangle \Rightarrow \langle S_{13}, \sigma_{1867} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7469} = \frac{\langle x := x + 1, \sigma_{1867} \rangle \Rightarrow \sigma_{1868} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1867} \rangle \Rightarrow \langle S_6, \sigma_{1868} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7468} = \frac{\mathcal{T}_{7469}}{\langle S_{13}, \sigma_{1867} \rangle \Rightarrow \langle S_{15}, \sigma_{1868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7470} = \frac{\langle S_6, \sigma_{1868} \rangle \Rightarrow \langle S_8, \sigma_{1868} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1868} \rangle \Rightarrow \langle S_9, \sigma_{1868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7472} = \langle S_8, \sigma_{1868} \rangle \Rightarrow \langle y := y + 1, \sigma_{1868} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7471} = \frac{\mathcal{T}_{7472}}{\langle S_9, \sigma_{1868} \rangle \Rightarrow \langle S_{10}, \sigma_{1868} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7473} = \frac{\langle y := y + 1, \sigma_{1868} \rangle \Rightarrow \sigma_{1869} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1868} \rangle \Rightarrow \langle S_{11}, \sigma_{1869} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7474} = \langle S_{11}, \sigma_{1869} \rangle \Rightarrow \langle S_{12}, \sigma_{1869} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7475} = \langle S_{12}, \sigma_{1869} \rangle \Rightarrow \langle S_{13}, \sigma_{1869} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7477} = \frac{\langle x := x + 1, \sigma_{1869} \rangle \Rightarrow \sigma_{1870} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1869} \rangle \Rightarrow \langle S_6, \sigma_{1870} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7476} = \frac{\mathcal{T}_{7477}}{\langle S_{13}, \sigma_{1869} \rangle \Rightarrow \langle S_{15}, \sigma_{1870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7478} = \frac{\langle S_6, \sigma_{1870} \rangle \Rightarrow \langle S_8, \sigma_{1870} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1870} \rangle \Rightarrow \langle S_9, \sigma_{1870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7480} = \langle S_8, \sigma_{1870} \rangle \Rightarrow \langle y := y + 1, \sigma_{1870} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7479} = \frac{\mathcal{T}_{7480}}{\langle S_9, \sigma_{1870} \rangle \Rightarrow \langle S_{10}, \sigma_{1870} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7481} = \frac{\langle y := y + 1, \sigma_{1870} \rangle \Rightarrow \sigma_{1871} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1870} \rangle \Rightarrow \langle S_{11}, \sigma_{1871} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7482} = \langle S_{11}, \sigma_{1871} \rangle \Rightarrow \langle S_{12}, \sigma_{1871} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7483} = \langle S_{12}, \sigma_{1871} \rangle \Rightarrow \langle S_{13}, \sigma_{1871} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7485} = \frac{\langle x := x + 1, \sigma_{1871} \rangle \Rightarrow \sigma_{1872} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1871} \rangle \Rightarrow \langle S_6, \sigma_{1872} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7484} = \frac{\mathcal{T}_{7485}}{\langle S_{13}, \sigma_{1871} \rangle \Rightarrow \langle S_{15}, \sigma_{1872} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7486} = \frac{\langle S_6, \sigma_{1872} \rangle \Rightarrow \langle S_8, \sigma_{1872} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1872} \rangle \Rightarrow \langle S_9, \sigma_{1872} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7488} = \langle S_8, \sigma_{1872} \rangle \Rightarrow \langle y := y + 1, \sigma_{1872} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7487} = \frac{\mathcal{T}_{7488}}{\langle S_9, \sigma_{1872} \rangle \Rightarrow \langle S_{10}, \sigma_{1872} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7489} = \frac{\langle y := y + 1, \sigma_{1872} \rangle \Rightarrow \sigma_{1873} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1872} \rangle \Rightarrow \langle S_{11}, \sigma_{1873} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7490} = \langle S_{11}, \sigma_{1873} \rangle \Rightarrow \langle S_{12}, \sigma_{1873} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7491} = \langle S_{12}, \sigma_{1873} \rangle \Rightarrow \langle S_{13}, \sigma_{1873} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7493} = \frac{\langle x := x + 1, \sigma_{1873} \rangle \Rightarrow \sigma_{1874} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1873} \rangle \Rightarrow \langle S_6, \sigma_{1874} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7492} = \frac{\mathcal{T}_{7493}}{\langle S_{13}, \sigma_{1873} \rangle \Rightarrow \langle S_{15}, \sigma_{1874} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7494} = \frac{\langle S_6, \sigma_{1874} \rangle \Rightarrow \langle S_8, \sigma_{1874} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1874} \rangle \Rightarrow \langle S_9, \sigma_{1874} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7496} = \langle S_8, \sigma_{1874} \rangle \Rightarrow \langle y := y + 1, \sigma_{1874} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7495} = \frac{\mathcal{T}_{7496}}{\langle S_9, \sigma_{1874} \rangle \Rightarrow \langle S_{10}, \sigma_{1874} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7497} = \frac{\langle y := y + 1, \sigma_{1874} \rangle \Rightarrow \sigma_{1875} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1874} \rangle \Rightarrow \langle S_{11}, \sigma_{1875} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7498} = \langle S_{11}, \sigma_{1875} \rangle \Rightarrow \langle S_{12}, \sigma_{1875} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7499} = \langle S_{12}, \sigma_{1875} \rangle \Rightarrow \langle S_{13}, \sigma_{1875} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7501} = \frac{\langle x := x + 1, \sigma_{1875} \rangle \Rightarrow \sigma_{1876} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1875} \rangle \Rightarrow \langle S_6, \sigma_{1876} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7500} = \frac{\mathcal{T}_{7501}}{\langle S_{13}, \sigma_{1875} \rangle \Rightarrow \langle S_{15}, \sigma_{1876} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7502} = \frac{\langle S_6, \sigma_{1876} \rangle \Rightarrow \langle S_8, \sigma_{1876} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1876} \rangle \Rightarrow \langle S_9, \sigma_{1876} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7504} = \langle S_8, \sigma_{1876} \rangle \Rightarrow \langle y := y + 1, \sigma_{1876} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7503} = \frac{\mathcal{T}_{7504}}{\langle S_9, \sigma_{1876} \rangle \Rightarrow \langle S_{10}, \sigma_{1876} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7505} = \frac{\langle y := y + 1, \sigma_{1876} \rangle \Rightarrow \sigma_{1877} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1876} \rangle \Rightarrow \langle S_{11}, \sigma_{1877} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7506} = \langle S_{11}, \sigma_{1877} \rangle \Rightarrow \langle S_{12}, \sigma_{1877} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7507} = \langle S_{12}, \sigma_{1877} \rangle \Rightarrow \langle S_{13}, \sigma_{1877} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7509} = \frac{\langle x := x + 1, \sigma_{1877} \rangle \Rightarrow \sigma_{1878} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1877} \rangle \Rightarrow \langle S_6, \sigma_{1878} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7508} = \frac{\mathcal{T}_{7509}}{\langle S_{13}, \sigma_{1877} \rangle \Rightarrow \langle S_{15}, \sigma_{1878} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7510} = \frac{\langle S_6, \sigma_{1878} \rangle \Rightarrow \langle S_8, \sigma_{1878} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1878} \rangle \Rightarrow \langle S_9, \sigma_{1878} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7512} = \langle S_8, \sigma_{1878} \rangle \Rightarrow \langle y := y + 1, \sigma_{1878} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7511} = \frac{\mathcal{T}_{7512}}{\langle S_9, \sigma_{1878} \rangle \Rightarrow \langle S_{10}, \sigma_{1878} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7513} = \frac{\langle y := y + 1, \sigma_{1878} \rangle \Rightarrow \sigma_{1879} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1878} \rangle \Rightarrow \langle S_{11}, \sigma_{1879} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7514} = \langle S_{11}, \sigma_{1879} \rangle \Rightarrow \langle S_{12}, \sigma_{1879} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7515} = \langle S_{12}, \sigma_{1879} \rangle \Rightarrow \langle S_{13}, \sigma_{1879} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7517} = \frac{\langle x := x + 1, \sigma_{1879} \rangle \Rightarrow \sigma_{1880} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1879} \rangle \Rightarrow \langle S_6, \sigma_{1880} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7516} = \frac{\mathcal{T}_{7517}}{\langle S_{13}, \sigma_{1879} \rangle \Rightarrow \langle S_{15}, \sigma_{1880} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7518} = \frac{\langle S_6, \sigma_{1880} \rangle \Rightarrow \langle S_8, \sigma_{1880} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1880} \rangle \Rightarrow \langle S_9, \sigma_{1880} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7520} = \langle S_8, \sigma_{1880} \rangle \Rightarrow \langle y := y + 1, \sigma_{1880} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7519} = \frac{\mathcal{T}_{7520}}{\langle S_9, \sigma_{1880} \rangle \Rightarrow \langle S_{10}, \sigma_{1880} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7521} = \frac{\langle y := y + 1, \sigma_{1880} \rangle \Rightarrow \sigma_{1881} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1880} \rangle \Rightarrow \langle S_{11}, \sigma_{1881} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7522} = \langle S_{11}, \sigma_{1881} \rangle \Rightarrow \langle S_{12}, \sigma_{1881} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7523} = \langle S_{12}, \sigma_{1881} \rangle \Rightarrow \langle S_{13}, \sigma_{1881} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7525} = \frac{\langle x := x + 1, \sigma_{1881} \rangle \Rightarrow \sigma_{1882} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1881} \rangle \Rightarrow \langle S_6, \sigma_{1882} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7524} = \frac{\mathcal{T}_{7525}}{\langle S_{13}, \sigma_{1881} \rangle \Rightarrow \langle S_{15}, \sigma_{1882} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7526} = \frac{\langle S_6, \sigma_{1882} \rangle \Rightarrow \langle S_8, \sigma_{1882} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1882} \rangle \Rightarrow \langle S_9, \sigma_{1882} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7528} = \langle S_8, \sigma_{1882} \rangle \Rightarrow \langle y := y + 1, \sigma_{1882} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7527} = \frac{\mathcal{T}_{7528}}{\langle S_9, \sigma_{1882} \rangle \Rightarrow \langle S_{10}, \sigma_{1882} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7529} = \frac{\langle y := y + 1, \sigma_{1882} \rangle \Rightarrow \sigma_{1883} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1882} \rangle \Rightarrow \langle S_{11}, \sigma_{1883} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7530} = \langle S_{11}, \sigma_{1883} \rangle \Rightarrow \langle S_{12}, \sigma_{1883} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7531} = \langle S_{12}, \sigma_{1883} \rangle \Rightarrow \langle S_{13}, \sigma_{1883} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7533} = \frac{\langle x := x + 1, \sigma_{1883} \rangle \Rightarrow \sigma_{1884} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1883} \rangle \Rightarrow \langle S_6, \sigma_{1884} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7532} = \frac{\mathcal{T}_{7533}}{\langle S_{13}, \sigma_{1883} \rangle \Rightarrow \langle S_{15}, \sigma_{1884} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7534} = \frac{\langle S_6, \sigma_{1884} \rangle \Rightarrow \langle S_8, \sigma_{1884} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1884} \rangle \Rightarrow \langle S_9, \sigma_{1884} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7536} = \langle S_8, \sigma_{1884} \rangle \Rightarrow \langle y := y + 1, \sigma_{1884} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7535} = \frac{\mathcal{T}_{7536}}{\langle S_9, \sigma_{1884} \rangle \Rightarrow \langle S_{10}, \sigma_{1884} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7537} = \frac{\langle y := y + 1, \sigma_{1884} \rangle \Rightarrow \sigma_{1885} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1884} \rangle \Rightarrow \langle S_{11}, \sigma_{1885} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7538} = \langle S_{11}, \sigma_{1885} \rangle \Rightarrow \langle S_{12}, \sigma_{1885} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7539} = \langle S_{12}, \sigma_{1885} \rangle \Rightarrow \langle S_{13}, \sigma_{1885} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7541} = \frac{\langle x := x + 1, \sigma_{1885} \rangle \Rightarrow \sigma_{1886} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1885} \rangle \Rightarrow \langle S_6, \sigma_{1886} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7540} = \frac{\mathcal{T}_{7541}}{\langle S_{13}, \sigma_{1885} \rangle \Rightarrow \langle S_{15}, \sigma_{1886} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7542} = \frac{\langle S_6, \sigma_{1886} \rangle \Rightarrow \langle S_8, \sigma_{1886} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1886} \rangle \Rightarrow \langle S_9, \sigma_{1886} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7544} = \langle S_8, \sigma_{1886} \rangle \Rightarrow \langle y := y + 1, \sigma_{1886} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7543} = \frac{\mathcal{T}_{7544}}{\langle S_9, \sigma_{1886} \rangle \Rightarrow \langle S_{10}, \sigma_{1886} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7545} = \frac{\langle y := y + 1, \sigma_{1886} \rangle \Rightarrow \sigma_{1887} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1886} \rangle \Rightarrow \langle S_{11}, \sigma_{1887} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7546} = \langle S_{11}, \sigma_{1887} \rangle \Rightarrow \langle S_{12}, \sigma_{1887} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7547} = \langle S_{12}, \sigma_{1887} \rangle \Rightarrow \langle S_{13}, \sigma_{1887} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7549} = \frac{\langle x := x + 1, \sigma_{1887} \rangle \Rightarrow \sigma_{1888} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1887} \rangle \Rightarrow \langle S_6, \sigma_{1888} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7548} = \frac{\mathcal{T}_{7549}}{\langle S_{13}, \sigma_{1887} \rangle \Rightarrow \langle S_{15}, \sigma_{1888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7550} = \frac{\langle S_6, \sigma_{1888} \rangle \Rightarrow \langle S_8, \sigma_{1888} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1888} \rangle \Rightarrow \langle S_9, \sigma_{1888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7552} = \langle S_8, \sigma_{1888} \rangle \Rightarrow \langle y := y + 1, \sigma_{1888} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7551} = \frac{\mathcal{T}_{7552}}{\langle S_9, \sigma_{1888} \rangle \Rightarrow \langle S_{10}, \sigma_{1888} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7553} = \frac{\langle y := y + 1, \sigma_{1888} \rangle \Rightarrow \sigma_{1889} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1888} \rangle \Rightarrow \langle S_{11}, \sigma_{1889} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7554} = \langle S_{11}, \sigma_{1889} \rangle \Rightarrow \langle S_{12}, \sigma_{1889} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7555} = \langle S_{12}, \sigma_{1889} \rangle \Rightarrow \langle S_{13}, \sigma_{1889} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7557} = \frac{\langle x := x + 1, \sigma_{1889} \rangle \Rightarrow \sigma_{1890} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1889} \rangle \Rightarrow \langle S_6, \sigma_{1890} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7556} = \frac{\mathcal{T}_{7557}}{\langle S_{13}, \sigma_{1889} \rangle \Rightarrow \langle S_{15}, \sigma_{1890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7558} = \frac{\langle S_6, \sigma_{1890} \rangle \Rightarrow \langle S_8, \sigma_{1890} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1890} \rangle \Rightarrow \langle S_9, \sigma_{1890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7560} = \langle S_8, \sigma_{1890} \rangle \Rightarrow \langle y := y + 1, \sigma_{1890} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7559} = \frac{\mathcal{T}_{7560}}{\langle S_9, \sigma_{1890} \rangle \Rightarrow \langle S_{10}, \sigma_{1890} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7561} = \frac{\langle y := y + 1, \sigma_{1890} \rangle \Rightarrow \sigma_{1891} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1890} \rangle \Rightarrow \langle S_{11}, \sigma_{1891} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7562} = \langle S_{11}, \sigma_{1891} \rangle \Rightarrow \langle S_{12}, \sigma_{1891} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7563} = \langle S_{12}, \sigma_{1891} \rangle \Rightarrow \langle S_{13}, \sigma_{1891} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7565} = \frac{\langle x := x + 1, \sigma_{1891} \rangle \Rightarrow \sigma_{1892} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1891} \rangle \Rightarrow \langle S_6, \sigma_{1892} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7564} = \frac{\mathcal{T}_{7565}}{\langle S_{13}, \sigma_{1891} \rangle \Rightarrow \langle S_{15}, \sigma_{1892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7566} = \frac{\langle S_6, \sigma_{1892} \rangle \Rightarrow \langle S_8, \sigma_{1892} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1892} \rangle \Rightarrow \langle S_9, \sigma_{1892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7568} = \langle S_8, \sigma_{1892} \rangle \Rightarrow \langle y := y + 1, \sigma_{1892} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7567} = \frac{\mathcal{T}_{7568}}{\langle S_9, \sigma_{1892} \rangle \Rightarrow \langle S_{10}, \sigma_{1892} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7569} = \frac{\langle y := y + 1, \sigma_{1892} \rangle \Rightarrow \sigma_{1893} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1892} \rangle \Rightarrow \langle S_{11}, \sigma_{1893} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7570} = \langle S_{11}, \sigma_{1893} \rangle \Rightarrow \langle S_{12}, \sigma_{1893} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7571} = \langle S_{12}, \sigma_{1893} \rangle \Rightarrow \langle S_{13}, \sigma_{1893} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7573} = \frac{\langle x := x + 1, \sigma_{1893} \rangle \Rightarrow \sigma_{1894} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1893} \rangle \Rightarrow \langle S_6, \sigma_{1894} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7572} = \frac{\mathcal{T}_{7573}}{\langle S_{13}, \sigma_{1893} \rangle \Rightarrow \langle S_{15}, \sigma_{1894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7574} = \frac{\langle S_6, \sigma_{1894} \rangle \Rightarrow \langle S_8, \sigma_{1894} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1894} \rangle \Rightarrow \langle S_9, \sigma_{1894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7576} = \langle S_8, \sigma_{1894} \rangle \Rightarrow \langle y := y + 1, \sigma_{1894} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7575} = \frac{\mathcal{T}_{7576}}{\langle S_9, \sigma_{1894} \rangle \Rightarrow \langle S_{10}, \sigma_{1894} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7577} = \frac{\langle y := y + 1, \sigma_{1894} \rangle \Rightarrow \sigma_{1895} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1894} \rangle \Rightarrow \langle S_{11}, \sigma_{1895} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7578} = \langle S_{11}, \sigma_{1895} \rangle \Rightarrow \langle S_{12}, \sigma_{1895} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7579} = \langle S_{12}, \sigma_{1895} \rangle \Rightarrow \langle S_{13}, \sigma_{1895} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7581} = \frac{\langle x := x + 1, \sigma_{1895} \rangle \Rightarrow \sigma_{1896} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1895} \rangle \Rightarrow \langle S_6, \sigma_{1896} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7580} = \frac{\mathcal{T}_{7581}}{\langle S_{13}, \sigma_{1895} \rangle \Rightarrow \langle S_{15}, \sigma_{1896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7582} = \frac{\langle S_6, \sigma_{1896} \rangle \Rightarrow \langle S_8, \sigma_{1896} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1896} \rangle \Rightarrow \langle S_9, \sigma_{1896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7584} = \langle S_8, \sigma_{1896} \rangle \Rightarrow \langle y := y + 1, \sigma_{1896} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7583} = \frac{\mathcal{T}_{7584}}{\langle S_9, \sigma_{1896} \rangle \Rightarrow \langle S_{10}, \sigma_{1896} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7585} = \frac{\langle y := y + 1, \sigma_{1896} \rangle \Rightarrow \sigma_{1897} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1896} \rangle \Rightarrow \langle S_{11}, \sigma_{1897} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7586} = \langle S_{11}, \sigma_{1897} \rangle \Rightarrow \langle S_{12}, \sigma_{1897} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7587} = \langle S_{12}, \sigma_{1897} \rangle \Rightarrow \langle S_{13}, \sigma_{1897} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7589} = \frac{\langle x := x + 1, \sigma_{1897} \rangle \Rightarrow \sigma_{1898} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1897} \rangle \Rightarrow \langle S_6, \sigma_{1898} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7588} = \frac{\mathcal{T}_{7589}}{\langle S_{13}, \sigma_{1897} \rangle \Rightarrow \langle S_{15}, \sigma_{1898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7590} = \frac{\langle S_6, \sigma_{1898} \rangle \Rightarrow \langle S_8, \sigma_{1898} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1898} \rangle \Rightarrow \langle S_9, \sigma_{1898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7592} = \langle S_8, \sigma_{1898} \rangle \Rightarrow \langle y := y + 1, \sigma_{1898} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7591} = \frac{\mathcal{T}_{7592}}{\langle S_9, \sigma_{1898} \rangle \Rightarrow \langle S_{10}, \sigma_{1898} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7593} = \frac{\langle y := y + 1, \sigma_{1898} \rangle \Rightarrow \sigma_{1899} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1898} \rangle \Rightarrow \langle S_{11}, \sigma_{1899} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7594} = \langle S_{11}, \sigma_{1899} \rangle \Rightarrow \langle S_{12}, \sigma_{1899} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7595} = \langle S_{12}, \sigma_{1899} \rangle \Rightarrow \langle S_{13}, \sigma_{1899} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7597} = \frac{\langle x := x + 1, \sigma_{1899} \rangle \Rightarrow \sigma_{1900} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1899} \rangle \Rightarrow \langle S_6, \sigma_{1900} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7596} = \frac{\mathcal{T}_{7597}}{\langle S_{13}, \sigma_{1899} \rangle \Rightarrow \langle S_{15}, \sigma_{1900} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7598} = \frac{\langle S_6, \sigma_{1900} \rangle \Rightarrow \langle S_8, \sigma_{1900} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1900} \rangle \Rightarrow \langle S_9, \sigma_{1900} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7600} = \langle S_8, \sigma_{1900} \rangle \Rightarrow \langle y := y + 1, \sigma_{1900} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7599} = \frac{\mathcal{T}_{7600}}{\langle S_9, \sigma_{1900} \rangle \Rightarrow \langle S_{10}, \sigma_{1900} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7601} = \frac{\langle y := y + 1, \sigma_{1900} \rangle \Rightarrow \sigma_{1901} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1900} \rangle \Rightarrow \langle S_{11}, \sigma_{1901} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7602} = \langle S_{11}, \sigma_{1901} \rangle \Rightarrow \langle S_{12}, \sigma_{1901} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7603} = \langle S_{12}, \sigma_{1901} \rangle \Rightarrow \langle S_{13}, \sigma_{1901} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7605} = \frac{\langle x := x + 1, \sigma_{1901} \rangle \Rightarrow \sigma_{1902} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1901} \rangle \Rightarrow \langle S_6, \sigma_{1902} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7604} = \frac{\mathcal{T}_{7605}}{\langle S_{13}, \sigma_{1901} \rangle \Rightarrow \langle S_{15}, \sigma_{1902} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7606} = \frac{\langle S_6, \sigma_{1902} \rangle \Rightarrow \langle S_8, \sigma_{1902} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1902} \rangle \Rightarrow \langle S_9, \sigma_{1902} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7608} = \langle S_8, \sigma_{1902} \rangle \Rightarrow \langle y := y + 1, \sigma_{1902} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7607} = \frac{\mathcal{T}_{7608}}{\langle S_9, \sigma_{1902} \rangle \Rightarrow \langle S_{10}, \sigma_{1902} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7609} = \frac{\langle y := y + 1, \sigma_{1902} \rangle \Rightarrow \sigma_{1903} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1902} \rangle \Rightarrow \langle S_{11}, \sigma_{1903} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7610} = \langle S_{11}, \sigma_{1903} \rangle \Rightarrow \langle S_{12}, \sigma_{1903} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7611} = \langle S_{12}, \sigma_{1903} \rangle \Rightarrow \langle S_{13}, \sigma_{1903} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7613} = \frac{\langle x := x + 1, \sigma_{1903} \rangle \Rightarrow \sigma_{1904} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1903} \rangle \Rightarrow \langle S_6, \sigma_{1904} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7612} = \frac{\mathcal{T}_{7613}}{\langle S_{13}, \sigma_{1903} \rangle \Rightarrow \langle S_{15}, \sigma_{1904} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7614} = \frac{\langle S_6, \sigma_{1904} \rangle \Rightarrow \langle S_8, \sigma_{1904} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1904} \rangle \Rightarrow \langle S_9, \sigma_{1904} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7616} = \langle S_8, \sigma_{1904} \rangle \Rightarrow \langle y := y + 1, \sigma_{1904} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7615} = \frac{\mathcal{T}_{7616}}{\langle S_9, \sigma_{1904} \rangle \Rightarrow \langle S_{10}, \sigma_{1904} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7617} = \frac{\langle y := y + 1, \sigma_{1904} \rangle \Rightarrow \sigma_{1905} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1904} \rangle \Rightarrow \langle S_{11}, \sigma_{1905} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7618} = \langle S_{11}, \sigma_{1905} \rangle \Rightarrow \langle S_{12}, \sigma_{1905} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7619} = \langle S_{12}, \sigma_{1905} \rangle \Rightarrow \langle S_{13}, \sigma_{1905} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7621} = \frac{\langle x := x + 1, \sigma_{1905} \rangle \Rightarrow \sigma_{1906} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1905} \rangle \Rightarrow \langle S_6, \sigma_{1906} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7620} = \frac{\mathcal{T}_{7621}}{\langle S_{13}, \sigma_{1905} \rangle \Rightarrow \langle S_{15}, \sigma_{1906} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7622} = \frac{\langle S_6, \sigma_{1906} \rangle \Rightarrow \langle S_8, \sigma_{1906} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1906} \rangle \Rightarrow \langle S_9, \sigma_{1906} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7624} = \langle S_8, \sigma_{1906} \rangle \Rightarrow \langle y := y + 1, \sigma_{1906} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7623} = \frac{\mathcal{T}_{7624}}{\langle S_9, \sigma_{1906} \rangle \Rightarrow \langle S_{10}, \sigma_{1906} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7625} = \frac{\langle y := y + 1, \sigma_{1906} \rangle \Rightarrow \sigma_{1907} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1906} \rangle \Rightarrow \langle S_{11}, \sigma_{1907} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7626} = \langle S_{11}, \sigma_{1907} \rangle \Rightarrow \langle S_{12}, \sigma_{1907} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7627} = \langle S_{12}, \sigma_{1907} \rangle \Rightarrow \langle S_{13}, \sigma_{1907} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7629} = \frac{\langle x := x + 1, \sigma_{1907} \rangle \Rightarrow \sigma_{1908} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1907} \rangle \Rightarrow \langle S_6, \sigma_{1908} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7628} = \frac{\mathcal{T}_{7629}}{\langle S_{13}, \sigma_{1907} \rangle \Rightarrow \langle S_{15}, \sigma_{1908} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7630} = \frac{\langle S_6, \sigma_{1908} \rangle \Rightarrow \langle S_8, \sigma_{1908} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1908} \rangle \Rightarrow \langle S_9, \sigma_{1908} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7632} = \langle S_8, \sigma_{1908} \rangle \Rightarrow \langle y := y + 1, \sigma_{1908} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7631} = \frac{\mathcal{T}_{7632}}{\langle S_9, \sigma_{1908} \rangle \Rightarrow \langle S_{10}, \sigma_{1908} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7633} = \frac{\langle y := y + 1, \sigma_{1908} \rangle \Rightarrow \sigma_{1909} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1908} \rangle \Rightarrow \langle S_{11}, \sigma_{1909} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7634} = \langle S_{11}, \sigma_{1909} \rangle \Rightarrow \langle S_{12}, \sigma_{1909} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7635} = \langle S_{12}, \sigma_{1909} \rangle \Rightarrow \langle S_{13}, \sigma_{1909} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7637} = \frac{\langle x := x + 1, \sigma_{1909} \rangle \Rightarrow \sigma_{1910} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1909} \rangle \Rightarrow \langle S_6, \sigma_{1910} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7636} = \frac{\mathcal{T}_{7637}}{\langle S_{13}, \sigma_{1909} \rangle \Rightarrow \langle S_{15}, \sigma_{1910} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7638} = \frac{\langle S_6, \sigma_{1910} \rangle \Rightarrow \langle S_8, \sigma_{1910} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1910} \rangle \Rightarrow \langle S_9, \sigma_{1910} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7640} = \langle S_8, \sigma_{1910} \rangle \Rightarrow \langle y := y + 1, \sigma_{1910} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7639} = \frac{\mathcal{T}_{7640}}{\langle S_9, \sigma_{1910} \rangle \Rightarrow \langle S_{10}, \sigma_{1910} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7641} = \frac{\langle y := y + 1, \sigma_{1910} \rangle \Rightarrow \sigma_{1911} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1910} \rangle \Rightarrow \langle S_{11}, \sigma_{1911} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7642} = \langle S_{11}, \sigma_{1911} \rangle \Rightarrow \langle S_{12}, \sigma_{1911} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7643} = \langle S_{12}, \sigma_{1911} \rangle \Rightarrow \langle S_{13}, \sigma_{1911} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7645} = \frac{\langle x := x + 1, \sigma_{1911} \rangle \Rightarrow \sigma_{1912} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1911} \rangle \Rightarrow \langle S_6, \sigma_{1912} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7644} = \frac{\mathcal{T}_{7645}}{\langle S_{13}, \sigma_{1911} \rangle \Rightarrow \langle S_{15}, \sigma_{1912} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7646} = \frac{\langle S_6, \sigma_{1912} \rangle \Rightarrow \langle S_8, \sigma_{1912} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1912} \rangle \Rightarrow \langle S_9, \sigma_{1912} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7648} = \langle S_8, \sigma_{1912} \rangle \Rightarrow \langle y := y + 1, \sigma_{1912} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7647} = \frac{\mathcal{T}_{7648}}{\langle S_9, \sigma_{1912} \rangle \Rightarrow \langle S_{10}, \sigma_{1912} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7649} = \frac{\langle y := y + 1, \sigma_{1912} \rangle \Rightarrow \sigma_{1913} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1912} \rangle \Rightarrow \langle S_{11}, \sigma_{1913} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7650} = \langle S_{11}, \sigma_{1913} \rangle \Rightarrow \langle S_{12}, \sigma_{1913} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7651} = \langle S_{12}, \sigma_{1913} \rangle \Rightarrow \langle S_{13}, \sigma_{1913} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7653} = \frac{\langle x := x + 1, \sigma_{1913} \rangle \Rightarrow \sigma_{1914} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1913} \rangle \Rightarrow \langle S_6, \sigma_{1914} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7652} = \frac{\mathcal{T}_{7653}}{\langle S_{13}, \sigma_{1913} \rangle \Rightarrow \langle S_{15}, \sigma_{1914} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7654} = \frac{\langle S_6, \sigma_{1914} \rangle \Rightarrow \langle S_8, \sigma_{1914} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1914} \rangle \Rightarrow \langle S_9, \sigma_{1914} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7656} = \langle S_8, \sigma_{1914} \rangle \Rightarrow \langle y := y + 1, \sigma_{1914} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7655} = \frac{\mathcal{T}_{7656}}{\langle S_9, \sigma_{1914} \rangle \Rightarrow \langle S_{10}, \sigma_{1914} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7657} = \frac{\langle y := y + 1, \sigma_{1914} \rangle \Rightarrow \sigma_{1915} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1914} \rangle \Rightarrow \langle S_{11}, \sigma_{1915} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7658} = \langle S_{11}, \sigma_{1915} \rangle \Rightarrow \langle S_{12}, \sigma_{1915} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7659} = \langle S_{12}, \sigma_{1915} \rangle \Rightarrow \langle S_{13}, \sigma_{1915} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7661} = \frac{\langle x := x + 1, \sigma_{1915} \rangle \Rightarrow \sigma_{1916} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1915} \rangle \Rightarrow \langle S_6, \sigma_{1916} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7660} = \frac{\mathcal{T}_{7661}}{\langle S_{13}, \sigma_{1915} \rangle \Rightarrow \langle S_{15}, \sigma_{1916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7662} = \frac{\langle S_6, \sigma_{1916} \rangle \Rightarrow \langle S_8, \sigma_{1916} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1916} \rangle \Rightarrow \langle S_9, \sigma_{1916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7664} = \langle S_8, \sigma_{1916} \rangle \Rightarrow \langle y := y + 1, \sigma_{1916} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7663} = \frac{\mathcal{T}_{7664}}{\langle S_9, \sigma_{1916} \rangle \Rightarrow \langle S_{10}, \sigma_{1916} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7665} = \frac{\langle y := y + 1, \sigma_{1916} \rangle \Rightarrow \sigma_{1917} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1916} \rangle \Rightarrow \langle S_{11}, \sigma_{1917} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7666} = \langle S_{11}, \sigma_{1917} \rangle \Rightarrow \langle S_{12}, \sigma_{1917} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7667} = \langle S_{12}, \sigma_{1917} \rangle \Rightarrow \langle S_{13}, \sigma_{1917} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7669} = \frac{\langle x := x + 1, \sigma_{1917} \rangle \Rightarrow \sigma_{1918} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1917} \rangle \Rightarrow \langle S_6, \sigma_{1918} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7668} = \frac{\mathcal{T}_{7669}}{\langle S_{13}, \sigma_{1917} \rangle \Rightarrow \langle S_{15}, \sigma_{1918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7670} = \frac{\langle S_6, \sigma_{1918} \rangle \Rightarrow \langle S_8, \sigma_{1918} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1918} \rangle \Rightarrow \langle S_9, \sigma_{1918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7672} = \langle S_8, \sigma_{1918} \rangle \Rightarrow \langle y := y + 1, \sigma_{1918} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7671} = \frac{\mathcal{T}_{7672}}{\langle S_9, \sigma_{1918} \rangle \Rightarrow \langle S_{10}, \sigma_{1918} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7673} = \frac{\langle y := y + 1, \sigma_{1918} \rangle \Rightarrow \sigma_{1919} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1918} \rangle \Rightarrow \langle S_{11}, \sigma_{1919} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7674} = \langle S_{11}, \sigma_{1919} \rangle \Rightarrow \langle S_{12}, \sigma_{1919} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7675} = \langle S_{12}, \sigma_{1919} \rangle \Rightarrow \langle S_{13}, \sigma_{1919} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7677} = \frac{\langle x := x + 1, \sigma_{1919} \rangle \Rightarrow \sigma_{1920} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1919} \rangle \Rightarrow \langle S_6, \sigma_{1920} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7676} = \frac{\mathcal{T}_{7677}}{\langle S_{13}, \sigma_{1919} \rangle \Rightarrow \langle S_{15}, \sigma_{1920} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7678} = \frac{\langle S_6, \sigma_{1920} \rangle \Rightarrow \langle S_8, \sigma_{1920} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1920} \rangle \Rightarrow \langle S_9, \sigma_{1920} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7680} = \langle S_8, \sigma_{1920} \rangle \Rightarrow \langle y := y + 1, \sigma_{1920} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7679} = \frac{\mathcal{T}_{7680}}{\langle S_9, \sigma_{1920} \rangle \Rightarrow \langle S_{10}, \sigma_{1920} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7681} = \frac{\langle y := y + 1, \sigma_{1920} \rangle \Rightarrow \sigma_{1921} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1920} \rangle \Rightarrow \langle S_{11}, \sigma_{1921} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7682} = \langle S_{11}, \sigma_{1921} \rangle \Rightarrow \langle S_{12}, \sigma_{1921} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7683} = \langle S_{12}, \sigma_{1921} \rangle \Rightarrow \langle S_{13}, \sigma_{1921} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7685} = \frac{\langle x := x + 1, \sigma_{1921} \rangle \Rightarrow \sigma_{1922} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1921} \rangle \Rightarrow \langle S_6, \sigma_{1922} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7684} = \frac{\mathcal{T}_{7685}}{\langle S_{13}, \sigma_{1921} \rangle \Rightarrow \langle S_{15}, \sigma_{1922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7686} = \frac{\langle S_6, \sigma_{1922} \rangle \Rightarrow \langle S_8, \sigma_{1922} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1922} \rangle \Rightarrow \langle S_9, \sigma_{1922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7688} = \langle S_8, \sigma_{1922} \rangle \Rightarrow \langle y := y + 1, \sigma_{1922} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7687} = \frac{\mathcal{T}_{7688}}{\langle S_9, \sigma_{1922} \rangle \Rightarrow \langle S_{10}, \sigma_{1922} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7689} = \frac{\langle y := y + 1, \sigma_{1922} \rangle \Rightarrow \sigma_{1923} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1922} \rangle \Rightarrow \langle S_{11}, \sigma_{1923} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7690} = \langle S_{11}, \sigma_{1923} \rangle \Rightarrow \langle S_{12}, \sigma_{1923} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7691} = \langle S_{12}, \sigma_{1923} \rangle \Rightarrow \langle S_{13}, \sigma_{1923} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7693} = \frac{\langle x := x + 1, \sigma_{1923} \rangle \Rightarrow \sigma_{1924} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1923} \rangle \Rightarrow \langle S_6, \sigma_{1924} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7692} = \frac{\mathcal{T}_{7693}}{\langle S_{13}, \sigma_{1923} \rangle \Rightarrow \langle S_{15}, \sigma_{1924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7694} = \frac{\langle S_6, \sigma_{1924} \rangle \Rightarrow \langle S_8, \sigma_{1924} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1924} \rangle \Rightarrow \langle S_9, \sigma_{1924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7696} = \langle S_8, \sigma_{1924} \rangle \Rightarrow \langle y := y + 1, \sigma_{1924} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7695} = \frac{\mathcal{T}_{7696}}{\langle S_9, \sigma_{1924} \rangle \Rightarrow \langle S_{10}, \sigma_{1924} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7697} = \frac{\langle y := y + 1, \sigma_{1924} \rangle \Rightarrow \sigma_{1925} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1924} \rangle \Rightarrow \langle S_{11}, \sigma_{1925} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7698} = \langle S_{11}, \sigma_{1925} \rangle \Rightarrow \langle S_{12}, \sigma_{1925} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7699} = \langle S_{12}, \sigma_{1925} \rangle \Rightarrow \langle S_{13}, \sigma_{1925} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7701} = \frac{\langle x := x + 1, \sigma_{1925} \rangle \Rightarrow \sigma_{1926} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1925} \rangle \Rightarrow \langle S_6, \sigma_{1926} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7700} = \frac{\mathcal{T}_{7701}}{\langle S_{13}, \sigma_{1925} \rangle \Rightarrow \langle S_{15}, \sigma_{1926} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7702} = \frac{\langle S_6, \sigma_{1926} \rangle \Rightarrow \langle S_8, \sigma_{1926} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1926} \rangle \Rightarrow \langle S_9, \sigma_{1926} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7704} = \langle S_8, \sigma_{1926} \rangle \Rightarrow \langle y := y + 1, \sigma_{1926} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7703} = \frac{\mathcal{T}_{7704}}{\langle S_9, \sigma_{1926} \rangle \Rightarrow \langle S_{10}, \sigma_{1926} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7705} = \frac{\langle y := y + 1, \sigma_{1926} \rangle \Rightarrow \sigma_{1927} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1926} \rangle \Rightarrow \langle S_{11}, \sigma_{1927} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7706} = \langle S_{11}, \sigma_{1927} \rangle \Rightarrow \langle S_{12}, \sigma_{1927} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7707} = \langle S_{12}, \sigma_{1927} \rangle \Rightarrow \langle S_{13}, \sigma_{1927} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7709} = \frac{\langle x := x + 1, \sigma_{1927} \rangle \Rightarrow \sigma_{1928} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1927} \rangle \Rightarrow \langle S_6, \sigma_{1928} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7708} = \frac{\mathcal{T}_{7709}}{\langle S_{13}, \sigma_{1927} \rangle \Rightarrow \langle S_{15}, \sigma_{1928} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7710} = \frac{\langle S_6, \sigma_{1928} \rangle \Rightarrow \langle S_8, \sigma_{1928} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1928} \rangle \Rightarrow \langle S_9, \sigma_{1928} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7712} = \langle S_8, \sigma_{1928} \rangle \Rightarrow \langle y := y + 1, \sigma_{1928} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7711} = \frac{\mathcal{T}_{7712}}{\langle S_9, \sigma_{1928} \rangle \Rightarrow \langle S_{10}, \sigma_{1928} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7713} = \frac{\langle y := y + 1, \sigma_{1928} \rangle \Rightarrow \sigma_{1929} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1928} \rangle \Rightarrow \langle S_{11}, \sigma_{1929} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7714} = \langle S_{11}, \sigma_{1929} \rangle \Rightarrow \langle S_{12}, \sigma_{1929} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7715} = \langle S_{12}, \sigma_{1929} \rangle \Rightarrow \langle S_{13}, \sigma_{1929} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7717} = \frac{\langle x := x + 1, \sigma_{1929} \rangle \Rightarrow \sigma_{1930} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1929} \rangle \Rightarrow \langle S_6, \sigma_{1930} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7716} = \frac{\mathcal{T}_{7717}}{\langle S_{13}, \sigma_{1929} \rangle \Rightarrow \langle S_{15}, \sigma_{1930} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7718} = \frac{\langle S_6, \sigma_{1930} \rangle \Rightarrow \langle S_8, \sigma_{1930} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1930} \rangle \Rightarrow \langle S_9, \sigma_{1930} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7720} = \langle S_8, \sigma_{1930} \rangle \Rightarrow \langle y := y + 1, \sigma_{1930} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7719} = \frac{\mathcal{T}_{7720}}{\langle S_9, \sigma_{1930} \rangle \Rightarrow \langle S_{10}, \sigma_{1930} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7721} = \frac{\langle y := y + 1, \sigma_{1930} \rangle \Rightarrow \sigma_{1931} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1930} \rangle \Rightarrow \langle S_{11}, \sigma_{1931} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7722} = \langle S_{11}, \sigma_{1931} \rangle \Rightarrow \langle S_{12}, \sigma_{1931} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7723} = \langle S_{12}, \sigma_{1931} \rangle \Rightarrow \langle S_{13}, \sigma_{1931} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7725} = \frac{\langle x := x + 1, \sigma_{1931} \rangle \Rightarrow \sigma_{1932} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1931} \rangle \Rightarrow \langle S_6, \sigma_{1932} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7724} = \frac{\mathcal{T}_{7725}}{\langle S_{13}, \sigma_{1931} \rangle \Rightarrow \langle S_{15}, \sigma_{1932} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7726} = \frac{\langle S_6, \sigma_{1932} \rangle \Rightarrow \langle S_8, \sigma_{1932} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1932} \rangle \Rightarrow \langle S_9, \sigma_{1932} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7728} = \langle S_8, \sigma_{1932} \rangle \Rightarrow \langle y := y + 1, \sigma_{1932} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7727} = \frac{\mathcal{T}_{7728}}{\langle S_9, \sigma_{1932} \rangle \Rightarrow \langle S_{10}, \sigma_{1932} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7729} = \frac{\langle y := y + 1, \sigma_{1932} \rangle \Rightarrow \sigma_{1933} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1932} \rangle \Rightarrow \langle S_{11}, \sigma_{1933} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7730} = \langle S_{11}, \sigma_{1933} \rangle \Rightarrow \langle S_{12}, \sigma_{1933} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7731} = \langle S_{12}, \sigma_{1933} \rangle \Rightarrow \langle S_{13}, \sigma_{1933} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7733} = \frac{\langle x := x + 1, \sigma_{1933} \rangle \Rightarrow \sigma_{1934} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1933} \rangle \Rightarrow \langle S_6, \sigma_{1934} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7732} = \frac{\mathcal{T}_{7733}}{\langle S_{13}, \sigma_{1933} \rangle \Rightarrow \langle S_{15}, \sigma_{1934} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7734} = \frac{\langle S_6, \sigma_{1934} \rangle \Rightarrow \langle S_8, \sigma_{1934} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1934} \rangle \Rightarrow \langle S_9, \sigma_{1934} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7736} = \langle S_8, \sigma_{1934} \rangle \Rightarrow \langle y := y + 1, \sigma_{1934} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7735} = \frac{\mathcal{T}_{7736}}{\langle S_9, \sigma_{1934} \rangle \Rightarrow \langle S_{10}, \sigma_{1934} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7737} = \frac{\langle y := y + 1, \sigma_{1934} \rangle \Rightarrow \sigma_{1935} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1934} \rangle \Rightarrow \langle S_{11}, \sigma_{1935} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7738} = \langle S_{11}, \sigma_{1935} \rangle \Rightarrow \langle S_{12}, \sigma_{1935} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7739} = \langle S_{12}, \sigma_{1935} \rangle \Rightarrow \langle S_{13}, \sigma_{1935} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7741} = \frac{\langle x := x + 1, \sigma_{1935} \rangle \Rightarrow \sigma_{1936} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1935} \rangle \Rightarrow \langle S_6, \sigma_{1936} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7740} = \frac{\mathcal{T}_{7741}}{\langle S_{13}, \sigma_{1935} \rangle \Rightarrow \langle S_{15}, \sigma_{1936} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7742} = \frac{\langle S_6, \sigma_{1936} \rangle \Rightarrow \langle S_8, \sigma_{1936} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1936} \rangle \Rightarrow \langle S_9, \sigma_{1936} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7744} = \langle S_8, \sigma_{1936} \rangle \Rightarrow \langle y := y + 1, \sigma_{1936} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7743} = \frac{\mathcal{T}_{7744}}{\langle S_9, \sigma_{1936} \rangle \Rightarrow \langle S_{10}, \sigma_{1936} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7745} = \frac{\langle y := y + 1, \sigma_{1936} \rangle \Rightarrow \sigma_{1937} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1936} \rangle \Rightarrow \langle S_{11}, \sigma_{1937} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7746} = \langle S_{11}, \sigma_{1937} \rangle \Rightarrow \langle S_{12}, \sigma_{1937} \rangle \text{ [for}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{7747} = \langle S_{12}, \sigma_{1937} \rangle \Rightarrow \langle S_{13}, \sigma_{1937} \rangle \text{ [if - tt}_{\text{sos}}]$$

 $\Downarrow$

$$\mathcal{T}_{7749} = \frac{\langle x := x + 1, \sigma_{1937} \rangle \Rightarrow \sigma_{1938} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1937} \rangle \Rightarrow \langle S_6, \sigma_{1938} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7748} = \frac{\mathcal{T}_{7749}}{\langle S_{13}, \sigma_{1937} \rangle \Rightarrow \langle S_{15}, \sigma_{1938} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7750} = \frac{\langle S_6, \sigma_{1938} \rangle \Rightarrow \langle S_8, \sigma_{1938} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1938} \rangle \Rightarrow \langle S_9, \sigma_{1938} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7752} = \langle S_8, \sigma_{1938} \rangle \Rightarrow \langle y := y + 1, \sigma_{1938} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7751} = \frac{\mathcal{T}_{7752}}{\langle S_9, \sigma_{1938} \rangle \Rightarrow \langle S_{10}, \sigma_{1938} \rangle} \text{ [comp}_{\text{sos}}^1]$$

 $\Downarrow$

$$\mathcal{T}_{7753} = \frac{\langle y := y + 1, \sigma_{1938} \rangle \Rightarrow \sigma_{1939} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1938} \rangle \Rightarrow \langle S_{11}, \sigma_{1939} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7754} = \langle S_{11}, \sigma_{1939} \rangle \Rightarrow \langle S_{12}, \sigma_{1939} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7755} = \langle S_{12}, \sigma_{1939} \rangle \Rightarrow \langle S_{13}, \sigma_{1939} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7757} = \frac{\langle x := x + 1, \sigma_{1939} \rangle \Rightarrow \sigma_{1940} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1939} \rangle \Rightarrow \langle S_6, \sigma_{1940} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7756} = \frac{\mathcal{T}_{7757}}{\langle S_{13}, \sigma_{1939} \rangle \Rightarrow \langle S_{15}, \sigma_{1940} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7758} = \frac{\langle S_6, \sigma_{1940} \rangle \Rightarrow \langle S_8, \sigma_{1940} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1940} \rangle \Rightarrow \langle S_9, \sigma_{1940} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7760} = \langle S_8, \sigma_{1940} \rangle \Rightarrow \langle y := y + 1, \sigma_{1940} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7759} = \frac{\mathcal{T}_{7760}}{\langle S_9, \sigma_{1940} \rangle \Rightarrow \langle S_{10}, \sigma_{1940} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7761} = \frac{\langle y := y + 1, \sigma_{1940} \rangle \Rightarrow \sigma_{1941} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1940} \rangle \Rightarrow \langle S_{11}, \sigma_{1941} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7762} = \langle S_{11}, \sigma_{1941} \rangle \Rightarrow \langle S_{12}, \sigma_{1941} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7763} = \langle S_{12}, \sigma_{1941} \rangle \Rightarrow \langle S_{13}, \sigma_{1941} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7765} = \frac{\langle x := x + 1, \sigma_{1941} \rangle \Rightarrow \sigma_{1942} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1941} \rangle \Rightarrow \langle S_6, \sigma_{1942} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7764} = \frac{\mathcal{T}_{7765}}{\langle S_{13}, \sigma_{1941} \rangle \Rightarrow \langle S_{15}, \sigma_{1942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7766} = \frac{\langle S_6, \sigma_{1942} \rangle \Rightarrow \langle S_8, \sigma_{1942} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1942} \rangle \Rightarrow \langle S_9, \sigma_{1942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7768} = \langle S_8, \sigma_{1942} \rangle \Rightarrow \langle y := y + 1, \sigma_{1942} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7767} = \frac{\mathcal{T}_{7768}}{\langle S_9, \sigma_{1942} \rangle \Rightarrow \langle S_{10}, \sigma_{1942} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7769} = \frac{\langle y := y + 1, \sigma_{1942} \rangle \Rightarrow \sigma_{1943} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1942} \rangle \Rightarrow \langle S_{11}, \sigma_{1943} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7770} = \langle S_{11}, \sigma_{1943} \rangle \Rightarrow \langle S_{12}, \sigma_{1943} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7771} = \langle S_{12}, \sigma_{1943} \rangle \Rightarrow \langle S_{13}, \sigma_{1943} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7773} = \frac{\langle x := x + 1, \sigma_{1943} \rangle \Rightarrow \sigma_{1944} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1943} \rangle \Rightarrow \langle S_6, \sigma_{1944} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7772} = \frac{\mathcal{T}_{7773}}{\langle S_{13}, \sigma_{1943} \rangle \Rightarrow \langle S_{15}, \sigma_{1944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7774} = \frac{\langle S_6, \sigma_{1944} \rangle \Rightarrow \langle S_8, \sigma_{1944} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1944} \rangle \Rightarrow \langle S_9, \sigma_{1944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7776} = \langle S_8, \sigma_{1944} \rangle \Rightarrow \langle y := y + 1, \sigma_{1944} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7775} = \frac{\mathcal{T}_{7776}}{\langle S_9, \sigma_{1944} \rangle \Rightarrow \langle S_{10}, \sigma_{1944} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7777} = \frac{\langle y := y + 1, \sigma_{1944} \rangle \Rightarrow \sigma_{1945} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1944} \rangle \Rightarrow \langle S_{11}, \sigma_{1945} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7778} = \langle S_{11}, \sigma_{1945} \rangle \Rightarrow \langle S_{12}, \sigma_{1945} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7779} = \langle S_{12}, \sigma_{1945} \rangle \Rightarrow \langle S_{13}, \sigma_{1945} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7781} = \frac{\langle x := x + 1, \sigma_{1945} \rangle \Rightarrow \sigma_{1946} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1945} \rangle \Rightarrow \langle S_6, \sigma_{1946} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7780} = \frac{\mathcal{T}_{7781}}{\langle S_{13}, \sigma_{1945} \rangle \Rightarrow \langle S_{15}, \sigma_{1946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7782} = \frac{\langle S_6, \sigma_{1946} \rangle \Rightarrow \langle S_8, \sigma_{1946} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1946} \rangle \Rightarrow \langle S_9, \sigma_{1946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7784} = \langle S_8, \sigma_{1946} \rangle \Rightarrow \langle y := y + 1, \sigma_{1946} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7783} = \frac{\mathcal{T}_{7784}}{\langle S_9, \sigma_{1946} \rangle \Rightarrow \langle S_{10}, \sigma_{1946} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7785} = \frac{\langle y := y + 1, \sigma_{1946} \rangle \Rightarrow \sigma_{1947} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1946} \rangle \Rightarrow \langle S_{11}, \sigma_{1947} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7786} = \langle S_{11}, \sigma_{1947} \rangle \Rightarrow \langle S_{12}, \sigma_{1947} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7787} = \langle S_{12}, \sigma_{1947} \rangle \Rightarrow \langle S_{13}, \sigma_{1947} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7789} = \frac{\langle x := x + 1, \sigma_{1947} \rangle \Rightarrow \sigma_{1948} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1947} \rangle \Rightarrow \langle S_6, \sigma_{1948} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7788} = \frac{\mathcal{T}_{7789}}{\langle S_{13}, \sigma_{1947} \rangle \Rightarrow \langle S_{15}, \sigma_{1948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7790} = \frac{\langle S_6, \sigma_{1948} \rangle \Rightarrow \langle S_8, \sigma_{1948} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1948} \rangle \Rightarrow \langle S_9, \sigma_{1948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7792} = \langle S_8, \sigma_{1948} \rangle \Rightarrow \langle y := y + 1, \sigma_{1948} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7791} = \frac{\mathcal{T}_{7792}}{\langle S_9, \sigma_{1948} \rangle \Rightarrow \langle S_{10}, \sigma_{1948} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7793} = \frac{\langle y := y + 1, \sigma_{1948} \rangle \Rightarrow \sigma_{1949} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1948} \rangle \Rightarrow \langle S_{11}, \sigma_{1949} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7794} = \langle S_{11}, \sigma_{1949} \rangle \Rightarrow \langle S_{12}, \sigma_{1949} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7795} = \langle S_{12}, \sigma_{1949} \rangle \Rightarrow \langle S_{13}, \sigma_{1949} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7797} = \frac{\langle x := x + 1, \sigma_{1949} \rangle \Rightarrow \sigma_{1950} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1949} \rangle \Rightarrow \langle S_6, \sigma_{1950} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7796} = \frac{\mathcal{T}_{7797}}{\langle S_{13}, \sigma_{1949} \rangle \Rightarrow \langle S_{15}, \sigma_{1950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7798} = \frac{\langle S_6, \sigma_{1950} \rangle \Rightarrow \langle S_8, \sigma_{1950} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1950} \rangle \Rightarrow \langle S_9, \sigma_{1950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7800} = \langle S_8, \sigma_{1950} \rangle \Rightarrow \langle y := y + 1, \sigma_{1950} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7799} = \frac{\mathcal{T}_{7800}}{\langle S_9, \sigma_{1950} \rangle \Rightarrow \langle S_{10}, \sigma_{1950} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7801} = \frac{\langle y := y + 1, \sigma_{1950} \rangle \Rightarrow \sigma_{1951} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1950} \rangle \Rightarrow \langle S_{11}, \sigma_{1951} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7802} = \langle S_{11}, \sigma_{1951} \rangle \Rightarrow \langle S_{12}, \sigma_{1951} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7803} = \langle S_{12}, \sigma_{1951} \rangle \Rightarrow \langle S_{13}, \sigma_{1951} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7805} = \frac{\langle x := x + 1, \sigma_{1951} \rangle \Rightarrow \sigma_{1952} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1951} \rangle \Rightarrow \langle S_6, \sigma_{1952} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7804} = \frac{\mathcal{T}_{7805}}{\langle S_{13}, \sigma_{1951} \rangle \Rightarrow \langle S_{15}, \sigma_{1952} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7806} = \frac{\langle S_6, \sigma_{1952} \rangle \Rightarrow \langle S_8, \sigma_{1952} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1952} \rangle \Rightarrow \langle S_9, \sigma_{1952} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7808} = \langle S_8, \sigma_{1952} \rangle \Rightarrow \langle y := y + 1, \sigma_{1952} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7807} = \frac{\mathcal{T}_{7808}}{\langle S_9, \sigma_{1952} \rangle \Rightarrow \langle S_{10}, \sigma_{1952} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7809} = \frac{\langle y := y + 1, \sigma_{1952} \rangle \Rightarrow \sigma_{1953} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1952} \rangle \Rightarrow \langle S_{11}, \sigma_{1953} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7810} = \langle S_{11}, \sigma_{1953} \rangle \Rightarrow \langle S_{12}, \sigma_{1953} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7811} = \langle S_{12}, \sigma_{1953} \rangle \Rightarrow \langle S_{13}, \sigma_{1953} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7813} = \frac{\langle x := x + 1, \sigma_{1953} \rangle \Rightarrow \sigma_{1954} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1953} \rangle \Rightarrow \langle S_6, \sigma_{1954} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7812} = \frac{\mathcal{T}_{7813}}{\langle S_{13}, \sigma_{1953} \rangle \Rightarrow \langle S_{15}, \sigma_{1954} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7814} = \frac{\langle S_6, \sigma_{1954} \rangle \Rightarrow \langle S_8, \sigma_{1954} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1954} \rangle \Rightarrow \langle S_9, \sigma_{1954} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7816} = \langle S_8, \sigma_{1954} \rangle \Rightarrow \langle y := y + 1, \sigma_{1954} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7815} = \frac{\mathcal{T}_{7816}}{\langle S_9, \sigma_{1954} \rangle \Rightarrow \langle S_{10}, \sigma_{1954} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7817} = \frac{\langle y := y + 1, \sigma_{1954} \rangle \Rightarrow \sigma_{1955} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1954} \rangle \Rightarrow \langle S_{11}, \sigma_{1955} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7818} = \langle S_{11}, \sigma_{1955} \rangle \Rightarrow \langle S_{12}, \sigma_{1955} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7819} = \langle S_{12}, \sigma_{1955} \rangle \Rightarrow \langle S_{13}, \sigma_{1955} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7821} = \frac{\langle x := x + 1, \sigma_{1955} \rangle \Rightarrow \sigma_{1956} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1955} \rangle \Rightarrow \langle S_6, \sigma_{1956} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7820} = \frac{\mathcal{T}_{7821}}{\langle S_{13}, \sigma_{1955} \rangle \Rightarrow \langle S_{15}, \sigma_{1956} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7822} = \frac{\langle S_6, \sigma_{1956} \rangle \Rightarrow \langle S_8, \sigma_{1956} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1956} \rangle \Rightarrow \langle S_9, \sigma_{1956} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7824} = \langle S_8, \sigma_{1956} \rangle \Rightarrow \langle y := y + 1, \sigma_{1956} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7823} = \frac{\mathcal{T}_{7824}}{\langle S_9, \sigma_{1956} \rangle \Rightarrow \langle S_{10}, \sigma_{1956} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7825} = \frac{\langle y := y + 1, \sigma_{1956} \rangle \Rightarrow \sigma_{1957} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1956} \rangle \Rightarrow \langle S_{11}, \sigma_{1957} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7826} = \langle S_{11}, \sigma_{1957} \rangle \Rightarrow \langle S_{12}, \sigma_{1957} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7827} = \langle S_{12}, \sigma_{1957} \rangle \Rightarrow \langle S_{13}, \sigma_{1957} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7829} = \frac{\langle x := x + 1, \sigma_{1957} \rangle \Rightarrow \sigma_{1958} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1957} \rangle \Rightarrow \langle S_6, \sigma_{1958} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7828} = \frac{\mathcal{T}_{7829}}{\langle S_{13}, \sigma_{1957} \rangle \Rightarrow \langle S_{15}, \sigma_{1958} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7830} = \frac{\langle S_6, \sigma_{1958} \rangle \Rightarrow \langle S_8, \sigma_{1958} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1958} \rangle \Rightarrow \langle S_9, \sigma_{1958} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7832} = \langle S_8, \sigma_{1958} \rangle \Rightarrow \langle y := y + 1, \sigma_{1958} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7831} = \frac{\mathcal{T}_{7832}}{\langle S_9, \sigma_{1958} \rangle \Rightarrow \langle S_{10}, \sigma_{1958} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7833} = \frac{\langle y := y + 1, \sigma_{1958} \rangle \Rightarrow \sigma_{1959} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1958} \rangle \Rightarrow \langle S_{11}, \sigma_{1959} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7834} = \langle S_{11}, \sigma_{1959} \rangle \Rightarrow \langle S_{12}, \sigma_{1959} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7835} = \langle S_{12}, \sigma_{1959} \rangle \Rightarrow \langle S_{13}, \sigma_{1959} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7837} = \frac{\langle x := x + 1, \sigma_{1959} \rangle \Rightarrow \sigma_{1960} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1959} \rangle \Rightarrow \langle S_6, \sigma_{1960} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7836} = \frac{\mathcal{T}_{7837}}{\langle S_{13}, \sigma_{1959} \rangle \Rightarrow \langle S_{15}, \sigma_{1960} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7838} = \frac{\langle S_6, \sigma_{1960} \rangle \Rightarrow \langle S_8, \sigma_{1960} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1960} \rangle \Rightarrow \langle S_9, \sigma_{1960} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7840} = \langle S_8, \sigma_{1960} \rangle \Rightarrow \langle y := y + 1, \sigma_{1960} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7839} = \frac{\mathcal{T}_{7840}}{\langle S_9, \sigma_{1960} \rangle \Rightarrow \langle S_{10}, \sigma_{1960} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7841} = \frac{\langle y := y + 1, \sigma_{1960} \rangle \Rightarrow \sigma_{1961} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1960} \rangle \Rightarrow \langle S_{11}, \sigma_{1961} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7842} = \langle S_{11}, \sigma_{1961} \rangle \Rightarrow \langle S_{12}, \sigma_{1961} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7843} = \langle S_{12}, \sigma_{1961} \rangle \Rightarrow \langle S_{13}, \sigma_{1961} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7845} = \frac{\langle x := x + 1, \sigma_{1961} \rangle \Rightarrow \sigma_{1962} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1961} \rangle \Rightarrow \langle S_6, \sigma_{1962} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7844} = \frac{\mathcal{T}_{7845}}{\langle S_{13}, \sigma_{1961} \rangle \Rightarrow \langle S_{15}, \sigma_{1962} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7846} = \frac{\langle S_6, \sigma_{1962} \rangle \Rightarrow \langle S_8, \sigma_{1962} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1962} \rangle \Rightarrow \langle S_9, \sigma_{1962} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7848} = \langle S_8, \sigma_{1962} \rangle \Rightarrow \langle y := y + 1, \sigma_{1962} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7847} = \frac{\mathcal{T}_{7848}}{\langle S_9, \sigma_{1962} \rangle \Rightarrow \langle S_{10}, \sigma_{1962} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7849} = \frac{\langle y := y + 1, \sigma_{1962} \rangle \Rightarrow \sigma_{1963} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1962} \rangle \Rightarrow \langle S_{11}, \sigma_{1963} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7850} = \langle S_{11}, \sigma_{1963} \rangle \Rightarrow \langle S_{12}, \sigma_{1963} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7851} = \langle S_{12}, \sigma_{1963} \rangle \Rightarrow \langle S_{13}, \sigma_{1963} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7853} = \frac{\langle x := x + 1, \sigma_{1963} \rangle \Rightarrow \sigma_{1964} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1963} \rangle \Rightarrow \langle S_6, \sigma_{1964} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7852} = \frac{\mathcal{T}_{7853}}{\langle S_{13}, \sigma_{1963} \rangle \Rightarrow \langle S_{15}, \sigma_{1964} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7854} = \frac{\langle S_6, \sigma_{1964} \rangle \Rightarrow \langle S_8, \sigma_{1964} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1964} \rangle \Rightarrow \langle S_9, \sigma_{1964} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7856} = \langle S_8, \sigma_{1964} \rangle \Rightarrow \langle y := y + 1, \sigma_{1964} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7855} = \frac{\mathcal{T}_{7856}}{\langle S_9, \sigma_{1964} \rangle \Rightarrow \langle S_{10}, \sigma_{1964} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7857} = \frac{\langle y := y + 1, \sigma_{1964} \rangle \Rightarrow \sigma_{1965} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1964} \rangle \Rightarrow \langle S_{11}, \sigma_{1965} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7858} = \langle S_{11}, \sigma_{1965} \rangle \Rightarrow \langle S_{12}, \sigma_{1965} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7859} = \langle S_{12}, \sigma_{1965} \rangle \Rightarrow \langle S_{13}, \sigma_{1965} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7861} = \frac{\langle x := x + 1, \sigma_{1965} \rangle \Rightarrow \sigma_{1966} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1965} \rangle \Rightarrow \langle S_6, \sigma_{1966} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7860} = \frac{\mathcal{T}_{7861}}{\langle S_{13}, \sigma_{1965} \rangle \Rightarrow \langle S_{15}, \sigma_{1966} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7862} = \frac{\langle S_6, \sigma_{1966} \rangle \Rightarrow \langle S_8, \sigma_{1966} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1966} \rangle \Rightarrow \langle S_9, \sigma_{1966} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7864} = \langle S_8, \sigma_{1966} \rangle \Rightarrow \langle y := y + 1, \sigma_{1966} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7863} = \frac{\mathcal{T}_{7864}}{\langle S_9, \sigma_{1966} \rangle \Rightarrow \langle S_{10}, \sigma_{1966} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7865} = \frac{\langle y := y + 1, \sigma_{1966} \rangle \Rightarrow \sigma_{1967} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1966} \rangle \Rightarrow \langle S_{11}, \sigma_{1967} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7866} = \langle S_{11}, \sigma_{1967} \rangle \Rightarrow \langle S_{12}, \sigma_{1967} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7867} = \langle S_{12}, \sigma_{1967} \rangle \Rightarrow \langle S_{13}, \sigma_{1967} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7869} = \frac{\langle x := x + 1, \sigma_{1967} \rangle \Rightarrow \sigma_{1968} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1967} \rangle \Rightarrow \langle S_6, \sigma_{1968} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7868} = \frac{\mathcal{T}_{7869}}{\langle S_{13}, \sigma_{1967} \rangle \Rightarrow \langle S_{15}, \sigma_{1968} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7870} = \frac{\langle S_6, \sigma_{1968} \rangle \Rightarrow \langle S_8, \sigma_{1968} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1968} \rangle \Rightarrow \langle S_9, \sigma_{1968} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7872} = \langle S_8, \sigma_{1968} \rangle \Rightarrow \langle y := y + 1, \sigma_{1968} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7871} = \frac{\mathcal{T}_{7872}}{\langle S_9, \sigma_{1968} \rangle \Rightarrow \langle S_{10}, \sigma_{1968} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7873} = \frac{\langle y := y + 1, \sigma_{1968} \rangle \Rightarrow \sigma_{1969} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1968} \rangle \Rightarrow \langle S_{11}, \sigma_{1969} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7874} = \langle S_{11}, \sigma_{1969} \rangle \Rightarrow \langle S_{12}, \sigma_{1969} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7875} = \langle S_{12}, \sigma_{1969} \rangle \Rightarrow \langle S_{13}, \sigma_{1969} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7877} = \frac{\langle x := x + 1, \sigma_{1969} \rangle \Rightarrow \sigma_{1970} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1969} \rangle \Rightarrow \langle S_6, \sigma_{1970} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7876} = \frac{\mathcal{T}_{7877}}{\langle S_{13}, \sigma_{1969} \rangle \Rightarrow \langle S_{15}, \sigma_{1970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7878} = \frac{\langle S_6, \sigma_{1970} \rangle \Rightarrow \langle S_8, \sigma_{1970} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1970} \rangle \Rightarrow \langle S_9, \sigma_{1970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7880} = \langle S_8, \sigma_{1970} \rangle \Rightarrow \langle y := y + 1, \sigma_{1970} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7879} = \frac{\mathcal{T}_{7880}}{\langle S_9, \sigma_{1970} \rangle \Rightarrow \langle S_{10}, \sigma_{1970} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7881} = \frac{\langle y := y + 1, \sigma_{1970} \rangle \Rightarrow \sigma_{1971} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1970} \rangle \Rightarrow \langle S_{11}, \sigma_{1971} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7882} = \langle S_{11}, \sigma_{1971} \rangle \Rightarrow \langle S_{12}, \sigma_{1971} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7883} = \langle S_{12}, \sigma_{1971} \rangle \Rightarrow \langle S_{13}, \sigma_{1971} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7885} = \frac{\langle x := x + 1, \sigma_{1971} \rangle \Rightarrow \sigma_{1972} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1971} \rangle \Rightarrow \langle S_6, \sigma_{1972} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7884} = \frac{\mathcal{T}_{7885}}{\langle S_{13}, \sigma_{1971} \rangle \Rightarrow \langle S_{15}, \sigma_{1972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7886} = \frac{\langle S_6, \sigma_{1972} \rangle \Rightarrow \langle S_8, \sigma_{1972} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1972} \rangle \Rightarrow \langle S_9, \sigma_{1972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7888} = \langle S_8, \sigma_{1972} \rangle \Rightarrow \langle y := y + 1, \sigma_{1972} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7887} = \frac{\mathcal{T}_{7888}}{\langle S_9, \sigma_{1972} \rangle \Rightarrow \langle S_{10}, \sigma_{1972} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7889} = \frac{\langle y := y + 1, \sigma_{1972} \rangle \Rightarrow \sigma_{1973} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1972} \rangle \Rightarrow \langle S_{11}, \sigma_{1973} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7890} = \langle S_{11}, \sigma_{1973} \rangle \Rightarrow \langle S_{12}, \sigma_{1973} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7891} = \langle S_{12}, \sigma_{1973} \rangle \Rightarrow \langle S_{13}, \sigma_{1973} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7893} = \frac{\langle x := x + 1, \sigma_{1973} \rangle \Rightarrow \sigma_{1974} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1973} \rangle \Rightarrow \langle S_6, \sigma_{1974} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7892} = \frac{\mathcal{T}_{7893}}{\langle S_{13}, \sigma_{1973} \rangle \Rightarrow \langle S_{15}, \sigma_{1974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7894} = \frac{\langle S_6, \sigma_{1974} \rangle \Rightarrow \langle S_8, \sigma_{1974} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1974} \rangle \Rightarrow \langle S_9, \sigma_{1974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7896} = \langle S_8, \sigma_{1974} \rangle \Rightarrow \langle y := y + 1, \sigma_{1974} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7895} = \frac{\mathcal{T}_{7896}}{\langle S_9, \sigma_{1974} \rangle \Rightarrow \langle S_{10}, \sigma_{1974} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7897} = \frac{\langle y := y + 1, \sigma_{1974} \rangle \Rightarrow \sigma_{1975} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1974} \rangle \Rightarrow \langle S_{11}, \sigma_{1975} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7898} = \langle S_{11}, \sigma_{1975} \rangle \Rightarrow \langle S_{12}, \sigma_{1975} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7899} = \langle S_{12}, \sigma_{1975} \rangle \Rightarrow \langle S_{13}, \sigma_{1975} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7901} = \frac{\langle x := x + 1, \sigma_{1975} \rangle \Rightarrow \sigma_{1976} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1975} \rangle \Rightarrow \langle S_6, \sigma_{1976} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7900} = \frac{\mathcal{T}_{7901}}{\langle S_{13}, \sigma_{1975} \rangle \Rightarrow \langle S_{15}, \sigma_{1976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7902} = \frac{\langle S_6, \sigma_{1976} \rangle \Rightarrow \langle S_8, \sigma_{1976} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1976} \rangle \Rightarrow \langle S_9, \sigma_{1976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7904} = \langle S_8, \sigma_{1976} \rangle \Rightarrow \langle y := y + 1, \sigma_{1976} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7903} = \frac{\mathcal{T}_{7904}}{\langle S_9, \sigma_{1976} \rangle \Rightarrow \langle S_{10}, \sigma_{1976} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7905} = \frac{\langle y := y + 1, \sigma_{1976} \rangle \Rightarrow \sigma_{1977} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1976} \rangle \Rightarrow \langle S_{11}, \sigma_{1977} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7906} = \langle S_{11}, \sigma_{1977} \rangle \Rightarrow \langle S_{12}, \sigma_{1977} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7907} = \langle S_{12}, \sigma_{1977} \rangle \Rightarrow \langle S_{13}, \sigma_{1977} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7909} = \frac{\langle x := x + 1, \sigma_{1977} \rangle \Rightarrow \sigma_{1978} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1977} \rangle \Rightarrow \langle S_6, \sigma_{1978} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7908} = \frac{\mathcal{T}_{7909}}{\langle S_{13}, \sigma_{1977} \rangle \Rightarrow \langle S_{15}, \sigma_{1978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7910} = \frac{\langle S_6, \sigma_{1978} \rangle \Rightarrow \langle S_8, \sigma_{1978} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1978} \rangle \Rightarrow \langle S_9, \sigma_{1978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7912} = \langle S_8, \sigma_{1978} \rangle \Rightarrow \langle y := y + 1, \sigma_{1978} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7911} = \frac{\mathcal{T}_{7912}}{\langle S_9, \sigma_{1978} \rangle \Rightarrow \langle S_{10}, \sigma_{1978} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7913} = \frac{\langle y := y + 1, \sigma_{1978} \rangle \Rightarrow \sigma_{1979} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1978} \rangle \Rightarrow \langle S_{11}, \sigma_{1979} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7914} = \langle S_{11}, \sigma_{1979} \rangle \Rightarrow \langle S_{12}, \sigma_{1979} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7915} = \langle S_{12}, \sigma_{1979} \rangle \Rightarrow \langle S_{13}, \sigma_{1979} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7917} = \frac{\langle x := x + 1, \sigma_{1979} \rangle \Rightarrow \sigma_{1980} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1979} \rangle \Rightarrow \langle S_6, \sigma_{1980} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7916} = \frac{\mathcal{T}_{7917}}{\langle S_{13}, \sigma_{1979} \rangle \Rightarrow \langle S_{15}, \sigma_{1980} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7918} = \frac{\langle S_6, \sigma_{1980} \rangle \Rightarrow \langle S_8, \sigma_{1980} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1980} \rangle \Rightarrow \langle S_9, \sigma_{1980} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7920} = \langle S_8, \sigma_{1980} \rangle \Rightarrow \langle y := y + 1, \sigma_{1980} \rangle \text{ [if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7919} = \frac{\mathcal{T}_{7920}}{\langle S_9, \sigma_{1980} \rangle \Rightarrow \langle S_{10}, \sigma_{1980} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7921} = \frac{\langle y := y + 1, \sigma_{1980} \rangle \Rightarrow \sigma_{1981} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1980} \rangle \Rightarrow \langle S_{11}, \sigma_{1981} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7922} = \langle S_{11}, \sigma_{1981} \rangle \Rightarrow \langle S_{12}, \sigma_{1981} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7923} = \langle S_{12}, \sigma_{1981} \rangle \Rightarrow \langle S_{13}, \sigma_{1981} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7925} = \frac{\langle x := x + 1, \sigma_{1981} \rangle \Rightarrow \sigma_{1982} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1981} \rangle \Rightarrow \langle S_6, \sigma_{1982} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7924} = \frac{\mathcal{T}_{7925}}{\langle S_{13}, \sigma_{1981} \rangle \Rightarrow \langle S_{15}, \sigma_{1982} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7926} = \frac{\langle S_6, \sigma_{1982} \rangle \Rightarrow \langle S_8, \sigma_{1982} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1982} \rangle \Rightarrow \langle S_9, \sigma_{1982} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7928} = \langle S_8, \sigma_{1982} \rangle \Rightarrow \langle y := y + 1, \sigma_{1982} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7927} = \frac{\mathcal{T}_{7928}}{\langle S_9, \sigma_{1982} \rangle \Rightarrow \langle S_{10}, \sigma_{1982} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7929} = \frac{\langle y := y + 1, \sigma_{1982} \rangle \Rightarrow \sigma_{1983} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1982} \rangle \Rightarrow \langle S_{11}, \sigma_{1983} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7930} = \langle S_{11}, \sigma_{1983} \rangle \Rightarrow \langle S_{12}, \sigma_{1983} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7931} = \langle S_{12}, \sigma_{1983} \rangle \Rightarrow \langle S_{13}, \sigma_{1983} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7933} = \frac{\langle x := x + 1, \sigma_{1983} \rangle \Rightarrow \sigma_{1984} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{14}, \sigma_{1983} \rangle \Rightarrow \langle S_6, \sigma_{1984} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\mathcal{T}_{7932} = \frac{\mathcal{T}_{7933}}{\langle S_{13}, \sigma_{1983} \rangle \Rightarrow \langle S_{15}, \sigma_{1984} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7934} = \frac{\langle S_6, \sigma_{1984} \rangle \Rightarrow \langle S_8, \sigma_{1984} \rangle \text{ [assert}_{\text{sos}}\text{]}}{\langle S_7, \sigma_{1984} \rangle \Rightarrow \langle S_9, \sigma_{1984} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7936} = \langle S_8, \sigma_{1984} \rangle \Rightarrow \langle y := y + 1, \sigma_{1984} \rangle \text{ [if - tt}_{\text{sos}}\text{]}$$

$$\mathcal{T}_{7935} = \frac{\mathcal{T}_{7936}}{\langle S_9, \sigma_{1984} \rangle \Rightarrow \langle S_{10}, \sigma_{1984} \rangle} \text{ [comp}_{\text{sos}}^1\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7937} = \frac{\langle y := y + 1, \sigma_{1984} \rangle \Rightarrow \sigma_{1985} \text{ [ass}_{\text{sos}}\text{]}}{\langle S_{10}, \sigma_{1984} \rangle \Rightarrow \langle S_{11}, \sigma_{1985} \rangle} \text{ [comp}_{\text{sos}}^2\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7938} = \langle S_{11}, \sigma_{1985} \rangle \Rightarrow \langle S_{12}, \sigma_{1985} \rangle \text{ [for}_{\text{sos}}\text{]}$$

$$\Downarrow$$

$$\mathcal{T}_{7939} = \langle S_{12}, \sigma_{1985} \rangle \Rightarrow \langle S_{13}, \sigma_{1985} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7941} = \frac{\langle x := x + 1, \sigma_{1985} \rangle \Rightarrow \sigma_{1986} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1985} \rangle \Rightarrow \langle S_6, \sigma_{1986} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7940} = \frac{\mathcal{T}_{7941}}{\langle S_{13}, \sigma_{1985} \rangle \Rightarrow \langle S_{15}, \sigma_{1986} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7942} = \frac{\langle S_6, \sigma_{1986} \rangle \Rightarrow \langle S_8, \sigma_{1986} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1986} \rangle \Rightarrow \langle S_9, \sigma_{1986} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7944} = \langle S_8, \sigma_{1986} \rangle \Rightarrow \langle y := y + 1, \sigma_{1986} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7943} = \frac{\mathcal{T}_{7944}}{\langle S_9, \sigma_{1986} \rangle \Rightarrow \langle S_{10}, \sigma_{1986} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7945} = \frac{\langle y := y + 1, \sigma_{1986} \rangle \Rightarrow \sigma_{1987} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1986} \rangle \Rightarrow \langle S_{11}, \sigma_{1987} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7946} = \langle S_{11}, \sigma_{1987} \rangle \Rightarrow \langle S_{12}, \sigma_{1987} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7947} = \langle S_{12}, \sigma_{1987} \rangle \Rightarrow \langle S_{13}, \sigma_{1987} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7949} = \frac{\langle x := x + 1, \sigma_{1987} \rangle \Rightarrow \sigma_{1988} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1987} \rangle \Rightarrow \langle S_6, \sigma_{1988} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7948} = \frac{\mathcal{T}_{7949}}{\langle S_{13}, \sigma_{1987} \rangle \Rightarrow \langle S_{15}, \sigma_{1988} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7950} = \frac{\langle S_6, \sigma_{1988} \rangle \Rightarrow \langle S_8, \sigma_{1988} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1988} \rangle \Rightarrow \langle S_9, \sigma_{1988} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7952} = \langle S_8, \sigma_{1988} \rangle \Rightarrow \langle y := y + 1, \sigma_{1988} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7951} = \frac{\mathcal{T}_{7952}}{\langle S_9, \sigma_{1988} \rangle \Rightarrow \langle S_{10}, \sigma_{1988} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7953} = \frac{\langle y := y + 1, \sigma_{1988} \rangle \Rightarrow \sigma_{1989} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1988} \rangle \Rightarrow \langle S_{11}, \sigma_{1989} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7954} = \langle S_{11}, \sigma_{1989} \rangle \Rightarrow \langle S_{12}, \sigma_{1989} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7955} = \langle S_{12}, \sigma_{1989} \rangle \Rightarrow \langle S_{13}, \sigma_{1989} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7957} = \frac{\langle x := x + 1, \sigma_{1989} \rangle \Rightarrow \sigma_{1990} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1989} \rangle \Rightarrow \langle S_6, \sigma_{1990} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7956} = \frac{\mathcal{T}_{7957}}{\langle S_{13}, \sigma_{1989} \rangle \Rightarrow \langle S_{15}, \sigma_{1990} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7958} = \frac{\langle S_6, \sigma_{1990} \rangle \Rightarrow \langle S_8, \sigma_{1990} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1990} \rangle \Rightarrow \langle S_9, \sigma_{1990} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7960} = \langle S_8, \sigma_{1990} \rangle \Rightarrow \langle y := y + 1, \sigma_{1990} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7959} = \frac{\mathcal{T}_{7960}}{\langle S_9, \sigma_{1990} \rangle \Rightarrow \langle S_{10}, \sigma_{1990} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7961} = \frac{\langle y := y + 1, \sigma_{1990} \rangle \Rightarrow \sigma_{1991} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1990} \rangle \Rightarrow \langle S_{11}, \sigma_{1991} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7962} = \langle S_{11}, \sigma_{1991} \rangle \Rightarrow \langle S_{12}, \sigma_{1991} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7963} = \langle S_{12}, \sigma_{1991} \rangle \Rightarrow \langle S_{13}, \sigma_{1991} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7965} = \frac{\langle x := x + 1, \sigma_{1991} \rangle \Rightarrow \sigma_{1992} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1991} \rangle \Rightarrow \langle S_6, \sigma_{1992} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7964} = \frac{\mathcal{T}_{7965}}{\langle S_{13}, \sigma_{1991} \rangle \Rightarrow \langle S_{15}, \sigma_{1992} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7966} = \frac{\langle S_6, \sigma_{1992} \rangle \Rightarrow \langle S_8, \sigma_{1992} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1992} \rangle \Rightarrow \langle S_9, \sigma_{1992} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7968} = \langle S_8, \sigma_{1992} \rangle \Rightarrow \langle y := y + 1, \sigma_{1992} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7967} = \frac{\mathcal{T}_{7968}}{\langle S_9, \sigma_{1992} \rangle \Rightarrow \langle S_{10}, \sigma_{1992} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7969} = \frac{\langle y := y + 1, \sigma_{1992} \rangle \Rightarrow \sigma_{1993} \text{ [ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1992} \rangle \Rightarrow \langle S_{11}, \sigma_{1993} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7970} = \langle S_{11}, \sigma_{1993} \rangle \Rightarrow \langle S_{12}, \sigma_{1993} \rangle \text{ [for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7971} = \langle S_{12}, \sigma_{1993} \rangle \Rightarrow \langle S_{13}, \sigma_{1993} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7973} = \frac{\langle x := x + 1, \sigma_{1993} \rangle \Rightarrow \sigma_{1994} \text{ [ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1993} \rangle \Rightarrow \langle S_6, \sigma_{1994} \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7972} = \frac{\mathcal{T}_{7973}}{\langle S_{13}, \sigma_{1993} \rangle \Rightarrow \langle S_{15}, \sigma_{1994} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7974} = \frac{\langle S_6, \sigma_{1994} \rangle \Rightarrow \langle S_8, \sigma_{1994} \rangle \text{ [assert}_{\text{sos}}]}{\langle S_7, \sigma_{1994} \rangle \Rightarrow \langle S_9, \sigma_{1994} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7976} = \langle S_8, \sigma_{1994} \rangle \Rightarrow \langle y := y + 1, \sigma_{1994} \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{7975} = \frac{\mathcal{T}_{7976}}{\langle S_9, \sigma_{1994} \rangle \Rightarrow \langle S_{10}, \sigma_{1994} \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7977} = \frac{\langle y := y + 1, \sigma_{1994} \rangle \Rightarrow \sigma_{1995} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1994} \rangle \Rightarrow \langle S_{11}, \sigma_{1995} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7978} = \langle S_{11}, \sigma_{1995} \rangle \Rightarrow \langle S_{12}, \sigma_{1995} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7979} = \langle S_{12}, \sigma_{1995} \rangle \Rightarrow \langle S_{13}, \sigma_{1995} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7981} = \frac{\langle x := x + 1, \sigma_{1995} \rangle \Rightarrow \sigma_{1996} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1995} \rangle \Rightarrow \langle S_6, \sigma_{1996} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7980} = \frac{\mathcal{T}_{7981}}{\langle S_{13}, \sigma_{1995} \rangle \Rightarrow \langle S_{15}, \sigma_{1996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7982} = \frac{\langle S_6, \sigma_{1996} \rangle \Rightarrow \langle S_8, \sigma_{1996} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1996} \rangle \Rightarrow \langle S_9, \sigma_{1996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7984} = \langle S_8, \sigma_{1996} \rangle \Rightarrow \langle y := y + 1, \sigma_{1996} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7983} = \frac{\mathcal{T}_{7984}}{\langle S_9, \sigma_{1996} \rangle \Rightarrow \langle S_{10}, \sigma_{1996} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7985} = \frac{\langle y := y + 1, \sigma_{1996} \rangle \Rightarrow \sigma_{1997} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1996} \rangle \Rightarrow \langle S_{11}, \sigma_{1997} \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_{7986} = \langle S_{11}, \sigma_{1997} \rangle \Rightarrow \langle S_{12}, \sigma_{1997} \rangle [\text{for}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7987} = \langle S_{12}, \sigma_{1997} \rangle \Rightarrow \langle S_{13}, \sigma_{1997} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_{7989} = \frac{\langle x := x + 1, \sigma_{1997} \rangle \Rightarrow \sigma_{1998} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1997} \rangle \Rightarrow \langle S_6, \sigma_{1998} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7988} = \frac{\mathcal{T}_{7989}}{\langle S_{13}, \sigma_{1997} \rangle \Rightarrow \langle S_{15}, \sigma_{1998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7990} = \frac{\langle S_6, \sigma_{1998} \rangle \Rightarrow \langle S_8, \sigma_{1998} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{1998} \rangle \Rightarrow \langle S_9, \sigma_{1998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$\Downarrow$

$$\mathcal{T}_{7992} = \langle S_8, \sigma_{1998} \rangle \Rightarrow \langle y := y + 1, \sigma_{1998} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7991} = \frac{\mathcal{T}_{7992}}{\langle S_9, \sigma_{1998} \rangle \Rightarrow \langle S_{10}, \sigma_{1998} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7993} = \frac{\langle y := y + 1, \sigma_{1998} \rangle \Rightarrow \sigma_{1999} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{1998} \rangle \Rightarrow \langle S_{11}, \sigma_{1999} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{7994} = \langle S_{11}, \sigma_{1999} \rangle \Rightarrow \langle S_{12}, \sigma_{1999} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7995} = \langle S_{12}, \sigma_{1999} \rangle \Rightarrow \langle S_{13}, \sigma_{1999} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{7997} = \frac{\langle x := x + 1, \sigma_{1999} \rangle \Rightarrow \sigma_{2000} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{1999} \rangle \Rightarrow \langle S_6, \sigma_{2000} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{7996} = \frac{\mathcal{T}_{7997}}{\langle S_{13}, \sigma_{1999} \rangle \Rightarrow \langle S_{15}, \sigma_{2000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{7998} = \frac{\langle S_6, \sigma_{2000} \rangle \Rightarrow \langle S_8, \sigma_{2000} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{2000} \rangle \Rightarrow \langle S_9, \sigma_{2000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{8000} = \langle S_8, \sigma_{2000} \rangle \Rightarrow \langle y := y + 1, \sigma_{2000} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{7999} = \frac{\mathcal{T}_{8000}}{\langle S_9, \sigma_{2000} \rangle \Rightarrow \langle S_{10}, \sigma_{2000} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{8001} = \frac{\langle y := y + 1, \sigma_{2000} \rangle \Rightarrow \sigma_{2001} [\text{ass}_{\text{sos}}]}{\langle S_{10}, \sigma_{2000} \rangle \Rightarrow \langle S_{11}, \sigma_{2001} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\Downarrow$$

$$\mathcal{T}_{8002} = \langle S_{11}, \sigma_{2001} \rangle \Rightarrow \langle S_{12}, \sigma_{2001} \rangle [\text{for}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{8003} = \langle S_{12}, \sigma_{2001} \rangle \Rightarrow \langle S_{13}, \sigma_{2001} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\Downarrow$$

$$\mathcal{T}_{8005} = \frac{\langle x := x + 1, \sigma_{2001} \rangle \Rightarrow \sigma_{2002} [\text{ass}_{\text{sos}}]}{\langle S_{14}, \sigma_{2001} \rangle \Rightarrow \langle S_6, \sigma_{2002} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{8004} = \frac{\mathcal{T}_{8005}}{\langle S_{13}, \sigma_{2001} \rangle \Rightarrow \langle S_7, \sigma_{2002} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{8006} = \frac{\langle S_6, \sigma_{2002} \rangle \Rightarrow \langle S_8, \sigma_{2002} \rangle [\text{assert}_{\text{sos}}]}{\langle S_7, \sigma_{2002} \rangle \Rightarrow \langle S_9, \sigma_{2002} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\Downarrow$$

$$\mathcal{T}_{8007} = \frac{\langle S_8, \sigma_{2002} \rangle \Rightarrow \langle \mathbf{abort}, \sigma_{2002} \rangle [\text{if} - \text{ff}_{\text{sos}}]}{\langle S_9, \sigma_{2002} \rangle \Rightarrow \langle S_{17}, \sigma_{2002} \rangle} [\text{comp}_{\text{sos}}^1]$$

↓

$$\mathcal{T}_{8008} = \frac{\langle \mathbf{abort}, \sigma_{2002} \rangle \Rightarrow \perp [\mathbf{abort}_{\text{sos}}]}{\langle S_{17}, \sigma_{2002} \rangle \Rightarrow \perp} [\text{comp} - \mathbf{abort}_{\text{sos}}]$$

Sentencias Abreviadas

$S_1 = (y := 0 ; \text{for } x := 0 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_2 = \text{for } x := 0 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1$

$S_3 = \text{if } 0 \leq 1000 \text{ then } x := 0 ; \text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1 \text{ else skip}$

$S_4 = (x := 0 ; \text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_5 = (x := 0 ; \text{assert } x \leq 999 \text{ before } y := y + 1)$

$S_6 = \text{assert } x \leq 999 \text{ before } y := y + 1$

$S_7 = (\text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_8 = \text{if } x \leq 999 \text{ then } y := y + 1 \text{ else abort}$

$S_9 = (\text{if } x \leq 999 \text{ then } y := y + 1 \text{ else abort} ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_{10} = (y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_{11} = \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1$

$S_{12} = \text{if } x + 1 \leq 1000 \text{ then } x := x + 1 ; \text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1 \text{ else skip}$

$S_{13} = (x := x + 1 ; \text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_{14} = (x := x + 1 ; \text{assert } x \leq 999 \text{ before } y := y + 1)$

$S_{15} = (\text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_{16} = (x := x + 1 ; \text{assert } x \leq 999 \text{ before } y := y + 1 ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

$S_{17} = (\text{abort} ; \text{for } x := x + 1 \text{ to } 1000 \text{ do assert } x \leq 999 \text{ before } y := y + 1)$

Estados

$$\begin{aligned}
\sigma_{181} &= \{x \mapsto 89, y \mapsto 90\} \\
\sigma_{182} &= \{x \mapsto 90, y \mapsto 90\} \\
\sigma_{183} &= \{x \mapsto 90, y \mapsto 91\} \\
\sigma_{184} &= \{x \mapsto 91, y \mapsto 91\} \\
\sigma_{185} &= \{x \mapsto 91, y \mapsto 92\} \\
\sigma_{186} &= \{x \mapsto 92, y \mapsto 92\} \\
\sigma_{187} &= \{x \mapsto 92, y \mapsto 93\} \\
\sigma_{188} &= \{x \mapsto 93, y \mapsto 93\} \\
\sigma_{189} &= \{x \mapsto 93, y \mapsto 94\} \\
\sigma_{190} &= \{x \mapsto 94, y \mapsto 94\} \\
\sigma_{191} &= \{x \mapsto 94, y \mapsto 95\} \\
\sigma_{192} &= \{x \mapsto 95, y \mapsto 95\} \\
\sigma_{193} &= \{x \mapsto 95, y \mapsto 96\} \\
\sigma_{194} &= \{x \mapsto 96, y \mapsto 96\} \\
\sigma_{195} &= \{x \mapsto 96, y \mapsto 97\} \\
\sigma_{196} &= \{x \mapsto 97, y \mapsto 97\} \\
\sigma_{197} &= \{x \mapsto 97, y \mapsto 98\} \\
\sigma_{198} &= \{x \mapsto 98, y \mapsto 98\} \\
\sigma_{199} &= \{x \mapsto 98, y \mapsto 99\} \\
\sigma_{200} &= \{x \mapsto 99, y \mapsto 99\} \\
\sigma_{201} &= \{x \mapsto 99, y \mapsto 100\} \\
\sigma_{202} &= \{x \mapsto 100, y \mapsto 100\} \\
\sigma_{203} &= \{x \mapsto 100, y \mapsto 101\} \\
\sigma_{204} &= \{x \mapsto 101, y \mapsto 101\} \\
\sigma_{205} &= \{x \mapsto 101, y \mapsto 102\} \\
\sigma_{206} &= \{x \mapsto 102, y \mapsto 102\} \\
\sigma_{207} &= \{x \mapsto 102, y \mapsto 103\} \\
\sigma_{208} &= \{x \mapsto 103, y \mapsto 103\} \\
\sigma_{209} &= \{x \mapsto 103, y \mapsto 104\} \\
\sigma_{210} &= \{x \mapsto 104, y \mapsto 104\} \\
\sigma_{211} &= \{x \mapsto 104, y \mapsto 105\} \\
\sigma_{212} &= \{x \mapsto 105, y \mapsto 105\} \\
\sigma_{213} &= \{x \mapsto 105, y \mapsto 106\} \\
\sigma_{214} &= \{x \mapsto 106, y \mapsto 106\} \\
\sigma_{215} &= \{x \mapsto 106, y \mapsto 107\} \\
\sigma_{216} &= \{x \mapsto 107, y \mapsto 107\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{289} &= \{x \mapsto 143, y \mapsto 144\} \\
\sigma_{290} &= \{x \mapsto 144, y \mapsto 144\} \\
\sigma_{291} &= \{x \mapsto 144, y \mapsto 145\} \\
\sigma_{292} &= \{x \mapsto 145, y \mapsto 145\} \\
\sigma_{293} &= \{x \mapsto 145, y \mapsto 146\} \\
\sigma_{294} &= \{x \mapsto 146, y \mapsto 146\} \\
\sigma_{295} &= \{x \mapsto 146, y \mapsto 147\} \\
\sigma_{296} &= \{x \mapsto 147, y \mapsto 147\} \\
\sigma_{297} &= \{x \mapsto 147, y \mapsto 148\} \\
\sigma_{298} &= \{x \mapsto 148, y \mapsto 148\} \\
\sigma_{299} &= \{x \mapsto 148, y \mapsto 149\} \\
\sigma_{300} &= \{x \mapsto 149, y \mapsto 149\} \\
\sigma_{301} &= \{x \mapsto 149, y \mapsto 150\} \\
\sigma_{302} &= \{x \mapsto 150, y \mapsto 150\} \\
\sigma_{303} &= \{x \mapsto 150, y \mapsto 151\} \\
\sigma_{304} &= \{x \mapsto 151, y \mapsto 151\} \\
\sigma_{305} &= \{x \mapsto 151, y \mapsto 152\} \\
\sigma_{306} &= \{x \mapsto 152, y \mapsto 152\} \\
\sigma_{307} &= \{x \mapsto 152, y \mapsto 153\} \\
\sigma_{308} &= \{x \mapsto 153, y \mapsto 153\} \\
\sigma_{309} &= \{x \mapsto 153, y \mapsto 154\} \\
\sigma_{310} &= \{x \mapsto 154, y \mapsto 154\} \\
\sigma_{311} &= \{x \mapsto 154, y \mapsto 155\} \\
\sigma_{312} &= \{x \mapsto 155, y \mapsto 155\} \\
\sigma_{313} &= \{x \mapsto 155, y \mapsto 156\} \\
\sigma_{314} &= \{x \mapsto 156, y \mapsto 156\} \\
\sigma_{315} &= \{x \mapsto 156, y \mapsto 157\} \\
\sigma_{316} &= \{x \mapsto 157, y \mapsto 157\} \\
\sigma_{317} &= \{x \mapsto 157, y \mapsto 158\} \\
\sigma_{318} &= \{x \mapsto 158, y \mapsto 158\} \\
\sigma_{319} &= \{x \mapsto 158, y \mapsto 159\} \\
\sigma_{320} &= \{x \mapsto 159, y \mapsto 159\} \\
\sigma_{321} &= \{x \mapsto 159, y \mapsto 160\} \\
\sigma_{322} &= \{x \mapsto 160, y \mapsto 160\} \\
\sigma_{323} &= \{x \mapsto 160, y \mapsto 161\} \\
\sigma_{324} &= \{x \mapsto 161, y \mapsto 161\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{397} &= \{x \mapsto 197, y \mapsto 198\} \\
\sigma_{398} &= \{x \mapsto 198, y \mapsto 198\} \\
\sigma_{399} &= \{x \mapsto 198, y \mapsto 199\} \\
\sigma_{400} &= \{x \mapsto 199, y \mapsto 199\} \\
\sigma_{401} &= \{x \mapsto 199, y \mapsto 200\} \\
\sigma_{402} &= \{x \mapsto 200, y \mapsto 200\} \\
\sigma_{403} &= \{x \mapsto 200, y \mapsto 201\} \\
\sigma_{404} &= \{x \mapsto 201, y \mapsto 201\} \\
\sigma_{405} &= \{x \mapsto 201, y \mapsto 202\} \\
\sigma_{406} &= \{x \mapsto 202, y \mapsto 202\} \\
\sigma_{407} &= \{x \mapsto 202, y \mapsto 203\} \\
\sigma_{408} &= \{x \mapsto 203, y \mapsto 203\} \\
\sigma_{409} &= \{x \mapsto 203, y \mapsto 204\} \\
\sigma_{410} &= \{x \mapsto 204, y \mapsto 204\} \\
\sigma_{411} &= \{x \mapsto 204, y \mapsto 205\} \\
\sigma_{412} &= \{x \mapsto 205, y \mapsto 205\} \\
\sigma_{413} &= \{x \mapsto 205, y \mapsto 206\} \\
\sigma_{414} &= \{x \mapsto 206, y \mapsto 206\} \\
\sigma_{415} &= \{x \mapsto 206, y \mapsto 207\} \\
\sigma_{416} &= \{x \mapsto 207, y \mapsto 207\} \\
\sigma_{417} &= \{x \mapsto 207, y \mapsto 208\} \\
\sigma_{418} &= \{x \mapsto 208, y \mapsto 208\} \\
\sigma_{419} &= \{x \mapsto 208, y \mapsto 209\} \\
\sigma_{420} &= \{x \mapsto 209, y \mapsto 209\} \\
\sigma_{421} &= \{x \mapsto 209, y \mapsto 210\} \\
\sigma_{422} &= \{x \mapsto 210, y \mapsto 210\} \\
\sigma_{423} &= \{x \mapsto 210, y \mapsto 211\} \\
\sigma_{424} &= \{x \mapsto 211, y \mapsto 211\} \\
\sigma_{425} &= \{x \mapsto 211, y \mapsto 212\} \\
\sigma_{426} &= \{x \mapsto 212, y \mapsto 212\} \\
\sigma_{427} &= \{x \mapsto 212, y \mapsto 213\} \\
\sigma_{428} &= \{x \mapsto 213, y \mapsto 213\} \\
\sigma_{429} &= \{x \mapsto 213, y \mapsto 214\} \\
\sigma_{430} &= \{x \mapsto 214, y \mapsto 214\} \\
\sigma_{431} &= \{x \mapsto 214, y \mapsto 215\} \\
\sigma_{432} &= \{x \mapsto 215, y \mapsto 215\}
\end{aligned}$$

$$\begin{aligned}\sigma_{505} &= \{x \mapsto 251, y \mapsto 252\} \\ \sigma_{506} &= \{x \mapsto 252, y \mapsto 252\} \\ \sigma_{507} &= \{x \mapsto 252, y \mapsto 253\} \\ \sigma_{508} &= \{x \mapsto 253, y \mapsto 253\} \\ \sigma_{509} &= \{x \mapsto 253, y \mapsto 254\} \\ \sigma_{510} &= \{x \mapsto 254, y \mapsto 254\} \\ \sigma_{511} &= \{x \mapsto 254, y \mapsto 255\} \\ \sigma_{512} &= \{x \mapsto 255, y \mapsto 255\} \\ \sigma_{513} &= \{x \mapsto 255, y \mapsto 256\} \\ \sigma_{514} &= \{x \mapsto 256, y \mapsto 256\} \\ \sigma_{515} &= \{x \mapsto 256, y \mapsto 257\} \\ \sigma_{516} &= \{x \mapsto 257, y \mapsto 257\} \\ \sigma_{517} &= \{x \mapsto 257, y \mapsto 258\} \\ \sigma_{518} &= \{x \mapsto 258, y \mapsto 258\} \\ \sigma_{519} &= \{x \mapsto 258, y \mapsto 259\} \\ \sigma_{520} &= \{x \mapsto 259, y \mapsto 259\} \\ \sigma_{521} &= \{x \mapsto 259, y \mapsto 260\} \\ \sigma_{522} &= \{x \mapsto 260, y \mapsto 260\} \\ \sigma_{523} &= \{x \mapsto 260, y \mapsto 261\} \\ \sigma_{524} &= \{x \mapsto 261, y \mapsto 261\} \\ \sigma_{525} &= \{x \mapsto 261, y \mapsto 262\} \\ \sigma_{526} &= \{x \mapsto 262, y \mapsto 262\} \\ \sigma_{527} &= \{x \mapsto 262, y \mapsto 263\} \\ \sigma_{528} &= \{x \mapsto 263, y \mapsto 263\} \\ \sigma_{529} &= \{x \mapsto 263, y \mapsto 264\} \\ \sigma_{530} &= \{x \mapsto 264, y \mapsto 264\} \\ \sigma_{531} &= \{x \mapsto 264, y \mapsto 265\} \\ \sigma_{532} &= \{x \mapsto 265, y \mapsto 265\} \\ \sigma_{533} &= \{x \mapsto 265, y \mapsto 266\} \\ \sigma_{534} &= \{x \mapsto 266, y \mapsto 266\} \\ \sigma_{535} &= \{x \mapsto 266, y \mapsto 267\} \\ \sigma_{536} &= \{x \mapsto 267, y \mapsto 267\} \\ \sigma_{537} &= \{x \mapsto 267, y \mapsto 268\} \\ \sigma_{538} &= \{x \mapsto 268, y \mapsto 268\} \\ \sigma_{539} &= \{x \mapsto 268, y \mapsto 269\} \\ \sigma_{540} &= \{x \mapsto 269, y \mapsto 269\}\end{aligned}$$

$$\begin{aligned}
\sigma_{613} &= \{x \mapsto 305, y \mapsto 306\} \\
\sigma_{614} &= \{x \mapsto 306, y \mapsto 306\} \\
\sigma_{615} &= \{x \mapsto 306, y \mapsto 307\} \\
\sigma_{616} &= \{x \mapsto 307, y \mapsto 307\} \\
\sigma_{617} &= \{x \mapsto 307, y \mapsto 308\} \\
\sigma_{618} &= \{x \mapsto 308, y \mapsto 308\} \\
\sigma_{619} &= \{x \mapsto 308, y \mapsto 309\} \\
\sigma_{620} &= \{x \mapsto 309, y \mapsto 309\} \\
\sigma_{621} &= \{x \mapsto 309, y \mapsto 310\} \\
\sigma_{622} &= \{x \mapsto 310, y \mapsto 310\} \\
\sigma_{623} &= \{x \mapsto 310, y \mapsto 311\} \\
\sigma_{624} &= \{x \mapsto 311, y \mapsto 311\} \\
\sigma_{625} &= \{x \mapsto 311, y \mapsto 312\} \\
\sigma_{626} &= \{x \mapsto 312, y \mapsto 312\} \\
\sigma_{627} &= \{x \mapsto 312, y \mapsto 313\} \\
\sigma_{628} &= \{x \mapsto 313, y \mapsto 313\} \\
\sigma_{629} &= \{x \mapsto 313, y \mapsto 314\} \\
\sigma_{630} &= \{x \mapsto 314, y \mapsto 314\} \\
\sigma_{631} &= \{x \mapsto 314, y \mapsto 315\} \\
\sigma_{632} &= \{x \mapsto 315, y \mapsto 315\} \\
\sigma_{633} &= \{x \mapsto 315, y \mapsto 316\} \\
\sigma_{634} &= \{x \mapsto 316, y \mapsto 316\} \\
\sigma_{635} &= \{x \mapsto 316, y \mapsto 317\} \\
\sigma_{636} &= \{x \mapsto 317, y \mapsto 317\} \\
\sigma_{637} &= \{x \mapsto 317, y \mapsto 318\} \\
\sigma_{638} &= \{x \mapsto 318, y \mapsto 318\} \\
\sigma_{639} &= \{x \mapsto 318, y \mapsto 319\} \\
\sigma_{640} &= \{x \mapsto 319, y \mapsto 319\} \\
\sigma_{641} &= \{x \mapsto 319, y \mapsto 320\} \\
\sigma_{642} &= \{x \mapsto 320, y \mapsto 320\} \\
\sigma_{643} &= \{x \mapsto 320, y \mapsto 321\} \\
\sigma_{644} &= \{x \mapsto 321, y \mapsto 321\} \\
\sigma_{645} &= \{x \mapsto 321, y \mapsto 322\} \\
\sigma_{646} &= \{x \mapsto 322, y \mapsto 322\} \\
\sigma_{647} &= \{x \mapsto 322, y \mapsto 323\} \\
\sigma_{648} &= \{x \mapsto 323, y \mapsto 323\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{721} &= \{x \mapsto 359, y \mapsto 360\} \\
\sigma_{722} &= \{x \mapsto 360, y \mapsto 360\} \\
\sigma_{723} &= \{x \mapsto 360, y \mapsto 361\} \\
\sigma_{724} &= \{x \mapsto 361, y \mapsto 361\} \\
\sigma_{725} &= \{x \mapsto 361, y \mapsto 362\} \\
\sigma_{726} &= \{x \mapsto 362, y \mapsto 362\} \\
\sigma_{727} &= \{x \mapsto 362, y \mapsto 363\} \\
\sigma_{728} &= \{x \mapsto 363, y \mapsto 363\} \\
\sigma_{729} &= \{x \mapsto 363, y \mapsto 364\} \\
\sigma_{730} &= \{x \mapsto 364, y \mapsto 364\} \\
\sigma_{731} &= \{x \mapsto 364, y \mapsto 365\} \\
\sigma_{732} &= \{x \mapsto 365, y \mapsto 365\} \\
\sigma_{733} &= \{x \mapsto 365, y \mapsto 366\} \\
\sigma_{734} &= \{x \mapsto 366, y \mapsto 366\} \\
\sigma_{735} &= \{x \mapsto 366, y \mapsto 367\} \\
\sigma_{736} &= \{x \mapsto 367, y \mapsto 367\} \\
\sigma_{737} &= \{x \mapsto 367, y \mapsto 368\} \\
\sigma_{738} &= \{x \mapsto 368, y \mapsto 368\} \\
\sigma_{739} &= \{x \mapsto 368, y \mapsto 369\} \\
\sigma_{740} &= \{x \mapsto 369, y \mapsto 369\} \\
\sigma_{741} &= \{x \mapsto 369, y \mapsto 370\} \\
\sigma_{742} &= \{x \mapsto 370, y \mapsto 370\} \\
\sigma_{743} &= \{x \mapsto 370, y \mapsto 371\} \\
\sigma_{744} &= \{x \mapsto 371, y \mapsto 371\} \\
\sigma_{745} &= \{x \mapsto 371, y \mapsto 372\} \\
\sigma_{746} &= \{x \mapsto 372, y \mapsto 372\} \\
\sigma_{747} &= \{x \mapsto 372, y \mapsto 373\} \\
\sigma_{748} &= \{x \mapsto 373, y \mapsto 373\} \\
\sigma_{749} &= \{x \mapsto 373, y \mapsto 374\} \\
\sigma_{750} &= \{x \mapsto 374, y \mapsto 374\} \\
\sigma_{751} &= \{x \mapsto 374, y \mapsto 375\} \\
\sigma_{752} &= \{x \mapsto 375, y \mapsto 375\} \\
\sigma_{753} &= \{x \mapsto 375, y \mapsto 376\} \\
\sigma_{754} &= \{x \mapsto 376, y \mapsto 376\} \\
\sigma_{755} &= \{x \mapsto 376, y \mapsto 377\} \\
\sigma_{756} &= \{x \mapsto 377, y \mapsto 377\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{829} &= \{x \mapsto 413, y \mapsto 414\} \\
\sigma_{830} &= \{x \mapsto 414, y \mapsto 414\} \\
\sigma_{831} &= \{x \mapsto 414, y \mapsto 415\} \\
\sigma_{832} &= \{x \mapsto 415, y \mapsto 415\} \\
\sigma_{833} &= \{x \mapsto 415, y \mapsto 416\} \\
\sigma_{834} &= \{x \mapsto 416, y \mapsto 416\} \\
\sigma_{835} &= \{x \mapsto 416, y \mapsto 417\} \\
\sigma_{836} &= \{x \mapsto 417, y \mapsto 417\} \\
\sigma_{837} &= \{x \mapsto 417, y \mapsto 418\} \\
\sigma_{838} &= \{x \mapsto 418, y \mapsto 418\} \\
\sigma_{839} &= \{x \mapsto 418, y \mapsto 419\} \\
\sigma_{840} &= \{x \mapsto 419, y \mapsto 419\} \\
\sigma_{841} &= \{x \mapsto 419, y \mapsto 420\} \\
\sigma_{842} &= \{x \mapsto 420, y \mapsto 420\} \\
\sigma_{843} &= \{x \mapsto 420, y \mapsto 421\} \\
\sigma_{844} &= \{x \mapsto 421, y \mapsto 421\} \\
\sigma_{845} &= \{x \mapsto 421, y \mapsto 422\} \\
\sigma_{846} &= \{x \mapsto 422, y \mapsto 422\} \\
\sigma_{847} &= \{x \mapsto 422, y \mapsto 423\} \\
\sigma_{848} &= \{x \mapsto 423, y \mapsto 423\} \\
\sigma_{849} &= \{x \mapsto 423, y \mapsto 424\} \\
\sigma_{850} &= \{x \mapsto 424, y \mapsto 424\} \\
\sigma_{851} &= \{x \mapsto 424, y \mapsto 425\} \\
\sigma_{852} &= \{x \mapsto 425, y \mapsto 425\} \\
\sigma_{853} &= \{x \mapsto 425, y \mapsto 426\} \\
\sigma_{854} &= \{x \mapsto 426, y \mapsto 426\} \\
\sigma_{855} &= \{x \mapsto 426, y \mapsto 427\} \\
\sigma_{856} &= \{x \mapsto 427, y \mapsto 427\} \\
\sigma_{857} &= \{x \mapsto 427, y \mapsto 428\} \\
\sigma_{858} &= \{x \mapsto 428, y \mapsto 428\} \\
\sigma_{859} &= \{x \mapsto 428, y \mapsto 429\} \\
\sigma_{860} &= \{x \mapsto 429, y \mapsto 429\} \\
\sigma_{861} &= \{x \mapsto 429, y \mapsto 430\} \\
\sigma_{862} &= \{x \mapsto 430, y \mapsto 430\} \\
\sigma_{863} &= \{x \mapsto 430, y \mapsto 431\} \\
\sigma_{864} &= \{x \mapsto 431, y \mapsto 431\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{937} &= \{x \mapsto 467, y \mapsto 468\} \\
\sigma_{938} &= \{x \mapsto 468, y \mapsto 468\} \\
\sigma_{939} &= \{x \mapsto 468, y \mapsto 469\} \\
\sigma_{940} &= \{x \mapsto 469, y \mapsto 469\} \\
\sigma_{941} &= \{x \mapsto 469, y \mapsto 470\} \\
\sigma_{942} &= \{x \mapsto 470, y \mapsto 470\} \\
\sigma_{943} &= \{x \mapsto 470, y \mapsto 471\} \\
\sigma_{944} &= \{x \mapsto 471, y \mapsto 471\} \\
\sigma_{945} &= \{x \mapsto 471, y \mapsto 472\} \\
\sigma_{946} &= \{x \mapsto 472, y \mapsto 472\} \\
\sigma_{947} &= \{x \mapsto 472, y \mapsto 473\} \\
\sigma_{948} &= \{x \mapsto 473, y \mapsto 473\} \\
\sigma_{949} &= \{x \mapsto 473, y \mapsto 474\} \\
\sigma_{950} &= \{x \mapsto 474, y \mapsto 474\} \\
\sigma_{951} &= \{x \mapsto 474, y \mapsto 475\} \\
\sigma_{952} &= \{x \mapsto 475, y \mapsto 475\} \\
\sigma_{953} &= \{x \mapsto 475, y \mapsto 476\} \\
\sigma_{954} &= \{x \mapsto 476, y \mapsto 476\} \\
\sigma_{955} &= \{x \mapsto 476, y \mapsto 477\} \\
\sigma_{956} &= \{x \mapsto 477, y \mapsto 477\} \\
\sigma_{957} &= \{x \mapsto 477, y \mapsto 478\} \\
\sigma_{958} &= \{x \mapsto 478, y \mapsto 478\} \\
\sigma_{959} &= \{x \mapsto 478, y \mapsto 479\} \\
\sigma_{960} &= \{x \mapsto 479, y \mapsto 479\} \\
\sigma_{961} &= \{x \mapsto 479, y \mapsto 480\} \\
\sigma_{962} &= \{x \mapsto 480, y \mapsto 480\} \\
\sigma_{963} &= \{x \mapsto 480, y \mapsto 481\} \\
\sigma_{964} &= \{x \mapsto 481, y \mapsto 481\} \\
\sigma_{965} &= \{x \mapsto 481, y \mapsto 482\} \\
\sigma_{966} &= \{x \mapsto 482, y \mapsto 482\} \\
\sigma_{967} &= \{x \mapsto 482, y \mapsto 483\} \\
\sigma_{968} &= \{x \mapsto 483, y \mapsto 483\} \\
\sigma_{969} &= \{x \mapsto 483, y \mapsto 484\} \\
\sigma_{970} &= \{x \mapsto 484, y \mapsto 484\} \\
\sigma_{971} &= \{x \mapsto 484, y \mapsto 485\} \\
\sigma_{972} &= \{x \mapsto 485, y \mapsto 485\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1045} &= \{x \mapsto 521, y \mapsto 522\} \\
\sigma_{1046} &= \{x \mapsto 522, y \mapsto 522\} \\
\sigma_{1047} &= \{x \mapsto 522, y \mapsto 523\} \\
\sigma_{1048} &= \{x \mapsto 523, y \mapsto 523\} \\
\sigma_{1049} &= \{x \mapsto 523, y \mapsto 524\} \\
\sigma_{1050} &= \{x \mapsto 524, y \mapsto 524\} \\
\sigma_{1051} &= \{x \mapsto 524, y \mapsto 525\} \\
\sigma_{1052} &= \{x \mapsto 525, y \mapsto 525\} \\
\sigma_{1053} &= \{x \mapsto 525, y \mapsto 526\} \\
\sigma_{1054} &= \{x \mapsto 526, y \mapsto 526\} \\
\sigma_{1055} &= \{x \mapsto 526, y \mapsto 527\} \\
\sigma_{1056} &= \{x \mapsto 527, y \mapsto 527\} \\
\sigma_{1057} &= \{x \mapsto 527, y \mapsto 528\} \\
\sigma_{1058} &= \{x \mapsto 528, y \mapsto 528\} \\
\sigma_{1059} &= \{x \mapsto 528, y \mapsto 529\} \\
\sigma_{1060} &= \{x \mapsto 529, y \mapsto 529\} \\
\sigma_{1061} &= \{x \mapsto 529, y \mapsto 530\} \\
\sigma_{1062} &= \{x \mapsto 530, y \mapsto 530\} \\
\sigma_{1063} &= \{x \mapsto 530, y \mapsto 531\} \\
\sigma_{1064} &= \{x \mapsto 531, y \mapsto 531\} \\
\sigma_{1065} &= \{x \mapsto 531, y \mapsto 532\} \\
\sigma_{1066} &= \{x \mapsto 532, y \mapsto 532\} \\
\sigma_{1067} &= \{x \mapsto 532, y \mapsto 533\} \\
\sigma_{1068} &= \{x \mapsto 533, y \mapsto 533\} \\
\sigma_{1069} &= \{x \mapsto 533, y \mapsto 534\} \\
\sigma_{1070} &= \{x \mapsto 534, y \mapsto 534\} \\
\sigma_{1071} &= \{x \mapsto 534, y \mapsto 535\} \\
\sigma_{1072} &= \{x \mapsto 535, y \mapsto 535\} \\
\sigma_{1073} &= \{x \mapsto 535, y \mapsto 536\} \\
\sigma_{1074} &= \{x \mapsto 536, y \mapsto 536\} \\
\sigma_{1075} &= \{x \mapsto 536, y \mapsto 537\} \\
\sigma_{1076} &= \{x \mapsto 537, y \mapsto 537\} \\
\sigma_{1077} &= \{x \mapsto 537, y \mapsto 538\} \\
\sigma_{1078} &= \{x \mapsto 538, y \mapsto 538\} \\
\sigma_{1079} &= \{x \mapsto 538, y \mapsto 539\} \\
\sigma_{1080} &= \{x \mapsto 539, y \mapsto 539\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1261} &= \{x \mapsto 629, y \mapsto 630\} \\
\sigma_{1262} &= \{x \mapsto 630, y \mapsto 630\} \\
\sigma_{1263} &= \{x \mapsto 630, y \mapsto 631\} \\
\sigma_{1264} &= \{x \mapsto 631, y \mapsto 631\} \\
\sigma_{1265} &= \{x \mapsto 631, y \mapsto 632\} \\
\sigma_{1266} &= \{x \mapsto 632, y \mapsto 632\} \\
\sigma_{1267} &= \{x \mapsto 632, y \mapsto 633\} \\
\sigma_{1268} &= \{x \mapsto 633, y \mapsto 633\} \\
\sigma_{1269} &= \{x \mapsto 633, y \mapsto 634\} \\
\sigma_{1270} &= \{x \mapsto 634, y \mapsto 634\} \\
\sigma_{1271} &= \{x \mapsto 634, y \mapsto 635\} \\
\sigma_{1272} &= \{x \mapsto 635, y \mapsto 635\} \\
\sigma_{1273} &= \{x \mapsto 635, y \mapsto 636\} \\
\sigma_{1274} &= \{x \mapsto 636, y \mapsto 636\} \\
\sigma_{1275} &= \{x \mapsto 636, y \mapsto 637\} \\
\sigma_{1276} &= \{x \mapsto 637, y \mapsto 637\} \\
\sigma_{1277} &= \{x \mapsto 637, y \mapsto 638\} \\
\sigma_{1278} &= \{x \mapsto 638, y \mapsto 638\} \\
\sigma_{1279} &= \{x \mapsto 638, y \mapsto 639\} \\
\sigma_{1280} &= \{x \mapsto 639, y \mapsto 639\} \\
\sigma_{1281} &= \{x \mapsto 639, y \mapsto 640\} \\
\sigma_{1282} &= \{x \mapsto 640, y \mapsto 640\} \\
\sigma_{1283} &= \{x \mapsto 640, y \mapsto 641\} \\
\sigma_{1284} &= \{x \mapsto 641, y \mapsto 641\} \\
\sigma_{1285} &= \{x \mapsto 641, y \mapsto 642\} \\
\sigma_{1286} &= \{x \mapsto 642, y \mapsto 642\} \\
\sigma_{1287} &= \{x \mapsto 642, y \mapsto 643\} \\
\sigma_{1288} &= \{x \mapsto 643, y \mapsto 643\} \\
\sigma_{1289} &= \{x \mapsto 643, y \mapsto 644\} \\
\sigma_{1290} &= \{x \mapsto 644, y \mapsto 644\} \\
\sigma_{1291} &= \{x \mapsto 644, y \mapsto 645\} \\
\sigma_{1292} &= \{x \mapsto 645, y \mapsto 645\} \\
\sigma_{1293} &= \{x \mapsto 645, y \mapsto 646\} \\
\sigma_{1294} &= \{x \mapsto 646, y \mapsto 646\} \\
\sigma_{1295} &= \{x \mapsto 646, y \mapsto 647\} \\
\sigma_{1296} &= \{x \mapsto 647, y \mapsto 647\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1369} &= \{x \mapsto 683, y \mapsto 684\} \\
\sigma_{1370} &= \{x \mapsto 684, y \mapsto 684\} \\
\sigma_{1371} &= \{x \mapsto 684, y \mapsto 685\} \\
\sigma_{1372} &= \{x \mapsto 685, y \mapsto 685\} \\
\sigma_{1373} &= \{x \mapsto 685, y \mapsto 686\} \\
\sigma_{1374} &= \{x \mapsto 686, y \mapsto 686\} \\
\sigma_{1375} &= \{x \mapsto 686, y \mapsto 687\} \\
\sigma_{1376} &= \{x \mapsto 687, y \mapsto 687\} \\
\sigma_{1377} &= \{x \mapsto 687, y \mapsto 688\} \\
\sigma_{1378} &= \{x \mapsto 688, y \mapsto 688\} \\
\sigma_{1379} &= \{x \mapsto 688, y \mapsto 689\} \\
\sigma_{1380} &= \{x \mapsto 689, y \mapsto 689\} \\
\sigma_{1381} &= \{x \mapsto 689, y \mapsto 690\} \\
\sigma_{1382} &= \{x \mapsto 690, y \mapsto 690\} \\
\sigma_{1383} &= \{x \mapsto 690, y \mapsto 691\} \\
\sigma_{1384} &= \{x \mapsto 691, y \mapsto 691\} \\
\sigma_{1385} &= \{x \mapsto 691, y \mapsto 692\} \\
\sigma_{1386} &= \{x \mapsto 692, y \mapsto 692\} \\
\sigma_{1387} &= \{x \mapsto 692, y \mapsto 693\} \\
\sigma_{1388} &= \{x \mapsto 693, y \mapsto 693\} \\
\sigma_{1389} &= \{x \mapsto 693, y \mapsto 694\} \\
\sigma_{1390} &= \{x \mapsto 694, y \mapsto 694\} \\
\sigma_{1391} &= \{x \mapsto 694, y \mapsto 695\} \\
\sigma_{1392} &= \{x \mapsto 695, y \mapsto 695\} \\
\sigma_{1393} &= \{x \mapsto 695, y \mapsto 696\} \\
\sigma_{1394} &= \{x \mapsto 696, y \mapsto 696\} \\
\sigma_{1395} &= \{x \mapsto 696, y \mapsto 697\} \\
\sigma_{1396} &= \{x \mapsto 697, y \mapsto 697\} \\
\sigma_{1397} &= \{x \mapsto 697, y \mapsto 698\} \\
\sigma_{1398} &= \{x \mapsto 698, y \mapsto 698\} \\
\sigma_{1399} &= \{x \mapsto 698, y \mapsto 699\} \\
\sigma_{1400} &= \{x \mapsto 699, y \mapsto 699\} \\
\sigma_{1401} &= \{x \mapsto 699, y \mapsto 700\} \\
\sigma_{1402} &= \{x \mapsto 700, y \mapsto 700\} \\
\sigma_{1403} &= \{x \mapsto 700, y \mapsto 701\} \\
\sigma_{1404} &= \{x \mapsto 701, y \mapsto 701\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1477} &= \{x \mapsto 737, y \mapsto 738\} \\
\sigma_{1478} &= \{x \mapsto 738, y \mapsto 738\} \\
\sigma_{1479} &= \{x \mapsto 738, y \mapsto 739\} \\
\sigma_{1480} &= \{x \mapsto 739, y \mapsto 739\} \\
\sigma_{1481} &= \{x \mapsto 739, y \mapsto 740\} \\
\sigma_{1482} &= \{x \mapsto 740, y \mapsto 740\} \\
\sigma_{1483} &= \{x \mapsto 740, y \mapsto 741\} \\
\sigma_{1484} &= \{x \mapsto 741, y \mapsto 741\} \\
\sigma_{1485} &= \{x \mapsto 741, y \mapsto 742\} \\
\sigma_{1486} &= \{x \mapsto 742, y \mapsto 742\} \\
\sigma_{1487} &= \{x \mapsto 742, y \mapsto 743\} \\
\sigma_{1488} &= \{x \mapsto 743, y \mapsto 743\} \\
\sigma_{1489} &= \{x \mapsto 743, y \mapsto 744\} \\
\sigma_{1490} &= \{x \mapsto 744, y \mapsto 744\} \\
\sigma_{1491} &= \{x \mapsto 744, y \mapsto 745\} \\
\sigma_{1492} &= \{x \mapsto 745, y \mapsto 745\} \\
\sigma_{1493} &= \{x \mapsto 745, y \mapsto 746\} \\
\sigma_{1494} &= \{x \mapsto 746, y \mapsto 746\} \\
\sigma_{1495} &= \{x \mapsto 746, y \mapsto 747\} \\
\sigma_{1496} &= \{x \mapsto 747, y \mapsto 747\} \\
\sigma_{1497} &= \{x \mapsto 747, y \mapsto 748\} \\
\sigma_{1498} &= \{x \mapsto 748, y \mapsto 748\} \\
\sigma_{1499} &= \{x \mapsto 748, y \mapsto 749\} \\
\sigma_{1500} &= \{x \mapsto 749, y \mapsto 749\} \\
\sigma_{1501} &= \{x \mapsto 749, y \mapsto 750\} \\
\sigma_{1502} &= \{x \mapsto 750, y \mapsto 750\} \\
\sigma_{1503} &= \{x \mapsto 750, y \mapsto 751\} \\
\sigma_{1504} &= \{x \mapsto 751, y \mapsto 751\} \\
\sigma_{1505} &= \{x \mapsto 751, y \mapsto 752\} \\
\sigma_{1506} &= \{x \mapsto 752, y \mapsto 752\} \\
\sigma_{1507} &= \{x \mapsto 752, y \mapsto 753\} \\
\sigma_{1508} &= \{x \mapsto 753, y \mapsto 753\} \\
\sigma_{1509} &= \{x \mapsto 753, y \mapsto 754\} \\
\sigma_{1510} &= \{x \mapsto 754, y \mapsto 754\} \\
\sigma_{1511} &= \{x \mapsto 754, y \mapsto 755\} \\
\sigma_{1512} &= \{x \mapsto 755, y \mapsto 755\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1585} &= \{x \mapsto 791, y \mapsto 792\} \\
\sigma_{1586} &= \{x \mapsto 792, y \mapsto 792\} \\
\sigma_{1587} &= \{x \mapsto 792, y \mapsto 793\} \\
\sigma_{1588} &= \{x \mapsto 793, y \mapsto 793\} \\
\sigma_{1589} &= \{x \mapsto 793, y \mapsto 794\} \\
\sigma_{1590} &= \{x \mapsto 794, y \mapsto 794\} \\
\sigma_{1591} &= \{x \mapsto 794, y \mapsto 795\} \\
\sigma_{1592} &= \{x \mapsto 795, y \mapsto 795\} \\
\sigma_{1593} &= \{x \mapsto 795, y \mapsto 796\} \\
\sigma_{1594} &= \{x \mapsto 796, y \mapsto 796\} \\
\sigma_{1595} &= \{x \mapsto 796, y \mapsto 797\} \\
\sigma_{1596} &= \{x \mapsto 797, y \mapsto 797\} \\
\sigma_{1597} &= \{x \mapsto 797, y \mapsto 798\} \\
\sigma_{1598} &= \{x \mapsto 798, y \mapsto 798\} \\
\sigma_{1599} &= \{x \mapsto 798, y \mapsto 799\} \\
\sigma_{1600} &= \{x \mapsto 799, y \mapsto 799\} \\
\sigma_{1601} &= \{x \mapsto 799, y \mapsto 800\} \\
\sigma_{1602} &= \{x \mapsto 800, y \mapsto 800\} \\
\sigma_{1603} &= \{x \mapsto 800, y \mapsto 801\} \\
\sigma_{1604} &= \{x \mapsto 801, y \mapsto 801\} \\
\sigma_{1605} &= \{x \mapsto 801, y \mapsto 802\} \\
\sigma_{1606} &= \{x \mapsto 802, y \mapsto 802\} \\
\sigma_{1607} &= \{x \mapsto 802, y \mapsto 803\} \\
\sigma_{1608} &= \{x \mapsto 803, y \mapsto 803\} \\
\sigma_{1609} &= \{x \mapsto 803, y \mapsto 804\} \\
\sigma_{1610} &= \{x \mapsto 804, y \mapsto 804\} \\
\sigma_{1611} &= \{x \mapsto 804, y \mapsto 805\} \\
\sigma_{1612} &= \{x \mapsto 805, y \mapsto 805\} \\
\sigma_{1613} &= \{x \mapsto 805, y \mapsto 806\} \\
\sigma_{1614} &= \{x \mapsto 806, y \mapsto 806\} \\
\sigma_{1615} &= \{x \mapsto 806, y \mapsto 807\} \\
\sigma_{1616} &= \{x \mapsto 807, y \mapsto 807\} \\
\sigma_{1617} &= \{x \mapsto 807, y \mapsto 808\} \\
\sigma_{1618} &= \{x \mapsto 808, y \mapsto 808\} \\
\sigma_{1619} &= \{x \mapsto 808, y \mapsto 809\} \\
\sigma_{1620} &= \{x \mapsto 809, y \mapsto 809\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1801} &= \{x \mapsto 899, y \mapsto 900\} \\
\sigma_{1802} &= \{x \mapsto 900, y \mapsto 900\} \\
\sigma_{1803} &= \{x \mapsto 900, y \mapsto 901\} \\
\sigma_{1804} &= \{x \mapsto 901, y \mapsto 901\} \\
\sigma_{1805} &= \{x \mapsto 901, y \mapsto 902\} \\
\sigma_{1806} &= \{x \mapsto 902, y \mapsto 902\} \\
\sigma_{1807} &= \{x \mapsto 902, y \mapsto 903\} \\
\sigma_{1808} &= \{x \mapsto 903, y \mapsto 903\} \\
\sigma_{1809} &= \{x \mapsto 903, y \mapsto 904\} \\
\sigma_{1810} &= \{x \mapsto 904, y \mapsto 904\} \\
\sigma_{1811} &= \{x \mapsto 904, y \mapsto 905\} \\
\sigma_{1812} &= \{x \mapsto 905, y \mapsto 905\} \\
\sigma_{1813} &= \{x \mapsto 905, y \mapsto 906\} \\
\sigma_{1814} &= \{x \mapsto 906, y \mapsto 906\} \\
\sigma_{1815} &= \{x \mapsto 906, y \mapsto 907\} \\
\sigma_{1816} &= \{x \mapsto 907, y \mapsto 907\} \\
\sigma_{1817} &= \{x \mapsto 907, y \mapsto 908\} \\
\sigma_{1818} &= \{x \mapsto 908, y \mapsto 908\} \\
\sigma_{1819} &= \{x \mapsto 908, y \mapsto 909\} \\
\sigma_{1820} &= \{x \mapsto 909, y \mapsto 909\} \\
\sigma_{1821} &= \{x \mapsto 909, y \mapsto 910\} \\
\sigma_{1822} &= \{x \mapsto 910, y \mapsto 910\} \\
\sigma_{1823} &= \{x \mapsto 910, y \mapsto 911\} \\
\sigma_{1824} &= \{x \mapsto 911, y \mapsto 911\} \\
\sigma_{1825} &= \{x \mapsto 911, y \mapsto 912\} \\
\sigma_{1826} &= \{x \mapsto 912, y \mapsto 912\} \\
\sigma_{1827} &= \{x \mapsto 912, y \mapsto 913\} \\
\sigma_{1828} &= \{x \mapsto 913, y \mapsto 913\} \\
\sigma_{1829} &= \{x \mapsto 913, y \mapsto 914\} \\
\sigma_{1830} &= \{x \mapsto 914, y \mapsto 914\} \\
\sigma_{1831} &= \{x \mapsto 914, y \mapsto 915\} \\
\sigma_{1832} &= \{x \mapsto 915, y \mapsto 915\} \\
\sigma_{1833} &= \{x \mapsto 915, y \mapsto 916\} \\
\sigma_{1834} &= \{x \mapsto 916, y \mapsto 916\} \\
\sigma_{1835} &= \{x \mapsto 916, y \mapsto 917\} \\
\sigma_{1836} &= \{x \mapsto 917, y \mapsto 917\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1945} &= \{x \mapsto 971, y \mapsto 972\} \\
\sigma_{1946} &= \{x \mapsto 972, y \mapsto 972\} \\
\sigma_{1947} &= \{x \mapsto 972, y \mapsto 973\} \\
\sigma_{1948} &= \{x \mapsto 973, y \mapsto 973\} \\
\sigma_{1949} &= \{x \mapsto 973, y \mapsto 974\} \\
\sigma_{1950} &= \{x \mapsto 974, y \mapsto 974\} \\
\sigma_{1951} &= \{x \mapsto 974, y \mapsto 975\} \\
\sigma_{1952} &= \{x \mapsto 975, y \mapsto 975\} \\
\sigma_{1953} &= \{x \mapsto 975, y \mapsto 976\} \\
\sigma_{1954} &= \{x \mapsto 976, y \mapsto 976\} \\
\sigma_{1955} &= \{x \mapsto 976, y \mapsto 977\} \\
\sigma_{1956} &= \{x \mapsto 977, y \mapsto 977\} \\
\sigma_{1957} &= \{x \mapsto 977, y \mapsto 978\} \\
\sigma_{1958} &= \{x \mapsto 978, y \mapsto 978\} \\
\sigma_{1959} &= \{x \mapsto 978, y \mapsto 979\} \\
\sigma_{1960} &= \{x \mapsto 979, y \mapsto 979\} \\
\sigma_{1961} &= \{x \mapsto 979, y \mapsto 980\} \\
\sigma_{1962} &= \{x \mapsto 980, y \mapsto 980\} \\
\sigma_{1963} &= \{x \mapsto 980, y \mapsto 981\} \\
\sigma_{1964} &= \{x \mapsto 981, y \mapsto 981\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1965} &= \{x \mapsto 981, y \mapsto 982\} \\
\sigma_{1966} &= \{x \mapsto 982, y \mapsto 982\} \\
\sigma_{1967} &= \{x \mapsto 982, y \mapsto 983\} \\
\sigma_{1968} &= \{x \mapsto 983, y \mapsto 983\} \\
\sigma_{1969} &= \{x \mapsto 983, y \mapsto 984\} \\
\sigma_{1970} &= \{x \mapsto 984, y \mapsto 984\} \\
\sigma_{1971} &= \{x \mapsto 984, y \mapsto 985\} \\
\sigma_{1972} &= \{x \mapsto 985, y \mapsto 985\} \\
\sigma_{1973} &= \{x \mapsto 985, y \mapsto 986\} \\
\sigma_{1974} &= \{x \mapsto 986, y \mapsto 986\} \\
\sigma_{1975} &= \{x \mapsto 986, y \mapsto 987\} \\
\sigma_{1976} &= \{x \mapsto 987, y \mapsto 987\} \\
\sigma_{1977} &= \{x \mapsto 987, y \mapsto 988\} \\
\sigma_{1978} &= \{x \mapsto 988, y \mapsto 988\} \\
\sigma_{1979} &= \{x \mapsto 988, y \mapsto 989\} \\
\sigma_{1980} &= \{x \mapsto 989, y \mapsto 989\} \\
\sigma_{1981} &= \{x \mapsto 989, y \mapsto 990\} \\
\sigma_{1982} &= \{x \mapsto 990, y \mapsto 990\} \\
\sigma_{1983} &= \{x \mapsto 990, y \mapsto 991\} \\
\sigma_{1984} &= \{x \mapsto 991, y \mapsto 991\}
\end{aligned}$$

$$\begin{aligned}
\sigma_{1985} &= \{x \mapsto 991, y \mapsto 992\} \\
\sigma_{1986} &= \{x \mapsto 992, y \mapsto 992\} \\
\sigma_{1987} &= \{x \mapsto 992, y \mapsto 993\} \\
\sigma_{1988} &= \{x \mapsto 993, y \mapsto 993\} \\
\sigma_{1989} &= \{x \mapsto 993, y \mapsto 994\} \\
\sigma_{1990} &= \{x \mapsto 994, y \mapsto 994\} \\
\sigma_{1991} &= \{x \mapsto 994, y \mapsto 995\} \\
\sigma_{1992} &= \{x \mapsto 995, y \mapsto 995\} \\
\sigma_{1993} &= \{x \mapsto 995, y \mapsto 996\} \\
\sigma_{1994} &= \{x \mapsto 996, y \mapsto 996\} \\
\sigma_{1995} &= \{x \mapsto 996, y \mapsto 997\} \\
\sigma_{1996} &= \{x \mapsto 997, y \mapsto 997\} \\
\sigma_{1997} &= \{x \mapsto 997, y \mapsto 998\} \\
\sigma_{1998} &= \{x \mapsto 998, y \mapsto 998\} \\
\sigma_{1999} &= \{x \mapsto 998, y \mapsto 999\} \\
\sigma_{2000} &= \{x \mapsto 999, y \mapsto 999\} \\
\sigma_{2001} &= \{x \mapsto 999, y \mapsto 1000\} \\
\sigma_{2002} &= \{x \mapsto 1000, y \mapsto 1000\}
\end{aligned}$$